Rockchip
RK3228H
Technical Reference Manual

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Chapter 1 Introduction

1.1 Overview

RK3228H is a high performance Quad-core application processor for Smart OTT/IPTV. It is a high-integration and cost efficient SOC for 4K HDR OTT/IPTV.
Quad-core Cortex-A53 is integrated with separate Neon and FPU coprocessor, also with shared L2 Cache. The Quad-core GPU supports high-resolution display and game.
Lots of high-performance interface to get very flexible solution, such as multi-channel display with HDMI2.0a, TV Encoder. TrustZone and crypto hardware are integrated for security. 32bits DDR3/DDR3L/DDR4/LPDDR3 provides high memory bandwidths.

1.2 Features

The features listed below which may or may not be present in actual product, may be subject to the third party licensing requirements. Please contact Rockchip for actual product feature configurations and licensing requirements.

1.2.1 Microprocessor

- Quad-core ARM Cortex-A53 MPCore processor, high-performance, low-power and cached application processor
- Full implementation of the ARM architecture v8-A instruction set, ARM Neon Advanced SIMD (single instruction, multiple data) support for accelerated media and signal processing computation
- ARMv8 Cryptography Extensions
- In-order pipeline with symmetric dual-issue of most instructions.
- Harvard Level 1 (L1) memory system with a Memory Management Unit (MMU).
- Level 2 (L2) memory system providing cluster memory coherency, including an L2 cache.
- Include VFP v3 hardware to support single and double-precision add, subtract, divide, multiply and accumulate, and square root operations
- SCU ensures memory coherency between the four CPUs for each cluster
- Integrated 32KB L1 instruction cache, 32KB L1 data cache with 4-way set associative
- Trustzone technology support
- One separate power domains for CPU core system to support internal power switch and externally turn on/off based on different application scenario
  - PD_A53: Cortex-A53 + Neon + FPU + L1 I/D Cache of core 1/2/3
- One isolated voltage domain to support DVFS

1.2.2 Memory Organization

- Internal on-chip memory
  - BootRom
  - Internal SRAM
- External off-chip memory①
  - DDR3/DDR3L/LPDDR3/DDR4
  - SPI Flash
  - eMMC

1.2.3 Internal Memory

- Internal BootRom
  - Size : 20KB
  - Support system boot from the following device :
    - SPI interface
    - eMMC interface
    - SDMMC interface
  - Support system code download by the following interface:
    - USB OTG interface
- Internal SRAM
  - Size : 36KB

1.2.4 External Memory or Storage device

- Dynamic Memory Interface (DDR3/DDR3L/LPDDR3/DDR4)
  - Compatible with JEDEC standard DDR3/DDR3L/LPDDR3/DDR4 SDRAM

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  - Supports 32 Bits data width, 2 ranks (chip selects), max 4GB addressing space per rank; totally addressing space is 4GB(max) also.
  - Programmable timing parameters to support DDR3/DDR3L/LPDDR3/DDR4 SDRAM from various vendor
  - Advanced command reordering and scheduling to maximize bus utilization
  - Low power modes, such as power-down and self-refresh for DDR3/DDR3L/LPDDR3/DDR4 SDRAM
  - Compensation for board delays and variable latencies through programmable pipelines
  - Programmable output and ODT impedance with dynamic PVT compensation
- eMMC Interface
  - Compatible with standard iNAND interface
  - Support MMC4.51 protocol
  - Provide eMMC boot sequence to receive boot data from external eMMC device
  - Support FIFO over-run and under-run prevention by stopping card clock automatically
  - Support CRC generation and error detection
  - Embedded clock frequency division control to provide programmable baud rate
  - Support block size from 1 to 65535Bytes
  - 8bits data bus width
  - Support up to HS200
- SD/MMC Interface
  - Compatible with SD3.0, MMC ver4.51
  - Support FIFO over-run and under-run prevention by stopping card clock automatically
  - Support CRC generation and error detection
  - Support block size from 1 to 65535Bytes
  - Data bus width is 4bits

1.2.5 System Component

- CRU (clock & reset unit)
  - Support clock gating control for individual components inside RK3228H
  - One oscillator with 24MHz clock input and 5 embedded PLLs
  - Support global soft-reset control for whole SOC, also individual soft-reset for every components
- Timer
  - 6 on-chip 64bits Timers in SoC with interrupt-based operation for non-secure operation
  - 2 on-chip 64bits Timers in SoC with interrupt-based operation for secure application
  - Provide two operation modes: free-running and user-defined count
  - Support timer work state checkable
  - Fixed 24MHz clock input
- PWM
  - Four on-chip PWMs with interrupt-based operation
  - Programmable pre-scaled operation to bus clock and then further scaled
  - Embedded 32-bit timer/counter facility
  - Support capture mode
  - Support continuous mode or one-shot mode
  - Provides reference mode and output various duty-cycle waveform
- WatchDog
  - 32 bits watchdog counter width
  - Counter clock is from apb bus clock
  - Counter counts down from a preset value to 0 to indicate the occurrence of a timeout
  - WDT can perform two types of operations when timeout occurs:
    - Generate a system reset
    - First generate an interrupt and if this is not cleared by the service routine by the time a second timeout occurs then generate a system reset
  - Programmable reset pulse length
  - Totally 16 defined-ranges of main timeout period
- Bus Architecture

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- 128-bit/64-bit/32-bit multi-layer AXI/AHB/APB composite bus architecture
- 5 embedded AXI interconnect
  - BUS interconnect with one 128-bits AXI masters, two 32-bit AXI Master, eight AHB slaves and two APB slaves
  - PERI interconnect with two 64bit AXI master, eight 32-bits AHB masters and lots of 32-bits AHB/APB slaves
  - GPU interconnect with two 128-bits AXI masters, one 32-bits APB Slave
  - Display interconnect with two 128-bits AXI master, seven 64-bits AXI masters and lots 32-bits AHB/APB slave
  - VENC interconnect with two 128-bits AXI masters, and one 32-bits AHB slaves
  - VDEC interconnect also with two 64-bits AXI master and two 32-bits AHB slaves
- Flexible different QoS solution to improve the utility of bus bandwidth

- Interrupt Controller
  - Support 3 PPI interrupt source and 128 SPI interrupt sources input from different components inside RK3228H
  - Support 16 softwre-triggered interrupts
  - Input interrupt level is fixed , only high-level sensitive
  - Two interrupt outputs (nFIQ and nIRQ)separatelyfor each Cortex-A53, both are low-level sensitive
  - Support different interrupt priority for each interrupt source, and they are always software-programmable
- DMAC
  - Micro-code programming based DMA
  - The specific instruction set provides flexibility for programming DMA transfers
  - Linked list DMA function is supported to complete scatter-gather transfer
  - Support internal instruction cache
  - Embedded DMA manager thread
  - Support data transfer types with memory-to-memory, memory-to-peripheral, peripheral-to-memory
  - Signals the occurrence of various DMA events using the interrupt output signals
  - Mapping relationship between each channel and different interrupt outputs is software-programmable
  - One embedded DMA controller for system
  - DMAC features:
    - 8 channels totally
    - 17 hardware request from peripherals
    - 2 interrupt output
    - Dual APB slave interface for register configuration, designated as secure and non-secure
    - Support trustzone technology and programmable secure state for each DMA channel
- Security system
  - Support TrustZone technology for the following components inside RK3228H
    - Cortex-A53, support security and non-security mode, switch by software
    - BUS_DMAC, support some dedicated channels work only in security mode
    - Secure OTP, only can be accessed by Cortex-A53 in security mode
    - Internal memory, part of space is addressed only in security mode, detailed size is software-programmable together with TZMA (TrustZone memory adapter)
  - Embedded encryption and decryption engine
    - Support AES 128/192/256 bits key mode, ECB/CBC/CTR chain mode, Slave/FIFO mode
    - Support DES/3DES (ECB and CBC chain mode) , 3DES (EDE/ EEE key mode), Slave/FIFO mode
    - Support SHA1/SHA256/MD5 (with hardware padding) HASH function, FIFO mode only
    - Support 160 bit Pseudo Random Number Generator (PRNG)
    - Support PKA 512/1024/2048 bit Exp Modulator

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1.2.6 Camera interface

- CIF
  - Support YCbCr422 input
  - Support Raw8bit input
  - Support CCIR656(PAL/NTSC) input
  - Support 1/2/4 channels mixed data of CCIR656 input
  - Support JPEG input
  - Support YCbCr422/420 output
  - Support UYVY/VYUY/YUYV/YVYU configurable
  - Support up to 8192x8192 resolution source
  - Support picture in picture
  - Support arbitrary size window crop
  - Support error/terminate interrupt and combined interrupt output
  - Support clk/vsync/href polarity configurable
  - Support one frame stop/ping-pong mode

1.2.7 Video CODEC

- Video Decoder
  - Real-time video decoder of
    - MPEG-1, MPEG-2, MPEG-4
    - H.263, H.264/AVC, H.265/HEVC, MVC
    - VC-1
    - AVS, AVS+
    - RM/RMVB
  - MMU Embedded
  - Supports frame timeout interrupt, frame finish interrupt and bit stream error interrupt
  - Error detection and concealment support for all video formats
  - Output data format YUV420 semi-planar, YUV400(monochrome), YUV422 is supported by H.264
  - H.264/AVC BP/MP/HP profile @ level 5.1; H.264/AVC MVC; up to 4Kx2K @ 30fps
  - H.265/HEVC Main/Main10 profile @ level 5.1 High-tier; up to 4Kx2K @ 60fps
  - MPEG-1, Main profile, up to 1080P @ 60fps
  - MPEG-2, SP@ML, MP@HL, up to 1080P @ 60fps
  - MPEG-4, ASP profile @ level 5, up to 1080P @ 60fps
  - AVS, Jizhun profile @ level 6.0; up to 1080P @ 60fps
  - AVS-P16 (AVS+), up to 1080P @ 60fps
  - H.263, up to 576P @ 60fps
  - VC-1, SP@ML, MP@HL, AP@L0-3, up to 1080P @ 60fps
  - For H.264, image cropping not supported
  - For MPEG-4, GMC (global motion compensation) not supported
  - For VC-1, upscaling and range mapping are supported in image post-processor
  - For MPEG-4 SP/H.263, using a modified H.264 in-loop filter to implement deblocking filter in post-processor unit
- H.264 Video Encoder
  - Support video encoder for H.264 UP to HP@level4.1, MVC
  - Only support I and P slices, not B slices
  - Support error resilience based on constrained intra prediction and slices
  - Input data format:
    - YCbCr 4:2:0 planar
    - YCbCr 4:2:0 semi-planar
    - YCbYCr 4:2:2
    - CbYCrY 4:2:2 interleaved
    - RGB444 and BGR444
    - RGB555 and BGR555
    - RGB565 and BGR565
    - RGB888 and BRG888
    - RGB101010 and BRG101010
  - Image size is from 96x96 to 1920x1080(Full HD)

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  - Maximum frame rate is up to 1920x1080 @ 30FPS②
- H.265 Video Encoder
  - Capable of encoding HEVC Main Profile @ L4.1 High-tier
  - Resolution and frame rate are up to 1920x1080 @ 30FPS②
  - Only support I and P slices, not B slices
  - High performance CABAC encoding
  - In-loop deblocking filtering
  - Loop filtering across slice
  - Noise reduction
  - Rate Control: VBR, CBR and ABR; ROI support

1.2.8 JPEG CODEC

- JPEG decoder
  - Input JPEG file : YCbCr 4:0:0, 4:2:0, 4:2:2, 4:4:0, 4:1:1 and 4:4:4 sampling formats
  - Output raw image : YCbCr 4:0:0, 4:2:0, 4:2:2, 4:4:0, 4:1:1 and 4:4:4 semi-planar
  - Decoder size is from 48x48 to 8176x8176(66.8Mpixels)
  - Support JPEG ROI(region of image) decode
  - Maximum data rate③ is up to 76million pixels per second
  - Embedded memory management unit(MMU)

1.2.9 Image Enhancement (IEP module)

- Image format support
  - Input data: YUV420/YUV422
  - Output data: YUV420/YUV422
  - YUV swap
  - UV SP/P
  - BT601_l/BT601_f/BT709_l/BT709_f color space conversion
  - YUV up/down sampling
- De-interlace
  - 3x5 Y motion detection matrix
  - Source width up to 1920
  - Configured high frequency de-interlace
  - I4O2 (Input 4 field, output 2 frame) /I4O1B/I4O1T/I2O1B/I2O1T mode
- Interface
  - 32bit AHB bus slave
  - 64bit AXI bus master
  - Combined interrupt output

1.2.10 Graphics Engine

RK3228H's GPU consists of quad core 3D Graphics engine and dual core 2D Graphics engine.

- Quad core 3D Graphics Engine:
  - High performance OpenGL ES1.1 and 2.0, OpenVG1.1 etc.
  - Separate vertex(geometry) and fragment(pixel) processing for maximum parallel throughput
  - Provide MMU and L2 Cache
- Dual-Core 2D Graphics Engine:
  - Source formats:
    - ABGR8888, XBGR888, ARGB8888, XRGB888
    - RGB888, RGB565
    - RGBA5551, RGBA4444
    - YUV420 planar, YUV420 semi-planar
    - YUV422 planar, YUV422 semi-planar
    - YUV 10-bit for YUV420/422 semi-planar
    - BPP8, BPP4, BPP2, BPP1
  - Destination formats:
    - ABGR8888, XBGR888, ARGB8888, XRGB888
    - RGB888, RGB565
    - RGBA5551, RGBA4444
    - YUV420 planar, YUV420 semi-planar
    - YUV422 planar, YUV422 semi-planar

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- Pixel Format conversion, BT.601/BT.709
- Max resolution: 8192x8192 source, 4096x4096 destination
- BitBLT
  - Two source BitBLT:
  - A+B=B only BitBLT, A support rotate and scale when B fixed
  - A+B=C second source (B) has same attribute with (C) plus rotation function
- Color fill with gradient fill, and pattern fill
- High-performance stretch and shrink
- Monochrome expansion for text rendering
- New comprehensive per-pixel alpha (color/alpha channel separately)
- Alpha blending modes including Java 2 Porter-Duff compositing blending rules, chroma key, pattern mask, fading
- Dither operation
- 0, 90, 180 and 270-degree rotation
- x-mirror and y-mirror rotation operation

1.2.11 Video OUT

- Display Interface
  - Support HDMI 2.0a output up to 4K@60Hz
  - TV interface: TV encoder 10bit out for DAC
  - HDMI interface:
    - 24 bit (RGB888 YCbCr444), 30 bit(RGB101010,YCbCr 420, YCbCr 444)
    - Max output resolution 4K for HDMI, 480i/576i for CVBS
  - 4 display layers:
    - Display layers of Win0, Win1, Win2, HWC
    - One background layer with programmable 24bits color
    - One video layer (win0/win1/win2)
      - RGB888, ARGB888, RGB565, YCbCr422, YCbCr420, YCbCr444,YCbCr 420 10bit，YCbCr 422 10bit，YCbCr 444 10bit
      - maximum resolution is 4096x2304, support virtual display
      - 1/8 to 8 scaling up/down engine with arbitrary non-integer ratio
      - 256 level alpha blending (pre-multiplied alpha support)
      - Support transparency color key
      - Support BG, RG, RB swap, x/y mirror
      - Support TV Encoder for PAL and NTSC
      - YCbCr2RGB(rec601-mpeg/rec601-jpeg/rec709)
      - YCbCr2RGB(BT2020)
      - RGB2YCbCr(BT601/BT709)
      - RGB2YCbCr(BT2020)
  - BT2020 and BT709/601 conversion
  - Win0,Win1,Win2 layer overlay exchangeable
  - Support replication (16bits to 24bits) and dithering (24bits to 16bits/ 18bits) operation
  - Blank and blank display
  - support HDR10
    - Win0 supports HDR2SDR and SDR2HDR
    - Other layers support SDR2HDR

1.2.12 HDMI

- Compliant HDMI 2.0a
- Support YUV420 4k x 2k @ 60fps
- Support for 4k x 2k and 3D video formats
- Support for up to 10.2bps bandwidth
- HPD input analog comparator
- Compliance HDMI compliance Test specification 1.4 and 2.x
- Support HDCP 1.4 and 2.2

1.2.13 Audio Interface

- I2S0/I2S1 with 8ch
  - supports up to 8 channels (8xTX, 8xRX)

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- Audio resolution from 16bits to 32bits
- Sample rate up to 192KHz
- Provides master and slave work mode, software configurable
- Support 3 I2S formats (normal, left-justified, right-justified)
- Support 4 PCM formats(early, late1, late2, late3)
- I2S and PCM mode cannot be used at the same time

- I2S2(PCM) with 2ch
  - Up to 2 channels (2xTX, 2xRX)
  - Audio resolution from 16bits to 32bits
  - Sample rate up to 192KHz
  - Provides master and slave work mode, software configurable
  - Support 3 I2S formats (normal , left-justified , right-justified)
  - Support 4 PCM formats(early , late1 , late2 , late3)
  - I2S and PCM cannot be used at the same time
- Audio CODEC
  - 24bit DAC
  - Support Line-out
  - Support Mono, Stereo, 5.1 HiFi channel performance
  - Integrated digital interpolation and decimation filter.
  - Sampling rate of 8kHz/12kHz/16kHz/24kHz/32kHz/44.1KHz/48KHz/96KHz
  - Optional fractional PLL available that support 6MHz to 20MHz clock input to any clock

1.2.14 Connectivity

- SDIO interface
  - Compatible with SDIO 3.0 protocol
  - 4bits data bus widths
- TS interface
  - Supports one TS input channels.
  - Supports 4 TS Input Mode: sync/valid mode in the case of serial TS input; nosync/valid mode, sync/valid, sync/burst mode in the case of parallel TS input.
  - Supports 2 TS sources: demodulators and local memory.
  - Supports 2 Built-in PTIs (Programmable Transport Interface) to process TS simultaneously, and Each PTI supports:
    - 64 PID filters.
    - TS descrambling with 16 sets of Control Word under CSA v2.0 standard, up to 104Mbps
    - 16 PES/ES filters with PTS/DTS extraction and ES start code detection.
    - 4/8 PCR extraction channels
    - 64 Section filters with CRC check, and three interrupt mode: stop per unit, full-stop, recycle mode with version number check
    - PID done and error interrupts for each channel
    - PCR/DTS/PTS extraction interrupt for each channel
  - Supports 1 PVR (Personal Video Recording) output channel.
  - 1 built-in multi-channel DMA Controller.
  - Support MMU

- Smart Card
  - support card activation and deactivation
  - support cold/warm reset
  - support Answer to Reset(ATR) response reception
  - support T0 for asynchronous half-duplex character transmission
  - support T1 for asynchronous half-duplex block transmission
  - support automatic operating voltage class selection
  - support adjustable clock rate and bit (baud) rate
  - support configurable automatic byte repetition

- GMAC 10/100/1000M Ethernet Controller
  - Supports 10/100/1000-Mbps data transfer rates with the RGMII interfaces

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- Supports 10/100-Mbps data transfer rates with the RMII interfaces
- There are 2 controllers, one is connected to internal FE PHY, the other is for external PHY device
- Supports both full-duplex and half-duplex operation
  - Supports CSMA/CD Protocol for half-duplex operation
  - Supports packet bursting and frame extension in 1000 Mbps half-duplex operation
  - Supports IEEE 802.3x flow control for full-duplex operation
  - Optional forwarding of received pause control frames to the user application in full-duplex operation
  - Back-pressure support for half-duplex operation
  - Automatic transmission of zero-quanta pause frame on de-assertion of flow control input in full-duplex operation
- Preamble and start-of-frame data (SFD) insertion in Transmit, and deletion in Receive paths
- Automatic CRC and pad generation controllable on a per-frame basis
- Options for Automatic Pad/CRC Stripping on receive frames
- Programmable Inter-Frame-Gap (40-96 bit times in steps of 8)
- Supports a variety of flexible address filtering modes
- Separate 32-bit status returned for transmission and reception packets
- Supports IEEE 802.1Q VLAN tag detection for reception frames
- Support detection of LAN wake-up frames and AMD Magic Packet frames
- Support checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame
- Support checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagram
- Comprehensive status reporting for normal operation and transfers with errors
- Automatic generation of PAUSE frame control or backpressure signal to the GMAC core based on Receive FIFO-fill (threshold configurable) level
- Handles automatic retransmission of Collision frames for transmission
- Discards frames on late collision, excessive collisions, excessive deferral and under-run conditions

- Ethernet PHY
  - Integrated IEEE 802.3/802.3u compliant 10/100Mbps Ethernet PHY
  - Supporting both full and half duplex for either 10 or 100 Mb/s data rate
  - Auto MDIX capable
  - Supports wake-on-LAN, EEE
  - 100Base-FX support
  - MII/RMII interface
  - Supports auto-negotiation
  - On-board diagnosis

- SPI Controller
  - Support serial-master and serial-slave mode, software-configurable
  - DMA-based or interrupt-based operation
  - Embedded two 32x16bits FIFO for TX and RX operation respectively
  - Support 2 chip-selects output in serial-master mode

- UART Controller
  - 3 on-chip UART controller inside RK3228H
  - DMA-based or interrupt-based operation
  - UART0/1/2Embedded two 64Bytes FIFO for TX and RX operation respectively
  - Support 5bit,6bit,7bit,8bit serial data transmit or receive
  - Standard asynchronous communication bits such as start, stop and parity
  - Support different input clock for UART operation to get up to 4Mbps or another special baud rate

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- Support non-integer clock divides for baud clock generation
- Support auto flow control mode

- I2C controller
  - 4 on-chip I2C controller in RK3228H
  - Multi-master I2C operation
  - Support 7bits and 10bits address mode
  - Software programmable clock frequency and transfer rate up to 400Kbit/s in the fast mode
  - Serial 8bits oriented and bidirectional data transfers can be made at up to 100Kbit/s in the standard mode

- GPIO
  - 4 groups of GPIO (GPIO0~GPIO3), 32 GPIOs per group in GPIO0~GPIO3, totally have 128 GPIOs
  - All of GPIOs can be used to generate interrupt to Cortex-A53
  - All of pull-up GPIOs are software-programmable for pull-up resistor or not
  - All of pull-down GPIOs are software-programmable for pull-down resistor
  - or not
  - All of GPIOs are always in input direction in default after power-on-reset

- USB Host2.0
  - Embedded 2 USB Host 2.0 interfaces
  - Compatible with USB Host2.0 specification
  - Supports high-speed(480Mbps), full-speed(12Mbps) and low-speed(1.5Mbps) mode
  - Provides 16 host mode channels
  - Support periodic out channel in host mode

- USB OTG2.0
  - Compatible with USB OTG2.0 specification
  - Supports high-speed(480Mbps), full-speed(12Mbps) and low-speed(1.5Mbps) mode
  - Support up to 9 device mode endpoints in addition to control endpoint 0
  - Support up to 6 device mode IN endpoints including control endpoint 0
  - Endpoints 1/3/5/7 can be used only as data IN endpoint
  - Endpoints 2/4/6 can be used only as data OUT endpoint
  - Endpoints 8/9 can be used as data OUT and IN endpoint
  - Provides 9 host mode channels
  -
- USB 3.0 xHCI Host Features
  - Support up to 64 devices
  - Support 1 interrupter
  - Support 1 USB2.0 port and 1 Super-Speed port
  - Concurrent USB3.0/USB2.0 traffic, up to 8.48Gbps bandwidth
  - Support standard or open-source xHCI and class driver
  - Support xHCI Debug Capability

- USB 3.0 Dual-Role Device (DRD) Features
  - Static Device operation
  - Static Host operation
  - USB3.0/USB2.0 OTG A device and B device basing on ID
  - UFP/DFP and Data Role Swap Defined in USB TypeC Specification
  - Not support USB3.0/USB2.0 OTG session request protocol(SRP), host negotiation protocol(HNP) and Role Swap Protocol(RSP)

1.2.15 Others

- Temperature Sensor(TS-ADC)
  - 10-bits ADC up to 50KS/s sampling rate
  - -40~125C temperature range and 5C temperature resolution

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- Successive Approximation ADC (SARADC)
  - GF 28nm process
  - 10-bit resolution
  - Up to 1MS/s sampling rate
  - 2 single-ended input channels
  - Current consumption: 0.5mA @ 1MS/s

- eFuse
  - One high-density electrical Fuse is integrated: 1024bits (32x32)
  - Support standby mode
  - Provide inactive mode, VP must be 0V or Floating in this mode

- OTP
  - Support 8192bits
  - Support Idle, Read, Program operation
  - Support boot function
  - Support Bist function

- Package Type
  - TFBGA395L (body: 14mm x 14mm ; ball size : 0.3mm ; ball pitch : 0.65mm)

Notes :
① *DDR3/DD4/LPDDR3 are not used simultaneously*
② *Actual maximum frame rate will depend on the clock frequency and system bus performance*
③ *Actual maximum data rate will depend on the clock frequency and JPEG compression rate*

1.3 Block Diagram

The following diagram shows the basic block diagram for RK3228H

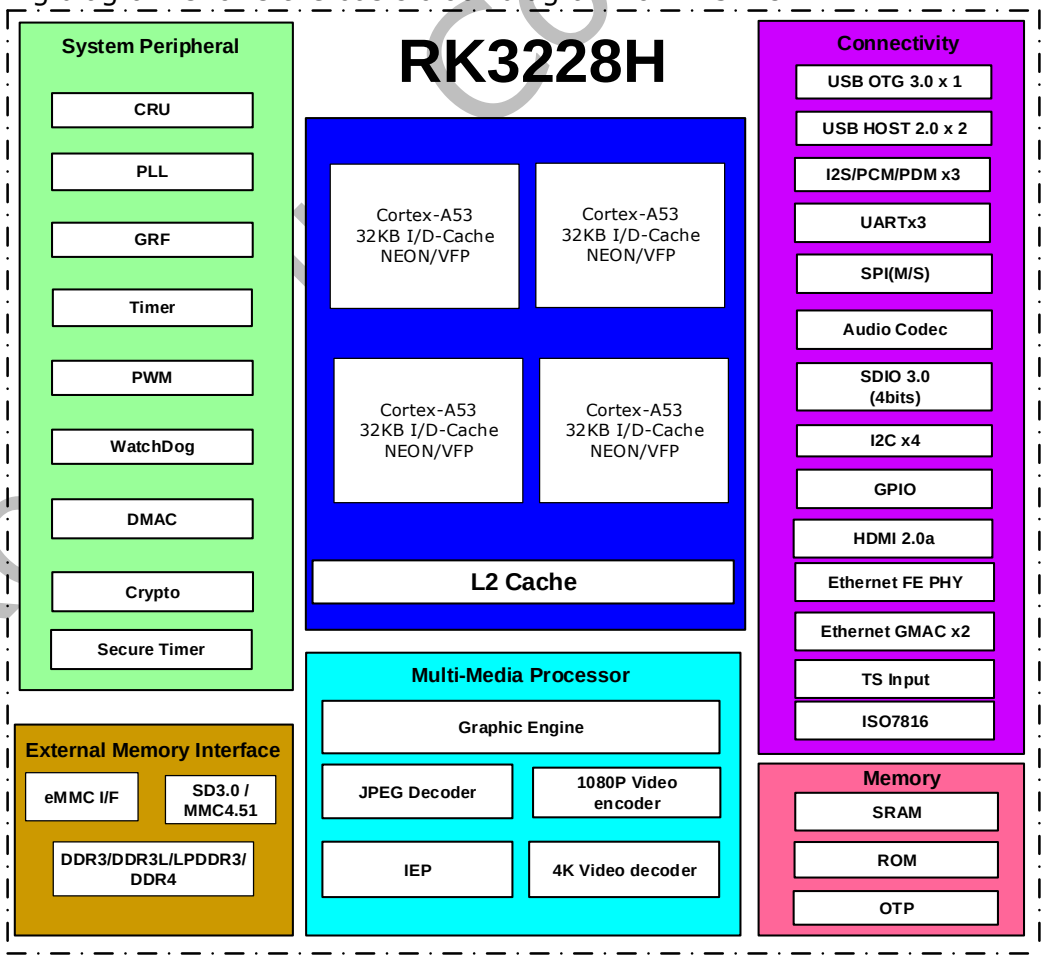

Fig. 1-1 RK3228H Block Diagram

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Chapter 2 System Overview

2.1 Address Mapping

RK3228H supports to boot from internal bootrom, which supports remap function by software programming. Remap is controlled by SGRF_SOC_CON2[10]. When remap is set to 1, the bootrom is mapped to address 0Xff080000 and internal memory is mapped to address 0Xffff0000.

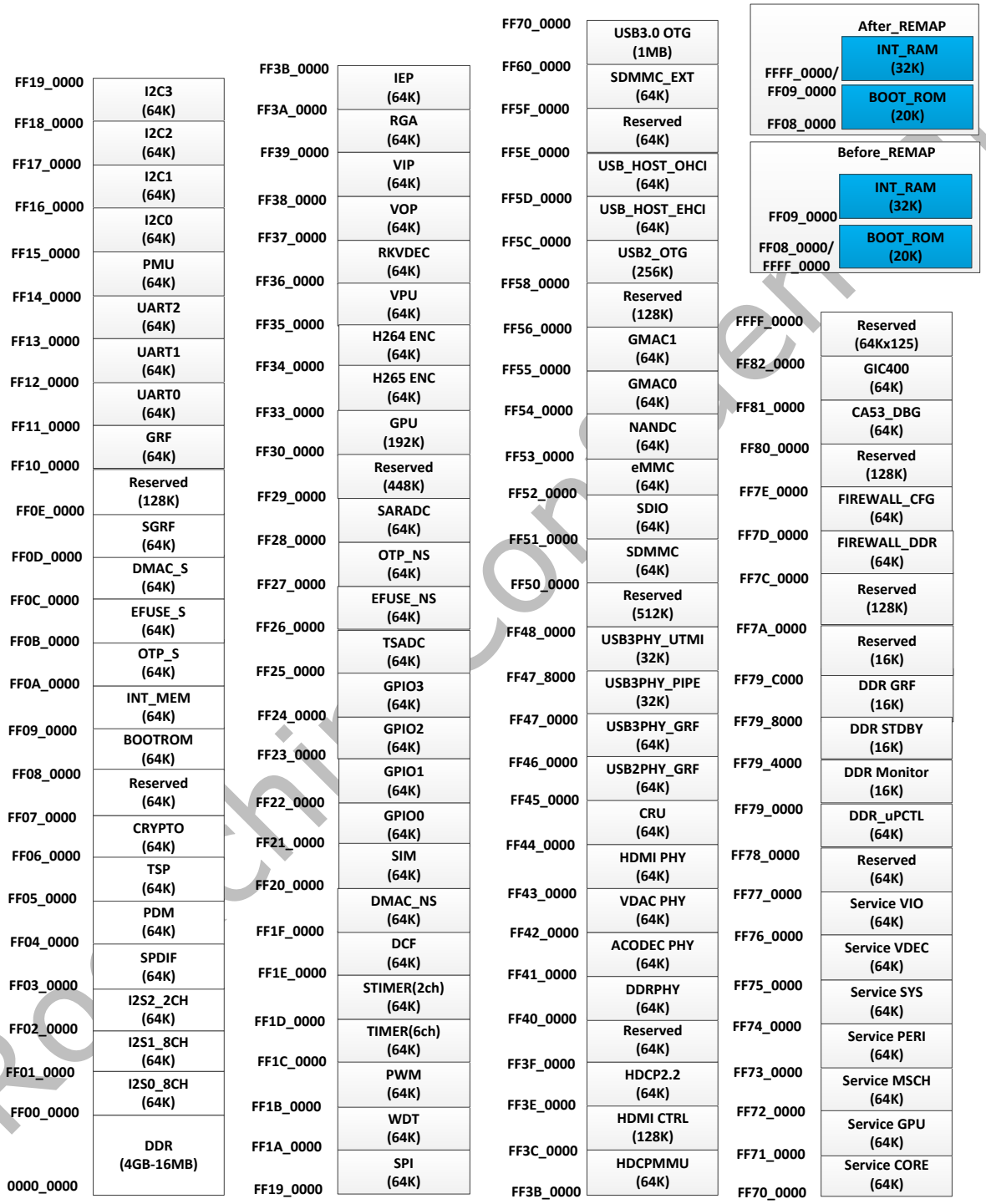

Fig. 2-1 RK3228H Address Mapping

2.2 System Boot

RK3228H provides system boot from off-chip devices such as SDMMC card, eMMC memory, serial nand or nor flash. When boot code is not ready in these devices, also provide system code download into them by USB OTG interface. All of the boot code will be stored in internal bootrom. The following is the whole boot procedure for boot code, which will be stored in

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bootrom in advance.
The following features are supports.

- Support system boot from the following device:
  - Serial Nor Flash, 1bit data width
  - eMMC Interface, 8bits data with
  - SDMMC Card, 4bits data with
- Support system code download by USB OTG

Following figure shows RK3228H boot procedure flow.

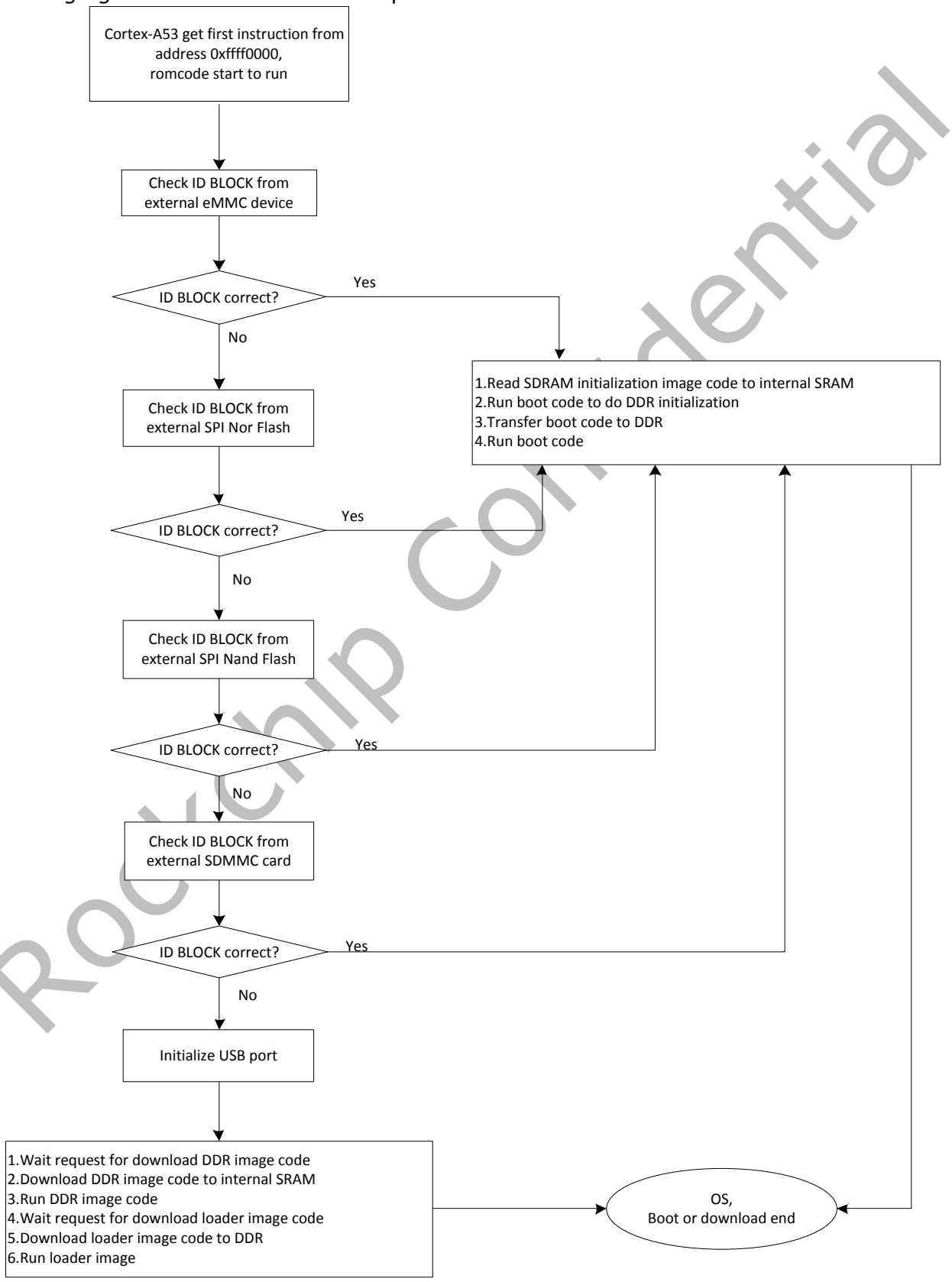

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Fig. 2-2 RK3228H boot procedure flow

2.3 System Interrupt connection

RK3228H provides an general interrupt controller(GIC) for CPU, which has 128 SPI (shared peripheral interrupts) interrupt sources and 3 PPI(Private peripheral interrupt) interrupt source and separately generates one nIRQ and one nFIQ to CPU. The triggered type for each interrupts is high level sensitive, not programmable. The detailed interrupt sources connection is in the following table. For detailed GIC setting, please refer to Chapter 9.

Table 2-1 RK3228H Interrupt connection list

IRQ Type	IRQ ID	Source(spi)	Polarity
SPI	32	(|bus_dmac_irq)	High level
	33	bus_dmac_irq_abort	High level
	34	dfi_alert_err_intr	High level
	35	upctl_awpoison_intr	High level
	36	sdmmc_ext_int	High level
	37	vop_intr_ddr	High level
	38	sdmmc_ext_dectn_in	High level
	39	rkvdec_m_dec_irq	High level
	40	upctl_arpoison_intr	High level
	41	vpu_xintdec_irq	High level
	42	sdmmc_ext_detectn_irq	High level
	43	vpu_mmu_irq	High level
	44	sdmmc_int	High level
	45	sdio_int	High level
	46	emmc_int	High level
	47	otp_int_ns	High level
	48	host0_ehci_int	High level
	49	host0_ohci_int	High level
	50	host0_arb_int	High level
	51	otp_int_s	High level
	52	ddrmon_int	High level
	53	gmac2phy_int	High level
	54	gmac2phy_pmt_int	High level
	55	otg_int	High level
	56	gmac2io_int	High level
	57	gmac2io_pmt_int	High level
	58	i2s0_8ch_intr	High level
	59	i2s1_8ch_intr	High level
	60	i2s2_2ch_intr	High level
	61	spdif_8ch_intr	High level
	62	crypto_int	High level
	63	iep_intr	High level
	64	vop_intr	High level
	65	rga_intr	High level
	66	hdcp_intr	High level
	67	hdmi_intr	High level
	68	rki2c0_int	High level
	69	rki2c1_int	High level
	70	rki2c2_int	High level
	71	rki2c3_int	High level
	72	wdt_intr	High level

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IRQ Type	IRQ ID	Source(spi)	Polarity
	73	stimer_intr0	High level
	74	stimer_intr1	High level
	75	timer_intr0	High level
	76	timer_intr1	High level
	77	timer_intr2	High level
	78	timer_intr3	High level
	79	timer_intr4	High level
	80	timer_intr5	High level
	81	spi0_intr	High level
	82	rkpwm_int	High level
	83	gpio0_intr	High level
	84	gpio1_intr	High level
	85	gpio2_intr	High level
	86	gpio3_intr	High level
	87	uart0_intr	High level
	88	uart1_intr	High level
	89	uart2_intr	High level
	90	tsadc_int	High level
	91	usbphy_otg_bvalid_irq	High level
	92	usbphy_otg_id_irq	High level
	93	usbphy_otg_linestate_irq	High level
	94	usbphy_host_linestate_irq	High level
	95	sdmmc_detectn_irq	High level
	96	cif_intr	High level
	97	sdmmc_dectn_in_flt	High level
	98	usb3otg_host_legacy_smi_interrupt	High level
	99	usb3otg_int	High level
	100	usb3otg_host_sys_err	High level
	101	usb3otg_pme_generation	High level
	102	macphy_int	High level
	103	hdmi_intr_wakeup	High level
	104	tsp_int	High level
	105	sim_int	High level
	106	rkvdec_m_mmu_irq	High level
	107	usb3phy_bvalid_irq	High level
	108	usb3phy_id_irq	High level
	109	usb3phy_linestate_irq	High level
	110	usb3phy_rxdet_irq	High level
	111	efuse_int	High level
	112	saradc_int	High level
	113	tsp_int_mmu	High level
	114	pdm_int	High level
	115	hdmiphy_irq	High level
	116	dcf_done_int	High level
	117	dcf_error_int	High level
	118	pmu_int	High level
	119	irq_gpu_gpmmu	High level
	120	irq_gpu_pp0	High level
	121	irq_gpu_ppmmu0	High level
	122	irq_gpu_gp	High level

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IRQ Type	IRQ ID	Source(spi)	Polarity
	123	irq_gpu_pp1	High level
	124	irq_gpu_ppmmu1	High level
	125	irq_gpu_pp	High level
	126	irq_gpu_pmu	High level
	127	rkvenc_h265_int	High level
	128	rkvenc_h265_mmu_int	High level
	129	rkvenc_h264_enc_int	High level
	130	rkvenc_h264_mmu_int	High level

2.4 System DMA hardware request connection

RK3228H provides one DMA controller inside the system. The trigger type for each of them is high level, not programmable. For detailed descriptions of DMAC, please refer to Chapter 8.

Table 2-2 RK3228H DMAC Hardware request connection list

Req Number	Source	Polarity
0	I2S2_2ch tx	High level
1	I2S2_2ch rx	High level
2	Uart0 tx	High level
3	Uart0 rx	High level
4	Uart1 tx	High level
5	Uart1 rx	High level
6	Uart2 tx	High level
7	Uart2 rx	High level
8	SPI tx	High level
9	SPI rx	High level
10	SPDIF	High level
11	I2S0_8ch tx	High level
12	I2S0_8ch rx	High level
13	pwm_tx	High level
14	I2S1_8ch_tx	High level
15	I2S1_8ch_rx	High level
16	pdm	High level

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Chapter 3 Clock & Reset Unit (CRU)

3.1 Overview

The CRU is an APB slave module that is designed for generating all of the internal and system clocks, resets of chip. CRU generates system clocks from PLL output clock or external clock source, and generates system reset from external power-on-reset, watchdog timer reset or software reset.

CRU supports the following features:

- Compliance to the AMBA APB interface
- Embedded 5 PLLs
- Flexible selection of clock source
- Supports the respective gating of all clocks
- Supports the respective software reset of all modules

3.2 Block Diagram

CRU comprises with:

- PLL
- Register configuration unit
- Clock generate unit
- Reset generate unit

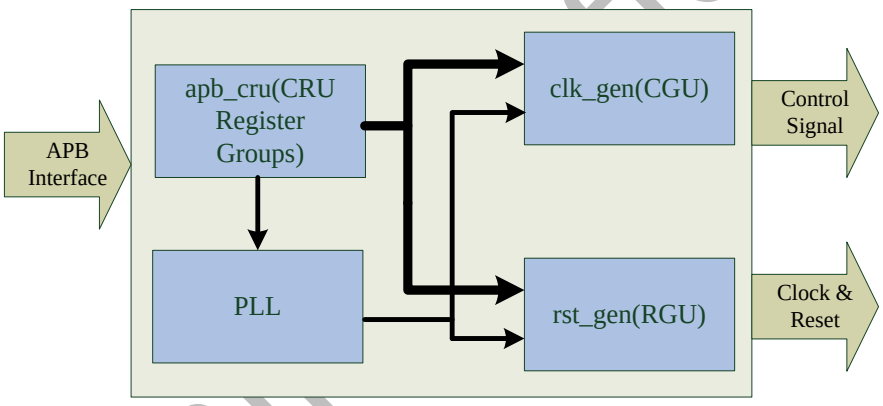

Fig. 3-1 CRU Block Diagram

3.3 System Clock Solution

The following diagrams show clock architecture (mux and divider information).

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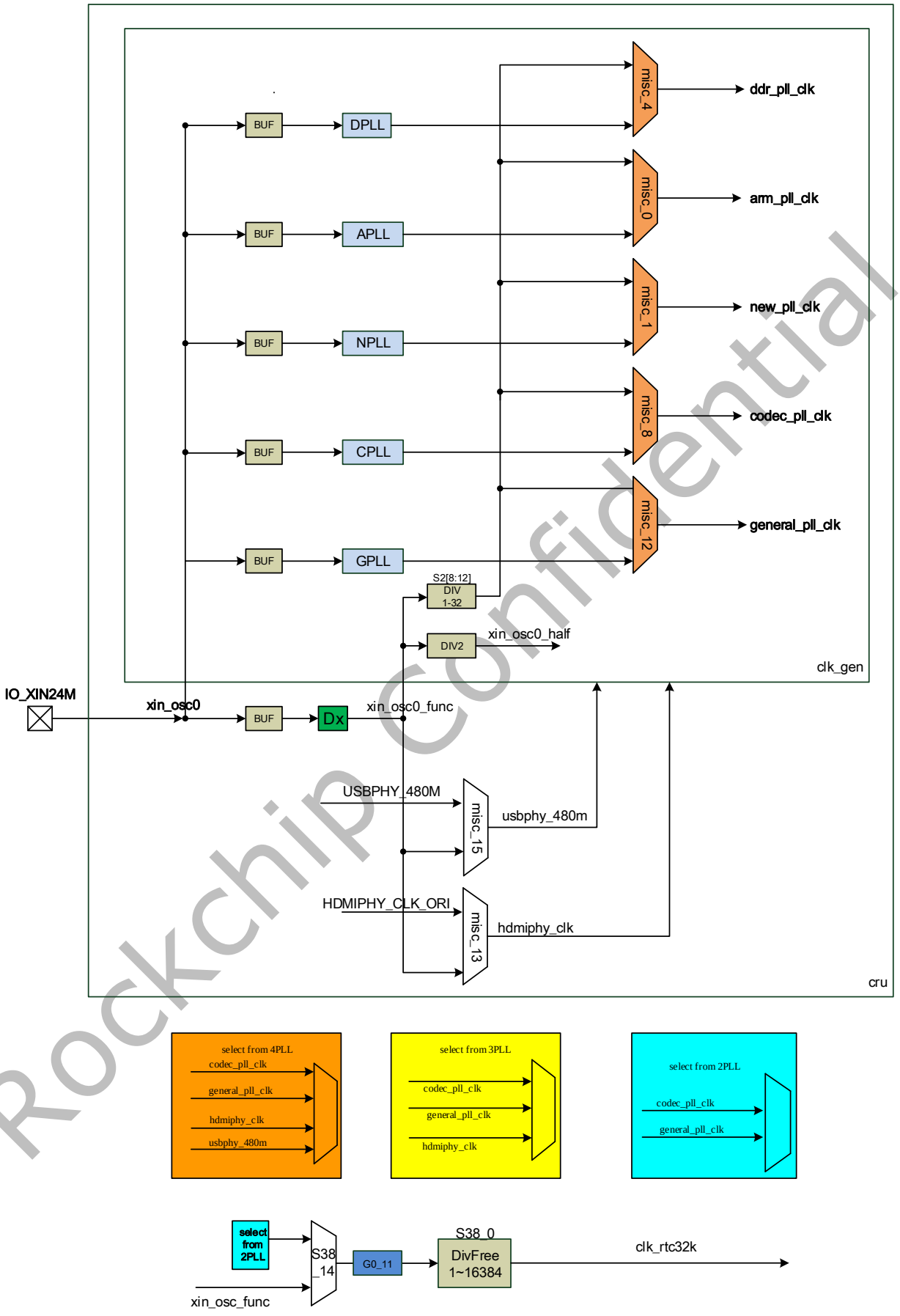

Fig. 3-2 RK3228H Clock Architecture Diagram 1

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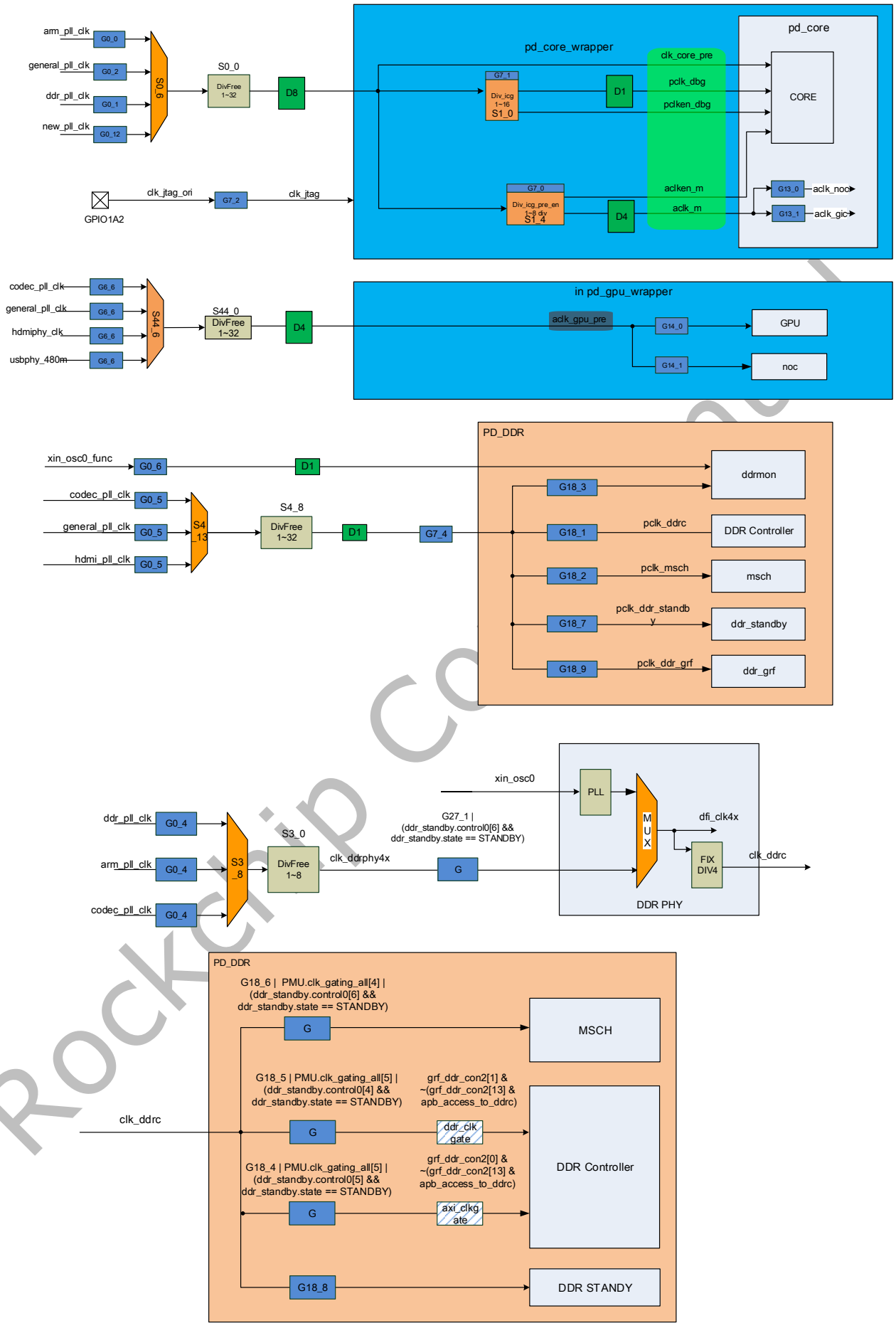

Fig. 3-3 RK3228H Clock Architecture Diagram 2

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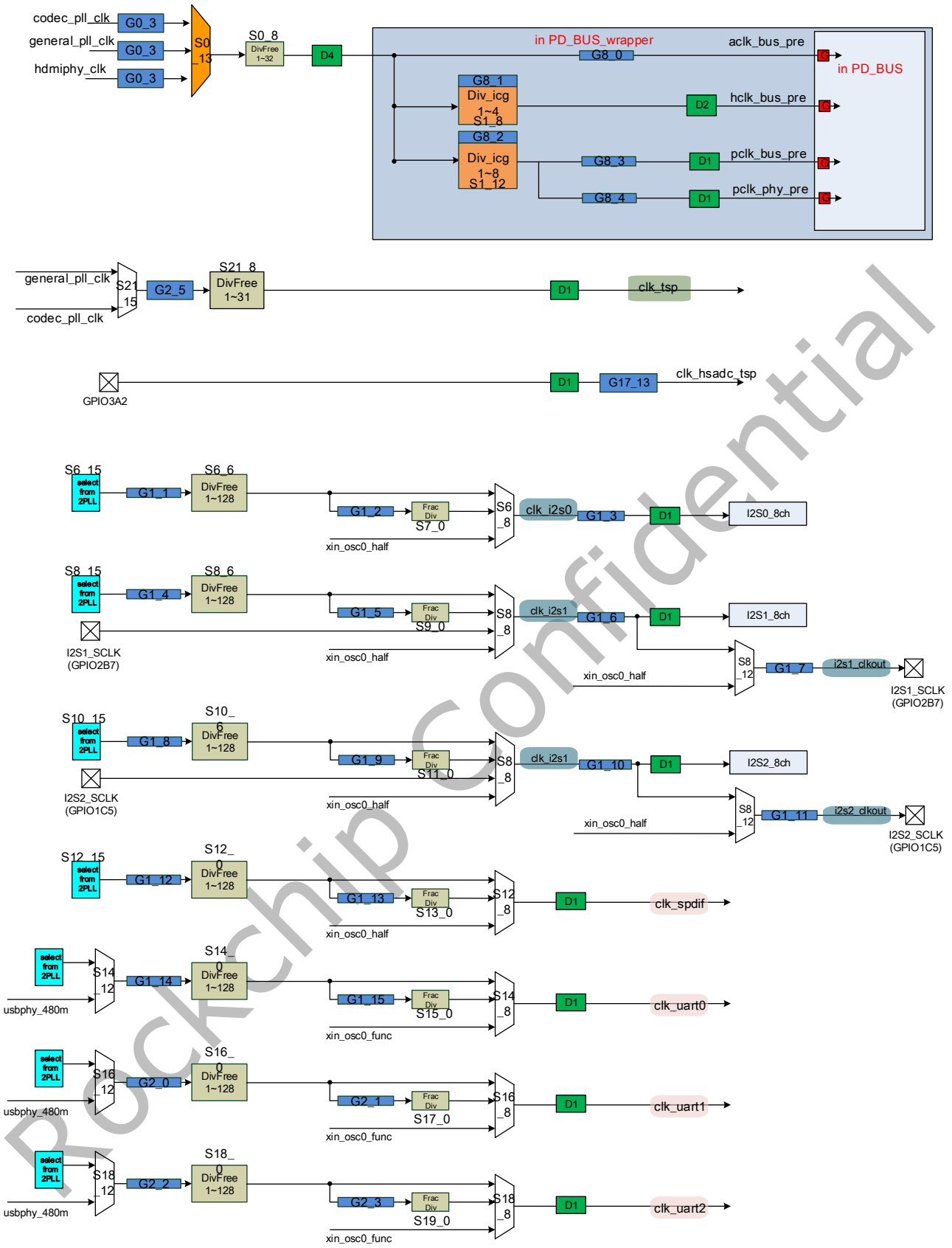

Fig. 3-4 RK3228H Clock Architecture Diagram 3

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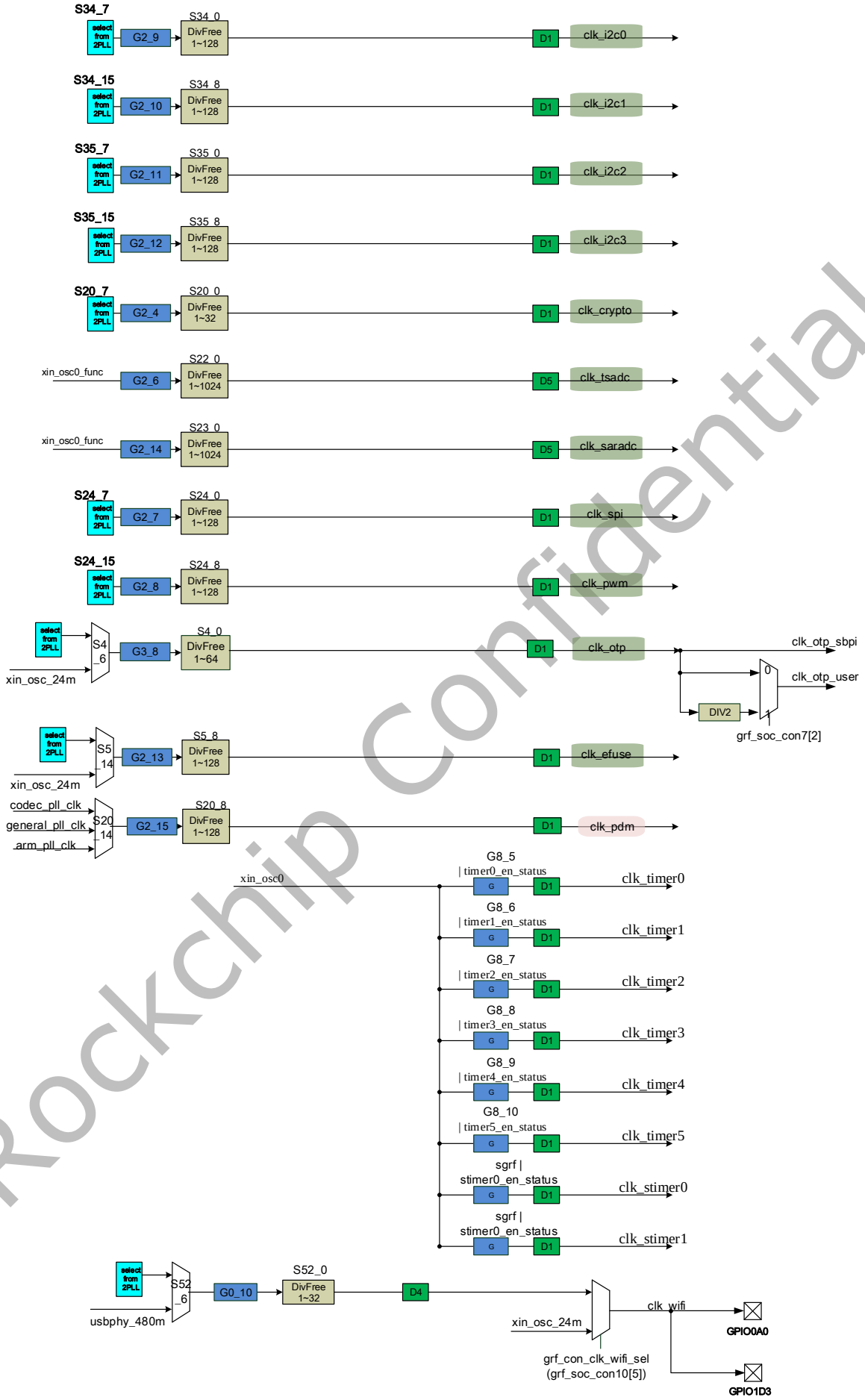

Fig. 3-5 RK3228H Clock Architecture Diagram 4

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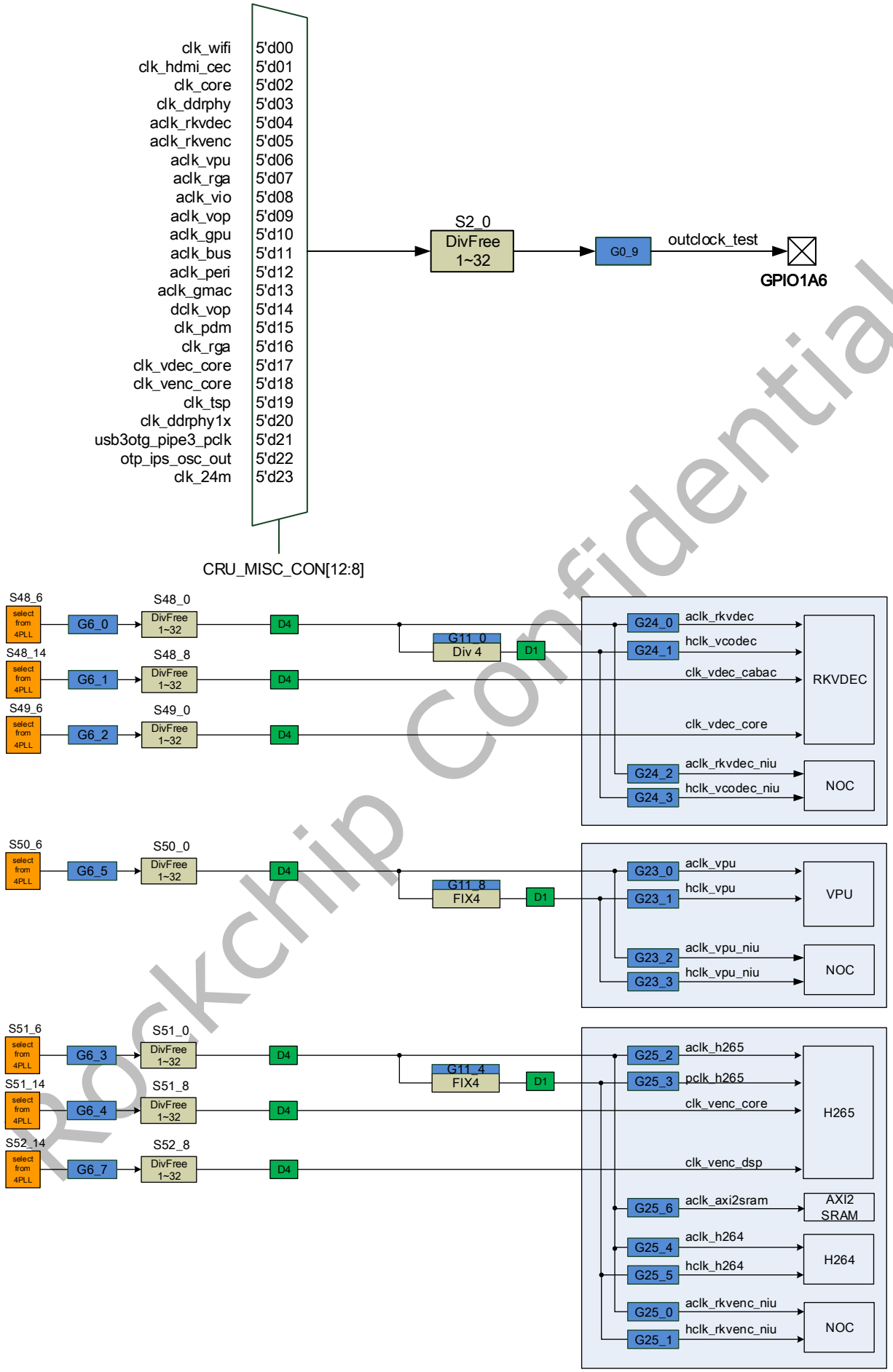

Fig. 3-6 RK3228H Clock Architecture Diagram 5

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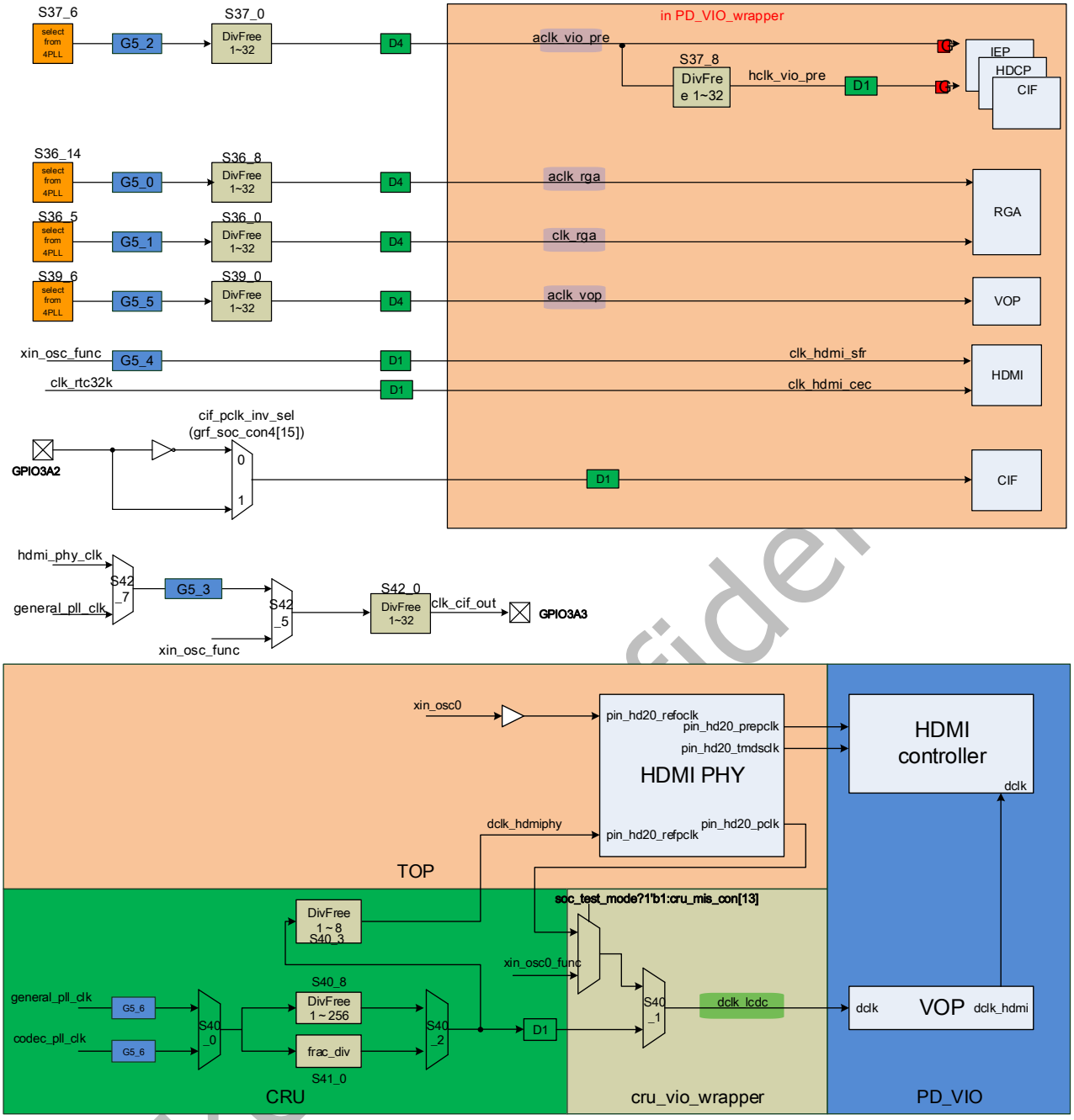

Fig. 3-7 RK3228H Clock Architecture Diagram 6

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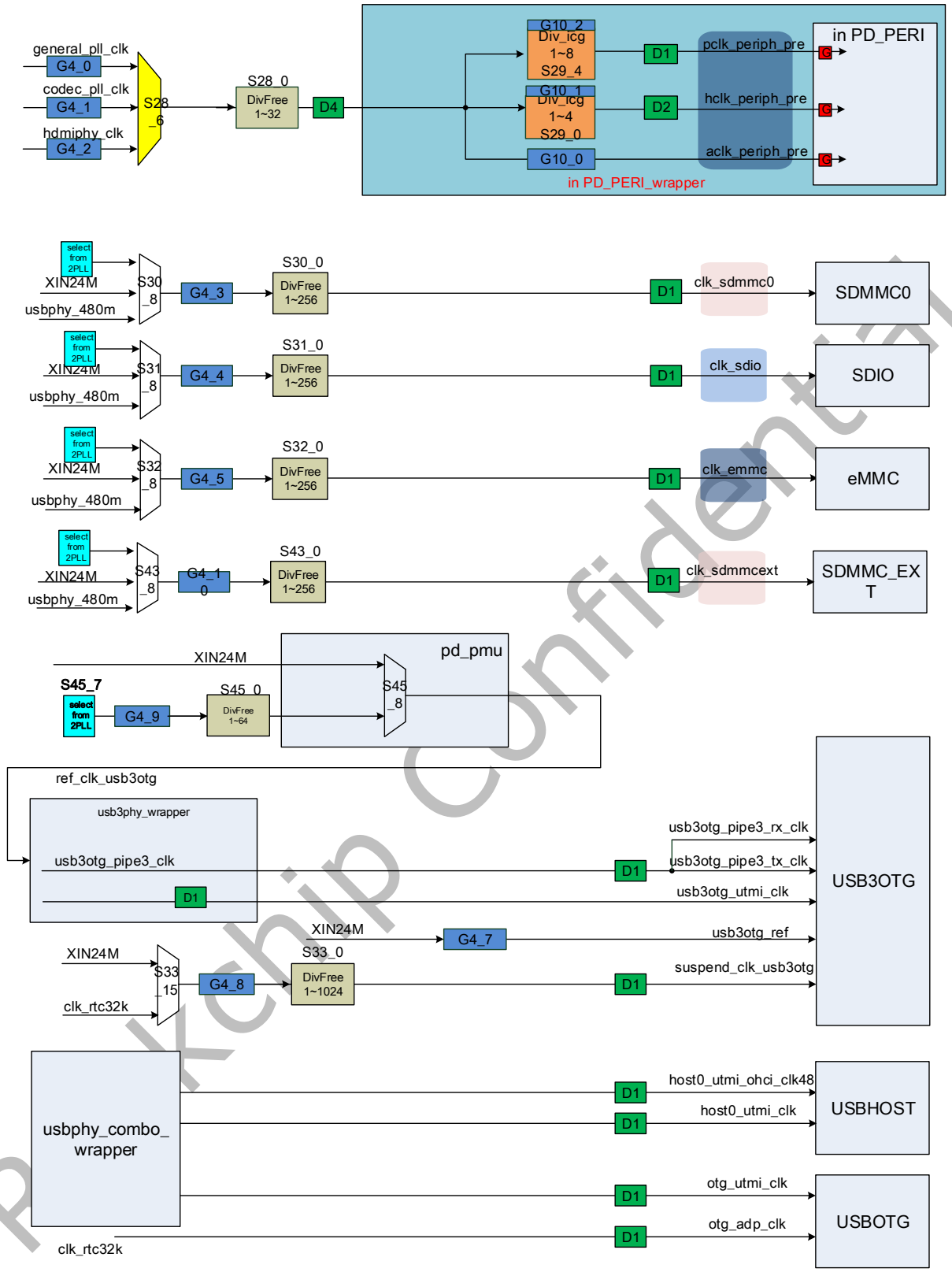

Fig. 3-8 RK3228H Clock Architecture Diagram 7

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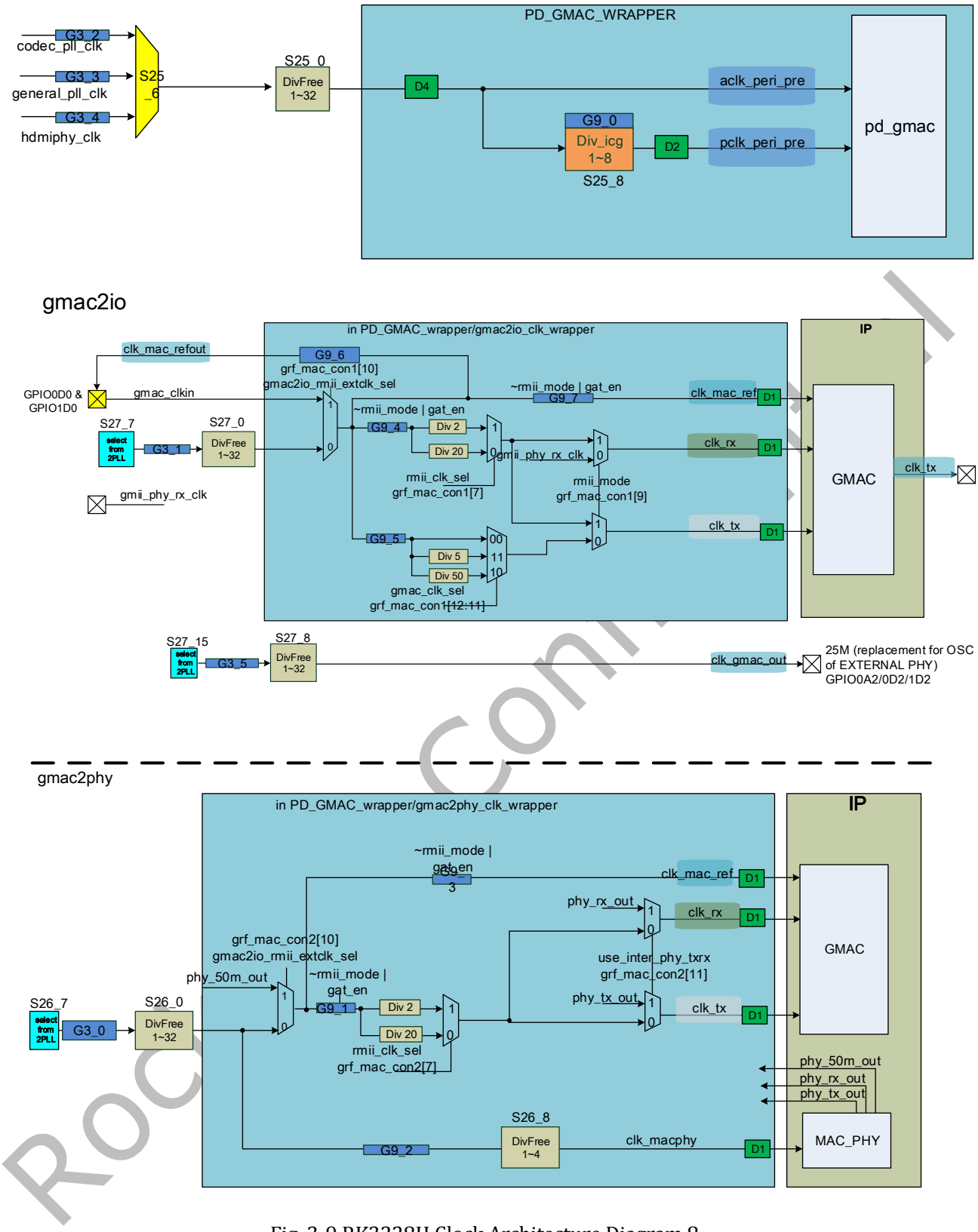

Fig. 3-9 RK3228H Clock Architecture Diagram 8

3.4 System Reset Solution

The following diagram shows reset architecture.

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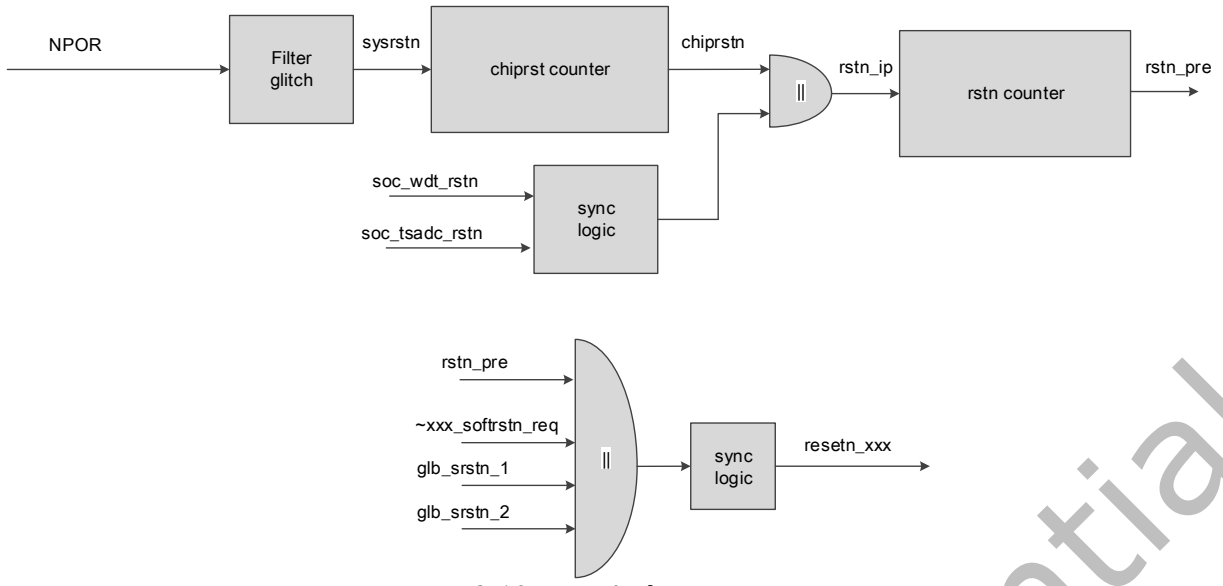

Fig. 3-10 Reset Architecture Diagram

Reset source of each reset signal includes hardware reset(NPOR), SoC watch dog reset(soc_wdt_rstn), SoC tsadc reset(soc_tsadc_rstn), software reset request(xxx_softrstn_req), global software reset1(glb_srstn_1), global software reset2(glb_srstn_2).

The 'xxx' of resetn_xxx and xxx_softrstn_req is the module name.

soc_wdt_rstn is the reset from watch-dog IP in the SoC.

glb_srstn_1 and glb_srstn_2 are the global software reset by programming CRU register. When writing register CRU_GLB_SRST_FST_VALUE as 0xfdb9, glb_srstn_1 will be asserted, and when writing register CRU_GLB_SRST_SND_VALUE as 0xeca8, glb_srstn_2 will be asserted. The two software resets will be self-cleared by hardware. glb_srstn_1 will reset the all logic, and glb_srstn_2 will reset the all logic except GRF and all GPIOs.

3.5 Function Description

There are four PLLs in the chip: ARM PLL, DDR PLL, CODEC PLL and GENERAL PLL, and it supports only one crystal oscillator: 24MHz. Each PLL can only receive 24MHz oscillator.

Four PLLs all can be set to slow mode or deep slow mode, directly output selectable 24MHz. When power on or changing PLL setting, we must force PLL into slow mode to ensure output stable clock.

To maximize the flexibility, some of clocks can select divider source from four PLLs.

To provide some specific frequency, another solution is integrated: fractional divider. In order to guarantee the performance for divided clock, there is some usage limit, we can only get low frequency and divider factor must be larger than 20.

All clocks can be software gated and all resets can be software generated.

3.6 PLL Introduction

3.6.1 Overview

The chip uses 3.2GHz PLL for all the PLLs. The 3.2GHz PLL is a general purpose, high-performance PLL-based clock generator. The PLL is a multi-function, general purpose frequency synthesizer. Ultra-wide input and output ranges along with best-in-class jitter performance allow the PLL to be used for almost any clocking application. With excellent supply noise immunity, the PLL is ideal for use in noisy mixed signal SoC environments. By combining ultra-low jitter output clocks into a low power, low area, widely programmable design, we can greatly simplify an SoC by enabling a single macro to be used for all clocking applications in the system.

3.2GHz PLL supports the following features:

- Input frequency range: 1MHz to 800MHz (Integer Mode) and 10MHz to 800MHz (Fractional Mode)
- Output Frequency Range: 16MHz to 3.2GHz
- 24 bit fractional accuracy, and fractional mode jitter performance to nearly match integer

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mode performance.

- 4:1 VCO frequency range allows PLL to be optimized for minimum jitter or minimum power.
- Isolated analog supply (1.8V) allows for excellent supply rejection in noisy SoC applications.
- Lock Detect Signal indicates when frequency lock has been achieved.

3.6.2 Block diagram

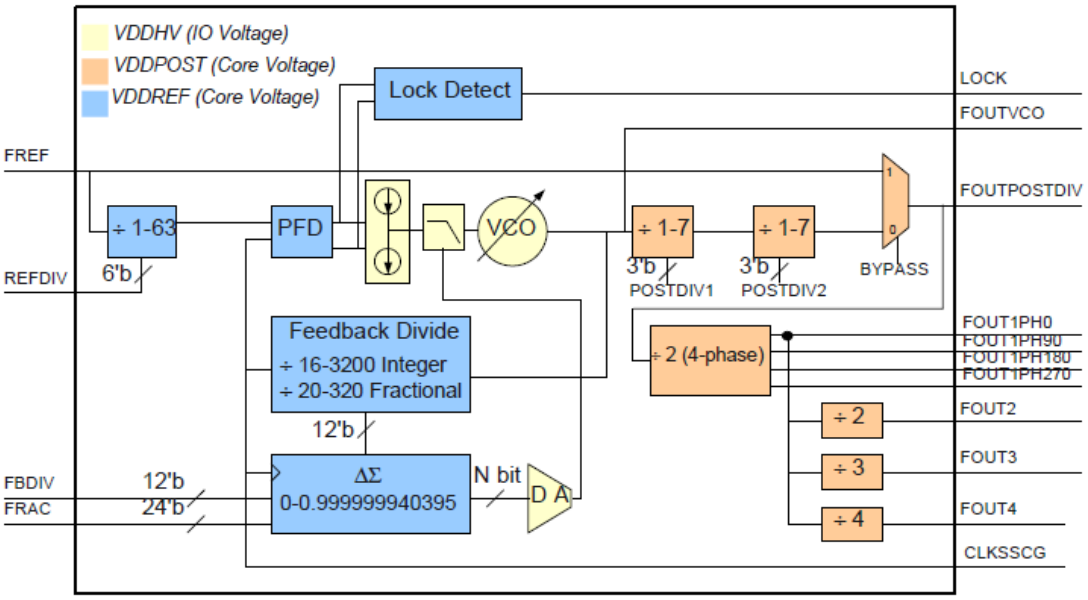

Fig. 3-11 PLL Block Diagram

How to calculate the PLL

The Fractional PLL output frequency can be calculated using some simple formulas. These formulas also embedded within the Fractional PLL Verilog model:

If DSMPD = 1 (DSM is disabled, "integer mode")

FOUTVCO = FREF / REFDIV * FBDIV

FOUTPOSTDIV = FOUTVCO / POSTDIV1 / POSTDIV2

If DSMPD = 0 (DSM is enabled, "fractional mode")

FOUTVCO = FREF / REFDIV * (FBDIV + FRAC / 224)

FOUTPOSTDIV = FOUTVCO / POSTDIV1 / POSTDIV2

Where:

FOUTVCO = Fractional PLL non-divided output frequency

FOUTPOSTDIV = Fractional PLL divided output frequency (output of second post divider)

FREF = Fractional PLL input reference frequency

REFDIV = Fractional PLL input reference clock divider

FVCO = Frequency of internal VCO

FBDIV = Integer value programmed into feedback divide

FRAC = Fractional value programmed into DSM

Changing the PLL Programming

In most cases the PLL programming can be changed on-the-fly and the PLL will simply slew to the new frequency. However, certain changes have the potential to cause glitches on the PLL output clocks. These changes include:

- Switching into or out of BYPASS mode may cause a glitch on FOUTPOSTDIV
- Changing POSTDIV1 or POSTDIV2 may cause a short pulse with width equal to as little as one VCO period on FOUTPOSTDIV
- Changing POSTDIV could cause a shortened pulse on FOUT1PH* or FOUT2/3/4
- Asserting PD or FOUTPOSTDIVPD may cause a glitch on FOUTPOSTDIV

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3.7 Register Description

3.7.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

3.7.2 Registers Summary

Name	Offset	Size	Reset Value	Description
CRU_APLL_CON0	0x0000	W	0x00003064	APLL configuration register0
CRU_APLL_CON1	0x0004	W	0x00001041	APLL configuration register1
CRU_APLL_CON2	0x0008	W	0x00000001	APLL configuration register2
CRU_APLL_CON3	0x000c	W	0x00000007	APLL configuration register3
CRU_APLL_CON4	0x0010	W	0x00007f00	APLL configuration register4
CRU_DPLL_CON0	0x0020	W	0x00001096	DPLL configuration register0
CRU_DPLL_CON1	0x0024	W	0x00001042	DPLL configuration register1
CRU_DPLL_CON2	0x0028	W	0x00000001	DPLL configuration register2
CRU_DPLL_CON3	0x002c	W	0x00000007	DPLL configuration register3
CRU_DPLL_CON4	0x0030	W	0x00007f00	DPLL configuration register4
CRU_CPLL_CON0	0x0040	W	0x000020c8	CPLL configuration register0
CRU_CPLL_CON1	0x0044	W	0x00001043	CPLL configuration register1
CRU_CPLL_CON2	0x0048	W	0x00000001	CPLL configuration register2
CRU_CPLL_CON3	0x004c	W	0x00000007	CPLL configuration register3
CRU_CPLL_CON4	0x0050	W	0x00007f00	CPLL configuration register4
CRU_GPLL_CON0	0x0060	W	0x00001051	GPLL configuration register0
CRU_GPLL_CON1	0x0064	W	0x00000042	GPLL configuration register1
CRU_GPLL_CON2	0x0068	W	0x00eb84f8	GPLL configuration register2
CRU_GPLL_CON3	0x006c	W	0x00000007	GPLL configuration register3
CRU_GPLL_CON4	0x0070	W	0x00007f00	GPLL configuration register4
CRU_CRU_MODE	0x0080	W	0x00000000	CRU_MODE
CRU_CRU_MISC	0x0084	W	0x0000a000	CRU_MISC
CRU_CRU_GLB_CNT_TH	0x0090	W	0x3a980064	CRU_GLB_CNT_TH
CRU_GLB_RST_ST	0x0094	W	0x00000000	GLB_RST_ST
CRU_GLB_SRST_SND_VALUE	0x0098	W	0x00000000	GLB_SRST_SND_VALUE
CRU_GLB_SRST_FST_VALUE	0x009c	W	0x00000000	GLB_SRST_FST_VALUE
CRU_NPLL_CON0	0x00a0	W	0x00003064	NPLL configuration register0
CRU_NPLL_CON1	0x00a4	W	0x00001041	NPLL configuration register1
CRU_NPLL_CON2	0x00a8	W	0x00000001	NPLL configuration register2
CRU_NPLL_CON3	0x00ac	W	0x00000007	NPLL configuration register3
CRU_NPLL_CON4	0x00b0	W	0x00007f00	NPLL configuration register4
CRU_CLKSEL_CON0	0x0100	W	0x00000300	Internal clock select and divide register0
CRU_CLKSEL_CON1	0x0104	W	0x00001113	Internal clock select and divide register1

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Name	Offset	Size	Reset Value	Description
CRU_CLKSEL_CON2	0x0108	W	0x00000003	Internal clock select and divide register2
CRU_CLKSEL_CON3	0x010c	W	0x00000000	Internal clock select and divide register3
CRU_CLKSEL_CON4	0x0110	W	0x00000780	Internal clock select and divide register4
CRU_CLKSEL_CON5	0x0114	W	0x00008000	Internal clock select and divide register5
CRU_CLKSEL_CON6	0x0118	W	0x0000000f	Internal clock select and divide register6
CRU_CLKSEL_CON7	0x011c	W	0x0bb8ea60	Internal clock select and divide register7
CRU_CLKSEL_CON8	0x0120	W	0x0000000f	Internal clock select and divide register8
CRU_CLKSEL_CON9	0x0124	W	0x0bb8ea60	Internal clock select and divide register9
CRU_CLKSEL_CON10	0x0128	W	0x0000000f	Internal clock select and divide register10
CRU_CLKSEL_CON11	0x012c	W	0x0bb8ea60	Internal clock select and divide register11
CRU_CLKSEL_CON12	0x0130	W	0x0000000f	Internal clock select and divide register12
CRU_CLKSEL_CON13	0x0134	W	0x0bb8ea60	Internal clock select and divide register13
CRU_CLKSEL_CON14	0x0138	W	0x00000007	Internal clock select and divide register14
CRU_CLKSEL_CON15	0x013c	W	0x0bb8ea60	Internal clock select and divide register15
CRU_CLKSEL_CON16	0x0140	W	0x00000007	Internal clock select and divide register16
CRU_CLKSEL_CON17	0x0144	W	0x0bb8ea60	Internal clock select and divide register17
CRU_CLKSEL_CON18	0x0148	W	0x00000007	Internal clock select and divide register18
CRU_CLKSEL_CON19	0x014c	W	0x0bb8ea60	Internal clock select and divide register19
CRU_CLKSEL_CON20	0x0150	W	0x00008f04	Internal clock select and divide register20
CRU_CLKSEL_CON21	0x0154	W	0x00000400	Internal clock select and divide register21
CRU_CLKSEL_CON22	0x0158	W	0x000001e0	Internal clock select and divide register22
CRU_CLKSEL_CON23	0x015c	W	0x000001e0	Internal clock select and divide register23
CRU_CLKSEL_CON24	0x0160	W	0x00000707	Internal clock select and divide register24
CRU_CLKSEL_CON25	0x0164	W	0x00000242	Internal clock select and divide register25
CRU_CLKSEL_CON26	0x0168	W	0x0000000f	Internal clock select and divide register26
CRU_CLKSEL_CON27	0x016c	W	0x00000705	Internal clock select and divide register27
CRU_CLKSEL_CON28	0x0170	W	0x00000042	Internal clock select and divide register28
CRU_CLKSEL_CON29	0x0174	W	0x00000022	Internal clock select and divide register29
CRU_CLKSEL_CON30	0x0178	W	0x00000003	Internal clock select and divide register30
CRU_CLKSEL_CON31	0x017c	W	0x00000001	Internal clock select and divide register31
CRU_CLKSEL_CON32	0x0180	W	0x00000001	Internal clock select and divide register32
CRU_CLKSEL_CON33	0x0184	W	0x0000030b	Internal clock select and divide register33
CRU_CLKSEL_CON34	0x0188	W	0x00000707	Internal clock select and divide register34
CRU_CLKSEL_CON35	0x018c	W	0x00000707	Internal clock select and divide register35
CRU_CLKSEL_CON36	0x0190	W	0x00004242	Internal clock select and divide register36
CRU_CLKSEL_CON37	0x0194	W	0x00000242	Internal clock select and divide register37
CRU_CLKSEL_CON38	0x0198	W	0x0000c2dc	Internal clock select and divide register38
CRU_CLKSEL_CON39	0x019c	W	0x00000001	Internal clock select and divide register39
CRU_CLKSEL_CON40	0x01a0	W	0x00003113	Internal clock select and divide register40
CRU_CLKSEL_CON41	0x01a4	W	0x0bb8ea60	Internal clock select and divide register41
CRU_CLKSEL_CON42	0x01a8	W	0x00000013	Internal clock select and divide register42
CRU_CLKSEL_CON43	0x01ac	W	0x00000003	Internal clock select and divide register43
CRU_CLKSEL_CON44	0x01b0	W	0x00000042	Internal clock select and divide register44

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Name	Offset	Size	Reset Value	Description
CRU_CLKSEL_CON45	0x01b4	W	0x0000001f	Internal clock select and divide register45
CRU_CLKSEL_CON46	0x01b8	W	0x00000000	Internal clock select and divide register46
CRU_CLKSEL_CON47	0x01bc	W	0x00000000	Internal clock select and divide register47
CRU_CLKSEL_CON48	0x01c0	W	0x00004201	Internal clock select and divide register48
CRU_CLKSEL_CON49	0x01c4	W	0x00000042	Internal clock select and divide register49
CRU_CLKSEL_CON50	0x01c8	W	0x00000042	Internal clock select and divide register50
CRU_CLKSEL_CON51	0x01cc	W	0x00000203	Internal clock select and divide register51
CRU_CLKSEL_CON52	0x01d0	W	0x0000021e	Internal clock select and divide register52
CRU_CLKGATE_CON0	0x0200	W	0x00000000	Internal clock gating register0
CRU_CLKGATE_CON1	0x0204	W	0x00000000	Internal clock gating register1
CRU_CLKGATE_CON2	0x0208	W	0x00000000	Internal clock gating register2
CRU_CLKGATE_CON3	0x020c	W	0x00000000	Internal clock gating register3
CRU_CLKGATE_CON4	0x0210	W	0x00000000	Internal clock gating register4
CRU_CLKGATE_CON5	0x0214	W	0x00000000	Internal clock gating register5
CRU_CLKGATE_CON6	0x0218	W	0x00000000	Internal clock gating register6
CRU_CLKGATE_CON7	0x021c	W	0x00000000	Internal clock gating register7
CRU_CLKGATE_CON8	0x0220	W	0x00000000	Internal clock gating register8
CRU_CLKGATE_CON9	0x0224	W	0x00000000	Internal clock gating register9
CRU_CLKGATE_CON10	0x0228	W	0x00000000	Internal clock gating register10
CRU_CLKGATE_CON11	0x022c	W	0x00000000	Internal clock gating register11
CRU_CLKGATE_CON12	0x0230	W	0x00000000	Internal clock gating register12
CRU_CLKGATE_CON13	0x0234	W	0x00000000	Internal clock gating register13
CRU_CLKGATE_CON14	0x0238	W	0x00000000	Internal clock gating register14
CRU_CLKGATE_CON15	0x023c	W	0x00000000	Internal clock gating register15
CRU_CLKGATE_CON16	0x0240	W	0x00000000	Internal clock gating register16
CRU_CLKGATE_CON17	0x0244	W	0x00000000	Internal clock gating register17
CRU_CLKGATE_CON18	0x0248	W	0x00000000	Internal clock gating register18
CRU_CLKGATE_CON19	0x024c	W	0x00000000	Internal clock gating register19
CRU_CLKGATE_CON20	0x0250	W	0x00000000	Internal clock gating register20
CRU_CLKGATE_CON21	0x0254	W	0x00000000	Internal clock gating register21
CRU_CLKGATE_CON22	0x0258	W	0x00000000	Internal clock gating register22
CRU_CLKGATE_CON23	0x025c	W	0x00000000	Internal clock gating register23
CRU_CLKGATE_CON24	0x0260	W	0x00000000	Internal clock gating register24
CRU_CLKGATE_CON25	0x0264	W	0x00000000	Internal clock gating register25
CRU_CLKGATE_CON26	0x0268	W	0x00000000	Internal clock gating register26
CRU_CLKGATE_CON27	0x026c	W	0x00000000	Internal clock gating register27
CRU_CLKGATE_CON28	0x0270	W	0x00000000	Internal clock gating register28
CRU_SSGTBL0_3	0x0280	W	0x00000000	SSMOD external wave table register0
CRU_SSGTBL4_7	0x0284	W	0x00000000	SSMOD external wave table register1
CRU_SSGTBL8_11	0x0288	W	0x00000000	SSMOD external wave table register2
CRU_SSGTBL12_15	0x028c	W	0x00000000	SSMOD external wave table register3
CRU_SSGTBL16_19	0x0290	W	0x00000000	SSMOD external wave table register4
CRU_SSGTBL20_23	0x0294	W	0x00000000	SSMOD external wave table register5

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Name	Offset	Size	Reset Value	Description
CRU_SSGTBL24_27	0x0298	W	0x00000000	SSMOD external wave table register6
CRU_SSGTBL28_31	0x029c	W	0x00000000	SSMOD external wave table register7
CRU_SSGTBL32_35	0x02a0	W	0x00000000	SSMOD external wave table register8
CRU_SSGTBL36_39	0x02a4	W	0x00000000	SSMOD external wave table register9
CRU_SSGTBL40_43	0x02a8	W	0x00000000	SSMOD external wave table register10
CRU_SSGTBL44_47	0x02ac	W	0x00000000	SSMOD external wave table register11
CRU_SSGTBL48_51	0x02b0	W	0x00000000	SSMOD external wave table register12
CRU_SSGTBL52_55	0x02b4	W	0x00000000	SSMOD external wave table register13
CRU_SSGTBL56_59	0x02b8	W	0x00000000	SSMOD external wave table register14
CRU_SSGTBL60_63	0x02bc	W	0x00000000	SSMOD external wave table register15
CRU_SSGTBL64_67	0x02c0	W	0x00000000	SSMOD external wave table register16
CRU_SSGTBL68_71	0x02c4	W	0x00000000	SSMOD external wave table register17
CRU_SSGTBL72_75	0x02c8	W	0x00000000	SSMOD external wave table register18
CRU_SSGTBL76_79	0x02cc	W	0x00000000	SSMOD external wave table register19
CRU_SSGTBL80_83	0x02d0	W	0x00000000	SSMOD external wave table register20
CRU_SSGTBL84_87	0x02d4	W	0x00000000	SSMOD external wave table register21
CRU_SSGTBL88_91	0x02d8	W	0x00000000	SSMOD external wave table register22
CRU_SSGTBL92_95	0x02dc	W	0x00000000	SSMOD external wave table register23
CRU_SSGTBL96_99	0x02e0	W	0x00000000	SSMOD external wave table register24
CRU_SSGTBL100_103	0x02e4	W	0x00000000	SSMOD external wave table register25
CRU_SSGTBL104_107	0x02e8	W	0x00000000	SSMOD external wave table register26
CRU_SSGTBL108_111	0x02ec	W	0x00000000	SSMOD external wave table register27
CRU_SSGTBL112_115	0x02f0	W	0x00000000	SSMOD external wave table register28
CRU_SSGTBL116_119	0x02f4	W	0x00000000	SSMOD external wave table register29
CRU_SSGTBL120_123	0x02f8	W	0x00000000	SSMOD external wave table register30
CRU_SSGTBL124_127	0x02fc	W	0x00000000	SSMOD external wave table register31
CRU_SOFTRST_CON0	0x0300	W	0x00000000	Internal software reset control register0
CRU_SOFTRST_CON1	0x0304	W	0x00000000	Internal software reset control register1
CRU_SOFTRST_CON2	0x0308	W	0x00000000	Internal software reset control register2
CRU_SOFTRST_CON3	0x030c	W	0x00000000	Internal software reset control register3
CRU_SOFTRST_CON4	0x0310	W	0x00000000	Internal software reset control register4
CRU_SOFTRST_CON5	0x0314	W	0x00000000	Internal software reset control register5
CRU_SOFTRST_CON6	0x0318	W	0x00000000	Internal software reset control register6
CRU_SOFTRST_CON7	0x031c	W	0x00000000	Internal software reset control register7
CRU_SOFTRST_CON8	0x0320	W	0x00000000	Internal software reset control register8
CRU_SOFTRST_CON9	0x0324	W	0x00000000	Internal software reset control register9
CRU_SOFTRST_CON10	0x0328	W	0x00000000	Internal software reset control register10
CRU_SOFTRST_CON11	0x032c	W	0x00000000	Internal software reset control register11
CRU_CRU_SDMMC_CON0	0x0380	W	0x00000004	sdmmc control0
CRU_CRU_SDMMC_CON1	0x0384	W	0x00000000	sdmmc control1

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Name	Offset	Size	Reset Value	Description
CRU_CRU_SDIO_CON0	0x0388	W	0x00000004	SDIO control0
CRU_CRU_SDIO_CON1	0x038c	W	0x00000000	SDIO control1
CRU_CRU_EMMC_CON0	0x0390	W	0x00000004	EMMC control0
CRU_CRU_EMMC_CON1	0x0394	W	0x00000000	EMMC control1
CRU_CRU_SDMMC_EXT_CON0	0x0398	W	0x00000004	SDMMC_EXT control0
CRU_CRU_SDMMC_EXT_CON1	0x039c	W	0x00000000	SDMMC_EXT control1

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

3.7.3 Detail Register Description

CRU_APLL_CON0
Address: Operational Base + offset (0x0000)
APLL configuration register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	bypass PLL Bypass. FREF bypasses PLL to FOUTPOSTDIV 1'b0: no bypass 1'b1: bypass
14:12	RW	0x3	postdiv1 First Post Divide Value (1-7)
11:0	RW	0x064	fbdiv Feedback Divide Value "Valid divider settings are: [16, 3200] in integer mode [20, 320] in fractional mode Tips: no plus one operation

CRU_APLL_CON1
Address: Operational Base + offset (0x0004)
APLL configuration register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	pllpdsel PLL global power down source selection "If pllpdsel == 1, PLL can be power down only by pllpd1, otherwise pll is power down when any one of refdiv/fbdiv/fracdiv is changed or pllpd0 is asserted.
14	RW	0x0	pllpd1 PLL global power down request 1'b0: no power down 1'b2: power down
13	RW	0x0	pllpd0 PLL global power down request 1'b0: no power down 1'b1: power down
12	RW	0x1	dsmpd PLL delta sigma modulator enable " 1'b0: modulator is enable, 1'b1: modulator is disabled
11	RO	0x0	reserved
10	RO	0x0	pll_lock PLL lock status 1'b0: unlock 1'b1: lock
9	RO	0x0	reserved
8:6	RW	0x1	postdiv2 Second Post Divide Value (1-7)
5:0	RW	0x01	refdiv Reference Clock Divide Value (1-63)

CRU_APLL_CON2
Address: Operational Base + offset (0x0008)
APLL configuration register2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	fout4phasepd "Power down 4-phase clocks and 2X, 3X, 4X clocks 1'b0: no power down 1'b1: power down
26	RW	0x0	foutvcopd Power down buffered VCO clock 1'b0: no power down 1'b1: power down

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Bit	Attr	Reset Value	Description
25	RW	0x0	foutpostdivpd Power down all outputs except for buffered VCO clock 1'b0: no power down 1'b1: power down
24	RW	0x0	dacpd Power down quantization noise cancellation DAC 1'b0: no power down 1'b1: power down
23:0	RW	0x000001	fracdiv Fractional part of feedback divide (fraction = FRAC/2^24)

CRU_APLL_CON3

Address: Operational Base + offset (0x000c)

APLL configuration register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	WO	0x00	ssmod_spread spread amplitude % = 0.1 * SPREAD[4:0]
7:4	WO	0x0	ssmod_divval Divider required to set the modulation frequency Divider required to set the modulation frequency
3	WO	0x0	ssmod_downspread Selects center spread or downs pread 1'b0: down spread 1'b1: center spread
2	WO	0x1	ssmod_reset Reset modulator state 1'b0: no reset 1'b1: reset
1	WO	0x1	ssmod_disable_sscg Bypass SSMOD by module 1'b0: no bypass 1'b1: bypass
0	WO	0x1	ssmod_bp Bypass SSMOD by integration 1'b0: no bypass 1'b1: bypass

CRU_APLL_CON4

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Address: Operational Base + offset (0x0010)
APLL configuration register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	WO	0x7f	ssmod_ext_maxaddr External wave table data inputs (0-255)
7:1	RO	0x0	reserved
0	WO	0x0	ssmod_sel_ext_wave select external wave 1'b0: no select ext_wave 1'b1: select ext_wave

CRU_DPLL_CON0
Address: Operational Base + offset (0x0020)
DPLL configuration register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	bypass PLL Bypass. FREF bypasses PLL to FOUTPOSTDIV 1'b0: no bypass 1'b1: bypass
14:12	RW	0x1	postdiv1 First Post Divide Value (1-7)
11:0	RW	0x096	fbdiv Feedback Divide Value "Valid divider settings are: [16, 3200] in integer mode [20, 320] in fractional mode Tips: no plus one operation

CRU_DPLL_CON1
Address: Operational Base + offset (0x0024)
DPLL configuration register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	pllpdsel PLL global power down source selection "If pllpdsel == 1, PLL can be power down only by pllpd1, otherwise pll is power down when any one of refdiv/fbdiv/fracdiv is changed or pllpd0 is asserted.
14	RW	0x0	pllpd1 PLL global power down request 1'b0: no power down 1'b2: power down
13	RW	0x0	pllpd0 PLL global power down request 1'b0: no power down 1'b1: power down
12	RW	0x1	dsmpd PLL delta sigma modulator enable " 1'b0: modulator is enable, 1'b1: modulator is disabled
11	RO	0x0	reserved
10	RO	0x0	pll_lock PLL lock status 1'b0: unlock 1'b1: lock
9	RO	0x0	reserved
8:6	RW	0x1	postdiv2 Second Post Divide Value (1-7)
5:0	RW	0x02	refdiv Reference Clock Divide Value (1-63)

CRU_DPLL_CON2
Address: Operational Base + offset (0x0028)
DPLL configuration register2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	fout4phasepd "Power down 4-phase clocks and 2X, 3X, 4X clocks 1'b0: no power down 1'b1: power down
26	RW	0x0	foutvcopd Power down buffered VCO clock 1'b0: no power down 1'b1: power down

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Bit	Attr	Reset Value	Description
25	RW	0x0	foutpostdivpd Power down all outputs except for buffered VCO clock 1'b0: no power down 1'b1: power down
24	RW	0x0	dacpd Power down quantization noise cancellation DAC 1'b0: no power down 1'b1: power down
23:0	RW	0x000001	fracdiv Fractional part of feedback divide (fraction = FRAC/2^24)

CRU_DPLL_CON3

Address: Operational Base + offset (0x002c)

DPLL configuration register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	WO	0x00	ssmod_spread spread amplitude % = 0.1 * SPREAD[4:0]
7:4	WO	0x0	ssmod_divval Divider required to set the modulation frequency Divider required to set the modulation frequency
3	WO	0x0	ssmod_downspread Selects center spread or downs pread 1'b0: down spread 1'b1: center spread
2	WO	0x1	ssmod_reset Reset modulator state 1'b0: no reset 1'b1: reset
1	WO	0x1	ssmod_disable_sscg Bypass SSMOD by module 1'b0: no bypass 1'b1: bypass
0	WO	0x1	ssmod_bp Bypass SSMOD by integration 1'b0: no bypass 1'b1: bypass

CRU_DPLL_CON4

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Address: Operational Base + offset (0x0030)
DPLL configuration register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	WO	0x7f	ssmod_ext_maxaddr External wave table data inputs (0-255)
7:1	RO	0x0	reserved
0	WO	0x0	ssmod_sel_ext_wave select external wave 1'b0: no select ext_wave 1'b1: select ext_wave

CRU_CPLL_CON0
Address: Operational Base + offset (0x0040)
CPLL configuration register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	bypass PLL Bypass. FREF bypasses PLL to FOUTPOSTDIV 1'b0: no bypass 1'b1: bypass
14:12	RW	0x2	postdiv1 First Post Divide Value (1-7)
11:0	RW	0x0c8	fbdiv Feedback Divide Value "Valid divider settings are: [16, 3200] in integer mode [20, 320] in fractional mode Tips: no plus one operation

CRU_CPLL_CON1
Address: Operational Base + offset (0x0044)
CPLL configuration register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	pllpdsel PLL global power down source selection "If pllpdsel == 1, PLL can be power down only by pllpd1, otherwise pll is power down when any one of refdiv/fbdiv/fracdiv is changed or pllpd0 is asserted.
14	RW	0x0	pllpd1 PLL global power down request 1'b0: no power down 1'b2: power down
13	RW	0x0	pllpd0 PLL global power down request 1'b0: no power down 1'b1: power down
12	RW	0x1	dsmpd PLL delta sigma modulator enable " 1'b0: modulator is enable, 1'b1: modulator is disabled
11	RO	0x0	reserved
10	RO	0x0	pll_lock PLL lock status 1'b0: unlock 1'b1: lock
9	RO	0x0	reserved
8:6	RW	0x1	postdiv2 Second Post Divide Value (1-7)
5:0	RW	0x03	refdiv Reference Clock Divide Value (1-63)

CRU_CPLL_CON2
Address: Operational Base + offset (0x0048)
CPLL configuration register2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	fout4phasepd "Power down 4-phase clocks and 2X, 3X, 4X clocks 1'b0: no power down 1'b1: power down
26	RW	0x0	foutvcopd Power down buffered VCO clock 1'b0: no power down 1'b1: power down

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Bit	Attr	Reset Value	Description
25	RW	0x0	foutpostdivpd Power down all outputs except for buffered VCO clock 1'b0: no power down 1'b1: power down
24	RW	0x0	dacpd Power down quantization noise cancellation DAC 1'b0: no power down 1'b1: power down
23:0	RW	0x000001	fracdiv Fractional part of feedback divide (fraction = FRAC/2^24)

CRU_CPLL_CON3

Address: Operational Base + offset (0x004c)

CPLL configuration register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	WO	0x00	ssmod_spread spread amplitude % = 0.1 * SPREAD[4:0]
7:4	WO	0x0	ssmod_divval Divider required to set the modulation frequency Divider required to set the modulation frequency
3	WO	0x0	ssmod_downspread Selects center spread or downs pread 1'b0: down spread 1'b1: center spread
2	WO	0x1	ssmod_reset Reset modulator state 1'b0: no reset 1'b1: reset
1	WO	0x1	ssmod_disable_sscg Bypass SSMOD by module 1'b0: no bypass 1'b1: bypass
0	WO	0x1	ssmod_bp Bypass SSMOD by integration 1'b0: no bypass 1'b1: bypass

CRU_CPLL_CON4

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Address: Operational Base + offset (0x0050)
CPLL configuration register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	WO	0x7f	ssmod_ext_maxaddr External wave table data inputs (0-255)
7:1	RO	0x0	reserved
0	WO	0x0	ssmod_sel_ext_wave select external wave 1'b0: no select ext_wave 1'b1: select ext_wave

CRU_GPLL_CON0
Address: Operational Base + offset (0x0060)
GPLL configuration register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	bypass PLL Bypass. FREF bypasses PLL to FOUTPOSTDIV 1'b0: no bypass 1'b1: bypass
14:12	RW	0x1	postdiv1 First Post Divide Value (1-7)
11:0	RW	0x051	fbdiv Feedback Divide Value "Valid divider settings are: [16, 3200] in integer mode [20, 320] in fractional mode Tips: no plus one operation

CRU_GPLL_CON1
Address: Operational Base + offset (0x0064)
GPLL configuration register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	pllpdsel PLL global power down source selection "If pllpdsel == 1, PLL can be power down only by pllpd1, otherwise pll is power down when any one of refdiv/fbdiv/fracdiv is changed or pllpd0 is asserted.
14	RW	0x0	pllpd1 PLL global power down request 1'b0: no power down 1'b2: power down
13	RW	0x0	pllpd0 PLL global power down request 1'b0: no power down 1'b1: power down
12	RW	0x0	dsmpd PLL delta sigma modulator enable " 1'b0: modulator is enable, 1'b1: modulator is disabled
11	RO	0x0	reserved
10	RO	0x0	pll_lock PLL lock status 1'b0: unlock 1'b1: lock
9	RO	0x0	reserved
8:6	RW	0x1	postdiv2 Second Post Divide Value (1-7)
5:0	RW	0x02	refdiv Reference Clock Divide Value (1-63)

CRU_GPLL_CON2
Address: Operational Base + offset (0x0068)
GPLL configuration register2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	fout4phasepd "Power down 4-phase clocks and 2X, 3X, 4X clocks 1'b0: no power down 1'b1: power down
26	RW	0x0	foutvcopd Power down buffered VCO clock 1'b0: no power down 1'b1: power down

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Bit	Attr	Reset Value	Description
25	RW	0x0	foutpostdivpd Power down all outputs except for buffered VCO clock 1'b0: no power down 1'b1: power down
24	RW	0x0	dacpd Power down quantization noise cancellation DAC 1'b0: no power down 1'b1: power down
23:0	RW	0xeb84f8	fracdiv Fractional part of feedback divide (fraction = FRAC/2^24)

CRU_GPLL_CON3

Address: Operational Base + offset (0x006c)

GPLL configuration register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	WO	0x00	ssmod_spread spread amplitude % = 0.1 * SPREAD[4:0]
7:4	WO	0x0	ssmod_divval Divider required to set the modulation frequency Divider required to set the modulation frequency
3	WO	0x0	ssmod_downspread Selects center spread or downs pread 1'b0: down spread 1'b1: center spread
2	WO	0x1	ssmod_reset Reset modulator state 1'b0: no reset 1'b1: reset
1	WO	0x1	ssmod_disable_sscg Bypass SSMOD by module 1'b0: no bypass 1'b1: bypass
0	WO	0x1	ssmod_bp Bypass SSMOD by integration 1'b0: no bypass 1'b1: bypass

CRU_GPLL_CON4

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Address: Operational Base + offset (0x0070)
GPLL configuration register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	WO	0x7f	ssmod_ext_maxaddr External wave table data inputs (0-255)
7:1	RO	0x0	reserved
0	WO	0x0	ssmod_sel_ext_wave select external wave 1'b0: no select ext_wave 1'b1: select ext_wave

CRU_CRU_MODE
Address: Operational Base + offset (0x0080)
CRU_MODE

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12	RW	0x0	gpll_work_mode PLL work mode select 1'b0: Slow mode, clock from external 24MHz/26MHz OSC (default) 1'b1: Normal mode, clock from PLL output
11:9	RO	0x0	reserved
8	RW	0x0	cpll_work_mode PLL work mode select 1'b0: Slow mode, clock from external 24MHz/26MHz OSC (default) 1'b1: Normal mode, clock from PLL output
7:5	RO	0x0	reserved
4	RW	0x0	dpll_work_mode PLL work mode select 1'b0: Slow mode, clock from external 24MHz/26MHz OSC (default) 1'b1: Normal mode, clock from PLL output
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	npll_work_mode PLL work mode select 1'b0: Slow mode, clock from external 24MHz/26MHz OSC (default) 1'b1: Normal mode, clock from PLL output
0	RW	0x0	apll_work_mode PLL work mode select 1'b0: Slow mode, clock from external 24MHz/26MHz OSC (default) 1'b1: Normal mode, clock from PLL output

CRU_CRU_MISC
Address: Operational Base + offset (0x0084)
CRU_MISC

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x1	usb480m_24m_sel USB PHY select 1'b1: when using USB480M as clock source, clock source freq is set to 24Mhz. 1'b0: when using USB480M as clock source, clock source freq is set to USBPHY480M output.
14	RO	0x0	reserved
13	RW	0x1	hdmiphy_24m_sel HDMI PHY select 1'b1: when using HDMIPHY as clock source, clock source freq is set to 24Mhz. 1'b0: when using HDMIPHY as clock source, clock source freq is set to HDMIPHY pixel output.

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Bit	Attr	Reset Value	Description
12:8	RW	0x00	testclk_sel Test clock out select 5'd00: clk_wifi 5'd01: clk_hdmi_cec 5'd02: clk_core 5'd03: clk_ddrphy 5'd04: aclk_rkvdec 5'd05: aclk_rkvenc 5'd06: aclk_vpu 5'd07: aclk_rga 5'd08: aclk_vio 5'd09: aclk_vop 5'd10: aclk_gpu 5'd11: aclk_bus 5'd12: aclk_peri 5'd13: aclk_gmac 5'd14: dclk_vop 5'd15: clk_pdm 5'd16: clk_rga 5'd17: clk_vdec_core 5'd18: clk_venc_core 5'd19: clk_tsp 5'd20: clk_ddrphy1x 5'd21: usb3otg_pipe3_pclk 5'd22: otp_ips_osc_out 5'd23: clk_24m default: buf_clk_wifi
7:3	RO	0x0	reserved
2	RW	0x0	core_wrst_wfien CPU warm reset by wfi enable 1'b1: cpu warm reset is valid when only when wfi is asserted. 1'b0: cpu warm reset is not
1	RW	0x0	core_srst_wfien CPU wfi reset enable 1'b1: cpu reseted when wfi and softrst0[4] are both asserted. 1'b0: cpu reseted only by softrst0[4]
0	RW	0x0	warmrst_en CPU warm reset enable 1'b1: enable cpu warm reset. 1'b0: disable cpu warm reset.

CRU_CRU_GLB_CNT_TH

Address: Operational Base + offset (0x0090)

CRU_GLB_CNT_TH

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x3a980064	global_reset_counter_threshold Global soft reset counter threshold Global soft reset counter threshold

CRU_GLB_RST_ST
Address: Operational Base + offset (0x0094)
GLB_RST_ST

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	W1C	0x0	snd_glb_tsadc_rst_st sencond global TSADC triggered reset flag 1'b0: last hot reset is not sencond global TSADC triggered reset 1'b1: last hot reset is sencond global TSADC triggered reset
4	W1C	0x0	fst_glb_tsadc_rst_st first global TSADC triggered reset flag 1'b0: last hot reset is not first global TSADC triggered reset 1'b1: last hot reset is first global TSADC triggered reset
3	W1C	0x0	snd_glb_wdt_rst_st sencond global WDT triggered reset flag 1'b0: last hot reset is not sencond global WDT triggered reset 1'b1: last hot reset is sencond global WDT triggered reset
2	W1C	0x0	fst_glb_wdt_rst_st first global WDT triggered reset flag 1'b0: last hot reset is not first global WDT triggered reset 1'b1: last hot reset is first global WDT triggered reset
1	W1C	0x0	snd_glb_rst_st second global rst flag 1'b0: last hot reset is not sencond global reset 1'b1: last hot reset is sencond global reset
0	W1C	0x0	fst_glb_rst_st first global rst flag 1'b0: last hot reset is not first global reset 1'b1: last hot reset is first global reset

CRU_GLB_SRST_SND_VALUE
Address: Operational Base + offset (0x0098)
GLB_SRST_SND_VALUE

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	GLB_SRST_SND_VALUE The second global software reset config value The second global software reset config value

CRU_GLB_SRST_FST_VALUE
Address: Operational Base + offset (0x009c)

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GLB_SRST_FST_VALUE

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	GLB_SRST_FST_VALUE The first global software reset config value The first global software reset config value

CRU_NPLL_CON0

Address: Operational Base + offset (0x00a0)

NPLL configuration register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	bypass PLL Bypass. FREF bypasses PLL to FOUTPOSTDIV 1'b0: no bypass 1'b1: bypass
14:12	RW	0x3	postdiv1 First Post Divide Value (1-7)
11:0	RW	0x064	fbdiv Feedback Divide Value "Valid divider settings are: [16, 3200] in integer mode [20, 320] in fractional mode Tips: no plus one operation

CRU_NPLL_CON1

Address: Operational Base + offset (0x00a4)

NPLL configuration register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	pllpdsel PLL global power down source selection "If pllpdsel == 1, PLL can be power down only by pllpd1, otherwise pll is power down when any one of refdiv/fbdiv/fracdiv is changed or pllpd0 is asserted.
14	RW	0x0	pllpd1 PLL global power down request 1'b0: no power down 1'b2: power down

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Bit	Attr	Reset Value	Description
13	RW	0x0	pllpd0 PLL global power down request 1'b0: no power down 1'b1: power down
12	RW	0x1	dsmpd PLL delta sigma modulator enable " 1'b0: modulator is enable, 1'b1: modulator is disabled
11	RO	0x0	reserved
10	RO	0x0	pll_lock PLL lock status 1'b0: unlock 1'b1: lock
9	RO	0x0	reserved
8:6	RW	0x1	postdiv2 Second Post Divide Value (1-7)
5:0	RW	0x01	refdiv Reference Clock Divide Value (1-63)

CRU_NPLL_CON2
Address: Operational Base + offset (0x00a8)
NPLL configuration register2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	fout4phasepd "Power down 4-phase clocks and 2X, 3X, 4X clocks 1'b0: no power down 1'b1: power down
26	RW	0x0	foutvcopd Power down buffered VCO clock 1'b0: no power down 1'b1: power down
25	RW	0x0	foutpostdivpd Power down all outputs except for buffered VCO clock 1'b0: no power down 1'b1: power down
24	RW	0x0	dacpd Power down quantization noise cancellation DAC 1'b0: no power down 1'b1: power down
23:0	RW	0x000001	fracdiv Fractional part of feedback divide (fraction = FRAC/2^24)

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CRU_NPLL_CON3
Address: Operational Base + offset (0x00ac)
NPLL configuration register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x00	ssmod_spread spread amplitude % = 0.1 * SPREAD[4:0]
7:4	RW	0x0	ssmod_divval Divider required to set the modulation frequency Divider required to set the modulation frequency
3	RW	0x0	ssmod_downspread Selects center spread or downs pread 1'b0: down spread 1'b1: center spread
2	RW	0x1	ssmod_reset Reset modulator state 1'b0: no reset 1'b1: reset
1	RW	0x1	ssmod_disable_sscg Bypass SSMOD by module 1'b0: no bypass 1'b1: bypass
0	RW	0x1	ssmod_bp Bypass SSMOD by integration 1'b0: no bypass 1'b1: bypass

CRU_NPLL_CON4
Address: Operational Base + offset (0x00b0)
NPLL configuration register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RW	0x7f	ssmod_ext_maxaddr External wave table data inputs (0-255)
7:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	ssmod_sel_ext_wave select external wave 1'b0: no select ext_wave 1'b1: select ext_wave

CRU_CLKSEL_CON0
Address: Operational Base + offset (0x0100)
Internal clock select and divide register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14:13	RW	0x0	bus_aclk_pll_sel bus_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMIPHY 2'b11:reserved
12:8	RW	0x03	bus_aclk_div_con bus_aclk integer divider control register clk=clk_src/(div_con+1)
7:6	RW	0x0	core_clk_pll_sel core_clk pll source selection register 2'b00:APLL 2'b01:GPLL 2'b10:DPLL 2'b11:NPLL
5	RO	0x0	reserved
4:0	RW	0x00	clk_core_div_con Core A53 clock divider frequency clk=clk_src/(div_con+1)

CRU_CLKSEL_CON1
Address: Operational Base + offset (0x0104)
Internal clock select and divide register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
14:12	RW	0x1	bus_pclk_div_con bus_pclk integer divider control register clk=clk_src/(div_con+1)
11:10	RO	0x0	reserved
9:8	RW	0x1	bus_hclk_div_con bus_hclk integer divider control register clk=clk_src/(div_con+1)
7	RO	0x0	reserved
6:4	RW	0x1	aclk_core_div_con aclk_core integer divider control register clk=clk_src/(div_con+1)
3:0	RW	0x3	clk_core_dbg_div_con clk_core_dbg integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON2
Address: Operational Base + offset (0x0108)
Internal clock select and divide register2

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x00	func_24m_div_con func_24m integer divider control register clk=clk_src/(div_con+1)
7:5	RO	0x0	reserved
4:0	RW	0x03	test_div_con test integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON3
Address: Operational Base + offset (0x010c)
Internal clock select and divide register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	ddr_clk_pll_sel ddr_clk pll source selection register 2'b00:DPLL 2'b01:APLL 2'b10:CPLL 2'b11:reserved
7:3	RO	0x0	reserved
2:0	RW	0x0	ddr_div_cnt ddrphy reference clock divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON4
Address: Operational Base + offset (0x0110)
Internal clock select and divide register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14:13	RW	0x0	ddrpdclk_clk_pll_sel pd_ddr pclk source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMIPHY 2'b11:reserved
12:8	RW	0x07	pd_ddr_div_con pd_ddr pclk divider control register clk=clk_src/(div_con+1)
7:6	RW	0x2	otp_pll_sel otp pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC input 2'b11:reserved
5:0	RW	0x00	otp_div_con otp integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON5
Address: Operational Base + offset (0x0114)
Internal clock select and divide register5

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x2	efuse_pll_sel efuse pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC
13	RO	0x0	reserved
12:8	RW	0x00	efuse_div_con efuse integer divider control register clk=clk_src/(div_con+1)
7:0	RO	0x0	reserved

CRU_CLKSEL_CON6
Address: Operational Base + offset (0x0118)
Internal clock select and divide register6

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2s0_pll_sel i2s0 pll source selection register 1'b0:CPLL 1'b1:GPLL
14:10	RO	0x0	reserved
9:8	RW	0x0	i2s0_clk_sel i2s0 clk source selection register 2'b00: divout 2'b01: frac_divout 2'b10: 12M clkin 2'b11: 12M clkin
7	RO	0x0	reserved
6:0	RW	0x0f	i2s0_pll_div_con i2s0 integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON7
Address: Operational Base + offset (0x011c)
Internal clock select and divide register7

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	i2s0_frac_div_con i2s0 fraction divider control register High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL_CON8
Address: Operational Base + offset (0x0120)
Internal clock select and divide register8

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2s1_pll_sel i2s1 pll source selection register 1'b0:CPLL 1'b1:GPLL
14:13	RO	0x0	reserved
12	RW	0x0	i2s1_out_sel i2s1 output clock selection register 1'b0: clk_i2s1 1'b1: 12M
11:10	RO	0x0	reserved
9:8	RW	0x0	i2s1_clk_sel i2s1 clk source selection register 2'b00: divout 2'b01: frac_divout 2'b10: IO I2S1 clkin 2'b11: 12M clkin
7	RO	0x0	reserved
6:0	RW	0x0f	i2s1_pll_div_con i2s1 integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON9
Address: Operational Base + offset (0x0124)
Internal clock select and divide register9

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	i2s1_frac_div_con i2s1 fraction divider control register High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL_CON10
Address: Operational Base + offset (0x0128)

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Internal clock select and divide register10

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2s2_pll_sel i2s2 pll source selection register 1'b0:CPLL 1'b1:GPLL
14:13	RO	0x0	reserved
12	RW	0x0	i2s2_out_sel i2s2 output clock selection register 1'b0: clk_i2s2 1'b1: 12M
11:10	RO	0x0	reserved
9:8	RW	0x0	i2s2_clk_sel i2s2 clk source selection register 2'b00: divout 2'b01: frac_divout 2'b10: IO I2S2 clkin 2'b11: 12M clkin
7	RO	0x0	reserved
6:0	RW	0x0f	i2s2_pll_div_con i2s2 integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON11

Address: Operational Base + offset (0x012c)

Internal clock select and divide register11

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	i2s2_frac_div_con i2s2 fraction divider control register High 16-bit for numerator Low 16-bit for denominator

CRU_CLKSEL_CON12

Address: Operational Base + offset (0x0130)

Internal clock select and divide register12

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	spdif_pll_sel spdif pll source selection register 1'b0:CPLL 1'b1:GPLL
14:10	RO	0x0	reserved
9:8	RW	0x0	spdif_clk_sel spdif clock source selection register 2'b00: divout 2'b01: frac_divout 2'b10: 12M clkin 2'b11: 12M clkin
7	RO	0x0	reserved
6:0	RW	0x0f	spdif_pll_div_con spdif pll divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON13
Address: Operational Base + offset (0x0134)
Internal clock select and divide register13

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	spdif_frac_div_con spdif fraction divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON14
Address: Operational Base + offset (0x0138)
Internal clock select and divide register14

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:12	RW	0x0	uart0_pll_sel clk_uart0 pll source select control register 2'b00: select codec pll clock 2'b01: select general pll clock 2'b10: select USBPHY 480M clock
11:10	RO	0x0	reserved
9:8	RW	0x0	uart0_clk_sel clk_uart0 clock source select control register 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b10: select 24MHz from osc input 2'b11: select 24MHz from osc input

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Bit	Attr	Reset Value	Description
7	RO	0x0	reserved
6:0	RW	0x07	uart0_pll_div_con clk_uart0 divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON15
Address: Operational Base + offset (0x013c)
Internal clock select and divide register15

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart0_frac_div_con Control UART0 fraction divider frequency. High 16-bit for numerator Low 16-bit for denominator "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

CRU_CLKSEL_CON16
Address: Operational Base + offset (0x0140)
Internal clock select and divide register16

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:12	RW	0x0	uart1_pll_sel clk_uart1 pll source select control register 2'b00: select codec pll clock 2'b01: select general pll clock 2'b10: select USBPHY 480M clock
11:10	RO	0x0	reserved
9:8	RW	0x0	uart1_clk_sel clk_uart1 clock source select control register 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b10: select 24MHz from osc input 2'b11: select 24MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x07	uart1_pll_div_con clk_uart1 divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON17
Address: Operational Base + offset (0x0144)
Internal clock select and divide register17

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Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart1_frac_div_con Control uart1 fraction divider frequency. High 16-bit for numerator Low 16-bit for denominator "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

CRU_CLKSEL_CON18
Address: Operational Base + offset (0x0148)
Internal clock select and divide register18

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RO	0x0	reserved
13:12	RW	0x0	uart2_pll_sel clk_uart2 pll source select control register 2'b00: select codec pll clock 2'b01: select general pll clock 2'b10: select USBPHY 480M clock
11:10	RO	0x0	reserved
9:8	RW	0x0	uart2_clk_sel clk_uart2 clock source select control register 2'b00: select divider ouput from pll divider 2'b01: select divider ouput from fraction divider 2'b10: select 24MHz from osc input 2'b11: select 24MHz from osc input
7	RO	0x0	reserved
6:0	RW	0x07	uart2_pll_div_con clk_uart2 divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON19
Address: Operational Base + offset (0x014c)
Internal clock select and divide register19

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	uart2_frac_div_con Control uart2 fraction divider frequency. High 16-bit for numerator Low 16-bit for denominator "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

CRU_CLKSEL_CON20

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Address: Operational Base + offset (0x0150)
Internal clock select and divide register20

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x2	pdm_pll_sel pdm pll source selection register 2'd0: CPLL 2'd1: GPLL 2'd2: APLL 2'd3: Reserved
13	RO	0x0	reserved
12:8	RW	0x0f	pdm_div_con pdm integer divider control register clk=clk_src/(div_con+1)
7	RW	0x0	crypto_pll_sel crypto pll source selection register 1'b0:CPLL 1'b1:GPLL
6:5	RO	0x0	reserved
4:0	RW	0x04	crypto_div_con crypto integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON21
Address: Operational Base + offset (0x0154)
Internal clock select and divide register21

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	tsp_pll_sel tsp pll source selection register 1'b0:CPLL 1'b1:GPLL
14:13	RO	0x0	reserved
12:8	RW	0x04	tsp_div_con tsp integer divider control register clk=clk_src/(div_con+1)
7:0	RO	0x0	reserved

CRU_CLKSEL_CON22
Address: Operational Base + offset (0x0158)

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Internal clock select and divide register22

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:0	RW	0x1e0	tsadc_div_con tsadc integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON23

Address: Operational Base + offset (0x015c)

Internal clock select and divide register23

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:0	RW	0x1e0	saradc_div_con saradc integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON24

Address: Operational Base + offset (0x0160)

Internal clock select and divide register24

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	clkpwm_pll_sel clkpwm pll source selection register 1'b0:CPLL 1'b1:GPLL
14:8	RW	0x07	pwm0_div_con pwm0 integer divider control register clk=clk_src/(div_con+1)
7	RW	0x0	clkspi_pll_sel clkspi pll source selection register 1'b0:CPLL 1'b1:GPLL
6:0	RW	0x07	spi0_div_con spi0 integer divider control register clk=clk_src/(div_con+1)

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CRU_CLKSEL_CON25
Address: Operational Base + offset (0x0164)
Internal clock select and divide register25

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10:8	RW	0x2	gmac_pclk_div_con gmac_pclk integer divider control register clk=clk_src/(div_con+1)
7:6	RW	0x1	gmac_aclk_pll_sel gmac_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMIPHY 2'b11:reserved
5	RO	0x0	reserved
4:0	RW	0x02	gmac_aclk_div_con gmac_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON26
Address: Operational Base + offset (0x0168)
Internal clock select and divide register26

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	clk_gmac2phy_div_con clk_gmac2phy integer divider control register clk=clk_src/(div_con+1)
7	RW	0x0	gmac2phy_pll_sel gmac2phy pll source selection register 1'b0:CPLL 1'b1:GPLL
6:5	RO	0x0	reserved
4:0	RW	0x0f	gmac2phy_div_con gmac2phy integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON27

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Address: Operational Base + offset (0x016c)
Internal clock select and divide register27

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	gmac2io_out_pll_sel gmac2io_out pll source selection register 1'b0:CPLL 1'b1:GPLL
14:13	RO	0x0	reserved
12:8	RW	0x07	gmac2io_out_div_con gmac2io_out integer divider control register clk=clk_src/(div_con+1)
7	RW	0x0	gmac2io_pll_sel gmac2io pll source selection register 1'b0:CPLL 1'b1:GPLL
6:5	RO	0x0	reserved
4:0	RW	0x05	gmac2io_div_con gmac2io integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON28
Address: Operational Base + offset (0x0170)
Internal clock select and divide register28

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x1	periph_pll_sel periph pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMIPHY 2'b11:reserved
5	RO	0x0	reserved
4:0	RW	0x02	periph_aclk_div_con periph_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON29
Address: Operational Base + offset (0x0174)

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Internal clock select and divide register29

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:7	RO	0x0	reserved
6:4	RW	0x2	periph_pclk_div_con periph_pclk integer divider control register clk=clk_src/(div_con+1)
3:2	RO	0x0	reserved
1:0	RW	0x2	periph_hclk_div_con periph_hclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON30

Address: Operational Base + offset (0x0178)

Internal clock select and divide register30

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	clksdmmc_pll_sel clksdmmc pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC input 2'b11:USBPHY 480M
7:0	RW	0x03	sdmmc0_div_con sdmmc0 integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON31

Address: Operational Base + offset (0x017c)

Internal clock select and divide register31

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	clksdio_pll_sel clksdio pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC input 2'b11:USBPHY 480M
7:0	RW	0x01	sdio_div_con sdio integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON32
Address: Operational Base + offset (0x0180)
Internal clock select and divide register32

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	clkemmc_pll_sel clkemmc pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC input 2'b11:USBPHY 480M
7:0	RW	0x01	emmc_div_con emmc integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON33
Address: Operational Base + offset (0x0184)
Internal clock select and divide register33

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	usb3_otg0_suspend_src_sel clk_usb3_otg0_suspend pll source selection register 1'b0: OSC input 1'b1: 32k clock
14:10	RO	0x0	reserved
9:0	RW	0x30b	clk_usb3_otg0_suspend_div_con clk_usb3_otg0_suspend integer divider control register clk=clk_src/(div_con+1)

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CRU_CLKSEL_CON34
Address: Operational Base + offset (0x0188)
Internal clock select and divide register34

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2c1_pll_sel i2c1 pll source selection register 1'b0:CPLL 1'b1:GPLL
14:8	RW	0x07	i2c1_div_con i2c1 integer divider control register clk=clk_src/(div_con+1)
7	RW	0x0	i2c0_pll_sel i2c0 pll source selection register 1'b0:CPLL 1'b1:GPLL
6:0	RW	0x07	i2c0_div_con i2c0 integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON35
Address: Operational Base + offset (0x018c)
Internal clock select and divide register35

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2c3_pll_sel i2c3 pll source selection register 1'b0:CPLL 1'b1:GPLL
14:8	RW	0x07	i2c3_div_con i2c3 integer divider control register clk=clk_src/(div_con+1)
7	RW	0x0	i2c2_pll_sel i2c2 pll source selection register 1'b0:CPLL 1'b1:GPLL
6:0	RW	0x07	i2c2_div_con i2c2 integer divider control register clk=clk_src/(div_con+1)

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CRU_CLKSEL_CON36
Address: Operational Base + offset (0x0190)
Internal clock select and divide register36

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x1	rga_aclk_pll_sel rga_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
13	RO	0x0	reserved
12:8	RW	0x02	rga_aclk_div_con rga_aclk integer divider control register clk=clk_src/(div_con+1)
7:6	RW	0x1	rga_clk_pll_sel rga_clk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x02	rga_clk_div_con rga_clk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON37
Address: Operational Base + offset (0x0194)
Internal clock select and divide register37

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12:8	RW	0x02	hclk_vio_div_con hclk_vio integer divider control register clk=clk_src/(div_con+1)

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Bit	Attr	Reset Value	Description
7:6	RW	0x1	vio_aclk_pll_sel vio_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x02	vio_aclk_div_con vio_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON38
Address: Operational Base + offset (0x0198)
Internal clock select and divide register38

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x3	rtc32k_clk_pll_sel rtc32k_clk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC input 2'b11:Reserved
13:0	RW	0x02dc	rtc32k_clk_div_con rtc32k_clk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON39
Address: Operational Base + offset (0x019c)
Internal clock select and divide register39

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x0	vop_aclk_pll_sel vop_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x01	vop_aclk_div_con vop_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON40
Address: Operational Base + offset (0x01a0)
Internal clock select and divide register40

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RW	0x31	vop_dclk_div_con vop_dclk integer divider control register clk=clk_src/(div_con+1)
7:6	RO	0x0	reserved
5:3	RW	0x2	hdmiphy_div_con hdmiphy integer divider control register clk=clk_src/(div_con+1)
2	RW	0x0	vop_dclk_frac_sel vop divider source selection register 1'b0: divout 1'b1: frac_divout
1	RW	0x1	vop_dclk_src_sel vop dclk source selection register 1'b0:HDMIPHY 1'b2:PLL
0	RW	0x1	vop_dclk_pll_src_sel vop dclk pll source selection register 1'b0:CPLL 1'b1:GPLL

CRU_CLKSEL_CON41
Address: Operational Base + offset (0x01a4)
Internal clock select and divide register41

Bit	Attr	Reset Value	Description
31:0	RW	0x0bb8ea60	dclk_vop_frac_div_con dclk_vop fraction divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON42
Address: Operational Base + offset (0x01a8)
Internal clock select and divide register42

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7	RW	0x0	cif_pll_sel cif pll source selection register 1'b0:HDMIPLL 1'b1:GPLL
6	RO	0x0	reserved
5	RW	0x0	cif_clk_sel cif clk source selection register 1'b0:PLL 1'b1:OSC input
4:0	RW	0x13	cif_div_con cif integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON43
Address: Operational Base + offset (0x01ac)
Internal clock select and divide register43

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:10	RO	0x0	reserved
9:8	RW	0x0	clksdmmcext_pll_sel clksdmmcext pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:OSC input 2'b11:USBPHY 480M
7:0	RW	0x03	sdmmcext_div_con sdmmcext integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON44
Address: Operational Base + offset (0x01b0)
Internal clock select and divide register44

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15:8	RO	0x0	reserved
7:6	RW	0x1	gpu_aclk_pll_sel gpu_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x02	gpu_aclk_div_con gpu_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON45
Address: Operational Base + offset (0x01b4)
Internal clock select and divide register45

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	clk_usb3phy_ref_sel usb3phy_ref clock source selection register 1'b0:OSC input 1'b1:PLL
7	RW	0x0	usb3phy_ref_pll_sel usb3phy_ref pll source selection register 1'b0:CPLL 1'b1:GPLL
6:0	RW	0x1f	usb3phy_ref_div_con usb3phy_ref integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON46
Address: Operational Base + offset (0x01b8)
Internal clock select and divide register46

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	Reserve write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:0	RO	0x0	reserved

CRU_CLKSEL_CON47
Address: Operational Base + offset (0x01bc)
Internal clock select and divide register47

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	Reserve write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:0	RO	0x0	reserved

CRU_CLKSEL_CON48
Address: Operational Base + offset (0x01c0)
Internal clock select and divide register48

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x1	cabac_clk_pll_sel cabac_clk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
13	RO	0x0	reserved
12:8	RW	0x02	cabac_clk_div_con cabac_clk integer divider control register clk=clk_src/(div_con+1)
7:6	RW	0x0	rkvdec_aclk_pll_sel rkvdec_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x01	rkvdec_aclk_div_con rkvdec_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON49
Address: Operational Base + offset (0x01c4)
Internal clock select and divide register49

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:6	RW	0x1	vdec_clk_pll_sel vdec_clk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x02	vdec_clk_div_con vdec_clk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON50
Address: Operational Base + offset (0x01c8)
Internal clock select and divide register50

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7:6	RW	0x1	vpu_aclk_pll_sel vpu_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x02	vpu_aclk_div_con vpu_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON51
Address: Operational Base + offset (0x01cc)
Internal clock select and divide register51

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x0	h265_core_clk_pll_sel h265_core_clk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M

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Bit	Attr	Reset Value	Description
13	RO	0x0	reserved
12:8	RW	0x02	h265_core_clk_div_con h265_core_clk integer divider control register clk=clk_src/(div_con+1)
7:6	RW	0x0	rkvenc_aclk_pll_sel rkvenc_aclk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
5	RO	0x0	reserved
4:0	RW	0x03	rkvenc_aclk_div_con rkvenc_aclk integer divider control register clk=clk_src/(div_con+1)

CRU_CLKSEL_CON52
Address: Operational Base + offset (0x01d0)
Internal clock select and divide register52

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:14	RW	0x0	h265_dsp_clk_pll_sel h265_dsp_clk pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b10:HDMI PHY 2'b11:USBPHY 480M
13	RO	0x0	reserved
12:8	RW	0x02	h265_dsp_clk_div_con h265_dsp_clk integer divider control register clk=clk_src/(div_con+1)
7:6	RW	0x0	wifi_pll_sel wifi pll source selection register 2'b00:CPLL 2'b01:GPLL 2'b11:USBPHY 480M 2'b11:Reserved
5:0	RW	0x1e	wifi_div_con wifi integer divider control register clk=clk_src/(div_con+1)

CRU_CLKGATE_CON0
Address: Operational Base + offset (0x0200)

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Internal clock gating register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:13	RO	0x0	reserved
12	RW	0x0	core_npll_clk_en core_npll clk gate enable register When HIGH, disable clock
11	RW	0x0	clk_rtc32k_src_en clk_rtc32k clk gate enable register "When HIGH, disable clock
10	RW	0x0	clk_wifi_src_en clk_wifi clk gate enable register "When HIGH, disable clock
9	RW	0x0	testclk_en tes clk gate enable register "When HIGH, disable clock
8:7	RO	0x0	reserved
6	RW	0x0	clk_ddrmon_en clk_ddrmon clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_ddrpd_src_en clk_ddrpd clk gate enable register When HIGH, disable clock
4	RW	0x0	clk_ddrphy_src_en clk_ddrphy clk gate enable register When HIGH, disable clock
3	RW	0x0	bus_src_clk_en bus_src clk gate enable register When HIGH, disable clock
2	RW	0x0	core_gpll_clk_en core_gpll clk gate enable register When HIGH, disable clock
1	RW	0x0	core_dpll_clk_en core_dpll clk gate enable register When HIGH, disable clock
0	RW	0x0	core_apll_clk_en core_apll clk gate enable register When HIGH, disable clock

CRU_CLKGATE_CON1

Address: Operational Base + offset (0x0204)

Internal clock gating register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	clk_uart0_frac_src_en clk_uart0_frac clk gate enable register "When HIGH, disable clock
14	RW	0x0	clk_uart0_src_en clk_uart0 clk gate enable register "When HIGH, disable clock
13	RW	0x0	clk_spdif_frac_src_en clk_spdif_frac clk gate enable register "When HIGH, disable clock
12	RW	0x0	clk_spdif_src_en clk_spdif clk gate enable register "When HIGH, disable clock
11	RW	0x0	clk_i2s2_out_en clk_i2s2_out clk gate enable register "When HIGH, disable clock
10	RW	0x0	clk_i2s2_en clk_i2s2 clk gate enable register "When HIGH, disable clock
9	RW	0x0	clk_i2s2_frac_src_en clk_i2s2_frac clk gate enable register "When HIGH, disable clock
8	RW	0x0	clk_i2s2_src_en clk_i2s2 clk gate enable register "When HIGH, disable clock
7	RW	0x0	clk_i2s1_out_en clk_i2s1_out clk gate enable register "When HIGH, disable clock
6	RW	0x0	clk_i2s1_en clk_i2s1 clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_i2s1_frac_src_en clk_i2s1_frac clk gate enable register "When HIGH, disable clock
4	RW	0x0	clk_i2s1_src_en clk_i2s1 clk gate enable register "When HIGH, disable clock
3	RW	0x0	clk_i2s0_en clk_i2s0 clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
2	RW	0x0	clk_i2s0_frac_src_en clk_i2s0_frac clk gate enable register "When HIGH, disable clock
1	RW	0x0	clk_i2s0_src_en clk_i2s0 clk gate enable register "When HIGH, disable clock
0	RO	0x0	reserved

CRU_CLKGATE_CON2

Address: Operational Base + offset (0x0208)

Internal clock gating register2

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	clk_pdm_src_en clk_pdm clk gate enable register "When HIGH, disable clock
14	RW	0x0	clk_saradc_src_en clk_saradc clk gate enable register "When HIGH, disable clock
13	RW	0x0	clk_efuse_src_en clk_efuse clk gate enable register "When HIGH, disable clock
12	RW	0x0	clk_i2c3_src_en clk_i2c3 clk gate enable register "When HIGH, disable clock
11	RW	0x0	clk_i2c2_src_en clk_i2c2 clk gate enable register "When HIGH, disable clock
10	RW	0x0	clk_i2c1_src_en clk_i2c1 clk gate enable register "When HIGH, disable clock
9	RW	0x0	clk_i2c0_src_en clk_i2c0 clk gate enable register "When HIGH, disable clock
8	RW	0x0	clk_pwm0_src_en clk_pwm0 clk gate enable register "When HIGH, disable clock
7	RW	0x0	clk_spi0_src_en clk_spi0 clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
6	RW	0x0	clk_tsadc_src_en clk_tsadc clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_tsp_src_en clk_tsp clk gate enable register "When HIGH, disable clock
4	RW	0x0	clk_crypto_src_en clk_crypto clk gate enable register "When HIGH, disable clock
3	RW	0x0	clk_uart2_frac_src_en clk_uart2_frac clk gate enable register "When HIGH, disable clock
2	RW	0x0	clk_uart2_src_en clk_uart2 clk gate enable register "When HIGH, disable clock
1	RW	0x0	clk_uart1_frac_src_en clk_uart1_frac clk gate enable register "When HIGH, disable clock
0	RW	0x0	clk_uart1_src_en clk_uart1 clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON3
Address: Operational Base + offset (0x020c)
Internal clock gating register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	clk_otp_src_en clk_otp clk gate enable register "When HIGH, disable clock
7:6	RO	0x0	reserved
5	RW	0x0	clk_gmac2io_out_en clk_gmac2io_out clk gate enable register "When HIGH, disable clock
4	RW	0x0	gmac_vpll_src_en gmac_vpll clk gate enable register "When HIGH, disable clock
3	RW	0x0	gmac_gpll_src_en gmac_gpll clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
2	RW	0x0	gmac_cpll_src_en gmac_cpll clk gate enable register "When HIGH, disable clock
1	RW	0x0	clk_gmac2io_src_en clk_gmac2io clk gate enable register "When HIGH, disable clock
0	RW	0x0	clk_gmac2phy_src_en clk_gmac2phy clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON4
Address: Operational Base + offset (0x0210)
Internal clock gating register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	RW	0x0	clk_sdmmcext_src_en clk_sdmmcext clk gate enable register "When HIGH, disable clock
9	RW	0x0	clk_usb3phy_ref_25m_en clk_usb3phy_ref_25m clk gate enable register "When HIGH, disable clock
8	RW	0x0	clk_usb3_otg0_suspend_en clk_usb3_otg0_suspend clk gate enable register "When HIGH, disable clock
7	RW	0x0	clk_usb3_otg0_ref_en clk_usb3_otg0_ref clk gate enable register "When HIGH, disable clock
6	RW	0x0	clk_otgphy0_en clk_otgphy0 clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_emmc_src_en clk_emmc clk gate enable register "When HIGH, disable clock
4	RW	0x0	clk_sdio_src_en clk_sdio clk gate enable register "When HIGH, disable clock
3	RW	0x0	clk_mmc0_src_en clk_mmc0 clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
2	RW	0x0	periph_vclk_src_en periph_vclk clk gate enable register "When HIGH, disable clock
1	RW	0x0	periph_cclk_src_en periph_cclk clk gate enable register "When HIGH, disable clock
0	RW	0x0	periph_gclk_src_en periph_gclk clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON5
Address: Operational Base + offset (0x0214)
Internal clock gating register5

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:7	RO	0x0	reserved
6	RW	0x0	dclk_vop_src_en dclk_vop clk gate enable register "When HIGH, disable clock
5	RW	0x0	aclk_vop_src_en aclk_vop clk gate enable register "When HIGH, disable clock
4	RW	0x0	clk_hdmi_sfr_en clk_hdmi_sfr clk gate enable register "When HIGH, disable clock
3	RW	0x0	clk_cif_out_src_en clk_cif_out clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_vio_src_en aclk_vio clk gate enable register "When HIGH, disable clock
1	RW	0x0	clk_rga_src_en clk_rga clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_rga_src_en aclk_rga clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON6
Address: Operational Base + offset (0x0218)
Internal clock gating register6

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7	RW	0x0	clk_venc_dsp_src_en clk_venc_dsp clk gate enable register "When HIGH, disable clock
6	RW	0x0	aclk_gpu_src_en aclk_gpu clk gate enable register "When HIGH, disable clock
5	RW	0x0	aclk_vpu_src_en aclk_vpu clk gate enable register "When HIGH, disable clock
4	RW	0x0	clk_venc_core_src_en clk_venc_core clk gate enable register "When HIGH, disable clock
3	RW	0x0	aclk_rkvenc_src_en aclk_rkvenc clk gate enable register "When HIGH, disable clock
2	RW	0x0	clk_vdec_core_src_en clk_vdec_core clk gate enable register "When HIGH, disable clock
1	RW	0x0	clk_cabac_src_en clk_cabac clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_rkvdec_src_en aclk_rkvdec clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON7
Address: Operational Base + offset (0x021c)
Internal clock gating register7

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:5	RO	0x0	reserved
4	RW	0x0	pclk_ddr_en pclk_ddr clk gate enable register "When HIGH, disable clock
3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x0	clk_jtag_en core jtag clock enable "When HIGH, disable clock
1	RW	0x0	clk_core_periph_en clk_core_periph clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_core_en aclk_core clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON8
Address: Operational Base + offset (0x0220)
Internal clock gating register8

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	RW	0x0	clk_timer5_en clk_timer5 clk gate enable register "When HIGH, disable clock
9	RW	0x0	clk_timer4_en clk_timer4 clk gate enable register "When HIGH, disable clock
8	RW	0x0	clk_timer3_en clk_timer3 clk gate enable register "When HIGH, disable clock
7	RW	0x0	clk_timer2_en clk_timer2 clk gate enable register "When HIGH, disable clock
6	RW	0x0	clk_timer1_en clk_timer1 clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_timer0_en clk_timer0 clk gate enable register "When HIGH, disable clock
4	RW	0x0	pclk_phy_en pclk_phy clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_bus_en pclk_bus clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
2	RW	0x0	pclk_bus_src_en pclk_bus clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_bus_en hclk_bus clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_bus_en aclk_bus clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON9

Address: Operational Base + offset (0x0224)

Internal clock gating register9

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7	RW	0x0	clk_gmac2io_ref_en clk_gmac2io_ref clk gate enable register "When HIGH, disable clock
6	RW	0x0	clk_gmac2io_refout_en clk_gmac2io_refout clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_gmac2io_tx_en clk_gmac2io_tx clk gate enable register "When HIGH, disable clock
4	RW	0x0	clk_gmac2io_rx_en clk_gmac2io_rx clk gate enable register "When HIGH, disable clock
3	RW	0x0	clk_gmac2phy_ref_en clk_gmac2phy_ref clk gate enable register "When HIGH, disable clock
2	RW	0x0	clk_macphy_en clk_macphy clk gate enable register "When HIGH, disable clock
1	RW	0x0	clk_gmac2phy_rx_en clk_gmac2phy_rx clk gate enable register "When HIGH, disable clock
0	RW	0x0	pclk_gmac_en pclk_gmac clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON10

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Address: Operational Base + offset (0x0228)
Internal clock gating register10

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:3	RO	0x0	reserved
2	RW	0x0	pclk_periph_en pclk_periph clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_periph_en hclk_periph clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_periph_en aclk_periph clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON11
Address: Operational Base + offset (0x022c)
Internal clock gating register11

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	hclk_vpu_en hclk_vpu clk gate enable register "When HIGH, disable clock
7:5	RO	0x0	reserved
4	RW	0x0	hclk_rkvenc_en hclk_rkvenc clk gate enable register "When HIGH, disable clock
3:1	RO	0x0	reserved
0	RW	0x0	hclk_rkvdec_en hclk_rkvdec clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON12
Address: Operational Base + offset (0x0230)
Internal clock gating register12

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Reserve write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

CRU_CLKGATE_CON13
Address: Operational Base + offset (0x0234)
Internal clock gating register13

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:2	RO	0x0	reserved
1	RW	0x0	aclk_gic400_en aclk_gic400 clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_core_niu_en aclk_core_niu clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON14
Address: Operational Base + offset (0x0238)
Internal clock gating register14

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:2	RO	0x0	reserved
1	RW	0x0	aclk_gpu_niu_en aclk_gpu_niu clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_gpu_en aclk_gpu clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON15
Address: Operational Base + offset (0x023c)
Internal clock gating register15

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

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Bit	Attr	Reset Value	Description
15	RW	0x0	pclk_phy_niu_en pclk_phy_niu clk gate enable register "When HIGH, disable clock
14	RW	0x0	pclk_bus_niu_en pclk_bus_niu clk gate enable register "When HIGH, disable clock
13	RW	0x0	hclk_bus_niu_en hclk_bus_niu clk gate enable register "When HIGH, disable clock
12	RW	0x0	aclk_bus_niu_en aclk_bus_niu clk gate enable register "When HIGH, disable clock
11	RW	0x0	aclk_dcf_en aclk_dcf clk gate enable register "When HIGH, disable clock
10	RW	0x0	pclk_i2c0_en pclk_i2c0 clk gate enable register "When HIGH, disable clock
9	RW	0x0	pclk_efuse_1024_en pclk_efuse_1024 clk gate enable register "When HIGH, disable clock
8	RW	0x0	sclk_crypto_en sclk_crypto clk gate enable register "When HIGH, disable clock
7	RW	0x0	mclk_crypto_en mclk_crypto clk gate enable register "When HIGH, disable clock
6	RW	0x0	hclk_spdif_8ch_en hclk_spdif_8ch clk gate enable register "When HIGH, disable clock
5	RW	0x0	hclk_i2s2_2ch_en hclk_i2s2_2ch clk gate enable register "When HIGH, disable clock
4	RW	0x0	hclk_i2s1_8ch_en hclk_i2s1_8ch clk gate enable register "When HIGH, disable clock
3	RW	0x0	hclk_i2s0_8ch_en hclk_i2s0_8ch clk gate enable register "When HIGH, disable clock
2	RW	0x0	hclk_rom_en hclk_rom clk gate enable register "When HIGH, disable clock
1	RW	0x0	aclk_dmac_bus_en aclk_dmac_bus clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
0	RW	0x0	aclk_intmem_en aclk_intmem clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON16
Address: Operational Base + offset (0x0240)
Internal clock gating register16

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	pclk_dcf_en pclk_dcf clk gate enable register "When HIGH, disable clock
14	RW	0x0	pclk_tsadc_en pclk_tsadc clk gate enable register "When HIGH, disable clock
13	RW	0x0	pclk_uart2_en pclk_uart2 clk gate enable register "When HIGH, disable clock
12	RW	0x0	pclk_uart1_en pclk_uart1 clk gate enable register "When HIGH, disable clock
11	RW	0x0	pclk_uart0_en pclk_uart0 clk gate enable register "When HIGH, disable clock
10	RW	0x0	pclk_gpio3_en pclk_gpio3 clk gate enable register "When HIGH, disable clock
9	RW	0x0	pclk_gpio2_en pclk_gpio2 clk gate enable register "When HIGH, disable clock
8	RW	0x0	pclk_gpio1_en pclk_gpio1 clk gate enable register "When HIGH, disable clock
7	RW	0x0	pclk_gpio0_en pclk_gpio0 clk gate enable register "When HIGH, disable clock
6	RW	0x0	pclk_rk_pwm_en pclk_rk_pwm clk gate enable register "When HIGH, disable clock
5	RW	0x0	pclk_spi0_en pclk_spi0 clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
4	RW	0x0	pclk_stimer_en pclk_stimer clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_timer0_en pclk_timer0 clk gate enable register "When HIGH, disable clock
2	RW	0x0	pclk_i2c3_en pclk_i2c3 clk gate enable register "When HIGH, disable clock
1	RW	0x0	pclk_i2c2_en pclk_i2c2 clk gate enable register "When HIGH, disable clock
0	RW	0x0	pclk_i2c1_en pclk_i2c1 clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON17
Address: Operational Base + offset (0x0244)
Internal clock gating register17

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	pclk_saradc_en pclk_saradc clk gate enable register "When HIGH, disable clock
14	RW	0x0	pclk_usb_grf_en pclk_usb_grf clk gate enable register "When HIGH, disable clock
13	RW	0x0	clk_hsadc_0_tsp_en clk_hsadc_0_tsp clk gate enable register "When HIGH, disable clock
12	RW	0x0	aclk_tsp_en aclk_tsp clk gate enable register "When HIGH, disable clock
11	RW	0x0	hclk_tsp_en hclk_tsp clk gate enable register "When HIGH, disable clock
10	RW	0x0	pclk_scr_en pclk_scr clk gate enable register "When HIGH, disable clock
9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8	RW	0x0	pclk_vdacphy_en pclk_vdacphy clk gate enable register "When HIGH, disable clock
7	RW	0x0	pclk_hdmiphy_en pclk_hdmiphy clk gate enable register "When HIGH, disable clock
6	RW	0x0	pclk_sgrf_en pclk_sgrf clk gate enable register "When HIGH, disable clock
5	RW	0x0	pclk_acodecphy_en pclk_acodecphy clk gate enable register "When HIGH, disable clock
4	RW	0x0	pclk_cru_en pclk_cru clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_ddrphy_en pclk_ddrphy clk gate enable register "When HIGH, disable clock
2	RW	0x0	pclk_usb3grf_en pclk_usb3grf clk gate enable register "When HIGH, disable clock
1	RO	0x0	reserved
0	RW	0x0	pclk_grf_en pclk_grf clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON18

Address: Operational Base + offset (0x0248)

Internal clock gating register18

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:8	RO	0x0	reserved
7	RW	0x0	pclk_ddrstdby_en pclk_ddrstdby clk gate enable register "When HIGH, disable clock
6	RW	0x0	clk_ddr_msch_en clk_ddr_msch clk gate enable register "When HIGH, disable clock
5	RW	0x0	clk_ddr_upctl_en clk_ddr_upctl clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
4	RW	0x0	aclk_ddr_upctl_en aclk_ddr_upctl clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_ddr_mon_en pclk_ddr_mon clk gate enable register "When HIGH, disable clock
2	RW	0x0	pclk_ddr_msch_en pclk_ddr_msch clk gate enable register "When HIGH, disable clock
1	RW	0x0	pclk_ddr_upctl_en pclk_ddr_upctl clk gate enable register "When HIGH, disable clock
0	RO	0x0	reserved

CRU_CLKGATE_CON19

Address: Operational Base + offset (0x024c)

Internal clock gating register19

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	hclk_sdmmc_ext_en hclk_sdmmc_ext hclk gate enable register "When HIGH, disable clock
14	RW	0x0	aclk_usb3otg_en aclk_usb3otg clk gate enable register "When HIGH, disable clock
13	RW	0x0	pclk_peri_niu_en pclk_peri_niu clk gate enable register "When HIGH, disable clock
12	RW	0x0	hclk_peri_niu_en hclk_peri_niu clk gate enable register "When HIGH, disable clock
11	RW	0x0	aclk_peri_niu_en aclk_peri_niu clk gate enable register "When HIGH, disable clock
10	RO	0x0	reserved
9	RW	0x0	hclk_otg_pmu_en hclk_otg_pmu clk gate enable register "When HIGH, disable clock
8	RW	0x0	hclk_otg_en hclk_otg clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
7	RW	0x0	hclk_host0_arb_en hclk_host0_arb clk gate enable register "When HIGH, disable clock
6	RW	0x0	hclk_host0_en hclk_host0 clk gate enable register "When HIGH, disable clock
5:3	RO	0x0	reserved
2	RW	0x0	hclk_emmc_en hclk_emmc clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_sdio_en hclk_sdio clk gate enable register "When HIGH, disable clock
0	RW	0x0	hclk_sdmmc_en hclk_sdmmc clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON20
Address: Operational Base + offset (0x0250)
Internal clock gating register20

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Reserve write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit

CRU_CLKGATE_CON21
Address: Operational Base + offset (0x0254)
Internal clock gating register21

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	aclk_hdcp_en aclk_hdcp clk gate enable register "When HIGH, disable clock
14	RW	0x0	hclk_h2p_en hclk_h2p clk gate enable register "When HIGH, disable clock
13	RW	0x0	pclk_h2p_en pclk_h2p clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
12	RW	0x0	hclk_ahb1tom_en hclk_ahb1tom clk gate enable register "When HIGH, disable clock
11	RW	0x0	hclk_rga_en hclk_rga clk gate enable register "When HIGH, disable clock
10	RW	0x0	aclk_rga_en aclk_rga clk gate enable register "When HIGH, disable clock
9	RW	0x0	hclk_cif_en hclk_cif clk gate enable register "When HIGH, disable clock
8	RW	0x0	aclk_cif_en aclk_cif clk gate enable register "When HIGH, disable clock
7	RW	0x0	hclk_iep_en hclk_iep clk gate enable register "When HIGH, disable clock
6	RW	0x0	aclk_iep_en aclk_iep clk gate enable register "When HIGH, disable clock
5	RW	0x0	hclk_vop_noc_en hclk_vop_noc clk gate enable register "When HIGH, disable clock
4	RW	0x0	aclk_vop_noc_en aclk_vop_noc clk gate enable register "When HIGH, disable clock
3	RW	0x0	hclk_vop_en hclk_vop clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_vop_en aclk_vop clk gate enable register "When HIGH, disable clock
1:0	RO	0x0	reserved

CRU_CLKGATE_CON22

Address: Operational Base + offset (0x0258)

Internal clock gating register22

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	pclk_hdcp_ctrl_en pclk_hdcp_ctrl clk gate enable register "When HIGH, disable clock
4	RW	0x0	pclk_hdmi_ctrl_en pclk_hdmi_ctrl clk gate enable register "When HIGH, disable clock
3	RW	0x0	aclk_rga_noc_en aclk_rga_noc clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_vio_noc_en aclk_vio_noc clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_vio_noc_en hclk_vio_noc clk gate enable register "When HIGH, disable clock
0	RW	0x0	hclk_hdcp_en hclk_hdcp clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON23
Address: Operational Base + offset (0x025c)
Internal clock gating register23

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:4	RO	0x0	reserved
3	RW	0x0	hclk_vpu_niu_en hclk_vpu_niu clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_vpu_niu_en aclk_vpu_niu clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_vpu_en hclk_vpu clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_vpu_en aclk_vpu clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON24
Address: Operational Base + offset (0x0260)
Internal clock gating register24

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:4	RO	0x0	reserved
3	RW	0x0	hclk_rkvdec_niu_en hclk_rkvdec_niu clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_rkvdec_niu_en aclk_rkvdec_niu clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_rkvdec_en hclk_rkvdec clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_rkvdec_en aclk_rkvdec clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON25
Address: Operational Base + offset (0x0264)
Internal clock gating register25

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:7	RO	0x0	reserved
6	RW	0x0	aclk_axi2sram_en axi2sram clk gate enable register "When HIGH, disable clock
5	RW	0x0	hclk_h264_en hclk_h264 clk gate enable register "When HIGH, disable clock
4	RW	0x0	aclk_h264_en aclk_h264 clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_h265_en pclk_h265 clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_h265_en aclk_h265 clk gate enable register "When HIGH, disable clock
1	RW	0x0	hclk_rkvenc_niu_en hclk_rkvenc_niu clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
0	RW	0x0	aclk_rkvenc_niu_en aclk_rkvenc_niu clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON26
Address: Operational Base + offset (0x0268)
Internal clock gating register26

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:6	RO	0x0	reserved
5	RW	0x0	pclk_gmac_niu_en pclk_gmac_niu clk gate enable register "When HIGH, disable clock
4	RW	0x0	aclk_gmac_niu_en aclk_gmac_niu clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_gmac2io_en pclk_gmac2io clk gate enable register "When HIGH, disable clock
2	RW	0x0	aclk_gmac2io_en aclk_gmac2io clk gate enable register "When HIGH, disable clock
1	RW	0x0	pclk_gmac2phy_en pclk_gmac2phy clk gate enable register "When HIGH, disable clock
0	RW	0x0	aclk_gmac2phy_en aclk_gmac2phy clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON27
Address: Operational Base + offset (0x026c)
Internal clock gating register27

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:2	RO	0x0	reserved
1	RW	0x0	clk4x_ddrphy_en clk4x_ddrphy clk gate enable register "When HIGH, disable clock

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Bit	Attr	Reset Value	Description
0	RW	0x0	clk_ddrphy_en clk_ddrphy clk gate enable register "When HIGH, disable clock

CRU_CLKGATE_CON28
Address: Operational Base + offset (0x0270)
Internal clock gating register28

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:5	RO	0x0	reserved
4	RW	0x0	pclk_otp_en pclk_otp clk gate enable register "When HIGH, disable clock
3	RW	0x0	pclk_pmu_en pclk_pmu clk gate enable register "When HIGH, disable clock
2	RW	0x0	pclk_usb3phy_pipe_en pclk_usb3phy_pipe clk gate enable register "When HIGH, disable clock
1	RW	0x0	pclk_usb3phy_otg_en pclk_usb3phy_otg clk gate enable register "When HIGH, disable clock
0	RW	0x0	hclk_pdm_en hclk_pdm clk gate enable register "When HIGH, disable clock

CRU_SSGTBL0_3
Address: Operational Base + offset (0x0280)
SSMOD external wave table register0

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl0_3 Extern wave table 0-3 7-0: table0 15-8: table1 23-16: table2 31-24: table3

CRU_SSGTBL4_7
Address: Operational Base + offset (0x0284)
SSMOD external wave table register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl4_7 Extern wave table 4-7 7-0: table4 15-8: table5 23-16: table6 31-24: table7

CRU_SSGTBL8_11
Address: Operational Base + offset (0x0288)
SSMOD external wave table register2

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl8_11 Extern wave table 8-11 7-0: table8 15-8: table9 23-16: table10 31-24: table11

CRU_SSGTBL12_15
Address: Operational Base + offset (0x028c)
SSMOD external wave table register3

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl12_15 Extern wave table 12-15 7-0: table12 15-8: table13 23-16: table14 31-24: table15

CRU_SSGTBL16_19
Address: Operational Base + offset (0x0290)
SSMOD external wave table register4

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl16_19 Extern wave table 16-19 7-0: table16 15-8: table17 23-16: table18 31-24: table19

CRU_SSGTBL20_23
Address: Operational Base + offset (0x0294)
SSMOD external wave table register5

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl20_23 Extern wave table 20-23 7-0: table20 15-8: table21 23-16: table22 31-24: table23

CRU_SSGTBL24_27
Address: Operational Base + offset (0x0298)
SSMOD external wave table register6

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl24_27 Extern wave table 24-27 7-0: table24 15-8: table25 23-16: table26 31-24: table27

CRU_SSGTBL28_31
Address: Operational Base + offset (0x029c)
SSMOD external wave table register7

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl28_31 Extern wave table 28-31 7-0: table28 15-8: table29 23-16: table30 31-24: table31

CRU_SSGTBL32_35
Address: Operational Base + offset (0x02a0)
SSMOD external wave table register8

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl32_35 Extern wave table 32-35 7-0: table32 15-8: table33 23-16: table34 31-24: table35

CRU_SSGTBL36_39
Address: Operational Base + offset (0x02a4)
SSMOD external wave table register9

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl36_39 Extern wave table 36-39 7-0: table36 15-8: table37 23-16: table38 31-24: table39

CRU_SSGTBL40_43
Address: Operational Base + offset (0x02a8)
SSMOD external wave table register10

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl40_43 Extern wave table 40-43 7-0: table40 15-8: table41 23-16: table42 31-24: table43

CRU_SSGTBL44_47
Address: Operational Base + offset (0x02ac)
SSMOD external wave table register11

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl44_47 Extern wave table 44-47 7-0: table44 15-8: table45 23-16: table46 31-24: table47

CRU_SSGTBL48_51
Address: Operational Base + offset (0x02b0)
SSMOD external wave table register12

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl48_51 Extern wave table 48-51 7-0: table48 15-8: table49 23-16: table50 31-24: table51

CRU_SSGTBL52_55
Address: Operational Base + offset (0x02b4)
SSMOD external wave table register13

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl52_55 Extern wave table 52-55 7-0: table52 15-8: table53 23-16: table54 31-24: table55

CRU_SSGTBL56_59
Address: Operational Base + offset (0x02b8)
SSMOD external wave table register14

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl56_59 Extern wave table 56-59 7-0: table56 15-8: table57 23-16: table58 31-24: table59

CRU_SSGTBL60_63
Address: Operational Base + offset (0x02bc)
SSMOD external wave table register15

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl60_63 Extern wave table 60-63 7-0: table60 15-8: table61 23-16: table62 31-24: table63

CRU_SSGTBL64_67
Address: Operational Base + offset (0x02c0)
SSMOD external wave table register16

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl64_67 Extern wave table 64-67 7-0: table64 15-8: table65 23-16: table66 31-24: table67

CRU_SSGTBL68_71
Address: Operational Base + offset (0x02c4)
SSMOD external wave table register17

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl68_71 Extern wave table 68-71 7-0: table68 15-8: table69 23-16: table70 31-24: table71

CRU_SSGTBL72_75
Address: Operational Base + offset (0x02c8)
SSMOD external wave table register18

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl72_75 Extern wave table 72-75 7-0: table72 15-8: table73 23-16: table74 31-24: table75

CRU_SSGTBL76_79
Address: Operational Base + offset (0x02cc)
SSMOD external wave table register19

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl76_79 Extern wave table 76-79 7-0: table76 15-8: table77 23-16: table78 31-24: table79

CRU_SSGTBL80_83
Address: Operational Base + offset (0x02d0)
SSMOD external wave table register20

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl80_83 Extern wave table 80-83 7-0: table80 15-8: table81 23-16: table82 31-24: table83

CRU_SSGTBL84_87
Address: Operational Base + offset (0x02d4)
SSMOD external wave table register21

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl84_87 Extern wave table 84-87 7-0: table84 15-8: table85 23-16: table86 31-24: table87

CRU_SSGTBL88_91
Address: Operational Base + offset (0x02d8)
SSMOD external wave table register22

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl88_91 Extern wave table 88-91 7-0: table88 15-8: table89 23-16: table90 31-24: table91

CRU_SSGTBL92_95
Address: Operational Base + offset (0x02dc)
SSMOD external wave table register23

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl92_95 Extern wave table 92-95 7-0: table92 15-8: table93 23-16: table94 31-24: table95

CRU_SSGTBL96_99
Address: Operational Base + offset (0x02e0)
SSMOD external wave table register24

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl96_99 Extern wave table 96-99 7-0: table96 15-8: table97 23-16: table98 31-24: table99

CRU_SSGTBL100_103
Address: Operational Base + offset (0x02e4)
SSMOD external wave table register25

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl100_103 Extern wave table 100-103 7-0: table100 15-8: table101 23-16: table102 31-24: table103

CRU_SSGTBL104_107
Address: Operational Base + offset (0x02e8)
SSMOD external wave table register26

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl104_107 Extern wave table 104-107 7-0: table104 15-8: table105 23-16: table106 31-24: table107

CRU_SSGTBL108_111
Address: Operational Base + offset (0x02ec)
SSMOD external wave table register27

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl108_111 Extern wave table 108-111 7-0: table108 15-8: table109 23-16: table110 31-24: table111

CRU_SSGTBL112_115
Address: Operational Base + offset (0x02f0)
SSMOD external wave table register28

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl112_115 Extern wave table 112-115 7-0: table112 15-8: table113 23-16: table114 31-24: table115

CRU_SSGTBL116_119
Address: Operational Base + offset (0x02f4)
SSMOD external wave table register29

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl116_119 Extern wave table 116-119 7-0: table116 15-8: table117 23-16: table118 31-24: table119

CRU_SSGTBL120_123
Address: Operational Base + offset (0x02f8)
SSMOD external wave table register30

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl120_123 Extern wave table 120-123 7-0: table120 15-8: table121 23-16: table122 31-24: table123

CRU_SSGTBL124_127
Address: Operational Base + offset (0x02fc)
SSMOD external wave table register31

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	ssgtbl124_127 Extern wave table 124-127 7-0: table124 15-8: table125 23-16: table126 31-24: table127

CRU_SOFTRST_CON0
Address: Operational Base + offset (0x0300)
Internal software reset control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	l2_srstn_req l2 reset request bit "When HIGH, reset relative logic
14	RW	0x0	strc_sys_asrstn_req strc_sys aresetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
13	RW	0x0	core_noc_srstn_req core_noc reset request bit "When HIGH, reset relative logic
12	RW	0x0	topdbg_srstn_req topdbg reset request bit "When HIGH, reset relative logic
11	RW	0x0	core3_dbg_srstn_req core3_dbg reset request bit "When HIGH, reset relative logic
10	RW	0x0	core2_dbg_srstn_req core2_dbg reset request bit "When HIGH, reset relative logic
9	RW	0x0	core1_dbg_srstn_req core1_dbg reset request bit "When HIGH, reset relative logic
8	RW	0x0	core0_dbg_srstn_req core0_dbg reset request bit "When HIGH, reset relative logic
7	RW	0x0	core3_srstn_req core3 reset request bit "When HIGH, reset relative logic
6	RW	0x0	core2_srstn_req core2 reset request bit "When HIGH, reset relative logic
5	RW	0x0	core1_srstn_req core1 reset request bit "When HIGH, reset relative logic
4	RW	0x0	core0_srstn_req core0 reset request bit "When HIGH, reset relative logic
3	RW	0x0	corepo3_srstn_req corepo3 reset request bit "When HIGH, reset relative logic
2	RW	0x0	corepo2_srstn_req corepo2 reset request bit "When HIGH, reset relative logic
1	RW	0x0	corepo1_srstn_req corepo1 reset request bit "When HIGH, reset relative logic
0	RW	0x0	corepo0_srstn_req corepo0 reset request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON1
Address: Operational Base + offset (0x0304)

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Internal software reset control register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	gpio3_srstn_req gpio3 reset request bit "When HIGH, reset relative logic
14	RW	0x0	gpio2_srstn_req gpio2 reset request bit "When HIGH, reset relative logic
13	RW	0x0	gpio1_srstn_req gpio1 reset request bit "When HIGH, reset relative logic
12	RW	0x0	gpio0_srstn_req gpio0 reset request bit "When HIGH, reset relative logic
11	RW	0x0	rom_srstn_req rom reset request bit "When HIGH, reset relative logic
10	RW	0x0	intmem_srstn_req intmem reset request bit "When HIGH, reset relative logic
9	RW	0x0	spdif_srstn_req spdif reset request bit "When HIGH, reset relative logic
8	RW	0x0	bussys_psrstn_req bussys presetn request bit "When HIGH, reset relative logic
7	RW	0x0	bussys_hsrstn_req bussys hresetn request bit "When HIGH, reset relative logic
6	RW	0x0	efuse_srstn_req efuse reset request bit "When HIGH, reset relative logic
5	RW	0x0	pmu_psrstn_req pmu presetn request bit "When HIGH, reset relative logic
4	RW	0x0	dap_sys_srstn_req dap_sys reset request bit "When HIGH, reset relative logic
3	RW	0x0	dap_srstn_req dap reset request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
2	RW	0x0	a53_gic_srstn_req a53_gic reset request bit "When HIGH, reset relative logic
1	RW	0x0	dbg_psrstn_req dbg presetn request bit "When HIGH, reset relative logic
0	RW	0x0	core_dbg_srstn_req core_dbg reset request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON2

Address: Operational Base + offset (0x0308)

Internal software reset control register2

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	i2c3_srstn_req i2c3 reset request bit "When HIGH, reset relative logic
14	RW	0x0	i2c2_srstn_req i2c2 reset request bit "When HIGH, reset relative logic
13	RW	0x0	i2c1_srstn_req i2c1 reset request bit "When HIGH, reset relative logic
12	RW	0x0	i2c0_srstn_req i2c0 reset request bit "When HIGH, reset relative logic
11	RW	0x0	uart2_psrstn_req uart2 presetn request bit "When HIGH, reset relative logic
10	RW	0x0	uart1_psrstn_req uart1 presetn request bit "When HIGH, reset relative logic
9	RW	0x0	uart0_psrstn_req uart0 presetn request bit "When HIGH, reset relative logic
8	RW	0x0	uart2_srstn_req uart2 reset request bit "When HIGH, reset relative logic
7	RW	0x0	uart1_srstn_req uart1 reset request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
6	RW	0x0	uart0_srstn_req uart0 reset request bit "When HIGH, reset relative logic
5	RW	0x0	i2s2_hsrstn_req i2s2 hresetn request bit "When HIGH, reset relative logic
4	RW	0x0	i2s1_hsrstn_req i2s1 hresetn request bit "When HIGH, reset relative logic
3	RW	0x0	i2s0_hsrstn_req i2s0 hresetn request bit "When HIGH, reset relative logic
2	RW	0x0	i2s2_srstn_req i2s2 reset request bit "When HIGH, reset relative logic
1	RW	0x0	i2s1_srstn_req i2s1 reset request bit "When HIGH, reset relative logic
0	RW	0x0	i2s0_srstn_req i2s0 reset request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON3
Address: Operational Base + offset (0x030c)
Internal software reset control register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RO	0x0	reserved
14	RW	0x0	dcf_psrstn_req dcf presetn request bit "When HIGH, reset relative logic
13	RW	0x0	dcf_asrstn_req dcf aresetn request bit "When HIGH, reset relative logic
12	RW	0x0	tsp_hsadc_srstn_req tsp_hsadc reset request bit "When HIGH, reset relative logic
11	RW	0x0	tsp_srstn_req tsp reset request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
10	RW	0x0	tsp_hsrstn_req tsp hresetn request bit "When HIGH, reset relative logic
9	RW	0x0	tsp_asrstn_req tsp aresetn request bit "When HIGH, reset relative logic
8	RW	0x0	dma_srstn_req dma reset request bit "When HIGH, reset relative logic
7	RW	0x0	pwm0_psrstn_req pwm0 presetn request bit "When HIGH, reset relative logic
6	RW	0x0	pwm0_srstn_req pwm0 reset request bit "When HIGH, reset relative logic
5	RW	0x0	efuse_ns_psrstn_req efuse_ns presetn request bit "When HIGH, reset relative logic
4	RW	0x0	efuse_se_psrstn_req efuse_se presetn request bit "When HIGH, reset relative logic
3	RW	0x0	i2c3_psrstn_req i2c3 presetn request bit "When HIGH, reset relative logic
2	RW	0x0	i2c2_psrstn_req i2c2 presetn request bit "When HIGH, reset relative logic
1	RW	0x0	i2c1_psrstn_req i2c1 presetn request bit "When HIGH, reset relative logic
0	RW	0x0	i2c0_psrstn_req i2c0 presetn request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON4
Address: Operational Base + offset (0x0310)
Internal software reset control register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	usb3grf_srstn_req usb3grf reset request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
14	RW	0x0	timer5_srstn_req timer5 reset request bit "When HIGH, reset relative logic
13	RW	0x0	timer4_srstn_req timer4 reset request bit "When HIGH, reset relative logic
12	RW	0x0	timer3_srstn_req timer3 reset request bit "When HIGH, reset relative logic
11	RW	0x0	timer2_srstn_req timer2 reset request bit "When HIGH, reset relative logic
10	RW	0x0	timer1_srstn_req timer1 reset request bit "When HIGH, reset relative logic
9	RW	0x0	timer0_srstn_req timer0 reset request bit "When HIGH, reset relative logic
8	RW	0x0	timer_6ch_psrstn_req timer_6ch presetn request bit "When HIGH, reset relative logic
7	RW	0x0	usb_grf_srstn_req usb_grf reset request bit "When HIGH, reset relative logic
6	RW	0x0	grf_srstn_req grf reset request bit "When HIGH, reset relative logic
5	RW	0x0	sgrf_srstn_req sgrf reset request bit "When HIGH, reset relative logic
4	RW	0x0	crypto_srstn_req crypto reset request bit "When HIGH, reset relative logic
3	RW	0x0	tsadc_psrstn_req tsadc presetn request bit "When HIGH, reset relative logic
2	RW	0x0	tsadc_srstn_req tsadc reset request bit "When HIGH, reset relative logic
1	RW	0x0	spi0_srstn_req spi0 reset request bit "When HIGH, reset relative logic
0	RW	0x0	scr_srstn_req scr reset request bit "When HIGH, reset relative logic

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CRU_SOFTRST_CON5
Address: Operational Base + offset (0x0314)
Internal software reset control register5

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	ddrphy_psrstn_req ddrphy presetn request bit "When HIGH, reset relative logic
14	RW	0x0	ddrphy_srstn_req ddrphy reset request bit "When HIGH, reset relative logic
13	RW	0x0	ddrctrl_psrstn_req ddrctrl presetn request bit "When HIGH, reset relative logic
12	RW	0x0	ddrctrl_srstn_req ddrctrl reset request bit "When HIGH, reset relative logic
11	RW	0x0	ddrmsch_srstn_req ddrmsch reset request bit "When HIGH, reset relative logic
10	RO	0x0	reserved
9	RW	0x0	msch_srstn_req msch reset request bit "When HIGH, reset relative logic
8	RW	0x0	dfimon_srstn_req dfimon reset request bit "When HIGH, reset relative logic
7	RW	0x0	grf_ddr_srstn_req grf_ddr reset request bit "When HIGH, reset relative logic
6	RW	0x0	saradc_psrstn_req saradc presetn request bit "When HIGH, reset relative logic
5	RW	0x0	saradc_srstn_req saradc reset request bit "When HIGH, reset relative logic
4	RO	0x0	reserved
3	RW	0x0	acodec_psrstn_req acodec presetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
2	RW	0x0	vdac_srstn_req vdac reset request bit "When HIGH, reset relative logic
1	RW	0x0	hdmiphy_srstn_req hdmiphy reset request bit "When HIGH, reset relative logic
0	RW	0x0	phyniu_srstn_req phyniu reset request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON6

Address: Operational Base + offset (0x0318)

Internal software reset control register6

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	emmc_srstn_req emmc reset request bit "When HIGH, reset relative logic
14	RW	0x0	sdio_srstn_req sdio reset request bit "When HIGH, reset relative logic
13	RW	0x0	mmc0_srstn_req mmc0 reset request bit "When HIGH, reset relative logic
12	RW	0x0	periphsys_hsrstn_req periphsys hresetn request bit "When HIGH, reset relative logic
11	RW	0x0	periph_niu_psrstn_req periph_niu presetn request bit "When HIGH, reset relative logic
10	RW	0x0	periph_niu_hsrstn_req periph_niu hresetn request bit "When HIGH, reset relative logic
9	RW	0x0	periph_niu_asrstn_req periph_niu aresetn request bit "When HIGH, reset relative logic
8	RW	0x0	sdmmcext_srstn_req sdmmcext reset request bit "When HIGH, reset relative logic
7	RW	0x0	gpu_niu_asrstn_req gpu_niu aresetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
6	RW	0x0	gpu_asrstn_req gpu aresetn request bit "When HIGH, reset relative logic
5	RW	0x0	otp_phy_srstn_req otp_phy reset request bit "When HIGH, reset relative logic
4	RW	0x0	macphy_srstn_req macphy reset request bit "When HIGH, reset relative logic
3	RW	0x0	gmac2io_asrstn_req gmac2io aresetn request bit "When HIGH, reset relative logic
2	RW	0x0	gmac2phy_asrstn_req gmac2phy aresetn request bit "When HIGH, reset relative logic
1	RW	0x0	gmac_niu_psrstn_req gmac_niu presetn request bit "When HIGH, reset relative logic
0	RW	0x0	gmac_niu_asrstn_req gmac_niu aresetn request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON7
Address: Operational Base + offset (0x031c)
Internal software reset control register7

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	usb3phy_pipe_srstn_req usb3phy_pipe reset request bit "When HIGH, reset relative logic
14	RW	0x0	usb3phy_u3_srstn_req usb3phy_u3 reset request bit "When HIGH, reset relative logic
13	RW	0x0	usb3phy_u2_srstn_req usb3phy_u2 reset request bit "When HIGH, reset relative logic
12	RW	0x0	usb3otg_utmi_srst_req usb3otg_utmi reset request bit "When HIGH, reset relative logic
11	RW	0x0	usb2host_utmi_srst_req usb2host_utmi reset request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
10	RW	0x0	usb2otg_utmi_srst_req usb2otg_utmi reset request bit "When HIGH, reset relative logic
9	RW	0x0	usbpor_srst_req usbpor reset request bit "When HIGH, reset relative logic
8	RW	0x0	usb3otg_srstn_req usb3otg reset request bit "When HIGH, reset relative logic
7	RW	0x0	usb2host_utmi_srstn_req usb2host_utmi reset request bit "When HIGH, reset relative logic
6	RW	0x0	usb2host_ehciphy_srstn_req usb2host_ehciphy reset request bit "When HIGH, reset relative logic
5	RW	0x0	usb2host_aux_srstn_req usb2host_aux reset request bit "When HIGH, reset relative logic
4	RW	0x0	usb2host_arb_srstn_req usb2host_arb reset request bit "When HIGH, reset relative logic
3	RW	0x0	usb2host_hsrstn_req usb2host hresetn request bit "When HIGH, reset relative logic
2	RW	0x0	usb2otg_adp_srstn_req usb2otg_adp reset request bit "When HIGH, reset relative logic
1	RW	0x0	usb2otg_srstn_req usb2otg reset request bit "When HIGH, reset relative logic
0	RW	0x0	usb2otg_hsrstn_req usb2otg hresetn request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON8

Address: Operational Base + offset (0x0320)

Internal software reset control register8

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	hdmi_psrstn_req hdmi presetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
14	RW	0x0	hdmi_srstn_req hdmi reset request bit "When HIGH, reset relative logic
13	RW	0x0	iep_hsrstn_req iep hresetn request bit "When HIGH, reset relative logic
12	RW	0x0	iep_asrstn_req iep aresetn request bit "When HIGH, reset relative logic
11	RW	0x0	rga_hsrstn_req rga hresetn request bit "When HIGH, reset relative logic
10	RW	0x0	rga_asrstn_req rga aresetn request bit "When HIGH, reset relative logic
9	RW	0x0	rga_niu_asrstn_req rga_niu aresetn request bit "When HIGH, reset relative logic
8	RW	0x0	rga_srstn_req rga reset request bit "When HIGH, reset relative logic
7	RW	0x0	vop_dsrstn_req vop dresetn request bit "When HIGH, reset relative logic
6	RW	0x0	vop_hsrstn_req vop hresetn request bit "When HIGH, reset relative logic
5	RW	0x0	vop_asrstn_req vop aresetn request bit "When HIGH, reset relative logic
4	RW	0x0	vop_niu_asrstn_req vop_niu aresetn request bit "When HIGH, reset relative logic
3	RW	0x0	vio_arbi_hsrstn_req vio_arbi hresetn request bit "When HIGH, reset relative logic
2	RW	0x0	vio_h2p_hsrstn_req vio_h2p hresetn request bit "When HIGH, reset relative logic
1	RW	0x0	vio_bus_hsrstn_req vio_bus hresetn request bit "When HIGH, reset relative logic
0	RW	0x0	vio_asrstn_req vio aresetn request bit "When HIGH, reset relative logic

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CRU_SOFTRST_CON9
Address: Operational Base + offset (0x0324)
Internal software reset control register9

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	usb3phy_pipe_psrstn_req usb3phy_pipe presetn request bit "When HIGH, reset relative logic
14	RW	0x0	usb3phy_otg_psrstn_req usb3phy_otg presetn request bit "When HIGH, reset relative logic
13	RW	0x0	pdm_srstn_req pdm reset request bit "When HIGH, reset relative logic
12	RW	0x0	pdm_hsrstn_req pdm hresetn request bit "When HIGH, reset relative logic
11	RW	0x0	ddrstdy_srstn_req ddrstdy reset request bit "When HIGH, reset relative logic
10	RW	0x0	ddrstdy_psrstn_req ddrstdy presetn request bit "When HIGH, reset relative logic
9	RW	0x0	ddrctrl_asrstn_req ddrctrl aresetn request bit "When HIGH, reset relative logic
8	RW	0x0	otp_user_srstn_req otp_user reset request bit "When HIGH, reset relative logic
7	RW	0x0	otp_sbpi_srstn_req otp_sbpi reset request bit "When HIGH, reset relative logic
6	RW	0x0	otp_psrstn_req otp presetn request bit "When HIGH, reset relative logic
5	RW	0x0	cif_psrstn_req cif presetn request bit "When HIGH, reset relative logic
4	RW	0x0	cif_hsrstn_req cif hresetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
3	RW	0x0	cif_asrstn_req cif aresetn request bit "When HIGH, reset relative logic
2	RW	0x0	hdcp_hsrstn_req hdcp hresetn request bit "When HIGH, reset relative logic
1	RW	0x0	hdcp_srstn_req hdcp reset request bit "When HIGH, reset relative logic
0	RW	0x0	hdcp_asrstn_req hdcp aresetn request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON10

Address: Operational Base + offset (0x0328)

Internal software reset control register10

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	ddrphydiv_srstn_req ddrphydiv reset request bit "When HIGH, reset relative logic
14:10	RO	0x0	reserved
9	RW	0x0	vdec_cabac_srstn_req vdec_cabac reset request bit "When HIGH, reset relative logic
8	RW	0x0	vdec_core_srstn_req vdec_core reset request bit "When HIGH, reset relative logic
7	RW	0x0	vdec_niu_hsrstn_req vdec_niu hresetn request bit "When HIGH, reset relative logic
6	RW	0x0	vdec_hsrstn_req vdec hresetn request bit "When HIGH, reset relative logic
5	RW	0x0	vdec_niu_asrstn_req vdec_niu aresetn request bit "When HIGH, reset relative logic
4	RW	0x0	vdec_asrstn_req vdec aresetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
3	RW	0x0	vcodec_niu_hsrstn_req vcodec_niu hresetn request bit "When HIGH, reset relative logic
2	RW	0x0	vcodec_hsrstn_req vcodec hresetn request bit "When HIGH, reset relative logic
1	RW	0x0	vcodec_niu_asrstn_req vcodec_niu aresetn request bit "When HIGH, reset relative logic
0	RW	0x0	vcodec_asrstn_req vcodec aresetn request bit "When HIGH, reset relative logic

CRU_SOFTRST_CON11
Address: Operational Base + offset (0x032c)
Internal software reset control register11

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	rkvenc_intmem_srstn_req rkvenc_intmem reset request bit "When HIGH, reset relative logic
7	RW	0x0	rkvenc_h264_hsrstn_req rkvenc_h264 hresetn request bit "When HIGH, reset relative logic
6	RW	0x0	rkvenc_h264_asrstn_req rkvenc_h264 aresetn request bit "When HIGH, reset relative logic
5	RW	0x0	rkvenc_h265_dsp_srstn_req rkvenc_h265_dsp reset request bit "When HIGH, reset relative logic
4	RW	0x0	rkvenc_h265_core_srstn_req rkvenc_h265_core reset request bit "When HIGH, reset relative logic
3	RW	0x0	rkvenc_h265_psrstn_req rkvenc_h265 presetn request bit "When HIGH, reset relative logic
2	RW	0x0	rkvenc_h265_asrstn_req rkvenc_h265 aresetn request bit "When HIGH, reset relative logic

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Bit	Attr	Reset Value	Description
1	RW	0x0	rkvenc_niu_hsrstn_req rkvenc_niu hresetn request bit "When HIGH, reset relative logic
0	RW	0x0	rkvenc_niu_asrstn_req rkvenc_niu aresetn request bit "When HIGH, reset relative logic

CRU_CRU_SDMMC_CON0
Address: Operational Base + offset (0x0380)
sdmmc control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	drv_sel drive select drive select
10:3	RW	0x00	drv_delaynum drive delay number drive delay number
2:1	RW	0x2	drv_degree drive degree drive degree
0	RW	0x0	init_state initial state initial state

CRU_CRU_SDMMC_CON1
Address: Operational Base + offset (0x0384)
sdmmc control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	RW	0x0	sample_sel sample select sample select
9:2	RW	0x00	sample_delaynum sample delay number sample delay number

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	sample_degree sample degree sample degree

CRU_CRU_SDIO_CON0
Address: Operational Base + offset (0x0388)
SDIO control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	drv_sel drive select drive select
10:3	RW	0x00	drv_delaynum drive delay number drive delay number
2:1	RW	0x2	drv_degree drive degree drive degree
0	RW	0x0	init_state initial state initial state

CRU_CRU_SDIO_CON1
Address: Operational Base + offset (0x038c)
SDIO control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	RW	0x0	sample_sel sample select sample select
9:2	RW	0x00	sample_delaynum sample delay number sample delay number
1:0	RW	0x0	sample_degree sample degree sample degree

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CRU_CRU_EMMC_CON0
Address: Operational Base + offset (0x0390)
EMMC control0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	drv_sel drive select drive select
10:3	RW	0x00	drv_delaynum drive delay number drive delay number
2:1	RW	0x2	drv_degree drive degree drive degree
0	RW	0x0	init_state initial state initial state

CRU_CRU_EMMC_CON1
Address: Operational Base + offset (0x0394)
EMMC control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	RW	0x0	sample_sel sample select sample select
9:2	RW	0x00	sample_delaynum sample delay number sample delay number
1:0	RW	0x0	sample_degree sample degree sample degree

CRU_CRU_SDMMC_EXT_CON0
Address: Operational Base + offset (0x0398)
SDMMC_EXT control0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:12	RO	0x0	reserved
11	RW	0x0	drv_sel drive select drive select
10:3	RW	0x00	drv_delaynum drive delay number drive delay number
2:1	RW	0x2	drv_degree drive degree drive degree
0	RW	0x0	init_state initial state initial state

CRU_CRU_SDMMC_EXT_CON1
Address: Operational Base + offset (0x039c)
SDMMC_EXT control1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_mask write mask bits "When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:11	RO	0x0	reserved
10	RW	0x0	sample_sel sample select sample select
9:2	RW	0x00	sample_delaynum sample delay number sample delay number
1:0	RW	0x0	sample_degree sample degree sample degree

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3.8 Timing Diagram

Power on reset timing is shown as follow:

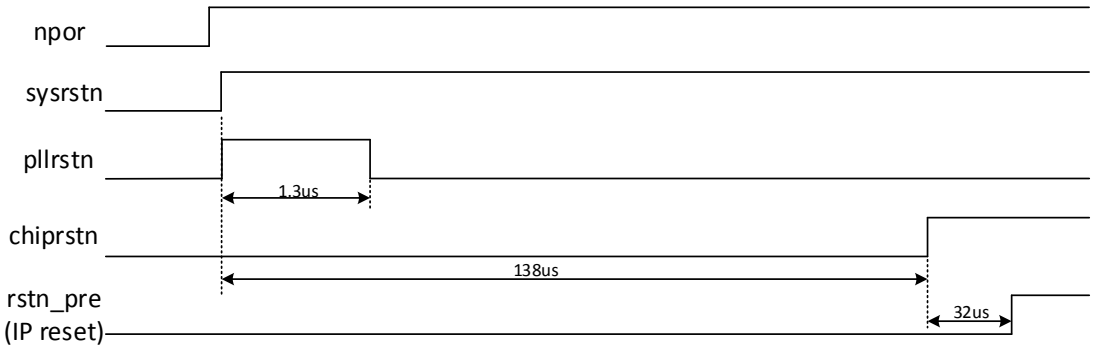

Fig. 3-12 Chip Power On Reset Timing Diagram

Npor is hardware reset signal from out-chip, which is filtered glitch to obtain signal sysrstn. To make PLLs work normally, the PLL reset signal (pllrstn) must maintain high for more than 1us, and PLLs start to lock when pllrstn de-assert, and the PLL max lock time is 1500 PLL REFCLK cycles. And then the system will wait about 138us, and then de-assert reset signal chiprstn. The signal chiprstn is used to generate output clocks in CRU. After CRU start output clocks, the system waits again for 768cycles (21.3us) to de-assert signal rstn_pre, which is used to generate power on reset of all IPs.

3.9 Application Notes

3.9.1 PLL usage

A. PLL output frequency configuration

FBDIV, POSTDIV1, BYPASS can be configured by programming CRU_APLL_CON0, CRU_DPLL_CON0 and CRU_GPLL_CON0.

DSMPD, REFDIV, POSTDIV2 can be configured by programming CRU_APLL_CON1, CRU_DPLL_CON1 and CRU_GPLL_CON1.

FRAC can be configured by programming CRU_APLL_CON2, CRU_DPLL_CON2 and CRU_GPLL_CON2.

If DSMPD = 1 (DSM is disabled, "integer mode")

FOUTVCO = FREF / REFDIV * FBDIV

FOUTPOSTDIV = FOUTVCO / POSTDIV1 / POSTDIV2

When FREF is 24MHz, and if 700MHz FOUTPOSTDIV is needed. The configuration can be:

DSMPD = 1
REFDIV = 6
FBDIV = 175
POSTDIV1=1
POSTDIV2=1

And then

FOUTVCO = FREF / REFDIV * FBDIV = 24/6*175=700

FOUTPOSTDIV = FOUTVCO / POSTDIV1 / POSTDIV2=700/1/1=700

If DSMPD = 0 (DSM is enabled, "fractional mode")

FOUTVCO = FREF / REFDIV * (FBDIV + FRAC / 224)

FOUTPOSTDIV = FOUTVCO / POSTDIV1 / POSTDIV2

When FREF is 24MHz, and if 491.52MHz FOUTPOSTDIV is needed. The configuration can be:

DSMPD = 0
REFDIV = 1
FBDIV = 40
FRAC = 24'hf5c28f
POSTDIV1=2
POSTDIV2=1

And then

FOUTVCO = FREF / REFDIV * (FBDIV + FRAC / 224) = 24/1*(40+24'hf5c28f /224)= 983.04

FOUTPOSTDIV = FOUTVCO / POSTDIV1 / POSTDIV2=983.04/2/1=491.52

B. PLL setting consideration

- If the POSTDIV value is changed during operation a short pulse (glitch) may occur on FOUTPOSTDIV. The minimum width of the short pulse will be equal to twice the period of the VCO. Therefore, if the circuitry clocked by the PLL is sensitive to short pulses, the new

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divide value should be re-timed so that it is synchronous with the rising edge of the output clock (FOUTPOSTDIV). Glitches cannot occur on any of the other outputs.

- For lowest power operation, the minimum VCO and FREF frequencies should be used. For minimum jitter operation, the highest VCO and FREF frequencies should be used. The normal operating range for the VCO is described above in .
- The supply rejection will be worse at the low end of the VCO range so care should be taken to keep the supply clean for low power applications.
- The feedback divider is not capable of dividing by all possible settings due to the use of a power-saving architecture. The following settings are valid for FBDIV:
- DSMPD=1 (Integer Mode)
- DSMPD=0 (Fractional Mode)
- The PD input places the PLL into the lowest power mode. In this case, all analog circuits are turned off and FREF will be "ignored". The FOUTPOSTDIV and FOUTVCO pins are forced to logic low (0V).
- The BYPASS pin controls a mux which selects FREF to be passed to the FOUTPOSTDIV when active high. However, the PLL continues to run as it normally would if bypass were low. This is a useful feature for PLL testing since the clock path can be verified without the PLL being required to work. Also, the effect that the PLL induced supply noise has on the output buffering can be evaluated. It is not recommended to switch between BYPASS mode and normal mode for regular chip operation since this may result in a glitch. Also, FOUTPOSTDIVPD should be set low if the PLL is to be used in BYPASS mode.

3.9.2 PLL frequency change and lock check

The PLL programming supports changed on-the-fly and the PLL will simply slew to the new frequency.

PLL lock state can be checked in CRU_APLL_CON1[10], CRU_DPLL_CON1[10], CRU_CPLL_CON1[10], CRU_GPLL_CON1[10] register. The lock state is high when both original hardware PLL lock and PLL counter lock are high. The PLL counter lock initial value is CRU_GLB_CNT_TH[31:16].

The max delay time is 500 REF_CLK.

PLL locking consists of three phases.

- Phase 1 is control voltage slewing. During this phase one of the clocks (reference or divide) is much faster than the other, and the PLL frequency adjusts almost continuously. When locking from power down, the divide clock is initially very slow and steadily increases frequency. It will take slightly longer for faster VCO settings when locking from power down, since the PLL must slew further.
- Phase 2 is small signal phase acquisition. During this phase, the internal up/down signals alternate semi-chaotically as the phase slowly adjusts until the two signals are aligned. The duration of this phase depends on the loop bandwidth and is faster with higher bandwidth. Bandwidth can be estimated as FREF / REFDIV / 20 for integer mode and FREF /REFDIV / 40 for fractional mode. The duration of small signal locking is about 1/Bandwidth.
- Phase 3 is the digital cycle count. After the last cycle slip is detected, an internal counter waits 256 FREF / REFDIV cycles before the lock signal goes high. This is frequently the dominant factor in lock time – especially for slower reference clock signals or large reference divide settings. This time can be calculated as 256*REFDIV/FREF.

3.9.3 Fractional divider usage

To get specific frequency, clocks of I2S, SPDIF, UART can be generated by fractional divider. Generally you must set that denominator is 20 times larger than numerator to generate precise clock frequency. So the fractional divider applies only to generate low frequency clock like I2S, UART.

3.9.4 Global software reset

Two global software resets are designed in the chip, you can program CRU_GLB_SRST_FST_VALUE[15:0] as 0xfdb9 to assert the first global software reset glb_srstn_1 and program CRU_GLB_SRST_SND_VALUE[15:0] as 0xeca8 to assert the second global software reset glb_srstn_2. These two software resets are self-deasserted by hardware.

Glb_srstn_1 resets almost all logic.

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Glb_srstn_2 resets almost all logic except GRF and GPIOs.

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Chapter 4 System Debug

4.1 Overview

The chip uses the DAPLITE Technology to support real-time debug.

4.1.1 Features

- Invasive debug with core halted
- SW-DP

4.1.2 Debug components address map

The following table shows the debug components address in memory map:

Module	Base Address
DAP_ROM	0xff800000

4.2 Block Diagram

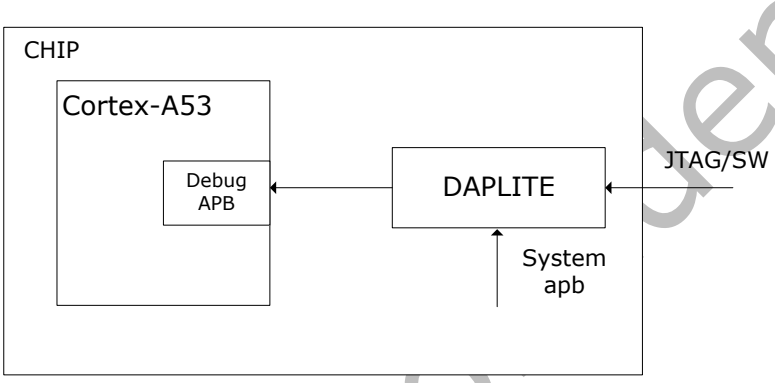

Fig. 4-1Debug system structure

4.3 Function Description

4.3.1 DAP

The DAP has following components:

- Serial Wire JTAG Debug Port(SWJ-DP)
- APB Access Port(APB-AP)
- ROM table

The debug port is the host tools interface to access the DAP-Lite. This interface controls any access ports provided within the DAP-Lite. The DAP-Lite supports a combined debug port which includes both JTAG and Serial Wire Debug(SWD), with a mechanism that supports switching between them.

The APB-AP acts as a bridge between SWJ-DP and APB bus which translate the Debug request to APB bus.

The DAP provides an internal ROM table connected to the master Debug APB port of the APB-Mux. The Debug ROM table is loaded at address 0x00000000 and 0x80000000 of this bus and is accessible from both APB-AP and the system APB input. Bit[31] of the address bus is not connected to the ROM Table, ensuring that both views read the same value. The ROM table stores the locations of the components on the Debug APB.

More information please refer to the document CoreSight_DAPLite_TRM.pdf for the debug detail description.

4.4 Register Description

Please refer to the documentCoreSight_DAPLite_TRM.pdf for the debug detail description.

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4.5 Interface Description

4.5.1 DAP SWJ-DP Interface

The following figure is the DAP SWJ-DP interface, the SWJ-DP is a combined JTAG-DP and SW-DP that enable you connect either a Serial Wire Debug(SWJ) to JTAG probe to a target.

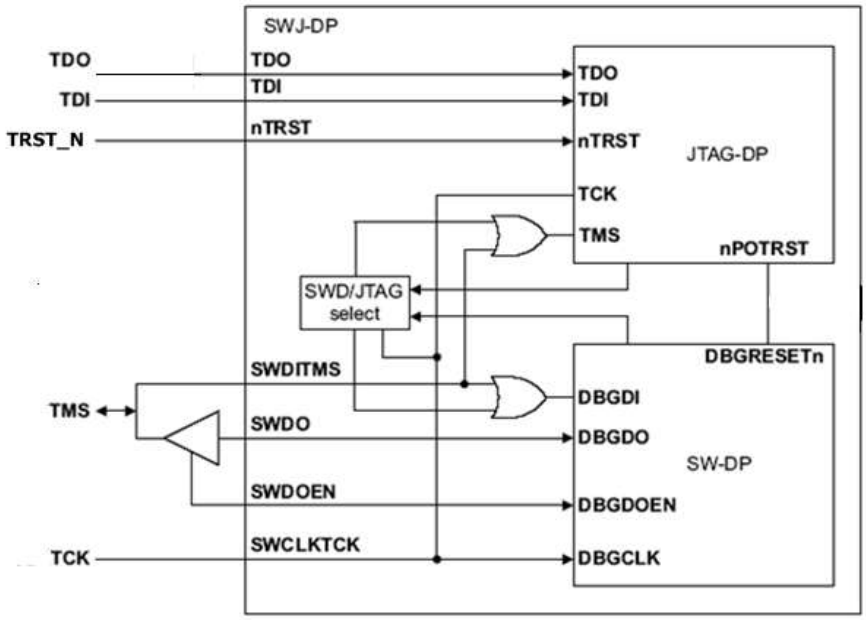

Fig. 4-2 DAP SWJ interface

4.5.2 DAP SW-DP Interface

This implementation is taken from ADIv5.1 and operates with a synchronous serial interface. This uses a single bidirectional data signal, and a clock signal.
The figure below describes the interaction between the timing of transactions on the serial wire interface, and the DAP internal bus transfers. It shows when the target responds with a WAIT acknowledgement.

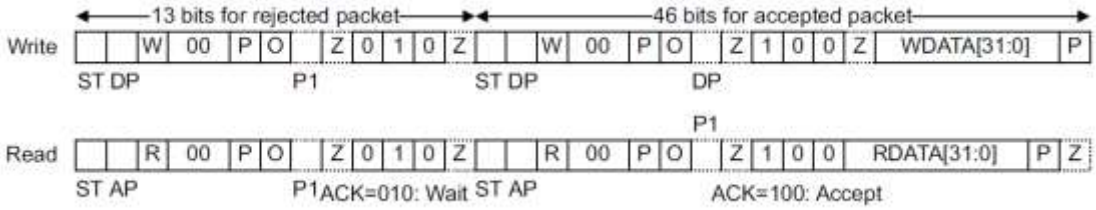

Fig. 4-3 SW-DP acknowledgement timing

Table 4-1 SW-DP Interface Description

Module pin	Direction	Pad name	IOMUX
jtag_tck	I	IO_SDMMC0d2_JTAGtck_GPIO1A2vccio3	GRF_GPIO1A_IOMUX[5:4]=2'b10& mmc0_detn
jtag_tms	I/O	IO_SDMMC0d3_JTAGtms_GPIO1A3vccio3	GRF_GPIO1A_IOMUX[7:6]=2'b10 & mmc0_detn

Note : mmc0_detn, when high, no sd card is used.

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Chapter 5 General Register Files (GRF)

5

5.1 Overview

The general register file will be used to do static set by software, which is composed of many registers for system control. The GRF is divided into four sections,

- GRF, used for general non-secure system,
- DDR_GRF, used for always on system
- USB2PHY_GRF, used for USB2 PHY control and query
- USB3PHY_GRF, used for USB3 PHY control and query

5.2 Function Description

The function of general register file is：

- IOMUX control
- Control the state of GPIO in power-down mode
- GPIO PAD pull down and pull up control
- Used for common system control
- Used to record the system state

5.3 GRF Register Description

5.3.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

5.3.2 Registers Summary

Name	Offset	Size	Reset Value	Description
GRF_GPIO0A_IOMUX	0x0000	W	0x00000000	GPIO0A iomux control
GRF_GPIO0B_IOMUX	0x0004	W	0x00000000	GPIO0B iomux control
GRF_GPIO0C_IOMUX	0x0008	W	0x00000000	GPIO0C iomux control
GRF_GPIO0D_IOMUX	0x000c	W	0x00000000	GPIO0D iomux control
GRF_GPIO1A_IOMUX	0x0010	W	0x000004aa	GPIO1A iomux control
GRF_GPIO1B_IOMUX	0x0014	W	0x00000000	GPIO1B iomux control
GRF_GPIO1C_IOMUX	0x0018	W	0x00000000	GPIO1C iomux control
GRF_GPIO1D_IOMUX	0x001c	W	0x00000000	GPIO1D iomux control
GRF_GPIO2A_IOMUX	0x0020	W	0x00000000	GPIO2A iomux control
GRF_GPIO2BL_IOMUX	0x0024	W	0x00000200	GPIO2BL iomux control
GRF_GPIO2BH_IOMUX	0x0028	W	0x00000000	GPIO2BH iomux control
GRF_GPIO2CL_IOMUX	0x002c	W	0x00000000	GPIO2CL iomux control
GRF_GPIO2CH_IOMUX	0x0030	W	0x00000000	GPIO2CH iomux control
GRF_GPIO2D_IOMUX	0x0034	W	0x00000000	GPIO2D iomux control
GRF_GPIO3AL_IOMUX	0x0038	W	0x00000000	GPIO3AL iomux control
GRF_GPIO3AH_IOMUX	0x003c	W	0x00000000	GPIO3AH iomux control
GRF_GPIO3BL_IOMUX	0x0040	W	0x00000000	GPIO3BL iomux control
GRF_GPIO3BH_IOMUX	0x0044	W	0x00000000	GPIO3BH iomux control
GRF_GPIO3C_IOMUX	0x0048	W	0x00000000	GPIO3C iomux control
GRF_GPIO3D_IOMUX	0x004c	W	0x00000000	GPIO3D iomux control
GRF_COM_IOMUX	0x0050	W	0x00000000	GRF common iomux control

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Name	Offset	Size	Reset Value	Description
GRF_GPIO0A_P	0x0100	W	0x0000566a	GPIO0A PU/PD control
GRF_GPIO0B_P	0x0104	W	0x0000aa6a	GPIO0B PU/PD control
GRF_GPIO0C_P	0x0108	W	0x0000aa6a	GPIO0C PU/PD control
GRF_GPIO0D_P	0x010c	W	0x0000aaaa	GPIO0D PU/PD control
GRF_GPIO1A_P	0x0110	W	0x0000a555	GPIO1A PU/PD control
GRF_GPIO1B_P	0x0114	W	0x000056a5	GPIO1B PU/PD control
GRF_GPIO1C_P	0x0118	W	0x00009a65	GPIO1C PU/PD control
GRF_GPIO1D_P	0x011c	W	0x0000aaaa	GPIO1D PU/PD control
GRF_GPIO2A_P	0x0120	W	0x00009556	GPIO2A PU/PD control
GRF_GPIO2B_P	0x0124	W	0x0000959a	GPIO2B PU/PD control
GRF_GPIO2C_P	0x0128	W	0x00005565	GPIO2C PU/PD control
GRF_GPIO2D_P	0x012c	W	0x000055a5	GPIO2D PU/PD control
GRF_GPIO3A_P	0x0130	W	0x000055a5	GPIO3A PU/PD control
GRF_GPIO3B_P	0x0134	W	0x00005aaa	GPIO3B PU/PD control
GRF_GPIO3C_P	0x0138	W	0x00006555	GPIO3C PU/PD control
GRF_GPIO3D_P	0x013c	W	0x0000555a	GPIO3D PU/PD control
GRF_GPIO0A_E	0x0200	W	0x00008011	GPIO0A drive strength control
GRF_GPIO0B_E	0x0204	W	0x0000aa2a	GPIO0B drive strength control
GRF_GPIO0C_E	0x0208	W	0x0000aa0a	GPIO0C drive strength control
GRF_GPIO0D_E	0x020c	W	0x0000005a	GPIO0D drive strength control
GRF_GPIO1A_E	0x0210	W	0x0000aaaa	GPIO1A drive strength control
GRF_GPIO1B_E	0x0214	W	0x0000aa2a	GPIO1B drive strength control
GRF_GPIO1C_E	0x0218	W	0x0000a88a	GPIO1C drive strength control
GRF_GPIO1D_E	0x021c	W	0x0000005a	GPIO1D drive strength control
GRF_GPIO2A_E	0x0220	W	0x00000000	GPIO2A drive strength control
GRF_GPIO2B_E	0x0224	W	0x00004145	GPIO2B drive strength control
GRF_GPIO2C_E	0x0228	W	0x00005515	GPIO2C drive strength control
GRF_GPIO2D_E	0x022c	W	0x0000aa01	GPIO2D drive strength control
GRF_GPIO3A_E	0x0230	W	0x0000aa22	GPIO3A drive strength control
GRF_GPIO3B_E	0x0234	W	0x00000000	GPIO3B drive strength control
GRF_GPIO3C_E	0x0238	W	0x0000aaaa	GPIO3C drive strength control
GRF_GPIO3D_E	0x023c	W	0x0000aaaa	GPIO3D drive strength control
GRF_GPIO0L_SR	0x0300	W	0x00000000	GPIO0 A/B SR control
GRF_GPIO0H_SR	0x0304	W	0x00000000	GPIO0 C/D SR control
GRF_GPIO1L_SR	0x0308	W	0x00000000	GPIO1 A/B SR control
GRF_GPIO1H_SR	0x030c	W	0x00000000	GPIO1 C/D SR control
GRF_GPIO2L_SR	0x0310	W	0x00000000	GPIO2 A/B SR control
GRF_GPIO2H_SR	0x0314	W	0x00000000	GPIO2 C/D SR control
GRF_GPIO3L_SR	0x0318	W	0x00000000	GPIO3 A/B SR control
GRF_GPIO3H_SR	0x031c	W	0x00000000	GPIO3 C/D SR control
GRF_GPIO0L_SMT	0x0380	W	0x00000000	GPIO0 A/B smitter control register
GRF_GPIO0H_SMT	0x0384	W	0x00000000	GPIO0 C/D smitter control register

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Name	Offset	Size	Reset Value	Description
GRF_GPIO1L_SMT	0x0388	W	0x00000000	GPIO1 A/B smitter control register
GRF_GPIO1H_SMT	0x038c	W	0x00000000	GPIO1 C/D smitter control register
GRF_GPIO2L_SMT	0x0390	W	0x00000000	GPIO2 A/B smitter control register
GRF_GPIO2H_SMT	0x0394	W	0x00000000	GPIO2 C/D smitter control register
GRF_GPIO3L_SMT	0x0398	W	0x00000000	GPIO3 A/B smitter control register
GRF_GPIO3H_SMT	0x039c	W	0x00000000	GPIO3 C/D smitter control register
GRF_SOC_CON0	0x0400	W	0x00000000	SOC control register0
GRF_SOC_CON1	0x0404	W	0x00000000	SOC control register1
GRF_SOC_CON2	0x0408	W	0x00001000	SOC control register2
GRF_SOC_CON3	0x040c	W	0x00000000	SOC control register3
GRF_SOC_CON4	0x0410	W	0x00000000	SOC control register4
GRF_SOC_CON5	0x0414	W	0x00000000	SOC control register5
GRF_SOC_CON6	0x0418	W	0x00000000	SOC control register6
GRF_SOC_CON7	0x041c	W	0x00000000	SOC control register7
GRF_SOC_CON8	0x0420	W	0x00000000	SOC control register8
GRF_SOC_CON9	0x0424	W	0x00000000	SOC control register9
GRF_SOC_CON10	0x0428	W	0x0000f800	SOC control register10
GRF_SOC_STATUS0	0x0480	W	0x00000000	SOC status register0
GRF_SOC_STATUS1	0x0484	W	0x00000000	SOC status register1
GRF_SOC_STATUS2	0x0488	W	0x00000000	SOC status register2
GRF_SOC_STATUS3	0x048c	W	0x00000000	SOC status register3
GRF_SOC_STATUS4	0x0490	W	0x00000000	SOC status register4
GRF_USB3OTG_CON0	0x04c0	W	0x00002000	USB3OTG control register0
GRF_USB3OTG_CON1	0x04c4	W	0x00001100	USB3OTG control register1
GRF_CPU_CON0	0x0500	W	0x00000060	CPU control register0
GRF_CPU_CON1	0x0504	W	0x0000000c	CPU control register1
GRF_CPU_STATUS0	0x0520	W	0x00000000	CPU status register0
GRF_CPU_STATUS1	0x0524	W	0x00000000	CPU status register1
GRF_OS_REG0	0x05c8	W	0x00000000	os register0
GRF_OS_REG1	0x05cc	W	0x00000000	os register1
GRF_OS_REG2	0x05d0	W	0x00000000	os register2
GRF_OS_REG3	0x05d4	W	0x00000000	os register3
GRF_OS_REG4	0x05d8	W	0x00000000	os register4
GRF_OS_REG5	0x05dc	W	0x00000000	os register5
GRF_OS_REG6	0x05e0	W	0x00000000	os register6
GRF_OS_REG7	0x05e4	W	0x00000000	os register7
GRF_SIG_DETECT_CON	0x0680	W	0x00000000	External signal detect configue register

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Name	Offset	Size	Reset Value	Description
GRF_SIG_DETECT_STATUS	0x0690	W	0x00000000	External signal detect status register
GRF_SIG_DETECT_STATUS_CLEAR	0x06a0	W	0x00000000	External signal detect status clear register
GRF_SDMMC_DET_COUNTER	0x06b0	W	0x00030100	SDMMC detect counter register
GRF_HOST0_CON0	0x0700	W	0x00000820	host0 control register0
GRF_HOST0_CON1	0x0704	W	0x000004bc	host0 control register1
GRF_HOST0_CON2	0x0708	W	0x00000019	host0 control register2
GRF_OTG_CON0	0x0880	W	0x00000000	OTG control register
GRF_HOST0_STATUS	0x0890	W	0x00000000	HOST0 status register
GRF_MAC_CON0	0x0900	W	0x00000000	MAC control register0
GRF_MAC_CON1	0x0904	W	0x00000000	MAC control register1
GRF_MAC_CON2	0x0908	W	0x00000000	MAC control register2
GRF_MACPHY_CON0	0x0b00	W	0x00002039	MACPHY control register0
GRF_MACPHY_CON1	0x0b04	W	0x00000000	MACPHY control register1
GRF_MACPHY_CON2	0x0b08	W	0x00000000	MACPHY control register2
GRF_MACPHY_CON3	0x0b0c	W	0x00000000	MACPHY control register3
GRF_MACPHY_STATUS	0x0b10	W	0x00000000	MACPHY status register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

5.3.3 Detail Register Description

GRF_GPIO0A_IOMUX
Address: Operational Base + offset (0x0000)
GPIO0A iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio0_a7_sel GPIO0A[7] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d0 2'b11: reserved

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Bit	Attr	Reset Value	Description
13:12	RW	0x0	gpio0_a6_sel GPIO0A[6] iomux select 2'b00: gpio 2'b01: hdmi_sda 2'b10: i2c3_sda 2'b11: reserved
11:10	RW	0x0	gpio0_a5_sel GPIO0A[5] iomux select 2'b00: gpio 2'b01: hdmi_scl 2'b10: i2c3_scl 2'b11: reserved
9:8	RW	0x0	gpio0_a4_sel GPIO0A[4] iomux select 2'b00: gpio 2'b01: hdmi_hdp 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio0_a3_sel GPIO0A[3] iomux select 2'b00: gpio 2'b01: hdmi_cec 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio0_a2_sel GPIO0A[2] iomux select 2'b00: gpio 2'b01: clk_out_gmacm0 2'b10: spdif_txm2 2'b11: reserved
3:2	RW	0x0	gpio0_a1_sel GPIO0A[1] iomux select 2'b00: gpio 2'b01: clk_32km0 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio0_a0_sel GPIO0A[0] iomux select 2'b00: gpio 2'b01: clk_out_wifim0 2'b10: reserved 2'b11: reserved

GRF_GPIO0B_IOMUX
Address: Operational Base + offset (0x0004)

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GPIO0B iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio0_b7_sel GPIO0B[7] iomux select 2'b00: gpio 2'b01: gmac_rxd0m0 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio0_b6_sel GPIO0B[6] iomux select 2'b00: gpio 2'b01: gmac_rxd1m0 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio0_b5_sel GPIO0B[5] iomux select 2'b00: gpio 2'b01: gmac_rxerm0 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio0_b4_sel GPIO0B[4] iomux select 2'b00: gpio 2'b01: gmac_txenm0 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio0_b3_sel GPIO0B[3] iomux select 2'b00: gpio 2'b01: gmac_mdiom0 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio0_b2_sel GPIO0B[2] iomux select 2'b00: gpio 2'b01: gmac_clkrxm0 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio0_b1_sel GPIO0B[1] iomux select 2'b00: gpio 2'b01: gmac_crsm0 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio0_b0_sel GPIO0B[0] iomux select 2'b00: gpio 2'b01: gmac_clktxm0 2'b10: reserved 2'b11: reserved

GRF_GPIO0C_IOMUX
Address: Operational Base + offset (0x0008)
GPIO0C iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio0_c7_sel GPIO0C[7] iomux select 2'b00: gpio 2'b01: gmac_txd3m0 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio0_c6_sel GPIO0C[6] iomux select 2'b00: gpio 2'b01: gmac_txd2m0 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio0_c5_sel GPIO0C[5] iomux select 2'b00: gpio 2'b01: gmac_rxd2m0 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio0_c4_sel GPIO0C[4] iomux select 2'b00: gpio 2'b01: gmac_rxd3m0 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio0_c3_sel GPIO0C[3] iomux select 2'b00: gpio 2'b01: gmac_mdcm0 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio0_c2_sel GPIO0C[2] iomux select 2'b00: gpio 2'b01: gmac_colm0 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio0_c1_sel GPIO0C[1] iomux select 2'b00: gpio 2'b01: gmac_txd0m0 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio0_c0_sel GPIO0C[0] iomux select 2'b00: gpio 2'b01: gmac_txd1m0 2'b10: reserved 2'b11: reserved

GRF_GPIO0D_IOMUX
Address: Operational Base + offset (0x000c)
GPIO0D iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio0_d7_sel GPIO0D[7] iomux select 2'b00: gpio 2'b01: fephyled_speed100 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio0_d6_sel GPIO0D[6] iomux select 2'b00: gpio 2'b01: fephyled_speed10 2'b10: fephyled_duplex 2'b11: sdmmc0_pwrenm1
11:10	RW	0x0	gpio0_d5_sel GPIO0D[5] iomux select 2'b00: gpio 2'b01: fephyled_rxm0 2'b10: fephyled_txm0 2'b11: reserved
9:8	RW	0x0	gpio0_d4_sel GPIO0D[4] iomux select 2'b00: gpio 2'b01: fephy_led_linkm0 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio0_d3_sel GPIO0D[3] iomux select 2'b00: gpio 2'b01: spdif_txm0 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio0_d2_sel GPIO0D[2] iomux select 2'b00: gpio 2'b01: clk_out_gmacm1 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio0_d1_sel GPIO0D[1] iomux select 2'b00: gpio 2'b01: gmac_rxdvm0 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	gpio0_d0_sel GPIO0D[0] iomux select 2'b00: gpio 2'b01: gmac_clkm0 2'b10: reserved 2'b11: reserved

GRF_GPIO1A_IOMUX
Address: Operational Base + offset (0x0010)
GPIO1A iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1_a7_sel GPIO1A[7] iomux select 2'b00: gpio 2'b01: sdmmc0_wp 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio1_a6_sel GPIO1A[6] iomux select 2'b00: gpio 2'b01: sdmmc0_clkout 2'b10: test_clk0 2'b11: reserved
11:10	RW	0x1	gpio1_a5_sel GPIO1A[5] iomux select 2'b00: gpio 2'b01: sdmmc0_detn 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio1_a4_sel GPIO1A[4] iomux select 2'b00: gpio 2'b01: sdmmc0_cmd 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
7:6	RW	0x2	gpio1_a3_sel GPIO1A[3] iomux select 2'b00: gpio 2'b01: sdmmc0_d3 2'b10: jtag_tms 2'b11: reserved
5:4	RW	0x2	gpio1_a2_sel GPIO1A[2] iomux select 2'b00: gpio 2'b01: sdmmc0_d2 2'b10: jtag_tck 2'b11: reserved
3:2	RW	0x2	gpio1_a1_sel GPIO1A[1] iomux select 2'b00: gpio 2'b01: sdmmc0_d1 2'b10: uart2dbg_rxm0 2'b11: reserved
1:0	RW	0x2	gpio1_a0_sel GPIO1A[0] iomux select 2'b00: gpio 2'b01: sdmmc0_d0 2'b10: uart2dbg_txm0 2'b11: reserved

GRF_GPIO1B_IOMUX
Address: Operational Base + offset (0x0014)
GPIO1B iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1_b7_sel GPIO1B[7] iomux select 2'b00: gpio 2'b01: sdmmc1_d1 2'b10: gmac_rxd2m1 2'b11: reserved

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Bit	Attr	Reset Value	Description
13:12	RW	0x0	gpio1_b6_sel GPIO1B[6] iomux select 2'b00: gpio 2'b01: sdmmc1_d0 2'b10: gmac_rxd3m1 2'b11: reserved
11:10	RW	0x0	gpio1_b5_sel GPIO1B[5] iomux select 2'b00: gpio 2'b01: sdmmc1_cmd 2'b10: gmac_rxclkm1 2'b11: reserved
9:8	RW	0x0	gpio1_b4_sel GPIO1B[4] iomux select 2'b00: gpio 2'b01: sdmmc1_clkout 2'b10: gmac_txclkm1 2'b11: reserved
7:6	RW	0x0	gpio1_b3_sel GPIO1B[3] iomux select 2'b00: gpio 2'b01: uart0_ctsn 2'b10: gmac_rxd0m1 2'b11: reserved
5:4	RW	0x0	gpio1_b2_sel GPIO1B[2] iomux select 2'b00: gpio 2'b01: uart0_rtsn 2'b10: gmac_rxd1m1 2'b11: reserved
3:2	RW	0x0	gpio1_b1_sel GPIO1B[1] iomux select 2'b00: gpio 2'b01: uart0_tx 2'b10: gmac_txd0m1 2'b11: reserved
1:0	RW	0x0	gpio1_b0_sel GPIO1B[0] iomux select 2'b00: gpio 2'b01: uart0_rx 2'b10: gmac_txd1m1 2'b11: reserved

GRF_GPIO1C_IOMUX
Address: Operational Base + offset (0x0018)

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GPIO1C iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1_c7_sel GPIO1C[7] iomux select 2'b00: gpio 2'b01: i2s2_lrcktxm0 2'b10: gmac_mdcm1 2'b11: pdm_sdi0m1
13:12	RW	0x0	gpio1_c6_sel GPIO1C[6] iomux select 2'b00: gpio 2'b01: i2s2_sclkm0 2'b10: gmac_rxdvm1 2'b11: pdm_clkm1
11:10	RW	0x0	gpio1_c5_sel GPIO1C[5] iomux select 2'b00: gpio 2'b01: i2s2_mclk 2'b10: gmac_clkm1 2'b11: reserved
9:8	RW	0x0	gpio1_c4_sel GPIO1C[4] iomux select 2'b00: gpio 2'b01: sdmmc1_wp 2'b10: gmac_colm1 2'b11: reserved
7:6	RW	0x0	gpio1_c3_sel GPIO1C[3] iomux select 2'b00: gpio 2'b01: sdmmc1_detn 2'b10: gmac_mdiom1 2'b11: pdm_fsyncm1
5:4	RW	0x0	gpio1_c2_sel GPIO1C[2] iomux select 2'b00: gpio 2'b01: sdmmc1_pwren 2'b10: gmac_crsm1 2'b11: reserved

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	gpio1_c1_sel GPIO1C[1] iomux select 2'b00: gpio 2'b01: sdmmc1_d3 2'b10: gmac_txd2m1 2'b11: reserved
1:0	RW	0x0	gpio1_c0_sel GPIO1C[0] iomux select 2'b00: gpio 2'b01: sdmmc1_d2 2'b10: gmac_txd3m1 2'b11: reserved

GRF_GPIO1D_IOMUX
Address: Operational Base + offset (0x001c)
GPIO1D iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio1_d7_sel GPIO1D[7] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio1_d6_sel GPIO1D[6] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio1_d5_sel GPIO1D[5] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio1_d4_sel GPIO1D[4] iomux select 2'b00: gpio 2'b01: clk32k_outm1 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio1_d3_sel GPIO1D[3] iomux select 2'b00: gpio 2'b01: clk_out_wifim1 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio1_d2_sel GPIO1D[2] iomux select 2'b00: gpio 2'b01: i2s2_lrckrxm0 2'b10: clk_out_gmacm2 2'b11: pdm_sdi3m1
3:2	RW	0x0	gpio1_d1_sel GPIO1D[1] iomux select 2'b00: gpio 2'b01: i2s2_sdom0 2'b10: gmac_txenm1 2'b11: pdm_sdi2m1
1:0	RW	0x0	gpio1_d0_sel GPIO1D[0] iomux select 2'b00: gpio 2'b01: i2s2_sdim0 2'b10: gmac_rxerm1 2'b11: pdm_sdi1m1

GRF_GPIO2A_IOMUX
Address: Operational Base + offset (0x0020)
GPIO2A iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio2_a7_sel GPIO2A[7] iomux select 2'b00: gpio 2'b01: sdmmc0_pwrenm0 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio2_a6_sel GPIO2A[6] iomux select 2'b00: gpio 2'b01: pwm_2 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio2_a5_sel GPIO2A[5] iomux select 2'b00: gpio 2'b01: pwm_1 2'b10: i2c1_scl 2'b11: reserved
9:8	RW	0x0	gpio2_a4_sel GPIO2A[4] iomux select 2'b00: gpio 2'b01: pwm_0 2'b10: i2c1_sda 2'b11: reserved
7:6	RW	0x0	gpio2_a3_sel GPIO2A[3] iomux select 2'b00: gpio 2'b01: efuse_pwren 2'b10: power_state3 2'b11: reserved
5:4	RW	0x0	gpio2_a2_sel GPIO2A[2] iomux select 2'b00: gpio 2'b01: pwm_ir 2'b10: power_state2 2'b11: reserved
3:2	RW	0x0	gpio2_a1_sel GPIO2A[1] iomux select 2'b00: gpio 2'b01: uart2dbg_rxm1 2'b10: power_state1 2'b11: reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	gpio2_a0_sel GPIO2A[0] iomux select 2'b00: gpio 2'b01: uart2dbg_txm1 2'b10: power_state0 2'b11: reserved

GRF_GPIO2BL_IOMUX
Address: Operational Base + offset (0x0024)
GPIO2BL iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RO	0x0	reserved
13:12	RW	0x0	gpio2_b6_sel GPIO2B[6] iomux select 2'b00: gpio 2'b01: i2c2_scl 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio2_b5_sel GPIO2B[5] iomux select 2'b00: gpio 2'b01: i2c2_sda 2'b10: tsadc_shut 2'b11: reserved
9:8	RW	0x2	gpio2_b4_sel GPIO2B[4] iomux select 2'b00: gpio 2'b01: spi_csn1m0 2'b10: flash_vol_sel 2'b11: reserved
7:6	RW	0x0	gpio2_b3_sel GPIO2B[3] iomux select 2'b00: gpio 2'b01: spi_csn0m0 2'b10: reserved 2'b11: reserved

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	gpio2_b2_sel GPIO2B[2] iomux select 2'b00: gpio 2'b01: spi_rxdm0 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio2_b1_sel GPIO2B[1] iomux select 2'b00: gpio 2'b01: spi_txdm0 2'b10: reserved 2'b11: reserved
1:0	RW	0x0	gpio2_b0_sel GPIO2B[0] iomux select 2'b00: gpio 2'b01: spi_clkm0 2'b10: reserved 2'b11: reserved

GRF_GPIO2BH_IOMUX
Address: Operational Base + offset (0x0028)
GPIO2BH iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:3	RO	0x0	reserved
2:0	RW	0x0	gpio2_b7_sel GPIO2B[7] iomux select 3'b000: gpio 3'b001: i2s1_mclk 3'b010: reserved 3'b011: tsp_syncm1 3'b100: cif_clkoutm1 3'b101: reserved 3'b110: reserved 3'b111: reserved

GRF_GPIO2CL_IOMUX

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Address: Operational Base + offset (0x002c)
GPIO2CL iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	reserved
14:12	RW	0x0	gpio2_c4_sel GPIO2C[4] iomux select 3'b000: gpio 3'b001: i2s1_sdio1 3'b010: pdm_sdi1m0 3'b011: card_rstm1 3'b100: reserved 3'b101: reserved 3'b110: reserved 3'b111: reserved
11:9	RW	0x0	gpio2_c3_sel GPIO2C[3] iomux select 3'b000: gpio 3'b001: i2s1_sdi 3'b010: pdm_sdi0m0 3'b011: card_clkm1 3'b100: reserved 3'b101: reserved 3'b110: reserved 3'b111: reserved
8:6	RW	0x0	gpio2_c2_sel GPIO2C[2] iomux select 3'b000: gpio 3'b001: i2s1_sclk 3'b010: pdm_clkm0 3'b011: tsp_d7m1 3'b100: cif_data7m1 3'b101: reserved 3'b110: reserved 3'b111: reserved

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Bit	Attr	Reset Value	Description
5:3	RW	0x0	gpio2_c1_sel GPIO2C[1] iomux select 3'b000: gpio 3'b001: i2s1_lrcktx 3'b010: spdif_txm1 3'b011: tsp_d6m1 3'b100: cif_data6m1 3'b101: reserved 3'b110: reserved 3'b111: reserved
2:0	RW	0x0	gpio2_c0_sel GPIO2C[0] iomux select 3'b000: gpio 3'b001: i2s1_lrckrx 3'b010: reserved 3'b011: tsp_d5m1 3'b100: cif_data5m1 3'b101: reserved 3'b110: reserved 3'b111: reserved

GRF_GPIO2CH_IOMUX
Address: Operational Base + offset (0x0030)
GPIO2CH iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio2_c7_sel GPIO2C[7] iomux select 2'b00: gpio 2'b01: i2s1_sdo 2'b10: pdm_fsyncm0 2'b11: reserved
13:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:3	RW	0x0	gpio2_c6_sel GPIO2C[6] iomux select 3'b000: gpio 3'b001: i2s1_sdio3 3'b010: pdm_sdi3m0 3'b011: card_iom1 3'b100: reserved 3'b101: reserved 3'b110: reserved 3'b111: reserved
2:0	RW	0x0	gpio2_c5_sel GPIO2C[5] iomux select 3'b000: gpio 3'b001: i2s1_sdio2 3'b010: pdm_sdi2m0 3'b011: card_detm1 3'b100: reserved 3'b101: reserved 3'b110: reserved 3'b111: reserved

GRF_GPIO2D_IOMUX
Address: Operational Base + offset (0x0034)
GPIO2D iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio2_d7_sel GPIO2D[7] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d4 2'b11: reserved

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Bit	Attr	Reset Value	Description
13:12	RW	0x0	gpio2_d6_sel GPIO2D[6] iomux select 2'b00: gpio 2'b01:reserved 2'b10: emmc_d3 2'b11: reserved
11:10	RW	0x0	gpio2_d5_sel GPIO2D[5] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d2 2'b11: reserved
9:8	RW	0x0	gpio2_d4_sel GPIO2D[4] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d1 2'b11: reserved
7:6	RW	0x0	gpio2_d3_sel GPIO2D[3] iomux select 2'b00: gpio 2'b01: usb3otg_drvbus 2'b10: reserved 2'b11: reserved
5:4	RW	0x0	gpio2_d2_sel GPIO2D[2] iomux select 2'b00: gpio 2'b01: usb2otg_drvbus 2'b10: reserved 2'b11: reserved
3:2	RW	0x0	gpio2_d1_sel GPIO2D[1] iomux select 2'b00: gpio 2'b01: i2c0_sda 2'b10: fephyled_rxm1 2'b11: fephyled_txm1
1:0	RW	0x0	gpio2_d0_sel GPIO2D[0] iomux select 2'b00: gpio 2'b01: i2c0_scl 2'b10: fephy_led_linkm1 2'b11: reserved

GRF_GPIO3AL_IOMUX
Address: Operational Base + offset (0x0038)

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GPIO3AL iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	reserved
14:12	RW	0x0	gpio3_a4_sel GPIO3A[4] iomux select 3'b000: gpio 3'b001: tsp_d0 3'b010: cif_data0 3'b011: sdmmc0ext_d0 3'b100: uart1_tx 3'b101: usb3phy_debug4 3'b110: reserved 3'b111: reserved
11:9	RW	0x0	gpio3_a3_sel GPIO3A[3] iomux select 3'b000: gpio 3'b001: tsp_syncm0 3'b010: cif_clkoutm0 3'b011: sdmmc0ext_wp 3'b100: reserved 3'b101: reserved 3'b110: reserved 3'b111: reserved
8:6	RW	0x0	gpio3_a2_sel GPIO3A[2] iomux select 3'b000: gpio 3'b001: tsp_clk 3'b010: cif_clkin 3'b011: sdmmc0ext_clkout 3'b100: spi_rxdm2 3'b101: usb3phy_debug3 3'b110: i2s2_sdim1 3'b111: reserved

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Bit	Attr	Reset Value	Description
5:3	RW	0x0	gpio3_a1_sel GPIO3A[1] iomux select 3'b000: gpio 3'b001: tsp_fail 3'b010: cif_href 3'b011: sdmmc0ext_det 3'b100: spi_txdm2 3'b101: usb3phy_debug2 3'b110: i2s2_sdom1 3'b111: reserved
2:0	RW	0x0	gpio3_a0_sel GPIO3A[0] iomux select 3'b000: gpio 3'b001: tsp_valid 3'b010: cif_vsync 3'b011: sdmmc0ext_cmd 3'b100: spi_clkm2 3'b101: usb3phy_debug1 3'b110: i2s2_sclkm1 3'b111: reserved

GRF_GPIO3AH_IOMUX
Address: Operational Base + offset (0x003c)
GPIO3AH iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:9	RO	0x0	reserved
8:6	RW	0x0	gpio3_a7_sel GPIO3A[7] iomux select 3'b000: gpio 3'b001: tsp_d3 3'b010: cif_data3 3'b011: sdmmc0ext_d3 3'b100: uart1_ctsn 3'b101: usb3phy_debug7 3'b110: reserved 3'b111: reserved

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Bit	Attr	Reset Value	Description
5:3	RW	0x0	gpio3_a6_sel GPIO3A[6] iomux select 3'b000: gpio 3'b001: tsp_d2 3'b010: cif_data2 3'b011: sdmmc0ext_d2 3'b100: uart1_rx 3'b101: usb3phy_debug6 3'b110: reserved 3'b111: reserved
2:0	RW	0x0	gpio3_a5_sel GPIO3A[5] iomux select 3'b000: gpio 3'b001: tsp_d1 3'b010: cif_data1 3'b011: sdmmc0ext_d1 3'b100: uart1_rtsn 3'b101: usb3phy_debug5 3'b110: reserved 3'b111: reserved

GRF_GPIO3BL_IOMUX
Address: Operational Base + offset (0x0040)
GPIO3BL iomux control

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:3	RO	0x0	reserved
2:0	RW	0x0	gpio3_b0_sel GPIO3B[0] iomux select 3'b000: gpio 3'b001: tsp_d4 3'b010: cif_data4 3'b011: spi_csn0m2 3'b100: i2s2_lrcktxm1 3'b101: usb3phy_debug8 3'b110: i2s2_lrckrxm1 3'b111: reserved

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GRF_GPIO3BH_IOMUX
Address: Operational Base + offset (0x0044)
GPIO3BH iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio3_b7_sel GPIO3B[7] iomux select 2'b00: gpio 2'b01: card_iom0 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio3_b6_sel GPIO3B[6] iomux select 2'b00: gpio 2'b01: card_detm0 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio3_b5_sel GPIO3B[5] iomux select 2'b00: gpio 2'b01: card_rstm0 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio3_b4_sel GPIO3B[4] iomux select 2'b00: gpio 2'b01: card_clkm0 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio3_b3_sel GPIO3B[3] iomux select 2'b00: gpio 2'b01: tsp_d7m0 2'b10: cif_data7m0 2'b11: reserved

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	gpio3_b2_sel GPIO3B[2] iomux select 2'b00: gpio 2'b01: tsp_d6m0 2'b10: cif_data6m0 2'b11: reserved
3:2	RW	0x0	gpio3_b1_sel GPIO3B[1] iomux select 2'b00: gpio 2'b01: tsp_d5m0 2'b10: cif_data5m0 2'b11: reserved
1:0	RO	0x0	reserved

GRF_GPIO3C_IOMUX
Address: Operational Base + offset (0x0048)
GPIO3C iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RW	0x0	gpio3_c7_sel GPIO3C[7] iomux select 2'b00: gpio 2'b01:reserved 2'b10: spi_clkm1 2'b11: reserved
13:12	RW	0x0	gpio3_c6_sel GPIO3C[6] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_pwren 2'b11: reserved
11:10	RW	0x0	gpio3_c5_sel GPIO3C[5] iomux select 2'b00: gpio 2'b01:reserved 2'b10: emmc_clkout 2'b11: reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x0	gpio3_c4_sel GPIO3C[4] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_rstn 2'b11: reserved
7:6	RW	0x0	gpio3_c3_sel GPIO3C[3] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_cmd 2'b11: reserved
5:4	RW	0x0	gpio3_c2_sel GPIO3C[2] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d7 2'b11: reserved
3:2	RW	0x0	gpio3_c1_sel GPIO3C[1] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d6 2'b11: reserved
1:0	RW	0x0	gpio3_c0_sel GPIO3C[0] iomux select 2'b00: gpio 2'b01: reserved 2'b10: emmc_d5 2'b11: reserved

GRF_GPIO3D_IOMUX
Address: Operational Base + offset (0x004c)
GPIO3D iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	gpio3_d7_sel GPIO3D[7] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
13:12	RW	0x0	gpio3_d6_sel GPIO3D[6] iomux select 2'b00: gpio 2'b01:reserved 2'b10: reserved 2'b11: reserved
11:10	RW	0x0	gpio3_d5_sel GPIO3D[5] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
9:8	RW	0x0	gpio3_d4_sel GPIO3D[4] iomux select 2'b00: gpio 2'b01: reserved 2'b10: reserved 2'b11: reserved
7:6	RW	0x0	gpio3_d3_sel GPIO3D[3] iomux select 2'b00: gpio 2'b01: reserved 2'b10: spi_csn1m1 2'b11: reserved
5:4	RW	0x0	gpio3_d2_sel GPIO3D[2] iomux select 2'b00: gpio 2'b01: reserved 2'b10: spi_csn0m1 2'b11: reserved
3:2	RW	0x0	gpio3_d1_sel GPIO3D[1] iomux select 2'b00: gpio 2'b01:reserved 2'b10: spi_txdm1 2'b11: reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	gpio3_d0_sel GPIO3D[0] iomux select 2'b00: gpio 2'b01: reserved 2'b10: spi_rxdm1 2'b11: reserved

GRF_COM_IOMUX
Address: Operational Base + offset (0x0050)
GRF common iomux control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12	RW	0x0	grf_clk_out_gmacm1_sel gmac m1 io select 0: before optimization 1: after optimization
11	RW	0x0	grf_clk_out_gmacm2_sel clk_out_gmacm2 select 0: before optimization 1: after optimization
10	RW	0x0	grf_gmac_m1_sel gmac m1 io select 0: before optimization 1: after optimization
9	RW	0x0	grf_cif_io_sel cif_io select 0: m0 mux solution 1: m1 mux solution
8	RW	0x0	grf_tsp_io_sel tsp_io select 0: m0 mux solution 1: m1 mux solution
7	RW	0x0	grf_card_io_sel card_io select 0: m0 mux solution 1: m1 mux solution

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Bit	Attr	Reset Value	Description
6	RW	0x0	grf_i2s2_io_sel i2s2_io select 0: m0 mux solution 1: m1 mux solution
5:4	RW	0x0	grf_con_spi_io_sel spi_io_sel bit control 2'b00: m0 mux solution 2'b01: m1 mux solution 2'b10: m2 mux solution 2'b11: reserved
3	RW	0x0	grf_con_pdm_iomux_sel pdm_iomux_sel bit control 0: m0 mux solution 1: m1 mux solution
2	RW	0x0	grf_con_gmac_iomux_sel gmac_iomux_sel bit control 0: m0 mux solution 1: m1 mux solution
1:0	RW	0x0	grf_uart_dbg_sel grf_con_iomux_uartdbgsel when grf_con_iomux_uartdbgena is 1, uartdbg source select 2'b00: m0 2'b01: m1 2'b10: usb2phy 2'b11: reserved

GRF_GPIO0A_P

Address: Operational Base + offset (0x0100)

GPIO0A PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x566a	gpio0_a_p gpio0a bit control GPIO0A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO0B_P
Address: Operational Base + offset (0x0104)
GPIO0B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaa6a	gpio0_b_p gpio0b bit control GPIO0B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO0C_P
Address: Operational Base + offset (0x0108)
GPIO0C PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xaa6a	gpio0_c_p gpio0c bit control GPIO0C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO0D_P
Address: Operational Base + offset (0x010c)
GPIO0D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio0_d_p gpio0d bit control GPIO0D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1A_P
Address: Operational Base + offset (0x0110)
GPIO1A PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xa555	gpio1_a_p gpio1a bit control GPIO1A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1B_P
Address: Operational Base + offset (0x0114)
GPIO1B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x56a5	gpio1_b_p gpio1b bit control GPIO1B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1C_P
Address: Operational Base + offset (0x0118)
GPIO1C PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x9a65	gpio1_c_p gpio1c bit control GPIO1C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO1D_P
Address: Operational Base + offset (0x011c)
GPIO1D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio1_d_p gpio1d bit control GPIO1D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2A_P
Address: Operational Base + offset (0x0120)
GPIO2A PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x9556	gpio2_a_p gpio2a bit control GPIO2A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2B_P
Address: Operational Base + offset (0x0124)
GPIO2B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x959a	gpio2_b_p gpio2b bit control GPIO2B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2C_P
Address: Operational Base + offset (0x0128)
GPIO2C PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x5565	gpio2_c_p gpio2c bit control GPIO2C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO2D_P
Address: Operational Base + offset (0x012c)
GPIO2D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x55a5	gpio2_d_p gpio2d bit control GPIO2D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3A_P
Address: Operational Base + offset (0x0130)
GPIO3A PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x55a5	gpio3_a_p gpio3a bit control GPIO3A PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3B_P
Address: Operational Base + offset (0x0134)
GPIO3B PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x5aaa	gpio3_b_p gpio3b bit control GPIO3B PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3C_P
Address: Operational Base + offset (0x0138)
GPIO3C PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x6555	gpio3_c_p gpio3c bit control GPIO3C PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO3D_P
Address: Operational Base + offset (0x013c)
GPIO3D PU/PD control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x555a	gpio3_d_p gpio3d bit control GPIO3D PU/PD programmation section, every GPIO bit corresponding to 2bits 2'b00: Z(Normal operation); 2'b01: weak 1(pull-up); 2'b10: weak 0(pull_down); 2'b11: Repeater(Bus keeper)

GRF_GPIO0A_E
Address: Operational Base + offset (0x0200)
GPIO0A drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x8011	gpio0_a_e gpio0a bit control GPIO0A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO0B_E
Address: Operational Base + offset (0x0204)
GPIO0B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaa2a	gpio0_b_e gpio0b bit control GPIO0B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO0C_E
Address: Operational Base + offset (0x0208)
GPIO0C drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xaa0a	gpio0_c_e gpio0c bit control GPIO0C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO0D_E
Address: Operational Base + offset (0x020c)
GPIO0D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x005a	gpio0_d_e gpio0d bit control GPIO0D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1A_E
Address: Operational Base + offset (0x0210)
GPIO1A drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xaaaa	gpio1_a_e gpio1a bit control GPIO1A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1B_E
Address: Operational Base + offset (0x0214)
GPIO1B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaa2a	gpio1_b_e gpio1b bit control GPIO1B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1C_E
Address: Operational Base + offset (0x0218)
GPIO1C drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xa88a	gpio1_c_e gpio1c bit control GPIO1C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO1D_E
Address: Operational Base + offset (0x021c)
GPIO1D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x005a	gpio1_d_e gpio1d bit control GPIO1D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2A_E
Address: Operational Base + offset (0x0220)
GPIO2A drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	gpio2_a_e gpio2a bit control GPIO2A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2B_E
Address: Operational Base + offset (0x0224)
GPIO2B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x4145	gpio2_b_e gpio2b bit control GPIO2B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2C_E
Address: Operational Base + offset (0x0228)
GPIO2C drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x5515	gpio2_c_e gpio2c bit control GPIO2C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO2D_E
Address: Operational Base + offset (0x022c)
GPIO2D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaa01	gpio2_d_e gpio2d bit control GPIO2D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3A_E
Address: Operational Base + offset (0x0230)
GPIO3A drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xaa22	gpio3_a_e gpio3a bit control GPIO3A drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3B_E
Address: Operational Base + offset (0x0234)
GPIO3B drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	gpio3_b_e gpio3b bit control GPIO3B drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3C_E
Address: Operational Base + offset (0x0238)
GPIO3C drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0xaaaa	gpio3_c_e gpio3c bit control GPIO3C drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO3D_E
Address: Operational Base + offset (0x023c)
GPIO3D drive strength control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0xaaaa	gpio3_d_e gpio3d bit control GPIO3D drive strength control, every GPIO bit corresponding to 2bits 2'b00: 2mA 2'b01: 4mA 2'b10: 8mA 2'b11: 12mA

GRF_GPIO0L_SR
Address: Operational Base + offset (0x0300)
GPIO0 A/B SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	grf_gpio0b_sr GPIO0B slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio0a_sr GPIO0A slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO0H_SR
Address: Operational Base + offset (0x0304)
GPIO0 C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio0d_sr GPIO0D slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio0c_sr GPIO0C slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO1L_SR
Address: Operational Base + offset (0x0308)
GPIO1 A/B SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	grf_gpio1b_sr GPIO1B slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio1a_sr GPIO1A slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO1H_SR
Address: Operational Base + offset (0x030c)
GPIO1 C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio1d_sr GPIO1D slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio1c_sr GPIO1C slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO2L_SR
Address: Operational Base + offset (0x0310)
GPIO2 A/B SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	grf_gpio2b_sr GPIO2B slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio2a_sr GPIO2A slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO2H_SR
Address: Operational Base + offset (0x0314)
GPIO2 C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio2d_sr GPIO2D slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio2c_sr GPIO2C slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO3L_SR
Address: Operational Base + offset (0x0318)
GPIO3 A/B SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	grf_gpio3b_sr GPIO3B slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio3a_sr GPIO3A slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO3H_SR
Address: Operational Base + offset (0x031c)
GPIO3 C/D SR control

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio3d_sr GPIO3D slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast
7:0	RW	0x00	grf_gpio3c_sr GPIO3C slew rate control for each bit 1'b0: slow(half frequency) 1'b1: fast

GRF_GPIO0L_SMT
Address: Operational Base + offset (0x0380)
GPIO0 A/B smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	grf_gpio0b_smt gpio0b_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio0a_smt gpio0a_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_GPIO0H_SMT
Address: Operational Base + offset (0x0384)
GPIO0 C/D smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio0d_smt gpio0d_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio0c_smt gpio0c_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_GPIO1L_SMT
Address: Operational Base + offset (0x0388)
GPIO1 A/B smitter control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio1b_smt gpio1b_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio1a_smt gpio1a_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_GPIO1H_SMT
Address: Operational Base + offset (0x038c)
GPIO1 C/D smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio1d_smt gpio1d_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio1c_smt gpio1c_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_GPIO2L_SMT

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Address: Operational Base + offset (0x0390)
GPIO2 A/B smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio2b_smt gpio2b_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio2a_smt gpio2a_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_GPIO2H_SMT
Address: Operational Base + offset (0x0394)
GPIO2 C/D smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio2d_smt gpio2d_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio2c_smt gpio2c_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

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GRF_GPIO3L_SMT
Address: Operational Base + offset (0x0398)
GPIO3 A/B smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio3b_smt gpio3b_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.
7:0	RW	0x00	grf_gpio3a_smt gpio3a_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_GPIO3H_SMT
Address: Operational Base + offset (0x039c)
GPIO3 C/D smitter control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:8	RW	0x00	grf_gpio3d_smt gpio3d_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	grf_gpio3c_smt gpio3c_smt bit control Schmitt trigger control. 0: No hysteresis 1: Schmitt trigger enabled.

GRF_SOC_CON0
Address: Operational Base + offset (0x0400)
SOC control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RO	0x0000	soc_con0 Reserved reserved

GRF_SOC_CON1
Address: Operational Base + offset (0x0404)
SOC control register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	soc_con1 Reserved reserved

GRF_SOC_CON2
Address: Operational Base + offset (0x0408)
SOC control register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	grf_con_i2s1_src_sel i2s1_src_sel bit control 1'b1 I2S1 is controll is connected with ACODEC PHY; 1'b0: I2S1 is connected with IO
14	RW	0x0	grf_con_i2s_acodec_en i2s_acodec_en bit control i2s_8ch iomux control 1:connect with acodec 0:connect with external io
13	RW	0x0	grf_con_ddrphy_bufferen_sel ddrphy_bufferen_sel bit control 1'b1: ddrphy_bufferen from grf_con_ddrphy_bufferen_core; 1'b0: ddrphy_bufferen from pmu
12	RW	0x1	grf_con_ddrphy_bufferen_core ddrphy_bufferen_core bit control 1'b1: enable ddrphy_bufferen; 1'b0: disable ddrphy_bufferen
11	RW	0x0	grf_con_hdmi_sdain_msk hdmi_sdain_msk bit control hdmi_sdain mask control 0: mask disable 1: mask enable
10	RW	0x0	grf_con_hdmi_sclin_msk hdmi_sclin_msk bit control hdmi_sclin mask control 0: mask disable 1: mask enable
9	RW	0x0	grf_con_hdmi_cecin_msk hdmi_cecin_msk bit control hdmi_cecin mask control 0: mask disable 1: mask enable

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Bit	Attr	Reset Value	Description
8	RW	0x0	grf_con_saradc_sel saradc_sel bit control SARADC controller selection 1'b1: select saradc auto controller 1'b0: select orignal saradc controller
7	RW	0x0	grf_con_hdmisda5v_gpio_iout hdmisda5v_gpio_iout bit control IO PAD output data 1'b0: set IO output to 0; 1'b1: set IO output to 1;
6	RW	0x0	grf_con_hdmisda5v_gpio_ioe_ hdmisda5v_gpio_ioe_ bit control IO Pad output enable bit control 1'b1: set IO as input; 1'b0: set IO as output;
5	RW	0x0	grf_con_hdmiscl5v_gpio_iout hdmiscl5v_gpio_iout bit control IO PAD output data 1'b0: set IO output to 0; 1'b1: set IO output to 1;
4	RW	0x0	grf_con_hdmiscl5v_gpio_ioe_ hdmiscl5v_gpio_ioe_ bit control IO Pad output enable bit control 1'b1: set IO as input; 1'b0: set IO as output;
3	RW	0x0	grf_con_hdmihpd5v_gpio_iout hdmihpd5v_gpio_iout bit control IO PAD output data 1'b0: set IO output to 0; 1'b1: set IO output to 1;
2	RW	0x0	grf_con_hdmihpd5v_gpio_ioe_ hdmihpd5v_gpio_ioe_ bit control IO Pad output enable bit control 1'b1: set IO as input; 1'b0: set IO as output;
1	RW	0x0	grf_con_hdmicec5v_gpio_iout hdmicec5v_gpio_iout bit control IO PAD output data 1'b0: set IO output to 0; 1'b1: set IO output to 1;
0	RW	0x0	grf_con_hdmicec5v_gpio_ioe_ hdmicec5v_gpio_ioe_ bit control IO Pad output enable bit control 1'b1: set IO as input; 1'b0: set IO as output;

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GRF_SOC_CON3
Address: Operational Base + offset (0x040c)
SOC control register3

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	grf_con_hdmisda5v_gpio_sel hdmisda5v_gpio_sel bit control "if grf_con_hdmisda5v_gpio_sel == 0 and grf_con_i2c3_sda5v == 0, SDA5V is controlled by HDMI controller; if grf_con_hdmisda5v_gpio_sel == 0 and grf_con_i2c3_sda5v == 1, SDA5V is controlled by I2C3 controller; if grf_con_hdmisda5v_gpio_sel == 1 and no matter grf_con_i2c3_sda5v what is, SDA5V is controlled by grf; "
14	RW	0x0	grf_con_hdmiscl5v_gpio_sel hdmiscl5v_gpio_sel bit control "if grf_con_hdmiscl5v_gpio_sel == 0 and grf_con_i2c3_scl5v == 0, SCL5V is controlled by HDMI controller; if grf_con_hdmiscl5v_gpio_sel == 0 and grf_con_i2c3_scl5v == 1, SCL5V is controlled by I2C3 controller; if grf_con_hdmiscl5v_gpio_sel == 1 and no matter grf_con_i2c3_scl5v what is, SCL5V is controlled by grf; "
13	RW	0x0	grf_con_hdmihpd5v_gpio_sel hdmihpd5v_gpio_sel bit control 1'b1: HPD5V io is controlled by grf_con_hdmihpd5v_gpio_ioe_ and grf_con_hdmihpd5v_gpio_iout; 1'b0: HPD5V is controlled by HDMI controller
12	RW	0x0	grf_con_hdmicec5v_gpio_sel hdmicec5v_gpio_sel bit control 1'b1: CEC5V io is controlled by grf_con_hdmicec5v_gpio_ioe_ and grf_con_hdmicec5v_gpio_iout; 1'b0: CEC5V is controlled by HDMI controller
11	RW	0x0	grf_con_h265enc_work_flag h265enc_work_flag bit control 1'b1: sram is controlled by h265 encoder 1'b0: sram is controlled by h264 encoder

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Bit	Attr	Reset Value	Description
10:9	RW	0x0	grf_vop_standby_sel vop_standby select dcf vop standby source 2'b00: from vop standby; 2'b01: from vop aclk en; 2'b10: from vop aclk en or vop standby; 2'b11 Reserved
8	RW	0x0	grf_hdmiphy_pll_pd hdmiphy_pll_pd hdmiphy pll power down, active high
7	RW	0x0	grf_hdmip_pdata_en hdmip_pdata enable hdmiphy input parallel data enable 1:enable 0:disable
6	RO	0x0	reserved
5:3	RW	0x0	grf_uart_rts_sel uart_rts select UART polarity selection for rts_n Every bit for one UART, bit2 is for UART2, bit1 is for UART1, bit0 is for UART0 1:cts_n is high active 0:cts_n is low active
2:0	RW	0x0	grf_uart_cts_sel uart_cts select UART polarity selection for cts_n Every bit for one UART, bit2 is for UART2, bit1 is for UART1, bit0 is for UART0 1:cts_n is high active 0:cts_n is low active

GRF_SOC_CON4

Address: Operational Base + offset (0x0410)

SOC control register4

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15	RW	0x0	cif_pclkin_inv_sel pclkin_inv select 0: inveter disable 1: inverter enable
14	RW	0x0	grf_con_gmac2io_extclk_sel gmac2io ext clock select 0: use external clock 1: use cru clock
13	RW	0x0	grf_con_hdmi_hpd_src_sel hdmi_hpd source select 0:from gpio of 3.3V or 5V 1:from SARADC CH0
12	RW	0x0	grf_force_jtag force jtag Force select jtag function from sdmmc0 IO 1:IO used for JTAG. 0:IO used for SDMMC
11	RW	0x0	grf_hdmi_cec_vsel grf_hdmi_cec_vsel hdmi cec port 3.3V/5V io select 1:IO is 3.3V 0:IO is 5V
10	RW	0x0	grf_hdmi_sda_vsel grf_hdmi_sda_vsel hdmi sda port 3.3V/5V io select 1:IO is 3.3V 0:IO is 5V
9	RW	0x0	grf_hdmi_scl_vsel grf_hdmi_scl_vsel hdmi scl port 3.3V/5V io select 1:IO is 3.3V 0:IO is 5V
8	RW	0x0	grf_hdmi_hdp_vsel grf_hdmi_hdp_vsel hdmi hpd port 3.3V/5V io select 1:IO is 3.3V 0:IO is 5V
7	RW	0x0	grf_vccio2_vsel_src grf_vccio2_vsel_src 1'b1: vccio2 vsel controlled by grf_vccio2_vsel; 1'b0: vccio2 vsel controlled by GPIO2B4 IO
6	RW	0x0	grf_pmuio_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V

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Bit	Attr	Reset Value	Description
5	RW	0x0	grf_vccio6_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V
4	RW	0x0	grf_vccio5_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V
3	RW	0x0	grf_vccio4_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V
2	RW	0x0	grf_vccio3_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V
1	RW	0x0	grf_vccio2_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V
0	RW	0x0	grf_vccio1_vsel VCC IO voltage select 1'b0:3.3V 1'b1:1.8V

GRF_SOC_CON5
Address: Operational Base + offset (0x0414)
SOC control register5

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	vpu_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response

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Bit	Attr	Reset Value	Description
14	RW	0x0	vop_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
13	RW	0x0	usb_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
12	RW	0x0	subvio_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
11	RW	0x0	rkvenc_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
10	RW	0x0	rkvdec_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
9	RW	0x0	vpu_pwr_IdleReq send idle request to vpu niu 0:disable 1:enable
8	RW	0x0	vio_pwr_IdleReq send idle request to vio niu 0:disable 1:enable
7	RW	0x0	sys_pwr_IdleReq send idle request to bus niu 0:disable 1:enable
6	RW	0x0	rkvenc_pwr_IdleReq rkvenc_pwr_IdleReq
5	RW	0x0	rkvdec_pwr_IdleReq send idle request to rkvdec niu 0:disable 1:enable
4	RW	0x0	peri_pwr_IdleReq send idle request to peri niu 0:disable 1:enable

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Bit	Attr	Reset Value	Description
3	RW	0x0	msch_pwr_IdleReq send idle request to msch niu 0:disable 1:enable
2	RW	0x0	msch_apb_pwr_IdleReq send idle request to mschapb niu 0:disable 1:enable
1	RW	0x0	gpu_pwr_IdleReq send idle request to gpu niu 0:disable 1:enable
0	RW	0x0	core_pwr_IdleReq send idle request to core niu 0:disable 1:enable

GRF_SOC_CON6
Address: Operational Base + offset (0x0418)
SOC control register6

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	peri_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
14	RW	0x0	nv_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
13	RW	0x0	msch_srv_fw_fwr_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response

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Bit	Attr	Reset Value	Description
12	RW	0x0	msch_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
11	RW	0x0	gpu_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
10	RW	0x0	gmac_fwr_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
9	RW	0x0	core_fwr_bus_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
8	RW	0x0	vcodec_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
7	RW	0x0	gpu_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
6	RW	0x0	core_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
5	RW	0x0	bus_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
4	RW	0x0	vop_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
3	RW	0x0	vio_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
2	RW	0x0	rkvenc_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response

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Bit	Attr	Reset Value	Description
1	RW	0x0	rkvdec_req_link_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response
0	RW	0x0	peri_req_pwrDiscTargPwrStall Response type when NIU is set Idle 0:error response 1:stall response

GRF_SOC_CON7
Address: Operational Base + offset (0x041c)
SOC control register7

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	grf_con_otp_usr_clk_mux otp user mode clock source mux 0: bypass clock 1: divide by 2
1	RW	0x0	grf_con_newpll_clamp_en newpll clamp enable 0: disable 1: enable
0	RW	0x0	grf_con_scr_sim_detect_inv_sel scr detect inveter select 0: inveter disable 1: inveter enable

GRF_SOC_CON8
Address: Operational Base + offset (0x0420)
SOC control register8

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	grf_tsadc_testbit_h tsadc_testbit_h bit register tsadc_testbit_h bit register

GRF_SOC_CON9

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Address: Operational Base + offset (0x0424)
SOC control register9

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	grf_tsadc_testbit_l tsadc_testbit_l bit register tsadc_testbit_l bit register

GRF_SOC_CON10
Address: Operational Base + offset (0x0428)
SOC control register10

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x1	grf_con_hdmi_sda5v_smt hdmi_sda5v_smt bit control hdmi_sda5v_smt bit control
14	RW	0x1	grf_con_hdmi_scl5v_smt hdmi_scl5v_smt bit control hdmi_scl5v_smt bit control
13	RW	0x1	grf_con_hdmi_hpd5v_smt hdmi_hpd5v_smt bit control hdmi_hpd5v_smt bit control
12	RW	0x1	grf_con_hdmi_cec5v_smt hdmi_cec5v_smt bit control hdmi_cec5v_smt bit control
11	RW	0x1	grf_con_gpiomut_pmuio_p2 gpiomut_pmuio_p2 bit control gpiomut_pmuio pull bit 2
10	RW	0x0	grf_con_gpiomut_pmuio_p1 gpiomut_pmuio_p1 bit control gpiomut_pmuio pull bit 1

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Bit	Attr	Reset Value	Description
9	RW	0x0	grf_con_sdmmc_pwren_sel iomux select GPIO2A7 sdmmc power selection 1'b0: from sdmmc_ext 1'b1: from sdmmc0
8	RW	0x0	grf_con_i2c3_sda5v iomux select "if grf_con_hdmisda5v_gpio_sel == 0 and grf_con_i2c3_sda5v == 0, SDA5V is controlled by HDMI controller; if grf_con_hdmisda5v_gpio_sel == 0 and grf_con_i2c3_sda5v == 1, SDA5V is controlled by I2C3 controller; if grf_con_hdmisda5v_gpio_sel == 1 and no matter grf_con_i2c3_sda5v what is, SDA5V is controlled by grf; "
7	RW	0x0	grf_con_i2c3_scl5v iomux select "if grf_con_hdmiscl5v_gpio_sel == 0 and grf_con_i2c3_scl5v == 0, SCL5V is controlled by HDMI controller; if grf_con_hdmiscl5v_gpio_sel == 0 and grf_con_i2c3_scl5v == 1, SCL5V is controlled by I2C3 controller; if grf_con_hdmiscl5v_gpio_sel == 1 and no matter grf_con_i2c3_scl5v what is, SCL5V is controlled by grf; "
6	RW	0x0	grf_con_tsadc_ch_inv tsadc_ch_inv bit control The enable signal of the clock inverter for the analog to digital interface 0:invert 1:don't invert
5	RW	0x0	grf_con_clk_wifi_sel clk_wifi_sel bit control clk_wifi (GPIO1D3) source selection 1'b1: from clk_wifi; 1'b0: from 24M OSC
4	RW	0x0	grf_con_i2s1_8ch_sdio3_oen i2s1_8ch_sdio3_oen bit control i2s1_8ch_sdio3_oen 1:output disable 2:output enable
3	RW	0x0	grf_con_i2s1_8ch_sdio2_oen i2s1_8ch_sdio2_oen bit control i2s1_8ch_sdio2_oen 1:output disable 1:output enable

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Bit	Attr	Reset Value	Description
2	RW	0x0	grf_con_i2s1_8ch_sdio1_oen i2s1_8ch_sdio1_oen bit control i2s1_8ch_sdio1_oen 1:output disable 0:output enable
1	RW	0x0	gpiomut_pmuio_iout gpiomut_pmuio_iout bit register gpiomut output value 1'b1: output 1; 1'b1: output 0
0	RW	0x0	gpiomut_pmuio_ioe_ gpiomut_pmuio_ioe_ bit register gpiomut output enable 1'b1: output disable; 1'b0: output enable

GRF_SOC_STATUS0
Address: Operational Base + offset (0x0480)
SOC status register0

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RO	0x0	h265enc_vpu_idle h265enc_vpu_idle bit register
26	RO	0x0	hdmicec5v_gpio_masked_pin hdmicec5v_gpio_masked_pin IO PAD input status
25	RO	0x0	hdmihpd5v_gpio_masked_pin hdmihpd5v_gpio_masked_pin IO PAD input status
24	RO	0x0	hdmisda5v_gpio_masked_pin IO PAD input status
23	RO	0x0	hdmiscl5v_gpio_masked_pin IO PAD input status
22	RO	0x0	gpiomut_pmuio_pin IO PAD input status
21	RO	0x0	grf_st_acodec_master_en st_acodec_master enable st_acodec_master enable
20	RO	0x0	gmac2phy_portselect gmac2phy_port select signal indicating the default PHY interface of MAC 1:MII 0:GMII

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Bit	Attr	Reset Value	Description
19	RO	0x0	grf_stat_vdac_dispdet grf_stat_vdac_dispdet bit register vdac cable detection output status
18	RO	0x0	vop_dma_finish vop_dma_finish bit register vop_dma_finish_status
17	RO	0x0	reserved
16	RO	0x0	timer_en_status5 timer_en_status5 bit register
15	RO	0x0	timer_en_status4 timer_en_status4 bit register
14	RO	0x0	timer_en_status3 timer_en_status3 bit register
13	RO	0x0	timer_en_status2 timer_en_status2 bit register
12	RO	0x0	timer_en_status1 timer_en_status1 bit register
11	RO	0x0	timer_en_status0 timer_en_status0 bit register
10	RO	0x0	gmac2io_portselect gmac2io_port select
9	RO	0x0	opt_sbpi_busy_ns opt_sbpi_busy_ns bit register
8	RO	0x0	opt_user_busy_ns opt_user_busy_ns bit register
7	RO	0x0	opt_sbpi_busy_s opt_sbpi_busy_s bit register
6	RO	0x0	opt_user_busy_s opt_user_busy_s bit register
5	RO	0x0	ddr_plllock ddr_plllock bit register DDRPLL of DDRPHY lock status.
4	RO	0x0	apll_lock pll_lock bit register APLL lock status.
3	RO	0x0	dpll_lock pll_lock bit register DPLL lock status.
2	RO	0x0	cpll_lock pll_lock bit register CPLL lock status.
1	RO	0x0	gpll_lock pll_lock bit register GPLL lock status.

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Bit	Attr	Reset Value	Description
0	WO	0x0	npll_lock pll_lock bit register NPLL lock status

GRF_SOC_STATUS1
Address: Operational Base + offset (0x0484)
SOC status register1

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19	RO	0x0	vpu_pwr_Idle vpu_pwr_Idle bit register idle status of vpu niu 0: idle is asserted 1: idle is de-asserted
18	RO	0x0	vio_pwr_Idle vio_pwr_Idle bit register idle status of vio niu 0: idle is asserted 1: idle is de-asserted
17	RO	0x0	sys_pwr_Idle sys_pwr_Idle bit register idle status of bus niu 0: idle is asserted 1: idle is de-asserted
16	RO	0x0	rkvenc_pwr_Idle rkvenc_pwr_Idle bit register idle status of rkvdec niu 0: idle is asserted 1: idle is de-asserted
15	RO	0x0	rkvdec_pwr_Idle rkvdec_pwr_Idle bit register rkvdec_pwr_Idle bit register 0: idle is asserted 1: idle is de-asserted
14	RO	0x0	peri_pwr_Idle peri_pwr_Idle bit register idle status of peri niu 0: idle is asserted 1: idle is de-asserted
13	RO	0x0	msch_pwr_Idle msch_pwr_Idle bit register idle status of msch niu 0: idle is asserted 1: idle is de-asserted

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Bit	Attr	Reset Value	Description
12	RO	0x0	msch_apb_pwr_Idle msch_apb_pwr_Idle bit register idle status of mschapb niu 0: idle is asserted 1: idle is de-asserted
11	RO	0x0	gpu_pwr_Idle gpu_pwr_Idle bit register idle status of gpu niu 0: idle is asserted 1: idle is de-asserted
10	RO	0x0	core_pwr_Idle core_pwr_Idle bit register idle status of core niu 0: idle is asserted 1: idle is de-asserted
9	RO	0x0	vpu_pwr_IdleAck vpu_pwr_IdleAck bit register idle acknowledge status from bus vpu 0: idle_ack asserted 1: idle_ack de-asserted
8	RO	0x0	vio_pwr_IdleAck vio_pwr_IdleAck bit register idle acknowledge status from bus vio 0: idle_ack asserted 1: idle_ack de-asserted
7	RO	0x0	sys_pwr_IdleAck sys_pwr_IdleAck bit register idle acknowledge status from bus niu 0: idle_ack asserted 1: idle_ack de-asserted
6	RO	0x0	rkvenc_pwr_IdleAck rkvenc_pwr_IdleAck bit register rkvenc_pwr_IdleAck bit register 0: idle_ack asserted 1: idle_ack de-asserted
5	RO	0x0	rkvdec_pwr_IdleAck rkvdec_pwr_IdleAck bit register idle acknowledge status from rkvdec niu 0: idle_ack asserted 1: idle_ack de-asserted
4	RO	0x0	peri_pwr_IdleAck peri_pwr_IdleAck bit register idle acknowledge status from peri niu 0: idle_ack asserted 1: idle_ack de-asserted

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Bit	Attr	Reset Value	Description
3	RO	0x0	msch_pwr_IdleAck msch_pwr_IdleAck bit register idle acknowledge status from msch niu 0: idle_ack asserted 1: idle_ack de-asserted
2	RO	0x0	msch_apb_pwr_IdleAck msch_apb_pwr_IdleAck bit register idle acknowledge status from mschapb niu 0: idle_ack asserted 1: idle_ack de-asserted
1	RO	0x0	gpu_pwr_IdleAck gpu_pwr_IdleAck bit register idle acknowledge status from gpu niu 0: idle_ack asserted 1: idle_ack de-asserted
0	RO	0x0	core_pwr_IdleAck core_pwr_IdleAck bit register idle acknowledge status from core niu 0: idle_ack asserted 1: idle_ack de-asserted

GRF_SOC_STATUS2
Address: Operational Base + offset (0x0488)
SOC status register2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	grf_sta_usb3otg_logic_analyzer_trace[31:0] usb3otg_logic_analyzer_trace[31:0] bit status

GRF_SOC_STATUS3
Address: Operational Base + offset (0x048c)
SOC status register3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	grf_sta_usb3otg_logic_analyzer_trace[63:32] usb3otg_logic_analyzer_trace[63:32] bit status

GRF_SOC_STATUS4
Address: Operational Base + offset (0x0490)
SOC status register4

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RO	0x000	grf_sta_usb3otg_host_current_belt[11:0] usb3otg_host_current_belt[11:0] bit status

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GRF_USB3OTG_CON0
Address: Operational Base + offset (0x04c0)
USB3OTG control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	grf_con_usb3otg_host_u2_port_disable USB2.0 Port Disable control. 0: Port Enabled 1: Port Disabled When 1, this signal stops reporting connect/disconnect events the port and keeps the port in disabled state.
14	RW	0x0	grf_con_usb3otg_host_port_power_control_present This indicates whether the host controller implementation includes port power control. 0: Indicates that the port does not have port power switches. 1: Indicates that the port has port power switches
13:8	RW	0x20	grf_con_usb3otg_fladj_30mhz_reg usb3otg_fladj_30mhz_reg bit control
7:6	RW	0x0	grf_con_usb3otg_hub_port_perm_attach Indicates if the device attached to a downstream port is permanently attached or not. 0: Not permanently attached 1: Permanently attached Bit0 is for USB2.0 port and bit1 are for USB 3.0 SS port.
5:4	RW	0x0	grf_con_usb3otg_hub_port_overcurrent This is the per port Overcurrent indication of the root-hub ports: 0: No Overcurrent 1: Overcurrent Bit0 is for USB 2.0 port and bit1 are for USB 3.0 SS port.

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	grf_con_usb3otg_bus_filter_bypass It is expected that this signal is set or reset at power-on reset and is not changed during the normal operation of the core. The function of each bit is: bus_filter_bypass[3]: Bypass the filter for utmiotg_iddig bus_filter_bypass[2]: Bypass the filters for utmisrp_bvalid and utmisrp_sessend bus_filter_bypass[1]: Bypass the filter for pipe3_PowerPresent all U3 ports bus_filter_bypass[0]: Bypass the filter for utmiotg_vbusvalid all U2 ports In non-OTG Host-only mode, internal bus filters are not needed. Values: 1'b0: Bus filter(s) enabled 1'b1: Bus filter(s) disabled (bypassed)

GRF_USB3OTG_CON1
Address: Operational Base + offset (0x04c4)
USB3OTG control register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:12	RW	0x1	grf_con_usb3otg_host_num_u3_port usb3otg_host_num_u3_port bit control xHCI usb3 port number, default as 1.
11:8	RW	0x1	grf_con_usb3otg_host_num_u2_port usb3otg_host_num_u2_port bit control xHCI host USB2 Port number, default as 1.
7:6	RO	0x0	reserved
5	RW	0x0	grf_con_usb3otg_host_legacy_smi_bar usb3otg_host_legacy_smi_bar bit control Use this register to support SMI on BAR defined in xHCI spec. SW must set this register, then clear this register to indicate Base Address Register written

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Bit	Attr	Reset Value	Description
4	RW	0x0	grf_con_usb3otg_host_legacy_smi_pci_cmd usb3otg_host_legacy_smi_pci_cmd bit control Use this register to support SMI on PCI Command defined in xHCI spec. SW must set this register, then clear this register to indicate PCI command register written.
3:2	RO	0x0	reserved
1	RW	0x0	grf_con_usb3otg_pme_en usb3otg_pme_en bit control Enable signal for the pme_generation. Enable the core to assert pme_generation.
0	RW	0x0	grf_con_usb3otg_host_u3_port_disable USB 3.0 SS Port Disable control. 0: Port Enabled 1: Port Disabled

GRF_CPU_CON0
Address: Operational Base + offset (0x0500)
CPU control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:12	RW	0x0	grf_con_cfgte cfgte bit control
11:8	RW	0x0	grf_con_cfgend cfgend bit control
7:5	RO	0x0	reserved
4	RW	0x0	grf_con_l2rstdisable l2rstdisable bit control
3:0	RW	0x0	grf_con_l1rstdisable l1rstdisable bit control

GRF_CPU_CON1
Address: Operational Base + offset (0x0504)
CPU control register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:6	RO	0x0	reserved
5	RW	0x0	grf_con_evento_clear vento_clear bit control
4	RW	0x0	grf_con_eventi eventi bit control
3	RW	0x1	grf_con_dbgselfaddrv dbgselfaddrv bit control
2	RW	0x1	grf_con_dbgromaddrv dbgromaddrv bit control
1	RW	0x0	grf_con_cfgsdisable cfgsdisable bit control
0	RW	0x0	grf_con_clrexmonreq clrexmonreq bit control

GRF_CPU_STATUS0
Address: Operational Base + offset (0x0520)
CPU status register0

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	grf_st_l2flushdone l2flushdone bit status
11	RO	0x0	grf_st_clrexmonack clrexmonack bit status
10	RO	0x0	grf_st_jtagnsw jtagnsw bit status
9	RO	0x0	grf_st_jtagtop jtagtop bit status
8	RO	0x0	evento_rising_edge evento_rising_edge bit status
7:4	RO	0x0	power_state power_state bit status
3:0	RO	0x0	grf_st_smpnamp smpnamp bit status

GRF_CPU_STATUS1
Address: Operational Base + offset (0x0524)

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CPU status register1

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	grf_st_standbywfil2 standbywfil2 bit status
11:8	RO	0x0	cpu_state cpu state status
7:4	RO	0x0	grf_st_standbywfi standbywfi bit status
3:0	RO	0x0	grf_st_standbywfe standbywfe bit status

GRF_OS_REG0

Address: Operational Base + offset (0x05c8)

os register0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg0 Reserved reserved

GRF_OS_REG1

Address: Operational Base + offset (0x05cc)

os register1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg1 Reserved reserved

GRF_OS_REG2

Address: Operational Base + offset (0x05d0)

os register2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg2 Reserved reserved

GRF_OS_REG3

Address: Operational Base + offset (0x05d4)

os register3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg3 Reserved reserved

GRF_OS_REG4

Address: Operational Base + offset (0x05d8)

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os register4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg4 Reserved reserved

GRF_OS_REG5
Address: Operational Base + offset (0x05dc)
os register5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg5 Reserved reserved

GRF_OS_REG6
Address: Operational Base + offset (0x05e0)
os register6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg6 Reserved reserved

GRF_OS_REG7
Address: Operational Base + offset (0x05e4)
os register7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	os_reg7 Reserved reserved

GRF_SIG_DETECT_CON
Address: Operational Base + offset (0x0680)
External signal detect configue register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	sdmmc_ext_detectn_neg_irq_en sdmmc_ext_detectn_neg_irq enable 1'b1: enable irq; 1'b0: disable irq.
2	RW	0x0	sdmmc_ext_detectn_pos_irq_en sdmmc_ext_detectn_pos_irq enable 1'b1: enable irq; 1'b0: disable irq.

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Bit	Attr	Reset Value	Description
1	RW	0x0	sdmmc_detectn_neg_irq_en sdmmc_detectn_neg_irq enable 1'b1: enable irq; 1'b0: disable irq.
0	RW	0x0	sdmmc_detectn_pos_irq_en sdmmc_detectn_pos_irq enable 1'b1: enable irq; 1'b0: disable irq.

GRF_SIG_DETECT_STATUS
Address: Operational Base + offset (0x0690)
External signal detect status register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	sdmmc_ext_detectn_neg_irq sdmmc_detectn_ext_neg irq status bit
2	RW	0x0	sdmmc_ext_detectn_pos_irq sdmmc_detectn_ext_pos irq status bit
1	RW	0x0	sdmmc_detectn_neg_irq sdmmc_detectn_neg irq status bit
0	RW	0x0	sdmmc_detectn_pos_irq sdmmc_detectn_pos irq status bit

GRF_SIG_DETECT_STATUS_CLEAR
Address: Operational Base + offset (0x06a0)
External signal detect status clear register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	WO	0x0	sdmmc_ext_detectn_neg_irq_clr sdmmc_ext_detectn_neg_irq clear bit
2	RW	0x0	sdmmc_ext_detectn_pos_irq_clr sdmmc_ext_detectn_pos_irq clear bit
1	RW	0x0	sdmmc_detectn_neg_irq_clr sdmmc_detectn_neg_irq clear bit
0	RW	0x0	sdmmc_detectn_pos_irq_clr sdmmc_detectn_pos_irq clear bit

GRF_SDMMC_DET_COUNTER
Address: Operational Base + offset (0x06b0)
SDMMC detect counter register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:0	RW	0x30100	sdmmc_detectn_count sdmmc_detectn_count bit register sdmmc_detectn_count bit register

GRF_HOST0_CON0
Address: Operational Base + offset (0x0700)
host0 control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:12	RO	0x0	reserved
11:6	RW	0x20	grf_con_host0_fladj_val_common host0_fladj_val_common bit control
5:0	RW	0x20	grf_con_host0_fladj_val host0_fladj_val bit control

GRF_HOST0_CON1
Address: Operational Base + offset (0x0704)
host0 control register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:14	RO	0x0	reserved
13	RW	0x0	grf_con_host0_arb_pause host0_arb_pause bit control
12	RW	0x0	grf_con_host0_ohci_susp_lgcy host0_ohci_susp_lgcy bit control
11	RW	0x0	grf_con_host0_ohci_cntsel host0_ohci_cntsel bit control
10	RW	0x1	grf_con_host0_ohci_clkcktrst host0_ohci_clkcktrst bit control

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Bit	Attr	Reset Value	Description
9	RW	0x0	grf_con_host0_app_prt_ovrcur host0_app_prt_ovrcur bit control
8	RW	0x0	grf_con_host0_autoppd_on_overcur_en host0_autoppd_on_overcur_en bit control
7	RW	0x1	grf_con_host0_word_if host0_word_if bit control
6	RW	0x0	grf_con_host0_sim_mode host0_sim_mode bit control
5	RW	0x1	grf_con_host0_incrx_en host0_incrx_en bit control
4	RW	0x1	grf_con_host0_incr8_en host0_incr8_en bit control
3	RW	0x1	grf_con_host0_incr4_en host0_incr4_en bit control
2	RW	0x1	grf_con_host0_incr16_en host0_incr16_en bit control
1	RW	0x0	grf_con_host0_hubsetup_min host0_hubsetup_min bit control
0	RW	0x0	grf_con_host0_app_start_clk host0_app_start_clk bit control

GRF_HOST0_CON2
Address: Operational Base + offset (0x0708)
host0 control register2

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

GRF_OTG_CON0
Address: Operational Base + offset (0x0880)
OTG control register

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:3	RO	0x0	reserved
2	RW	0x0	otg_dbnce_fltr_bypass otg_dbnce_fltr_bypass bit control
1:0	RW	0x0	otg_scaledown_mode otg_scaledown_mode bit control

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GRF_HOST0_STATUS
Address: Operational Base + offset (0x0890)
HOST0 status register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	host0_ehci_power_state_ack host0_ehci_power_state_ack bit status
29	RO	0x0	host0_ehci_pme_status host0_ehci_pme_status bit status
28	RO	0x0	grf_stat_host0_ehci_bufacc host0_ehci_bufacc bit status
27	RO	0x0	grf_stat_host0_ehci_xfer_prdc host0_ehci_xfer_prdc bit status
26	RO	0x0	grf_stat_host0_ohci_ccs host0_ohci_ccs bit status
25	RO	0x0	grf_stat_host0_ohci_rwe host0_ohci_rwe bit status
24	RO	0x0	grf_stat_host0_ohci_drwe host0_ohci_drwe bit status
23	RO	0x0	grf_stat_host0_ohci_globalsuspend host0_ohci_globalsuspend bit status
22	RO	0x0	grf_stat_host0_ohci_bufacc host0_ohci_bufacc bit status
21	RO	0x0	grf_stat_host0_ohci_rmtwkp host0_ohci_rmtwkp bit status
20:17	RO	0x0	grf_stat_host0_ehci_lpsmc_state host0_ehci_lpsmc_state bit status
16:11	RO	0x00	grf_stat_host0_ehci_usbsts host0_ehci_usbsts bit status
10:0	RO	0x000	grf_stat_host0_ehci_xfer_cnt host0_ehci_xfer_cnt bit status

GRF_MAC_CON0
Address: Operational Base + offset (0x0900)
MAC control register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:14	RO	0x0	reserved
13:7	RW	0x00	gmac2io_clk_rx_dl_cfg gmac2io_clk_rx_dl_cfg bit control
6:0	RO	0x00	gmac2io_clk_tx_dl_cfg gmac2io_clk_tx_dl_cfg bit control

GRF_MAC_CON1

Address: Operational Base + offset (0x0904)

MAC control register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12:11	RW	0x0	gmac2io_gmii_clk_sel gmac2io_gmii_clk_sel bit control GMII clock selection 2'b00:125MHz 2'b11:25MHz 2'b10:2.5MHz
10	RW	0x0	gmac2io_rmii_extclk_sel gmac2io_rmii_extclk_sel bit control
9	RW	0x0	gmac2io_rmii_mode gmac2io_rmii_mode bit control RMII mode selection 2'b11:RMII mode 2'b00:MII mode 2'b01:reserved 2'b10:reserved
8	RO	0x0	reserved
7	RW	0x0	gmac2io_rmii_clk_sel gmac2io_rmii_clk_sel bit control RMII clock selection 1'b1:25MHz 1'b0:2.5MHz

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Bit	Attr	Reset Value	Description
6:4	RW	0x0	gmac2io_phy_intf_sel gmac2io_phy_intf_sel bit control PHY interface select 3'b001:RGMII 3'b100:RMII All others:Reserved
3	RW	0x0	gmac2io_flowctrl gmac2io_flowctrl bit control GMAC transmit flow control When set high, instructs the GMAC to transmit PAUSE Control frame in Full-duplex mode. In Half-duplex mode, the GMAC enables the Back-pressure function until this signal is made low again
2	RW	0x0	gmac2io_mac_speed gmac2io_mac_speed bit control MAC speed 1'b1:100-Mbps 1'b0:10-Mbps
1	RW	0x0	gmac2io_rxclk_dly_ena gmac2io_rxclk_dly_ena bit control RGMII RX clock delayline enable 1'b1:enable 1'b0:disable
0	RW	0x0	gmac2io_txclk_dly_ena gmac2io_txclk_dly_ena bit control RGMII TX clock delayline enable 1'b1:enable 1'b0:disable

GRF_MAC_CON2
Address: Operational Base + offset (0x0908)
MAC control register2

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11	RW	0x0	gmac2phy_use_inter_phy_txrx gmac2phy_use_inter_phy_txrx bit control gmac2phy_use_inter_phy_txrx bit control
10	RW	0x0	gmac2phy_rmii_extclk_sel gmac2phy_rmii_extclk_sel bit control gmac2phy_rmii_extclk_sel bit control
9	RW	0x0	gmac2phy_rmii_mode gmac2phy_rmii_mode bit control RMII mode selection 2'b11:RMII mode 2'b00:MII mode 2'b01:reserved 2'b10:reserved
8	RO	0x0	reserved
7	RW	0x0	gmac2phy_rmii_clk_sel gmac2phy_rmii_clk_sel bit control PHY interface select 3'b001:RGMII 3'b100:RMII All others:Reserved
6:4	RW	0x0	gmac2phy_phy_intf_sel gmac2phy_phy_intf_sel bit control PHY interface select 3'b001:RGMII 3'b100:RMII All others:Reserved
3	RW	0x0	gmac2phy_flowctrl gmac2phy_flowctrl bit control GMAC transmit flow control When set high, instructs the GMAC to transmit PAUSE Control frame in Full-duplex mode. In Half-duplex mode, the GMAC enables the Back-pressure function until this signal is made low again
2	RW	0x0	gmac2phy_mac_speed gmac2phy_mac_speed bit control MAC speed 1'b1:100-Mbps 1'b0:10-Mbps
1:0	RO	0x0	reserved

GRF_MACPHY_CON0

Address: Operational Base + offset (0x0b00)

MACPHY control register0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Reserved reserved
15	RW	0x0	macphy_ref_clk_sel mac phy reference clock source select
14	RW	0x0	macphy_clk_freq 礫'for 25 MHz clock input; 礆'for 50 MHz clock input.
13	RW	0x1	macphy_automodix_en Enables auto-detection of MDI/MDIX mode. Refer to "cfg_mode"
12	RW	0x0	macphy_en_high Defines polarity of output enable signals. ''0'' for active low output enable signal. "mdio_dir, rxdz,miiz,rxerz" signal polarity control.
11	RW	0x0	macphy_fx_mode Enables FX mode
10	RW	0x0	macphy_adc_bp Puts the ADC by default in bypass mode
9	RW	0x0	macphy_pll_bp Puts the PLL by default in bypass mode
8	RW	0x0	macphy_smii_soure_sync smii source sync register field. Only relevant for SMII mode
7:6	RW	0x0	macphy_mii_mode MII mode register field. ''00'' for MII mode, "01" for RMII mode, "10" for SMII, "11" reserved
5:3	RW	0x7	macphy_mode MODE register file. "000" - 10BaseT, Half Duplex, Auto negotiation disabled "001" - 10Base-T, Full Duplex, Auto negotiation disabled "010" - 100Base-TX, Half Duplex, Auto-negotiation disabled "011" - 100Base-TX, Full Duplex, Auto-negotiation disabled "100" - 100Base-Tx, Half Duplex, Auto-negotaition Enabled "101" - Repeater mode, 100Base-Tx, Half Duplex, Auto-negotiation Enabled "110" - Power down mode, In this mode phy wake up in power fown mode "111" - All capable, Full Duplex, 10 & 100 BT, Auto negotiation enabled, AutoMDIX enable
2	RW	0x0	macphy_powerup_reset Power Up Reset bit. Default value of powerup_reset bit 0 - Power up reset disabled by default 1- Power up reset enabled by default

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Bit	Attr	Reset Value	Description
1	RW	0x0	macphy_power_down Power Down bit. Default value of True power down bit 1 - True power down is active by default 0 - True power down is not active by default
0	RW	0x1	macphy_enable PHY enable signal (active high). 1 = Enable MACHY IP 0 = Disable MACHY IP

GRF_MACPHY_CON1
Address: Operational Base + offset (0x0b04)
MACPHY control register1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Reserved reserved
15	RW	0x0	polarity_stat_tx polarity control of tx status
14	RW	0x0	polarity_stat_rx polarity control of rx status
13	RW	0x0	polarity_stat_duplex polarity control of duplex status
12	RW	0x0	polarity_stat_link polarity control of link status
11	RW	0x0	polarity_stat_speed10 polarity control of speed10 status
10	RW	0x0	polarity_stat_speed100 polarity control of speed100 status
9	RW	0x0	grf_con_rmii_mode rmii_mode bit control
8	RW	0x0	macphy_speed_sel 0: speed 100 1: speed 10
7:3	RW	0x00	macphy_phy_addr PHY ADD register field. Must be unique in multi-PHY environment (like repeater).
2:0	RW	0x0	macphy_np_msg_code Next Page Message Code. Automatic generation of Next page with fault code

GRF_MACPHY_CON2
Address: Operational Base + offset (0x0b08)
MACPHY control register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Reserved reserved
15:0	RW	0x0000	macphy_id PHY ID Number,macphy_cfg_phy_id[15:0]

GRF_MACPHY_CON3
Address: Operational Base + offset (0x0b0c)
MACPHY control register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Reserved reserved
15:12	RW	0x0	macphy_cfg_rev_nr Manufacturer's Revision Number
11:6	RW	0x00	macphy_model_nr Manufacturer's Model Number
5:0	RW	0x00	macphy_id PHY ID Number,macphy_cfg_phy_id[21:16]

GRF_MACPHY_STATUS
Address: Operational Base + offset (0x0b10)
MACPHY status register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RO	0x0	macphy_stat_speed100 macphy_stat_speed100 bit status Speed100 indication. Output driven low
5	RO	0x0	macphy_stat_speed10 macphy_stat_speed10 bit status Speed10 indication. Output is driven low
4	RO	0x0	macphy_stat_duplex macphy_stat_duplex bit status Duplex indication (low = full-duplex mode).Output is driven low
3	RO	0x0	macphy_stat_rx macphy_stat_rx bit status RX activity indication.Output is driven low
2	RO	0x0	macphy_stat_link macphy_stat_link bit status Link ON indication. Output is driven low
1	RO	0x0	macphy_stat_tx macphy_stat_tx bit status TX activity indication.Output is driven low

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Bit	Attr	Reset Value	Description
0	RO	0x0	macphy_stat_powerup_reset macphy_stat_powerup_reset bit status Power up reset state signal. To signal to the system that PHY is out of power down mode

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5.4 DDR_GRF Register Description

5.4.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

5.4.2 Registers Summary

Name	Offset	Size	Reset Value	Description
DDR_GRF_CON0	0x0000	W	0x000001fd	DDR Control Register0
DDR_GRF_CON1	0x0004	W	0x00000000	DDR Control Register1
DDR_GRF_CON2	0x0008	W	0x00000000	DDR Control Register2
DDR_GRF_CON3	0x000c	W	0x00000000	DDR Control Register3
DDR_GRF_STATUS0	0x0100	W	0x00000000	DDR Status Register0
DDR_GRF_STATUS1	0x0104	W	0x00000000	DDR Status Register1
DDR_GRF_STATUS2	0x0108	W	0x00000000	DDR Status Register2
DDR_GRF_STATUS3	0x010c	W	0x00000000	DDR Status Register3
DDR_GRF_STATUS4	0x0110	W	0x00000000	DDR Status Register4
DDR_GRF_STATUS5	0x0114	W	0x00000000	DDR Status Register5
DDR_GRF_STATUS6	0x0118	W	0x00000000	DDR Status Register6
DDR_GRF_STATUS7	0x011c	W	0x00000000	DDR Status Register7
DDR_GRF_STATUS8	0x0120	W	0x00000000	DDR Status Register8
DDR_GRF_STATUS9	0x0124	W	0x00000000	DDR Status Register9
DDR_GRF_STATUS10	0x0128	W	0x00000000	DDR Status Register10

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

5.4.3 Detail Register Description

DDR_GRF_CON0
Address: Operational Base + offset (0x0000)
DDR Control Register0

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	grf_con_csysreq_upctl_ddrstdby 1'b0: override csysreq from ddrc and send valid csysreq signal to PMU. 1'b1: csysreq is controlled by ddrc only
14	RW	0x0	grf_con_csysreq_upctl_pmu 1'b0: override csysreq from ddrc and send valid csysreq signal to ddr standby module. 1'b1: csysreq is controlled by ddrc only
13:12	RW	0x0	grf_con_dfi_phymstr_type dfi_phymstr_type bit control
11	RW	0x0	grf_con_dfi_phymstr_state_sel dfi_phymstr_state_sel bit control

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Bit	Attr	Reset Value	Description
10:9	RW	0x0	grf_con_dfi_phymstr_cs_state dfi_phymstr_cs_state bit control
8	RW	0x1	grf_con_dfi_phymstr_req dfi_phymstr_req bit control
7	RW	0x1	grf_con_upctl_axi upctl_axi bit control
6	RW	0x1	grf_con_upctl_arurgent_0 upctl_arurgent_0 bit control
5	RW	0x1	grf_con_upctl_arposion upctl_arposion bit control
4	RW	0x1	grf_con_upctl_awposion upctl_awposion bit control
3	RW	0x1	grf_con_upctl_awurgent upctl_awurgent bit control
2	RW	0x1	grf_con_pa_wmask pa_wmask bit control
1:0	RW	0x1	grf_con_pa_rmask pa_rmask bit control

DDR_GRF_CON1
Address: Operational Base + offset (0x0004)
DDR Control Register1

Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:12	RW	0x0	grf_con_upctl_awregion upctl_awregion bit control
11:8	RW	0x0	grf_con_upctl_arregion upctl_arregion bit control
7:4	RW	0x0	grf_con_upctl_arqos upctl_arqos bit control
3:0	RW	0x0	grf_con_upctl_awqos upctl_awqos bit control

DDR_GRF_CON2
Address: Operational Base + offset (0x0008)
DDR Control Register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RW	0x0	grf_dfi_init_start dfi_init_start control
14	RW	0x0	grf_dfi_init_start_sel dfi_init_start_sel control 1: set ddrphy dfi init start controlled by grf_dfi_init_start 0: set ddrphy dif init start by DDRC
13	RW	0x0	grf_upctl_apb_gate_en 1: enable ddrc axi clock or working clock when accessing DDRC apb register.
12	RW	0x0	grf_ddrc_idle_sel ddrc_idle_sel control
11:9	RO	0x0	reserved
8	RW	0x0	grf_con_dfi_lp_ack dfi_lp_ack bit control
7	RW	0x0	grf_con_dfi_lp_req dfi_lp_req bit control
6	RW	0x0	grf_con_dfi_phyupd_req dfi_phyupd_req bit control
5:2	RW	0x0	grf_con_ddrc_auto_sr_dly ddrc_auto_sr_dly bit control
1	RW	0x0	grf_con_ddrc_cg_en ddrc_cg_en bit control 1'b1: ddrc clock gate enable by apb access enable 1'b0: ddrc clock gate enable by apb access disable
0	RW	0x0	grf_con_axi_cg_en axi_cg_en bit control 1'b1: ddrc axi clock gate enable by apb access enable 1'b0: ddrc axi clock gate enable by apb access disable

DDR_GRF_CON3

Address: Operational Base + offset (0x000c)

DDR Control Register3

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	WO	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12	RW	0x0	dfi_ctrlupd_ack2 dfi_ctrlupd_ack2 bit control
11	RW	0x0	dfi_ctrlupd_ack dfi_ctrlupd_ack bit control
10	RW	0x0	dfi_phyupd_req dfi_phyupd_req bit control
9:8	RW	0x0	dfi_phyupd_type dfi_phyupd_type bit control
7:6	RW	0x0	dfi_wrlvl_mode dfi_wrlvl_mode bit control
5:4	RW	0x0	dfi_rdlvl_gate_mode dfi_rdlvl_gate_mode bit control
3:2	RW	0x0	dfi_rdlvl_mode dfi_rdlvl_mode bit control
1:0	RW	0x0	dfi_alert_n dfi_alert_n bit control

DDR_GRF_STATUS0
Address: Operational Base + offset (0x0100)
DDR Status Register0

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RO	0x0	ctl_idle_upctl ctl_idle_upctl bit status upctl_raq_pop_0 bit status
27	RO	0x0	grf_st_dfi_phymstr_ack upctl_raq_pop_0 bit status
26	RO	0x0	grf_st_upctl_raq_pop_0 upctl_raq_pop_0 bit status
25	RO	0x0	grf_st_upctl_raq_push_0 upctl_waq_split_0 bit status
24	RO	0x0	grf_st_upctl_raq_split_0 upctl_raq_split_0 bit status upctl_waq_split_0 bit status

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Bit	Attr	Reset Value	Description
23	RO	0x0	grf_st_upctl_waq_pop_0 upctl_waq_split_0 bit status
22	RO	0x0	grf_st_upctl_waq_push_0 upctl_waq_split_0 bit status
21	RO	0x0	grf_st_upctl_waq_split_0 upctl_waq_split_0 bit status
20:14	RO	0x00	grf_st_lpr_credit_cnt hpr_credit_cnt bit status
13:7	RO	0x00	grf_st_hpr_credit_cnt hpr_credit_cnt bit status
6:0	RO	0x00	grf_st_wr_credit_cnt wr_credit_cnt bit status

DDR_GRF_STATUS1
Address: Operational Base + offset (0x0104)
DDR Status Register1

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:14	RO	0x0	hif_refresh_req_bank hif_refresh_req_bank bit status
13:12	RO	0x0	stat_upctl_reg_selfref_type stat_upctl_reg_selfref_type bit status
11	RO	0x0	csysack_upctl_axi csysack_upctl_axi bit status
10	RO	0x0	cactive_upctl_axi cactive_upctl_axi bit status
9:8	RO	0x0	reserved
7:4	RO	0x0	grf_st_upctl_raq_wcount_0 upctl_raq_wcount_0 bit status
3:0	RO	0x0	grf_st_upctl_waq_wcount_0 upctl_waq_wcount_0 bit status

DDR_GRF_STATUS2
Address: Operational Base + offset (0x0108)
DDR Status Register2

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:14	RO	0x00	grf_st_wr_credit_cnt wr_credit_cnt bit status
13:7	RO	0x00	grf_st_hpr_credit_cnt hif_refresh_req_bank bit status
6:0	RO	0x00	grf_st_lpr_credit_cnt hif_refresh_req_bank bit status

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DDR_GRF_STATUS3
Address: Operational Base + offset (0x010c)
DDR Status Register3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data0[31:0] mrr_data0[31:0] data status

DDR_GRF_STATUS4
Address: Operational Base + offset (0x0110)
DDR Status Register4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data0[63:32] mrr_data0[63:32] data status

DDR_GRF_STATUS5
Address: Operational Base + offset (0x0114)
DDR Status Register5

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data0[95:64] mrr_data0[95:64] data status

DDR_GRF_STATUS6
Address: Operational Base + offset (0x0118)
DDR Status Register6

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data0[127:96] mrr_data0[127:96] data status

DDR_GRF_STATUS7
Address: Operational Base + offset (0x011c)
DDR Status Register7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data1[31:0] mrr_data1[31:0] data status

DDR_GRF_STATUS8
Address: Operational Base + offset (0x0120)
DDR Status Register8

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data1[63:32] mrr_data1[63:32] data status

DDR_GRF_STATUS9
Address: Operational Base + offset (0x0124)
DDR Status Register9

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data1[95:64] mrr_data1[95:64] data status

DDR_GRF_STATUS10
Address: Operational Base + offset (0x0128)
DDR Status Register10

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	mrr_data1[127:96] mrr_data1[127:96] data status

5.5 USB2PHY_GRF Register Description

5.5.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

5.5.2 Registers Summary

Name	Offset	Size	Reset Value	Description
USBPHY_REG0	0x0000	W	0x00002146	USB PHY Register0
USBPHY_REG1	0x0004	W	0x00000000	USB PHY Register1
USBPHY_REG2	0x0008	W	0x00000002	USB PHY Register2
USBPHY_REG3	0x000c	W	0x000000c8	USB PHY Register3
USBPHY_REG4	0x0010	W	0x000015b4	USB PHY Register4
USBPHY_REG5	0x0014	W	0x000011cb	USB PHY Register5
USBPHY_REG6	0x0018	W	0x0000022b	USB PHY Register6
USBPHY_REG7	0x001c	W	0x00000044	USB PHY Register7
USBPHY_REG8	0x0020	W	0x00000000	USB PHY Register8
USBPHY_REG9	0x0024	W	0x00000000	USB PHY Register9
USBPHY_REG10	0x0028	W	0x00000000	USB PHY Register10
USBPHY_REG11	0x002c	W	0x00000000	USB PHY Register11
USBPHY_REG12	0x0030	W	0x00002146	USB PHY Register12
USBPHY_REG13	0x0034	W	0x00000000	USB PHY Register13
USBPHY_REG14	0x0038	W	0x00000002	USB PHY Register14
USBPHY_REG15	0x003c	W	0x000000c8	USB PHY Register15
USBPHY_REG16	0x0040	W	0x000015b4	USB PHY Register16
USBPHY_REG17	0x0044	W	0x000011cb	USB PHY Register17
USBPHY_REG18	0x0048	W	0x00000005	USB PHY Register18
USBPHY_REG19	0x004c	W	0x00000044	USB PHY Register19
USBPHY_REG20	0x0050	W	0x00000000	USB PHY Register20
USBPHY_REG21	0x0054	W	0x00000000	USB PHY Register21
USBPHY_REG22	0x0058	W	0x00000000	USB PHY Register22
USBPHY_REG23	0x005c	W	0x00000000	USB PHY Register23
USBPHY_CON0	0x0100	W	0x00000052	USB PHY control register0
USBPHY_CON1	0x0104	W	0x000001d2	USB PHY control register1

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Name	Offset	Size	Reset Value	Description
USBPHY_CON2	0x0108	W	0x00000000	USB PHY control register2
USBPHY_CON3	0x010c	W	0x00000019	USB PHY control register3
SIG_DETECT_USB2PHY_CON0	0x0110	W	0x00000000	SIG DETECT USB2PHY control register0

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

5.5.3 Detail Register Description

USBPHY_REG0
Address: Operational Base + offset (0x0000)
USB PHY Register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x2146	usbphy_reg0 usbcomb phy control reg. BIT15 to 0 usbcomb phy control reg. BIT15 to 0

USBPHY_REG1
Address: Operational Base + offset (0x0004)
USB PHY Register1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg1 usbcomb phy control reg. BIT31 to 16 usbcomb phy control reg. BIT31 to 16

USBPHY_REG2
Address: Operational Base + offset (0x0008)
USB PHY Register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0002	usbphy_reg2 usbcomb phy control reg. BIT47 to 32 usbcomb phy control reg. BIT47 to 32

USBPHY_REG3
Address: Operational Base + offset (0x000c)
USB PHY Register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x00c8	usbphy_reg3 usbcomb phy control reg. BIT63 to 48 usbcomb phy control reg. BIT63 to 48

USBPHY_REG4
Address: Operational Base + offset (0x0010)
USB PHY Register4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x15b4	usbphy_reg4 usbcomb phy control reg. BIT79 to 64 usbcomb phy control reg. BIT79 to 64

USBPHY_REG5
Address: Operational Base + offset (0x0014)
USB PHY Register5

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x11cb	usbphy_reg5 usbcomb phy control reg. BIT95 to 80 usbcomb phy control reg. BIT95 to 80

USBPHY_REG6
Address: Operational Base + offset (0x0018)
USB PHY Register6

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x022b	usbphy_reg6 usbcomb phy control reg. BIT111 to 96 usbcomb phy control reg. BIT111 to 96

USBPHY_REG7
Address: Operational Base + offset (0x001c)
USB PHY Register7

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0044	usbphy_reg7 usbcomb phy control reg. BIT127 to 112 usbcomb phy control reg. BIT127 to 112

USBPHY_REG8
Address: Operational Base + offset (0x0020)
USB PHY Register8

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg8 usbcomb phy control reg. BIT143 to 128 usbcomb phy control reg. BIT143 to 128

USBPHY_REG9
Address: Operational Base + offset (0x0024)
USB PHY Register9

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	usbphy_reg9 usbcomb phy control reg. BIT159 to 144 usbcomb phy control reg. BIT159 to 144

USBPHY_REG10
Address: Operational Base + offset (0x0028)
USB PHY Register10

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg10 usbcomb phy control reg. BIT175 to 160 usbcomb phy control reg. BIT175 to 160

USBPHY_REG11
Address: Operational Base + offset (0x002c)
USB PHY Register11

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg11 usbcomb phy control reg. BIT191 to 176 usbcomb phy control reg. BIT191 to 176

USBPHY_REG12
Address: Operational Base + offset (0x0030)
USB PHY Register12

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x2146	usbphy_reg12 usbcomb phy control reg. BIT207 to 192 usbcomb phy control reg. BIT207 to 192

USBPHY_REG13
Address: Operational Base + offset (0x0034)
USB PHY Register13

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg13 usbcomb phy control reg. BIT223 to 208 usbcomb phy control reg. BIT223 to 208

USBPHY_REG14
Address: Operational Base + offset (0x0038)
USB PHY Register14

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x0002	usbphy_reg14 usbcomb phy control reg. BIT239 to 224 usbcomb phy control reg. BIT239 to 224

USBPHY_REG15
Address: Operational Base + offset (0x003c)
USB PHY Register15

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x00c8	usbphy_reg15 usbcomb phy control reg. BIT255 to 240 usbcomb phy control reg. BIT255 to 240

USBPHY_REG16
Address: Operational Base + offset (0x0040)
USB PHY Register16

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x15b4	usbphy_reg16 usbcomb phy control reg. BIT271 to 256 usbcomb phy control reg. BIT271 to 256

USBPHY_REG17
Address: Operational Base + offset (0x0044)
USB PHY Register17

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x11cb	usbphy_reg17 usbcomb phy control reg. BIT287 to 272 usbcomb phy control reg. BIT287 to 272

USBPHY_REG18
Address: Operational Base + offset (0x0048)
USB PHY Register18

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0005	usbphy_reg18 usbcomb phy control reg. BIT303 to 288 usbcomb phy control reg. BIT303 to 288

USBPHY_REG19
Address: Operational Base + offset (0x004c)
USB PHY Register19

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:0	RW	0x0044	usbphy_reg19 usbcomb phy control reg. BIT319 to 304 usbcomb phy control reg. BIT319 to 304

USBPHY_REG20
Address: Operational Base + offset (0x0050)
USB PHY Register20

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg20 usbcomb phy control reg. BIT335 to 320 usbcomb phy control reg. BIT335 to 320

USBPHY_REG21
Address: Operational Base + offset (0x0054)
USB PHY Register21

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg21 usbcomb phy control reg. BIT351 to 336 usbcomb phy control reg. BIT351 to 336

USBPHY_REG22
Address: Operational Base + offset (0x0058)
USB PHY Register22

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg22 usbcomb phy control reg. BIT367 to 352 usbcomb phy control reg. BIT367 to 352

USBPHY_REG23
Address: Operational Base + offset (0x005c)
USB PHY Register23

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:0	RW	0x0000	usbphy_reg23 usbcomb phy control reg. BIT383 to 368 usbcomb phy control reg. BIT383 to 368

USBPHY_CON0
Address: Operational Base + offset (0x0100)
USB PHY control register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	usbotg_utmi_iddig usbotg_utmi_iddig bit control USB Plug Indicator Ooutput
8	RW	0x0	usbotg_utmi_dmpulldown usbotg_utmi_dmpulldown bit control Enable DMINUS Pull Down resistor
7	RW	0x0	usbotg_utmi_dppulldown usbotg_utmi_dppulldown bit control Enable DPLUS Pull Down resistor
6	RW	0x1	usbotg_utmi_termselect usbotg_utmi_termselect bit control Termination select between FS/LS and HS Terminations
5:4	RW	0x1	usbotg_utmi_xcvrselect usbotg_utmi_xcvrselect bit control Transceiver Select between FS/LS and HS Transceivers
3:2	RW	0x0	usbotg_utmi_opmode usbotg_utmi_opmode bit control Operational mode selector between various modes
1	RW	0x1	usbotg_utmi_suspend_n usbotg_utmi_suspend_n bit control Suspend Mode enable 1'b0:suspend 1'b1:normal
0	RO	0x0	reserved

USBPHY_CON1
Address: Operational Base + offset (0x0104)
USB PHY control register1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:9	RO	0x0	reserved
8	RW	0x1	usbhost_utmi_dmpulldown usbhost_utmi_dmpulldown bit control Enable DMINUS Pull Down resistor
7	RW	0x1	usbhost_utmi_dppulldown usbhost_utmi_dppulldown bit control Enable DPLUS Pull Down resistor

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Bit	Attr	Reset Value	Description
6	RW	0x1	usbhost_utmi_termselect usbhost_utmi_termselect bit control Termination select between FS/LS and HS Terminations
5:4	RW	0x1	usbhost_utmi_xcvrselect usbhost_utmi_xcvrselect bit control Transceiver Select between FS/LS and HS Transceivers
3:2	RW	0x0	usbhost_utmi_opmode usbhost_utmi_opmode bit control Operational mode selector between various modes
1	RW	0x1	usbhost_utmi_suspend_n usbhost_utmi_suspend_n bit control Suspend Mode enable 1'b0: suspend 1'b1: normal
0	RO	0x0	reserved

USBPHY_CON2
Address: Operational Base + offset (0x0108)
USB PHY control register2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12	RW	0x0	vdm_src_en_usbotg vdm_src_en_usbotg bit control open dm voltage source
11	RW	0x0	vdp_src_en_usbotg vdp_src_en_usbotg bit control open dp voltage source
10	RW	0x0	rdm_pdwn_en_usbotg rdm_pdwn_en_usbotg bit control open dm pull down resistor
9	RW	0x0	idp_src_en_usbotg idp_src_en_usbotg bit control open dm source current
8	RW	0x0	idm_sink_en_usbotg idm_sink_en_usbotg bit control open dm sink current

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Bit	Attr	Reset Value	Description
7	RW	0x0	idp_sink_en_usbotg idp_sink_en_usbotg bit control open dp sink current
6:5	RO	0x0	reserved
4	RW	0x0	usbphy_commononn usbphy_commononn bit control configure PLL clock output in suspend mode
3	RW	0x0	bypasssel_usbotg bypasssel_usbotg bit control bypass select
2	RW	0x0	bypassdmen_usbotg bypassdmen_usbotg bit control bypass dm enable
1	RW	0x0	usbotg_disable_1 usbotg_disable_1 bit control bypass OTG function
0	RW	0x0	usbotg_disable_0 usbotg_disable_0 bit control bypass OTG function

USBPHY_CON3
Address: Operational Base + offset (0x010c)
USB PHY control register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:12	RO	0x0	reserved
11	RW	0x0	usbhost_utmi_drvvbus usbhost_utmi_drvvbus bit control USB HOST utmi_fs_drvvbus bit control
10	RW	0x0	usbhost_utmi_drvvbus_sel usbhost_utmi_drvvbus_sel bit control USB HOST utmi_drvvbus_sel bit control
9	RW	0x0	usbhost_utmi_fs_se0 usbhost_utmi_fs_se0 bit control USB HOST utmi_fs_se0 bit control

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Bit	Attr	Reset Value	Description
8	RW	0x0	usbhost_utmi_fs_data usbhost_utmi_fs_data bit control USB HOST utmi_fs_data bit control
7	RW	0x0	usbhost_utmi_fs_oe usbhost_utmi_fs_oe bit control USB HOST utmi_fs_oe bit control
6	RW	0x0	usbhost_utmi_fs_xver_own usbhost_utmi_fs_xver_own bit control USB HOST utmi_fs_xver_own bit control
5	RW	0x0	usbhost_utmi_idpullup usbhost_utmi_idpullup bit control USB HOST utmi_idpullup bit control
4	RW	0x1	usbhost_utmi_dmpulldown usbhost_utmi_dmpulldown bit control Enable DMINUS Pull Down resistor
3	RW	0x1	usbhost_utmi_dppulldown usbhost_utmi_dppulldown bit control Enable DPLUS Pull Down resistor
2	RW	0x0	usbhost_utmi_dischrgvbus usbhost_utmi_dischrgvbus bit control USB HOST utmi_dischrgvbus bit control
1	RW	0x0	usbhost_utmi_chrgvbus usbhost_utmi_chrgvbus bit control USB HOST utmi_chrgvbus bit control
0	RW	0x1	usbhost_utmi_drvvbus usbhost_utmi_drvvbus bit control USB HOST utmi_drvvbus bit control

SIG_DETECT_USB2PHY_CON0
Address: Operational Base + offset (0x0110)
SIG DETECT USB2PHY control register0

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RO	0x0	grf_stat_usbphy_dp_detected grf_stat_usbphy_dp_detected bit status grf_stat_usbphy_dp_detected bit status
24	RO	0x0	grf_stat_usbphy_cp_detected grf_stat_usbphy_cp_detected bit status grf_stat_usbphy_cp_detected bit status
23	RO	0x0	grf_stat_usbphy_dcp_detected grf_stat_usbphy_dcp_detected bit status grf_stat_usbphy_dcp_detected bit status
22	RO	0x0	usbhost_phy_ls_fs_rcv usbhost_phy_ls_fs_rcv bit status host_phy_ls_fs_rcv status

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Bit	Attr	Reset Value	Description
21	RO	0x0	usbhost_utmi_avalid usbhost_utmi_avalid bit status host_utmi_avalid status
20	RO	0x0	usbhost_utmi_bvalid usbhost_utmi_bvalid bit status host_utmi_bvalid status
19	RO	0x0	usbhost_utmi_hostdisconnect usbhost_utmi_hostdisconnect bit status host_utmi_hostdisconnect status
18	RO	0x0	usbhost_utmi_iddig_o usbhost_utmi_iddig_o bit status host_utmi_iddig_o status
17:16	RO	0x0	usbhost_utmi_linestate usbhost_utmi_linestate bit status host_utmi_linestate status
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	usbhost_utmi_sessend usbhost_utmi_sessend bit status host_utmi_sessend status
14	RO	0x0	usbhost_utmi_vbusvalid usbhost_utmi_vbusvalid bit status host_utmi_vbusvalid status
13	RO	0x0	usbhost_utmi_vmi usbhost_utmi_vmi bit status host_utmi_vmi status
12	RO	0x0	usbhost_utmi_vpi usbhost_utmi_vpi bit status host_utmi_vpi status
13:12	RW	0x0	host0_ls_filter_time_sel host0_ls_filter_time_sel bit control host0_ls_lfiter time select 00:100us 01:500us 10:1ms 11:10ms

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Bit	Attr	Reset Value	Description
11	RO	0x0	usbotg_phy_ls_fs_rcv usbotg_phy_ls_fs_rcv bit status utmi_phy_ls_fs_rcv_out status
10	RO	0x0	usbotg_utmi_avalid usbotg_utmi_avalid bit status otg_utmi avalid bit status
11:10	RW	0x0	otg0_ls_filter_time_sel otg0_ls_filter_time_sel bit control otg0_ls_lfiter time select 00:100us 01:500us 10:1ms 11:10ms
9	RO	0x0	usbotg_utmi_bvalid usbotg_utmi_bvalid bit status otg_utmi bvalid bit status
8	RO	0x0	usbotg_utmi_fs_xver_own usbotg_utmi_fs_xver_own bit status OTG utmi_fs_xver_own bit control
9:8	RW	0x0	otg0_id_filter_time_sel otg0_id_filter_time_sel bit control otg0_id_filter_time select 00:5ms 01:15ms 10:35ms
7	RO	0x0	usbotg_utmi_hostdisconnect usbotg_utmi_hostdisconnect bit status otg_utmi_hostdisconnect status
6	RO	0x0	usbotg_utmi_iddig usbotg_utmi_iddig bit status usbotg_utmi_iddig select between grf and phy 1:from grf 0:from phy
5:4	RO	0x0	usbotg_utmi_linestate usbotg_utmi_linestate bit status otg_utmi_linestate bit status
5:4	RO	0x0	otg0_id_irq otg0_id_irq bit status otg0_id bit status
5:4	RW	0x0	otg0_id_irq otg0_id_irq bit control otg0_id bit status
3	RO	0x0	usbotg_utmi_sessend usbotg_utmi_sessend bit status otg_utmi_sessend bit status

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Bit	Attr	Reset Value	Description
2	RO	0x0	usbotg_utmi_vbusvalid usbotg_utmi_vbusvalid bit status otg_utmi_vbusvalid bit status
3:2	RO	0x0	otg0_bvalid_irq otg0_bvalid_irq bit status otg0_bvalid bit status
3:2	RW	0x0	otg0_bvalid_irq otg0_bvalid_irq bit control otg0_bvalid bit status
1	RO	0x0	usbotg_utmi_vmi usbotg_utmi_vmi bit status otg_utmi_vmi bit status
1	RW	0x0	host0_linestate_irq host0_linestate_irq bit control host0_linestate bit status
1	RO	0x0	host0_linestate_irq host0_linestate_irq bit status host0_linestate bit status
0	RO	0x0	usbotg_utmi_vpi usbotg_utmi_vpi bit status otg_utmi_vpi bit status
0	RO	0x0	otg0_linestate_irq otg0_linestate_irq bit status otg0_linestate bit status
0	RW	0x0	otg0_linestate_irq otg0_linestate_irq bit control otg0_linestate bit status

5.6 USB3PHY_GRF Register Description

5.6.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

5.6.2 Registers Summary

Name	Offset	Size	Reset Value	Description
USB3PHY_CON0	0x0000	W	0x00000000	USB3 PHY Control Register0
USB3PHY_CON1	0x0004	W	0x00000000	USB3 PHY Control Register1
USB3PHY_CON2	0x0008	W	0x00000000	USB3 PHY Control Register2
USB3PHY_CON3	0x000c	W	0x00000001	USB3 PHY Control Register3
USB3PHY_CON4	0x0010	W	0x00000000	USB3 PHY Control Register4
USB3PHY_CON5	0x0014	W	0x00000000	USB3 PHY Control Register5
USB3PHY_CON6	0x0018	W	0x00000000	USB3 PHY Control Register6
USB3PHY_CON7	0x001c	W	0x00000000	USB3 PHY Control Register7
USB3PHY_CON8	0x0020	W	0x00000014	USB3 PHY Control Register8
USB3PHY_CON9	0x0024	W	0x00000000	USB3 PHY Control Register9

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Name	Offset	Size	Reset Value	Description
USB3PHY_SIG_DETECT_CON0	0x0028	W	0x00000000	USB3 PHY SIG DETECT Control Register0
USB3PHY_STATUS1	0x0034	W	0x00000000	USB3 PHY STATUS1 Register1
USB3_WAKEUP_CON0	0x0040	W	0x00000000	USB3 WAKEUP Control Register0

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

5.6.3 Detail Register Description

USB3PHY_CON0

Address: Operational Base + offset (0x0000)

USB3 PHY Control Register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:13	RO	0x0	reserved
12	RW	0x0	vdm_src_en_usb3otg vdm_src_en_usb3otg bit control open dm voltage source
11	RW	0x0	vdp_src_en_usb3otg vdp_src_en_usb3otg bit control open dp voltage source
10	RW	0x0	rdm_pdwn_en_usb3otg rdm_pdwn_en_usb3otg bit control open dm pull down resistor
9	RW	0x0	idp_src_en_usb3otg idp_src_en_usb3otg bit control open dm source current
8	RW	0x0	idm_sink_en_usb3otg idm_sink_en_usb3otg bit control open dm sink current
7	RW	0x0	idp_sink_en_usb3otg idp_sink_en_usb3otg bit control open dp sink current
6:0	RO	0x0	reserved

USB3PHY_CON1

Address: Operational Base + offset (0x0004)

USB3 PHY Control Register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:1	RO	0x0	reserved
0	RW	0x0	usb3otg_utmi_iddig usb3otg_utmi_iddig bit control usb3otg_utmi_iddig bit control
0	RW	0x0	usb3otg_utmi_dmpulldown usb3otg_utmi_dmpulldown bit control usb3otg_utmi_dmpulldown bit control
0	RW	0x0	usb3otg_utmi_dppulldown usb3otg_utmi_dppulldown bit control usb3otg_utmi_dppulldown bit control
0	RW	0x0	usb3otg_utmi_suspend_n usb3otg_utmi_suspend_n bit control usb3otg_utmi_suspend_n bit control
0	RW	0x0	usb3otg_utmi_opmode usb3otg_utmi_opmode bit control usb3otg_utmi_opmode bit control
0	RW	0x0	usb3otg_utmi_xcvrselect usb3otg_utmi_xcvrselect bit control usb3otg_utmi_xcvrselect bit control
0	RW	0x0	usb3otg_utmi_termselect usb3otg_utmi_termselect bit control usb3otg_utmi_termselect bit control

USB3PHY_CON2

Address: Operational Base + offset (0x0008)

USB3 PHY Control Register2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15:1	RO	0x0	reserved
0	RW	0x0	usb3phy_con2 Reserved reserved

USB3PHY_CON3
Address: Operational Base + offset (0x000c)
USB3 PHY Control Register3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:10	RO	0x0	reserved
9	RW	0x0	usb3otg_utmi_fs_se0 usb3otg_utmi_fs_se0 bit control OTG utimi_fs_xver_own bit control
8	RW	0x0	usb3otg_utmi_fs_data usb3otg_utmi_fs_data bit control OTG utimi_fs_xver_own bit control
7	RW	0x0	usb3otg_utmi_fs_oe usb3otg_utmi_fs_oe bit control OTG utmi_fs_xver_own bit control
6	RW	0x0	usb3otg_utmi_fs_xver_own usb3otg_utmi_fs_xver_own bit control OTG utmi_fs_xver_own bit control
5	RO	0x0	reserved
4	RW	0x0	usb3otg_utmi_dischrgvbus usb3otg_utmi_dischrgvbus bit control USB3 OTG utmi_dischrgvbus bit control
3	RW	0x0	usb3otg_utmi_chrgvbus usb3otg_utmi_chrgvbus bit control USB3 OTG utmi_chrgvbus bit control
2	RW	0x0	usb3otg_utmi_drvvbus usb3otg_utmi_drvvbus bit control USB3 OTG utmi_drvvbus bit control
1	RW	0x0	usb3otg_utmi_drvvbus_sel usb3otg_utmi_drvvbus_sel bit control USB3 OTG utmi_drvvbus_sel bit control

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Bit	Attr	Reset Value	Description
0	RW	0x1	usb3otg_utmi_idpullup usb3otg_utmi_idpullup bit control USB3 OTG utmi_idpullup bit control

USB3PHY_CON4
Address: Operational Base + offset (0x0010)
USB3 PHY Control Register4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:1	RO	0x0	reserved
0	RW	0x0	usb3phy_con4 usb3phy_con4 bit control reserved

USB3PHY_CON5
Address: Operational Base + offset (0x0014)
USB3 PHY Control Register5

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:1	RO	0x0	reserved
0	RW	0x0	usb3phy_con5 usb3phy_con5 bit control reserved

USB3PHY_CON6
Address: Operational Base + offset (0x0018)
USB3 PHY Control Register6

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:1	RO	0x0	reserved
0	RW	0x0	usb3phy_con6 usb3phy_con6 bit control reserved

USB3PHY_CON7
Address: Operational Base + offset (0x001c)
USB3 PHY Control Register7

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:1	RO	0x0	reserved
0	RW	0x0	usb3phy_con7 usb3phy_con7 bit control reserved

USB3PHY_CON8
Address: Operational Base + offset (0x0020)
USB3 PHY Control Register8

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;

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Bit	Attr	Reset Value	Description
15	RW	0x0	usb3phy_usb2only usb3phy_usb2only bit control usb3phy_usb2only bit control
14	RW	0x0	usb3otg_pipe3_powerpresent usb3otg_pipe3_powerpresent bit control usb3otg_pipe3_powerpresent bit control
13:6	RO	0x0	reserved
5	RW	0x0	usb3otg_pipe3_txdetectrxloopbk usb3otg_pipe3_txdetectrxloopbk bit control usb3otg_pipe3_txdetectrxloopbk bit control
4:3	RW	0x2	usb3otg_pipe3_powerdown usb3otg_pipe3_powerdown bit control usb3otg_pipe3_powerdown bit control
2	RW	0x1	usb3otg_pipe3_txelecidle usb3otg_pipe3_txelecidle bit control usb3otg_pipe3_txelecidle bit control
1	RW	0x0	usb3otg_pipe3_rxtermination usb3otg_pipe3_rxtermination bit control usb3otg_pipe3_rxtermination bit control
0	RW	0x0	grf_con_usb3_sftsel grf_con_usb3_sftsel bit control grf_con_usb3_sftsel bit control

USB3PHY_CON9
Address: Operational Base + offset (0x0024)
USB3 PHY Control Register9

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:1	RO	0x0	reserved
0	RW	0x0	usb3phy_con9 Reserved reserved

USB3PHY_SIG_DETECT_CON0
Address: Operational Base + offset (0x0028)
USB3 PHY SIG DETECT Control Register0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15	RO	0x0	reserved
14	RO	0x0	grf_stat_usb3phy_dp_detected grf_stat_usb3phy_dp_detected bit status grf_stat_usb3phy_dp_detected bit status
13	RO	0x0	grf_stat_usb3phy_cp_detected grf_stat_usb3phy_cp_detected bit status grf_stat_usb3phy_cp_detected bit status
12	RO	0x0	grf_stat_usb3phy_dcp_detected grf_stat_usb3phy_dcp_detected bit status grf_stat_usb3phy_dcp_detected bit status
11	RO	0x0	usb3otg_utmireset usb3otg_utmireset bit status usb3otg_utmireset bit status
10	RO	0x0	usb3otg_phy_ls_fs_rcv usb3otg_phy_ls_fs_rcv bit status usb3otg_phy_ls_fs_rcv bit status
9	RO	0x0	usb3otg_utmi_avalid usb3otg_utmi_avalid bit status usb3otg_utmi_avalid bit status
8	RO	0x0	usb3otg_utmi_bvalid usb3otg_utmi_bvalid bit status usb3otg_utmi_bvalid bit status
7	RO	0x0	usb3otg_utmi_hostdisconnect usb3otg_utmi_hostdisconnect bit status usb3otg_utmi_hostdisconnect bit status
6	RO	0x0	usb3otg_utmi_iddig usb3otg_utmi_iddig bit status usb3otg_utmi_iddig bit status
5:4	RO	0x0	usb3otg_utmi_linestate usb3otg_utmi_linestate bit status usb3otg_utmi_linestate bit status
3	RO	0x0	usb3otg_utmi_sessend usb3otg_utmi_sessend bit status usb3otg_utmi_sessend bit status

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Bit	Attr	Reset Value	Description
2	RO	0x0	usb3otg_utmi_vbusvalid usb3otg_utmi_vbusvalid bit status usb3otg_utmi_vbusvalid bit status
3:2	RW	0x0	otg0_ls_filter_time_sel otg0_ls_filter_time_sel bit control otg_ls filter time select 00:100us 01:500us 10:1ms 11:10ms
1	RO	0x0	usb3otg_utmi_vmi usb3otg_utmi_vmi bit status usb3otg_utmi_vmi bit status
0	RO	0x0	usb3otg_utmi_vpi usb3otg_utmi_vpi bit status usb3otg_utmi_vpi bit status
1:0	RW	0x0	otg0_id_filter_time_sel otg0_id_filter_time_sel bit control otg_id_filter time select 00:5ms 01:15ms 10:35ms

USB3PHY_STATUS1

Address: Operational Base + offset (0x0034)

USB3 PHY STATUS1 Register1

Bit	Attr	Reset Value	Description
31	RO	0x0	usb3phy_tx_pll_lock usb3phy_tx_pll_lock bit status usb3phy_tx_pll_lock bit status
30	RO	0x0	usb3otg_pipe3_reset_n usb3otg_pipe3_reset_n bit status usb3otg_pipe3_reset_n bit status
29:24	RO	0x0	reserved
23:16	RO	0x00	usb3_phy_obs usb3_phy_obs bit status usb3_phy_obs bit status
15	RO	0x0	usb3otg_pipe3_elasbuffermode usb3otg_pipe3_elasbuffermode bit status usb3otg_pipe3_elasbuffermode bit status
14:13	RO	0x0	usb3otg_pipe3_powerdown usb3otg_pipe3_powerdown bit status usb3otg_pipe3_powerdown bit status

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Bit	Attr	Reset Value	Description
12	RO	0x0	usb3otg_pipe3_rxeqtrain usb3otg_pipe3_rxeqtrain bit status usb3otg_pipe3_rxeqtrain bit status
11	RO	0x0	usb3otg_pipe3_rxpolarity usb3otg_pipe3_rxpolarity bit status usb3otg_pipe3_rxpolarity bit status
10	RO	0x0	usb3otg_pipe3_rxtermination usb3otg_pipe3_rxtermination bit status usb3otg_pipe3_rxtermination bit status
9	RO	0x0	usb3otg_pipe3_txdetectrxloopbk usb3otg_pipe3_txdetectrxloopbk bit status usb3otg_pipe3_txdetectrxloopbk bit status
8	RO	0x0	usb3otg_pipe3_compliance usb3otg_pipe3_compliance bit status usb3otg_pipe3_compliance bit status
7	RO	0x0	usb3otg_pipe3_txoneszeros usb3otg_pipe3_txoneszeros bit status usb3otg_pipe3_txoneszeros bit status
6	RO	0x0	usb3otg_pipe3_phystatus usb3otg_pipe3_phystatus bit status usb3otg_pipe3_phystatus bit status
5	RO	0x0	usb3otg_pipe3_rxelecidle usb3otg_pipe3_rxelecidle bit status usb3otg_pipe3_rxelecidle bit status
4:2	RO	0x0	usb3otg_pipe3_rxstatus usb3otg_pipe3_rxstatus bit status usb3otg_pipe3_rxstatus bit status
1	RO	0x0	usb3otg_pipe3_rxvalid usb3otg_pipe3_rxvalid bit status usb3otg_pipe3_rxvalid bit status
0	RO	0x0	usb3otg_pipe3_powerpresent usb3otg_pipe3_powerpresent bit status usb3otg_pipe3_powerpresent bit status

USB3_WAKEUP_CON0

Address: Operational Base + offset (0x0040)

USB3 WAKEUP Control Register0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_enable Bit0~15 write enable "When bit16=1, bit0 can be written by software. When bit16=0, bit 0 cannot be written by software; When bit 17=1, bit 1 can be written by software. When bit 17=0, bit 1 cannot be written by software; …… When bit 31=1, bit 15 can be written by software. When bit 31=0, bit 15 cannot be written by software;
15:6	RO	0x0	reserved
5:4	RO	0x0	usb3_id_irq usb3_id_irq bit status usb3_id_irq bit status
4	RW	0x0	usb3_rxdet_en usb3_rxdet_en bit control usb3_rxdet_en bit control
5:4	RW	0x0	usb3_id_irq usb3_id_irq bit control usb3_id_irq bit control
3:2	RO	0x0	usb3_bvalid_irq usb3_bvalid_irq bit status usb3_bvalid_irq bit status
3:2	RW	0x0	usb3_bvalid_irq usb3_bvalid_irq bit control usb3_bvalid_irq bit control
1	RO	0x0	usb3_rxdet_irq usb3_rxdet_irq bit status usb3_rxdet_irq bit status
1	RW	0x0	usb3_rxdet_irq usb3_rxdet_irq bit control usb3_rxdet_irq bit control
0	RO	0x0	usb3_linestate_irq usb3_linestate_irq bit status usb3_linestate_irq bit status
0	RW	0x0	usb3_linestate_irq usb3_linestate_irq bit control usb3_linestate_irq bit control

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Chapter 6 Cortex-A53

6.1 Overview

The RK3228H has a quad-core Cortex-A53 cluster with 256K L2 memory. Cortex-A53 processor, which is a mid-range, low-power processor that implements the ARMv8-A architecture.

The Cortex-A53 processor includes following features:

- Full implementation of the ARMv8-A architecture instruction set
- Support for both AArch32 and AArch64 Execution status.
- Support for all exception levels, EL0, EL1, EL2, and EL3, in each execution states.
- Support A32 instruction set, previously called the ARM instruction set.
- Support T32 instruction set, previously called the Thumb instruction set.
- Support A64 instruction set.
- In-order pipeline with symmetric dual-issue of most instructions.
- Harvard Level 1(L1) memory system with a Memory Management Unit (MMU).
- Level 2(L2) memory system providing cluster memory coherency, with L2 cache.
- Support advanced SIMD and Floating-point Extension for integer and floating-point vector operations.
- Support ARMv8 Cryptography Extensions.
- Support AMBA 4 ACE bus architecture.

The configuration details of little cluster and big cluster are shown in following tables

Table 1-1 CPU Configuration

Number of CPU	**4**
L1 I cache size	32K
L1 D cache size	32K
L2 cache size	256K
L2 data RAM output latency	3 cycles
L2 data RAM input latency	2 cycles
CPU cache protection	No
SCU L2 cache protection	No
BUS master interface	ACE
NEON and floating point support	Yes
Cryptography extension	Yes

6.2 Block Diagram

The Cortex-A53 sub system is shown in Figure 1-1. As illustrated, dual-core Cortex-A53 connects to system bus through asynchronous bridges which can handle with CDC(clock domain crossing) issue.

The Cortex-A53 is connected with system counter, which can run under a constant frequency clock, for PPI interrupt generation.

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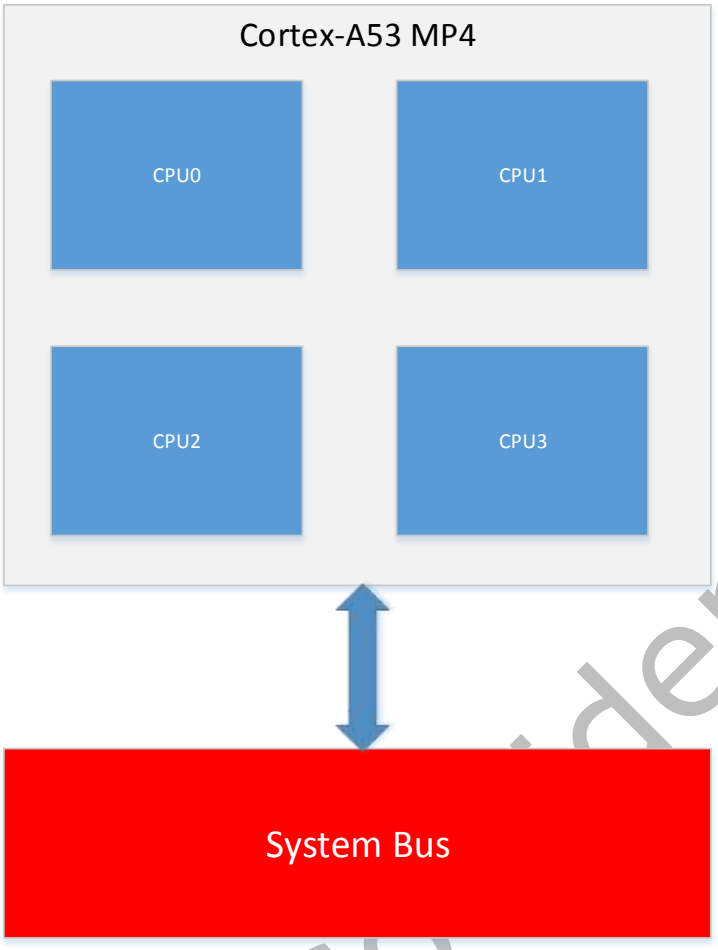

Fig.1-1 Block Diagram

6.3 Function Description

Please refer to the document cortex_a53_r0p4_trm.pdf for the detail function description.

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Chapter 7 Interconnect

7.1 Overview

The chip-level interconnect consists of one interconnect. It enables communication among the modules and subsystems in the device.

The interconnect handles many types of data transfers. It transfers data with a maximum width of 128 bits from the initiator to the target. It is a little-endian platform.

7.2 Block Diagram

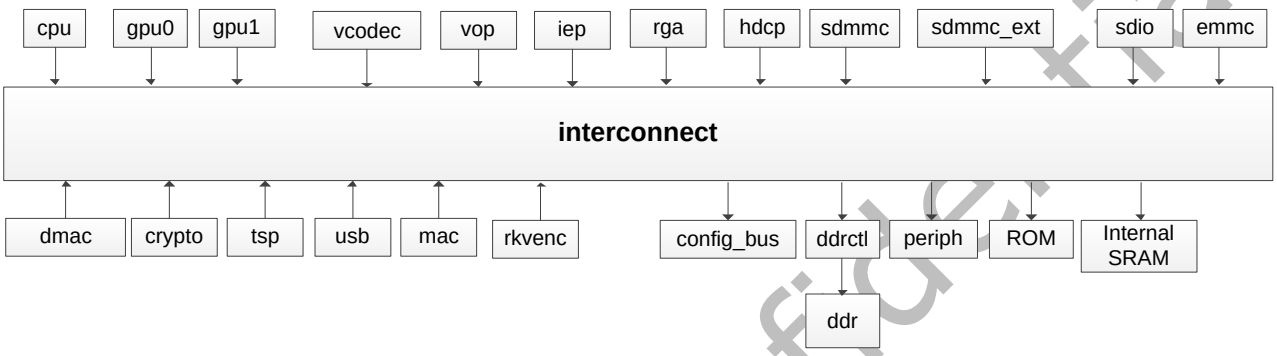

Fig. 7-1 Interconnect diagram

7.3 Function Description

7.3.1 Master & Slave

The main interconnect is connected with all the related IPs of the system, the interface between the IP and the interconnect is called as NIU(native interface unit). All the connected NIU are list as bellowing:

Table 7-1 Master NIU

Master NIU	Description
cpu	CPU port, access to any peripheral device
dmac	DMAC master
crypto	Crypto master, used in trustzone technolgy
tsp	TSP master
gpu0	GPU master0
gpu1	GPU master1
vcodec	Video codec master
vop	VOP master
rkvenc	rk video encoder
iep	IEP master
rga	RGA master

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hdcp	HDCP master
sdmmc	SDMMC master
sdmmc_ext	SDMMC_ext master
sdio	SDIO master
emmc	eMMC master
mac	GMAC master
usb	USB master

7.3.2 QoS management

The interconnect offers 4 modes of qos management:

- None, QoSGenerator is disabled, and priority information are stuck at 0.
- Fixed, QoSGenerator drives applies a fixed urgency to read transactions, and a (possibly different) urgency to write transactions.
- Limiter, QoSGenerator behaves as in fixed mode, but limits the traffic bandwidth coming from that socket, possibly stalling requests if the initiator attempts to exceed its budget.
- Regulator, QoSGenerator promotes are demotes hurry, depending the bandwidth obtained by the initiator is below or beyond a bandwidth budget. As transactions exceeding the bandwidth limit are sent (even though demoted), the regulator mode may be considered as a softer version of the limiter mode.

Limiter Behavior

When configured in bandwidth limiter, the unit uses a 23 bit counter to measure the average bandwidth. This counter has a 1/256 byte resolution and works as follows:

- Adds the number of byte rounded up to 16 (1 -> 16) and then multiplied by 256 to the current value, each time a request is sent.
- Subtracts the Bandwidth register value every cycle. If the Counter becomes negative, force it to 0.
- If the Counter value is greater than the Saturation register value multiplied by 16*256, any incoming request is stalled until this condition disappears. Note that the Counter cannot wrap-around because the maximum value it can reach is: SaturationMax*16*256 + BurstMax*256 = 1023*4K+4K*256 = 5116K or 223 = 8192K.

The following example will show the Counter behavior: 32 byte bursts, F=400MHz, BW=200MB/s, T=0.32us. The Bandwidth register will be set to 256*200/400 = 128, and the Saturation register to 128*0.32*400/4096 = 4 (which corresponds to 64 bytes).

Regulator Behavior

When configured in bandwidth regulator, the unit uses a 23 bit counter to measure the average bandwidth. This counter has a 1/256 byte resolution and works as follows:

- Adds the number of byte rounded up to 16 and then multiplied by 256 to the current value, each time a response is received. If the result is greater than the Saturation register value multiplied by 16*256, saturation to this value is applied.
- Subtracts the Bandwidth register value every cycle. If the Counter becomes negative, force it to 0.
- If the Counter value is less than or equal to the Saturation register value multiplied by 16*256/2, the SocketMst Hurry signal will be set to the HurryHigh register, and HurryLow otherwise. Note that Urgency and Press will be also set to the same value.

The following example will show the Counter behavior: 1Kbyte bursts, F=500MHz, BW=2GB/s,T=2.048us. The Bandwidth register will be set to 256*2000/500 = 1024, and the

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Saturation register to 1024*2.048*500/4096 = 256 (which corresponds to 4 Kbytes).

QoS Generator Programming

Bandwidth: This log2(socket.wData/8)+8 bits register defines the bandwidth in 1/256th byte per cycle unit. This allows a 2 MByte/s resolution at 500MHz. When thebandwidth is given in MByte/s, the value of this register will

be equal to 256*BWMB/s / FMHz.

Saturation: This 10 bits register defines the number of byte used for bandwidth measurement. It is expressed in 16bytes unit (up to 16 Kbyte). Usually the integration window is given in us or in cycle: the value of this register will beequal to Bandwidh*Tus*FMHz/(256*16)or Bandwidh*Ncycle/(256*16).

7.3.3 Memory Scheduler

Memory scheduler is a special NIU of the interconnect, it mainly deal with the transaction inside the interconnect and convert it to the transaction which the ddr protocol controller can recognize.

Following table shows the software configurable setting for the memory scheduler when the system connected to different size of ddr device.

The DDRCONF[3:0] is a configurable register inside the interconnect.

R: indicates Row bits

B: indicates Bank bits

C: indicates Column bits

D: indicates Chip selects bits

G: indicates Bank Group bits for DDR4

Table 7-2 DDR configuration

0	CDRRRRRRRRRRRRRRRBBBCCCCCCC----
1	DRRRRRRRRRRRRRRRRBBBCCCCCCCC----
2	DRRRRRRRRRRRRRRRRBBBCCCCCCCCC----
3	RRRRRRRRRRRRRRRRBBBCCCCCCCCC----
4	RRRRRRRRRRRRRRRRBBBCCCCCCCCC----
5	CRRRRRRRRRRRRRRRRDBBBCCCCCCC----
6	RRRRRRRRRRRRRRRRDBBBCCCCCCCC----
7	RRRRRRRRRRRRRRRRDBBBCCCCCCCC----
8	CDRRRRRRRRRRRRRRRRBBCCCCCCCC----
9	DCRRRRRRRRRRRRRRRRBBBCCCCCCCC----
10	RRRRRRRRRRRRRRRRBBGCCCCCCCGC----
11	DRRRRRRRRRRRRRRRRBBGCCCCCCCGC----
12	RRRRRRRRRRRRRRRRRBBGCCCCCCGC----
13	DRRRRRRRRRRRRRRRRBBGCCCCCCGC----
14	RRRRRRRRRRRRRRRRRBBCCCCCCCGC----
15	DRRRRRRRRRRRRRRRRBBCCCCCCCGC----
16	DRRRRRRRRRRRRRRRRRBBCCCCCCGC----
17	RRRRRRRRRRRRRRRRDBBGCCCCCCCGC----
18	RRRRRRRRRRRRRRRRDBBGCCCCCCGC----

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19	RRRRRRRRRRRRRRRRDBBCCCCCCCGC----
20	RRRRRRRRRRRRRRRRDBBCCCCCCGC----
21	RRRRRRRRRRRRRRRDBBCCCCCCCGGC----
22	DRRRRRRRRRRRRRRRRBBCCCCCCCBC----
23	RRRRRRRRRRRRRRRRRBBGCCCCCCCC----

7.4 Register Description

7.4.1 Address Mapping

Table 7-3 Service Module Address

service Module	Base address
service_msch	0xFF72_0000
service_rkvdec	0xFF75_0000
service_rkvenc	0xFF77_0000
service_gpu	0xFF71_0000
service_vpu	0xFF77_8000
service_vio	0xFF76_0000
service_sys	0xFF74_0000
service_peri	0xFF73_0000
service_core	0xFF70_0000

7.4.2 Registers Summary

service_msch:

Name	Offset	Size	Reset Value	Description
coreid	0x0000	W	0xe5a68c02	Core ID register
revisionid	0x0004	W	0x00018100	Revision ID register
ddrconf	0x0008	W	0x00000000	ddr configuration definition.
ddrtiming	0x000c	W	0x1c514256	ddr timing definition.
ddrmode	0x0010	W	0x00000000	ddr mode definition.
readlatency	0x0014	W	0x00000028	read latency register
Aging0	0x0018	W	0x000000FF	AGING slice threshold for port 0
Aging1	0x001c	W	0x000000FF	AGING slice threshold for port 1
Aging2	0x0020	W	0x000000FF	AGING slice threshold for port 2
Aging3	0x0024	W	0x000000FF	AGING slice threshold for port 3
Aging4	0x0028	W	0x000000FF	AGING slice threshold for port 4

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Name	Offset	Size	Reset Value	Description
Aging5	0x002c	W	0x000000FF	AGING slice threshold for port 5
activate	0x0038	W	0x00000400	activate register
devtodev	0x003c	W	0x0000002A	devtodev register
Ddr4Timing	0x0040	W	0x0000033C	DDR4 timing

service_core:

Name	Offset	Size	Reset Value	Description
Cpu_qos_Priority	0x0008	W	0x00000005	Priority register
Cpu_qos_Mode	0x000c	W	0x00000003	Mode register
Cpu_qos_Bandwidth	0x0010	W	0x00000040	Bandwidth register
Cpu_qos_Saturation	0x0014	W	0x00000040	Saturation register
Cpu_qos_ExtControl	0x0018	W	0x00000000	External inputs control

service_rkvdec:

Name	Offset	Size	Reset Value	Description
service_RKvdec_r_Priority	0x0008	W	0x00000005	Priority register
service_RKvdec_r_Mode	0x000c	W	0x00000003	Mode register
service_RKvdec_r_Bandwidth	0x0010	W	0x00000000	Bandwidth register
service_RKvdec_r_Saturation	0x0014	W	0x00000040	Saturation register
service_RKvdec_r_ExtControl	0x0018	W	0x00000000	External inputs control
service_RKvdec_w_Priority	0x0088	W	0x00000005	Priority register
service_RKvdec_w_Mode	0x008c	W	0x00000003	Mode register
service_RKvdec_w_Bandwidth	0x0090	W	0x0000004c	Bandwidth register
service_RKvdec_w_Saturation	0x0094	W	0x00000040	Saturation register
service_RKvdec_w_ExtControl	0x0098	W	0x00000000	External inputs control

service_rkvenc:

Name	Offset	Size	Reset Value	Description
service_RKvenc_h265_Priority	0x0008	W	0x00000005	Priority register
service_ RKvenc_h265_Mode	0x000c	W	0x00000003	Mode register
service_ RKvenc_h265_Bandwidth	0x0010	W	0x00000000	Bandwidth register

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Name	Offset	Size	Reset Value	Description
service_ RKvenc_h265_Saturation	0x0014	W	0x00000040	Saturation register
service_ RKvenc_h265_ExtControl	0x0018	W	0x00000000	External inputs control
service_ RKvenc_h264_Priority	0x0088	W	0x00000005	Priority register
service_ RKvenc_h264_Mode	0x008c	W	0x00000003	Mode register
service_ RKvenc_h264_Bandwidth	0x0090	W	0x0000004c	Bandwidth register
service_ RKvenc_h264_Saturation	0x0094	W	0x00000040	Saturation register
service_ RKvenc_h264_ExtControl	0x0098	W	0x00000000	External inputs control

service_vop:

Name	Offset	Size	Reset Value	Description
service_vop_Priority	0x0008	W	0x00000005	Priority register
service_vop_Mode	0x000c	W	0x00000003	Mode register
service_vop_Bandwidth	0x0010	W	0x00000166	Bandwidth register
service_vop_Saturation	0x0014	W	0x00000040	Saturation register
service_vop_ExtControl	0x0018	W	0x00000000	External inputs control

service_gpu:

Name	Offset	Size	Reset Value	Description
service_gpu0_Priority	0x0008	W	0x80000005	Priority register
service_gpu0_Mode	0x000c	W	0x00000003	Mode register
service_gpu0_Bandwidth	0x0010	W	0x0000018a	Bandwidth register
service_gpu0_Saturation	0x0014	W	0x00000040	Saturation register
service_gpu0_ExtControl	0x0018	W	0x00000000	External inputs control
service_gpu1_Priority	0x0088	W	0x80000005	Priority register
service_gpu1_Mode	0x008c	W	0x00000003	Mode register
service_gpu1_Bandwidth	0x0090	W	0x0000018a	Bandwidth register
service_gpu1_Saturation	0x0094	W	0x00000040	Saturation register

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Name	Offset	Size	Reset Value	Description
service_gpu1_ExtControl	0x0098	W	0x00000000	External inputs control

service_vpu:

Name	Offset	Size	Reset Value	Description
service_vpu_Priority	0x0008	W	0x00000005	Priority register
service_vpu_Mode	0x000c	W	0x00000003	Mode register
service_vpu_Bandwidth	0x0010	W	0x00000066	Bandwidth register
service_vpu_Saturation	0x0014	W	0x00000040	Saturation register
service_vpu_ExtControl	0x0018	W	0x00000000	External inputs control

service_vio:

Name	Offset	Size	Reset Value	Description
service_vio_Iep_Priority	0x0088	W	0x00000005	Priority register
service_vio_Iep_Mode	0x008c	W	0x00000003	Mode register
service_vio_Iep_Bandwidth	0x0090	W	0x0000020c	Bandwidth register
service_vio_Iep_Saturation	0x0094	W	0x00000040	Saturation register
service_vio_Iep_ExtControl	0x0098	W	0x00000000	External inputs control
service_vio_Rga_w_Priority	0x0108	W	0x00000005	Priority register
service_vio_Rga_w_Mode	0x010c	W	0x00000003	Mode register
service_vio_Rga_w_Bandwidth	0x0110	W	0x000002bb	Bandwidth register
service_vio_Rga_w_Saturation	0x0114	W	0x00000040	Saturation register
service_vio_Rga_w_ExtControl	0x0118	W	0x00000000	External inputs control
service_vio_Hdcp_Priority	0x0188	W	0x00000005	Priority register
service_vio_Hdcp_Mode	0x018c	W	0x00000003	Mode register
service_vio_Hdcp_Bandwidth	0x0190	W	0x000002bb	Bandwidth register
service_vio_Hdcp_Saturation	0x0194	W	0x00000040	Saturation register
service_vio_Hdcp_ExtControl	0x0198	W	0x00000000	External inputs control
service_vio_Rga_r_Priority	0x0208	W	0x00000005	Priority register

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Name	Offset	Size	Reset Value	Description
service_vio_Rga_r_Mode	0x020c	W	0x00000003	Mode register
service_vio_Rga_r_Bandwidth	0x0210	W	0x000002bb	Bandwidth register
service_vio_Rga_r_Saturation	0x0214	W	0x00000040	Saturation register
service_vio_Rga_r_ExtControl	0x0218	W	0x00000000	External inputs control

service_bus:

Name	Offset	Size	Reset Value	Description
service_sys_Dma_Priority	0x0088	W	0x00000005	DMA Priority register
service_sys_Dma_Mode	0x008c	W	0x00000003	DMA Mode register
service_sys_Dma_Bandwidth	0x0090	W	0x00000066	DMA Bandwidth register
service_sys_Dma_Saturation	0x0094	W	0x00000040	DMA Saturation register
service_sys_Dma_ExtControl	0x0098	W	0x00000000	DMA External inputs control
service_sys_Crypto_Priority	0x0108	W	0x00000005	Crypto Priority register
service_sys_Crypto_Mode	0x010c	W	0x00000000	Crypto Mode register
service_sys_Crypto_ExtControl	0x0118	W	0x00000000	Crypto External inputs control
service_sys_Tsp_Priority	0x0188	W	0x00000005	Priority register
service_sys_Tsp_Mode	0x018c	W	0x00000000	Mode register
service_sys_Tsp_ExtControl	0x0198	W	0x00000000	External inputs control

service_peri:

Name	Offset	Size	Reset Value	Description
service_peri_Emmc_Priority	0x0008	W	0x00000005	Priority register
service_peri_Emmc_Mode	0x000c	W	0x00000000	Mode register
service_peri_Emmc_ExtControl	0x0018	W	0x00000000	External inputs control
service_peri_Gmac_Priority	0x0088	W	0x00000005	Priority register
service_peri_Gmac_Mode	0x008c	W	0x00000000	Mode register
service_peri_Gmac_ExtControl	0x0098	W	0x00000000	External inputs control
service_peri_Nandc_Priority	0x0108	W	0x00000005	Priority register
service_peri_Nandc_Mode	0x010c	W	0x00000000	Mode register
service_peri_Nandc_ExtControl	0x0118	W	0x00000000	External inputs control

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Name	Offset	Size	Reset Value	Description
service_peri_Sdio_Priority	0x0188	W	0x00000005	Priority register
service_peri_Sdio_Mode	0x018c	W	0x00000000	Mode register
service_peri_Sdio_ExtControl	0x0198	W	0x00000000	External inputs control
service_peri_Sdmmc_Priority	0x0208	W	0x80000005	Priority register
service_peri_Sdmmc_Mode	0x020c	W	0x00000000	Mode register
service_peri_Sdmmc_ExtControl	0x0218	W	0x00000000	External inputs control
service_peri_Usbhost0_Priority	0x0288	W	0x00000005	Priority register
service_peri_Usbhost0_Mode	0x028c	W	0x00000000	Mode register
service_peri_Usbhost0_ExtControl	0x0298	W	0x00000000	External inputs control
service_peri_Usbhost1_Priority	0x0308	W	0x00000005	Priority register
service_peri_Usbhost1_Mode	0x030c	W	0x00000000	Mode register
service_peri_Usbhost1_ExtControl	0x0318	W	0x00000000	External inputs control
service_peri_Usbhost2_Priority	0x0388	W	0x00000005	Priority register
service_peri_Usbhost2_Mode	0x038c	W	0x00000000	Mode register
service_peri_Usbhost2_ExtControl	0x0398	W	0x00000000	External inputs control
service_peri_Usbotg_Priority	0x0408	W	0x00000005	Priority register
service_peri_Usbotg_Mode	0x040c	W	0x00000000	Mode register
service_peri_Usbotg_ExtControl	0x0418	W	0x00000000	External inputs control

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

other master's qos register is same with service_core:cpup_qos

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7.4.3 Detail Register Description

memory scheduler coreid

Address: Operational Base + offset (0x0000)

Bit	Attr	Reset Value	Description
31:0	RO	0xe5a68c02	Core type id

memory scheduler revisionid

Address: Operational Base + offset (0x0004)

Bit	Attr	Reset Value	Description
31:0	RO	0x00018100	msch' revision id

memory scheduler ddrconf

Address: Operational Base + offset (0x0008)

Memory scheduler configuration register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	DdrConf select the ddr rank,row,bank,col sequence

memory scheduler ddrtiming

Address: Operational Base + offset (0x000c)

Memory scheduler timing register

Bit	Attr	Reset Value	Description
31	RW	0x0	BwRatio When set to zero, one DRAM clock cycle (two DDR transfers) is used to transfer each word of data. When set to one, two DRAM clock cycles (four DDR transfers) are used to transfer each word of data. This is applicable when half of a DRAM data bus width is used.
30:26	RW	0x07	WrToRd The minimum number of scheduler clock cycles between the last DRAM Write command and a Read command (WL x tCkD + tWTR). tCkD is the DRAM clock period.
25:21	RW	0x02	RdToWr The minimum number of scheduler clock cycles between the last DRAM Read command and a Write command (DDR2: 2 x tCkD, DDR3: (RL - WL + 2) x tCkD). tCkD is the DRAM clock period.
20:18	RW	0x4	BurstLen The DRAM burst duration on the DRAM data bus in scheduler clock cycles. Also equal to scheduler clock cycles between two DRAM commands (BL / 2 x tCkD). tCkD is the DRAM clock period.

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Bit	Attr	Reset Value	Description
17:12	RW	0x14	WrToMiss The minimum number of scheduler clock cycles between the last DRAM Write command and a new Read or Write command in another page of the same bank (WL x tCkD + tWR + tRP + tRCD). tCkD is the DRAM clock period.
11:6	RW	0x09	RdToMiss The minimum number of scheduler clock cycles between the last DRAM Read command and a new Read or Write command in another page of the same bank (tRTP + tRP + tRCD - BL x tCkD / 2). tCkD is the DRAM clock period.
5:0	RW	0x16	ActToAct The minimum number of scheduler clock cycles between two consecutive DRAM Activate commands on the same bank (tRC/ tCkG). tCkG is the clock period of the SoC DRAM scheduler.

memory scheduler ddrmode

Address: Operational Base + offset (0x0010)

Memory scheduler mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	BwRatioExtended When set to 1, support for 4x Bwratio.
0	RW	0x0	AutoPrecharge When set to one, pages are automatically closed after each access, when set to zero, pages are left opened until an access in a different page occurs

memory scheduler readlatency

Address: Operational Base + offset (0x0014)

Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x28	ReadLatency The DRAM type-specific number of cycles from a scheduler request to a protocol controller response. This is a fixed value depending on the type of DRAM memory. <See SoC-specific memory controller documentation>.

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memory scheduler aging0

Address: Operational Base + offset (0x0018)

Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xff	Aging0 AGING slice threshold for port 0

memory scheduler aging1

Address: Operational Base + offset (0x001c)

Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xff	Aging1 AGING slice threshold for port 1

memory scheduler aging2

Address: Operational Base + offset (0x0020)

Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xff	Aging2 AGING slice threshold for port 2

memory scheduler aging3

Address: Operational Base + offset (0x0024)

Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xff	Aging3 AGING slice threshold for port 3

memory scheduler aging4

Address: Operational Base + offset (0x0028)

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Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xff	Aging4 AGING slice threshold for port 4

memory scheduler aging5

Address: Operational Base + offset (0x002c)

Memory scheduler read latency register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xff	Aging5 AGING slice threshold for port 5

memory scheduler activate

Address: Operational Base + offset (0x0038)

Memory scheduler activate register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10	RW	0x1	FawBank The number of Banks of a given device involved in the FAW period. Set to zero for 2-bank memories (WideIO). Set to one for memories with 4 banks or more (DDR).
9:4	RW	0x00	Faw The number of cycles for the four bank activate (FAW) period (tFAW).
3:0	RW	0x0	Rrd 'The number of cycles between two consecutive Activate commands on different Banks of the same device (tRRD).

memory scheduler devtodev

Address: Operational Base + offset (0x003c)

Memory scheduler devtodev register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	BusWrToRd Field0001 Abstract The number of cycles between the last write data to a device and the first read data of another device of a memory array with multiple ranks (2 x tCkD). tCkD is the DRAM clock period.
3:2	RW	0x0	BusRdToWr Field0000 Abstract The number of cycles between the last read data of a device and the first write data to another device of a memory array with multiple ranks (2 x tCkD). tCkD is the DRAM clock period.
1:0	RW	0x0	BusRdToRd The number of cycles between the last read data of a device and the first read data of another device of a memory array with multiple ranks (tCkD). tCkD is the DRAM clock period.

memory scheduler ddr4timing

Address: Operational Base + offset (0x0040)

Memory scheduler devtodev register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	RrdL Register field RrdL stores the minimum number of scheduler clock cycles on the DRAM command bus, between two Activate commands to different banks of the same bank group in the same device.
7:3	RW	0x0	WrToRdL Register field WrToRdL stores the result of nMin – tData, where nMin is the minimum number of scheduler clock cycles on the DRAM command bus in the interval between a Write command and the next Read command to an already opened page in the same bank group.
2:0	RW	0x0	CcdL Register field CcdL stores the minimum number of scheduler clock cycles, on the DRAM command bus, between column access commands, such as Read or Write operations, to an already opened page in the same bank group.

Following is CPU port's QoS register detail description. Other ports have the same register. The only different is the base address's offset.

Cpu_qos_Priority

Address: Operational Base + offset (0x0008)

Priority register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

Cpu_qos_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

Cpu_qos_Bandwidth
Address: Operational Base + offset (0x0010)
Bandwidth register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0040	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

Cpu_qos_Saturation
Address: Operational Base + offset (0x0014)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

cpu_qos ExtControl

Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	Replace the External reference by the local clock.
1	RW	0x0	ExtThr input controls Low/High priority instead of bandwidth threshold.
0	RW	0x0	Combines the Socket QoS with Regulator QoS

servicec_RKvdec_r_Priority
Address: Operational Base + offset (0x0008)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

servicec_RKvdec_r_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

servicec_RKvdec_r_Bandwidth
Address: Operational Base + offset (0x0010)
Bandwidth register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

servicec_RKvdec_r_Saturation
Address: Operational Base + offset (0x0014)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

servicec_RKvdec_r_ExtControl
Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

servicec_RKvdec_w_Priority
Address: Operational Base + offset (0x0088)

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Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

servicec_RKvdec_w_Mode
Address: Operational Base + offset (0x008c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

servicec_RKvdec_w_Bandwidth
Address: Operational Base + offset (0x0090)
Bandwidth register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x04c	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

servicec_RKvdec_w_Saturation
Address: Operational Base + offset (0x0094)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

servicec_RKvdec_w_ExtControl
Address: Operational Base + offset (0x0098)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

servicec_RKvenc_h265_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

servicec_RKvenc_h265_Bandwidth
Address: Operational Base + offset (0x0010)
Bandwidth register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0000	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

servicec_RKvenc_h265_Saturation
Address: Operational Base + offset (0x0014)
Saturation register

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Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

servicec_RKvenc_h265_ExtControl
Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

servicec_RKvenc_h264_Priority
Address: Operational Base + offset (0x0088)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

servicec_RKvenc_h264_Mode
Address: Operational Base + offset (0x008c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

servicec_RKvenc_h264_Bandwidth
Address: Operational Base + offset (0x0090)
Bandwidth register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x04c	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

servicec_RKvenc_h264_Saturation
Address: Operational Base + offset (0x0094)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

servicec_RKvenc_h264_ExtControl
Address: Operational Base + offset (0x0098)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_gpu0_Priority

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Address: Operational Base + offset (0x0008)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x1	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_gpu0_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_gpu0_Bandwidth
Address: Operational Base + offset (0x0010)
Bandwidth register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x18a	Bandwidth Field0000 Abstract In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_gpu0_Saturation
Address: Operational Base + offset (0x0014)
Saturation register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_gpu0_ExtControl
Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_gpu1_Priority
Address: Operational Base + offset (0x0088)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x1	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_gpu1_Mode
Address: Operational Base + offset (0x008c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_gpu1_Bandwidth
Address: Operational Base + offset (0x0090)
Bandwidth register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x18a	Bandwidth Field0000 Abstract In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_gpu1_Saturation
Address: Operational Base + offset (0x0094)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_gpu1_ExtControl
Address: Operational Base + offset (0x0098)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_vpu_Priority

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Address: Operational Base + offset (0x0008)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_vpu_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_vpu_Bandwidth
Address: Operational Base + offset (0x0010)
Bandwidth register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x066	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_vpu_Saturation
Address: Operational Base + offset (0x0014)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_vpu_ExtControl
Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_vio_vop_Priority
Address: Operational Base + offset (0x0088)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_ vio_vop_Mode
Address: Operational Base + offset (0x008c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_ vio_vop_Bandwidth
Address: Operational Base + offset (0x0090)
Bandwidth register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0166	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_ vio_vop_Saturation
Address: Operational Base + offset (0x0094)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_ vio_vop_ExtControl
Address: Operational Base + offset (0x0098)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_vio_Iep_Priority
Address: Operational Base + offset (0x0108)

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Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_vio_Iep_Mode

Address: Operational Base + offset (0x010c)

Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_vio_Iep_Bandwidth

Address: Operational Base + offset (0x0110)

Bandwidth register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x020c	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_vio_Iep_Saturation

Address: Operational Base + offset (0x0114)

Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_vio_Iep_ExtControl
Address: Operational Base + offset (0x0118)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_vio_Rga_w_Priority
Address: Operational Base + offset (0x0288)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_vio_Rga_w_Mode
Address: Operational Base + offset (0x028c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_vio_Rga_w_Bandwidth
Address: Operational Base + offset (0x0290)
Bandwidth register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x2bb	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_vio_Rga_w_Saturation
Address: Operational Base + offset (0x0294)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_vio_Rga_w_ExtControl
Address: Operational Base + offset (0x0298)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_vio_Hdcp_Priority
Address: Operational Base + offset (0x0008)

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Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_vio_Hdcp_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_vio_Hdcp_Bandwidth
Address: Operational Base + offset (0x0010)
Bandwidth register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x2bb	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_vio_Hdcp_Saturation
Address: Operational Base + offset (0x0014)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_vio_Hdcp_ExtControl
Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_vio_Rga_r_Priority
Address: Operational Base + offset (0x0208)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_vio_Rga_r_Mode
Address: Operational Base + offset (0x020c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_vio_Rga_r_Bandwidth
Address: Operational Base + offset (0x0210)
Bandwidth register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x2bb	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_vio_Rga_r_Saturation
Address: Operational Base + offset (0x0214)
Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_vio_Rga_r_ExtControl
Address: Operational Base + offset (0x0218)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_sys_Dma_Priority
Address: Operational Base + offset (0x0008)
DMA Priority register

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Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_sys_Dma_Mode
Address: Operational Base + offset (0x000c)
DMA Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x3	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_sys_Dma_Bandwidth
Address: Operational Base + offset (0x0010)
DMA Bandwidth register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RW	0x0066	Bandwidth In Bandwidth Limiter or Bandwidth Regulator mode, the bandwidth threshold in units of 1/256th bytes per cycle. For example, 80 MBps on a 250 MHz interface is value 0x0052.

service_sys_Dma_Saturation
Address: Operational Base + offset (0x0014)
DMA Saturation register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x040	Saturation In Bandwidth Limiter or Bandwidth Regulator mode, the maximum data count value, in units of 16 bytes. This determines the window of time over which bandwidth is measured. For example, to measure bandwidth within a 1000 cycle window on a 64-bit interface is value 0x1F4.

service_sys_Dma_ExtControl
Address: Operational Base + offset (0x0018)
DMA External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_sys_Crypto_Priority
Address: Operational Base + offset (0x0008)
Crypto Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_sys_Crypto_Mode
Address: Operational Base + offset (0x000c)
Crypto Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_sys_Crypto_ExtControl
Address: Operational Base + offset (0x0018)
Crypto External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_sys_Tsp_Priority
Address: Operational Base + offset (0x0188)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_sys_Tsp_Mode
Address: Operational Base + offset (0x018c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_sys_Tsp_ExtControl
Address: Operational Base + offset (0x0198)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Emmc_Priority
Address: Operational Base + offset (0x0008)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Emmc_Mode
Address: Operational Base + offset (0x000c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Emmc_ExtControl
Address: Operational Base + offset (0x0018)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Gmac2io_Priority
Address: Operational Base + offset (0x0088)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Gmac2io_Mode
Address: Operational Base + offset (0x008c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Gmac2io_ExtControl
Address: Operational Base + offset (0x0098)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Gmach2phy_Priority
Address: Operational Base + offset (0x0488)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Gmach2phy_Mode
Address: Operational Base + offset (0x048c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Gmach2phy_ExtControl
Address: Operational Base + offset (0x0498)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Sdio_Priority
Address: Operational Base + offset (0x0188)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Sdio_Mode
Address: Operational Base + offset (0x018c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Sdio_ExtControl
Address: Operational Base + offset (0x0198)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Sdmmc_Priority
Address: Operational Base + offset (0x0208)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x1	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Sdmmc_Mode
Address: Operational Base + offset (0x020c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Sdmmc_ExtControl
Address: Operational Base + offset (0x0218)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Usbhost_Priority
Address: Operational Base + offset (0x0288)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Usbhost_Mode
Address: Operational Base + offset (0x028c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Usbhost_ExtControl
Address: Operational Base + offset (0x0298)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Usbotg_Priority
Address: Operational Base + offset (0x0408)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x0	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Usbotg_Mode
Address: Operational Base + offset (0x040c)
Mode register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Usbotg_ExtControl
Address: Operational Base + offset (0x0418)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

service_peri_Sdmmc_ext_Priority
Address: Operational Base + offset (0x0208)
Priority register

Bit	Attr	Reset Value	Description
31	RO	0x1	Mark Backward compatibility marker when 0.
30:4	RO	0x0	reserved
3:2	RW	0x1	P1 In Programmable or Bandwidth Limiter mode, the priority level for read transactions. In Bandwidth regulator mode, the priority level when the used throughput is below the threshold. In Bandwidth Regulator mode, P1 should have a value equal or greater than P0.
1:0	RW	0x1	P0 In Programmable or Bandwidth Limiter mode, the priority level for write transactions. In Bandwidth Regulator mode, the priority level when the used throughput is above the threshold. In Bandwidth Regulator mode, P0 should have a value equal or lower than P1.

service_peri_Sdmmc_ext_Mode
Address: Operational Base + offset (0x020c)
Mode register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	Mode 0 = Programmable mode: a programmed priority is assigned to each read or write, 1 = Bandwidth Limiter Mode: a hard limit restricts throughput, 2 = Bypass mode: (<See SoC-specific QoS generator documentation>), 3 = Bandwidth Regulator mode: priority decreases when throughput exceeds a threshold.

service_peri_Sdmmc_ext_ExtControl
Address: Operational Base + offset (0x0218)
External inputs control

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	IntClkEn n/a
1	RW	0x0	ExtThrEn n/a
0	RW	0x0	SocketQosEn n/a

7.5 Application Notes

7.5.1 QoS setting

The CPU read channel, VOP read channel, GPU write channel have the external QoS control.

After reset each master port both have priority setting as 1. It's recommended that field 0 of QoS. ExtControl set to 1 to enable the external qos control. And priority setting of each master kept at 1.

7.5.2 Idle request

The main interconnect supports flushing the ongoing transaction when the software needed to do so.

If the GPU power domain need to disconnect from the main interconnect, Idle request has to be sent to GPU NIU, the NIU will respond a ack, and when it's ready to be disconnect, one Idle signal will be send out . Then, if GPU still have transaction to be sent to the memory scheduler, it will be stalled by the NIU.

If the GPU system power domain is disconnected as the above flow, then CPU want to access to the GPU system, it will response error to CPU.

The sequence is like following figure shows:

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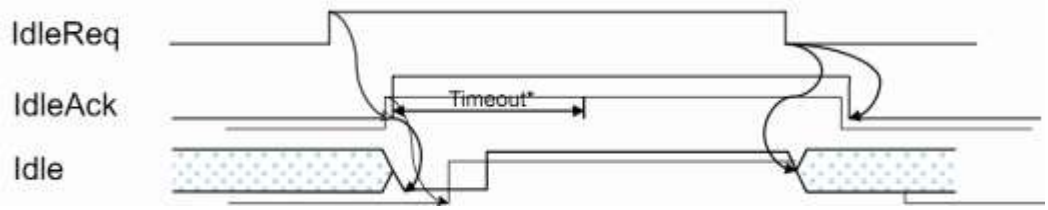

Fig. 7-2 Idle request

The idle request is set by GRF register.

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Chapter 8 DMA Controller (DMAC)

8.1 Overview

This device supports 1 Direct Memory Access (DMA) Controllers. It (DMAC) supports transfers between memory and memory, peripheral and memory. DMAC is under Non-secure state after reset, and the secure state can be changed by configuring SGRF module.
DMAC supports the following features:

- Supports Trustzone technology
- Supports 17 peripheral request
- Up to 64bits data size
- 8 channel at the same time
- Up to burst 16
- 16 interrupts output and 1 abort output
- Supports 128 MFIFO depth

Following table shows the DMAC request mapping scheme.

Table 8-1 DMAC Request Mapping Table

Req number	Source	Polarity
0	I2S2_2CH_TX	High level
1	I2S2_2CH_RX	High level
2	UART0_TX	High level
3	UART0_RX	High level
4	UART1_TX	High level
5	UART1_RX	High level
6	UART2_TX	High level
7	UART2_RX	High level
8	SPI0_TX	High level
9	SPI0_RX	High level
10	SPDIF_8CH_TX	High level
11	I2S0_8CH_TX	High level
12	I2S0_8CH_RX	High level
13	PWM_TX	High level
14	I2S1_8CH_TX	High level
15	I2S1_8CH_RX	High level
16	PDM_TX	High level

DMAC support incrementing-address burst and fixed-address burst. But in the case of access SPI and UART at byte or halfword size, DMAC only support fixed-address burst and the address must be aligned to word.

8.2 Block Diagram

Following figure shows the block diagram of DMAC.

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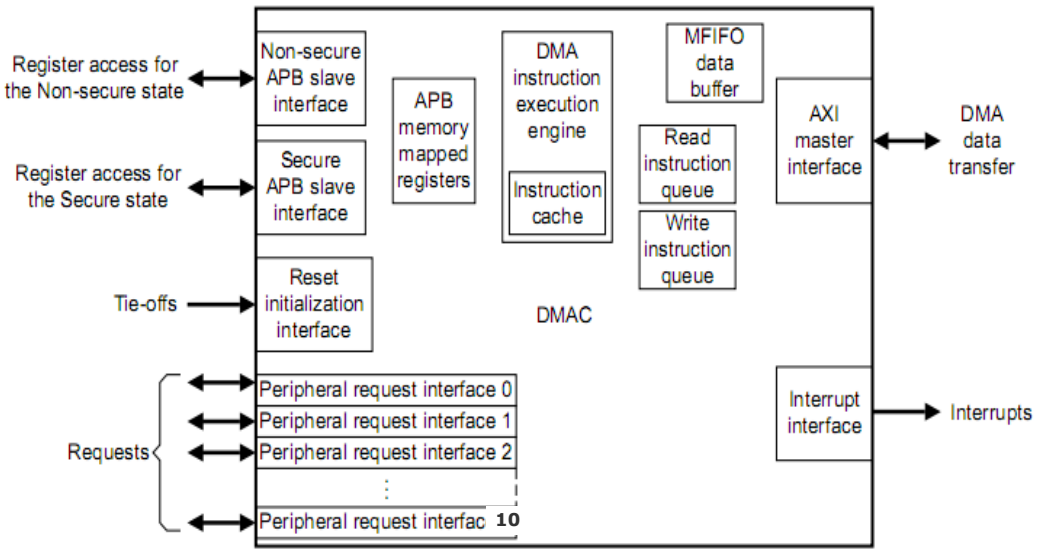

Fig. 8-1 Block diagram of DMAC

As the DMAC supports Trustzone technology, so dual APB interfaces enable the operation of the DMAC to be partitioned into the secure state and Non-secure state. You can use the APB interfaces to access status registers and also directly execute instructions in the DMAC. The default interface after reset is Non-secure apb interface.

8.3 Function Description

8.3.1 Introduction

The DMAC contains an instruction processing block that enables it to process program code that controls a DMA transfer. The program code is stored in a region of system memory that the DMAC accesses using its AXI interface. The DMAC stores instructions temporarily in a cache. It supports 8 channels, each channel capable of supporting a single concurrent thread of DMA operation. In addition, a single DMA manager thread exists, and you can use it to initialize the DMA channel threads. The DMAC executes up to one instruction for each AXI clock cycle. To ensure that it regularly executes each active thread, it alternates by processing the DMA manager thread and then a DMA channel thread. It uses a round-robin process when selecting the next active DMA channel thread to execute.

The DMAC uses variable-length instructions that consist of one to six bytes. It provides a separate Program Counter (PC) register for each DMA channel. When a thread requests an instruction from an address, the cache performs a look-up. If a cache hit occurs, then the cache immediately provides the data. Otherwise, the thread is stalled while the DMAC uses the AXI interface to perform a cache line fill. If an instruction is greater than 4 bytes, or spans the end of a cache line, the DMAC performs multiple cache accesses to fetch the instruction. When a cache line fill is in progress, the DMAC enables other threads to access the cache, but if another cache miss occurs, this stalls the pipeline until the first line fill is complete.

When a DMA channel thread executes a load or store instruction, the DMAC adds the instruction to the relevant read or write queue. The DMAC uses these queues as an instruction storage buffer prior to it issuing the instructions on the AXI bus. The DMAC also contains a Multi First-In-First-Out (MFIFO) data buffer that it uses to store data that it reads, or writes, during a DMA transfer.

8.3.2 Operating states

Following figure shows the operating states for the DMA manager thread and DMA channel threads.

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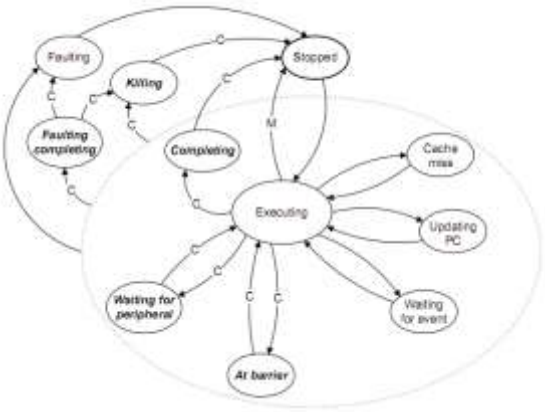

Fig. 8-2 DMAC operation states

Notes:arcs with no letter designator indicate state transitions for the DMA manager and DMA channel threads, otherwise use is restricted as follows:
C DMA channel threads only.
M DMA manager thread only.

After the DMAC exits from reset, it sets all DMA channel threads to the stopped state, and DMA manager thread moves to the Stopped state.

8.4 Register Description

8.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
DMAC_DSR	0x0000	W	0x00000000	DMA Manager Status Register
DMAC_DPC	0x0004	W	0x00000000	DMA Program Counter Register
DMAC_INTEN	0x0020	W	0x00000000	Interrupt Enable Register
DMAC_EVENT_RIS	0x0024	W	0x00000000	Event-Interrupt Raw Status Register
DMAC_INTMIS	0x0028	W	0x00000000	Interrupt Status Register
DMAC_INTCLR	0x002c	W	0x00000000	Interrupt Clear Register
DMAC_FSRD	0x0030	W	0x00000000	Fault Status DMA Manager Register
DMAC_FSRC	0x0034	W	0x00000000	Fault Status DMA Channel Register
DMAC_FTRD	0x0038	W	0x00000000	Fault Type DMA Manager Register
DMAC_FTR0	0x0040	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR1	0x0044	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR2	0x0048	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR3	0x004c	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR4	0x0050	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR5	0x0054	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR6	0x0058	W	0x00000000	Fault Type DMA Channel Register
DMAC_FTR7	0x005c	W	0x00000000	Fault Type DMA Channel Register
DMAC_CSR0	0x0100	W	0x00000000	Channel Status Registers
DMAC_CPC0	0x0104	W	0x00000000	Channel Program Counter Registers
DMAC_CSR1	0x0108	W	0x00000000	Channel Status Registers
DMAC_CPC1	0x010c	W	0x00000000	Channel Program Counter Registers

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Name	Offset	Size	Reset Value	Description
DMAC_CSR2	0x0110	W	0x00000000	Channel Status Registers
DMAC_CPC2	0x0114	W	0x00000000	Channel Program Counter Registers
DMAC_CSR3	0x0118	W	0x00000000	Channel Status Registers
DMAC_CPC3	0x011c	W	0x00000000	Channel Program Counter Registers
DMAC_CSR4	0x0120	W	0x00000000	Channel Status Registers
DMAC_CPC4	0x0124	W	0x00000000	Channel Program Counter Registers
DMAC_CSR5	0x0128	W	0x00000000	Channel Status Registers
DMAC_CPC5	0x012c	W	0x00000000	Channel Program Counter Registers
DMAC_CSR6	0x0130	W	0x00000000	Channel Status Registers
DMAC_CPC6	0x0134	W	0x00000000	Channel Program Counter Registers
DMAC_CSR7	0x0138	W	0x00000000	Channel Status Registers
DMAC_CPC7	0x013c	W	0x00000000	Channel Program Counter Registers
DMAC_SAR0	0x0400	W	0x00000000	Source Address Registers
DMAC_DAR0	0x0404	W	0x00000000	Destination Address Registers
DMAC_CCR0	0x0408	W	0x00000000	Channel Control Registers
DMAC_LC0_0	0x040c	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_0	0x0410	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR1	0x0420	W	0x00000000	Source Address Registers
DMAC_DAR1	0x0424	W	0x00000000	Destination Address Registers
DMAC_CCR1	0x0428	W	0x00000000	Channel Control Registers
DMAC_LC0_1	0x042c	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_1	0x0430	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR2	0x0440	W	0x00000000	Source Address Registers
DMAC_DAR2	0x0444	W	0x00000000	Destination Address Registers
DMAC_CCR2	0x0448	W	0x00000000	Channel Control Registers
DMAC_LC0_2	0x044c	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_2	0x0450	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR3	0x0460	W	0x00000000	Source Address Registers
DMAC_DAR3	0x0464	W	0x00000000	Destination Address Registers
DMAC_CCR3	0x0468	W	0x00000000	Channel Control Registers
DMAC_LC0_3	0x046c	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_3	0x0470	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR4	0x0480	W	0x00000000	Source Address Registers
DMAC_DAR4	0x0484	W	0x00000000	Destination Address Registers
DMAC_CCR4	0x0488	W	0x00000000	Channel Control Registers
DMAC_LC0_4	0x048c	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_4	0x0490	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR5	0x04a0	W	0x00000000	Source Address Registers

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Name	Offset	Size	Reset Value	Description
DMAC_DAR5	0x04a4	W	0x00000000	Destination Address Registers
DMAC_CCR5	0x04a8	W	0x00000000	Channel Control Registers
DMAC_LC0_5	0x04ac	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_5	0x04b0	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR6	0x04c0	W	0x00000000	Source Address Registers
DMAC_DAR6	0x04c4	W	0x00000000	Destination Address Registers
DMAC_CCR6	0x04c8	W	0x00000000	Channel Control Registers
DMAC_LC0_6	0x04cc	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_6	0x04d0	W	0x00000000	Loop Counter 1 Registers
DMAC_SAR7	0x04e0	W	0x00000000	Source Address Registers
DMAC_DAR7	0x04e4	W	0x00000000	Destination Address Registers
DMAC_CCR7	0x04e8	W	0x00000000	Channel Control Registers
DMAC_LC0_7	0x04ec	W	0x00000000	Loop Counter 0 Registers
DMAC_LC1_7	0x04f0	W	0x00000000	Loop Counter 1 Registers
DMAC_DBGSTATUS	0x0d00	W	0x00000000	Debug Status Register
DMAC_DBGCMD	0x0d04	W	0x00000000	Debug Command Register
DMAC_DBGINST0	0x0d08	W	0x00000000	Debug Instruction-0 Register
DMAC_DBGINST1	0x0d0c	W	0x00000000	Debug Instruction-1 Register
DMAC_CR0	0x0e00	W	0x00047051	Configuration Register 0
DMAC_CR1	0x0e04	W	0x00000057	Configuration Register 1
DMAC_CR2	0x0e08	W	0x00000000	Configuration Register 2
DMAC_CR3	0x0e0c	W	0x00000000	Configuration Register 3
DMAC_CR4	0x0e10	W	0x00000006	Configuration Register 4
DMAC_CRDn	0x0e14	W	0x02094733	DMA Configuration Register
DMAC_WD	0x0e80	W	0x00000000	DMA Watchdog Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access.*
For DMAC0 channel register, only the channel 0~5 is valid.

8.4.2 Detail Register Description

DMAC_DSR
Address: Operational Base + offset (0x0000)
DMA Manager Status Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RO	0x0	Provides the security status of the DMA manager thread: 0 = DMA manager operates in the Secure state 1 = DMA manager operates in the Non-secure state.
8:4	RO	0x00	When the DMA manager thread executes a DMAWFE instruction, it waits for the following event to occur: b00000 = event[0] b00001 = event[1] b00010 = event[2] … b11111 = event[31].

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Bit	Attr	Reset Value	Description
3:0	RO	0x0	The operating state of the DMA manager: b0000 = Stopped b0001 = Executing b0010 = Cache miss b0011 = Updating PC b0100 = Waiting for event b0101-b1110 = reserved b1111 = Faulting.

DMAC_DPC
Address: Operational Base + offset (0x0004)
DMA Program Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Program counter for the DMA manager thread

DMAC_INTEN
Address: Operational Base + offset (0x0020)
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Program the appropriate bit to control how the DMAC responds when it executes DMASEV: Bit [N] = 0 If the DMAC executes DMASEV for the event-interrupt resource N then the DMAC signals event N to all of the threads. Set bit [N] to 0 if your system design does not use irq[N] to signal an interrupt request. Bit [N] = 1 If the DMAC executes DMASEV for the event-interrupt resource N then the DMAC sets irq[N] HIGH. Set bit [N] to 1 if your system designer requires irq[N] to signal an interruptrequest.

DMAC_EVENT_RIS
Address: Operational Base + offset (0x0024)
Event-Interrupt Raw Status Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Returns the status of the event-interrupt resources: Bit [N] = 0 Event N is inactive or irq[N] is LOW. Bit [N] = 1 Event N is active or irq[N] is HIGH.

DMAC_INTMIS
Address: Operational Base + offset (0x0028)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Provides the status of the interrupts that are active in the DMAC: Bit [N] = 0 Interrupt N is inactive and therefore irq[N] is LOW. Bit [N] = 1 Interrupt N is active and therefore irq[N] is HIGH

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DMAC_INTCLR
Address: Operational Base + offset (0x002c)
Interrupt Clear Register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	Controls the clearing of the irq outputs: Bit [N] = 0 The status of irq[N] does not change. Bit [N] = 1 The DMAC sets irq[N] LOW if the INTEN Register programs the DMAC to signal an interrupt. Otherwise, the status of irq[N] does not change.

DMAC_FSRD
Address: Operational Base + offset (0x0030)
Fault Status DMA Manager Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Provides the fault status of the DMA manager. Read as: 0 = the DMA manager thread is not in the Faulting state 1 = the DMA manager thread is in the Faulting state.

DMAC_FSRC
Address: Operational Base + offset (0x0034)
Fault Status DMA Channel Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Each bit provides the fault status of the corresponding channel. Read as: Bit [N] = 0 No fault is present on DMA channel N. Bit [N] = 1 DMA channel N is in the Faulting or Faulting completing state.

DMAC_FTRD
Address: Operational Base + offset (0x0038)
Fault Type DMA Manager Register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	If the DMA manager aborts, this bit indicates if the erroneous instruction was read from the system memory or from the debug interface: 0 = instruction that generated an abort was read from system memory 1 = instruction that generated an abort was read from the debug interface.
29:17	RO	0x0	reserved
16	RO	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA manager performs an instruction fetch: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response

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Bit	Attr	Reset Value	Description
15:6	RO	0x0	reserved
5	RO	0x0	Indicates if the DMA manager was attempting to execute DMAWFE or DMASEV with inappropriate security permissions: 0 = DMA manager has appropriate security to execute DMAWFE or DMASEV 1 = a DMA manager thread in the Non-secure state attempted to execute either: DMAWFE to wait for a secure event DMASEV to create a secure event or secure interrupt
4	RO	0x0	Indicates if the DMA manager was attempting to execute DMAGO with inappropriate security permissions: 0 = DMA manager has appropriate security to execute DMAGO 1 = DMA manager thread in the Non-secure state attempted to execute DMAGO to create a DMA channel operating in the Secure state.
3:2	RO	0x0	reserved
1	RO	0x0	Indicates if the DMA manager was attempting to execute an instruction operand that was not valid for the configuration of the DMAC: 0 = valid operand 1 = invalid operand.
0	RW	0x0	Indicates if the DMA manager was attempting to execute an undefined instruction: 0 = defined instruction 1 = undefined instruction.

DMAC_FTR0~DMAC_FTR7

Address: Operational Base + offset (0x0040)
Operational Base+0x44
Operational Base+0x48
Operational Base+0x4C
Operational Base+0x50
Operational Base+0x54
Operational Base+0x58
Operational Base+0x5C

Fault Type DMA Channel Register

Bit	Attr	Reset Value	Description
31	RO	0x0	Indicates if the DMA channel has locked-up because of resource starvation: 0 = DMA channel has adequate resources 1 = DMA channel has locked-up because of insufficient resources. This fault is an imprecise abort

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Bit	Attr	Reset Value	Description
30	RO	0x0	If the DMA channel aborts, this bit indicates if the erroneous instruction was read from the system memory or from the debug interface: 0 = instruction that generated an abort was read from system memory 1 = instruction that generated an abort was read from the debug interface. This fault is an imprecise abort but the bit is only valid when a precise abort occurs.
29:19	RO	0x0	reserved
18	RO	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA channel thread performs a data read: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is an imprecise abort
17	RO	0x0	Indicates the AXI response that the DMAC receives on the BRESP bus, after the DMA channel thread performs a data write: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is an imprecise abort.
16	RO	0x0	Indicates the AXI response that the DMAC receives on the RRESP bus, after the DMA channel thread performs an instruction fetch: 0 = OKAY response 1 = EXOKAY, SLVERR, or DECERR response. This fault is a precise abort.
15:14	RO	0x0	reserved
13	RO	0x0	Indicates if the MFIFO did not contain the data to enable the DMAC to perform the DMAST: 0 = MFIFO contains all the data to enable the DMAST to complete 1 = previous DMALDs have not put enough data in the MFIFO to enable the DMAST to complete. This fault is a precise abort.
12	RO	0x0	Indicates if the MFIFO prevented the DMA channel thread from executing DMALD or DMAST. Depending on the instruction: DMALD 0 = MFIFO contains sufficient space 1 = MFIFO is too small to hold the data that DMALD requires. DMAST 0 = MFIFO contains sufficient data 1 = MFIFO is too small to store the data to enable DMAST to complete. This fault is an imprecise abort
11:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RO	0x0	Indicates if a DMA channel thread, in the Non-secure state, attempts to program the CCRn Register to perform a secure read or secure write: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to perform a secure read or secure write. This fault is a precise abort
6	RO	0x0	Indicates if a DMA channel thread, in the Non-secure state, attempts to execute DMAWFP, DMALDP, DMASTP, or DMAFLUSHP with inappropriate security permissions: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to execute either: o DMAWFP to wait for a secure peripheral o DMALDP or DMASTP to notify a secure peripheral o DMAFLUSHP to flush a secure peripheral. This fault is a precise abort.
5	RO	0x0	Indicates if the DMA channel thread attempts to execute DMAWFE or DMASEV with inappropriate security permissions: 0 = a DMA channel thread in the Non-secure state is not violating the security permissions 1 = a DMA channel thread in the Non-secure state attempted to execute either: DMAWFE to wait for a secure event DMASEV to create a secure event or secure interrupt. This fault is a precise abort.
4:2	RO	0x0	reserved
1	RO	0x0	Indicates if the DMA channel thread was attempting to execute an instruction operand that was not valid for the configuration of the DMAC: 0 = valid operand 1 = invalid operand. This fault is a precise abort.
0	RO	0x0	Indicates if the DMA channel thread was attempting to execute an undefined instruction: 0 = defined instruction 1 = undefined instruction. This fault is a precise abort

DMAC_CSR0~DMAC_CSR7

Address:Operational Base+0x100
Operational Base+0x108
Operational Base+0x110
Operational Base+0x118
Operational Base+0x120

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Operational Base+0x128
Operational Base+0x130
Operational Base+0x138

Channel Status Registers

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RO	0x0	The channel non-secure bit provides the security of the DMA channel: 0 = DMA channel operates in the Secure state 1 = DMA channel operates in the Non-secure state
20:16	RO	0x0	reserved
15	RO	0x0	When the DMA channel thread executes DMAWFP this bit indicates if the periph operand was set: 0 = DMAWFP executed with the periph operand not set 1 = DMAWFP executed with the periph operand set
14	RO	0x0	When the DMA channel thread executes DMAWFP this bit indicates if the burst or single operand were set: 0 = DMAWFP executed with the single operand set 1 = DMAWFP executed with the burst operand set.
13:9	RO	0x0	reserved
8:4	RO	0x00	If the DMA channel is in the Waiting for event state or the Waiting for peripheral state then these bits indicate the event or peripheral number that the channel is waiting for: b00000 = DMA channel is waiting for event, or peripheral, 0 b00001 = DMA channel is waiting for event, or peripheral, 1 b00010 = DMA channel is waiting for event, or peripheral, 2 … b11111 = DMA channel is waiting for event, or peripheral, 31
3:0	RO	0x0	The channel status encoding is: b0000 = Stopped b0001 = Executing b0010 = Cache miss b0011 = Updating PC b0100 = Waiting for event b0101 = At barrier b0110 = reserved b0111 = Waiting for peripheral b1000 = Killing b1001 = Completing b1010-b1101 = reserved b1110 = Faulting completing b1111 = Faulting

DMAC_CPC0~DMAC_CPC7

Address:Operational Base+0x104
Operational Base+0x10C
Operational Base+0x114

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Operational Base+0x11c
Operational Base+0x124
Operational Base+0x12C
Operational Base+0x134
Operational Base+0x13C

Channel Program Counter Registers

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Program counter for the DMA channel 0 thread

DMAC_SAR0~DMAC_SAR7

Address:Operational Base+0x400
Operational Base+0x420
Operational Base+0x440
Operational Base+0x460
Operational Base+0x480
Operational Base+0x4A0
Operational Base+0x4C0
Operational Base+0x4E0

Source Address Registers

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Address of the source data for DMA channel 0

DMAC_DAR0~DMAC_DAR7

Address:Operational Base+0x404
Operational Base+0x424
Operational Base+0x444
Operational Base+0x464
Operational Base+0x484
Operational Base+0x4A4
Operational Base+0x4C4
Operational Base+0x4E4

DestinationAddress Registers

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Address of the Destination data for DMA channel 0

DMAC_CCR0~DMAC_CCR7

Address:Operational Base+0x408
Operational Base+0x428
Operational Base+0x448
Operational Base+0x468
Operational Base+0x488
Operational Base+0x4A8
Operational Base+0x4C8
Operational Base+0x4E8

Channel Control Registers

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:25	RO	0x0	Programs the state of AWCACHE[3,1:0]a when the DMAC writes the destination data. Bit [27] 0 = AWCACHE[3] is LOW 1 = AWCACHE[3] is HIGH. Bit [26] 0 = AWCACHE[1] is LOW 1 = AWCACHE[1] is HIGH. Bit [25] 0 = AWCACHE[0] is LOW 1 = AWCACHE[0] is HIGH
24:22	RO	0x0	Programs the state of AWPROT[2:0]a when the DMAC writes the destination data. Bit [24] 0 = AWPROT[2] is LOW 1 = AWPROT[2] is HIGH. Bit [23] 0 = AWPROT[1] is LOW 1 = AWPROT[1] is HIGH. Bit [22] 0 = AWPROT[0] is LOW 1 = AWPROT[0] is HIGH
21:18	RO	0x0	For each burst, these bits program the number of data transfers that the DMAC performs when it writes the destination data: b0000 = 1 data transfer b0001 = 2 data transfers b0010 = 3 data transfers … b1111 = 16 data transfers. The total number of bytes that the DMAC writes out of the MFIFO when it executes a DMAST instruction is the product of dst_burst_len and dst_burst_size
17:15	RO	0x0	For each beat within a burst, it programs the number of bytes that the DMAC writes to the destination: b000 = writes 1 byte per beat b001 = writes 2 bytes per beat b010 = writes 4 bytes per beat b011 = writes 8 bytes per beat b100 = writes 16 bytes per beat b101-b111 = reserved. The total number of bytes that the DMAC writes out of the MFIFO when it executes a DMAST instruction is the product of dst_burst_len and dst_burst_size.
14	RO	0x0	Programs the burst type that the DMAC performs when it writes the destination data: 0 = Fixed-address burst. The DMAC signals AWBURST[0] LOW. 1 = Incrementing-address burst. The DMAC signals AWBURST[0] HIGH.

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Bit	Attr	Reset Value	Description
13:11	RO	0x0	Set the bits to control the state of ARCACHE[2:0]a when the DMAC reads the source data. Bit [13] 0 = ARCACHE[2] is LOW 1 = ARCACHE[2] is HIGH. Bit [12] 0 = ARCACHE[1] is LOW 1 = ARCACHE[1] is HIGH. Bit [11] 0 = ARCACHE[0] is LOW 1 = ARCACHE[0] is HIGH.
10:8	RO	0x0	Programs the state of ARPROT[2:0]a when the DMAC reads the source data. Bit [10] 0 = ARPROT[2] is LOW 1 = ARPROT[2] is HIGH. Bit [9] 0 = ARPROT[1] is LOW 1 = ARPROT[1] is HIGH. Bit [8] 0 = ARPROT[0] is LOW 1 = ARPROT[0] is HIGH.
7:4	RO	0x0	For each burst, these bits program the number of data transfers that the DMAC performs when it reads the source data: b0000 = 1 data transfer b0001 = 2 data transfers b0010 = 3 data transfers … b1111 = 16 data transfers. The total number of bytes that the DMAC reads into the MFIFO when it executes a DMALD instruction is the product of src_burst_len and src_burst_size
3:1	RO	0x0	For each beat within a burst, it programs the number of bytes that the DMAC reads from the source: b000 = reads 1 byte per beat b001 = reads 2 bytes per beat b010 = reads 4 bytes per beat b011 = reads 8 bytes per beat b100 = reads 16 bytes per beat b101-b111 = reserved. The total number of bytes that the DMAC reads into the MFIFO when it executes a DMALD instruction is the product of src_burst_len and src_burst_size
0	RO	0x0	Programs the burst type that the DMAC performs when it reads the source data: 0 = Fixed-address burst. The DMAC signals ARBURST[0] LOW. 1 = Incrementing-address burst. The DMAC signals ARBURST[0] HIGH

DMAC_LC0_0~DMAC_LC0_7
Address: Operational Base+0x40c
Operational Base+0x42C

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Operational Base+0x44C
Operational Base+0x46C
Operational Base+0x48C
Operational Base+0x4AC
Operational Base+0x4CC
Operational Base+0x4EC

Loop Counter 0 Registers

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	Loop counter 0 iterations

DMAC_LC1_0~DMAC_LC1_7

Address:Operational Base+0x410
Operational Base+0x430
Operational Base+0x450
Operational Base+0x470
Operational Base+0x490
Operational Base+0x4B0
Operational Base+0x4D0
Operational Base+0x4F0

Loop Counter 1 Registers

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	Loop counter 1 iterations

DMAC_DBGSTATUS

Address: Operational Base + offset (0x0d00)

Debug Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RO	0x0	The debug encoding is as follows: b00 = execute the instruction that the DBGINST [1:0] Registers contain b01 = reserved b10 = reserved b11 = reserved.

DMAC_DBGCMD

Address: Operational Base + offset (0x0d04)

Debug Command Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	WO	0x0	The debug encoding is as follows: b00 = execute the instruction that the DBGINST [1:0] Registers contain b01 = reserved b10 = reserved b11 = reserved

DMAC_DBGINST0

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Address: Operational Base + offset (0x0d08)
Debug Instruction-0 Register

Bit	Attr	Reset Value	Description
31:24	WO	0x00	Instruction byte 1
23:16	WO	0x00	Instruction byte 0
15:11	RO	0x0	reserved
10:8	WO	0x0	DMA channel number: b000 = DMA channel 0 b001 = DMA channel 1 b010 = DMA channel 2 … b111 = DMA channel 7
7:1	RO	0x0	reserved
0	WO	0x0	The debug thread encoding is as follows: 0 = DMA manager thread 1 = DMA channel.

DMAC_DBGINST1
Address: Operational Base + offset (0x0d0c)
Debug Instruction-1 Register

Bit	Attr	Reset Value	Description
31:24	WO	0x00	Instruction byte 5
23:16	WO	0x00	Instruction byte 4
15:8	WO	0x00	Instruction byte 3
7:0	WO	0x00	Instruction byte 2

DMAC_CR0
Address: Operational Base + offset (0x0e00)
Configuration Register 0

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:17	RO	0x02	Number of interrupt outputs that the DMAC provides: b00000 = 1 interrupt output, irq[0] b00001 = 2 interrupt outputs, irq[1:0] b00010 = 3 interrupt outputs, irq[2:0] … b11111 = 32 interrupt outputs, irq[31:0].
16:12	RO	0x07	Number of peripheral request interfaces that the DMAC provides: b00000 = 1 peripheral request interface b00001 = 2 peripheral request interfaces b00010 = 3 peripheral request interfaces … b11111 = 32 peripheral request interfaces.
11:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:4	RO	0x5	Number of DMA channels that the DMAC supports: b000 = 1 DMA channel b001 = 2 DMA channels b010 = 3 DMA channels … b111 = 8 DMA channels.
3	RO	0x0	reserved
2	RO	0x0	Indicates the status of the boot_manager_ns signal when the DMAC exited from reset: 0 = boot_manager_ns was LOW 1 = boot_manager_ns was HIGH.
1	RO	0x0	Indicates the status of the boot_from_pc signal when the DMAC exited from reset: 0 = boot_from_pc was LOW 1 = boot_from_pc was HIGH
0	RO	0x1	Supports peripheral requests: 0 = the DMAC does not provide a peripheral request interface 1 = the DMAC provides the number of peripheral request interfaces that the num_periph_req field specifies.

DMAC_CR1
Address: Operational Base + offset (0x0e04)
Configuration Register 1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x5	[7:4] num_i-cache_lines Number of i-cache lines: b0000 = 1 i-cache line b0001 = 2 i-cache lines b0010 = 3 i-cache lines … b1111 = 16 i-cache lines.
3	RO	0x0	reserved
2:0	RO	0x7	The length of an i-cache line: b000-b001 = reserved b010 = 4 bytes b011 = 8 bytes b100 = 16 bytes b101 = 32 bytes b110-b111 = reserved

DMAC_CR2
Address: Operational Base + offset (0x0e08)
Configuration Register 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Provides the value of boot_addr[31:0] when the DMAC exited from reset

DMAC_CR3
Address: Operational Base + offset (0x0e0c)
Configuration Register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	Provides the security state of an event-interrupt resource: Bit [N] = 0 Assigns event<N> or irq[N] to the Secure state. Bit [N] = 1 Assigns event<N> or irq[N] to the Non-secure state.

DMAC_CR4
Address: Operational Base + offset (0x0e10)
Configuration Register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000006	Provides the security state of the peripheral request interfaces: Bit [N] = 0 Assigns peripheral request interface N to the Secure state. Bit [N] = 1 Assigns peripheral request interface N to the Non-secure state

DMAC_CRDn
Address: Operational Base + offset (0x0e14)
DMA Configuration Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RO	0x020	The number of lines that the data buffer contains: b000000000 = 1 line b000000001 = 2 lines … b111111111 = 1024 lines
19:16	RO	0x9	The depth of the read queue: b0000 = 1 line b0001 = 2 lines … b1111 = 16 lines.
15	RO	0x0	reserved
14:12	RO	0x4	Read issuing capability that programs the number of outstanding read transactions: b000 = 1 b001 = 2 … b111 = 8

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Bit	Attr	Reset Value	Description
11:8	RO	0x7	The depth of the write queue: b0000 = 1 line b0001 = 2 lines … b1111 = 16 lines.
7	RO	0x0	reserved
6:4	RO	0x3	Write issuing capability that programs the number of outstanding write transactions: b000 = 1 b001 = 2 … b111 = 8
3	RO	0x0	reserved
2:0	RO	0x3	The data bus width of the AXI interface: b000 = reserved b001 = reserved b010 = 32-bit b011 = 64-bit b100 = 128-bit b101-b111 = reserved.

DMAC_WD
Address: Operational Base + offset (0x0e80)
DMA Watchdog Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Controls how the DMAC responds when it detects a lock-up condition: 0 = the DMAC aborts all of the contributing DMA channels and sets irq_abort HIGH 1 = the DMAC sets irq_abort HIGH.

8.5 Timing Diagram

Following picture shows the relationship between dma_req and dma_ack.

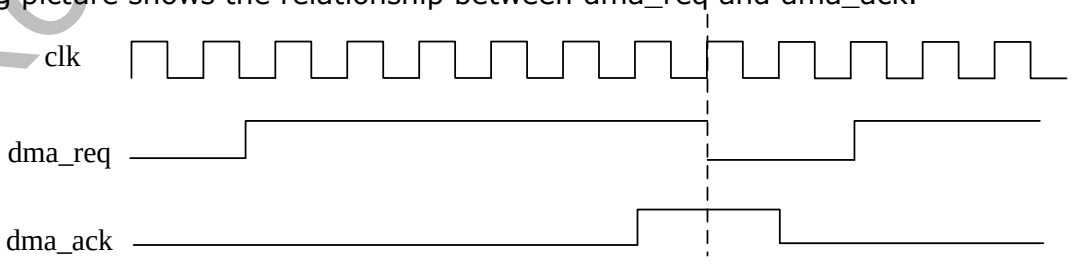

Fig. 8-3 DMAC request and acknowledge timing

8.6 Interface Description

DMAC has the following tie-off signals. It can be configured by SGRF register. (Please refer to the chapter to find how to configure)

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Table 8-2 DMAC boot interface

Interface	Reset value	Control source
boot_manager_ns	0x0	sgrf_dmac1_con5[15]
boot_irq_ns	0x0	sgrf_dmac1_con3[15:0]
boot_periph_ns	0x0	{sgrf_dmac1_con5[3:0],sgrf_dmac1_con4[15:0]}
grf_drtype_uart0_tx	0x1	sgrf_dmac1_con0[1:0]
grf_drtype_uart0_rx	0x1	sgrf_dmac1_con0[3:2]
grf_drtype_uart1_tx	0x1	sgrf_dmac1_con0[5:4]
grf_drtype_uart1_rx	0x1	sgrf_dmac1_con0[7:6]
grf_drtype_uart2_tx	0x1	sgrf_dmac1_con0[9:8]
grf_drtype_uart2_rx	0x1	sgrf_dmac1_con0[11:10]
grf_drtype_spi0_tx	0x1	sgrf_dmac1_con0[13:12]
grf_drtype_spi0_rx	0x1	sgrf_dmac1_con0[15:14]
grf_drtype_i2s0_8ch_tx	0x1	sgrf_dmac1_con1[1:0]
grf_drtype_i2s0_8ch_rx	0x1	sgrf_dmac1_con1[3:2]
grf_drtype_i2s1_8ch_tx	0x1	sgrf_dmac1_con1[5:4]
grf_drtype_i2s1_8ch_rx	0x1	sgrf_dmac1_con1[7:6]
grf_drtype_i2s2_2ch_tx	0x1	sgrf_dmac1_con1[9:8]
grf_drtype_i2s2_2ch_rx	0x1	sgrf_dmac1_con1[11:10]
grf_drtype_spdif	0x1	sgrf_dmac1_con1[13:12]
grf_drtype_pwm	0x1	sgrf_dmac1_con1[15:14]
grf_drtype_pdm	0x1	sgrf_dmac1_con2[1:0]

boot_manager_ns
When the DMAC exits from reset, this signal controls the security state of the DMA manager thread:
0 = assigns DMA manager to the Secure state
1 = assigns DMA manager to the Non-secure state.
boot_irq_ns
Controls the security state of an event-interrupt resource, when the DMAC exits from reset:
boot_irq_ns[x] is LOW
The DMAC assigns event<x> or irq[x] to the Secure state.
boot_irq_ns[x] is HIGH
The DMAC assigns event<x> or irq[x] to the Non-secure state.
boot_periph_ns
Controls the security state of a peripheral request interface, when the DMAC exits from reset:
boot_periph_ns[x] is LOW
The DMAC assigns peripheral request interface x to the Secure state.
boot_periph_ns[x] is HIGH
The DMAC assigns peripheral request interface x to the Non-secure state.
grf_drtype_<x>
The DMAC sets the state of the request_type flag:
grf_drtype_<x>[1:0]=b00: request_type<x> = Single.
grf_drtype_<x>[1:0]=b01: request_type<x> = Burst.

8.7 Application Notes

8.7.1 Using the APB slave interfaces

You must ensure that you use the appropriate APB interface, depending on the security state

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in which the boot_manager_ns initializes the DMAC to operate. For example, if the DMAC is in the secure state, you must issue the instruction using the secure APB interface, otherwise the DMAC ignores the instruction. You can use the secure APB interface, or the non-secure APB interface, to start or restart a DMA channel when the DMAC is in the Non-secure state.
The necessary steps to start a DMA channel thread using the debug instruction registers as following:
1. Create a program for the DMA channel.
2. Store the program in a region of system memory.
3. Poll the DBGSTATUS Register to ensure that debug is idle, that is, the dbgstatus bit is 0.
4. Write to the DBGINST0 Register and enter the:
- Instruction byte 0 encoding for DMAGO.
- Instruction byte 1 encoding for DMAGO.
- Debug thread bit to 0. This selects the DMA manager thread.

5. Write to the DBGINST1 Register with the DMAGO instruction byte [5:2] data, see Debug Instruction-1 Register o. You must set these four bytes to the address of the first instruction in the program, that was written to system memory in step 2.
6. Writing zero to the DBGCMD Register. The DMAC starts the DMA channel thread and sets the dbgstatus bit to 1.

8.7.2 Security usage

DMA manager thread is in the secure state
If the DNS bit is 0, the DMA manager thread operates in the secure state and it only performs secure instruction fetches. When a DMA manager thread in the secure state processes:
DMAGO
It uses the status of the ns bit, to set the security state of the DMA channel thread by writing to the CNS bit for that channel.
DMAWFE
It halts execution of the thread until the event occurs. When the event occurs, the DMAC continues execution of the thread, irrespective of the security state of the corresponding INS bit.
DMASEV
It sets the corresponding bit in the INT_EVENT_RIS Register, irrespective of the security state of the corresponding INS bit.
DMA manager thread is in the Non-secure state
If the DNS bit is 1, the DMA manager thread operates in the Non-secure state, and it only performs non-secure instruction fetches. When a DMA manager thread in the Non-secure state processes:
DMAGO
The DMAC uses the status of the ns bit, to control if it starts a DMA channel thread. If:
ns = 0
The DMAC does not start a DMA channel thread and instead it:
1. Executes a NOP.
2. Sets the FSRD Register, see Fault Status DMA Manager
3. Sets the dmago_err bit in the FTRD Register, see Fault Type DMA Manager Register.
4. Moves the DMA manager to the Faulting state.

ns = 1
The DMAC starts a DMA channel thread in the Non-secure state and programs the CNS bit to be non-secure.
DMAWFE
The DMAC uses the status of the corresponding INS bit, in the CR3 Register, to control if it waits for the event. If:
INS = 0
The event is in the Secure state. The DMAC:
1. Executes a NOP.
2. Sets the FSRD Register, see Fault Status DMA Manager Register.
3. Sets the mgr_evnt_err bit in the FTRD Register, see Fault Type DMA Manager Register.
4. Moves the DMA manager to the Faulting state.

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INS = 1
The event is in the Non-secure state. The DMAC halts execution of the thread and waits for the event to occur.

DMASEV

The DMAC uses the status of the corresponding INS bit, in the CR3Register, to control if it creates the event-interrupt. If:
INS = 0
The event-interrupt resource is in the secure state. The DMAC:
1. Executes a NOP.
2. Sets the FSRD Register, see Fault Status DMA Manager Register.
3. Sets the mgr_evnt_err bit in the FTRD Register, see Fault Type DMA Manager Register.
4. Moves the DMA manager to the Faulting state.

INS = 1
The event-interrupt resource is in the Non-secure state. The DMAC creates the event-interrupt.

DMA channel thread is in the secure state

When the CNS bit is 0, the DMA channel thread is programmed to operate in the Secure state and it only performs secure instruction fetches.
When a DMA channel thread in the secure state processes the following instructions:

DMAWFE

The DMAC halts execution of the thread until the event occurs. When the event occurs, the DMAC continues execution of the thread, irrespective of the security state of the corresponding INS bit, in the CR3 Register.

DMASEV

The DMAC creates the event-interrupt, irrespective of the security state of the corresponding INS bit, in the CR3 Register.

DMAWFP

The DMAC halts execution of the thread until the peripheral signals a DMA request. When this occurs, the DMAC continues execution of the thread, irrespective of the security state of the corresponding PNS bit, in the CR4 Register.

DMALDP, DMASTP

The DMAC sends a message to the peripheral to communicate that data transfer is complete, irrespective of the security state of the corresponding PNS bit, in the CR4 Register.

DMAFLUSHP

The DMAC clears the state of the peripheral and sends a message to the peripheral to resend its level status, irrespective of the security state of the corresponding PNS bit, in the CR4 Register.
When a DMA channel thread is in the Secure state, it enables the DMAC to perform secure and non-secure AXI accesses

DMA channel thread is in the Non-secure state

When the CNS bit is 1, the DMA channel thread is programmed to operate in the Non-secure state and it only performs non-secure instruction fetches.
When a DMA channel thread in the Non-secure state processes the following instructions:

DMAWFE

The DMAC uses the status of the corresponding INS bit, in the CR3 Register, to control if it waits for the event. If:
INS = 0
The event is in the Secure state. The DMAC:
1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_evnt_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.

INS = 1
The event is in the Non-secure state. The DMAC halts execution of the thread and waits for the event to occur.

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DMASEV

The DMAC uses the status of the corresponding INS bit, in the CR3 Register, to control if it creates the event. If:

INS = 0

The event-interrupt resource is in the Secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_evnt_err bit in the FTRn Register, see Fault Type DMA Channel Registers .
4. Moves the DMA channel to the Faulting completing state.

INS = 1

The event-interrupt resource is in the Non-secure state. The DMAC creates the event-interrupt.

DMAWFP

The DMAC uses the status of the corresponding PNS bit, in the CR4 Register, to control if it waits for the peripheral to signal a request. If:

PNS = 0

The peripheral is in the Secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_periph_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.

PNS = 1

The peripheral is in the Non-secure state. The DMAC halts execution of the thread and waits for the peripheral to signal a request.

DMALDP, DMASTP

The DMAC uses the status of the corresponding PNS bit, in the CR4 Register, to control if it sends an acknowledgement to the peripheral. If:

PNS = 0

The peripheral is in the secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_periph_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.

PNS = 1

The peripheral is in the Non-secure state. The DMAC sends a message to the peripheral to communicate when the data transfer is complete.

DMAFLUSHP

The DMAC uses the status of the corresponding PNS bit, in the CR4 Register, to control if it sends a flush request to the peripheral. If:

PNS = 0

The peripheral is in the secure state. The DMAC:

1. Executes a NOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.
3. Sets the ch_periph_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel to the Faulting completing state.

PNS = 1

The peripheral is in the Non-secure state. The DMAC clears the state of the peripheral and sends a message to the peripheral to resend its level status.

When a DMA channel thread is in the Non-secure state, and a DMAMOV CCR instruction attempts to program the channel to perform a secure AXI transaction, the DMAC:

1. Executes a DMANOP.
2. Sets the appropriate bit in the FSRC Register that corresponds to the DMA channel number. See Fault Status DMA Channel Register.

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3. Sets the ch_rdwr_err bit in the FTRn Register, see Fault Type DMA Channel Registers.
4. Moves the DMA channel thread to the Faulting completing state.

8.7.3 Programming restrictions

Fixed unaligned bursts

The DMAC does not support fixed unaligned bursts. If you program the following conditions, the DMAC treats this as a programming error:

Unaligned read

- src_inc field is 0 in the CCRn Register
- the SARn Register contains an address that is not aligned to the size of data that the src_burst_size field contain

Unaligned write

- dst_inc field is 0 in the CCRn Register
- the DARn Register contains an address that is not aligned to the size of data that the dst_burst_size field contains

Endian swap size restrictions

If you program the endian_swap_size field in the CCRn Register, to enable a DMA channel to perform an endian swap then you must set the corresponding SARn Register and the corresponding DARn Register to contain an address that is aligned to the value that the endian_swap_size field contains.

Updating DMA channel control registers during a DMA cycle restrictions

Prior to the DMAC executing a sequence of DMALD and DMAST instructions, the values you program in to the CCRn Register, SARn Register, and DARn Register control the data byte lane manipulation that the DMAC performs when it transfers the data from the source address to the destination address. You'd better not update these registers during a DMA cycle.

Resource sharing between DMA channels

DMA channel programs share the MFIFO data storage resource. You must not start a set of concurrently running DMA channel programs with a resource requirement that exceeds the configured size of the MFIFO. If you exceed this limit then the DMAC might lock up and generate a Watchdog abort.

8.7.4 Unaligned transfers may be corrupted

For a configuration with more than one channel, if any of channels 1 to 7 is performing transfers between certain types of misaligned source and destination addresses, then the output data may be corrupted by the action of channel 0.

Data corruption might occur if all of the following are true:

1. Two beats of AXI read data are received for one of channels 1 to 7.
2. Source and destination address alignments mean that each read data beat is splited across two lines in the data buffer (see Splitting data, below).
3. There is one idle cycle between the two read data beats.
4. Channel 0 performs an operation that updates channel control information during this idle cycle (see Updates to channel control information, below)

Splitting data

Depending upon the programmed values for the DMA transfer, one beat of read data from the AXI interface need to be splited across two lines in the internal data buffer. This occurs when the read data beat contains data bytes which will be written to addresses that wrap around at the AXI interface data width, so that these bytes could not be transferred by a single AXI write data beat of the full interface width.

Most applications of DMA-330 do not split data in this way, so are NOT vulnerable to data corruption from this defect.

The following cases are NOT vulnerable to data corruption because they do not split data:

- Byte lane offset between source and destination addresses is 0 when source and destination addresses have the same byte lane alignment, the offset is 0 and a wrap operation that splits data cannot occur.
- Byte lane offset between source and destination addresses is a multiple of source size

Table 8-3 Source size in CCRn

Source size in CCRn	Allowed offset between SARn and DARn
SS8	any offset allowed.

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SS16	0,2,4,6,8,10,12,14
SS32	0,4,8,12
SS64	0,8

8.7.5 Interrupt shares between channel

As the DMAC does not record which channel (or list of channels) have asserted an interrupt. So it will depend on your program and whether any of the visible information for that program can be used to determine progress, and help identify the interrupt source.
There are 4 likely information sources that can be used to determine the progress made by a program:
- Program counter (PC)
- Source address
- Destination address
- Loop counters (LC)

For example, a program might emit an interrupt each time that it iterates around a loop. In this case, the interrupt service routine (ISR) would need to store the loop value of each channel when it is called, and then compare against the new value when it is next called. A change in value would indicate that the program has progressed.
The ISR must be carefully written to ensure that no interrupts are lost. The sequence of operations is as follows:
1. Disable interrupts
2. Immediately clear the interrupt in DMA-330
3. Check the relevant registers for both channels to determine which must be serviced
4. Take appropriate action for the channels
5. Re-enable interrupts and exit ISR

8.7.6 Instruction sets

Table 8-4 DMAC Instruction sets

Mnemonic	Instruction	Thread usage
DMAADDH	Add Halfword	C
DMAEND	End	M/C
DMAFLUSHP	Flush and notify Peripheral	C
DMAGO	Go	M
DMAKILL	Kill	C
DMALD	Load	C
DMALDP	Load Peripheral	C
DMALP	Loop	C
DMALPEND	Loop End	C
DMALPFE	Loop Forever	C
DMAMOV	Move	C
DMANOP	No operation	M/C
DMARMB	Read Memory Barrier	C
DMASEV	Send Event	M/C
DMAST	Store	C
DMASTP	Store and notify Peripheral	C
DMASTZ	Store Zero	C
DMAWFE	Wait For Event M	M/C
DMAWFP	Wait For Peripheral	C
DMAWMB	Write Memory Barrier	C
DMAADNH	Add Negative Halfword	C

Notes: Thread usage: C=DMA channel, M=DMA manager

8.7.7 Assembler directives

In this document, only DMMADNH instruction is took as an example to show the way the instruction assembled. For the other instructions, please refer to pl330_trm.pdf.

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DMAADNH

Add Negative Halfword adds an immediate negative 16-bit value to the SARn Register or DARn Register, for the DMA channel thread. This enables the DMAC to support 2D DMA operations, or reading or writing an area of memory in a different order to naturally incrementing addresses. See Source Address Registers and Destination Address Registers.

The immediate unsigned 16-bit value is one-extended to 32 bits, to create a value that is the two's complement representation of a negative number between -65536 and -1, before the DMAC adds it to the address using 32-bit addition. The DMAC discards the carry bit so that addresses wrap from 0xFFFFFFFF to 0x00000000. The net effect is to subtract between 65536 and 1 from the current value in the Source or Destination Address Register.

Following table shows the instruction encoding.

Table 8-5 DMAC instruction encoding

Imm[15:8]	Imm[7:0]	0	1	0	1	1	1	ra	0

Assembler syntax

DMAADNH <address_register>, <16-bit immediate>

where:

<address_register>

Selects the address register to use. It must be either:

SAR

SARn Register and sets ra to 0.

DAR

DARn Register and sets ra to 1.

<16-bit immediate>

The immediate value to be added to the <address_register>.

You should specify the 16-bit immediate as the number that is to be represented in the instruction encoding. For example, DMAADNH DAR, 0xFFF0 causes the value 0xFFFFFFF0 to be added to the current value of the Destination Address Register, effectively subtracting 16 from the DAR.

You can only use this instruction in a DMA channel thread.

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Chapter 9 Generic Interrupt Controller (GIC)

13

13.1 Overview

There is a generic interrupt controller(GIC400) in RK3228H which generates physical interrupts to Cortex-A53. It has two interfaces, the distributor interface connects to the interrupt source, and the CPU interface connects to Cortex-A53. The details of CPU interface connectivity are shown in the following table.

Table 1-1 CPU interface connectivity

CPU Interface Number	Connectivity
CPU interface 0	CPU0
CPU interface 1	CPU1
CPU interface 2	CPU2
CPU interface 3	CPU3

It supports the following features:

- Supports 128 hardware interrupt inputs
- Masking of any interrupts
- Prioritization of interrupts
- Distribution of the interrupts to the target Cortex-A53 processor(s)
- Generation of interrupts by software
- Supports Security Extensions

13.2 Block Diagram

The generic interrupt controller comprises with:

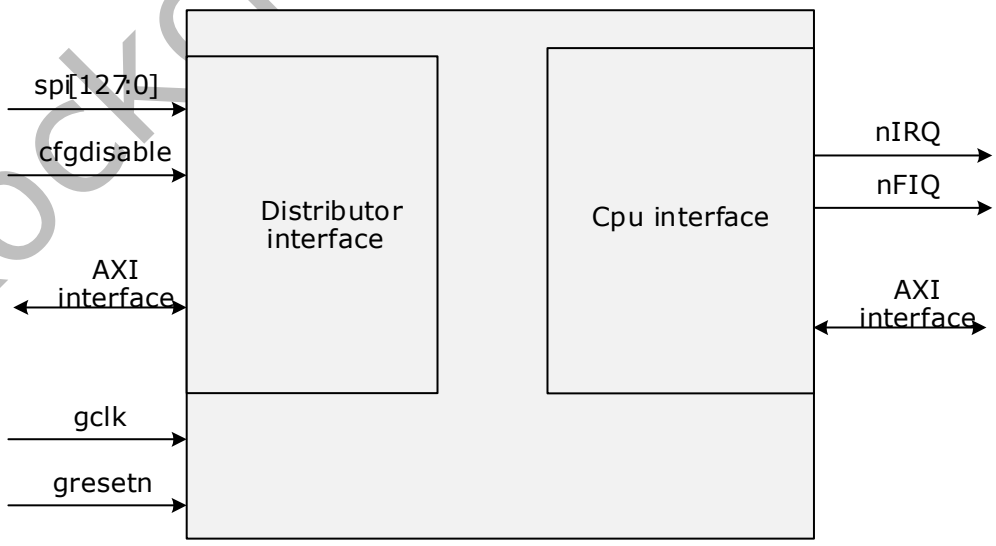

Fig.1-1 Block Diagram

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13.3 Function Description

Please refer to the document IHI0048B_gic_architecture_specification.pdf for the detail function description.

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Chapter 10 DMC (Dynamic Memory Interface)

10

10.1 Overview

The DMC includes DDR protocol controller (PCTL) and DDR PHY which are a complete memory interface solution for DDR memory subsystems.

The PCTL SoC application bus interface supports AXI interface, with a flexible address mapper logic allow application-specific mapping of row, column, bank, bank group and rank bits to achieve industry leading high-efficiency, low-latency and high-performance from memory interface.

The DDR PHY provides control features to ease the customer implementation of digitally controlled features of the PHY such as initialization, DQS gate training, write leveling training and programmable configuration controls.

The DMC supports the following features:

- Support DDR3/DDR4/LPDDR3
- Support up to 2 ranks and up to 4GB capacity
- Support 32-bit and 16-bit DDR data bus width
- Support up to 21-type address mapping
- Support up to 32-bank (including bank group)
- Support DDR burst8 only
- Support different CL/WL latency
- Support DDR3/DDR4/LPDDR3 SW frequency change and SW fast frequency change
- Support DDR4 DBI, PDA
- Support auto gated clock through DDRC and AXI low power interface
- Support auto put DDR PHY entry or exit self-refresh by DFI lower power interface
- Support auto or SW issue entry or exit clock stop/power-down/self-refresh/deep power-down/max power saving mode
- Support SW or PMU auto let DDR PHY entry or exit retention/self-refresh
- Support open, close, intelligent pre-charge paging policy
- Support Advance refresh control
- Support APB interface for PCTL and PHY software-accessible registers
- Support automatic DQS gate training and automatic write leveling training
- Support TX DLL and RX DLL adjustment
- Support Internal PLL with SSC or external PLL

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10.2 Block Diagram

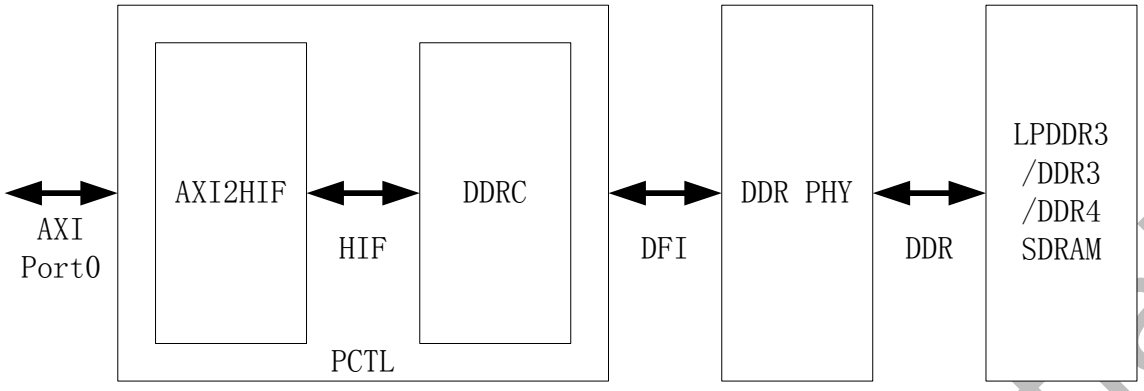

Fig. 10-1DMC Block Diagram

10.3 Function Description

10.3.1 PCTL

PCTL supports only one AXI Port0, it receives AXI transactions from memory schedule of interconnect. These transactions are queued internally and scheduled for access in order to the SDRAM while satisfying SDRAM protocol timing requirements. It in turn issues commands on the DFI interface to the DDR PHY block which launches and captures data to and from the SDRAM. PCTL contains the following main components:

- AXI2HIF block: This block provides the AXI interface to system level and HIF interface to DDRC block. It provides bus protocol handing, data buffering, data bus size conversion and memory burst alignment. Read is stored in a SRAM, read re-order buffer and return in order to the AXI Port.
- DDRC block: This block issues the read/write commands in order, carries out the DRAM page management, issues DRAM maintenance commands, and implement the DFI interface. Write data is stored in an SRAM until its associated command is issued to the PHY. Read data is handled by the response engine in the DDRC and is returned in order on the HIF.

10.3.2 DDR PHY

DDR PHY supports DDR3/DDR4/LPDDR2/LPDDR3 SDRAM and provides turnkey physical interface solutions for ICs requiring access to JEDEC compatible SDRAM devices. It is optimized for low power and high speed (up to 1066Mbps for DDR2/LPDDR2 and up to 1866Mbps for DDR3/LPDDR3/DDR4) applications with robust timing and small silicon area in 28nm process. It supports all JEDEC DDR3/DDR4/LPDDR2/LPDDR3 SDRAM components in the market. The PHY components contain DDR specialized functional and utility SSTL and HSUL I/Os up to 1866MHz in 28nm, critical timing synchronization module (TSM) and a low power/jitter DLLs with programmable fine-grain control for any SDRAM interface.

10.4 Register Description

Slave address can be divided into different length for different usage, which is shown as follows.

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10.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
DDRC_MSTR	0x0000	W	0x03040001	MSTR
DDRC_STAT	0x0004	W	0x00000000	STAT
DDRC_MRCTRL0	0x0010	W	0x00000000	MRCTRL0
DDRC_MRCTRL1	0x0014	W	0x00000000	MRCTRL1
DDRC_MRSTAT	0x0018	W	0x00000000	MRSTAT
DDRC_MRCTRL2	0x001c	W	0x00000000	MRCTRL2
DDRC_DERATEEN	0x0020	W	0x00000000	DERATEEN
DDRC_DERATEINT	0x0024	W	0x00800000	DERATEINT
DDRC_PWRCTL	0x0030	W	0x00000000	PWRCTL
DDRC_PWRTMG	0x0034	W	0x00402010	PWRTMG
DDRC_HWLPCTL	0x0038	W	0x00000003	HWLPCTL
DDRC_HWFFCCTL	0x003c	W	0x00000010	HWFFCCTL
DDRC_HWFFCSTAT	0x0040	W	0x00000000	HWFFCSTAT
DDRC_RFSHCTL0	0x0050	W	0x00210000	RFSHCTL0
DDRC_RFSHCTL1	0x0054	W	0x00000000	RFSHCTL1
DDRC_RFSHCTL3	0x0060	W	0x00000000	RFSHCTL3
DDRC_RFSHTMG	0x0064	W	0x0062008c	RFSHTMG
DDRC_CRCPARCTL0	0x00c0	W	0x00000000	CRCPARCTL0
DDRC_CRCPARCTL1	0x00c4	W	0x00001000	CRCPARCTL1
DDRC_CRCPARSTAT	0x00cc	W	0x00000000	CRCPARSTAT
DDRC_INIT0	0x00d0	W	0x0002004e	INIT0
DDRC_INIT1	0x00d4	W	0x00000000	INIT1
DDRC_INIT2	0x00d8	W	0x00000d05	INIT2
DDRC_INIT3	0x00dc	W	0x00000510	INIT3
DDRC_INIT4	0x00e0	W	0x00000000	INIT4
DDRC_INIT5	0x00e4	W	0x00100004	INIT5
DDRC_INIT6	0x00e8	W	0x00000000	INIT6
DDRC_INIT7	0x00ec	W	0x00000000	INIT7
DDRC_DIMMCTL	0x00f0	W	0x00000000	DIMMCTL
DDRC_RANKCTL	0x00f4	W	0x0000066f	RANKCTL
DDRC_DRAMTMG0	0x0100	W	0x0f101b0f	DRAMTMG0
DDRC_DRAMTMG1	0x0104	W	0x00080414	DRAMTMG1
DDRC_DRAMTMG2	0x0108	W	0x0305060d	DRAMTMG2
DDRC_DRAMTMG3	0x010c	W	0x0050400c	DRAMTMG3
DDRC_DRAMTMG4	0x0110	W	0x05040405	DRAMTMG4
DDRC_DRAMTMG5	0x0114	W	0x05050403	DRAMTMG5
DDRC_DRAMTMG6	0x0118	W	0x02020005	DRAMTMG6
DDRC_DRAMTMG7	0x011c	W	0x00000202	DRAMTMG7
DDRC_DRAMTMG8	0x0120	W	0x03034405	DRAMTMG8
DDRC_DRAMTMG9	0x0124	W	0x0004040d	DRAMTMG9
DDRC_DRAMTMG10	0x0128	W	0x001c180a	DRAMTMG10

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Name	Offset	Size	Reset Value	Description
DDRC_DRAMTMG11	0x012c	W	0x440c021c	DRAMTMG11
DDRC_DRAMTMG12	0x0130	W	0x00020010	DRAMTMG12
DDRC_DRAMTMG13	0x0134	W	0x1c200004	DRAMTMG13
DDRC_DRAMTMG14	0x0138	W	0x000000a0	DRAMTMG14
DDRC_DRAMTMG15	0x013c	W	0x00000000	DRAMTMG15
DDRC_DRAMTMG17	0x0144	W	0x00000000	DRAMTMG17
DDRC_ZQCTL0	0x0180	W	0x02000040	ZQCTL0
DDRC_ZQCTL1	0x0184	W	0x02000100	ZQCTL1
DDRC_ZQCTL2	0x0188	W	0x00000000	ZQCTL2
DDRC_ZQSTAT	0x018c	W	0x00000000	ZQSTAT
DDRC_DFITMG0	0x0190	W	0x07020002	DFITMG0
DDRC_DFITMG1	0x0194	W	0x00000404	DFITMG1
DDRC_DFILPCFG0	0x0198	W	0x07000000	DFILPCFG0
DDRC_DFILPCFG1	0x019c	W	0x00000000	DFILPCFG1
DDRC_DFIUPD0	0x01a0	W	0x00400003	DFIUPD0
DDRC_DFIUPD1	0x01a4	W	0x00000001	DFIUPD1
DDRC_DFIUPD2	0x01a8	W	0x00000001	DFIUPD2
DDRC_DFIMISC	0x01b0	W	0x00000001	DFIMISC
DDRC_DFITMG2	0x01b4	W	0x00000202	DFITMG2
DDRC_DFITMG3	0x01b8	W	0x00000000	DFITMG3
DDRC_DFISTAT	0x01bc	W	0x00000000	DFISTAT
DDRC_DBICTL	0x01c0	W	0x00000001	DBICTL
DDRC_DFIPHYMSTR	0x01c4	W	0x00000001	DFIPHYMSTR
DDRC_ADDRMAP0	0x0200	W	0x00000000	ADDRMAP0
DDRC_ADDRMAP1	0x0204	W	0x00000000	ADDRMAP1
DDRC_ADDRMAP2	0x0208	W	0x00000000	ADDRMAP2
DDRC_ADDRMAP3	0x020c	W	0x00000000	ADDRMAP3
DDRC_ADDRMAP4	0x0210	W	0x00000000	ADDRMAP4
DDRC_ADDRMAP5	0x0214	W	0x00000000	ADDRMAP5
DDRC_ADDRMAP6	0x0218	W	0x00000000	ADDRMAP6
DDRC_ADDRMAP7	0x021c	W	0x00000000	ADDRMAP7
DDRC_ADDRMAP8	0x0220	W	0x00000000	ADDRMAP8
DDRC_ADDRMAP9	0x0224	W	0x00000000	ADDRMAP9
DDRC_ADDRMAP10	0x0228	W	0x00000000	ADDRMAP10
DDRC_ADDRMAP11	0x022c	W	0x00000000	ADDRMAP11
DDRC_ODTCFG	0x0240	W	0x04000400	ODTCFG
DDRC_ODTMAP	0x0244	W	0x00002211	ODTMAP
DDRC_SCHED	0x0250	W	0x00000804	SCHED
DDRC_SCHED1	0x0254	W	0x00000000	SCHED1
DDRC_PERFLPR1	0x0264	W	0x0f00007f	PERFLPR1
DDRC_PERFWR1	0x026c	W	0x0f00007f	PERFWR1
DDRC_DBG0	0x0300	W	0x00000000	DBG0
DDRC_DBG1	0x0304	W	0x00000000	DBG1

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Name	Offset	Size	Reset Value	Description
DDRC_DBGCAM	0x0308	W	0x00000000	DBGCAM
DDRC_DBGCMD	0x030c	W	0x00000000	DBGCMD
DDRC_DBGSTAT	0x0310	W	0x00000000	DBGSTAT
DDRC_SWCTL	0x0320	W	0x00000001	SWCTL
DDRC_SWSTAT	0x0324	W	0x00000001	SWSTAT
DDRC_POISONCFG	0x036c	W	0x00110011	POISONCFG
DDRC_POISONSTAT	0x0370	W	0x00000000	POISONSTAT
DDRC_PSTAT	0x03fc	W	0x00000000	PSTAT
DDRC_PCCFG	0x0400	W	0x00000000	PCCFG
DDRC_PCFGR_0	0x0404	W	0x00000000	PCFGR_0
DDRC_PCFGW_0	0x0408	W	0x00004000	PCFGW_0
DDRC_PCTRL_0	0x0490	W	0x00000000	PCTRL_0
DDRC_FREQ1_DERATEEN	0x2020	W	0x00000000	DERATEEN
DDRC_FREQ1_DERATEINT	0x2024	W	0x00800000	DERATEINT
DDRC_FREQ1_RFSHCTL0	0x2050	W	0x00210000	RFSHCTL0
DDRC_FREQ1_RFSHTMG	0x2064	W	0x0062008c	RFSHTMG
DDRC_FREQ1_INIT3	0x20dc	W	0x00000510	INIT3
DDRC_FREQ1_INIT4	0x20e0	W	0x00000000	INIT4
DDRC_FREQ1_INIT6	0x20e8	W	0x00000000	INIT6
DDRC_FREQ1_INIT7	0x20ec	W	0x00000000	INIT7
DDRC_FREQ1_DRAMTMG0	0x2100	W	0x0f101b0f	DRAMTMG0
DDRC_FREQ1_DRAMTMG1	0x2104	W	0x00080414	DRAMTMG1
DDRC_FREQ1_DRAMTMG2	0x2108	W	0x0305060d	DRAMTMG2
DDRC_FREQ1_DRAMTMG3	0x210c	W	0x0050400c	DRAMTMG3
DDRC_FREQ1_DRAMTMG4	0x2110	W	0x05040405	DRAMTMG4
DDRC_FREQ1_DRAMTMG5	0x2114	W	0x05050403	DRAMTMG5
DDRC_FREQ1_DRAMTMG6	0x2118	W	0x02020005	DRAMTMG6
DDRC_FREQ1_DRAMTMG7	0x211c	W	0x00000202	DRAMTMG7
DDRC_FREQ1_DRAMTMG8	0x2120	W	0x03034405	DRAMTMG8
DDRC_FREQ1_DRAMTMG9	0x2124	W	0x0004040d	DRAMTMG9
DDRC_FREQ1_DRAMTMG10	0x2128	W	0x001c180a	DRAMTMG10
DDRC_FREQ1_DRAMTMG11	0x212c	W	0x440c021c	DRAMTMG11
DDRC_FREQ1_DRAMTMG12	0x2130	W	0x00020010	DRAMTMG12
DDRC_FREQ1_DRAMTMG13	0x2134	W	0x1c200004	DRAMTMG13
DDRC_FREQ1_DRAMTMG14	0x2138	W	0x000000a0	DRAMTMG14
DDRC_FREQ1_DRAMTMG15	0x213c	W	0x00000000	DRAMTMG15
DDRC_FREQ1_DRAMTMG17	0x2144	W	0x00000000	DRAMTMG17
DDRC_FREQ1_ZQCTL0	0x2180	W	0x02000040	ZQCTL0
DDRC_FREQ1_DFITMG0	0x2190	W	0x07020002	DFITMG0
DDRC_FREQ1_DFITMG1	0x2194	W	0x00000404	DFITMG1
DDRC_FREQ1_DFITMG2	0x21b4	W	0x00000202	DFITMG2
DDRC_FREQ1_DFITMG3	0x21b8	W	0x00000000	DFITMG3
DDRC_FREQ1_ODTCFG	0x2240	W	0x04000400	ODTCFG

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*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

Name	Offset	Size	Reset Value	Description
DDRPHY_REG0	0x0000	W	0x0000000f	DDR PHY register 00
DDRPHY_REG1	0x0004	W	0x00000008	DDR PHY register 01
DDRPHY_REG2	0x0008	W	0x00000000	DDR PHY register 02
DDRPHY_REG3	0x000c	W	0x00000022	DDR PHY register 03
DDRPHY_REG4	0x0010	W	0x00000022	DDR PHY register 04
DDRPHY_REG5	0x0014	W	0x00000000	DDR PHY register 05
DDRPHY_REG6	0x0018	W	0x00000002	DDR PHY register 06
DDRPHY_REG9	0x0024	W	0x00000000	DDR PHY register 09
DDRPHY_REGA	0x0028	W	0x00000000	DDR PHY register 0A
DDRPHY_REGB	0x002c	W	0x00000000	DDR PHY register 0B
DDRPHY_REGC	0x0030	W	0x00000000	DDR PHY register 0C
DDRPHY_REG11	0x0044	W	0x00000014	DDR PHY register 11
DDRPHY_REG12	0x0048	W	0x000000a2	DDR PHY register 12
DDRPHY_REG13	0x004c	W	0x0000000c	DDR PHY register 13
DDRPHY_REG14	0x0050	W	0x00000008	DDR PHY register 13
DDRPHY_REG16	0x0058	W	0x00000014	DDR PHY register 16
DDRPHY_REG17	0x005c	W	0x00000044	DDR PHY register 17
DDRPHY_REG18	0x0060	W	0x00000014	DDR PHY register 17
DDRPHY_REG20	0x0080	W	0x00000014	DDR PHY register 20
DDRPHY_REG21	0x0084	W	0x00000004	DDR PHY register 21
DDRPHY_REG26	0x0098	W	0x0000000c	DDR PHY register 26
DDRPHY_REG27	0x009c	W	0x00000000	DDR PHY register 27
DDRPHY_REG28	0x00a0	W	0x00000001	DDR PHY register 28
DDRPHY_REG29	0x00a4	W	0x00000002	DDR PHY register 29
DDRPHY_REG2B	0x00ac	W	0x00000081	DDR PHY register 2B
DDRPHY_REG2E	0x00b8	W	0x00000004	DDR PHY register 2E
DDRPHY_REG2F	0x00bc	W	0x00000014	DDR PHY register 2F
DDRPHY_REG30	0x00c0	W	0x00000014	DDR PHY register 30
DDRPHY_REG31	0x00c4	W	0x00000004	DDR PHY register 31
DDRPHY_REG36	0x00d8	W	0x00000004	DDR PHY register 36
DDRPHY_REG37	0x00dc	W	0x00000000	DDR PHY register 37
DDRPHY_REG38	0x00e0	W	0x00000001	DDR PHY register 38
DDRPHY_REG39	0x00e4	W	0x00000002	DDR PHY register 39
DDRPHY_REG3B	0x00ec	W	0x00000081	DDR PHY register 3B
DDRPHY_REG3E	0x00f8	W	0x00000004	DDR PHY register 3E
DDRPHY_REG3F	0x00fc	W	0x00000014	DDR PHY register 3F
DDRPHY_REG40	0x0100	W	0x00000014	DDR PHY register 40
DDRPHY_REG41	0x0104	W	0x00000004	DDR PHY register 41
DDRPHY_REG46	0x0118	W	0x0000000c	DDR PHY register 46

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Name	Offset	Size	Reset Value	Description
DDRPHY_REG47	0x011c	W	0x00000000	DDR PHY register 47
DDRPHY_REG48	0x0120	W	0x00000001	DDR PHY register 48
DDRPHY_REG49	0x0124	W	0x00000002	DDR PHY register 49
DDRPHY_REG4B	0x012c	W	0x00000091	DDR PHY register 4B
DDRPHY_REG4E	0x0138	W	0x00000004	DDR PHY register 4E
DDRPHY_REG4F	0x013c	W	0x00000014	DDR PHY register 4F
DDRPHY_REG50	0x0140	W	0x00000014	DDR PHY register 50
DDRPHY_REG51	0x0144	W	0x00000004	DDR PHY register 51
DDRPHY_REG56	0x0158	W	0x0000000c	DDR PHY register 56
DDRPHY_REG57	0x015c	W	0x00000000	DDR PHY register 57
DDRPHY_REG58	0x0160	W	0x00000001	DDR PHY register 58
DDRPHY_REG59	0x0164	W	0x00000002	DDR PHY register 59
DDRPHY_REG5B	0x016c	W	0x00000091	DDR PHY register 5B
DDRPHY_REG5E	0x0178	W	0x00000004	DDR PHY register 5E
DDRPHY_REG5F	0x017c	W	0x00000014	DDR PHY register 5F
DDRPHY_REG70	0x01c0	W	0x00000077	DDR PHY register 70
DDRPHY_REG71	0x01c4	W	0x00000077	DDR PHY register 71
DDRPHY_REG72	0x01c8	W	0x00000077	DDR PHY register 72
DDRPHY_REG73	0x01cc	W	0x00000077	DDR PHY register 73
DDRPHY_REG74	0x01d0	W	0x00000077	DDR PHY register 74
DDRPHY_REG75	0x01d4	W	0x00000077	DDR PHY register 75
DDRPHY_REG76	0x01d8	W	0x00000077	DDR PHY register 76
DDRPHY_REG77	0x01dc	W	0x00000077	DDR PHY register 77
DDRPHY_REG78	0x01e0	W	0x00000077	DDR PHY register 78
DDRPHY_REG79	0x01e4	W	0x00000077	DDR PHY register 79
DDRPHY_REG7A	0x01e8	W	0x00000077	DDR PHY register 7A
DDRPHY_REG7B	0x01ec	W	0x00000077	DDR PHY register 7B
DDRPHY_REG7C	0x01f0	W	0x00000077	DDR PHY register 7C
DDRPHY_REG7D	0x01f4	W	0x00000077	DDR PHY register 7D
DDRPHY_REG7E	0x01f8	W	0x00000077	DDR PHY register 7E
DDRPHY_REG7F	0x01fc	W	0x00000077	DDR PHY register 7F
DDRPHY_REG80	0x0200	W	0x00000007	DDR PHY register 80
DDRPHY_REG81	0x0204	W	0x00000077	DDR PHY register 81
DDRPHY_REG82	0x0208	W	0x00000077	DDR PHY register 82
DDRPHY_REG83	0x020c	W	0x00000077	DDR PHY register 83
DDRPHY_REG84	0x0210	W	0x00000077	DDR PHY register 84
DDRPHY_REG85	0x0214	W	0x00000077	DDR PHY register 85
DDRPHY_REG86	0x0218	W	0x00000077	DDR PHY register 86
DDRPHY_REG87	0x021c	W	0x00000077	DDR PHY register 87
DDRPHY_REG88	0x0220	W	0x00000077	DDR PHY register 88
DDRPHY_REG89	0x0224	W	0x00000077	DDR PHY register 89
DDRPHY_REG8A	0x0228	W	0x00000077	DDR PHY register 8A
DDRPHY_REG8B	0x022c	W	0x00000077	DDR PHY register 8B

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Name	Offset	Size	Reset Value	Description
DDRPHY_REG8C	0x0230	W	0x00000077	DDR PHY register 8C
DDRPHY_REG8D	0x0234	W	0x00000077	DDR PHY register 8D
DDRPHY_REG8E	0x0238	W	0x00000077	DDR PHY register 8E
DDRPHY_REG8F	0x023c	W	0x00000077	DDR PHY register 8F
DDRPHY_REG90	0x0240	W	0x00000007	DDR PHY register 90
DDRPHY_REG91	0x0244	W	0x00000077	DDR PHY register 91
DDRPHY_REG92	0x0248	W	0x00000077	DDR PHY register 92
DDRPHY_REG93	0x024c	W	0x00000077	DDR PHY register 93
DDRPHY_REG94	0x0250	W	0x00000077	DDR PHY register 94
DDRPHY_REG95	0x0254	W	0x00000077	DDR PHY register 95
DDRPHY_REG96	0x0258	W	0x00000077	DDR PHY register 96
DDRPHY_REG97	0x025c	W	0x00000077	DDR PHY register 97
DDRPHY_REG98	0x0260	W	0x00000077	DDR PHY register 98
DDRPHY_REG99	0x0264	W	0x00000077	DDR PHY register 99
DDRPHY_REG9A	0x0268	W	0x00000077	DDR PHY register 9A
DDRPHY_REG9B	0x026c	W	0x00000077	DDR PHY register 9B
DDRPHY_REGB0	0x02c0	W	0x00000077	DDR PHY register B0
DDRPHY_REGB1	0x02c4	W	0x00000077	DDR PHY register B1
DDRPHY_REGB2	0x02c8	W	0x00000077	DDR PHY register B2
DDRPHY_REGB3	0x02cc	W	0x00000077	DDR PHY register B3
DDRPHY_REGB4	0x02d0	W	0x00000077	DDR PHY register B4
DDRPHY_REGB5	0x02d4	W	0x00000077	DDR PHY register B5
DDRPHY_REGB6	0x02d8	W	0x00000077	DDR PHY register B6
DDRPHY_REGB7	0x02dc	W	0x00000077	DDR PHY register B7
DDRPHY_REGB8	0x02e0	W	0x00000077	DDR PHY register B8
DDRPHY_REGB9	0x02e4	W	0x00000077	DDR PHY register B9
DDRPHY_REGBA	0x02e8	W	0x00000077	DDR PHY register BA
DDRPHY_REGBB	0x02ec	W	0x00000077	DDR PHY register BB
DDRPHY_REGBC	0x02f0	W	0x00000077	DDR PHY register BC
DDRPHY_REGBD	0x02f4	W	0x00000077	DDR PHY register BD
DDRPHY_REGBE	0x02f8	W	0x00000077	DDR PHY register BE
DDRPHY_REGC0	0x0300	W	0x00000077	DDR PHY register C0
DDRPHY_REGC1	0x0304	W	0x00000077	DDR PHY register C1
DDRPHY_REGC2	0x0308	W	0x00000077	DDR PHY register C2
DDRPHY_REGC3	0x030c	W	0x00000077	DDR PHY register C3
DDRPHY_REGC4	0x0310	W	0x00000077	DDR PHY register C4
DDRPHY_REGC5	0x0314	W	0x00000077	DDR PHY register C5
DDRPHY_REGC6	0x0318	W	0x00000077	DDR PHY register C6
DDRPHY_REGC7	0x031c	W	0x00000077	DDR PHY register C7
DDRPHY_REGC8	0x0320	W	0x00000077	DDR PHY register C8
DDRPHY_REGC9	0x0324	W	0x00000077	DDR PHY register C9
DDRPHY_REGCA	0x0328	W	0x00000007	DDR PHY register CA
DDRPHY_REGCB	0x032c	W	0x00000077	DDR PHY register CB

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Name	Offset	Size	Reset Value	Description
DDRPHY_REGCC	0x0330	W	0x00000077	DDR PHY register CC
DDRPHY_REGCD	0x0334	W	0x00000077	DDR PHY register CD
DDRPHY_REGCE	0x0338	W	0x00000077	DDR PHY register CE
DDRPHY_REGCF	0x033c	W	0x00000077	DDR PHY register CF
DDRPHY_REGD0	0x0340	W	0x00000077	DDR PHY register D0
DDRPHY_REGD1	0x0344	W	0x00000077	DDR PHY register D1
DDRPHY_REGD2	0x0348	W	0x00000077	DDR PHY register D2
DDRPHY_REGD3	0x034c	W	0x00000077	DDR PHY register D3
DDRPHY_REGD4	0x0350	W	0x00000077	DDR PHY register D4
DDRPHY_REGD5	0x0354	W	0x00000007	DDR PHY register D5
DDRPHY_REGD6	0x0358	W	0x00000077	DDR PHY register D6
DDRPHY_REGD7	0x035c	W	0x00000077	DDR PHY register D7
DDRPHY_REGD8	0x0360	W	0x00000077	DDR PHY register D8
DDRPHY_REGD9	0x0364	W	0x00000077	DDR PHY register D9
DDRPHY_REGDA	0x0368	W	0x00000077	DDR PHY register DA
DDRPHY_REGDB	0x036c	W	0x00000077	DDR PHY register DB
DDRPHY_REGDC	0x0370	W	0x00000077	DDR PHY register DC
DDRPHY_REGDD	0x0374	W	0x00000077	DDR PHY register DD
DDRPHY_REGDE	0x0378	W	0x00000070	DDR PHY register DE
DDRPHY_REGDF	0x037c	W	0x00000077	DDR PHY register DF
DDRPHY_REGE0	0x0380	W	0x00000007	DDR PHY register E0
DDRPHY_REGE1	0x0384	W	0x00000077	DDR PHY register E1
DDRPHY_REGE2	0x0388	W	0x00000077	DDR PHY register E2
DDRPHY_REGE3	0x038c	W	0x00000077	DDR PHY register E3
DDRPHY_REGE4	0x0390	W	0x00000077	DDR PHY register E4
DDRPHY_REGE5	0x0394	W	0x00000077	DDR PHY register E5
DDRPHY_REGE6	0x0398	W	0x00000077	DDR PHY register E6
DDRPHY_REGE7	0x039c	W	0x00000077	DDR PHY register E7
DDRPHY_REGE8	0x03a0	W	0x00000077	DDR PHY register E8
DDRPHY_REGE9	0x03a4	W	0x00000077	DDR PHY register E9
DDRPHY_REGEA	0x03a8	W	0x00000077	DDR PHY register EA
DDRPHY_REGEB	0x03ac	W	0x00000007	DDR PHY register EB
DDRPHY_REGEC	0x03b0	W	0x00000064	DDR PHY register EC
DDRPHY_REGED	0x03b4	W	0x0000001a	DDR PHY register ED
DDRPHY_REGEE	0x03b8	W	0x00000002	DDR PHY register EE
DDRPHY_REGEF	0x03bc	W	0x00000000	DDR PHY register EF
DDRPHY_REGF0	0x03c0	W	0x00000000	DDR PHY register F0
DDRPHY_REGF1	0x03c4	W	0x00000000	DDR PHY register F1
DDRPHY_REGF2	0x03c8	W	0x00000000	DDR PHY register F2
DDRPHY_REGFA	0x03e8	W	0x00000000	DDR PHY register FA
DDRPHY_REGFB	0x03ec	W	0x00000000	DDR PHY register FB
DDRPHY_REGFC	0x03f0	W	0x00000000	DDR PHY register FC
DDRPHY_REGFD	0x03f4	W	0x00000000	DDR PHY register FD

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Name	Offset	Size	Reset Value	Description
DDRPHY_REGFE	0x03f8	W	0x00000000	DDR PHY register FE
DDRPHY_REGFF	0x03fc	W	0x00000000	DDR PHY register FF
DDRPHY_REG100	0x0400	W	0x00000006	DDR PHY register 100
DDRPHY_REG101	0x0404	W	0x00000000	DDR PHY register 101
DDRPHY_REG102	0x0408	W	0x00000000	DDR PHY register 102
DDRPHY_REG103	0x040c	W	0x00000000	DDR PHY register 103
DDRPHY_REG104	0x0410	W	0x0000007f	DDR PHY register 104
DDRPHY_REG105	0x0414	W	0x00000000	DDR PHY register 105
DDRPHY_REG106	0x0418	W	0x00000030	DDR PHY register 106
DDRPHY_REG107	0x041c	W	0x00000064	DDR PHY register 107
DDRPHY_REG108	0x0420	W	0x00000008	DDR PHY register 108

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

10.4.2 Detail Register Description

DDRC_MSTR

Address: Operational Base + offset (0x0000)

MSTR

Bit	Attr	Reset Value	Description
31:30	RW	0x0	device_config Configuration of device used 00: x4 device 01: x8 device 10: x16 device 11: x32 device Programming Mode: Static
29	RW	0x0	frequency_mode Choose wich register are used 0: original registers 1: FREQ1 registers Programming Mode: Quasi-dynamic Group 2
28:26	RO	0x0	reserved
25:24	RW	0x3	active_ranks 01: One Rank 11: Two Ranks Programming Mode: Static
23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22	RW	0x0	frequency_ratio selects the frequency ratio 0: 1:2 mode 1: 1:1 mode Programming Mode: Static
21:20	RW	0x0	active_logical_ranks Number of logical ranks for DDR4 3DS 0: monolithic (no stack) 1: 2H stack 2: 4H stack 3: 8H stack Programming Mode: Static
19:16	RW	0x4	burst_rdwr SDRAM burst length used: 0001: burst length of 2 0010: burst length of 4 0100: burst length of 8 1000: burst length of 16 Programming Mode: Static
15	RW	0x0	dll_off_mode 1: dll-off mode for lower frequency operation 0: dll-on mode for normal frequency operation Programming Mode: Quasi-dynamic Group 2
14	RO	0x0	reserved
13:12	RW	0x0	data_bus_width 00: full DQ bus width to SDRAM 01: half DQ bus width to SDRAM 1x: reserved Programming Mode: Static
11	RW	0x0	geardown_mode 1: enable geardow mode 0: normal mode Programming Mode: Quasi-dynamic Group 2
10	RW	0x0	en_2t_timing_mode 1: use 2T timing 0: use 1T timing Programming Mode: Static
9	RW	0x0	burstchop 1: enable burst-chop Programming Mode: Static
8	RW	0x0	burst_mode 0: sequential burst mode 1: interleaved burst mode Programming Mode: Static

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Bit	Attr	Reset Value	Description
7:6	RO	0x0	reserved
5	RW	0x0	lpddr4 1: LPDDR4 0: Non-LPDDR4 Programming Mode: Static
4	RW	0x0	ddr4 1: DDR4 0: Non-DDR4 Programming Mode: Static
3	RW	0x0	lpddr3 1: LPDDR3 0: Non-LPDDR3 Programming Mode: Static
2	RW	0x0	lpddr2 1: LPDDR2 0: Non-LPDDR2 Programming Mode: Static
1	RW	0x0	mobile 1: Mobile/LPDDR 0: Non-Mobile Programming Mode: Static
0	RW	0x1	ddr3 1: DDR3 0: Non-DDR3 Programming Mode: Static

DDRC_STAT

Address: Operational Base + offset (0x0004)

STAT

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	selfref_cam_not_empty Self refresh with CAMs not empty. Set to 1 when Self Refresh is entered but CAMs are not drained. Cleared after exiting Self Refresh Programming Mode: Dynamic
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:8	RO	0x0	selfref_state Self refresh state. This indicates self refresh or self refresh power down state for LPDDR4. This register is used for frequency change and MRR/MRW access during self refresh. 00:SDRAM is not in Self Refresh. 01:Self refresh 1 10:Self refresh power down 11:Self refresh 2 Programming Mode: Dynamic
7:6	RO	0x0	reserved
5:4	RO	0x0	selfref_type Flags if Self Refresh (except LPDDR4) or SR-Powerdown (LPDDR4) is entered and if it was under Automatic Self Refresh control only or not. 00:SDRAM is not in Self Refresh (except LPDDR4) or SR-Powerdown (LPDDR4). If retry is enabled by CRCPARCTRL1.crc_parity_retry_enable, this also indicates SRE command is still in parity error window or retry is in-progress. 11:SDRAM is in Self Refresh (except LPDDR4) or SRPowerdown (LPDDR4), which was caused by Automatic Self Refresh only. If retry is enabled, this guarantees SRE command is executed correctly without parity error. 10:SDRAM is in Self Refresh (except LPDDR4) or SRPowerdown (LPDDR4), which was not caused solely under Automatic Self Refresh control. It could have been caused by Hardware Low Power Interface and/or Software (PWRCTL.selfref_sw). If retry is enabled, this guarantees SRE command is executed correctly without parity error. 01: SDRAM is in Self Refresh, which was caused by PHY Master Request. Programming Mode: Dynamic
3	RO	0x0	reserved
2:0	RO	0x0	operating_mode Operating mode. This is 3-bits wide in configurations with mDDR/LPDDR2/LPDDR3/LPDDR4/DDR4 support and 2-bits in all other configurations. non-mDDR/LPDDR2/LPDDR3/LPDDR4 and non-DDR4 designs: 00: Init; 01:Normal; 10:Power-down; 11:Self refresh mDDR/LPDDR2/LPDDR3 or DDR4 designs: 000:Init; 001:Normal; 010:Power-down; 011:Self refresh; 1XX:Deep power-down / Maximum Power Saving Mode LPDDR4 designs: 000:Init; 001:Normal; 010:Power-down; 011:Self refresh / Self refresh power-down Programming Mode: Dynamic

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DDRC_MRCTRL0

Address: Operational Base + offset (0x0010)

MRCTRL0

Bit	Attr	Reset Value	Description
31	RW	0x0	mr_wr Setting this register bit to 1 triggers a mode register read or write operation. When the MR operation is complete, the uMCTL2 automatically clears this bit. The other register fields of this register must be written in a separate APB transaction, before setting this mr_wr bit. It is recommended NOT to set this signal if in Init, Deep power-down or MPSM operating modes. Programming Mode: Dynamic
30	RW	0x0	pba_mode Indicates whether PBA access is executed. When setting this bit to 1 along with setting pda_en to 1, uMCTL2 initiates PBA access instead of PDA access. 0: Per DRAM Addressability mode; 1: Per Buffer Addressability mode The completion of PBA access is confirmed by MRSTAT.pda_done in the same way as PDA. Programming Mode: Dynamic
29:16	RO	0x0	reserved
15:12	RW	0x0	mr_addr Address of the mode register that is to be written to. 0000: MR0; 0001: MR1; 0010: MR2; 0011: MR3; 0100: MR4; 0101: MR5; 0110: MR6; 0111: MR7. Don't Care for LPDDR2/LPDDR3/LPDDR4 (see MRCTRL1.mr_data for mode register addressing in LPDDR2/LPDDR3/LPDDR4) In case of DDR4, the bit[3:2] corresponds to the bank group bits. Programming Mode: Dynamic
11:6	RO	0x0	reserved
5:4	RW	0x0	mr_rank Controls which rank is accessed by MRCTRL0.mr_wr. Normally, it is desired to access all ranks, so all bits should be set to 1. 0x1: select rank 0 only 0x2: select rank 1 only 0x3: select rank 0 and 1 Programming Mode: Dynamic

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_init_int Indicates whether Software intervention is allowed via MRCTRL0/MRCTRL1 before automatic SDRAM initialization routine or not. For DDR4, this bit can be used to initialize the DDR4 RCD (MR7) before automatic SDRAM initialization. For LPDDR4, this bit can be used to program additional mode registers before automatic SDRAM initialization if necessary. Note that this must be cleared to 0 after completing Software operation. Otherwise, SDRAM initialization routine will not re-start. 0: Software intervention is not allowed; 1: Software intervention is allowed Programming Mode: Dynamic
2	RW	0x0	pda_en Indicates whether the mode register operation is MRS in PDA mode or not 0: MRS; 1: MRS in Per DRAM Addressability mode Note that when pba_mode=1, PBA access is initiated instead of PDA access. Programming Mode: Dynamic
1	RW	0x0	mpr_en Indicates whether the mode register operation is MRS or WR/RD for MPR (only supported for DDR4) 0: MRS; 1: WR/RD for MPR Programming Mode: Dynamic
0	RW	0x0	mr_type Indicates whether the mode register operation is read or write. Only used for LPDDR2/LPDDR3/LPDDR4/DDR4. 0: Write; 1: Read Programming Mode: Dynamic

DDRC_MRCTRL1

Address: Operational Base + offset (0x0014)

MRCTRL1

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
17:0	RW	0x00000	mr_data Mode register write data for all non-LPDDR2/non-LPDDR3/non-LPDDR4 modes. For LPDDR2/LPDDR3/LPDDR4, MRCTRL1[15:0] are interpreted as [15:8] MR Address [7:0] MR data for writes, don't care for reads. This is 18-bits wide in configurations with DDR4 support and 16-bits in all other configurations. Programming Mode: Dynamic

DDRC_MRSTAT

Address: Operational Base + offset (0x0018)

MRSTAT

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RO	0x0	pda_done The SoC core may initiate a MR write operation in PDA/PBA mode only if this signal is low. This signal goes high when three consecutive MRS commands related to the PDA/PBA mode are issued to the SDRAM. This signal goes low when MRCTRL0.pda_en becomes 0. Therefore, it is recommended to write MRCTRL0.pda_en to 0 after this signal goes high in order to prepare to perform PDA operation next time. 0: Indicates that mode register write operation related to PDA/PBA is in progress or has not started yet. 1: Indicates that mode register write operation related to PDA/PBA has competed. Programming Mode: Dynamic
7:1	RO	0x0	reserved
0	RO	0x0	mr_wr_busy The SoC core may initiate a MR write operation only if this signal is low. This signal goes high in the clock after the uMCTL2 accepts the MRW/MRR request. It goes low when the MRW/MRR command is issued to the SDRAM. It is recommended not to perform MRW/MRR commands when MRSTAT.mr_wr_busy is high. 0: Indicates that the SoC core can initiate a mode register write operation 1: Indicates that mode register write operation is in progress Programming Mode: Dynamic

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DDRC_MRCTRL2

Address: Operational Base + offset (0x001c)

MRCTRL2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mr_device_sel Indicates the device(s) to be selected during the MRS that happens in PDA mode. Each bit is associated with one device. For example, bit[0] corresponds to Device 0, bit[1] to Device 1 etc. A '1' should be programmed to indicate that the MRS command should be applied to that device. A '0' indicates that the MRS commands should be skipped for that device. Programming Mode: Dynamic

DDRC_DERATEEN

Address: Operational Base + offset (0x0020)

DERATEEN

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:8	RW	0x0	rc_derate_value Derate value of tRC for LPDDR4 0: Derating uses +1. 1: Derating uses +2. 2: Derating uses +3. 3: Derating uses +4. Present only in designs configured to support LPDDR4. The required number of cycles for derating can be determined by dividing 3.75ns by the core_ddrc_core_clk period, and rounding up the next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:4	RW	0x0	derate_byte Derate byte Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4 Indicates which byte of the MRR data is used for derating. Programming Mode: Static
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	derate_value Derate value 0: Derating uses +1. 1: Derating uses +2. Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4 Set to 0 for all LPDDR2 speed grades as derating value of +1.875 ns is less than a core_ddrc_core_clk period. For LPDDR3/4, if the period of core_ddrc_core_clk is less than 1.875ns, this register field should be set to 1; otherwise it should be set to 0. Programming Mode: Quasi-dynamic Group 2 and Group 4
0	RW	0x0	derate_enable Enables derating 0: Timing parameter derating is disabled 1: Timing parameter derating is enabled using MR4 read value. Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4 This field must be set to 0 for non-LPDDR2/LPDDR3/LPDDR4 mode. Programming Mode: Dynamic

DDRC_DERATEINT

Address: Operational Base + offset (0x0024)

DERATEINT

Bit	Attr	Reset Value	Description
31:0	RW	0x00800000	mr4_read_interval Interval between two MR4 reads, used to derate the timing parameters. Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4. This register must not be set to zero. Unit: DFI clock cycle. Programming Mode: Static

DDRC_PWRCTL

Address: Operational Base + offset (0x0030)

PWRCTL

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	lpddr4_sr_allowed Indicates whether transition from SR-PD to SR and back to SR-PD is allowed. This register field cannot be modified while PWRCTL.selfref_sw==1. 0: SR-PD -> SR -> SR-PD not allowed 1: SR-PD -> SR -> SR-PD allowed Programming Mode: Dynamic
7	RW	0x0	dis_cam_drain_selfref Indicates whether skipping CAM draining is allowed when entering Self-Refresh. This register field cannot be modified while PWRCTL.selfref_sw==1. 0: CAMs must be empty before entering SR 1: CAMs are not emptied before entering SR Programming Mode: Dynamic
6	RW	0x0	stay_in_selfref Self refresh state is an intermediate state to enter to Selfrefresh power down state or exit Self refresh power down state for LPDDR4. This register controls transition from the Self refresh state. 1: Prohibit transition from Self refresh state 0: Allow transition from Self refresh state Programming Mode: Dynamic
5	RW	0x0	selfref_sw A value of 1 to this register causes system to move to Self Refresh state immediately, as long as it is not in INIT or DPD/MPSM operating_mode. This is referred to as Software Entry/Exit to Self Refresh. 1: Software Entry to Self Refresh 0: Software Exit from Self Refresh Programming Mode: Dynamic
4	RW	0x0	mpsm_en When this is 1, the uMCTL2 puts the SDRAM into maximum power saving mode when the transaction store is empty. This register must be reset to '0' to bring uMCTL2 out of maximum power saving mode. Present only in designs configured to support DDR4. For non-DDR4, this register should not be set to 1. Programming Mode: Dynamic

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Bit	Attr	Reset Value	Description
3	RW	0x0	en_dfi_dram_clk_disable Enable the assertion of dfi_dram_clk_disable whenever a clock is not required by the SDRAM. If set to 0, dfi_dram_clk_disable is never asserted. Assertion of dfi_dram_clk_disable is as follows: In DDR2/DDR3, can only be asserted in Self Refresh. In DDR4, can be asserted in the Self Refresh and Maximum Power Saving Mode In mDDR/LPDDR2/LPDDR3, can be asserted in the Self Refresh, Power Down, Deep Power Down and Normal operation (Clock Stop) In LPDDR4, can be asserted in the Self Refresh Power Down, Power Down and Normal operation (Clock Stop) Programming Mode: Dynamic
2	RW	0x0	deeppowerdown_en When this is 1, uMCTL2 puts the SDRAM into deep powerdown mode when the transaction store is empty. This register must be reset to '0' to bring uMCTL2 out of deep power-down mode. Controller performs automatic SDRAM initialization on deep power-down exit. Present only in designs configured to support mDDR or LPDDR2 or LPDDR3. For non-mDDR/non-LPDDR2/nonLPDDR3, this register should not be set to 1. Programming Mode: Dynamic
1	RW	0x0	powerdown_en If true then the uMCTL2 goes into power-down after a programmable number of cycles "maximum idle clocks before power down" PWRTMG.powerdown_to_x32). This register bit may be re-programmed during the course of normal operation. Programming Mode: Dynamic
0	RW	0x0	selfref_en If true then the uMCTL2 puts the SDRAM into Self Refresh after a programmable number of cycles "maximum idle clocks before Self Refresh (PWRTMG.selfref_to_x32)". This register bit may be re-programmed during the course of normal operation. Programming Mode: Dynamic

DDRC_PWRTMG

Address: Operational Base + offset (0x0034)

PWRTMG

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Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x40	selfref_to_x32 After this many clocks of the DDRC command channel being idle the uMCTL2 automatically puts the SDRAM into Self Refresh. The DDRC command channel is considered idle when there are no HIF commands outstanding. This must be enabled in the PWRCTL.selfref_en. Unit: Multiples of 32 DFI clocks. Programming Mode: Static
15:8	RW	0x20	t_dpd_x4096 Minimum deep power-down time. For mDDR, value from the JEDEC specification is 0 as mDDR exits from deep power-down mode immediately after PWRCTL.deeppowerdown_en is de-asserted. For LPDDR2/LPDDR3, value from the JEDEC specification is 500us. Unit: Multiples of 4096 DFI clocks. Programming Mode: Static
7:5	RO	0x0	reserved
4:0	RW	0x10	powerdown_to_x32 After this many clocks of the DDRC command channel being idle the uMCTL2 automatically puts the SDRAM into powerdown. The DDRC command channel is considered idle when there are no HIF commands outstanding. This must be enabled in the PWRCTL.powerdown_en. Unit: Multiples of 32 DFI clocks Programming Mode: Static

DDRC_HWLPCTL

Address: Operational Base + offset (0x0038)

HWLPCTL

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x000	hw_lp_idle_x32 Hardware idle period. The cactive_ddrc output is driven low if the DDRC command channel is idle for hw_lp_idle * 32 cycles if not in INIT or DPD/MPSM operating_mode. The DDRC command channel is considered idle when there are no HIF commands outstanding. The hardware idle function is disabled when hw_lp_idle_x32=0. Unit: Multiples of 32 DFI clocks. Programming Mode: Static

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Bit	Attr	Reset Value	Description
15:2	RO	0x0	reserved
1	RW	0x1	hw_lp_exit_idle_en When this bit is programmed to 1 the cactive_in_ddrc pin of the DDRC can be used to exit from the automatic clock stop, automatic power down or automatic self-refresh modes. Note, it will not cause exit of Self-Refresh that was caused by Hardware Low Power Interface and/or Software (PWRCTL.selfref_sw). Programming Mode: Static
0	RW	0x1	hw_lp_en Enable for Hardware Low Power Interface. Programming Mode: Quasi-dynamic Group 3

DDRC_HWFFCCTL

Address: Operational Base + offset (0x003c)

HWFFCCTL

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	target_vrcg Set target value of VRCG (MR13 OP[3]). This field value is used when HWFFC request has been issued. Programming Mode: Static
5	RW	0x0	init_vrcg Set initial value of VRCG (MR13 OP[3]). This field value is used when HWFFCCTL.hwffc_en has been changed to 11. Programming Mode: Static
4	RW	0x1	init_fsp Set initial value of FSP-OP (MR13 OP[7]). This field value is used when HWFFCCTL.hwffc_en has been changed to 11. Programming Mode: Static
3:1	RO	0x0	reserved
0	RW	0x0	hwffc_en Enable HWFFC through Hardware Low Power Interface. The other fields of this register is used only when changing this field to 11. 00: Disable HWFFC 10: Intermediate, set only when disabling HWFFC 11: Enable HWFFC 01: Not allowed Programming Mode: Dynamic

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DDRC_HWFFCSTAT

Address: Operational Base + offset (0x0040)

HWFFCSTAT

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RO	0x0	current_vrcg Indicates current value of VRCG (MR13 OP[3]). Programming Mode: Dynamic
8	RO	0x0	current_fsp Indicates current value of FSP-OP (MR13 OP[7]). Programming Mode: Dynamic
7:5	RO	0x0	reserved
4	RO	0x0	current_frequency Indicates the current frequency. 0: Frequency 0/Normal 1: Frequency 1/FREQ1 Programming Mode: Dynamic
3:2	RO	0x0	reserved
1	RO	0x0	hwffc_operating_mode Operating mode of HWFFC. 0: Normal 1: Self Refresh or SR-Powerdown Programming Mode: Dynamic
0	RO	0x0	hwffc_in_progress Indicates HWFFC is in progress. Programming Mode: Dynamic

DDRC_RFSHCTL0

Address: Operational Base + offset (0x0050)

RFSHCTL0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:20	RW	0x2	refresh_margin Threshold value in number of DFI clock cycles before the critical refresh or page timer expires. A critical refresh is to be issued before this threshold is reached. It is recommended that this not be changed from the default value, currently shown as 0x2. It must always be less than internally used t_rfc_nom_x32. Note that, in LPDDR2/LPDDR3/LPDDR4, internally used t_rfc_nom_x32 may be equal to RFSHTMG.t_rfc_nom_x32>>2 if derating is enabled (DERATEEN.derate_enable=1). Otherwise, internally used t_rfc_nom_x32 will be equal to RFSHTMG.t_rfc_nom_x32. Unit: Multiples of 32 DFI clocks. Programming Mode: Dynamic - Refresh Related
19:17	RO	0x0	reserved
16:12	RW	0x10	refresh_to_x32 If the refresh timer (tRFCnom, also known as tREFI) has expired at least once, but it has not expired (RFSHCTL0.refresh_burst+1) times yet, then a speculative refresh may be performed. A speculative refresh is a refresh performed at a time when refresh would be useful, but before it is absolutely required. When the SDRAM bus is idle for a period of time determined by this RFSHCTL0.refresh_to_x32 and the refresh timer has expired at least once since the last refresh, then a speculative refresh is performed. Speculative refreshes continues successively until there are no refreshes pending or until new reads or writes are issued to the uMCTL2. Unit: Multiples of 32 DFI clocks. Programming Mode: Dynamic - Refresh Related
11:9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8:4	RW	0x00	refresh_burst The programmed value + 1 is the number of refresh timeouts that is allowed to accumulate before traffic is blocked and the refreshes are forced to execute. Closing pages to perform a refresh is a one-time penalty that must be paid for each group of refreshes. Therefore, performing refreshes in a burst reduces the per-refresh penalty of these page closings. Higher numbers for RFSHCTL.refresh_burst slightly increases utilization; lower numbers decreases the worstcase latency associated with refreshes. 0: single refresh 1: burst-of-2 refresh 7: burst-of-8 refresh For DDR2/3, the refresh is always per-rank and not perbank. The rank refresh can be accumulated over 8*tREFI cycles using the burst refresh feature. In DDR4 mode, according to Fine Granularity feature, 8 refreshes can be postponed in 1X mode, 16 refreshes in 2X mode and 32 refreshes in 4X mode. If using PHY-initiated updates, care must be taken in the setting of RFSHCTL0.refresh_burst, to ensure that tRFCmax is not violated due to a PHY-initiated update occurring shortly before a refresh burst was due. In this situation, the refresh burst will be delayed until the PHYinitiated update is complete. Programming Mode: Dynamic - Refresh Related
3	RO	0x0	reserved
2	RW	0x0	per_bank_refresh 1: Per bank refresh; 0: All bank refresh. Per bank refresh allows traffic to flow to other banks. Per bank refresh is not supported by all LPDDR2 devices but should be supported by all LPDDR3/LPDDR4 devices. Programming Mode: Static
1:0	RO	0x0	reserved

DDRC_RFSHCTL1

Address: Operational Base + offset (0x0054)

RFSHCTL1

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:16	RW	0x000	refresh_timer1_start_value_x32 Refresh timer start for rank 1 (only present in multi-rank configurations). This is useful in staggering the refreshes to multiple ranks to help traffic to proceed. This is explained in Refresh Controls section of architecture chapter. Unit: Multiples of 32 DFI clock cycles. Programming Mode: Dynamic - Refresh Related
15:12	RO	0x0	reserved
11:0	RW	0x000	refresh_timer0_start_value_x32 Refresh timer start for rank 0 (only present in multi-rank configurations). This is useful in staggering the refreshes to multiple ranks to help traffic to proceed. This is explained in Refresh Controls section of architecture chapter. Unit: Multiples of 32 DFI clock cycles. Programming Mode: Dynamic - Refresh Related

DDRC_RFSHCTL3

Address: Operational Base + offset (0x0060)

RFSHCTL3

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x0	refresh_mode Fine Granularity Refresh Mode 000: Fixed 1x (Normal mode) 001: Fixed 2x 010: Fixed 4x 101: Enable on the fly 2x (not supported) 110: Enable on the fly 4x (not supported) Everything else - reserved Note: Only Fixed 1x mode is supported if RFSHCTL3.dis_auto_refresh = 1. Note: The on-the-fly modes are not supported in this version of the uMCTL2. Note: This must be set up while the Controller is in reset or while the Controller is in self-refresh mode. Changing this during normal operation is not allowed. Making this a dynamic register will be supported in future version of the uMCTL2. Note: This register field has effect only if a DDR4 SDRAM device is in use (MSTR.ddr4 = 1). Programming Mode: Quasi-dynamic Group 2

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Bit	Attr	Reset Value	Description
3:2	RO	0x0	reserved
1	RW	0x0	refresh_update_level Toggle this signal (either from 0 to 1 or from 1 to 0) to indicate that the refresh register(s) have been updated. refresh_update_level must not be toggled when the DDRC is in reset (core_ddrc_rstn = 0). The refresh register(s) are automatically updated when exiting reset. Programming Mode: Dynamic
0	RW	0x0	dis_auto_refresh When '1', disable auto-refresh generated by the uMCTL2. When auto-refresh is disabled, the SoC core must generate refreshes using the registers DBGCMD.rankn_refresh. When dis_auto_refresh transitions from 0 to 1, any pending refreshes are immediately scheduled by the uMCTL2. If DDR4 CRC/parity retry is enabled (CRCPARCTL1.crc_parity_retry_enable = 1), disable autorefresh is not supported, and this bit must be set to '0'. (DDR4 only) If FGR mode is enabled (RFSHCTL3.refresh_mode > 0), disable auto-refresh is not supported, and this bit must be set to '0'. This register field is changeable on the fly. Programming Mode: Dynamic - Refresh Related

DDRC_RFSHTMG

Address: Operational Base + offset (0x0064)

RFSHTMG

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:16	RW	0x062	t_rfc_nom_x32 tREFI: Average time interval between refreshes per rank (Specification: 7.8us for DDR2, DDR3 and DDR4. See JEDEC specification for mDDR, LPDDR2, LPDDR3 and LPDDR4). For LPDDR2/LPDDR3/LPDDR4: if using all-bank refreshes (RFSHCTL0.per_bank_refresh= 0), this register should be set to tREFIab if using per-bank refreshes (RFSHCTL0.per_bank_refresh = 1), this register should be set to tREFIpb When the controller is operating in 1:2 frequency ratio mode, program this to (tREFI/2), no rounding up. In DDR4 mode, tREFI value is different depending on the refresh mode. The user should program the appropriate value from the spec based on the value programmed in the refresh mode register. Note that RFSHTMG.t_rfc_nom_x32 * 32 must be greater than RFSHTMG.t_rfc_min, and RFSHTMG.t_rfc_nom_x32 must be greater than 0x1. Non-DDR4 or DDR4 Fixed 1x mode:RFSHTMG.t_rfc_nom_x32 must be less than or equal to 0xFFE. DDR4 Fixed 2x mode: RFSHTMG.t_rfc_nom_x32 must be less than or equal to 0x7FF. DDR4 Fixed 4x mode: RFSHTMG.t_rfc_nom_x32 must be less than or equal to 0x3FF. Unit: Multiples of 32 DFI clocks. Programming Mode: Dynamic - Refresh Related
15	RW	0x0	lpddr3_trefbw_en Used only when LPDDR3 memory type is connected. Should only be changed when uMCTL2 is in reset. Specifies whether to use the tREFBW parameter (required by some LPDDR3 devices which comply with earlier versions of the LPDDR3 JEDEC specification) or not: 0: tREFBW parameter not used 1: tREFBW parameter used Programming Mode: Static
14:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x08c	t_rfc_min tRFC (min): Minimum time from refresh to refresh or activate. t_rfc_min should be set to RoundUp(RoundUp(tRFCmin/tCK)/2). In LPDDR2/LPDDR3/LPDDR4 mode: if using all-bank refreshes, the tRFCmin value in the above equations is equal to tRFCab if using per-bank refreshes, the tRFCmin value in the above equations is equal to tRFCpb In DDR4 mode, the tRFCmin value in the above equations is different depending on the refresh mode (fixed 1X,2X,4X) and the device density. The user should program the appropriate value from the spec based on the 'refresh_mode' and the device density that is used. Unit: DFI Clocks. Programming Mode: Dynamic - Refresh Related

DDRC_CRCPARCTL0

Address: Operational Base + offset (0x00c0)

CRCPARCTL0

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	W1C	0x0	dfi_alert_err_cnt_clr DFI alert error count clear. Clear bit for DFI alert error counter. Asserting this bit will clear the DFI alert error counter, CRCPARSTAT.dfi_alert_err_cnt. When the clear operation is complete, the uMCTL2 automatically clears this bit. Programming Mode: Dynamic
1	W1C	0x0	dfi_alert_err_int_clr Interrupt clear bit for DFI alert error. If this bit is set, the alert error interrupt on CRCPARSTAT.dfi_alert_err_int will be cleared. When the clear operation is complete, the uMCTL2 automatically clears this bit. Programming Mode: Dynamic
0	RW	0x0	dfi_alert_err_int_en Interrupt enable bit for DFI alert error. If this bit is set, any parity/CRC error detected on the dfi_alert_n input will result in an interrupt being set on CRCPARSTAT.dfi_alert_err_int. Programming Mode: Dynamic

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DDRC_CRCPARCTL1

Address: Operational Base + offset (0x00c4)

CRCPARCTL1

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	retry_add_rd_lat Retry additional read latency value. Delay value used is retry_add_rd_lat+1. Only used if CRCPARCTL1.retry_add_rd_lat_en is enabled. Selects the number of pipeline stages to dfi_rddata_valid/dfi_rddata/dfi_rddata_dbi before rest of internal uMCTL2 logic observes it. Required to compensate for fact delay in PHY/PCB for generating dfi_alert_n for retry may be more than the delay in PHY/PCB on read data path. Recommended settings (in terms of core_ddrc_core_clk): (Maximum Alert delay through PHY/PCB from erroneous read command including tPAR_UNKNOWN) - (Minimum Read data delay through PHY/PCB from erroneous read command) + (PHY's max granularity of dfi_rddata beats that may be corrupted before erroneous Read) Note: This calculation depends on various items such as RL, tPAR_ALERT_ON/tPAR_UNKNOWN/RCD/PHY/PCB behavior. Unit: DFI clock cycles. Programming Mode: Static
15	RW	0x0	retry_add_rd_lat_en Retry additional read latency enable. Number of pipeline stages selected is defined as CRCPARCTL1.retry_add_lat+1. Only set if CRCPARCTL1.crc_parity_retry_enable = 1 Programming Mode: Static
14:13	RO	0x0	reserved
12	RW	0x1	caparity_disable_before_sr If DDR4-SDRAM's CA parity is enabled by INIT6.mr5[2:0]!=0 and this register is set to 1, CA parity is automatically disabled before Self-Refresh entry and enabled after SelfRefresh exit by issuing MR5. 1: CA parity is disabled before Self-Refresh entry 0: CA parity is not disabled before Self-Refresh entry If Geardown is used by MSTR.geardown_mode=1, this register must be set to 1. If this register set to 0, DRAMTMG5.t_ckesr and DRAMTMG5.t_cksre must be increased by PL(Parity latency) Programming Mode: Static
11:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RW	0x0	crc_inc_dm CRC Calculation setting register 1: CRC includes DM signal 0: CRC not includes DM signal Present only in designs configured to support DDR4. Programming Mode: Static
6:5	RO	0x0	reserved
4	RW	0x0	crc_enable CRC enable Register 1: Enable generation of CRC 0: Disable generation of CRC The setting of this register should match the CRC mode register setting in the DRAM. Programming Mode: Static
3:1	RO	0x0	reserved
0	RW	0x0	parity_enable C/A Parity enable register 1: Enable generation of C/A parity and detection of C/A parity error 0: Disable generation of C/A parity and disable detection of C/A parity error If RCD's parity error detection or SDRAM's parity detection is enabled, this register should be 1. Programming Mode: Static

DDRC_CRCPARSTAT

Address: Operational Base + offset (0x00cc)

CRCPARSTAT

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	RO	0x0	dfi_alert_err_fatl_int Fatal parity error interrupt. One or more these situation below happens, this interrupt bit is set: MPSMX caused parity error. (RCD's parity error detection only) Parity error happens again during software intervention time MRS was in retry_fifo_max_hold_timer_x4 window from alert_n=0 or STAT.operating_mode is Init. It remains set until cleared by CRCPARCTL0.dfi_alert_err_fatl_clr. If this interrupt is asserted, system reset is strongly recommended. Programming Mode: Static

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Bit	Attr	Reset Value	Description
16	RO	0x0	dfi_alert_err_int DFI alert error interrupt. If a parity/CRC error is detected on dfi_alert_n, and the interrupt is enabled by CRCPARCTL0.dfi_alert_err_int_en, this interrupt bit will be set. It will remain set until cleared by CRCPARCTL0.dfi_alert_err_int_clr Programming Mode: Static
15:0	RO	0x0000	dfi_alert_err_cnt DFI alert error count. If a parity/CRC error is detected on dfi_alert_n, this counter be incremented. This is independent of the setting of CRCPARCTL0.dfi_alert_err_int_en. It will saturate at 0xFFFF, and can be cleared by asserting CRCPARCTL0.dfi_alert_err_cnt_clr. Programming Mode: Static

DDRC_INIT0

Address: Operational Base + offset (0x00d0)

INIT0

Bit	Attr	Reset Value	Description
31:30	RW	0x0	skip_dram_init If lower bit is enabled the SDRAM initialization routine is skipped. The upper bit decides what state the controller starts up in when reset is removed 00: SDRAM Intialization routine is run after power-up 01: SDRAM Intialization routine is skipped after powerup. Controller starts up in Normal Mode 11: SDRAM Intialization routine is skipped after powerup. Controller starts up in Self-refresh Mode 10: SDRAM Intialization routine is run after power-up. Programming Mode: Quasi-dynamic Group 2
29:26	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
25:16	RW	0x002	post_cke_x1024 Cycles to wait after driving CKE high to start the SDRAM initialization sequence. Unit: 1024 DFI clock cycles. DDR2 typically requires a 400 ns delay, requiring this value to be programmed to 2 at all clock speeds. LPDDR2/LPDDR3 typically requires this to be programmed for a delay of 200 us. LPDDR4 typically requires this to be programmed for a delay of 2 us. When the controller is operating in 1:2 frequency ratio mode, program this to JEDEC spec value divided by 2, and round it up to the next integer value. Programming Mode: Static
15:12	RO	0x0	reserved
11:0	RW	0x04e	pre_cke_x1024 Cycles to wait after reset before driving CKE high to start the SDRAM initialization sequence. Unit: 1024 DFI clock cycles. DDR2 specifications typically require this to be programmed for a delay of >= 200 us. LPDDR2/LPDDR3: tINIT1 of 100 ns (min) LPDDR4: tINIT3 of 2 ms (min) When the controller is operating in 1:2 frequency ratio mode, program this to JEDEC spec value divided by 2, and round it up to the next integer value. Programming Mode: Static

DDRC_INIT1

Address: Operational Base + offset (0x00d4)

INIT1

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24:16	RW	0x000	dram_rstn_x1024 Number of cycles to assert SDRAM reset signal during init sequence. This is only present for designs supporting DDR3, DDR4 or LPDDR4 devices. When the controller is operating in 1:2 frequency ratio mode, program this to JEDEC spec value divided by 2, and round it up to the next integer value. Unit: 1024 DFI clock cycles. Programming Mode: Static
15:4	RO	0x0	reserved
3:0	RW	0x0	pre_ocd_x32 Wait period before driving the OCD complete command to SDRAM. Unit: Counts of a global timer that pulses every 32 DFI clock cycles. There is no known specific requirement for this; it may be set to zero. Programming Mode: Static

DDRC_INIT2

Address: Operational Base + offset (0x00d8)

INIT2

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x0d	idle_after_reset_x32 Idle time after the reset command, tINIT4. Present only in designs configured to support LPDDR2. When the controller is operating in 1:2 frequency ratio mode, program this to JEDEC spec value divided by 2, and round it up to the next integer value. Unit: 32 DFI clock cycles. Programming Mode: Static
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x5	min_stable_clock_x1 Time to wait after the first CKE high, tINIT2. Present only in designs configured to support LPDDR2/LPDDR3. LPDDR2/LPDDR3 typically requires 5 x tCK delay. When the controller is operating in 1:2 frequency ratio mode, program this to JEDEC spec value divided by 2, and round it up to the next integer value. Unit: DFI clock cycles. Programming Mode: Static

DDRC_INIT3

Address: Operational Base + offset (0x00dc)

INIT3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mr DDR2: Value to write to MR register. Bit 8 is for DLL and the setting here is ignored. The uMCTL2 sets this bit appropriately. DDR3/DDR4: Value loaded into MR0 register. mDDR: Value to write to MR register. LPDDR2/LPDDR3/LPDDR4 - Value to write to MR1 register Programming Mode: Quasi-dynamic Group 1 and Group 4
15:0	RW	0x0510	emr DDR2: Value to write to EMR register. Bits 9:7 are for OCD and the setting in this register is ignored. The uMCTL2 sets those bits appropriately. DDR3/DDR4: Value to write to MR1 register Set bit 7 to 0. If PHY-evaluation mode training is enabled, this bit is set appropriately by the uMCTL2 during write leveling. mDDR: Value to write to EMR register. LPDDR2/LPDDR3/LPDDR4 - Value to write to MR2 register Programming Mode: Quasi-dynamic Group 4

DDRC_INIT4

Address: Operational Base + offset (0x00e0)

INIT4

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	emr2 DDR2: Value to write to EMR2 register. DDR3/DDR4: Value to write to MR2 register LPDDR2/LPDDR3/LPDDR4: Value to write to MR3 register mDDR: Unused Programming Mode: Quasi-dynamic Group 4
15:0	RW	0x0000	emr3 DDR2: Value to write to EMR3 register. DDR3/DDR4: Value to write to MR3 register mDDR/LPDDR2/LPDDR3: Unused LPDDR4: Value to write to MR13 register Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_INIT5

Address: Operational Base + offset (0x00e4)

INIT5

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x10	dev_zqinit_x32 ZQ initial calibration, tZQINIT. Present only in designs configured to support DDR3 or DDR4 or LPDDR2/LPDDR3. DDR3 typically requires 512 SDRAM clock cycles. DDR4 requires 1024 SDRAM clock cycles. LPDDR2/LPDDR3 requires 1 us. When the controller is operating in 1:2 frequency ratio mode, program this to JEDEC spec value divided by 2, and round it up to the next integer value. Unit: 32 DFI clock cycles. Programming Mode: Static
15:10	RO	0x0	reserved
9:0	RW	0x004	max_auto_init_x1024 Maximum duration of the auto initialization, tINIT5. Present only in designs configured to support LPDDR2/LPDDR3. LPDDR2/LPDDR3 typically requires 10 us. Unit: 1024 DFI clock cycles. Programming Mode: Static

DDRC_INIT6

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Address: Operational Base + offset (0x00e8)

INIT6

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mr4 DDR4- Value to be loaded into SDRAM MR4 registers. LPDDR4- Value to be loaded into SDRAM MR11 registers. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:0	RW	0x0000	mr5 DDR4- Value to be loaded into SDRAM MR5 registers. LPDDR4- Value to be loaded into SDRAM MR12 registers. Programming Mode: Quasi-dynamic Group 1 and Group 4

DDRC_INIT7

Address: Operational Base + offset (0x00ec)

INIT7

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mr6 DDR4- Value to be loaded into SDRAM MR6 registers. LPDDR4- Value to be loaded into SDRAM MR14 registers. Programming Mode: Quasi-dynamic Group 4
15:0	RW	0x0000	mr22 LPDDR4- Value to be loaded into SDRAM MR22 registers. Used in LPDDR4 designs only. Programming Mode: Quasi-dynamic Group 4

DDRC_DIMMCTL

Address: Operational Base + offset (0x00f0)

DIMMCTL

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	mrs_bg1_en Enable for BG1 bit of MRS command. BG1 bit of the mode register address is specified as RFU (Reserved for Future Use) and must be programmed to 0 during MRS. In case where DRAMs which do not have BG1 are attached and both the CA parity and the Output Inversion are enabled, this must be set to 0, so that the calculation of CA parity will not include BG1 bit. Note: This has no effect on the address of any other memory accesses, or of software-driven mode register accesses. If address mirroring is enabled, this is applied to BG1 of even ranks and BG0 of odd ranks. 1: Enabled; 0: Disabled Programming Mode: Static
3	RW	0x0	mrs_a17_en Enable for A17 bit of MRS command. A17 bit of the mode register address is specified as RFU (Reserved for Future Use) and must be programmed to 0 during MRS. In case where DRAMs which do not have A17 are attached and the Output Inversion are enabled, this must be set to 0, so that the calculation of CA parity will not include A17 bit. Note: This has no effect on the address of any other memory accesses, or of software-driven mode register accesses. 1: Enabled; 0: Disabled Programming Mode: Static
2:0	RO	0x0	reserved

DDRC_RANKCTL

Address: Operational Base + offset (0x00f4)

RANKCTL

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:8	RW	0x6	diff_rank_wr_gap Only present for multi-rank configurations. Indicates the number of clocks of gap in data responses when performing consecutive writes to different ranks. This is used to switch the delays in the PHY to match the rank requirements. This value should consider both PHY requirement and ODT requirement. PHY requirement: tphy_wrcsgap (see PHY databook for value of tphy_wrcsgap) If CRC feature is enabled, should be increased by 1. If write preamble is set to 2tCK(DDR4 only), should be increased by 1. If write postamble is set to 1.5tCK(LPDDR4 only), should be increased by 1. ODT requirement: The value programmed in this register takes care of the ODT switch off timing requirement when switching ranks during writes. For LPDDR4, the requirement is ODTLoff - ODTLon - BL/2 + 1 When the controller is operating in 1:2 mode, program this to the larger value divided by two and round it up to the next integer. Programming Mode: Quasi-dynamic Group 2
7:4	RW	0x6	diff_rank_rd_gap Only present for multi-rank configurations. Indicates the number of clocks of gap in data responses when performing consecutive reads to different ranks. This is used to switch the delays in the PHY to match the rank requirements. This value should consider both PHY requirement and ODT requirement. PHY requirement: tphy_rdcsgap (see PHY databook for value of tphy_rdcsgap) If read preamble is set to 2tCK(DDR4 only), should be increased by 1. If read postamble is set to 1.5tCK(LPDDR4 only), should be increased by 1. ODT requirement: The value programmed in this register takes care of the ODT switch off timing requirement when switching ranks during reads. When the controller is operating in 1:2 mode, program this to the larger value divided by two and round it up to the next integer. Programming Mode: Static

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Bit	Attr	Reset Value	Description
3:0	RW	0xf	max_rank_rd Only present for multi-rank configurations. Background: Reads to the same rank can be performed back-to-back. Reads to different ranks require additional gap dictated by the register RANKCTL.diff_rank_rd_gap. This is to avoid possible data bus contention as well as to give PHY enough time to switch the delay when changing ranks. The uMCTL2 arbitrates for bus access on a cycle-by-cycle basis; therefore after a read is scheduled, there are few clock cycles (determined by the value on RANKCTL.diff_rank_rd_gap register) in which only reads from the same rank are eligible to be scheduled. This prevents reads from other ranks from having fair access to the data bus. This parameter represents the maximum number of reads that can be scheduled consecutively to the same rank. After this number is reached, a delay equal to RANKCTL.diff_rank_rd_gap is inserted by the scheduler to allow all ranks a fair opportunity to be scheduled. Higher numbers increase bandwidth utilization, lower numbers increase fairness. This feature can be DISABLED by setting this register to 0. When set to 0, the Controller will stay on the same rank as long as commands are available for it. Minimum programmable value is 0 (feature disabled) and maximum programmable value is 0xF. Programming Mode: Static

DDRC_DRAMTMG0

Address: Operational Base + offset (0x0100)

DRAMTMG0

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30:24	RW	0x0f	wr2pre Minimum time between write and precharge to same bank. Unit: DFI Clocks Specifications: WL + BL/2 + tWR = approximately 8 cycles + 15 ns = 14 clocks @400MHz and less for lower frequencies where: WL = write latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM. BST (burst terminate) is not supported at present. tWR = Write recovery time. This comes directly from the SDRAM specification. Add one extra cycle for LPDDR2/LPDDR3/LPDDR4 for this parameter. When the controller is operating in 1:2 frequency ratio mode, 1T mode, divide the above value by 2. No rounding up. When the controller is operating in 1:2 frequency ratio mode, 2T mode or LPDDR4 mode, divide the above value by 2 and round it up to the next integer value. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
23:22	RO	0x0	reserved
21:16	RW	0x10	t_faw tFAW Valid only when 8 or more banks(or banks x bank groups) are present. In 8-bank design, at most 4 banks must be activated in a rolling window of tFAW cycles. When the controller is operating in 1:2 frequency ratio mode, program this to (tFAW/2) and round up to next integer value. In a 4-bank design, set this register to 0x1 independent of the 1:1/1:2 frequency mode. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
15	RO	0x0	reserved
14:8	RW	0x1b	t_ras_max tRAS(max): Maximum time between activate and precharge to same bank. This is the maximum time that a page can be kept open Minimum value of this register is 1. Zero is invalid. When the controller is operating in 1:2 frequency ratio mode, program this to (tRAS(max)-1)/2. No rounding up. Unit: Multiples of 1024 DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x0f	t_ras_min tRAS(min): Minimum time between activate and precharge to the same bank. When the controller is operating in 1:2 frequency mode, 1T mode, program this to tRAS(min)/2. No rounding up. When the controller is operating in 1:2 frequency ratio mode, 2T mode or LPDDR4 mode, program this to (tRAS(min)/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG1

Address: Operational Base + offset (0x0104)

DRAMTMG1

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:16	RW	0x08	t_xp tXP: Minimum time after power-down exit to any operation. For DDR3, this should be programmed to tXPDLL if slow powerdown exit is selected in MR0[12]. If C/A parity for DDR4 is used, set to (tXP+PL) instead. If LPDDR4 is selected and its spec has tCKELPD parameter, set to the larger of tXP and tCKELPD instead. When the controller is operating in 1:2 frequency ratio mode, program this to (tXP/2) and round it up to the next integer value. Units: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x04	rd2pre tRTP: Minimum time from read to precharge of same bank. DDR2: tAL + BL/2 + max(tRTP, 2) - 2 DDR3: tAL + max (tRTP, 4) DDR4: Max of following two equations: tAL + max (tRTP, 4) or, RL + BL/2 - tRP (*). mDDR: BL/2 LPDDR2: Depends on if it's LPDDR2-S2 or LPDDR2-S4: LPDDR2-S2: BL/2 + tRTP - 1. LPDDR2-S4: BL/2 + max(tRTP,2) - 2. LPDDR3: BL/2 + max(tRTP,4) - 4 LPDDR4: BL/2 + max(tRTP,8) - 8 (*) When both DDR4 SDRAM and ST-MRAM are used simultaneously, use SDRAM's tRP value for calculation. When the controller is operating in 1:2 mode, 1T mode, divide the above value by 2. No rounding up. When the controller is operating in 1:2 mode, 2T mode or LPDDR4 mode, divide the above value by 2 and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
7	RO	0x0	reserved
6:0	RW	0x14	t_rc tRC: Minimum time between activates to same bank. When the controller is operating in 1:2 frequency ratio mode, program this to (tRC/2) and round up to next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG2

Address: Operational Base + offset (0x0108)

DRAMTMG2

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:24	RW	0x03	write_latency Set to WL. Time from write command to write data on SDRAM interface. For mDDR, it should normally be set to 1. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. This register field is not required for DDR2 and DDR3 (except if MEMC_TRAINING is set), as the DFI read and write latencies defined in DFITMG0 and DFITMG1 are sufficient for those protocols Unit: DFI clocks Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
23:22	RO	0x0	reserved
21:16	RW	0x05	read_latency Set to RL. Time from read command to read data on SDRAM interface. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. This register field is not required for DDR2 and DDR3 (except if MEMC_TRAINING is set), as the DFI read and write latencies defined in DFITMG0 and DFITMG1 are sufficient for those protocols Unit: DFI clocks Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x06	rd2wr DDR2/3/mDDR: RL + BL/2 + 2 - WL DDR4: RL + BL/2 + 1 + WR_PREAMBLE - WL LPDDR2/LPDDR3: RL + BL/2 + RU(tDQSCKmax/tCK) + 1 - WL LPDDR4(DQ ODT is Disabled): RL + BL/2 + RU(tDQSCKmax/tCK) + WR_PREAMBLE + RD_POSTAMBLE - WL LPDDR4(DQ ODT is Enabled) : RL + BL/2 + RU(tDQSCKmax/tCK) + RD_POSTAMBLE - ODTLon - RU(tODTon(min)/tCK) Minimum time from read command to write command. Include time for bus turnaround and all per-bank, per-rank, and global constraints. Please see the relevant PHY databook for details of what should be included here. Unit: DFI Clocks. Where: WL = write latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM RL = read latency = CAS latency WR_PREAMBLE = write preamble. This is unique to DDR4 and LPDDR4. RD_POSTAMBLE = read postamble. This is unique to LPDDR4. For LPDDR2/LPDDR3/LPDDR4, if derating is enabled (DERATEEN.derate_enable=1), derated tDQSCKmax should be used. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x0d	wr2rd DDR4: CWL + PL + BL/2 + tWTR_L LPDDR2/3/4: WL + BL/2 + tWTR + 1 Others: CWL + BL/2 + tWTR In DDR4, minimum time from write command to read command for same bank group. In others, minimum time from write command to read command. Includes time for bus turnaround, recovery times, and all per-bank, per-rank, and global constraints. Unit: DFI Clocks. Where: CWL = CAS write latency WL = Write latency PL = Parity latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM tWTR_L = internal write to read command delay for same bank group. This comes directly from the SDRAM specification. tWTR = internal write to read command delay. This comes directly from the SDRAM specification. Add one extra cycle for LPDDR2/LPDDR3/LPDDR4 operation. When the controller is operating in 1:2 mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4

DDRC_DRAMTMG3

Address: Operational Base + offset (0x010c)

DRAMTMG3

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:20	RW	0x005	t_mrw Time to wait after a mode register write or read (MRW or MRR). Present only in designs configured to support LPDDR2, LPDDR3 or LPDDR4. LPDDR2 typically requires value of 5. LPDDR3 typically requires value of 10. LPDDR4: Set this to the larger of tMRW and tMRWCKEL. For LPDDR2, this register is used for the time from a MRW/MRR to all other commands. When the controller is operating in 1:2 frequency ratio mode, program this to the above values divided by 2 and round it up to the next integer value. For LDPDR3, this register is used for the time from a MRW/MRR to a MRW/MRR. Programming Mode: Quasi-dynamic Group 2 and Group 4
19:18	RO	0x0	reserved
17:12	RW	0x04	t_mrd tMRD: Cycles to wait after a mode register write or read. Depending on the connected SDRAM, tMRD represents: DDR2/mDDR: Time from MRS to any command DDR3/4: Time from MRS to MRS command LPDDR2: not used LPDDR3/4: Time from MRS to non-MRS command. When the controller is operating in 1:2 frequency ratio mode, program this to (tMRD/2) and round it up to the next integer value. If C/A parity for DDR4 is used, set to tMRD_PAR(tMOD+PL) instead. Programming Mode: Quasi-dynamic Group 2 and Group 4
11:10	RO	0x0	reserved
9:0	RW	0x00c	t_mod tMOD: Parameter used only in DDR3 and DDR4. Cycles between load mode command and following non-load mode command. If C/A parity for DDR4 is used, set to tMOD_PAR(tMOD+PL) instead. If MPR writes for DDR4 are used, set to tMOD + AL (or tMPD_PAR + AL if C/A parity is also used). Set to tMOD/2 (rounded up to next integer) if controller is operating in 1:2 frequency ratio mode. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG4

Address: Operational Base + offset (0x0110)

DRAMTMG4

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Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RW	0x05	t_rcd tRCD - tAL: Minimum time from activate to read or write command to same bank. When the controller is operating in 1:2 frequency ratio mode, program this to ((tRCD - tAL)/2) and round it up to the next integer value. Minimum value allowed for this register is 1, which implies minimum (tRCD - tAL) value to be 2 when the controller is operating in 1:2 frequency ratio mode. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
23:20	RO	0x0	reserved
19:16	RW	0x4	t_ccd DDR4: tCCD_L: This is the minimum time between two reads or two writes for same bank group. Others: tCCD: This is the minimum time between two reads or two writes. When the controller is operating in 1:2 frequency ratio mode, program this to (tCCD_L/2 or tCCD/2) and round it up to the next integer value. Unit: DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:12	RO	0x0	reserved
11:8	RW	0x4	t_rrd DDR4: tRRD_L: Minimum time between activates from bank "a" to bank "b" for same bank group. Others: tRRD: Minimum time between activates from bank "a" to bank "b" When the controller is operating in 1:2 frequency ratio mode, program this to (tRRD_L/2 or tRRD/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:5	RO	0x0	reserved
4:0	RW	0x05	t_rp tRP: Minimum time from precharge to activate of same bank. When the controller is operating in 1:2 frequency ratio mode, t_rp should be set to RoundDown(RoundUp(tRP/tCK)/2) + 1. When the controller is operating in 1:2 frequency ratio mode in LPDDR4, t_rp should be set to RoundUp(RoundUp(tRP/tCK)/2). Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

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DDRC_DRAMTMG5

Address: Operational Base + offset (0x0114)

DRAMTMG5

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x5	t_cksrx This is the time before Self Refresh Exit that CK is maintained as a valid clock before issuing SRX. Specifies the clock stable time before SRX. Recommended settings: mDDR: 1 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKCKEH DDR2: 1 DDR3: tCKSRX DDR4: tCKSRX When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:20	RO	0x0	reserved
19:16	RW	0x5	t_cksre This is the time after Self Refresh Down Entry that CK is maintained as a valid clock. Specifies the clock disable delay after SRE. Recommended settings: mDDR: 0 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKELCK DDR2: 1 DDR3: max (10 ns, 5 tCK) DDR4: max (10 ns, 5 tCK) (+ PL(parity latency)(*)) (*)Only if CRCPARCTL1.caparity_disable_before_sr=0, this register should be increased by PL. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x04	t_ckesr Minimum CKE low width for Self refresh or Self refresh power down entry to exit timing in memory clock cycles. Recommended settings: mDDR: tRFC LPDDR2: tCKESR LPDDR3: tCKESR LPDDR4: max(tCKE, tSR) DDR2: tCKE DDR3: tCKE + 1 DDR4: tCKE + 1 (+ PL(parity latency)(*)) (*)Only if CRCPARCTL1.caparity_disable_before_sr=0, this register should be increased by PL. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:5	RO	0x0	reserved
4:0	RW	0x03	t_cke Minimum number of cycles of CKE HIGH/LOW during power-down and self refresh. LPDDR2/LPDDR3 mode: Set this to the larger of tCKE or tCKESR LPDDR4 mode: Set this to the larger of tCKE or tSR. Non-LPDDR2/non-LPDDR3/non-LPDDR4 designs: Set this to tCKE value. When the controller is operating in 1:2 frequency ratio mode, program this to (value described above)/2 and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG6

Address: Operational Base + offset (0x0118)

DRAMTMG6

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RW	0x2	t_ckdpde This is the time before Self Refresh Exit that CK is maintained as a valid clock before issuing SRX. Specifies the clock stable time before SRX. Recommended settings: mDDR: 1 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKCKEH DDR2: 1 DDR3: tCKSRX DDR4: tCKSRX When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:20	RO	0x0	reserved
19:16	RW	0x2	t_ckdpdx This is the time before Deep Power Down Exit that CK is maintained as a valid clock before issuing DPDX. Specifies the clock stable time before DPDX. Recommended settings: mDDR: 1 LPDDR2: 2 LPDDR3: 2 When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x5	t_ckcsx This is the time before Clock Stop Exit that CK is maintained as a valid clock before issuing Clock Stop Exit. Specifies the clock stable time before next command after Clock Stop Exit. Recommended settings: mDDR: 1 LPDDR2: tXP + 2 LPDDR3: tXP + 2 LPDDR4: tXP + 2 When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG7

Address: Operational Base + offset (0x011c)

DRAMTMG7

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x2	t_ckpde This is the time after Power Down Entry that CK is maintained as a valid clock. Specifies the clock disable delay after PDE. Recommended settings: mDDR: 0 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKELCK When using DDR2/3/4 SDRAM, this register should be set to the same value as DRAMTMG5.t_cksre. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x2	t_ckpdx This is the time before Power Down Exit that CK is maintained as a valid clock before issuing PDX. Specifies the clock stable time before PDX. Recommended settings: mDDR: 0 LPDDR2: 2 LPDDR3: 2 LPDDR4: 2 When using DDR2/3/4 SDRAM, this register should be set to the same value as DRAMTMG5.t_cksrx. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG8

Address: Operational Base + offset (0x0120)

DRAMTMG8

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x03	t_xs_fast_x32 tXS_FAST: Exit Self Refresh to ZQCL, ZQCS and MRS (only CL, WR, RTP and Geardown mode). When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: This is applicable to only ZQCL/ZQCS commands. Note: Ensure this is less than or equal to t_xs_x32. Programming Mode: Quasi-dynamic Group 2 and Group 4
23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:16	RW	0x03	t_xs_abort_x32 tXS_ABORT: Exit Self Refresh to commands not requiring a locked DLL in Self Refresh Abort. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: Ensure this is less than or equal to t_xs_x32. Programming Mode: Quasi-dynamic Group 2 and Group 4
15	RO	0x0	reserved
14:8	RW	0x44	t_xs_dll_x32 tXSDLL: Exit Self Refresh to commands requiring a locked DLL. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: Used only for DDR2, DDR3 and DDR4 SDRAMs. Programming Mode: Quasi-dynamic Group 2 and Group 4
7	RO	0x0	reserved
6:0	RW	0x05	t_xs_x32 tXS: Exit Self Refresh to commands not requiring a locked DLL. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: Used only for DDR2, DDR3 and DDR4 SDRAMs. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG9

Address: Operational Base + offset (0x0124)

DRAMTMG9

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	ddr4_wr_preamble DDR4 Write preamble mode 0: 1tCK preamble 1: 2tCK preamble Programming Mode: Quasi-dynamic Group 2 and Group 4
29:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:16	RW	0x4	t_ccd_s tCCD_S: This is the minimum time between two reads or two writes for different bank group. For bank switching (from bank "a" to bank "b"), the minimum time is this value + 1. When the controller is operating in 1:2 frequency ratio mode, program this to (tCCD_S/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:12	RO	0x0	reserved
11:8	RW	0x4	t_rrd_s tRRD_S: Minimum time between activates from bank "a" to bank "b" for different bank group. When the controller is operating in 1:2 frequency ratio mode, program this to (tRRD_S/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:6	RO	0x0	reserved
5:0	RW	0x0d	wr2rd_s CWL + PL + BL/2 + tWTR_S Minimum time from write command to read command for different bank group. Includes time for bus turnaround, recovery times, and all per-bank, per-rank, and global constraints. Present only in designs configured to support DDR4. Unit: DFI Clocks. Where: CWL = CAS write latency PL = Parity latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM tWTR_S = internal write to read command delay for different bank group. This comes directly from the SDRAM specification. When the controller is operating in 1:2 mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4

DDRC_DRAMTMG10

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Address: Operational Base + offset (0x0128)

DRAMTMG10

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:16	RW	0x1c	t_sync_gear Indicates the time between MRS command and the sync pulse time. This must be even number of clocks. For DDR4-2666 and DDR4-3200, this parameter is defined as tMOD(min)+4nCK tMOD(min) is greater of 24nCK or 15ns 15ns / .625ns = 24 Max value for this register is 24+4 = 28 When the controller is operating in 1:2 mode, program this to (tSYNC_GEAR/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
15:13	RO	0x0	reserved
12:8	RW	0x18	t_cmd_gear Sync pulse to first valid command. For DDR4-2666 and DDR4-3200, this parameter is defined as tMOD(min). tMOD(min) is greater of 24nCK or 15ns 15ns / .625ns = 24 Max value for this register is 24 When the controller is operating in 1:2 mode, program this to (tCMD_GEAR/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
7:4	RO	0x0	reserved
3:2	RW	0x2	t_gear_setup Geardown setup time. Minimum value of this register is 1. Zero is invalid. For DDR4-2666 and DDR4-3200, this parameter is defined as 2 clks When the controller is operating in 1:2 frequency ratio mode, program this to (tGEAR_setup/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4

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Bit	Attr	Reset Value	Description
1:0	RW	0x2	t_gear_hold Geardown hold time. Minimum value of this register is 1. Zero is invalid. For DDR4-2666 and DDR4-3200, this parameter is defined as 2 clks When the controller is operating in 1:2 frequency ratio mode, program this to (tGEAR_hold/2) and round it up to the next integer value. Unit: DFI Clocks Value After Reset: 0x2 Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG11

Address: Operational Base + offset (0x012c)

DRAMTMG11

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x44	post_mpsm_gap_x32 tXMPDLL: This is the minimum Exit MPSM to commands requiring a locked DLL. When the controller is operating in 1:2 frequency ratio mode, program this to (tXMPDLL/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: Multiples of 32 DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:21	RO	0x0	reserved
20:16	RW	0x0c	t_mpx_lh tMPX_LH: This is the minimum CS_n Low hold time to CKE rising edge. When the controller is operating in 1:2 frequency ratio mode, program this to RoundUp(tMPX_LH/2)+1. Present only in designs configured to support DDR4. Unit: DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x2	t_mpx_s tMPX_S: Minimum time CS setup time to CKE. When the controller is operating in 1:2 frequency ratio mode, program this to (tMPX_S/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:5	RO	0x0	reserved
4:0	RW	0x1c	t_ckmpe tCKMPE: Minimum valid clock requirement after MPSM entry. Present only in designs configured to support DDR4. Unit: DFI Clocks. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG12

Address: Operational Base + offset (0x0130)

DRAMTMG12

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:16	RW	0x2	t_cmdcke tCMDCKE: Delay from valid command to CKE input LOW. Set this to the larger of tESCKE or tCMDCKE When the controller is operating in 1:2 frequency ratio mode, program this to (max(tESCKE, tCMDCKE)/2) and round it up to the next integer value. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:5	RO	0x0	reserved
4:0	RW	0x10	t_mrd_pda tMRD_PDA: This is the Mode Register Set command cycle time in PDA mode. When the controller is operating in 1:2 frequency ratio mode, program this to (tMRD_PDA/2) and round it up to the next integer value. Programming Mode: Quasi-dynamic Group 2 and Group 4

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DDRC_DRAMTMG13

Address: Operational Base + offset (0x0134)

DRAMTMG13

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x1c	odtloff LPDDR4: tODTLoff: This is the latency from CAS-2 command to tODToff reference. When the controller is operating in 1:2 frequency ratio mode, program this to (tODTLoff/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:22	RO	0x0	reserved
21:16	RW	0x20	t_ccd_mw LPDDR4: tCCDMW: This is the minimum time from write or masked write to masked write command for same bank. When the controller is operating in 1:2 frequency ratio mode, program this to (tCCDMW/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:3	RO	0x0	reserved
2:0	RW	0x4	t_ppd LPDDR4: tPPD: This is the minimum time from precharge to precharge command. When the controller is operating in 1:2 frequency ratio mode, program this to (tPPD/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG14

Address: Operational Base + offset (0x0138)

DRAMTMG14

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x0a0	t_xsr tXSR: Exit Self Refresh to any command. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Note: Used only for mDDR/LPDDR2/LPDDR3/LPDDR4 mode. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG15

Address: Operational Base + offset (0x013c)

DRAMTMG15

Bit	Attr	Reset Value	Description
31	RW	0x0	en_dfi_lp_t_stab 1: Enable using tSTAB when exiting DFI LP. Needs to be set when the PHY is stopping the clock during DFI LP to save maximum power. 0: Disable using tSTAB when exiting DFI LP Programming Mode: Quasi-dynamic Group 2 and Group 4
30:8	RO	0x0	reserved
7:0	RW	0x00	t_stab_x32 tSTAB: Stabilization time. It is required in the following two cases for DDR3/DDR4 RDIMM : when exiting power saving mode, if the clock was stopped, after re-enabling it the clock must be stable for a time specified by tSTAB in the case of input clock frequency change (DDR4) after issuing control words that refers to clock timing (Specification: 6us for DDR3, 5us for DDR4) When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Unit: Multiples of 32 DFI clock cycles. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DRAMTMG17

Address: Operational Base + offset (0x0144)

DRAMTMG17

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Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	t_vrcg_enable LPDDR4: tVRCG_ENABLE: VREF high current mode enable time. When the controller is operating in 1:2 frequency ratio mode, program this to (tVRCG_ENABLE/2) and round it up to the next integer value. Unit: DFI clocks Programming Mode: Quasi-dynamic Group 4
15:7	RO	0x0	reserved
6:0	RW	0x00	t_vrcg_disable LPDDR4: tVRCG_ENABLE: VREF high current mode disable time. When the controller is operating in 1:2 frequency ratio mode, program this to (tVRCG_DISABLE/2) and round it up to the next integer value. Unit: DFI clocks Programming Mode: Quasi-dynamic Group 4

DDRC_ZQCTL0

Address: Operational Base + offset (0x0180)

ZQCTL0

Bit	Attr	Reset Value	Description
31	RW	0x0	dis_auto_zq 1: Disable uMCTL2 generation of ZQCS/MPC(ZQ calibration) command. Register DBGCMD.zq_calib_short can be used instead to issue ZQ calibration request from APB module. 0: Internally generate ZQCS/MPC(ZQ calibration) commands based on ZQCTL1.t_zq_short_interval_x1024. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Dynamic

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Bit	Attr	Reset Value	Description
30	RW	0x0	dis_srx_zqcl 1: Disable issuing of ZQCL/MPC(ZQ calibration) command at Self-Refresh/SR-Powerdown exit. Only applicable when run in DDR3 or DDR4 or LPDDR2 or LPDDR3 or LPDDR4 mode. 0: Enable issuing of ZQCL/MPC(ZQ calibration) command at Self-Refresh/SR-Powerdown exit. Only applicable when run in DDR3 or DDR4 or LPDDR2 or LPDDR3 or LPDDR4 mode. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4
29	RW	0x0	zq_resistor_shared 1: Denotes that ZQ resistor is shared between ranks. Means ZQinit/ZQCL/ZQCS/MPC(ZQ calibration) commands are sent to one rank at a time with tZQinit/tZQCL/tZQCS/tZQCAL/tZQLAT timing met between commands so that commands to different ranks do not overlap. 0: ZQ resistor is not shared. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static
28	RW	0x0	dis_mpsmx_zqcl 1: Disable issuing of ZQCL command at Maximum Power Saving Mode exit. Only applicable when run in DDR4 mode. 0: Enable issuing of ZQCL command at Maximum Power Saving Mode exit. Only applicable when run in DDR4 mode. This is only present for designs supporting DDR4 devices. Note: Do not issue ZQCL command at Maximum Power Save Mode exit if the UMCTL2_SHARED_AC configuration parameter is set. Program it to 1'b1. The software can send ZQCS after exiting MPSM mode. Programming Mode: Static
27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26:16	RW	0x200	t_zq_long_nop tZQoper for DDR3/DDR4, tZQCL for LPDDR2/LPDDR3, tZQCAL for LPDDR4: Number of DFI clock cycles of NOP required after a ZQCL (ZQ calibration long)/MPC(ZQ Start) command is issued to SDRAM. When the controller is operating in 1:2 frequency ratio mode: DDR3/DDR4: program this to tZQoper/2 and round it up to the next integer value. LPDDR2/LPDDR3: program this to tZQCL/2 and round it up to the next integer value. LPDDR4: program this to tZQCAL/2 and round it up to the next integer value. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static
15:10	RO	0x0	reserved
9:0	RW	0x040	t_zq_short_nop tZQCS for DDR3/DD4/LPDDR2/LPDDR3, tZQLAT for LPDDR4: Number of DFI clock cycles of NOP required after a ZQCS (ZQ calibration short)/MPC(ZQ Latch) command is issued to SDRAM. When the controller is operating in 1:2 frequency ratio mode, program this to tZQCS/2 and round it up to the next integer value. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static

DDRC_ZQCTL1

Address: Operational Base + offset (0x0184)

ZQCTL1

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RW	0x020	t_zq_reset_nop tZQReset: Number of DFI clock cycles of NOP required after a ZQReset (ZQ calibration Reset) command is issued to SDRAM. When the controller is operating in 1:2 frequency ratio mode, program this to tZQReset/2 and round it up to the next integer value. This is only present for designs supporting LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static

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Bit	Attr	Reset Value	Description
19:0	RW	0x00100	t_zq_short_interval_x1024 Average interval to wait between automatically issuing ZQCS (ZQ calibration short)/MPC(ZQ calibration) commands to DDR3/DDR4/LPDDR2/LPDDR3/LPDDR4 devices. Meaningless, if ZQCTL0.dis_auto_zq=1. Unit: 1024 DFI clock cycles. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static

DDRC_ZQCTL2

Address: Operational Base + offset (0x0188)

ZQCTL2

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	zq_reset Setting this register bit to 1 triggers a ZQ Reset operation. When the ZQ Reset operation is complete, the uMCTL2 automatically clears this bit. It is recommended NOT to set this signal if in Init, Self-Refresh(except LPDDR4) or SRPowerdown(LPDDR4) or Deep power-down operating modes. This is only present for designs supporting LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Dynamic

DDRC_ZQSTAT

Address: Operational Base + offset (0x018c)

ZQSTAT

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	zq_reset_busy SoC core may initiate a ZQ Reset operation only if this signal is low. This signal goes high in the clock after the uMCTL2 accepts the ZQ Reset request. It goes low when the ZQ Reset command is issued to the SDRAM and the associated NOP period is over. It is recommended not to perform ZQ Reset commands when this signal is high. 0: Indicates that the SoC core can initiate a ZQ Reset operation 1: Indicates that ZQ Reset operation is in progress Programming Mode: Dynamic

DDRC_DFITMG0

Address: Operational Base + offset (0x0190)

DFITMG0

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RW	0x07	dfi_t_ctrl_delay Specifies the number of DFI clock cycles after an assertion or de-assertion of the DFI control signals that the control signals at the PHY-DRAM interface reflect the assertion or de-assertion. If the DFI clock and the memory clock are not phase-aligned, this timing parameter should be rounded up to the next integer value. Programming Mode: Quasi-dynamic Group 4
23	RW	0x0	dfi_rddata_use_dfi_phy_clk Defines whether dfi_rddata_en/dfi_rddata/dfi_rddata_valid is generated using HDR (DFI clock) or SDR (DFI PHY clock) values. Selects whether value in DFITMG0.dfi_t_rddata_en is in terms of HDR (DFI clock) or SDR (DFI PHY clock) cycles: 0: in terms of HDR (DFI clock) cycles, Only support HDR DFI clock. 1: in terms of SDR (DFI PHY clock) cycles Programming Mode: Static
22:16	RW	0x02	dfi_t_rddata_en Time from the assertion of a read command on the DFI interface to the assertion of the dfi_rddata_en signal. Refer to PHY specification for correct value. This corresponds to the DFI parameter trddata_en. Unit: DFI clock cycles Programming Mode: Quasi-dynamic Group 1 and Group 4

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Bit	Attr	Reset Value	Description
15	RW	0x0	dfi_wrdata_use_dfi_phy_clk Defines whether dfi_wrdata_en/dfi_wrdata/dfi_wrdata_mask is generated using HDR (DFI clock) or SDR (DFI PHY clock) values Selects whether value in DFITMG0.dfi_tphy_wrlat is in terms of HDR (DFI clock) or SDR (DFI PHY clock) cycles Selects whether value in DFITMG0.dfi_tphy_wrdata is in terms of HDR (DFI clock) or SDR (DFI PHY clock) cycles 0: in terms of HDR (DFI clock) cycles, Only support HDR DFI clock. 1: in terms of SDR (DFI PHY clock) cycles Programming Mode: Static
14	RO	0x0	reserved
13:8	RW	0x00	dfi_tphy_wrdata Specifies the number of clock cycles between when dfi_wrdata_en is asserted to when the associated write data is driven on the dfi_wrdata signal. This corresponds to the DFI timing parameter tphy_wrdata. Refer to PHY specification for correct value. Note, max supported value is 8. Unit: DFI clock cycles. Programming Mode: Quasi-dynamic Group 4
7:6	RO	0x0	reserved
5:0	RW	0x02	dfi_tphy_wrlat Write latency Number of clocks from the write command to write data enable (dfi_wrdata_en). This corresponds to the DFI timing parameter tphy_wrlat. Refer to PHY specification for correct value. For LPDDR4, dfi_tphy_wrlat>60 is not supported. Unit: DFI clock cycles Programming Mode: Quasi-dynamic Group 1 and Group 4

DDRC_DFITMG1

Address: Operational Base + offset (0x0194)

DFITMG1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:28	RW	0x0	dfi_t_cmd_lat Specifies the number of DFI PHY clock cycles between when the dfi_cs signal is asserted and when the associated command is driven. This field is used for CAL mode, should be set to '0' or the value which matches the CAL mode register setting in the DRAM. If the PHY can add the latency for CAL mode, this should be set to '0'. Valid Range: 0, 3, 4, 5, 6, and 8 Programming Mode: Quasi-dynamic Group 2 and Group 4
27:26	RO	0x0	reserved
25:24	RW	0x0	dfi_t_parin_lat Specifies the number of DFI PHY clock cycles between when the dfi_cs signal is asserted and when the associated dfi_parity_in signal is driven. Programming Mode: Quasi-dynamic Group 4
23:21	RO	0x0	reserved
20:16	RW	0x00	dfi_t_wrdata_delay Specifies the number of DFI clock cycles between when the dfi_wrdata_en signal is asserted and when the corresponding write data transfer is completed on the DRAM bus. This corresponds to the DFI timing parameter twrdata_delay. Refer to PHY specification for correct value. Value to be programmed is in terms of DFI clocks, not PHY clocks. In FREQ_RATIO=2, divide PHY's value by 2 and round up to next integer. Programming Mode: Quasi-dynamic Group 4
15:13	RO	0x0	reserved
12:8	RW	0x04	dfi_t_dram_clk_disable Specifies the number of DFI clock cycles from the assertion of the dfi_dram_clk_disable signal on the DFI until the clock to the DRAM memory devices, at the PHY-DRAM boundary, maintains a low value. If the DFI clock and the memory clock are not phase aligned, this timing parameter should be rounded up to the next integer value. Programming Mode: Quasi-dynamic Group 4
7:5	RO	0x0	reserved
4:0	RW	0x04	dfi_t_dram_clk_enable Specifies the number of DFI clock cycles from the deassertion of the dfi_dram_clk_disable signal on the DFI until the first valid rising edge of the clock to the DRAM memory devices, at the PHY-DRAM boundary. If the DFI clock and the memory clock are not phase aligned, this timing parameter should be rounded up to the next integer value. Programming Mode: Quasi-dynamic Group 4

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DDRC_DFILPCFG0

Address: Operational Base + offset (0x0198)

DFILPCFG0

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RW	0x07	dfi_tlp_resp Setting in DFI clock cycles for DFI's tlp_resp time. Same value is used for both Power Down, Self Refresh, Deep Power Down and Maximum Power Saving modes. DFI 2.1 specification onwards, recommends using a fixed value of 7 always. Programming Mode: Static
23:20	RW	0x0	dfi_lp_wakeup_dpd Value in DFI clock cycles to drive on dfi_lp_wakeup signal when Deep Power Down mode is entered. Determines the DFI's tlp_wakeup time: 0x0 - 16 cycles 0x1 - 32 cycles 0x2 - 64 cycles 0x3 - 128 cycles 0x4 - 256 cycles 0x5 - 512 cycles 0x6 - 1024 cycles 0x7 - 2048 cycles 0x8 - 4096 cycles 0x9 - 8192 cycles 0xA - 16384 cycles 0xB - 32768 cycles 0xC - 65536 cycles 0xD - 131072 cycles 0xE - 262144 cycles 0xF - Unlimited This is only present for designs supporting mDDR or LPDDR2/LPDDR3 devices. Programming Mode: Static
19:17	RO	0x0	reserved
16	RW	0x0	dfi_lp_en_dpd Enables DFI Low Power interface handshaking during Deep Power Down Entry/Exit. 0 - Disabled; 1 - Enabled This is only present for designs supporting mDDR or LPDDR2/LPDDR3 devices. Programming Mode: Static

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Bit	Attr	Reset Value	Description
15:12	RW	0x0	dfi_lp_wakeup_sr Value in DFI clpck cycles to drive on dfi_lp_wakeup signal when Self Refresh mode is entered. Determines the DFI's tlp_wakeup time: 0x0 - 16 cycles 0x1 - 32 cycles 0x2 - 64 cycles 0x3 - 128 cycles 0x4 - 256 cycles 0x5 - 512 cycles 0x6 - 1024 cycles 0x7 - 2048 cycles 0x8 - 4096 cycles 0x9 - 8192 cycles 0xA - 16384 cycles 0xB - 32768 cycles 0xC - 65536 cycles 0xD - 131072 cycles 0xE - 262144 cycles 0xF - Unlimited Programming Mode: Static
11:9	RO	0x0	reserved
8	RW	0x0	dfi_lp_en_sr Enables DFI Low Power interface handshaking during Self Refresh Entry/Exit. 0 - Disabled; 1 - Enabled Programming Mode: Static

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Bit	Attr	Reset Value	Description
7:4	RW	0x0	dfi_lp_wakeup_pd Value in DFI clock cycles to drive on dfi_lp_wakeup signal when Power Down mode is entered. Determines the DFI's tlp_wakeup time: 0x0 - 16 cycles 0x1 - 32 cycles 0x2 - 64 cycles 0x3 - 128 cycles 0x4 - 256 cycles 0x5 - 512 cycles 0x6 - 1024 cycles 0x7 - 2048 cycles 0x8 - 4096 cycles 0x9 - 8192 cycles 0xA - 16384 cycles 0xB - 32768 cycles 0xC - 65536 cycles 0xD - 131072 cycles 0xE - 262144 cycles 0xF - Unlimited Programming Mode: Static
3:1	RO	0x0	reserved
0	RW	0x0	dfi_lp_en_pd Enables DFI Low Power interface handshaking during Power Down Entry/Exit. 0: Disabled; 1: Enabled Programming Mode: Static

DDRC_DFILPCFG1

Address: Operational Base + offset (0x019c)

DFILPCFG1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x0	dfi_lp_wakeup_mpsm Value in DFI clock cycles to drive on dfi_lp_wakeup signal when Maximum Power Saving Mode is entered. Determines the DFI's tlp_wakeup time: 0x0 - 16 cycles 0x1 - 32 cycles 0x2 - 64 cycles 0x3 - 128 cycles 0x4 - 256 cycles 0x5 - 512 cycles 0x6 - 1024 cycles 0x7 - 2048 cycles 0x8 - 4096 cycles 0x9 - 8192 cycles 0xA - 16384 cycles 0xB - 32768 cycles 0xC - 65536 cycles 0xD - 131072 cycles 0xE - 262144 cycles 0xF - Unlimited This is only present for designs supporting DDR4 devices. Programming Mode: Static
3:1	RO	0x0	reserved
0	RW	0x0	dfi_lp_en_mpsm Enables DFI Low Power interface handshaking during Maximum Power Saving Mode Entry/Exit. 0 - Disabled; 1 - Enabled This is only present for designs supporting DDR4 devices. Programming Mode: Static

DDRC_DFIUPD0

Address: Operational Base + offset (0x01a0)

DFIUPD0

Bit	Attr	Reset Value	Description
31	RW	0x0	dis_auto_ctrlupd When '1', disable the automatic dfi_ctrlupd_req generation by the uMCTL2. The core must issue the dfi_ctrlupd_req signal using register DBGCMD.ctrlupd. When '0', uMCTL2 issues dfi_ctrlupd_req periodically. Programming Mode: Quasi-dynamic Group 3

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Bit	Attr	Reset Value	Description
30	RW	0x0	dis_auto_ctrlupd_srx When '1', disable the automatic dfi_ctrlupd_req generation by the uMCTL2 at self-refresh exit. When '0', uMCTL2 issues a dfi_ctrlupd_req before or after exiting self-refresh, depending on DFIUPD0.ctrlupd_pre_srx. Programming Mode: Static
29	RW	0x0	ctrlupd_pre_srx Selects dfi_ctrlupd_req requirements at SRX: 0 : send ctrlupd after SRX 1 : send ctrlupd before SRX If DFIUPD0.dis_auto_ctrlupd_srx=1, this register has no impact, because no dfi_ctrlupd_req will be issued when SRX. Programming Mode: Static
28:26	RO	0x0	reserved
25:16	RW	0x040	dfi_t_ctrlup_max Specifies the maximum number of DFI clock cycles that the dfi_ctrlupd_req signal can assert. Lowest value to assign to this variable is 0x40. Programming Mode: Static
15:10	RO	0x0	reserved
9:0	RW	0x003	dfi_t_ctrlup_min Specifies the minimum number of DFI clock cycles that the dfi_ctrlupd_req signal must be asserted. The uMCTL2 expects the PHY to respond within this time. If the PHY does not respond, the uMCTL2 will de-assert dfi_ctrlupd_req after dfi_t_ctrlup_min + 2 cycles. Lowest value to assign to this variable is 0x3. Programming Mode: Static

DDRC_DFIUPD1

Address: Operational Base + offset (0x01a4)

DFIUPD1

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	dfi_t_ctrlupd_interval_min_x1024 This is the minimum amount of time between uMCTL2 initiated DFI update requests (which is executed whenever the uMCTL2 is idle). Set this number higher to reduce the frequency of update requests, which can have a small impact on the latency of the first read request when the uMCTL2 is idle. Minimum allowed value for this field is 1. Unit: 1024 DFI clock cycles Programming Mode: Static
15:8	RO	0x0	reserved
7:0	RW	0x01	dfi_t_ctrlupd_interval_max_x1024 This is the maximum amount of time between uMCTL2 initiated DFI update requests. This timer resets with each update request; when the timer expires dfi_ctrlupd_req is sent and traffic is blocked until the dfi_ctrlupd_ackx is received. PHY can use this idle time to recalibrate the delay lines to the DLLs. The DFI controller update is also used to reset PHY FIFO pointers in case of data capture errors. Updates are required to maintain calibration over PVT, but frequent updates may impact performance. Minimum allowed value for this field is 1. Note: Value programmed for DFIUPD1.dfi_t_ctrlupd_interval_max_x1024 must be greater than DFIUPD1.dfi_t_ctrlupd_interval_min_x1024. Unit: 1024 DFI clock cycles Programming Mode: Static

DDRC_DFIUPD2

Address: Operational Base + offset (0x01a8)

DFIUPD2

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	dfi_phyupd_en Enables the support for acknowledging PHY-initiated updates: 0 - Disabled; 1 - Enabled Programming Mode: Static

DDRC_DFIMISC

Address: Operational Base + offset (0x01b0)

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DFIMISC

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	dfi_frequency Indicates the operating frequency of the system. The number of supported frequencies and the mapping of signal values to clock frequencies are defined by the PHY. Programming Mode: Quasi-dynamic Group 1
7:6	RO	0x0	reserved
5	RW	0x0	dfi_init_start PHY init start request signal.When asserted it triggers the PHY init start request Programming Mode: Quasi-dynamic Group 3
4	RW	0x0	ctl_idle_en Enables support of ctl_idle signal, which is non-DFI related pin specific to certain Synopsys PHYs. See signal description of ctl_idle signal for further details of ctl_idle functionality. Programming Mode: Static
3	RW	0x0	share_dfi_dram_clk_disable Indicate dfi_dram_clk_disable is shared for two channels or not. 1: Share mode 0: Not share. In Shared mode, Controller does not request PHY stop clock while any rank of any channel has not disable clock. Note: when dfi_dram_clk_disable is shared by two channels, an additional DFF is inserted, that will cause dram clock enable is delay one cycle. Suggest set dfi_t_dram_clk_enable value to Tdram_clk_enable+1. Tdram_clk_enable value is from PHY, which indicate how many cycles from dfi_dram_clk_disable de-assert to dram clock output. Programming Mode: Static
2	RW	0x0	dfi_data_cs_polarity Defines polarity of dfi_wrdata_cs and dfi_rddata_cs signals. 0: Signals are active low; 1: Signals are active high Programming Mode: Static
1	RW	0x0	phy_dbi_mode DBI implemented in DDRC or PHY. 0 - DDRC implements DBI functionality. 1 - PHY implements DBI functionality. Present only in designs configured to support DDR4 and LPDDR4. Programming Mode: Static
0	RW	0x1	dfi_init_complete_en PHY initialization complete enable signal. When asserted the dfi_init_complete signal can be used to trigger SDRAM initialisation Programming Mode: Quasi-dynamic Group 3

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DDRC_DFITMG2

Address: Operational Base + offset (0x01b4)

DFITMG2

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:8	RW	0x02	dfi_tphy_rdcslat Number of DFI PHY clock cycles between when a read command is sent on the DFI control interface and when the associated dfi_rddata_cs signal is asserted. This corresponds to the DFI timing parameter tphy_rdcslat. Refer to PHY specification for correct value. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:6	RO	0x0	reserved
5:0	RW	0x02	dfi_tphy_wrcslat Number of DFI PHY clock cycles between when a write command is sent on the DFI control interface and when the associated dfi_wrdata_cs signal is asserted. This corresponds to the DFI timing parameter tphy_wrcslat. Refer to PHY specification for correct value. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_DFITMG3

Address: Operational Base + offset (0x01b8)

DFITMG3

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	dfi_t_geardown_delay The delay from dfi_geardown_en assertion to the time of the PHY being ready to receive commands. Refer to PHY specification for correct value. When the controller is operating in 1:2 frequency ratio mode, program this to (tgeardown_delay/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Static

DDRC_DFISTAT

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Address: Operational Base + offset (0x01bc)

DFISTAT

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	dfi_lp_ack Stores the value of the dfi_lp_ack input to the controller. Programming Mode: Dynamic
0	RO	0x0	dfi_init_complete The status flag register which announces when the DFI initialization has been completed. The DFI INIT triggered by dfi_init_start signal and then the dfi_init_complete flag is polled to know when the initialization is done. Programming Mode: Dynamic

DDRC_DBICTL

Address: Operational Base + offset (0x01c0)

DBICTL

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	rd_dbi_en Read DBI enable signal in DDRC. 0 - Read DBI is disabled. 1 - Read DBI is enabled. This signal must be set the same value as DRAM's mode register. DDR4: MR5 bit A12. When x4 devices are used, this signal must be set to 0. LPDDR4: MR3[6] Programming Mode: Quasi-dynamic Group 1
1	RW	0x0	wr_dbi_en Write DBI enable signal in DDRC. 0 - Write DBI is disabled. 1 - Write DBI is enabled. This signal must be set the same value as DRAM's mode register. DDR4: MR5 bit A11. When x4 devices are used, this signal must be set to 0. LPDDR4: MR3[7] Programming Mode: Quasi-dynamic Group 1

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Bit	Attr	Reset Value	Description
0	RW	0x1	dm_en DM enable signal in DDRC. 0 - DM is disabled; 1 - DM is enabled. This signal must be set the same logical value as DRAM's mode register. DDR4: Set this to same value as MR5 bit A10. When x4 devices are used, this signal must be set to 0. LPDDR4: Set this to inverted value of MR13[5] which is opposite polarity from this signal Programming Mode: Quasi-dynamic Group 3

DDRC_DFIPHYMSTR

Address: Operational Base + offset (0x01c4)

DFIPHYMSTR

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	dfi_phymstr_en Enables the PHY Master Interface: 0 - Disabled; 1 - Enabled Programming Mode: Dynamic

DDRC_ADDRMAP0

Address: Operational Base + offset (0x0200)

ADDRMAP0

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:16	RW	0x00	addrmap_dch_bit0 Selects the HIF address bit used as data channel address bit 0. Valid Range: 0 to 30, and 31 (Traffic constraints apply based on the register value when UMCTL2_EXCL_ACCESS>0. See Exclusive Access section for details.) Internal Base: 2 The selected address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then channel bit is set to 0. Programming Mode: Static

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Bit	Attr	Reset Value	Description
15:13	RO	0x0	reserved
12:8	RW	0x00	addrmap_cs_bit1 Selects the HIF address bit used as rank address bit 1. Valid Range: 0 to 28, and 31 Internal Base: 7 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then rank address bit 1 is set to 0. Programming Mode: Static
7:5	RO	0x0	reserved
4:0	RW	0x00	addrmap_cs_bit0 Selects the HIF address bit used as rank address bit 0. Valid Range: 0 to 29, and 31 Internal Base: 6 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then rank address bit 0 is set to 0. Programming Mode: Static

DDRC_ADDRMAP1

Address: Operational Base + offset (0x0204)

ADDRMAP1

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:16	RW	0x00	addrmap_bank_b2 Selects the HIF address bit used as bank address bit 2. Valid Range: 0 to 31 and 63 Internal Base: 4 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 63 and then bank address bit 2 is set to 0. Programming Mode: Static
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x00	addrmap_bank_b1 Selects the HIF address bits used as bank address bit 1. Valid Range: 0 to 32 and 63 Internal Base: 3 The selected HIF address bit for each of the bank address bits is determined by adding the internal base to the value of this field. If unused, set to 63 and then bank address bit 1 is set to 0. Programming Mode: Static
7:6	RO	0x0	reserved
5:0	RW	0x00	addrmap_bank_b0 Selects the HIF address bits used as bank address bit 0. Valid Range: 0 to 32 and 63 Internal Base: 2 The selected HIF address bit for each of the bank address bits is determined by adding the internal base to the value of this field. If unused, set to 63 and then bank address bit 0 is set to 0. Programming Mode: Static

DDRC_ADDRMAP2

Address: Operational Base + offset (0x0208)

ADDRMAP2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	addrmap_col_b5 Full bus width mode: Selects the HIF address bit used as column address bit 5. Half bus width mode: Selects the HIF address bit used as column address bit 6. Valid Range: 0 to 7, and 15 Internal Base: 5 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then this column address bit is set to 0. Programming Mode: Static
23:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:16	RW	0x0	addrmap_col_b4 Full bus width mode: Selects the HIF address bit used as column address bit 4. Half bus width mode: Selects the HIF address bit used as column address bit 5. Valid Range: 0 to 7, and 15 Internal Base: 4 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then this column address bit is set to 0. Programming Mode: Static
15:12	RO	0x0	reserved
11:8	RW	0x0	addrmap_col_b3 Full bus width mode: Selects the HIF address bit used as column address bit 3. Half bus width mode: Selects the HIF address bit used as column address bit 4. Valid Range: 0 to 7 Internal Base: 3 The selected HIF address bit is determined by adding the internal base to the value of this field. Note, if UMCTL2_INCL_ARB=1, MEMC_BURST_LENGTH=16, Full bus width (MSTR.data_bus_width=00) and BL16 (MSTR.burst_rdwr=1000), it is recommended to program this to 0. Programming Mode: Static
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	addrmap_col_b2 Full bus width mode: Selects the HIF address bit used as column address bit 2. Half bus width mode: Selects the HIF address bit used as column address bit 3. Valid Range: 0 to 7 Internal Base: 2 The selected HIF address bit is determined by adding the internal base to the value of this field. Note, if UMCTL2_INCL_ARB=1 and MEMC_BURST_LENGTH=8, it is required to program this to 0 unless: in Half or Quarter bus width (MSTR.data_bus_width!=00) and PCCFG.bl_exp_mode==1 and either In DDR4 and ADDRMAP8.addrmap_bg_b0==0 or In LPDDR4 and ADDRMAP1.addrmap_bank_b0==0 If UMCTL2_INCL_ARB=1 and MEMC_BURST_LENGTH=16, it is required to program this to 0 unless: in Half or Quarter bus width (MSTR.data_bus_width!=00) and PCCFG.bl_exp_mode==1 and In DDR4 and ADDRMAP8.addrmap_bg_b0==0 Otherwise, if MEMC_BURST_LENGTH=8 and Full Bus Width (MSTR.data_bus_width==00), it is recommended to program this to 0 so that HIF[2] maps to column address bit 2. If MEMC_BURST_LENGTH=16 and Full Bus Width (MSTR.data_bus_width==00), it is recommended to program this to 0 so that HIF[2] maps to column address bit 2. If MEMC_BURST_LENGTH=16 and Half Bus Width (MSTR.data_bus_width==01), it is recommended to program this to 0 so that HIF[2] maps to column address bit 3. Programming Mode: Static

DDRC_ADDRMAP3

Address: Operational Base + offset (0x020c)

ADDRMAP3

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
28:24	RW	0x00	addrmap_col_b9 Full bus width mode: Selects the HIF address bit used as column address bit 9. Half bus width mode: Selects the HIF address bit used as column address bit 11 (10 in LPDDR2/LPDDR3 mode). Valid Range: 0 to 7, and 31. Internal Base: 9 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then this column address bit is set to 0. Note: Per JEDEC DDR2/3/mDDR specification, column address bit 10 is reserved for indicating auto-precharge, and hence no source address bit can be mapped to column address bit 10. In LPDDR2/LPDDR3, there is a dedicated bit for autoprecharge in the CA bus and hence column bit 10 is used. Programming Mode: Static
23:21	RO	0x0	reserved
20:16	RW	0x00	addrmap_col_b8 Full bus width mode: Selects the HIF address bit used as column address bit 8. Half bus width mode: Selects the HIF address bit used as column address bit 9. Valid Range: 0 to 7, and 31. Internal Base: 8 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then this column address bit is set to 0. Note: Per JEDEC DDR2/3/mDDR specification, column address bit 10 is reserved for indicating auto-precharge, and hence no source address bit can be mapped to column address bit 10. In LPDDR2/LPDDR3, there is a dedicated bit for autoprecharge in the CA bus and hence column bit 10 is used. Programming Mode: Static
15:13	RO	0x0	reserved
12:8	RW	0x00	addrmap_col_b7 Full bus width mode: Selects the HIF address bit used as column address bit 7. Half bus width mode: Selects the HIF address bit used as column address bit 8. Valid Range: 0 to 7, and 31. Internal Base: 7 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then this column address bit is set to 0. Programming Mode: Static

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Bit	Attr	Reset Value	Description
7:4	RO	0x0	reserved
3:0	RW	0x0	addrmap_col_b6 Full bus width mode: Selects the HIF address bit used as column address bit 6. Half bus width mode: Selects the HIF address bit used as column address bit 7. Valid Range: 0 to 7, and 15 Internal Base: 6 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then this column address bit is set to 0. Programming Mode: Static

DDRC_ADDRMAP4

Address: Operational Base + offset (0x0210)

ADDRMAP4

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	addrmap_col_b11 Full bus width mode: Selects the HIF address bit used as column address bit 13 (11 in LPDDR2/LPDDR3 mode). Half bus width mode: Unused. To make it unused, this should be tied to 4'hF. Valid Range: 0 to 7, and 31. Internal Base: 11 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then this column address bit is set to 0. Note: Per JEDEC DDR2/3/mDDR specification, column address bit 10 is reserved for indicating auto-precharge, and hence no source address bit can be mapped to column address bit 10. In LPDDR2/LPDDR3, there is a dedicated bit for autoprecharge in the CA bus and hence column bit 10 is used. Programming Mode: Static
7:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	addrmap_col_b10 Full bus width mode: Selects the HIF address bit used as column address bit 11 (10 in LPDDR2/LPDDR3 mode). Half bus width mode: Selects the HIF address bit used as column address bit 13 (11 in LPDDR2/LPDDR3 mode). Valid Range: 0 to 7, and 31. Internal Base: 10 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 31 and then this column address bit is set to 0. Note: Per JEDEC DDR2/3/mDDR specification, column address bit 10 is reserved for indicating auto-precharge, and hence no source address bit can be mapped to column address bit 10. In LPDDR2/LPDDR3, there is a dedicated bit for autoprecharge in the CA bus and hence column bit 10 is used. Programming Mode: Static

DDRC_ADDRMAP5

Address: Operational Base + offset (0x0214)

ADDRMAP5

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	addrmap_row_b11 Selects the HIF address bit used as row address bit 11. Valid Range: 0 to 11, and 15 Internal Base: 17 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 11 is set to 0. Programming Mode: Static
23:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:16	RW	0x0	addrmap_row_b2_10 Selects the HIF address bits used as row address bits 2 to 10. Valid Range: 0 to 11, and 15 Internal Base: 8 (for row address bit 2), 9 (for row address bit 3), 10 (for row address bit 4) etc increasing to 16 (for row address bit 10) The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. When value 15 is used the values of row address bits 2 to 10 are defined by registers ADDRMAP9, ADDRMAP10, ADDRMAP11. Programming Mode: Static
15:12	RO	0x0	reserved
11:8	RW	0x0	addrmap_row_b1 Selects the HIF address bits used as row address bit 1. Valid Range: 0 to 11 Internal Base: 7 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. Programming Mode: Static
7:4	RO	0x0	reserved
3:0	RW	0x0	addrmap_row_b0 Selects the HIF address bits used as row address bit 0. Valid Range: 0 to 11 Internal Base: 6 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. Programming Mode: Static

DDRC_ADDRMAP6

Address: Operational Base + offset (0x0218)

ADDRMAP6

Bit	Attr	Reset Value	Description
31	RW	0x0	lpddr3_6gb_12gb Set this to 1 if there is an LPDDR3 SDRAM 6Gb or 12Gb device in use. 1 - LPDDR3 SDRAM 6Gb/12Gb device in use. Every address having row[14:13]==2'b11 is considered as invalid 0 - non-LPDDR3 6Gb/12Gb device in use. All addresses are valid Present only in designs configured to support LPDDR3. Programming Mode: Static

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	lpddr4_6gb_12gb_24gb Indicates what type of LPDDR4 SDRAM device is in use. 2'b00: No LPDDR4 SDRAM 6Gb/12Gb/24Gb device in use. All addresses are valid 2'b01: LPDDR4 SDRAM 6Gb device in use. Every address having row[14:13]==2'b11 is considered as invalid 2'b10: LPDDR4 SDRAM 12Gb device in use. Every address having row[15:14]==2'b11 is considered as invalid 2'b11: LPDDR4 SDRAM 24Gb device in use. Unsupported Present only in designs configured to support LPDDR4. Programming Mode: Static
28	RO	0x0	reserved
27:24	RW	0x0	addrmap_row_b15 Selects the HIF address bit used as row address bit 15. Valid Range: 0 to 11, and 15 Internal Base: 21 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 15 is set to 0. Programming Mode: Static
23:20	RO	0x0	reserved
19:16	RW	0x0	addrmap_row_b14 Selects the HIF address bit used as row address bit 14. Valid Range: 0 to 11, and 15 Internal Base: 20 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 14 is set to 0. Programming Mode: Static
15:12	RO	0x0	reserved
11:8	RW	0x0	addrmap_row_b13 Selects the HIF address bit used as row address bit 13. Valid Range: 0 to 11, and 15 Internal Base: 19 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 13 is set to 0. Programming Mode: Static
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	addrmap_row_b12 Selects the HIF address bit used as row address bit 12. Valid Range: 0 to 11, and 15 Internal Base: 18 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 12 is set to 0. Programming Mode: Static

DDRC_ADDRMAP7

Address: Operational Base + offset (0x021c)

ADDRMAP7

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	addrmap_row_b17 Selects the HIF address bit used as row address bit 17. Valid Range: 0 to 11, and 15 Internal Base: 23 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 17 is set to 0. Programming Mode: Static
7:4	RO	0x0	reserved
3:0	RW	0x0	addrmap_row_b16 Selects the HIF address bit used as row address bit 16. Valid Range: 0 to 11, and 15 Internal Base: 22 The selected HIF address bit is determined by adding the internal base to the value of this field. If unused, set to 15 and then row address bit 16 is set to 0. Programming Mode: Static

DDRC_ADDRMAP8

Address: Operational Base + offset (0x0220)

ADDRMAP8

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:8	RW	0x00	addrmap_bg_b1 Selects the HIF address bits used as bank group address bit 1. Valid Range: 0 to 32, and 63 Internal Base: 3 The selected HIF address bit for each of the bank group address bits is determined by adding the internal base to the value of this field. If unused, set to 63 and then bank group address bit 1 is set to 0. Programming Mode: Static
7:6	RO	0x0	reserved
5:0	RW	0x00	addrmap_bg_b0 Selects the HIF address bits used as bank group address bit 0. Valid Range: 0 to 32 and 63 Internal Base: 2 The selected HIF address bit for each of the bank group address bits is determined by adding the internal base to the value of this field. If unused, set to 63 and then bank group address bit 0 is set to 0. Programming Mode: Static

DDRC_ADDRMAP9

Address: Operational Base + offset (0x0224)

ADDRMAP9

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	addrmap_row_b5 Selects the HIF address bits used as row address bit 5. Valid Range: 0 to 11 Internal Base: 11 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static
23:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:16	RW	0x0	addrmap_row_b4 Selects the HIF address bits used as row address bit 4. Valid Range: 0 to 11 Internal Base: 10 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static
15:12	RO	0x0	reserved
11:8	RW	0x0	addrmap_row_b3 Selects the HIF address bits used as row address bit 3. Valid Range: 0 to 11 Internal Base: 9 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static
7:4	RO	0x0	reserved
3:0	RW	0x0	addrmap_row_b2 Selects the HIF address bits used as row address bit 2. Valid Range: 0 to 11 Internal Base: 8 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static

DDRC_ADDRMAP10

Address: Operational Base + offset (0x0228)

ADDRMAP10

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RW	0x0	addrmap_row_b9 Selects the HIF address bits used as row address bit 9. Valid Range: 0 to 11 Internal Base: 15 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static
23:20	RO	0x0	reserved
19:16	RW	0x0	addrmap_row_b8 Selects the HIF address bits used as row address bit 8. Valid Range: 0 to 11 Internal Base: 14 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static
15:12	RO	0x0	reserved
11:8	RW	0x0	addrmap_row_b7 Selects the HIF address bits used as row address bit 7. Valid Range: 0 to 11 Internal Base: 13 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static
7:4	RO	0x0	reserved
3:0	RW	0x0	addrmap_row_b6 Selects the HIF address bits used as row address bit 6. Valid Range: 0 to 11 Internal Base: 12 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static

DDRC_ADDRMAP11

Address: Operational Base + offset (0x022c)

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ADDRMAP11

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	addrmap_row_b10 Selects the HIF address bits used as row address bit 10. Valid Range: 0 to 11 Internal Base: 16 The selected HIF address bit for each of the row address bits is determined by adding the internal base to the value of this field. This register field is used only when ADDRMAP5.addrmap_row_b2_10 is set to value 15. Programming Mode: Static

DDRC_ODTCFG

Address: Operational Base + offset (0x0240)

ODTCFG

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x4	wr_odt_hold DFI PHY clock cycles to hold ODT for a write command. The minimum supported value is 2. Recommended values: DDR2: BL8: 0x5 (DDR2-400/533/667), 0x6 (DDR2-800), 0x7(DDR2-1066); BL4: 0x3 (DDR2-400/533/667), 0x4 (DDR2-800), 0x5 (DDR2-1066) DDR3: BL8: 0x6 DDR4: BL8: 5 + WR_PREAMBLE + CRC_MODE. WR_PREAMBLE = 1 (1tCK write preamble), 2 (2tCK write preamble). CRC_MODE = 0 (not CRC mode), 1 (CRC mode) LPDDR3: BL8: 7 + RU(tODTon(max)/tCK) Programming Mode: Quasi-dynamic Group 1 and Group 4
23:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20:16	RW	0x00	wr_odt_delay The delay, in DFI PHY clock cycles, from issuing a write command to setting ODT values associated with that command. ODT setting must remain constant for the entire time that DQS is driven by the uMCTL2. Recommended values: DDR2: CWL + AL - 3 (DDR2-400/533/667), CWL + AL - 4 (DDR2-800), CWL + AL - 5 (DDR2-1066). If (CWL + AL - 3 < 0), uMCTL2 does not support ODT for write operation. DDR3: 0x0 DDR4: DFITMG1.dfi_t_cmd_lat (to adjust for CAL mode) LPDDR3: WL - 1 - RU(tODTon(max)/tCK)) Programming Mode: Quasi-dynamic Group 1 and Group 4
15:12	RO	0x0	reserved
11:8	RW	0x4	rd_odt_hold DFI PHY clock cycles to hold ODT for a read command. The minimum supported value is 2. Recommended values: DDR2: BL8: 0x6 (not DDR2-1066), 0x7 (DDR2-1066); BL4: 0x4 (not DDR2-1066), 0x5 (DDR2-1066) DDR3: BL8 - 0x6 DDR4: BL8: 5 + RD_PREAMBLE. RD_PREAMBLE = 1 (1tCK write preamble), 2 (2tCKwrite preamble) LPDDR3: BL8: 5 + RU(tDQSCK(max)/tCK) - RD(tDQSCK(min)/tCK) + RU(tODTon(max)/tCK) Programming Mode: Quasi-dynamic Group 1 and Group 4
7	RO	0x0	reserved
6:2	RW	0x00	rd_odt_delay The delay, in DFI PHY clock cycles, from issuing a read command to setting ODT values associated with that command. ODT setting must remain constant for the entire time that DQS is driven by the uMCTL2. Recommended values: DDR2: CL + AL - 4 (not DDR2-1066), CL + AL - 5 (DDR2-1066). If (CL + AL - 4 < 0), uMCTL2 does not support ODT for read operation. DDR3: CL - CWL DDR4: CL - CWL - RD_PREAMBLE + WR_PREAMBLE + DFITMG1.dfi_t_cmd_lat (to adjust for CAL mode). WR_PREAMBLE = 1 (1tCK write preamble), 2 (2tCK write preamble). RD_PREAMBLE = 1 (1tCK write preamble), 2 (2tCK write preamble). If (CL - CWL - RD_PREAMBLE + WR_PREAMBLE) < 0, uMCTL2 does not support ODT for read operation. LPDDR3: RL + RD(tDQSCK(min)/tCK) - 1 - RU(tODTon(max)/tCK) Programming Mode: Quasi-dynamic Group 1 and Group 4
1:0	RO	0x0	reserved

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DDRC_ODTMAP

Address: Operational Base + offset (0x0244)

ODTMAP

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:12	RW	0x2	rank1_rd_odt Indicates which remote ODTs must be turned on during a read from rank 1. Each rank has a remote ODT (in the SDRAM) which can be turned on by setting the appropriate bit here. Rank 0 is controlled by the LSB; rank 1 is controlled by bit next to the LSB, etc. For each rank, set its bit to 1 to enable its ODT. Programming Mode: Static
11:10	RO	0x0	reserved
9:8	RW	0x2	rank1_wr_odt Indicates which remote ODTs must be turned on during a write to rank 1. Each rank has a remote ODT (in the SDRAM) which can be turned on by setting the appropriate bit here. Rank 0 is controlled by the LSB; rank 1 is controlled by bit next to the LSB, etc. For each rank, set its bit to 1 to enable its ODT. Programming Mode: Static
7:6	RO	0x0	reserved
5:4	RW	0x1	rank0_rd_odt Indicates which remote ODTs must be turned on during a read from rank 0. Each rank has a remote ODT (in the SDRAM) which can be turned on by setting the appropriate bit here. Rank 0 is controlled by the LSB; rank 1 is controlled by bit next to the LSB, etc. For each rank, set its bit to 1 to enable its ODT. Programming Mode: Static
3:2	RO	0x0	reserved
1:0	RW	0x1	rank0_wr_odt Indicates which remote ODTs must be turned on during a write to rank 0. Each rank has a remote ODT (in the SDRAM) which can be turned on by setting the appropriate bit here. Rank 0 is controlled by the LSB; rank 1 is controlled by bit next to the LSB, etc. For each rank, set its bit to 1 to enable its ODT. Programming Mode: Static

DDRC_SCHED

Address: Operational Base + offset (0x0250)

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SCHED

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	rdwr_idle_gap When the preferred transaction store is empty for these many clock cycles, switch to the alternate transaction store if it is non-empty. The read transaction store (both high and low priority) is the default preferred transaction store and the write transaction store is the alternative store. When prefer write over read is set this is reversed. 0x0 is a legal value for this register. When set to 0x0, the transaction store switching will happen immediately when the switching conditions become true. Programming Mode: Static
23:12	RO	0x0	reserved
11:8	RW	0x8	lpr_num_entries Number of entries in the low priority transaction store is this value + 1. (MEMC_NO_OF_ENTRY - (SCHED.lpr_num_entries + 1)) is the number of entries available for the high priority transaction store. Setting this to maximum value allocates all entries to low priority transaction store. Setting this to 0 allocates 1 entry to low priority transaction store and the rest to high priority transaction store. Programming Mode: Static
7:3	RO	0x0	reserved
2	RW	0x1	pageclose If true, bank is kept open only while there are page hit transactions available in the CAM to that bank. The last read or write command in the CAM with a bank and page hit will be executed with auto-precharge if SCHED1.pageclose_timer=0. Even if this register set to 1 and SCHED1.pageclose_timer is set to 0, explicit precharge (and not auto-precharge) may be issued in some cases where there is a mode switch between Write and Read or between LPR and HPR. The Read and Write commands that are executed as part of the ECC scrub requests are also executed without auto-precharge. If false, the bank remains open until there is a need to close it (to open a different page, or for page timeout or refresh timeout) - also known as open page policy. The pageclose feature provids a midway between Open and Close page policies. Programming Mode: Quasi-dynamic Group 3
1:0	RO	0x0	reserved

DDRC_SCHED1

Address: Operational Base + offset (0x0254)

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SCHED1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	pageclose_timer This field works in conjunction with SCHED.pageclose. It only has meaning if SCHED.pageclose==1. If SCHED.pageclose==1 and pageclose_timer==0, then an auto-precharge may be scheduled for last read or write command in the CAM with a bank and page hit. Note, sometimes an explicit precharge is scheduled instead of the auto-precharge. See SCHED.pageclose for details of when this may happen. If SCHED.pageclose==1 and pageclose_timer>0, then an auto-precharge is not scheduled for last read or write command in the CAM with a bank and page hit. Instead, a timer is started, with pageclose_timer as the initial value. There is a timer on a per bank basis. The timer decrements unless the next read or write in the CAM to a bank is a page hit. It gets reset to pageclose_timer value if the next read or write in the CAM to a bank is a page hit. Once the timer has reached zero, an explcit precharge will be attempted to be scheduled. Programming Mode: Static

DDRC_PERFLPR1

Address: Operational Base + offset (0x0264)

PERFLPR1

Bit	Attr	Reset Value	Description
31:24	RW	0x0f	lpr_xact_run_length Number of transactions that are serviced once the LPR queue goes critical is the smaller of: (a) This number (b) Number of transactions available. Unit: Transaction. Programming Mode: Quasi-dynamic Group 3
23:16	RO	0x0	reserved
15:0	RW	0x007f	lpr_max_starve Number of DFI clocks that the LPR queue can be starved before it goes critical. The minimum valid functional value for this register is 0x1. Programming it to 0x0 will disable the starvation functionality; during normal operation, this function should not be disabled as it will cause excessive latencies. Programming Mode: Quasi-dynamic Group 3

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DDRC_PERFWR1

Address: Operational Base + offset (0x026c)

PERFWR1

Bit	Attr	Reset Value	Description
31:24	RW	0x0f	w_xact_run_length Number of transactions that are serviced once the WR queue goes critical is the smaller of: (a) This number (b) Number of transactions available. Unit: Transaction. Programming Mode: Quasi-dynamic Group 3
23:16	RO	0x0	reserved
15:0	RW	0x007f	w_max_starve Number of DFI clocks that the WR queue can be starved before it goes critical. The minimum valid functional value for this register is 0x1. Programming it to 0x0 will disable the starvation functionality; during normal operation, this function should not be disabled as it will cause excessive latencies. Programming Mode: Quasi-dynamic Group 3

DDRC_DBG0

Address: Operational Base + offset (0x0300)

DBG0

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	dis_collision_page_opt When this is set to '0', auto-precharge is disabled for the flushed command in a collision case. Collision cases are write followed by read to same address, read followed by write to same address, or write followed by write to same address with DBG0.dis_wc bit = 1 (where same address comparisons exclude the two address bits representing critical word). Programming Mode: Static
3:0	RO	0x0	reserved

DDRC_DBG1

Address: Operational Base + offset (0x0304)

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DBG1

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	dis_hif When 1, uMCTL2 asserts the HIF command signal hif_cmd_stall. uMCTL2 will ignore the hif_cmd_valid and all other associated request signals. This bit is intended to be switched on-the-fly. Programming Mode: Dynamic
0	RW	0x0	dis_dq When 1, uMCTL2 will not de-queue any transactions from the CAM. Bypass is also disabled. All transactions are queued in the CAM. No reads or writes are issued to SDRAM as long as this is asserted. This bit may be used to prevent reads or writes being issued by the uMCTL2, which makes it safe to modify certain register fields associated with reads and writes (see User Guide for details). After setting this bit, it is strongly recommended to poll DBGCAM.wr_data_pipeline_empty and DBGCAM.rd_data_pipeline_empty, before making changes to any registers which affect reads and writes. This will ensure that the relevant logic in the DDRC is idle. This bit is intended to be switched on-the-fly. Programming Mode: Dynamic

DDRC_DBGCAM

Address: Operational Base + offset (0x0308)

DBGCAM

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RO	0x0	wr_data_pipeline_empty When 1, indicates that the write data pipeline on the DFI interface is empty. This register is intended to be polled at least twice after setting DBG1.dis_dq, to ensure that all remaining commands/data have completed. Programming Mode: Dynamic
28	RO	0x0	rd_data_pipeline_empty When 1, indicates that the read data pipeline on the DFI interface is empty. This register is intended to be polled at least twice after setting DBG1.dis_dq, to ensure that all remaining commands/data have completed. Programming Mode: Dynamic
27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26	RO	0x0	dbg_wr_q_empty When 1, all the Write command queues and Write data buffers inside DDRC are empty. This register is to be used for debug purpose. An example use-case scenario: When Controller enters SelfRefresh using the Low-Power entry sequence, Controller is expected to have executed all the commands in its queues and the write and read data drained. Hence this register should be 1 at that time. Programming Mode: Dynamic
25	RO	0x0	dbg_rd_q_empty When 1, all the Read command queues and Read data buffers inside DDRC are empty. This register is to be used for debug purpose. An example use-case scenario: When Controller enters SelfRefresh using the Low-Power entry sequence, Controller is expected to have executed all the commands in its queues and the write and read data drained. Hence this register should be 1 at that time. Programming Mode: Dynamic
24	RO	0x0	dbg_stall Stall Programming Mode: Dynamic
23:20	RO	0x0	reserved
19:16	RO	0x0	dbg_w_q_depth Write queue depth The last entry of WR queue is reserved for ECC SCRUB operation. This entry is not included in the calculation of the queue depth. Programming Mode: Dynamic
15:12	RO	0x0	reserved
11:8	RO	0x0	dbg_lpr_q_depth Low priority read queue depth. The last entry of Lpr queue is reserved for ECC SCRUB operation. This entry is not included in the calculation of the queue depth. Programming Mode: Dynamic
7:0	RO	0x0	reserved

DDRC_DBGCMD

Address: Operational Base + offset (0x030c)

DBGCMD

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	ctrlupd Setting this register bit to 1 indicates to the uMCTL2 to issue a dfi_ctrlupd_req to the PHY. When this request is stored in the uMCTL2, the bit is automatically cleared. This operation must only be performed when DFIUPD0.dis_auto_ctrlupd=1. Programming Mode: Dynamic
4	RW	0x0	zq_calib_short Setting this register bit to 1 indicates to the uMCTL2 to issue a ZQCS (ZQ calibration short)/MPC(ZQ calibration) command to the SDRAM. When this request is stored in the uMCTL2, the bit is automatically cleared. This operation can be performed only when ZQCTL0.dis_auto_zq=1. It is recommended NOT to set this register bit if in Init operating mode. This register bit is ignored when in SelfRefresh(except LPDDR4) and SR-Powerdown(LPDDR4) and Deep power-down operating modes and Maximum Power Saving Mode. Programming Mode: Dynamic
3:2	RO	0x0	reserved
1	RW	0x0	rank1_refresh Setting this register bit to 1 indicates to the uMCTL2 to issue a refresh to rank 1. Writing to this bit causes DBGSTAT.rank1_refresh_busy to be set. When DBGSTAT.rank1_refresh_busy is cleared, the command has been stored in uMCTL2. This operation can be performed only when RFSHCTL3.dis_auto_refresh=1. It is recommended NOT to set this register bit if in Init or Deep power-down operating modes or Maximum Power Saving Mode. Programming Mode: Dynamic
0	RW	0x0	rank0_refresh Setting this register bit to 1 indicates to the uMCTL2 to issue a refresh to rank 0. Writing to this bit causes DBGSTAT.rank0_refresh_busy to be set. When DBGSTAT.rank0_refresh_busy is cleared, the command has been stored in uMCTL2. This operation can be performed only when RFSHCTL3.dis_auto_refresh=1. It is recommended NOT to set this register bit if in Init or Deep power-down operating modes or Maximum Power Saving Mode. Programming Mode: Dynamic

DDRC_DBGSTAT

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Address: Operational Base + offset (0x0310)

DBGSTAT

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RO	0x0	ctrlupd_busy SoC core may initiate a ctrlupd operation only if this signal is low. This signal goes high in the clock after the uMCTL2 accepts the ctrlupd request. It goes low when the ctrlupd operation is initiated in the uMCTL2. It is recommended not to perform ctrlupd operations when this signal is high. 0 - Indicates that the SoC core can initiate a ctrlupd operation 1 - Indicates that ctrlupd operation has not been initiated yet in the uMCTL2 Programming Mode: Dynamic
4	RO	0x0	zq_calib_short_busy SoC core may initiate a ZQCS (ZQ calibration short) operation only if this signal is low. This signal goes high in the clock after the uMCTL2 accepts the ZQCS request. It goes low when the ZQCS operation is initiated in the uMCTL2. It is recommended not to perform ZQCS operations when this signal is high. 0 - Indicates that the SoC core can initiate a ZQCS operation 1 - Indicates that ZQCS operation has not been initiated yet in the uMCTL2 Programming Mode: Dynamic
3:2	RO	0x0	reserved
1	RO	0x0	rank1_refresh_busy SoC core may initiate a rank1_refresh operation (refresh operation to rank 1) only if this signal is low. This signal goes high in the clock after DBGCMD.rank1_refresh is set to one. It goes low when the rank1_refresh operation is stored in the uMCTL2. It is recommended not to perform rank1_refresh operations when this signal is high. 0 - Indicates that the SoC core can initiate a rank1_refresh operation 1 - Indicates that rank1_refresh operation has not been stored yet in the uMCTL2 Programming Mode: Dynamic

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Bit	Attr	Reset Value	Description
0	RO	0x0	rank0_refresh_busy SoC core may initiate a rank0_refresh operation (refresh operation to rank 0) only if this signal is low. This signal goes high in the clock after DBGCMD.rank0_refresh is set to one. It goes low when the rank0_refresh operation is stored in the uMCTL2. It is recommended not to perform rank0_refresh operations when this signal is high. 0 - Indicates that the SoC core can initiate a rank0_refresh operation 1 - Indicates that rank0_refresh operation has not been stored yet in the uMCTL2 Programming Mode: Dynamic

DDRC_SWCTL

Address: Operational Base + offset (0x0320)

SWCTL

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	sw_done Enable quasi-dynamic register programming outside reset. Program register to 0 to enable quasi-dynamic programming. Set back register to 1 once programming is done. Programming Mode: Dynamic

DDRC_SWSTAT

Address: Operational Base + offset (0x0324)

SWSTAT

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x1	sw_done_ack Register programming done. This register is the echo of SWCTL.sw_done. Wait for sw_done value 1 to propagate to sw_done_ack at the end of the programming sequence to ensure that the correct registers values are propagated to the destination clock domains. Programming Mode: Static

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DDRC_POISONCFG

Address: Operational Base + offset (0x036c)

POISONCFG

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	rd_poison_intr_clr Interrupt clear for read transaction poisoning. Allow 2/3 clock cycles for correct value to propagate to core logic and clear the interrupts. Programming Mode: Dynamic
23:21	RO	0x0	reserved
20	RW	0x1	rd_poison_intr_en If set to 1, enables interrupts for read transaction poisoning Programming Mode: Dynamic
19:17	RO	0x0	reserved
16	RW	0x1	rd_poison_slverr_en If set to 1, enables SLVERR response for read transaction poisoning Programming Mode: Dynamic
15:9	RO	0x0	reserved
8	W1C	0x0	wr_poison_intr_clr Interrupt clear for write transaction poisoning. Allow 2/3 clock cycles for correct value to propagate to core logic and clear the interrupts. Programming Mode: Dynamic
7:5	RO	0x0	reserved
4	RW	0x1	wr_poison_intr_en If set to 1, enables interrupts for write transaction poisoning Programming Mode: Dynamic
3:1	RO	0x0	reserved
0	RW	0x1	wr_poison_slverr_en If set to 1, enables SLVERR response for write transaction poisoning Programming Mode: Dynamic

DDRC_POISONSTAT

Address: Operational Base + offset (0x0370)

POISONSTAT

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Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RO	0x0	rd_poison_intr_0 Read transaction poisoning error interrupt for port 0. This register is a APB clock copy (double register synchronizer) of the interrupt asserted when a transaction is poisoned on the corresponding AXI port's read address channel. Bit 0 corresponds to Port 0, and so on. Interrupt is cleared by register rd_poison_intr_clr, then value propagated to APB clock. Programming Mode: Dynamic
15:1	RO	0x0	reserved
0	RO	0x0	wr_poison_intr_0 Write transaction poisoning error interrupt for port 0. This register is a APB clock copy (double register synchronizer) of the interrupt asserted when a transaction is poisoned on the corresponding AXI port's write address channel. Bit 0 corresponds to Port 0, and so on. Interrupt is cleared by register wr_poison_intr_clr, then value propagated to APB clock. Programming Mode: Dynamic

DDRC_PSTAT

Address: Operational Base + offset (0x03fc)

PSTAT

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RO	0x0	wr_port_busy_0 Indicates if there are outstanding writes for AXI port 0. Programming Mode: Dynamic
15:1	RO	0x0	reserved
0	RO	0x0	rd_port_busy_0 Indicates if there are outstanding reads for AXI port 0. Programming Mode: Dynamic

DDRC_PCCFG

Address: Operational Base + offset (0x0400)

PCCFG

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	bl_exp_mode Burst length expansion mode. By default (i.e. bl_exp_mode==0) XPI expands every AXI burst into multiple HIF commands, using the memory burst length as a unit. If set to 1, then XPI will use half of the memory burst length as a unit. This applies to both reads and writes. When MSTR.data_bus_width==00, setting bl_exp_mode to 1 has no effect. This can be used in cases where Partial Writes is enabled (UMCTL2_PARTIAL_WR=1), in order to avoid or minimize t_ccd_l penalty in DDR4 and t_ccd_mw penalty in LPDDR4. Hence, bl_exp_mode=1 is only recommended if DDR4 or LPDDR4. Note that if DBICTL.dm_en=0, functionality is not supported in the following cases: UMCTL2_PARTIAL_WR=0 UMCTL2_PARTIAL_WR=1, MSTR.data_bus_width=01, MEMC_BURST_LENGTH=8 and MSTR.burst_rdwr=1000 (LPDDR4 only) UMCTL2_PARTIAL_WR=1, MSTR.data_bus_width=01, MEMC_BURST_LENGTH=4 and MSTR.burst_rdwr=0100 (DDR4 only), with either MSTR.burstchop=0 or CRCPARCTL1.crc_enable=1 Programming Mode: Static
7:5	RO	0x0	reserved
4	RW	0x0	pagematch_limit Page match four limit. If set to 1, limits the number of consecutive same page DDRC transactions that can be granted by the Port Arbiter to four when Page Match feature is enabled. If set to 0, there is no limit imposed on number of consecutive same page DDRC transactions. Programming Mode: Static
3:1	RO	0x0	reserved
0	RW	0x0	go2critical_en If set to 1 (enabled), sets co_gs_go2critical_wr and co_gs_go2critical_lpr / co_gs_go2critical_hpr signals going to DDRC based on urgent input (awurgent, arurgent) coming from AXI master. If set to 0 (disabled), co_gs_go2critical_wr and co_gs_go2critical_lpr / co_gs_go2critical_hpr signals at DDRC are driven to 1b'0. Programming Mode: Static

DDRC_PCFGR_0

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Address: Operational Base + offset (0x0404)

PCFGR_0

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	rd_port_pagematch_en If set to 1, enables the Page Match feature. If enabled, once a requesting port is granted, the port is continued to be granted if the following immediate commands are to the same memory page (same bank and same row). See also related PCCFG.pagematch_limit register. Programming Mode: Static
13	RW	0x0	rd_port_urgent_en If set to 1, enables the AXI urgent sideband signal (arurgent). When enabled and arurgent is asserted by the master, that port becomes the highest priority and co_gs_go2critical_lpr/co_gs_go2critical_hpr signal to DDRC is asserted if enabled in PCCFG.go2critical_en register. Note that arurgent signal can be asserted anytime and as long as required which is independent of address handshaking (it is not associated with any particular command). Programming Mode: Static
12	RW	0x0	rd_port_aging_en If set to 1, enables aging function for the read channel of the port. Programming Mode: Static
11:10	RO	0x0	reserved
9:0	RW	0x000	rd_port_priority Determines the initial load value of read aging counters. These counters will be parallel loaded after reset, or after each grant to the corresponding port. The aging counters down-count every clock cycle where the port is requesting but not granted. The higher significant 5-bits of the read aging counter sets the priority of the read channel of a given port. Port's priority will increase as the higher significant 5-bits of the counter starts to decrease. When the aging counter becomes 0, the corresponding port channel will have the highest priority level (timeout condition -Priority0). For single port configurations, the aging counters are only used when they timeout (become 0) to force read-write direction switching. In this case, external dynamic priority input, arqos (for reads only) can still be used to set the DDRC read priority (2 priority levels: low priority read - LPR, high priority read - HPR) on a command by command basis. Note: The two LSBs of this register field are tied internally to 2'b00. Programming Mode: Static

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DDRC_PCFGW_0

Address: Operational Base + offset (0x0408)

PCFGW_0

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x1	wr_port_pagematch_en If set to 1, enables the Page Match feature. If enabled, once a requesting port is granted, the port is continued to be granted if the following immediate commands are to the same memory page (same bank and same row). See also related PCCFG.pagematch_limit register. Programming Mode: Static
13	RW	0x0	wr_port_urgent_en If set to 1, enables the AXI urgent sideband signal (awurgent). When enabled and awurgent is asserted by the master, that port becomes the highest priority and co_gs_go2critical_wr signal to DDRC is asserted if enabled in PCCFG.go2critical_en register. Note that awurgent signal can be asserted anytime and as long as required which is independent of address handshaking (it is not associated with any particular command). Programming Mode: Static
12	RW	0x0	wr_port_aging_en If set to 1, enables aging function for the write channel of the port. Programming Mode: Static
11:10	RO	0x0	reserved
9:0	RW	0x000	wr_port_priority Determines the initial load value of write aging counters. These counters will be parallel loaded after reset, or after each grant to the corresponding port. The aging counters down-count every clock cycle where the port is requesting but not granted. The higher significant 5-bits of the write aging counter sets the initial priority of the write channel of a given port. Port's priority will increase as the higher significant 5-bits of the counter starts to decrease. When the aging counter becomes 0, the corresponding port channel will have the highest priority level. For single port configurations, the aging counters are only used when they timeout (become 0) to force read-write direction switching. Note: The two LSBs of this register field are tied internally to 2'b00. Programming Mode: Static

DDRC_PCTRL_0

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Address: Operational Base + offset (0x0490)

PCTRL_0

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	port_en Enables AXI port 0. Programming Mode: Dynamic

DDRC_FREQ1_DERATEEN

Address: Operational Base + offset (0x2020)

DERATEEN

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:8	RW	0x0	rc_derate_value Derate value of tRC for LPDDR4 0: Derating uses +1. 1: Derating uses +2. 2: Derating uses +3. 3: Derating uses +4. Present only in designs configured to support LPDDR4. The required number of cycles for derating can be determined by dividing 3.75ns by the core_ddrc_core_clk period, and rounding up the next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:4	RW	0x0	derate_byte Derate byte Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4 Indicates which byte of the MRR data is used for derating. Programming Mode: Static
3:2	RO	0x0	reserved
1	RW	0x0	derate_value Derate value 0: Derating uses +1. 1: Derating uses +2. Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4 Set to 0 for all LPDDR2 speed grades as derating value of +1.875 ns is less than a core_ddrc_core_clk period. For LPDDR3/4, if the period of core_ddrc_core_clk is less than 1.875ns, this register field should be set to 1; otherwise it should be set to 0. Programming Mode: Quasi-dynamic Group 2 and Group 4

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Bit	Attr	Reset Value	Description
0	RW	0x0	derate_enable Enables derating 0: Timing parameter derating is disabled 1: Timing parameter derating is enabled using MR4 read value. Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4 This field must be set to 0 for non-LPDDR2/LPDDR3/LPDDR4 mode. Programming Mode: Dynamic

DDRC_FREQ1_DERATEINT

Address: Operational Base + offset (0x2024)

DERATEINT

Bit	Attr	Reset Value	Description
31:0	RW	0x00800000	mr4_read_interval Interval between two MR4 reads, used to derate the timing parameters. Present only in designs configured to support LPDDR2/LPDDR3/LPDDR4. This register must not be set to zero. Unit: DFI clock cycle. Programming Mode: Static

DDRC_FREQ1_RFSHCTL0

Address: Operational Base + offset (0x2050)

RFSHCTL0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:20	RW	0x2	refresh_margin Threshold value in number of DFI clock cycles before the critical refresh or page timer expires. A critical refresh is to be issued before this threshold is reached. It is recommended that this not be changed from the default value, currently shown as 0x2. It must always be less than internally used t_rfc_nom_x32. Note that, in LPDDR2/LPDDR3/LPDDR4, internally used t_rfc_nom_x32 may be equal to RFSHTMG.t_rfc_nom_x32>>2 if derating is enabled (DERATEEN.derate_enable=1). Otherwise, internally used t_rfc_nom_x32 will be equal to RFSHTMG.t_rfc_nom_x32. Unit: Multiples of 32 DFI clocks. Programming Mode: Dynamic - Refresh Related
19:17	RO	0x0	reserved
16:12	RW	0x10	refresh_to_x32 If the refresh timer (tRFCnom, also known as tREFI) has expired at least once, but it has not expired (RFSHCTL0.refresh_burst+1) times yet, then a speculative refresh may be performed. A speculative refresh is a refresh performed at a time when refresh would be useful, but before it is absolutely required. When the SDRAM bus is idle for a period of time determined by this RFSHCTL0.refresh_to_x32 and the refresh timer has expired at least once since the last refresh, then a speculative refresh is performed. Speculative refreshes continues successively until there are no refreshes pending or until new reads or writes are issued to the uMCTL2. Unit: Multiples of 32 DFI clocks. Programming Mode: Dynamic - Refresh Related
11:9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8:4	RW	0x00	refresh_burst The programmed value + 1 is the number of refresh timeouts that is allowed to accumulate before traffic is blocked and the refreshes are forced to execute. Closing pages to perform a refresh is a one-time penalty that must be paid for each group of refreshes. Therefore, performing refreshes in a burst reduces the per-refresh penalty of these page closings. Higher numbers for RFSHCTL.refresh_burst slightly increases utilization; lower numbers decreases the worstcase latency associated with refreshes. 0: single refresh 1: burst-of-2 refresh 7: burst-of-8 refresh For DDR2/3, the refresh is always per-rank and not perbank. The rank refresh can be accumulated over 8*tREFI cycles using the burst refresh feature. In DDR4 mode, according to Fine Granularity feature, 8 refreshes can be postponed in 1X mode, 16 refreshes in 2X mode and 32 refreshes in 4X mode. If using PHY-initiated updates, care must be taken in the setting of RFSHCTL0.refresh_burst, to ensure that tRFCmax is not violated due to a PHY-initiated update occurring shortly before a refresh burst was due. In this situation, the refresh burst will be delayed until the PHYinitiated update is complete. Programming Mode: Dynamic - Refresh Related
3	RO	0x0	reserved
2	RW	0x0	per_bank_refresh 1: Per bank refresh; 0: All bank refresh. Per bank refresh allows traffic to flow to other banks. Per bank refresh is not supported by all LPDDR2 devices but should be supported by all LPDDR3/LPDDR4 devices. Programming Mode: Static
1:0	RO	0x0	reserved

DDRC_FREQ1_RFSHTMG

Address: Operational Base + offset (0x2064)

RFSHTMG

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:16	RW	0x062	t_rfc_nom_x32 tREFI: Average time interval between refreshes per rank (Specification: 7.8us for DDR2, DDR3 and DDR4. See JEDEC specification for mDDR, LPDDR2, LPDDR3 and LPDDR4). For LPDDR2/LPDDR3/LPDDR4: if using all-bank refreshes (RFSHCTL0.per_bank_refresh= 0), this register should be set to tREFIab if using per-bank refreshes (RFSHCTL0.per_bank_refresh = 1), this register should be set to tREFIpb When the controller is operating in 1:2 frequency ratio mode, program this to (tREFI/2), no rounding up. In DDR4 mode, tREFI value is different depending on the refresh mode. The user should program the appropriate value from the spec based on the value programmed in the refresh mode register. Note that RFSHTMG.t_rfc_nom_x32 * 32 must be greater than RFSHTMG.t_rfc_min, and RFSHTMG.t_rfc_nom_x32 must be greater than 0x1. Non-DDR4 or DDR4 Fixed 1x mode:RFSHTMG.t_rfc_nom_x32 must be less than or equal to 0xFFE. DDR4 Fixed 2x mode: RFSHTMG.t_rfc_nom_x32 must be less than or equal to 0x7FF. DDR4 Fixed 4x mode: RFSHTMG.t_rfc_nom_x32 must be less than or equal to 0x3FF. Unit: Multiples of 32 DFI clocks. Programming Mode: Dynamic - Refresh Related
15	RW	0x0	lpddr3_trefbw_en Used only when LPDDR3 memory type is connected. Should only be changed when uMCTL2 is in reset. Specifies whether to use the tREFBW parameter (required by some LPDDR3 devices which comply with earlier versions of the LPDDR3 JEDEC specification) or not: 0: tREFBW parameter not used 1: tREFBW parameter used Programming Mode: Static
14:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x08c	t_rfc_min tRFC (min): Minimum time from refresh to refresh or activate. t_rfc_min should be set to RoundUp(RoundUp(tRFCmin/tCK)/2). In LPDDR2/LPDDR3/LPDDR4 mode: if using all-bank refreshes, the tRFCmin value in the above equations is equal to tRFCab if using per-bank refreshes, the tRFCmin value in the above equations is equal to tRFCpb In DDR4 mode, the tRFCmin value in the above equations is different depending on the refresh mode (fixed 1X,2X,4X) and the device density. The user should program the appropriate value from the spec based on the 'refresh_mode' and the device density that is used. Unit: DFI Clocks. Programming Mode: Dynamic - Refresh Related

DDRC_FREQ1_INIT3

Address: Operational Base + offset (0x20dc)

INIT3

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mr DDR2: Value to write to MR register. Bit 8 is for DLL and the setting here is ignored. The uMCTL2 sets this bit appropriately. DDR3/DDR4: Value loaded into MR0 register. mDDR: Value to write to MR register. LPDDR2/LPDDR3/LPDDR4 - Value to write to MR1 register Programming Mode: Quasi-dynamic Group 1 and Group 4
15:0	RW	0x0510	emr DDR2: Value to write to EMR register. Bits 9:7 are for OCD and the setting in this register is ignored. The uMCTL2 sets those bits appropriately. DDR3/DDR4: Value to write to MR1 register Set bit 7 to 0. If PHY-evaluation mode training is enabled, this bit is set appropriately by the uMCTL2 during write leveling. mDDR: Value to write to EMR register. LPDDR2/LPDDR3/LPDDR4 - Value to write to MR2 register Programming Mode: Quasi-dynamic Group 4

DDRC_FREQ1_INIT4

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Address: Operational Base + offset (0x20e0)

INIT4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	emr2 DDR2: Value to write to EMR2 register. DDR3/DDR4: Value to write to MR2 register LPDDR2/LPDDR3/LPDDR4: Value to write to MR3 register mDDR: Unused Programming Mode: Quasi-dynamic Group 4
15:0	RW	0x0000	emr3 DDR2: Value to write to EMR3 register. DDR3/DDR4: Value to write to MR3 register mDDR/LPDDR2/LPDDR3: Unused LPDDR4: Value to write to MR13 register Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_INIT6

Address: Operational Base + offset (0x20e8)

INIT6

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mr4 DDR4- Value to be loaded into SDRAM MR4 registers. LPDDR4- Value to be loaded into SDRAM MR11 registers. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:0	RW	0x0000	mr5 DDR4- Value to be loaded into SDRAM MR5 registers. LPDDR4- Value to be loaded into SDRAM MR12 registers. Programming Mode: Quasi-dynamic Group 1 and Group 4

DDRC_FREQ1_INIT7

Address: Operational Base + offset (0x20ec)

INIT7

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mr6 DDR4- Value to be loaded into SDRAM MR6 registers. LPDDR4- Value to be loaded into SDRAM MR14 registers. Programming Mode: Quasi-dynamic Group 4

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	mr22 LPDDR4- Value to be loaded into SDRAM MR22 registers. Used in LPDDR4 designs only. Programming Mode: Quasi-dynamic Group 4

DDRC_FREQ1_DRAMTMG0

Address: Operational Base + offset (0x2100)

DRAMTMG0

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x0f	wr2pre Minimum time between write and precharge to same bank. Unit: DFI Clocks Specifications: WL + BL/2 + tWR = approximately 8 cycles + 15 ns = 14 clocks @400MHz and less for lower frequencies where: WL = write latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM. BST (burst terminate) is not supported at present. tWR = Write recovery time. This comes directly from the SDRAM specification. Add one extra cycle for LPDDR2/LPDDR3/LPDDR4 for this parameter. When the controller is operating in 1:2 frequency ratio mode, 1T mode, divide the above value by 2. No rounding up. When the controller is operating in 1:2 frequency ratio mode, 2T mode or LPDDR4 mode, divide the above value by 2 and round it up to the next integer value. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
23:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21:16	RW	0x10	t_faw tFAW Valid only when 8 or more banks(or banks x bank groups) are present. In 8-bank design, at most 4 banks must be activated in a rolling window of tFAW cycles. When the controller is operating in 1:2 frequency ratio mode, program this to (tFAW/2) and round up to next integer value. In a 4-bank design, set this register to 0x1 independent of the 1:1/1:2 frequency mode. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
15	RO	0x0	reserved
14:8	RW	0x1b	t_ras_max tRAS(max): Maximum time between activate and precharge to same bank. This is the maximum time that a page can be kept open Minimum value of this register is 1. Zero is invalid. When the controller is operating in 1:2 frequency ratio mode, program this to (tRAS(max)-1)/2. No rounding up. Unit: Multiples of 1024 DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:6	RO	0x0	reserved
5:0	RW	0x0f	t_ras_min tRAS(min): Minimum time between activate and precharge to the same bank. When the controller is operating in 1:2 frequency mode, 1T mode, program this to tRAS(min)/2. No rounding up. When the controller is operating in 1:2 frequency ratio mode, 2T mode or LPDDR4 mode, program this to (tRAS(min)/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG1

Address: Operational Base + offset (0x2104)

DRAMTMG1

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20:16	RW	0x08	t_xp tXP: Minimum time after power-down exit to any operation. For DDR3, this should be programmed to tXPDLL if slow powerdown exit is selected in MR0[12]. If C/A parity for DDR4 is used, set to (tXP+PL) instead. If LPDDR4 is selected and its spec has tCKELPD parameter, set to the larger of tXP and tCKELPD instead. When the controller is operating in 1:2 frequency ratio mode, program this to (tXP/2) and round it up to the next integer value. Units: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
15:14	RO	0x0	reserved
13:8	RW	0x04	rd2pre tRTP: Minimum time from read to precharge of same bank. DDR2: tAL + BL/2 + max(tRTP, 2) - 2 DDR3: tAL + max (tRTP, 4) DDR4: Max of following two equations: tAL + max (tRTP, 4) or, RL + BL/2 - tRP (*). mDDR: BL/2 LPDDR2: Depends on if it's LPDDR2-S2 or LPDDR2-S4: LPDDR2-S2: BL/2 + tRTP - 1. LPDDR2-S4: BL/2 + max(tRTP,2) - 2. LPDDR3: BL/2 + max(tRTP,4) - 4 LPDDR4: BL/2 + max(tRTP,8) - 8 (*) When both DDR4 SDRAM and ST-MRAM are used simultaneously, use SDRAM's tRP value for calculation. When the controller is operating in 1:2 mode, 1T mode, divide the above value by 2. No rounding up. When the controller is operating in 1:2 mode, 2T mode or LPDDR4 mode, divide the above value by 2 and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
7	RO	0x0	reserved
6:0	RW	0x14	t_rc tRC: Minimum time between activates to same bank. When the controller is operating in 1:2 frequency ratio mode, program this to (tRC/2) and round up to next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG2

Rockchip Confidential

Address: Operational Base + offset (0x2108)

DRAMTMG2

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RW	0x03	write_latency Set to WL. Time from write command to write data on SDRAM interface. For mDDR, it should normally be set to 1. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. This register field is not required for DDR2 and DDR3 (except if MEMC_TRAINING is set), as the DFI read and write latencies defined in DFITMG0 and DFITMG1 are sufficient for those protocols Unit: DFI clocks Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
23:22	RO	0x0	reserved
21:16	RW	0x05	read_latency Set to RL. Time from read command to read data on SDRAM interface. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. This register field is not required for DDR2 and DDR3 (except if MEMC_TRAINING is set), as the DFI read and write latencies defined in DFITMG0 and DFITMG1 are sufficient for those protocols Unit: DFI clocks Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x06	rd2wr DDR2/3/mDDR: RL + BL/2 + 2 - WL DDR4: RL + BL/2 + 1 + WR_PREAMBLE - WL LPDDR2/LPDDR3: RL + BL/2 + RU(tDQSCKmax/tCK) + 1 - WL LPDDR4(DQ ODT is Disabled): RL + BL/2 + RU(tDQSCKmax/tCK) + WR_PREAMBLE + RD_POSTAMBLE - WL LPDDR4(DQ ODT is Enabled) : RL + BL/2 + RU(tDQSCKmax/tCK) + RD_POSTAMBLE - ODTLon - RU(tODTon(min)/tCK) Minimum time from read command to write command. Include time for bus turnaround and all per-bank, per-rank, and global constraints. Please see the relevant PHY databook for details of what should be included here. Unit: DFI Clocks. Where: WL = write latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM RL = read latency = CAS latency WR_PREAMBLE = write preamble. This is unique to DDR4 and LPDDR4. RD_POSTAMBLE = read postamble. This is unique to LPDDR4. For LPDDR2/LPDDR3/LPDDR4, if derating is enabled (DERATEEN.derate_enable=1), derated tDQSCKmax should be used. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x0d	wr2rd DDR4: CWL + PL + BL/2 + tWTR_L LPDDR2/3/4: WL + BL/2 + tWTR + 1 Others: CWL + BL/2 + tWTR In DDR4, minimum time from write command to read command for same bank group. In others, minimum time from write command to read command. Includes time for bus turnaround, recovery times, and all per-bank, per-rank, and global constraints. Unit: DFI Clocks. Where: CWL = CAS write latency WL = Write latency PL = Parity latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM tWTR_L = internal write to read command delay for same bank group. This comes directly from the SDRAM specification. tWTR = internal write to read command delay. This comes directly from the SDRAM specification. Add one extra cycle for LPDDR2/LPDDR3/LPDDR4 operation. When the controller is operating in 1:2 mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4

DDRC_FREQ1_DRAMTMG3

Address: Operational Base + offset (0x210c)

DRAMTMG3

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:20	RW	0x005	t_mrw Time to wait after a mode register write or read (MRW or MRR). Present only in designs configured to support LPDDR2, LPDDR3 or LPDDR4. LPDDR2 typically requires value of 5. LPDDR3 typically requires value of 10. LPDDR4: Set this to the larger of tMRW and tMRWCKEL. For LPDDR2, this register is used for the time from a MRW/MRR to all other commands. When the controller is operating in 1:2 frequency ratio mode, program this to the above values divided by 2 and round it up to the next integer value. For LDPDR3, this register is used for the time from a MRW/MRR to a MRW/MRR. Programming Mode: Quasi-dynamic Group 2 and Group 4
19:18	RO	0x0	reserved
17:12	RW	0x04	t_mrd tMRD: Cycles to wait after a mode register write or read. Depending on the connected SDRAM, tMRD represents: DDR2/mDDR: Time from MRS to any command DDR3/4: Time from MRS to MRS command LPDDR2: not used LPDDR3/4: Time from MRS to non-MRS command. When the controller is operating in 1:2 frequency ratio mode, program this to (tMRD/2) and round it up to the next integer value. If C/A parity for DDR4 is used, set to tMRD_PAR(tMOD+PL) instead. Programming Mode: Quasi-dynamic Group 2 and Group 4
11:10	RO	0x0	reserved
9:0	RW	0x00c	t_mod tMOD: Parameter used only in DDR3 and DDR4. Cycles between load mode command and following non-load mode command. If C/A parity for DDR4 is used, set to tMOD_PAR(tMOD+PL) instead. If MPR writes for DDR4 are used, set to tMOD + AL (or tMPD_PAR + AL if C/A parity is also used). Set to tMOD/2 (rounded up to next integer) if controller is operating in 1:2 frequency ratio mode. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG4

Address: Operational Base + offset (0x2110)

DRAMTMG4

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Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RW	0x05	t_rcd tRCD - tAL: Minimum time from activate to read or write command to same bank. When the controller is operating in 1:2 frequency ratio mode, program this to ((tRCD - tAL)/2) and round it up to the next integer value. Minimum value allowed for this register is 1, which implies minimum (tRCD - tAL) value to be 2 when the controller is operating in 1:2 frequency ratio mode. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4
23:20	RO	0x0	reserved
19:16	RW	0x4	t_ccd DDR4: tCCD_L: This is the minimum time between two reads or two writes for same bank group. Others: tCCD: This is the minimum time between two reads or two writes. When the controller is operating in 1:2 frequency ratio mode, program this to (tCCD_L/2 or tCCD/2) and round it up to the next integer value. Unit: DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:12	RO	0x0	reserved
11:8	RW	0x4	t_rrd DDR4: tRRD_L: Minimum time between activates from bank "a" to bank "b" for same bank group. Others: tRRD: Minimum time between activates from bank "a" to bank "b" When the controller is operating in 1:2 frequency ratio mode, program this to (tRRD_L/2 or tRRD/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:5	RO	0x0	reserved
4:0	RW	0x05	t_rp tRP: Minimum time from precharge to activate of same bank. When the controller is operating in 1:2 frequency ratio mode, t_rp should be set to RoundDown(RoundUp(tRP/tCK)/2) + 1. When the controller is operating in 1:2 frequency ratio mode in LPDDR4, t_rp should be set to RoundUp(RoundUp(tRP/tCK)/2). Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

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DDRC_FREQ1_DRAMTMG5

Address: Operational Base + offset (0x2114)

DRAMTMG5

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x5	t_cksrx This is the time before Self Refresh Exit that CK is maintained as a valid clock before issuing SRX. Specifies the clock stable time before SRX. Recommended settings: mDDR: 1 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKCKEH DDR2: 1 DDR3: tCKSRX DDR4: tCKSRX When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:20	RO	0x0	reserved
19:16	RW	0x5	t_cksre This is the time after Self Refresh Down Entry that CK is maintained as a valid clock. Specifies the clock disable delay after SRE. Recommended settings: mDDR: 0 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKELCK DDR2: 1 DDR3: max (10 ns, 5 tCK) DDR4: max (10 ns, 5 tCK) (+ PL(parity latency)(*)) (*)Only if CRCPARCTL1.caparity_disable_before_sr=0, this register should be increased by PL. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x04	t_ckesr Minimum CKE low width for Self refresh or Self refresh power down entry to exit timing in memory clock cycles. Recommended settings: mDDR: tRFC LPDDR2: tCKESR LPDDR3: tCKESR LPDDR4: max(tCKE, tSR) DDR2: tCKE DDR3: tCKE + 1 DDR4: tCKE + 1 (+ PL(parity latency)(*)) (*)Only if CRCPARCTL1.caparity_disable_before_sr=0, this register should be increased by PL. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:5	RO	0x0	reserved
4:0	RW	0x03	t_cke Minimum number of cycles of CKE HIGH/LOW during power-down and self refresh. LPDDR2/LPDDR3 mode: Set this to the larger of tCKE or tCKESR LPDDR4 mode: Set this to the larger of tCKE or tSR. Non-LPDDR2/non-LPDDR3/non-LPDDR4 designs: Set this to tCKE value. When the controller is operating in 1:2 frequency ratio mode, program this to (value described above)/2 and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG6

Address: Operational Base + offset (0x2118)

DRAMTMG6

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RW	0x2	t_ckdpde This is the time before Self Refresh Exit that CK is maintained as a valid clock before issuing SRX. Specifies the clock stable time before SRX. Recommended settings: mDDR: 1 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKCKEH DDR2: 1 DDR3: tCKSRX DDR4: tCKSRX When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:20	RO	0x0	reserved
19:16	RW	0x2	t_ckdpdx This is the time before Deep Power Down Exit that CK is maintained as a valid clock before issuing DPDX. Specifies the clock stable time before DPDX. Recommended settings: mDDR: 1 LPDDR2: 2 LPDDR3: 2 When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x5	t_ckcsx This is the time before Clock Stop Exit that CK is maintained as a valid clock before issuing Clock Stop Exit. Specifies the clock stable time before next command after Clock Stop Exit. Recommended settings: mDDR: 1 LPDDR2: tXP + 2 LPDDR3: tXP + 2 LPDDR4: tXP + 2 When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG7

Address: Operational Base + offset (0x211c)

DRAMTMG7

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x2	t_ckpde This is the time after Power Down Entry that CK is maintained as a valid clock. Specifies the clock disable delay after PDE. Recommended settings: mDDR: 0 LPDDR2: 2 LPDDR3: 2 LPDDR4: tCKELCK When using DDR2/3/4 SDRAM, this register should be set to the same value as DRAMTMG5.t_cksre. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x2	t_ckpdx This is the time before Power Down Exit that CK is maintained as a valid clock before issuing PDX. Specifies the clock stable time before PDX. Recommended settings: mDDR: 0 LPDDR2: 2 LPDDR3: 2 LPDDR4: 2 When using DDR2/3/4 SDRAM, this register should be set to the same value as DRAMTMG5.t_cksrx. When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. This is only present for designs supporting mDDR or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG8

Address: Operational Base + offset (0x2120)

DRAMTMG8

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x03	t_xs_fast_x32 tXS_FAST: Exit Self Refresh to ZQCL, ZQCS and MRS (only CL, WR, RTP and Geardown mode). When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: This is applicable to only ZQCL/ZQCS commands. Note: Ensure this is less than or equal to t_xs_x32. Programming Mode: Quasi-dynamic Group 2 and Group 4
23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:16	RW	0x03	t_xs_abort_x32 tXS_ABORT: Exit Self Refresh to commands not requiring a locked DLL in Self Refresh Abort. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: Ensure this is less than or equal to t_xs_x32. Programming Mode: Quasi-dynamic Group 2 and Group 4
15	RO	0x0	reserved
14:8	RW	0x44	t_xs_dll_x32 tXSDLL: Exit Self Refresh to commands requiring a locked DLL. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: Used only for DDR2, DDR3 and DDR4 SDRAMs. Programming Mode: Quasi-dynamic Group 2 and Group 4
7	RO	0x0	reserved
6:0	RW	0x05	t_xs_x32 tXS: Exit Self Refresh to commands not requiring a locked DLL. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Unit: Multiples of 32 DFI clocks. Note: Used only for DDR2, DDR3 and DDR4 SDRAMs. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG9

Address: Operational Base + offset (0x2124)

DRAMTMG9

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	ddr4_wr_preamble DDR4 Write preamble mode 0: 1tCK preamble 1: 2tCK preamble Programming Mode: Quasi-dynamic Group 2 and Group 4
29:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18:16	RW	0x4	t_ccd_s tCCD_S: This is the minimum time between two reads or two writes for different bank group. For bank switching (from bank "a" to bank "b"), the minimum time is this value + 1. When the controller is operating in 1:2 frequency ratio mode, program this to (tCCD_S/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:12	RO	0x0	reserved
11:8	RW	0x4	t_rrd_s tRRD_S: Minimum time between activates from bank "a" to bank "b" for different bank group. When the controller is operating in 1:2 frequency ratio mode, program this to (tRRD_S/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:6	RO	0x0	reserved
5:0	RW	0x0d	wr2rd_s CWL + PL + BL/2 + tWTR_S Minimum time from write command to read command for different bank group. Includes time for bus turnaround, recovery times, and all per-bank, per-rank, and global constraints. Present only in designs configured to support DDR4. Unit: DFI Clocks. Where: CWL = CAS write latency PL = Parity latency BL = burst length. This must match the value programmed in the BL bit of the mode register to the SDRAM tWTR_S = internal write to read command delay for different bank group. This comes directly from the SDRAM specification. When the controller is operating in 1:2 mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 1 and Group 2 and Group 4

DDRC_FREQ1_DRAMTMG10

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Address: Operational Base + offset (0x2128)

DRAMTMG10

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:16	RW	0x1c	t_sync_gear Indicates the time between MRS command and the sync pulse time. This must be even number of clocks. For DDR4-2666 and DDR4-3200, this parameter is defined as tMOD(min)+4nCK tMOD(min) is greater of 24nCK or 15ns 15ns / .625ns = 24 Max value for this register is 24+4 = 28 When the controller is operating in 1:2 mode, program this to (tSYNC_GEAR/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
15:13	RO	0x0	reserved
12:8	RW	0x18	t_cmd_gear Sync pulse to first valid command. For DDR4-2666 and DDR4-3200, this parameter is defined as tMOD(min). tMOD(min) is greater of 24nCK or 15ns 15ns / .625ns = 24 Max value for this register is 24 When the controller is operating in 1:2 mode, program this to (tCMD_GEAR/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4
7:4	RO	0x0	reserved
3:2	RW	0x2	t_gear_setup Geardown setup time. Minimum value of this register is 1. Zero is invalid. For DDR4-2666 and DDR4-3200, this parameter is defined as 2 clks When the controller is operating in 1:2 frequency ratio mode, program this to (tGEAR_setup/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Quasi-dynamic Group 2 and Group 4

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Bit	Attr	Reset Value	Description
1:0	RW	0x2	t_gear_hold Geardown hold time. Minimum value of this register is 1. Zero is invalid. For DDR4-2666 and DDR4-3200, this parameter is defined as 2 clks When the controller is operating in 1:2 frequency ratio mode, program this to (tGEAR_hold/2) and round it up to the next integer value. Unit: DFI Clocks Value After Reset: 0x2 Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG11

Address: Operational Base + offset (0x212c)

DRAMTMG11

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x44	post_mpsm_gap_x32 tXMPDLL: This is the minimum Exit MPSM to commands requiring a locked DLL. When the controller is operating in 1:2 frequency ratio mode, program this to (tXMPDLL/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: Multiples of 32 DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:21	RO	0x0	reserved
20:16	RW	0x0c	t_mpx_lh tMPX_LH: This is the minimum CS_n Low hold time to CKE rising edge. When the controller is operating in 1:2 frequency ratio mode, program this to RoundUp(tMPX_LH/2)+1. Present only in designs configured to support DDR4. Unit: DFI clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:8	RW	0x2	t_mpx_s tMPX_S: Minimum time CS setup time to CKE. When the controller is operating in 1:2 frequency ratio mode, program this to (tMPX_S/2) and round it up to the next integer value. Present only in designs configured to support DDR4. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:5	RO	0x0	reserved
4:0	RW	0x1c	t_ckmpe tCKMPE: Minimum valid clock requirement after MPSM entry. Present only in designs configured to support DDR4. Unit: DFI Clocks. When the controller is operating in 1:2 frequency ratio mode, divide the value calculated using the above equation by 2, and round it up to next integer. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG12

Address: Operational Base + offset (0x2130)

DRAMTMG12

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:16	RW	0x2	t_cmdcke tCMDCKE: Delay from valid command to CKE input LOW. Set this to the larger of tESCKE or tCMDCKE When the controller is operating in 1:2 frequency ratio mode, program this to (max(tESCKE, tCMDCKE)/2) and round it up to the next integer value. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:5	RO	0x0	reserved
4:0	RW	0x10	t_mrd_pda tMRD_PDA: This is the Mode Register Set command cycle time in PDA mode. When the controller is operating in 1:2 frequency ratio mode, program this to (tMRD_PDA/2) and round it up to the next integer value. Programming Mode: Quasi-dynamic Group 2 and Group 4

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DDRC_FREQ1_DRAMTMG13

Address: Operational Base + offset (0x2134)

DRAMTMG13

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x1c	odtloff LPDDR4: tODTLoff: This is the latency from CAS-2 command to tODToff reference. When the controller is operating in 1:2 frequency ratio mode, program this to (tODTLoff/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
23:22	RO	0x0	reserved
21:16	RW	0x20	t_ccd_mw LPDDR4: tCCDMW: This is the minimum time from write or masked write to masked write command for same bank. When the controller is operating in 1:2 frequency ratio mode, program this to (tCCDMW/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4
15:3	RO	0x0	reserved
2:0	RW	0x4	t_ppd LPDDR4: tPPD: This is the minimum time from precharge to precharge command. When the controller is operating in 1:2 frequency ratio mode, program this to (tPPD/2) and round it up to the next integer value. Unit: DFI Clocks. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG14

Address: Operational Base + offset (0x2138)

DRAMTMG14

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x0a0	t_xsr tXSR: Exit Self Refresh to any command. When the controller is operating in 1:2 frequency ratio mode, program this to the above value divided by 2 and round up to next integer value. Note: Used only for mDDR/LPDDR2/LPDDR3/LPDDR4 mode. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG15

Address: Operational Base + offset (0x213c)

DRAMTMG15

Bit	Attr	Reset Value	Description
31	RW	0x0	en_dfi_lp_t_stab 1: Enable using tSTAB when exiting DFI LP. Needs to be set when the PHY is stopping the clock during DFI LP to save maximum power. 0: Disable using tSTAB when exiting DFI LP Programming Mode: Quasi-dynamic Group 2 and Group 4
30:8	RO	0x0	reserved
7:0	RW	0x00	t_stab_x32 tSTAB: Stabilization time. It is required in the following two cases for DDR3/DDR4 RDIMM : when exiting power saving mode, if the clock was stopped, after re-enabling it the clock must be stable for a time specified by tSTAB in the case of input clock frequency change (DDR4) after issuing control words that refers to clock timing (Specification: 6us for DDR3, 5us for DDR4) When the controller is operating in 1:2 frequency ratio mode, program this to recommended value divided by two and round it up to next integer. Unit: Multiples of 32 DFI clock cycles. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DRAMTMG17

Address: Operational Base + offset (0x2144)

DRAMTMG17

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Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	t_vrcg_enable LPDDR4: tVRCG_ENABLE: VREF high current mode enable time. When the controller is operating in 1:2 frequency ratio mode, program this to (tVRCG_ENABLE/2) and round it up to the next integer value. Unit: DFI clocks Programming Mode: Quasi-dynamic Group 4
15:7	RO	0x0	reserved
6:0	RW	0x00	t_vrcg_disable LPDDR4: tVRCG_ENABLE: VREF high current mode disable time. When the controller is operating in 1:2 frequency ratio mode, program this to (tVRCG_DISABLE/2) and round it up to the next integer value. Unit: DFI clocks Programming Mode: Quasi-dynamic Group 4

DDRC_FREQ1_ZQCTL0

Address: Operational Base + offset (0x2180)

ZQCTL0

Bit	Attr	Reset Value	Description
31	RW	0x0	dis_auto_zq 1: Disable uMCTL2 generation of ZQCS/MPC(ZQ calibration) command. Register DBGCMD.zq_calib_short can be used instead to issue ZQ calibration request from APB module. 0: Internally generate ZQCS/MPC(ZQ calibration) commands based on ZQCTL1.t_zq_short_interval_x1024. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Dynamic

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Bit	Attr	Reset Value	Description
30	RW	0x0	dis_srx_zqcl 1: Disable issuing of ZQCL/MPC(ZQ calibration) command at Self-Refresh/SR-Powerdown exit. Only applicable when run in DDR3 or DDR4 or LPDDR2 or LPDDR3 or LPDDR4 mode. 0: Enable issuing of ZQCL/MPC(ZQ calibration) command at Self-Refresh/SR-Powerdown exit. Only applicable when run in DDR3 or DDR4 or LPDDR2 or LPDDR3 or LPDDR4 mode. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Quasi-dynamic Group 2 and Group 4
29	RW	0x0	zq_resistor_shared 1: Denotes that ZQ resistor is shared between ranks. Means ZQinit/ZQCL/ZQCS/MPC(ZQ calibration) commands are sent to one rank at a time with tZQinit/tZQCL/tZQCS/tZQCAL/tZQLAT timing met between commands so that commands to different ranks do not overlap. 0: ZQ resistor is not shared. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static
28	RW	0x0	dis_mpsmx_zqcl 1: Disable issuing of ZQCL command at Maximum Power Saving Mode exit. Only applicable when run in DDR4 mode. 0: Enable issuing of ZQCL command at Maximum Power Saving Mode exit. Only applicable when run in DDR4 mode. This is only present for designs supporting DDR4 devices. Note: Do not issue ZQCL command at Maximum Power Save Mode exit if the UMCTL2_SHARED_AC configuration parameter is set. Program it to 1'b1. The software can send ZQCS after exiting MPSM mode. Programming Mode: Static
27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26:16	RW	0x200	t_zq_long_nop tZQoper for DDR3/DDR4, tZQCL for LPDDR2/LPDDR3, tZQCAL for LPDDR4: Number of DFI clock cycles of NOP required after a ZQCL (ZQ calibration long)/MPC(ZQ Start) command is issued to SDRAM. When the controller is operating in 1:2 frequency ratio mode: DDR3/DDR4: program this to tZQoper/2 and round it up to the next integer value. LPDDR2/LPDDR3: program this to tZQCL/2 and round it up to the next integer value. LPDDR4: program this to tZQCAL/2 and round it up to the next integer value. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static
15:10	RO	0x0	reserved
9:0	RW	0x040	t_zq_short_nop tZQCS for DDR3/DD4/LPDDR2/LPDDR3, tZQLAT for LPDDR4: Number of DFI clock cycles of NOP required after a ZQCS (ZQ calibration short)/MPC(ZQ Latch) command is issued to SDRAM. When the controller is operating in 1:2 frequency ratio mode, program this to tZQCS/2 and round it up to the next integer value. This is only present for designs supporting DDR3/DDR4 or LPDDR2/LPDDR3/LPDDR4 devices. Programming Mode: Static

DDRC_FREQ1_DFITMG0

Address: Operational Base + offset (0x2190)

DFITMG0

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RW	0x07	dfi_t_ctrl_delay Specifies the number of DFI clock cycles after an assertion or de-assertion of the DFI control signals that the control signals at the PHY-DRAM interface reflect the assertion or de-assertion. If the DFI clock and the memory clock are not phase-aligned, this timing parameter should be rounded up to the next integer value. Programming Mode: Quasi-dynamic Group 4

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Bit	Attr	Reset Value	Description
23	RW	0x0	dfi_rddata_use_dfi_phy_clk Defines whether dfi_rddata_en/dfi_rddata/dfi_rddata_valid is generated using HDR (DFI clock) or SDR (DFI PHY clock) values. Selects whether value in DFITMG0.dfi_t_rddata_en is in terms of HDR (DFI clock) or SDR (DFI PHY clock) cycles: 0: in terms of HDR (DFI clock) cycles, Only support HDR DFI clock. 1: in terms of SDR (DFI PHY clock) cycles Programming Mode: Static
22:16	RW	0x02	dfi_t_rddata_en Time from the assertion of a read command on the DFI interface to the assertion of the dfi_rddata_en signal. Refer to PHY specification for correct value. This corresponds to the DFI parameter trddata_en. Unit: DFI clock cycles Programming Mode: Quasi-dynamic Group 1 and Group 4
15	RW	0x0	dfi_wrdata_use_dfi_phy_clk Defines whether dfi_wrdata_en/dfi_wrdata/dfi_wrdata_mask is generated using HDR (DFI clock) or SDR (DFI PHY clock) values Selects whether value in DFITMG0.dfi_tphy_wrlat is in terms of HDR (DFI clock) or SDR (DFI PHY clock) cycles Selects whether value in DFITMG0.dfi_tphy_wrdata is in terms of HDR (DFI clock) or SDR (DFI PHY clock) cycles 0: in terms of HDR (DFI clock) cycles, Only support HDR DFI clock. 1: in terms of SDR (DFI PHY clock) cycles Programming Mode: Static
14	RO	0x0	reserved
13:8	RW	0x00	dfi_tphy_wrdata Specifies the number of clock cycles between when dfi_wrdata_en is asserted to when the associated write data is driven on the dfi_wrdata signal. This corresponds to the DFI timing parameter tphy_wrdata. Refer to PHY specification for correct value. Note, max supported value is 8. Unit: DFI clock cycles. Programming Mode: Quasi-dynamic Group 4
7:6	RO	0x0	reserved
5:0	RW	0x02	dfi_tphy_wrlat Write latency Number of clocks from the write command to write data enable (dfi_wrdata_en). This corresponds to the DFI timing parameter tphy_wrlat. Refer to PHY specification for correct value. For LPDDR4, dfi_tphy_wrlat>60 is not supported. Unit: DFI clock cycles Programming Mode: Quasi-dynamic Group 1 and Group 4

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DDRC_FREQ1_DFITMG1

Address: Operational Base + offset (0x2194)

DFITMG1

Bit	Attr	Reset Value	Description
31:28	RW	0x0	dfi_t_cmd_lat Specifies the number of DFI PHY clock cycles between when the dfi_cs signal is asserted and when the associated command is driven. This field is used for CAL mode, should be set to '0' or the value which matches the CAL mode register setting in the DRAM. If the PHY can add the latency for CAL mode, this should be set to '0'. Valid Range: 0, 3, 4, 5, 6, and 8 Programming Mode: Quasi-dynamic Group 2 and Group 4
27:26	RO	0x0	reserved
25:24	RW	0x0	dfi_t_parin_lat Specifies the number of DFI PHY clock cycles between when the dfi_cs signal is asserted and when the associated dfi_parity_in signal is driven. Programming Mode: Quasi-dynamic Group 4
23:21	RO	0x0	reserved
20:16	RW	0x00	dfi_t_wrdata_delay Specifies the number of DFI clock cycles between when the dfi_wrdata_en signal is asserted and when the corresponding write data transfer is completed on the DRAM bus. This corresponds to the DFI timing parameter twrdata_delay. Refer to PHY specification for correct value. Value to be programmed is in terms of DFI clocks, not PHY clocks. In FREQ_RATIO=2, divide PHY's value by 2 and round up to next integer. Programming Mode: Quasi-dynamic Group 4
15:13	RO	0x0	reserved
12:8	RW	0x04	dfi_t_dram_clk_disable Specifies the number of DFI clock cycles from the assertion of the dfi_dram_clk_disable signal on the DFI until the clock to the DRAM memory devices, at the PHY-DRAM boundary, maintains a low value. If the DFI clock and the memory clock are not phase aligned, this timing parameter should be rounded up to the next integer value. Programming Mode: Quasi-dynamic Group 4
7:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x04	dfi_t_dram_clk_enable Specifies the number of DFI clock cycles from the deassertion of the dfi_dram_clk_disable signal on the DFI until the first valid rising edge of the clock to the DRAM memory devices, at the PHY-DRAM boundary. If the DFI clock and the memory clock are not phase aligned, this timing parameter should be rounded up to the next integer value. Programming Mode: Quasi-dynamic Group 4

DDRC_FREQ1_DFITMG2

Address: Operational Base + offset (0x21b4)

DFITMG2

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:8	RW	0x02	dfi_tphy_rdcslat Number of DFI PHY clock cycles between when a read command is sent on the DFI control interface and when the associated dfi_rddata_cs signal is asserted. This corresponds to the DFI timing parameter tphy_rdcslat. Refer to PHY specification for correct value. Programming Mode: Quasi-dynamic Group 2 and Group 4
7:6	RO	0x0	reserved
5:0	RW	0x02	dfi_tphy_wrcslat Number of DFI PHY clock cycles between when a write command is sent on the DFI control interface and when the associated dfi_wrdata_cs signal is asserted. This corresponds to the DFI timing parameter tphy_wrcslat. Refer to PHY specification for correct value. Programming Mode: Quasi-dynamic Group 2 and Group 4

DDRC_FREQ1_DFITMG3

Address: Operational Base + offset (0x21b8)

DFITMG3

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	dfi_t_geardown_delay The delay from dfi_geardown_en assertion to the time of the PHY being ready to receive commands. Refer to PHY specification for correct value. When the controller is operating in 1:2 frequency ratio mode, program this to (tgeardown_delay/2) and round it up to the next integer value. Unit: DFI Clocks Programming Mode: Static

DDRC_FREQ1_ODTCFG

Address: Operational Base + offset (0x2240)

ODTCFG

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x4	wr_odt_hold DFI PHY clock cycles to hold ODT for a write command. The minimum supported value is 2. Recommended values: DDR2: BL8: 0x5 (DDR2-400/533/667), 0x6 (DDR2-800), 0x7(DDR2-1066); BL4: 0x3 (DDR2-400/533/667), 0x4 (DDR2-800), 0x5 (DDR2-1066) DDR3: BL8: 0x6 DDR4: BL8: 5 + WR_PREAMBLE + CRC_MODE. WR_PREAMBLE = 1 (1tCK write preamble), 2 (2tCK write preamble). CRC_MODE = 0 (not CRC mode), 1 (CRC mode) LPDDR3: BL8: 7 + RU(tODTon(max)/tCK) Programming Mode: Quasi-dynamic Group 1 and Group 4
23:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20:16	RW	0x00	wr_odt_delay The delay, in DFI PHY clock cycles, from issuing a write command to setting ODT values associated with that command. ODT setting must remain constant for the entire time that DQS is driven by the uMCTL2. Recommended values: DDR2: CWL + AL - 3 (DDR2-400/533/667), CWL + AL - 4 (DDR2-800), CWL + AL - 5 (DDR2-1066). If (CWL + AL - 3 < 0), uMCTL2 does not support ODT for write operation. DDR3: 0x0 DDR4: DFITMG1.dfi_t_cmd_lat (to adjust for CAL mode) LPDDR3: WL - 1 - RU(tODTon(max)/tCK)) Programming Mode: Quasi-dynamic Group 1 and Group 4
15:12	RO	0x0	reserved
11:8	RW	0x4	rd_odt_hold DFI PHY clock cycles to hold ODT for a read command. The minimum supported value is 2. Recommended values: DDR2: BL8: 0x6 (not DDR2-1066), 0x7 (DDR2-1066); BL4: 0x4 (not DDR2-1066), 0x5 (DDR2-1066) DDR3: BL8 - 0x6 DDR4: BL8: 5 + RD_PREAMBLE. RD_PREAMBLE = 1 (1tCK write preamble), 2 (2tCKwrite preamble) LPDDR3: BL8: 5 + RU(tDQSCK(max)/tCK) - RD(tDQSCK(min)/tCK) + RU(tODTon(max)/tCK) Programming Mode: Quasi-dynamic Group 1 and Group 4
7	RO	0x0	reserved
6:2	RW	0x00	rd_odt_delay The delay, in DFI PHY clock cycles, from issuing a read command to setting ODT values associated with that command. ODT setting must remain constant for the entire time that DQS is driven by the uMCTL2. Recommended values: DDR2: CL + AL - 4 (not DDR2-1066), CL + AL - 5 (DDR2-1066). If (CL + AL - 4 < 0), uMCTL2 does not support ODT for read operation. DDR3: CL - CWL DDR4: CL - CWL - RD_PREAMBLE + WR_PREAMBLE + DFITMG1.dfi_t_cmd_lat (to adjust for CAL mode). WR_PREAMBLE = 1 (1tCK write preamble), 2 (2tCK write preamble). RD_PREAMBLE = 1 (1tCK write preamble), 2 (2tCK write preamble). If (CL - CWL - RD_PREAMBLE + WR_PREAMBLE) < 0, uMCTL2 does not support ODT for read operation. LPDDR3: RL + RD(tDQSCK(min)/tCK) - 1 - RU(tODTon(max)/tCK) Programming Mode: Quasi-dynamic Group 1 and Group 4
1:0	RO	0x0	reserved

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DDRPHY_REG0

Address: Operational Base + offset (0x0000)

DDR PHY register 00

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	channel select 4'b0011: DQ 16bit 4'b1111: DQ 32bit
3	RW	0x1	soft reset 1, active low
2	RW	0x1	soft reset 0, active low
1	RW	0x1	
0	RW	0x1	

DDRPHY_REG1

Address: Operational Base + offset (0x0004)

DDR PHY register 01

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:3	RW	0x01	
2:0	RW	0x0	PHY working condition select 0x0: ddr2 PHY mode 0x1: lpddr2 PHY mode 0x2: ddr3 PHY mode 0x3: lpddr3 PHY mode 0x4: ddr4 PHY mode

DDRPHY_REG2

Address: Operational Base + offset (0x0008)

DDR PHY register 02

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:6	RW	0x0	Write leveling CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
5:4	RW	0x0	DQS gating calibration CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
3	RW	0x0	Write leveling calibration bypass mode, active high
2	RW	0x0	Write leveling calibration control, active high
1	RW	0x0	DQS gating calibration bypass mode, active high
0	RW	0x0	DQS gating calibration control, active high

DDRPHY_REG3

Address: Operational Base + offset (0x000c)

DDR PHY register 03

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x2	right channel A read ODT delay by read odt configure bypass mode
3	RO	0x0	reserved
2:0	RW	0x2	left channel A read ODT delay by read odt configure bypass mode

DDRPHY_REG4

Address: Operational Base + offset (0x0010)

DDR PHY register 04

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x2	right channel B read ODT delay by read odt configure bypass mode
3	RO	0x0	reserved
2:0	RW	0x2	left channel B read ODT delay by read odt configure bypass mode

DDRPHY_REG5

Address: Operational Base + offset (0x0014)

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DDR PHY register 05

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	Write leveling load mode[7:0]

DDRPHY_REG6

Address: Operational Base + offset (0x0018)

DDR PHY register 06

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	Write leveling load mode select[1:0]
5:0	RW	0x02	Write leveling load mode[13:8]

DDRPHY_REG9

Address: Operational Base + offset (0x0024)

DDR PHY register 09

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	lower power bypass 0: Normal mode; 1: Bypass
5:0	RO	0x0	reserved

DDRPHY_REGA

Address: Operational Base + offset (0x0028)

DDR PHY register 0A

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	CL DDR2 /DDR3/DDR4 CAS Latency; LPDDR2/3 RL value

DDRPHY_REGB

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Address: Operational Base + offset (0x002c)

DDR PHY register 0B

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	AL DDR2/DDR3/DDR4 additive latency

DDRPHY_REGC

Address: Operational Base + offset (0x0030)

DDR PHY register 0C

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	CWL DDR2/DDR3/DDR4 WRITE CAS Latency; LPDDR2/3 WL value

DDRPHY_REG11

Address: Operational Base + offset (0x0044)

DDR PHY register 11

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	CMD NRCOMP, except for CK/CKB. The larger the value, the stronger the drive strength.

DDRPHY_REG12

Address: Operational Base + offset (0x0048)

DDR PHY register 12

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:3	RW	0x14	CMD PRCOMP, except for CK/CKB. The larger the value, the stronger the driver strength.
2	RO	0x0	reserved
1	RW	0x1	CMD weak pull up enable, active low

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Bit	Attr	Reset Value	Description
0	RW	0x0	CMD weak pull down enable, active high

DDRPHY_REG13

Address: Operational Base + offset (0x004c)

DDR PHY register 13

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	CMD DLL clock phase select in bypass mode 0: no delay 1: 90°delay
3	RW	0x1	CMD DLL enable 0: disable 1: enable
2:0	RW	0x4	CMD AND ADDRESS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG14

Address: Operational Base + offset (0x0050)

DDR PHY register 13

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x1	CK DLL clock phase select in bypass mode 0: no delay 1: 90°delay
2:0	RW	0x0	CK DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG16

Address: Operational Base + offset (0x0058)

DDR PHY register 16

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Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	CK/CKB NRCOMP. The larger the value, the stronger the drive strength.

DDRPHY_REG17

Address: Operational Base + offset (0x005c)

DDR PHY register 17

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x4	CMD/CK falling edge slew rate control, larger value means larger falling slew rate
3	RO	0x0	reserved
2:0	RW	0x4	CMD/CK rising edge slew rate control, larger vale means larger rising slew rate

DDRPHY_REG18

Address: Operational Base + offset (0x0060)

DDR PHY register 17

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	CK/CKB PRCOMP. The larger the value, the stronger the drive strength.

DDRPHY_REG20

Address: Operational Base + offset (0x0080)

DDR PHY register 20

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Left channel A NRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ0 to A_DQ7.

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DDRPHY_REG21

Address: Operational Base + offset (0x0084)

DDR PHY register 21

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Left channel A read pull-up ODT. The larger the value, the smaller the pull-up resistance in the scope from A_DQ0 to A_DQ7.

DDRPHY_REG26

Address: Operational Base + offset (0x0098)

DDR PHY register 26

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Left channel A write DQ DLL phase select in bypass mode. 0: no delay 1: 90°delay
3	RW	0x1	Left channel A write DQ DLL enable, active HIGH.
2:0	RW	0x4	Left channel A write DQ DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG27

Address: Operational Base + offset (0x009c)

DDR PHY register 27

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	Left channel A write DQS DLL phase select in bypass mode. 0: no delay 1: 90°delay
2:0	RW	0x0	Left channel A write DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

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DDRPHY_REG28

Address: Operational Base + offset (0x00a0)

DDR PHY register 28

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	Left channel A read DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay

DDRPHY_REG29

Address: Operational Base + offset (0x00a4)

DDR PHY register 29

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x1	Weak pull up of the A_DQS[0].
0	RW	0x0	Weak pull down of the A_DQS[0]. REG29[1:0]=2'b00. Pull Up REG29[1:0]=2'b01. Middle Level REG29[1:0]=2'b10. High-Z REG29[1:0]=2'b11. Pull Down

DDRPHY_REG2B

Address: Operational Base + offset (0x00ac)

DDR PHY register 2B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x4	Left channel A falling edge slew rate control, larger value means larger falling slew rate
4:2	RW	0x0	Left channel A rising edge slew rate control, larger value means larger rising slew rate.
1:0	RW	0x1	

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DDRPHY_REG2E

Address: Operational Base + offset (0x00b8)

DDR PHY register 2E

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Left channel A read pull-up ODT. The larger the value, the smaller the pull-up resistance in the scope from A_DQ0 to A_DQ7.

DDRPHY_REG2F

Address: Operational Base + offset (0x00bc)

DDR PHY register 2F

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Left channel A PRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ0 to A_DQ7.

DDRPHY_REG30

Address: Operational Base + offset (0x00c0)

DDR PHY register 30

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Right channel A NRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ8 to A_DQ15.

DDRPHY_REG31

Address: Operational Base + offset (0x00c4)

DDR PHY register 31

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Right channel A read pull-down ODT. The larger the value, the smaller the pull-down resistance in the scope from A_DQ8 to A_DQ15.

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DDRPHY_REG36

Address: Operational Base + offset (0x00d8)

DDR PHY register 36

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Right channel A write DQ DLL phase select in bypass mode 0: no delay 1: 90°delay
3	RW	0x0	Right channel A write DQ DLL enable, active high
2:0	RW	0x4	Right channel A write DQ DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG37

Address: Operational Base + offset (0x00dc)

DDR PHY register 37

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	Right channel A write DQS DLL phase select in bypass mode. 0: no delay 1: 90°delay
2:0	RW	0x0	Right channel A write DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG38

Address: Operational Base + offset (0x00e0)

DDR PHY register 38

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x1	Right channel A read DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay

DDRPHY_REG39

Address: Operational Base + offset (0x00e4)

DDR PHY register 39

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x1	Weak pull up of the A_DQS[1].
0	RW	0x0	Weak pull down of the A_DQS[1]. REG39[1:0] = 2'b00. Pull Up. REG39[1:0] =2'b01. Middle level. REG39[1:0] = 2'b10. High-Z. REG39[1:0] = 2'b11. Pull Down.

DDRPHY_REG3B

Address: Operational Base + offset (0x00ec)

DDR PHY register 3B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x4	Right channel A falling edge slew rate control, larger value means larger falling slew rate
4:2	RW	0x0	Right channel A rising edge slew rate control, larger value means larger rising slew rate.
1:0	RW	0x1	

DDRPHY_REG3E

Address: Operational Base + offset (0x00f8)

DDR PHY register 3E

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x04	Right channel A read pull-up ODT. The larger the value, the smaller the pull-up resistance in the scope from A_DQ8 to A_DQ15.

DDRPHY_REG3F

Address: Operational Base + offset (0x00fc)

DDR PHY register 3F

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Right channel A PRCOMP. The larger the value, the stronger the drive strength in the scope from A_DQ8 to A_DQ15.

DDRPHY_REG40

Address: Operational Base + offset (0x0100)

DDR PHY register 40

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Left channel B NRCOMP. The larger the value, the stronger the drive strength in the scope from B_DQ0 to B_DQ7.

DDRPHY_REG41

Address: Operational Base + offset (0x0104)

DDR PHY register 41

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Left channel B read pull-down ODT. The larger the value, the smaller the pull-down resistance in the scope from B_DQ0 to B_DQ7.

DDRPHY_REG46

Address: Operational Base + offset (0x0118)

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DDR PHY register 46

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Left channel B write DQ DLL phase select in bypass mode. 0: no delay 1: 90°delay
3	RW	0x1	Field0001 Abstract Left channel B write DQ DLL enable, active HIGH.
2:0	RW	0x4	Left channel B write DQ DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG47

Address: Operational Base + offset (0x011c)

DDR PHY register 47

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	Left channel B write DQS DLL phase select in bypass mode. 0: no delay 1: 90°delay
2:0	RW	0x0	Left channel B write DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG48

Address: Operational Base + offset (0x0120)

DDR PHY register 48

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	Left channel B read DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay

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DDRPHY_REG49

Address: Operational Base + offset (0x0124)

DDR PHY register 49

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x1	Weak pull up of the B_DQS[0].
0	RW	0x0	Weak pull down of the B_DQS[0]. REG49[1:0] = 2'b00. Pull Up. REG49[1:0] =2'b01. Middle level. REG49[1:0] = 2'b10. High-Z. REG49[1:0] = 2'b11. Pull Down.

DDRPHY_REG4B

Address: Operational Base + offset (0x012c)

DDR PHY register 4B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x4	Left channel B falling edge slew rate control, larger value means larger falling slew rate
4:2	RW	0x4	Left channel B rising edge slew rate control, larger value means larger rising slew rate
1:0	RW	0x1	

DDRPHY_REG4E

Address: Operational Base + offset (0x0138)

DDR PHY register 4E

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Left channel B read pull-up ODT. The larger the value, the smaller the pull-up resistance in the scope from B_DQ0 to B_DQ7.

DDRPHY_REG4F

Address: Operational Base + offset (0x013c)

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DDR PHY register 4F

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Left channel B PRCOMP. The larger the value, the stronger the drive strength in the scope from B_DQ0 to B_DQ7.

DDRPHY_REG50

Address: Operational Base + offset (0x0140)

DDR PHY register 50

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Right channel B NRCOMP. The larger the value, the stronger the drive strength in the scope from B_DQ8 to B_DQ15.

DDRPHY_REG51

Address: Operational Base + offset (0x0144)

DDR PHY register 51

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Right channel B read pull-down ODT. The larger the value, the smaller the pull-down resistance in the scope from B_DQ8 to B_DQ15.

DDRPHY_REG56

Address: Operational Base + offset (0x0158)

DDR PHY register 56

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Right channel B write DQ DLL phase select in bypass mode. 0: no delay 1: 90°delay
3	RW	0x1	Right channel B write DQ DLL enable, active HIGH.

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Bit	Attr	Reset Value	Description
2:0	RW	0x4	Right channel B write DQ DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG57

Address: Operational Base + offset (0x015c)

DDR PHY register 57

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	Right channel B write DQS DLL phase select in bypass mode. 0: no delay 1: 90°delay
2:0	RW	0x0	Right channel B write DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay 4: 90°delay 5: 112.5°delay 6: 135°delay 7: 157.5°delay

DDRPHY_REG58

Address: Operational Base + offset (0x0160)

DDR PHY register 58

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	Right channel B read DQS DLL delay 0: no delay 1: 22.5°delay 2: 45°delay 3: 67.5°delay

DDRPHY_REG59

Address: Operational Base + offset (0x0164)

DDR PHY register 59

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x1	Weak pull up of the B_DQS[1].
0	RW	0x0	Weak pull down of the B_DQS[1]. REG59[1:0] = 2'b00. Pull Up. REG59[1:0] =2'b01. Middle level. REG59[1:0] = 2'b10. High-Z. REG59[1:0] = 2'b11. Pull Down

DDRPHY_REG5B

Address: Operational Base + offset (0x016c)

DDR PHY register 5B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x4	Right channel B falling edge slew rate control, larger value means larger falling slew rate
4:2	RW	0x4	Field0001 Abstract Right channel B rising edge slew rate control, larger value means larger rising slew rate
1:0	RW	0x1	

DDRPHY_REG5E

Address: Operational Base + offset (0x0178)

DDR PHY register 5E

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x04	Right channel B read pull-up ODT. The larger the value, the smaller the pull-up resistance in the scope from B_DQ8 to B_DQ15.

DDRPHY_REG5F

Address: Operational Base + offset (0x017c)

DDR PHY register 5F

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x14	Right channel B PRCOMP. The larger the value, the stronger the drive strength in the scope from B_DQ8 to B_DQ15.

DDRPHY_REG70

Address: Operational Base + offset (0x01c0)

DDR PHY register 70

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DM0 RX de-skew
3:0	RW	0x7	CS0 A_DM0 TX de-skew

DDRPHY_REG71

Address: Operational Base + offset (0x01c4)

DDR PHY register 71

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ0 RX de-skew
3:0	RW	0x7	CS0 A_DQ0 TX de-skew

DDRPHY_REG72

Address: Operational Base + offset (0x01c8)

DDR PHY register 72

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ1 RX de-skew
3:0	RW	0x7	CS0 A_DQ1 TX de-skew

DDRPHY_REG73

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Address: Operational Base + offset (0x01cc)

DDR PHY register 73

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ2 RX de-skew
3:0	RW	0x7	CS0 A_DQ2 TX de-skew

DDRPHY_REG74

Address: Operational Base + offset (0x01d0)

DDR PHY register 74

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ3 RX de-skew
3:0	RW	0x7	CS0 A_DQ3 TX de-skew

DDRPHY_REG75

Address: Operational Base + offset (0x01d4)

DDR PHY register 75

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ4 RX de-skew
3:0	RW	0x7	CS0 A_DQ4 TX de-skew

DDRPHY_REG76

Address: Operational Base + offset (0x01d8)

DDR PHY register 76

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ5 RX de-skew
3:0	RW	0x7	CS0 A_DQ5 TX de-skew

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DDRPHY_REG77

Address: Operational Base + offset (0x01dc)

DDR PHY register 77

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ6 RX de-skew
3:0	RW	0x7	CS0 A_DQ6 TX de-skew

DDRPHY_REG78

Address: Operational Base + offset (0x01e0)

DDR PHY register 78

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ7 RX de-skew
3:0	RW	0x7	CS0 A_DQ7 TX de-skew

DDRPHY_REG79

Address: Operational Base + offset (0x01e4)

DDR PHY register 79

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQS0 RX de-skew
3:0	RW	0x7	CS0 A_DQS0 TX de-skew

DDRPHY_REG7A

Address: Operational Base + offset (0x01e8)

DDR PHY register 7A

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS0 A_DQSB0 TX de-skew

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DDRPHY_REG7B

Address: Operational Base + offset (0x01ec)

DDR PHY register 7B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DM1 RX de-skew
3:0	RW	0x7	CS0 A_DM1 TX de-skew

DDRPHY_REG7C

Address: Operational Base + offset (0x01f0)

DDR PHY register 7C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ8 RX de-skew
3:0	RW	0x7	CS0 A_DQ8 TX de-skew

DDRPHY_REG7D

Address: Operational Base + offset (0x01f4)

DDR PHY register 7D

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ9 RX de-skew
3:0	RW	0x7	CS0 A_DQ9 TX de-skew

DDRPHY_REG7E

Address: Operational Base + offset (0x01f8)

DDR PHY register 7E

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x7	CS0 A_DQ10 RX de-skew
3:0	RW	0x7	CS0 A_DQ10 TX de-skew

DDRPHY_REG7F

Address: Operational Base + offset (0x01fc)

DDR PHY register 7F

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ11 RX de-skew
3:0	RW	0x7	CS0 A_DQ11 TX de-skew

DDRPHY_REG80

Address: Operational Base + offset (0x0200)

DDR PHY register 80

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ12 RX de-skew
3:0	RW	0x7	CS0 A_DQ12 TX de-skew

DDRPHY_REG81

Address: Operational Base + offset (0x0204)

DDR PHY register 81

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ13 RX de-skew
3:0	RW	0x7	CS0 A_DQ13 TX de-skew

DDRPHY_REG82

Address: Operational Base + offset (0x0208)

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DDR PHY register 82

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQ14 RX de-skew
3:0	RW	0x7	CS0 A_DQ14 TX de-skew

DDRPHY_REG83

Address: Operational Base + offset (0x020c)

DDR PHY register 83

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0_A_DQ15 RX de-skew
3:0	RW	0x7	CS0 A_DQ15 TX de-skew

DDRPHY_REG84

Address: Operational Base + offset (0x0210)

DDR PHY register 84

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 A_DQS1 RX de-skew
3:0	RW	0x7	CS0 A_DQS1 TX de-skew

DDRPHY_REG85

Address: Operational Base + offset (0x0214)

DDR PHY register 85

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS0 A_DQSB1 TX de-ske

DDRPHY_REG86

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Address: Operational Base + offset (0x0218)

DDR PHY register 86

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DM0 RX de-skew
3:0	RW	0x7	CS0 B_DM0 TX de-skew

DDRPHY_REG87

Address: Operational Base + offset (0x021c)

DDR PHY register 87

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ0 RX de-skew
3:0	RW	0x7	CS0 B_DQ0 TX de-skew

DDRPHY_REG88

Address: Operational Base + offset (0x0220)

DDR PHY register 88

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ1 RX de-skew
3:0	RW	0x7	CS0 B_DQ1 TX de-skew

DDRPHY_REG89

Address: Operational Base + offset (0x0224)

DDR PHY register 89

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ2 RX de-skew
3:0	RW	0x7	FCS0 B_DQ2 TX de-skew

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DDRPHY_REG8A

Address: Operational Base + offset (0x0228)

DDR PHY register 8A

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ3 RX de-skew
3:0	RW	0x7	CS0 B_DQ3 TX de-skew

DDRPHY_REG8B

Address: Operational Base + offset (0x022c)

DDR PHY register 8B

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ4 RX de-skew
3:0	RW	0x7	CS0 B_DQ4 TX de-skew

DDRPHY_REG8C

Address: Operational Base + offset (0x0230)

DDR PHY register 8C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ5 RX de-skew
3:0	RW	0x7	CS0 B_DQ5 TX de-skew

DDRPHY_REG8D

Address: Operational Base + offset (0x0234)

DDR PHY register 8D

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ6 RX de-skew

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Bit	Attr	Reset Value	Description
3:0	RW	0x7	CS0 B_DQ6 TX de-skew

DDRPHY_REG8E

Address: Operational Base + offset (0x0238)

DDR PHY register 8E

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ7 RX de-skew
3:0	RW	0x7	CS0 B_DQ7 TX de-skew

DDRPHY_REG8F

Address: Operational Base + offset (0x023c)

DDR PHY register 8F

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQS0 RX de-skew
3:0	RW	0x7	CS0 B_DQS0 TX de-skew

DDRPHY_REG90

Address: Operational Base + offset (0x0240)

DDR PHY register 90

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS0 B_DQSB0 TX de-skew

DDRPHY_REG91

Address: Operational Base + offset (0x0244)

DDR PHY register 91

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DM1 RX de-skew
3:0	RW	0x7	CS0 B_DM1 TX de-skew

DDRPHY_REG92

Address: Operational Base + offset (0x0248)

DDR PHY register 92

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ8 RX de-skew
3:0	RW	0x7	CS0 B_DQ8 TX de-skew

DDRPHY_REG93

Address: Operational Base + offset (0x024c)

DDR PHY register 93

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ9 RX de-skew
3:0	RW	0x7	CS0 B_DQ9 TX de-skew

DDRPHY_REG94

Address: Operational Base + offset (0x0250)

DDR PHY register 94

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ10 RX de-skew
3:0	RW	0x7	CS0 B_DQ10 TX de-skew

DDRPHY_REG95

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Address: Operational Base + offset (0x0254)

DDR PHY register 95

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ11 RX de-skew
3:0	RW	0x7	CS0 B_DQ11 TX de-skew

DDRPHY_REG96

Address: Operational Base + offset (0x0258)

DDR PHY register 96

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ12 RX de-skew
3:0	RW	0x7	CS0 B_DQ12 TX de-skew

DDRPHY_REG97

Address: Operational Base + offset (0x025c)

DDR PHY register 97

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ13 RX de-skew
3:0	RW	0x7	CS0 B_DQ13 TX de-skew

DDRPHY_REG98

Address: Operational Base + offset (0x0260)

DDR PHY register 98

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ14 RX de-skew
3:0	RW	0x7	CS0 B_DQ14 TX de-skew

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DDRPHY_REG99

Address: Operational Base + offset (0x0264)

DDR PHY register 99

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQ15 RX de-skew
3:0	RW	0x7	CS0 B_DQ15 TX de-skew

DDRPHY_REG9A

Address: Operational Base + offset (0x0268)

DDR PHY register 9A

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS0 B_DQS1 RX de-skew
3:0	RW	0x7	Field0000 Abstract Field0000 Description

DDRPHY_REG9B

Address: Operational Base + offset (0x026c)

DDR PHY register 9B

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS0 B_DQSB1 TX de-skew

DDRPHY_REGB0

Address: Operational Base + offset (0x02c0)

DDR PHY register B0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A1 de-skew

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Bit	Attr	Reset Value	Description
3:0	RW	0x7	A0 de-skew

DDRPHY_REGB1

Address: Operational Base + offset (0x02c4)

DDR PHY register B1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A3 de-skew
3:0	RW	0x7	A2 de-skew

DDRPHY_REGB2

Address: Operational Base + offset (0x02c8)

DDR PHY register B2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A5 de-skew
3:0	RW	0x7	A4 de-skew

DDRPHY_REGB3

Address: Operational Base + offset (0x02cc)

DDR PHY register B3

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A7 de-skew
3:0	RW	0x7	A6 de-skew

DDRPHY_REGB4

Address: Operational Base + offset (0x02d0)

DDR PHY register B4

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A9 de-skew
3:0	RW	0x7	A8 de-skew

DDRPHY_REGB5

Address: Operational Base + offset (0x02d4)

DDR PHY register B5

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A11 de-skew
3:0	RW	0x7	A10 de-skew

DDRPHY_REGB6

Address: Operational Base + offset (0x02d8)

DDR PHY register B6

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A13 de-skew
3:0	RW	0x7	A12 de-skew

DDRPHY_REGB7

Address: Operational Base + offset (0x02dc)

DDR PHY register B7

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	A15 de-skew
3:0	RW	0x7	A14 de-skew

DDRPHY_REGB8

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Address: Operational Base + offset (0x02e0)

DDR PHY register B8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	B1 de-skew
3:0	RW	0x7	B0 de-skew

DDRPHY_REGB9

Address: Operational Base + offset (0x02e4)

DDR PHY register B9

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	RAS# de-skew
3:0	RW	0x7	B2 de-skew

DDRPHY_REGBA

Address: Operational Base + offset (0x02e8)

DDR PHY register BA

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	WE# de-skew
3:0	RW	0x7	CAS# de-skew

DDRPHY_REGBB

Address: Operational Base + offset (0x02ec)

DDR PHY register BB

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CKB de-skew
3:0	RW	0x7	CK de-skew

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DDRPHY_REGBC

Address: Operational Base + offset (0x02f0)

DDR PHY register BC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CKE de-skew
3:0	RW	0x7	ODT0 de-skew

DDRPHY_REGBD

Address: Operational Base + offset (0x02f4)

DDR PHY register BD

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CSB0 de-skew
3:0	RW	0x7	RESETN de-skew

DDRPHY_REGBE

Address: Operational Base + offset (0x02f8)

DDR PHY register BE

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CSB1 de-skew
3:0	RW	0x7	ODT1 de-skew

DDRPHY_REGC0

Address: Operational Base + offset (0x0300)

DDR PHY register C0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DM0 RX de-skew

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Bit	Attr	Reset Value	Description
3:0	RW	0x7	CS1 A_DM0 TX de-skew

DDRPHY_REGC1

Address: Operational Base + offset (0x0304)

DDR PHY register C1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ0 RX de-skew
3:0	RW	0x7	CS1 A_DQ0 TX de-skew

DDRPHY_REGC2

Address: Operational Base + offset (0x0308)

DDR PHY register C2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ1 RX de-skew
3:0	RW	0x7	CS1 A_DQ1 TX de-skew

DDRPHY_REGC3

Address: Operational Base + offset (0x030c)

DDR PHY register C3

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ2 RX de-skew
3:0	RW	0x7	CS1 A_DQ2 TX de-skew

DDRPHY_REGC4

Address: Operational Base + offset (0x0310)

DDR PHY register C4

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ3 RX de-skew
3:0	RW	0x7	CS1 A_DQ3 TX de-skew

DDRPHY_REGC5

Address: Operational Base + offset (0x0314)

DDR PHY register C5

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ4 RX de-skew
3:0	RW	0x7	CS1 A_DQ4 TX de-skew

DDRPHY_REGC6

Address: Operational Base + offset (0x0318)

DDR PHY register C6

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ5 RX de-skew
3:0	RW	0x7	CS1 A_DQ5 TX de-skew

DDRPHY_REGC7

Address: Operational Base + offset (0x031c)

DDR PHY register C7

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ6 RX de-skew
3:0	RW	0x7	CS1 A_DQ6 TX de-skew

DDRPHY_REGC8

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Address: Operational Base + offset (0x0320)

DDR PHY register C8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ7 RX de-skew
3:0	RW	0x7	CS1 A_DQ7 TX de-skew

DDRPHY_REGC9

Address: Operational Base + offset (0x0324)

DDR PHY register C9

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQS0 RX de-skew
3:0	RW	0x7	CS1 A_DQS0 TX de-skew

DDRPHY_REGCA

Address: Operational Base + offset (0x0328)

DDR PHY register CA

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS1 A_DQSB0 TX de-skew

DDRPHY_REGCB

Address: Operational Base + offset (0x032c)

DDR PHY register CB

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DM1 RX de-skew
3:0	RW	0x7	CS1 A_DM1 TX de-skew

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DDRPHY_REGCC

Address: Operational Base + offset (0x0330)

DDR PHY register CC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ8 RX de-skew
3:0	RW	0x7	CS1 A_DQ8 TX de-skew

DDRPHY_REGCD

Address: Operational Base + offset (0x0334)

DDR PHY register CD

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ9 RX de-skew
3:0	RW	0x7	CS1 A_DQ9 TX de-skew

DDRPHY_REGCE

Address: Operational Base + offset (0x0338)

DDR PHY register CE

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ10 RX de-skew
3:0	RW	0x7	CS1 A_DQ10 TX de-skew

DDRPHY_REGCF

Address: Operational Base + offset (0x033c)

DDR PHY register CF

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ11 RX de-skew
3:0	RW	0x7	CS1 A_DQ11 TX de-skew

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DDRPHY_REGD0

Address: Operational Base + offset (0x0340)

DDR PHY register D0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ12 RX de-skew
3:0	RW	0x7	CS1 A_DQ12 TX de-skew

DDRPHY_REGD1

Address: Operational Base + offset (0x0344)

DDR PHY register D1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ13 RX de-skew
3:0	RW	0x7	CS1 A_DQ13 TX de-skew

DDRPHY_REGD2

Address: Operational Base + offset (0x0348)

DDR PHY register D2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQ14 RX de-skew
3:0	RW	0x7	CS1 A_DQ14 TX de-skew

DDRPHY_REGD3

Address: Operational Base + offset (0x034c)

DDR PHY register D3

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x7	A_DQ15 RX de-skew
3:0	RW	0x7	CS1 A_DQ15 TX de-skew

DDRPHY_REGD4

Address: Operational Base + offset (0x0350)

DDR PHY register D4

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 A_DQS1 RX de-skew
3:0	RW	0x7	CS1 A_DQS1 TX de-skew

DDRPHY_REGD5

Address: Operational Base + offset (0x0354)

DDR PHY register D5

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS1 A_DQSB1 TX de-skew

DDRPHY_REGD6

Address: Operational Base + offset (0x0358)

DDR PHY register D6

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DM0 RX de-skew
3:0	RW	0x7	CS1 B_DM0 TX de-skew

DDRPHY_REGD7

Address: Operational Base + offset (0x035c)

DDR PHY register D7

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ0 RX de-skew
3:0	RW	0x7	CS1 B_DQ0 TX de-skew

DDRPHY_REGD8

Address: Operational Base + offset (0x0360)

DDR PHY register D8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ1 RX de-skew
3:0	RW	0x7	CS1 B_DQ1 TX de-skew

DDRPHY_REGD9

Address: Operational Base + offset (0x0364)

DDR PHY register D9

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ2 RX de-skew
3:0	RW	0x7	CS1 B_DQ2 TX de-skew

DDRPHY_REGDA

Address: Operational Base + offset (0x0368)

DDR PHY register DA

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ3 RX de-skew
3:0	RW	0x7	CS1 B_DQ3 TX de-skew

DDRPHY_REGDB

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Address: Operational Base + offset (0x036c)

DDR PHY register DB

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ4 RX de-skew
3:0	RW	0x7	CS1 B_DQ4 TX de-skew

DDRPHY_REGDC

Address: Operational Base + offset (0x0370)

DDR PHY register DC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ5 RX de-skew
3:0	RW	0x7	CS1 B_DQ5 TX de-skew

DDRPHY_REGDD

Address: Operational Base + offset (0x0374)

DDR PHY register DD

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ6 RX de-skew
3:0	RW	0x7	CS1 B_DQ6 TX de-skew

DDRPHY_REGDE

Address: Operational Base + offset (0x0378)

DDR PHY register DE

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ7 RX de-skew
3:0	RW	0x0	CS1 B_DQ7 TX de-skew

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DDRPHY_REGDF

Address: Operational Base + offset (0x037c)

DDR PHY register DF

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQS0 RX de-skew
3:0	RW	0x7	CS1 B_DQS0 TX de-skew

DDRPHY_REGE0

Address: Operational Base + offset (0x0380)

DDR PHY register E0

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS1 B_DQSB0 TX de-skew

DDRPHY_REGE1

Address: Operational Base + offset (0x0384)

DDR PHY register E1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DM1 RX de-skew
3:0	RW	0x7	CS1 B_DM1 TX de-skew

DDRPHY_REGE2

Address: Operational Base + offset (0x0388)

DDR PHY register E2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ8 RX de-skew
3:0	RW	0x7	CS1 B_DQ8 TX de-skew

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DDRPHY_REGE3

Address: Operational Base + offset (0x038c)

DDR PHY register E3

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ9 RX de-skew
3:0	RW	0x7	CS1 B_DQ9 TX de-skew

DDRPHY_REGE4

Address: Operational Base + offset (0x0390)

DDR PHY register E4

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ10 RX de-skew
3:0	RW	0x7	CS1 B_DQ10 TX de-skew

DDRPHY_REGE5

Address: Operational Base + offset (0x0394)

DDR PHY register E5

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ11 RX de-skew
3:0	RW	0x7	CS1 B_DQ11 TX de-ske

DDRPHY_REGE6

Address: Operational Base + offset (0x0398)

DDR PHY register E6

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x7	CS1 B_DQ12 RX de-skew
3:0	RW	0x7	CS1 B_DQ12 TX de-skew

DDRPHY_REGE7

Address: Operational Base + offset (0x039c)

DDR PHY register E7

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ13 RX de-skew
3:0	RW	0x7	CS1 B_DQ13 TX de-skew

DDRPHY_REGE8

Address: Operational Base + offset (0x03a0)

DDR PHY register E8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ14 RX de-skew
3:0	RW	0x7	CS1 B_DQ14 TX de-skew

DDRPHY_REGE9

Address: Operational Base + offset (0x03a4)

DDR PHY register E9

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQ15 RX de-skew
3:0	RW	0x7	CS1 B_DQ15 TX de-skew

DDRPHY_REGEA

Address: Operational Base + offset (0x03a8)

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DDR PHY register EA

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	CS1 B_DQS1 RX de-skew
3:0	RW	0x7	CS1 B_DQS1 TX de-skew

DDRPHY_REGEB

Address: Operational Base + offset (0x03ac)

DDR PHY register EB

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x7	CS1 B_DQSB1 TX de-skew

DDRPHY_REGEC

Address: Operational Base + offset (0x03b0)

DDR PHY register EC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x64	bit7~0 of PLL Feedback divide, bit8 see REGED

DDRPHY_REGED

Address: Operational Base + offset (0x03b4)

DDR PHY register ED

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x1	PLL clock out enable 1: out enable 0: close clock out
3:2	RW	0x2	
1	RW	0x1	PLL power down enable 1: power down 0: power up
0	RW	0x0	bit8 of PLL Feedback divide

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DDRPHY_REGEE

Address: Operational Base + offset (0x03b8)

DDR PHY register EE

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x02	PLL Pre-divide [4:0]

DDRPHY_REGEF

Address: Operational Base + offset (0x03bc)

DDR PHY register EF

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	Select clock source 1: select DDR PHY internal PLL 0: select System PLL
6:0	RO	0x0	reserved

DDRPHY_REGF0

Address: Operational Base + offset (0x03c0)

DDR PHY register F0

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	Channel B High 8bit write leveling done
2	RO	0x0	Channel B Low 8bit write leveling done
1	RO	0x0	Channel A High 8bit write leveling done
0	RO	0x0	Channel A Low 8bit write leveling done

DDRPHY_REGF1

Address: Operational Base + offset (0x03c4)

DDR PHY register F1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	Channel A High 8bit write leveling dqs value
3:0	RO	0x0	Channel A Low 8bit write leveling dqs value

DDRPHY_REGF2

Address: Operational Base + offset (0x03c8)

DDR PHY register F2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RO	0x0	Channel B High 8bit write leveling dqs value
3:0	RO	0x0	Channel B Low 8bit write leveling dqs value

DDRPHY_REGFA

Address: Operational Base + offset (0x03e8)

DDR PHY register FA

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	Channel B High 8bit dqs gate sample dqs value(idqs) (3)
2	RO	0x0	Channel B Low 8bit dqs gate sample dqs value(idqs) (3)
1	RO	0x0	Channel A High 8bit dqs gate sample dqs value(idqs) (3)
0	RO	0x0	Channel A Low 8bit dqs gate sample dqs value(idqs) (3)

DDRPHY_REGFB

Address: Operational Base + offset (0x03ec)

DDR PHY register FB

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:5	RO	0x0	Calibration get the dll configure channel A low 8bit(3)
4:3	RO	0x0	Calibration get the ophsel configure channel A low 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel A low 8bit(3)

DDRPHY_REGFC

Address: Operational Base + offset (0x03f0)

DDR PHY register FC

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RO	0x0	Calibration get the dll configure channel A high 8bit(3)
4:3	RO	0x0	Calibration get the ophsel configure channel A high 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel A high 8bit(3)

DDRPHY_REGFD

Address: Operational Base + offset (0x03f4)

DDR PHY register FD

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RO	0x0	Calibration get the dll configure channel B low 8bit(3)
4:3	RO	0x0	Calibration get the ophsel configure channel B low 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel B low 8bit(3)

DDRPHY_REGFE

Address: Operational Base + offset (0x03f8)

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DDR PHY register FE

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RO	0x0	Calibration get the dll configure channel B high 8bit(3)
4:3	RO	0x0	Calibration get the ophsel configure channel B high 8bit(3)
2:0	RO	0x0	Calibration get the cyclesel configure channel B high 8bit(3)

DDRPHY_REGFF

Address: Operational Base + offset (0x03fc)

DDR PHY register FF

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	Channel B High 8bit Calibration done
2	RO	0x0	Channel B Low 8bit Calibration done
1	RO	0x0	Channel A High 8bit Calibration done
0	RO	0x0	Channel A Low 8bit Calibration done

DDRPHY_REG100

Address: Operational Base + offset (0x0400)

DDR PHY register 100

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	Selects external wave table for Spread Spectrum Modulation 1: select external wave table 0: select internal wave table Note, only effective when Spread Spectrum Modulation enabled
6:4	RO	0x0	reserved
3:0	RW	0x6	Divider for the modulation frequency

DDRPHY_REG101

Address: Operational Base + offset (0x0404)

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DDR PHY register 101

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	REG101[7:0], REG102[7:0], REG103[7:0] is the Pre-PLL fractional feedback divider value. Note, if enabled fractional divider or Spread Spectrum Modulation, the Pre-PLL total feedback divider equals this value divide by 2^24 and add the integer feedback value.

DDRPHY_REG102

Address: Operational Base + offset (0x0408)

DDR PHY register 102

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	see REG101

DDRPHY_REG103

Address: Operational Base + offset (0x040c)

DDR PHY register 103

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	See REG101

DDRPHY_REG104

Address: Operational Base + offset (0x0410)

DDR PHY register 104

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x7f	External wave table depth, range 0~127. Every time before write/read external wave table, it should write/read this address to clear the base address for internal wave values. Note, only effective when Spread Spectrum Modulation enabled and selected external wave table.

DDRPHY_REG105

Address: Operational Base + offset (0x0414)

DDR PHY register 105

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	External wave values, the offset address auto increased every 1 write/read time up to the external wave table depth.

DDRPHY_REG106

Address: Operational Base + offset (0x0418)

DDR PHY register 106

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x1	Selects center spread or down spread 0: Center Spread 1: Down Spread Note, only effective when Spread Spectrum Modulation enabled
4	RW	0x1	Spread Spectrum Modulation enable 1'b0: enable 1'b1: disable
3:0	RW	0x0	Pre-PLL feedback divider bit[11:8]

DDRPHY_REG107

Address: Operational Base + offset (0x041c)

DDR PHY register 107

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x64	Pre-PLL feedback divider bit[7:0]

DDRPHY_REG108

Address: Operational Base + offset (0x0420)

DDR PHY register 108

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x08	Sets modulation depth (Spread%) Spread% = 0.049% * REG108[6:0]. Eg. 7'b000_0000 = 0 7'b000_0001 = 0.049% 7'b000_0010 = 0.098% 7'b000_0011 = 0.147% … 7'b100_0000 = 3.136%(MAX) Note, only effective when Spread Spectrum Modulation enabled

10.5 Interface Description

DDR IOs are listed as following Table.

Table 10-1DDR IO description

Pin Name	Description
CK	Positive differential clock
CKB	Negative differential clock
CKE	Active-high clock enable signal for two chip select.
CSBi (i=0,1)	Active-low chip select signal. There are two chip select.
RASB	Active-low row address strobe
CASB	Active-low column address strobe
WEB	Active-low write enable strobe

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BA[2:0]	Bank address signal
A[15:0]	Address signal
DQ[31:0]	Bidirectional data line
DQS[3:0]	Positive differential bidirectional data strobes
DQSB[3:0]	Negative differential bidirectional data strobes.
DM[3:0]	Active-low data mask signal.
ODTi (i=0,1)	On-Die Termination output signal for two chip select.
RESETN	Reset signal.

The DDR4 Interface of DDR PHY is reused with DDR3, so the mapping relation between the DDR4 and DDR3 show in the following table

Table 10-2DDR4 IO Mapping Table

DDR4	DDR3
A0	A9
A1	A14
A2	A13
A3	A11
A4	A2
A5	A4
A6	A3
A7	A6
A8	A5
A9	A1
A10	A0
A11	A7
A12	CASB

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A13	A8
WEB/A14	ODT0
CASB/A15	BA1
RASB/A16	CKE
BA0	BA2
BA1	A12
BG0	BA0
CK	CK
CKB	CKB
CKE	RASB
CSB0	A10
ODT0	A15
ACTN	CSB0
RESETN	RESETN
ODT1	ODT1
CSB1	CSB1
BG1	WEB

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10.6 Application Notes

10.6.1 Initialization

PCTL & DDR PHY Initialization

1. Assert the resets (presentn, core_ddrc_rstn and aresetn_0 of PCTL)
2. Configure PLL of DDR PHY and wait PLL lock.
3. De-assert presetn if clocks are active and stable.
4. Initial PCTL register and PHY Register
5. Start PHY initialization with DFIMIS[5] register of PCTL
6. De-assert the remaining resets (core_ddrc_rstn and aresetn_0 of PCTL)
7. Wait PCTL initialization done (STAT.operation_mode==normal)

8. Start PHY dqs calibration with PHYREG02 register and wait calibration finish with PHYREGFF register.

9. (optional) After dqs calibration, start write leveling training with PHYREG02 register and wait write leveling training finish with PHYJREGF0 register.

10. Start Write and Read.

DDR3 Initialization Sequence

The initialization steps for DDR3 SDRAMs are as follows:

1. Power-up.

2. Maintain dfi_reset_n low for duration specified by INIT1.dram_rstn_x1024. Specification requires at least 200 us with stable power.

3. Issue NOP/deselect for duration specified by INIT0.pre_cke_x1024. Specification requires at least 500 us.

4. Assert CKE and issue NOP/deselect for INIT0.post_cke_x1024 (specification requires at least tXPR).

5. Issue MRS (mode register set) command to load MR2 with INIT4.emr2 value followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

6. Issue MRS command to load MR3 with INIT4.emr3 followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

7. Issue MRS command to load MR1 with INIT4.emr followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

8. Issue MRS command to load MR0 with INIT3.mr followed by NOP/deselect for duration of DRAMTMG3.t_mod.

9. Issue ZQCL command to start ZQ calibration and wait for INIT5.dev_zqinit_x32.

10. Wait for INIT5.dev_zqinit_x32 counting to finish. Ensure wait from step 8 is larger than tDLLK.

11. The PCTL controller is now ready for normal operation.

DDR4 Initialization Sequence

The initialization steps for DDR4 SDRAMs are as follows:

1. Power-up.

2. Maintain dfi_reset_n low for duration specified by INIT1.dram_rstn_x1024. Specification requires at least 200 us with stable power.

3. Issue NOP/deselect for duration specified by INIT0.pre_cke_x1024. Specification requires at least 500 us.

4. Assert CKE and issue NOP/deselect for INIT0.post_cke_x1024 (specification requires at least tXPR).

5. Issue MRS (mode register set) command to load MR3 with INIT4.emr3 value followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

6. Issue MRS (mode register set) command to load MR6 with INIT7.mr6 value followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

7. Issue MRS (mode register set) command to load MR5 with INIT6.mr5 value followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

8. Issue MRS (mode register set) command to load MR4 with INIT7.mr4 value followed by

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NOP/deselect for duration of DRAMTMG3.t_mrd.

9. Issue MRS command to load MR2 with INIT4.emr2 followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

10. Issue MRS command to load MR1 with INIT4.emr followed by NOP/deselect for duration of DRAMTMG3.t_mrd.

11. Issue MRS command to load MR0 with INIT3.mr followed by NOP/deselect for duration of DRAMTMG3.t_mod.

12. Issue ZQCL command to start ZQ calibration and wait for INIT5.dev_zqinit_x32.

13. Wait for INIT5.dev_zqinit_x32 counting to finish. Ensure wait from step 8 is larger than tDLLK.

14. The PCTL controller is now ready for normal operation.

LPDDR2/3 Initialization Sequence

The initialization steps for LPDDR2/3 SDRAMs are as follows:

1. Power-up.

2. CKE is held low for a duration specified by INIT0.pre_cke_x1024. The clock is checked to be stable for duration specified by INIT2.min_stable_clock_x1 (minimum of 5 clock cycles) prior to the first low to high transition of CKE.

3. Assert CKE for INIT0.post_cke_x1024 (specification requires at least 200 us).

4. A MRW (Reset) command is issued to MRW63 register. Values of MA<7:0> = 3FH and OP<7:0> =00H is used for this command. The MRW reset command brings the device to the device auto initialization (resetting) state in the power-on initialization sequence.

5. Issue NOP/deselect for duration specified by INIT2.idle_after_reset_x32 (specification requires 1 us minimum) and INIT5.max_auto_init_x1024 (specification requires maximum time of 10 us).

6. An MRW ZQ initialization calibration command is issued to the memory to register MR10 to initiate the ZQ calibration. Values of MA<7:0> = 0AH and OP<7:0> = FFH is used for this command.

7. Issue NOP/deselect for duration specified by INIT5.dev_zqinit_x32 (specification requires a minimum time of 1us).

8. Program MR2 register by setting MR2 register to INIT3.emr followed by a NOP/deselect for a duration specified by DRAMTMG3.t_mrw (typical value of 5 clock cycles).

9. Program MR1 register by setting MR1 register to INIT3.mr followed by a NOP/deselect for a duration specified by DRAMTMG3.t_mrw (typical value of 5 clock cycles).

10. Program MR3 register by setting MR3 register to INIT4.emr2 followed by a NOP/deselect for a duration specified by DRAMTMG3.t_mrw (typical value of 5 clock cycles).

11. Schedule multiple all bank refresh.

12. The PCTL controller is now ready for normal operation.

10.6.2 TX DLLs

All high speed IO signals' phase can be adjusted by TX DLLs. See the following table.

Table 10-3DDR PHY TX DLLs Delay Step

Offset	Bit	Control Signal Phase	Default	Description

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0x4c	2~0	CMD	0x4	CMD DLL delay step
0x50	2~0	CK	0x0	CK DLL delay step
0x98	2~0	A_DM0, A_DQ7~A_DQ0	0x4	DM and DQ DLL Signal delay step
0xd8	2~0	A_DM1, A_DQ15~A_DQ8	0x4	
0x118	2~0	B_DM0, B_DQ7~B_DQ0	0x4	
0x158	2~0	B_DM1, B_DQ15~B_DQ8	0x4	
0x9c	2~0	A_DQS0, A_DQSB0	0x0	TX DQS DLL Signal delay step
0xdc	2~0	A_DQS1, A_DQSB1	0x0	
0x11c	2~0	B_DQS0, B_DQSB0	0x0	
0x15c	2~0	B_DQS1, B_DQSB1	0x0	

Step 0x0 values means no phase delay, and 0x4 increases delay phase to 90 deg, 0x7 values corresponds to maximum phase delay. All DLLs having 8 delay steps which can get 90 deg phase delay by setting 0x4.

10.6.3 RX DLLs

The RX DLLs are used for sample RX DQS signals with proper phase delay and pulse edges. The DQS squelch (Rx Squelch) signal opens a window for passing RX DQS pulses, both RX DQS and DQS squelch signal phase can be adjusted by corresponding DLLs.

Table 10-4DDR PHY RX DQS Delay Step

Offset	Bit	Control Signal Phase	Default	Description
0x0a0	1~0	A_DQS0,A_DQSB0	0x1	Read DQS DLL delay phase
0x0e0	1~0	A_DQS1,A_DQSB1	0x1	
0x120	1~0	A_DQS0,A_DQSB0	0x1	
0x160	1~0	A_DQS1,A_DQSB1	0x1	
0x0b0	7~5	left channel A DQS gating	0x1	CS0 DQS gating delay, unit x1 clock cycle
0x0f0	7~5	right channel A DQS gating	0x1	
0x130	7~5	left channel B DQS gating	0x1	
0x170	7~5	reft channel B DQS gating	0x1	
0x0b4	7~5	left channel A DQS gating	0x1	CS1 DQS gating delay, unit x1 clock

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0x0f4	7~5	right channel A DQS gating	0x1	cycle
0x134	7~5	left channel B DQS gating	0x1	
0x174	7~5	reft channel B DQS gating	0x1	
0x0b0	4~3	left channel A DQS gating	0x1	CS0 additive and accumulative DQS gating delay, unit 4x clock cycle
0x0f0	4~3	right channel A DQS gating	0x1	
0x130	4~3	left channel B DQS gating	0x1	
0x170	4~3	reft channel B DQS gating	0x1	
0x0b4	4~3	left channel A DQS gating	0x1	CS1 additive and accumulative DQS gating delay, unit 4x clock cycle
0x0f4	4~3	right channel A DQS gating	0x1	
0x134	4~3	left channel B DQS gating	0x1	
0x174	4~3	reft channel B DQS gating	0x1	
0x0b0	2~0	left channel A DQS gating	0x4	CS0 additive and accumulative DQS gating delay, unit per DLL step
0x0f0	2~0	right channel A DQS gating	0x4	
0x130	2~0	left channel B DQS gating	0x4	
0x170	2~0	reft channel B DQS gating	0x4	
0x0b4	2~0	left channel A DQS gating	0x4	CS1 additive and accumulative DQS gating delay, unit per DLL step
0x0f4	2~0	right channel A DQS gating	0x4	
0x134	2~0	left channel B DQS gating	0x4	
0x174	2~0	reft channel B DQS gating	0x4	

10.6.4 High Speed IO Drive Strength

The tuning rage of driver resistance is 19.6ohm to to +∞. By default, 0x14 is 37.6ohm for DDR3 DQ and CMD driver. When the control bit is set to be larger, the drive strength becomes stronger.

Table 10-5CK/CMD Driver output resistance

Offset	Bit	Default	Description
0x44	4~0	0x14	adjustable CMD pull-down resistance
0x48	7~3	0x14	adjustable CMD pull-up resistance

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0x58	4~0	0x14	adjustable CK pull-down resistance
0x60	4~0	0x14	adjustable CK pull-up resistance
0x5c	6~4	0x04	Adjust CMD/CK falling edge slew rate
0x5c	2~0	0x04	Adjust CMD/CK rising edge slew rate

Table 10-6DM, DQ Signal Drive Strength Register

Offset	Bit	Default	Description
0x80	4~0	0x14	pull-down driving resistance for A_DQ0~A_DQ7
0xbc	4~0	0x14	pull-up driving resistance for A_DQ0~A_DQ7
0xc0	4~0	0x14	pull-down driving resistance for A_DQ8~A_DQ15
0xfc	4~0	0x14	pull-up driving resistance for A_DQ8~A_DQ15
0x100	4~0	0x14	pull-down driving resistance for B_DQ0~B_DQ7
0x13c	4~0	0x14	pull-up driving resistance for B_DQ0~B_DQ7
0x140	4~0	0x14	pull-down driving resistance for B_DQ8~B_DQ15
0x17c	4~0	0x14	pull-up driving resistance for B_DQ8~B_DQ15
0x0ac	7~5	0x4	Falling edge slew rate control for A_DQ0~A_DQ7
	4~2	0x4	Rising edge slew rate control for A_DQ0~A_DQ7
0x0ec	7~5	0x4	Falling edge slew rate control for A_DQ8~A_DQ15
	4~2	0x4	Rising edge slew rate control for A_DQ8~A_DQ15
0x12c	7~5	0x4	Falling edge slew rate control for B_DQ0~B_DQ7
	4~2	0x4	Rising edge slew rate control for B_DQ0~B_DQ7
0x16c	7~5	0x4	Falling edge slew rate control for B_DQ8~B_DQ15
	4~2	0x4	Rising edge slew rate control for B_DQ8~B_DQ15
0x084	4~0	0x4	pull-down ODT resistance for A_DQ0~A_DQ7
0x0b8	4~0	0x4	pull-up ODT resistance for A_DQ0~A_DQ7
0x0c4	4~0	0x4	pull-down ODT resistance for A_DQ8~A_DQ15

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0x0f8	4~0	0x4	pull-up ODT resistance for A_DQ8~A_DQ15
0x104	4~0	0x4	pull-down ODT resistance for B_DQ0~B_DQ7
0x138	4~0	0x4	pull-up ODT resistance for B_DQ0~B_DQ7
0x144	4~0	0x4	pull-down ODT resistance for B_DQ8~B_DQ15
0x178	4~0	0x4	pull-up ODT resistance for B_DQ8~B_DQ15

The value is larger, the drive strength is stronger.

DDR3 1.5V DQ/DQS/CMD driver and ODT strength table

Table 10-7DDR3 1.5V DQ/DQS/CMD Driver and ODT resistance

Control bit	**5'b00000**	**5'b00001**	**5'b00010**	**5'b00011**
Pull-up/down	+∞	451.6ohm	225.8ohm	150.5ohm
Control bit	**5'b00100**	**5'b00101**	**5'b00110**	**5'b00111**
Pull-up/down	112.9ohm	90.3ohm	75.3ohm	64.5ohm
Control bit	**5'b01000**	**5'b01001**	**5'b01010**	**5'b01011**
Pull-up/down	56.5ohm	50.2ohm	45.2ohm	41.1ohm
Control bit	**5'b01100**	**5'b01101**	**5'b01110**	**5'b01111**
Pull-up/down	37.6ohm	34.7ohm	33.3ohm	30.1ohm
Control bit	**5'b10000**	**5'b10001**	**5'b10010**	**5'b10011**
Pull-up/down	56.5ohm	50.2ohm	45.2ohm	41.1ohm
Control bit	**5'b10100**	**5'b10101**	**5'b10110**	**5'b10111**
Pull-up/down	37.6ohm	34.7ohm	33.3ohm	30.1ohm
Control bit	**5'b11000**	**5'b11001**	**5'b11010**	**5'b11011**
Pull-up/down	28.2ohm	26.6ohm	25.1ohm	23.8ohm
Control bit	**5'b11100**	**5'b11101**	**5'b11110**	**5'b11111**
Pull-up/down	22.6ohm	21.5ohm	20.5ohm	19.6ohm

Table 10-8LPDDR2/3 1.2V DQ/DQS/CMD driver and ODT resistance

Control bit	**5'b00000**	**5'b00001**	**5'b00010**	**5'b00011**

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Pull-up/down	+∞	480.4ohm	240.2ohm	160.1ohm
Control bit	**5'b00100**	**5'b00101**	**5'b00110**	**5'b00111**
Pull-up/down	120.1ohm	96.1ohm	80.1ohm	68.6ohm
Control bit	**5'b01000**	**5'b01001**	**5'b01010**	**5'b01011**
Pull-up/down	60.0ohm	53.4ohm	48.0ohm	43.7ohm
Control bit	**5'b01100**	**5'b01101**	**5'b01110**	**5'b01111**
Pull-up/down	40.0ohm	37.0ohm	34.3ohm	32.0ohm
Control bit	**5'b10000**	**5'b10001**	**5'b10010**	**5'b10011**
Pull-up/down	60.0ohm	53.4ohm	48.0ohm	43.7ohm
Control bit	**5'b10100**	**5'b10101**	**5'b10110**	**5'b10111**
Pull-up/down	40.0ohm	37.0ohm	34.3ohm	32.0ohm
Control bit	**5'b11000**	**5'b11001**	**5'b11010**	**5'b11011**
Pull-up/down	30.0ohm	28.3ohm	26.7ohm	25.3ohm
Control bit	**5'b11100**	**5'b11101**	**5'b11110**	**5'b11111**
Pull-up/down	24.0ohm	22.9ohm	21.8ohm	20.9ohm

10.6.5 PHY Low Speed Mode (200MHz)

DDR PHY supports low speed to high speed DDR3 by using two operating mode: normal delay line mode up to 800Mbps or more, low power mode where we support any speed up to 533Mbps. If all TX DLLs are bypassed, the PHY will enter low power state.

The DDR PHY entries low power mode when setting DLLs into Bypass mode. The following table illustrates related register settings.

Table 10-9Low Power DLL Setting

Offset	Bit	Default	Low power Setting	Description
0x290	4	0x0	0x1	right channel B TX DQ DLL in bypass mode
	3	0x0	0x1	left channel B TX DQ DLL in bypass mode
	2	0x0	0x1	right channel A TX DQ DLL in bypass mode
	1	0x0	0x1	left channel A TX DQ DLL in bypass mode

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	0	0x0	0x1	CMD/CK DLL in bypass mode
0x4c	4	0x0	0x1	CMD DLL phase select
0x50	3	0x0	0x0	CK DLL phase select
0x98	4	0x0	0x1	A_DQ0~A_DQ7 TX DLL phase select
0x9c	3	0x0	0x0	A_DQS0/A_DQSB0 TX DLL phase select
0xd8	4	0x0	0x1	A_DQ8~A_DQ15 TX DLL phase select
0xdc	3	0x0	0x0	A_DQS1/A_DQSB1 TX DLL phase select
0x118	4	0x1	0x1	B_DQ0~B_DQ7 TX DLL phase select
0x11c	3	0x0	0x0	B_DQS0/B_DQSB0 TX DLL phase select
0x158	4	0x1	0x1	B_DQ8~B_DQ15 TX DLL phase select
0x15c	3	0x0	0x0	B_DQS1/B_DQSB1 TX DLL phase select

10.6.6 Per bit de-skew tuning

Per-bit de-skew is designed for compensating PCB trace mismatch, DDR PHY support skew individually adjustable for all PHY signals. There are eight steps for each bit de-skew adjusting, and the adjust resolution under different corners is shown below:

Table 10-10per-bit de-skew tuning resolution

	ff	**tt**	**ss**
de-skew resolution	15ps	20ps	32ps

Pre-bit de-skew is realized with inverter chain delay, per-bit de-skew control signals select how much inverters are connected to data path, the minimum resolution is determined by the two inverters minimum delay.

TX path deskew and RX path deskew employ same delay line, and they have same deskew tuning resolution. Minimum RX deskew tuning resolution can be about 28ps with SMIC55ll tt corner process, and we can re-design tuning resolution according to system and customer requirement.

10.6.7 DDR PHY Calibration

DDR PHY auto dqs calibration function has been implemented in the PHY. The entire training processes only need to configure the register to start andwait for finish.

The entire training process is as follows:

1. PHY’s register is reset, the setup is complete.

2. Send the initial command to dram and complete dram initialization.

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3. Set the PHY's register beginning calibration.

Offset	Bit	Default	Description
0x8	5~4	0x0	DQS gating calibration CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
	1	0x0	set calibration bypass mode(1:bypass mode; 0:nomal)
	0	0x0	set calibration start (1: start; 0: stop)

4. Wait for the calibration finish by PHYREGFF.

5. Normal read and writes operation can begin.

10.6.8 DDR PHY Write Leveling Training

DDR PHY auto write leveling training function has been implemented in the PHY. The entire training processes only need to configure the register to start and wait for finish。

The entire training process is as follows:

1. PHY's register is reset, the setup is complete.

2. Send the initial command to dram and complete dram initialization.

3. Set the PHYREG05 and PHYREG06 to configure the dram mode register which used to enable dram write leveling training function.

4. Set the PHY's register to begintraining.

Offset	Bit	Default	Description
0x8	7~6	0x0	Write leveling CS select signal 2'b00: select CS0 and CS1 2'b01: select CS1 2'b10: select CS0
	3	0x0	Write leveling calibration bypass mode, active high
	2	0x0	Write leveling calibration control, active high

5. Wait for the calibration finish by PHYREGF0.

6. Normal read and writes operation can begin.

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Chapter 11 Crypto

11.1 Overview

Crypto is a hardware accelerator of encrypting or decrypting. It supports the most commonly used algorithm: DES/3DES, AES, SHA1, SHA256, MD5 and RSA.

The Crypto supports following features:

- Support AES 128/192/256 bits key mode, ECB/CBC/CTR/XTS chain mode, Slave/FIFO mode
- Support DES/3DES (ECB and CBC chain mode) , 3DES (EDE/ EEE key mode), Slave/FIFO mode
- Support SHA1/SHA256/MD5 (with hardware padding) HASH function, FIFO mode only
- Support 160 bit Pseudo Random Number Generator (PRNG)
- Support 256 bit True Random Number Generator (TRNG)
- Support PKA 512/1024/2048 bit Exp Modulator
- Support up to 150M clock frequency

11.2 Block Diagram

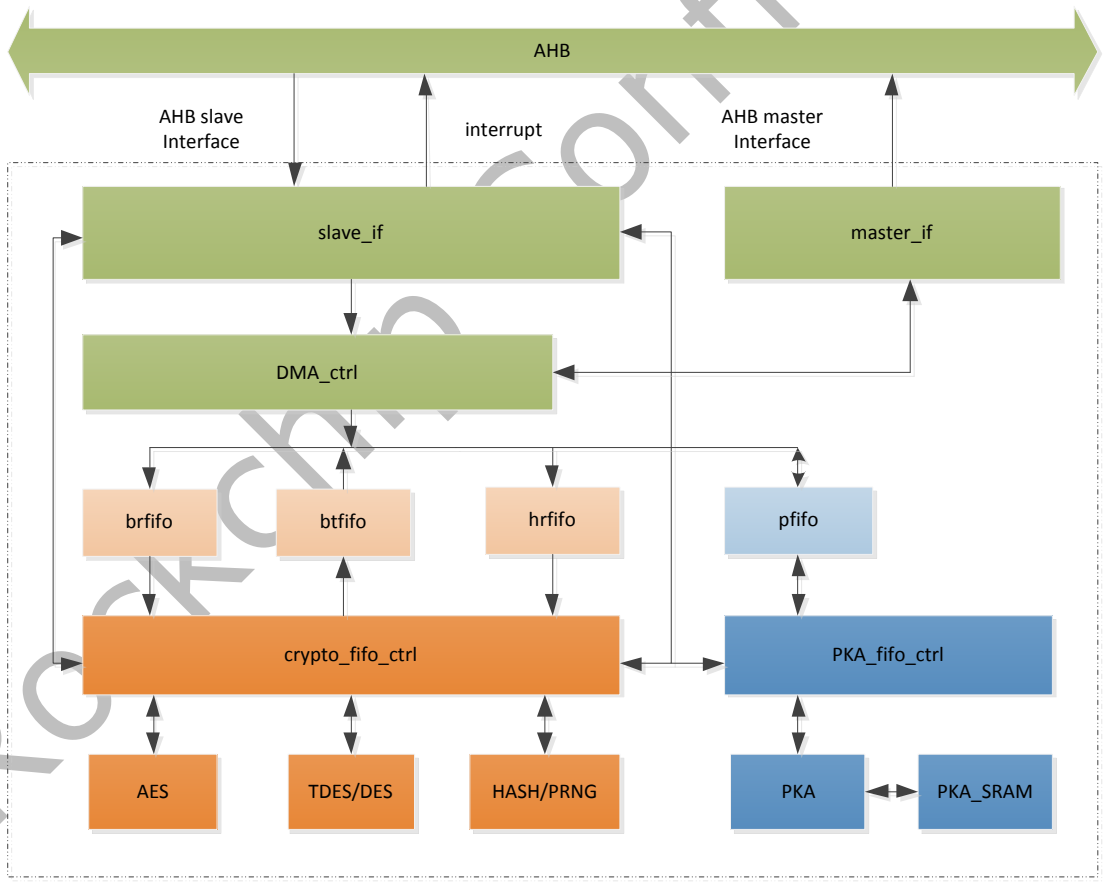

Fig.11-1Crypto Architecture

11.3 Register description

11.3.1 Register Summary

Name	Offset	Size	Reset Value	Description

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Name	Offset	Size	Reset Value	Description
Crypto_INTSTS	0x00000	W	0x00000000	Interrupt Status Register
Crypto_INTENA	0x00004	W	0x00000000	Interrupt Set Register
Crypto_CTRL	0x00008	W	0x00000000	Control Register
Crypto_CONF	0x0000c	W	0x00000000	Configure Register
Crypto_BRDMAS	0x00010	W	0x00000000	Block Receiving DMA Start Address Register
Crypto_BTDMAS	0x00014	W	0x00000000	Block Transmitting DMA Start Address Register
Crypto_BRDMAL	0x00018	W	0x00000000	Block Receiving DMA Length Register
Crypto_HRDMAS	0x0001c	W	0x00000000	Hash Receiving DMA Start Address Register
Crypto_HRDMAL	0x00020	W	0x00000000	Hash Receiving DMA Length Register
Crypto_AES_CTRL	0x00080	W	0x00000000	AES Control Register
Crypto_AES_STS	0x00084	W	0x00000000	Status Register
Crypto_AES_DIN_0	0x00088	W	0x00000000	AES Input Data 0 Register
Crypto_AES_DIN_1	0x0008c	W	0x00000000	AES Input Data 1 Register
Crypto_AES_DIN_2	0x00090	W	0x00000000	AES Input Data 2 Register
Crypto_AES_DIN_3	0x00094	W	0x00000000	AES Input Data 3 Register
Crypto_AES_DOUT_0	0x00098	W	0x00000000	AES Output Data 0 Register
Crypto_AES_DOUT_1	0x0009c	W	0x00000000	AES Output Data 1 Register
Crypto_AES_DOUT_2	0x000a0	W	0x00000000	AES Output Data 2 Register
Crypto_AES_DOUT_3	0x000a4	W	0x00000000	AES Output Data 3 Register
Crypto_AES_IV_0	0x000a8	W	0x00000000	AES IV data 0 Register
Crypto_AES_IV_1	0x000ac	W	0x00000000	AES IV data 1 Register
Crypto_AES_IV_2	0x000b0	W	0x00000000	AES IV data 2 Register
Crypto_AES_IV_3	0x000b4	W	0x00000000	AES IV data 3 Register
Crypto_AES_KEY_0	0x000b8	W	0x00000000	AES Key data 0 Register
Crypto_AES_KEY_1	0x000bc	W	0x00000000	AES Key data 1 Register
Crypto_AES_KEY_2	0x000c0	W	0x00000000	AES Key data 2 Register
Crypto_AES_KEY_3	0x000c4	W	0x00000000	AES Key data 3 Register
Crypto_AES_KEY_4	0x000c8	W	0x00000000	AES Key data 4 Register
Crypto_AES_KEY_5	0x000cc	W	0x00000000	AES Key data 5 Register
Crypto_AES_KEY_6	0x000d0	W	0x00000000	AES Key data 6 Register
Crypto_AES_KEY_7	0x000d4	W	0x00000000	AES Key data 7 Register
Crypto_AES_CNT_0	0x000d8	W	0x00000000	AES Input Counter 0 Register
Crypto_AES_CNT_1	0x000dc	W	0x00000000	AES Input Counter 1 Register
Crypto_AES_CNT_2	0x000e0	W	0x00000000	AES Input Counter 2 Register
Crypto_AES_CNT_3	0x000e4	W	0x00000000	AES Input Counter 3 Register
Crypto_TDES_CTRL	0x00100	W	0x00000000	TDES Control Register
Crypto_TDES_STS	0x00104	W	0x00000000	Status Register
Crypto_TDES_DIN_0	0x00108	W	0x00000000	TDES Input Data 0 Register

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Name	Offset	Size	Reset Value	Description
Crypto_TDES_DIN_1	0x0010c	W	0x00000000	TDES Input Data 1 Register
Crypto_TDES_DOUT_0	0x00110	W	0x00000000	TDES Output Data 0 Register
Crypto_TDES_DOUT_1	0x00114	W	0x00000000	TDES Output Data 1 Register
Crypto_TDES_IV_0	0x00118	W	0x00000000	TDES IV data 0 Register
Crypto_TDES_IV_1	0x0011c	W	0x00000000	TDES IV data 1 Register
Crypto_TDES_KEY1_0	0x00120	W	0x00000000	TDES Key1 data 1 Register
Crypto_TDES_KEY1_1	0x00124	W	0x00000000	TDES Key1 data 1 Register
Crypto_TDES_KEY2_0	0x00128	W	0x00000000	TDES Key2 data 0 Register
Crypto_TDES_KEY2_1	0x0012c	W	0x00000000	TDES Key2 data 1 Register
Crypto_TDES_KEY3_0	0x00130	W	0x00000000	TDES Key3 data 0 Register
Crypto_TDES_KEY3_1	0x00134	W	0x00000000	TDES Key3 data 1 Register
Crypto_HASH_CTRL	0x00180	W	0x00000000	Hash Control Register
Crypto_HASH_STS	0x00184	W	0x00000000	Hash Status Register
Crypto_HASH_MSG_LEN	0x00188	W	0x00000000	Hash Message Len
Crypto_HASH_DOUT_0	0x0018c	W	0x00000000	Hash Result Register 0
Crypto_HASH_DOUT_1	0x00190	W	0x00000000	Hash Result Register 1
Crypto_HASH_DOUT_2	0x00194	W	0x00000000	Hash Result Register 2
Crypto_HASH_DOUT_3	0x00198	W	0x00000000	Hash Result Register 3
Crypto_HASH_DOUT_4	0x0019c	W	0x00000000	Hash Result Register 4
Crypto_HASH_DOUT_5	0x001a0	W	0x00000000	Hash Result Register 5
Crypto_HASH_DOUT_6	0x001a4	W	0x00000000	Hash Result Register 6
Crypto_HASH_DOUT_7	0x001a8	W	0x00000000	Hash Result Register 7
Crypto_HASH_SEED_0	0x001ac	W	0x00000000	PRNG Seed/HMAC Key Register 0
Crypto_HASH_SEED_1	0x001b0	W	0x00000000	PRNG Seed/HMAC Key Register 1
Crypto_HASH_SEED_2	0x001b4	W	0x00000000	PRNG Seed/HMAC Key Register 2
Crypto_HASH_SEED_3	0x001b8	W	0x00000000	PRNG Seed/HMAC Key Register 3
Crypto_HASH_SEED_4	0x001bc	W	0x00000000	PRNG Seed/HMAC Key Register 4
Crypto_TRNG_CTRL	0x00200	W	0x00000000	TRNG Control register
Crypto_TRNG_DOUT_0	0x00204	W	0x00000000	TRNG output register 0
Crypto_TRNG_DOUT_1	0x00208	W	0x00000000	TRNG output register 1
Crypto_TRNG_DOUT_2	0x0020c	W	0x00000000	TRNG output register 2
Crypto_TRNG_DOUT_3	0x00210	W	0x00000000	TRNG output register 3
Crypto_TRNG_DOUT_4	0x00214	W	0x00000000	TRNG output register 4
Crypto_TRNG_DOUT_5	0x00218	W	0x00000000	TRNG output register 5
Crypto_TRNG_DOUT_6	0x0021c	W	0x00000000	TRNG output register 6
Crypto_TRNG_DOUT_7	0x00220	W	0x00000000	TRNG output register 7
Crypto_PKA_CTRL	0x00280	W	0x00000000	PKA Control Register
Crypto_AES_TKEY_0	0x00300	W	0x00000000	AES Tweak Key data 0 Register
Crypto_AES_TKEY_1	0x00304	W	0x00000000	AES Tweak Key data 1 Register
Crypto_AES_TKEY_2	0x00308	W	0x00000000	AES Tweak Key data 2 Register
Crypto_AES_TKEY_3	0x0030c	W	0x00000000	AES Tweak Key data 3 Register
Crypto_AES_TKEY_4	0x00310	W	0x00000000	AES Tweak Key data 4 Register
Crypto_AES_TKEY_5	0x00314	W	0x00000000	AES Tweak Key data 5 Register

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Name	Offset	Size	Reset Value	Description
Crypto_AES_TKEY_6	0x00318	W	0x00000000	AES Tweak Key data 6 Register
Crypto_AES_TKEY_7	0x0031c	W	0x00000000	AES Tweak Key data 7 Register
Crypto_AES_TWK_0	0x00320	W	0x00000000	AES Tweak data 0 Register
Crypto_AES_TWK_1	0x00324	W	0x00000000	AES Tweak data 1 Register
Crypto_AES_TWK_2	0x00328	W	0x00000000	AES Tweak data 2 Register
Crypto_AES_TWK_3	0x0032c	W	0x00000000	AES Tweak data 3 Register
Crypto_KEY_SECURE	0x00330	W	0x00000000	Key Secure Control Register
Crypto_CLK_GATE	0x00334	W	0x0000000f	Clock Gate Control Register
Crypto_CRYPTO_VER	0x003f0	W	0x00000300	Crypto Version register
Crypto_PKA_M	0x0400	W	0x00000000	PKA input/output data
Crypto_PKA_C	0x0500	W	0x00000000	PKA C value
Crypto_PKA_N	0x0600	W	0x00000000	PKA modular
Crypto_PKA_E	0x0700	W	0x00000000	PKA exponent

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

11.3.2 Detail Register Description

Crypto_INTSTS

Address: Operational Base + offset (0x0000)

Interrupt Status Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	PKA_DONE_INT PKA Done Interrupt
4	W1C	0x0	HASH_DONE_INT Hash Done Interrupt
3	W1C	0x0	HRDMA_ERR_INT Specifies the interrupt of hash receiving DMA Error
2	W1C	0x0	HRDMA_DONE_INT Specifies the interrupt of hash receiving DMA Done
1	W1C	0x0	BCDMA_ERR_INT Specifies the interrupt of block cipher Error
0	W1C	0x0	BCDMA_DONE_INT Specifies the interrupt of block cipher Done

Crypto_INTENA

Address: Operational Base + offset (0x0004)

Interrupt Set Register

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Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	PKA_DONE_ENA Set the interrupt Enable of PKA done 1'b1: enable 1'b0: disable
4	RW	0x0	HASH_DONE_ENA Set the interrupt Enable of hash done 1'b1: enable 1'b0: disable
3	RW	0x0	HRDMA_ERR_ENA Set the interrupt Enable of hash receiving DMA Error 1'b1: enable 1'b0: disable
2	RW	0x0	HRDMA_DONE_ENA Set the interrupt Enable of hash receiving DMA DONE 1'b1: enable 1'b0: disable
1	RW	0x0	BCDMA_ERR_ENA Set the interrupt Enable of block cipher DMA Error 1'b1: enable 1'b0: disable
0	RW	0x0	BCDMA_DONE_ENA Set the interrupt Enable of block cipher DMA DONE 1'b1: enable 1'b0: disable

Crypto_CTRL

Address: Operational Base + offset (0x0008)

Control Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	Write mask
15:10	RO	0x0	reserved
9	RW	0x0	TRNG_FLUSH Flush TRNG Software write 1 to start. When finishes, the core will clear it.
8	R/WSC	0x0	TRNG_START Start TRNG Software write 1 to start. When finishes, the core will clear it.

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Bit	Attr	Reset Value	Description
7	R/W SC	0x0	PKA_FLUSH Software write 1 to start Flush Process. The process will clear BRFIFO, BTFIFO, and state machine. Then Software should write 0 to end FLUSH Process .
6	RW	0x0	HASH_FLUSH Software write 1 to start Flush Process. The process will clear BRFIFO, BTFIFO, and state machine. Then Software should write 0 to end FLUSH Process .
5	RW	0x0	BLOCK_FLUSH Software write 1 to start Flush Process. The process will clear BRFIFO, BTFIFO, and state machine. Then software should write 0 to end FLUSH Process. It must last for at least 20 cycles to clean registers and FSM .
4	R/W SC	0x0	PKA_START Starts/initializes PKA Software write 1 to start. When finishes, the core will clear it.
3	R/W SC	0x0	HASH_START Starts/initializes HASH/PRNG/HMAC Software write 1 to start. When finishes, the core will clear it.
2	R/W SC	0x0	BLOCK_START Starts/initializes Block Cipher Software write 1 to start. When finishes, the core will clear it.
1	R/W SC	0x0	TDES_START Starts/initializes TDES Software write 1 to start. When finishes, the core will clear it.
0	R/W SC	0x0	AES_START Starts/initializes AES Software write 1 to start. When finishes, the core will clear it.

Crypto_CONF

Address: Operational Base + offset (0x000c)

Control Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	HR_ADDR_MODE Hash Receive DMA Address Mode 1'b1: fix 1'b0: increment

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Bit	Attr	Reset Value	Description
7	RW	0x0	BT_ADDR_MODE Block Transmit DMA Address Mode 1'b1: fix 1'b0: increment
6	RW	0x0	BR_ADDR_MODE Block Receive DMA Address Mode 1'b1: fix 1'b0: increment
5	RW	0x0	Byteswap_HRFIFO If this bit is high, then the data read from the bus is byte-swapped in a word boundary. If this bit is low (default), then the data is handed over to the FIFO without byte-swap. For little endian bus, this bit should be 1'b1.
4	RW	0x0	Byteswap_BTFIFO If this bit is high, then the data read from the bus is byte-swapped in a word boundary. If this bit is low (default), then the data is handed over to the FIFO without byte-swap. For little endian bus, this bit should be 1'b1.
3	RW	0x0	Byteswap_BRFIFO If this bit is high, then the data read from the bus is byte-swapped in a word boundary. If this bit is low (default), then the data is handed over to the FIFO without byte-swap. For little endian bus, this bit should be 1'b1.
2	RW	0x0	DESSEL Specifies the Destination block cipher of FIFO. 1'b0:AES; 1'b1:DES.
1:0	RW	0x0	HASHINSEL Specifies the following 2'b00:Data from independent source 2'b01:Data from block cipher input 2'b10:Data from block cipher output 2'b11:Reserved.

Crypto_BRDMAS

Address: Operational Base + offset (0x0010)

Block Receiving DMA Start Address Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STARTADDR Specifies the Start Address of DMA The address should be aligned by 32-bit.

Crypto_BTDMAS

Address: Operational Base + offset (0x0014)

Block Transmiting DMA Start Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STARTADDR Specifies the Start Address of DMA The address needs to be aligned by 32-bit.

Crypto_BRDMAL

Address: Operational Base + offset (0x0018)

Block Receiving DMA Length Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	LENGTH Specifies the Block length of DMA. The length unit is WORD.

Crypto_HRDMAS

Address: Operational Base + offset (0x001c)

Hash Receiving DMA Start Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STARTADDR Specifies the Start Address of DMA The address needs to be aligned by 32-bit.

Crypto_HRDMAL

Address: Operational Base + offset (0x0020)

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Hash Receiving DMA Length Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	LENGTH Specifies the Block length of DMA. The length unit is BYTE.

Crypto_AES_CTRL

Address: Operational Base + offset (0x0080)

AES Control Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	AES_ByteSwap_Tweak Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Tweak Value byte swap 1 : Enables Tweak Value byte swap
12	RW	0x0	AES_ByteSwap_Tkey Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Tweak Key byte swap 1 : Enables Tweak Key byte swap
11	RW	0x0	AES_ByteSwap_CNT Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Counter data byte swap 1 : Enables Counter data byte swap
10	RW	0x0	AES_ByteSwap_Key Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Key byte swap 1 : Enables Key byte swap
9	RW	0x0	AES_ByteSwap_IV Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Initial value byte swap 1 : Enables Initial value byte swap
8	RW	0x0	AES_ByteSwap_DO Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Output data byte swap 1 : Enables Output data byte swap

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Bit	Attr	Reset Value	Description
7	RW	0x0	AES_ByteSwap_DI Change the Big-endian and Little-endian by swapping the byte order. 0 : Disables Input data byte swap 1 : Enables Input data byte swap
6	RW	0x0	AES_KeyChange Specifies the AES key change mode selection signal. When the bit is asserted, it will not do key-expansion function to calculate new sub-key. So it is a faster way, when several times of calculation use the same key. But if the keys are different, asserting this bit will have the wrong result. 0 : Key is not changed 1 : Key is changed
5:4	RW	0x0	AES_ChainMode Specifies AES chain mode selection 00 : ECB mode 01 : CBC mode 10 : CTR mode 11 : XTS mode
3:2	RW	0x0	AES_KeySize Specifies the AES key size selection signal 00 : 128-bit key 01 : 192-bit key 10 : 256-bit key
1	RW	0x0	AES_FifoMode Specify AES Fifo Mode 0 : Slave mode 1 : fifo mode
0	RW	0x0	AES_Enc Specifies the Encryption/ Decryption mode selection signal 0 : Encryption 1 : Decryption

Crypto_AES_STS

Address: Operational Base + offset (0x0084)

Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	AES_DONE When AES finish, it will be HIGH, And it will not be LOW until it restart . 1 : done 0 : not done

Crypto_AES_DIN_0

Address: Operational Base + offset (0x0088)

AES Input Data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_0 Specifies AES Input data [127:96].

Crypto_AES_DIN_1

Address: Operational Base + offset (0x008c)

AES Input Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_1 Specifies AES Input data [95:64].

Crypto_AES_DIN_2

Address: Operational Base + offset (0x0090)

AES Input Data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_2 Specifies AES Input data [63:32]

Crypto_AES_DIN_3

Address: Operational Base + offset (0x0094)

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AES Input Data 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_DIN_3 Specifies AES Input data [31:0]

Crypto_AES_DOUT_0

Address: Operational Base + offset (0x0098)

AES Output Data 0 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_0 Specifies AES Output data [127:96].

Crypto_AES_DOUT_1

Address: Operational Base + offset (0x009c)

AES Output Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_1 Specifies the Output data [95:64].

Crypto_AES_DOUT_2

Address: Operational Base + offset (0x00a0)

AES Output Data 2 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_2 Specifies AES Output data [63:32].

Crypto_AES_DOUT_3

Address: Operational Base + offset (0x00a4)

AES Output Data 3 Register

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	AES_DOUT_3 Specifies AES Output data [31:0].

Crypto_AES_IV_0

Address: Operational Base + offset (0x00a8)

AES IV data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_0 Specifies AES Initialization vector [127:96]

Crypto_AES_IV_1

Address: Operational Base + offset (0x00 ac)

AES IV data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_1 Specifies AES Initialization vector [95:64]

Crypto_AES_IV_2

Address: Operational Base + offset (0x00b0)

AES IV data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_2 Specifies AES Initialization vector [63:32]

Crypto_AES_IV_3

Address: Operational Base + offset (0x00b4)

AES IV data 3 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_IV_3 Specifies AES Initialization vector [31:0]

Crypto_AES_KEY_0

Address: Operational Base + offset (0x00b8)

AES Key data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_0 Specifies AES key data [255:224]

Crypto_AES_KEY_1

Address: Operational Base + offset (0x00bc)

AES Key data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_1 Specifies AES key data [223:192]

Crypto_AES_KEY_2

Address: Operational Base + offset (0x00c0)

AES Key data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_2 Specifies AES key data [191:160]

Crypto_AES_KEY_3

Address: Operational Base + offset (0x00c4)

AES Key data 3 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_3 Specifies AES key data [159:128]

Crypto_AES_KEY_4

Address: Operational Base + offset (0x00c8)

AES Key data 4 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_4 Specifies AES key data [127:96]

Crypto_AES_KEY_5

Address: Operational Base + offset (0x00cc)

AES Key data 5 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_5 Specifies the key data [95:64]

Crypto_AES_KEY_6

Address: Operational Base + offset (0x00d0)

AES Key data 6 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_6 Specifies AES key data [63:32]

Crypto_AES_KEY_7

Address: Operational Base + offset (0x00d4)

AES Key data 7 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY_7 Specifies the key data [31:0]

Crypto_AES_CNT_0

Address: Operational Base + offset (0x00d8)

AES Input Counter 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_0 Specifies AES Input Counter [127:96].

Crypto_AES_CNT_1

Address: Operational Base + offset (0x00dc)

AES Input Counter 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_1 Specifies AES Input Counter [95:64].

Crypto_AES_CNT_2

Address: Operational Base + offset (0x00e0)

AES Input Counter 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_2 Specifies AES Input Counter[63:32]

Crypto_AES_CNT_3

Address: Operational Base + offset (0x00e4)

AES Input Counter 3 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_CNT_3 Specifies AES Input Counter [31:0]

Crypto_TDES_CTRL

Address: Operational Base + offset (0x00100)

TDES Control Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	TDES_ByteSwap_Key 0 = Disables Key byte swap 1 = Enables Key byte swap
7	RW	0x0	TDES_ByteSwap_IV 0 = Disables Initial value byte swap 1 = Enables Initial value byte swap
6	RW	0x0	TDES_ByteSwap_DO 0 = Disables Output data byte swap 1 = Enables Output data byte swap
5	RW	0x0	TDES_ByteSwap_DI 0 = Disables Input data byte swap 1 = Enables Input data byte swap
4	RW	0x0	TDES_ChainMode Specifies TDES chain mode selection 0 : ECB mode 1 : CBC mode
3	RW	0x0	TDES_EEE Specifies the TDES key mode selection 1'b0 : EDE 1'b1 : EEE
2	RW	0x0	TDES_Select Specify DES or TDES cipher 1'b0 : DES 1'b1 : TDES
1	RW	0x0	TDES_FifoMode Specify TDES Fifo Mode 1'b0: Slave mode 1'b1: Fifo mode
0	RW	0x0	TDES_Enc Specifies the Encryption/ Decryption mode selection signal 0 : Encryption 1 : Decryption

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Crypto_TDES_STS

Address: Operational Base + offset (0x00104)

Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	TDES_DONE When DES/TDES finishes, it will be HIGH, And it will not be LOW until it restart . 1: done 0: not done

Crypto_TDES_DIN_0

Address: Operational Base + offset (0x00108)

TDES Input Data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_DIN_0 Specifies TDES Input data [63:32].

Crypto_TDES_DIN_1

Address: Operational Base + offset (0x0010c)

TDES Input Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_DIN_1 Specifies TDES Input data [31:0].

Crypto_TDES_DOUT_0

Address: Operational Base + offset (0x00110)

TDES Output Data 0 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TDES_DOUT_0 Specifies TDES Output data [63:32].

Crypto_TDES_DOUT_1

Address: Operational Base + offset (0x00114)

TDES Output Data 1 Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TDES_DOUT_1 Specifies TDES Output data [31:0].

Crypto_TDES_IV_0

Address: Operational Base + offset (0x00118)

TDES IV data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_IV_0 Specifies TDES Initialization vector [63:32]

Crypto_TDES_IV_1

Address: Operational Base + offset (0x0011c)

TDES IV data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_IV_1 Specifies TDES Initialization vector [31:0]

Crypto_TDES_KEY1_0

Address: Operational Base + offset (0x00120)

TDES Key1 data 1 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY1_0 Specifies TDES key1 data [63:32]

Crypto_TDES_KEY1_1

Address: Operational Base + offset (0x00124)

TDES Key1 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY1_1 Specifies TDES key1 data [31:0]

Crypto_TDES_KEY2_0

Address: Operational Base + offset (0x00128)

TDES Key2 data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY2_0 Specifies TDES key2 data [63:32]

Crypto_TDES_KEY2_1

Address: Operational Base + offset (0x0012c)

TDES Key2 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY_1 Specifies TDES key data [31:0]

Crypto_TDES_KEY3_0

Address: Operational Base + offset (0x00130)

TDES Key3 data 0 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TDES_KEY3_0 Specifies TDES key3 data [63:32]

Crypto_TDES_KEY3_1

Address: Operational Base + offset (0x00134)

TDES Key3 data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_KEY3_1 Specifies TDES key3 data [31:0]

Crypto_HASH_CTRL

Address: Operational Base + offset (0x00180)

Hash Control Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	HASH_SWAP_DO Specifies the Byte swap of data output (hash result) 0 = Does not swap (default) 1 = Swap
2	RW	0x0	reserved
1:0	RW	0x0	Engine_Selection 2'b00: SHA1_HASH 2'b01: MD5_HASH 2'b10: SHA256_HASH 2'b11: PRNG

Crypto_HASH_STS

Address: Operational Base + offset (0x00184)

Hash Status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	HASH_DONE Hash Done Signal When HASH finish, it will be HIGH, And it will not be LOW until it restart 1'b1 : done 1'b0 : not done

Crypto_HASH_MSG_LEN

Address: Operational Base + offset (0x00188)

Hash Message Len

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	Msg_size Hash total byte.

Crypto_HASH_DOUT_0

Address: Operational Base + offset (0x0018c)

Hash Result Register 0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_0 Specifies the HASH Result [159:128]

Crypto_HASH_DOUT_1

Address: Operational Base + offset (0x00190)

Hash Result Register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_1 Specifies the HASH Result [127:96]

Crypto_HASH_DOUT_2

Address: Operational Base + offset (0x00194)

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Hash Result Register 2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_2 Specifies the HASH Result [95:64]

Crypto_HASH_DOUT_3

Address: Operational Base + offset (0x00198)

Hash Result Register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_3 Specifies the HASH Result [63:32]

Crypto_HASH_DOUT_4

Address: Operational Base + offset (0x0019c)

Hash Result Register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_4 Specifies the HASH Result [31:0]

Crypto_HASH_DOUT_5

Address: Operational Base + offset (0x001a0)

Hash Result Register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_5 Specifies the HASH Result [31:0]

Crypto_HASH_DOUT_6

Address: Operational Base + offset (0x001a4)

Hash Result Register 6

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_6 Specifies the HASH Result [31:0]

Crypto_HASH_DOUT_7

Address: Operational Base + offset (0x001a8)

Hash Result Register 7

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HASH_RESULT_7 Specifies the HASH Result [31:0]

Crypto_HASH_SEED_0

Address: Operational Base + offset (0x001ac)

PRNG Seed/HMAC Key Register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_0 Specifies PRNG Seed/HMAC Key buffer [159:128]

Crypto_HASH_SEED_1

Address: Operational Base + offset (0x001b0)

PRNG Seed/HMAC Key Register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_1 Specifies PRNG Seed/HMAC Key buffer [127:96]

Crypto_HASH_SEED_2

Address: Operational Base + offset (0x001b4)

PRNG Seed/HMAC Key Register 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_2 Specifies PRNG Seed/HMAC Key buffer [95:64]

Crypto_HASH_SEED_3

Address: Operational Base + offset (0x001b8)

PRNG Seed/HMAC Key Register 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_3 Specifies PRNG Seed/HMAC Key buffer [63:32]

Crypto_HASH_SEED_4

Address: Operational Base + offset (0x001bc)

PRNG Seed/HMAC Key Register 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HASH_SEED_4 Specifies PRNG Seed/HMAC Key buffer [31:0]

Crypto_TRNG_CTRL

Address: Operational Base + offset (0x00200)

crypto TRNG control

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	reserved

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Bit	Attr	Reset Value	Description
16	RW	0x0000	osc_enable osc_ring enable It controls the running of osc_ring. And it is independent of clock and flush signal. This mean that it can run even when clock is gating or flush is asserted as long as osc_enable is asserted. Before it is used to get TRNG result , please run osc_ring first to get enough entropy. 1'b1: Enable ; 1'b0: Disable ;
15:0	RW	0x0000	period sample period TRNG use clock_crypto to sample ring osc output, this parameter is specify how many cycles to generate 1 bit random data.

Crypto_TRNG_DOUT_0

Address: Operational Base + offset (0x00204)

TRNG output register0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_TRNG_DOUT_1

Address: Operational Base + offset (0x00204)

TRNG output register1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_TRNG_DOUT_2

Address: Operational Base + offset (0x00208)

TRNG output register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_TRNG_DOUT_3

Address: Operational Base + offset (0x00210)

TRNG output register3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_TRNG_DOUT_4

Address: Operational Base + offset (0x00214)

TRNG output register4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_TRNG_DOUT_5

Address: Operational Base + offset (0x00218)

TRNG output register5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_TRNG_DOUT_6

Address: Operational Base + offset (0x0021c)

TRNG output register6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

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Crypto_TRNG_DOUT_7

Address: Operational Base + offset (0x00220)

TRNG output register7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	TRNG output

Crypto_PKA_CTRL

Address: Operational Base + offset (0x00280)

PKA Control Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	block_size PKA Size It specifies the bits of N in PKA calculation. 2'b00: 512 bit 2'b01: 1024 bit 2'b10: 2048 bit

Crypto_AES_TKEY_0

Address: Operational Base + offset (0x0300)

AES Tweak Key data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_0 Specifies AES-XTS tweak key data [255:224]

Crypto_AES_TKEY_1

Address: Operational Base + offset (0x0304)

AES Tweak Key data 1 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_1 Specifies AES-XTS tweak key data [223:192]

Crypto_AES_TKEY_2

Address: Operational Base + offset (0x0308)

AES Tweak Key data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_2 Specifies AES-XTS tweak key data [191:160]

Crypto_AES_TKEY_3

Address: Operational Base + offset (0x030c)

AES Tweak Key data 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_3 Specifies AES-XTS tweak key data [159:128]

Crypto_AES_TKEY_4

Address: Operational Base + offset (0x0310)

AES Tweak Key data 4 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_4 Specifies AES-XTS tweak key data [127:96]

Crypto_AES_TKEY_5

Address: Operational Base + offset (0x0314)

AES Tweak Key data 5 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_5 Specifies AES-XTS tweak key data [95:64]

Crypto_AES_TKEY_6

Address: Operational Base + offset (0x0318)

AES Tweak Key data 6 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_6 Specifies AES-XTS tweak key data [63:32]

Crypto_AES_TKEY_7

Address: Operational Base + offset (0x031c)

AES Tweak Key data 7 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TKEY_7 Specifies AES-XTS tweak key data [31:0]

Crypto_AES_TWK_0

Address: Operational Base + offset (0x0320)

AES Tweak data 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TWK_0 Specifies AES-XTS tweak date[127:96]

Crypto_AES_TWK_1

Address: Operational Base + offset (0x0324)

AES Tweak data 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TWK_1 Specifies AES-XTS tweak date[95:64]

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Crypto_AES_TWK_2

Address: Operational Base + offset (0x0328)

AES Tweak data 2 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TWK_2 Specifies AES-XTS tweak date[63:32]

Crypto_AES_TWK_3

Address: Operational Base + offset (0x032c)

AES Tweak data 3 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	AES_TWK_3 Specifies AES-XTS tweak date[31:0]

Crypto_KEY_SECURE

Address: Operational Base + offset (0x0330)

Key secure register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	KEY_SECURE 0 : AES key ,tkey and DES key are non-secure 1 : AES key ,tkey and DES key are secure

Crypto_CLK_GATE

Address: Operational Base + offset (0x0334)

Clock Gate Control register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x1	AES_CLK_GATE 0: AES clock is gated 1: AES clock is not gated
2	RW	0x1	HASH_CLK_GATE 0: HASH clock is gated 1: HASH clock is not gated
1	RW	0x1	TDES_CLK_GATE 0: TDES clock is gated 1: TDES clock is not gated
0	RW	0x1	RSA_CLK_GATE 0: RSA clock is gated 1: RSA clock is not gated

Crypto_CRYPTO_VER

Address: Operational Base + offset (0x03f0)

Key secure register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000300	CRYPTO_VER Crypto version

Crypto_PKA_M

Address: Operational Base + offset (0x0400)

PKA input or output data

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PKA input or output data. PKA result = (M ^ E) mod N. When it finishes, the result data is in M position. Start from PKA_M base address, and may contain 512/1024/2048 bits data.

Crypto_PKA_C

Address: Operational Base + offset (0x0500)

PKA C value

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	PKA pre-calculate data, C = 2 ^(2n+2) mod N

Crypto_PKA_N

Address: Operational Base + offset (0x0600)

PKA modular

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PKA modular

Crypto_PKA_E

Address: Operational Base + offset (0x0700)

PKA exponent

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PKA exponent.

11.4 Application Note

11.4.1 Reset port

CRU.crypto_srstn_req is used to do a soft reset to crypto . Please refer to "Chapter CRU" for more details.

11.4.2 Overall Performance

Use CRU.clk_crypto_div_con to select crypto frequency: F_{crypto} = F_{aclk} / (div + 1).

Make sure F_{crypto} do not exceed 150M.

The performance of crypto FIFO mode is list below.

Table 11-1Crypto Performance Description

algorithm	cycle	block size	frequency	throughput rate
DES	17	64 bit	150M	<=565 M bps
TDES	51	64 bit	150M	<=188 M bps

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AES	11/13/15	128 bit	150M	<=1745/1477/1280Mbps
SHA-1	81	512 bit	150M	<=948 Mbps
MD5	65	512 bit	150M	<=1182 Mbps

11.4.3 Usage

1. Symmetric algorithm

DES/3DES, AES are symmetric algorithms. There are two ways of using these algorithms: Slave mode and FIFO mode, AES_XTS only support FIFO mode.

In Slave mode, you can calculate 1 block size of data by starting the engine. Take AES-128 for example, you should

- Program Input 128 bit Data to AES_DIN_0~AES_DIN_3;
- Program Input 128 bit Key to AES_KEY_0~AES_KEY_3;
- Program control mode to AES_CTRL to run in different mode;
- Program CTRL.AES_START to run;
- Wait AES_STS.DONE high;
- Read AES_DOUT_0 ~ AES_DOUT_3 to get result.

In FIFO mode,

- Program the source address to BRDMAS, the destination address to BTDMAS, program the length in word unit to BRDMAL;
- Program Input 128 bit Key to AES_KEY_0~AES_KEY_3;
- Program control mode to AES_CTRL to run in different mode;
- Program INTENA to enable interrupt;
- Program CTRL.BLOCK_START to start;
- Wait interrupt asserted;
- Program INTSTS to clear interrupt status;
- Read the destination address which BTDMA points to.

FIFO mode get much higher throughput rate.

2. HASH

HASH is used to get digest of data. Only support FIFO mode.

There are three source: (1) hr_fifo; (2) br_fifo; (3) bt_fifo.

Take hr_fifo for example

- Program CTRL.HASH_FLUSH 1'b1 to clear, wait several cycle (>= 10 cycles) , and program CTRL.HASH_FLUSH 1'b0;
- Program data source address to HRDMAS, program 1 time data length in word unit to HRDMAL, program total length in byte unit to HASH_MSG_LEN;
- Program HASH_CTRL to choose algorithm, for example SHA-256;
- Program INTENA to enable interrupt;
- Program CTRL.HASH_START 1'b1 to start;
- Wait interrupt asserted; Only if HRDMAL length meets can this interrupt be asserted. If you have another section of data to hash, program the new data source address to HRDMAS, HASH_MSG_LEN need not to be programmed;
- Wait HASH_STS.done asserted. Only if Hash_MSG_LEN meet can this bit status register asserted;
- Read HASH_DOUT_0 – HASH_DOUT_7 to get result.

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3. Asymmetric Algorithm

Support 512/1024/2048 bit RSA calculation. It provide the big number calculation. Result = ME mod N

- Program CTRL.PKA_FLUSH 1'b1 to flush RSA module;
- Wait CTRL.PKA_FLUSH to be LOW. It is self-cleared;
- Program input_data(M) to PKA_M;
- Program pre_caculated C to PKA_C;
- Program Key(N) to PKA_N;
- Program Key(E) to PKA_E. C = 2 (2n+2) mod N. n is the required bit of N. For example 2048 bit N, n = 2048;
- Program PKA_CTRL to select RSA size: 512/1024/2048
- Program INTENA to enable interrupt;
- Program CTRL.PKA_START to start;
- Wait interrupt asserted.
- Read PKA_M to get results.

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Chapter 12 Mobile Storage Host Controller

12.1 Overview

The Mobile Storage Host Controller is designed to support Secure Digital memory (SD-max version 3.01) with 1 bits or 4 bits data width, Multimedia Card(MMC-max version 4.51) with 1 bits or 4 bits or 8 bits data width.

The Host Controller is instantiated for SDMMC, SDMMC_EXT, SDIO and EMMC. The interface difference between these instances is shown in "Interface Description".

The Host Controller supports following features:

- Bus Interface Features:
  - Support AMBA AHB interface for master and slave
  - Supports internal DMA interface(IDMAC)
    - Supports 16/32-bit data transfers
    - Single-channel; single engine used for Transmit and Receive, which are mutually exclusive
    - Dual-buffer and chained descriptor linked list
    - Each descriptor can transfer up to 4KB of data in chained mode and 8KB of data in dual-buffer mode
    - Programmable burst size for optimal host bus utilization
  - Support combined single FIFO for both transmit and receive operations
  - Support FIFO size of 256x32
  - Support FIFO over-run and under-run prevention by stopping card clock
- Card Interface Features:
  - Support Secure Digital memory protocol commands
  - Support Secure Digital I/O protocol commands
  - Support Multimedia Card protocol commands
  - Support Command Completion Signal and interrupts to host
  - Support CRC generation and error detection
  - Support programmable baud rate
  - Support power management and power switch
  - Support card detection
  - Support write protection
  - Support hardware reset
  - Support SDIO interrupts in 1-bit and 4-bit modes
  - Support 4-bit mode in SDIO3.0
  - Support SDIO suspend and resume operation
  - Support SDIO read wait
  - Support block size of 1 to 65,535 bytes
  - Support 1-bit, 4-bit and 8-bit SDR modes
  - Support boot in 1-bit, 4-bit and 8-bit SDR modes
  - Support Packed Commands, CMD21, CMD49
- Clock Interface Features:
  - Support 0/90/180/270-degree phase shift operation for sample clock (cclk_in_sample) and drive clock(cclk_in_drv) relative to function clock(cclk_in) respectively

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- Support phase tuning using delay line for sample clock(cclk_in_sample) and drive clock(cclk_in_drv) relative to function clock (cclk_in) respectively. The max number of delay element number is 256

12.2 Block Diagram

The Host Controller consists of the following main functional blocks.

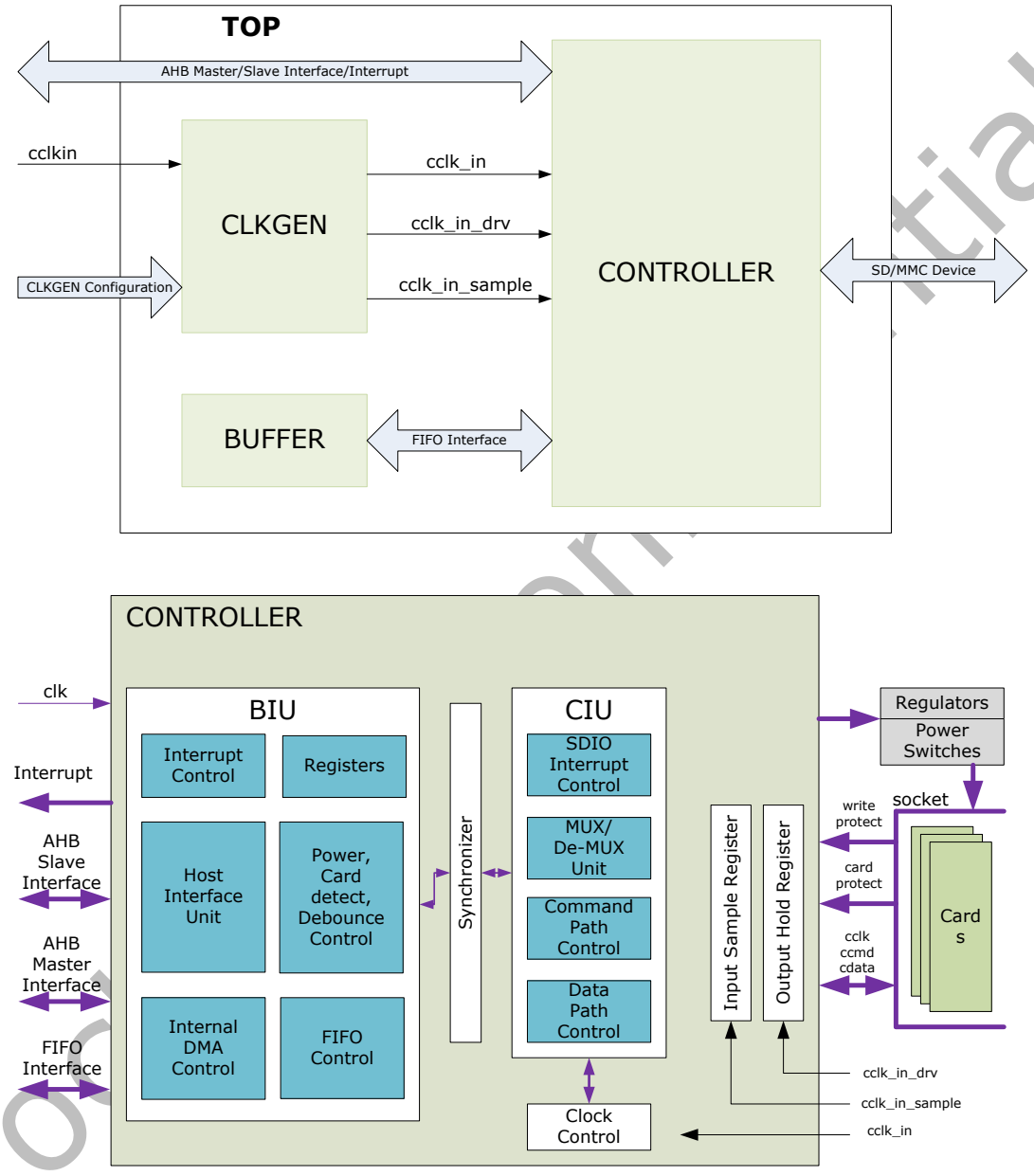

Fig. 12-1 Host Controller Block Diagram

- Clock Generate Unit(CLKGEN): generates card interface clock cclk_in/ cclk_sample/cclk_drv based on cclkin and configuration information.
- Asynchronous dual-port memory(BUFFER): Uses a two-clock synchronous read and synchronous write dual-port RAM. One of the ports is connected to the host clock, and the second port is connected to the card clock.
- Bus Interface Unit (BIU): Provides AMBA AHB interfaces for register and data read/writes.
- Card Interface Unit (CIU): Takes care of the SD/MMC protocols and provides clock management.

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12.3 Function Description

12.3.1 Bus Interface Unit

The Bus Interface Unit provides the following functions:

- Host interface
- Interrupt control
- Register access
- External FIFO access
- Power control and card detection

1. Host Interface Unit

The Host Interface Unit is an AHB slave interface, which provides the interface between the SD/MMC card and the host bus.

2. Register Unit

The register unit is part of the bus interface unit; it provides read and write access to the registers.

All registers reside in the Bus Interface Unit clock domain. When a command is sent to a card by setting the start_bit, which is bit[31] of the CMD register, all relevant registers needed for the CIU operation are transferred to the CIU block. During this time, the registers that are transferred from the BIU to the CIU should not be written. The software should wait for the hardware to clear the start bit before writing to these registers again. The register unit has a hardware locking feature to prevent illegal writes to registers. The lock is necessary in order to avoid metastability violations, both because the host and card clock domains are different and to prevent illegal software operations.

Once a command start is issued by setting the start_bit of the CMD register, the following registers cannot be reprogrammed until the command is accepted by the card interface unit:

- CMD – Command
- CMDARG – Command Argument
- BYTCNT – Byte Count
- BLKSIZ – Block Size
- CLKDIV – Clock Divider
- CLKENA – Clock Enable
- CLKSRC – Clock Source
- TMOUT – Timeout
- CTYPE – Card Type

The hardware resets the start_bit once the CIU accepts the command. If a host write to any of these registers is attempted during this locked time, then the write is ignored and the hardware lock error bit is set in the raw interrupt status register. Additionally, if the interrupt is enabled and not masked for a hardware lock error, then an interrupt is sent to the host.

When the Card Interface Unit is in an idle state, it typically takes the following number of clocks for the command handshake, where clk is the BIU clock and cclk_in is the CIU clock:

3 (clk) + 3 (cclk_in)

Once a command is accepted, you can send another command to the CIU-which has a one-deep command queue-under the following conditions:

- If the previous command was not a data transfer command, the new command is sent to the SD/MMC card once the previous command completes.

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- If the previous command is a data transfer command and if wait_prvdata_complete (bit[13]) of the Command register is set for the new command, the new command is sent to the SD/MMC card only when the data transfer completes.
- If the wait_prvdata_complete is 0, then the new command is sent to the SD/MMC card as soon as the previous command is sent. Typically, you should use this only to stop or abort a previous data transfer or query the card status in the middle of a data transfer.

3. Interrupt Controller Unit

The interrupt controller unit generates an interrupt that depends on the controller raw interrupt status, the interrupt-mask register, and the global interrupt-enable register bit. Once an interrupt condition is detected, it sets the corresponding interrupt bit in the raw interrupt status register. The raw interrupt status bit stays on until the software clears the bit by writing a 1 to the interrupt bit; a 0 leaves the bit untouched.

The interrupt port, int, is an active-high, level-sensitive interrupt. The interrupt port is active only when any bit in the raw interrupt status register is active, the corresponding interrupt mask bit is 1, and the global interrupt enable bit is 1. The interrupt port is registered in order to avoid any combinational glitches.

The int_enable is reset to 0 on power-on, and the interrupt mask bits are set to 32'h0, which masks all the interrupts.

Notes:

Before enabling the interrupt, it is always recommended that you write 32'hffff_ffff to the raw interrupt status register in order to clear any pending unserviced interrupts. When clearing interrupts during normal operation, ensure that you clear only the interrupt bits that you serviced.

The SDIO Interrupts, Receive FIFO Data Request (RXDR), and Transmit FIFO Data Request (TXDR) are set by level-sensitive interrupt sources. Therefore, the interrupt source should be first cleared before you can clear the interrupt bit of the Raw Interrupt register. For example, on seeing the Receive FIFO Data Request (RXDR) interrupt, the FIFO should be emptied so that the "FIFO count greater than the RX-Watermark" condition, which triggers the interrupt, becomes inactive. The rest of the interrupts are triggered by a single clock-pulse-width source.

Table 12-1 Bits in Interrupt Status Register

Bits	Interrupt	Description
24	sdio_interrupt	Interrupt from SDIO card. In MMC-Ver3.3-only mode, these bits are always 0
16	Card no-busy	If card exit busy status, the interrupt happened
15	End Bit Error (read) /Write no CRC (EBE)	Error in end-bit during read operation, or no data CRC received during write operation. For MMC CMD19, there may be no CRC status returned by the card. Hence, EBE is set for CMD19. The application should not treat this as an error.
14	Auto Command Done (ACD)	Stop/abort commands automatically sent by card unit and not initiated by host; similar to Command Done (CD) interrupt. Recommendation: Software typically need not enable this for non CE-ATA accesses; Data Transfer Over (DTO) interrupt that comes after this interrupt determines whether data

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Bits	Interrupt	Description
		transfer has correctly competed.
13	Start Bit Error (SBE)	Error in data start bit when data is read from a card. In 4-bit mode, if DAT[0] line indicates start bit-that is, 0-and any of the other data bits do not have start bit, then this error is set. Busy Complete Interrupt when data is written to the card. This interrupt is generated after completion of busy driven by the card after the last data block is written into the card.
12	Hardware Locked write Error (HLE)	During hardware-lock period, write attempted to one of locked registers. When software sets the start_cmd bit in the CMD register, the Host Controller tries to load the command. If the command buffer is already filled with a command, this error is raised. The software then has to reload the command.
11	FIFO Underrun/ Overrun Error (FRUN)	Host tried to push data when FIFO was full, or host tried to read data when FIFO was empty. Typically this should not happen, except due to error in software. Card unit never pushes data into FIFO when FIFO is full, and pop data when FIFO is empty. If IDMAC is enabled, FIFO underrun/overrun can occur due to a programming error on MSIZE and watermark values in FIFOTH register.
10	Data Starvation by Host Timeout (HTO)	To avoid data loss, card clock out is stopped if FIFO is empty when writing to card, or FIFO is full when reading from card. Whenever card clock is stopped to avoid data loss, data-starvation timeout counter is started with data-timeout value. This interrupt is set if host does not fill data into FIFO during write to card, or does not read from FIFO during read from card before timeout period. Even after timeout, card clock stays in stopped state, with CIU state machines waiting. It is responsibility of host to push or pop data into FIFO upon interrupt, which automatically restarts cclk_out and card state machines. Even if host wants to send stop/abort command, it still needs to ensure it has to push or pop FIFO so that clock starts in order for stop/abort command to send on cmd signal along with data that is sent or received on data line.
9	Data Read Timeout (DRTO)	In Normal functioning mode: Data read timeout (DRTO) Data timeout occurred. Data Transfer Over (DTO) also set if data timeout occurs.

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Bits	Interrupt	Description
		In Boot Mode: Boot Data Start (BDS) When set, indicates that Host Controller has started to receive boot data from the card. A write to this register with a value of 1 clears this interrupt.
8	Response Timeout (RTO)	In Normal functioning mode: Response timeout (RTO) Response timeout occurred. Command Done (CD) also set if response timeout occurs. If command involves data transfer and when response times out, no data transfer is attempted by Host Controller. In Boot Mode: Boot Ack Received (BAR) When expect_boot_ack is set, on reception of a boot acknowledge pattern—0-1-0—this interrupt is asserted. A write to this register with a value of 1 clears this interrupt.
7	Data CRC Error (DCRC)	Received Data CRC does not match with locally-generated CRC in CIU. Can also occur if the Write CRC status is incorrectly sampled by the Host.
6	Response CRC Error (RCRC)	Response CRC does not match with locally-generated CRC in CIU.
5	Receive FIFO Data Request (RXDR)	Interrupt set during read operation from card when FIFO level is greater than Receive-Threshold level. Recommendation: In DMA modes, this interrupt should not be enabled. In non-DMA mode: pop RX_WMark + 1 data from FIFO.

4	Transmit FIFO Data Request (TXDR)	Interrupt set during write operation to card when FIFO level reaches less than or equal to Transmit-Threshold level.
		Recommendation:
		In DMA modes, this interrupt should not be enabled.
		In non-DMA mode:
		if (pending_bytes > (FIFO_DEPTH - TX_WMark))
		push (FIFO_DEPTH - TX_WMark) data into FIFO
		else
		push pending_bytes data into FIFO

3	Data Transfer Over (DTO)	Indicates Data transfer completed. Though on detection of errors-Start Bit Error, Data CRC error, and so on, DTO may or may not be set; the application must issue

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Bits	Interrupt	Description
		CMD12, which ensures that DTO is set. Recommendation: In non-DMA mode, when data is read from card, on seeing interrupt, host should read any pending data from FIFO. In DMA mode, DMA controllers guarantee FIFO is flushed before interrupt. DTO bit is set at the end of the last data block, even if the device asserts MMC busy after the last data block.
2	Command Done(CD)	Command sent to card and got response from card, even if Response Error or CRC error occurs.
1	Response Error (RE)	Error in received response set if one of following occurs: ● Transmission bit != 0 ● Command index mismatch ● End-bit != 1
0	Card-Detect (CDT)	When one or more cards inserted or removed, this interrupt occurs. Software should read card-detect register to determine current card status. Recommendation: After power-on and before enabling interrupts, software should read card detect register and store it in memory. When interrupt occurs, it should read card detect register and compare it with value stored in memory to determine which card(s) were removed/inserted. Before exiting ISR, software should update memory with new card-detect value.

4. FIFO Controller Unit

The FIFO controller interfaces the external FIFO to the host interface and the card controller unit. When FIFO overrun and under-run conditions occur, the card clock stops in order to avoid data loss.

The FIFO uses a two-clock synchronous read and synchronous write dual-port RAM. One of the ports is connected to the host clock, clk, and the second port is connected to the card clock, cclk_in.

Notes: The FIFO controller does not support simultaneous read/write access from the same port. For debugging purposes, the software may try to write into the FIFO and read back the data; results are indeterminate, since the design does not support read/write access from the same port.

5. Power Control and Card Detection Unit

The register unit has registers that control the power. Power to each card can be selectively turned on or off.

The card detection unit looks for any changes in the card-detect signals for card insertion or card removal. It filters out the debounces associated with mechanical insertion or removal, and generates one interrupt to the host. You can program the debounce filter value.

On power-on, the controller should read in the card_detect port and store the value in the memory. Upon receiving a card-detect interrupt, it should again read the card_detect port and

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XOR with the previous card-detect status to find out which card has interrupted. If more than one card is simultaneously removed or inserted, there is only one card-detect interrupt; the XOR value indicates which cards have been disturbed. The memory should be updated with the new card-detect value.

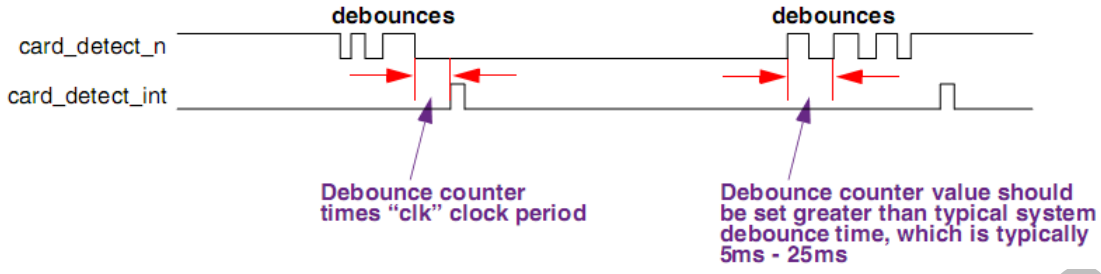

Fig. 12-2 SD/MMC Card-Detect Signal

6. DMA Interface Unit

DMA signals interface the Host Controller to an external DMA controller to reduce the software overhead during FIFO data transfers. The DMA request/acknowledge handshake is used for only data transfers. The DMA interface provides a connection to the DMA Controller.

On seeing the DMA request, the DMA controller initiates accesses through the host interface to read or write into the data FIFO. The Host Controller has FIFO transmit/receive watermark registers that you can set, depending on system latency. The DMA interface asserts the request in the following cases:

- Read from a card when the data FIFO word count exceeds the Rx-Watermark level
- Write to a card when the FIFO word count is less than or equal to the Tx-Watermark level

When the DMA interface is enabled, you can use normal host read/write to access the data FIFO.

12.3.2 Card Interface Unit

The Card Interface Unit (CIU) interfaces with the Bus Interface Unit (BIU) and the devices. The host writes command parameters to the BIU control registers, and these parameters are then passed to the CIU. Depending on control register values, the CIU generates SD/MMC command and data traffic on a selected card bus according to SD/MMC protocol. The Host Controller accordingly controls the command and data path.

The following software restrictions should be met for proper CIU operation:

- Only one data transfer command can be issued at a time.
- During an open-ended card write operation, if the card clock is stopped because the FIFO is empty, the software must first fill the data into the FIFO and start the card clock. It can then issue only a stop/abort command to the card.
- When issuing card reset commands (CMD0, CMD15 or CMD52_reset) while a card data transfer is in progress, the software must set the stop_abort_cmd bit in the Command register so that the Host Controller can stop the data transfer after issuing the card reset command.
- When the data end bit error is set in the RINTSTS register, the Host Controller does not guarantee SDIO interrupts. The software should ignore the SDIO interrupts and issue the stop/abort command to the card, so that the card stops sending the read data.
- If the card clock is stopped because the FIFO is full during a card read, the software should read at least two FIFO locations to start the card clock.

The CIU block consists of the following primary functional blocks:

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- Command path
- Data path
- SDIO interrupt control
- Clock control
- Mux/demux unit

1. Command Path

The command path performs the following functions:

- Loads clock parameters
- Loads card command parameters
- Sends commands to card bus (ccmd_out line)
- Receives responses from card bus (ccmd_in line)
- Sends responses to BIU
- Drives the P-bit on command line

A new command is issued to the Host Controller by programming the BIU registers and setting the start_cmd bit in the Command register. The BIU asserts start_cmd, which indicates that a new command is issued to the SD/MMC device. The command path loads this new command (command, command argument, timeout) and sends acknowledge to the BIU by asserting cmd_taken.

Once the new command is loaded, the command path state machine sends a command to the device bus-including the internally generated CRC7-and receives a response, if any. The state machine then sends the received response and signals to the BIU that the command is done, and then waits for eight clocks before loading a new command.

Load Command Parameters

One of the following commands or responses is loaded in the command path:

- New command from BIU – When start_cmd is asserted, then the start_cmd bit is set in the Command register.
- Internally-generated auto-stop command – When the data path ends, the stop command request is loaded.
- IRQ response with RCA 0x000 – When the command path is waiting for an IRQ response from the MMC card and a "send irq response" request is signaled by the BIU, then the send_irq_response bit is set in the control register.

Loading a new command from the BIU in the command path depends on the following Command register bit settings:

- update_clock_registers_only – If this bit is set in the Command register, the command path updates only the clock enable, clock divider, and clock source registers. If this bit is not set, the command path loads the command, command argument, and timeout registers; it then starts processing the new command.
- wait_prvdata_complete – If this bit is set, the command path loads the new command under one of the following conditions:
  - Immediately, if the data path is free (that is, there is no data transfer in progress), or if an open-ended data transfer is in progress (byte_count = 0).
  - After completion of the current data transfer, if a predefined data transfer is in progress.

Send Command and Receive Response

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Once a new command is loaded in the command path, update_clock_registers_only bit is unset – the command path state machine sends out a command on the device bus; the command path state machine is illustrated in following figure.

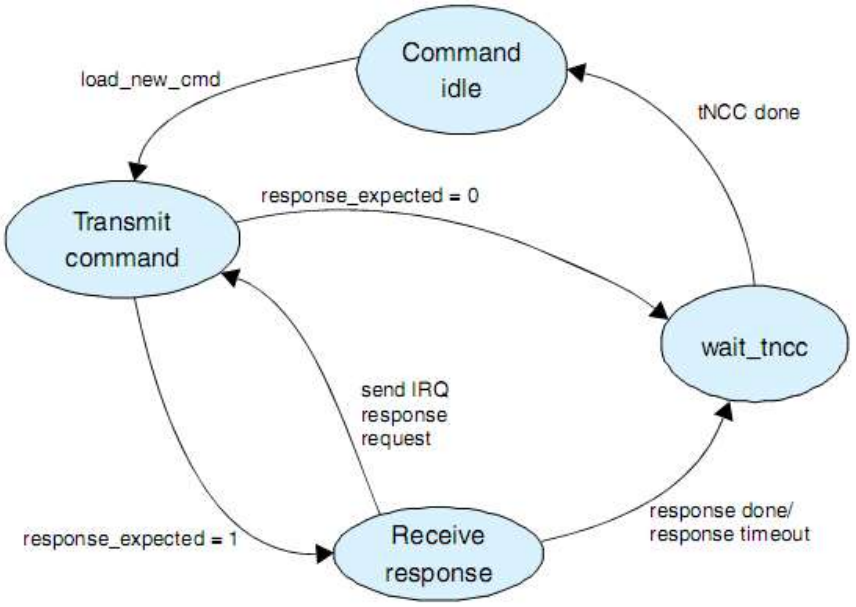

Fig. 12-3 Host Controller Command Path State Machine

The command path state machine performs the following functions, according to Command register bit values:

- send_initialization – Initialization sequence of 80 clocks is sent before sending the command.
- response_expected – Response is expected for the command. After the command is sent out, the command path state machine receives a 48-bit or 136-bit response and sends it to the BIU. If the start bit of the card response is not received within the number of clocks programmed in the timeout register, then the response timeout and command done bit is set in the Raw Interrupt Status register as a signal to the BIU. If the response-expected bit is not set, the command path sends out a command and signals a response done to the BIU; that is, the command done bit is set in the Raw Interrupt Status register.
- response_length – If this bit is set, a 136-bit response is received; if it is not set, a 48-bit response is received.
- check_response_crc – If this bit is set, the command path compares CRC7 received in the response with the internally-generated CRC7. If the two do not match, the response CRC error is signaled to the BIU; that is, the response CRC error bit is set in the Raw Interrupt Status register.

Send Response to BIU

If the response_expected bit is set in the Command register, the received response is sent to the BIU. The Response0 register is updated for a short response, and the Response3, Response2, Response1, and Response0 registers are updated on a long response, after which the Command Done bit is set. If the response is for an auto_stop command sent by the CIU, the response is saved in the Response1 register, after which the Auto Command Done bit is set.

Additionally, the command path checks for the following:

- Transmission bit = 0
- Command index matches command index of the sent command

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- End bit = 1 in received card response

The command index is not checked for a 136-bit response or if the check_response_crc bit is unset. For a 136-bit response and reserved CRC 48-bit responses, the command index is reserved-that is, 111111.

Polling Command Completion Signal

The device generates the Command Completion Signal in order to notify the host controller of the normal command completion or command termination.

Command Completion Signal Detection and Interrupt to Host Processor

If the ccs_expected bit is set in the Command register, the Command Completion Signal (CCS) from the device is indicated by setting the Data Transfer Over (DTO) bit in the RINTSTS register. The Host Controller generates a Data Transfer Over (DTO) interrupt if this interrupt is not masked.

Command Completion Signal Timeout

If the command expects a CCS from the device—if the ccs_expected bit is set in the Command register—the command state machine waits for the CCS and remains in a wait_CCSS state. If the device fails to send out the CCS, the host software should implement a timeout mechanism to free the command and data path. The host controller does not implement a hardware timer; it is the responsibility of the host software to maintain a software timer.

In the event of a CCS timeout, the host should issue a CCSD by setting the send_ccsd bit in the CTRL register. The host controller command state machine sends the CCSD to the device and exits to an idle state. After sending the CCSD, the host should also send a CMD12 to the device in order to abort the outstanding command.

Send Command Completion Signal Disable

If the send_ccsd bit is set in the CTRL register, the host sends a Command Completion Signal Disable (CCSD) pattern on the CMD line. The host can send the CCSD while waiting for the CCS or after a CCS timeout happens.

After sending the CCSD pattern, the host sets the Command Done (CD) bit in RINTSTS and also generates an interrupt to the host if the Command Done interrupt is not masked.

2. Data Path

The data path block pops the data FIFO and transmits data on cdata_out during a write data transfer, or it receives data on cdata_in and pushes it into the FIFO during a read data transfer. The data path loads new data parameters—that is, data expected, read/write data transfer, stream/block transfer, block size, byte count, card type, timeout registers—whenever a data transfer command is not in progress.

If the data_expected bit is set in the Command register, the new command is a data transfer command and the data path starts one of the following:

- Transmit data if the read/write bit = 1
- Data receive if read/write bit = 0

Data Transmit

The data transmit state machine, illustrated in following figure, starts data transmission two clocks after a response for the data write command is received; this occurs even if the command path detects a response error or response CRC error. If a response is not received from the card because of a response timeout, data is not transmitted. Depending upon the value of the transfer_mode bit in the Command register, the data transmit state machine puts data on the card data bus in a stream or in block(s).

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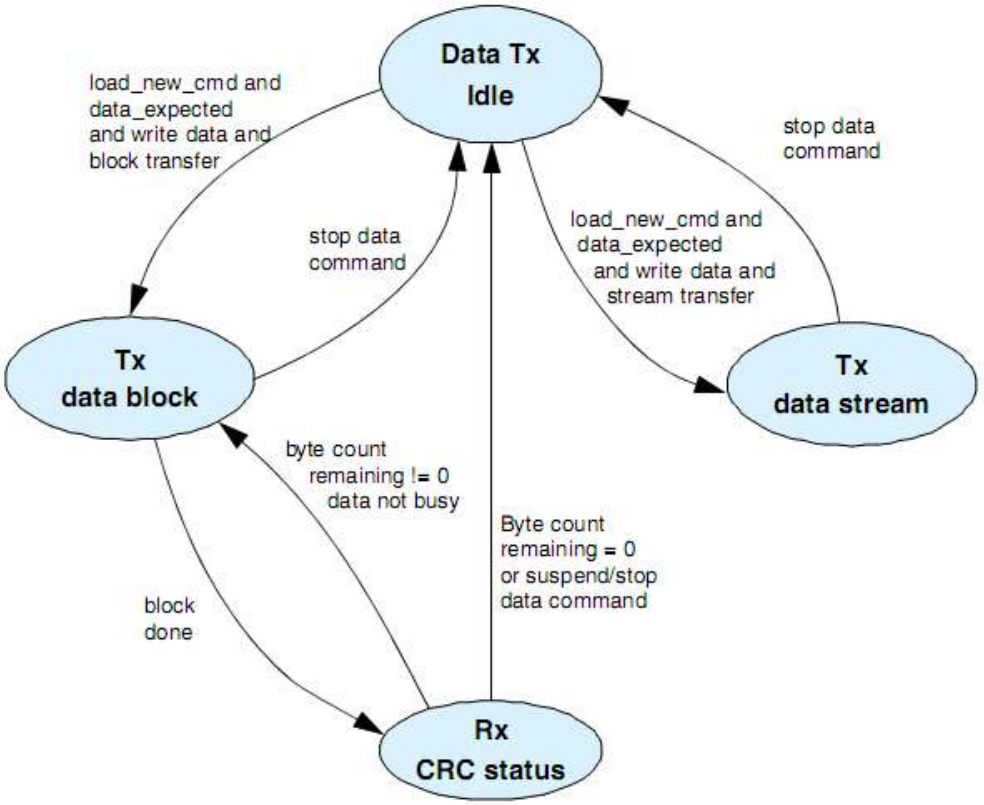

Fig. 12-4 Host Controller Data Transmit State Machine

Stream Data Transmit

If the transfer_mode bit in the Command register is set to 1, it is a stream-write data transfer. The data path pops the FIFO from the BIU and transmits in a stream to the card data bus. If the FIFO becomes empty, the card clock is stopped and restarted once data is available in the FIFO.

If the byte_count register is programmed to 0, it is an open-ended stream-write data transfer. During this data transfer, the data path continuously transmits data in a stream until the host software issues a stop command. A stream data transfer is terminated when the end bit of the stop command and end bit of the data match over two clocks.

If the byte_count register is programmed with a non-zero value and the send_auto_stop bit is set in the Command register, the stop command is internally generated and loaded in the command path when the end bit of the stop command occurs after the last byte of the stream write transfer matches.

This data transfer can also terminate if the host issues a stop command before all the data bytes are transferred to the card bus.

Single Block Data

If the transfer_mode bit in the Command register is set to 0 and the byte_count register value is equal to the value of the block_size register, a single-block write-data transfer occurs. The data transmit state machine sends data in a single block, where the number of bytes equals the block size, including the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set for a 1-bit, 4-bit, or 8-bit data transfer, the data is transmitted on 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and transmitted for 1, 4, or 8 data lines, respectively.

After a single data block is transmitted, the data transmit state machine receives the CRC status from the card and signals a data transfer to the BIU; this happens when the

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data-transfer-over bit is set in the RINTSTS register.

If a negative CRC status is received from the card, the data path signals a data CRC error to the BIU by setting the data CRC error bit in the RINTSTS register.

Additionally, if the start bit of the CRC status is not received by two clocks after the end of the data block, a CRC status start bit error is signaled to the BIU by setting the write-no-CRC bit in the RINTSTS register.

Multiple Block Data

A multiple-block write-data transfer occurs if the transfer_mode bit in the Command register is set to 0 and the value in the byte_count register is not equal to the value of the block_size register. The data transmit state machine sends data in blocks, where the number of bytes in a block equals the block size, including the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set to 1-bit, 4-bit, or 8-bit data transfer, the data is transmitted on 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and transmitted on 1, 4, or 8 data lines, respectively.

After one data block is transmitted, the data transmit state machine receives the CRC status from the card. If the remaining byte_count becomes 0, the data path signals to the BIU that the data transfer is done; this happens when the data-transfer-over bit is set in the RINTSTS register.

If the remaining data bytes are greater than 0, the data path state machine starts to transmit another data block.

If a negative CRC status is received from the card, the data path signals a data CRC error to the BIU by setting the data CRC error bit in the RINTSTS register, and continues further data transmission until all the bytes are transmitted.

Additionally, if the CRC status start bit is not received by two clocks after the end of a data block, a CRC status start bit error is signaled to the BIU by setting the write-no-CRC bit in the RINTSTS register; further data transfer is terminated.

If the send_auto_stop bit is set in the Command register, the stop command is internally generated during the transfer of the last data block, where no extra bytes are transferred to the card. The end bit of the stop command may not exactly match the end bit of the CRC status in the last data block.

If the block size is less than 4, 16, or 32 for card data widths of 1 bit, 4 bits, or 8 bits, respectively, the data transmit state machine terminates the data transfer when all the data is transferred, at which time the internally generated stop command is loaded in the command path.

If the byte_count is 0 – the block size must be greater than 0 – it is an open-ended block transfer. The data transmit state machine for this type of data transfer continues the block-write data transfer until the host software issues a stop or abort command.

Data Receive

The data-receive state machine, illustrated in following figure, receives data two clock cycles after the end bit of a data read command, even if the command path detects a response error or response CRC error. If a response is not received from the card because a response timeout occurs, the BIU does not receive a signal that the data transfer is complete; this happens if the command sent by the Host Controller is an illegal operation for the card, which keeps the card from starting a read data transfer.

If data is not received before the data timeout, the data path signals a data timeout to the BIU and an end to the data transfer done. Based on the value of the transfer_mode bit in the Command register, the data-receive state machine gets data from the card data bus in a

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stream or block(s).

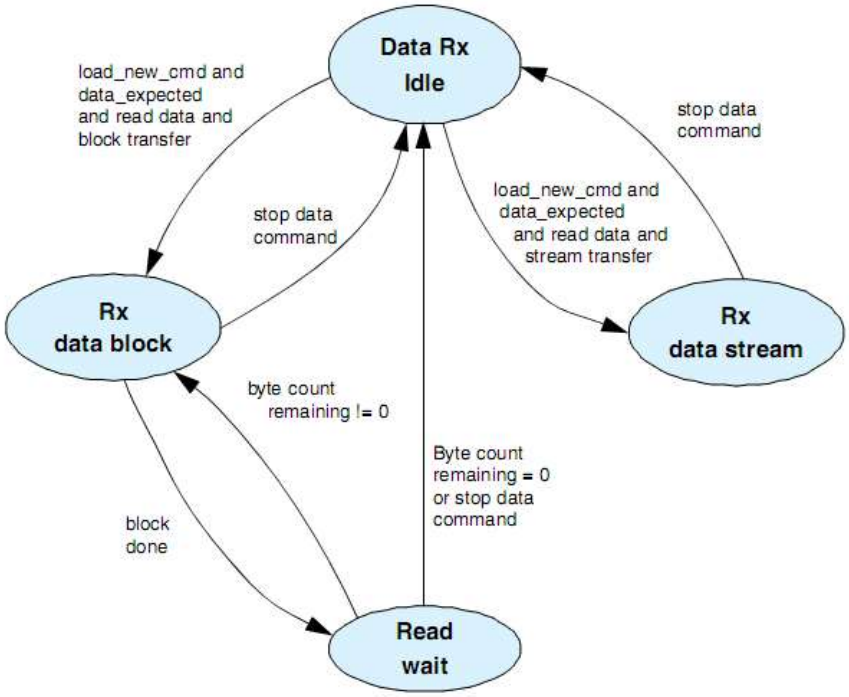

Fig. 12-5 Host Controller Data Receive State Machine

Stream Data Read

A stream-read data transfer occurs if the transfer_mode bit in the Command register equals 1, at which time the data path receives data from the card and pushes it to the FIFO. If the FIFO becomes full, the card clock stops and restarts once the FIFO is no longer full.

An open-ended stream-read data transfer occurs if the byte_count register equals 0. During this type of data transfer, the data path continuously receives data in a stream until the host software issues a stop command. A stream data transfer terminates two clock cycles after the end bit of the stop command.

If the byte_count register contains a non-zero value and the send_auto_stop bit is set in the Command register, a stop command is internally generated and loaded into the command path, where the end bit of the stop command occurs after the last byte of the stream data transfer is received. This data transfer can terminate if the host issues a stop or abort command before all the data bytes are received from the card.

Single-Block Data Read

A single-block read-data transfer occurs if the transfer_mode bit in the Command register is set to 0 and the value of the byte_count register is equal to the value of the block_size register. When a start bit is received before the data times out, data bytes equal to the block size and CRC16 are received and checked with the internally-generated CRC16.

If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set to a 1-bit, 4-bit, or 8-bit data transfer, data is received from 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and checked for 1, 4, or 8 data lines, respectively. If there is a CRC16 mismatch, the data path signals a data CRC error to the BIU. If the received end bit is not 1, the BIU receives an end-bit error.

Multiple-Block Data Read

If the transfer_mode bit in the Command register is set to 0 and the value of the byte_count register is not equal to the value of the block_size register, it is a multiple-block read-data transfer. The data-receive state machine receives data in blocks, where the number of bytes in a block is equal to the block size, including the internally-generated CRC16.

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If the CTYPE register bit for the selected card – indicated by the card_num value in the Command register – is set to a 1-bit, 4-bit, or 8-bit data transfer, data is received from 1, 4, or 8 data lines, respectively, and CRC16 is separately generated and checked for 1, 4, or 8 data lines, respectively.

After a data block is received, if the remaining byte_count becomes 0, the data path signals a data transfer to the BIU.

If the remaining data bytes are greater than 0, the data path state machine causes another data block to be received. If CRC16 of a received data block does not match the internally-generated CRC16, a data CRC error to the BIU and data reception continue further data transmission until all bytes are transmitted.

Additionally, if the end of a received data block is not 1, data on the data path signals terminate the bit error to the CIU and the data-receive state machine terminates data reception, waits for data timeout, and signals to the BIU that the data transfer is complete.

If the send_auto_stop bit is set in the Command register, the stop command is internally generated when the last data block is transferred, where no extra bytes are transferred from the card; the end bit of the stop command may not exactly match the end bit of the last data block.

If the requested block size for data transfers to cards is less than 4, 16, or 32 bytes for 1-bit, 4-bit, or 8-bit data transfer modes, respectively, the data-transmit state machine terminates the data transfer when all data is transferred, at which point the internally-generated stop command is loaded in the command path. Data received from the card after that are then ignored by the data path.

If the byte_count is 0—the block size must be greater than 0—it is an open-ended block transfer. For this type of data transfer, the data-receive state machine continues the block-read data transfer until the host software issues a stop or abort command.

Auto-Stop

The Host Controller internally generates a stop command and is loaded in the command path when the send_auto_stop bit is set in the Command register.

The software should set the send_auto_stop bit according to details listed in following table.

Table 12-2 Auto-Stop Generation

Card type	Transfer type	Byte Count	send_auto_stop bit set	Comments
MMC	Stream read	0	No	Open-ended stream
MMC	Stream read	>0	Yes	Auto-stop after all bytes transfer
MMC	Stream write	0	No	Open-ended stream
MMC	Stream write	>0	Yes	Auto-stop after all bytes transfer
MMC	Single-block read	>0	No	Byte count =0 is illegal
MMC	Single-block write	>0	No	Byte count =0 is illegal
MMC	Multiple-block read	0	No	Open-ended multiple block

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Card type	Transfer type	Byte Count	send_auto_stop bit set	Comments
MMC	Multiple-block read	>0	Yes①	Pre-defined multiple block
MMC	Multiple-block write	0	No	Open-ended multiple block
MMC	Multiple-block write	>0	Yes①	Pre-defined multiple block
SDMEM	Single-block read	>0	No	Byte count =0 is illegal
SDMEM	Single-block write	>0	No	Byte count =0 illegal
SDMEM	Multiple-block read	0	No	Open-ended multiple block
SDMEM	Multiple-block read	>0	Yes	Auto-stop after all bytes transfer
SDMEM	Multiple-block write	0	No	Open-ended multiple block
SDMEM	Multiple-block write	>0	Yes	Auto-stop after all bytes transfer
SDIO	Single-block read	>0	No	Byte count =0 is illegal
SDIO	Single-block write	>0	No	Byte count =0 illegal
SDIO	Multiple-block read	0	No	Open-ended multiple block
SDIO	Multiple-block read	>0	No	Pre-defined multiple block
SDIO	Multiple-block write	0	No	Open-ended multiple block
SDIO	Multiple-block write	>0	No	Pre-defined multiple block

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①:*The condition under which the transfer mode is set to block transfer and byte_count is equal to block size is treated as a single-block data transfer command for both MMC and SD cards. If byte_count = n*block_size (n = 2, 3, …), the condition is treated as a predefined multiple-block data transfer command. In the case of an MMC card, the host software can perform a predefined data transfer in two ways: 1) Issue the CMD23 command before issuing CMD18/CMD25 commands to the card – in this case, issue MD18/CMD25 commands without setting the send_auto_stop bit. 2) Issue CMD18/CMD25 commands without issuing CMD23 command to the card, with the send_auto_stop bit set. In this case, the multiple-block data transfer is terminated by an internally-generated auto-stop command after the programmed byte count.*

The following list conditions for the auto-stop command.

- Stream read for MMC card with byte count greater than 0 – The Host Controller generates an internal stop command and loads it into the command path so that the end bit of the stop command is sent out when the last byte of data is read from the card and no extra data byte is received. If the byte count is less than 6 (48 bits), a few extra data bytes are received from the card before the end bit of the stop command is sent.
- Stream write for MMC card with byte count greater than 0 - The Host Controller generates an internal stop command and loads it into the command path so that the end bit of the stop command is sent when the last byte of data is transmitted on the card bus and no extra data byte is transmitted. If the byte count is less than 6 (48 bits), the data path transmits the data last in order to meet the above condition.
- Multiple-block read memory for SD card with byte count greater than 0 – If the block size is less than 4 (single-bit data bus), 16 (4-bit data bus), or 32 (8-bit data bus), the auto-stop command is loaded in the command path after all the bytes are read. Otherwise, the top command is loaded in the command path so that the end bit of the stop command is sent after the last data block is received.
- Multiple-block write memory for SD card with byte count greater than 0 – If the block size is less than 3 (single-bit data bus), 12 (4-bit data bus), or 24 (8-bit data bus), the auto-stop command is loaded in the command path after all data blocks are transmitted. Otherwise, the stop command is loaded in the command path so that the end bit of the stop command is sent after the end bit of the CRC status is received.
- Precaution for host software during auto-stop – Whenever an auto-stop command is issued, the host software should not issue a new command to the SD/MMC device until the auto-stop is sent by the Host Controller and the data transfer is complete. If the host issues a new command during a data transfer with the auto-stop in progress, an auto-stop command may be sent after the new command is sent and its response is received; this can delay sending the stop command, which transfers extra data bytes. For a stream write, extra data bytes are erroneous data that can corrupt the card data. If the host wants to terminate the data transfer before the data transfer is complete, it can issue a stop or abort command, in which case the Host Controller does not generate an auto-stop command.

3. Non-Data Transfer Commands that Use Data Path

Some non-data transfer commands (non-read/write commands) also use the data path. Following table lists the commands and register programming requirements for them.

Table 12-3 Non-data Transfer Commands and Requirements

Base Address [12:8]	CMD 27	CMD 30	CMD 42	ACMD 13	ACMD 22	ACMD 51
Command register programming						
cmd_index	6'h1B	6'h1E	6'h2A	6'h0D	6'h16	6'h33
response_expect	1	1	1	1	1	1
rResponse_length	0	0	0	0	0	0

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Base Address [12:8]	CMD 27	CMD 30	CMD 42	ACMD 13	ACMD 22	ACMD 51
check_response_crc	1	1	1	1	1	1
data_expected	1	1	1	1	1	1
read/write	1	0	1	0	0	0
transfer_mode	0	0	0	0	0	0
send_auto_stop	0	0	0	0	0	0
wait_prevdata_complete	0	0	0	0	0	0
stop_abort_cmd	0	0	0	0	0	0
Command Argument register programming						
	stuff bits	32-bit write protect data address	stuff bits	stuff bits	stuff bits	stuff bits
Block Size register programming						
	16	4	Num_bytes①	64	4	8
Byte Count register programming						
	16	4	Num_bytes①	64	4	8

①: *Num_bytes = No. of bytes specified as per the lock card data structure (Refer to the SD specification and the MMC specification)*

4. SDIO Interrupt Control

Interrupts for SD cards are reported to the BIU by asserting an interrupt signal for two clock cycles. SDIO cards signal an interrupt by asserting cdata_in low during the interrupt period; an interrupt period for the selected card is determined by the interrupt control state machine. An interrupt period is always valid for non-active or non-selected cards, and 1-bit data mode for the selected card. An interrupt period for a wide-bus active or selected card is valid for the following conditions:

- Card is idle
- Non-data transfer command in progress
- Third clock after end bit of data block between two data blocks
- From two clocks after end bit of last data until end bit of next data transfer command

Bear in mind that, in the following situations, the controller does not sample the SDIO interrupt of the selected card when the card data width is 4 bits. Since the SDIO interrupt is level-triggered, it is sampled in a further interrupt period and the host does not lose any SDIO

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interrupt from the card.

- Read/Write Resume – The CIU treats the resume command as a normal data transfer command. SDIO interrupts during the resume command are handled similarly to other data commands. According to the SDIO specification, for the normal data command the interrupt period ends after the command end bit of the data command; for the resume command, it ends after the response end bit. In the case of the resume command, the Controller stops the interrupt sampling period after the resume command end bit, instead of stopping after the response end bit of the resume command.
- Suspend during read transfer – If the read data transfer is suspended by the host, the host sets the abort_read_data bit in the controller to reset the data state machine. In the CIU, the SDIO interrupts are handled such that the interrupt sampling starts after the abort_read_data bit is set by the host. In this case the controller does not sample SDIO interrupts between the period from response of the suspend command to setting the abort_read_data bit, and starts sampling after setting the abort_read_data bit.

5. Clock Control

The clock control block provides different clock frequencies required for SD/MMC cards. The cclk_in signal is the source clock (cclk_in >= card max operating frequency) for clock divider of the clock control block. This source clock (cclk_in) is used to generate different card clock frequencies (cclk_out). The card clock can have different clock frequencies, since the card can be a low-speed card or a full-speed card. The Host Controller provides one clock signal (cclk_out).

The clock frequency of a card depends on the following clock control registers:

- Clock Divider register – Internal clock dividers are used to generate different clock frequencies required for card. The division factor for each clock divider can be programmed by writing to the Clock Divider register. The clock divider is an 8-bit value that provides a clock division factor from 1 to 510; a value of 0 represents a clock-divider bypass, a value of 1 represents a divide by 2, a value of 2 represents a divide by 4, and so on.
- Clock Control register – cclk_out can be enabled or disabled for each card under the following conditions:
  - clk_enable – cclk_out for a card is enabled if the clk_enable bit for a card in the Clock Control register is programmed (set to 1) or disabled (set to 0).
  - Low-power mode – Low-power mode of a card can be enabled by setting the low-power mode bit of the Clock Control register to 1. If low-power mode is enabled to save card power, the cclk_out is disabled when the card is idle for at least 8 card clock cycles. It is enabled when a new command is loaded and the command path goes to a non-idle state.

Additionally, cclk_out is disabled when an internal FIFO is full – card read (no more data can be received from card) – or when the FIFO is empty – card write (no data is available for transmission). This helps to avoid FIFO overrun and underrun conditions. It is used by the command and data path to qualify cclk_in for driving outputs and sampling inputs at the programmed clock frequency for the selected card, according to the Clock Divider and Clock Source register values.

Under the following conditions, the card clock is stopped or disabled, along with the active clk_en, for the selected card:

- Clock can be disabled by writing to Clock Enable register (clk_en bit = 1).

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- If low-power mode is selected and card is idle, or not selected for 8 clocks.
- FIFO is full and data path cannot accept more data from the card and data transfer is incomplete –to avoid FIFO overrun.
- FIFO is empty and data path cannot transmit more data to the card and data transfer is incomplete – to avoid FIFO underrun.

6. Error Detection

- Response
  - Response timeout – Response expected with response start bit is not received within programmed number of clocks in timeout register.
  - Response CRC error – Response is expected and check response CRC requested; response CRC7 does not match with the internally-generated CRC7.
  - Response error – Response transmission bit is not 0, command index does not match with the command index of the send command, or response end bit is not 1.
- Data transmit
  - No CRC status – During a write data transfer, if the CRC status start bit is not received two clocks after the end bit of the data block is sent out, the data path does the following:
    - Signals no CRC status error to the BIU
    - Terminates further data transfer
    - Signals data transfer done to the BIU
  - Negative CRC – If the CRC status received after the write data block is negative (that is, not 010), a data CRC error is signaled to the BIU and further data transfer is continued.
  - Data starvation due to empty FIFO – If the FIFO becomes empty during a write data transmission, or if the card clock is stopped and the FIFO remains empty for data timeout clocks, then a data-starvation error is signaled to the BIU and the data path continues to wait for data in the FIFO.
- Data receive
  - Data timeout – During a read-data transfer, if the data start bit is not received before the number of clocks that were programmed in the timeout register, the data path does the following:
    - Signals data-timeout error to the BIU
    - Terminates further data transfer
    - Signals data transfer done to BIU
  - Data start bit error – During a 4-bit or 8-bit read-data transfer, if the all-bit data line does not have a start bit, the data path signals a data start bit error to the BIU and waits for a data timeout, after which it signals that the data transfer is done.
  - Data CRC error – During a read-data-block transfer, if the CRC16 received does not match with the internally generated CRC16, the data path signals a data CRC error to the BIU and continues further data transfer.
  - Data end-bit error – During a read-data transfer, if the end bit of the received data is not 1, the data path signals an end-bit error to the BIU, terminates further data transfer, and signals to the BIU that the data transfer is done.
  - Data starvation due to FIFO full – During a read data transmission and when the FIFO becomes full, the card clock is stopped. If the FIFO remains full for data timeout clocks, a data starvation error is signaled to the BIU (Data Starvation by Host Timeout bit is

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set in RINTSTS Register) and the data path continues to wait for the FIFO to start to empty.

12.3.3 Internal Direct Memory Access Controller (IDMAC)

The Internal Direct Memory Access Controller (IDMAC) has a Control and Status Register (CSR) and a single Transmit/Receive engine, which transfers data from host memory to the device port and vice versa. The controller utilizes a descriptor to efficiently move data from source to destination with minimal Host CPU intervention. You can program the controller to interrupt the Host CPU in situations such as data Transmit and Receive transfer completion from the card, as well as other normal or error conditions.

The IDMAC and the Host driver communicate through a single data structure. CSR addresses 0x80 to 0x98 are reserved for host programming.

The IDMAC transfers the data received from the card to the Data Buffer in the Host memory, and it transfers Transmit data from the Data Buffer in the Host memory to the FIFO. Descriptors that reside in the Host memory act as pointers to these buffers.

A data buffer resides in physical memory space of the Host and consists of complete data or partial data. Buffers contain only data, while buffer status is maintained in the descriptor. Data chaining refers to data that spans multiple data buffers. However, a single descriptor cannot span multiple data.

A single descriptor is used for both reception and transmission. The base address of the list is written into Descriptor List Base Address Register (DBADDR @0x88). A descriptor list is forward linked. The Last Descriptor can point back to the first entry in order to create a ring structure. The descriptor list resides in the physical memory address space of the Host. Each descriptor can point to a maximum of two data buffers.

1. IDMAC CSR Access

When an IDMAC is introduced, an additional CSR space resides in the IDMAC that controls the IDMAC functionality. The host accesses the new CSR space in addition to the existing control register set in the BIU. The IDMAC CSR primarily contains descriptor information. For a write operation to the CSR, the respective CSR logic of the IDMAC and BIU decodes the address before accepting. For a read operation from the CSR, the appropriate CSR read path is enabled.

You can enable or disable the IDMAC operation by programming bit[25] in the CTRL register of the BIU. This allows the data transfer by accessing the slave interface on the AMBA bus if the IDMAC is present but disabled. When IDMAC is enabled, the FIFO cannot be accessed through the slave interface.

2. Descriptors

- Descriptor structures

The IDMAC uses these types of descriptor structures:

  - Dual-Buffer Structure – The distance between two descriptors is determined by the Skip Length value programmed in the Descriptor Skip Length (DSL) field of the Bus Mode Register (BMOD @0x80).

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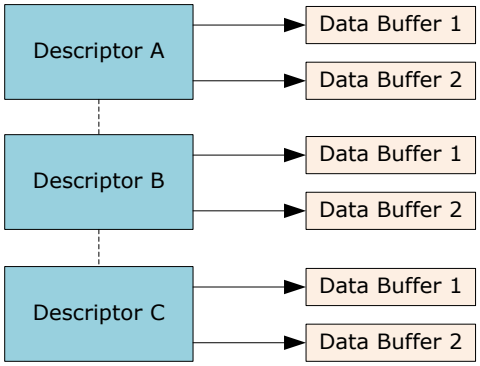

Fig. 12-6 Dual-Buffer Descriptor Structure

- Chain Structure – Each descriptor points to a unique buffer and the next descriptor.

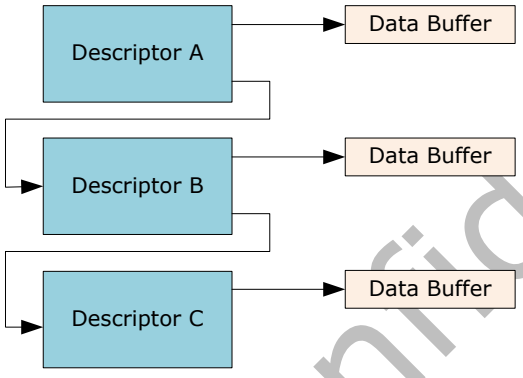

Fig. 12-7 Chain Descriptor Structure

- Descriptor formats

Following figure illustrates the internal formats of a descriptor. The descriptor addresses must be aligned to the bus width used for 32-bit AHB data buses. Each descriptor contains 16 bytes of control and status information. DES0 is a notation used to denote the [31:0] bits, DES1 to denote [63:32] bits, DES2 to denote [95:64] bits, DES3 to denote [127:96] bits.

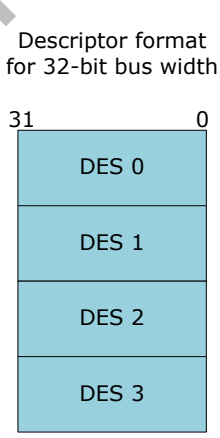

Fig. 12-8 Descriptor Formats for 32-bit AHB Address Bus Width

- The DES0 element in the IDMAC contains control and status information.

Table 12-4 Bits in IDMAC DES0 Element

Bit	Name	Description
31	OWN	When set, this bit indicates that the descriptor is owned by the

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Bit	Name	Description
		IDMAC. When this bit is reset, it indicates that the descriptor is owned by the Host. The IDMAC clears this bit when it completes the data transfer.
30	Card Error Summary (CES)	These error bits indicate the status of the transaction to or from the card. These bits are also present in RINTSTS Indicates the logical OR of the following bits: ● EBE: End Bit Error ● RTO: Response Time out ● RCRC: Response CRC ● SBE: Start Bit Error ● DRTO: Data Read Timeout ● DCRC: Data CRC for Receive ● RE: Response Error
29:6	Reserved	-
5	End of Ring (ER)	When set, this bit indicates that the descriptor list reached its final descriptor. The IDMAC returns to the base address of the list, creating a Descriptor Ring. This is meaningful for only a dual-buffer descriptor structure.
4	Second Address Chained (CH)	When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When this bit is set, BS2 (DES1[25:13]) should be all zeros.
3	First Descriptor (FS)	When set, this bit indicates that this descriptor contains the first buffer of the data. If the size of the first buffer is 0, next Descriptor contains the beginning of the data.
2	Last Descriptor (LD)	This bit is associated with the last block of a DMA transfer. When set, the bit indicates that the buffers pointed to by this descriptor are the last buffers of the data. After this descriptor is completed, the remaining byte count is 0. In other words, after the descriptor with the LD bit set is completed, the remaining byte count should be 0.
1	Disable Interrupt on Completion (DIC)	When set, this bit will prevent the setting of the TI/RI bit of the IDMAC Status Register (IDSTS) for the data that ends in the buffer pointed to by this descriptor.
0	Reserved	-

■ The DES1 element contains the buffer size.

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Table 12-5 Bits in IDMAC DES1 Element

Bit	Name	Description
31:26	Reserved	-
25:13	Buffer 2 Size (BS2)	These bits indicate the second data buffer byte size. The buffer size must be a multiple of 2, 4, or 8, depending upon the bus widths—16, 32, and 64 respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and proceeds to the next buffer in case of a dual-buffer structure. This field is not valid for chain structure; that is, if DES0[4] is set.
12:0	Buffer 1 Size (BS1)	Indicates the data buffer byte size, which must be a multiple of 2, 4, or 8 bytes, depending upon the bus widths—16, 32, and 64, respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. This field should not be zero. Note: If there is only one buffer to be programmed, you need to use only the Buffer 1, and not Buffer 2.

- The DES2 element contains the address pointer to the data buffer.

Table 12-6 Bits in IDMAC DES2 Element

Bit	Name	Description
31:26	Reserved	
25:13	Buffer 2 Size (BS2)	These bits indicate the second data buffer byte size. The buffer size must be a multiple of 2, 4, or 8, depending upon the bus widths—16, 32, and 64 respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and proceeds to the next buffer in case of a dual-buffer structure. This field is not valid for chain structure; that is, if DES0[4] is set.
12:0	Buffer 1 Size (BS1)	Indicates the data buffer byte size, which must be a multiple of 2, 4, or 8 bytes, depending upon the bus widths—16, 32, and 64, respectively. In the case where the buffer size is not a multiple of 2, 4, or 8, the resulting behavior is undefined. This field should not be zero. Note: If there is only one buffer to be programmed, you need to use only the Buffer 1, and not Buffer 2.

- The DES3 element contains the address pointer to the next descriptor if the present descriptor is not the last descriptor in a chained descriptor structure or the second buffer address for a dual-buffer structure.

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Table 12-7 Bits in IDMAC DES3 Element

Bit	Name	Description
31:0	Buffer Address Pointer 2/ Next Descriptor Address (BAP2)	These bits indicate the physical address of the second buffer when the dual-buffer structure is used. If the Second Address Chained (DES0[4]) bit is set, then this address contains the pointer to the physical memory where the Next Descriptor is present. If this is not the last descriptor, then the Next Descriptor address pointer must be bus-width aligned.

3. Initialization

IDMAC initialization occurs as follows:

1) Write to IDMAC Bus Mode Register—BMOD to set Host bus access parameters.
2) Write to IDMAC Interrupt Enable Register—IDINTEN to mask unnecessary interrupt causes.
3) The software driver creates either the Transmit or the Receive descriptor list. Then it writes to IDMAC Descriptor List Base Address Register (DBADDR), providing the IDMAC with the starting address of the list.
4) The IDMAC engine attempts to acquire descriptors from the descriptor lists.

- Host Bus Burst Access

The IDMAC attempts to execute fixed-length burst transfers on the AHB Master interface if configured using the FB bit of the IDMAC Bus Mode register. The maximum burst length is indicated and limited by the PBL field. The descriptors are always accessed in the maximum possible burst-size for the 16-bytes to be read— 16*8/bus-width.

The IDMAC initiates a data transfer only when sufficient space to accommodate the configured burst is available in the FIFO or the number of bytes to the end of data, when less than the configured burst-length.

The IDMAC indicates the start address and the number of transfers required to the AHB Master Interface. When the AHB Interface is configured for fixed-length bursts, then it transfers data using the best combination of INCR4/8/16 and SINGLE transactions. Otherwise, in no fixed-length bursts, it transfers data using INCR (undefined length) and SINGLE transactions.

- Host Data Buffer Alignment

The Transmit and Receive data buffers in host memory must be aligned, depending on the data width.

- Buffer Size Calculations

The driver knows the amount of data to transmit or receive. For transmitting to the card, the IDMAC transfers the exact number of bytes to the FIFO, indicated by the buffer size field of DES1.

If a descriptor is not marked as last-LS bit of DES0-then the corresponding buffer(s) of the descriptor are full, and the amount of valid data in a buffer is accurately indicated by its buffer size field. If a descriptor is marked as last, then the buffer cannot be full, as indicated by the buffer size in DES1. The driver is aware of the number of locations that are valid in this case.

- Transmission

IDMAC transmission occurs as follows:

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1) The Host sets up the elements (DES0-DES3) for transmission and sets the OWN bit (DES0[31]). The Host also prepares the data buffer.
2) The Host programs the write data command in the CMD register in BIU.
3) The Host will also program the required transmit threshold level (TX_WMark field in FIFOTH register).
4) The IDMAC determines that a write data transfer needs to be done as a consequence of step 2.
5) The IDMAC engine fetches the descriptor and checks the OWN bit. If the OWN bit is not set, it means that the host owns the descriptor. In this case the IDMAC enters suspend state and asserts the Descriptor Unable interrupt in the IDSTS register. In such a case, the host needs to release the IDMAC by writing any value to the poll demand register.
6) It will then wait for Command Done (CD) bit and no errors from BIU which indicates that a transfer can be done.
7) The IDMAC engine will now wait for a DMA interface request from BIU. This request will be generated based on the programmed transmit threshold value. For the last bytes of data which can't be accessed using a burst, SINGLE transfers are performed on AHB Master Interface.
8) The IDMAC fetches the Transmit data from the data buffer in the Host memory and transfers to the FIFO for transmission to card.
9) When data spans across multiple descriptors, the IDMAC will fetch the next descriptor and continue with its operation with the next descriptor. The Last Descriptor bit in the descriptor indicates whether the data spans multiple descriptors or not.
10) When data transmission is complete, status information is updated in IDSTS register by setting Transmit Interrupt, if enabled. Also, the OWN bit is cleared by the IDMAC by performing a write transaction to DES0.

- Reception

IDMAC reception occurs as follows:

1) The Host sets up the element (DES0-DES3) for reception, sets the OWN (DES0[31]).
2) The Host programs the read data command in the CMD register in BIU.
3) The Host will program the required receive threshold level (RX_WMark field in FIFOTH register).
4) The IDMAC determines that a read data transfer needs to be done as a consequence of step 2.
5) The IDMAC engine fetches the descriptor and checks the OWN bit. If the OWN bit is not set, it means that the host owns the descriptor. In this case the DMA enters suspend state and asserts the Descriptor Unable interrupt in the IDSTS register. In such a case, the host needs to release the IDMAC by writing any value to the poll demand register.
6) It will then wait for Command Done (CD) bit and no errors from BIU which indicates that a transfer can be done.
7) The IDMAC engine will now wait for a DMA interface request from BIU. This request will be generated based on the programmed receive threshold value. For the last bytes of data which can't be accessed using a burst, SINGLE transfers are performed on AHB.
8) The IDMAC fetches the data from the FIFO and transfer to Host memory.
9) When data spans across multiple descriptors, the IDMAC will fetch the next descriptor and continue with its operation with the next descriptor. The Last Descriptor bit in the descriptor indicates whether the data spans multiple descriptors or not.

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10) When data reception is complete, status information is updated in IDSTS register by setting Receive Interrupt, if enabled. Also, the OWN bit is cleared by the IDMAC by performing a write transaction to DES0.

- Interrupts

Interrupts can be generated as a result of various events. IDSTS register contains all the bits that might cause an interrupt. IDINTEN register contains an Enable bit for each of the events that can cause an interrupt.

There are two groups of summary interrupts-Normal and Abnormal-as outlined in IDSTS register. Interrupts are cleared by writing a 1 to the corresponding bit position. When all the enabled interrupts within a group are cleared, the corresponding summary bit is cleared. When both the summary bits are cleared, the interrupt signal dmac_intr_o is de-asserted.

Interrupts are not queued and if the interrupt event occurs before the driver has responded to it, no additional interrupts are generated. For example, Receive Interrupt—IDSTS[1] indicates that one or more data was transferred to the Host buffer.

An interrupt is generated only once for simultaneous, multiple events. The driver must scan IDSTS register for the interrupt cause.

12.4 Register Description

12.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
SDMMC_CTRL	0x0000	W	0x01000000	Control register
SDMMC_PWREN	0x0004	W	0x00000000	Power-enable register
SDMMC_CLKDIV	0x0008	W	0x00000000	Clock-divider register
SDMMC_CLKSRC	0x000c	W	0x00000000	SD clock source register
SDMMC_CLKENA	0x0010	W	0x00000000	Clock-enable register
SDMMC_TMOUT	0x0014	W	0xffffff40	Time-out register
SDMMC_CTYPE	0x0018	W	0x00000000	Card-type register
SDMMC_BLKSIZ	0x001c	W	0x00000200	Block-size register
SDMMC_BYTCNT	0x0020	W	0x00000200	Byte-count register
SDMMC_INTMASK	0x0024	W	0x00000000	Interrupt-mask register
SDMMC_CMDARG	0x0028	W	0x00000000	Command-argument register
SDMMC_CMD	0x002c	W	0x00000000	Command register
SDMMC_RESP0	0x0030	W	0x00000000	Response-0 register
SDMMC_RESP1	0x0034	W	0x00000000	Response-1 register
SDMMC_RESP2	0x0038	W	0x00000000	Response-2 register
SDMMC_RESP3	0x003c	W	0x00000000	Response-3 register
SDMMC_MINTSTS	0x0040	W	0x00000000	Masked interrupt-status register
SDMMC_RINTSTS	0x0044	W	0x00000000	Raw interrupt-status register
SDMMC_STATUS	0x0048	W	0x00000406	Status register
SDMMC_FIFOTH	0x004c	W	0x00000000	FIFO threshold register
SDMMC_CDETECT	0x0050	W	0x00000000	Card-detect register
SDMMC_WRTPRT	0x0054	W	0x00000000	Write-protect register
SDMMC_TCBCNT	0x005c	W	0x00000000	Transferred CIU card byte count

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Name	Offset	Size	Reset Value	Description
SDMMC_TBBCNT	0x0060	W	0x00000000	Transferred host/DMA to/from BIU-FIFO byte count
SDMMC_DEBNCE	0x0064	W	0x00ffffff	Card detect debounce register
SDMMC_USRID	0x0068	W	0x07967797	User ID register
SDMMC_VERID	0x006c	W	0x5342270a	Version ID register
SDMMC_HCON	0x0070	W	0x00000000	Hardware configuration register
SDMMC_UHS_REG	0x0074	W	0x00000000	UHS-1 register
SDMMC_RST_n	0x0078	W	0x00000001	Hardware reset register
SDMMC_BMOD	0x0080	W	0x00000000	Bus mode register
SDMMC_PLDMND	0x0084	W	0x00000000	Poll demand register
SDMMC_DBADDR	0x0088	W	0x00000000	Descriptor list base address register
SDMMC_IDSTS	0x008c	W	0x00000000	Internal DMAC status register
SDMMC_IDINTEN	0x0090	W	0x00000000	Internal DMAC interrupt enable register
SDMMC_DSCADDR	0x0094	W	0x00000000	Current host descriptor address register
SDMMC_BUFADDR	0x0098	W	0x00000000	Current buffer descriptor address register
SDMMC_CARDTHRCTL	0x0100	W	0x00000000	Card read threshold enable register
SDMMC_BACK_END_POWER	0x0104	W	0x00000000	Back-end power register
SDMMC_EMMC_DDR_REG	0x010c	W	0x00000000	eMMC4.5 DDR start bit detection control register
SDMMC_FIFO_BASE	0x0200	W	0x00000000	FIFO base address register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

12.4.2 Detail Register Description

SDMMC_CTRL

Address: Operational Base + offset (0x0000)

Control register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	use_internal_dmac Present only for the Internal DMAC configuration; else, it is reserved. 0: The host performs data transfers through the slave interface 1: Internal DMAC used for data transfe
24:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11	RW	0x0	ceata_device_interrupt_status 0: Interrupts not enabled in CE-ATA device (nIEN = 1 in ATA control register) 1: Interrupts are enabled in CE-ATA device (nIEN = 0 in ATA control register) Software should appropriately write to this bit after power-on reset or any other reset to CE-ATA device. After reset, usually CE-ATA device interrupt is disabled (nIEN = 1). If the host enables CE-ATA device interrupt, then software should set this bit.
10	RW	0x0	send_auto_stop_ccsd 0: Clear bit if Mobile Storage Host Controller does not reset the bit. 1: Send internally generated STOP after sending CCSD to CE-ATA device. NOTE: Always set send_auto_stop_ccsd and send_ccsd bits together send_auto_stop_ccsd should not be set independent of send_ccsd. When set, Mobile Storage Host Controller automatically sends internally- generated STOP command (CMD12) to CE-ATA device. After sending internally-generated STOP command, Auto Command Done (ACD) in RINTSTS is set and generates interrupt to host if Auto Command Done interrupt is not masked. After sending the CCSD, Mobile Storage Host Controller automatically clears send_auto_stop_ccsd bit.
9	RW	0x0	send_ccsd 0: Clear bit if Mobile Storage Host Controller does not reset the bit. 1: Send Command Completion Signal Disable (CCSD) to CE-ATA device When set, Mobile Storage Host Controller sends CCSD to CE-ATA device. Software sets this bit only if current command is expecting CCS (that is, RW_BLK) and interrupts are enabled in CE-ATA device. Once the CCSD pattern is sent to device, Mobile Storage Host Controller automatically clears send_ccsd bit. It also sets Command Done (CD) bit in RINTSTS register and generates interrupt to host if Command Done interrupt is not masked. NOTE: Once send_ccsd bit is set, it takes two card clock cycles to drive the CCSD on the CMD line. Due to this, during the boundary conditions it may happen that CCSD is sent to the CE-ATA device, even if the device signalled CCS

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Bit	Attr	Reset Value	Description
8	RW	0x0	abort_read_data 0: no change 1: after suspend command is issued during read-transfer, software polls card to find when suspend happened. Once suspend occurs, software sets bit to reset data state-machine, which is waiting for next block of data. Bit automatically clears once data state machine resets to idle. Used in SDIO card suspend sequence.
7	RW	0x0	send_irq_response 0: no change 1: send auto IRQ response Bit automatically clears once response is sent. To wait for MMC card interrupts, host issues CMD40, and SDMMC Controller waits for interrupt response from MMC card(s). In meantime, if host wants SDMMC Controller to exit waiting for interrupt state, it can set this bit, at which time SDMMC Controller command state-machine sends CMD40 response on bus and returns to idle state.
6	RW	0x0	read_wait 0: clear read wait 1: assert read wait For sending read-wait to SDIO cards
5	RW	0x0	dma_enable 0: disable DMA transfer mode 1: enable DMA transfer mode Even when DMA mode is enabled, host can still push/pop data into or from FIFO; this should not happen during the normal operation. If there is simultaneous FIFO access from host/DMA, the data coherency is lost. Also, there is no arbitration inside SDMMC Controller to prioritize simultaneous host/DMA access.
4	RW	0x0	int_enable Global interrupt enable/disable bit: 0: disable interrupts 1: enable interrupts The int port is 1 only when this bit is 1 and one or more unmasked interrupts are set.
3	RO	0x0	reserved
2	W1C	0x0	dma_reset 0: no change 1: reset internal DMA interface control logic To reset DMA interface, firmware should set bit to 1. This bit is auto-cleared after two AHB clocks.

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Bit	Attr	Reset Value	Description
1	W1C	0x0	fifo_reset 0: no change 1: reset to data FIFO To reset FIFO pointers To reset FIFO, firmware should set bit to 1. This bit is auto-cleared after completion of reset operation
0	W1C	0x0	controller_reset 0: no change 1: reset SDMMC controller To reset controller, firmware should set bit to 1. This bit is auto-cleared after two AHB and two cclk_in clock cycles. This resets: a. BIU/CIU interface b. CIU and state machines c. abort_read_data, send_irq_response, and read_wait bits of Control register d. start_cmd bit of Command register Does not affect any registers or DMA interface, or FIFO or host interrupts

SDMMC_PWREN

Address: Operational Base + offset (0x0004)

Power-enable register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	power_enable Power on/off switch for the card. Once power is turned on, firmware should wait for regulator/switch ramp-up time before trying to initialize card. 0: power off 1: power on Bit values output to card_power_en port.

SDMMC_CLKDIV

Address: Operational Base + offset (0x0008)

Clock-divider register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	clk_divider0 Clock divider-0 value. Clock division is 2*n. For example, value of 0 means divide by 2*0 = 0 (no division, bypass), value of 1 means divide by 2*1 = 2, value of "ff"means divide by 2*255 = 510, and so on

SDMMC_CLKSRC

Address: Operational Base + offset (0x000c)

SD clock source register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	clk_source Clock divider source for up to 16 SD cards supported. Each card has two bits assigned to it. For example, bits[1:0] assigned for card-0, which maps and internally routes clock divider[3:0] outputs to cclk_out[15:0] pins, depending on bit value. 00: Clock divider 0 The cclk_out is always from clock divider 0, and this register is not implemented.

SDMMC_CLKENA

Address: Operational Base + offset (0x0010)

Clock-enable register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	cclk_low_power Low-power control for SD card clock and MMC card clock supported. 0: non-low-power mode 1: low-power mode; stop clock when card in IDLE (should be normally set to only MMC and SD memory cards; for SDIO cards, if interrupts must be detected, clock should not be stopped).
15:1	RO	0x0	reserved
0	RW	0x0	cclk_enable Clock-enable control for SD card clock and MMC card clock supported. 0: clock disabled 1: clock enabled

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SDMMC_TMOUT

Address: Operational Base + offset (0x0014)

Time-out register

Bit	Attr	Reset Value	Description
31:8	RW	0xffffff	data_timeout Value for card Data Read Timeout; same value also used for Data Starvation by Host timeout. Value is in number of card output clocks cclk_out of selected card. Note: The software timer should be used if the timeout value is in the order of 100 ms. In this case, read data timeout interrupt needs to be disabled.
7:0	RW	0x40	response_timeout Response timeout value. Value is in number of card output clocks –cclk_out.

SDMMC_CTYPE

Address: Operational Base + offset (0x0018)

Card-type register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	card_width_8 Indicates if card is 8-bit: 0: non 8-bit mode 1: 8-bit mode
15:1	RO	0x0	reserved
0	RW	0x0	card_width Indicates if card is 1-bit or 4-bit: 0: 1-bit mode 1: 4-bit mode

SDMMC_BLKSIZ

Address: Operational Base + offset (0x001c)

Block-size register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0200	block_size Block size

SDMMC_BYTCNT

Address: Operational Base + offset (0x0020)

Byte-count register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000200	byte_count Number of bytes to be transferred; should be integer multiple of Block Size for block transfers. For undefined number of byte transfers, byte count should be set to 0. When byte count is set to 0, it is responsibility of host to explicitly send stop/abort command to terminate data transfer.

SDMMC_INTMASK

Address: Operational Base + offset (0x0024)

Interrupt-mask register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	sdio_int_mask Mask SDIO interrupts. When masked, SDIO interrupt detection for that card is disabled. A 0 masks an interrupt, and 1 enables an interrupt.
23:17	RO	0x0	reserved
16	RW	0x0	data_nobusy_int_mask 0: data no busy interrupt not masked 1: data no busy interrupt masked

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	int_mask Bits used to mask unwanted interrupts. Value of 0 masks interrupt; value of 1 enables interrupt. [15]: End-bit error (read)/Write no CRC (EBE) [14]: Auto command done (ACD) [13]: Start-bit error (SBE) [12]: Hardware locked write error (HLE) [11]: FIFO underrun/overrun error (FRUN) [10]: Data starvation-by-host timeout (HTO) /Volt_switch_int [9]: Data read timeout (DRTO) [8]: Response timeout (RTO) [7]: Data CRC error (DCRC) [6]: Response CRC error (RCRC) [5]: Receive FIFO data request (RXDR) [4]: Transmit FIFO data request (TXDR) [3]: Data transfer over (DTO) [2]: Command done (CD) [1]: Response error (RE) [0]: Card detect (CD)

SDMMC_CMDARG

Address: Operational Base + offset (0x0028)

Command-argument register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cmd_arg Value indicates command argument to be passed to card.

SDMMC_CMD

Address: Operational Base + offset (0x002c)

Command register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	start_cmd Start command. Once command is taken by CIU, bit is cleared. When bit is set, host should not attempt to write to any command registers. If write is attempted, hardware lock error is set in raw interrupt register. Once command is sent and response is received from SD_MMC cards, Command Done bit is set in raw interrupt register.
30	RO	0x0	reserved
29	RW	0x0	use_hold_reg Use Hold Register 0: CMD and DATA sent to card bypassing HOLD Register 1: CMD and DATA sent to card through the HOLD Register Note: a. Set to 1'b1 for SDR12 and SDR25 (with non-zero phase-shifted cclk_in_drv); zero phase shift is not allowed in these modes. b. Set to 1'b0 for SDR50, SDR104, and DDR50 (with zero phase-shifted cclk_in_drv). c. Set to 1'b1 for SDR50, SDR104, and DDR50 (with non-zero phase-shifted cclk_in_drv).
28	RW	0x0	volt_switch Voltage switch bit. 0: no voltage switching 1: voltage switching enabled; must be set for CMD11 only
27	RW	0x0	boot_mode Boot Mode. 0: mandatory Boot operation 1: alternate Boot operation
26	RW	0x0	disable_boot Disable Boot. When software sets this bit along with start_cmd, CIU terminates the boot operation. Do NOT set disable_boot and enable_boot together.
25	RW	0x0	expect_boot_ack Expect Boot Acknowledge. When Software sets this bit along with enable_boot, CIU expects a boot acknowledge start pattern of 0-1-0 from the selected card.
24	RW	0x0	enable_boot Enable Boot—this bit should be set only for mandatory boot mode.When Software sets this bit along with start_cmd, CIU starts the boot sequence for the corresponding card by asserting the CMD line low. Do NOT set disable_boot and enable_boot together.

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Bit	Attr	Reset Value	Description
23	RW	0x0	ccs_expected 0: Interrupts are not enabled in CE-ATA device (nIEN = 1 in ATA control register), or command does not expect CCS from device 1: Interrupts are enabled in CE-ATA device (nIEN = 0), and RW_BLK command expects command completion signal from CE-ATA device. If the command expects Command Completion Signal (CCS) from the CE-ATA device, the software should set this control bit. Mobile Storage Host Controller sets Data Transfer Over (DTO) bit in RINTSTS register and generates interrupt to host if Data Transfer Over interrupt is not masked.
22	RW	0x0	read_ceata_device 0: Host is not performing read access (RW_REG or RW_BLK) towards CE-ATA device 1: Host is performing read access (RW_REG or RW_BLK) towards CE-ATA device Software should set this bit to indicate that CE-ATA device is being accessed for read transfer. This bit is used to disable read data timeout indication while performing CE-ATA read transfers. Maximum value of I/O transmission delay can be no less than 10 seconds. Mobile Storage Host Controller should not indicate read data timeout while waiting for data from CE-ATA device.
21	RW	0x0	update_clock_registers_only 0: normal command sequence 1: do not send commands, just update clock register value into card clock domain Following register values transferred into card clock domain: CLKDIV, CLRSRC, CLKENA. Changes card clocks (change frequency, truncate off or on, and set low-frequency mode); provided in order to change clock frequency or stop clock without having to send command to cards. During normal command sequence, when update_clock_registers_only = 0, following control registers are transferred from BIU to CIU: CMD, CMDARG, TMOUT, CTYPE, BLKSIZ, BYTCNT. CIU uses new register values for new command sequence to card. When bit is set, there are no Command Done interrupts because no command is sent to SD_MMC_CEATA cards.
20:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15	RW	0x0	send_initialization 0: do not send initialization sequence (80 clocks of 1) before sending this command 1: send initialization sequence before sending this command After power on, 80 clocks must be sent to card for initialization before sending any commands to card. Bit should be set while sending first command to card so that controller will initialize clocks before sending command to card. This bit should not be set for either of the boot modes (alternate or mandatory).
14	RW	0x0	stop_abort_cmd 0: neither stop nor abort command to stop current data transfer in progress. If abort is sent to function-number currently selected or not in data-transfer mode, then bit should be set to 0. 1: stop or abort command intended to stop current data transfer in progress. When open-ended or predefined data transfer is in progress, and host issues stop or abort command to stop data transfer, bit should be set so that command/data state-machines of CIU can return correctly to idle state. This is also applicable for Boot mode transfers. To Abort boot mode, this bit should be set along with CMD[26] = disable_boot.
13	RW	0x0	wait_prvdata_complete 0: send command at once, even if previous data transfer has not completed 1: wait for previous data transfer completion before sending command The wait_prvdata_complete = 0 option typically used to query status of card during data transfer or to stop current data transfer; card_number should be same as in previous command.
12	RW	0x0	send_auto_stop 0: no stop command sent at end of data transfer 1: send stop command at end of data transfer When set, SDMMC Controller sends stop command to SD_MMC cards at end of data transfer. a. when send_auto_stop bit should be set, since some data transfers do not need explicit stop commands b. open-ended transfers that software should explicitly send to stop command Additionally, when "resume" is sent to resume –suspended memory access of SD-Combo card –bit should be set correctly if suspended data transfer needs send_auto_stop. Don't care if no data expected from card.

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Bit	Attr	Reset Value	Description
11	RW	0x0	transfer_mode 0: block data transfer command 1: stream data transfer command Don't care if no data expected.
10	RW	0x0	wr 0: read from card 1: write to card Don't care if no data expected from card.
9	RW	0x0	data_expected 0: no data transfer expected (read/write) 1: data transfer expected (read/write)
8	RW	0x0	check_response_crc 0: do not check response CRC 1: check response CRC Some of command responses do not return valid CRC bits. Software should disable CRC checks for those commands in order to disable CRC checking by controller
7	RW	0x0	response_length 0: short response expected from card 1: long response expected from card
6	RW	0x0	response_expect 0: no response expected from card 1: response expected from card
5:0	RW	0x00	cmd_index Command index

SDMMC_RESP0

Address: Operational Base + offset (0x0030)

Response-0 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response0 Bit[31:0] of response

SDMMC_RESP1

Address: Operational Base + offset (0x0034)

Response-1 register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response Register represents bit[63:32] of long response. When CIU sends auto-stop command, then response is saved in register. Response for previous command sent by host is still preserved in Response 0 register. Additional auto-stop issued only for data transfer commands, and response type is always "short" for them.

SDMMC_RESP2

Address: Operational Base + offset (0x0038)

Response-2 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response2 Bit[95:64] of long response

SDMMC_RESP3

Address: Operational Base + offset (0x003c)

Response-3 register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	response3 Bit[127:96] of long response

SDMMC_MINTSTS

Address: Operational Base + offset (0x0040)

Masked interrupt-status register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RO	0x0	sdio_interrupt Interrupt from SDIO card; SDIO interrupt for card enabled only if corresponding sdio_int_mask bit is set in Interrupt mask register (mask bit 1 enables interrupt; 0 masks interrupt). 0: no SDIO interrupt from card 1: SDIO interrupt from card

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Bit	Attr	Reset Value	Description
23:17	RO	0x0	reserved
16	RW	0x0	data_nobusy_int_status Data no busy Interrupt Status
15:0	RO	0x0000	int_status Interrupt enabled only if corresponding bit in interrupt mask register is set. [15]: End-bit error (read)/Write no CRC (EBE) [14]: Auto command done (ACD) [13]: Start-bit error (SBE) [12]: Hardware locked write error (HLE) [11]: FIFO underrun/overrun error (FRUN) [10]: Data starvation-by-host timeout (HTO) /Volt_switch_int [9]: Data read timeout (DRTO) [8]: Response timeout (RTO) [7]: Data CRC error (DCRC) [6]: Response CRC error (RCRC) [5]: Receive FIFO data request (RXDR) [4]: Transmit FIFO data request (TXDR) [3]: Data transfer over (DTO) [2]: Command done (CD) [1]: Response error (RE) [0]: Card detect (CD)

SDMMC_RINTSTS

Address: Operational Base + offset (0x0044)

Raw interrupt-status register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RO	0x0	sdio_interrupt Interrupt from SDIO card; Writes to these bits clear them. Value of 1 clears bit and 0 leaves bit intact. 0: no SDIO interrupt from card 1: SDIO interrupt from card
23:17	RO	0x0	reserved
16	RW	0x0	data_nobusy_int_status Data no busy interrupt status

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	int_status Writes to bits clear status bit. Value of 1 clears status bit, and value of 0 leaves bit intact. Bits are logged regardless of interrupt mask status. [15]: End-bit error (read)/Write no CRC (EBE) [14]: Auto command done (ACD) [13]: Start-bit error (SBE) [12]: Hardware locked write error (HLE) [11]: FIFO underrun/overrun error (FRUN) [10]: Data starvation-by-host timeout (HTO) /Volt_switch_int [9]: Data read timeout (DRTO) [8]: Response timeout (RTO) [7]: Data CRC error (DCRC) [6]: Response CRC error (RCRC) [5]: Receive FIFO data request (RXDR) [4]: Transmit FIFO data request (TXDR) [3]: Data transfer over (DTO) [2]: Command done (CD) [1]: Response error (RE) [0]: Card detect (CD)

SDMMC_STATUS

Address: Operational Base + offset (0x0048)

Status register

Bit	Attr	Reset Value	Description
31	RO	0x0	dma_req DMA request signal state
30	RO	0x0	dma_ack DMA acknowledge signal state
29:17	RO	0x0000	fifo_count Number of filled locations in FIFO
16:11	RO	0x00	response_index Index of previous response, including any auto-stop sent by core
10	RO	0x1	data_state_mc_busy Data transmit or receive state-machine is busy
9	RO	0x0	data_busy Inverted version of raw selected card_data[0] 0: card data not busy 1: card data busy default value is 1 or 0 depending on cdata_in

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Bit	Attr	Reset Value	Description
8	RO	0x0	data_3_status Raw selected card_data[3]; checks whether card is present 0: card not present 1: card present default value is 1 or 0 depending on cdata_in
7:4	RO	0x0	command_fsm_states Command FSM states: 0: idle 1: send init sequence 2: Tx cmd start bit 3: Tx cmd tx bit 4: Tx cmd index + arg 5: Tx cmd crc7 6: Tx cmd end bit 7: Rx resp start bit 8: Rx resp IRQ response 9: Rx resp tx bit 10: Rx resp cmd idx 11: Rx resp data 12: Rx resp crc7 13: Rx resp end bit 14: Cmd path wait NCC 15: Wait; CMD-to-response turnaround The command FSM state is represented using 19 bits. The STATUS Register[7:4] has 4 bits to represent the command FSM states. Using these 4 bits, only 16 states can be represented. Thus three states cannot be represented in the STATUS[7:4] register. The three states that are not represented in the STATUS Register[7:4] are: a. Bit 16 –Wait for CCS b. Bit 17 –Send CCSD c. Bit 18 –Boot Mode Due to this, while command FSM is in "Wait for CCS state" or "Send CCSD" or "Boot Mode", the Status register indicates status as 0 for the bit field [7:4].
3	RO	0x0	fifo_full FIFO is full status
2	RO	0x1	fifo_empty FIFO is empty status
1	RO	0x1	fifo_tx_watermark FIFO reached Transmit watermark level; not qualified with data transfer
0	RO	0x0	fifo_rx_watermark FIFO reached Receive watermark level; not qualified with data transfer

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SDMMC_FIFOTH

Address: Operational Base + offset (0x004c)

FIFO threshold register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:28	RW	0x0	dma_mutiple_transaction_size Burst size of multiple transaction; should be programmed same as DMA controller multiple-transaction-size SRC/DEST_MSIZE. 0: 1 transfers 1: 4 2: 8 3: 16 4: 32 5: 64 6: 128 7: 256 The unit for transfer is the H_DATA_WIDTH parameter. A single transfer (dw_dma_single assertion in case of Non DW DMA interface) would be signalled based on this value. Value should be sub-multiple of (RX_WMark + 1)* (F_DATA_WIDTH/H_DATA_WIDTH) and (FIFO_DEPTH - TX_WMark)* (F_DATA_WIDTH/ H_DATA_WIDTH) For example, if FIFO_DEPTH = 16, FDATA_WIDTH == H_DATA_WIDTH Allowed combinations for MSize and TX_WMark are: MSize = 1, TX_WMARK = 1-15 MSize = 4, TX_WMark = 8 MSize = 4, TX_WMark = 4 MSize = 4, TX_WMark = 12 MSize = 8, TX_WMark = 8 MSize = 8, TX_WMark = 4 Allowed combinations for MSize and RX_WMark are: MSize = 1, RX_WMARK = 0-14 MSize = 4, RX_WMark = 3 MSize = 4, RX_WMark = 7 MSize = 4, RX_WMark = 11 MSize = 8, RX_WMark = 7 Recommended: MSize = 8, TX_WMark = 8, RX_WMark = 7

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Bit	Attr	Reset Value	Description
27:16	RW	0x000	rx_wmark FIFO threshold watermark level when receiving data to card. When FIFO data count reaches greater than this number, DMA/FIFO request is raised. During end of packet, request is generated regardless of threshold programming in order to complete any remaining data. In non-DMA mode, when receiver FIFO threshold (RXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, interrupt is not generated if threshold programming is larger than any remaining data. It is responsibility of host to read remaining bytes on seeing Data Transfer Done interrupt. In DMA mode, at end of packet, even if remaining bytes are less than threshold, DMA request does single transfers to flush out any remaining bytes before Data Transfer Done interrupt is set. 12 bits-1 bit less than FIFO-count of status register, which is 13 bits. Limitation: RX_WMark <= FIFO_DEPTH-2 Recommended: (FIFO_DEPTH/2) - 1; (means greater than (FIFO_DEPTH/2) - 1) *NOTE: In DMA mode during CCS time-out, the DMA does not generate the request at the end of packet, even if remaining bytes are less than threshold. In this case, there will be some data left in the FIFO. It is the responsibility of the application to reset the FIFO after the CCS timeout.*
15:12	RO	0x0	reserved
11:0	RW	0x000	tx_wmark FIFO threshold watermark level when transmitting data to card. When FIFO data count is less than or equal to this number, DMA/FIFO request is raised. If Interrupt is enabled, then interrupt occurs. During end of packet, request or interrupt is generated, regardless of threshold programming. In non-DMA mode, when transmit FIFO threshold (TXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, on last interrupt, host is responsible for filling FIFO with only required remaining bytes (not before FIFO is full or after CIU completes data transfers, because FIFO may not be empty). In DMA mode, at end of packet, if last transfer is less than burst size, DMA controller does single cycles until required bytes are transferred. 12 bits -1 bit less than FIFO-count of status register, which is 13 bits. Limitation: TX_WMark >= 1; Recommended: FIFO_DEPTH/2; (means less than or equal to FIFO_DEPTH/2)

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SDMMC_CDETECT

Address: Operational Base + offset (0x0050)

Card-detect register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	card_detect_n Value on card_detect_n input ports; read-only bits. 0 represents presence of card.

SDMMC_WRTPRT

Address: Operational Base + offset (0x0054)

Write-protect register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	write_protect Value on card_write_prt input port. 1 represents write protection.

SDMMC_TCBCNT

Address: Operational Base + offset (0x005c)

Transferred CIU card byte count

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	trans_card_byte_count Number of bytes transferred by CIU unit to card. In 32-bit or 64-bit AMBA data-bus-width modes, register should be accessed in full to avoid read-coherency problems. In 16-bit AMBA data-bus-width mode, internal 16-bit coherency register is implemented. User should first read lower 16 bits and then higher 16 bits. When reading lower 16 bits, higher 16 bits of counter are stored in temporary register. When higher 16 bits are read, data from temporary register is supplied. Both TCBCNT and TBBCNT share same coherency register. When AREA_OPTIMIZED parameter is 1, register should be read only after data transfer completes; during data transfer, register returns 0.

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SDMMC_TBBCNT

Address: Operational Base + offset (0x0060)

Transferred host/DMA to/from BIU-FIFO byte count

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	trans_fifo_byte_count Number of bytes transferred between Host/DMA memory and BIU FIFO. In 32-bit or 64-bit AMBA data-bus-width modes, register should be accessed in full to avoid read-coherency problems. In 16-bit AMBA data-bus-width mode, internal 16-bit coherency register is implemented. User should first read lower 16 bits and then higher 16 bits. When reading lower 16 bits, higher 16 bits of counter are stored in temporary register. When higher 16 bits are read, data from temporary register is supplied. Both TCBCNT and TBBCNT share same coherency register.

SDMMC_DEBNCE

Address: Operational Base + offset (0x0064)

Card detect debounce register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0xffffff	debounce_count Number of host clocks (clk) used by debounce filter logic; typical debounce time is 5-25 ms.

SDMMC_USRID

Address: Operational Base + offset (0x0068)

User ID register

Bit	Attr	Reset Value	Description
31:0	RW	0x07967797	usrid User identification register. The default value is determined by Configuration Value.

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SDMMC_VERID

Address: Operational Base + offset (0x006c)

Version ID register

Bit	Attr	Reset Value	Description
31:0	RO	0x5342270a	verid Version identification register; register value is hard-wired. Can be read by firmware to support different versions of core.

SDMMC_HCON

Address: Operational Base + offset (0x0070)

Hardware configuration register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HCON Configuration Dependent. Hardware configurations selected by user before synthesizing core. Register values can be used to develop configuration-independent software drivers. [0]: CARD_TYPE 0: MMC_ONLY 1: SD_MMC [5:1]: NUM_CARDS - 1 [6]: H_BUS_TYPE 0: APB 1: AHB [9:7]: H_DATA_WIDTH 0: 16 bits 1: 32 bits 2: 64 bits others: reserved [15:10]: H_ADDR_WIDTH 0 to 7: reserved 8: 9 bits 9: 10 bits … 31: 32 bits 32 to 63: reserved [17:16]: DMA_INTERFACE 0: none 1: DMA 1 2: DMA 2 3: DMA 3 [20:18]: GE_DMA_DATA_WIDTH 0: 16 bits 1: 32 bits 2: 64 bits others: reserved [21]: FIFO_RAM_INSIDE 0: outside 1: inside [22]: IMPLEMENT_HOLD_REG 0: no hold register 1: hold register [23]: SET_CLK_FALSE_PATH 0: no false path 1: false path set [25:24]: NUM_CLK_DIVIDER-1 [26]: AREA_OPTIMIZED 0: no area optimization 1: Area optimization

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SDMMC_UHS_REG

Address: Operational Base + offset (0x0074)

UHS-1 register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	ddr_reg DDR mode. Determines the voltage fed to the buffers by an external voltage regulator. 0: non-DDR mode 1: DDR mode UHS_REG [16] should be set for card.
15:0	RO	0x0	reserved

SDMMC_RST_n

Address: Operational Base + offset (0x0078)

Hardware reset register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	card_reset Hardware reset. 0: active mode 1: reset These bits cause the cards to enter pre-idle state, which requires them to be re-initialized. CARD_RESET[0] should be set to 1'b1 to reset card.

SDMMC_BMOD

Address: Operational Base + offset (0x0080)

Bus mode register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10:8	RO	0x0	PBL Programmable Burst Length. These bits indicate the maximum number of beats to be performed in one IDMAC transaction. The IDMAC will always attempt to burst as specified in PBL each time it starts a Burst transfer on the host bus. The permissible values are 1, 4, 8, 16, 32, 64, 128 and 256. This value is the mirror of MSIZE of FIFOTH register. In order to change this value, write the required value to FIFOTH register. This is an encode value as follows. 0: 1 transfers 1: 4 transfers 2: 8 transfers 3: 16 transfers 4: 32 transfers 5: 64 transfers 6: 128 transfers 7: 256 transfers Transfer unit is either 16, 32, or 64 bits, based on HDATA_WIDTH. PBL is a read-only value and is applicable only for Data Access; it does not apply to descriptor accesses.
7	RW	0x0	DE IDMAC Enable. When set, the IDMAC is enabled.
6:2	RW	0x00	DSL Descriptor Skip Length. Specifies the number of HWord/Word/Dword (depending on 16/32/64-bit bus) to skip between two unchained descriptors. This is applicable only for dual buffer structure.
1	RW	0x0	FB Fixed Burst. Controls whether the AHB Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 or INCR16 during start of normal burst transfers. When reset, the AHB will use SINGLE and INCR burst transfer operations.
0	RW	0x0	SWR Software Reset. When set, the DMA Controller resets all its internal registers. It is automatically cleared after 1 clock cycle.

SDMMC_PLDMND

Address: Operational Base + offset (0x0084)

Poll demand register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	PD Poll Demand. If the OWN bit of a descriptor is not set, the FSM goes to the Suspend state. The host needs to write any value into this register for the IDMAC FSM to resume normal descriptor fetch operation. This is a write only register.

SDMMC_DBADDR

Address: Operational Base + offset (0x0088)

Descriptor list base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	SDL Start of Descriptor List. Contains the base address of the First Descriptor. The LSB bits [0/1/2:0] for 16/32/64-bit bus-width) are ignored and taken as all-zero by the IDMAC internally. Hence these LSB bits are read-only.

SDMMC_IDSTS

Address: Operational Base + offset (0x008c)

Internal DMAC status register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:13	RO	0x0	FSM DMAC FSM present state. 0: DMA_IDLE 1: DMA_SUSPEND 2: DESC_RD 3: DESC_CHK 4: DMA_RD_REQ_WAI 5: DMA_WR_REQ_WAI 6: DMA_RD 7: DMA_WR 8: DESC_CLOSE

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Bit	Attr	Reset Value	Description
12:10	RO	0x0	EB Error Bits. Indicates the type of error that caused a Bus Error. Valid only with atal Bus Error bit—IDSTS[2] (IDSTS64[2], in case of 64-bit address configuration) set. This field does not generate an interrupt. 1: Host Abort received during transmission 2: Host Abort received during reception Others: Reserved
9	RW	0x0	AIS Abnormal Interrupt Summary. Logical OR of the following: IDSTS[2] Fatal Bus Interrupt IDSTS[4] DU bit Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared. Writing a 1 clears this bit.
8	RW	0x0	NIS Normal Interrupt Summary. Logical OR of the following: IDSTS[0] Transmit Interrupt IDSTS[1] Receive Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes NIS to be set is cleared. Writing a 1 clears this bit.
7:6	RO	0x0	reserved
5	RW	0x0	CES Card Error Summary. Indicates the status of the transaction to/from the card; also present in RINTSTS. Indicates the logical OR of the following bits: EBE: End Bit Error RTO: Response Timeout/Boot Ack Timeout RCRC: Response CRC SBE: Start Bit Error DRTO: Data Read Timeout/BDS timeout DCRC: Data CRC for Receive RE: Response Error Writing a 1 clears this bit. The abort condition of the IDMAC depends on the setting of this CES bit. If the CES bit is enabled, then the IDMAC aborts on a "response error"; however, it will not abort if the CES bit is cleared.
4	RW	0x0	DU Descriptor Unavailable Interrupt. This bit is set when the descriptor is unavailable due to OWN bit = 0 (DES0[31] =0). Writing a 1 clears this bit.
3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x0	FBE Fatal Bus Error Interrupt. Indicates that a Bus Error occurred (IDSTS[12:10]) (IDSTS64[12:10], in case of 64-bit address configuration). When this bit is set, the DMA disables all its bus accesses. Writing a 1 clears this bit.
1	RW	0x0	RI Receive Interrupt. Indicates the completion of data reception for a descriptor. Writing a 1 clears this bit.
0	RW	0x0	TI Transmit Interrupt. Indicates that data transmission is finished for a descriptor. Writing 1 clears this bit.

SDMMC_IDINTEN

Address: Operational Base + offset (0x0090)

Internal DMAC interrupt enable register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	AI Abnormal Interrupt Summary Enable. When set, an abnormal interrupt is enabled. This bit enables the following bits: IDINTEN[2] Fatal Bus Error Interrupt IDINTEN[4] DU Interrupt
8	RW	0x0	NI Normal Interrupt Summary Enable. When set, a normal interrupt is enabled. When reset, a normal interrupt is disabled. This bit enables the following bits: IDINTEN[0] Transmit Interrupt IDINTEN[1] Receive Interrupt
7:6	RO	0x0	reserved
5	RW	0x0	CES Card Error summary Interrupt Enable. When set, it enables the Card Interrupt summary.
4	RW	0x0	DU Descriptor Unavailable Interrupt. When set along with Abnormal Interrupt Summary Enable, the DU interrupt is enabled.
3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x0	FBE Fatal Bus Error Enable. When set with Abnormal Interrupt Summary Enable, the Fatal Bus Error Interrupt is enabled. When reset, Fatal Bus Error Enable Interrupt is disabled.
1	RW	0x0	RI Receive Interrupt Enable. When set with Normal Interrupt Summary Enable, Receive Interrupt is enabled. When reset, Receive Interrupt is disabled.
0	RW	0x0	TI Transmit Interrupt Enable. When set with Normal Interrupt Summary Enable, Transmit Interrupt is enabled. When reset, Transmit Interrupt is disabled.

SDMMC_DSCADDR

Address: Operational Base + offset (0x0094)

Current host descriptor address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HDA Host Descriptor Address Pointer. Cleared on reset. Pointer updated by IDMAC during operation. This register points to the start address of the current descriptor read by the IDMAC.

SDMMC_BUFADDR

Address: Operational Base + offset (0x0098)

Current buffer descriptor address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HBA Host Buffer Address Pointer. Cleared on Reset. Pointer updated by IDMAC during operation. This register points to the current Data Buffer Address being accessed by the IDMAC.

SDMMC_CARDTHRCTL

Address: Operational Base + offset (0x0100)

Card read threshold enable register

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Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x000	CardRdThreshold Card Read Threshold size
15:2	RO	0x0	reserved
1	RW	0x0	BsyClrIntEn Busy Clear Interrupt generation: 0: Busy Clear Interrupt disabled 1: Busy Clear Interrupt enabled Note: The application can disable this feature if it does not want to wait for a Busy Clear Interrupt. For example, in a multi-card scenario, the application can switch to the other card without waiting for a busy to be completed. In such cases, the application can use the polling method to determine the status of busy. By default this feature is disabled and backward-compatible to the legacy drivers where polling is used.
0	RW	0x0	CardRdThrEn Card Read Threshold Enable. 0: Card Read Threshold disabled 1: Card Read Threshold enabled. Host Controller initiates Read Transfer only if CardRdThreshold amount of space is available in receive FIFO.

SDMMC_BACK_END_POWER

Address: Operational Base + offset (0x0104)

Back-end power register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	back_end_power Back end power 0: Off; Reset 1: Back-end Power supplied to card application

SDMMC_EMMC_DDR_REG

Address: Operational Base + offset (0x010c)

eMMC4.5 DDR start bit detection control register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	HALF_START_BIT Control for start bit detection mechanism inside Mobile Storage Host Controller based on duration of start bit; each bit refers to one slot. For eMMC 4.5, start bit can be: 0: Full cycle (HALF_START_BIT = 0) 1: Less than one full cycle (HALF_START_BIT = 1) Set HALF_START_BIT=1 for eMMC 4.5 and above; set to 0 for SD applications.

SDMMC_FIFO_BASE

Address: Operational Base + offset (0x0200)

FIFO base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fifo_base_addr FIFO base address

12.5 Interface Description

The interface and IOMUX setting for SDMMC, SDMMC_EXT, SDIO, EMMC are shown as follows.

Table 12-8 SDMMC Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
sdmmc_cclk	O	IO_SDMMC0clkout_TESTclk0_GPIO1A6vccio3	GRF_GPIO1A_IOMUX[13:12]=2’b01
sdmmc_ccmd	I/O	IO_SDMMC0cmd_GPIO1A4vccio3	GRF_GPIO1A_IOMUX[9:8]=2’b01
sdmmc_cdata0	I/O	IO_SDMMC0d0_UART2DBGtxm0_GPIO1A0vccio3	GRF_GPIO1A_IOMUX[1:0]=2’b01
sdmmc_cdata1	I/O	IO_SDMMC0d1_UART2DBGrxm0_GPIO1A1vccio3	GRF_GPIO1A_IOMUX[3:2]=2’b01
sdmmc_cdata2	I/O	IO_SDMMC0d2_JTAGtck_GPIO1A2vccio3	GRF_GPIO1A_IOMUX[5:4]=2’b01
sdmmc_cdata3	I/O	IO_SDMMC0d3_JTAGtms_GPIO1A3vccio3	GRF_GPIO1A_IOMUX[7:6]=2’b01

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sdmmc_cdetectn	I	IO_SDMMC0detn_GPIO1A5vccio3	GRF_GPIO1A_IOMUX[11:10]=2'b01
sdmmc_wprt	I	IO_SDMMC0wp_GPIO1A7vccio3	GRF_GPIO1A_IOMUX[15:14]=2'b01
sdmmc_pwren	O	IO_SDMMC0pwrenm0_GPIO2A7vccio5	GRF_SOC_CON10[9]=0 && GRF_GPIO2A_IOMUX[15:14]=2'b01
		IO_FEPHYLEDspeed10_FEPHYLEDduplex_SDMMC0pwrenm1_GPIO0D6vccio1	GRF_SOC_CON10[9]=0 && GRF_GPIO0D_IOMUX[13:12]=2'b11

Notes: I=input, O=output, I/O=input/output, bidirectional

Table 12-9 SDMMC_EXT Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
sdmmc_ext_cclk	O	IO_TSPclk_CIFclkin_SDMMC0EXTclkout_SPIrxdm2_USB3PHYdebug3_I2S2sdim1_GPIO3A2vccio6	GRF_GPIO3AL_IOMUX[8:6]=3'b011
sdmmc_ext_ccmd	I/O	IO_TSPvalid_CIFvsync_SDMMC0EXTcmd_SPIclkm2_USB3PHYdebug1_I2S2sclkm1_GPIO3A0vccio6	GRF_GPIO3AL_IOMUX[2:0]=3'b011
sdmmc_ext_cdata0	I/O	IO_TSPd0_CIFda0_SDMMC0EXTd0_UART1tx_USB3PHYdebug4_GPIO3A4vccio6	GRF_GPIO3AL_IOMUX[14:12]=3'b011
sdmmc_ext_cdata1	I/O	IO_TSPd1_CIFdata1_SDMMC0EXTd1_UART1rtsn_USB3PHYdebug5_GPIO3A5vccio6	GRF_GPIO3AH_IOMUX[2:0]=3'b011
sdmmc_ext_cdata2	I/O	IO_TSPd2_CIFdata2_SDMMC0EXTd2_UART1rx_USB3PHYdebug6_GPIO3A6vccio6	GRF_GPIO3AH_IOMUX[5:3]=3'b011
sdmmc_ext_cdata3	I/O	IO_TSPd3_CIFdata3_SDMMC0EXTd3_UART1ctsn_USB3PHYdebug7_GPIO3A7vccio6	GRF_GPIO3AH_IOMUX[8:6]=3'b011
sdmmc_ext_wprt	I	IO_TSPsyncm0_CIFclkoutm0_SDMMC0EXTwp_GPIO3A3vccio6	GRF_GPIO3AL_IOMUX[11:9]=3'b011

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sdmmc_ext_cdetectn	I	IO_TSPfail_CIFhref_SDMMC0EXTdet_SPItxdm2_USB3PHYdebug2_I2S2sdom1_GPIO3A1vccio6	GRF_GPIO3AL_IOMUX[5:3]=3'b011
sdmmc_ext_pwren	O	IO_SDMMC0pwrenm0_GPIO2A7vccio5	GRF_SOC_CON10[9]=1 && GRF_GPIO2A_IOMUX[15:14]=2'b01
		IO_FEPHYLEDspeed10_FEPHYLEDduplex_SDMMC0pwrenm1_GPIO0D6vccio1	GRF_SOC_CON10[9]=1 && GRF_GPIO0D_IOMUX[13:12]=2'b11

Notes: I=input, O=output, I/O=input/output, bidirectional

Table 12-10 SDIO Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
sdio_cclk	O	IO_SDMMC1clkout_GMACtxclkm1_GPIO1B4vccio4	GRF_GPIO1B_IOMUX[9:8]=2'b01
sdio_ccmd	I/O	IO_SDMMC1cmd_GMACrxclkm1_GPIO1B5vccio4	GRF_GPIO1B_IOMUX[11:10]=2'b01
sdio_cdata0	I/O	IO_SDMMC1d0_GMACrxd3m1_GPIO1B6vccio4	GRF_GPIO1B_IOMUX[13:12]=2'b01
sdio_cdata1	I/O	IO_SDMMC1d1_GMACrxd2m1_GPIO1B7vccio4	GRF_GPIO1B_IOMUX[15:14]=2'b01
sdio_cdata2	I/O	IO_SDMMC1d2_GMACtxd3m1_GPIO1C0vccio4	GRF_GPIO1C_IOMUX[1:0]=2'b01
sdio_cdata3	I/O	IO_SDMMC1d3_GMACtxd2m1_GPIO1C1vccio4	GRF_GPIO1C_IOMUX[3:2]=2'b01
sdio_cdetectn	I	IO_SDMMC1detn_GMACmdiom1_PDMfsyncm1_GPIO1C3vccio4	GRF_GPIO1C_IOMUX[7:6]=2'b01
sdio_wprt	I	IO_SDMMC1wp_GMACcolm1_GPIO1C4vccio4	GRF_GPIO1C_IOMUX[9:8]=2'b01
sdio_pwren	O	IO_SDMMC1pwren_GMACcrsm1_GPIO1C2vccio4	GRF_GPIO1C_IOMUX[5:4]=2'b01

Notes: I=input, O=output, I/O=input/output, bidirectional

Table 12-11 EMMC Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting

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Module Pin	Direction	Pad Name	IOMUX Setting
emmc_cclk	O	IO_FLASHrdy_EMMCclkout_GPIO3C5vccio2	GRF_GPIO3C_IOMUX[11:10]=2'b10
emmc_ccmd	I/O	IO_FLASHcs2_EMMCcmd_GPIO3C3vccio2	GRF_GPIO3C_IOMUX[7:6]=2'b10
emmc_cdata0	I/O	IO_FLASHd0_EMMCd0_GPIO0A7vccio2	GRF_GPIO0A_IOMUX[15:14]=2'b10
emmc_cdata1	I/O	IO_FLASHd1_EMMCd1_GPIO2D4vccio2	GRF_GPIO2D_IOMUX[9:8]=2'b10
emmc_cdata2	I/O	IO_FLASHd2_EMMCd2_GPIO2D5vccio2	GRF_GPIO2D_IOMUX[11:10]=2'b10
emmc_cdata3	I/O	IO_FLASHd3_EMMCd3_GPIO2D6vccio2	GRF_GPIO2D_IOMUX[13:12]=2'b10
emmc_cdata4	I/O	IO_FLASHd4_EMMCd4_GPIO2D7vccio2	GRF_GPIO2D_IOMUX[15:14]=2'b10
emmc_cdata5	I/O	IO_FLASHd5_EMMCd5_GPIO3C0vccio2	GRF_GPIO3C_IOMUX[1:0]=2'b10
emmc_cdata6	I/O	IO_FLASHd6_EMMCd6_GPIO3C1vccio2	GRF_GPIO3C_IOMUX[3:2]=2'b10
emmc_cdata7	I/O	IO_FLASHd7_EMMCd7_GPIO3C2vccio2	GRF_GPIO3C_IOMUX[5:4]=2'b10
emmc_pwren	O	IO_FLASHwp_EMMCpwren_GPIO3C6vccio2	GRF_GPIO3C_IOMUX[13:12]=2'b10
emmc_rstn	O	IO_FLASHcs3_EMMCrstn_GPIO3C4vccio2	GRF_GPIO3C_IOMUX[9:8]=2'b10

Notes: I=input, O=output, I/O=input/output, bidirectional

12.6 Application Notes

12.6.1 Card-Detect and Write-Protect Mechanism

Following figure illustrates how the SD/MMC card detection and write-protect signals are connected. Most of the SD/MMC sockets have card-detect pins. When no card is present, card_detect_n is 1 due to the pull-up. When the card is inserted, the card-detect pin is shorted to ground, which makes card_detect_n go to 0. Similarly in SD cards, when the write-protect switch is toward the left, it shorts the write_protect port to ground.

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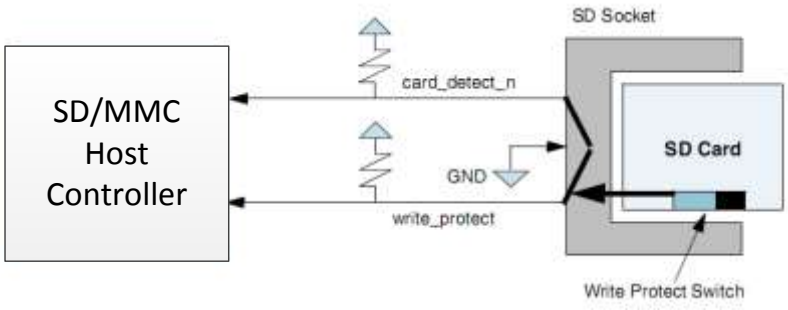

Fig. 12-9 SD/MMC Card-Detect and Write-Protect

12.6.2 SD/MMC Termination Requirement

Following Figure illustrates the SD/MMC termination requirements, which is required to pull up ccmd and cdata lines on the device bus. The recommended specification for pull-up on the ccmd line (Rcmd) is 4.7K - 100K for MMC, and 10K - 100K for an SD. The recommended pull-up on the cdata line (Rdat) is 50K - 100K.

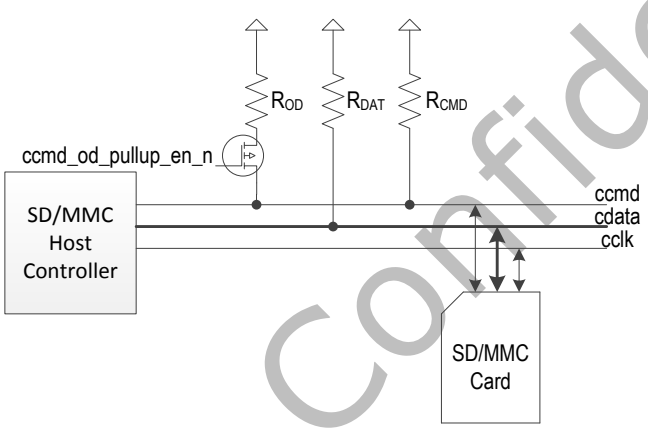

Fig. 12-10 SD/MMC Card Termination

1. Rcmd and Rod Calculation

The SD/MMC card enumeration happens at a very low frequency – 100-400KHz. Since the MMC bus is a shared bus between multiple cards, during enumeration open-drive mode is used to avoid bus conflict. Cards that drive 0 win over cards that drive "z". The pull-up in the command line pulls the bus to 1 when all cards drive "z". During normal data transfer, the host chooses only one card and the card driver switches to push-pull mode.

For example, if enumeration is done at 400KHz and the total bus capacitance is 200 pf, the pull-up needed during enumeration is:

2.2 RC = rise-time = 1/400KHz

R = 1/(2.2 * C * 100KHz)

= 1/(2.2 x 200 x 10**-12 x 400 x 10**3)

= 1/(17.6 x 10**-5)

= 5.68K

The ROD and RCMD should be adjusted in such a way that the effective pull-up is at the maximum 5.68K during enumeration. If there are only a few cards in the bus, a fixed RCMD resister is sufficient and there is no need for an additional ROD pull-up during enumeration. You should also ensure the effective pull-up will not violate the Iol rating of the drivers.

In SD mode, since each card has a separate bus, the capacitance is less, typically in the order

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of 20-30pf (host capacitance + card capacitance + trace + socket capacitance). For example, if enumeration is done at 400KHz and the total bus capacitance is 20pf, the pull-up needed during enumeration is:

2.2 RC = rise-time = 1/400KHz

R = 1/(2.2 * C * 100KHz)

= 1/(2.2 x 20 x 10**-12 x 400 x 10**3)

= 1/(1.76 x 10**-5)

= 56.8K

Therefore, a fixed 56.8K permanent Rcmd is sufficient in SD mode to enumerate the cards.

The driver of the SD/MMC on the "command" port needs to be only a push-pull driver. During enumeration, the SD/MMC emulates an open-drain driver by driving only a 0 or a"z" by controlling the ccmd_out and ccmd_out_en signals.

12.6.3 Software/Hardware Restriction

Before issuing a new data transfer command, the software should ensure that the card is not busy due to any previous data transfer command. Before changing the card clock frequency, the software must ensure that there are no data or command transfers in progress.

If the card is enumerated in SDR50, or DDR50 mode, then the application must program the use_hold_reg bit[29] in the CMD register to 1'b0 (phase shift of cclk_in_drv = 0) or 1'b1 (phase shift of cclk_in_drv>0). If the card is enumerated in SDR12 or SDR25 mode, the application must program the use_hold_reg bit[29] in the CMD register to 1'b1.

This programming should be done for all data transfer commands and non-data commands that are sent to the card. When the use_hold_reg bit is programmed to 1'b0, the Host Controller bypasses the Hold Registers in the transmit path. The value of this bit should not be changed when a Command or Data Transfer is in progress. For more details on using use_hold_reg and the implementation requirements for meeting the Card input hold time, refer to "Recommended Usage" in following table.

Table 12-12 Recommended Usage of use_hold_reg

No.	Speed Mode	use_hold_reg	cclk_in (MHz)	clk_in_drv (MHz)	clk_divider	Phase shift
1	SDR104	1'b0	200	200	0	0
2	SDR104	1'b1	200	200	0	Tunable> 0
3	SDR50	1'b0	100	100	0	0
4	SDR50	1'b1	100	100	0	Tunable> 0
5	DDR50 (8bit)	1'b0	100	100	1	0
6	DDR50 (8bit)	1'b1	100	100	1	Tunable> 0

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No.	Speed Mode	use_hold_reg	cclk_in (MHz)	clk_in_drv (MHz)	clk_divider	Phase shift
7	DDR50 (4bit)	1'b0	50	50	0	0
8	DDR50 (4bit)	1'b1	50	50	0	Tunable> 0
9	SDR25	1'b1	50	50	0	Tunable> 0
10	SDR12	1'b1	50	50	1	Tunable> 0

To avoid glitches in the card clock outputs, the software should use the following steps when changing the card clock frequency:

1) Before disable the clocks, ensure that the card is not busy due to any previous data command. To determine this, check for 0 in bit9 of STATUS register.

2) Update the Clock Enable register to disable all clocks. To ensure completion of any previous command before this update, send a command to the CIU to update the clock registers by setting:

- start_cmd bit
- "update clock registers only" bits
- "wait_previous data complete" bit

Wait for the CIU to take the command by polling for 0 on the start_cmd bit.

3) Set the start_cmd bit to update the Clock Divider and/or Clock Source registers, and send a command to the CIU in order to update the clock registers; wait for the CIU to take the command.

4) Set start_cmd to update the Clock Enable register in order to enable the required clocks and send a command to the CIU to update the clock registers; wait for the CIU to take the command.

In non-DMA mode, while reading from a card, the Data Transfer Over (RINTSTS[3]) interrupt occurs as soon as the data transfer from the card is over. There still could be some data left in the FIFO, and the RX_WMark interrupt may or may not occur, depending on the remaining bytes in the FIFO. Software should read any remaining bytes upon seeing the Data Transfer Over (DTO) interrupt. While using the external DMA interface for reading from a card, the DTO interrupt occurs only after all the data is flushed to memory by the DMA interface unit.

While writing to a card in external DMA mode, if an undefined-length transfer is selected by setting the Byte Count Register to 0, the DMA logic will likely request more data than it will send to the card, since it has no way of knowing at which point the software will stop the transfer. The DMA request stops as soon as the DTO is set by the CIU.

If the software issues a controller_reset command by setting control register bit[0] to 1, all the CIU state machines are reset; the FIFO is not cleared. The DMA sends all remaining bytes to the host. In addition to a card-reset, if a FIFO reset is also issued, then:

- Any pending DMA transfer on the bus completes correctly
- DMA data read is ignored
- Write data is unknown(x)

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Additionally, if dma_reset is also issued, any pending DMA transfer is abruptly terminated. When the DMA is used, the DMA controller channel should also be reset and reprogrammed.

If any of the previous data commands do not properly terminate, then the software should issue the FIFO reset in order to remove any residual data, if any, in the FIFO. After asserting the FIFO reset, you should wait until this bit is cleared.

One data-transfer requirement between the FIFO and host is that the number of transfers should be a multiple of the FIFO data width (32bits). For example, you want to write only 15 bytes to an SD/MMC card (BYTCNT), the host should write 16 bytes to the FIFO or program the DMA to do 16-byte transfers. The software can still program the Byte Count register to only 15, at which point only 15 bytes will be transferred to the card. Similarly, when 15 bytes are read from a card, the host should still read all 16 bytes from the FIFO.

It is recommended that you not change the FIFO threshold register in the middle of data transfers.

12.6.4 Programming Sequence

1. Initialization

Following figure illustrates the initialization flow.

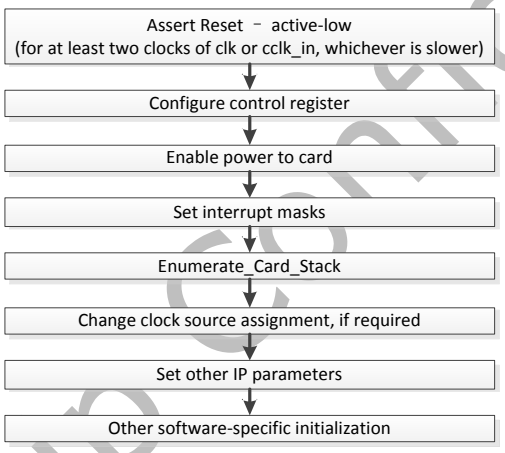

Fig. 12-11 Host Controller Initialization Sequence

Once the power and clocks are stable, reset_n should be asserted(active-low) for at least two clocks of clk or cclk_in, whichever is slower. The reset initializes the registers, ports, FIFO-pointers, DMA interface controls, and state-machines in the design. After power-on reset, the software should do the following:

1) Configure control register – For MMC mode, enable the open-drain pullup by setting enable_OD_pullup(bit24) in the control register.
2) Enable power to cards – Before enabling the power, confirm that the voltage setting to the voltage regulators is correct. Enable power to the connected cards by setting the corresponding bit to 1 in the Power Enable register. Wait for the power ramp-up time.
3) Set masks for interrupts by clearing appropriate bits in the Interrupt Mask register. Set the global int_enable bit of the Control register. It is recommended that you write 0xffff_ffff to the Raw Interrupt register in order to clear any pending interrupts before setting the int_enable bit.
4) Enumerate card stack – Each card is enumerated according to card type; for details, refer to "Enumerated Card Stack". For enumeration, you should restrict the clock frequency to 400KHz.

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5) Changing clock source assignment – set the card frequency using the clock-divider and clock-source registers; for details, refer to "Clock Programming". MMC cards operate at a maximum of 20MHz (at maximum of 52MHz in high-speed mode). SD mode operates at a maximum of 25MHz (at maximum of 50MHz in high-speed mode).
6) Set other parameters, which normally do not need to be changed with every command, with a typical value such as timeout values in cclk_out according to SD/MMC specifications.

- ResponseTimeOut = 0x64
- DataTimeOut = highest of one of the following:
- (10*((TAAC*Fop)+(100*NSAC))
- Host FIFO read/write latency from FIFO empty/full
- Set the debounce value to 25ms(default:0x0fffff) in host clock cycle units in the DEBNCE register.
- FIFO threshold value in bytes in the FIFOTH register.

2. Enumerated Card Stack

The card stack does the following:

- Enumerates all connected cards
- Sets the RCA for the connected cards
- Reads card-specific information
- Stores card-specific information locally

Enumeration depends on the operating mode of the SD/MMC card; the card type is first identified and the appropriate card enumeration routine is called.

1) Check if the card is connected.
2) Clear the card type register to set the card width as a single bit. For the given card number, clear the corresponding bits in the card_type register. Clear the register bit for a 1-bit, 4-bit bus width. For example, for card number=1, clear bit 0 and bit 16 of the card_type register.
3) Set clock frequency to FOD=400KHz, maximum – Program clock divider0 (bits 0-7 in the CLKDIV register) value to one-half of the cclk_in frequency divided by 400KHz. For example, if cclk_in is 20MHz, then the value is 20, 000/(2*400)=25.
4) Identify the card type; that is, SD, MMC, or SDIO.
   a. Send CMD5 first. If a response is received, then the card is SDIO
   b. If not, send CMD8 with the following Argument

      Bit[31:12] = 20'h0 //reserved bits

      Bit[11:8] = 4'b0001 //VHS value

      Bit[7:0] = 8'b10101010 //Preferred Check Pattern by SD2.0

   c. If Response is received the card supports High Capacity SD2.0 then send ACMD41 with the following Argument

      Bit[31] = 1'b0; //Reserved bits

      Bit[30] = 1'b1; //High Capacity Status

      Bit[29:24] = 6'h0; //Reserved bits

      Bit[23:0] = Supported Voltage Range

   d. If Response is received for ACMD41 then the card is SD. Otherwise the card is MMC.

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e. If response is not received for initial CMD8 then card does not support High Capacity SD2.0, then issue CMD0 followed by ACMD41 with the following Argument

Bit[31] = 1'b0; //Reserved bits

Bit[30] = 1'b0; //High Capacity Status

Bit[29:24] = 6'h0; //Reserved bits

Bit[23:0] = Supported Voltage Range

5) Enumerate the card according to the card type.
6) Use a clock source with a frequency = Fod (that is, 400KHz) and use the following enumeration command sequence:

- SD card – Send CMD0, CMD8, ACMD41, CMD2, CMD3.
- MMC – Send CMD0, CMD1, CMD2, CMD3.

3. Power Control

You can implement power control using the following registers, along with external circuitry:

- Control register bits card_voltage_a and card_voltage_b – Status of these bits is reflected at the IO pins. The bits can be used to generate or control the supply voltage that the memory cards require.
- Power enable register – Control power to individual cards.

Programming these two register depends on the implemented external circuitry. While turning on or off the power enable, you should confirm that power supply settings are correct. Power to all cards usually should be disabled while switching off the power.

4. Clock Programming

The Host Controller supports one clock sources. The clock to an individual card can be enabled or disabled. Registers that support this are:

- CLKDIV – Programs individual clock source frequency. CLKDIV limited to 0 or 1 is recommended.
- CLKSRC – Assign clock source for each card.
- CLKENA – Enables or disables clock for individual card and enables low-power mode, which automatically stops the clock to a card when the card is idle for more than 8 clocks.

The Host Controller loads each of these registers only when the start_cmd bit and the Update_clk_regs_only bit in the CMD register are set. When a command is successfully loaded, the Host Controller clears this bit, unless the Host Controller already has another command in the queue, at which point it gives an HLE(Hardware Locked Error).

Software should look for the start_cmd and the Update_clk_regs_only bits, and should also set the wait_prvdata_complete bit to ensure that clock parameters do not change during data transfer. Note that even though start_cmd is set for updating clock registers, the Host Controller does not raise a command_done signal upon command completion.

The following shows how to program these registers:

1) Confirm that no card is engaged in any transaction; if there is a transaction, wait until it finishes.
2) Stop all clocks by writing xxxx0000 to the CLKENA register. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.

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3) Program the CLKDIV and CLKSRC registers, as required. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.
4) Re-enable all clocks by programming the CLKENA register. Set the start_cmd, Update_clk_regs_only, and wait_prvdata_complete bits in the CMD register. Wait until start_cmd is cleared or an HLE is set; in case of an HLE, repeat the command.

5. No-Data Command With or Without Response Sequence

To send any non-data command, the software needs to program the CMD register @0x2C and the CMDARG register @0x28 with appropriate parameters. Using these two registers, the Host Controller forms the command and sends it to the command bus. The Host Controller reflects the errors in the command response through the error bits of the RINTSTS register.

When a response is received – either erroneous or valid – the Host Controller sets the command_done bit in the RINTSTS register. A short response is copied in Response Register0, while along response is copied to all four response registers @0x30, 0x34, 0x38, and 0x3C. The Response3 register bit 31 represents the MSB, and the Response0 register bit 0 represents the LSB of a long response.

For basic commands or non-data commands, follow these steps:

1) Program the Command register @0x28 with the appropriate command argument parameter.
2) Program the Command register @0x2C with the settings in following table.

Table 12-13 Command Settings for No-Data Command

Parameter	Value	Description
Default		
start_cmd	1	-
use_hold_reg	1/0	Choose value based on speed mode being used;ref to "use_hold_reg" on CMD register
update_clk_regs_only	0	No clock parameters update command
data_expected	0	No data command
card number	0	Actual card number(one controller only connect one card, the num is No. 0)
cmd_index	command-index	-
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop

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Parameter	Value	Description
		data transfer, such as CMD12
response_length	0	Can be 1 for R2(long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4, CMD15, and so on
User-selectable		
wait_prvdata_complete	1	Before sending command on command line, host should wait for completion of any data command in process, if any (recommended to always set this bit, unless the current command is to query status or stop data transfer when transfer is in progress)
check_response_crc	1	If host should crosscheck CRC of response received

1) Wait for command acceptance by host. The following happens when the command is loaded into the Host Controller:
   - Host Controller accepts the command for execution and clears the start_cmd bit in the CMD register, unless one command is in process, at which point the Host Controller can load and keep the second command in the buffer.
   - If the Host Controller is unable to load the command – that is, a command is already in progress, a second command is in the buffer, and a third command is attempted – then it generates an HLE (hardware-locked error).
2) Check if there is an HLE.
3) Wait for command execution to complete. After receiving either a response from a card or response timeout, the Host Controller sets the command_done bit in the RINTSTS register. Software can either poll for this bit or respond to a generated interrupt.
4) Check if response_timeout error, response_CRC error, or response error is set. This can be done either by responding to an interrupt raised by these errors or by polling bits 1, 6, and 8 from the RINTSTS register @0x44. If no response error is received, then the response is valid. If required, the software can copy the response from the response registers @0x30-0x3C.

Software should not modify clock parameters while a command is being executed.

6. Data Transfer Commands

Data transfer commands transfer data between the memory card and the Host Controller. To send a data command, the Host Controller needs a command argument, total data size, and block size. Software can receive or send data through the FIFO.

Before a data transfer command, software should confirm that the card is not busy and is in a transfer state, which can be done using the CMD13 and CMD7 commands, respectively.

For the data transfer commands, it is important that the same bus width that is programmed

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in the card should be set in the card type register @0x18.

The Host Controller generates an interrupt for different conditions during data transfer, which are reflected in the RINTSTS register @0x44 as:

1) Data_Transfer_Over (bit 3) – When data transfer is over or terminated. If there is a response timeout error, then the Host Controller does not attempt any data transfer and the "Data Transfer Over" bit is never set.
2) Transmit_FIFO_Data_request (bit 4) – FIFO threshold for transmitting data was reached; software is expected to write data, if available, in FIFO.
3) Receive_FIFO_Data_request (bit 5) – FIFO threshold for receiving data was reached; software is expected to read data from FIFO.
4) Data starvation by Host timeout (bit 10) – FIFO is empty during transmission or is full during reception. Unless software writes data for empty condition or reads data for full condition, the Host Controller cannot continue with data transfer. The clock to the card has been stopped.
5) Data read timeout error (bit 9) – Card has not sent data within the timeout period.
6) Data CRC error (bit 7) – CRC error occurred during data reception.
7) Start bit error (bit 13) – Start bit was not received during data reception.
8) End bit error (bit 15) – End bit was not received during data reception or for a write operation; a CRC error is indicated by the card.

Conditions 6, 7, and 8 indicate that the received data may have errors. If there was a response timeout, then no data transfer occurred.

7. Single-Block or Multiple-Block Read

Steps involved in a single-block or multiple-block read are:

1) Write the data size in bytes in the BYTCNT register @0x20.
2) Write the block size in bytes in the BLKSIZ register @0x1C. The Host Controller expects data from the card in blocks of size BLKSIZ each.
3) Program the CMDARG register @0x28 with the data address of the beginning of a data read.
4) Program the Command register with the parameters listed in following table. For SD and MMC cards, use CMD17 for a single-block read and CMD18 for a multiple-block read. For SDIO cards, use CMD53 for both single-block and multiple-block transfers.

Table 12-14 Command Setting for Single or Multiple-Block Read

Parameter	Value	Description
Default		
start_cmd	1	-
use_hold_reg	1/0	Choose value based on speed mode being used;ref to "use_hold_reg" on CMD register
update_clk_regs_only	0	No clock parameters update command

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Parameter	Value	Description
card number	0	Actual card number(one controller only connect one card, the num is No.0)
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
send_auto_stop	0/1	-
transfer_mode	0	Block transfer
read_write	0	Read from card
data_expected	1	Data command
response_length	0	Can be 1 for R2(long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4, CMD15, and so on
User-selectable		
cmd_index	command-index	-
wait_prvdata_complete	1	0- Sends command immediately 1- Sends command after previous data transfer ends
check_response_crc	1	0- Host Controller should not check response CRC 1- Host Controller should check response CRC

After writing to the CMD register, the Host Controller starts executing the command; when the command is sent to the bus, the command_done interrupt is generated.

- Software should look for data error interrupts; that is, bits 7, 9, 13, and 15 of the RINTSTS register. If required, software can terminate the data transfer by sending a STOP command.
- Software should look for Receive_FIFO_Data_request and/or data starvation by host timeout conditions. In both cases, the software should read data from the FIFO and make space in the FIFO for receiving more data.
- When a Data_Transfer_Over interrupt is received, the software should read the remaining data from the FIFO.

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8. Single-Block or Multiple-Block Write

Steps involved in a single-block or multiple-block write are:

1) Write the data size in bytes in the BYTCNT register @0x20.
2) Write the block size in bytes in the BLKSIZ register @0x1C; the Host Controller sends data in blocks of size BLKSIZ each.
3) Program CMDARG register @0x28 with the data address to which data should be written.
4) Write data in the FIFO; it is usually best to start filling data the full depth of the FIFO.
5) Program the Command register with the parameters listed in following table.

Table 12-15 Command Settings for Single or Multiple-Block Write

Parameter	Value	Description
Default		
start_cmd	1	-
use_hold_reg	1/0	Choose value based on speed mode being used;ref to "use_hold_reg" on CMD register
update_clk_regs_only	0	No clock parameters update command
card number	0	Actual card number(one controller only connect one card, the num is No. 0)
send_initialization	0	Can be 1, but only for card reset commands, such as CMD0
stop_abort_cmd	0	Can be 1 for commands to stop data transfer, such as CMD12
send_auto_stop	0/1	-
transfer_mode	0	Block transfer
read_write	1	Write to card
data_expected	1	Data command
response_length	0	Can be 1 for R2(long) response
response_expect	1	Can be 0 for commands with no response; for example, CMD0, CMD4, CMD15, and so on
User-selectable		

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Parameter	Value	Description
cmd_index	command-index	-
wait_prvdata_complete	1	0- Sends command immediately 1- Sends command after previous data transfer ends
check_response_crc	1	0- Host Controller should not check response CRC 1- Host Controller should check response CRC

After writing to the CMD register, Host Controller starts executing a command; when the command is sent to the bus, a command_done interrupt is generated.

- Software should look for data error interrupts; that is, for bits 7, 9, and 15 of the RINTSTS register. If required, software can terminate the data transfer by sending the STOP command.
- Software should look for Transmit_FIFO_Data_Request and/or timeout conditions from data starvation by the host. In both cases, the software should write data into the FIFO.
- When a Data_Transfer_Over interrupt is received, the data command is over. For an open-ended block transfer, if the byte count is 0, the software must send the STOP command. If the byte count is not 0, then upon completion of a transfer of a given number of bytes, the Host Controller should send the STOP command, if necessary. Completion of the AUTO-STOP command is reflected by the Auto_command_done interrupt – bit 14 of the RINTSTS register. A response to AUTO_STOP is stored in RESP1 @0x34.

9. Stream Read

A stream read is like the block read mentioned in "Single-Block or Multiple-Block Read", except for the following bits in the Command register:

transfer_mode = 1; //Stream transfer

cmd_index = CMD20;

A stream transfer is allowed for only a single-bit bus width.

10. Stream Write

A stream write is exactly like the block write mentioned in "Single-Block or Multiple-Block Write", except for the following bits in the Command register:

transfer_mode = 1;//Stream transfer

cmd_index = CMD11;

In a stream transfer, if the byte count is 0, then the software must send the STOP command. If the byte count is not 0, then when a given number of bytes completes a transfer, the Host Controller sends the STOP command. Completion of this AUTO_STOP command is reflected by the Auto_command_done interrupt. A response to an AUTO_STOP is stored in the RESP1 register@0x34.

A stream transfer is allowed for only a single-bit bus width.

11. Packed Commands

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In order to reduce overhead, read and write commands can be packed in groups of commands—either all read or all write—that transfer the data for all commands in the group in one transfer on the bus.

Packed commands can be of two types:

- Packed Write: CMD23 →CMD25
- Packed Read: CMD23 → CMD25 → CMD23 → CMD18

Packed commands are put in packets by the application software and are transparent to the core.

12. Sending Stop or Abort in Middle of Transfer

The STOP command can terminate a data transfer between a memory card and the Controller, while the ABORT command can terminate an I/O data transfer for only the SDIO_IOONLY and SDIO_COMBO cards.

- Send STOP command – Can be sent on the command line while a data transfer is in progress; this command can be sent at any time during a data transfer.

You can also use an additional setting for this command in order to set the Command register bits (5-0) to CMD12 and set bit 14 (stop_abort_cmd) to 1. If stop_abort_cmd is not set to 1, the Controller does not know that the user stopped a data transfer. Reset bit 13 of the Command register (wait_prvdata_complete) to 0 in order to make the Controller send the command at once, even though there is a data transfer in progress.

- Send ABORT command – Can be used with only an SDIO_IOONLY or SDIO_COMBO card. To abort the function that is transferring data, program the function number in ASx bits (CCCR register of card, address 0x06, bits (0-2) using CMD52.

13. Suspend or Resume Sequence

In an SDIO card, the data transfer between an I/O function and the Controller can be temporarily halted using the SUSPEND command; this may be required in order to perform a high-priority data transfer with another function. When desired, the data transfer can be resumed using the RESUME command.

The following functions can be implemented by programming the appropriate bits in the CCCR register (Function 0) of the SDIO card. To read from or write to the CCCR register, use the CMD52 command.

- SUSPEND data transfer – Non-data command

1) Check if the SDIO card supports the SUSPEND/RESUME protocol; this can be done through the SBS bit in the CCCR register @0x08 of the card.
2) Check if the data transfer for the required function number is in process; the function number that is currently active is reflected in bits 0-3 of the CCCR register @0x0D. Note that if the BS bit (address 0xc::bit 0) is 1, then only the function number given by the FSx bits is valid.
3) To suspend the transfer, set BR (bit 2) of the CCCR register @0x0C.
4) Poll for clear status of bits BR (bit 1) and BS (bit 0) of the CCCR @0x0C. The BS (Bus Status) bit is 1 when the currently-selected function is using the data bus; the BR (Bus Release) bit remains 1 until the bus release is complete. When the BR and BS bits are 0, the data transfer from the selected function has been suspended.

- RESUME data transfer – This is a data command

1) Check that the card is not in a transfer state, which confirms that the bus is free for data transfer.

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2) If the card is in a disconnect state, select it using CMD7. The card status can be retrieved in response to CMD52/CMD53 commands.
3) Check that a function to be resumed is ready for data transfer; this can be confirmed by reading the RFx flag in CCCR @0x0F. If RF = 1, then the function is ready for data transfer.
4) To resume transfer, use CMD52 to write the function number at FSx bits (0-3) in the CCCR register @0x0D. Form the command argument for CMD52 and write it in CMDARG @0x28.
5) Write the block size in the BLKSIZ register @0x1C; data will be transferred in units of this block size.
6) Write the byte count in the BYTCNT register @0x20. This is the total size of the data; that is, the remaining bytes to be transferred. It is the responsibility of the software to handle the data.
7) Program Command register; similar to a block transfer.
8) When the Command register is programmed, the command is sent and the function resumes data transfer. Read the DF flag (Resume Data Flag). If it is 1, then the function has data for the transfer and will begin a data transfer as soon as the function or memory is resumed. If it is 0, then the function has no data for the transfer.
9) If the DF flag is 0, then in case of a read, the Host Controller waits for data. After the data timeout period, it gives a data timeout error.

6. Read_Wait Sequence

Read_wait is used with only the SDIO card and can temporarily stall the data transfer—either from function or memory—and allow the host to send commands to any function within the SDIO device. The host can stall this transfer for as long as required. The Host Controller provides the facility to signal this stall transfer to the card. The steps for doing this are:

1) Check if the card supports the read_wait facility; read SRW (bit 2) of the CCCR register @0x08. If this bit is 1, then all functions in the card support the read_wait facility. Use CMD52 to read this bit.
2) If the card supports the read_wait signal, then assert it by setting the read_wait (bit 6) in the CTRL register @0x00.
3) Clear the read_wait bit in the CTRL register.

14. Controller/DMA/FIFO Reset Usage

- Controller reset – Resets the controller by setting the controller_reset bit (bit 0) in the CTRL register; this resets the CIU and state machines, and also resets the BIU-to-CIU interface. Since this reset bit is self-clearing, after issuing the reset, wait until this bit is cleared.
- FIFO reset - Resets the FIFO by setting the fifo_reset bit (bit 1) in the CTRL register; this resets the FIFO pointers and counters of the FIFO. Since this reset bit is self-clearing, after issuing the reset, wait until this bit is cleared.

In external DMA transfer mode, even when the FIFO pointers are reset, if there is a DMA transfer in progress, it could push or pop data to or from the FIFO; the DMA itself completes correctly. In order to clear the FIFO, the software should issue an additional FIFO reset and clear any FIFO underrun or overrun errors in the RAWINTS register caused by the DMA transfers after the FIFO was reset.

15. Card Read Threshold

When an application needs to perform a Single or Multiple Block Read command, the application must program the CardThrCtl register with the appropriate Card Read Threshold

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size (CardRdThreshold) and set the Card Read Threshold Enable (CardRdThrEnable) bit to 1'b1. This additional programming ensures that the Host controller sends a Read Command only if there is space equal to the CardRDThreshold available in the Rx FIFO. This in turn ensures that the card clock is not stopped in the middle a block of data being transmitted from the card. The Card Read Threshold can be set to the block size of the transfer, which guarantees that there is a minimum of one block size of space in the RxFIFO before the controller enables the card clock.The Card Read Threshold is required when the Round Trip Delay is greater than 0.5cclk_in period.

16. Error Handling

The Host Controller implements error checking; errors are reflected in the RAWINTS register@0x44 and can be communicated to the software through an interrupt, or the software can poll for these bits. Upon power-on, interrupts are disabled (int_enable in the CTRL register is 0), and all the interrupts are masked (bits 0-31 of the INTMASK register; default is 0).

Error handling:

- Response and data timeout errors – For response timeout, software can retry the command. For data timeout, the Host Controller has not received the data start bit – either for the first block or the intermediate block – within the timeout period, so software can either retry the whole data transfer again or retry from a specified block onwards. By reading the contents of the TCBCNT later, the software can decide how many bytes remain to be copied.
- Response errors – Set when an error is received during response reception. In this case, the response that copied in the response registers is invalid. Software can retry the command.
- Data errors – Set when error in data reception are observed; for example, data CRC, start bit not found, end bit not found, and so on. These errors could be set for any block-first block, intermediate block, or last block. On receipt of an error, the software can issue a STOP or ABORT command and retry the command for either whole data or partial data.
- Hardware locked error – Set when the Host Controller cannot load a command issued by software. When software sets the start_cmd bit in the CMD register, the Host Controller tries to load the command. If the command buffer is already filled with a command, this error is raised. The software then has to reload the command.
- FIFO underrun/overrun error – If the FIFO is full and software tries to write data in the FIFO, then an overrun error is set. Conversely, if the FIFO is empty and the software tries to read data from the FIFO, an underrun error is set. Before reading or writing data in the FIFO, the software should read the fifo_empty or fifo_full bits in the Status register.
- Data starvation by host timeout – Raised when the Host Controller is waiting for software intervention to transfer the data to or from the FIFO, but the software does not transfer within the stipulated timeout period. Under this condition and when a read transfer is in process, the software should read data from the FIFO and create space for further data reception. When a transmit operation is in process, the software should fill data in the FIFO in order to start transferring data to the card.
- CRC Error on Command – If a CRC error is detected for a command, the CE-ATA device does not send a response, and a response timeout is expected from the Host Controller. The ATA layer is notified that an MMC transport layer error occurred.

Notes: During a multiple-block data transfer, if a negative CRC status is received from the device, the data path signals a data CRC error to the BIU by setting the data CRC error bit in the RINTSTS register. It then continues further data transmission until all the bytes are transmitted.

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12.6.5 Voltage Switching

The Host Controller supports SD 3.0 Ultra High Speed (UHS-1) and is capable of voltage switching in SD-mode, which can be applied to SD High-Capacity (SDHC) and SD Extended Capacity (SDXC) cards. UHS-1 supports only 4-bit mode.

However, whether the IO voltage of 1.8v supported or not is depended on the SoC design.

SD 3.0 UHS-1 supports the following transfer speed modes for UHS-50 and/or UHS-104 cards:

- DS – default-speed up to 25MHz, 3.3V signaling
- HS – high-speed up to 50MHz, 3.3V signaling
- SDR12 – SDR up to SDR 25MHz, 1.8V signaling
- SDR25 – SDR up to 50MHz, 1.8V signaling
- SDR50 – SDR up to 100MHz, 1.8V signaling
- DDR50 – DDR up to 50MHz, 1.8V signaling

Voltage selection can be done in only SD mode. The first CMD0 selects the bus mode-either SD mode or SPI mode. The card must be in SD mode in order for 1.8V signaling mode to apply, during which time the card cannot be switched to SPI mode or 3.3V signaling without a power cycle.

If the System BIOS in an embedded system already knows that it is connected to an SD 3.0 card, then the driver programs the Controller to initiate ACMD41. The software knows from the response of ACMD41 whether or not the card supports voltage switching to 1.8V.

- If bit 32 of ACMD41 response is 1'b1: card supports voltage switching and next command-CMD11-invokes voltage switching sequence. After CMD11 is started, the software must program the IO voltage selection register based on the soc architecture.
- If bit 32 of ACMD41 response is 1'b0: card does not support voltage switching and CMD11 should not be started.

If the card and host controller accept voltage switching, then they support UHS-1 modes of data transfer. After the voltage switch to 1.8V, SDR12 is the default speed.

Since the UHS-1 can be used in only 4-bit mode, the software must start ACMD6 and change the card data width to 4-bit mode; ACMD6 is driven in any of the UHS-1 speeds. If the host wants to select the DDR mode of data transfer, then the software must program the DDR_REG register in the CSR space with the appropriate card number.

To choose from any of the SDR or DDR modes, appropriate values should be programmed in the CLKDIV register.

1. Voltage Switch Operation

The Voltage Switch operation must be performed in SD mode only.

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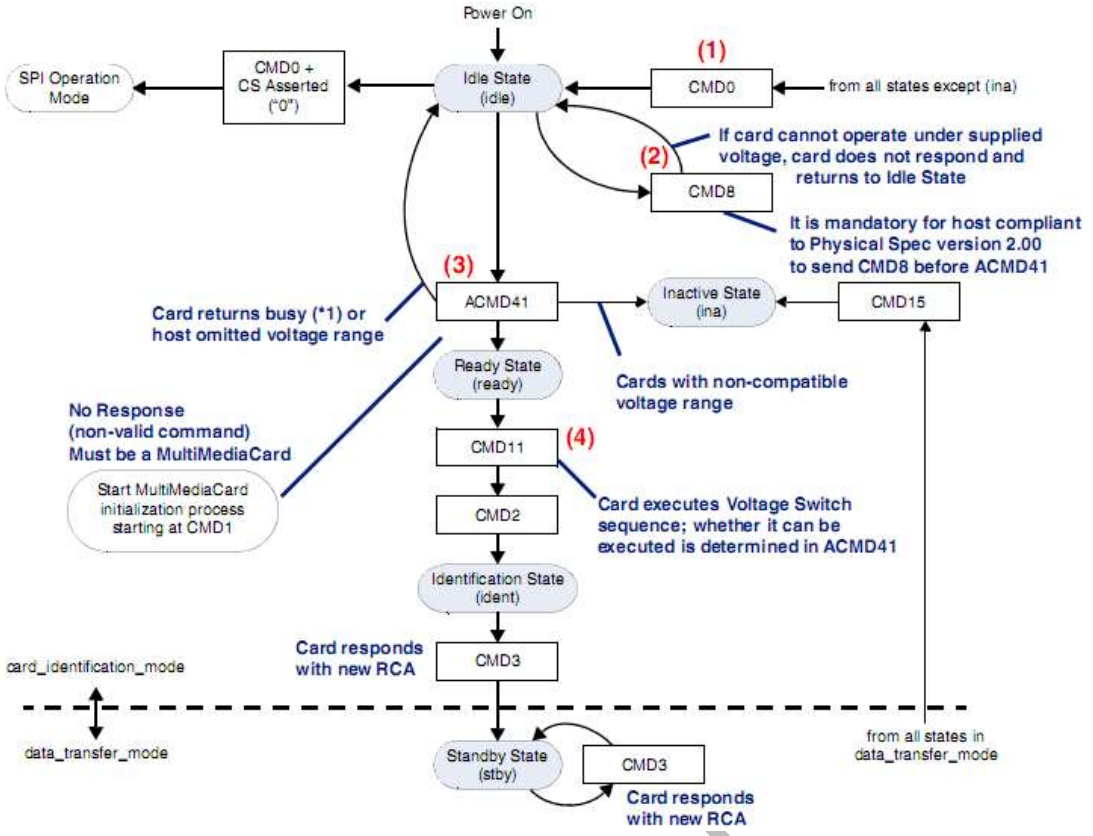
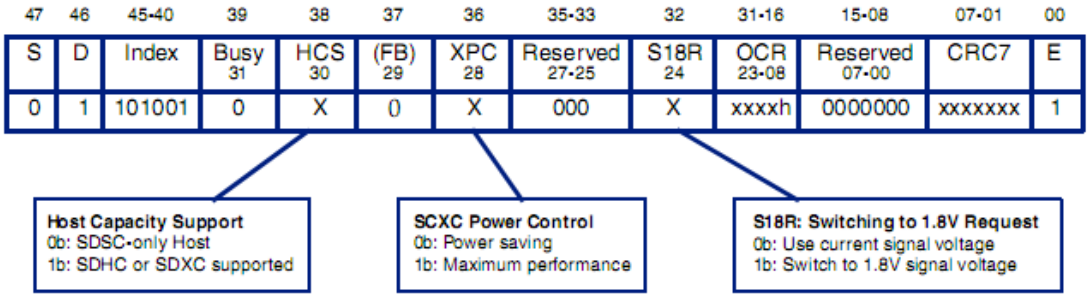


Fig. 12-12 Voltage Switching Command Flow Diagram

The following outlines the steps for the voltage switch programming sequence

1) Software Driver starts CMD0, which selects the bus mode as SD.
2) After the bus is in SD card mode, CMD8 is started in order to verify if the card is compatible with the SD Memory Card Specification, Version 2. 00. CMD8 determines if the card is capable of working within the host supply voltage specified in the VHS (19:16) field of the CMD; the card supports the current host voltage if a response to CMD8 is received.
3) ACMD 41 is started. The response to this command informs the software if the card supports voltage switching; bits 38, 36, and 32 are checked by the card argument of ACMD41; refer to following figure.



47	46	45-40	39	38	37	36	35-33	32	31-16	15-08	07-01	00
S	D	Index	Busy 31	HCS 30	(FB) 29	XPC 28	Reserved 27-25	S18R 24	OCR 23-08	Reserved 07-00	CRC7	E
0	1	101001	0	X	0	X	000	X	xxxxh	0000000	xxxxxxx	1

Host Capacity Support
0b: SDSC-only Host
1b: SDHC or SDXC supported

SCXC Power Control
0b: Power saving
1b: Maximum performance

S18R: Switching to 1.8V Request
0b: Use current signal voltage
1b: Switch to 1.8V signal voltage

Fig. 12-13 ACMD41 Argument

- Bit 30 informs the card if host supports SDHC/SDXC or not; this bit should be set to 1'b1.
- Bit 28 can be either 1 or 0.
- Bit 24 should be set to 1'b1, indicating that the host is capable of voltage switching; refer to following figure.

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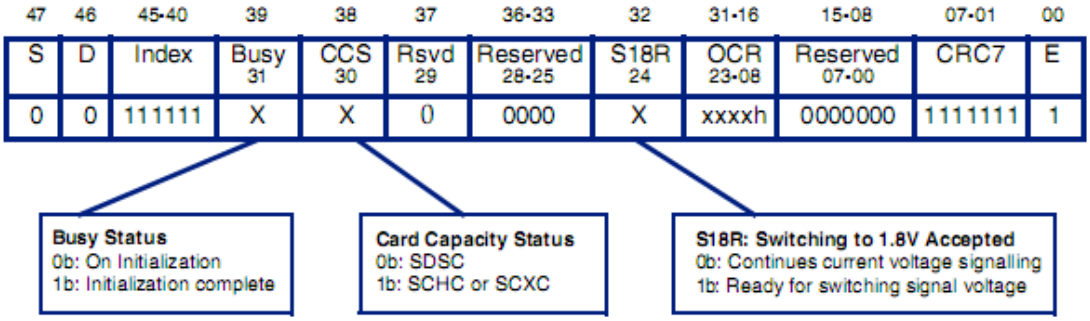

Fig. 12-14 ACMD41 Response(R3)

- Bit 30 – If set to 1'b1, card supports SDHC/SDXC; if set to 1'b0, card supports only SDSC
- Bit 24 – If set to 1'b1, card supports voltage switching and is ready for the switch
- Bit 31 – If set to 1'b1, initialization is over; if set to 1'b0, means initialization in process

4) If the card supports voltage switching, then the software must perform the steps discussed for either the "Voltage Switch Normal Scenario" or the "Voltage Switch Error Scenario".

2. Voltage Switch Normal Scenario

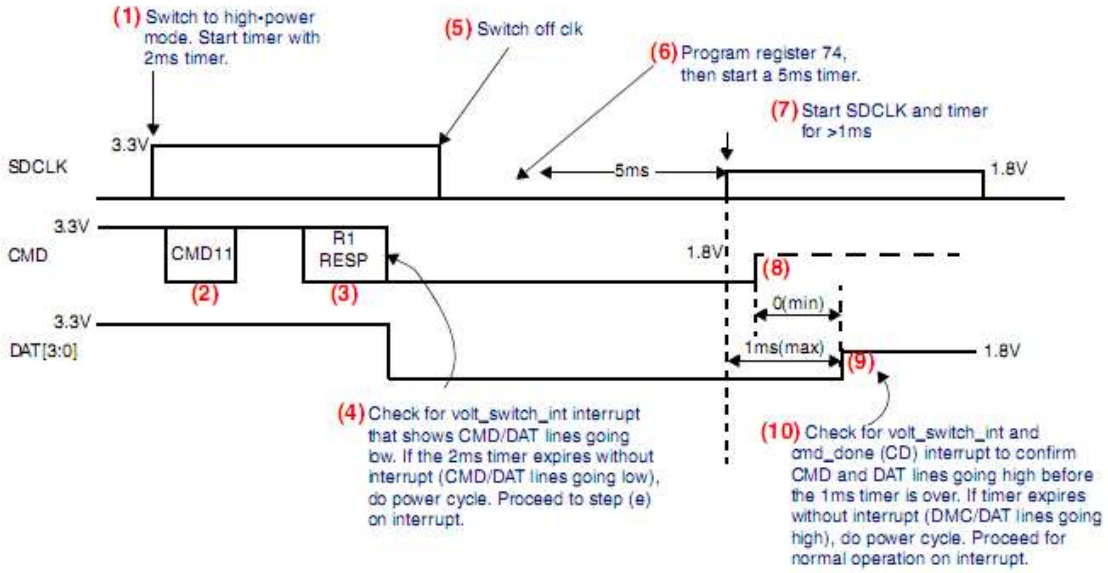

Fig. 12-15 Voltage Switch Normal Scenario

1) The host programs CLKENA—cclk_low_power register—with zero (0) for the corresponding card, which makes the host controller move to high-power mode. The application should start a timer with a recommended value of 2ms; this value of 2 ms is determined as below:Total clk required for CMD11 = 48 clks

   Total clk required for RESP R1 = 48 clks

   Maximum clk delay between MCD11 end to start of RESP1 = 60 clks

   Total = 48+48 + 60 = 160

   Minimum frequency during enumeration is 100 KHz; that is, 10us

   Total time = 160 * 10us = 1600us = 1. 6ms ~ 2ms

2) The host issues CMD11 to start the voltage switch sequence. Set bit 28 to 1'b1 in CMD when setting CMD11; for more information on setting bits, refer to "Boot Operation".

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3) The card returns R1 response; the host controller does not generate cmd_done interrupt on receiving R1 response.
4) The card drives CMD and DAT [3:0] to low immediately after the response. The host controller generates interrupt (VOLT_SWITCH_INT) once the CMD or DAT [3:0] line goes low. The application should wait for this interrupt. If the 2ms timer expires without an interrupt (CMD/DAT lines going low), do a power cycle.

Note: Before doing a power cycle, switch off the card clock by programming CLKENA register

Proceed to step (5) on getting an interrupt (VOLT_SWITCH_INT).

Note: This interrupt must be cleared once this interrupt is received. Additionally, this interrupt should not be masked during the voltage switch sequence.

If the timer expires without interrupt (CMD/DAT lines going low), perform a power cycle. Proceed to step (5) on interrupt.

1) Program the CLKENA, cclk_enable register, with 0 for the corresponding card; the host stops supplying SDCLK.
2) Program Voltage register to the required values for the corresponding card. The application should start a timer > 5ms.
3) After the 5ms timer expires, the host voltage regulator is stable. Program CLKENA, cclk_enable register, with 1 for the corresponding card; the host starts providing SDCLK at 1. 8V; this can be at zero time after Voltage register has been programmed. When the CLKENA register is programmed, the application should start another timer > 1ms.
4) By detecting SDCLK, the card drives CMD to high at 1. 8V for at least one clock and then stops driving (tri-state); CMD is triggered by the rising edge of SDCLK (SDR timing).
5) If switching to 1. 8V signaling is completed successfully, the card drives DAT [3:0] to high at 1. 8V for at least one clock and then stops driving (tri-state); DAT [3:0] is triggered by the rising edge of SDCLK (SDR timing). DAT[3:0] must be high within 1ms from the start of SDCLK.
6) The host controller generates a voltage switch interrupt (VOLT_SWITCH_INT) and a command done (CD) interrupt once the CMD and DAT[3:0] lines go high. The application should wait for this interrupt to confirm CMD and DAT lines going high before the 1ms timer is done.

If the timer expires without the voltage switch interrupt (VOLT_SWITCH_INT), a power cycle should be performed. Program the CLKENA register to stop the clock for the corresponding card number. Wait for the cmd_done (CD) interrupt. Proceed for normal operation on interrupt. After the sequence is completed, the host and the card start communication in SDR12 timing.

3. Voltage Switch Error Scenario

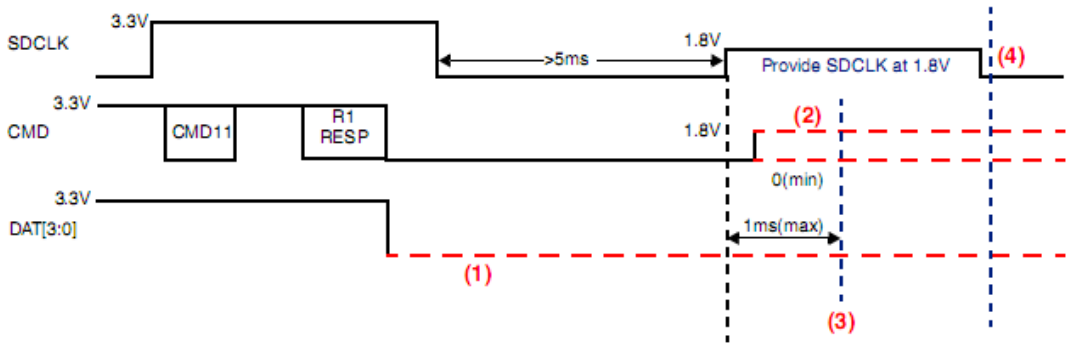

Fig. 12-16 Voltage Switch Error Scenario

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1) If the interrupt (VOLT_SWITCH_INT) does not come, then the 2 ms timer should time out and a power cycle should be initiated.

Note: Before performing a power cycle, switch off the card clock by programming CLKENA register; no cmd_done (CD) interrupt is generated.

Additionally, if the card detects a voltage error at any point in between steps (5) and (7) in the card keeps driving DAT[3:0] to low until card power off.

2) CMD can be low or tri-state.
3) The host controller generates a voltage switch interrupt once the CMD and DAT[3:0] lines go high. The application should check for an interrupt to confirm CMD and DAT lines going high before the 1 ms timer is done.

If the 1 ms timer expires without interrupt (VOLT_SWITCH_INT) and cmd_done (CD), a power cycle should be performed. Program the CLKENA register to stop SDCLK of the corresponding card. Wait for the cmd_done interrupt. Proceed for normal operation on interrupt.

4) If DAT[3:0] is low, the host drives SDCLK to low and then stops supplying the card power.

Note: The card checks voltages of its own regulator output and host signals to ensure they are less than 2. 5V. Errors are indicated by (1) and (2).

- If voltage switching is accepted by the card, the default speed is SDR12.
- Command Done is given:
  - If voltage switching is properly done, CMD and DAT line goes high.
  - If switching is not complete, the 1ms timer expires, and the card clk is switched off.

Note: No other CMD should be driven before the voltage switching operation is completed and Command Done is received.

- The application should use CMD6 to check and select the particular function; the function appropriate-speed should be selected.

After the function switches, the application should program the correct value in the CLKDIV register, depending on the function chosen. Additionally, if Function 0x4 of the Access mode is chosen—that is, DDR50, then the application should also program 1'b1 in DDR_REG for the card number that has been selected for DDR50 mode.

12.6.6 Back-End Power

Each device needs one bit to control the back-end power supply for an embedded device; this bit does not control the VDDH of the host controller. A back_end_power register enables software programming for back-end power. The value on this register is output to the back_end_power signal, which can be used to switch power on and off the embedded device.

12.6.7 DDR Operation

1. 4-bit DDR Programming Sequence

DDR programming should be done only after the voltage switch operation has completed. The following outlines the steps for the DDR programming sequence:

1) Once the voltage switch operation is complete, the user must program voltage selection register to the required values for the corresponding card.

- To start a card to work in DDR mode, the application must program a bit of the newly defined UHS_REG[16] register with a value of 1'b1.
- The bit that the user programs depends on which card is to be accessed in DDR mode.

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2) To move back to SDR mode, a power cycle should be run on the card—putting the card in SDR12 mode—and only then should UHS_REG[16]be set back to 1'b0 for the appropriate card.

2. 8-bit DDR Programming Sequence

The following outlines the steps for the 8-bit DDR programming sequence:

1) The cclk_in signal should be twice the speed of the required cclk_out. Thus, if the cclk_out signal is required to be 50 MHz, the cclk_in signal should be 100 MHz.
2) The CLKDIV register should always be programmed with a value higher than zero (0); that is, a clock divider should always be used for 8-bit DDR mode.
3) The application must program the UHS_REG[16] register (DDR_REG bits) by assigning it with a value of 1 for the bit corresponding to the card number; this causes the selected card to start working in DDR mode.
4) Depending on the card number, the CTYPE [31:16] bits should be set in order to make the host work in the 8-bit mode.

3. eMMC4.5 DDR START Bit

The eMMC4.5 changes the START bit definition in the following manner:

1) Receiver samples the START bit on the rising edge.
2) On the next rising edge after sampling the START bit, the receiver must sample the data.
3) Removes requirement of the START bit and END bit to be high for one full cycle.

Notes: The Host Controller does not support a START bit duration higher than one clock cycle. START bit durations of one or less than one clock cycle are supported and can be defined at the time of startup by programming the EMMC_DDR_REG register.

Following figure illustrates cases for the definition change of the START bit with eMMC4.5; it also illustrates how some of these cases can fail in sampling when higher-value delays are considered for I/O PADs.

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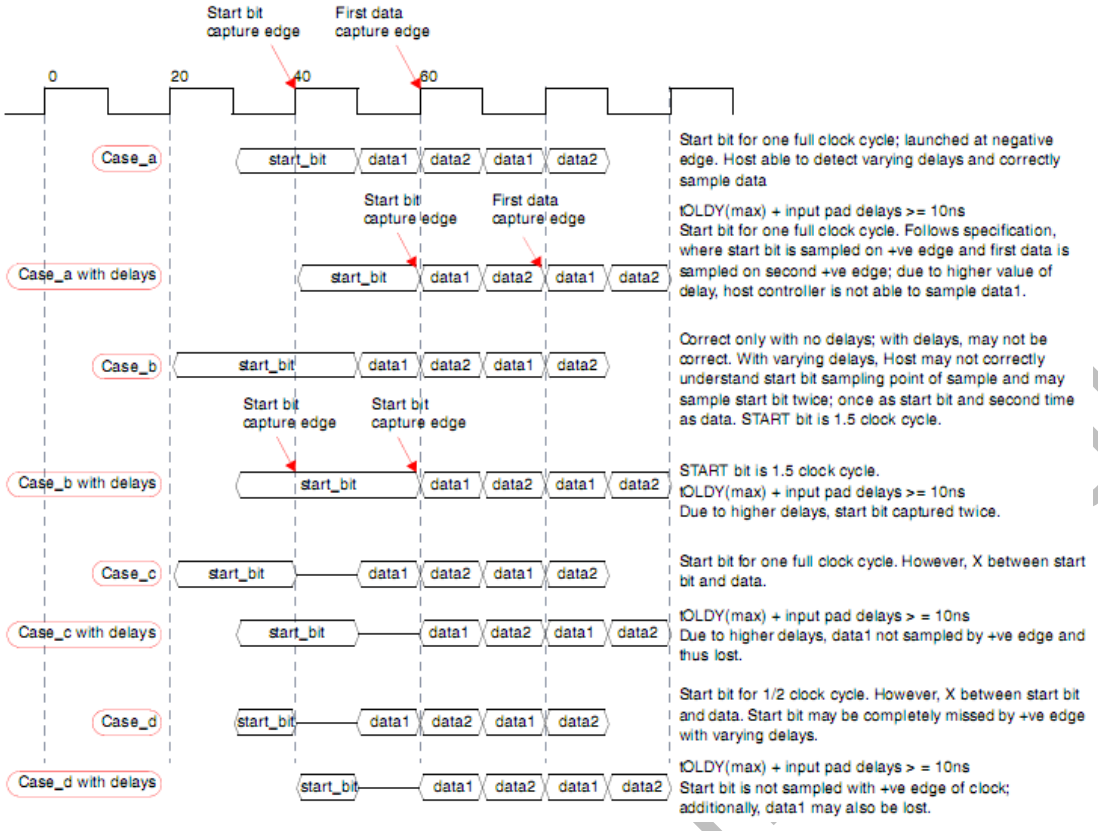

Fig. 12-17 CASES for eMMC 4.5 START bit

4. Reset Command/Moving from DDR50 to SDR12

To reset the mode of operation from DDR50 to SDR12, the following sequence of operations has to be done by the application:

1) Issue CMD0.

When CMD0 is received, the card changes from DDR50 to SDR12.

2) Program the CLKDIV register with an appropriate value.
3) Set DDR_REG to 0.

Note: The Voltage register should not be programmed to 0 while switching from DDR50 to SDR12, since the card is still operating in 1.8V mode after receiving CMD0.

12.6.8 H/W Reset Operation

When the RST_n signal goes low, the card enters a pre-idle state from any state other than the inactive state.

H/W Reset Programming Sequence

The following outlines the steps for the H/W reset programming sequence:

11) Program CMD12 to end any transfer in process.
12) Wait for DTO, even if no response is sent back by the card.
13) Set the following resets:

- DMA reset– CTRL[2]
- FIFO reset – CTRL[1] bits

Note: The above steps are required only if a transfer is in process.

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14) Program the CARD_RESET register with a value of 0; this can be done at any time when the card is connected to the controller. This programming asserts the RST_n signal and resets the card.
15) Wait for minimum of 1 µs or cclk_in period, whichever is greater
16) After a minimum of 1 µs, the application should program a value of 0 into the CARD_RESET register. This de-asserts the RST_n signal and takes the card out of reset.
17) The application can program a new CMD only after a minimum of 200 µs after the de-assertion of the RST_n signal, as per the MMC 4.41 standard.

Note: For backward compatibility, the RST_n signal is temporarily disabled in the card by default. The host may need to set the signal as either permanently enabled or permanently disabled before it uses the card.

12.6.9 FBE Scenarios

An FBE occurs due to an AHB error response on the AHB bus. This is a system error, so the software driver should not perform any further programming to the Host. The only recovery mechanism from such scenarios is to do one of the following:

- Issue a hard reset by asserting the reset_n signal
- Do a program controller reset by writing to the CTRL[0] register

1. FIFO Overflow and Underflow

During normal data transfer conditions, FIFO overflow and underflow will not occur. However if there is a programming error, then FIFO overflow/underflow can result. For example, consider the following scenarios.

- For transmit: PBL=4, Tx watermark = 1. For the above programming values, if the FIFO has only one location empty, it issues a dma_req to IDMAC FSM. Due to PBL value=4, the IDMAC FSM performs 4 pushes into the FIFO. This will result in a FIFO overflow interrupt.
- For receive: PBL=4, Rx watermark = 1. For the above programming values, if the FIFO has only one location filled, it issues a dma_req to IDMAC FSM. Due to PBL value=4, the IDMAC FSM performs 4 pops to the FIFO. This will result in a FIFO underflow interrupt.

The driver should ensure that the number of bytes to be transferred as indicated in the descriptor should be a multiple of 4bytes with respect to H_DATA_WIDTH=32. For example, if the BYTCNT = 13, the number of bytes indicated in the descriptor should be 16 for H_DATA_WIDTH=32.

2. Programming of PBL and Watermark Levels

The DMAC performs data transfers depending on the programmed PBL and threshold values.

Table 12-16 PBL and Watermark Levels

PBL (Number of transfers)	Tx/Rx Watermark Value
1	greater than or equal to 1
4	greater than or equal to 4
8	greater than or equal to 8
16	greater than or equal to 16

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PBL (Number of transfers)	Tx/Rx Watermark Value
32	greater than or equal to 32
64	greater than or equal to 64
128	greater than or equal to 128
256	greater than or equal to 256

12.6.10 Variable Delay/Clock Generation

Variable delay mechanism for the cclk_in_drv is optional, but it can be useful in order to meet a range of hold-time requirements across modes. Variable delay mechanism for the cclk_in_sample is mandatory and is required to achieve the correct sampling point for data.

cclk_in/cclk_in_sample/ cclk_in_drv is generated by Clock Generation Unit (CLKGEN) with variable delay mechanism, which includes Phase Shift Unit and Delay Line Unit selectable.

The Phase Shift Unit can shift cclk_in_sample/cclk_in_drv by 0/90/180/270-degree relative to cclk_in, controlled by sample_degree/drv_degree.

The Delay Line Unit can shift cclk_in_sample/cclk_in_drv in the unit of 40ps~80ps for every delay element. The delay unit number is determined by sample_delaynum/drv_delaynum, and enabled by sample_sel/drv_sel.

cclk_in is generated by cclkin divided by 2. cclk_in_drv and cclk_in_sample clocks are phase-shifted with delayed versions of cclk_in. All clocks are recommended to have a 50% duty cycle; DDR modes must have 50% duty cycles.

The architecture is as follows.

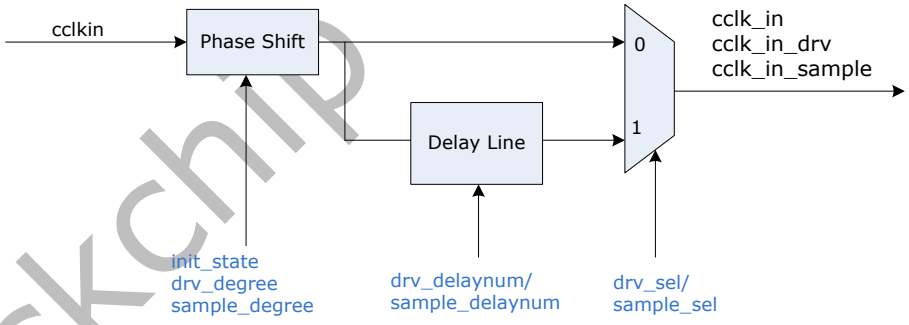

Fig. 12-18 Clock Generation Unit

The control signals for different Host Controller instance are shown as follows:

Table 12-17 Configuration for SDMMC Clock Generation

Signal Name	Source	Default	Description
init_state	CRU_SDMMC_CON0[0]	0	Soft initial state for phase shift.
drv_degree[1:0]	CRU_SDMMC_CON0[2:1]	2	Phase shift for cclk_in_drv. 0: 0-degree 1: 90-degree 2: 180-degree

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Signal Name	Source	Default	Description
			3: 270-degree
drv_delaynum [7:0]	CRU_SDMMC_CON0[10:3]	0	Element number in delay line for cclk_in_drv
drv_sel	CRU_SDMMC_CON0[11]	0	cclk_in_drv source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line
sample_degree[1:0]	CRU_SDMMC_CON1[2:1]	0	Phase shift for cclk_in_sample. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
sample_delaynum[7:0]	CRU_SDMMC_CON1[10:3]	0	Element number in delay line for cclk_in_sample
sample_sel	CRU_SDMMC_CON1[11]	0	cclk_in_sample source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line

Table 12-18 Configuration for SDMMC_EXT Clock Generation

Signal Name	Source	Default	Description
init_state	CRU_SDMMC_EXT_CON0[0]	0	Soft initial state for phase shift.
drv_degree[1:0]	CRU_SDMMC_EXT_CON0[2:1]	2	Phase shift for cclk_in_drv. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
drv_delaynum [7:0]	CRU_SDMMC_EXT_CON0[10:3]	0	Element number in delay line for cclk_in_drv
drv_sel	CRU_SDMMC_EXT_CON0[11]	0	cclk_in_drv source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line
sample_degree[1:0]	CRU_SDMMC_EXT_CON1[2:1]	0	Phase shift for cclk_in_sample. 0: 0-degree

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Signal Name	Source	Default	Description
			1: 90-degree 2: 180-degree 3: 270-degree
sample_delaynum[7:0]	CRU_SDMMC_EXT_CON1[10:3]	0	Element number in delay line for cclk_in_sample
sample_sel	CRU_SDMMC_EXT_CON1[11]	0	cclk_in_sample source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line

Table 12-19 Configuration for SDIO Clock Generation

Signal Name	Source	Default	Description
init_state	CRU_SDIO_CON0[0]	0	Soft initial state for phase shift.
drv_degree[1:0]	CRU_SDIO_CON0[2:1]	2	Phase shift for cclk_in_drv. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
drv_delaynum[7:0]	CRU_SDIO_CON0[10:3]	0	Element number in delay line for cclk_in_drv
drv_sel	CRU_SDIO_CON0[11]	0	cclk_in_drv source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line
sample_degree[1:0]	CRU_SDIO_CON1[2:1]	0	Phase shift for cclk_in_sample. 0: 0-degree 1: 90-degree 2: 180-degree 3: 270-degree
sample_delaynum[7:0]	CRU_SDIO_CON1[10:3]	0	Element number in delay line for cclk_in_sample
sample_sel	CRU_SDIO_CON1[11]	0	cclk_in_sample source selection: 0: use clock after phase_shift 1: use clock after phase_shift and delay line

The following outlines the steps for clock generation sequence:

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1) Assert init_state to soft reset the CLKGEN.
2) Configure drv_degree/sample_degree.
3) If fine adjustment required, delay line can be used by configuring drv_delaynum/sample_delaynum and drv_sel/sample_sel.
4) Dis-assert init_state to start CLKGEN.

12.6.11 Variable Delay Tuning

Tuning is defined by SD and MMC cards to determine the correct sampling point required for the host, especially for the speed modes SDR104 and HS200 where the output delays from the cards can be up to 2 UI. Tuning is required for other speed modes-such as DDR50-even though the output delay from the card is less than one cycle.

Command for tuning is different for different cards.

- SD Memory Card:
  - CMD19 – SD card for SDR50 and SDR104 speed modes. Tuning data is defined by card specifications.
  - CMD6 – SD card for speed modes not supporting CMD19. Tuning data is the 64byte SD status.
- Multimedia Card:
  - CMD21 – MMC card for HS200 speed mode. Tuning data is defined by card specifications.
  - CMD8 – MMC card for speed modes not supporting CMD21. Tuning data is 512 byte ExtCSD data.

The following is the procedure for variable delay tuning:

1) Set a phase shift of 0-degree on cclk_in_sample.
2) Send the Tuning command to the card; the card in turn sends an R1 response on the CMD line and tuning data on the DAT line.
3) If the host sees any of the errors—start bit error, data crc error, end bit error, data read time-out, response crc error, response error—then the sampling point is incorrect.
4) Send CMD12 to bring the host controller state machines to idle.
- The card may treat CMD12 as an invalid command because the card has successfully sent the tuning data, and it cannot send a response.
- The host controller may generate a response time-out interrupt that must be cleared by software.
5) Repeat steps 2) to 4) by increasing the phase shift value or delay element number on cclk_in_sample until the correct sampling point is received such that the host does not see any of the errors.
6) Mark this phase shift value as the starting point of the sampling window.
7) Repeat steps 2 to 4 by increasing the phase shift value or delay element number on cclk_in_sample until the host sees the errors starting to come again or the phase shift value reaches 360-degree.
8) Mark the last successful phase shift value as the ending point of the sampling window.

A window is established where the tuning block is matched. For example, for a scenario where the tuning block is received correctly for a phase shift window of 90-degree and 180-degree, then an appropriate sampling point is established as 135-degree. Once a sampling point is established, no errors should be visible in the tuning block.

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12.6.12 Package Command

In order to reduce overhead, read and write commands can be packed in groups of commands-either all read or all write-that transfer the data for all commands in the group in one transfer on the bus.

Packed commands can be of two types:

- Packed Write: CMD23 → CMD25
- Packed Read: CMD23 → CMD25 → CMD23 → CMD18

Packed commands are put in packets by the application software and are transparent to the core. For more information on packed commands, refer to the eMMC specification.

12.6.13 Card Detection Method

There are many methods for SDMMC/SDMMC_EXT/SDIO card detection.

- Method1: Using SDMMC_CDETECT register, which is value on card_detect_n input port. 0 represents presence of card.
- Method2: Using card detection unit in Host Controller, outputting host interrupt. The card detection unit looks for any changes in the card-detect signals for card insertion or card removal. It filters out the debounces associated with mechanical insertion or removal, and generates one interrupt to the host. You can program the debounce filter value in SDMMC_DEBNCE [23:0].Following figure illustrates the timing for card-detect signals.

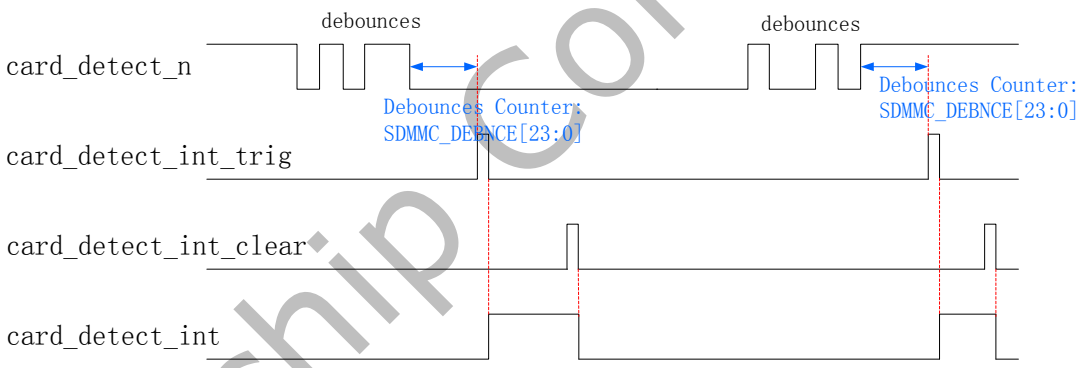

Fig. 12-19 Card Detection Method 2

- Method3: Using card detection unit in GRF, outputting sdmmc_detect_dual edge_int(IRQ_ID[63] for SDMMC, IRQ_ID[10] for SDMMC_EXT). Similar to Method2, except that the debounce time is configurable by GRF_SDMMC_DET; and the insertion/removal detection interrupt can be enabled or cleared respectively. The detailed register information is:

Table 12-20 Register for SDMMC Card Detection Method 3

Signal Name	Source	Default	Description
sd_detectn_rise_edge_irq_en	GRF_SIG_DETECT_CON[0]	0	sdmmc detect_n signal rise edge interrupt enable. 1: enable 0: disable
sd_detect_fall_edge_detect_en	GRF_SIG_DETECT_CON[1]	0	sd_detect_falling_edge enable 0: disable 1: enable
sd_detect_rising_ed	SIG_DETECT_STATU	0	sd_detect_rising_edge status

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Signal Name	Source	Default	Description
ge_dectect_status	S[0]		0: disable 1: enable
sd_detect_fall_edge_detect_status	SIG_DETECT_STATUS[1]	0	sd_detect_falling_edge status 0: disable 1: enable
sd_detect_rising_edge_dectect_clr	SIG_DETECT_CLR[0]	0	sd_detect_rising_edge clear 0: disable 1: enable
sd_detect_fall_edge_detect_clr	SIG_DETECT_CLR[1]	0	sd_detect_falling_edge clear 0: disable 1: enable

Table 12-21 Register for SDMMC_EXT Card Detection Method 3

Signal Name	Source	Default	Description
sd_detectn_rise_edge_irq_en	GRF_SIG_DETECT_CON[2]	0	sdmmc_ext detect_n signal rise edge interrupt enable. 1: enable 0: disable
sd_detect_fall_edge_detect_en	GRF_SIG_DETECT_CON[3]	0	sd_detect_falling_edge enable 0: disable 1: enable
sd_detect_rising_edge_dectect_status	SIG_DETECT_STATUS[2]	0	sd_detect_rising_edge status 0: disable 1: enable
sd_detect_fall_edge_detect_status	SIG_DETECT_STATUS[3]	0	sd_detect_falling_edge status 0: disable 1: enable
sd_detect_rising_edge_dectect_clr	SIG_DETECT_CLR[2]	0	sd_detect_rising_edge clear 0: disable 1: enable
sd_detect_fall_edge_detect_clr	SIG_DETECT_CLR[3]	0	sd_detect_falling_edge clear 0: disable 1: enable

- Method4: Using filtered card_detect_n with the debounce time GRF_SDMMC_DET for interrupt source, connecting to *IRQ*(IRQ_ID[65] for SDMMC, IRQ_ID[6] for SDMMC_EXT) directly.

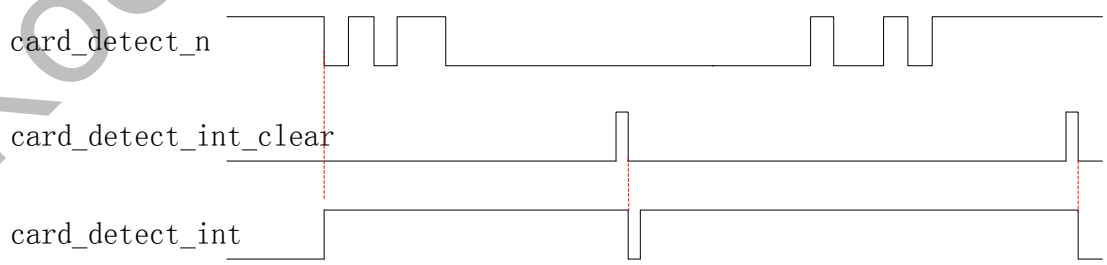

Fig. 12-20 Card Detection Method 4

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Chapter 13 Temperature-Sensor ADC(TS-ADC)

14

13.1 Overview

TS-ADC Controller module supports user-defined mode and automatic mode. User-defined mode refers, TSADC all the control signals entirely by software writing to register for direct control. Automatic mode refers to the module automatically poll TSADC output, and the results were checked. If you find that the temperature High in a period of time, an interrupt is generated to the processor down-measures taken; if the temperature over a period of time High, the resulting TSHUT gave CRU module, let it reset the entire chip, or via GPIO give PMIC.
TS-ADC Controller supports the following features:

- Support User-Defined Mode and Automatic Mode
- In User-Defined Mode, start_of_conversion can be controlled completely by software, and also can be generated by hardware.
- In Automatic Mode, the temperature of alarm(high/low temperature) interrupt can be configurable
- In Automatic Mode, the temperature of system reset can be configurable
- Support to 1 channel TS-ADC, the temperature criteria can be configurable
- In Automatic Mode, the time interval of temperature detection can be configurable
- In Automatic Mode, when detecting a high temperature, the time interval of temperature detection can be configurable
- High temperature denounce can be configurable
- -40~125°C temperature range and 5°C temperature resolution
- 10-bit SARADC up to 50KS/s sampling rate

13.2 Block Diagram

TS-ADC controller comprises with:

- APB Interface
- TS-ADC control logic

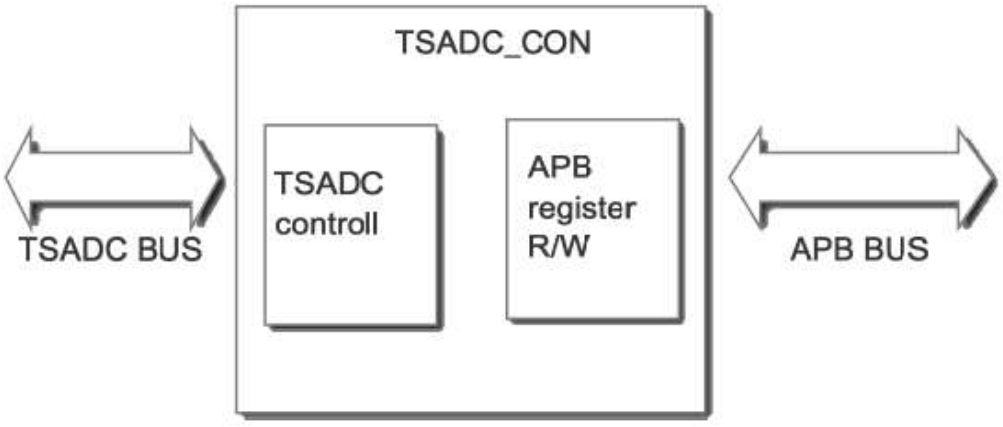

Fig.13-1 TS-ADC Controller Block Diagram

13.3 Function Description

13.3.1 APB Interface

There is an APB Slave interface in TS-ADC Controller, which is used to configure the TS-ADC Controller registers and look up the temperature from the temperature sensor.

13.3.2 TS-ADC Controller

This block is exploited to realize binary search algorithm, storing the intermediate result and generate control signal for analog block. This block compares the analog input with the voltage generated from D/A Converter, and output the comparison result to SAR and Control

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Logic Block for binary search. Three level amplifiers are employed in this comparator to provide enough gain.

13.4 Register description

13.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
TSADC_USER_CON	0x0000	W	0x00000200	The control register of A/D Converter.
TSADC_AUTO_CON	0x0004	W	0x00000000	TSADC auto mode control register
TSADC_INT_EN	0x0008	W	0x00000000	
TSADC_INT_PD	0x000c	W	0x00000000	
TSADC_DATA0	0x0020	W	0x00000000	This register contains the data after A/D Conversion.
TSADC_DATA1	0x0024	W	0x00000000	This register contains the data after A/D Conversion.
TSADC_COMP0_INT	0x0030	W	0x00000000	TSADC high temperature level for source 0
TSADC_COMP1_INT	0x0034	W	0x00000000	TSADC high temperature level for source 1
TSADC_COMP0_SHUT	0x0040	W	0x00000000	TSADC high temperature level for source 0
TSADC_COMP1_SHUT	0x0044	W	0x00000000	TSADC high temperature level for source 1
TSADC_HIGHT_INT_DEBOUNCE	0x0060	W	0x00000003	high temperature debounce
TSADC_HIGHT_TSHUT_DEBOUNCE	0x0064	W	0x00000003	high temperature debounce
TSADC_AUTO_PERIOD	0x0068	W	0x00010000	TSADC auto access period
TSADC_AUTO_PERIOD_HT	0x006c	W	0x00010000	TSADC auto access period when temperature is high
TSADC_COMP0_LOW_INT	0x0080	W	0x00000000	TSADC low temperature level for source 0
TSADC_COMP1_LOW_INT	0x0084	W	0x00000000	TSADC low temperature level for source 1

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

13.4.2 Detail Register Description

TSADC_USER_CON

Address: Operational Base + offset (0x0000)

The control register of A/D Converter.

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RO	0x0	adc_status ADC status (EOC) 0: ADC stop 1: Conversion in progress
11:6	RW	0x08	inter_pd_soc interleave between power down and start of conversion

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Bit	Attr	Reset Value	Description
5	RW	0x0	start When software write 1 to this bit , start_of_conversion will be assert. This bit will be cleared after TSADC access finishing. When TSADC_USER_CON[4] = 1'b1 take effect.
4	RW	0x0	start_mode start mode. 0: tsadc controller will asert start_of_conversion after "inter_pd_soc" cycles. 1: the start_of_conversion will be controlled by TSADC_USER_CON[5].
3	RW	0x0	adc_power_ctrl ADC power down control bit 0: ADC power down 1: ADC power up and reset
2:0	RW	0x0	adc_input_src_sel ADC input source selection(CH_SEL[2:0]). 000 : Input source 0 (SARADC_AIN[0]) 001 : Input source 1 (SARADC_AIN[1]) Others : Reserved

TSADC_AUTO_CON
Address: Operational Base + offset (0x0004)
TSADC auto mode control register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	last_tshut_2cru last_tshut_2cru for cru first/second reset TSHUT status. This bit will set to 1 when tshut is valid, and only be cleared when application write 1 to it. This bit will not be cleared by system reset.
24	RW	0x0	last_tshut_2gpio last_tshut_2gpio for hardware reset TSHUT status. This bit will set to 1 when tshut is valid, and only be cleared when application write 1 to it. This bit will not be cleared by system reset.
23:18	RO	0x0	reserved
17	RO	0x0	sample_dly_sel 0: AUTO_PERIOD is used 1: AUTO_PERIOD_HT is used
16	RO	0x0	auto_status 0: auto mode stop; 1: auto mode in progress.

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Bit	Attr	Reset Value	Description
15:14	RO	0x0	reserved
13	RW	0x0	src1_lt_en 0: do not care low temperature of source 0 1: enable the low temperature monitor of source 0
12	RW	0x0	src0_lt_en 0: do not care low temperature of source 0 1: enable the low temperature monitor of source 0
11:9	RO	0x0	reserved
8	RW	0x0	tshut_prolarity 0: low active 1: high active
7:6	RO	0x0	reserved
5	RW	0x0	src1_en 0: do not care the temperature of source 1 1: if the temperature of source 0 is too high , TSHUT will be valid
4	RW	0x0	src0_en 0: do not care the temperature of source 0 1: if the temperature of source 0 is too high , TSHUT will be valid
3:2	RO	0x0	reserved
1	RW	0x0	tsadc_q_sel temperature coefficient 1'b0:use tsadc_q as output(positive temperature coefficient) 1'b1:use(1024 - tsadc_q) as output (negative temperature coefficient) RK3399 is negative temprature coefficient, so please set this bit as 1'b1
0	RW	0x0	auto_en 0: TSADC controller works at user-define mode 1: TSADC controller works at auto mode

TSADC_INT_EN

Address: Operational Base + offset (0x0008)

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	eoc_int_en eoc_Interrupt enable. eoc_interrupt enable in user defined mode 0: disable 1: enable
15:14	RO	0x0	reserved
13	RW	0x0	lt_inten_src1 low temperature interrupt enable for src1 0: disable 1: enable

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Bit	Attr	Reset Value	Description
12	RW	0x0	lt_inten_src0 low temperature interrupt enable for src0 0: disable 1: enable
11:10	RO	0x0	reserved
9	RW	0x0	tshut_2cru_en_src1 0: TSHUT output to cru disabled. TSHUT output will always keep low . 1: TSHUT output works.
8	RW	0x0	tshut_2cru_en_src0 0: TSHUT output to cru disabled. TSHUT output will always keep low . 1: TSHUT output works.
7:6	RO	0x0	reserved
5	RW	0x0	tshut_2gpio_en_src1 0: TSHUT output to gpio disabled. TSHUT output will always keep low . 1: TSHUT output works.
4	RW	0x0	tshut_2gpio_en_src0 0: TSHUT output to gpio disabled. TSHUT output will always keep low . 1: TSHUT output works.
3:2	RO	0x0	reserved
1	RW	0x0	ht_inten_src1 high temperature interrupt enable for src1 0: disable 1: enable
0	RW	0x0	ht_inten_src0 high temperature interrupt enable for src0 0: disable 1: enable

TSADC_INT_PD

Address: Operational Base + offset (0x000c)

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	eoc_int_pd Interrupt status. This bit will be set to 1 when end-of-conversion. Set 0 to clear the interrupt.
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13	RW	0x0	lt_irq_src1 When TSADC output is lower than COMP_INT_LOW, this bit will be valid, which means temperature is low, and the application should in charge of this. write 1 to it , this bit will be cleared.
12	RW	0x0	lt_irq_src0 When TSADC output is lower than COMP_INT_LOW, this bit will be valid, which means temperature is low, and the application should in charge of this. write 1 to it , this bit will be cleared.
11:6	RO	0x0	reserved
5	RW	0x0	tshut_o_src1 TSHUT output status When TSADC output is bigger than COMP_SHUT, this bit will be valid, which means temperature is VERY high, and the application should in charge of this. write 1 to it , this bit will be cleared.
4	RW	0x0	tshut_o_src0 TSHUT output status When TSADC output is bigger than COMP_SHUT, this bit will be valid, which means temperature is VERY high, and the application should in charge of this. write 1 to it , this bit will be cleared.
3:2	RO	0x0	reserved
1	RW	0x0	ht_irq_src1 When TSADC output is bigger than COMP_INT, this bit will be valid, which means temperature is high, and the application should in charge of this. write 1 to it , this bit will be cleared.
0	RW	0x0	ht_irq_src0 When TSADC output is bigger than COMP_INT, this bit will be valid, which means temperature is high, and the application should in charge of this. write 1 to it , this bit will be cleared.

TSADC_DATA0

Address: Operational Base + offset (0x0020)

This register contains the data after A/D Conversion.

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RO	0x000	adc_data A/D value of the channel 0 last conversion (DOUT[9:0]).

TSADC_DATA1

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Address: Operational Base + offset (0x0024)
This register contains the data after A/D Conversion.

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RO	0x000	adc_data A/D value of the channel 0 last conversion (DOUT[9:0]).

TSADC_COMP0_INT
Address: Operational Base + offset (0x0030)
TSADC high temperature level for source 0

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src0 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is high. TSADC_INT will be valid.

TSADC_COMP1_INT
Address: Operational Base + offset (0x0034)
TSADC high temperature level for source 1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src1 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is high. TSADC_INT will be valid.

TSADC_COMP0_SHUT
Address: Operational Base + offset (0x0040)
TSADC high temperature level for source 0

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src0 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is too high. TSHUT will be valid.

TSADC_COMP1_SHUT
Address: Operational Base + offset (0x0044)
TSADC high temperature level for source 1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x000	tsadc_comp_src1 TSADC high temperature level. TSADC output is bigger than tsadc_comp, means the temperature is too high. TSHUT will be valid.

TSADC_HIGHT_INT_DEBOUNCE
Address: Operational Base + offset (0x0060)
high temperature debounce

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x03	debounce TSADC controller will only generate interrupt or TSHUT when temperature is higher than COMP_INT for "debounce" times.

TSADC_HIGHT_TSHUT_DEBOUNCE
Address: Operational Base + offset (0x0064)
high temperature debounce

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x03	debounce TSADC controller will only generate interrupt or TSHUT when temperature is higher than COMP_SHUT for "debounce" times.

TSADC_AUTO_PERIOD
Address: Operational Base + offset (0x0068)
TSADC auto access period

Bit	Attr	Reset Value	Description
31:0	RW	0x00010000	auto_period when auto mode is enabled, this register controls the interleave between every two accessing of TSADC.

TSADC_AUTO_PERIOD_HT
Address: Operational Base + offset (0x006c)
TSADC auto access period when temperature is high

Bit	Attr	Reset Value	Description
31:0	RW	0x00010000	auto_period This register controls the interleave between every two accessing of TSADC after the temperature is higher than COMP_SHUT or COMP_INT

TSADC_COMP0_LOW_INT
Address: Operational Base + offset (0x0080)
TSADC low temperature level for source 0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src0 TSADC low temperature level. TSADC output is lower than tsadc_comp, means the temperature is low. TSADC_LOW_INT will be valid.

TSADC_COMP1_LOW_INT
Address: Operational Base + offset (0x0084)
TSADC low temperature level for source 1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	tsadc_comp_src1 TSADC low temperature level. TSADC output is lower than tsadc_comp, means the temperature is low. TSADC_LOW_INT will be valid.

13.5 Application Notes

13.5.1 Single-sample conversion

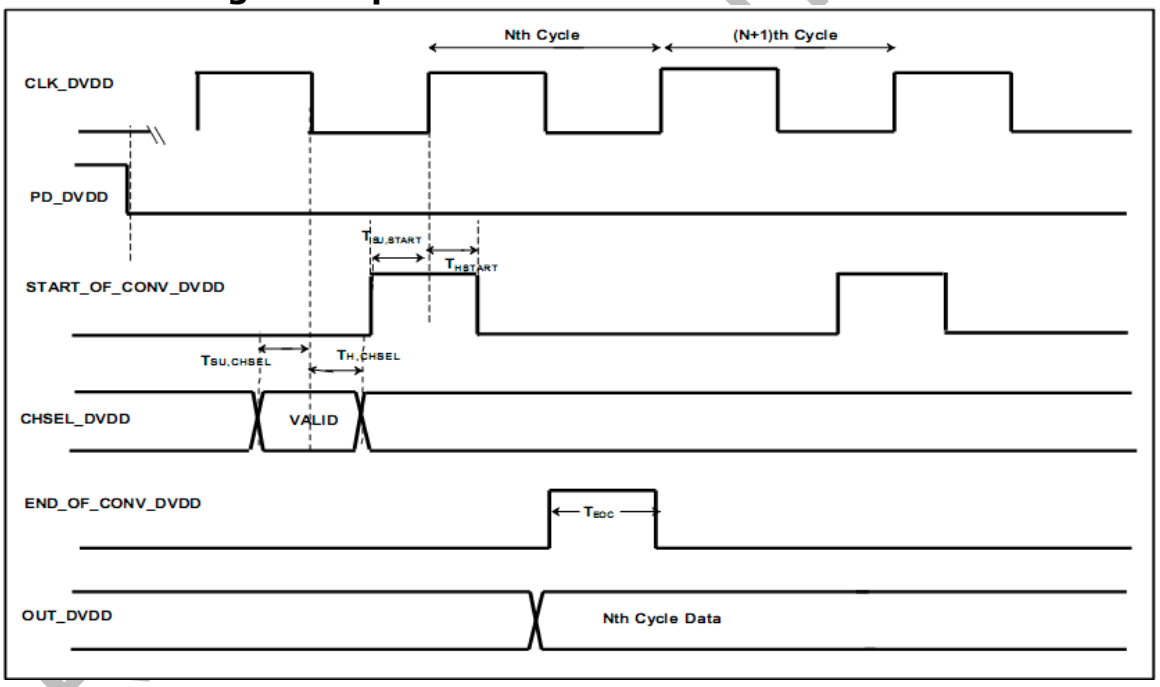

Fig. 13-2the start flow to enable the sensor and adc

13.5.2 Temperature-to-code mapping

Table 13-1 Temperature Code Mapping

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temp (C)	Code
-40	3800
-35	3792
-30	3783
-25	3774
-20	3765
-15	3756
-10	3747
-5	3737
0	3728
5	3718
10	3708
15	3698
20	3688
25	3678
30	3667
35	3656
40	3645
45	3634
50	3623
55	3611
60	3600
65	3588
70	3575
75	3563
80	3550
85	3537
90	3524
95	3510
100	3496
105	3482
110	3467
115	3452
120	3437
125	3421

Note:
Code to Temperature mapping of the Temperature sensor is a piece wise linear curve. Any temperature, code faling between to 2 give temperatures can be linearly interpolated.
Code to Temperature mapping should be updated based on sillcon results.

13.5.3 User-Define Mode

- In user-define mode, the PD_DVDD and CHSEL_DVDD are generate by setting register TSADC_USER_CON, bit[3] and bit[2:0]. In order to ensure timing between PD_DVDD and CHSEL_DVDD, the CHSEL_DVDD must be set before the PD_DVDD.
- In user-define mode, you can choose the method to control the START_OF_CONVERSION by setting bit[4] of TSADC_USER_CON. If set to 0, the start_of_conversion will be assert after "inter_pd_soc" cycles, which could be set by bit[11:6] of TSADC_USER_CON. And if start_mode was set 1, the start_of_conversion will be controlled by bit[5] of TSADC_USER_CON.
- Software can get the four channel temperature from TSADC_DATAn (n=0,1,2,3).

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13.5.4 Automatic Mode

You can use the automatic mode with the following step:

- Set TSADC_AUTO_PERIOD，configure the interleave between every two accessing of TSADC in normal operation.
- Set TSADC_AUTO_PERIOD_HT. configure the interleave between every two accessing of TSADC after the temperature is higher than COMP_SHUT or COMP_INT.
- Set TSADC_COMPn_INT(n=0,1), configure the high temperature level, if tsadc output is smaller than the value, means the temperature is high, tsadc_int will be asserted.
- Set TSADC_COMPn_SHUT(n=0,1), configure the super high temperature level, if tsadc output is smaller than the value, means the temperature is too high, TSHUT will be asserted.
- Set TSADC_INT_EN, you can enable the high temperature interrupt for all channel; and you can also set TSHUT output to gpio to reset the whole chip; and you can set TSHUT output to cru to reset the whole chip.
- Set TSADC_HIGHT_INT_DEBOUNCE and TSADC_HIGHT_TSHUT_DEBOUNCE, if the temperature is higher than COMP_INT or COMP_SHUT for "debounce" times, TSADC controller will generate interrupt or TSHUT.
- Set TSADC_AUTO_CON, enable the TSADC controller.

Chapter 14 Embedded SRAM

13.1 Overview

The Embedded SRAM is the AXI slave device, which supports read and write access to provide system fast access data storage

13.1.1 Features supported

- Provide 36KB access space
- Support security and non-security access
- Security or non-security space is software programmable
- Security space is nx4KB(up to whole memory space)
- Support 64bit AXI bus

13.1.2 Features not supported

- Don't support AXI lock transaction
- Don't support AXI exclusive transaction
- Don't support AXI cache function
- Don't support AXI protection function

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13.2 Block Diagram

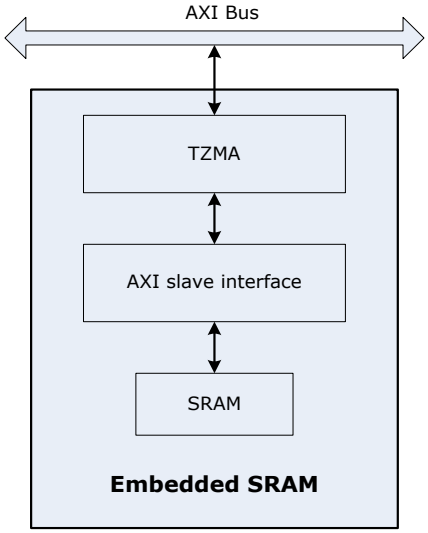

Fig.14-1 Embedded SRAM block diagram

13.3 Function Description

13.3.1 TZMA

Please refer to 7.3.3 for TZMA functional description

13.3.2 AXI slave interface

The AXI slave interface is bridge which translate AXI bus access to SRAM interface.

13.3.3 Embedded SRAM access path

The Embedded SRAM can only be accessed by Cortex-A53, DMAC_BUS and CRYPTO

13.3.4 Remap

The Embedded SRAM support remap.
Before remap, the Embedded SRAM address range is 0xff09_0000~0xff09_8fff,
After set remap, (ref Security GRF register SGRF_SCON0, bit[10]), the system can still access the Embedded SRAM by the old address. at same time, the system also can access the Embedded SRAM by the new address 0xffff_0000 ~ 0xffff_8fff (include the bootaddr)

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Chapter 15 GPU (Graphics Process Unit)

14.1 Overview

The GPU is a hardware accelerator for 2D and 3D graphics systems. Its triangle rate can be 30 Mtris/s, pixel rate can be 300Mpix/s@300MHz.
The GPU supports the following graphics standards:

- OpenGL ES 2.0
- OpenGL ES 1.1
- OpenVG 1.1
- EGL 1.5

The GPU consists of:

- 2 Pixel Processors (PPs)
- 1 geometry Processor (GP)
- 2 Level2 Cache controller (L2)
- 1 Memory Management Unit (MMU) for each GP and PP included in the GPU

The GPU contains a 32-bit APB bus and 2 128-bit AXI bus. CPU configures GPU through APB bus, GPU read and write data through AXI bus.

14.2 Block Diagram

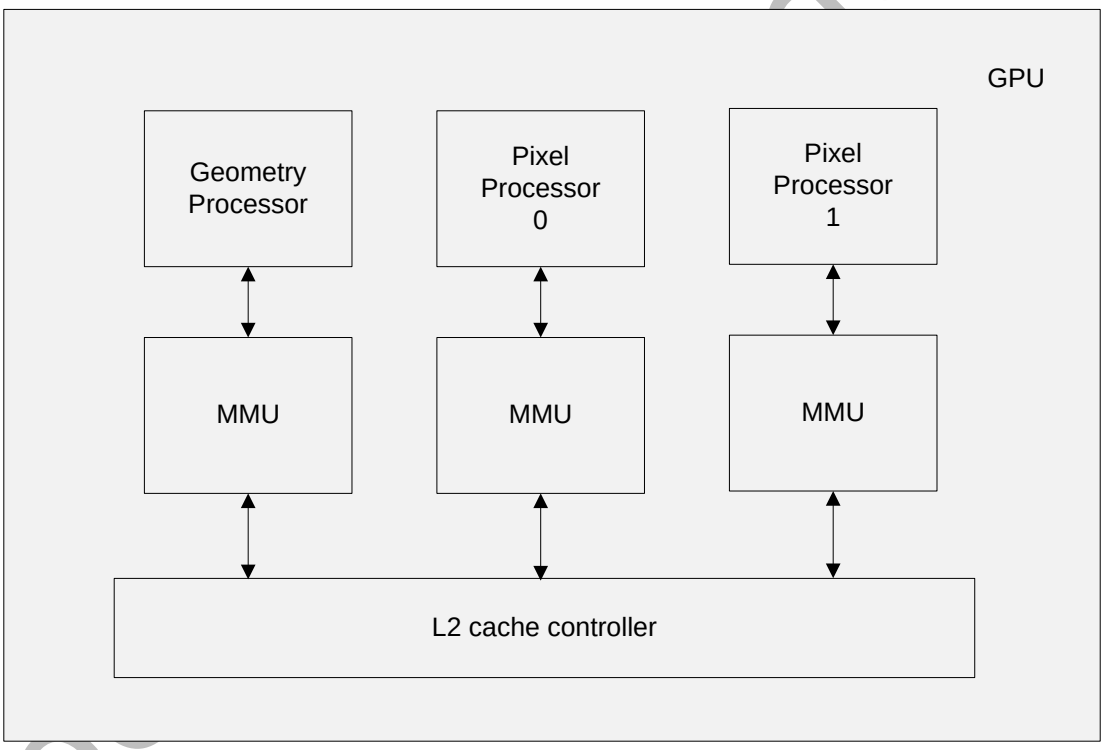

Fig. 15-1GPU block diagram

The GPU contains 1 geometry processor, 2 pixel processors, 3 MMU and 2 L2 cache controller.
The pixel processor features are:

- each pixel processor used processes a different tile, enabling a faster turnaround
- programmable fragment shader
- alpha blending
- complete non-power-of-2 texture support
- cube mapping
- fast dynamic branching
- fast trigonometric functions, including arctangent
- framebuffer blend with destination Alpha
- indexable texture samplers
- line, quad, triangle and point sprites
- no limit on program length
- perspective correct texturing

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- point sampling, bilinear, and trilinear filtering
- programmable mipmap level-of-detail biasing and replacement
- stencil buffering, 8-bit
- two-sided stencil
- unlimited dependent texture reads
- 4-level hierarchical Z and stencil operations
- 4-bit per texel compressed texture format
- Up to 512 times Full Scene Anti-Aliasing (FSAA). 4x multisampling by 128x supersampling.

The geometry processor features are:

- two programmable vertex shaders
- flexible input and output formats
- autonomous operation tile list generation
- indexed and non-indexed geometry input
- primitive constructions with points, lines, triangles and quads.

The L2 cache controller features are:

- 64KB
- 4-way set-associative
- supports up to 32 outstanding AXI transactions
- implements a standard pseudo-LRU algorithm
- cache line and line fill burst size is 64 bytes
- supports eight to 64bytes uncached read bursts and write bursts
- 128-bit interface to memory sub-system
- support for hit-under-miss and miss-under-miss with the only limitation of AXI ordering rules.

The MMU features are:

- accesses control registers through the bus infrastructure to configure the memory system.
- each processor has its own MMU to control and translate memory accesses that the GPU initiates.

APB broadcast features are:

- configuration of multiple PPs in parallel
- the ability to use a single read to poll multiple PP interrupts.

DMA features are:

- The register DMA reduces the number of required APB transactions by configuring the rest of the GPU using configuration data stored in main memory. The driver writes the configuration data for each frame to main memory while the previous frame is rendered. The register DMA unit performs the setup after the previous frame is completed. This reduces the system overhead between frames, and reduces the workload for the CPU. The DMA simplifies transfer of GPU commands and data from memory to the pixel processors. A counter in the DMA determines how many register write packages are processed.

Load balancing features are:

- The address of the tile lists and the number of tiles in the framebuffer is programmable. The dynamic load balancing unit assigns a new tile to the different pixel processors ecause they complete the previous tile. The dynamic load balancing unit iterates over the frame in a Z-order pattern starting at the first tile for pixel processor 0-3 and the last tile for pixel processor 4-7. This ensures that the pixel processors connected to the same level 2 cache process nearby tiles. This improves cache efficiency. This also balances the workload for the different pixel processors regardless of the frame content.

14.3 Register Description

The GPU base addressis 0XFF30_0000.

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14.4 Interface Description

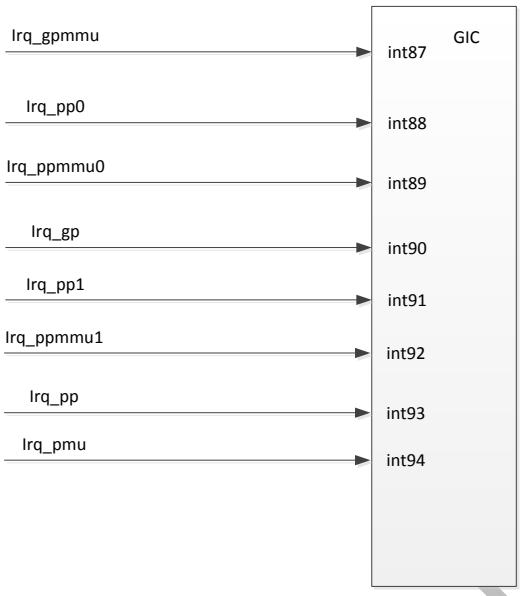

Fig. 15-2 GPU interrupt connection

Pmu interrupt keeps 0 because GPU is not configured to support PMU function.

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Chapter 16 Multi-format Video Encoder And Decoder

15.1 Overview

The multi-format video codec (referred as codec in the following paragraph) supports three main world-wide advanced video coding standard (H264/VP9/HEVC) with high decoding performance @4K and h264/jpeg encoding.
The codec is connected to the AHB bus through an AHB slave and the AXI bus through an AXI master. The register configuration is fed into the decoder through the AHB slave interface while the large data such as stream data are transacted between DDR and the decoder through the AXI master interface.
In order to improve large data transaction performance, codec embeds MMU (memory management unit) and supports the cacheable bus operation.
The codec supports the following features:

- Supports HEVC decoding
  - Main and Main10 Profile up to Level 5.1 High Tier: 4096x2304@60 fps
  - Supports frame timeout interrupt , frame finish interrupt, bus error interrupt and bitstream error interrupt
  - Supports RLC write mode, RLC mode and Normal Mode
- Supports H264 decoding
  - The following profiles up to Level 5.1 : 4096x2304 @30fps
    - Baseline Profile
    - Main Profile
    - High Profile
    - High 10 profile
    - High 4:2:2 Profile(the MBAFF feature is not supported)
  - Supports frame timeout interrupt, frame finish interrupt and bitstream error interrupt, buffer empty interrupt
  - Supports slice by slice or random size stream decoding
  - Supports error-mode decoding and error info output
  - Supports RLC write mode, RLC mode and Normal Mode
- Supports VP9 decoding
  - Profile 0: 4096x2304 @30fps
  - Supports frame timeout interrupt, frame finish interrupt and bitstream error interrupt
  - Supports RLC mode and Normal Mode
- MMU embedded with MMU interrupt support
- Supports MPEG-4 decoding
  - 60fps at 1920x1088
  - Simple Profile, levels 0-6
  - Advanced Simple Profile, levels 0-5
- MPEG-2/ MPEG-1
  - 60fps at 1920x1088
  - Main Profile, low, medium and high levels
- AVS/ AVSPLUS
  - 60fps at 1920x1080
- Supports JPEG decoding
  - 48x48 to 8176x8176(66.8 Mpixels)，Step size 8 pixels
  - Baseline interleaved, and supports ROI (region of image) decode
- Built-in post processor in H.264(1080p) decoder supports
  - Stand-alone mode: rotation, RGB conversion, scaling, dithering
  - Pipe-ling mode:, RGB conversion, scaling, dithering and alpha blending
- Supports encoding of the following standards:
  - H.264: up to HP level 4.1
    - Built-in pre-processor in common video H.264 encoder supports rotation, YCbCr conversion
  - HEVC encoder : Main Profile @ L4.1
    - Main Profile @ L4.1 High-tier
    - Performance is 30 fps at 1920 x1080
    - I/P slices

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- Supported Prediction Unit(PU) size : 16x16, 8x8
- Supported Transform Unit(TU) size : 16x16 to 8x8
- Multi slice : Independent slice segment and dependent slice segment
- High performance CABAC encoding
- In-loop deblocking filtering
- Loop filtering across slice
- Max resolution: 8192x4096
- Min resolution: 256x128
- Handling multi-instances
- Lossless coding
- Noise reduction
- Rate Contro:VBR, CBR and ABR,ROI support

- JPEG: Baseline (DCT sequential)

15.2 Block Diagram

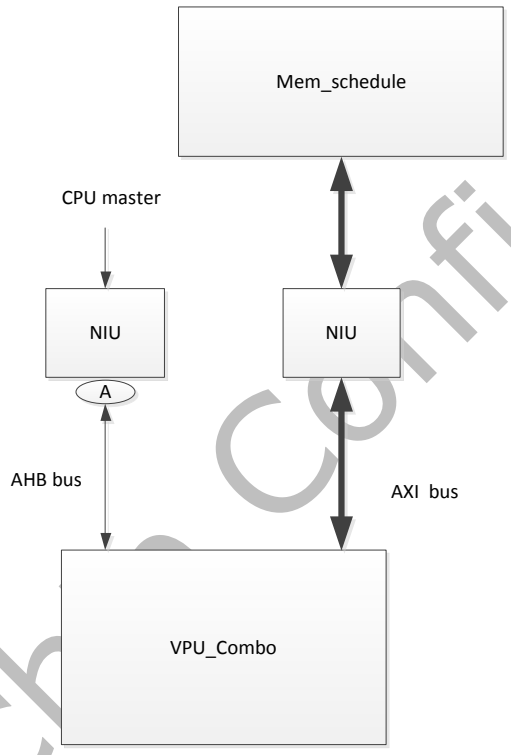

Fig. 16-1 Decoder in SOC

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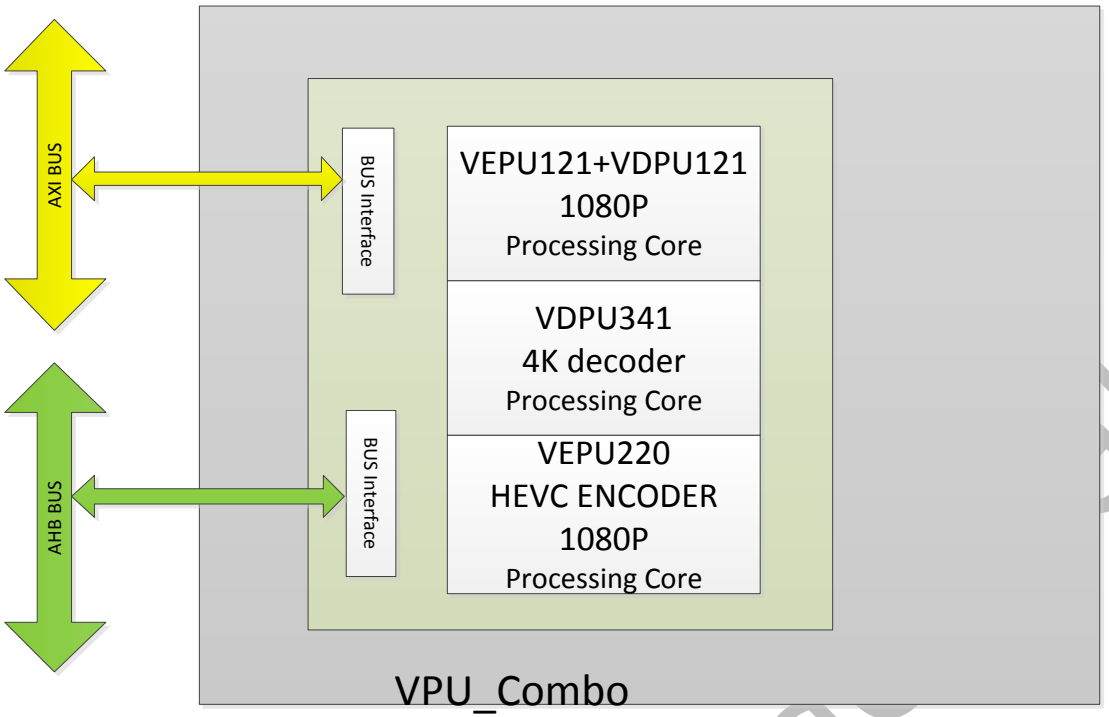

Fig. 16-2 VPU Block Diagram

As shown in the figures above, CPU accesses to the decoder register bank through AHB bus. Bitstream and other necessary data are fed into processing core though AXI read channel, and after several steps of decoding process, decoded pictures and other information data are transferred to designated location in the DDR through AXI write channel.

VDPU121 1080P processing core support multi-format decoder,such as mpeg4,avsplus and so on.

VEPU121 1080P processing core support h.264 and jpeg encoder.

VDPU341 4K decoder processing core can support h.265/vp9/h.264 video standard decoder,max solution size is 4096x2304.

VEPU220 HEVC encoder processing core can support hevc 1080p encoder.

Please note that VEPU220 share some of memory with VEPU121,so this two processing core can’t be work together.

15.3 Video frame format

This chapter describes different input and output video frame formats supported by VCODEC. Each function module has its own supported video frame formats, and this chapter describes all the video frame formats.

15.3.1 YCbCr 4:2:0 Planar Format

In the planar format, each video sample component forms one memory plane. Within one plane, the data has to be stored linearly and contiguously in the memory as shown in Fig. 16-3. The luminance samples are stored in raster-scan order (Y0Y1 Y2Y3 Y4….). The chrominance samples are stored in two planes also in raster scan order (Cb0Cb1 Cb2Cb3 Cb4…. and Cr0Cr1 Cr2Cr3 Cr4….). In this format each pixel takes 12 bits of memory.

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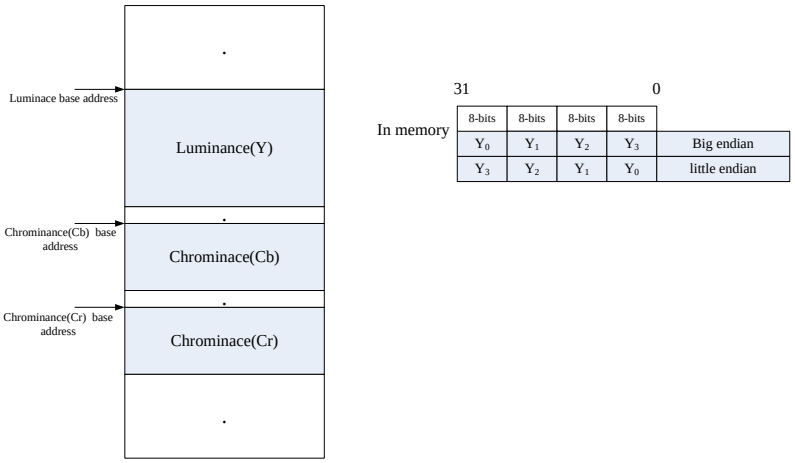

Fig. 16-3 VCODEC YCbCr 4:2:0 planar format

15.3.2 YCbCr 4:2:0 Semi-Planar format

In semi-planar YcbCr4:2:0 format the luminance samples from one plane in memory, and chrominance samples from another. Within one plane, the data has to be stored linearly and contiguously in the memory. The luminance pixels are stored in raster-scan order (Y0Y1 Y2Y3 Y4….). The interleaved chrominance CbCr samples are stored in raster-scan order in memory as Cb0Cr0 Cb1Cr1 Cb2 Cr2 Cb3Cr3 Cb4 Cr4….

Semi-Planar format supports both progressive and interlaced format as presented in Fig. 16-4.The interlaced format may be alternative line or each line.

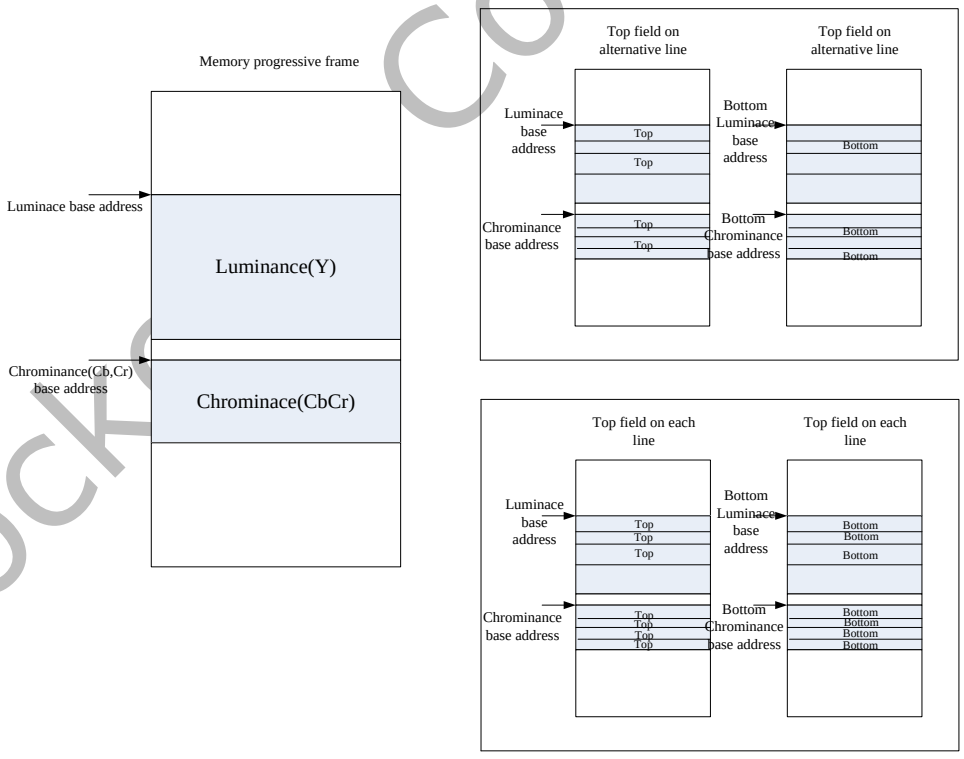

Fig. 16-4 VCODEC YCbCr 4:2:0 Semi-planar format

15.3.3 YCbCr 4:2:0 Tiled Semi-Planar Format

Like the YCbCr 4:2:0 semi-planar format, the tiled semi-planar format is also organized in the memory on two separate planes. The difference between these formats is that in tiled format

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the pixel samples are not anymore in raster-scan order but are stored macroblock(16x16 pixels) by macroblock. The samples of each macroblock are stored in consecutive addresses and the macroblocks are ordered from left to right and from top to down as Fig. 16-5. When this format used as input data format, it causes the lowest bus load to the system as there is minimal amount of non-sequential memory addressing required when reading the input data to the post-processor.

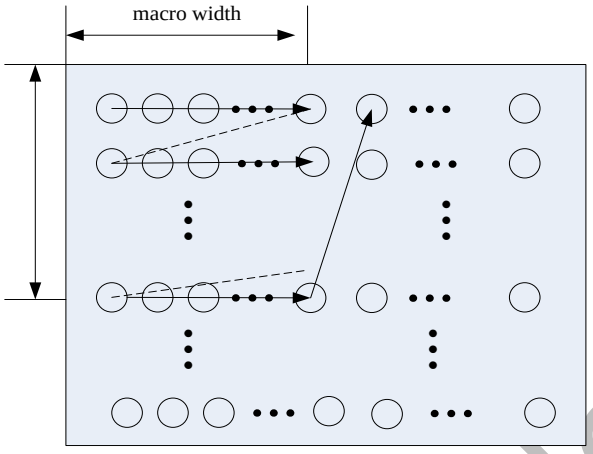

Fig. 16-5 VCODEC Tile scan mode

15.3.4 YCbCr 4:2:2 Interleaved Format

In the interleaved YCbCr 4:2:2 format the pixel samples from a single plane in which the data has to be stored linearly and contiguously as shown in Fig. 16-6. The pixel data is in raster scan order and the chrominance samples are interleaved between the luminance samples as Y0Cb0 Y1Cr0 Y2 Cb1 Y3Cr1 Y4 Cb2…. YCrCb, CbYCrY and CrYCbY component orders are supported also. In this format, each pixel takes 16 bits in the memory.

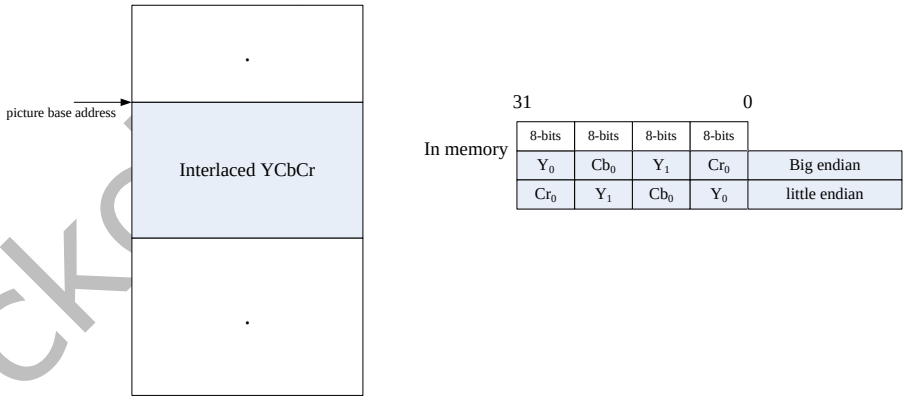

Fig. 16-6 VCODEC YCbCr4:2:2 Interleaved format

15.3.5 AYCbCr 4:4:4 Interleaved Format

In the interleaved YcbCr 4:2:2 format, the pixel samples from a single plane in which the data has to be stored linearly and contiguously as show in Fig. 16-7. The pixel data is in raster scan order and the chrominance and alpha channel samples are interleaved between the luminance samples as A0Y0 Cb0Cr0 A1 Y1 Cb1Cr1….

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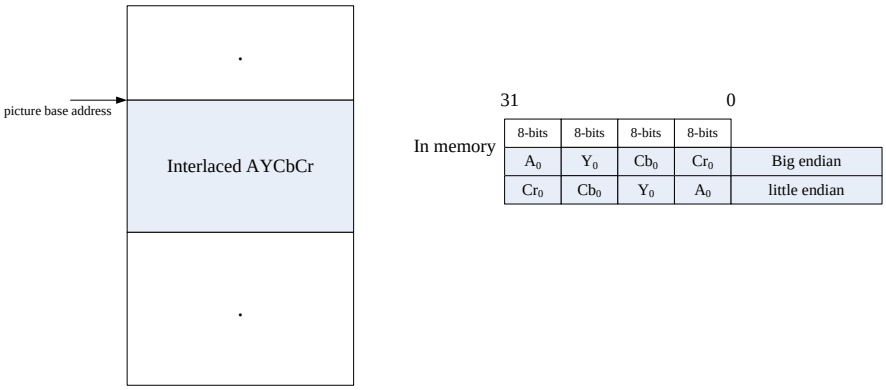

Fig. 16-7 VCODEC AYCbCr 4:4:4 Interleaved format

15.3.6 RGB 16bpp Format

In this format each pixel is represented by 16 or less bits containing the red, blue and green samples. There are several 16bpp formats which use different number of bits for each sample. For example the RGB 5-5-5 format uses 5 bits for each sample and 1 bit is left unused or can represent a transparency flag, where RGB 5-6-5 uses 6 bits for the G sample and 5 bits for R and B samples. Common for all 16bpp types is that two pixels fit into one 32-bit space.

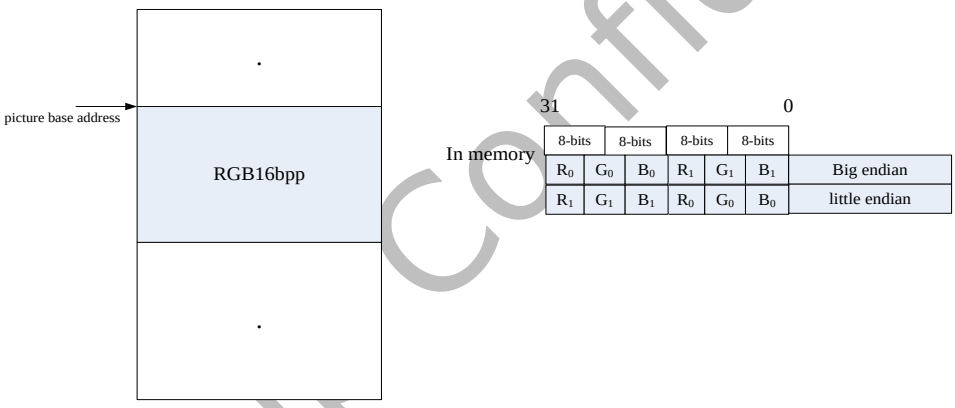

Fig. 16-8 VCODEC RGB 16bpp format

15.3.7 RGB 32bpp Format

Any RGB format that has its pixels represented by more than 16bits each is considered to be of 32bpp type. Typically in this format each pixel is represented by three bytes containing a red, blue and green sample and a 4th byte which can be empty or hold an alpha blending value. Common for all 32bpp types is that only one pixel fit into one 32-bit space. The data has to be stored linearly and contiguously in the memory.

15.3.8 HEVC encoder frame format

A frame buffer is configured with the following parameters:

- Alignment
  - Luminance frame buffer base address in 16-byte alignment
  - Cb frame buffer base address in 16-byte alignment
  - Cr fame buffer base address in 16-byte alignment: In case of NV12 or NV21, a base address for Cr is meaningless because a base address for Cb is used to store or load the interleaved Cb/Cr samples.
  - In 2D mode, the height of frame buffer is the multiple of 32.

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- Stride
  - A stride should be multiple of 16 for each luma and chroma.
- Planar Format

There are three memory in this format - Y, Cb and Cr.

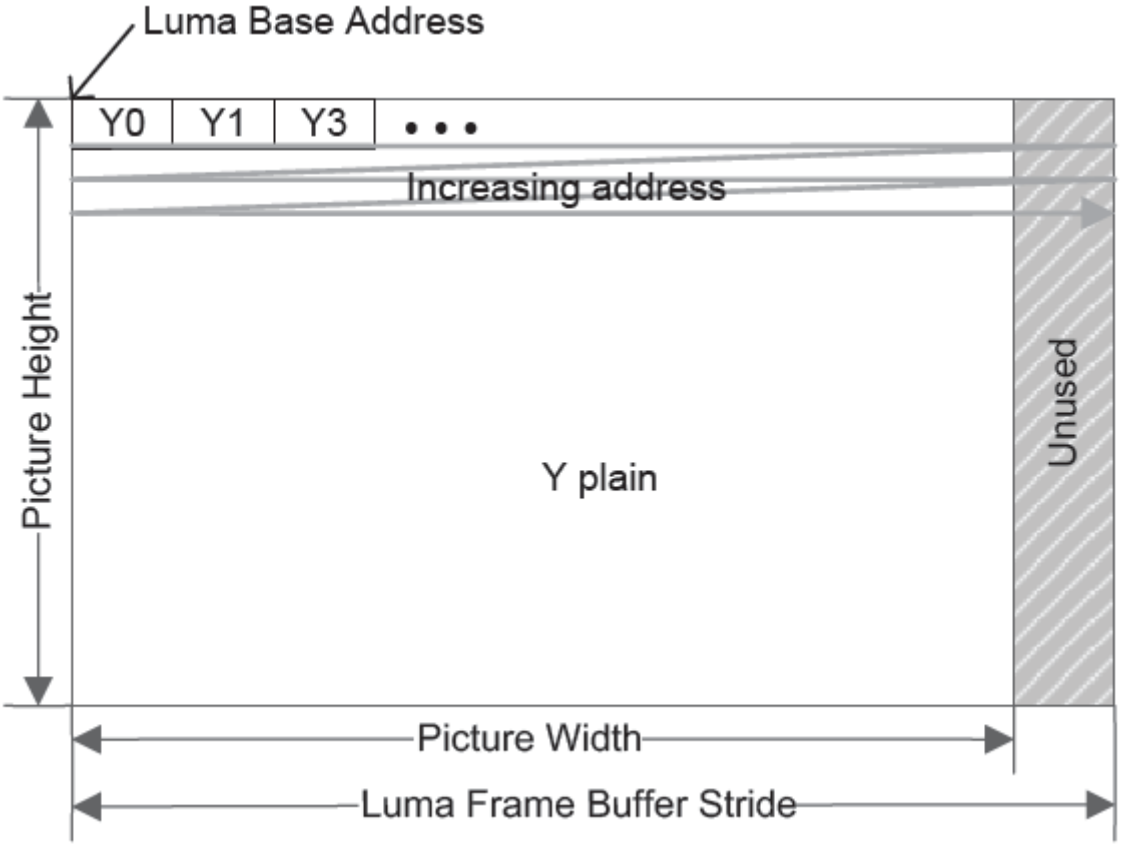

Fig. 16-9 Luma in Planar Format

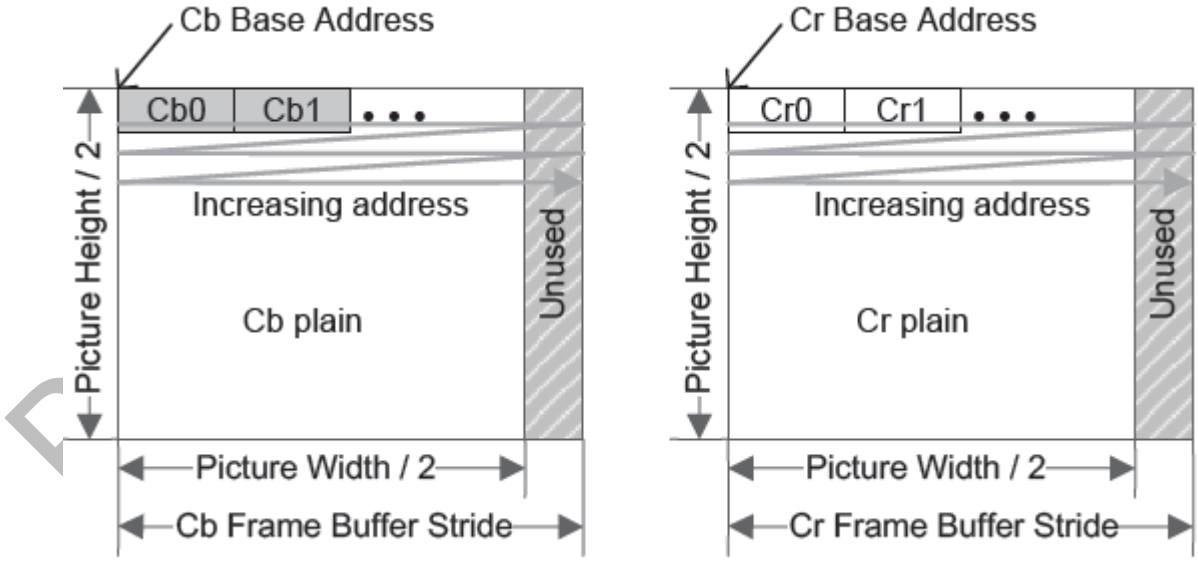

Fig. 16-10 Cb and Cr in Planar Format

- NV12 and NV21 Format

These have 2 plains which are luma and chroma memory. The luma plain of NV12 and NV21 is the same as the YCbCr 4:2:0 Planar Format. The chroma plain has interleaved Cb and Cr in chroma plain. The order of chroma sample in NV12 is Cb, Cr, Cb, Cr, … and NV21 has the opposite sample order such as Cr, Cb, Cr, Cb, …

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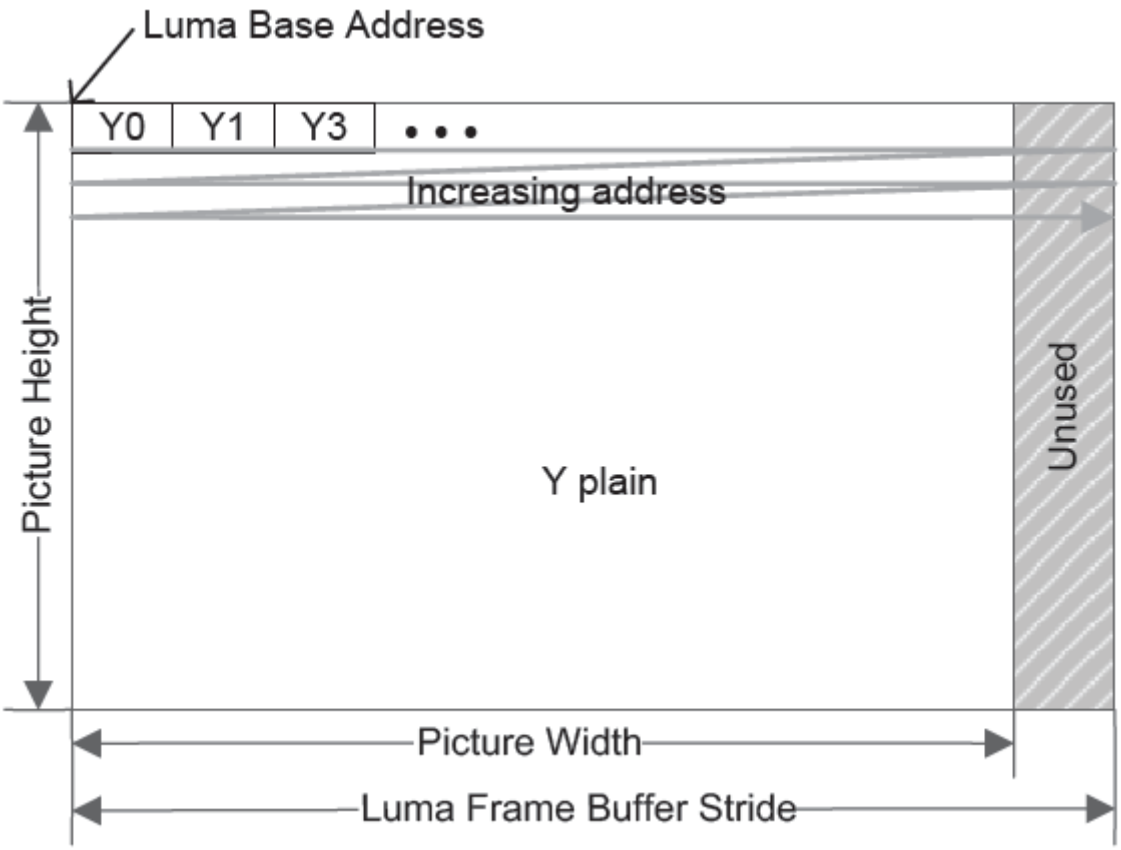

Fig. 16-11 Luma in NV12 and NV21 Format

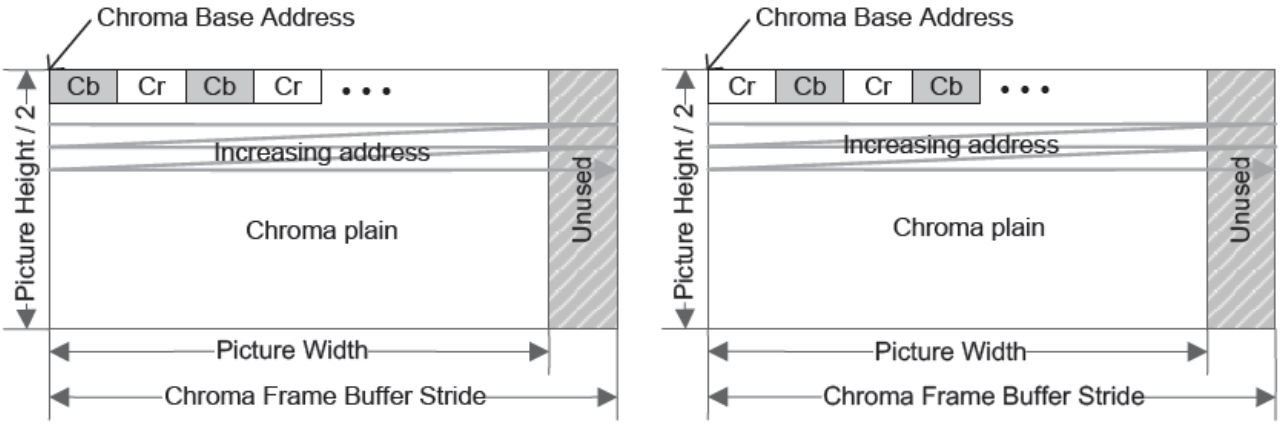

Fig. 16-12 Cb and Cr in Planar Mode

- Packed Format

This is YUV4:2:2 format and has only 1 memory for Y, Cb and Cr. The order of samples in packed mode is Y,Cb,Y,Cr,… The width of plain in packed mode is double of picture width because Y and C exist in same plain. Even though packed mode has YUV4:2:2 image, hevc encodes YUV4:2:0 image by ignoring the even line chroma format.

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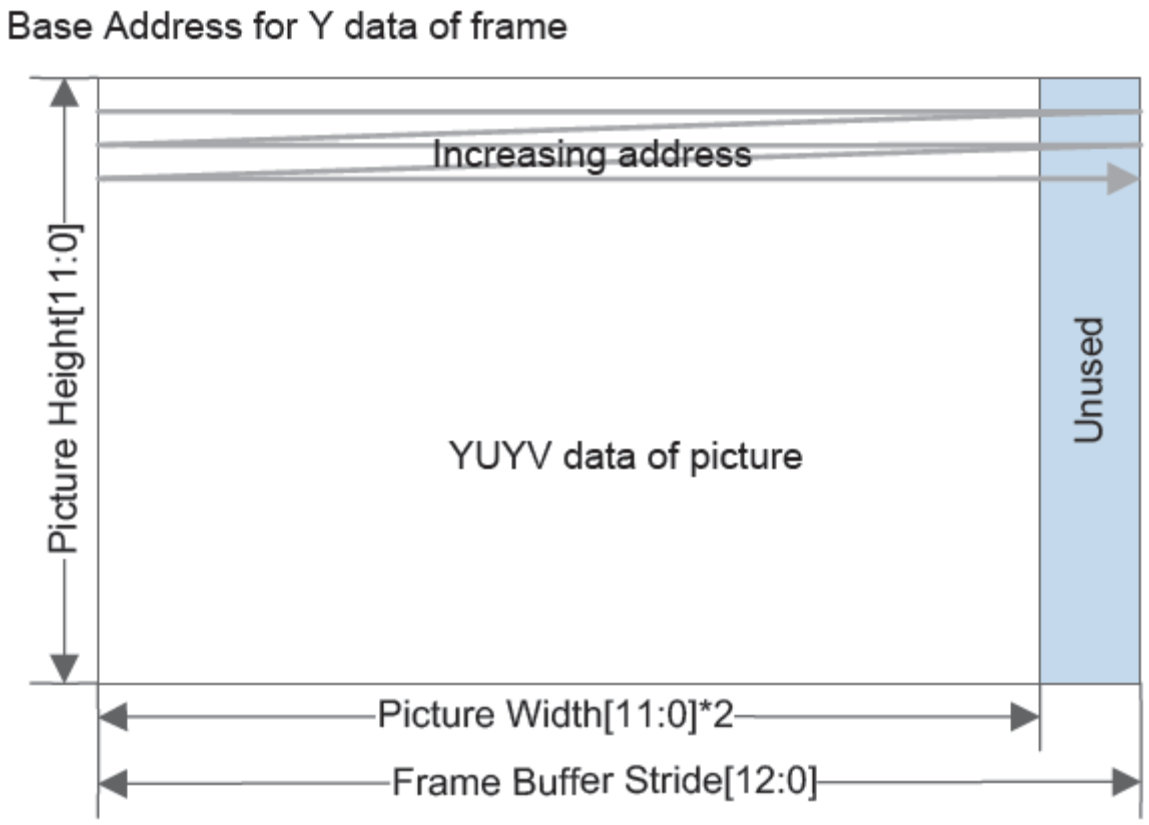

Fig. 16-13 Packed Format

- 2D Format

2D mode has 2 plains, Y and C. In the Y plain, the address increases in vertical direction until 32 lines as shown in Figure 3.7, "2D Format". Long burst (128-byte) access of 2D mode can improve bandwidth efficiency.

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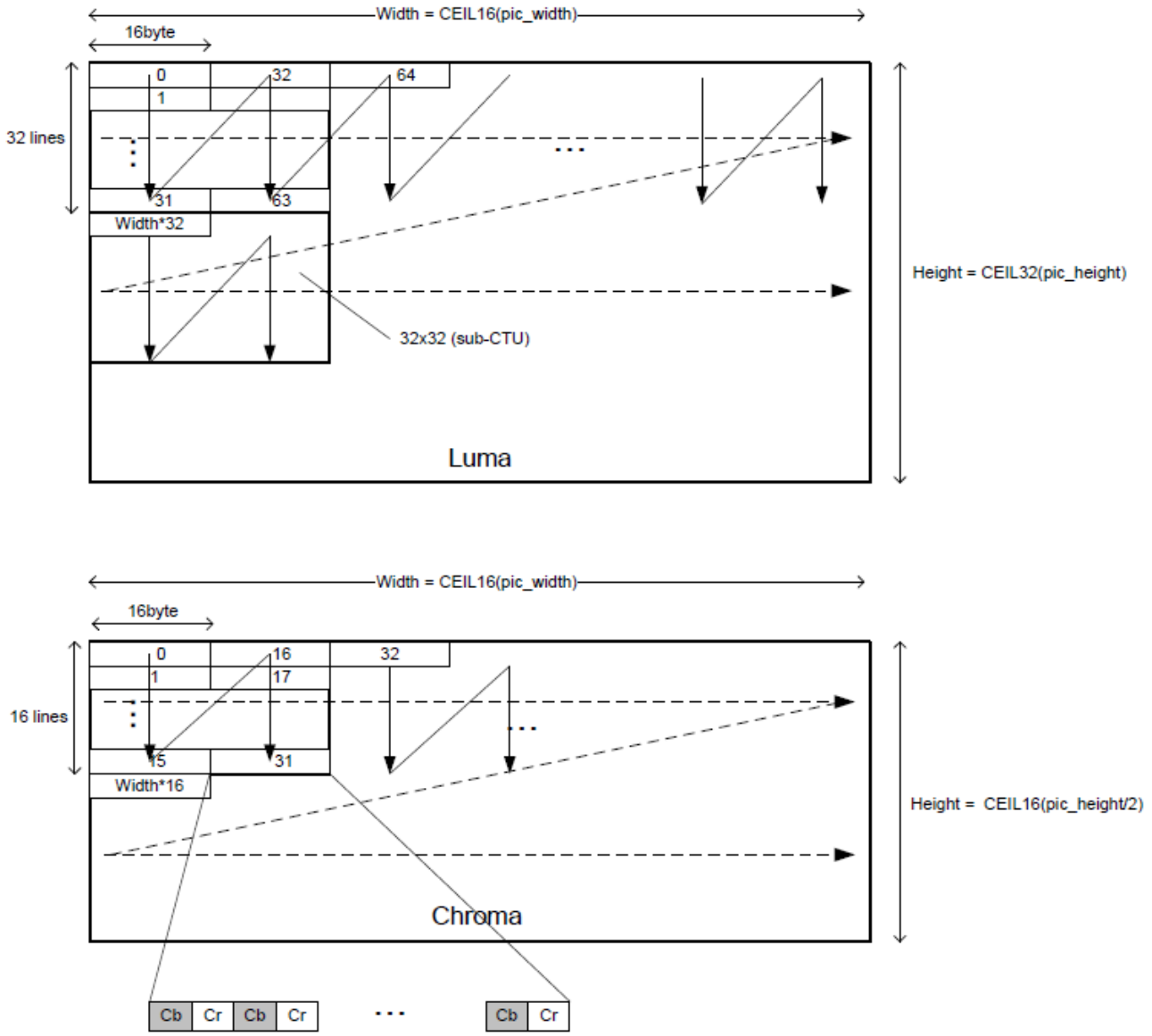

Fig. 16-14 2D Format

15.4 Function Description

15.4.1 HEVC Standard

High Efficiency Video Coding (HEVC) is a video compression standard, a successor to H.264/MPEG-4 AVC (Advanced Video Coding), that was jointly developed by the ISO/IEC Moving Picture Experts Group (MPEG) and ITU-T Video Coding Experts Group (VCEG) as ISO/IEC 23008-2 MPEG-H Part 2 and ITU-T H.265. MPEG and VCEG established a Joint Collaborative Team on Video Coding (JCT-VC) to develop the HEVC standard.
HEVC was designed to substantially improve coding efficiency compared to H.264/MPEG-4 AVC HP, i.e. to reduce bitrate requirements by half with comparable image quality, at the expense of increased computational complexity. HEVC was designed with the goal of allowing video content to have a data compression ratio of up to 1000:1. Depending on the application requirements HEVC encoders can trade off computational complexity, compression rate, robustness to errors, and encoding delay time.Two of the key features where HEVC was improved compared to H.264/MPEG-4 AVC was support for higher resolution video and improved parallel processing methods.

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15.4.2 H264 Standard

H.264 or MPEG-4 Part 10, Advanced Video Coding (MPEG-4 AVC) is a video compression format that is currently one of the most commonly used formats for the recording, compression, and distribution of video content. The final drafting work on the first version of the standard was completed in May 2003, and various extensions of its capabilities have been added in subsequent editions.
H.264/MPEG-4 AVC is a block-oriented motion-compensation-based video compression standard developed by the ITU-T Video Coding Experts Group (VCEG) together with the ISO/IEC JTC1 Moving Picture Experts Group (MPEG). The project partnership effort is known as the Joint Video Team (JVT). The ITU-T H.264 standard and the ISO/IEC MPEG-4 AVC standard (formally, ISO/IEC 14496-10 – MPEG-4 Part 10, Advanced Video Coding) are jointly maintained so that they have identical technical content.

15.4.3 VP9 Standard

VP9 is an open and royalty free video coding format being developed by Google. VP9 is a successor to VP8. Chromium, Chrome, Firefox, and Opera support playing VP9 video format in the HTML5 video tag. Development of VP9 started in Q3 2011. One of the goals for VP9 is to reduce the bit rate by 50% compared to VP8 while having the same video quality. Another goal for VP9 is to improve it to the point where it would have better compression efficiency than High Efficiency Video Coding.

15.4.4 MMU

The MMU divides memory into 4KB pages, where each page can be individually configured. For each page the following parameters are specified:

- Address translation of virtual memory, this enables the processor to work using address that differ from the physical address in the memory system.
- The permitted types of accesses to that page. Each page can permit read, write, both, or none.

The MMU use 2-level page table structure:
1. The first level, the page directory consists of 1024 directory table entries(DTEs), each pointing to a page table.
2. The second level, the page table consists of 1024 page table entries(PTEs), each pointing to a page in memory.

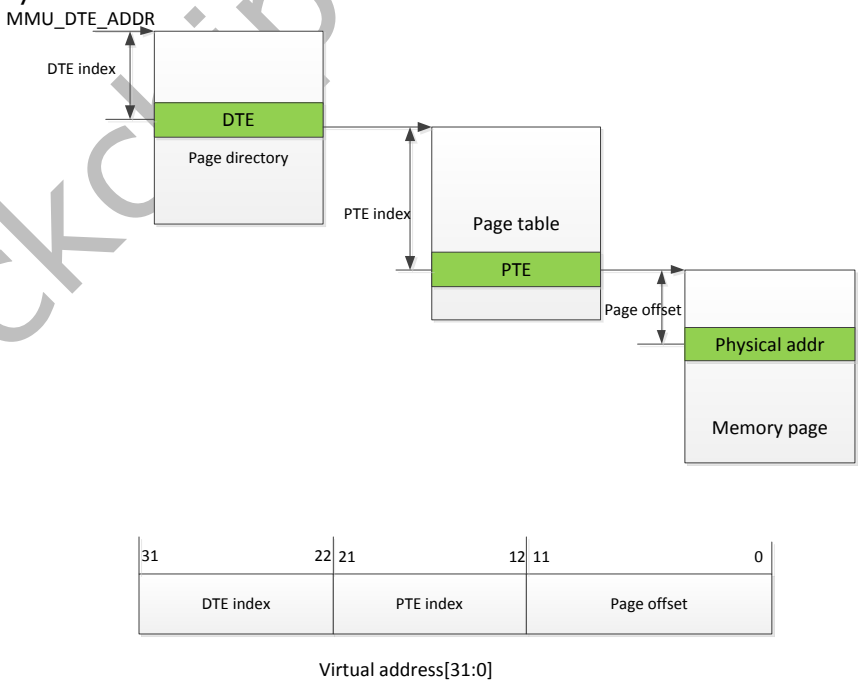

Fig. 16-15 structure of two-level page table

15.4.5 HEVC Working Mode

There are there working modes to be selected for HEVC decoder: RLC Mode, RLC Write Mode, and Normal Mode.

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The key differences among three working modes are whether CABAC module and Post-CABAC module are involved into the hardware decoding process.
For RLC mode, CABAC are bypassed and the input bitstream to the Post-CABAC module should be already decoded.
For RLC write mode, the decoded results by CABAC are output to the DDR, and the following decoding processes are stopped.
As for the normal mode, all the modules are involved into the decoding process, and complete decoding results are output. Normally, this mode should be selected.

15.4.6 H264 Working Mode

There are there working modes which can be chosen when decoding HEVC: RLC Mode, RLC Write Mode, and Normal Mode.
The key differences among three working modes are whether stream-parse module and post-stream-parse modules are involved into the hardware decoding process.
For RLC mode, stream-parse module is bypassed and the input data to the Post-CABAC module should be already parsed which can be recognized by the post-stream-parse modules.
For RLC write mode, the parsed results by stream-parse module are output to the DDR, and the following decoding processes are stopped.
As for the normal mode, all the modules are involved into the decoding process, and complete decoding results are output. Normally, this mode should be selected.

15.4.7 VP9 Working Mode

There are two working modes to be selected for VP9 decoder: RLC Mode, Normal Mode.
The key differences among these two working modes are whether strmd module is involved into the hardware decoding process.
For RLC mode, strmd is bypassed and the input dec_ctrl cmd and inter cmd to the Post-strmd module should be already decoded. The dec_ctrl cmd should be put on the address swreg4_strm_rlc_base while the inter cmd should be put on the address swreg34_vp9_intercmd_base.
As for the normal mode, all the modules are involved into the decoding process, and complete decoding results are output. Normally, this mode should be selected.

15.4.8 Input Data Format for HEVC

When the decoder works in RLC mode, the parsed stream info (which can be called dec_ctrl commands) must be needed, which should be 128bit aligned.
When the decoder works in normal mode or RLC write mode, the bitstream, tbl ,pps and rps are needed for decoding.
1. bitstream
The bitstream must be 128bit align.
2. hevc table
hevc table is used for hevc context initialization, and it contains 156 units of 160-byte data.
3. hevc pps
Hevc pps contains 64 units of 80-byte data.

Table 16-1 sps format

video_parameter_set_id	4bit
seq_parameter_set_id	4bit
chroma_format_idc	2bit
pic_width_in_luma_samples	13bit
pic_height_in_luma_samples	13bit
bit_depth_luma	4bit
bit_depth_chroma	4bit
log2_max_pic_order_cnt_lsb	5bit
log2_diff_max_min_luma_coding_block_size	3bit
log2_min_luma_coding_block_size	3bit
log2_min_transform_block_size	3bit

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log2_diff_max_min_transform_block_size	2bit
max_transform_hierarchy_depth_inter	3bit
max_transform_hierarchy_depth_intra	3bit
scaling_list_enabled_flag	1bit
amp_enabled_flag	1bit
sample_adaptive_offset_enabled_flag	1bit
pcm_enabled_flag	1bit
pcm_sample_bit_depth_luma	4bit
pcm_sample_bit_depth_chroma	4bit
pcm_loop_filter_disabled_flag	1bit
log2_diff_max_min_pcm_luma_coding_block_size	3bit
log2_min_pcm_luma_coding_block_size	3bit
num_short_term_ref_pic_sets	7bit
long_term_ref_pics_present_flag	1bit
num_long_term_ref_pics_sps	6bit
sps_temporal_mvp_enabled_flag	1bit
strong_intra_smoothing_enabled_flag	1bit
Transform_skip_rotation_enabled_flag	1bit
Transform_skip_context_enabled_flag	1bit
intra_block_copy_enabled_flag	1bit
residual_dpcm_intra_enabled_flag	1bit
residual_dpcm_inter_enabled_flag	1bit
extended_precision_processing_flag	1bit
intra_smoothing_disabled_flag	1bit
Reserved	32bit

Table 16-2 pps format

pps_pic_parameter_set_id	6bit
pps_seq_parameter_set_id	4bit
dependent_slice_segments_enabled_flag	1bit
output_flag_present_flag	1bit
num_extra_slice_header_bits	3bit//?
sign_data_hiding_flag	1bit
cabac_init_present_flag	1bit
num_ref_idx_l0_default_active	4bit
num_ref_idx_l1_default_active	4bit
init_qp_minus26	6bit
constrained_intra_pred_flag	1bit
transform_skip_enabled_flag	1bit
cu_qp_delta_enabled_flag	1bit
Log2MinCuQpDeltaSize	3bit
pps_cb_qp_offset	5bit
pps_cr_qp_offset	5bit
pps_slice_chroma_qp_offsets_present_flag	1bit
weighted_pred_flag	1bit
weighted_bipred_flag	1bit
transquant_bypass_enabled_flag	1bit

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tiles_enabled_flag	1bit
entropy_coding_sync_enabled_flag	1bit
pps_loop_filter_across_slices_enabled_flag	1bit
loop_filter_across_tiles_enabled_flag	1bit
deblocking_filter_override_enabled_flag	1bit
pps_deblocking_filter_disabled_flag	1bit
pps_beta_offset_div2	4bit
pps_tc_offset_div2	4bit
lists_modification_present_flag	1bit
log2_parallel_merge_level	3bit
slice_segment_header_extension_present_flag	1bit
log2_transform_skip_max_size_minus2	3bit
If(tiles_enable_flag) {	
num_tile_columns	5bit
num_tile_rows	5bit
column_width[20]	8 * 20bits
column_height[22]	8 * 22 bit
}	
Scaleing_address	32bit

4. HEVC rps
HEVC rps contains a number of slice data composed by 2 units of 32-byte data.

15.4.9 Input Data for H264

When in RLC mode, dec_ctrl cmds are needed, and they all should be 128bit aligned.
When in normal mode or direct write mode, bitstream, tbl ,pps and rps are need, which should be 128bit aligned.
1. bitstream
The bitstream data should be put on the address, which are set as swreg4_strm_rlc_base. The bitstream data should be 128bit aligned, and there are three modes to fetch bitstream: frame by frame, slice by slice, and random.
For frame by frame mode, swreg2_sysctrl[25] must be configured as 1'b0,and swreg2_sysctrl[28] must be configured as 1'b0. The bitstream of whole frame must be send to the decoder completely.
For slice by slice mode, swreg2_sysctrl[25] must be configured as 1'b0, swreg2_sysctrl[28] must be configured as 1'b1, and swreg1_int[6] must be configured as 1'b1. The bitstream will be split into one or more slices. When swreg1_int[16] is 1,we must send the next slice pack to the decoder, and then configure swreg1_int[10] as 1'b1. Repeat step mention above until the whole slice pack is sent completely.
For random mode, swreg2_sysctrl[25] must be configured as 1'b1, and swreg1_int[6] must be configured as 1'b1. The bitstream will be split into random pack .when swreg1_int[16] is 1,we must send the next pack to the decoder, and then configure swreg1_int[10] as 1'b1. Repeat the operation until the whole pack is sent completely. When sending a packet at the end of the bitstream, we must configure config[26] as 1'b1.
2. H264 table
H264 table is used for h264 context initialization. It contains four parts, and each part is 128bit align.

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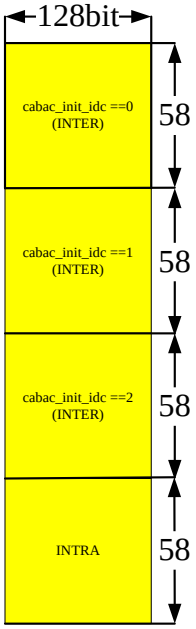

Fig. 16-16 h264 table

3. h264 pps
There are 256 units of 256bit data.

Table 16-3 pps format

seq_parameter_set_id	4bit
profile_idc	8bit
constraint_set3_flag	1bit
chroma_format_idc	2bit
bit_depth_luma_minus	3bit
bit_depth_chroma_minus	3bit
qpprime_y_zero_transform_bypass_flag	1bit
log2_max_frame_num_minus4	4bit
max_num_ref_frames	4bit
pic_order_cnt_type	2bit
log2_max_pic_order_cnt_lsb_minus4	4bit
delta_pic_order_always_zero_flag	1bit
pic_width_in_mbs	9bit
pic_height_in_mbs	9bit
frame_mbs_only_flag	1bit
mb_adaptive_frame_field_flag	1bit
direct_8x8_inference_flag	1bit
mvc_extension_enable	1bit
num_views	2bit
view_id[0]	10bit
view_id[1]	10bit
num_anchor_refs_l0	1bit
anchor_ref_l0	10bit
num_anchor_refs_l1	1bit
anchor_ref_l1	10bit
num_non_anchor_refs_l0	1bit
non_anchor_ref_l0	10bit

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num_non_anchor_refs_l1	1bit
non_anchor_ref_l1	10bit
Align(128)	3bit
pps_pic_parameter_set_id	8bit
pps_seq_parameter_set_id	5bit
entropy_coding_mode_flag	1bit
bottom_field_pic_order_in_frame_present_flag	1bit
num_ref_idx_l0_default_active_minus1	5bit
num_ref_idx_l1_default_active_minus1	5bit
weighted_pred_flag	1bit
weighted_bipred_idc	2bit
pic_init_qp_minus26	7bit
pic_init_qs_minus26	6bit
chroma_qp_index_offset	5bit
deblocking_filter_control_present_flag	1bit
constrained_intra_pred_flag	1bit
redundant_pic_cnt_present_flag	1bit
transform_8x8_mode_flag	1bit
second_chroma_qp_index_offset	5bit
Scaling_list_enable_flag	1bit
Scaleing_list_address	32bit
Is_long_term[16]	16bit
Voidx	16bit
Align(128)	8bit

4. h264 rps

The data should be 128bit aligned, and there are two modes to reorder ref pic: software and hardware.

For hardware mode, swreg2_sysctrl[24] must be configured as 1'b1, and there is 8 data (the unit is 128bit) for rps initialization in one frame. For software mode, swreg2_sysctrl[24] must be configured as 1'b0, there is 3 data(the unit is 128bit) in one slice, So a frame including the slices number multiplied by 3 the amount of data.

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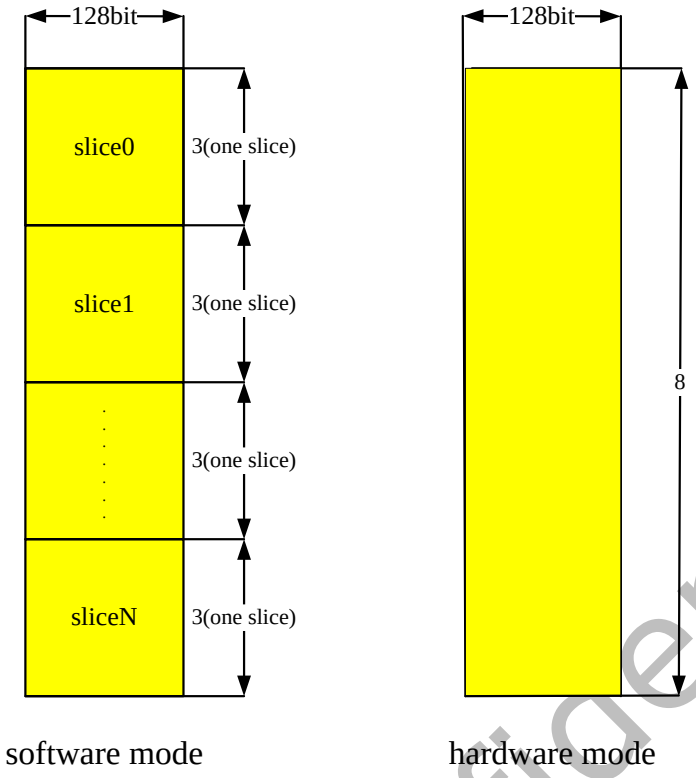

Fig. 16-17 h264 rps data format

For software mode,Each Specific data are listed in the following table:

Table 16-4 software rps format

for(j = 0; j < 2; j++)	
{	
for(i = 0; i < 32; i++)	
{	
dbp_idx[m][j][i]	5bit
bottom_flag[m][j][i]	1bit
View_id	1bit
}	
}	
Align(128)	

For hardware mode, Each Specific data are listed in the following table:

Table 16-5 hardware rps format

Framenum_wrap	256bit
No used	16bit
No used	16bit
for(j = 0; j <3; j++)	list p,list b0,listb1
{	
for(i = 0; i < 32; i++)	
{	
dbp_idx[j][i]	5bit
bottom_flag[j][i]	1bit
view_id[j][i]	1bit
}	
}	

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Align(128)	

15.4.10 Input Data for VP9

When in RLC mode, two data inputs are needed, which are dec_ctrl cmd and inter cmd, and they all should be 128bit aligned.
When in normal mode, three data inputs are needed, which are bitstream, prob and segid from last frame.
1. vp9 prob data format:
For intra_only frame, there are 149 units of 128bit data; for inter frame, there are 153 units of 128bit data.

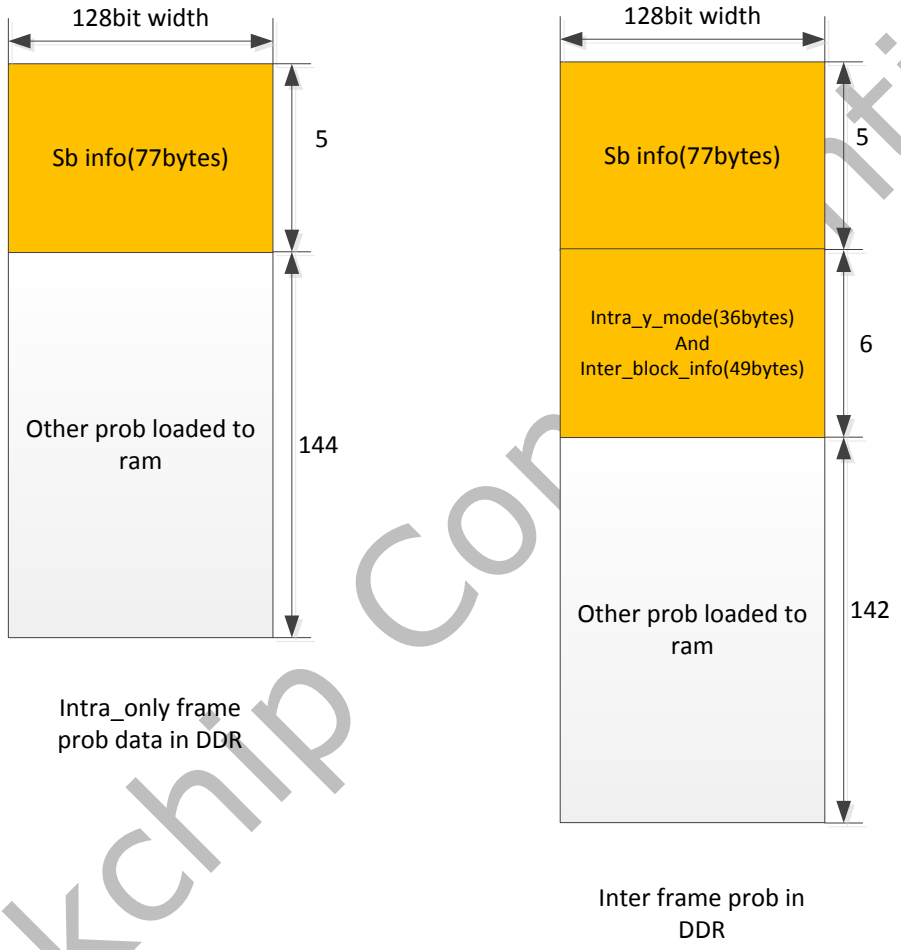

Fig. 16-18 vp9 prob data format

2. the format of segid from last frame data format
The segid data is store with CTU raster scan mode, every CTU contain two 128bits. In the CTU, every 8x8 block has a segid data (Smn) which is 4 bits.

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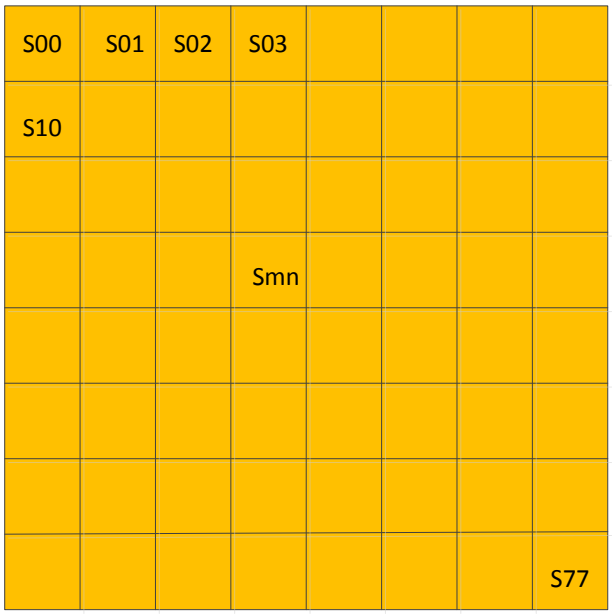


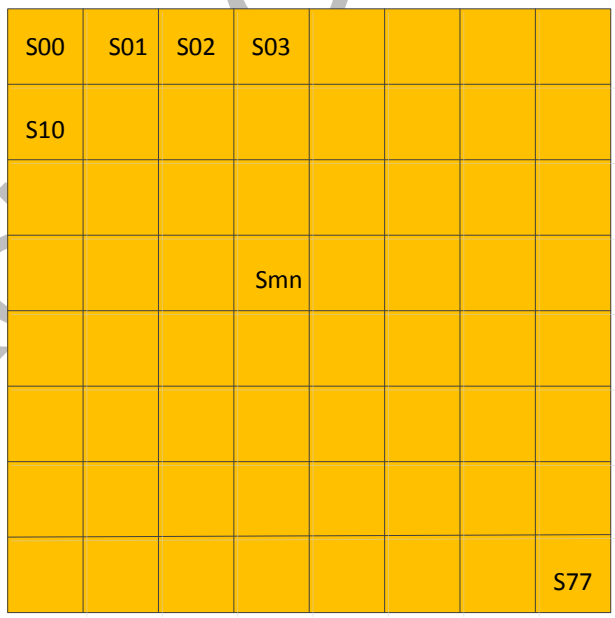
Segid data format in a CTU

Fig. 16-19 vp9 segid data format

15.4.11 Output Data Format

1. Decoded frame data format

The decoded frame data are stored in the location with raster scan order where designated in the register configuration. Y component of the video data are stored first, and then UV component of the video data are stored.



Segid data format in a CTU

Fig. 16-20 Data format in the DDR of Dec_out_base vp9

2. COLMV data

COLMV data, which are needed for inter prediction , are in the DDR space following the YUV data.

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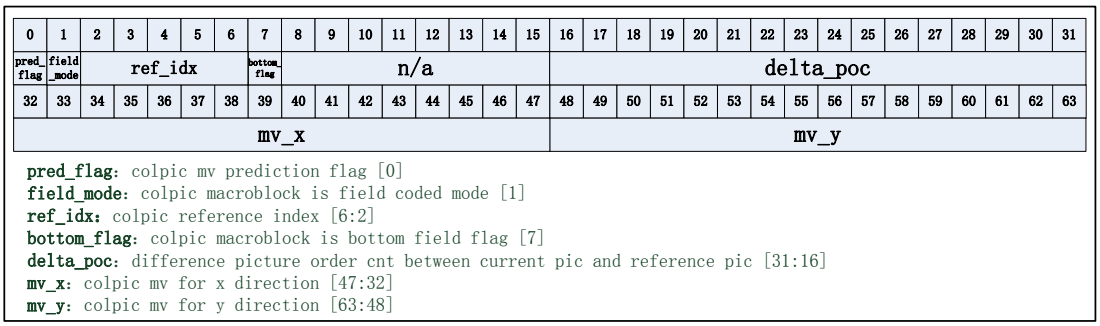

Fig. 16-21 H264 colmv output format

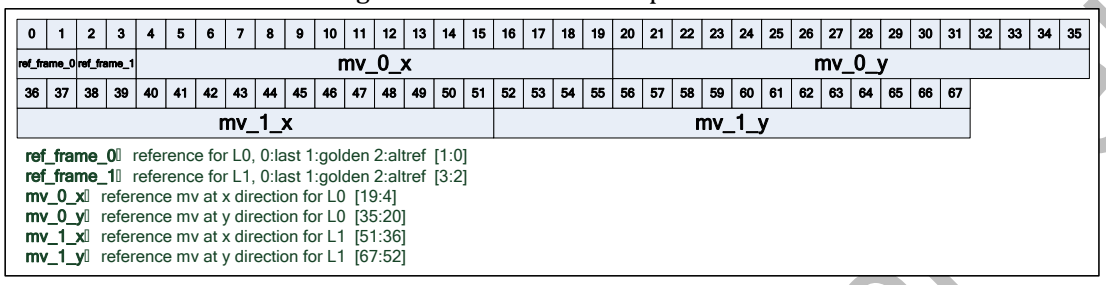

Fig. 16-22 VP9 colmv output format

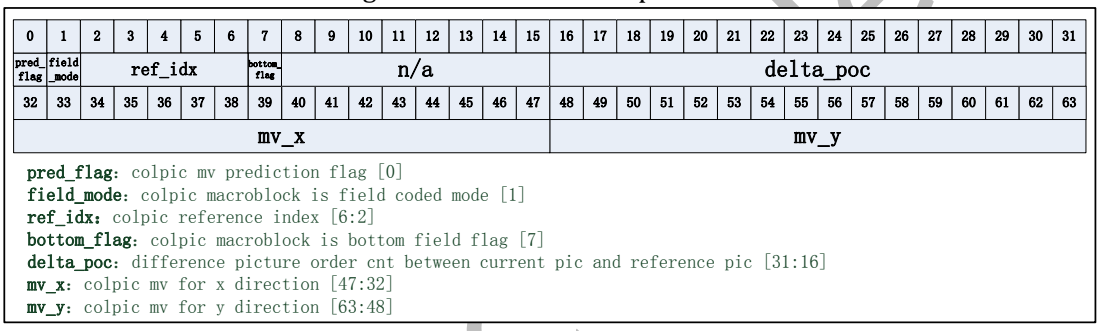

Fig. 16-23 HEVC colmv output format

3. Only for VP9
(1) Cur segid output:
The cur segid output data format is the same with segid from last frame.
(2) Cur count output:
For intra only frame, there are 98 words for no coeff count and 1440 words for coeff count.
For inter frame, there are 426 words for no coeff count and 2880 words for coeff count.
For every count, its real valid bits are 24, but it used 32bits in the DDR.
Segid data are output only when the decoder is decoding VP9.

15.4.12 H264 Error Process

For h264, it has two types of error. One is the normal error, and the other one is the a logic error. Whether it is a normal error or a logic error, the swreg76_h264_errorinfo_num[13:0] will record the number of slices in frame, swreg76_h264_errorinfo_num[15] will be set to 1,and swreg76_h264_errorinfo_num[29:16] will record the number of error slices in frame.
1. normal error
For the normal error, it is recorded by the detection of syntax that is beyond the scope. When swreg1_int[19] is configured as 1'd1, the hardware will wait the end signal of deblocking and then reset itself. And for normal error, the error info will be put on the address swreg4_strm_rlc_base. Specific data are listed in the following table:

Table 16-6 error info format

start_ctu_address	16bit	When it is inter error,it is error ctu assr,or it is the first ctu of current slice.
end_ctu_address	16bit	The last ctu of the current slice
error_type	16bit	When error_type[8] is equal to 1,it means that the

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		slice is wrong. the error_type[7:0] see the 3-2 table
slice_byte_offset	24bit	The first byte of current slice in the bitstream

When The corresponding register shown in the following table is configured as 1'b0,it will not detect The corresponding error.

Table 16-7 normal error table

error_type	Syntax	enable
0	start_mb_nr(syntax)	swreg44_strmd_error_en[0]
1	slice_type	swreg44_strmd_error_en[1]
2	pic_parameter_set_id	swreg44_strmd_error_en[2]
3	Framenum	swreg44_strmd_error_en[3]
4	idr_pic_id	swreg44_strmd_error_en[4]
5	delta_pic_order_cnt_bottom	swreg44_strmd_error_en[5]
6	delta_pic_order_cnt	swreg44_strmd_error_en[6]
7	num_ref_idx_lx_active_minus1	swreg44_strmd_error_en[7]
8	Cabac_init_idc	swreg44_strmd_error_en[8]
9	Slice_qp_delta	swreg44_strmd_error_en[9]
10	disable_deblocking_filter_idc	swreg44_strmd_error_en[10]
11	slice_alpha_c0&beta_offset_div2	swreg44_strmd_error_en[11]
12	modification_of_pic_nums_idc	swreg44_strmd_error_en[12]
13	modification_of_pic_num	swreg44_strmd_error_en[13]
14	abs_diff_pic_num_minus1	swreg44_strmd_error_en[14]
15	abs_diff_pic_num_nonexist	swreg44_strmd_error_en[15]
16	long_term_pic_num	swreg44_strmd_error_en[16]
17	long_term_pic_num_nonexist	swreg44_strmd_error_en[17]
18	abs_diff_view_idx_minus1	swreg44_strmd_error_en[18]
19	memory_management_control_operation	swreg44_strmd_error_en[19]
20	difference_of_pic_nums_minus1	swreg44_strmd_error_en[20]
21	long_term_frame_idx	swreg44_strmd_error_en[21]

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22	max_long_term_frame_idx_plus1	swreg44_strmd_error_en[22]
23	log2_weight_denom	swreg44_strmd_error_en[23]
24	pred_weight_offset	swreg44_strmd_error_en[24]
25	View_id	swreg44_strmd_error_en[25]
26	Skip_run	swreg77_h264_error_e[13]
27	Mb_type	swreg77_h264_error_e[14]
28	Coded_block_pattern	swreg77_h264_error_e[15]
29	Mb_qp_delta	swreg77_h264_error_e[16]
30	Ref_idx_lX	swreg77_h264_error_e[17]
31	Mvd_x	swreg77_h264_error_e[18]
32	Mvd_y	swreg77_h264_error_e[19]
33	Sub_mb_type	swreg77_h264_error_e[20]
34	Coeff_token	swreg77_h264_error_e[21]
35	Level_prefix	swreg77_h264_error_e[22]
36	Total_zero	swreg77_h264_error_e[23]
37	Run_before	swreg77_h264_error_e[24]
38	Coeff_abs_level_minus1	swreg77_h264_error_e[25]
39	Reserved	swreg77_h264_error_e[26]
40	Data_runout	swreg77_h264_error_e[28]
255	Inter error	

For normal error except inter error, When swreg1_int[19] is configured as 1'd0, the hardware will stop the decoder and reset itself.And the error_type in the table 3-2 will put on the sw_strmd_error_status[27:0],the current ctu_x will put on the swreg46_strmd_error_ctu[7:0], the current ctu_x will put on the swreg46_strmd_error_ctu[15:8]. The Inter error be put on the address swreg4_strm_rlc_base any ways.

2. logic error

For logic error, it is recorded by the detection of different slice of syntax elements that are the same.

When swreg1_int[19] is configured as 1'd1, when there is any stream error, the hardware will wait the end signal of deblocking and then reset itself. And the error_type in the table 3-3 will put on the sw_strmd_error_status[27:0],the current ctu_x will put on the swreg46_strmd_error_ctu[7:0], the current ctu_x will put on the swreg46_strmd_error_ctu[15:8].

Table 16-8 logic error table

Error_type	syntax	enable
43	Delta_pic_order_cnt	swreg44_strmd_error_en[27]
44	Delta_pic_order_cnt_bottom	swreg44_strmd_error_en[28]
45	Pic_order_cnt_lsb	swreg44_strmd_error_en[29]

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46	idr_pic_id	swreg44_strmd_error_en[30]
47	Bottom_field_flag	swreg44_strmd_error_en[31]
48	Field_pic_flag	swreg77_h264_error_e[0]
49	Frame_num	swreg77_h264_error_e[1]
50	pps_id_logic	swreg77_h264_error_e[2]
51	Slice_type_logic	swreg77_h264_error_e[3]
52	First_mb_in_slice_logic	swreg77_h264_error_e[4]
53	Nal_ref_idc_logic	swreg77_h264_error_e[5]
54	Idr_flag_logic	swreg77_h264_error_e[6]
55	Inter_view_flag_logic	swreg77_h264_error_e[7]
56	Anchor_pic_flag_logic	swreg77_h264_error_e[8]
57	View_id_logic	swreg77_h264_error_e[9]

For logic error, When swreg1_int[19] is configured as 1'd0, the hardware will stop the decoder and reset itself.And the error_type shown above will put on the sw_strmd_error_status[27:0],the current ctu_x will put on the swreg46_strmd_error_ctu[7:0], the current ctu_x will put on the swreg46_strmd_error_ctu[15:8].

15.4.13 HEVC Error Process

For hevc, it has only one types of error. When an error is detected, the hardware will stop the decoder and reset itself. And the error_type in the table 3-3 will put on the sw_strmd_error_status[27:0],the current ctu_x will put on the swreg46_strmd_error_ctu[7:0], the current ctu_x will put on the swreg46_strmd_error_ctu[15:8].

15.4.14 HEVC 2-ratio scale down work mode

For HEVC decoder mode,we support 2-ratio scale down work mode.It will most be used in hevc decoder 4k size but display just support 1080p,it will reduce the DDR bandwidth request for vop,that is the decoder will have more DDR bandwidth can be used,and it will improve our decoder performance.
In this mode,we also support un-referenced frame auto detect,so we allow not to output current decodered frame which will not be used for reference frame but just to output 2-ratio scale down frame data for display.Use this method,it will improve out decoder performance more.
User can set swreg[24] 1'b1 to enable 2-ratio scale down work mode,and you should config swreg96~swreg98 also.
If 2-ratio scale down work mode be enabled,User can set swreg[25] 1'b1 to enable just only output 2-ratio scale down frame data for display when decoder detect current frame will not be used for reference frame.
To used this mode have some limitations.First,it just support luma and chroma have the same bitdepth(y10c10 or y8c8).Second,this mode can't be used when decoder frame have tile split.

15.4.15 Write decoder frame align mode

This mode is just used in vdpu341 verision(4k decoder),it is very useful to save DDR bandwidth used when user need to decoder hevc source which bitdepth is 10bit,but not all 10bit hevc source can benefit,in frame tile split source,it does't have any useful,Our decoder will auto dectect tile split,and the disable this mode auto.
So,for user,you can always set swreg[23] 1'b1 to enable this mode.and we strongly suggest enable this mode.

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15.4.16 Register config link table mode

VDPU34x(4K decoder) Video decoder support link table mode to improve decoder continuity and reduce CPU participate and interrupt response time.In this mode,the register which used for config will be prepare into DDR,after that user enable link table mode,and then our decoder will fetch the register config information through AXI bus frame by frame.On link table mode,the decoder will auto finsh all the frame decoder which be prepared in DDR and not need CPU participate.

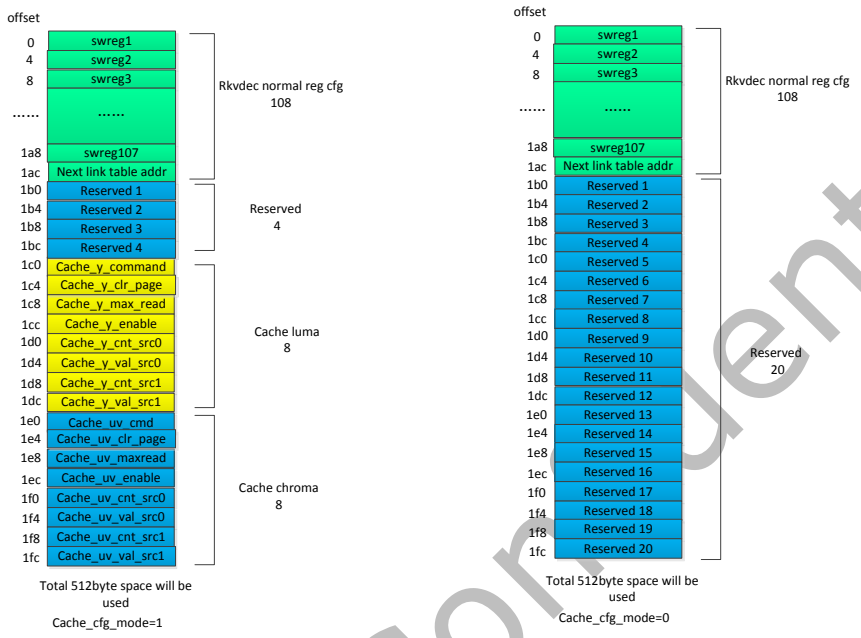

Fig. 16-24 Link tbale register prepare in DDR format

As show in the Fig- Link tbale register prepare in DDR format,each frame config information need 512bytes in DDR.We have two cache config mode,you can use link_table_swreg0[1] to select,if cache_cfg_mode=1,user should prepare cache config information to DDR, otherwise,user can doesn't care cache config,hardware will auto select the best to config. After hardware decoder one frame,will write back some information to DDR where the write back base address is current config storage addr base.All the write back information are listed in the fellows table.

Table 16-9 Link table write back register

Write back register	Description	offset
swreg1	interrupt and enable register	0x0
swreg4	the stream or rlc read data base address	0x4
swreg41	the base address of rlcwrite base address	0x8
swreg45	cabac error status	0xc
swreg46	strmd error ctu	0x10
swreg47	when there is any error, it is for the position of sao decode output to busifd	0x14
swreg64	hevc performnace cycle	0x18
swreg65	axi ddr read data num	0x1c
swreg66	axi ddr write data num	0x20
swreg68	performance cnt sel	0x24
swreg69	performance_cnt0	0x28

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Write back register	Description	offset
swreg70	performance_cnt1	0x2c
swreg71	performance_cnt2	0x30
swreg75	error info base addr	0x34
swreg76	for vp9 16 tile cols, every cols contains 4bits	0x38
swreg100	Axi performance latency module contrl register0	0x3c
swreg101	swreg101_PERF_LATENCY_CTRL1	0x40
swreg102	swreg102_PERF_RD_MAX_LATENCY_NUM0	0x44
swreg103	swreg103_PERF_RD_LATENCY_SAMP_NUM	0x48
swreg104	swreg104_PERF_RD_LATENCY_ACC_SUM	0x4c
swreg105	swreg105_PERF_RD_AXI_TOTAL_BYTE	0x50
swreg106	swreg106_PERF_WR_AXI_TOTAL_BYTE	0x54
swreg107	swreg107_PERF_WORKING_CNT	0x58
PERFCNT_SRC0	performance counter 0 source register,luma maybe not need to write	0x5c
PERFCNT_VAL0	performance counter 0 value register,luma	0x60
PERFCNT_SRC1	performance counter 1 source register,luma maybe not need to write	0x64
PERFCNT_VAL1	performance counter 1 value register,luma	0x68
PERFCNT_SRC0	performance counter 0 source register,chrm maybe not need to write	0x6c
PERFCNT_VAL0	performance counter 0 value register,chrm	0x70
PERFCNT_SRC1	performance counter 1 source register,chrm maybe not need to write	0x74
PERFCNT_VAL1	performance counter 1 value register,chrm	0x78
reversed		0x7c

The base address of config link table mode is RKVDEC_BASE+0x300,you can get more information at 15.7.5.

15.4.17 QOS mode

In vdpu341 hardware,we support QOS mode,in this mode,we can request high priority to fetch data,you can control QOS work by config swreg99 and swreg108.For more detail,you can see the register description.
User should accord to the soc system to select appropriate level.

15.4.18 MPEG-4 decoder

The features that video decoder supports about MPEG-4/H.263 shows as below.

Table 16-10 MPEG-4/H.263 feature

Feature	Decoder support
Input data format	MPEG-4/H.263 elementary video stream
Decoding scheme	Frame by frame(or field by field)

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	Video packet by video packet
Output data format	YCbCr 4:2:0 semi-planar
Supported image size	48x48 to 1920x1088(MPEG-4, Sorenson Spark) 48x48 to 720x576(H.263) Step size 16 pixels
Maximum frame rate	60fps at 1920x1088
Maximum bit rate	As specified by MPEG-4 ASP level5
Error detection and concealment	Supported

The decoder of MPEG-4/H.263 has two operating modes: in the primary mode the HW performs entropy decoding, and in the secondary mode SW performs entropy decoding. Secondary mode is used in MPEG-4 data partitioned stream decoding.

15.4.19 MPEG-2/MPEG-1 decoder

The features of MPEG-2/MPEG-1 supported by decoder are shown as Table 16-11.

Table 16-11 MPEG-2/MPEG-1 features

Feature	Decoder support
Input data format	MPEG-2/MPEG-1 elementary video stream
Decoding scheme	Frame by frame(or field by field) Video packet by video packet
Output data format	YCbCr 4:2:0 semi-planar
Supported image size	48x48 to 1920x1088 Step size 16 pixels
Maximum frame rate	60fps at 1920x1088
Maximum bit rate	As specified by MPEG-2 MP high level
Error detection and concealment	Supported

The dataflow of MPEG-2/MPEG-1 is the same of H.264 HW performs entropy decoding as 错误!未找到引用源。.

15.4.20 VC-1 decoder

The features of VC-1 supported by decoder are shown as Table 16-12.

Table 16-12 VC-1 features

Feature	Decoder support
Input data format	VC-1
Decoding scheme	Frame by frame(or field by field) Slice by slice
Output data format	YCbCr 4:2:0 semi-planar
Supported image size	48x48 to 1920x1088 Step size 16 pixels
Maximum frame rate	30fps at 1920x1088
Maximum bit rate	As specified by VC-1 AP level3
Error detection and concealment	Supported

The VC-1 decoder has only one operating mode in which the HW performs entropy decoding.

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15.4.21 AVS/AVS+ decoder

The features of AVS/AVS+ supported by decoder are shown as Table 1-13.

Table 16-13 AVS/AVS+ features

Feature	Decoder support
Input data format	AVS/AVS+ stream
Decoding scheme	Frame by frame(or field by field) Slice by slice
Output data format	YCbCr 4:2:0 semi-planar raster-scan YCbCr 4:2:0 semi-planar 8x4 tile
Supported image size	48x48 to 1920x1088 Step size 16 pixels
Maximum frame rate	60fps at 1920x1088
Maximum bit rate	As specified by AVS/AVS+ standard
Error detection and concealment	Supported

The decoder has only one operating mode in which the HW performs entropy decoding.

15.4.22 JPEG Decoder

JPEG features supported by decoder are as shown in Table 16-14.

Table 16-14 JPEG features

Feature	Decoder support
Input data format	JFIF file format 1.02 YCbCr 4:0:0, 4:2:0, 4:2:2, 4:4:0, 4:1:1 and 4:4:4 semi-planar
Decoding scheme	Input: buffer by buffer, from 5Kb to 8MB at a time① Output: from 1 MB row to 16 Mpixels at a time②
Output data format	YCbCr 4:0:0, 4:2:0, 4:2:2, 4:4:0, 4:1:1 and 4:4:4 semi-planar
Supported image size	48x48 to 8176x8176(66.8 Mpixels) Step size 8 pixels③
Maximum frame rate	Up to 76 million pixels pre second
Maximum bit rate	As specified by the Divx specification
Thumbnail decoding	JPEG compressed thumbnails supported
Error detection	Supported

①Programmable buffer size for optimizing performance and memory consumption. Interrupt will be issued when buffer runs empty, and the control software will load more streams to external memory.
②Programmable output slice for optimizing performance and memory consumption. Interrupt will be issued when the requested area decoded. The control software can be used to switch the decoder output address each time.
③Non-16x16 dividable resolutions will be filled to 16 pixel boundary.

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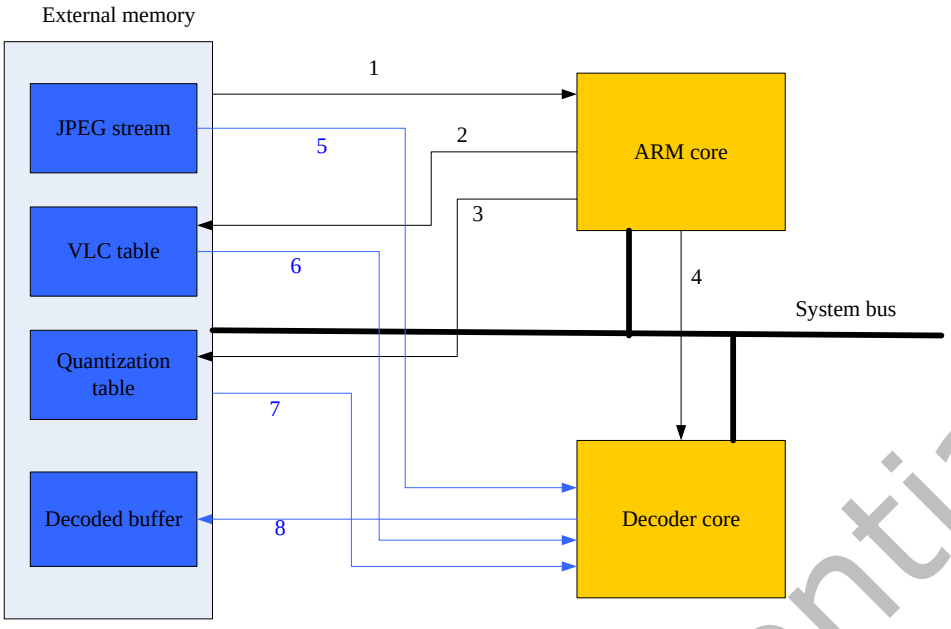

Fig. 16-25 The dataflow of JPEG decoder

The data flow of jpeg decoder is as Fig. 16-25shown. The decoder software starts decoding the picture by parsing the stream headers (1) and then writes the following items to external memory:
VLC tables (2)
Quantization tables (3)
Last step for the software is to write the hardware control registers and to enable the hardware (4). After starting hardware, SW waits interrupt from HW.
Hardware decodes the picture by reading stream (5), VLC (6) and QP (7) tables. Hardware writes the decoded output picture memory one macroblock at a time (8). When the picture has been fully decoded, or the hardware has run out of stream data, it gives an interrupt with a proper status flag and provides stream end address for software to continue and returns to initial state.

15.4.23 Image Post-processor

The features supported by Post-processor are as show in Table 16-15.

Table 16-15 Post-processor features

Feature	Post-processor support
Input data format	Any format generated by the decoder in combined mode YCbCr 4:2:0 semi-planar YCbCr 4:2:0 planar YCbYCr 4:2:2 YCrYCb 4:2:2 CbYCrY 4:2:2 CrYCbY 4:2:2
Post-processor scheme	Frame by frame. Post-processor handles the image macroblock by macroblock, also in standalone mode.
Input image source	Internal source(combined mode) External source(standalone mode): e.g. a software decoder or camera interface
Output data format	YCbCr 4:2:0 semi-planar YCbCr 4:2:2

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	YCrYCb 4:2:2 CbYCrY 4:2:2 CrYCbY 4:2:2 Fully configurable ARGB channel lengths and locations inside 32 bits, e.g. ARGB 32-bit (8-8-8-8), RGB 16-bit(5-6-5), ARGB 16-ibt(4-4-4-4).
Input image size (combined mode)	48x48 to 8176x8176(66.8 Mpixels) Step size 16 pixels
Input image size (stand-alone mode)	Width from 48 to 8176 Height from 48 to 8176 Maximum size limited to 16.7 Mpixels Step size 16 pixels
Output image size	16x16 to 1920x1088 Horizontal step size 8 Vertical step size 2
Image up-scaling①	Bicubic polynomial interpolation with a four-tap horizontal kernel and a two-tap vertical kernel Arbitrary, non-integer scaling ratio, separately for both dimensions. Maximum output width is 3x the input width(within the maximum output image size limit) Maximum output height is 3x the input height -2 pixels (within the maximum output image size limit) Maximum output height is 2.5x the input height – 2 pixels (within the maximum output image size limit) when running RealVideo, VP8 format in pipeline
Image down-scaling①	Proprietary averaging filter Arbitrary, non-integer scaling ratio separately for both dimensions Unlimited down-scaling ratio
YCbCr to RGB color conversion	BT.601-5 compliant BT.709 compliant User definable conversion coefficient
Dithering	2x2 ordered spatial dithering for 4,5 and 6 bit RGB channel precision
Programmable alpha channel	Constant eight bit value can be set to the alpha channel of the 24-bit RGB output data to control the transparency of the output picture. The resulting 32-bit ARGB data can be used as input data for later alpha blending.
Alpha blending	Output image can be alpha blended with two rectangular areas. YCbCr semi-planar 4:2:0 PP output format is not supported when performing alpha blending. The supported overlay input formats are following. 8 bit alpha value + YCbCr 4:4:4, big endian channel order being A-Y-Cb-Cr, 8 bits each. 8 bit alpha value + 24 bit RGB, big endian channel order being

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	A-R-G-B, 8 bits each
Deinterlacing	Conditional spatial deinterlace filtering. Supports only YCbCr 4:2:0 input format. Usable in stand-alone post-processing mode only.
RGB image contrast adjustment	Segmented linear
RGB image brightness adjustment	Linear
RGB image color saturation adjustment	Linear
De-blocking filter for MPEG-4 simple profile /H.263 /Sorenson	Using a modified H.264 in-loop filter as a post-processing filter. Filtering has to be performed in combined mode
Image cropping / digital zoom	User definable start position, height and width. Can be used with scaling to perform digital zoom. Usable only for JPEG or stand-alone mode.
Picture in picture	Output image can be written to any location inside video memory. Up to 1920x1088 sized displays supported.
Output image masking	Output image writing can be prevented on two rectangular areas in the image. The masking feature is exclusive with alpha blending; however it is possible to have one masking area and one blending area.
Image rotation	Rotation 90,180, or 270 degrees Horizontal flip Vertical flip

①It is not allowed to perform horizontal up-scaling and vertical down-scaling (or vice versa) at the same. If needed, this kind of operation can be performed in two phases.
The PP has two modes: standalone mode and pipe-line mode. In standalone mode, picture processing is performed to any external source. The processing is done independently and asynchronously from the video decoder. The dataflow block gram is as Fig. 16-26 shows.

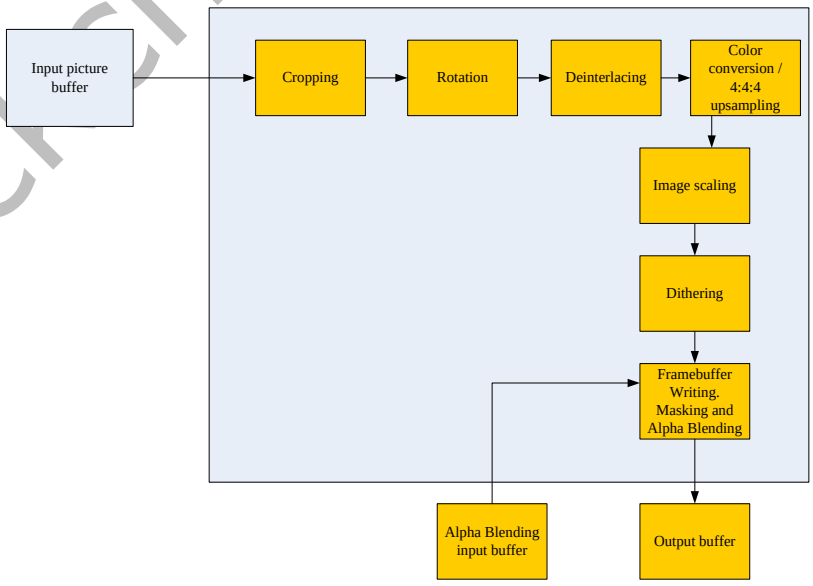

Fig. 16-26 Post-process standalone dataflow

In pipe-line mode, the post-processor works together with the multi-format decoder. The PP will take its input directly from the decoder. The post-processor doesn't have cropping function

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in pipe-line mode other than combined with jpeg decoder. The dataflow is as Fig. 16-27 show. In the pipe-line mode, most decoder will also put the data to the decoder out buffer other than JPEG decoder. So, JPEG decoder with pipe-line mode will save bus bandwidth when it crops the input picture to a smaller picture.

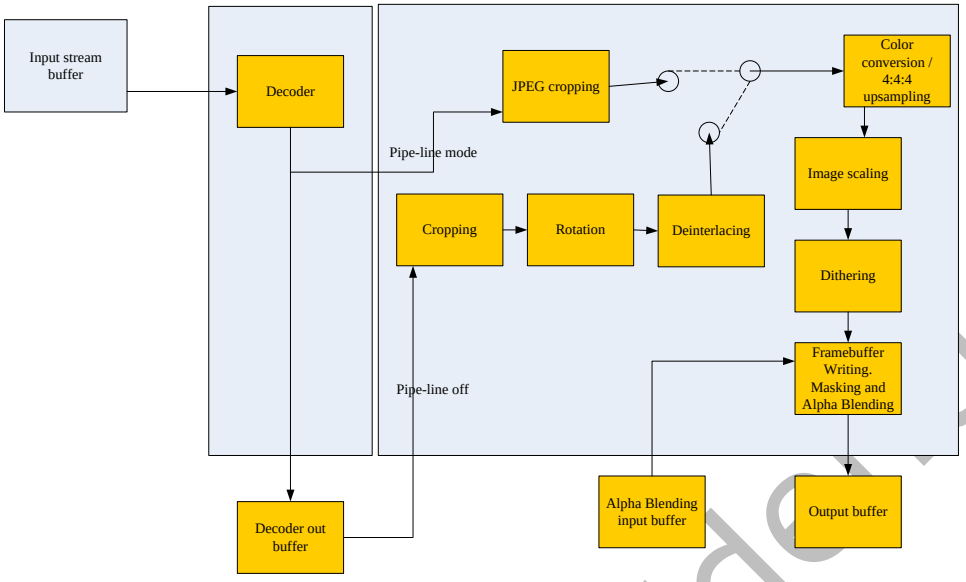

Fig. 16-27 Post-process Pipe-line Mode Dataflow

The post-processor has some restrictions in the input and output picture size. Table 16-16 presents the divisibility requirements for all the post-processor parameters.

Table 16-16 Requirements for post-processor

Output format / parameters	YCbCr 4:2:0	YCbCr 4:2:2	RGB16bpp	RGB32bpp
Input picture width and height	16	16	16	16
Cropped picture width and height	8	8	8	8
Cropping start coordinates(x,y)	16	16	16	16
Output picture width	8	8	8	8
Output picture height	2	2	2	2
Masks width and origin X	8	4	4	2
Masks width and origin Y	2	1	1	1
Frame buffer width and origin X	8	4	4	2
Frame buffer height and origin Y	2	1	1	1

15.4.24 Image Pre-processor

Pre-processor is pipelined with the encoder and it can be used only with the encoder.

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Pre-processor features are presented in Table 16-17.

Table 16-17 Post-processor features

Feature	Encoder support
RGB to YCbCr 4:2:0 color space conversion	BT.601, BT.709 or user defined coefficients conversion for RGB: RGB444 and BGR444 RGB555 and BRG555 RGB565 and BGR565 RGB888 and BRG888 RGB101010 and BRG101010
YCbCr 4:2:2 to YCbCr 4:2:0 color space conversion	YCbCr formats: YCbCr 4:2:0 planar YCbCr 4:2:0 semi-planar YCbYCr 4:2:2 CbYCrY 4:2:2 interleaved
Cropping	Video – from 8192x8192 to any supported encoding size
Rotation	90 or 270 degrees

15.4.25 H.264 Encoder

The H.264 features supported by encoder are as shown in Table 16-18 .

Table 16-18 Video encoder H.264 feature

Feature	Encoder support
Input data format	YCbCr formats: YCbCr 4:2:0 planar YCbCr 4:2:0 semi-planar YCbYCr 4:2:2① CbYCrY 4:2:2 Interleaved① RGB formats:① RGB444 to BGR444 RGB555 to BGR555 RGB565 to BGR565 RGB888 to BRG888 RGB101010 and BRG 101010
Output data format	H.264: Byte unit stream NAL unit stream
Supported image size	96x96 to 1920x1080(Full HD) Step size 4 pixels
Maximum frame rate	30 fps at 1920 x1080
Bit rate	Maximum 20Mbps Minimum

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	10kbps

①internally encoder handles image only in 4:2:0 format

Figure Fig. 16-28 illustrates the encoder data flow in H.264 encoding mode. The numbers present the following transactions:
Memory-mapped register writes and reads
Input image read
Reference image write
Reference image read
NAL sizes write from HW
NAL sizes read to SW
Output byte or NAL unit stream write from HW
Output byte or NAL unit stream headers write from SW

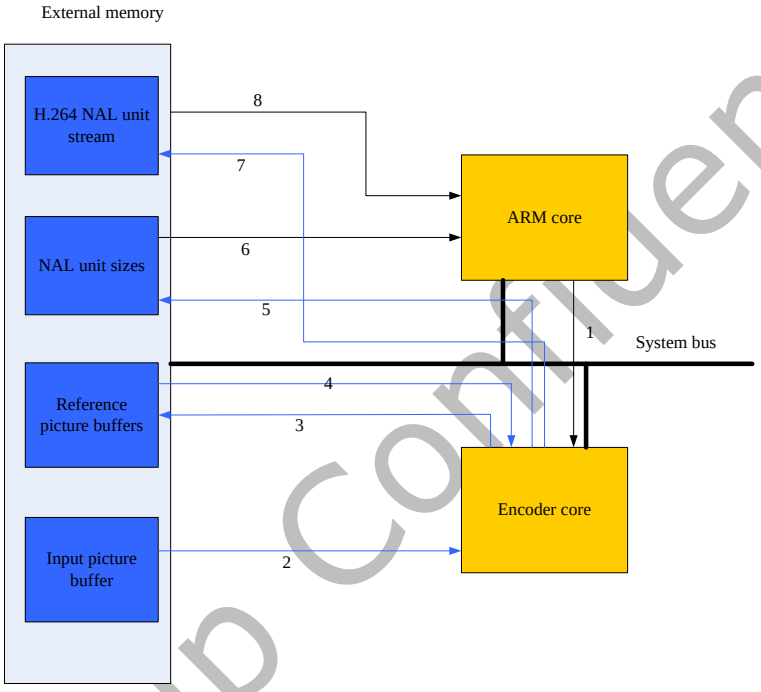

Fig. 16-28 Video Encoder Dataflow

The encoder software starts encoding the first picture by initializing hardware and writing the stream headers. After HW has encoded the image, SW calculates new quantization values for HW, and initializes HW again.

15.4.26 HEVC Encoder

HEVC ENCODER is low-cost H.265/HEVC hardware codec IP that is capable of encoding FHD H.265/HEVC main profile L4.1 . It contains a 32-bit processor called V-CPU, which is responsible for encoding bitstream syntax in encoder from sequence to slice header unit, prescanning slice data, controlling the underlying video hardware blocks called V-CORE, and communicating with host CPU through host register interface. The V-CORE performs actual processing of coded slice data including mode decision, motion estimation/compensation, transform, entropy encoding, and loop filtering. This software and hardware combined architecture can provide flexibility and high throughput at the same time. It can be connected through the industry standard interfaces - 32-bit AMBA 3 APB bus for host CPU system control and 128-bit AMBA3 AXI for data transfer. There are two 128bit AXI buses available - one for accessing external memory and the other for on-chip SRAM memory.
HEVC ENCODER supports the following features:

- Capable of encoding HEVC Main Profile @ L4.1 High-tier
  - I/P slices
  - Supported Prediction Unit(PU) size : 16x16, 8x8

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  - Supported Transform Unit(TU) size : 16x16 to 8x8
  - Multi slice : Independent slice segment and dependent slice segment
  - High performance CABAC encoding
  - In-loop deblocking filtering
  - Loop filtering across slice
- Max resolution: 8192x4096
- Min resolution: 256x128
- Handling multi-instances
- Lossless coding
- Noise reduction
- Rate Contro
  - VBR, CBR and ABR
  - ROI support
- 30 fps at 1920 x1080

15.5 Register Description

15.5.1 Internal Address Mapping

This section describes the control/status registers of the codec.
If vdpu341 decoder is chosen to work, the register base address is vdpu341_base.
If vdpu121 decoder is chosen to work, the register base address is vdpu121_base.
If vepu121 encoder is chosen to work, the register base address is vepu121_base.
All the register config base are listed as fellows:

Table 16-19 Base address of config

Config Register	Base addr
vdpu341 function config base	vdpu341_base
vdpu341 mmu config base	vdpu341_base+0x480 for read channel vdpu341_base+0x4c0 for write channel
vdpu341 cache config base	vdpu341_base+0x400 for luma channel vdpu341_base+0x440 for chroma channel
vdpu341 link table config base	vdpu341_base+0x300
vdpu121 function config base	vdpu121_base
vdpu121 mmu config base	vdpu121_base+0x400
vdpu121 cache config base	vdpu121_base+0x800
vdpu121 avs/avs+ config base	vdpu121_base+0xc00
vepu121 function config base	vepu121_base
vepu121 mmu config base	vepu121_base+0x800

Table 16-20 Base address value

Base addr	value
vdpu341_base	0xff36_0000
vdpu121_base	0xFF35_0400
vepu121_base	0xFF34_0000
vepu220_base	0xFF33_0000

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15.5.2 VDPU341 Function Config Register Summary

Name	Offset	Size	Reset Value	Description
rkvdec_swreg0_id	0x00000	W	0x68761f00	ID register (read only)
rkvdec_swreg1_int	0x00004	W	0x00200022	interrupt and decoder enable register
rkvdec_swreg2_sysctrl	0x00008	W	0x00000000	Data input and output endian setting and sys ctrl
rkvdec_swreg3_picpar	0x0000c	W	0x00000000	picture parameters
rkvdec_swreg4_strm_rlc_base	0x00010	W	0x00000000	the stream or rlc data base address
rkvdec_swreg5_stream_rlc_len	0x00014	W	0x00000000	amount of stream bytes or rlc data byte in the input buffer or the
rkvdec_swreg6_cabactbl_prob_base	0x00018	W	0x00000000	the base address of cabac table
rkvdec_swreg7_decout_base	0x0001c	W	0x00000000	base address of decoder output picture base address
rkvdec_swreg8_y_virstride	0x00020	W	0x00000000	the ouput picture y virtual stride
rkvdec_swreg9_yuv_virstride	0x00024	W	0x00000000	the ouput picture yuv virtual stride
rkvdec_swreg10_h264_refer0_base	0x00028	W	0x00000000	base address for reference picture index 0
rkvdec_swreg10_vp9_cprheader_offset	0x00028	W	0x00000000	vp9 compressed header offset
rkvdec_swreg10_hevc_refer0_base	0x00028	W	0x00000000	base address for reference picture index 0

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg11_h264_refer1_base	0x0002c	W	0x00000000	base address for reference picture index 1
rkvdec_swreg11_vp9_referlast_base	0x0002c	W	0x00000000	base address for last frame
rkvdec_swreg11_hevc_refer1_base	0x0002c	W	0x00000000	base address for reference picture index 1
rkvdec_swreg12_h264_refer2_base	0x00030	W	0x00000000	base address for reference picture index 2
rkvdec_swreg12_vp9_refergolden_base	0x00030	W	0x00000000	base address for golden frame
rkvdec_swreg12_hevc_refer2_base	0x00030	W	0x00000000	base address for reference picture index 2
rkvdec_swreg13_h264_refer3_base	0x00034	W	0x00000000	base address for reference picture index 3
rkvdec_swreg13_vp9_referalfter_base	0x00034	W	0x00000000	base address for referalfter frame
rkvdec_swreg13_hevc_refer3_base	0x00034	W	0x00000000	base address for reference picture index 3
rkvdec_swreg14_h264_refer4_base	0x00038	W	0x00000000	base address for reference picture index 4
rkvdec_swreg14_vp9count_base	0x00038	W	0x00000000	vp9 count base addr
rkvdec_swreg14_hevc_refer4_base	0x00038	W	0x00000000	base address for reference picture index 4
rkvdec_swreg15_vp9_segidlast_base	0x0003c	W	0x00000000	base address for last frame segment id
rkvdec_swreg15_hevc_refer5_base	0x0003c	W	0x00000000	base address for reference picture index 5

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg15_h264_refer5_base	0x0003c	W	0x00000000	base address for reference picture index 5
rkvdec_swreg16_vp9_segidcur_base	0x00040	W	0x00000000	base address for cur frame segment id
rkvdec_swreg16_hevc_refer6_base	0x00040	W	0x00000000	base address for reference picture index 6
rkvdec_swreg16_h264_refer6_base	0x00040	W	0x00000000	base address for reference picture index 6
rkvdec_swreg17_vp9_frame_size_last	0x00044	W	0x00000000	vp9 last frame size
rkvdec_swreg17_hevc_refer7_base	0x00044	W	0x00000000	base address for reference picture index 7
rkvdec_swreg17_h264_refer7_base	0x00044	W	0x00000000	base address for reference picture index 7
rkvdec_swreg18_vp9_frame_size_golden	0x00048	W	0x00000000	vp9 golden frame size
rkvdec_swreg18_hevc_refer8_base	0x00048	W	0x00000000	base address for reference picture index 8
rkvdec_swreg18_h264_refer8_base	0x00048	W	0x00000000	base address for reference picture index 8
rkvdec_swreg19_vp9_frame_size_altref	0x0004c	W	0x00000000	vp9 alfter frame size
rkvdec_swreg19_hevc_refer9_base	0x0004c	W	0x00000000	base address for reference picture index 9
rkvdec_swreg19_h264_refer9_base	0x0004c	W	0x00000000	base address for reference picture index 9
rkvdec_swreg20_vp9_segid_grp0	0x00050	W	0x00000000	vp9 segid syntax grp0

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg20_hevc_refer10_base	0x00050	W	0x00000000	base address for reference picture index 10
rkvdec_swreg20_h264_refer10_base	0x00050	W	0x00000000	base address for reference picture index 10
rkvdec_swreg21_vp9_segid_grp1	0x00054	W	0x00000000	vp9 segid syntax grp1
rkvdec_swreg21_hevc_refer11_base	0x00054	W	0x00000000	base address for reference picture index 11
rkvdec_swreg21_h264_refer11_base	0x00054	W	0x00000000	base address for reference picture index 11
rkvdec_swreg22_vp9_segid_grp2	0x00058	W	0x00000000	vp9 segid syntax grp2
rkvdec_swreg22_hevc_refer12_base	0x00058	W	0x00000000	base address for reference picture index 12
rkvdec_swreg22_h264_refer12_base	0x00058	W	0x00000000	base address for reference picture index 12
rkvdec_swreg23_vp9_segid_grp3	0x0005c	W	0x00000000	vp9 segid syntax grp3
rkvdec_swreg23_hevc_refer13_base	0x0005c	W	0x00000000	base address for reference picture index 13
rkvdec_swreg23_h264_refer13_base	0x0005c	W	0x00000000	base address for reference picture index 13

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg24_vp9_segid_grp4	0x00060	W	0x00000000	vp9 segid syntax grp4
rkvdec_swreg24_hevc_refer14_base	0x00060	W	0x00000000	base address for reference picture index 14
rkvdec_swreg24_h264_refer14_base	0x00060	W	0x00000000	base address for reference picture index 14
rkvdec_swreg25_vp9_segid_grp5	0x00064	W	0x00000000	vp9 segid syntax grp5
rkvdec_swreg25_refer0_poc	0x00064	W	0x00000000	the poc of reference picture index 0
rkvdec_swreg26_vp9_segid_grp6	0x00068	W	0x00000000	vp9 segid syntax grp6
rkvdec_swreg26_refer1_poc	0x00068	W	0x00000000	the poc of reference picture index 1
rkvdec_swreg27_vp9_segid_grp7	0x0006c	W	0x00000000	vp9 segid syntax grp7
rkvdec_swreg27_refer2_poc	0x0006c	W	0x00000000	the poc of reference picture index 2
rkvdec_swreg28_vp9_cprheader_config	0x00070	W	0x00000000	vp9 compressed header config info
rkvdec_swreg28_refer3_poc	0x00070	W	0x00000000	the poc of reference picture index 3
rkvdec_swreg29_vp9_lref_scale	0x00074	W	0x00000000	scaling factor for last reference picture
rkvdec_swreg29_refer4_poc	0x00074	W	0x00000000	the poc of reference picture index 4

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg30_vp9_gref_scale	0x00078	W	0x00000000	scaling factor for golden reference picture
rkvdec_swreg30_refer5_poc	0x00078	W	0x00000000	the poc of reference picture index 5
rkvdec_swreg31_vp9_aref_scale	0x0007c	W	0x00000000	scaling factor for alfter reference picture
rkvdec_swreg31_refer6_poc	0x0007c	W	0x00000000	the poc of reference picture index 6
rkvdec_swreg32_vp9_ref_deltas_lastframe	0x00080	W	0x00000000	vp9 ref deltas
rkvdec_swreg32_refer7_poc	0x00080	W	0x00000000	the poc of reference picture index 7
rkvdec_swreg33_vp9_info_lastframe	0x00084	W	0x00000000	vp9 info for lastframe
rkvdec_swreg33_refer8_poc	0x00084	W	0x00000000	the poc of reference picture index 8
rkvdec_swreg34_vp9_intercmd_base	0x00088	W	0x00000000	inter cmd base addr
rkvdec_swreg34_refer9_poc	0x00088	W	0x00000000	the poc of reference picture index 9
rkvdec_swreg35_vp9_intercmd_num	0x0008c	W	0x00000000	vp9 intercmd num
rkvdec_swreg35_refer10_poc	0x0008c	W	0x00000000	the poc of reference picture index 10
rkvdec_swreg36_vp9_lasttile_size	0x00090	W	0x00000000	vp9 lasttile size
rkvdec_swreg36_refer11_poc	0x00090	W	0x00000000	the poc of reference picture index 11

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg37_vp9_lastf_hor_virstride	0x00094	W	0x00000000	Register0000 Abstract
rkvdec_swreg37_refer12_poc	0x00094	W	0x00000000	the poc of reference picture index 12
rkvdec_swreg38_vp9_goldenf_hor_virstride	0x00098	W	0x00000000	vp9 golden frame horizontal virstride
rkvdec_swreg38_refer13_poc	0x00098	W	0x00000000	the poc of reference picture index 13
rkvdec_swreg39_vp9_altreff_hor_virstride	0x0009c	W	0x00000000	vp9 altref frame horizontal virstride
rkvdec_swreg39_refer14_poc	0x0009c	W	0x00000000	the poc of reference picture index 14
rkvdec_swreg40_cur_poc	0x000a0	W	0x00000000	the poc of cur picture
rkvdec_swreg41_rlcwrite_base	0x000a4	W	0x00000000	the base address or rlcwrite base addr
rkvdec_swreg42_pps_base	0x000a8	W	0x00000000	the base address of pps
rkvdec_swreg43_rps_base	0x000ac	W	0x00000000	the base address of rps
rkvdec_swreg44_strmd_error_en	0x000b0	W	0x00000000	cabac error enable config
rkvdec_swreg45_strmd_error_status	0x000b4	W	0x00000000	cabac error status
rkvdec_swreg45_vp9_error_info0	0x000b4	W	0x00000000	vp9 error info
rkvdec_swreg46_strmd_error_ctu	0x000b8	W	0x00400000	strmd error ctu

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg47_sao_ctu_position	0x000bc	W	0x00000000	sao ctu position
rkvdec_swreg48_h264_refer15_base	0x000c0	W	0x00000000	base address for reference picture index 15
rkvdec_swreg48_vp9_last_ystride	0x000c0	W	0x00000000	last ref ystride
rkvdec_swreg49_h264_refer15_poc	0x000c4	W	0x00000000	the poc of reference picture index 15
rkvdec_swreg49_vp9_golden_ystride	0x000c4	W	0x00000000	golden ref ystride
rkvdec_swreg50_h264_refer16_poc	0x000c8	W	0x00000000	the poc of reference picture index 16
rkvdec_swreg50_vp9_altrefy_ystride	0x000c8	W	0x00000000	altref ref ystride
rkvdec_swreg51_h264_refer17_poc	0x000cc	W	0x00000000	the poc of reference picture index 17
rkvdec_swreg51_vp9_lastref_yuvstride	0x000cc	W	0x00000000	vp9 lastref yuv stride
rkvdec_swreg52_vp9_refcolmv_base	0x000d0	W	0x00000000	vp9 colmv ref base
rkvdec_swreg52_h264_refer18_poc	0x000d0	W	0x00000000	the poc of reference picture index 18
rkvdec_swreg53_h264_refer19_poc	0x000d4	W	0x00000000	the poc of reference picture index 19
rkvdec_swreg54_h264_refer20_poc	0x000d8	W	0x00000000	the poc of reference picture index 20
rkvdec_swreg55_h264_refer21_poc	0x000dc	W	0x00000000	the poc of reference picture index 21

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg56_h264_refer22_poc	0x000e0	W	0x00000000	the poc of reference picture index 22
rkvdec_swreg57_h264_refer23_poc	0x000e4	W	0x00000000	the poc of reference picture index 23
rkvdec_swreg58_h264_refer24_poc	0x000e8	W	0x00000000	the poc of reference picture index 24
rkvdec_swreg59_h264_refer25_poc	0x000ec	W	0x00000000	the poc of reference picture index 25
rkvdec_swreg60_h264_refer26_poc	0x000f0	W	0x00000000	the poc of reference picture index 26
rkvdec_swreg61_h264_refer27_poc	0x000f4	W	0x00000000	the poc of reference picture index 27
rkvdec_swreg62_h264_refer28_poc	0x000f8	W	0x00000000	the poc of reference picture index 28
rkvdec_swreg63_h264_refer29_poc	0x000fc	W	0x00000000	the poc of reference picture index 29
rkvdec_swreg64_performance_cycle	0x00100	W	0x00000000	hevc performance cycle
rkvdec_swreg65_axi_ddr_rdata	0x00104	W	0x00000000	axi ddr read data num
rkvdec_swreg66_axi_ddr_wdata	0x00108	W	0x00000000	axi ddr write data number
rkvdec_swreg67_fpgadebug_reset	0x0010c	W	0x000000ff	fpga debug reset signals
rkvdec_swreg68_performance_sel	0x00110	W	0x00000000	
rkvdec_swreg69_performance_cnt0	0x00114	W	0x00000000	Register0000 Abstract

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg70_performance_cnt1	0x00118	W	0x00000000	Register0000 Abstract
rkvdec_swreg71_performance_cnt2	0x0011c	W	0x00000000	Register0000 Abstract
rkvdec_swreg72_h264_refer30_poc	0x00120	W	0x00000000	the poc of reference picture index 30
rkvdec_swreg73_h264_refer31_poc	0x00124	W	0x00000000	the poc of reference picture index 31
rkvdec_swreg74_h264_cur_poc1	0x00128	W	0x00000000	h264 cur poc for bottom filed
rkvdec_swreg75_h264_errorinfo_base	0x0012c	W	0x00000000	h264 error info base addr
rkvdec_swreg75_vp9_error_info1	0x0012c	W	0x00000000	vp9 error info1
rkvdec_swreg76_h264_errorinfo_num	0x00130	W	0x00000000	h264 error info num
rkvdec_swreg76_vp9_error_ctu1	0x00130	W	0x00000000	vp9 error ctu1
rkvdec_swreg77_h264_error_e	0x00134	W	0x00000000	h264 error enable high bits
rkvdec_swreg78_colmv_cur_base	0x00138	W	0x00000000	cur frame colmv output addr
rkvdec_swreg79_colmv_ref0_base	0x0013c	W	0x00000000	ref0 frame colmv base addr
rkvdec_swreg80_colmv_ref1_base	0x00140	W	0x00000000	ref1 frame colmv base addr
rkvdec_swreg81_colmv_ref2_base	0x00144	W	0x00000000	ref2 frame colmv base addr

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg82_colmv_ref3_base	0x00148	W	0x00000000	ref3 frame colmv base addr
rkvdec_swreg83_colmv_ref4_base	0x0014c	W	0x00000000	ref4 frame colmv base addr
rkvdec_swreg84_colmv_ref5_base	0x00150	W	0x00000000	ref5 frame colmv base addr
rkvdec_swreg85_colmv_ref6_base	0x00154	W	0x00000000	ref6 frame colmv base addr
rkvdec_swreg86_colmv_ref7_base	0x00158	W	0x00000000	ref7 frame colmv base addr
rkvdec_swreg87_colmv_ref8_base	0x0015c	W	0x00000000	ref8 frame colmv base addr
rkvdec_swreg88_colmv_ref9_base	0x00160	W	0x00000000	ref9 frame colmv base addr
rkvdec_swreg89_colmv_ref10_base	0x00164	W	0x00000000	ref10 frame colmv base addr
rkvdec_swreg90_colmv_ref11_base	0x00168	W	0x00000000	ref11 frame colmv base addr
rkvdec_swreg91_colmv_ref12_base	0x0016c	W	0x00000000	ref12 frame colmv base addr
rkvdec_swreg92_colmv_ref13_base	0x00170	W	0x00000000	ref13 frame colmv base addr
rkvdec_swreg93_colmv_ref14_base	0x00174	W	0x00000000	ref14 frame colmv base addr
rkvdec_swreg94_colmv_ref15_base	0x00178	W	0x00000000	ref15 frame colmv base addr
rkvdec_swreg95_scanlist_addr	0x0017c	W	0x00000000	scanlist addr

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg96_sd_decout_base	0x00180	W	0x00000000	base address of decoder output picture base address
rkvdec_swreg97_sd_y_virstride	0x00184	W	0x00000000	the ouput picture y virtual stride
rkvdec_swreg98_sd_hor_stride	0x00188	W	0x00000000	picture parameters
rkvdec_swreg99_qos_ctrl	0x0018c	W	0x00000000	rkvdec rd/wr
rkvdec_swreg100_PERF_LATENCY_CTRL0	0x00190	W	0x00000000	Axi performance latency module contrl register0
rkvdec_swreg101_PERF_LATENCY_CTRL1	0x00194	W	0x00000000	PERF_LATENCY_CTRL1
rkvdec_swreg102_PERF_RD_MAX_LATENCY_NUM0	0x00198	W	0x00000000	Read max latency number
rkvdec_swreg103_PERF_RD_LATENCY_SAMP_NUM	0x0019c	W	0x00000000	The number of bigger than configed threshold value
rkvdec_swreg104_PERF_RD_LATENCY_ACC_SUM	0x001a0	W	0x00000000	Total sample number
rkvdec_swreg105_PERF_RD_AXI_TOTAL_BYTE	0x001a4	W	0x00000000	perf_rd_axi_total_byte
rkvdec_swreg106_PERF_WR_AXI_TOTAL_BYTE	0x001a8	W	0x00000000	perf_wr_axi_total_byte
rkvdec_swreg107_PERF_WORKING_CNT	0x001ac	W	0x00000000	perf_working_cnt

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Name	Offset	Size	Reset Value	Description
rkvdec_swreg108_qos1	0x001b0	W	0x00000000	perf_working_cnt

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.3 VDPU341 Function Config Detail Register Description

rkvdec_swreg0_id

Address: Operational Base + offset (0x00000)

ID register (read only)

Bit	Attr	Reset Value	Description
31:16	RO	0x6876	prod_num product number The ascii code of "hv", which is 0x6876
15	RO	0x0	reserved
14	RO	0x0	codec_flag codec flag 0: only dec 1: dec + enc
13	RO	0x0	reserved
12	RW	0x1	profile hevc profile 0: Main 1: Main10

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Bit	Attr	Reset Value	Description
11:9	RO	0x7	dec_support dec support bits bit0: HEVC support or not, when it is 1'b1, support bit1: H264 support or not bit2: VP9 support or not
8	RO	0x1	level level 0: FHD 1: UHD
7:0	RO	0x00	minor_ver minor version minor version

rkvdec_swreg1_int

Address: Operational Base + offset (0x00004)

interrupt and decoder enable register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	sw_allow_not_wr_unref_bframe 1:if dec not referance b frame,will not write to ddr if wan't to set this bit to 1,should set sw_scl_down_en=1 also. it only used in hevc and unsupport tile mode

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Bit	Attr	Reset Value	Description
24	RW	0x0	sw_scl_down_en 1:enable 2ratio scaler down enable,it only used in hevc and unsupport tile mode
23	RW	0x0	sw_wr_ddr_align_en 1:set to 1 only when hevc mode decoder,and also request tile mode is disable. it only used in hevc and unsupport tile mode
22	RW	0x0	sw_softreset_rdy when it is 1'b1, it says that softreset has been done
21	RW	0x1	sw_force_softreset_valid when sw_force_softreset_valid is 1'b1, sw_softrst_en will always be valid to the system no matter that whether the axi bus is idle; when sw_force_softreset_valid is 1'b0, sw_softrst_en will only be valid when the axi bus is idle.
20	RW	0x0	sw_softrst_en_p softreset enable softreset enable signal write 1 to soft reset, write 0 invalid puls register

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Bit	Attr	Reset Value	Description
19	RW	0x0	sw_h264orvp9_error_mode h264 or vp9 error mode for VP9: 1'b0: when there is any stream or colmv error, the hardware will stop the decode and reset itself 1'b1: when there is any stream or colmv error, the hardware will still decode the next slice it is recommend that when vp9 , it is configed to 1'b0 for H264: 1'b0: when there is any stream , the hardware will stop the decoder and reset itself 1'b1: when there is any stream error, the hardware will wait the end signal of deblocking and then reset itself
18	RW	0x0	sw_cabu_end_sta cabac decode end status hevc: cabac decode end status h264& vp9 : streamd decode status
17	RW	0x0	sw_colmv_ref_error_sta colmv ref error status colmv ref error status hevc&vp9: when it is 1'b1, it means that inter module read the invalid dpb frame It will self reset the hardware h264: when it is 1'b1, it means that inter module read the invalid dpb frame. when sw_h264_error_mode is 1'b0, it will self reset the hardware, otherwise it will not

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Bit	Attr	Reset Value	Description
16	RW	0x0	sw_buf_empty_sta buffer empty sta buffer empty status, only when sw_buf_empty_en is 1'b1 , this bit is valid, now is for no valid
15	RW	0x0	sw_dec_timeout_sta decoder timeout interrupt status When high the decoder has been idling for too long. it will self reset the hardware only when sw_dec_timeout_e is 1'b1, this bit is valid
14	RW	0x0	sw_dec_error_sta status bit of input stream error hevc & vp9:when high, an error is found in input data stream decoding. It will self reset the hardware. h264: when high, an error is found in input data stream decoding.when sw_h264_error_mode is 1'b0, it will self reset the hardware, otherwise it will not
13	RW	0x0	sw_dec_bus_sta bus error status When this bit is high, there is error on the axi bus, it will self reset hardware
12	RW	0x0	sw_dec_rdy_sta decoder ready status when this bit is high, decoder has decoded a picture(the loop filter module send out a frame rdy)

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Bit	Attr	Reset Value	Description
11	RW	0x0	sw_dec_commonirq_mode dec irq common mode 1'b0: in H264 and vp9 mode, the interrrupt will wait strmd end pulse 1'b1: in H264 and vp9 mode, the interrupt will not wait strmd end plus
10	WO	0x0	sw_dec_e_rewrite_valid sw_de_e rewrite valid signal sw_dec_e rewrite valid signal maybe for only when buffer empty, restart the decoder use
9	RW	0x0	sw_dec_irq_raw the raw status of sw_dec_irq the raw status of sw_dec_irq,SW should reset this bit after interrupt is handled
8	RO	0x0	sw_dec_irq decoder IRQ when high, decoder requests an interrrupt. sw_dec_irq = sw_dec_irq_raw && (sw_dec_irq_dis == 1'b0)
7	RW	0x0	sw_stmerror_waitdecfifo_empty whether the stream error process wait the decfifo empty when it is 1'b0, the stream error process will no wait the ca2decfifo empty when it is 1'b1, the stream error process will wait the ca2decfifo empty when sw_dec_mode is HEVC and VP9, it always take effect; when sw_dec_mode is H264, it only take effect when sw_h264_error_mode is 1'b0

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Bit	Attr	Reset Value	Description
6	RW	0x0	sw_buf_empty_en buffer empty int enable buffer empty interrupt enable, now is for no use
5	RW	0x1	sw_dec_timeout_e Timeout interrupt enable If enabled HW may return timeout interrupt in case HW gets stucked while decoding picture.
4	RW	0x0	sw_dec_irq_dis decoder IRQ disable When hight, there are no interrupts concerning decoder from HW. Polling must be used to see the interrupt status
3	RW	0x0	sw_timeout_mode time out mode timeout mode select 1'b0: TIMEOUT_CYCLES is 241'b1 1'b1: TIMEOUT_CYCLES is 181'b1
2	RW	0x0	sw_dec_e_strmd_clkgate_dis strmd clkgate disable in streamd module, there contains HEVC,H264,VP9 modules, when it is 1'b1, these modules will no auto clkgate
1	RW	0x1	sw_dec_clkgate_e decoder dynamic clock gating enable 0 = clock is running for all structures 1 = clock is gated for decoder structures that are not used

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_dec_e decoder enable Decoder enable. Setting this bit high will start the decoding operation. HW will reset this when the picture is decoded ready or bus error or time out interrupt is given for all decode format. HW will reset this when picture is processed stream error for vp9 & hevc & (h264 when sw_h264_error_mode is 1'b0)

rkvdec_swreg2_sysctrl

Address: Operational Base + offset (0x00008)

Data input and output endian setting and sys ctrl

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_ycacherd_prior sw_ycacherd_prior 1'b0: y cacherd prior is higher than uv 1'b1: y cache prior is equal than uv
30	RW	0x0	sw_colmv_mode colmv mode 1'b0: default colmv storage mode, the colmv is put in sw_decout_base + sw_yuv_virstride 1'b1: the colmv is put in swreg78_colmv_cur_base
29	RW	0x0	sw_buspr_slot_disable slot disable 1'b1:bus prefetch slot manage disable

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Bit	Attr	Reset Value	Description
28	RW	0x0	sw_h264_frame_orslice h264 frame or slice for H264 use 1'b0: frame 1'b1: slice when sw_h264_streamd_mode is 1'b0, this register is valid
27	RW	0x0	sw_h264_firstslice_flag firstslice flag 1'b1: first packet in the frame, for h264 decode to read rps/pps data because the first_mb_in_slice may be wrong, so need this syntax
26	RW	0x0	sw_h264_stream_lastpacket stream last packet flag when sw_h264_stream_mode is 1'b1 sw_h264_stream_lastpacket 1'b0: this packet is not the last packet of frame 1'b1: the packet is the last packet of frame
25	RW	0x0	sw_h264_stream_mode h264 stream mode 1'b0: stream packet is slice by slice or frame by frame, should use sw_h264_frame_orslice 1'b1: stream packet is random, should use sw_h264_stream_last
24	RW	0x0	sw_h264_rps_mode h264 rps mode 1'b0: hardware parse rps mode 1'b1: software parse rps mode
23:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21:20	RW	0x0	sw_dec_mode dec mode 2'd0: hevc 2'd1: h264 2'd2: vp9
19	RO	0x0	reserved
18:12	RW	0x00	sw_strm_start_bit exact bit of stream start exact bit of streamd start word where decoding can be started (asosiates with sw_str_rlc_base)
11	RW	0x0	sw_rlc_mode rlc mode enable 0 = HW decodes video from bit stream 1 = HW decodes video from RLC input data
10	RW	0x0	sw_rlc_mode_direct_write cabac decode output direct write cabac decode output direct write enable when this bit is enable , all the module other than cabac and busifd are not work
9	RO	0x0	reserved
8	RW	0x0	sw_out_cbcr_swap output cbcr swap 1'b0: cb(u) is in the lower address, cr(v) is in the higher address 1'b1: cb(u) is in the higher address,cr(v) is in the lower address sw_in_cbcr_swap is the same with sw_out_cbcr_swap

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Bit	Attr	Reset Value	Description
7	RW	0x0	sw_out_swap32_e decoder output data and dpb input data 32bit swap may be used for 64 or 128 bit environment 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped
6	RW	0x0	sw_out_endian dec output data and colmv , dpb data and colmv input endian 0 = little endian 1 = big endian for litter enadian , a data 0x12345678, 0x78 is stored in lower address, 0x12 is stored in higher address
5	RW	0x0	sw_str_swap64_e stream 64bit data swap may be used for 128 bit environment 0 = no swapping of 64 bit words 1 = 64 bit data words are swapped
4	RW	0x0	sw_str_swap32_e stream 32bit data swap may be used for 64 or 128 bit environment 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_str_endian stream data input endian mode 0 = little endian 1 = big endian for litter enadian , a data 0x12345678, 0x78 is stored in lower address, 0x12 is stored in higher address
2	RW	0x0	sw_in_swap64_e input 64bit data swap for other than stream and dpb data may be used for 128 bit environment 0 = no swapping of 64 bit words 1 = 64 bit data words are swapped
1	RW	0x0	sw_in_swap32_e input 32bit data swap for other than stream and dpb data may be used for 64 or 128 bit environment 0 = no swapping of 32 bit words 1 = 32 bit data words are swapped
0	RW	0x0	sw_in_endian decoder input endian mode for other than stream and dpb data 0 = little endian 1 = big endian for litter enadian , a data 0x12345678, 0x78 is stored in lower address, 0x12 is stored in higher address

rkvdec_swreg3_picpar

Address: Operational Base + offset (0x0000c)

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picture parameters

Bit	Attr	Reset Value	Description
31:21	RW	0x000	sw_slice_num_lowbits slice number in a frame hevc:slice number in a frame (0~199, when it is 0, it real means 1 slice in a frame) just only used for rps read. 2013.11.27 change the meaning from count from 1, so it will be in 1~200 2013.11.30 sw_slice_num max value is change to 600, so sw_slice_num expand to 10bit h264:slice number in a frame (0~4095, when it is 1, it real means 1 slice in a frame), for H264, it means sw_slice_num_lowbits vp9: no use
20:12	RW	0x000	sw_uv_hor_virstride uv horizontal virstride picture horizontal virtual stride (the unit is 128bit) the max is (4096x1.5 + 128) /16 = 0x188 suggest this register to config to even for advance ddr performance
11	RW	0x0	sw_slice_num_highbit the highest bit of sw_slice_num the highest bit of sw_slice_num
10	RW	0x0	sw_uv_hor_virstride_highbit Field0000 Abstract sw_uv_hor_virstride high bit

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	sw_y_hor_virstride picture horizontal virtual stride picture horizontal virtual stride (the unit is 128bit) the max is (4096x1.5 + 128) /16 = 0x188 suggest this register to config to even for advance ddr performance 2015.10.23, change from 9 bit to 10bit

rkvdec_swreg4_strm_rlc_base

Address: Operational Base + offset (0x00010)

the stream or rlc data base address

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_strm_rlc_base the stream or rlc data base address when swreg2.sw_rlc_mode =1, it is base address for rlc data when swreg2.sw_rlc_mode =0, it is base address for stream , after a frame is decoded ready or error (stream error , time out , bus error) , it is the last address of the stream the address should 128bit align
3:0	RO	0x0	reserved

rkvdec_swreg5_stream_rlc_len

Address: Operational Base + offset (0x00014)

amount of stream bytes or rlc data byte in the input buffer or the

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Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:0	RW	0x0000000	sw_stream_len amount of stream (unit is 8bit) in the input buffer amount of stream 8bits in the input buffer the max of sw_stream_len : 4096x2304x1.5x1.5 = 0x1440000 128bits unit: 0x1440000/16 = 0x144000 it is count from 0 2013.10.15 change to 23bit for zty's suggestion 2013.10.28, amount of stream data bytes in input buffer. it is count from 1, change to 27bits

rkvdec_swreg6_cabactbl_prob_base

Address: Operational Base + offset (0x00018)

the base address of cabac table

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_cabactbl_base the base address of cabac table the base address of cabac table the address should 128bit align
3:0	RO	0x0	reserved

rkvdec_swreg7_decout_base

Address: Operational Base + offset (0x0001c)

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base address of decoder output picture base address

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_decout_base base address of decoder output picture addr base address of decoder output picture the address should be 128bit align in H264 decode format, the top field and bottom field are the same addr
3:0	RO	0x0	reserved

rkvdec_swreg8_y_virstride

Address: Operational Base + offset (0x00020)

the ouput picture y virtual stride

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:0	RW	0x000000	sw_y_virstride the output picture y virtual stride the output picture y virtual stride (the unit is 128bit) the max: (4096x1.5 +128) x 2304 = 0xdc8000 we can know the sw_uvout_base = sw_decout_base + (sw_y_virstride <<4) 2015.10.23, change from 20bits to 21bits

rkvdec_swreg9_yuv_virstride

Address: Operational Base + offset (0x00024)

the ouput picture yuv virtual stride

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Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:0	RW	0x000000	sw_yuv_virstride the ouput picture yuv virtual stride the output picture yuv virtual stride (the unit is 128bit) the max : (4096x1.5 +128) x 2304 x1.5 = 0x14ac000 we can know the sw_mvout_base = sw_decout_base + (sw_yuv_virstride <<4) for yuv422: 4096x2304x2x1.25= 0x1680000 2015.10.23, change from 21bits to 22bits

rkvdec_swreg10_h264_refer0_base

Address: Operational Base + offset (0x00028)

base address for reference picture index 0

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer0_base base address for reference picture index0 base address for reference picture index0 (the address should be 128bit align)
3	RW	0x0	sw_ref0_colmv_use_flag ref0 colmv use flag ref0 colmv use flag
2	RW	0x0	sw_ref0_botfield_used bottom field is used bottom field is used the same meaning with ref_valid

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Bit	Attr	Reset Value	Description
1	RW	0x0	sw_ref0_topfield_used top field is used top field is used the same meaning with ref_valid
0	RW	0x0	sw_ref0_field reference 0 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg10_vp9_cprheader_offset

Address: Operational Base + offset (0x00028)

vp9 compressed header offset

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	sw_vp9_cprheader_offset vp9 compressed header offset vp9 compressed header offset, at most 2000 probs, 10bit per prob, 20000 bit at most now is for no use, because it can read from the last syntax of the uncompressed header

rkvdec_swreg10_hevc_refer0_base

Address: Operational Base + offset (0x00028)

base address for reference picture index 0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer0_base base address for reference picture index0 base address for reference picture index 0 (the address should be 128bit align)
3:0	RW	0x0	sw_ref_valid_0_3 valid flag for picture index 0 ~3 valid flag for picture index 0 ~3

rkvdec_swreg11_h264_refer1_base

Address: Operational Base + offset (0x0002c)

base address for reference picture index 1

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer1_base base address for reference picture index1 base address for reference picture index1 (the address should be 128bit align)
3	RW	0x0	sw_ref1_colmv_use_flag sw_ref1_colmv_use_flag sw_ref1_colmv_use_flag
2	RW	0x0	sw_ref1_botfield_used ref1 bottom field is used ref1 bottom field is used
1	RW	0x0	sw_ref1_topfield_used ref1 topfield is used ref1 topfield is used

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_ref1_field reference 1 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg11_vp9_referlast_base

Address: Operational Base + offset (0x0002c)

base address for last frame

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9last_base base address for last frame base address for last (the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg11_hevc_refer1_base

Address: Operational Base + offset (0x0002c)

base address for reference picture index 1

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer1_base base address for reference picture index 1 base address for reference picture index 1 (the address should be 128bit align)
3:0	RW	0x0	sw_ref_valid_4_7 valid flag for picture index 4 ~7 valid flag for picture index 4 ~7

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rkvdec_swreg12_h264_refer2_base

Address: Operational Base + offset (0x00030)

base address for reference picture index 2

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer2_base base address for reference picture index2 base address for reference picture index2 (the address should be 128bit align)
3	RW	0x0	sw_ref2_colmv_use_flag sw_ref2_colmv_use_flag sw_ref2_colmv_use_flag
2	RW	0x0	sw_ref2_botfield_used ref2 bottom field is used ref2 bottom field is used
1	RW	0x0	sw_ref2_topfield_used ref2 topfield is used ref2 topfield is used
0	RW	0x0	sw_ref2_field reference 2 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg12_vp9_refergolden_base

Address: Operational Base + offset (0x00030)

base address for golden frame

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9golden_base base address for golden frame base address for golden (the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg12_hevc_refer2_base

Address: Operational Base + offset (0x00030)

base address for reference picture index 2

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer2_base base address for reference picture index 2 base address for reference picture index 2 (the address should be 128bit align)
3:0	RW	0x0	sw_ref_valid_8_11 valid flag for picture index 8~11 valid flag for picture index 8~11

rkvdec_swreg13_h264_refer3_base

Address: Operational Base + offset (0x00034)

base address for reference picture index 3

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer3_base base address for reference picture index3 base address for reference picture index3 (the address should be 128bit align)

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_ref3_colmv_use_flag sw_ref3_colmv_use_flag sw_ref3_colmv_use_flag
2	RW	0x0	sw_ref3_botfield_used ref3 bottom field is used ref3 bottom field is used
1	RW	0x0	sw_ref3_topfield_used ref3 topfield is used ref3 topfield is used
0	RW	0x0	sw_ref3_field reference 3 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg13_vp9_referalfter_base

Address: Operational Base + offset (0x00034)

base address for referalfter frame

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9alfter_base base address for alfter frame base address for alfter (the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg13_hevc_refer3_base

Address: Operational Base + offset (0x00034)

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base address for reference picture index 3

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer3_base base address for reference picture index 3 base address for reference picture index 3 (the address should be 128bit align)
3	RO	0x0	reserved
2:0	RW	0x0	sw_ref_valid_12_14 valid flag for picture index 12~14 valid flag for picture index 12~14

rkvdec_swreg14_h264_refer4_base

Address: Operational Base + offset (0x00038)

base address for reference picture index 4

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer4_base base address for reference picture index4 base address for reference picture index4 (the address should be 128bit align)
3	RW	0x0	sw_ref4_colmv_use_flag sw_ref4_colmv_use_flag sw_ref4_colmv_use_flag
2	RW	0x0	sw_ref4_botfield_used ref4 bottom field is used ref4 bottom field is used

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Bit	Attr	Reset Value	Description
1	RW	0x0	sw_ref4_topfield_used ref4 topfield is used ref4 topfield is used
0	RW	0x0	sw_ref4_field reference 4 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg14_vp9count_base

Address: Operational Base + offset (0x00038)

vp9 count base addr

Bit	Attr	Reset Value	Description
31:3	RW	0x00000000	sw_vp9count_base vp9 count base addr vp9 count base addr
2:1	RO	0x0	reserved
0	RW	0x0	sw_vp9count_update_en vp9 count update en when 1'b1, the hardware will always update count when 1'b0, the hardware will auto check whether update the count

rkvdec_swreg14_hevc_refer4_base

Address: Operational Base + offset (0x00038)

base address for reference picture index 4

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer4_base base address for reference picture index 4 base address for reference picture index 4(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg15_vp9_segidlast_base

Address: Operational Base + offset (0x0003c)

base address for last frame segment id

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9segidlast_base base address for vp9 last frame segment id base address for vp9 last frame segment id (the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg15_hevc_refer5_base

Address: Operational Base + offset (0x0003c)

base address for reference picture index 5

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer5_base base address for reference picture index 5 base address for reference picture index 5(the address should be 128bit align)
3:0	RO	0x0	reserved

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rkvdec_swreg15_h264_refer5_base

Address: Operational Base + offset (0x0003c)

base address for reference picture index 5

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer5_base base address for reference picture index5 base address for reference picture index5 (the address should be 128bit align)
3	RW	0x0	sw_ref5_colmv_use_flag sw_ref5_colmv_use_flag sw_ref5_colmv_use_flag
2	RW	0x0	sw_ref5_botfield_used ref5 bottom field is used ref5 bottom field is used
1	RW	0x0	sw_ref5_topfield_used ref5 topfield is used ref5 topfield is used
0	RW	0x0	sw_ref5_field reference 5 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg16_vp9_segidcur_base

Address: Operational Base + offset (0x00040)

base address for cur frame segment id

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9segidcur_base base address for vp9 cur frame segment id base address for vp9 cur frame segment id (the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg16_hevc_refer6_base

Address: Operational Base + offset (0x00040)

base address for reference picture index 6

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer6_base base address for reference picture index 6 base address for reference picture index 6(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg16_h264_refer6_base

Address: Operational Base + offset (0x00040)

base address for reference picture index 6

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer6_base base address for reference picture index6 base address for reference picture index6 (the address should be 128bit align)

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_ref6_colmv_use_flag sw_ref6_colmv_use_flag sw_ref6_colmv_use_flag
2	RW	0x0	sw_ref6_botfield_used ref6 botfield is used ref6 botfield is used
1	RW	0x0	sw_ref6_topfield_used ref6 topfield is used ref6 topfield is used
0	RW	0x0	sw_ref6_field reference 6 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg17_vp9_frame_size_last

Address: Operational Base + offset (0x00044)

vp9 last frame size

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_frameheight_last last frame frame_size_height last frame frame_size_height
15:0	RW	0x0000	sw_framewidth_last last frame frame_size_width last frame frame_size_width

rkvdec_swreg17_hevc_refer7_base

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Address: Operational Base + offset (0x00044)

base address for reference picture index 7

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer7_base base address for reference picture index 7 base address for reference picture index 7(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg17_h264_refer7_base

Address: Operational Base + offset (0x00044)

base address for reference picture index 7

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer7_base base address for reference picture index7 base address for reference picture index7 (the address should be 128bit align)
3	RW	0x0	sw_ref7_colmv_use_flag sw_ref7_colmv_use_flag sw_ref7_colmv_use_flag
2	RW	0x0	sw_ref7_botfield_used ref7 bottom field is used ref7 bottom field is used
1	RW	0x0	sw_ref7_topfield_used ref7 topfield is used ref7 topfield is used

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_ref7_field reference 7 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg18_vp9_frame_size_golden

Address: Operational Base + offset (0x00048)

vp9 golden frame size

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_frameheight_golden golden frame_size_height golden frame_size_height
15:0	RW	0x0000	sw_framewidth_golden golden frame_size_width golden frame_size_width

rkvdec_swreg18_hevc_refer8_base

Address: Operational Base + offset (0x00048)

base address for reference picture index 8

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer8_base base address for reference picture index 8 base address for reference picture index 8(the address should be 128bit align)
3:0	RO	0x0	reserved

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rkvdec_swreg18_h264_refer8_base

Address: Operational Base + offset (0x00048)

base address for reference picture index 8

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer8_base base address for reference picture index8 base address for reference picture index8 (the address should be 128bit align)
3	RW	0x0	sw_ref8_colmv_use_flag sw_ref8_colmv_use_flag sw_ref8_colmv_use_flag
2	RW	0x0	sw_ref8_botfield_used ref8 bottom field is used ref8 bottom field is used
1	RW	0x0	sw_ref8_topfiled_used ref8 topfield is used ref8 topfield is used
0	RW	0x0	sw_ref8_field reference 8 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg19_vp9_frame_size_altref

Address: Operational Base + offset (0x0004c)

vp9 alfter frame size

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_frameheight_alfter alfter frame_size_height alfter frame_size_height
15:0	RW	0x0000	sw_framewidth_alfter alfter frame_size_width alfter frame_size_width

rkvdec_swreg19_hevc_refer9_base

Address: Operational Base + offset (0x0004c)

base address for reference picture index 9

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer9_base base address for reference picture index 9 base address for reference picture index 9(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg19_h264_refer9_base

Address: Operational Base + offset (0x0004c)

base address for reference picture index 9

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer9_base base address for reference picture index9 base address for reference picture index9 (the address should be 128bit align)

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_ref9_colmv_use_flag sw_ref9_colmv_use_flag sw_ref9_colmv_use_flag
2	RW	0x0	sw_ref9_botfield_used ref9 bottom field is used ref9 bottom field is used
1	RW	0x0	sw_ref9_topfield_used ref9 topfield is used ref9 topfield is used
0	RW	0x0	sw_ref9_field reference 9 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg20_vp9_segid_grp0

Address: Operational Base + offset (0x00050)

vp9 segid syntax grp0

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid0_frame_skip_en frame skip feature enable frame skip feature enable
21:20	RW	0x0	sw_vp9segid0_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]

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Bit	Attr	Reset Value	Description
19	RW	0x0	sw_vp9segid0_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid0_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level
11	RW	0x0	sw_vp9segid0_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid0_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid0_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RW	0x0	sw_vp9segid_abs_delta abs delta used to decide quant and loopfilter param

rkvdec_swreg20_hevc_refer10_base

Address: Operational Base + offset (0x00050)

base address for reference picture index 10

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer10_base base address for reference picture index 10 base address for reference picture index 10(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg20_h264_refer10_base

Address: Operational Base + offset (0x00050)

base address for reference picture index 10

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer10_base base address for reference picture index10 base address for reference picture index10 (the address should be 128bit align)
3	RW	0x0	sw_ref10_colmv_use_flag sw_ref10_colmv_use_flag sw_ref10_colmv_use_flag
2	RW	0x0	sw_ref10_botfield_used ref10 bottom field is used ref10 bottom field is used
1	RW	0x0	sw_ref10_topfield_used ref10 topfield is used ref10 topfield is used

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_ref10_field reference 10 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg21_vp9_segid_grp1

Address: Operational Base + offset (0x00054)

vp9 segid syntax grp1

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid1_frame_skip_en frame skip feature enable frame skip feature enable
21:20	RW	0x0	sw_vp9segid1_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid1_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid1_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level

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Bit	Attr	Reset Value	Description
11	RW	0x0	sw_vp9segid1_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid1_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid1_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

rkvdec_swreg21_hevc_refer11_base

Address: Operational Base + offset (0x00054)

base address for reference picture index 11

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer11_base base address for reference picture index 11 base address for reference picture index 11(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg21_h264_refer11_base

Address: Operational Base + offset (0x00054)

base address for reference picture index 11

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer11_base base address for reference picture index11 base address for reference picture index11 (the address should be 128bit align)
3	RW	0x0	sw_ref11_colmv_use_flag sw_ref11_colmv_use_flag sw_ref11_colmv_use_flag
2	RW	0x0	sw_ref11_botfield_used ref11 bottom field is used ref11 bottom field is used
1	RW	0x0	sw_ref11_topfield_used ref11 topfield is used ref11 topfield is used
0	RW	0x0	sw_ref11_field reference 11 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg22_vp9_segid_grp2

Address: Operational Base + offset (0x00058)

vp9 segid syntax grp2

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid2_frame_skip_en frame skip feature enable frame skip feature enable

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Bit	Attr	Reset Value	Description
21:20	RW	0x0	sw_vp9segid2_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid2_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid2_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level
11	RW	0x0	sw_vp9segid2_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid2_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid2_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

rkvdec_swreg22_hevc_refer12_base

Address: Operational Base + offset (0x00058)

base address for reference picture index 12

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer12_base base address for reference picture index 12 base address for reference picture index 12(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg22_h264_refer12_base

Address: Operational Base + offset (0x00058)

base address for reference picture index 12

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer12_base base address for reference picture index12 base address for reference picture index12 (the address should be 128bit align)
3	RW	0x0	sw_ref12_colmv_use_flag sw_ref12_colmv_use_flag sw_ref12_colmv_use_flag
2	RW	0x0	sw_ref12_botfield_used ref12 bottom field is used ref12 bottom field is used
1	RW	0x0	sw_ref12_topfield_used ref12 topfield is used ref12 topfield is used

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_ref12_field reference 12 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg23_vp9_segid_grp3

Address: Operational Base + offset (0x0005c)

vp9 segid syntax grp3

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid3_frame_skip_en frame skip feature enable frame skip feature enable
21:20	RW	0x0	sw_vp9segid3_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid3_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid3_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level

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Bit	Attr	Reset Value	Description
11	RW	0x0	sw_vp9segid3_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid3_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid3_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

rkvdec_swreg23_hevc_refer13_base

Address: Operational Base + offset (0x0005c)

base address for reference picture index 13

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer13_base base address for reference picture index 13 base address for reference picture index 13(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg23_h264_refer13_base

Address: Operational Base + offset (0x0005c)

base address for reference picture index 13

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer13_base base address for reference picture index13 base address for reference picture index13 (the address should be 128bit align)
3	RW	0x0	sw_ref13_colmv_use_flag sw_ref13_colmv_use_flag sw_ref13_colmv_use_flag
2	RW	0x0	sw_ref13_botfield_used ref13 bottom field is used ref13 bottom field is used
1	RW	0x0	sw_ref13_topfield_used ref13 topfield is used ref13 topfield is used
0	RW	0x0	sw_ref13_field reference 13 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg24_vp9_segid_grp4

Address: Operational Base + offset (0x00060)

vp9 segid syntax grp4

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid4_frame_skip_en frame skip feature enable frame skip feature enable

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Bit	Attr	Reset Value	Description
21:20	RW	0x0	sw_vp9segid4_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid4_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid4_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level
11	RW	0x0	sw_vp9segid4_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid4_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid4_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

rkvdec_swreg24_hevc_refer14_base

Address: Operational Base + offset (0x00060)

base address for reference picture index 14

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer14_base base address for reference picture index 14 base address for reference picture index 14(the address should be 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg24_h264_refer14_base

Address: Operational Base + offset (0x00060)

base address for reference picture index 14

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer14_base base address for reference picture index14 base address for reference picture index14 (the address should be 128bit align)
3	RW	0x0	sw_ref14_colmv_use_flag sw_ref14_colmv_use_flag sw_ref14_colmv_use_flag
2	RW	0x0	sw_ref14_botfield_used ref14 bottom field is used ref14 bottom field is used
1	RW	0x0	sw_ref14_topfield_used ref14 topfield is used ref14 topfield is used

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_ref14_field reference 14 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg25_vp9_segid_grp5

Address: Operational Base + offset (0x00064)

vp9 segid syntax grp5

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid5_frame_skip_en frame skip feature enable frame skip feature enable
21:20	RW	0x0	sw_vp9segid5_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid5_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid5_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level

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Bit	Attr	Reset Value	Description
11	RW	0x0	sw_vp9segid5_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid5_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid5_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

rkvdec_swreg25_refer0_poc

Address: Operational Base + offset (0x00064)

the poc of reference picture index 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer0_poc the poc of reference picture index 0 the poc of reference picture index 0

rkvdec_swreg26_vp9_segid_grp6

Address: Operational Base + offset (0x00068)

vp9 segid syntax grp6

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22	RW	0x0	sw_vp9segid6_frame_skip_en frame skip feature enable frame skip feature enable
21:20	RW	0x0	sw_vp9segid6_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid6_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid6_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level
11	RW	0x0	sw_vp9segid6_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid6_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid6_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

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rkvdec_swreg26_refer1_poc

Address: Operational Base + offset (0x00068)

the poc of reference picture index 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer1_poc the poc of reference picture index 1 the poc of reference picture index 1

rkvdec_swreg27_vp9_segid_grp7

Address: Operational Base + offset (0x0006c)

vp9 segid syntax grp7

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RW	0x0	sw_vp9segid7_frame_skip_en frame skip feature enable frame skip feature enable
21:20	RW	0x0	sw_vp9segid7_referinfo specifies segment i's reference_info which is used to get ref_frame[0] specifies segment i's reference_info which is used to get ref_frame[0]
19	RW	0x0	sw_vp9segid7_referinfo_en frame reference info enable frame reference info enable
18:12	RW	0x00	sw_vp9segid7_frame_loopfilter_value frame loopfitler value specifies segment i's loopfilter_delta value which is used to calculate filter level

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Bit	Attr	Reset Value	Description
11	RW	0x0	sw_vp9segid7_frame_loopfitler_value_en frame_loopfilter_value feature enable frame_loopfilter_value feature enable
10:2	RW	0x000	sw_vp9segid7_frame_qp_delta frame qp delta specifies segment i's qp_delta value which is used to calculate y_dequant and uv_dequant
1	RW	0x0	sw_vp9segid7_frame_qp_delta_en frame_qp_delta feature enable frame_qp_delta feature enable
0	RO	0x0	reserved

rkvdec_swreg27_refer2_poc

Address: Operational Base + offset (0x0006c)

the poc of reference picture index 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer2_poc the poc of reference picture index 2 the poc of reference picture index 2

rkvdec_swreg28_vp9_cprheader_config

Address: Operational Base + offset (0x00070)

vp9 compressed header config info

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:3	RW	0x0	sw_vp9_frame_reference_mode frame_reference_mode frame_reference_mode specifies frame reference mode. SINGLE_REFERENCE = 0, COMPOUND_REFERENCE = 1, REFERENCE_MODE_SELECT = 2, REFERENCE_MODES = 3, When frame_reference_mode_flag0 is not present ,it equal to 0 by default. When frame_reference_mode_flag1 is not present ,it equal to 0 by default. frame_reference_mode = frame_reference_mode_flag0 == 0 ? frame_reference_mode_flag1 == 0 ? REFERENCE_MODE_SELECT : COMPOUND_REFERENCE) : SINGLE_REFERENCE

2:0	RW	0x0	sw_vp9_tx_mode
			tx_mode
			tx_mode specifies frame transform mode.
			ONLY_4X4 = 0, // only 4x4 transform used
			ALLOW_8X8 = 1, // allow block transform size up to 8x8
			ALLOW_16X16 = 2, // allow block transform size up to 16x16
			ALLOW_32X32 = 3, // allow block transform size up to 32x32
			TX_MODE_SELECT = 4, // transform specified for each block

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rkvdec_swreg28_refer3_poc

Address: Operational Base + offset (0x00070)

the poc of reference picture index 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer3_poc the poc of reference picture index 3 the poc of reference picture index 3

rkvdec_swreg29_vp9_lref_scale

Address: Operational Base + offset (0x00074)

scaling factor for last reference picture

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_vp9_lref_ver_scale vertical scaling factor for last reference picture vertical scaling factor for last reference picture
15:0	RW	0x0000	sw_vp9_lref_hor_scale horizontal scaling factor for last reference picture horizontal scaling factor for last reference picture sw_vp9_lref_hor_scale = (last_ref_width / cur_width) * 0x4000

rkvdec_swreg29_refer4_poc

Address: Operational Base + offset (0x00074)

the poc of reference picture index 4

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer4_poc the poc of reference picture index 4 the poc of reference picture index 4

rkvdec_swreg30_vp9_gref_scale

Address: Operational Base + offset (0x00078)

scaling factor for golden reference picture

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_vp9_gref_ver_scale vertical scaling factor for golden reference picture vertical scaling factor for golden reference picture
15:0	RW	0x0000	sw_vp9_gref_hor_scale horizontal scaling factor for golden reference picture horizontal scaling factor for golden reference picture sw_vp9_gref_hor_scale = (golden_ref_width / cur_width) * 0x4000

rkvdec_swreg30_refer5_poc

Address: Operational Base + offset (0x00078)

the poc of reference picture index 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer5_poc the poc of reference picture index 5 the poc of reference picture index 5

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rkvdec_swreg31_vp9_aref_scale

Address: Operational Base + offset (0x0007c)

scaling factor for alfter reference picture

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_vp9_aref_ver_scale vertical scaling factor for alfter reference picture vertical scaling factor for alfter reference picture
15:0	RW	0x0000	sw_vp9_aref_hor_scale horizontal scaling factor for alfter reference picture horizontal scaling factor for alfter reference picture sw_vp9_gref_hor_scale = (alfter_ref_width / cur_width) * 0x4000

rkvdec_swreg31_refer6_poc

Address: Operational Base + offset (0x0007c)

the poc of reference picture index 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer6_poc the poc of reference picture index 6 the poc of reference picture index 6

rkvdec_swreg32_vp9_ref_deltas_lastframe

Address: Operational Base + offset (0x00080)

vp9 ref deltas

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Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_vp9_ref_deltas_lastframe vp9 ref deltas vp9 ref deltas of lastframe, for cal loopfilter filter type use

rkvdec_swreg32_refer7_poc

Address: Operational Base + offset (0x00080)

the poc of reference picture index 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer7_poc the poc of reference picture index 7 the poc of reference picture index 7

rkvdec_swreg33_vp9_info_lastframe

Address: Operational Base + offset (0x00084)

vp9 info for lastframe

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:20	RW	0x0	sw_vp9_color_space_lastkeyframe vp9 last keyframe color_space vp9 last keyframe color_space
19	RW	0x0	sw_vp9_last_widhheight_eqcur last width and height equal cur last width and height equal cur frame

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Bit	Attr	Reset Value	Description
18	RW	0x0	sw_vp9_last_intra_only last frame intra only flag vp9 last frame intra only flag to give inter command use it is for last_dec_frame
17	RW	0x0	sw_vp9_last_show_frame last show frame flag for cal the flag use_prev_in_find_mv_refs which is to inter cmd it is for last_dec_frame
16	RW	0x0	sw_segmentation_enable_lstframe segmentation_enable for last frame 1'b1:sw_segmentation_enable for last frame it is for last_dec_frame
15:14	RO	0x0	reserved
13:0	RW	0x0000	sw_vp9_mode_deltas_lastframe vp9 mode deltas vp9 mode deltas it is for last dec frame

rkvdec_swreg33_refer8_poc

Address: Operational Base + offset (0x00084)

the poc of reference picture index 8

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer8_poc the poc of reference picture index 8 the poc of reference picture index 8

rkvdec_swreg34_vp9_intercmd_base

Address: Operational Base + offset (0x00088)

inter cmd base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9_intercmd_base vp9 intercmd base addr vp9 inter command base addr, when sw_rlc_mode is 1'b1; when sw_dec_mode is VP9 and sw_rlc_mode is 1'b1, when read this register, after a frame is decoded ready or error (stream error , time out , bus error) , it is the end address of the intercmd
3:0	RO	0x0	reserved

rkvdec_swreg34_refer9_poc

Address: Operational Base + offset (0x00088)

the poc of reference picture index 9

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer9_poc the poc of reference picture index 9 the poc of reference picture index 9

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rkvdec_swreg35_vp9_intercmd_num

Address: Operational Base + offset (0x0008c)

vp9 intercmd num

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	sw_vp9_intercmd_num sw_vp9_intercmd_num when rlc_mode is 1'b1, for sw_vp9_intercmd_num it's unit is 128bit it count from 1

rkvdec_swreg35_refer10_poc

Address: Operational Base + offset (0x0008c)

the poc of reference picture index 10

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer10_poc the poc of reference picture index 10 the poc of reference picture index 10

rkvdec_swreg36_vp9_lasttile_size

Address: Operational Base + offset (0x00090)

vp9 lasttile size

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:0	RW	0x000000	sw_vp9_lasttile_size vp9 last tile size vp9 last tile size ofr cur frame its unit is byte The real meaning the current frame size

rkvdec_swreg36_refer11_poc

Address: Operational Base + offset (0x00090)

the poc of reference picture index 11

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer11_poc the poc of reference picture index 11 the poc of reference picture index 11

rkvdec_swreg37_vp9_lastf_hor_virstride

Address: Operational Base + offset (0x00094)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_vp9_lastfuv_hor_virstride sw_vp9_lastfuv_hor_virstride vp9 last frame uv horizontal virstride 2015.10.23, change from 9bits to 10bits
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	sw_vp9_lastfy_hor_virstride vp9 last frame y horizontal virstride vp9 last frame y horizontal virstride 2015.10.23, change from 9bits to 10bits

rkvdec_swreg37_refer12_poc

Address: Operational Base + offset (0x00094)

the poc of reference picture index 12

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer12_poc the poc of reference picture index 12 the poc of reference picture index 12

rkvdec_swreg38_vp9_goldenf_hor_virstride

Address: Operational Base + offset (0x00098)

vp9 golden frame horizontal virstride

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_vp9_goldenuv_hor_virstride vp9 goldenuv horizontal virstride vp9 golden uv horizontal virstirde 2015.10.23, change from 9bits to 10bits
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	sw_vp9_goldenfy_hor_virstride sw_vp9_goldenfy_hor_virstirde vp9 golden frame y horizontal virstride 2015.10.23, change from 9bits to 10bits

rkvdec_swreg38_refer13_poc

Address: Operational Base + offset (0x00098)

the poc of reference picture index 13

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer13_poc the poc of reference picture index 13 the poc of reference picture index 13

rkvdec_swreg39_vp9_altreff_hor_virstride

Address: Operational Base + offset (0x0009c)

vp9 altref frame horizontal virstride

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_vp9_altreffuv_hor_virstride vp9 altreffuv horizontal virstride vp9 altreff uv horizontal virstirde 2015.10.23, change from 9bits to 10bits
15:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	sw_vp9_altreffy_hor_virstride sw_vp9_altreffy_hor_virstirde vp9 altref frame y horizontal virstride 2015.10.23, change from 9bits to 10bits

rkvdec_swreg39_refer14_poc

Address: Operational Base + offset (0x0009c)

the poc of reference picture index 14

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer14_poc the poc of reference picture index 14 the poc of reference picture index 14

rkvdec_swreg40_cur_poc

Address: Operational Base + offset (0x000a0)

the poc of cur picture

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_cur_poc the poc of the cur picture the poc of the cur picture for H264, it may be cur frame poc or cur top field poc

rkvdec_swreg41_rlcwrite_base

Address: Operational Base + offset (0x000a4)

the base address or rlcwrite base addr

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Bit	Attr	Reset Value	Description
31:3	RW	0x00000000	sw_rlcwrite_base the base address of rlcwrite the base address of rlcwrite(the address should 64bit align) cabac output write to this rlcwrite base address when sw_rlc_mode_direct_write in swreg2_sysctrl is valid
2:0	RO	0x0	reserved

rkvdec_swreg42_pps_base

Address: Operational Base + offset (0x000a8)

the base address of pps

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_pps_base the base address of pps the base address of pps (the address should 128bit align) it is for storing sps(sequence parameter set) and pps(picture parameter set)
3:0	RO	0x0	reserved

rkvdec_swreg43_rps_base

Address: Operational Base + offset (0x000ac)

the base address of rps

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_rps_base rps base address rps(reference picture set) base address (the address should 128bit align)
3:0	RO	0x0	reserved

rkvdec_swreg44_strmd_error_en

Address: Operational Base + offset (0x000b0)

cabac error enable config

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_strmd_error_e strmd error enable regs strmd error enable regs in HEVC,it is called sw_cabac_error_e for VP9, it use sw_strmd_error_e[3:0] sw_strmd_error_e[0] is for sw_vp9_tilesize_error sw_strmd_error_e[1] is for sw_vp9_segskip_error sw_strmd_error_e[2] is for sw_vp9_error_init_error sw_strmd_error_e[3] is for sw_vp9_uncpr sw_strmd_error_e[4] is for sw_vp9_refscale_erro, sw_vp9_refscale_erro now is no use

rkvdec_swreg45_strmd_error_status

Address: Operational Base + offset (0x000b4)

cabac error status

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Bit	Attr	Reset Value	Description
31:28	RW	0x0	sw_colmv_error_ref_picidx colmv error ref picidx when sw_colmv_ref_error_sta is 1'b1, these bits are used for tell which dpb frame is invalid but is read by inter module it is for H264 and HEVC
27:0	RW	0x0000000	sw_strmd_error_status cabac error status strmd error status in HEVC & H264, it is called cabac error status

rkvdec_swreg45_vp9_error_info0

Address: Operational Base + offset (0x000b4)

vp9 error info

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	vp9_error_info0 vp9_error_info0 for vp9 16 tile cols, every cols contains 4bits vp9_error_info0[3:0] is for col 0 vp9_error_info0[1:0] is to tell tile_rows_cnt[1:0] vp9_error_info0[3:2] is to tell the error type 2'b00: no error 2'b01: tilesize error 2'b10: seg skip error 2'b11: ref scale error

rkvdec_swreg46_strmd_error_ctu

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Address: Operational Base + offset (0x000b8)

strmd error ctu

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	sw_vp9_error_ctu0_en to tell whether is any error of vp9 1'b1: there is aleast a error in vp9 strmd now ,is no for use
23	RO	0x0	reserved
22:16	RW	0x40	sw_streamfifo_space2full stream fifo space to full It is for debug use, to tell the stream fifo space to full for HEVC , H264 and VP9
15:8	RW	0x00	sw_strmd_error_ctu_yoffset strmd error ctu yoffset strmd error ctu yoffset for HEVC , H264 and VP9 for VP9, it is the value of stsw_vp9_error_ctu0_y
7:0	RW	0x00	sw_strmd_error_ctu_xoffset strmd error ctu xoffset strmd error ctu xoffset for all HEVC and H264 and VP9 for vp9, it is the value of stsw_vp9_error_ctu0_x

rkvdec_swreg47_sao_ctu_position

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Address: Operational Base + offset (0x000bc)

sao ctu position

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_saowr_yoffset saowr y offset saowr y offset , its unit is 4 line pixel 0:sao begin to write 0~3 pic line 1:sao begin to write 4-7 pic line ..... only valid when ip has any error.
15:9	RO	0x0	reserved
8:0	RW	0x000	sw_saowr_xoffet saowr x address offset saowr x address offset, its unit is 128bit 0:sao begin to write cur line first 128bit 1:sao begin to write cur line second 128bit ..... only valid when ip has any error.

rkvdec_swreg48_h264_refer15_base

Address: Operational Base + offset (0x000c0)

base address for reference picture index 15

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_refer15_base base address for reference picture index15 base address for reference picture index15 (the address should be 128bit align)

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_ref15_colmv_use_flag sw_ref15_colmv_use_flag sw_ref15_colmv_use_flag
2	RW	0x0	sw_ref15_botfield_used ref15 bottom field is used ref15 bottom field is used
1	RW	0x0	sw_ref15_topfield_used ref15 topfield is used ref15 topfield is used
0	RW	0x0	sw_ref15_field reference 15 picture field flag 1'b0: frame 1'b1: field

rkvdec_swreg48_vp9_last_ystride

Address: Operational Base + offset (0x000c0)

last ref ystride

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:0	RW	0x000000	sw_vp9_lastfy_virstride last frame y virstride vp9 last frame y stride 2015.10.23, change from 20bits to 21bits

rkvdec_swreg49_h264_refer15_poc

Address: Operational Base + offset (0x000c4)

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the poc of reference picture index 15

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer15_poc the poc of reference picture index 15 the poc of reference picture index 15

rkvdec_swreg49_vp9_golden_ystride

Address: Operational Base + offset (0x000c4)

golden ref ystride

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:0	RW	0x000000	sw_vp9_goldeny_virstride golden frame y virstride vp9 golden frame y stride 2015.10.23, change from 20bits to 21bits

rkvdec_swreg50_h264_refer16_poc

Address: Operational Base + offset (0x000c8)

the poc of reference picture index 16

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer16_poc the poc of reference picture index 16 the poc of reference picture index 16

rkvdec_swreg50_vp9_altrefy_ystride

Address: Operational Base + offset (0x000c8)

altref ref ystride

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Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:0	RW	0x000000	sw_vp9_altrefy_virstride altref frame y virstride vp9 altref frame y stride change from 20bits to 21bits

rkvdec_swreg51_h264_refer17_poc

Address: Operational Base + offset (0x000cc)

the poc of reference picture index 17

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer17_poc the poc of reference picture index 17 the poc of reference picture index 17

rkvdec_swreg51_vp9_lastref_yuvstride

Address: Operational Base + offset (0x000cc)

vp9 lastref yuv stride

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:0	RW	0x000000	sw_vp9_lastref_yuv_virstride lastref frame yuv virstride vp9 lastref frame yuv vir stride 2015.10.23, change from 21bits to 22bits

rkvdec_swreg52_vp9_refcolmv_base

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Address: Operational Base + offset (0x000d0)

vp9 colmv ref base

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_vp9_refcolmv_base vp9 ref colmv base vp9 ref colmv base addr
3:0	RO	0x0	reserved

rkvdec_swreg52_h264_refer18_poc

Address: Operational Base + offset (0x000d0)

the poc of reference picture index 18

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer18_poc the poc of reference picture index 18 the poc of reference picture index 18

rkvdec_swreg53_h264_refer19_poc

Address: Operational Base + offset (0x000d4)

the poc of reference picture index 19

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer19_poc the poc of reference picture index 19 the poc of reference picture index 19

rkvdec_swreg54_h264_refer20_poc

Address: Operational Base + offset (0x000d8)

the poc of reference picture index 20

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer20_poc the poc of reference picture index 20 the poc of reference picture index 20

rkvdec_swreg55_h264_refer21_poc

Address: Operational Base + offset (0x000dc)

the poc of reference picture index 21

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer21_poc the poc of reference picture index 21 the poc of reference picture index 21

rkvdec_swreg56_h264_refer22_poc

Address: Operational Base + offset (0x000e0)

the poc of reference picture index 22

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer22_poc the poc of reference picture index 22 the poc of reference picture index 22

rkvdec_swreg57_h264_refer23_poc

Address: Operational Base + offset (0x000e4)

the poc of reference picture index 23

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer23_poc the poc of reference picture index 23 the poc of reference picture index 23

rkvdec_swreg58_h264_refer24_poc

Address: Operational Base + offset (0x000e8)

the poc of reference picture index 24

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer24_poc the poc of reference picture index 24 the poc of reference picture index 24

rkvdec_swreg59_h264_refer25_poc

Address: Operational Base + offset (0x000ec)

the poc of reference picture index 25

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer25_poc the poc of reference picture index 25 the poc of reference picture index 25

rkvdec_swreg60_h264_refer26_poc

Address: Operational Base + offset (0x000f0)

the poc of reference picture index 26

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer26_poc the poc of reference picture index 26 the poc of reference picture index 26

rkvdec_swreg61_h264_refer27_poc

Address: Operational Base + offset (0x000f4)

the poc of reference picture index 27

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer27_poc the poc of reference picture index 27 the poc of reference picture index 27

rkvdec_swreg62_h264_refer28_poc

Address: Operational Base + offset (0x000f8)

the poc of reference picture index 28

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer28_poc the poc of reference picture index 28 the poc of reference picture index 28

rkvdec_swreg63_h264_refer29_poc

Address: Operational Base + offset (0x000fc)

the poc of reference picture index 29

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer29_poc the poc of reference picture index 29 the poc of reference picture index 29

rkvdec_swreg64_performance_cycle

Address: Operational Base + offset (0x00100)

hevc performance cycle

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_performance_cycle hevc running cycle hevc running cycle if just want to analys a frame performance cycle, should set the register 0 before start a frame

rkvdec_swreg65_axi_ddr_rdata

Address: Operational Base + offset (0x00104)

axi ddr read data num

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_axi_ddr_rdata axi ddr rdata num axi ddr rdata num, the unit is byte

rkvdec_swreg66_axi_ddr_wdata

Address: Operational Base + offset (0x00108)

axi ddr write data number

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_axi_ddr_wdata hevc write data byte num hevc write data byte num

rkvdec_swreg67_fpgadebug_reset

Address: Operational Base + offset (0x0010c)

fpga debug reset signals

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	sw_filer_resetn Field0000 Abstract Field0000 Description
6	RW	0x1	sw_recon_resetn Field0000 Abstract Field0000 Description
5	RW	0x1	sw_inter_resetn Field0000 Abstract Field0000 Description
4	RW	0x1	sw_intra_resetn Field0000 Abstract Field0000 Description
3	RW	0x1	sw_transd_resetn Field0000 Abstract Field0000 Description

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Bit	Attr	Reset Value	Description
2	RW	0x1	sw_dec_ctrl_resetn Field0000 Abstract Field0000 Description
1	RW	0x1	sw_cabac_resetn cabac resetn cabac resetn
0	RW	0x1	sw_busifd_resetn busifd reset n when write to 1'b0, reset the busifd module when write to 1'b1, unreset the busifd module

rkvdec_swreg68_performance_sel

Address: Operational Base + offset (0x00110)

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
			perf_cnt2_sel Field0000 Abstract 0 : don't work; 1 : cycles counter for cabac in buffer empty; 2 : cycles counter for cabac in buffer full; 3 : cycles counter for cabac out buffer empty; 4 : cycles counter for cabac out buffer full; 5 : cycles counter for transd input data ready; 6 : cycles counter for transd write data to recon allow; 7 : cycles counter for dec2transd cmd empty; 8 : cycles counter for dec2transd cmd full; 9 : cycles counter for transd2dblk bs fifo empty; 10: cycles counter for transd2dblk bs fifo full; 11: cycles counter for dec2intra cmd fifo empty; 12: cycles counter for dec2intra cmd fifo full; 13: cycles counter for mc2recon cmd fifo empty; 14: cycles counter for mc2recon cmd fifo full; 15: cycles counter for mc2recon data fifo empty; 16: cycles counter for mc2recon data fifo full; 17: cycles counter for recon2filter data write allow; 18: cycles counter for inter2busifd cmd fifo empty; 19: cycles counter for inter2busifd cmd fifo
21:16	RW	0x00	

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Bit	Attr	Reset Value	Description
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
			perf_cnt1_sel Field0000 Abstract 0 : don't work; 1 : cycles counter for cabac in buffer empty; 2 : cycles counter for cabac in buffer full; 3 : cycles counter for cabac out buffer empty; 4 : cycles counter for cabac out buffer full; 5 : cycles counter for transd input data ready; 6 : cycles counter for transd write data to recon allow; 7 : cycles counter for dec2transd cmd empty; 8 : cycles counter for dec2transd cmd full; 9 : cycles counter for transd2dblk bs fifo empty; 10: cycles counter for transd2dblk bs fifo full; 11: cycles counter for dec2intra cmd fifo empty; 12: cycles counter for dec2intra cmd fifo full; 13: cycles counter for mc2recon cmd fifo empty; 14: cycles counter for mc2recon cmd fifo full; 15: cycles counter for mc2recon data fifo empty; 16: cycles counter for mc2recon data fifo full; 17: cycles counter for recon2filter data write allow; 18: cycles counter for inter2busifd cmd fifo empty; 19: cycles counter for inter2busifd cmd fifo
13:8	RW	0x00	

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Bit	Attr	Reset Value	Description
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x00	perf_cnt0_sel sel counter0 to cal which signal 0 : don't work; 1 : cycles counter for cabac in buffer empty; 2 : cycles counter for cabac in buffer full; 3 : cycles counter for cabac out buffer empty; 4 : cycles counter for cabac out buffer full; 5 : cycles counter for transd input data ready; 6 : cycles counter for transd write data to recon allow; 7 : cycles counter for dec2transd cmd empty; 8 : cycles counter for dec2transd cmd full; 9 : cycles counter for transd2dblk bs fifo empty; 10: cycles counter for transd2dblk bs fifo full; 11: cycles counter for dec2intra cmd fifo empty; 12: cycles counter for dec2intra cmd fifo full; 13: cycles counter for mc2recon cmd fifo empty; 14: cycles counter for mc2recon cmd fifo full; 15: cycles counter for mc2recon data fifo empty; 16: cycles counter for mc2recon data fifo full; 17: cycles counter for recon2filter data write allow; 18: cycles counter for inter2busifd cmd fifo empty; 19: cycles counter for inter2busifd cmd fifo

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rkvdec_swreg69_performance_cnt0

Address: Operational Base + offset (0x00114)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_cnt0 Field0000 Abstract Field0000 Description

rkvdec_swreg70_performance_cnt1

Address: Operational Base + offset (0x00118)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_cnt1

rkvdec_swreg71_performance_cnt2

Address: Operational Base + offset (0x0011c)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_cnt2 Field0000 Abstract Field0000 Description

rkvdec_swreg72_h264_refer30_poc

Address: Operational Base + offset (0x00120)

the poc of reference picture index 30

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer30_poc the poc of reference picture index 30 the poc of reference picture index 30

rkvdec_swreg73_h264_refer31_poc

Address: Operational Base + offset (0x00124)

the poc of reference picture index 31

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_refer31_poc the poc of reference picture index 31 the poc of reference picture index 31

rkvdec_swreg74_h264_cur_poc1

Address: Operational Base + offset (0x00128)

h264 cur poc for bottom filed

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_h264_cur_poc1 h264 cur poc for bottom field h264 cur poc for bottom field

rkvdec_swreg75_h264_errorinfo_base

Address: Operational Base + offset (0x0012c)

h264 error info base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:3	RW	0x00000000	sw_errorinfo_base error info base addr error info base addr every slice contains 256 bits error info
2:0	RO	0x0	reserved

rkvdec_swreg75_vp9_error_info1

Address: Operational Base + offset (0x0012c)

vp9 error info1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	vp9_error_info1 vp9_error_info1 for vp9 16 tile cols, every cols contains 4bits vp9_error_info0[3:0] is for col 8 vp9_error_info0[1:0] is to tell tile_rows_cnt[1:0] vp9_error_info0[3:2] is to tell the error type 2'b00: no error 2'b01: tilesize error 2'b10: seg skip error

rkvdec_swreg76_h264_errorinfo_num

Address: Operational Base + offset (0x00130)

h264 error info num

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:16	RW	0x0000	sw_error_packet_num error packet number error packet number
15	RW	0x0	sw_strmd_detect_error_flag strmd error detect flag streamd detect error flag, it is only for vp9 and h264 format
14	RO	0x0	reserved
13:0	RW	0x0000	sw_slicedec_num slice dec num h264 decoded num, the max slice num for H264 is 4096

rkvdec_swreg76_vp9_error_ctu1

Address: Operational Base + offset (0x00130)

vp9 error ctu1

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	vp9_error_ctu1_en ctu1 is error 1'b1: there are at least two ctus error
14	RO	0x0	reserved
13:8	RW	0x00	vp9_error_ctu1_y vp9 error ctu1 y to tell the error ctu1 y position
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:0	RW	0x00	vp9_error_ctu1_x vp9 error ctu1 x to tell the error ctu1 x position

rkvdec_swreg77_h264_error_e

Address: Operational Base + offset (0x00134)

h264 error enable high bits

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:28	RW	0x0	sw_strmd_error_en stream error enable sw_strmd_error_en[0]: HEVC, H264 slice end error enable sw_strmd_error_en[1]: HEVC,H264, VP9 frame end error enable
27:0	RW	0x0000000	sw_h264_error_en_highbits h264 error enable high bits h264 error enable bits

rkvdec_swreg78_colmv_cur_base

Address: Operational Base + offset (0x00138)

cur frame colmv output addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_cur_base cur frame colmv base only valid when sw_colmv_mode is 1'b1 cur frame colmv base addr for HEVC,H264 and vp9
3:0	RO	0x0	reserved

rkvdec_swreg79_colmv_ref0_base

Address: Operational Base + offset (0x0013c)

ref0 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref0_base ref0 frame colmv base only valid when sw_colmv_mode is 1'b1 ref0 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg80_colmv_ref1_base

Address: Operational Base + offset (0x00140)

ref1 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref1_base ref1 frame colmv base only valid when sw_colmv_mode is 1'b1 ref1 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg81_colmv_ref2_base

Address: Operational Base + offset (0x00144)

ref2 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref2_base ref2 frame colmv base only valid when sw_colmv_mode is 1'b1 ref2 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg82_colmv_ref3_base

Address: Operational Base + offset (0x00148)

ref3 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref3_base ref3 frame colmv base only valid when sw_colmv_mode is 1'b1 ref3 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg83_colmv_ref4_base

Address: Operational Base + offset (0x0014c)

ref4 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref4_base ref4 frame colmv base only valid when sw_colmv_mode is 1'b1 ref4 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg84_colmv_ref5_base

Address: Operational Base + offset (0x00150)

ref5 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref5_base ref0 frame colmv base only valid when sw_colmv_mode is 1'b1 ref5 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg85_colmv_ref6_base

Address: Operational Base + offset (0x00154)

ref6 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref6_base ref6 frame colmv base only valid when sw_colmv_mode is 1'b1 ref6 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg86_colmv_ref7_base

Address: Operational Base + offset (0x00158)

ref7 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref7_base ref7 frame colmv base only valid when sw_colmv_mode is 1'b1 ref7 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg87_colmv_ref8_base

Address: Operational Base + offset (0x0015c)

ref8 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref8_base ref8 frame colmv base only valid when sw_colmv_mode is 1'b1 ref8 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg88_colmv_ref9_base

Address: Operational Base + offset (0x00160)

ref9 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref9_base ref9 frame colmv base only valid when sw_colmv_mode is 1'b1 ref9 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg89_colmv_ref10_base

Address: Operational Base + offset (0x00164)

ref10 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref10_base ref10 frame colmv base only valid when sw_colmv_mode is 1'b1 ref10 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg90_colmv_ref11_base

Address: Operational Base + offset (0x00168)

ref11 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref11_base ref11 frame colmv base only valid when sw_colmv_mode is 1'b1 ref11 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg91_colmv_ref12_base

Address: Operational Base + offset (0x0016c)

ref12 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref12_base ref12 frame colmv base only valid when sw_colmv_mode is 1'b1 ref12 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg92_colmv_ref13_base

Address: Operational Base + offset (0x00170)

ref13 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref13_base ref13 frame colmv base only valid when sw_colmv_mode is 1'b1 ref13 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg93_colmv_ref14_base

Address: Operational Base + offset (0x00174)

ref14 frame colmv base addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref14_base ref14 frame colmv base only valid when sw_colmv_mode is 1'b1 ref14 frame colmv base addr for H264 and HEVC
3:0	RO	0x0	reserved

rkvdec_swreg94_colmv_ref15_base

Address: Operational Base + offset (0x00178)

ref15 frame colmv base addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_colmv_ref15_base ref15 frame colmv base only valid when sw_colmv_mode is 1'b1 ref15 frame colmv base addr only for HEVC
3:0	RO	0x0	reserved

rkvdec_swreg95_scanlist_addr

Address: Operational Base + offset (0x0017c)

scanlist addr

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	scanlist_addr the addr for scanlist table should be align to 16byte,and will be used for h264 and hevc
3:1	RO	0x0	reserved
0	RW	0x0	scanlist_addr_valid_en enable valid flag for scanlist addr 0:scanlsit addr get from pps table 1:scanlist addr get from config

rkvdec_swreg96_sd_decout_base

Address: Operational Base + offset (0x00180)

base address of decoder output picture base address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	sw_sd_decout_base base address of decoder output picture addr base address of decoder output picture the address should be 128bit align in H264 decode format, the top field and bottom field are the same addr
3:0	RO	0x0	reserved

rkvdec_swreg97_sd_y_virstride

Address: Operational Base + offset (0x00184)

the ouput picture y virtual stride

Bit	Attr	Reset Value	Description
31	RO	0x0	sw_no_ref_bframe_dec 1:find this frame is not referace B frame
30:21	RO	0x0	reserved
20:0	RW	0x000000	sw_sd_y_virstride the output picture y virtual stride the output picture y virtual stride (the unit is 128bit) the max: (4096x1.5 +128) x 2304 = 0xdc8000 we can know the sw_uvout_base = sw_decout_base + (sw_y_virstride <<4) 2015.10.23, change from 20bits to 21bits

rkvdec_swreg98_sd_hor_stride

Address: Operational Base + offset (0x00188)

picture parameters

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Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_sd_uv_hor_virstride uv horizontal virstride picture horizontal virtual stride (the unit is 128bit) the max is (4096x1.5 + 128) /16 = 0x188 suggest this register to config to even for advance ddr performance
15:10	RO	0x0	reserved
9:0	RW	0x000	sw_sd_y_hor_virstride picture horizontal virtual stride picture horizontal virtual stride (the unit is 128bit) the max is (4096x1.5 + 128) /16 = 0x188 suggest this register to config to even for advance ddr performance 2015.10.23, change from 9 bit to 10bit

rkvdec_swreg99_qos_ctrl

Address: Operational Base + offset (0x0018c)

rkvdec rd/wr

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:26	RW	0x0	sw_axi_wr_hurry_level axi wr hurry level 00: hurry off 01~11:hurry level

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Bit	Attr	Reset Value	Description
25:24	RW	0x0	sw_axi_rd_hurry_level axi rd hurry level 00: hurry off 01~11:hurry level
23:16	RW	0x00	sw_bus2mc_buffer_qos_level bus to mc buffer qos level range is :0~255 ths value is means that left space <= sw_bus2mc_buffer_qos_level,it will give hurry
15:0	RW	0x0000	swreg_block_gating_e Field0000 Abstract

rkvdec_swreg100_PERF_LATENCY_CTRL0

Address: Operational Base + offset (0x00190)

Axi performance latency module contrl register0

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:8	RW	0x000	sw_rd_latency_thr sw_rd_latency_thr
7:4	RW	0x0	sw_rd_latency_id sw_rd_latency_id

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_axi_cnt_type sw_axi_cnt_type sw_axi_cnt_type
2	RW	0x0	sw_axi_perf_frm_type sw_axi_perf_frm_type 1'b0: clear by frame end 1'b1: clear by software configuration
1	RW	0x0	sw_axi_perf_clr_e sw_axi_perf_clr_e 1'b0: software clear disable 1'b1: software clear enalbe
0	RW	0x0	sw_axi_perf_work_e sw_axi_perf_work_e 1'b0: disable 1'b1: enable

rkvdec_swreg101_PERF_LATENCY_CTRL1

Address: Operational Base + offset (0x00194)

PERF_LATENCY_CTRL1

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	sw_rd_total_bytes_mode 0:cal all id 1:cal sw_ar_count_id

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Bit	Attr	Reset Value	Description
11:8	RW	0x0	sw_aw_count_id sw_aw_count_id sw_aw_count_id
7:4	RW	0x0	sw_ar_count_id sw_ar_count_id sw_ar_count_id
3	RW	0x0	sw_aw_cnt_id_type sw_aw_cnt_id_type sw_aw_cnt_id_type
2	RW	0x0	sw_ar_cnt_id_type sw_ar_cnt_id_type sw_ar_cnt_id_type
1:0	RW	0x0	sw_addr_align_type sw_addr_align_type sw_addr_align_type

rkvdec_swreg102_PERF_RD_MAX_LATENCY_NUM0

Address: Operational Base + offset (0x00198)

Read max latency number

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	rd_max_latency_num_ch0 read max latency value of channel 0

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rkvdec_swreg103_PERF_RD_LATENCY_SAMP_NUM

Address: Operational Base + offset (0x0019c)

The number of bigger than configed threshold value

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_thr_num_ch0 read latency thr number channel 0

rkvdec_swreg104_PERF_RD_LATENCY_ACC_SUM

Address: Operational Base + offset (0x001a0)

Total sample number

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_acc_sum

rkvdec_swreg105_PERF_RD_AXI_TOTAL_BYTE

Address: Operational Base + offset (0x001a4)

perf_rd_axi_total_byte

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_rd_axi_total_byte perf_rd_axi_total_byte perf_rd_axi_total_byte

rkvdec_swreg106_PERF_WR_AXI_TOTAL_BYTE

Address: Operational Base + offset (0x001a8)

perf_wr_axi_total_byte

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_wr_axi_total_byte perf_wr_axi_total_byte perf_wr_axi_total_byte

rkvdec_swreg107_PERF_WORKING_CNT

Address: Operational Base + offset (0x001ac)

perf_working_cnt

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_working_cnt perf_working_cnt perf_working_cnt

rkvdec_swreg108_qos1

Address: Operational Base + offset (0x001b0)

perf_working_cnt

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_wr_wait_cycle_qos hw find sw_wr_wait_cycle_qos cycle can't wr to ddr,it will give hurry

15.5.4 VDPU341 MMU Register Summary

Name	Offset	Size	Reset Value	Description
rkvdec_mmu_DTE_ADDR	0x00000	W	0x00000000	MMU current page Table address
rkvdec_mmu_STATUS	0x00004	W	0x00000018	MMU status register

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Name	Offset	Size	Reset Value	Description
rkvdec_mmu_COMMAND	0x00008	W	0x00000000	MMU command register
rkvdec_mmu_PAGE_FAULT_ADDR	0x0000c	W	0x00000000	MMU logical address of last page fault
rkvdec_mmu_ZAP_ONE_LINE	0x00010	W	0x00000000	MMU Zap cache line register
rkvdec_mmu_INT_RAWSTAT	0x00014	W	0x00000000	MMU raw interrupt status register
rkvdec_mmu_INT_CLEAR	0x00018	W	0x00000000	MMU raw interrupt status register
rkvdec_mmu_INT_MASK	0x0001c	W	0x00000000	MMU raw interrupt status register
rkvdec_mmu_INT_STATUS	0x00020	W	0x00000000	MMU raw interrupt status register
rkvdec_mmu_AUTO_GATING	0x00024	W	0x00000001	mmu auto gating

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.5 VDPU341 MMU Detail Register Description

rkvdec_mmu_DTE_ADDR

Address: Operational Base + offset (0x00000)

MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR mmu dte base addr mmu dte base addr , the address must be 4kb aligned

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rkvdec_mmu_STATUS

Address: Operational Base + offset (0x00004)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Field0000 Abstract Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE Field0000 Abstract The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x1	REPLAY_BUFFER_EMPTY Field0000 Abstract The MMU replay buffer is empty
3	RO	0x1	MMU_IDLE Field0000 Abstract The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE Field0001 Abstract MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE Field0000 Abstract MMU page fault mode currently enabled . The mode is enabled by command.

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Bit	Attr	Reset Value	Description
0	RO	0x0	PAGING_ENABLED Field0000 Abstract Paging is enabled

rkvdec_mmu_COMMAND

Address: Operational Base + offset (0x00008)

MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD Field0000 Abstract MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

rkvdec_mmu_PAGE_FAULT_ADDR

Address: Operational Base + offset (0x0000c)

MMU logical address of last page fault

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR Field0000 Abstract address of last page fault

rkvdec_mmu_ZAP_ONE_LINE

Address: Operational Base + offset (0x00010)

MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE Field0000 Abstract address to be invalidated from the page table cache

rkvdec_mmu_INT_RAWSTAT

Address: Operational Base + offset (0x00014)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault

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rkvdec_mmu_INT_CLEAR

Address: Operational Base + offset (0x00018)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	WO	0x0	PAGE_FAULT Field0000 Abstract page fault

rkvdec_mmu_INT_MASK

Address: Operational Base + offset (0x0001c)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error enable an interrupt source if the corresponding mask bit is set to 1
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault enable an interrupt source if the corresponding mask bit is set to 1

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rkvdec_mmu_INT_STATUS

Address: Operational Base + offset (0x00020)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RO	0x0	PAGE_FAULT Field0000 Abstract page fault

rkvdec_mmu_AUTO_GATING

Address: Operational Base + offset (0x00024)

mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_gating mmu auto gating when it is 1'b1, the mmu will auto gating it self

15.5.6 VDPU341 Cache Register Summary

Name	Offset	Size	Reset Value	Description
pref_cache_VERSION	0x00000	W	0xcac20101	VERSION register
pref_cache_SIZE	0x00004	W	0x07100206	L2 cache SIZE
pref_cache_STATUS	0x00008	W	0x00000000	Status register

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Name	Offset	Size	Reset Value	Description
pref_cache_COMMAND	0x00010	W	0x00000000	Command setting register
pref_cache_CLEAR_PAGE	0x00014	W	0x00000000	clear page register
pref_cache_MAX_READS	0x00018	W	0x0000001c	maximum read register
pref_cache_ENABLE	0x0001c	W	0x00000003	enables cacheable accesses and cache read allocation
pref_cache_PERFCNT_SRC0	0x00020	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL0	0x00024	W	0x00000000	performance counter 0 value register
pref_cache_PERFCNT_SRC1	0x00028	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL1	0x0002c	W	0x00000000	performance counter 1 value register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.7 VDPU341 Cache Detail Register Description

pref_cache_VERSION

Address: Operational Base + offset (0x00000)

VERSION register

Bit	Attr	Reset Value	Description
31:16	RO	0xcac2	PRODUCT_ID Field0000 Abstract Field0000 Description

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Bit	Attr	Reset Value	Description
15:8	RO	0x01	VERSION_MAJOR Field0000 Abstract Field0000 Description
7:0	RO	0x01	VERSION_MINOR Field0000 Abstract Field0000 Description

pref_cache_SIZE

Address: Operational Base + offset (0x00004)

L2 cache SIZE

Bit	Attr	Reset Value	Description
31:24	RO	0x07	External_bus_width Field0000 Abstract Log2 external bus width in bits
23:16	RO	0x10	CACHE_SIZE Field0000 Abstract Log2 cache size in bytes for Y channel , its value is 0x10 for UV channel, its value is 0xf
15:8	RO	0x02	ASSOCIATIVITY Field0000 Abstract Log2 associativity
7:0	RO	0x06	LINE_SIZE Field0000 Abstract Log2 line size in bytes

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pref_cache_STATUS

Address: Operational Base + offset (0x00008)

Status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	DATA_BUSY Field0000 Abstract set when the cache is busy handling data
0	RW	0x0	CMD_BUSY Field0000 Abstract set when the cache is busy handling commands

pref_cache_COMMAND

Address: Operational Base + offset (0x00010)

Command setting register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	sw_addrb_sel Field0000 Abstract 2'b00: to sel b[14:6] 2'b01: to sel b[15:9], b[7:6] 2'b10: to sel b[16:10], b[7:6] 2'b11: to sel b[17:11], b[7:6]
3:0	RW	0x0	COMMAND Field0000 Abstract The possible command is 1 = Clear entire cache

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pref_cache_CLEAR_PAGE

Address: Operational Base + offset (0x00014)

clear page register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	CLEAR_PAGE Field0000 Abstract writing an address, invlidates all lines in that page from the cache

pref_cache_MAX_READS

Address: Operational Base + offset (0x00018)

maximum read register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x1c	MAX_READS Field0000 Abstract Limit the number of outstanding read transactions to this amount

pref_cache_ENABLE

Address: Operational Base + offset (0x0001c)

enables cacheable accesses and cache read allocation

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	sw_cache_linsize cache linesize 1'b0: the cache line is 32bytes 1'b1: the cache line is 64bytes
3	RW	0x0	sw_cache_clk_disgate cache clk disgate cache clk disgate when it is 1'b0, enable cache clk auto clkgating when it is 1'b1, disable cache clk auto clkgating
2	RW	0x0	sw_readbuffer_counter_reject_en counter reject enable default is 1'b0, for enhance cacheable read performnace in readbuffer. 1'b1: normal origin counter reject
1	RW	0x1	permit_cache_read_allocate cache read allocate 1'b1: permit cache read allocate
0	RW	0x1	permit_cacheable_access cacheable access 1'b1: permit cacheable access

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pref_cache_PERFCNT_SRC0

Address: Operational Base + offset (0x00020)

performance counter 0 source register

Bit	Attr	Reset Value	Description

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	PERFCNT_SRC0 Field0000 Abstract This register holds all the possible source values for Performance Counter 0 0: disabled 1: total clock cycles 2: active clock cycles 3: read transactions, master 4: word reads, master 5: read transactions, slave 6: word reads, slave 7: read hit, slave 8: read misses, slave 9: read invalidates, slave 10: cacheable read transactions, slave 11: bad hit nmber, slave

pref_cache_PERFCNT_VAL0

Address: Operational Base + offset (0x00024)

performance counter 0 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL0 Field0000 Abstract Performance counter 0 value

pref_cache_PERFCNT_SRC1

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Address: Operational Base + offset (0x00028)

performance counter 0 source register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	PERFCNT_SRC1 Field0000 Abstract This register holds all the possible source values for Performance Counter 1 0: disabled 1: total clock cycles 2: active clock cycles 3: read transactions, master 4: word reads, master 5: read transactions, slave 6: word reads, slave 7: read hit, slave 8: read misses, slave 9: read invalidates, slave 10: cacheable read transactions, slave 11: bad hit nmber, slave

pref_cache_PERFCNT_VAL1

Address: Operational Base + offset (0x0002c)

performance counter 1 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL1 Field0000 Abstract Performance counter 1 value

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15.5.8 VDPU341 Link Table Register Summary

Name	Offset	Size	Reset Value	Description
rkvdec_link_swreg0_link_mode_en	0x00000	W	0x00000000	Register0000 Abstract
rkvdec_link_swreg1_cfg_start_addr	0x00004	W	0x00000000	
rkvdec_link_swreg2_link_mode	0x00008	W	0x00000000	Register0000 Abstract
rkvdec_link_swreg3_config_done	0x0000c	W	0x00000000	Register0000 Abstract
rkvdec_link_swreg4_decodered_num	0x00010	W	0x00000000	Register0000 Abstract
rkvdec_link_swreg5_dec_total_num	0x00014	W	0x00000000	Register0000 Abstract

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.9 VDPU341 Link Table Detail Register Description

rkvdec_link_swreg0_link_mode_en

Address: Operational Base + offset (0x00000)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	link_table_irq when high, decoder requests an interrrupt. link table irq = sw_dec_irq_raw && (sw_dec_irq_dis == 1'b0)
7:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	cache_cfg_mode_sel cache cfg mode sel 0:use rtl default value to config cache, ip will auto clr cache when it begin to start dec a frame. 1:use config in ddr to config cache .
0	RW	0x0	link_mode_en Field0000 Abstract when error see by hw,it will auto reset to 0

rkvdec_link_swreg1_cfg_start_addr

Address: Operational Base + offset (0x00004)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	reg_cfg_addr Field0000 Abstract it should be align to 32 byte
3:0	RO	0x0	reserved

rkvdec_link_swreg2_link_mode

Address: Operational Base + offset (0x00008)

Register0000 Abstract

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	link_mode 0: normal mode,the first to start link mode 1:add extra ready frame to decoder
30	RO	0x0	reserved
29:0	RW	0x00000000	pre_frame_num Field0000 Abstract Field0000 Description

rkvdec_link_swreg3_config_done

Address: Operational Base + offset (0x0000c)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	config_done Field0000 Abstract after config okay,config this bit to 1

rkvdec_link_swreg4_decodered_num

Address: Operational Base + offset (0x00010)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31	RW	0x0	decoder_error_flag Field0000 Abstract 0:no error 1:error,you will need to see swreg1 to check error type

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Bit	Attr	Reset Value	Description
30	RO	0x0	reserved
29:0	RO	0x00000000	decoder_num Field0000 Abstract Field0000 Description

rkvdec_link_swreg5_dec_total_num

Address: Operational Base + offset (0x00014)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:0	RO	0x00000000	dec_total_num Field0000 Abstract Field0000 Description

15.5.10 VDPU121 Function Config Register Summary

Name	Offset	Size	Reset Value	Description
VDPU_SWREG0	0x00000	W	0x00010100	axi control
VDPU_SWREG1	0x00004	W	0x00000000	color coeff register
VDPU_SWREG2	0x00008	W	0x00000000	color coeff register
VDPU_SWREG3	0x0000c	W	0x00000000	color coeff register
VDPU_SWREG4	0x00010	W	0x00000000	scl ctrl register
VDPU_SWREG5	0x00014	W	0x00000000	scl ctrl register
VDPU_SWREG6	0x00018	W	0x00000000	scl ctrl register

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG7	0x0001c	W	0x00000000	Amount of pixels beyond border
VDPU_SWREG8	0x00020	W	0x00000000	Amount of pixels beyond border
VDPU_SWREG9	0x00024	W	0x00000000	Rmask register
VDPU_SWREG10	0x00028	W	0x00000000	Gmask register
VDPU_SWREG11	0x0002c	W	0x00000000	Bmask register
VDPU_SWREG12	0x00030	W	0x00000000	PP input picture base address for Y bottom field
VDPU_SWREG13	0x00034	W	0x00000000	PP input picture base for Ch bottom field
VDPU_SWREG14	0x00038	W	0x00000000	coordinate used in macroblock crop
VDPU_SWREG15	0x0003c	W	0x00000000	range map register
VDPU_SWREG16	0x00040	W	0x00000000	total num of padded for RGB
VDPU_SWREG17	0x00044	W	0x00000000	hw support informan,read only
VDPU_SWREG18	0x00048	W	0x00000000	base address for reading post-processing input picture uminan
VDPU_SWREG19	0x0004c	W	0x00000000	Base address for reading post-processing input picture Cb/Ch
VDPU_SWREG20	0x00050	W	0x00000000	input cr component address

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG21	0x00054	W	0x00000000	Base address for writing post-processed picture luminance/RGB
VDPU_SWREG22	0x00058	W	0x00000000	Base address for writing post-processed picture Ch
VDPU_SWREG23	0x0005c	W	0x00000000	Display width and PP input size extension register
VDPU_SWREG24	0x00060	W	0x00000000	alpha blending base address
VDPU_SWREG25	0x00064	W	0x00000000	ablend of pixels scanline
VDPU_SWREG26	0x00068	W	0x00000000	x-coordinate of mask area 1 for Horizontal start pixel
VDPU_SWREG27	0x0006c	W	0x00000000	y-coordinate of mask area 1 for Horizontal start pixel
VDPU_SWREG28	0x00070	W	0x00000000	x-coordinate of mask area 2 for Horizontal start pixel
VDPU_SWREG29	0x00074	W	0x00000000	y-coordinate of mask area 2 for Horizontal start pixel
VDPU_SWREG30	0x00078	W	0x00000000	register for deinterlace ctrl
VDPU_SWREG31	0x0007c	W	0x00000000	contrast adjust threshold
VDPU_SWREG32	0x00080	W	0x00000000	contrast adjust offset

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG33	0x00084	W	0xfc874780	Synthesis configuration register post-processor (read only)
VDPU_SWREG34	0x00088	W	0x00000000	PP input pic size register
VDPU_SWREG35	0x0008c	W	0x00000000	PP output pic size register
VDPU_SWREG36	0x00090	W	0x00000000	the dither mode for RGB
VDPU_SWREG37	0x00094	W	0x00000000	PP input/output data format
VDPU_SWREG38	0x00098	W	0x00000000	PP input/output data format
VDPU_SWREG39	0x0009c	W	0x00000000	Register0000 Abstract
VDPU_SWREG40	0x000a0	W	0x00000000	pp int register
VDPU_SWREG41	0x000a4	W	0x00000008	enable ctrl flag
VDPU_SWREG50	0x000c8	W	0x00000000	video decoder ctrl register
VDPU_SWREG51	0x000cc	W	0x00000000	the stream length
VDPU_SWREG52	0x000d0	W	0x00000000	error concealment case related
VDPU_SWREG53	0x000d4	W	0x00000000	decoder format
VDPU_SWREG54	0x000d8	W	0x00000000	endian for input/output data
VDPU_SWREG55	0x000dc	W	0x00000000	decoder int register
VDPU_SWREG56	0x000e0	W	0x00200101	axi ctrl for decoder
VDPU_SWREG57	0x000e4	W	0x00000010	enable flag for decoder
VDPU_SWREG58	0x000e8	W	0x00000000	soft reset register

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG59	0x000ec	W	0x00000000	H264, MPEG4, VC1, VP6 Prediction filter tap
VDPU_SWREG60	0x000f0	W	0x00000000	additional chrominance address
VDPU_SWREG61	0x000f4	W	0x00000000	standard dependent tables start address
VDPU_SWREG62	0x000f8	W	0x00000000	Direct mode motion vector write/read start address
VDPU_SWREG63	0x000fc	W	0x00000000	write decoder output picture or field start address
VDPU_SWREG64	0x00100	W	0x00000000	rlc or vlc mode input data start addr
VDPU_SWREG65	0x00104	W	0x00000000	refbufferd related
VDPU_SWREG66	0x00108	W	0x67312688	ID register
VDPU_SWREG67	0x0010c	W	0xe5da0000	Synthesis configuration register decoder 1(read only)
VDPU_SWREG68	0x00110	W	0x00000000	sum of partitions(read only)
VDPU_SWREG69	0x00114	W	0x00000000	sum information (read only)
VDPU_SWREG70	0x00118	W	0x00000000	sum of the decoded motion vector y-components(read only)
VDPU_SWREG71	0x0011c	W	0xfbb56f80	information for read only register
VDPU_SWREG72	0x00120	W	0x00000000	debug0

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG73	0x00124	W	0x00000000	debug registers
VDPU_SWREG74	0x00128	W	0x00000000	MV address for h264
VDPU_SWREG75	0x0012c	W	0x00000000	H.264 Intra prediction 4x4 mode start address
VDPU_SWREG76	0x00130	W	0x00000000	the number of referance pic
VDPU_SWREG77	0x00134	W	0x00000000	the number of referance pic
VDPU_SWREG78	0x00138	W	0x00000000	the number of referance pic
VDPU_SWREG79	0x0013c	W	0x00000000	the number of referance pic
VDPU_SWREG80	0x00140	W	0x00000000	the number of referance pic
VDPU_SWREG81	0x00144	W	0x00000000	the number of referance pic
VDPU_SWREG82	0x00148	W	0x00000000	the number of referance pic
VDPU_SWREG83	0x0014c	W	0x00000000	the number of referance pic
VDPU_SWREG84	0x00150	W	0x00000000	referance frame0 address for h264
VDPU_SWREG85	0x00154	W	0x00000000	referance frame1 address for h264
VDPU_SWREG86	0x00158	W	0x00000000	referance frame2 address for h264
VDPU_SWREG87	0x0015c	W	0x00000000	referance frame3 address for h264

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG88	0x00160	W	0x00000000	referance frame4 address for h264
VDPU_SWREG89	0x00164	W	0x00000000	referance frame5 address for h264
VDPU_SWREG90	0x00168	W	0x00000000	referance frame6 address for h264
VDPU_SWREG91	0x0016c	W	0x00000000	referance frame7 address for h264
VDPU_SWREG92	0x00170	W	0x00000000	referance frame8 address for h264
VDPU_SWREG93	0x00174	W	0x00000000	referance frame9 address for h264
VDPU_SWREG94	0x00178	W	0x00000000	referance frame10 address for h264
VDPU_SWREG95	0x0017c	W	0x00000000	referance frame11 address for h264
VDPU_SWREG96	0x00180	W	0x00000000	referance frame12 address for h264
VDPU_SWREG97	0x00184	W	0x00000000	referance frame13 address for h264
VDPU_SWREG98	0x00188	W	0x00000000	referance frame14 address for h264
VDPU_SWREG99	0x0018c	W	0x00000000	referance frame15 address for h264
VDPU_SWREG100	0x00190	W	0x00000000	initial reference picture list related
VDPU_SWREG101	0x00194	W	0x00000000	initial reference picture list related

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG102	0x00198	W	0x00000000	initial reference picture list related
VDPU_SWREG103	0x0019c	W	0x00000000	initial reference picture list related
VDPU_SWREG104	0x001a0	W	0x00000000	initial reference picture list related
VDPU_SWREG105	0x001a4	W	0x00000000	initial reference picture list related
VDPU_SWREG106	0x001a8	W	0x00000000	initial reference picture list related
VDPU_SWREG107	0x001ac	W	0x00000000	long term flag for reference pictuure index
VDPU_SWREG108	0x001b0	W	0x00000000	valid flag for reference picture index
VDPU_SWREG109	0x001b4	W	0x00000000	the stream start word for decoder
VDPU_SWREG110	0x001b8	W	0x00000000	h264 pic mb size
VDPU_SWREG111	0x001bc	W	0x00000000	h264 ctrl related
VDPU_SWREG112	0x001c0	W	0x00000000	current frame related
VDPU_SWREG113	0x001c4	W	0x00000000	reference picture related
VDPU_SWREG114	0x001c8	W	0x00000000	maximum reference
VDPU_SWREG115	0x001cc	W	0x00000000	enable flag
VDPU_SWREG120	0x001e0	W	0x00000000	multi format reuse register0
VDPU_SWREG121	0x001e4	W	0x00000000	multi format reuse register1

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG122	0x001e8	W	0x00000000	multi format reuse register2
VDPU_SWREG123	0x001ec	W	0x00000000	multi format reuse register3
VDPU_SWREG124	0x001f0	W	0x00000000	multi format reuse register4
VDPU_SWREG125	0x001f4	W	0x00000000	multi format reuse register5
VDPU_SWREG126	0x001f8	W	0x00000000	multi format reuse register6
VDPU_SWREG127	0x001fc	W	0x00000000	multi format reuse register7
VDPU_SWREG128	0x00200	W	0x00000000	multi format reuse register8
VDPU_SWREG129	0x00204	W	0x00000000	multi format reuse register9
VDPU_SWREG130	0x00208	W	0x00000000	multi format reuse register10
VDPU_SWREG131	0x0020c	W	0x00000000	multi format reuse register11
VDPU_SWREG132	0x00210	W	0x00000000	multi format reuse register12
VDPU_SWREG133	0x00214	W	0x00000000	multi format reuse register13
VDPU_SWREG134	0x00218	W	0x00000000	multi format reuse register14
VDPU_SWREG135	0x0021c	W	0x00000000	multi format reuse register15

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG136	0x00220	W	0x00000000	multi format reuse register16
VDPU_SWREG137	0x00224	W	0x00000000	multi format reuse register17
VDPU_SWREG138	0x00228	W	0x00000000	multi format reuse register18
VDPU_SWREG139	0x0022c	W	0x00000000	multi format reuse register19
VDPU_SWREG140	0x00230	W	0x00000000	multi format reuse register20
VDPU_SWREG141	0x00234	W	0x00000000	multi format reuse register21
VDPU_SWREG142	0x00238	W	0x00000000	multi format reuse register22
VDPU_SWREG143	0x0023c	W	0x00000000	multi format reuse register23
VDPU_SWREG144	0x00240	W	0x00000000	multi format reuse register24
VDPU_SWREG145	0x00244	W	0x00000000	multi format reuse register25
VDPU_SWREG146	0x00248	W	0x00000000	multi format reuse register26
VDPU_SWREG147	0x0024c	W	0x00000000	multi format reuse register27
VDPU_SWREG148	0x00250	W	0x00000000	multi format reuse register28
VDPU_SWREG149	0x00254	W	0x00000000	multi format reuse register29

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG150	0x00258	W	0x00000000	multi format reuse register30
VDPU_SWREG151	0x0025c	W	0x00000000	multi format reuse register31
VDPU_SWREG152	0x00260	W	0x00000000	multi format reuse register32
VDPU_SWREG153	0x00264	W	0x00000000	multi format reuse register33
VDPU_SWREG154	0x00268	W	0x00000000	multi format reuse register34
VDPU_SWREG155	0x0026c	W	0x00000000	multi format reuse register35
VDPU_SWREG156	0x00270	W	0x00000000	multi format reuse register36
VDPU_SWREG157	0x00274	W	0x00000000	multi format reuse register37
VDPU_SWREG158	0x00278	W	0x00000000	multi format reuse register38
VDPU_SWREG164_PERF_LATENCY_CTRL0	0x00290	W	0x00000000	Axi performance latency module contrl register0
VDPU_SWREG165_PERF_LATENCY_CTRL1	0x00294	W	0x00000000	PERF_LATENCY_CTRL1
VDPU_SWREG166_PERF_RD_MAX_LATENCY_NUM0	0x00298	W	0x00000000	Read max latency number
VDPU_SWREG167_PERF_RD_LATENCY_SAMP_NUM	0x0029c	W	0x00000000	The number of bigger than configed threshold value

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Name	Offset	Size	Reset Value	Description
VDPU_SWREG168_PERF_RD_LATENCY_ACC_SUM	0x002a0	W	0x00000000	Total sample number
VDPU_SWREG169_PERF_RD_AXI_TOTAL_BYTE	0x002a4	W	0x00000000	perf_rd_axi_total_byte
VDPU_SWREG170_PERF_WR_AXI_TOTAL_BYTE	0x002a8	W	0x00000000	perf_wr_axi_total_byte
VDPU_SWREG171_PERF_WORKING_CNT	0x002ac	W	0x00000000	perf_working_cnt

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.11 VDPU121 Function Config Detail Register Description

VDPU_SWREG0

Address: Operational Base + offset (0x00000)

axi control

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x01	sw_axi_id_wr AXI Write ID if you config 0,will modify as 1 by hw
15:8	RW	0x01	sw_axi_id_rd AXI Read ID if you config 0,will modify as 1 by hw
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	sw_scmd_off on-off for AXI Single Command Multiple Data on-off for AXI Single Command Multiple Data 0:on 1:off
4:0	RW	0x00	sw_max_burst_len the max burst length can be used by axi bus range : 1-16

VDPU_SWREG1

Address: Operational Base + offset (0x00004)

color coeff register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RW	0x000	sw_coe_2st used in red color components calculate,used together with cr pix used for red color components calculate,used together with cr pix
19:10	RW	0x000	sw_coe_1st_1 used in all color components calculate,used together with y pix used for all color components calculate,used together with y pix
9:0	RW	0x000	sw_coe_1st_0 used for all color components calculate,used together with y pix

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VDPU_SWREG2

Address: Operational Base + offset (0x00008)

color coeff register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RW	0x000	sw_coe_5st used for blue components calculate,used together with cb pix
19:10	RW	0x000	sw_coe_4st used for green color components calculate,used together with cb pix
9:0	RW	0x000	sw_coe_3st used for green color components calculate,used together with cr pix

VDPU_SWREG3

Address: Operational Base + offset (0x0000c)

color coeff register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	sw_coe_6st used for burightness adjust,used together with y pix

VDPU_SWREG4

Address: Operational Base + offset (0x00010)

scl ctrl register

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Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:22	RW	0x0	sw_scl_mode_vrt 0 = no scl 1 = up scl 2 = down scl
21:20	RW	0x0	sw_scl_mode_hrz to sellect scaling mode for Horizontal 0 = no scl 1 = up scl 2 = down scl
19:18	RO	0x0	reserved
17:0	RW	0x00000	sw_scl_fct_w scaling factor of width value = (output_width-1)/(input_width-1)

VDPU_SWREG5

Address: Operational Base + offset (0x00014)

scl ctrl register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:0	RW	0x00000	sw_scl_fct_h scaling factor of height value = (output_width-1)/(input_width-1)

VDPU_SWREG6

Address: Operational Base + offset (0x00018)

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scl ctrl register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_scl_fct_h_inv scaling factor of height and cv value =(inputw-1) / (outputw-1)
15:0	RW	0x0000	sw_scl_fct_w_inv scaling factor of width and ch value =(inputw-1) / (outputw-1)

VDPU_SWREG7

Address: Operational Base + offset (0x0001c)

Amount of pixels beyond border

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:16	RW	0x000	sw_pixnum_down_byd the Amount of vertical pixels beyond the down border Range : 0-dst_height
15:11	RO	0x0	reserved
10:0	RW	0x000	sw_pixnum_up_byd the Amount of vertical pixels beyond the up border Range : 0-dst_height

VDPU_SWREG8

Address: Operational Base + offset (0x00020)

Amount of pixels beyond border

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Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:16	RW	0x000	sw_pixnum_right_byd the Amount of vertical pixels beyond the right border Range : 0~dst_width
15:11	RO	0x0	reserved
10:0	RW	0x000	sw_pixnum_left_byd the Amount of vertical pixels beyond the left border Range : 0~dst_width

VDPU_SWREG9

Address: Operational Base + offset (0x00024)

Rmask register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_mask_r color R/(alpha channnel) component 's bit mask

VDPU_SWREG10

Address: Operational Base + offset (0x00028)

Gmask register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_mask_g color G/(alpha channnel) component 's bit mask

VDPU_SWREG11

Address: Operational Base + offset (0x0002c)

Bmask register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_mask_b color B/(alpha channnel) component 's bit mask

VDPU_SWREG12

Address: Operational Base + offset (0x00030)

PP input picture base address for Y bottom field

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_botfld_y_st_adr input bottom field pp start address for y component input bottom field pp start address for y component
1:0	RO	0x0	reserved

VDPU_SWREG13

Address: Operational Base + offset (0x00034)

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PP input picture base for Ch bottom field

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_botfld_c_st_adr input bottom field pp start address for c component
1:0	RO	0x0	reserved

VDPU_SWREG14

Address: Operational Base + offset (0x00038)

coordinate used in macroblock crop

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	sw_mrmb_8pix_flag the most right unrotated MB of input picture just 8 lines pix data 829PP input picture width is not 16 pixels multiple. Only 8 pixels of the most right MB of the unrotated input picture is used for PP input.
28	RW	0x0	sw_mdmb_8pix_flag the most down unrotated MB of input picture just 8 rows pix data
27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26:24	RW	0x0	sw_mbcrop_crdty_ext in order to support jpeg to extend coordinate y bits
23:16	RW	0x00	sw_mbcrop_crdty coordinate y used in macroblock crop
15:12	RO	0x0	reserved
11:9	RW	0x0	sw_mbcrop_crdtx_ext in order to support jpeg to extend bits
8:0	RW	0x000	sw_mbcrop_crdtx coordinate x used in macroblock crop

VDPU_SWREG15

Address: Operational Base + offset (0x0003c)

range map register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	sw_rangemap_coef_c the value of chrominance component range map VC- 1:c range map value +9
7:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	sw_yuv_conv_range to declaration the range of YCbCr when RGB conversion Y: 0:16~235 1:0~255 C: 0:16~240 1:0~255
4:0	RW	0x00	sw_rangemap_y the value of Y component range map VC- 1:y range map value +9

VDPU_SWREG16

Address: Operational Base + offset (0x00040)

total num of padded for RGB

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20:16	RW	0x00	sw_padd_b the total num of padded in front of B componet
15:13	RO	0x0	reserved
12:8	RW	0x00	sw_padd_g the total num of padded in front of G componet the total num of padded in front of G componet

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Bit	Attr	Reset Value	Description
7:5	RO	0x0	reserved
4:0	RW	0x00	sw_padd_r the total num of padded in front of R componet

VDPU_SWREG17

Address: Operational Base + offset (0x00044)

hw support informan,read only

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	pp_deinterl_en on-off for pp deinterlance 0: off 1 : on
29	RO	0x0	pp_abled_en on-off for pp Alpha Blending 0: off 1 : on
28	RO	0x0	pp_work_en pp work allow flag 0 : off 1: on
27:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RO	0x0	pp_outw_1920_en on-off for pp output width up to 1920 1st priority used
2	RO	0x0	pp_outw_1280_en on-off for pp output width up to 1280 2st priority used
1	RO	0x0	pp_outw_720_en on-off for pp output width up to 720 3st priority used
0	RO	0x0	pp_outw_352_en on-off for pp output width up to 352 4st priority used

VDPU_SWREG18

Address: Operational Base + offset (0x00048)

base address for reading post-processing input picture uminan

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_y_in_st_adr input y component start address The start address of topfield of the picture when data come from fields.extemal mode support only
1:0	RO	0x0	reserved

VDPU_SWREG19

Address: Operational Base + offset (0x0004c)

Base address for reading post-processing input picture Cb/Ch

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_cb_in_st_adr input cb component start address The start address of topfield of the picture when data come from fields,extemal mode support only
1:0	RO	0x0	reserved

VDPU_SWREG20

Address: Operational Base + offset (0x00050)

input cr component address

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_cr_in_st_adr input cr component start address The start address of topfield of the picture when data come from fields,extemal mode support only
1:0	RO	0x0	reserved

VDPU_SWREG21

Address: Operational Base + offset (0x00054)

Base address for writing post-processed picture luminance/RGB

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_y_out_st_adr output y component start address also the start address of YUYV and RGB

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VDPU_SWREG22

Address: Operational Base + offset (0x00058)

Base address for writing post-processed picture Ch

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_c_out_st_adr output chrominace component start address format is uvuvuv....

VDPU_SWREG23

Address: Operational Base + offset (0x0005c)

Display width and PP input size extension register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_abled_st_adr_1st 1st alpha blending start address 1.valid when mask1 is used in alpha blending mode 2.Format of data the same as in PP input. 3.Amount of data is related to mask 1 size or ablend1_scanline informed with mask 1 size or with ablend1_scanline if ablend when crop flag valid

VDPU_SWREG24

Address: Operational Base + offset (0x00060)

alpha blending base address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_abled_st_adr_2st 2st alpha blending start address 1.valid when mask2 is used in alpha blending mode 2.Format of data the same as in PP input. 3.Amount of data is related to mask 2 size or ablend1_scanline informed with mask 1 size or with ablend1_scanline if ablend when crop flag valid

VDPU_SWREG25

Address: Operational Base + offset (0x00064)

ablend of pixels scanline

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	sw_scanl_abld2 ablend 2 of pixels scanline corresponding function should be enabled
15:13	RO	0x0	reserved
12:0	RW	0x0000	sw_scanl_abld1 ablend 1 of pixels scanline corresponding function should be enabled

VDPU_SWREG26

Address: Operational Base + offset (0x00068)

x-coordinate of mask area 1 for Horizontal start pixel

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:12	RW	0x000	sw_end_coordx_ma1 the end x-coordinate of mask area 1 of Horizontal start pixel range:sw_st_coordx_ma1-dst width
11	RO	0x0	reserved
10:0	RW	0x000	sw_st_coordx_ma1 the start x-coordinate of mask area 1 of Horizontal start pixel

VDPU_SWREG27

Address: Operational Base + offset (0x0006c)

y-coordinate of mask area 1 for Horizontal start pixel

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:12	RW	0x000	sw_end_coordy_ma1 the start y-coordinate of mask area 1 of Vertical start pixel range:sw_st_coordy_ma1~dst width
11	RO	0x0	reserved
10:0	RW	0x000	sw_st_coordy_ma1 the start y-coordinate of mask area 1 of Vertical start pixel

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VDPU_SWREG28

Address: Operational Base + offset (0x00070)

x-coordinate of mask area 2 for Horizontal start pixel

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:11	RW	0x000	sw_end_coordx_ma2 the end x-coordinate of mask area 2 of Horizontal start pixel range:sw_st_coordx_ma2~dst width
10:0	RW	0x000	sw_st_coordx_ma2 the start x-coordinate of mask area 2 of Horizontal start pixel

VDPU_SWREG29

Address: Operational Base + offset (0x00074)

y-coordinate of mask area 2 for Horizontal start pixel

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:11	RW	0x000	sw_end_coordy_ma2 the start y-coordinate of mask area 2 of Vertical start pixel range:sw_st_coordy_ma2-dst width
10:0	RW	0x000	sw_st_coordy_ma2 the start y-coordinate of mask area 2 of Vertical start pixel

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VDPU_SWREG30

Address: Operational Base + offset (0x00078)

register for deinterlace ctrl

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:16	RW	0x0000	sw_deinterl_edge the edge detect value of deinterlacing Edge detect value used for deinterlacing
15:14	RO	0x0	reserved
13:0	RW	0x0000	sw_deinterl_thr the threshold value of deinterlace

VDPU_SWREG31

Address: Operational Base + offset (0x0007c)

contrast adjust threshold

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	sw_cont_thr1 the threshold value 1 for contrast adjust
7:0	RW	0x00	sw_cont_thr0 the threshold value 0 for contrast adjust

VDPU_SWREG32

Address: Operational Base + offset (0x00080)

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contrast adjust offset

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	sw_cont_offset1 the offset value 1 for contrast adjust
15:10	RO	0x0	reserved
9:0	RW	0x000	sw_cont_offset0 the offset value 0 for contrast adjust

VDPU_SWREG33

Address: Operational Base + offset (0x00084)

Synthesis configuration register post-processor (read only)

Bit	Attr	Reset Value	Description
31	RO	0x1	abld_crop_flag 0: unsupport crop,the exact image of the area being alpha blended should exist in the external memory 1: supprot crop,one picture in external memory which come from blended area can be cropped
30	RO	0x1	accut_out_exist_flag Pixel Accurate PP output mode exists flag PIP: 0 : use 8 pixels (width) or 2 pixels (height) steps to adjust Scaling and masks 1 : use 1 pixel for RGB and 2 pixels for subsampled chroma formats to adjust Scaling and masks

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Bit	Attr	Reset Value	Description
29	RO	0x1	tile_exist_flag the output of yuv422 tiled exist flag 0 : no exist 1 : exist
28	RO	0x1	dither_exist_flag Dithering exists flag 0 : no exist 1 : exist
27:26	RO	0x3	scl_perf_sel scaling perfomance sel 0 : without scaling 1 : low perfomance scaling 2 : high perfomance scaling 3 : high and fast perfomance scaling
25	RO	0x0	deinterl_exist_flag Deinterlacing exits flag 0 : no exist 1 : exist
24	RO	0x0	abld_exist_flag alpha blending exists flag 0 : no exist 1 : exist
23	RO	0x1	pp_in_buf_sel PP input buffering select 0 : output buffering is 1 MB 1 : output buffering is 4 MB
22:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18	RO	0x1	pp_endian_mode PP output endian mode select 0 : Endian mode supported except RGB 1 : Endian mode supported for all format
17	RO	0x1	pp_out_buf_sel PP output buffering select 0 : output buffering is 1 unit 1 : output buffering is 4 unit
16	RO	0x1	ppd_exist_flag PPD exists flag 0 : no exist 1 : exist
15:14	RO	0x1	pp_tile_in_mode PD input tiled mode 0 : unsupport 1 : 8x4 tile be used
13:11	RO	0x0	reserved
10:0	RO	0x780	ppd_max_outw the max pixels width allow for pp output

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VDPU_SWREG34

Address: Operational Base + offset (0x00088)

PP input pic size register

Bit	Attr	Reset Value	Description

Bit	Attr	Reset Value	Description
31:29	RW	0x0	sw_pp_inh_ext PP input extended height in order to support jpeg
28:21	RW	0x00	sw_pp_inh the picture height of PP input with in macro blocks which can be cropped from a bigger picture when in the condtion of external mode
20:12	RW	0x000	sw_org_inw_ext the orginal width of pp input pic in MBS
11:9	RW	0x0	sw_pp_inw_ext PP input extended width in order to support jpeg
8:0	RW	0x000	sw_pp_inw the picture width of PP input with in macro blocks which can be cropped from a bigger picture when in the condtion of external mode

VDPU_SWREG35

Address: Operational Base + offset (0x0008c)

PP output pic size register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	sw_pp_outh_ext the extension height of pp output

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Bit	Attr	Reset Value	Description
26:16	RW	0x000	sw_pp_outh (output width = 2*n (n=1,2,.....) || output width =(configurationed Pixel Accurate PP output configuration)*n) && (pp output width < 1920 || pp output width< 3*(sw_pp_inh-8))
15:12	RO	0x0	reserved
11	RW	0x0	sw_pp_outw_ext the extension width of pp output
10:0	RW	0x000	sw_pp_outw the pp output width (output width = 8*n (n=1,2,.....) || output width =(configurationed Pixel Accurate PP output configuration)*n) && (pp output width < 1920 || pp output width< 3*(sw_pp_inw-8))

VDPU_SWREG36

Address: Operational Base + offset (0x00090)

the dither mode for RGB

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	sw_dither_mode_b the dither mode for B-component 0 : no use dithering 1 : 4-bits dither matrix be used 2 : 5-bits dither matrix be used 3 : 6-bits dither matrix be used
3:2	RW	0x0	sw_dither_mode_g the dither mode for G-component 0 : no use dithering 1 : 4-bits dither matrix be used 2 : 5-bits dither matrix be used 3 : 6-bits dither matrix be used
1:0	RW	0x0	sw_dither_mode_r the dither mode for R-component 0 : no use dithering 1 : 4-bits dither matrix be used 2 : 5-bits dither matrix be used 3 : 6-bits dither matrix be used

VDPU_SWREG37

Address: Operational Base + offset (0x00094)

PP input/output data format

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26:24	RW	0x0	sw_pp_in_data_strc the data structure of pp input picture 0 : Top field 1 : Bottom field 2 : Interlaced field 3 : Interlaced frame 4 : Ripped top field 5 : Ripped bottom field if value=0/1/2,then should read every line from the base address,if value=3/4/5,then should read every second line from the base address
23:20	RO	0x0	reserved
19	RW	0x0	sw_pp_out_crbf_en output yuv422 or yuv420,cr before cb 0 : Y0CbY0Cr / CbY0CrY0 1 : Y0CrY0Cb / CrY0CbY0
18	RW	0x0	sw_pp_in_crbf_en in yuv422 or yuv420,cr before cb yuv422: 0 : Y0CbY0Cr / CbY0CrY0 1 : Y0CrY0Cb / CrY0CbY0 yuv420 semiplanar chrominance: 0 : CbCrCbCr 1:CrCbCrCb
17	RW	0x0	sw_pp_out_yuv_order the output yuv order 0 : Y0CbY0Cr / Y0CrY0Cb 1 : CbY0CrY0 / CrY0CbY0

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Bit	Attr	Reset Value	Description
16	RW	0x0	sw_pp_in_yuv_order the input yuv order 0 : Y0CbY0Cr / Y0CrY0Cb 1 : CbY0CrY0 / CrY0CbY0
15:12	RO	0x0	reserved
11	RW	0x0	sw_pp_out_wordsp the 32bit data swap for pp output data, it will be used in 64 bit environment 0 : no swapping 1 : swapping high and low 32bit data
10	RW	0x0	sw_pp_out_hfwordsp the half word swap inside of word 0:no swap 1:swap also be used as change pixel orders for 16 bit RGB,support all output format require pp_endian_mode=1
9	RW	0x0	sw_pp_abld1_in_wordsp the 32bit data swap for the input 32bit data swap of Alpha blend for Alpha blend source 1 0 : no swapping 1 : swapping high and low 32bit data
8	RW	0x0	sw_pp_in_wordsp the 32bit data swap for pp input data, it will be used in 64 bit environment 0 : no swapping 1 : swapping high and low 32bit data

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Bit	Attr	Reset Value	Description
7:5	RO	0x0	reserved
4	RW	0x0	sw_rgb_pix_bits rga bits used sel 0: every word have only one rga pixel 1:every word have two rga pixel
3	RW	0x0	sw_pp_out_endian the endian mode of pp output for all yuv output endian mode or any data when pp_endian_mode=1 0 : big endian 1 : little endian if pp_endian_mode=0: 16 bit RGB: this bit used as pixel swapping bit 32 bit RGB: no used
2	RW	0x0	sw_pp_abld2_in_endian the endian select for input data of Alpha blend source 2 0: same with sw_pp_in_endian 1: same with sw_pp_abld1_in_endian '0' = Use PP in endian/swap definitions (sw_pp_in_endian, sw_pp_in_swap) '1' = Use Ablend source 1 endian/swap definitions
1	RW	0x0	sw_pp_abld1_in_endian the endian mode of input data for Alpha blend source 1 0 : big endian (0-1-2-3) 1 : little endian (3-2-1-0)

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_pp_in_endian the endian mode of pp input picture when PP in standalone mode this bit will no to be used when PP is running pipelined with the decoder 0 : big endian (0-1-2-3) 1 : little endian (3-2-1-0)

VDPU_SWREG38

Address: Operational Base + offset (0x00098)

PP input/output data format

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:20	RW	0x0	sw_pp_in_tilmod the tiled mode of pp input data only support yuv420 input data,can be as pipeline or external mode 0 : Tiled mode not be actived 1 : 8x4 sized tiles be used 2,3 : reserved
19	RO	0x0	reserved
18:16	RW	0x0	sw_pp_in_fmt_ecp Escape PP in format be actived when sw_pp_in_fmt = 3'b111 0 : YCbCr 4:4:4 1 : YCbCr 4:1:1

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Bit	Attr	Reset Value	Description
15:14	RO	0x0	reserved
13:11	RW	0x0	sw_pp_out_fmt 0 : RGB 1 : YCbCr 4:2:0 ; planar (Not supported) 2 : YCbCr 4:2:2 ; planar (Not supported) 3 : YUYV 4:2:2 ; interleaved 4 : YCbCr 4:4:4 ; planar (Not supported) 5 : YCh 4:2:0 ; chrominance interleaved 6 : YCh 4:2:2 ; (Not supported) 7 : YCh 4:4:4 (Not supported)
10:8	RW	0x0	sw_pp_in_fmt the input format of pp input data 0 : YUYV 4:2:2 ; interleaved and it only supported in external mode 1 : YCbCr 4:2:0 ; the format of Semi-planar in linear raster-scan 2 : YCbCr 4:2:0 ; planar and it only supported in external mode 3 : YCbCr 4:0:0 ; it only supported in pipelined mode 4 : YCbCr 4:2:2 ; Semi-planar and it only supported only in pipelined mode 5 : YCbCr 4:2:0 ; Semi-planar in tiled format and it only supported in external mode 6 : YCbCr 4:4:0 ; Semi-planar and it only supported for jpeg in pipelined mode 7 : same as sw_pp_in_fmt_ecp
7:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	sw_rot_mode Rotation mode 0 : rotation disabled 1 : rotate + 90 2 : rotate –90 3 : horizontal flip (mirror) 4 : vertical flip 5 : rotate 180

VDPU_SWREG39

Address: Operational Base + offset (0x0009c)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	sw_display_w the display width Max support 1920

VDPU_SWREG40

Address: Operational Base + offset (0x000a0)

pp int register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_pp_bus_sts the Interrupt status bit for tell sw bus have some error
2	RW	0x0	sw_pp_rdy_sts the Interrupt status bit for tell sw processed a picture
1	RW	0x0	sw_pp_irq_dis the pp finish interrupt request diable flag 1 : use polling to see the interrupt 0 : use sw_pp_irq
0	RW	0x0	sw_pp_irq the pp finish interrupt request after sw query this interrupt,shoud write 0 to reset. this bit will no used in pipeline mode

VDPU_SWREG41

Address: Operational Base + offset (0x000a4)

enable ctrl flag

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	sw_pp_ahb_hlock_en the enable flag for AHB master HLOCK the service is locked to pp as long as it needs the bus

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Bit	Attr	Reset Value	Description
27	RW	0x0	sw_rightwd_cross_en the enable flag for Right side overcross 0 : disable, 1 : enable
26	RW	0x0	sw_leftsd_cross_en the enable flag for Left side overcross 0 : disable, 1 : enable
25	RW	0x0	sw_downwd_cross_en the enable flag for Downward overcross 0 : disable, 1 : enable
24	RW	0x0	sw_upwd_cross_en the enable flag for Upward overcross 0 : disable, 1 : enable
23	RW	0x0	sw_mask2_abld_en the enable flag for Mask 2 alpha blending alpha blending for the output picture , only be supported when data format is RGB/YUYV422 Alpha blending read data from alpha blend 2 base address
22	RW	0x0	sw_mask1_abld_en the enable flag for Mask 1 alpha blending alpha blending for the output picture , only be supported when data format is RGB/YUYV422 Alpha blending read data from alpha blend 1 base address

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Bit	Attr	Reset Value	Description
21	RW	0x0	sw_mask2_en the enable flag for mask 2 0 : disable, 1 : enable
20	RW	0x0	sw_mask1_en the enable flag for mask 1 0 : disable, 1 : enable
19:17	RO	0x0	reserved
16	RW	0x0	sw_pp_discd_en the enable flag for PP data discard the burst length will be fix after sw_pp_discd_en=1,and extra read data will auto be discarded by HW
15:12	RO	0x0	reserved
11	RW	0x0	sw_pp_out_tiled_en the enable flag for pp output tiled mode only used in YCbYCr format . Tile size : 4x4 pixels.
10	RW	0x0	sw_pp_fdscl_en the enable flag for fast downscaling 0 : disable 1 : enabled. it will inprove the performance but will decrease the quality of the pic
9	RW	0x0	sw_rangemap_c_en the enable flag for C component Range map

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Bit	Attr	Reset Value	Description
8	RW	0x0	sw_rangemap_y_en the enable flag for Y component Range map VC1: used as range expansion enable
7:5	RO	0x0	reserved
4	RW	0x0	sw_pp_pipl_en pp pipeline width Decoder enable 0 : disable, external mode 1 : enable, pipeline mode,Post-processing pipeline with decoder
3	RW	0x1	sw_pp_clkgate_en pp auto clock gating: default is 1 0 : don't auto gating 1 : audo gating PP dynamic clock gating enable: 1 = Clock is gated from PP structures that are not used 0 = Clock is running for all PP structures Note: Clock gating value can be changed only when PP is not enabled
2	RW	0x0	sw_deint_en Deinterlace enable flag the input data should be interlaced format
1	RW	0x0	sw_deint_bld_en on-off Blend for deinterlacing

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_pp_dec_st post-processing start flag after config other register,write 1 to start post-processing operation,and hw will reset to 0 after it decodered a picture should be under External mode.

VDPU_SWREG50

Address: Operational Base + offset (0x000c8)

video decoder ctrl register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	sw_refbuf_pid the pic id for reference buffer The used reference picture ID for reference buffer usage
24:13	RW	0x000	sw_refbuf_thrd Reference buffer threshold value Used shut down buffer
12	RW	0x0	sw_dec_tiled_lsb the enable for lsb tiled mode 0 : Tiled mode disable 1 : Tiled mode enabled for 8x4 tile size
11	RW	0x0	sw_adv_pref_dis disable for Advanced PREFETCH mode

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Bit	Attr	Reset Value	Description
10	RW	0x0	sw_dec_ascmd0_dis the disable for AXI Single Command Multiple Data0
9	RW	0x0	sw_skip_sel AVS format: 0 : skip mbs use special MB type 1 : avs skip mbs have the same skip run syntax element as h264 VP8 format: 0:HW will decoder mb_coe_skip_flag 1:HW don't decoder that MPEG4: Valid when divx be enabled
8	RW	0x0	sw_dblk_flt_dis the disable for current pic deblock filtering 1: disable 0: enable
7	RW	0x0	sw_dec_fixed_quant h.264: this bit is for the enable of multi view coding other format(VC1 or VP8) 0: it can be different inside pic for Quantization parameter 1: it is fixed for Quantization parameter
6:1	RW	0x00	sw_adtion_latency the additional latency for decoder master interface Can be used to slow down 8*sw_dec_latency cycles of IDLE between services,so if sw_dec_latency =0,that is no latency

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_dec_tiled_msb the enable for msb tiled mode 0 : Tiled mode disable 1 : Tiled mode enabled for 8x4 tile size

VDPU_SWREG51

Address: Operational Base + offset (0x000cc)

the stream length

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:25	RW	0x00	sw_qp_init_val the qp(quantization parameter)'s Initial value
24	RW	0x0	sw_strm_len_ext The extension bit of sw_strm_len

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Bit	Attr	Reset Value	Description
23:0	RW	0x000000	sw_strm_len the stream data bytes number in input buffer. if the buffer size be given small than it required,hw will give an interrupt,and then you should config again,and the stream start address should be config also. VC1/VP6: one picture/slice of the picture's should be included in the input buffer H264/H263/MPEG*: one slice of the picture's should be included in the input buffer JPEG: 256bytes or onepicture should be included in the input buffer

VDPU_SWREG52

Address: Operational Base + offset (0x000d0)

error concealment case related

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:17	RW	0x0000	sw_adv_pref_thrd the threshold value for advanced prefetch when current MB num > this threshold value,then advanced mode will be closed
16:8	RW	0x000	sw_xdim_mbst the X dimension value for Start MB from SW it may be used in error concealment case

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	sw_ydim_mbst the Y dimension value for Start MB from SW it may be used in error concealment case

VDPU_SWREG53

Address: Operational Base + offset (0x000d4)

decoder format

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	sw_dec_fmt_sel the dec format select 0 : H.264, 1 : MPEG-4, 2 : H.263, 3 : JPEG, 4 : VC-1, 5 : MPEG-2, 6 : MPEG-1, 7 : VP6, 8 : RV, 9 : VP7, 10 : VP8, 11 : AVS, others : reserved

VDPU_SWREG54

Address: Operational Base + offset (0x000d8)

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endian for input/output data

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	sw_dec_strendian_e the endian mode of stream data 0 : Big endian (0-1-2-3 order) 1 : Little endian (3-2-1-0 order)
4	RW	0x0	sw_dec_strm_wordsp the 32bit data swap for stream data it will be used in 64 bit environment 0 : no swapping 1 : swapping high and low 32bit data
3	RW	0x0	sw_dec_out_wordsp the 32bit data swap for dec output data, it will be used in 64 bit environment 0 : no swapping 1 : swapping high and low 32bit data
2	RW	0x0	sw_dec_in_wordsp the 32bit data swap for dec input data, it will be used in 64 bit environment 0 : no swapping 1 : swapping high and low 32bit data note : it no used for stream data
1	RW	0x0	sw_dec_out_endian the endian mode of dec output Decoder output endian mode: 0 : Big endian (0-1-2-3 order) 1 : Little endian (3-2-1-0 order)

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_dec_in_endian the endian mode of dec input 0 : Big endian (0-1-2-3 order) 1 : Little endian (3-2-1-0 order) note : it no used for stream data

VDPU_SWREG55

Address: Operational Base + offset (0x000dc)

decoder int register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	sw_timeout_det_sts the Interrupt status bit for tell us timeout detected AXI in IDLE status too long
12	RW	0x0	sw_error_det_sts the Interrupt status bit for tell us error detected Interrupt status bit input stream error. When high, an error is found in input data stream decoding. HW will self reset. (1,2,3,6,48,55,57)
11	RO	0x0	reserved
10	RW	0x0	sw_bslice_det_sts the Interrupt status bit for tell us B slice be detected DIVX3:the value extension header flag

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Bit	Attr	Reset Value	Description
9	RW	0x0	sw_slice_det_sts the Interrupt status bit for tell us slice be decoded
8	RW	0x0	sw_aso_det_sts the Interrupt status bit for tell us ASO detectet ASO:Arbitrary Slice Ordering
7	RO	0x0	reserved
6	RW	0x0	sw_buf_emt_sts the Interrupt status bit for tell input buffer empty
5	RW	0x0	sw_bus_sts the Interrupt status bit for tell sw bus have some error
4	RW	0x0	sw_dec_rdy_sts the Interrupt status bit for tell sw processed a picture
3:2	RO	0x0	reserved
1	RW	0x0	sw_dec_irq_dis the decoder finish interrupt request diable flag 1 : use polling to see the interrupt 0 : use sw_pp_irq

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_dec_irq the decoder finish interrupt request after sw query this interrupt,shoud write 0 to reset.

VDPU_SWREG56

Address: Operational Base + offset (0x000e0)

axi ctrl for decoder

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x0	sw_axi_sel axi signals selected for encoder or decoder 0: auto sel for encoder or decoder 1: sel decoder (only used in the middle decoder frame to set bus_dec_en to 0)
22	RW	0x0	sw_dec_data_discd_en enable for Data discard the fixed burst length will be used ,and the more read datas will be auto discarded by hw
21	RW	0x1	sw_bus_pos_sel the parallel or serial mode for axi read and write 0:serial 1:parallel
20:16	RW	0x00	sw_dec_max_burlen the max burst length can be used by axi bus range : 1-16

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Bit	Attr	Reset Value	Description
15:8	RW	0x01	sw_dec_axi_id_wr AXI Write ID if you config 0,will modify as 1 by hw
7:0	RW	0x01	sw_dec_axi_id_rd AXI Read ID if you config 0/5,will modify as 1 by hw

VDPU_SWREG57

Address: Operational Base + offset (0x000e4)

enable flag for decoder

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_dec_timeout_mode dec timeout mode selset when 1'b0 , timeout cycle is 181'b1 when 1'b1, timeout cycle is 221'b1
30	RO	0x0	reserved
29	RW	0x0	sw_cache_en cache enable 1'b1: cache enable 1'b0: cache disable when sw_cache_en is 1'b1, sw_pref_sigchan should also be 1'b1
28	RW	0x0	sw_pref_sigchan prefetch single channel enable 1'b1: prefetch single channel enable

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Bit	Attr	Reset Value	Description
27	RW	0x0	sw_intra_dbl3t In chroma dc intra prediction, when this bit is enable, there will 3 cycle enhance for every block
26	RW	0x0	sw_intra_dblspeed intra double speed enable Intra double speed enable
25	RW	0x0	sw_inter_dblspeed inter double speed enable Inter double speed enable
24:23	RO	0x0	reserved
22	RW	0x0	sw_st_code_exist existence flag for stream start code 0 : not exist 1 : exist
21	RW	0x0	sw_addit_ch_fmt_wen Enable for additional chrominance data format writing tiled mode should be disable,when this bit be used decoder writes chrominance: group of 8 pixels of Cb then corresponding 8 pixels of Cr Data is written to sw_dec_ch8pix_st_adr.
20	RW	0x0	sw_rlc_mode_en enable for RLC mode 0:decoder data come from bit stream(VLC mode),side information (bitplane data in VC-1) 1:decoder data come from RLC input data,only h.264 and MPEG4 sp be valid

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Bit	Attr	Reset Value	Description
19	RW	0x0	sw_divx3_en enable for divx3 used sw_dec_fmt_sle= MPEG4: 0 : disabled 1 : enabled
18	RW	0x0	sw_prog_jpeg_en enable flag for Progressive JPEG 0 : baseline JPEG 1 : progressive JPEG
17	RW	0x0	sw_curpic_code_sel the current picture coding mode select 0 : progressive 1 : interlaced
16	RW	0x0	sw_curpic_stru_sel the current picture Structure selected 0 : frame structure, (that isMBAFF structured picture is interlaced) 1 : field structure
15	RW	0x0	sw_pic_type_sel1 pic type sel1 flag 0:desided by sw_pic_type_sel0 1: picture type is BI/D/B note: BI is for vc1 D is for mpeg1 B is for h264

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Bit	Attr	Reset Value	Description
14	RW	0x0	sw_pic_type_sel0 pic type sel0 flag should need sw_pic_type_sel1=0 0: Intra type (I) 1:Inter type (P)
13	RW	0x0	sw_pic_decfield_sel select which field will be decodered 0 : bottom field 1 : top field
12	RW	0x0	sw_fwd_refpic_mode_sel progressive and interlaced for coding mode used for forward reference picture: 0 : progressive 1 : interlaced the backward reference picture is the same as current picture
11	RW	0x0	sw_sorspa_en enable flag for Sorenson Sparc valid when sw_dec_fmt_sel= MPEG- 4 0 = disabled 1 = enable if enable,will use Sorenson escape coding to compatible stream for h.263

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Bit	Attr	Reset Value	Description
10	RW	0x0	sw_dmmv_wr_en enable flag for Direct mode motion vector write current picture 0:disable 1:enable this bit used in MPEG2 or VP6 is for the purpose error concealment case. this bit used in h264 is for the purpose write DPB case with the corresponding reference picture. this bit used in other decoder format is for the purpose writing to external memory starting from mv start address
9	RW	0x0	sw_reftop_en enable flag for reference top field sw_dec_fmt_sel =VC-1 and sw_ref_frm = 0 , 0 = bottom field 1 = top field
8	RW	0x0	sw_first_reftop_en enable flag for FWD reference top field have been decoded first 0 : FWD reference bottom field 1 : FWD reference top field
7	RW	0x0	sw_sequ_mbaff_en the enable flag for Sequence includes MBAFF coded pictures 0:disable 1:enable

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Bit	Attr	Reset Value	Description
6	RW	0x0	sw_rd_cnt_tab_en the enable flag for reading Picture order count table read data from memory used 0:disable 1:enable (hw will read pic order counts)
5	RW	0x0	sw_timeout_sts_en the enable flag for Timeout interrupt 0:disable 1:enable (if hw can be working status too long,you will get an timeout interrupt)
4	RW	0x1	sw_dec_clkgate_en the enable flag for Decoder auto clock gating default hw will reset to 1 0:disable 1:enable
3	RO	0x0	reserved
2	RW	0x0	sw_dec_wr_extmem_dis disable flag for wiriting decoder output data to external memory 0 : enable 1:disable(no write to external memory)
1	RW	0x0	sw_refpic_buf2_en enable flag for Refer picture buffer 2 0:disable 1:enable (should : pic size > QVGA)

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_dec_st_work enable flag for decoder to start working hw will auto reset this be after a frame be decodered no matter it right or have some error

VDPU_SWREG58

Address: Operational Base + offset (0x000e8)

soft reset register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	sw_soft_rst the soft reset for decoder or pp or encoder write 1 to reset,and it will auto reset to 0 after one cycle

VDPU_SWREG59

Address: Operational Base + offset (0x000ec)

H264, MPEG4, VC1, VP6 Prediction filter tap

Bit	Attr	Reset Value	Description
31:22	RW	0x000	sw_pflt_set0_tap0 Prediction filter 0, tap 0
21:12	RW	0x000	sw_pflt_set0_tap1 Prediction filter set 0, tap 1

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Bit	Attr	Reset Value	Description
11:2	RW	0x000	sw_pflt_set0_tap2 Prediction filter set 0, tap 2
1:0	RO	0x0	reserved

VDPU_SWREG60

Address: Operational Base + offset (0x000f0)

additional chrominance address

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_addit_ch_st_adr the start address for additional chrominance data format The usage is enabled by sw_addit_ch_fmt_wen
1:0	RO	0x0	reserved

VDPU_SWREG61

Address: Operational Base + offset (0x000f4)

standard dependent tables start address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_qtable_st_adr standard dependent tables start address JPEG : AC,DC, QP tables MPEG4/2 : QP table H.264 :various tables VP6/7/8 : stream decoding tables RV : picture slice sizes
1:0	RO	0x0	reserved

VDPU_SWREG62

Address: Operational Base + offset (0x000f8)

Direct mode motion vector write/read start address

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_dmmv_st_adr Direct mode motion vector write/read start address H264: Direct mode motion vector write/read start address Progressive JPEG: the start address for ACDC coefficient read/write If current round is for DC components : this start address is pointing to luminance AC component rounds: this start address is used for current type
1:0	RO	0x0	reserved

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VDPU_SWREG63

Address: Operational Base + offset (0x000fc)

write decoder output picture or field start address

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_dec_out_st_adr write decoder output picture or field start address video: write decoder output picture or field start address JPEG/VP8 snapshot: wirete decoder output luminance picture start address
1:0	RO	0x0	reserved

VDPU_SWREG64

Address: Operational Base + offset (0x00100)

rlc or vlc mode input data start addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	sw_rlc_vlc_st_adr rlc or vlc mode input data start addr RLC mode: RLC data start address VLC mode: Stream start address HW return value of last_byte_address by this register to tell where stream has been read when you get some abnormality interrupt,may be used for debug VP7/VP8: DCT stream for MB rows 0,2n start address
1:0	RO	0x0	reserved

VDPU_SWREG65

Address: Operational Base + offset (0x00104)

refbufferd related

Bit	Attr	Reset Value	Description
31	RW	0x0	sw_refbu_e 0 : disable 1 : enable
30:19	RW	0x000	sw_refbu_thr_level Reference buffer disable threshold value (cache miss amount). Used to buffer shut down (if more misses than allowed)

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Bit	Attr	Reset Value	Description
18:14	RW	0x00	sw_refbuf_picid the picture id for refference buffer
13	RW	0x0	sw_refbuf_idcal_e Enable for HW internal reference ID calculation If given threshold level is reached by any picture_id after first MB row, that picture_id is used for reference buffer fill for rest of the picture
12	RW	0x0	sw_refbuf_fildpar_mod_e the mode enable for Field parity mode enable. 0 : the result field of the evaluation be used 1 : the parity mode field be used
11:9	RO	0x0	reserved
8:0	RW	0x000	sw_refbuf_y_ofset the y offset for refbufferd if hw should compensate the global motion of the video for better buffer hit rate will use this coordinate

VDPU_SWREG66

Address: Operational Base + offset (0x00108)

ID register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x6731	prod_id product number
15:12	RO	0x2	major_num
11:4	RO	0x68	minor_num
3	RO	0x1	ascii_id_en enable for ASCII product ID
2:0	RO	0x0	build_ver

VDPU_SWREG67

Address: Operational Base + offset (0x0010c)

Synthesis configuration register decoder 1(read only)

Bit	Attr	Reset Value	Description
31	RO	0x1	jpeg_allow_flag JPEG sampling support 16Mpixel~67Mpixel be sampled and supported by 411 and 444
30	RO	0x1	refbuf_allow_flag ref buffer support 0:no support 1:support

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Bit	Attr	Reset Value	Description
29	RO	0x1	divx_allow_flag divx support
28	RO	0x0	refbuf2_allow_flag refbuffer2 support
27:26	RO	0x1	rv_allow_flag
25	RO	0x0	rom_imp_type rom implementation type 0: from actual ROM units 1: from RTL
24	RO	0x1	vp7_allow_flag vp7 support
23	RO	0x1	vp8_allow_flag vp8 support
22	RO	0x1	avs_allow_flag avs support
21:20	RO	0x1	mvc_allow_flag mvc support

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Bit	Attr	Reset Value	Description
19	RO	0x1	vp8_snpt_en vp8 snapshot,decoder format support
18:17	RO	0x1	tile_mode_sel tile mode support 0:no support 1: 8x4 support 2,3: no used
16:0	RO	0x0	reserved

VDPU_SWREG68

Address: Operational Base + offset (0x00110)

sum of partitions(read only)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	sw_refbuf_sum_top sum of the top partitions
15:0	RO	0x0000	sw_refbuf_sum_bot sum of the bottom partitions

VDPU_SWREG69

Address: Operational Base + offset (0x00114)

sum information (read only)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0000	sw_refbuf_sum_hit sum of the refbufferd hits of the picture
15:0	RO	0x0000	sw_luma_sum_intra sum of the luminance 8x8 intra partitons of the picture.

VDPU_SWREG70

Address: Operational Base + offset (0x00118)

sum of the decoded motion vector y-components(read only)

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:0	RO	0x000000	sw_ycomp_mv_sum sum of the decoded motion vector y-components

VDPU_SWREG71

Address: Operational Base + offset (0x0011c)

information for read only register

Bit	Attr	Reset Value	Description
31	RO	0x1	dec_mpeg2_allow Decoding format support, for MPEG-2 / MPEG-1 0:no support 1:support

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Bit	Attr	Reset Value	Description
30:29	RO	0x3	dec_vc1_allow Decoding format support, VC-1 0 :not supported 1 :simple profile be supported 2 :main profile be supported 3 :advanced profile be supported
28	RO	0x1	dec_jpeg_allow Decoding format support for JPEG 0:no support 1:support
27:26	RO	0x2	dec_mpeg4_allow Decoding format support for MPEG-4 / H.263 0 :not supported 1 :simple profile be supported 2 :advanced simple profile be supported
25:24	RO	0x3	dec_h264_allow h264 be support 0:no support 1:baseline profile be supported 2:high profile be supported
23	RO	0x1	dec_vp6_allow Decoding format support for VP6 0:no support 1:support
22	RO	0x0	dec_prog_jpeg_allow support for progressive jpeg 0:no support 1:support

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Bit	Attr	Reset Value	Description
21	RO	0x1	outbuf_sel output buffer selected 0 : 1MB buffer be used 1 : 4MB buffer be used
20	RO	0x1	refbuf_exist reference buffer support 0 : not supported 1 : support
19:16	RO	0x5	dec_std_bus 0 : error 1 : AHB master, AHB slave 2 : OCP master, OCP slave 3 : AXI master, AXI slave 4 : AXI master, APB slave 5 : AXI master, AHB slave
15:14	RO	0x1	rtl_lan_sel 0: no used 1:vhdl 2:verilog
13:12	RO	0x2	bus_w 0 : error 1 : word bus 2 : double word bus 3 : quadruple word bus
11	RO	0x1	dec_srson_allow Decoding format support for Sorenson 0:no support 1:support

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Bit	Attr	Reset Value	Description
10:0	RO	0x780	sw_dec_max_allow_w the max width can be decoder

VDPU_SWREG72

Address: Operational Base + offset (0x00120)

debug0

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RO	0x00	debug_service debug_service signals this value[6:0]=service_wr[2:0], service_rd[3:0]
23:0	RO	0x0	reserved

VDPU_SWREG73

Address: Operational Base + offset (0x00124)

debug registers

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x0	debug_mv_req mvst_mv_req signal value
29	RO	0x0	debug_rlc_req prtr_res_y_req signal value

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Bit	Attr	Reset Value	Description
28	RO	0x0	debug_res_y_req prtr_res_y_req signal value
27	RO	0x0	debug_res_c_req
26	RO	0x0	debug_strm_da_e
25	RO	0x0	debug_frm_rdy
24	RO	0x0	debug_flt_req
23	RO	0x0	debug_ref0_req
22	RO	0x0	debug_ref1_req
21	RO	0x0	reserved
20:0	RO	0x000000	debug_mb_cnt

VDPU_SWREG74

Address: Operational Base + offset (0x00128)

MV address for h264

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_diff_mv_st_adr the Differential motion vector start address Differential motion vector base address used for h264 only it also reuse used as: [29:25] : 9st forward picid of inital reference pic list [24:20] : 8st forward picid of inital reference pic list [19:15] : 7st forward picid of inital reference pic list [14:10] : 6st forward picid of inital reference pic list [9:5] : 5st forward picid of inital reference pic list [4:0] : 4st forward picid of inital reference pic list
1:0	RO	0x0	reserved

VDPU_SWREG75

Address: Operational Base + offset (0x0012c)

H.264 Intra prediction 4x4 mode start address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_pred4x4_st_adr H.264 Intra prediction 4x4 mode start address also be used as: [29:25] : 15st forward picid of inital reference pic list [24:20] : 14st forward picid of inital reference pic list [19:15] : 13st forward picid of inital reference pic list [14:10] : 12st forward picid of inital reference pic list [9:5] : 11st forward picid of inital reference pic list [4:0] : 10st forward picid of inital reference pic list
1:0	RO	0x0	reserved

VDPU_SWREG76

Address: Operational Base + offset (0x00130)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx1 the number of referance pic index0
15:0	RW	0x0000	h264_num_ref_idx0 the number of referance pic index0

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VDPU_SWREG77

Address: Operational Base + offset (0x00134)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx3 the number of referance pic index3
15:0	RW	0x0000	h264_num_ref_idx2 the number of referance pic index2

VDPU_SWREG78

Address: Operational Base + offset (0x00138)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx5 the number of referance pic index5
15:0	RW	0x0000	h264_num_ref_idx4 the number of referance pic index4

VDPU_SWREG79

Address: Operational Base + offset (0x0013c)

the number of referance pic

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx7 the number of referance pic index7
15:0	RW	0x0000	h264_num_ref_idx6 the number of referance pic index6

VDPU_SWREG80

Address: Operational Base + offset (0x00140)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx9 the number of referance pic index9
15:0	RW	0x0000	h264_num_ref_idx8 the number of referance pic index8

VDPU_SWREG81

Address: Operational Base + offset (0x00144)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx11 the number of referance pic index11

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	h264_num_ref_idx10 the number of referance pic index10

VDPU_SWREG82

Address: Operational Base + offset (0x00148)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx13 the number of referance pic index13
15:0	RW	0x0000	h264_num_ref_idx12 the number of referance pic index12

VDPU_SWREG83

Address: Operational Base + offset (0x0014c)

the number of referance pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_num_ref_idx15 the number of referance pic index15
15:0	RW	0x0000	h264_num_ref_idx14 the number of referance pic index14

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VDPU_SWREG84

Address: Operational Base + offset (0x00150)

referance frame0 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref0_st_addr the start address of referance frame0
1	RW	0x0	h264_ref0_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref0_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG85

Address: Operational Base + offset (0x00154)

referance frame1 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref1_st_addr the start address of referance frame1
1	RW	0x0	h264_ref1_field_en the type Refer picture consist of 0 : frame 1 : field

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Bit	Attr	Reset Value	Description
0	RW	0x0	h264_ref1_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG86

Address: Operational Base + offset (0x00158)

referance frame2 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref2_st_addr the start address of referance frame2
1	RW	0x0	h264_ref2_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref2_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG87

Address: Operational Base + offset (0x0015c)

referance frame3 address for h264

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref3_st_addr the start address of referance frame3
1	RW	0x0	h264_ref3_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref3_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG88

Address: Operational Base + offset (0x00160)

referance frame4 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref4_st_addr the start address of referance frame4
1	RW	0x0	h264_ref4_field_en the type Refer picture consist of 0 : frame 1 : field

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Bit	Attr	Reset Value	Description
0	RW	0x0	h264_ref4_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG89

Address: Operational Base + offset (0x00164)

referance frame5 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref5_st_addr the start address of referance frame5
1	RW	0x0	h264_ref5_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref5_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG90

Address: Operational Base + offset (0x00168)

referance frame6 address for h264

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref6_st_addr the start address of referance frame6
1	RW	0x0	h264_ref6_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref6_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG91

Address: Operational Base + offset (0x0016c)

referance frame7 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref7_st_addr the start address of referance frame7
1	RW	0x0	h264_ref7_field_en the type Refer picture consist of 0 : frame 1 : field

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Bit	Attr	Reset Value	Description
0	RW	0x0	h264_ref7_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG92

Address: Operational Base + offset (0x00170)

referance frame8 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref8_st_addr the start address of referance frame8
1	RW	0x0	h264_ref8_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref8_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG93

Address: Operational Base + offset (0x00174)

referance frame9 address for h264

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref9_st_addr the start address of referance frame9
1	RW	0x0	h264_ref9_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref9_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG94

Address: Operational Base + offset (0x00178)

referance frame10 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref10_st_addr the start address of referance frame10
1	RW	0x0	h264_ref10_field_en the type Refer picture consist of 0 : frame 1 : field

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Bit	Attr	Reset Value	Description
0	RW	0x0	h264_ref10_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG95

Address: Operational Base + offset (0x0017c)

referance frame11 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref11_st_addr the start address of referance frame11
1	RW	0x0	h264_ref11_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref11_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG96

Address: Operational Base + offset (0x00180)

referance frame12 address for h264

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref12_st_addr the start address of referance frame12
1	RW	0x0	h264_ref12_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref12_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG97

Address: Operational Base + offset (0x00184)

referance frame13 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref13_st_addr the start address of referance frame13
1	RW	0x0	h264_ref13_field_en the type Refer picture consist of 0 : frame 1 : field

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Bit	Attr	Reset Value	Description
0	RW	0x0	h264_ref13_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG98

Address: Operational Base + offset (0x00188)

referance frame14 address for h264

Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref14_st_addr the start address of referance frame14
1	RW	0x0	h264_ref14_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref14_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG99

Address: Operational Base + offset (0x0018c)

referance frame15 address for h264

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RW	0x00000000	h264_ref15_st_addr the start address of referance frame15
1	RW	0x0	h264_ref15_field_en the type Refer picture consist of 0 : frame 1 : field
0	RW	0x0	h264_ref15_closer_sel Which field is more closer to current picture 0 : bottom field be selected 1 : top field be selected

VDPU_SWREG100

Address: Operational Base + offset (0x00190)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	h264_init_reflist_df5 5st initial reference picture list for direct forward picid used for h264
24:20	RW	0x00	h264_init_reflist_df4 4st initial reference picture list for direct forward picid used for h264

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Bit	Attr	Reset Value	Description
19:15	RW	0x00	h264_init_reflist_df3 3st initial reference picture list for direct forward picid used for h264
14:10	RW	0x00	h264_init_reflist_df2 2st initial reference picture list for direct forward picid used for h264
9:5	RW	0x00	h264_init_reflist_df1 1st initial reference picture list for direct forward picid used for h264
4:0	RW	0x00	h264_init_reflist_df0 0st initial reference picture list for direct forward picid used for h264

VDPU_SWREG101

Address: Operational Base + offset (0x00194)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	h264_init_reflist_df11 11st initial reference picture list for direct forward picid used for h264

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Bit	Attr	Reset Value	Description
24:20	RW	0x00	h264_init_reflist_df10 10st initial reference picture list for direct forward picid used for h264
19:15	RW	0x00	h264_init_reflist_df9 9st initial reference picture list for direct forward picid used for h264
14:10	RW	0x00	h264_init_reflist_df8 8st initial reference picture list for direct forward picid used for h264
9:5	RW	0x00	h264_init_reflist_df7 7st initial reference picture list for direct forward picid used for h264
4:0	RW	0x00	h264_init_reflist_df6 6st initial reference picture list for direct forward picid used for h264

VDPU_SWREG102

Address: Operational Base + offset (0x00198)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:15	RW	0x00	h264_init_reflist_df15 15st initial reference picture list for direct forward picid used for h264
14:10	RW	0x00	h264_init_reflist_df14 14st initial reference picture list for direct forward picid used for h264
9:5	RW	0x00	h264_init_reflist_df13 13st initial reference picture list for direct forward picid used for h264
4:0	RW	0x00	h264_init_reflist_df12 12st initial reference picture list for direct forward picid used for h264

VDPU_SWREG103

Address: Operational Base + offset (0x0019c)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	h264_init_reflist_db5 5st initial reference picture list for direct backward picid used for h264

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Bit	Attr	Reset Value	Description
24:20	RW	0x00	h264_init_reflist_db4 4st initial reference picture list for direct backward picid used for h264
19:15	RW	0x00	h264_init_reflist_db3 3st initial reference picture list for direct backward picid used for h264
14:10	RW	0x00	h264_init_reflist_db2 2st initial reference picture list for direct backward picid used for h264
9:5	RW	0x00	h264_init_reflist_db1 1st initial reference picture list for direct backward picid used for h264
4:0	RW	0x00	h264_init_reflist_db0 0st initial reference picture list for direct backward picid used for h264

VDPU_SWREG104

Address: Operational Base + offset (0x001a0)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29:25	RW	0x00	h264_init_reflist_db11 11st initial reference picture list for direct backward picid used for h264
24:20	RW	0x00	h264_init_reflist_db10 10st initial reference picture list for direct backward picid used for h264
19:15	RW	0x00	h264_init_reflist_db9 9st initial reference picture list for direct backward picid used for h264
14:10	RW	0x00	h264_init_reflist_db8 8st initial reference picture list for direct backward picid used for h264
9:5	RW	0x00	h264_init_reflist_db7 7st initial reference picture list for direct backward picid used for h264
4:0	RW	0x00	h264_init_reflist_db6 6st initial reference picture list for direct backward picid used for h264

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VDPU_SWREG105

Address: Operational Base + offset (0x001a4)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:15	RW	0x00	h264_init_reflist_db15 15st initial reference picture list for direct backward picid used for h264
14:10	RW	0x00	h264_init_reflist_db14 14st initial reference picture list for direct backward picid used for h264
9:5	RW	0x00	h264_init_reflist_db13 13st initial reference picture list for direct backward picid used for h264
4:0	RW	0x00	h264_init_reflist_db12 12st initial reference picture list for direct backward picid used for h264

VDPU_SWREG106

Address: Operational Base + offset (0x001a8)

initial reference picture list related

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:15	RW	0x00	h264_init_reflist_pf3 3st initial reference picture list for P forward picid Initial reference picture list for P forward picid 3

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Bit	Attr	Reset Value	Description
14:10	RW	0x00	h264_init_reflist_pf2 2st initial reference picture list for P forward picid Initial reference picture list for P forward picid 2
9:5	RW	0x00	h264_init_reflist_pf1 1st initial reference picture list for P forward picid Initial reference picture list for P forward picid 1
4:0	RW	0x00	h264_init_reflist_pf0 0st initial reference picture list for P forward picid used in 264

VDPU_SWREG107

Address: Operational Base + offset (0x001ac)

long term flag for reference pictuure index

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	h264_refpic_term_flag long term flag for reference pictuure index

VDPU_SWREG108

Address: Operational Base + offset (0x001b0)

valid flag for reference picture index

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	h264_refpic_valid_flag valid flag for reference picture index

VDPU_SWREG109

Address: Operational Base + offset (0x001b4)

the stream start word for decoder

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	h264_strm_start_bit the stream start word for decoder assosiates with sw_rlc_vlc_st_adr

VDPU_SWREG110

Address: Operational Base + offset (0x001b8)

h264 pic mb size

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:22	RW	0x00	h264_flt_offset_cr_qp filter offset of cr qp
21:17	RW	0x00	h264_flt_offset_cb_qp filter offset of cb qp

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Bit	Attr	Reset Value	Description
16:9	RW	0x00	h264_pic_mb_h Picture height in macroblocks value =((pixel height+15)/16). used for frame or single field size being decoded
8:0	RW	0x000	h264_pic_mb_w Picture width in macroblocks value = ((pixel width + 15) /16)

VDPU_SWREG111

Address: Operational Base + offset (0x001bc)

h264 ctrl related

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:16	RW	0x0	h264_wp_bslice_sel wp(weighted prediction) specification for B slice 0 : default wp be used 1 : explicit wp be used 2 : implicit wp be used
15:5	RO	0x0	reserved
4:0	RW	0x00	h264_max_refnum short and long term reference frames's maximum number this value is for decoded picture buffer

VDPU_SWREG112

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Address: Operational Base + offset (0x001c0)

current frame related

Bit	Attr	Reset Value	Description
31	RW	0x0	h264_dblk_ctrl_flag the control present flag of deblocking filter to indicates if the slice header will have the deblocking filter's extra variables controlling characteristics
30	RW	0x0	h264_rpcp_flag redundant picture count present flag to specifies whether redundant picture count syntax elements
29:21	RO	0x0	reserved
20:16	RW	0x00	h264_curfrm_len the bit length of input data stream's frame num H.264: Bit length of frame_num in data stream
15:0	RW	0x0000	h264_curfrm_num the current frame number for h264 it may be use for reference picture reordering and identify short-term reference frames

VDPU_SWREG113

Address: Operational Base + offset (0x001c4)

reference picture related

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
26:16	RW	0x000	h264_mk_len the length of marking bits use for decoded reference picture
15:0	RW	0x0000	h264_idrp_id instantaneous decoding refresh picture id

VDPU_SWREG114

Address: Operational Base + offset (0x001c8)

maximum reference

Bit	Attr	Reset Value	Description
31:24	RW	0x00	h264_pps_id the id for picture parameter set it identifies the slice header will have the id of picture parameter set
23:19	RW	0x00	h264_max_refidx1 the maximum reference index 1 it will be used in decoding inter predicted macro blocks
18:14	RW	0x00	h264_max_refidx0 the maximum reference index 0 it will be used in decoding inter predicted macro blocks
13:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	h264_pocf_len the length of picture order count field in stream

VDPU_SWREG115

Address: Operational Base + offset (0x001cc)

enable flag

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	h264_idr_pic_flag instantaneous decoding refresh picture flag
7	RW	0x0	h264_dlmv_method_en the method to use to derive luma motion vectors with B_skip, B_Direct_16x16 and B_direct_8x8_inference_flag
6	RW	0x0	h264_monochr_en monochromatic enable sampling format , 0 : 4:2:0 1 : 4:0:0
5	RW	0x0	h264_cabac_en enable for cabac

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Bit	Attr	Reset Value	Description
4	RW	0x0	h264_pslice_wp_en enable flag of Weighted prediction for P slices
3	RW	0x0	h264_nimb_intra_en if intra prediction uses only neighbouring intra macroblocks 0 : neighbouring inter macroblocks are used in intra prediction process 1 : neighbouring intra macroblocks are used
2	RW	0x0	h264_tranf_flag_en_8x8 8x8 transform flag enable for stream decoding
1	RW	0x0	h264_scl_matrix_en scaling matrix enable 0 : normal transform 1 : use scaling matrix for transform
0	RW	0x0	h264_fieldpic_flag_exist Flag for streamd that field_pic_flag exists in stream Flag for streamd that field_pic_flag exists in stream

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VDPU_SWREG120

Address: Operational Base + offset (0x001e0)

multi format reuse register0

Bit	Attr	Reset Value	Description

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg0 multi format reuse register0 except h264 MPEG4/JPEG/VC-1/MPEG2/VP6/RV/VP7/VP: [31:2] RLC mode: Base address for RLC data VLC mode: Stream start address VP7/VP8: [31:2] This base address is used as sw_dct_strm0_base including DCT stream for MB rows 0,2n

VDPU_SWREG121

Address: Operational Base + offset (0x001e4)

multi format reuse register1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg1 multi format reuse register1 except h264 RV: [20:16] : frame size length vc-1: [31] : bitplane mode enable for corresponding element0 [30] : bitplane mode enable for corresponding element1 [29] : bitplane mode enable for corresponding element2 [28:24] : alemative PQUANT. [23:20] : weather dq_profile will be set to picture edges [19] : TTMB or TTFRM sel [18:14] : qpindex value [12] : enable for bilinear motion compensation ena [11] : enable for iform quantizer e [10] : enable for HALFQP [9:8] : frame level transform type sel [7] : the 2st byte of the stream emulation byte [6] : enable for antization parameter change inside frame [5] : enable for VC-1 advanced profile JPEG: [26:0] : progressive JPEG MPEG2 : [12] : enable for bilinear motion compensation VP7/VP8: [31:26] : DCT stream partition index 1 of start bit [25:20] : DCT stream partition index 2 of start bit

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VDPU_SWREG122

Address: Operational Base + offset (0x001e8)

multi format reuse register2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg2 multi format reuse register2 except h264 MPEG4: [31:26] : Exact bit of stream start word [25] : enable for sync markers [24] : enable for Type 1 quantization [23:19] : the offset of Qp filter [18:14] : the offset of Qp filter for cr [0] : filed_pic_flag exists in stream JPEG : [31:26] : Exact bit of stream start word [25] : enable for sync markers [12:11] : total of Quantization tables [10:8] : the sampling format for input pic [7] : JPEG width [6] : weather current strem buffer contain the end of a JPEG image [5:0] : vlc table VC-1 : [31:26] : Exact bit of stream start word [25] : enable for sync markers [24] : Quantisation profile [23] : each mb take on quantization step size or not [22] : range reduced [20] : chrominance interpolation accuracy information [17:13] : select tables which be used to dec [12:10] : select mode syntax element table [9:7] : select mv table [6:4] : select CPBCY table [3:0] : block pattern table select

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VDPU_SWREG123

Address: Operational Base + offset (0x001ec)

multi format reuse register3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg3 multi format reuse register3 except h264 JPEG: [15:0] : start marker frequency. vc-1: [31:24] : B picture scl factor [23:19] : FWD direction reference distance [18:14] : BWD direction reference distance vp6: [17:14] : loop filter limit value [13] : enable for variance test [12:10] : filter MV size threshold [9:0] : filter variance threshold RV : [31:30] : rv8 for rv9 sel [29:28] : OSV quant. [27:14] : direct mv and wp scl value for FWD dir [13:0] : direct mv and wp scl value for BWD dir VP7/VP : [31:16] : value 0 for inital dc predictor [15:0] : value 1 for inital dc predictor

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VDPU_SWREG124

Address: Operational Base + offset (0x001f0)

multi format reuse register4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg4 multi format reuse register4 except h264 MPEG4: [31:2] : MB ctrl start address VC-1: [24] : enable for 0st intensity compensation [23:16] : iscale value [15:0] : ishift value VP6: [23:0] : total of CTRL stream data RV : [12:0] : total of slices for one picture VP7/VP : [27:24] : coeffient partitions number [23:0] : total of CTRL stream data

VDPU_SWREG125

Address: Operational Base + offset (0x001f4)

multi format reuse register5

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg5 multi format reuse register5 except h264 JPEG: [31:2] : Cb ACDC coeff start address VP6/VP7/vp: [31:22] : prediction filter with set 5 and tap 1 [21:12] : prediction filter with set 5 and tap 2 [11:2] : prediction filter with set 5 and tap 3

VDPU_SWREG126

Address: Operational Base + offset (0x001f8)

multi format reuse register6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg6 multi format reuse register6 except h264 JPEG: [31:2] : Cr ACDC coeff start address VP6/VP7/vp: [31:22] : prediction filter with set 6 and tap 0 [21:12] : prediction filter with set 6 and tap 1 [11:2] : prediction filter with set 6 and tap 2

VDPU_SWREG127

Rockchip Confidential

Address: Operational Base + offset (0x001fc)

multi format reuse register7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg7 multi format reuse register7 except h264 VP6/VP7/vp: [31:22] : prediction filter with set 6 and tap 3 [21:12] : prediction filter with set 7 and tap 0 [11:2] : prediction filter with set 7 and tap 1

VDPU_SWREG128

Address: Operational Base + offset (0x00200)

multi format reuse register8

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg8 multi format reuse register8 except h264 VP6: [31:22] : prediction filter with set 7 and tap 2 [21:12] : prediction filter with set 7 and tap 3 VP7/VP: [31:22] : prediction filter with set 7 and tap 2 [21:12] : prediction filter with set 7 and tap 3 [11:10] : extra prediction filter with set 2 and tap -1 [9:8] : extra prediction filter with set 2 and tap 4 [7:6] : extra prediction filter with set 4 and tap -1 [5:4] : extra prediction filter with set 4 and tap 4 [3:2] : extra prediction filter with set 6 and tap -1 [1:0] : extra prediction filter with set 6 and tap 4

VDPU_SWREG129

Address: Operational Base + offset (0x00204)

multi format reuse register9

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg9 multi format reuse register9 except h264 VP6: [29:24] : 56st coef of scan read index [23:18] : 57st coef of scan read index [17:12] : 58st coef of scan read index [11:6] : 59st coef of scan read index [5:0] : 60st coef of scan read index

VDPU_SWREG130

Address: Operational Base + offset (0x00208)

multi format reuse register10

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg10 multi format reuse register10 except h264 VP6: [29:24] : 61st coef of scan read index [23:18] : 62st coef of scan read index [17:12] : 63st coef of scan read index VP7/VP8: [31:27] : quantisizer delta0 for VP8 used [26:22] : quantisizer delta1 for VP8 used [21:11] : QP0 for VP7 and quantisizer value for LUT used for VP8 [10:0] : QP1 for VP7 and quantisizer value for LUT used for VP8

VDPU_SWREG131

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Address: Operational Base + offset (0x0020c)

multi format reuse register11

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg11 multi format reuse register11 except h264 MPEG4/H263/VC-1/vp6/rv/VP7/VP8: [31:2] : reference pic0 start address JPEG: [31:2] : the ch decoder output start address

VDPU_SWREG132

Address: Operational Base + offset (0x00210)

multi format reuse register12

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg12 multi format reuse register12 except h264 VP7/VP8: [31] : type of loop filter [30:28] : sharpness of loop filter [27:21] : MB type0 adjustment of filter level [20:14] : MB type1 adjustment of filter level [13:7] : MB type2 adjustment of filter level [6:0] : MB type3 adjustment of filter level

VDPU_SWREG133

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Address: Operational Base + offset (0x00214)

multi format reuse register13

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg13 multi format reuse register13 except h264 VP6: [29:24] : 51st coef of scan read index [23:18] : 52st coef of scan read index [17:12] : 53st coef of scan read index [11:6] : 54st coef of scan read index [5:0] : 55st coef of scan read index VP7/VP8: [27:21] : reference frame type0 adjustment of filter level [20:14] : reference frame type1 adjustment of filter level [13:7] : reference frame type2 adjustment of filter level [6:0] : reference frame type3 adjustment of filter level

VDPU_SWREG134

Address: Operational Base + offset (0x00218)

multi format reuse register14

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg14 multi format reuse register14 except h264 MPEG4/vc1/MEPG2/RV: [31:2] : reference pic2 start address JPEG: [30:24] : code words of length 6 [21:16] : code words of length 5 [15:11] : code words of length 4 [10:7] : code words of length 3 [5:3] : code words of length 2 [1:0] : code words of length 1

VDPU_SWREG135

Address: Operational Base + offset (0x0021c)

multi format reuse register15

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg15 multi format reuse register15 except h264 MPEG4/vc1/MEPG2/RV: [31:2] : reference pic3 start address JPEG: [30:24] : code words of length 10 [23:16] : code words of length 9 [15:8] : code words of length 8 [7:0] : code words of length 7

VDPU_SWREG136

Rockchip Confidential

Address: Operational Base + offset (0x00220)

multi format reuse register16

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg16 multi format reuse register16 except h264 VP6/VP7/VP8 [31:2] : golden reference pic start address(PIC_ID 4) [0] : golden reference pic siggn bias(VP7/VP8) VC-1: [31:16] : length of picture header [13] : 1/4 MV/MB sel [11] : enable for ref pic range reduce [10:9] : max different mv length [7:6] : select range for mv [5] : enable for overlap smoothing [4:3] : MB overlap smoothing method MPEG4/MPEG2: [19] : alternalte scan flag [18:15] : HRZ AXI's bit amount for representing FWD MV [14:11] : VRZ AXI's bit amount for representing FWD MV [10:7] : HRZ AXI's bit amount for representing BWD MV [6:3] : VRZ AXI's bit amount for representing BWD MV [2] : FWD MV Y resolution [1] : the ctrl bit for rounding(MPEG4),BWD MV Y resolution(MPEG2) [0] : pic type of previous anchor(MPEG4) JPEG: [30:24] : code words of length 14 [23:16] : code words of length 13 [15:8] : code words of length 12

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VDPU_SWREG137

Address: Operational Base + offset (0x00224)

multi format reuse register17

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg17 multi format reuse register17 except h264 MPEG4: [26:0] : reference distance syntax for delta value0 be used JPEG: [31:27] : tab2:code words of length 4 [26:23] : tab2:code words of length 3 [21:19] : tab2:code words of length 2 [17:16] : tab2:code words of length 1 [15:8] : tab1:code words of length 16 [7:0] : tab1:code words of length 15 VC-1: [24] : enable for intensity compensation 3 [23:16] : intensity compensation's iscale value [15:0] : intensity compensation's ishift value VP6/VP7: [29:24] : 1st coef of scan read index [23:18] : 2st coef of scan read index [17:12] : 3st coef of scan read index [11:6] : 4st coef of scan read index [5:0] : 5st coef of scan read index VP8: [31:2] : alternate ref pic start addres(PIC ID5) [0] : alternate ref pic sign bias

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VDPU_SWREG138

Address: Operational Base + offset (0x00228)

multi format reuse register18

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg18 multi format reuse register18 except h264 MPEG4: [26:0] : reference distance syntax for delta value -1 be used JPEG: [31:24] : tab2:code words of length 8 [23:16] : tab2:code words of length 7 [14:8] : tab2:code words of length 6 [5:0] : tab2:code words of length 5 VC-1: [24] : enable for intensity compensation 4 [23:16] : intensity compensation's iscale value [15:0] : intensity compensation's shift value VP6/VP7: [29:24] : 6st coef of scan read index [23:18] : 7st coef of scan read index [17:12] : 8st coef of scan read index [11:6] : 9st coef of scan read index [5:0] : 10st coef of scan read index

VDPU_SWREG139

Address: Operational Base + offset (0x0022c)

multi format reuse register19

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg19 multi format reuse register19 except h264 MPEG4: [26:0] : reference distance syntax for delta value1 be used JPEG: [31:24] : tab2:code words of length 12 [23:16] : tab2:code words of length 11 [15:8] : tab2:code words of length 10 [7:0] : tab2:code words of length 9 VP6/VP7: [29:24] : 11st coef of scan read index [23:18] : 12st coef of scan read index [17:12] : 13st coef of scan read index [11:6] : 14st coef of scan read index [5:0] : 15st coef of scan read index

VDPU_SWREG140

Address: Operational Base + offset (0x00230)

multi format reuse register20

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg20 multi format reuse register20 except h264 JPEG: [31:24] : tab2:code words of length 16 [23:16] : tab2:code words of length 15 [15:8] : tab2:code words of length 14 [7:0] : tab2:code words of length 13 VP6: [29:24] : 16st coef of scan read index [23:18] : 17st coef of scan read index [17:12] : 18st coef of scan read index [11:6] : 19st coef of scan read index [5:0] : 20st coef of scan read index VP7/VP8: [31:2] : DCT stream MB row 2,2n+1 start address

VDPU_SWREG141

Address: Operational Base + offset (0x00234)

multi format reuse register21

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg21 multi format reuse register21 except h264 JPEG: [31:28] : tab1:code words of length 8 [27:24] : tab1:code words of length 7 [23:20] : tab1:code words of length 6 [19:16] : tab1:code words of length 5 [15:12] : tab1:code words of length 4 [11:8] : tab1:code words of length 3 [6:4] : tab1:code words of length 2 [1:0] : tab1:code words of length 1 VP6: [29:24] : 21st coef of scan read index [23:18] : 22st coef of scan read index [17:12] : 23st coef of scan read index [11:6] : 24st coef of scan read index [5:0] : 25st coef of scan read index VP7/VP8 [31:2] : DCT stream MB row 2,2n+2 start address

VDPU_SWREG142

Address: Operational Base + offset (0x00238)

multi format reuse register22

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg22 multi format reuse register22 except h264 JPEG: [31:28] : tab1:code words of length 16 [27:24] : tab1:code words of length 15 [23:20] : tab1:code words of length 14 [19:16] : tab1:code words of length 13 [15:12] : tab1:code words of length 12 [11:8] : tab1:code words of length 11 [6:4] : tab1:code words of length 10 [1:0] : tab1:code words of length 9 VP6: [29:24] : 26st coef of scan read index [23:18] : 27st coef of scan read index [17:12] : 28st coef of scan read index [11:6] : 29st coef of scan read index [5:0] : 30st coef of scan read index VP7/VP8 [31:2] : DCT stream MB row 2,2n+3 start address

VDPU_SWREG143

Address: Operational Base + offset (0x0023c)

multi format reuse register23

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg23 multi format reuse register23 except h264 JPEG: [31:28] : tab2:code words of length 8 [27:24] : tab2:code words of length 7 [23:20] : tab2:code words of length 6 [19:16] : tab2:code words of length 5 [15:12] : tab2:code words of length 4 [11:8] : tab2:code words of length 3 [6:4] : tab2:code words of length 2 [1:0] : tab2:code words of length 1 VP6: [29:24] : 31st coef of scan read index [23:18] : 32st coef of scan read index [17:12] : 33st coef of scan read index [11:6] : 34st coef of scan read index [5:0] : 35st coef of scan read index VP7/VP8 [31:2] : DCT stream MB row 2,2n+4 start address

VDPU_SWREG144

Address: Operational Base + offset (0x00240)

multi format reuse register24

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg24 multi format reuse register24 except h264 JPEG: [31:28] : tab2:code words of length 16 [27:24] : tab2:code words of length 15 [23:20] : tab2:code words of length 14 [19:16] : tab2:code words of length 13 [15:12] : tab2:code words of length 12 [11:8] : tab2:code words of length 11 [6:4] : tab2:code words of length 10 [1:0] : tab2:code words of length 9 VP6: [29:24] : 36st coef of scan read index [23:18] : 37st coef of scan read index [17:12] : 38st coef of scan read index [11:6] : 39st coef of scan read index [5:0] : 40st coef of scan read index VP7/VP8 [31:2] : DCT stream MB row 2,2n+5 start address

VDPU_SWREG145

Address: Operational Base + offset (0x00244)

multi format reuse register25

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg25 multi format reuse register25 except h264 JPEG: [31:28] : tab3:code words of length 8 [27:24] : tab3:code words of length 7 [23:20] : tab3:code words of length 6 [19:16] : tab3:code words of length 5 [15:12] : tab3:code words of length 4 [11:8] : tab3:code words of length 3 [6:4] : tab2:code words of length 2 [1:0] : tab3:code words of length 1 VC-1: [31:2] : bitplane mb ctrl start address VP6/VP7/VP8: [31:2] : ctrl data stream start address

VDPU_SWREG146

Address: Operational Base + offset (0x00248)

multi format reuse register26

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg26 multi format reuse register26 except h264 JPEG: [31:28] : tab3:code words of length 16 [27:24] : tab3:code words of length 15 [23:20] : tab3:code words of length 14 [19:16] : tab3:code words of length 13 [15:12] : tab3:code words of length 12 [11:8] : tab3:code words of length 11 [6:4] : tab3:code words of length 10 [1:0] : tab3:code words of length 9 VP6: [29:24] : 41st coef of scan read index [23:18] : 42st coef of scan read index [17:12] : 43st coef of scan read index [11:6] : 44st coef of scan read index [5:0] : 45st coef of scan read index VP7/VP8 [31:2] : DCT stream MB row 2,2n+6 start address

VDPU_SWREG147

Address: Operational Base + offset (0x0024c)

multi format reuse register27

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg27 multi format reuse register27 except h264 VP6: [29:24] : 46st coef of scan read index [23:18] : 47st coef of scan read index [17:12] : 48st coef of scan read index [11:6] : 49st coef of scan read index [5:0] : 50st coef of scan read index VP7/VP8 [31:2] : DCT stream MB row 2,2n+7 start address

VDPU_SWREG148

Address: Operational Base + offset (0x00250)

multi format reuse register28

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg28 multi format reuse register28 except h264 MPEG4/VC-1/MPEG2/RV/VP7/VP8: [31:2] : ref pic index 1 start address JPEG/VP8: [7:0] : snapshot

VDPU_SWREG149

Address: Operational Base + offset (0x00254)

multi format reuse register29

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg29 multi format reuse register29 except h264 VC-1: [24] : enable for intensity compensation 1 [23:16] : intensity compensation iscale value [15:0] : intensity compensation ishift value VP7/VP8: [31:2] : the segmentation map value start address [1] : enable for segmentation map update [0] : enable for segmentation

VDPU_SWREG150

Address: Operational Base + offset (0x00258)

multi format reuse register30

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg30 multi format reuse register30 except h264 VC-1: [24] : enable for intensity compensation 2 [23:16] : intensity compensation iscale value [15:0] : intensity compensation ishift value VP7/VP8: [29:24] : DCT stream partition index 3 of start bit [23:18] : DCT stream partition index 4 of start bit [17:12] : DCT stream partition index 5 of start bit [11:6] : DCT stream partition index 6 of start bit [5:0] : DCT stream partition index 7 of start bit

VDPU_SWREG151

Address: Operational Base + offset (0x0025c)

multi format reuse register31

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg31 multi format reuse register31 except h264 VP7/VP8: [31:27] : quantisizer delta2 for VP8 used [26:22] : quantisizer delta3 for VP8 used [21:11] : QP2 for VP7 and quantisizer value for LUT used for VP8 [10:0] : QP3 for VP7 and quantisizer value for LUT used for VP8

VDPU_SWREG152

Address: Operational Base + offset (0x00260)

multi format reuse register32

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg32 multi format reuse register32 except h264 VP7/VP8: [31:27] : quantisizer delta4 for VP8 used [21:11] : QP4 for VP7 and quantisizer value for LUT used for VP8 [10:0] : QP5 for VP7 and quantisizer value for LUT used for VP8

VDPU_SWREG153

Address: Operational Base + offset (0x00264)

multi format reuse register33

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg33 multi format reuse register33 except h264 VC-1/VP6/RV/VP7/VP8: [31:22] : prediction filter with set 0,tap3 (also for mpeg4) [21:12] : prediction filter with set 1,tap0 [11:2] : prediction filter with set 1,tap1

VDPU_SWREG154

Address: Operational Base + offset (0x00268)

multi format reuse register34

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg34 multi format reuse register34 except h264 VC-1/VP6/RV/VP7/VP8: [31:22] : prediction filter with set 1,tap2 [21:12] : prediction filter with set 1,tap3 [11:2] : prediction filter with set 2,tap0(no for vc-1)

VDPU_SWREG155

Address: Operational Base + offset (0x0026c)

multi format reuse register35

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg35 multi format reuse register35 except h264 VC-1/VP6/VP7/VP8: [31:22] : prediction filter with set 2,tap1 [21:12] : prediction filter with set 2,tap2 [11:2] : prediction filter with set 2,tap3

VDPU_SWREG156

Address: Operational Base + offset (0x00270)

multi format reuse register36

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg36 multi format reuse register36 except h264 VP6/VP7/VP8: [31:22] : prediction filter with set 3,tap0 [21:12] : prediction filter with set 3,tap1 [11:2] : prediction filter with set 3,tap2

VDPU_SWREG157

Address: Operational Base + offset (0x00274)

multi format reuse register37

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg37 multi format reuse register37 except h264 VP6/VP7/VP8: [31:22] : prediction filter with set 3,tap3 [21:12] : prediction filter with set 4,tap0 [11:2] : prediction filter with set 4,tap1

VDPU_SWREG158

Address: Operational Base + offset (0x00278)

multi format reuse register38

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mfr_reg38 multi format reuse register38 except h264 VP6/VP7/VP8: [31:22] : prediction filter with set 4,tap2 [21:12] : prediction filter with set 4,tap3 [11:2] : prediction filter with set 5,tap0

VDPU_SWREG164_PERF_LATENCY_CTRL0

Address: Operational Base + offset (0x00290)

Axi performance latency module contrl register0

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:8	RW	0x000	sw_rd_latency_thr sw_rd_latency_thr

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Bit	Attr	Reset Value	Description
7:4	RW	0x0	sw_rd_latency_id sw_rd_latency_id
3	RW	0x0	sw_axi_cnt_type sw_axi_cnt_type sw_axi_cnt_type
2	RW	0x0	sw_axi_perf_frm_type sw_axi_perf_frm_type 1'b0: clear by frame end 1'b1: clear by software configuration
1	W1C	0x0	sw_axi_perf_clr_e sw_axi_perf_clr_e 1'b0: software clear disable 1'b1: software clear enalbe
0	RW	0x0	sw_axi_perf_work_e sw_axi_perf_work_e 1'b0: disable 1'b1: enable

VDPU_SWREG165_PERF_LATENCY_CTRL1

Address: Operational Base + offset (0x00294)

PERF_LATENCY_CTRL1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:8	RW	0x0	sw_aw_count_id sw_aw_count_id sw_aw_count_id
7:4	RW	0x0	sw_ar_count_id sw_ar_count_id sw_ar_count_id
3	RW	0x0	sw_aw_cnt_id_type sw_aw_cnt_id_type sw_aw_cnt_id_type
2	RW	0x0	sw_ar_cnt_id_type sw_ar_cnt_id_type sw_ar_cnt_id_type
1:0	RW	0x0	sw_addr_align_type sw_addr_align_type sw_addr_align_type

VDPU_SWREG166_PERF_RD_MAX_LATENCY_NUM0

Address: Operational Base + offset (0x00298)

Read max latency number

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	rd_max_latency_num_ch0 read max latency value of channel 0

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VDPU_SWREG167_PERF_RD_LATENCY_SAMP_NUM

Address: Operational Base + offset (0x0029c)

The number of bigger than configed threshold value

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_thr_num_ch0 read latency thr number channel 0

VDPU_SWREG168_PERF_RD_LATENCY_ACC_SUM

Address: Operational Base + offset (0x002a0)

Total sample number

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_acc_sum

VDPU_SWREG169_PERF_RD_AXI_TOTAL_BYTE

Address: Operational Base + offset (0x002a4)

perf_rd_axi_total_byte

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_rd_axi_total_byte perf_rd_axi_total_byte perf_rd_axi_total_byte

VDPU_SWREG170_PERF_WR_AXI_TOTAL_BYTE

Address: Operational Base + offset (0x002a8)

perf_wr_axi_total_byte

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_wr_axi_total_byte perf_wr_axi_total_byte perf_wr_axi_total_byte

VDPU_SWREG171_PERF_WORKING_CNT

Address: Operational Base + offset (0x002ac)

perf_working_cnt

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_working_cnt perf_working_cnt perf_working_cnt

15.5.12 VDPU121 MMU Register Summary

Name	Offset	Size	Reset Value	Description
VCODEC_MMU_DTE_ADDR	0x00000	W	0x00000000	MMU current page Table address
VCODEC_MMU_STATUS	0x00004	W	0x00000018	MMU status register
VCODEC_MMU_COMMAND	0x00008	W	0x00000000	MMU command register
VCODEC_MMU_PAGE_FAULT_ADDR	0x0000c	W	0x00000000	MMU logical address of last page fault
VCODEC_MMU_ZAP_ONE_LINE	0x00010	W	0x00000000	MMU Zap cache line register
VCODEC_MMU_INT_RAWSTAT	0x00014	W	0x00000000	MMU raw interrupt status register

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Name	Offset	Size	Reset Value	Description
VCODEC_MMU_INT_CLEAR	0x00018	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_MASK	0x0001c	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_STATUS	0x00020	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_AUTO_GATING	0x00024	W	0x00000001	mmu auto gating

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.13 VDPU121 MMU Detail Register Description

VCODEC_MMU_DTE_ADDR

Address: Operational Base + offset (0x00000)

MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR MMU_DTE_ADDR MMU current page Table address

VCODEC_MMU_STATUS

Address: Operational Base + offset (0x00004)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10:6	RO	0x00	PAGE_FAULT_BUS_ID page fault bus id Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE page fault acess The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x1	REPLAY_BUFFER_EMPTY replay buffer empty status 1'b1:The MMU replay buffer is empty
3	RO	0x1	MMU_IDLE mmu idle status The MMU is idle when accesses are being translated and there are no unfinished translated accesses. 1'b1: MMU is idle
2	RO	0x0	STAIL_ACTIVE stall active status MMU stall mode currently enabled. The mode is enabled by command 1'b1: MMU is in stall active status
1	RO	0x0	PAGE_FAULT_ACTIVE page fault active status MMU page fault mode currently enabled . The mode is enabled by command. 1'b1: page fault is active

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Bit	Attr	Reset Value	Description
0	RO	0x0	PAGING_ENABLED Paging enabled status 1'b0: paging is disabled 1'b1: Paging is enabled

VCODEC_MMU_COMMAND

Address: Operational Base + offset (0x00008)

MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD mmu cmd MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

VCODEC_MMU_PAGE_FAULT_ADDR

Address: Operational Base + offset (0x0000c)

MMU logical address of last page fault

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR page fault addr address of last page fault

VCODEC_MMU_ZAP_ONE_LINE

Address: Operational Base + offset (0x00010)

MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE zap one line address to be invalidated from the page table cache

VCODEC_MMU_INT_RAWSTAT

Address: Operational Base + offset (0x00014)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error read bus error status
0	RW	0x0	PAGE_FAULT page fault page fault status

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VCODEC_MMU_INT_CLEAR

Address: Operational Base + offset (0x00018)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR read bus error write 1 to clear read bus error
0	WO	0x0	PAGE_FAULT page fault clear write 1 to page fault clear

VCODEC_MMU_INT_MASK

Address: Operational Base + offset (0x0001c)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error enable the read bus interrupt souce when this bit is set to 1'b1
0	RW	0x0	PAGE_FAULT page fault mask enable the page fault interrupt souce when this bit is set to 1'b1

VCODEC_MMU_INT_STATUS

Address: Operational Base + offset (0x00020)

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MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR read bus error status 1'b1:read bus error status
0	RO	0x0	PAGE_FAULT page fault status 1'b1:page fault

VCODEC_MMU_AUTO_GATING

Address: Operational Base + offset (0x00024)

mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_clkgating mmu_auto_clkgating when it is 1'b1, the mmu will auto gating it self

15.5.14 VDPU121 Cache Register Summary

Name	Offset	Size	Reset Value	Description
pref_cache_VERSION	0x00000	W	0xcac20101	VERSION register
pref_cache_SIZE	0x00004	W	0x06110206	L2 cache SIZE
pref_cache_STATUS	0x00008	W	0x00000000	Status register

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Name	Offset	Size	Reset Value	Description
pref_cache_COMMAND	0x00010	W	0x00000000	Command setting register
pref_cache_CLEAR_PAGE	0x00014	W	0x00000000	clear page register
pref_cache_MAX_READS	0x00018	W	0x0000001c	maximum read register
pref_cache_PERFCNT_SRC0	0x00020	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL0	0x00024	W	0x00000000	performance counter 0 value register
pref_cache_PERFCNT_SRC1	0x00028	W	0x00000000	performance counter 0 source register
pref_cache_PERFCNT_VAL1	0x0002c	W	0x00000000	performance counter 1 value register

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.15 VDPU121 Cache Detail Register Description

pref_cache_VERSION

Address: Operational Base + offset (0x00000)

VERSION register

Bit	Attr	Reset Value	Description
31:16	RO	0xcac2	PRODUCT_ID Field0000 Abstract Field0000 Description
15:8	RO	0x01	VERSION_MAJOR Field0000 Abstract Field0000 Description

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Bit	Attr	Reset Value	Description
7:0	RO	0x01	VERSION_MINOR Field0000 Abstract Field0000 Description

pref_cache_SIZE

Address: Operational Base + offset (0x00004)

L2 cache SIZE

Bit	Attr	Reset Value	Description
31:24	RO	0x06	External_bus_width Field0000 Abstract Log2 external bus width in bits
23:16	RO	0x11	CACHE_SIZE Field0000 Abstract Log2 cache size in bytes
15:8	RO	0x02	ASSOCIATIVITY Field0000 Abstract Log2 associativity
7:0	RO	0x06	LINE_SIZE Field0000 Abstract Log2 line size in bytes

pref_cache_STATUS

Address: Operational Base + offset (0x00008)

Status register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	DATA_BUSY Field0000 Abstract set when the cache is busy handling data
0	RO	0x0	CMD_BUSY Field0000 Abstract set when the cache is busy handling commands

pref_cache_COMMAND

Address: Operational Base + offset (0x00010)

Command setting register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	sw_addrb_sel Field0000 Abstract 2'b00: to sel b[14:6] 2'b01: to sel b[15:9], b[7:6] 2'b10: to sel b[16:10], b[7:6] 2'b11: to sel b[17:11], b[7:6]
3	RO	0x0	reserved
2:0	WO	0x0	COMMAND Field0000 Abstract The possible command is 1 = Clear entire cache

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pref_cache_CLEAR_PAGE

Address: Operational Base + offset (0x00014)

clear page register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	CLEAR_PAGE Field0000 Abstract writing an address, invlidates all lines in that page from the cache

pref_cache_MAX_READS

Address: Operational Base + offset (0x00018)

maximum read register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x1c	MAX_READS Field0000 Abstract Limit the number of outstanding read transactions to this amount

pref_cache_PERFCNT_SRC0

Address: Operational Base + offset (0x00020)

performance counter 0 source register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	PERFCNT_SRC0 Field0000 Abstract This register holds all the possible source values for Performance Counter 0 0: total clock cycles 1: active clock cycles 2: read transactions, master 3: word reads, master 4: read transactions, slave 5: word reads, slave 6: read hit, slave 7: read misses, slave 8: read invalidates, slave 9: cacheable read transactions, slave 10: bad hit nmber, slave

pref_cache_PERFCNT_VAL0

Address: Operational Base + offset (0x00024)

performance counter 0 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL0 Field0000 Abstract Performance counter 0 value

pref_cache_PERFCNT_SRC1

Address: Operational Base + offset (0x00028)

performance counter 0 source register

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	PERFCNT_SRC1 Field0000 Abstract This register holds all the possible source values for Performance Counter 1 0: total clock cycles 1: active clock cycles 2: read transactions, master 3: word reads, master 4: read transactions, slave 5: word reads, slave 6: read hit, slave 7: read misses, slave 8: read invalidates, slave 9: cacheable read transactions, slave 10: bad hit nmber, slave

pref_cache_PERFCNT_VAL1

Address: Operational Base + offset (0x0002c)

performance counter 1 value register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERFCNT_VAL1 Field0000 Abstract Performance counter 1 value

15.5.16 VEPU121 Function Config Register Summary

Name	Offset	Size	Reset Value	Description

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_0	0x00000	W	0x00000000	1st quantization for jpeg lumin table
VEPU_swreg_1	0x00004	W	0x00000000	2st quantization for jpeg lumin table
VEPU_swreg_2	0x00008	W	0x00000000	3st quantization for jpeg lumin table
VEPU_swreg_3	0x0000c	W	0x00000000	4st quantization for jpeg lumin table
VEPU_swreg_4	0x00010	W	0x00000000	5st quantization for jpeg lumin table
VEPU_swreg_5	0x00014	W	0x00000000	6st quantization for jpeg lumin table/part 1 for qp round
VEPU_swreg_6	0x00018	W	0x00000000	7st quantization for jpeg lumin table
VEPU_swreg_7	0x0001c	W	0x00000000	8st quantization for jpeg lumin table
VEPU_swreg_8	0x00020	W	0x00000000	9st quantization for jpeg lumin table
VEPU_swreg_9	0x00024	W	0x00000000	10st quantization for jpeg lumin table
VEPU_swreg_10	0x00028	W	0x00000000	11st quantization for jpeg lumin table
VEPU_swreg_11	0x0002c	W	0x00000000	12st quantization for jpeg lumin table
VEPU_swreg_12	0x00030	W	0x00000000	13st quantization for jpeg lumin table
VEPU_swreg_13	0x00034	W	0x00000000	14st quantization for jpeg lumin table

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_14	0x00038	W	0x00000000	15st quantization for jpeg lumin table
VEPU_swreg_15	0x0003c	W	0x00000000	16st quantization for jpeg lumin table
VEPU_swreg_16	0x00040	W	0x00000000	1st quantization for jpeg chroma table
VEPU_swreg_17	0x00044	W	0x00000000	2st quantization for jpeg chroma table
VEPU_swreg_18	0x00048	W	0x00000000	3st quantization for jpeg chroma table
VEPU_swreg_19	0x0004c	W	0x00000000	4st quantization for jpeg chroma table
VEPU_swreg_20	0x00050	W	0x00000000	5st quantization for jpeg chroma table
VEPU_swreg_21	0x00054	W	0x00000000	6st quantization for jpeg chroma table
VEPU_swreg_22	0x00058	W	0x00000000	7st quantization for jpeg chroma table
VEPU_swreg_23	0x0005c	W	0x00000000	8st quantization for jpeg chroma table/part 3 for qp round
VEPU_swreg_24	0x00060	W	0x00000000	9st quantization for jpeg chroma table
VEPU_swreg_25	0x00064	W	0x00000000	10st quantization for jpeg chroma table
VEPU_swreg_26	0x00068	W	0x00000000	11st quantization for jpeg chroma table
VEPU_swreg_27	0x0006c	W	0x00000000	12st quantization for jpeg chroma

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_28	0x00070	W	0x00000000	13st quantization for jpeg chroma
VEPU_swreg_29	0x00074	W	0x00000000	14st quantization for jpeg chroma
VEPU_swreg_30	0x00078	W	0x00000000	15st quantization for jpeg chroma
VEPU_swreg_31	0x0007c	W	0x00000000	16st quantization for jpeg chroma
VEPU_swreg_44	0x000b0	W	0x00000000	Intra slice bitmap
VEPU_swreg_45	0x000b4	W	0x00000000	Intra slice bitmap
VEPU_swreg_46	0x000b8	W	0x00000000	intra macro block sellect register
VEPU_swreg_47	0x000bc	W	0x00000000	CIR intra control register
VEPU_swreg_48	0x000c0	W	0x00000000	input luma start address
VEPU_swreg_49	0x000c4	W	0x00000000	input cb start address
VEPU_swreg_50	0x000c8	W	0x00000000	input cr start address
VEPU_swreg_51	0x000cc	W	0x00000000	stream header bits left register
VEPU_swreg_52	0x000d0	W	0x00000000	stream header bits left register
VEPU_swreg_53	0x000d4	W	0x00000000	stream buffer register
VEPU_swreg_54	0x000d8	W	0x01010000	axi control register
VEPU_swreg_55	0x000dc	W	0x00000000	qp related
VEPU_swreg_56	0x000e0	W	0x00000000	the luma reference frame start address

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_57	0x000e4	W	0x00000000	the chroma reference frame start address
VEPU_swreg_58	0x000e8	W	0x00000000	the result of qp sum div2
VEPU_swreg_59	0x000ec	W	0x00000000	Register0000 Abstract
VEPU_swreg_60	0x000f0	W	0x00000000	Register0001 Abstract
VEPU_swreg_61	0x000f4	W	0x00000000	input luminance information
VEPU_swreg_62	0x000f8	W	0x00000000	rlc_sum
VEPU_swreg_63	0x000fc	W	0x00000000	the reconstructed luma start address
VEPU_swreg_64	0x00100	W	0x00000000	the reconstructed chroma start address
VEPU_swreg_65_reuse	0x00104	W	0x00000000	checkpoint 1 and 2
VEPU_swreg_66_reuse	0x00108	W	0x00000000	checkpoint 3 and 4
VEPU_swreg_67_reuse	0x0010c	W	0x00000000	checkpoint 5 and 6
VEPU_swreg_68_reuse	0x00110	W	0x00000000	checkpoint 7 and 8
VEPU_swreg_69_reuse	0x00114	W	0x00000000	checkpoint 9 and 10
VEPU_swreg_70_reuse	0x00118	W	0x00000000	checkpoint word error 1 and 2
VEPU_swreg_71_reuse	0x0011c	W	0x00000000	checkpoint word error 1 and 2

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_72_reuse	0x00120	W	0x00000000	checkpoint word error 1 and 2
VEPU_swreg_73_reuse	0x00124	W	0x00000000	checkpoint delta QP register
VEPU_swreg_74	0x00128	W	0x00000000	input image format
VEPU_swreg_75	0x0012c	W	0x00000000	intra/inter mode
VEPU_swreg_76_reuse	0x00130	W	0x00000000	encoder control regsiter 0
VEPU_swreg_77	0x00134	W	0x00000000	output stream start address
VEPU_swreg_78	0x00138	W	0x00000000	output control start address
VEPU_swreg_79	0x0013c	W	0x00000000	next picture luminance start address
VEPU_swreg_80	0x00140	W	0x00000000	Base address for MV output
VEPU_swreg_81	0x00144	W	0x00000000	the cabac table start address
VEPU_swreg_82	0x00148	W	0x00000000	the first of ROI area register
VEPU_swreg_83	0x0014c	W	0x00000000	the second of ROI area register
VEPU_swreg_84	0x00150	W	0x00000000	Stabilization matrix
VEPU_swreg_85	0x00154	W	0x00000000	Stabilization matrix
VEPU_swreg_86	0x00158	W	0x00000000	Stabilization matrix
VEPU_swreg_87	0x0015c	W	0x00000000	Stabilization matrix

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_88	0x00160	W	0x00000000	Stabilization matrix
VEPU_swreg_89	0x00164	W	0x00000000	Stabilization matrix
VEPU_swreg_90	0x00168	W	0x00000000	Stabilization matrix
VEPU_swreg_91	0x0016c	W	0x00000000	Stabilization matrix
VEPU_swreg_92	0x00170	W	0x00000000	Stabilization matrix
VEPU_swreg_93	0x00174	W	0x00000000	the output of Stabilization motion sum
VEPU_swreg_94	0x00178	W	0x00000000	output of Stabilization
VEPU_swreg_95	0x0017c	W	0x00000000	RGB to YUV conversion coefficient register
VEPU_swreg_96	0x00180	W	0x00000000	RGB to YUV conversion coefficient register
VEPU_swreg_97	0x00184	W	0x00000000	RGB to YUV conversion coefficient register
VEPU_swreg_98	0x00188	W	0x00000000	RGA MASK
VEPU_swreg_99	0x0018c	W	0x00000000	mv related
VEPU_swreg_100_reuse	0x00190	W	0x00000000	QP register
VEPU_swreg_101_read	0x00194	W	0x1f522780	hw config reg
VEPU_swreg_102	0x00198	W	0x00000000	mvc related
VEPU_swreg_103	0x0019c	W	0x00000000	encoder start
VEPU_swreg_104	0x001a0	W	0x00000000	mb control register
VEPU_swreg_105	0x001a4	W	0x00000000	SWAP

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Name	Offset	Size	Reset Value	Description
VEPU_swreg_106_reuse	0x001a8	W	0x00000000	encoder control register 1
VEPU_swreg_107_reuse	0x001ac	W	0x00000000	JPEG control regsiter
VEPU_swreg_108_reuse	0x001b0	W	0x00000000	intra_slice_bmp2
VEPU_swreg_109	0x001b4	W	0x00001000	encoder status
VEPU_swreg_110_read	0x001b8	W	0x48311220	product ID
VEPU_swreg_120_183	0x001e0	W	0x00000000	DMV_4p_1p_penalty

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

15.5.17 VEPU121 Function Config Detail Register Description

VEPU_swreg_0

Address: Operational Base + offset (0x00000)

1st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant1 jpeg luma quantiztion 1 jpeg luma quantization 1

VEPU_swreg_1

Address: Operational Base + offset (0x00004)

2st quantization for jpeg lumin table

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant2 jpeg luma quantization 2 jpeg luma quantization 2

VEPU_swreg_2

Address: Operational Base + offset (0x00008)

3st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant3 jpeg luma quantization 3 jpeg luma quantization 3

VEPU_swreg_3

Address: Operational Base + offset (0x0000c)

4st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant4 jpeg luma quantization 4 jpeg luma quantization 4

VEPU_swreg_4

Address: Operational Base + offset (0x00010)

5st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant5 jpeg luma quantization 5 jpeg luma quantization 5

VEPU_swreg_5

Address: Operational Base + offset (0x00014)

6st quantization for jpeg lumin table/part 1 for qp round

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant6 jpeg luma quantization 6 jpeg luma quantization 6

VEPU_swreg_6

Address: Operational Base + offset (0x00018)

7st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant7 jpeg luma quantization 7 jpeg luma quantization 7

VEPU_swreg_7

Address: Operational Base + offset (0x0001c)

8st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant8 jpeg luma quantization 8 jpeg luma quantization 8

VEPU_swreg_8

Address: Operational Base + offset (0x00020)

9st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant9 jpeg luma quantization 9 jpeg luma quantization 9

VEPU_swreg_9

Address: Operational Base + offset (0x00024)

10st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant10 jpeg luma quantization 10 jpeg luma quantization 10

VEPU_swreg_10

Address: Operational Base + offset (0x00028)

11st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant11 jpeg luma quantization 11 jpeg luma quantization 11

VEPU_swreg_11

Address: Operational Base + offset (0x0002c)

12st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant12 jpeg luma quantization 12 jpeg luma quantization 12

VEPU_swreg_12

Address: Operational Base + offset (0x00030)

13st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant13 jpeg luma quantization 13 jpeg luma quantization 13

VEPU_swreg_13

Address: Operational Base + offset (0x00034)

14st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant14 jpeg luma quantization 14 jpeg luma quantization 14

VEPU_swreg_14

Address: Operational Base + offset (0x00038)

15st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_luma_quant15 jpeg luma quantization 15 jpeg luma quantization 15

VEPU_swreg_15

Address: Operational Base + offset (0x0003c)

16st quantization for jpeg lumin table

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	sw_jpeg_luma_quant16 jpeg luma quantization 16 jpeg luma quantization 16

VEPU_swreg_16

Address: Operational Base + offset (0x00040)

1st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant1 jpeg chroma quantization 1 jpeg chroma quantization 1

VEPU_swreg_17

Address: Operational Base + offset (0x00044)

2st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant2 jpeg chroma quantization 2 jpeg chroma quantization 2

VEPU_swreg_18

Address: Operational Base + offset (0x00048)

3st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant3 jpeg chroma quantization 3 jpeg chroma quantization 3

VEPU_swreg_19

Address: Operational Base + offset (0x0004c)

4st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant4 jpeg chroma quantization 4 jpeg chroma quantization 4

VEPU_swreg_20

Address: Operational Base + offset (0x00050)

5st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant5 jpeg chroma quantization 5 jpeg chroma quantization 5

VEPU_swreg_21

Address: Operational Base + offset (0x00054)

6st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant6 jpeg chroma quantization 6 jpeg chroma quantization 6

VEPU_swreg_22

Address: Operational Base + offset (0x00058)

7st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant7 jpeg chroma quantization 7 jpeg chroma quantization 7

VEPU_swreg_23

Address: Operational Base + offset (0x0005c)

8st quantization for jpeg chroma table/part 3 for qp round

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant8 jpeg chroma quantization 8 jpeg chroma quantization 8

VEPU_swreg_24

Address: Operational Base + offset (0x00060)

9st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant9 jpeg chroma quantization 9 jpeg chroma quantization 9

VEPU_swreg_25

Address: Operational Base + offset (0x00064)

10st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant10 jpeg chroma quantization 10 jpeg chroma quantization 10

VEPU_swreg_26

Address: Operational Base + offset (0x00068)

11st quantization for jpeg chroma table

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x000	sw_jpeg_chroma_quant11 jpeg chroma quantization 11 jpeg chroma quantization 11

VEPU_swreg_27

Address: Operational Base + offset (0x0006c)

12st quantization for jpeg chroma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant11 jpeg chroma quantization 11 jpeg chroma quantization 11

VEPU_swreg_28

Address: Operational Base + offset (0x00070)

13st quantization for jpeg chroma

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	sw_jpeg_chroma_quant13 jpeg chroma quantization 13 jpeg chroma quantization 13

VEPU_swreg_29

Address: Operational Base + offset (0x00074)

14st quantization for jpeg chroma

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant14 jpeg chroma quantization 14 jpeg chroma quantization 14

VEPU_swreg_30

Address: Operational Base + offset (0x00078)

15st quantization for jpeg chroma

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	sw_jpeg_chroma_quant15 jpeg chroma quantization 15 jpeg chroma quantization 15

VEPU_swreg_31

Address: Operational Base + offset (0x0007c)

16st quantization for jpeg chroma

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_jpeg_chroma_quant16 jpeg chroma quantization 16 jpeg chroma quantization 16

VEPU_swreg_44

Address: Operational Base + offset (0x000b0)

Intra slice bitmap

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	intra_slice_bmp0 Intra slice bitmap for slices0 to slices31 bit0 : slices0 bit1 : slices1 bit2 : slices2 ..... bit31 : slices31

VEPU_swreg_45

Address: Operational Base + offset (0x000b4)

Intra slice bitmap

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	intra_slice_bmp1 Intra slice bitmap for slices32 to slices63 bit0 : slices32 bit1 : slices33 bit2 : slices34 ..... bit31 : slices63

VEPU_swreg_46

Address: Operational Base + offset (0x000b8)

intra macro block sellect register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	intra_up_mb_area The up intra macro block's area used in row The top intra macro block's area used in row
23:16	RW	0x00	intra_down_mb_area The down intra macro block's area used in row The bottom intra macro block's area used in row
15:8	RW	0x00	intra_left_mb_area The left intra macro block's area used in column The left intra macro block's area used in column

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	intra_right_mb_area The right intra macro block's area used in column The right intra macro block's area used in column

VEPU_swreg_47

Address: Operational Base + offset (0x000bc)

CIR intra control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	cir_first_intra the first macro block sellected for cir 0:disable other:enable and be set
15:0	RW	0x0000	cir_intra_mb_itvl the interval for cir intra macro block 0: disable other: enable and be set

VEPU_swreg_48

Address: Operational Base + offset (0x000c0)

input luma start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_in_st_adr input luma start address input luma start address

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VEPU_swreg_49

Address: Operational Base + offset (0x000c4)

input cb start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_in_st_adr input cb start address input cb start address

VEPU_swreg_50

Address: Operational Base + offset (0x000c8)

input cr start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_in_st_adr input cr start address input cr start address

VEPU_swreg_51

Address: Operational Base + offset (0x000cc)

stream header bits left register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	strm_header_left_hbits the high 32 bit of stram header be left the high 32 bit of stram header be left

VEPU_swreg_52

Address: Operational Base + offset (0x000d0)

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stream header bits left register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	strm_header_left_lbits the low 32 bit of stram header be left the low 32 bit of stram header be left

VEPU_swreg_53

Address: Operational Base + offset (0x000d4)

stream buffer register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	strm_bufsize_lmt the limit size of steam buffer the limit size of steam buffer

VEPU_swreg_54

Address: Operational Base + offset (0x000d8)

axi control register

Bit	Attr	Reset Value	Description
31:24	RW	0x01	axi_rd_id axi read id if config 0,it will be modify as 1 by HW auto
23:16	RW	0x01	axi_wr_id axi write id if config 0,it will be modify as 1 by HW auto
15:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:8	RW	0x00	burst_len burst length burst length
7:3	RO	0x0	reserved
2	RW	0x0	burst_incr_mod_sel burst increment mode select 0: single burst selected 1: incr burst selected
1	RW	0x0	burst_discard on-off burst data dicard 0:disable ,off 1:enable,on
0	RW	0x0	burst_disable disable burst mode for AXI 0: enable 1:disable

VEPU_swreg_55

Address: Operational Base + offset (0x000dc)

qp related

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:12	RW	0x0	roi_dlt_qp1 1st for delta qp for roi 1st for delta qp for roi

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Bit	Attr	Reset Value	Description
11:8	RW	0x0	roi_dlt_qp2 2st for delta qp for roi 2st for delta qp for roi
7:4	RO	0x0	reserved
3:0	RW	0x0	qp_adjst QP adjustment for mad signed register; range from -8 to 7

VEPU_swreg_56

Address: Operational Base + offset (0x000e0)

the luma reference frame start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_ref_st_adr the luma reference frame start address the luma reference frame start address

VEPU_swreg_57

Address: Operational Base + offset (0x000e4)

the chroma reference frame start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	chroma_ref_st_adr the chroma reference frame start address the chroma reference frame start address

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VEPU_swreg_58

Address: Operational Base + offset (0x000e8)

the result of qp sum div2

Bit	Attr	Reset Value	Description
31:11	RW	0x000000	qp_sum_div2 the result of (qp sum)/2 the result of (qp sum)/2
10:0	RO	0x0	reserved

VEPU_swreg_59

Address: Operational Base + offset (0x000ec)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	h264_qurt_pixmv_dis disable the function of quarter pixel MVs used in h264 1:disable 0:default,enable
27:26	RO	0x0	reserved
25:24	RW	0x0	dblking_flt_mode deblocking filter mode select 0 : enabled 1 : disabled 2 : disabled on slice
23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:21	RW	0x0	h264_cabac_idc the initial idc for cabac used in h264 0,1,2: used 3:no use
20	RW	0x0	entry_code_fmt the format of stream entropy coding h.264: 0: cavlc 1: cabac
19:18	RO	0x0	reserved
17	RW	0x0	h264_trfmod_8x8 on-off for 8x8 transform used in h264 on-off for 8x8 transform used in h264
16	RW	0x0	h264_res_intermod_4x4 the restriction inter mode selected in 4x4 block the restriction inter mode selected in 4x4 block
15	RW	0x0	h264_strm_mod_sel used to select stream mode 0 : NAL unit ; 1 : BYTE

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Bit	Attr	Reset Value	Description
14:8	RW	0x00	h264_slice_num the h264 slice numeber in on picture 0=one slice in currect picture 1=two slice in currect picture .....
7:0	RO	0x0	reserved

VEPU_swreg_60

Address: Operational Base + offset (0x000f0)

Register0001 Abstract

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:16	RW	0x00	strm_st_offset the start offset for stream
15:8	RW	0x00	skip_mb_mode H.264:SKIP macroblock mode
7:6	RO	0x0	reserved
5:4	RW	0x0	right_spill the right edge of image for spill pixels div4 value range:0~3

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	bot_spill the bottom edge of image for spill pixels the bottom edge of image for spill pixels

VEPU_swreg_61

Address: Operational Base + offset (0x000f4)

input luminance information

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22:20	RW	0x0	offset_in_chroma then offset of input chroma byte unit
19	RO	0x0	reserved
18:16	RW	0x0	offset_in_luma then offset of input luminance byte unit
15:14	RO	0x0	reserved
13:0	RW	0x0000	row_len_in_luma the row length of input luminance

VEPU_swreg_62

Address: Operational Base + offset (0x000f8)

rlc_sum

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:0	RW	0x000000	rlc_sum rlc_sum rlc_sum

VEPU_swreg_63

Address: Operational Base + offset (0x000fc)

the reconstructed luma start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	recon_luma_st_adr the reconstructed luma start address the reconstructed luma start address

VEPU_swreg_64

Address: Operational Base + offset (0x00100)

the reconstructed chroma start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	recon_chroma_st_adr the reconstructed chroma start address the reconstructed chroma start address

VEPU_swreg_65_reuse

Address: Operational Base + offset (0x00104)

checkpoint 1 and 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_chkpt_1 1st word used for check point used in h.264 1st word used for check point used in h.264
15:0	RW	0x0000	h264_chkpt_2 2st word used for check point used in h.264 2st word used for check point used in h.264

VEPU_swreg_66_reuse

Address: Operational Base + offset (0x00108)

checkpoint 3 and 4

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_chkpt_3 3st word used for check point used in h.264 3st word used for check point used in h.264
15:0	RW	0x0000	h264_chkpt_4 4st word used for check point used in h.264 4st word used for check point used in h.264

VEPU_swreg_67_reuse

Address: Operational Base + offset (0x0010c)

checkpoint 5 and 6

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_chkpt_5 5st word used for check point used in h.264 5st word used for check point used in h.264

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	h264_chkpt_6 6st word used for check point used in h.264 6st word used for check point used in h.264

VEPU_swreg_68_reuse

Address: Operational Base + offset (0x00110)

checkpoint 7 and 8

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_chkpt_7 7st word used for check point used in h.264 7st word used for check point used in h.264
15:0	RW	0x0000	h264_chkpt_8 8st word used for check point used in h.264 8st word used for check point used in h.264

VEPU_swreg_69_reuse

Address: Operational Base + offset (0x00114)

checkpoint 9 and 10

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_chkpt_9 9st word used for check point used in h.264 9st word used for check point used in h.264
15:0	RW	0x0000	h264_chkpt_10 10st word used for check point used in h.264 10st word used for check point used in h.264

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VEPU_swreg_70_reuse

Address: Operational Base + offset (0x00118)

checkpoint word error 1 and 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_errchkpt_1 1st word error check point used in h.264 1st word error check point used in h.264
15:0	RW	0x0000	h264_errchkpt_2 2st word error check point used in h.264 2st word error check point used in h.264

VEPU_swreg_71_reuse

Address: Operational Base + offset (0x0011c)

checkpoint word error 1 and 2

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_errchkpt_3 3st word error check point used in h.264 3st word error check point used in h.264
15:0	RW	0x0000	h264_errchkpt_4 4st word error check point used in h.264 4st word error check point used in h.264

VEPU_swreg_72_reuse

Address: Operational Base + offset (0x00120)

checkpoint word error 1 and 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	h264_errchkpt_5 5st word error check point used in h.264 5st word error check point used in h.264
15:0	RW	0x0000	h264_errchkpt_6 6st word error check point used in h.264 6st word error check point used in h.264

VEPU_swreg_73_reuse

Address: Operational Base + offset (0x00124)

checkpoint delta QP register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	chkqp_1 1st for delta qp check point 1st for delta qp check point
23:20	RW	0x0	chkqp_2 2st for delta qp check point 2st for delta qp check point
19:16	RW	0x0	chkqp_3 3st for delta qp check point 3st for delta qp check point
15:12	RW	0x0	chkqp_4 4st for delta qp check point 4st for delta qp check point

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Bit	Attr	Reset Value	Description
11:8	RW	0x0	chkqp_5 5st for delta qp check point 5st for delta qp check point
7:4	RW	0x0	chkqp_6 6st for delta qp check point 6st for delta qp check point
3:0	RW	0x0	chkqp_7 7st for delta qp check point 7st for delta qp check point

VEPU_swreg_74

Address: Operational Base + offset (0x00128)

input image format

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RW	0x00	mad_thsld MAD threshold value = (MAD threshold)/256
23:16	RW	0x00	encodered_slices the number of encoder slices which used in h.264 the number of encoder slices which used in h.264
15:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x0	img_fmt_in input image format. YUV420P YUV420SP YUV422 UYVY422 RGB565 RGB444 RGB888 RGB101010
3:2	RW	0x0	img_in_rot the input image rotation 0 : no rotation 1 : rotate right 90 degress 2 : rotate left 90 degress
1	RO	0x0	reserved
0	RW	0x0	nal_mode the output of NAL size to base control the output of NAL size to base control

VEPU_swreg_75

Address: Operational Base + offset (0x0012c)

intra/inter mode

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	intramod_16x16 the Intra prediction for 16x16 mode favor

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	intermod the intra/inter selection for inter macro block mode favor the intra/inter selection for inter macro block mode favor

VEPU_swreg_76_reuse

Address: Operational Base + offset (0x00130)

encoder control regsiter 0

Bit	Attr	Reset Value	Description
31:26	RW	0x00	pps_init_qp pps init qp in picture used in h264 pps init qp in picture used in h264 range : 0~51
25:22	RW	0x0	slice_flt_alpha the offset of slice filter alpha c0 used in h264 offset div2 range : -6~6
21:18	RW	0x0	slice_flt_beta the filter beta offset for h264 slice config value = (real value)/2 signed register range : -6 ~6
17:13	RW	0x00	qp_offset_ch the qp index offset for chroma qp used in h264 signed register range : -12~12

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Bit	Attr	Reset Value	Description
12:9	RO	0x0	reserved
8	RW	0x0	sw_qpass jpeg enc quant bypass
7:5	RO	0x0	reserved
4:1	RW	0x0	idr_picid IDR pic ID IDR pic ID
0	RW	0x0	constr_intra_pred constrained intra prediction constrained intra prediction

VEPU_swreg_77

Address: Operational Base + offset (0x00134)

output stream start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	output_strm_st_adr output stream start address output stream start address

VEPU_swreg_78

Address: Operational Base + offset (0x00138)

output control start address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	output_ctrl_st_adr output control start address output control start address

VEPU_swreg_79

Address: Operational Base + offset (0x0013c)

next picture luminance start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	next_luma_st_adr next picture luminance start address next picture luminance start address

VEPU_swreg_80

Address: Operational Base + offset (0x00140)

Base address for MV output

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mv_out_st_adr MV wr start address

VEPU_swreg_81

Address: Operational Base + offset (0x00144)

the cabac table start address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cabac_table_st_adr the cabac table start address H264: cabac table

VEPU_swreg_82

Address: Operational Base + offset (0x00148)

the first of ROI area register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	first_roi_tmb the top mb column for first roi area (inside area)
23:16	RW	0x00	first_roi_bmb the bottom mb column for first roi area (outside area)
15:8	RW	0x00	first_roi_lmb the left mb column for first roi area qp=qp + roi1_Delta_Qp (inside area)
7:0	RW	0x00	first_roi_rmb the right mb column for first roi area qp=qp - roi1_Delta_Qp (outside area)

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VEPU_swreg_83

Address: Operational Base + offset (0x0014c)

the second of ROI area register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	second_roi_rmb (inside area)
23:16	RW	0x00	second_roi_bmb (outside area)
15:8	RW	0x00	second_roi_lmb qp=qp + roi1_Delta_Qp (inside area)
7:0	RW	0x00	second_roi_tmb qp=qp - roi1_Delta_Qp (outside area)

VEPU_swreg_84

Address: Operational Base + offset (0x00150)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix1 the 1st output of Stabilization matrix (position@ up-left)

VEPU_swreg_85

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Address: Operational Base + offset (0x00154)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix2 the 2st output of Stabilization matrix (position @ up)

VEPU_swreg_86

Address: Operational Base + offset (0x00158)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix3 the 3st output of Stabilization matrix (position @up-right)

VEPU_swreg_87

Address: Operational Base + offset (0x0015c)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23:0	RW	0x000000	stab_matrix4 the 4st output of Stabilization matrix (position @ left)

VEPU_swreg_88

Address: Operational Base + offset (0x00160)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix5 the 5st output of Stabilization matrix (position @GMV)

VEPU_swreg_89

Address: Operational Base + offset (0x00164)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix6 the 6st output of Stabilization matrix (position@right)

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VEPU_swreg_90

Address: Operational Base + offset (0x00168)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix7 the 7st output of Stabilization matrix (position@down-left)

VEPU_swreg_91

Address: Operational Base + offset (0x0016c)

Stabilization matrix

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix8 the 8st output of Stabilization matrix (position@down)

VEPU_swreg_92

Address: Operational Base + offset (0x00170)

Stabilization matrix

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:26	RW	0x00	stab_gmv_vrtl the output of Stabilization GMV vertical signed register range : -16~16
25:24	RO	0x0	reserved
23:0	RW	0x000000	stab_matrix9 the 9st output of Stabilization matrix (position@down- right)

VEPU_swreg_93

Address: Operational Base + offset (0x00174)

the output of Stabilization motion sum

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	stab_motion_sum the output of Stabilization motion sum read value = (real value)/8 range : 0~1089*253*255*53/8

VEPU_swreg_94

Address: Operational Base + offset (0x00178)

output of Stabilization

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:8	RW	0x000000	stab_min_value the minimum value output of Stabilization range : 0~255*253*253
7:6	RW	0x0	stab_mod_sel the mode select of Stabilization 0 : disabled 1 : stab only 2 : stab+encode
5:0	RW	0x00	stab_hor_gmv the horizontal output of Stabilization GMV signed register range : -16~16

VEPU_swreg_95

Address: Operational Base + offset (0x0017c)

RGB to YUV conversion coefficient register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	rgb2yuv_coe2 the 2st conversion coefficien for RGB to YUV the 2st conversion coefficien for RGB to YUV
15:0	RW	0x0000	rgb2yuv_coe1 the 1st conversion coefficien for RGB to YUV the 1st conversion coefficien for RGB to YUV

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VEPU_swreg_96

Address: Operational Base + offset (0x00180)

RGB to YUV conversion coefficient register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	rgb2yuv_coe5 the 5st conversion coefficien for RGB to YUV the 5st conversion coefficien for RGB to YUV
15:0	RW	0x0000	rgb2yuv_coe3 the 3st conversion coefficien for RGB to YUV the 3st conversion coefficien for RGB to YUV

VEPU_swreg_97

Address: Operational Base + offset (0x00184)

RGB to YUV conversion coefficient register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	rgb2yuv_coe6 the 6st conversion coefficien for RGB to YUV the 6st conversion coefficien for RGB to YUV

VEPU_swreg_98

Address: Operational Base + offset (0x00188)

RGA MASK

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20:16	RW	0x00	bcmpt_mask_postition the mask msb bit position of rgb B-component range：0~31
15:13	RO	0x0	reserved
12:8	RW	0x00	gcmpt_mask_postition the mask msb bit position of rgb G-component range：0~31
7:5	RO	0x0	reserved
4:0	RW	0x00	rcmpt_mask_postition the mask msb bit position of rgb R-component range：0~31

VEPU_swreg_99

Address: Operational Base + offset (0x0018c)

mv related

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:21	RW	0x000	mv_1p_ply 1p of differential MV penalty differential MV penalty for 1p
20:11	RW	0x000	mv_1p_4p_ply 1p or 4p of differential MV penalty ME. DMVPenaltyQp

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Bit	Attr	Reset Value	Description
10:1	RW	0x000	mv_4p_ply 4p of differential MV penalty
0	RW	0x0	mutimv_en on-off flag for using exceed one mv every mb on-off flag for using exceed one mv every mb

VEPU_swreg_100_reuse

Address: Operational Base + offset (0x00190)

QP register

Bit	Attr	Reset Value	Description
31:26	RW	0x00	h264_init_luma_qp Initial luma qp used in h264 range: 0~51
25:20	RW	0x00	h264_max_qp H.264 Minimum QP range : 0~51
19:14	RW	0x00	h264_min_qp Minimum QP range:0~51
13	RO	0x0	reserved
12:0	RW	0x0000	h264_chkpt_distance checkpoint distance for macro block checkpoint distance for macro block

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VEPU_swreg_101_read

Address: Operational Base + offset (0x00194)

hw config reg

Bit	Attr	Reset Value	Description
31:12	RO	0x1f522	HW_CONFIG Field0000 Description
11:0	RO	0x780	MAX_VID_WIDTH Field0000 Description

VEPU_swreg_102

Address: Operational Base + offset (0x00198)

mvc related

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:20	RW	0x0	mv_favor_16x16 Zero 16x16 MV favor value = (real value)/2.
19:11	RW	0x000	mv_ply_4x4 4x4 Mv Penalty
10:8	RW	0x0	mvc_view_id MVC view_id

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Bit	Attr	Reset Value	Description
7	RW	0x0	mvc_anchor_pic_flag to specifie picture is one part of anchor access unit
6:4	RW	0x0	mvc_priority_id MVC priority_id
3:1	RW	0x0	mvc_temporal_id MVC temporal_id
0	RW	0x0	mvc_inter_view_flag the inter-view prediction of picture MVC inter_view_flag.

VEPU_swreg_103

Address: Operational Base + offset (0x0019c)

encoder start

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:20	RW	0x000	enc_height encoderd height lum height (macroblock unit) H264: [6..255] JPEG: [6..511]
19:17	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
16:8	RW	0x000	enc_width the encoder width lum width (macroblock unit) H264: range : 9~255 JPEG: range : 6~511
7:6	RW	0x0	enc_frame_type frame type selected for current frame 0: INTER 1: INTRA(IDR) 2: MVC-INTER
5:4	RW	0x0	enc_fmt encoding format selected 2 : JPEG 3 : H264
3:1	RO	0x0	reserved
0	RW	0x0	enc_en encoder enable flag encoder enable

VEPU_swreg_104

Address: Operational Base + offset (0x001a0)

mb control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	mb_count_out mb_count_out mb_count_out
15:0	RW	0x0000	mb_cnt macroblock_count macroblock_count

VEPU_swreg_105

Address: Operational Base + offset (0x001a4)

SWAP

Bit	Attr	Reset Value	Description
31	RW	0x0	swap8_in input swap 8-bits or not flag 0: no swap 1: swap 8bit
30	RW	0x0	swap16_in input swap 16-bits or not flag 0: no swap 1: swap 16bit
29	RW	0x0	swap32_in input swap 32-bits or not flag 0: no swap 1: swap 32bit

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Bit	Attr	Reset Value	Description
28	RW	0x0	swap8_out output swap 8-bits or not flag 0: no swap 1: swap 8bit
27	RW	0x0	swap16_out output swap 16-bits or not flag 0: no swap 1: swap 16bit
26	RW	0x0	swap32_out output swap 32-bits or not flag 0: no swap 1: swap 32bit
25	RO	0x0	reserved
24	RW	0x0	test_irq test irq
23:20	RW	0x0	test_counter test counter test counter
19	RW	0x0	coher_test_reg test register coherency test register coherency
18	RW	0x0	coher_test_mem test memory coherency test memory coherency
17:0	RW	0x00000	test_len test data length

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VEPU_swreg_106_reuse

Address: Operational Base + offset (0x001a8)

encoder control register 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	pic_para_id H.264 picture parameter id set
23:16	RW	0x00	intra_pred_mode intra prediction previous fpr 4x4 mode favor used in h264 H.264 intra prediction previous 4x4 mode favor
15:0	RW	0x0000	frame_num H.264 frame number

VEPU_swreg_107_reuse

Address: Operational Base + offset (0x001ac)

JPEG control regsiter

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:20	RW	0x000	mv_ply_16x8_8x16 Penalty for using 16x8 or 8x16 MV. Penalty for using 16x8 or 8x16 MV
19:10	RW	0x000	mv_ply_8x8 Penalty for using 8x8 MV Penalty for using 8x8 MV

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Bit	Attr	Reset Value	Description
9:0	RW	0x000	mv_ply_8x4_4x8 Penalty for using 8x4 or 4x8 MV. Penalty for using 8x4 or 4x8 MV.

VEPU_swreg_108_reuse

Address: Operational Base + offset (0x001b0)

intra_slice_bmp2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	intra_slice_bmp2 Field0000 Abstract bit0 : slices64 bit1 : slices65 bit2 : slices66 ..... bit31 : slices95

VEPU_swreg_109

Address: Operational Base + offset (0x001b4)

encoder status

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	int_non Field0000 Abstract Field0000 Description

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Bit	Attr	Reset Value	Description
27:25	RO	0x0	reserved
24	RW	0x0	mv_sad_wren the each MB MV and SAD be writed to mv_wr_st_adr enable
23:21	RO	0x0	reserved
20	RW	0x0	rocon_write_dis write reconstructed image disable flag
19:17	RO	0x0	reserved
16	RW	0x0	slice_rdyint_en enable slice ready interruupt enable slice ready interruupt
15:13	RO	0x0	reserved
12	RW	0x1	clk_gating_en clock gating enable flag default clk_gating_en =1'b1
11	RO	0x0	reserved
10	RW	0x0	int_timeout_en enable interrupt for timeout
9	RW	0x0	irq_clr irq clear
8	RW	0x0	irq_dis irq disable

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Bit	Attr	Reset Value	Description
7	RO	0x0	reserved
6	RW	0x0	irq_timeout HW wait timeout flag
5	RW	0x0	irq_buffer_full buffer full flag
4	RW	0x0	irq_bus_error bus error irq
3	RW	0x0	fuse_int Field0000 Abstract Field0000 Description
2	RW	0x0	irq_slice_ready slice ready flag
1	RW	0x0	irq_frame_rdy one frame encoder sucess flag
0	RW	0x0	enc_irq enc interrupt

VEPU_swreg_110_read

Address: Operational Base + offset (0x001b8)

product ID

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Bit	Attr	Reset Value	Description
31:16	RO	0x4831	prod_id Product ID
15:12	RO	0x1	major_num Major number
11:4	RO	0x22	minor_num Minor number
3:0	RO	0x0	synthesis

VEPU_swreg_120_183

Address: Operational Base + offset (0x001e0)

DMV_4p_1p_penalty

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	dmv_ply_table DMV 4p/1p penalty table values addr range : 0x01e0~0x02dc swreg120: DMV 4p/1p penalty table values swreg121: DMV 4p/1p penalty table values swreg122: DMV 4p/1p penalty table values swreg123: DMV 4p/1p penalty table values ..... swreg183: DMV 4p/1p penalty table values

15.5.18 VEPU121 MMU Register Summary

Name	Offset	Size	Reset Value	Description

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Name	Offset	Size	Reset Value	Description
VCODEC_MMU_DTE_ADDR	0x00000	W	0x00000000	MMU current page Table address
VCODEC_MMU_STATUS	0x00004	W	0x00000018	MMU status register
VCODEC_MMU_COMMAND	0x00008	W	0x00000000	MMU command register
VCODEC_MMU_PAGE_FAULT_ADDR	0x0000c	W	0x00000000	MMU logical address of last page fault
VCODEC_MMU_ZAP_ONE_LINE	0x00010	W	0x00000000	MMU Zap cache line register
VCODEC_MMU_INT_RAWSTAT	0x00014	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_CLEAR	0x00018	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_MASK	0x0001c	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_INT_STATUS	0x00020	W	0x00000000	MMU raw interrupt status register
VCODEC_MMU_AUTO_GATING	0x00024	W	0x00000001	mmu auto gating

*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

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15.5.19 VEPU121 MMU Detail Register Description

VCODEC_MMU_DTE_ADDR

Address: Operational Base + offset (0x00000)

MMU current page Table address

Bit	Attr	Reset Value	Description

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR MMU_DTE_ADDR MMU current page Table address

VCODEC_MMU_STATUS

Address: Operational Base + offset (0x00004)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID page fault bus id Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE page fault acess The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x1	REPLAY_BUFFER_EMPTY replay buffer empty status 1'b1:The MMU replay buffer is empty
3	RO	0x1	MMU_IDLE mmu idle status The MMU is idle when accesses are being translated and there are no unfinished translated accesses. 1'b1: MMU is idle

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Bit	Attr	Reset Value	Description
2	RO	0x0	STAIL_ACTIVE stall active status MMU stall mode currently enabled. The mode is enabled by command 1'b1: MMU is in stall active status
1	RO	0x0	PAGE_FAULT_ACTIVE page fault active status MMU page fault mode currently enabled . The mode is enabled by command. 1'b1: page fault is active
0	RO	0x0	PAGING_ENABLED Paging enabled status 1'b0: paging is disabled 1'b1: Paging is enabled

VCODEC_MMU_COMMAND

Address: Operational Base + offset (0x00008)

MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	WO	0x0	MMU_CMD mmu cmd MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

VCODEC_MMU_PAGE_FAULT_ADDR

Address: Operational Base + offset (0x0000c)

MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR page fault addr address of last page fault

VCODEC_MMU_ZAP_ONE_LINE

Address: Operational Base + offset (0x00010)

MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE zap one line address to be invalidated from the page table cache

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VCODEC_MMU_INT_RAWSTAT

Address: Operational Base + offset (0x00014)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error read bus error status
0	RW	0x0	PAGE_FAULT page fault page fault status

VCODEC_MMU_INT_CLEAR

Address: Operational Base + offset (0x00018)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR read bus error write 1 to clear read bus error
0	WO	0x0	PAGE_FAULT page fault clear write 1 to page fault clear

VCODEC_MMU_INT_MASK

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Address: Operational Base + offset (0x0001c)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error enable the read bus interrupt souce when this bit is set to 1'b1
0	RW	0x0	PAGE_FAULT page fault mask enable the page fault interrupt souce when this bit is set to 1'b1

VCODEC_MMU_INT_STATUS

Address: Operational Base + offset (0x00020)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR read bus error status 1'b1:read bus error status
0	RO	0x0	PAGE_FAULT page fault status 1'b1:page fault

VCODEC_MMU_AUTO_GATING

Address: Operational Base + offset (0x00024)

mmu auto gating

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_clkgating mmu_auto_clkgating when it is 1'b1, the mmu will auto gating it self

15.5.20 VEPU220 Register Summary

Name	Offset	Size	Reset Value	Description
hevc_enc_swreg0_vpu_po_conf	0x00000	W	0x00000000	vpu_po_conf
hevc_enc_swreg1_vcpu_cur_pc	0x00004	W	0x00000000	vcpu_cur_pc
hevc_enc_swreg3_vpu_pdbg_step_mask	0x0000c	W	0x00000000	Debugger control
hevc_enc_swreg4_vpu_pdbg_ctrl	0x00010	W	0x00000000	vpu_pdbg_ctrl
hevc_enc_swreg5_vpu_pdbg_idx_reg	0x00014	W	0x00000000	vpu pdbg idx reg
hevc_enc_swreg6_vpu_pdbg_wdate_reg	0x00018	W	0x00000000	vpu pdbg wdata(write only)
hevc_enc_swreg7_vpu_pdbg_rdata_reg	0x0001c	W	0x00000000	read data register (read only)
hevc_enc_swreg8_vpu_fio_ctrl_addr	0x00020	W	0x00000000	vpu fio ctrl addr
hevc_enc_swreg9_vpu_fio_data	0x00024	W	0x00000000	vpu fio data
hevc_enc_swreg12_vpu_vint_reason_usr	0x00030	W	0x00000000	vpu init reason usr
hevc_enc_swreg13_vpu_vint_reason_clr	0x00034	W	0x00000000	Interrupt Reason Clear
hevc_enc_swreg14_vpu_host_int_req	0x00038	W	0x00000000	Interrupt request sent from host processor to VPU for the comma
hevc_enc_swreg15_vpu_vint_clear	0x0003c	W	0x00000000	VPU interrupt clear
hevc_enc_swreg16_vpu_hint_clear	0x00040	W	0x00000000	Host interrupt clear
hevc_enc_swreg17_vpu_int_sts	0x00044	W	0x00000000	Interrupt status

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg18_vpu_vint_enable	0x00048	W	0x00000000	Interrupt Enable
hevc_enc_swreg19_vpu_vint_reason	0x0004c	W	0x00000000	VPU interrupt reason
hevc_enc_swreg20_vpu_reset_req	0x00050	W	0x00000000	VPU reset request for each block of clock domain
hevc_enc_swreg21_vpu_reset_status	0x00054	W	0x00000000	Reset status for each clock domain
hevc_enc_swreg22_vpu_restart	0x00058	W	0x00000000	V-CPU restart request
hevc_enc_swreg23_vpu_clk_mask	0x0005c	W	0x00000000	Register0000 Abstract
hevc_enc_swreg24_vpu_remap_ctrl	0x00060	W	0x00000000	Vpu Remap Ctrol
hevc_enc_swreg25_vpu_remap_vaddr	0x00064	W	0x00000000	Remap region base address in virtual address space
hevc_enc_swreg26_vpu_remap_paddr	0x00068	W	0x00000000	Remap region base address in physical address space
hevc_enc_swreg27_vpu_remap_core_start	0x0006c	W	0x00000000	VPU Start Request
hevc_enc_swreg28_vpu_busy_status	0x00070	W	0x00000000	VPU Busy Status
hevc_enc_swreg29_vpu_halt_status	0x00074	W	0x00000000	Reserved for Report VPU status (not implemented yet)
hevc_enc_swreg30_vpu_vcore_parse_status	0x00078	W	0x00000000	Reserved for VPU status reproting (Not implemented yet)
hevc_enc_swreg31_vpu_vcore_dec_status	0x0007c	W	0x00000000	Reserved for VPU status reproting (Not implemented yet)
hevc_enc_swreg64_command	0x00100	W	0x00000000	Command to run VPU
hevc_enc_swreg65_core_index	0x00104	W	0x00000000	V-CORE Index
hevc_enc_swreg66_inst_index_cod_std	0x00108	W	0x00000000	Codec Standard and Instance Index
hevc_enc_swreg67_cmd_enc_set_param_option	0x0010c	W	0x00000000	Host should set this as one of the following while using SET_PA

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg67_init_option	0x0010c	W	0x00000000	Reserved
hevc_enc_swreg67_sfb_option	0x0010c	W	0x00000000	Set Frame Buffer option
hevc_enc_swreg68_ret_success	0x00110	W	0x00000000	Result of the run command
hevc_enc_swreg69_ret_fail_reason	0x00114	W	0x00000000	Fail reason of the run command
hevc_enc_swreg70_addr_code_base	0x00118	W	0x00000000	Code Buffer Base Address
hevc_enc_swreg70_ret_fw_version	0x00118	W	0x00000000	get fw version
hevc_enc_swreg71_ret_product_name	0x0011c	W	0x00000000	ret_product_name
hevc_enc_swreg71_code_size	0x0011c	W	0x00000000	Code Buffer Size
hevc_enc_swreg72_ret_product_version	0x00120	W	0x00000000	ret_product_version
hevc_enc_swreg72_bs_start_addr	0x00120	W	0x00000000	Parameter for code buffer
hevc_enc_swreg72_common_pic_info	0x00120	W	0x00000000	Dpb Information
hevc_enc_swreg72_code_param	0x00120	W	0x00000000	Parameter for code buffer
hevc_enc_swreg73_ret_std_def0	0x00124	W	0x00000000	ret_std_def0
hevc_enc_swreg73_pic_size	0x00124	W	0x00000000	pic_size
hevc_enc_swreg73_bs_size	0x00124	W	0x00000000	Bitstream buffer size
hevc_enc_swreg73_hw_option	0x00124	W	0x00000000	VPU hardware options
hevc_enc_swreg73_ret_cur_sp	0x00124	W	0x00000000	Current Stack Pointer
hevc_enc_swreg73_cur_sp	0x00124	W	0x00000000	Current Stack Pointer
hevc_enc_swreg74_set_fb_num	0x00128	W	0x00000000	set fb num
hevc_enc_swreg74_ret_std_def1	0x00128	W	0x00000000	ret_std_def0
hevc_enc_swreg74_bs_param	0x00128	W	0x00000000	Bitstream buffer parameters
hevc_enc_swreg75_ret_codec_std	0x0012c	W	0x00000000	ret_codec_std

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg75_bs_options	0x0012c	W	0x00000000	Bistream buffer option
hevc_enc_swreg76_bs_rd_ptr	0x00130	W	0x00000000	Bistream Buffer Read Pointer
hevc_enc_swreg76_hw_date	0x00130	W	0x00000000	hw_date
hevc_enc_swreg77_timeout_cnt	0x00134	W	0x00000000	Timeout Count(RSVD)
hevc_enc_swreg77_bs_wr_ptr	0x00134	W	0x00000000	Bistream Buffer Write Pointer
hevc_enc_swreg77_ret_conf_revision	0x00134	W	0x00000000	ret_conf_revision
hevc_enc_swreg78_addr_work_base	0x00138	W	0x00000000	Work Buffer Base Address
hevc_enc_swreg79_work_size	0x0013c	W	0x00000000	work_size
hevc_enc_swreg79_common_work_size	0x0013c	W	0x00000000	Work Buffer Size
hevc_enc_swreg80_work_param	0x00140	W	0x00000000	Work Buffer parameter
hevc_enc_swreg81_addr_temp_base	0x00144	W	0x00000000	Temporal Buffer Base Address
hevc_enc_swreg82_temp_size	0x00148	W	0x00000000	Temporal Buffer Size
hevc_enc_swreg83_temp_param	0x0014c	W	0x00000000	Temporal Buffer Parameter
hevc_enc_swreg84_addr_sec_axi	0x00150	W	0x00000000	Secondary AXI Base Address
hevc_enc_swreg85_sec_axi_size	0x00154	W	0x00000000	Seconary AXI Memory Size
hevc_enc_swreg85_fbc_stride	0x00154	W	0x00000000	fbc_stride
hevc_enc_swreg86_use_sec_axi	0x00158	W	0x00000000	Secondary AXI Usage option
hevc_enc_swreg86_addr_sub_sampled_fb_base	0x00158	W	0x00000000	addr_sub_sampled_fb_base
hevc_enc_swreg87_cmd_enc_set_custom_gop_enable	0x0015c	W	0x00000000	Custom GOP setting
hevc_enc_swreg87_cmd_enc_vui_param_flags	0x0015c	W	0x00000000	vui flags

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg87_sub_sampled_one_fb_size	0x0015c	W	0x00000000	sub_sampled_one_fb_size
hevc_enc_swreg87_cmd_enc_addr_report_base	0x0015c	W	0x00000000	addr report base
hevc_enc_swreg87_cmd_enc_set_param_enable	0x0015c	W	0x00000000	Encoder Parameter Change Setting
hevc_enc_swreg88_cmd_enc_report_size	0x00160	W	0x00000000	addr report size
hevc_enc_swreg88_cmd_enc_vui_aspect_ratio_idc	0x00160	W	0x00000000	enc vui aspect ratio idc
hevc_enc_swreg88_addr_luma_base0	0x00160	W	0x00000000	addr_luma_base0
hevc_enc_swreg88_cmd_enc_seq_src_size	0x00160	W	0x00000000	A size of Source Picture
hevc_enc_swreg88_cmd_enc_custom_gop_param	0x00160	W	0x00000000	custom gop param
hevc_enc_swreg89_cmd_enc_vui_sar_size	0x00164	W	0x00000000	A size of vui sar
hevc_enc_swreg89_cmd_enc_custom_gop_pic_param_0	0x00164	W	0x00000000	custom gop pic param0
hevc_enc_swreg89_cmd_enc_report_param	0x00164	W	0x00000000	report param
hevc_enc_swreg89_addr_cb_base0	0x00164	W	0x00000000	addr_cb_base0
hevc_enc_swreg90_cmd_enc_code_option	0x00168	W	0x00000000	code option
hevc_enc_swreg90_addr_cr_base0	0x00168	W	0x00000000	addr_cr_base0
hevc_enc_swreg90_addr_fbc_y_offset0	0x00168	W	0x00000000	addr_fbc_y_offset0
hevc_enc_swreg90_cmd_enc_custom_gop_pic_param_1	0x00168	W	0x00000000	enc_custom_gop_pic_param_1

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg90_cmd_enc_vui_overscan_appropriate	0x00168	W	0x00000000	vui_overscan_appropriate
hevc_enc_swreg91_cmd_enc_custom_gop_pic_param_2	0x0016c	W	0x00000000	custom gop param2
hevc_enc_swreg91_cmd_enc_pic_parm	0x0016c	W	0x00000000	enc_pic_parm
hevc_enc_swreg91_fbc_c_offset0	0x0016c	W	0x00000000	fbc_c_offset0
hevc_enc_swreg91_cmd_enc_vui_video_signal	0x0016c	W	0x00000000	vui video signal
hevc_enc_swreg91_cmd_enc_seq_param	0x0016c	W	0x00000000	It enables to set SEQ_PARAM
hevc_enc_swreg92_cmd_enc_src_pic_idx	0x00170	W	0x00000000	enc_src_pic_idx
hevc_enc_swreg92_cmd_enc_seq_gop_param	0x00170	W	0x00000000	cmd encode seq_gop_param
hevc_enc_swreg92_addr_luma_base1	0x00170	W	0x00000000	addr_luma_base1
hevc_enc_swreg92_cmd_enc_custom_gop_pic_param_3	0x00170	W	0x00000000	custom gop param3
hevc_enc_swreg92_cmd_enc_vui_chroma_sample_loc	0x00170	W	0x00000000	vui_chroma_sample_loc
hevc_enc_swreg93_cmd_enc_custom_gop_pic_param_4	0x00174	W	0x00000000	custom gop param
hevc_enc_swreg93_cmd_enc_vui_disp_win_left_right	0x00174	W	0x00000000	vui_disp_win_left_right field
hevc_enc_swreg93_addr_cb_base1	0x00174	W	0x00000000	addr_cb_base1
hevc_enc_swreg93_cmd_enc_src_addr_y	0x00174	W	0x00000000	enc_src_addr_y
hevc_enc_swreg93_cmd_enc_seq_intra_param	0x00174	W	0x00000000	enable seq_intra_param
hevc_enc_swreg94_addr_cr_base1	0x00178	W	0x00000000	addr_cr_base1

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg94_cmd_enc_seq_conf_win_top_bot	0x00178	W	0x00000000	It enables cmd enc_seq_config_win_top_bot
hevc_enc_swreg94_cmd_enc_custom_gop_pic_param_5	0x00178	W	0x00000000	custom gop param5
hevc_enc_swreg94_cmd_enc_src_addr_u	0x00178	W	0x00000000	enc_src_addr_u
hevc_enc_swreg95_cmd_enc_custom_gop_pic_param_6	0x0017c	W	0x00000000	custom gop param6
hevc_enc_swreg95_cmd_seq_conf_win_left_right	0x0017c	W	0x00000000	cmd SEQ_CONF_WIN_RIGHT
hevc_enc_swreg95_cmd_enc_src_addr_v	0x0017c	W	0x00000000	enc_src_addr_v
hevc_enc_swreg95_addr_fbc_c_offset1	0x0017c	W	0x00000000	FBC chroma offset base of index1
hevc_enc_swreg96_cmd_enc_seq_frame_rate	0x00180	W	0x00000000	ENC_SEQ_FRAME_RATE
hevc_enc_swreg96_cmd_enc_custom_gop_pic_param_7	0x00180	W	0x00000000	custom gop param7
hevc_enc_swreg96_cmd_enc_src_stride	0x00180	W	0x00000000	enc_src_stride
hevc_enc_swreg96_addr_luma_base2	0x00180	W	0x00000000	luma base of index2
hevc_enc_swreg97_addr_cb_base2	0x00184	W	0x00000000	Cb base of index2
hevc_enc_swreg97_cmd_enc_src_format	0x00184	W	0x00000000	enc_src_format
hevc_enc_swreg97_cmd_enc_seq_independent_slice	0x00184	W	0x00000000	CMD_ENC_SEQ_INDEPENDENT_SLICE
hevc_enc_swreg98_cmd_enc_custom_gop_pic_lambda_0	0x00188	W	0x00000000	custom gop lambda0
hevc_enc_swreg98_addr_cr_base2	0x00188	W	0x00000000	Cr base of index2
hevc_enc_swreg98_cmd_enc_seq_dependent_slice	0x00188	W	0x00000000	cmd_enc_seq_dependent_slice

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg98_cmd_enc_prefix_sei_nal_addr	0x00188	W	0x00000000	enc_prefix_sei_nal_addr
hevc_enc_swreg99_addr_fbc_c_offset2	0x0018c	W	0x00000000	FBC chroma offset base of index2
hevc_enc_swreg99_cmd_enc_prefix_sei_info	0x0018c	W	0x00000000	enc_prefix_sei_info
hevc_enc_swreg99_cmd_enc_custom_gop_pic_lambda_1	0x0018c	W	0x00000000	custom gop lambda1
hevc_enc_swreg99_cmd_enc_seq_intra_refresh	0x0018c	W	0x00000000	cmd_enc_seq_intra_refresh
hevc_enc_swreg100_cmd_enc_custom_gop_pic_lambda_2	0x00190	W	0x00000000	custom gop lambda2
hevc_enc_swreg100_addr_luma_base3	0x00190	W	0x00000000	Luma base of index3
hevc_enc_swreg100_cmd_enc_suffix_sei_nal_addr	0x00190	W	0x00000000	suffix_sei_nal_addr
hevc_enc_swreg100_cmd_enc_param	0x00190	W	0x00000000	Display flag
hevc_enc_swreg100_cmd_enc_src_timestamp_low	0x00190	W	0x00000000	src_timestamp_low
hevc_enc_swreg101_addr_cb_base3	0x00194	W	0x00000000	Cb base of index3
hevc_enc_swreg101_cmd_enc_suffix_sei_info	0x00194	W	0x00000000	enc_suffix_sei_info
hevc_enc_swreg101_cmd_enc_src_timestamp_high	0x00194	W	0x00000000	src_timestamp_high
hevc_enc_swreg101_cmd_enc_custom_gop_pic_lambda_3	0x00194	W	0x00000000	custom gop lambda3
hevc_enc_swreg101_cmd_enc_reserved	0x00194	W	0x00000000	reserved
hevc_enc_swreg102_cmd_enc_rc_param	0x00198	W	0x00000000	cmd_enc_rc_param

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg102_cmd_enc_custom_gop_pic_lambda_4	0x00198	W	0x00000000	custom gop lambda4
hevc_enc_swreg102_addr_cr_base3	0x00198	W	0x00000000	Cr base of index3
hevc_enc_swreg102_cmd_enc_longterm_pic	0x00198	W	0x00000000	cmd_enc_longterm_pic
hevc_enc_swreg102_addr_fbc_y_offset3	0x00198	W	0x00000000	FBC luma offset base of index3
hevc_enc_swreg103_addr_fbc_c_offset3	0x0019c	W	0x00000000	FBC chroma offset base of index3
hevc_enc_swreg103_cmd_enc_rc_min_max_qp	0x0019c	W	0x00000000	It enables enc_rc_min_max_qp
hevc_enc_swreg103_cmd_enc_custom_gop_pic_lambda_5	0x0019c	W	0x00000000	custom gop lambda5
hevc_enc_swreg104_cmd_enc_custom_gop_pic_lambda_6	0x001a0	W	0x00000000	custom gop lambda6
hevc_enc_swreg104_cmd_enc_rc_bit_ratio_layer_0_3	0x001a0	W	0x00000000	enc_rc_bit_ratio_layer_0_3
hevc_enc_swreg104_addr_luma_base4	0x001a0	W	0x00000000	Luma base of index4
hevc_enc_swreg104_cmd_enc_roi_param	0x001a0	W	0x00000000	enc_roi_param
hevc_enc_swreg105_cmd_enc_custom_gop_pic_lambda_7	0x001a4	W	0x00000000	custom gop lambda7
hevc_enc_swreg105_cmd_enc_roi_map_addr	0x001a4	W	0x00000000	roi_map_addr
hevc_enc_swreg105_addr_cb_base4	0x001a4	W	0x00000000	Cb base of index4
hevc_enc_swreg105_cmd_enc_rc_bit_ratio_layer_4_7	0x001a4	W	0x00000000	cmd_enc_rc_bit_ratio_layer_4_7
hevc_enc_swreg106_addr_cr_base4	0x001a8	W	0x00000000	Cr base of index4
hevc_enc_swreg106_ret_enc_pic_idx	0x001a8	W	0x00000000	enc_pic_ind

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg106_cmd_enc_nr_param	0x001a8	W	0x00000000	cmd enc nr_param
hevc_enc_swreg106_cmd_enc_ctu_mode_map_addr	0x001a8	W	0x00000000	ctu_mode_map_addr
hevc_enc_swreg106_addr_fbc_y_offset4	0x001a8	W	0x00000000	Cr base of index4
hevc_enc_swreg107_cmd_enc_ctu_qp_map_addr	0x001ac	W	0x00000000	enc_ctu_qp_map_addr
hevc_enc_swreg107_addr_fbc_c_offset4	0x001ac	W	0x00000000	FBC Chroma offset base of index4
hevc_enc_swreg107_ret_enc_pic_slice_num	0x001ac	W	0x00000000	pic_slice_num
hevc_enc_swreg107_cmd_enc_nr_weight	0x001ac	W	0x00000000	cmd_enc_nr_weight
hevc_enc_swreg108_ret_enc_skip	0x001b0	W	0x00000000	ret_enc_skip
hevc_enc_swreg108_addr_luma_base5	0x001b0	W	0x00000000	Luma base of index5
hevc_enc_swreg108_cmd_enc_num_units_in_tick	0x001b0	W	0x00000000	cmd_enc_num_units_in_tick
hevc_enc_swreg109_cmd_enc_time_scale	0x001b4	W	0x00000000	cmd_enc_time_scale
hevc_enc_swreg109_addr_cb_base5	0x001b4	W	0x00000000	Cb base of index5
hevc_enc_swreg109_ret_enc_pic_num_intra	0x001b4	W	0x00000000	pic_num_intra
hevc_enc_swreg110_cmd_enc_num_ticks_poc_diff_one	0x001b8	W	0x00000000	cmd_enc_num_ticks_poc_diff_one
hevc_enc_swreg110_ret_enc_pic_num_merge	0x001b8	W	0x00000000	pic_num_merge
hevc_enc_swreg110_addr_cr_base5	0x001b8	W	0x00000000	Cr base of index5
hevc_enc_swreg110_addr_fbc_y_offset5	0x001b8	W	0x00000000	FBC luma offset base of index5

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg111_addr_fbc_c_offset5	0x001bc	W	0x00000000	FBC chroma offset base of index5
hevc_enc_swreg111_cmd_enc_rc_trans_rate	0x001bc	W	0x00000000	cmd_enc_rc_trans_rate
hevc_enc_swreg112_addr_luma_base6	0x001c0	W	0x00000000	Luma base of index6
hevc_enc_swreg112_cmd_enc_rc_target_rate	0x001c0	W	0x00000000	CMD_ENC_RC_TARGET_RATE
hevc_enc_swreg112_ret_enc_pic_num_skip	0x001c0	W	0x00000000	pic_num_skip
hevc_enc_swreg113_addr_cb_base6	0x001c4	W	0x00000000	Cb base of index6
hevc_enc_swreg113_ret_enc_pic_avg_cu_qp	0x001c4	W	0x00000000	avg_cu_qp
hevc_enc_swreg114_addr_fbc_y_offset6	0x001c8	W	0x00000000	FBC luma offset base of index6
hevc_enc_swreg114_ret_enc_pic_byte	0x001c8	W	0x00000000	pic_byte
hevc_enc_swreg114_addr_cr_base6	0x001c8	W	0x00000000	Cr base of index6
hevc_enc_swreg115_ret_enc_pic_idx	0x001cc	W	0x00000000	pic_idx
hevc_enc_swreg115_ret_enc_min_fb_num	0x001cc	W	0x00000000	ret_enc_min_fb_num
hevc_enc_swreg115_addr_fbc_c_offset6	0x001cc	W	0x00000000	FBC chroma offset base of index6
hevc_enc_swreg116_ret_enc_pic_poc	0x001d0	W	0x00000000	pic_poc
hevc_enc_swreg116_ret_enc_nal_info_to_be_encoded	0x001d0	W	0x00000000	user data flag
hevc_enc_swreg116_addr_luma_base7	0x001d0	W	0x00000000	Luma base of index7
hevc_enc_swreg117_addr_cb_base7	0x001d4	W	0x00000000	Cb base of index7
hevc_enc_swreg117_ret_frame_cycle	0x001d4	W	0x00000000	a frame processing cycle
hevc_enc_swreg118_addr_fbc_y_offset7	0x001d8	W	0x00000000	FBC luma offset base of index7

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg118_ret_enc_used_src_idx	0x001d8	W	0x00000000	enc_used_src_idx
hevc_enc_swreg118_ret_min_src_buf_num	0x001d8	W	0x00000000	DPB out flag
hevc_enc_swreg118_addr_cr_base7	0x001d8	W	0x00000000	Cr base of index7
hevc_enc_swreg119_ret_enc_pic_num	0x001dc	W	0x00000000	ret_enc_pic_num
hevc_enc_swreg119_addr_fbc_c_offset7	0x001dc	W	0x00000000	FBC chroma offset base of index7
hevc_enc_swreg120_addr_mv_col0	0x001e0	W	0x00000000	Colocated mv buffer base of index 0
hevc_enc_swreg120_ret_enc_pic_type	0x001e0	W	0x00000000	ret_enc_pic_type
hevc_enc_swreg121_addr_mv_col1	0x001e4	W	0x00000000	Colocated mv buffer base of index 1
hevc_enc_swreg122_addr_mv_col2	0x001e8	W	0x00000000	Colocated mv buffer base of index 2
hevc_enc_swreg123_addr_mv_col3	0x001ec	W	0x00000000	Colocated mv buffer base of index 3
hevc_enc_swreg124_addr_mv_col4	0x001f0	W	0x00000000	Colocated mv buffer base of index 4
hevc_enc_swreg125_addr_mv_col5	0x001f4	W	0x00000000	Colocated mv buffer base of index 5
hevc_enc_swreg126_addr_mv_col6	0x001f8	W	0x00000000	mv buffer base of index 6
hevc_enc_swreg127_addr_mv_col7	0x001fc	W	0x00000000	Colocated mv buffer base of index 7
hevc_enc_swreg128_mmu_DTE_ADDR	0x00200	W	0x00000000	MMU current page Table address
hevc_enc_swreg129_mmu_STATUS	0x00204	W	0x00000000	MMU status register
hevc_enc_swreg130_mmu_COMMAND	0x00208	W	0x00000000	MMU command register
hevc_enc_swreg131_mmu_PAGE_FAULT_ADDR	0x0020c	W	0x00000000	MMU logical address of last page fault
hevc_enc_swreg132_mmu_ZAP_ONE_LINE	0x00210	W	0x00000000	MMU Zap cache line register

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg133_mmu_INT_RAWSTAT	0x00214	W	0x00000000	MMU raw interrupt status register
hevc_enc_swreg134_mmu_INT_CLEAR	0x00218	W	0x00000000	MMU raw interrupt status register
hevc_enc_swreg135_mmu_INT_MASK	0x0021c	W	0x00000000	MMU raw interrupt status register
hevc_enc_swreg136_mmu_INT_STATUS	0x00220	W	0x00000000	MMU raw interrupt status register
hevc_enc_swreg137_mmu_AUTO_GATING	0x00224	W	0x00000000	mmu auto gating
hevc_enc_swreg138_Reserved	0x00228	W	0x00000000	Register0000 Abstract
hevc_enc_swreg139_Reserved	0x0022c	W	0x00000000	Register0001 Abstract
hevc_enc_swreg140_Reserved	0x00230	W	0x00000000	Register0002 Abstract
hevc_enc_swreg141_Reserved	0x00234	W	0x00000000	Register0003 Abstract
hevc_enc_swreg142_Reserved	0x00238	W	0x00000000	Register0004 Abstract
hevc_enc_swreg143_Reserved	0x0023c	W	0x00000000	Register0005 Abstract
hevc_enc_swreg144_AXI_CTRL	0x00240	W	0x00000000	Register0006 Abstract
hevc_enc_swreg145_PERF_LATENCY_CTRL0	0x00244	W	0x00000000	Axi performance latency module contrl register0
hevc_enc_swreg146_PERF_LATENCY_CTRL1	0x00248	W	0x00000000	PERF_LATENCY_CTRL1
hevc_enc_swreg147_PERF_RD_MAX_LATENCY_NUM0	0x0024c	W	0x00000000	Read max latency number
hevc_enc_swreg148_PERF_RD_LATENCY_SAMP_NUM	0x00250	W	0x00000000	The number of bigger than configed threshold value
hevc_enc_swreg149_PERF_RD_LATENCY_ACC_SUM	0x00254	W	0x00000000	Total sample number
hevc_enc_swreg150_PERF_RD_AXI_TOTAL_BYTE	0x00258	W	0x00000000	perf_rd_axi_total_byte

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Name	Offset	Size	Reset Value	Description
hevc_enc_swreg151_PERF_WR_AXI_TOTAL_BYTE	0x0025c	W	0x00000000	perf_wr_axi_total_byte
hevc_enc_swreg152_PERF_WORKING_CNT	0x00260	W	0x00000000	perf_working_cnt

15.5.21 VEPU220 Detail Register Description

hevc_enc_swreg0_vpu_po_conf
Address: Operational Base + offset (0x00000)
vpu_po_conf

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	WO	0x0	USE_PO_CONF USE_PO_CONF(WO) Host processor should set 0 for this field which is required when VPU initialization
2:1	RO	0x0	reserved
0	WO	0x0	DEBUGMODE DEBUGMODE(Write Only) DEBUGMODE

hevc_enc_swreg1_vcpu_cur_pc
Address: Operational Base + offset (0x00004)
vcpu_cur_pc

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	cur_pc cur_pc PC value represents the address of instruction which is executed in V-CPU. (for debugging purpose only)

hevc_enc_swreg3_vpu_pdbg_step_mask
Address: Operational Base + offset (0x0000c)
Debugger control

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	step_mask_enable step_mask_enable It is V-CPU interrupt mask for STEP mode

hevc_enc_swreg4_vpu_pdbg_ctrl
Address: Operational Base + offset (0x00010)

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vpu_pdbg_ctrl

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	WO	0x0	immbrk immediate break Immediate break It forces to stop V-CPU and enter in a debug mode to analysis hang-up situation. V-CPU cannot resume from the break point.
2	WO	0x0	stablebrk stable break Stable break It is an external break request when V-CPU is in stable (breakable) status.
1	WO	0x0	resume resume Resume It resumes from the breakpoint.
0	WO	0x0	step step Step It runs V-CPU in step instruction mode.

hevc_enc_swreg5_vpu_pdbg_idx_reg
Address: Operational Base + offset (0x00014)
vpu pdbg idx reg

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	WO	0x0	rddbg Read Operation Request Read Operation Request RDDBG and WRDBG cannot be 1 at the same time.
8	WO	0x0	wrdbg Write Opration Request Write Operation Request RDDBG and WRDBG cannot be 1 at the same time.
7:0	WO	0x00	dbgidx Debug Index Debug Index Debugger Register Index

hevc_enc_swreg6_vpu_pdbg_wdate_reg
Address: Operational Base + offset (0x00018)

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vpu pdbg wdata(write only)

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	vpu_pdbg_wdata_reg Write Data to Debugger To write data to the debugger, 1. Write some data to this register. 2. Write VCPU_PDBG_IDX_REG with WRDBG as 1 and DBGIDX as this register address. 3. After writing is completed, VPU_PDBG_WDATA_REG will be cleared.

hevc_enc_swreg7_vpu_pdbg_rdata_reg

Address: Operational Base + offset (0x0001c)

read data register (read only)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	vpu_pdbg_rdata_reg Read data to the debugger To read data to the debugger, 1. Write VCPU_PDBG_IDX_REG with RDDBG as 1 and DBGIDX as the register address to be read. 2. Read from the specified register to this register.

hevc_enc_swreg8_vpu_fio_ctrl_addr

Address: Operational Base + offset (0x00020)

vpu fio ctrl addr

Bit	Attr	Reset Value	Description
31	RW	0x0	ready ready for transaction Ready for the transaction When writing, it should be 0.
30:17	RW	0x0000	reserved Field0000 Abstract Field0000 Description
16	WO	0x0	rw_flag Read/Write transaction control Read/Write transaction control 0: read 1: write
15:0	WO	0x0000	fio_addr Fio Addr FIO Address

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hevc_enc_swreg9_vpu_fio_data
Address: Operational Base + offset (0x00024)
vpu fio data

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fio_data fio data When writing, FIO data should be written to this register firstly. When reading, 1. Write data to this register. 2. Check whether READY flag of VPU_FIO_CTRL_ADDR register is 1. 3. When READY flag is asserted, VPU reads data from this VPU_FIO_DATA.

hevc_enc_swreg12_vpu_vint_reason_usr
Address: Operational Base + offset (0x00030)
vpu init reason usr

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	bsempty_intr_user Bitstream empty interrupt Bitstream empty(feeding request) interrupt
14	RW	0x0	cmde_intr_user CREATE_INSTANCE command done interrupt CREATE_INSTANCE command done interrupt
13	RO	0x0	reserved
12	RW	0x0	cmdc_intr_user CHANGE_INST command done interrupt CHANGE_INST command done interrupt
11	RW	0x0	cmdb_intr_user WAKEUP_VPU command done interrupt WAKEUP_VPU command done interrupt
10	RW	0x0	cmda_intr_user SLEEP_VPU command done interrupt SLEEP_VPU command done interrupt
9	RW	0x0	cmd9_intr_user QUERY_DECODER command done interrupt QUERY_DECODER command done interrupt
8	RW	0x0	cmd8_intr_user GET_FW_VERSION command done interrupt GET_FW_VERSION command done interrupt
7:6	RO	0x0	reserved
5	RW	0x0	cmd5_intr_user FLUSH_DECODER command done interrupt FLUSH_DECODER command done interrupt

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Bit	Attr	Reset Value	Description
4	RW	0x0	cmd4_intr_user INIT_VPU command done interrupt SET_FRAMEBUF command done interrupt
3	RW	0x0	cmd3_intr_user DEC_PIC command done interrupt DEC_PIC command done interrupt
2	RW	0x0	cmd2_intr_user FINI_SEQ command done interrupt FINI_SEQ command done interrupt
1	RW	0x0	cmd1_intr_user DEC_PIC_HDR command done interrupt DEC_PIC_HDR command done interrupt
0	RW	0x0	cmd0_intr_user INIT_VPU command done interrupt INIT_VPU command done interrupt

hevc_enc_swreg13_vpu_vint_reason_clr
Address: Operational Base + offset (0x00034)
Interrupt Reason Clear

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	besempty_clr Bitstream empty(feeding request) interrupt clear Bitstream empty(feeding request) interrupt clear
14	RW	0x0	cmde_clr CREATE_INSTANCE command done interrupt clear CREATE_INSTANCE command done interrupt clear
13	RO	0x0	reserved
12	RW	0x0	cmdc_clr CHANGE_INST command done interrupt clear CHANGE_INST command done interrupt clear
11	RW	0x0	cmdb_clr WAKEUP_VPU command done interrupt clear WAKEUP_VPU command done interrupt clear
10	RW	0x0	cmda_clr SLEEP_VPU command done interrupt clear SLEEP_VPU command done interrupt clear

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Bit	Attr	Reset Value	Description
9	RW	0x0	cmd9_clr QUERY_DECODER command done interrupt clear QUERY_DECODER command done interrupt clear
8	RW	0x0	cmd8_clr GET_FW_VERSION command done interrupt clear GET_FW_VERSION command done interrupt clear
7:6	RO	0x0	reserved
5	RW	0x0	cmd5_clr FLUSH_DECODER command done interrupt clear FLUSH_DECODER command done interrupt clear
4	RW	0x0	cmd4_clr SET_FRAMEBUF command done interrupt clear SET_FRAMEBUF command done interrupt clear
3	RW	0x0	cmd3_clr DEC_PIC command done interrupt clear DEC_PIC command done interrupt clear
2	RW	0x0	cmd2_clr FINI_SEQ command done interrupt clear FINI_SEQ command done interrupt clear
1	RW	0x0	cmd1_clr DEC_PIC_HDR command done interrupt clear DEC_PIC_HDR command done interrupt clear
0	RW	0x0	cmd0_clr INIT_VPU command done interrupt clear INIT_VPU command done interrupt clear

hevc_enc_swreg14_vpu_host_int_req
Address: Operational Base + offset (0x00038)
Interrupt request sent from host processor to VPU for the comma

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	hintreq Hint Req If this is set to 1, an interrupt named HOST interrupt is sent to VPU.

hevc_enc_swreg15_vpu_vint_clear

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Address: Operational Base + offset (0x0003c)
VPU interrupt clear

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	vintclr Vint Clear Host processor can clear the VPU interrupt which has been pending through this register. As mentioned in VPU_VINT_REASON_CLR, this register setting definetely come after VPU_INT_REASON setting. Also, setting this VPU_VINT_CLEAR to 1 forces VPU_VPU_INT_STS INT_STS to be cleared automatically.

hevc_enc_swreg16_vpu_hint_clear
Address: Operational Base + offset (0x00040)
Host interrupt clear

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	hintclr Hint Clear VPU can clear the host interrupt which has been pending through this register. When VPU operation for the host interrupt is done, VPU sets this reigster to clear the host interrupt.

hevc_enc_swreg17_vpu_int_sts
Address: Operational Base + offset (0x00044)
Interrupt status

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	vpu_vpu_int_sts VPU Interrupt status which comes from VPU to Host VPU Interrupt status which comes from VPU to Host This register turns 1 if VPU_VINT_REASON is not zero (any bit of VPU_VINT_REASON is on), and it becomes to be clear by setting VPU_VINT_CLEAR register. This is a way of interrupt check by register polling.

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hevc_enc_swreg18_vpu_vint_enable
Address: Operational Base + offset (0x00048)
Interrupt Enable

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	cmdf_en UPDATE_BS command done interrupt enable UPDATE_BS command done interrupt enable
14	RW	0x0	cmde_en CREATE_INSTANCE command done interrupt enable CREATE_INSTANCE command done interrupt enable
13	RO	0x0	reserved
12	RW	0x0	cmdc_en Field0010 Abstract CHANGE_INST command done interrupt enable
11	RW	0x0	cmdb_en WAKEUP_VPU command done interrupt enable WAKEUP_VPU command done interrupt enable
10	RW	0x0	cmda_en SLEEP_VPU command done interrupt enable SLEEP_VPU command done interrupt enable
9	RW	0x0	cmd9_en QUERY_DECODER command done interrupt enable QUERY_DECODER command done interrupt enable
8	RW	0x0	cmd8_en GET_FW_VERSION command done interrupt enable GET_FW_VERSION command done interrupt enable
7:6	RO	0x0	reserved
5	RW	0x0	cmd5_en FLUSH_DECODER command done interrupt enable FLUSH_DECODER command done interrupt enable

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Bit	Attr	Reset Value	Description
4	RW	0x0	cmd4_en SET_FRAMEBUF command done interrupt enable SET_FRAMEBUF command done interrupt enable
3	RW	0x0	cmd3_en DEC_PIC command done interrupt enable DEC_PIC command done interrupt enable
2	RW	0x0	cmd2_en FINI_SEQ command done interrupt enable FINI_SEQ command done interrupt enable
1	RW	0x0	cmd1_en DEC_PIC_HDR command done interrupt enable DEC_PIC_HDR command done interrupt enable
0	RW	0x0	cmd0_en INIT_VPU command done interrupt enable INIT_VPU command done interrupt enable

hevc_enc_swreg19_vpu_vint_reason
Address: Operational Base + offset (0x0004c)
VPU interrupt reason

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	bsempty_intr Bitstream empty Bitstream empty(feeding request)
14	RW	0x0	cmde_intr CREATE_INSTANCE command done CREATE_INSTANCE command done
13	RO	0x0	reserved
12	RW	0x0	cmdc_intr CHANGE_INST command done CHANGE_INST command done
11	RW	0x0	cmdb_intr WAKEUP_VPU command done WAKEUP_VPU command done
10	RW	0x0	cmda_intr SLEEP_VPU command done SLEEP_VPU command done
9	RW	0x0	cmd9_intr QUERY_DECODER command done QUERY_DECODER command done

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Bit	Attr	Reset Value	Description
8	RW	0x0	cmd8_intr GET_FW_VERSION command done GET_FW_VERSION command done
7:6	RO	0x0	reserved
5	RW	0x0	cmd5_intr FLUSH_DECODER command done FLUSH_DECODER command done
4	RW	0x0	cmd4_intr INIT_VPU command done SET_FRAMEBUF command done
3	RW	0x0	cmd3_intr DEC_PIC command done DEC_PIC command done
2	RW	0x0	cmd2_intr FINI_SEQ command done FINI_SEQ command done
1	RW	0x0	cmd1_intr DEC_PIC_HDR command done DEC_PIC_HDR command done
0	RW	0x0	cmd0_intr INIT_VPU command don INIT_VPU command done

hevc_enc_swreg20_vpu_reset_req
Address: Operational Base + offset (0x00050)
VPU reset request for each block of clock domain

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	mrst_req Reset request for MCLK (low-latency processor AXI bus clock for Reset request for MCLK (low-latency processor AXI bus clock for V-CPU) domain
25	RW	0x0	vrst_req Reset rquest for VCLK (V-CPU clock) domain Reset rquest for VCLK (V-CPU clock) domain
24	RW	0x0	varst_req Reset rquest for ACLK (high bandwidth AXI bus clock for V-CPU) Reset rquest for ACLK (high bandwidth AXI bus clock for V-CPU) domain

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	arst_req Reset request for ACLK (high bandwidth AXI bus clock for each V Reset request for ACLK (high bandwidth AXI bus clock for each VCORE) domain [23] : V-CORE7 [22] : V-CORE6 [21] : V-CORE5 [20] : V-CORE4 [19] : V-CORE3 [18] : V-CORE2 [17] : V-CORE1 [16] : V-CORE0 0x0 [15:8] BRST_REQ RW Reset request for BCLK
15:8	RW	0x00	brst_req Reset request for BCLK (BPU clock for each V-CORE) domain Reset request for BCLK (BPU clock for each V-CORE) domain [15] : V-CORE7 [14] : V-CORE6 [13] : V-CORE5 [12] : V-CORE4 [11] : V-CORE3 [10] : V-CORE2 [9] : V-CORE1 [8] : V-CORE0
7:0	RW	0x00	crst_req Reset request for CCLK (VCE clock for each V-CORE) domain Reset request for CCLK (VCE clock for each V-CORE) domain [7] : V-CORE7 [6] : V-CORE6 [5] : V-CORE5 [4] : V-CORE4 [3] : V-CORE3 [2] : V-CORE2 [1] : V-CORE1 [0] : V-CORE0

hevc_enc_swreg21_vpu_reset_status

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Address: Operational Base + offset (0x00054)
Reset status for each clock domain

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RO	0x0	mrst_sts MCLK domain reset status MCLK domain reset status
25	RO	0x0	vrst_sts VCLK domain reset status VCLK domain reset status
24	RO	0x0	varst_sts ACLK domain (for vCPU) reset status ACLK domain (for vCPU) reset status
23:16	RO	0x00	arst_sts ACLK domain (for each vCORE) reset status ACLK domain (for each vCORE) reset status
15:8	RO	0x00	brst_sts BCLK domain (for each vCORE) reset status BCLK domain (for each vCORE) reset status
7:0	RO	0x00	crst_sts CCLK domain (for each vCORE) reset status CCLK domain (for each vCORE) reset status

hevc_enc_swreg22_vpu_restart
Address: Operational Base + offset (0x00058)
V-CPU restart request

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	WO	0x0	vcpu_restart This register restarts V-CPU from the reset vector without clearin This register restarts V-CPU from the reset vector without clearing H/W logic. 0: DO NOTHING 1: RESTART REQUEST

hevc_enc_swreg23_vpu_clk_mask
Address: Operational Base + offset (0x0005c)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	mclk_en MCLK domain gating MCLK domain gating

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Bit	Attr	Reset Value	Description
25	RW	0x0	vclk_en VCLK domain gating VCLK domain gating
24	RW	0x0	aclk_cpu_en ACLK domain (for V-CPU) gating ACLK domain (for V-CPU) gating
23:16	RW	0x00	aclk_en ACLK domain (for each V-CORE) gating ACLK domain (for each V-CORE) gating
15:8	RW	0x00	bclk_en BCLK domain (for each V-CORE) gating BCLK domain (for each V-CORE) gating
7:0	RW	0x00	cclk_en CCLK domain (for each V-CORE) gating CCLK domain (for each V-CORE) gating

hevc_enc_swreg24_vpu_remap_ctrl
Address: Operational Base + offset (0x00060)
Vpu Remap Ctrol

Bit	Attr	Reset Value	Description
31	RW	0x0	remap_globen remap global configuration enable Remap global configuration enable [23:16] is valid only if this flag is 1. Thus, global configurations such as AXIID_PROC and ENDIAN is valid only if REMAP_GLOBEN is 1. If not, the value in the fields are ignored.
30:24	RO	0x0	reserved
23:20	RW	0x0	axiid_proc Upper AXI-ID for processor bus to distinguish guest Upper AXI-ID for processor bus to distinguish guest OS For virtualization only. Use this value from higher bit of the 4 bits.
19:16	RW	0x0	endian Endianness for memory access Endianness for memory access Endian control for processor memory transaction (except stream)

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Bit	Attr	Reset Value	Description
15:12	RW	0x0	remap_idx Remap index Remap index Only 0 to 3 are valid.
11	RW	0x0	remap_page_size_en Enable to update a Remap Page Size. Enable to update a Remap Page Size. In other words, Remap Page Size can change only if this bit is 1.
10	RW	0x0	remap_attr Region Attribute - Bypass Region Attribute - Bypass 0: Normal 1: Bypass region
9	RW	0x0	remap_berr Region Attribute - Bus Error Region Attribute - Bus Error 0: Normal 1: Make Bus Error for the region
8:0	RW	0x000	remap_psize Remap Page Size (page base should be aligned in 4K Remap Page Size (page base should be aligned in 4K region) 0x001: 4K 0x002: 8K 0x004: 16K 0x008: 32K 0x010 : 64K 0x020 : 128K 0x040 : 256K 0x080: 512K 0x100: 1M

hevc_enc_swreg25_vpu_remap_vaddr
Address: Operational Base + offset (0x00064)
Remap region base address in virtual address space

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:12	RW	0x00000	vpu_remap_vaddr Virtual address space is an address generated by V-CPU. Virtual address space is an address generated by V-CPU. Section address should be aligned to 4KB boundary. CAUTION> In case of CODE section (which has REMAP IDX 0), virtual address for the section should be 0x0, since V-CPU always start booting up from virtual address 0. If not, VPU can not boot-up.
11:0	RO	0x0	reserved

hevc_enc_swreg26_vpu_remap_paddr
Address: Operational Base + offset (0x00068)
Remap region base address in physical address space

Bit	Attr	Reset Value	Description
31:12	RW	0x00000	vpu_remap_paddr Real address(physical address) as a pair of virtual address. Real address(physical address) as a pair of virtual address. It should aligned to 4KB boundary.
11:0	RO	0x0	reserved

hevc_enc_swreg27_vpu_remap_core_start
Address: Operational Base + offset (0x0006c)
VPU Start Request

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	vpu_remap_core_start It starts VPU after initial setting has been done. It starts VPU after initial setting has been done. After setting up remap or initial configuartion, host should set this register to init VPU

hevc_enc_swreg28_vpu_busy_status
Address: Operational Base + offset (0x00070)
VPU Busy Status

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	vpu_busy_status Vpu Busy Status Command status check - command reentrance check This flag should be set as 1 just before send a command. If this flag is 1, another command can not sent to VPU. This flag is cleared by VPU.

hevc_enc_swreg29_vpu_halt_status
Address: Operational Base + offset (0x00074)
Reserved for Report VPU status (not implemented yet)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RO	0x00	vpu_halt_status For power control / status checking of V-CPU For power control / status checking of V-CPU Remappable status

hevc_enc_swreg30_vpu_vcore_parse_status
Address: Operational Base + offset (0x00078)
Reserved for VPU status reproting (Not implemented yet)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	vpu_vcore_parse_status V-CORE pasring status - busy V-CORE pasring status - busy

hevc_enc_swreg31_vpu_vcore_dec_status
Address: Operational Base + offset (0x0007c)
Reserved for VPU status reproting (Not implemented yet)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	vpu_vcore_dec_status V-CORE decoding status - LF write-back V-CORE decoding status - LF write-back

hevc_enc_swreg64_command
Address: Operational Base + offset (0x00100)
Command to run VPU

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	command command œ x0001 : INIT_VPU Boot up vCPU œ x0002 : SET_PARAM / DEC_PIC_HDR Encode/decode sequence initialize. In encode case, VPU analysis encoding parameter and encode sequence header. In decode case, VPU decode sequence header and report sequence header information. œ x0004 : FINI_SEQ Terminate encoding/decoding of video sequence. œ x0008 : ENC_PIC / DEC_PIC Encode/decode one picture. VPU encodes/decodes one picture. œ x0010 : SET_FRAMEBUF Set decoded/reconstructed frame buffer SDRAM address and maximum frame buffer number. Before decode picture run command, host must inform framebuffer SDRAM address to VPU then BIT processor arrange frame buffer for decoded/reconstructed image and return frame buffer index to host at end of decoding picture. œ x0020 : FLUSH_DECODER Return framebuffer indexes remaining in DPB in display order and initialize the DPB. This command is used when a sequence changes or random access to a sequence is required in the middle of decoding. 0x0100 : GET_FW_VERSION Return thre revision information of the firmware in 32bit integer. œ x0200 : QUERY_DECODER Reserved for future œ x0400 : SLEEP_VPU Save context (all internal variables, which are needed for the recovery) to the working buffer. œ x0800 : WAKEUP_VPU Load the context from the working buffer which have been saved after decoding the previous picture. œ x1000 : CHANGE_INST Reserved for future œ x4000 : CREATE_INSTANCE

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hevc_enc_swreg65_core_index
Address: Operational Base + offset (0x00104)
V-CORE Index

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	vcore_index V-CORE index - reserved for future use, so please set this as 0. V-CORE index - reserved for future use, so please set this as 0.

hevc_enc_swreg66_inst_index_cod_std
Address: Operational Base + offset (0x00108)
Codec Standard and Instance Index

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	codec_std Codec standard to run Codec standard to run HEVC_DEC = 0x0 (currently, only HEVC_DEC is available)
15:0	RW	0x0000	inst_index Instance Index Instance Index VPU can decode more than one decoding instance simultaneously. If multiple instances are running, each instance must have their own process index that is assigned by this register. For example, two instances are running simultaneously, the first instance has the process index 0, the second instance has the process index 1. Instance index shall be in the range of 0 to 65535, inclusive.

hevc_enc_swreg67_cmd_enc_set_param_option
Address: Operational Base + offset (0x0010c)
Host should set this as one of the following while using SET_PA

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	set_param_option Run command status 0: It enables to set COMMON registers. 1: It enables to set CUSTOM_GOP registers. 2: It enables to set SEI registers. 3: It enables to set VUI registers.

hevc_enc_swreg67_init_option
Address: Operational Base + offset (0x0010c)
Reserved

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	init_option Init Option VPU initialization option

hevc_enc_swreg67_sfb_option
Address: Operational Base + offset (0x0010c)
Set Frame Buffer option

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:20	RW	0x00	fbc_mode fbc_mode FBC prediction control 0x00: Best predection (best for bandwidth, some performance overhead) 0x0C: Normal prediction (good for bandwidth and performance) 0x3C: Basic predcition (best for performance) WARNING) Do not set this as 0x3F, which might cause unpredictable operation.
19:16	RW	0x0	fb_endian Framebuffer endianness Framebuffer endianness
15:5	RO	0x0	reserved
4	RW	0x0	setup_done setup_done Setup Done Please set this flag as 1 when setup for frame buffer is done

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Bit	Attr	Reset Value	Description
3	RW	0x0	setup_start setup_start Setup Start Please set this flag as 1 when host issues SET_FRAME_BUF command for the first time. In other words, this is 0 when giving VPU the second SET_FRAME_BUF command for unregistered frame buffers. (only 8 framebuffers possible to register at once)
2:0	RW	0x0	sfb_option sfb_option 0x0 : SET_FRAME_BUF 0x1~0x7 : Reserved

hevc_enc_swreg68_ret_success
Address: Operational Base + offset (0x00110)
Result of the run command

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	run_cmd_status Run command status 00: FAIL 01: SUCCESS 10: SUCCESS_WITH_WARNIG

hevc_enc_swreg69_ret_fail_reason
Address: Operational Base + offset (0x00114)
Fail reason of the run command

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fail_reason fail reason Please refer to Section A.1, "System Error"defining errors that VPU might have.

hevc_enc_swreg70_addr_code_base
Address: Operational Base + offset (0x00118)
Code Buffer Base Address

Bit	Attr	Reset Value	Description
31:12	RW	0x00000	code_buf_base Code Buf Base Base address of V-CPU micro code buffer It should be aligned to 4KB range.

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Bit	Attr	Reset Value	Description
11:0	RO	0x0	reserved

hevc_enc_swreg70_ret_fw_version
Address: Operational Base + offset (0x00118)
get fw version

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	ret_fw_version ret_fw_version Firmware version (internal use only)

hevc_enc_swreg71_ret_product_name
Address: Operational Base + offset (0x0011c)
ret_product_name

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RO	0x0	hw_name hwc_name VPU hardware product name always returns "WAVE" for WAVE product series.

hevc_enc_swreg71_code_size
Address: Operational Base + offset (0x0011c)
Code Buffer Size

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	code_buf_size Code Buf Size Size of CODE buffer It should be aligned to 4KB range.

hevc_enc_swreg72_ret_product_version
Address: Operational Base + offset (0x00120)
ret_product_version

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RO	0x0	hw_version hw_version VPU hardware product version always returns "4200" for WAVE420 product.

hevc_enc_swreg72_bs_start_addr
Address: Operational Base + offset (0x00120)
Parameter for code buffer

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	bs_start_addr Start Address of Bitstream Buffer (fixed in ring buffer mode) Start Address of Bitstream Buffer (fixed in ring buffer mode) It should be aligned in bus width (128bit/16 byte).
3:0	RO	0x0	reserved

hevc_enc_swreg72_common_pic_info
Address: Operational Base + offset (0x00120)
Dpb Information

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	nv21_enable nv21_enable This field enables NV21 format for BWB output. If this field is 0, BWB output color format is NV12. Otherwise, BWB output color format is NV21. The value of NV21 is valid when both BWB and Cb/Cr interleave are enabled. This field is valid only if WTL_ENABLE = 1
28	RW	0x0	wtl_enable wtl_enable WTL_ENABLE This fields indicates target frame buffers are used for uncompressed linear frame. When this field is 0, target frame buffers are used for CFrame - frame buffer compression.
27:24	RW	0x0	axi_id axi_id AXI_ID to distinguish guest OS For virtualization only. Set this value in highest bit order.

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Bit	Attr	Reset Value	Description
23	RW	0x0	wtl_align_mode wtl_align_mode Pixel Alignment Mode (Not Implemented Yet) 0: place lower position pixel in MSB part in the bus - MSB like pixel alignment. 1: place lower position pixel in LSB part in the bus - LSB like pixel alignment This field is valid only if WTL_ENABLE = 1
22:20	RW	0x0	bwb_output_format bwb_output_format BWB output format 000: 8bit down sampling (1 byte/pixel) 001: A pixel is stored in 16 bit and aligned to MSB part (2 bytes/pixel) 010: 3 pixels are stored in 32 bit and aligned to MSB part (4 bytes/3 pixels) 011: Reserved 100: 8bit down sampling (1 byte/pixel) 101: A pixel is stored in 16 bit and aligned to LSB part (2 bytes/pixel) 110: 3 pixels are stored in 32 bit and aligned to LSB part (4 bytes/3 pixels) 111: Reserved This field is valid only if sequence is MAIN10 profile and bit-depth is bigger than 8. This field is valid only if WTL_ENABLE = 1.
19:17	RW	0x0	dpb_color_format dpb_color_format Color format 000: YUV420 This field is valid only if WTL_ENABLE = 1. This field ALWAYS should have 000.

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Bit	Attr	Reset Value	Description
16	RW	0x0	dpb_cbcr_interleave dpb_cbcr_interleave Cb/Cr interleaved 1 : set linear frame buffer output as Chroma interleaved format (semi-planar format) 0 : set linear frame buffer output as planar format (separated format) This flag is valid only if WTL_ENABLE = 1
15:0	RW	0x0000	dpb_stride dpb_stride Stride for the storing alignment

hevc_enc_swreg72_code_param
Address: Operational Base + offset (0x00120)
Parameter for code buffer

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	code_axiid AXI-ID for the V-CPU part (for virtualization) AXI-ID for the V-CPU part (for virtualization)
3:0	RW	0x0	code_endian Code Endian Endianness

hevc_enc_swreg73_ret_std_def0
Address: Operational Base + offset (0x00124)
ret_std_def0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rsvd Configuration Information #0 (internal use only) Configuration Information #0 (internal use only)

hevc_enc_swreg73_pic_size
Address: Operational Base + offset (0x00124)
pic_size

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	dpb_width dpb_width Dpb Picture Width

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	dpb_height dpb_height Dpb Picture Height

hevc_enc_swreg73_bs_size
Address: Operational Base + offset (0x00124)
Bitstream buffer size

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	bs_buf_size Size of Bistream buffer (fixed in ring buffer mode) Size of Bistream buffer (fixed in ring buffer mode) It should be aligned in bus width (128bit/16byte).
3:0	RO	0x0	reserved

hevc_enc_swreg73_hw_option
Address: Operational Base + offset (0x00124)
VPU hardware options

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	uart_option Reserved for debug UART option Reserved for debug UART option
7:2	RO	0x0	reserved
1	RW	0x0	cache_disable it disables use of cache it disables use of cache
0	RW	0x0	use_debug It uses debug mode It uses debug mode

hevc_enc_swreg73_ret_cur_sp
Address: Operational Base + offset (0x00124)
Current Stack Pointer

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ret_cur_sp ret_cur_sp Current stack pointer to backup When VPU power is going off by SLEEP_VPU command, the stack pointer should be backed up. This register returns the current supervisor stack pointer for backing up.

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hevc_enc_swreg73_cur_sp
Address: Operational Base + offset (0x00124)
Current Stack Pointer

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cur_sp cur_sp It restores the stack pointer back to the previous one that was backedup. When VPU resumes by WAKEUP_VPU command, the stack pointer should be restored.

hevc_enc_swreg74_set_fb_num
Address: Operational Base + offset (0x00128)
set fb num

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	fb_num_start Endianness of bitstream buffer The number of start frame buffer to set using SET_FRAME_BUFFER comamnd. This value is valid only if SFB_OPTION is set as 0 (SET_FRAME_BUF).
7:5	RO	0x0	reserved
4:0	RW	0x00	fb_num_end fb_num_end The number of end frame buffer to set using SET_FRAME_BUFFER comamnd. This value is valid only if SFB_OPTION is set as 0 (SET_FRAME_BUF).

hevc_enc_swreg74_ret_std_def1
Address: Operational Base + offset (0x00128)
ret_std_def0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rsvd Configuration Information #1 (internal use only) Configuration Information #1 (internal use only)

hevc_enc_swreg74_bs_param
Address: Operational Base + offset (0x00128)
Bitstream buffer parameters

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	bs_slice_wr_ptr_interrupt_flag It updates wr_ptr each slice. It updates wr_ptr each slice.
4	RW	0x0	bs_wrap_around_flag A wrap around flag A wrap around flag
3:0	RW	0x0	bs_endian Endianness of bitstream buffer Endianness of bitstream buffer

hevc_enc_swreg75_ret_codec_std
Address: Operational Base + offset (0x0012c)
ret_codec_std

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	codec_std Configuration Information #1 (internal use only) General Configuration Information (internal use only)

hevc_enc_swreg75_bs_options
Address: Operational Base + offset (0x0012c)
Bistream buffer option

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	stream_end This field makes VPU assumes Stream End when there is no stream This field makes VPU assumes Stream End when there is no stream to feed in the buffer.

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Bit	Attr	Reset Value	Description
0	RW	0x0	explict_end Explicit End Explicit End When this field is set to 1, VPU assumes that bitstream buffer has one NAL unit or more for decoding a signle frame - it works based on the BS_NAL_LEVEL_PUMP register on. Then it follows the tasks below. 1. VPU decodes until the end of bitstream buffer (WR_PTR) even if the last 3 bytes are all 0. 2. VPU returns success if it has decoded a frame successfuly. If bitstream is insufficient to complete decoding a frame, VPU performs what it is supposed to do with the specified task in BS_SHORTAGE_OPTION of BS_PARAM. If this flag is 0, 1. VPU decodes to the almost end of bitstream buffer (WR_PTR), but not to some bytes (less than 3). It intentionally does not comsume some a few bytes, because VPU is not sure if the last bytes are the start code of next NAL or they might not fill the offset in the CABAC. 2. VPU returns success if it has decoded a frame successfuly. If bitstream is insufficient to complete decoding a frame, VPU stops decoding and waits for more bitstream to be filled. Then host processor can do either feed bitstream more or set EXPLICIT_END to complete decoding anyhow. Bitstream_empty interrupt is asserted when EXPLITCT_END = 0 or when bitstream buffer is near empty not real empty for seamless decoding. Caution Host processor can set this register any time, but cannot clear during command processing.

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hevc_enc_swreg76_bs_rd_ptr
Address: Operational Base + offset (0x00130)
Bistream Buffer Read Pointer

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rd_ptr rd_ptr Start address of bitstream for handling current command Host processor cannot update this register in the middle of decoding. Updating before start decoding is only allowed. VPU updates this RD_PTR as the scheme described below. When BS_NAL_LEVEL_PUMP of BS_PARAM is 1, VPU updates RET_ADDR_BS_RD_PTR to the end address of the NAL. When BS_NAL_LEVEL_PUMP of BS_PARAM is 0, VPU updates RET_ADDR_BS_RD_PTR as end address of the last NAL for a frame.

hevc_enc_swreg76_hw_date
Address: Operational Base + offset (0x00130)
hw_date

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	hw_date hw_date The date that the hardware has been configured (YYYYmmdd in digit)

hevc_enc_swreg77_timeout_cnt
Address: Operational Base + offset (0x00134)
Timeout Count(RSVD)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	time_out_cnt Time out Count Watchdog timer setting value

hevc_enc_swreg77_bs_wr_ptr
Address: Operational Base + offset (0x00134)
Bistream Buffer Write Pointer

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	wr_ptr End address of bitstream for handling current command End address of bitstream for handling current command Host can update this register anytime. If a bitstream_empty interrupt is asserted, host processor should do either feed more bitstream and update WR_PTR or set EXPLICIT_END to complete decoding anyhow.

hevc_enc_swreg77_ret_conf_revision
Address: Operational Base + offset (0x00134)
ret_conf_revision

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	hw_revisison hw_revision The revision number when the hardware has been configured

hevc_enc_swreg78_addr_work_base
Address: Operational Base + offset (0x00138)
Work Buffer Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	work_buf_base Base address of work buffer for each instance Base address of work buffer for each instance This address should be aligned in 4K boundary.

hevc_enc_swreg79_work_size
Address: Operational Base + offset (0x0013c)
work_size

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x000	work_buf_size work_buf_size Size of work buffer (4K boundary) The size of work buffer should be larger than required minimum work buffer size. This address should be aligned in 4K boundary

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hevc_enc_swreg79_common_work_size
Address: Operational Base + offset (0x0013c)
Work Buffer Size

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	work_buf_size Size of work buffer(4k boundary) Size of work buffer (4K boundary) The size of work buffer should be larger than required minimum work buffer size. This address should be aligned in 4K boundary.

hevc_enc_swreg80_work_param
Address: Operational Base + offset (0x00140)
Work Buffer parameter

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	work_buf_endian Endianness of work buffer Endianness of work buffer

hevc_enc_swreg81_addr_temp_base
Address: Operational Base + offset (0x00144)
Temporal Buffer Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	temp_buf_base Base address of temporal buffer for this frame Base address of temporal buffer for this frame Note Each V-CORE should set this field for its own temporal buffer.

hevc_enc_swreg82_temp_size
Address: Operational Base + offset (0x00148)
Temporal Buffer Size

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	temp_buf_size Temporal buffer size for this frame Temporal buffer size for this frame

hevc_enc_swreg83_temp_param
Address: Operational Base + offset (0x0014c)
Temporal Buffer Parameter

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	temp_buf_endian Endianness of temporal buffer Endianness of temporal buffer Note Each V-CORE should set this register for its own temporal buffer.

hevc_enc_swreg84_addr_sec_axi
Address: Operational Base + offset (0x00150)
Secondary AXI Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sec_axi_base The base address of secondary AXI memory The base address of secondary AXI memory

hevc_enc_swreg85_sec_axi_size
Address: Operational Base + offset (0x00154)
Seconary AXI Memory Size

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sec_axi_mem_size Seconary AXI Memory Size Seconary AXI Memory Size. It is valid only when USE_SEC_AXI is not 0.

hevc_enc_swreg85_fbc_stride
Address: Operational Base + offset (0x00154)
fbc_stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	fbc_y_stride fbc_y_stride A stride of fbc for luma component
15:0	RW	0x0000	fbc_c_stride fbc_c_stride A stride of fbc for chroma component

hevc_enc_swreg86_use_sec_axi
Address: Operational Base + offset (0x00158)
Secondary AXI Usage option

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15	RW	0x0	sec_axi_lf_row Use 2nd AXI temp buffer for Read channel of LF row Use 2nd AXI temp buffer for Read channel of LF row buffer.
14:12	RO	0x0	reserved
11	RW	0x0	sec_axi_rdo_enable Use 2nd AXI temp buffer for Read channel of RDO row Use 2nd AXI temp buffer for Read channel of RDO row buffer.
10:0	RO	0x0	reserved

hevc_enc_swreg86_addr_sub_sampled_fb_base
Address: Operational Base + offset (0x00158)
addr_sub_sampled_fb_base

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_sub_sampled_fb addr_sub_sampled_fb A base address of frame buffer for sub sampled decoded frames

hevc_enc_swreg87_cmd_enc_set_custom_gop_enable
Address: Operational Base + offset (0x0015c)
Custom GOP setting

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	RW	0x0	custom_gop_pic_lambda_7_en It enables to set CUSTOM_GOP_PIC_LAMBDA_7 It enables to set CUSTOM_GOP_PIC_LAMBDA_7
16	RW	0x0	custom_gop_pic_lambda_6_en It enables to set CUSTOM_GOP_PIC_LAMBDA_6. It enables to set CUSTOM_GOP_PIC_LAMBDA_6.
15	RW	0x0	custom_gop_pic_lambda_5_en It enables to set custom_gop_pic_lambda_5_en It enables to set custom_gop_pic_lambda_5_en

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Bit	Attr	Reset Value	Description
14	RW	0x0	custom_gop_pic_lambda_4_en It enables to set custom_gop_pic_lambda_4_en It enables to set custom_gop_pic_lambda_4_en
13	RW	0x0	custom_gop_pic_lambda_3_en It enables to set custom_gop_pic_lambda_3_en It enables to set custom_gop_pic_lambda_3_en
12	RW	0x0	custom_gop_pic_lambda_2_en It enables to setcustom_gop_pic_lambda_2_en It enables to set custom_gop_pic_lambda_2_en
11	RW	0x0	custom_gop_pic_lambda_1_en It enables to set custom_gop_pic_lambda_1_en It enables to set custom_gop_pic_lambda_1_en
10	RW	0x0	custom_gop_pic_lambda_0_en It enables to set custom_gop_pic_lambda_0_en It enables to set custom_gop_pic_lambda_0_en
9	RO	0x0	reserved
8	RW	0x0	custom_gop_pic_param_7_en It enables to set custom_gop_pic_param_7_en It enables to set custom_gop_pic_param_7_en.
7	RW	0x0	custom_gop_pic_param_6_en It enables to set custom_gop_pic_param_6_en It enables to set custom_gop_pic_param_6_en
6	RW	0x0	custom_gop_pic_param_5_en It enables to set custom_gop_pic_param_5_en It enables to set custom_gop_pic_param_5_en

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Bit	Attr	Reset Value	Description
5	RW	0x0	custom_gop_pic_param_4_en It enables to set custom_gop_pic_param_4_en It enables to set custom_gop_pic_param_4_en
4	RW	0x0	custom_gop_pic_param_3_en It enables to set custom_gop_pic_param_3_en It enables to set custom_gop_pic_param_3_en
3	RW	0x0	custom_gop_pic_param_2_en It enables to set custom_gop_pic_param_2_en It enables to set custom_gop_pic_param_2_en
2	RW	0x0	custom_gop_pic_param_1_en It enables to set custom_gop_pic_param_1_en It enables to set custom_gop_pic_param_1_en
1	RW	0x0	custom_gop_pic_param_0_en It enables to set custom_gop_pic_param_0_en It enables to set custom_gop_pic_param_0_en
0	RW	0x0	custom_gop_param_en It enables to set CUSTOM_GOP_PARAM It enables to set CUSTOM_GOP_PARAM

hevc_enc_swreg87_cmd_enc_vui_param_flags
Address: Operational Base + offset (0x0015c)
vui flags

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	bitstream_restriction_flag A flag whether to insert bitstream_restriction_flag syntax A flag whether to insert bitstream_restriction_flag syntax of VUI parameters

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Bit	Attr	Reset Value	Description
8	RW	0x0	default_display_window_flag A flag whether to insert default_display_window_flag syntax A flag whether to insert default_display_window_flag syntax of VUI parameters
7	RW	0x0	chroma_loc_info_present_flag A flag whether to insert chroma_loc_info_present_flag A flag whether to insert chroma_loc_info_present_flag syntax of VUI parameters
6	RW	0x0	colour_description_present_flag A flag whether to insert colour_description_present_flag syntax of VUI parameters A flag whether to insert colour_description_present_flag syntax of VUI parameters
5	RW	0x0	video_signal_type_present_flag It enables to set custom_gop_pic_param_4_en A flag whether to insert video_signal_type_present_flag syntax of VUI parameters
4	RW	0x0	overscan_info_present_flag A flag whether to insert overscan_info_present_flag syntax of VUI parameters A flag whether to insert overscan_info_present_flag syntax of VUI parameters
3	RW	0x0	aspect_ratio_info_present_flag A flag whether to insert aspect_ratio_info_present_flag A flag whether to insert aspect_ratio_info_present_flag syntax of VUI parameters
2:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	neutral_chroma_indication_flag A flag whether to insert neutral_chroma_indication_flag syntax of VUI paramaters A flag whether to insert neutral_chroma_indication_flag syntax of VUI paramaters

hevc_enc_swreg87_sub_sampled_one_fb_size
Address: Operational Base + offset (0x0015c)
sub_sampled_one_fb_size

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sub_sampled_one_fb_size sub_sampled_one_fb_size A size of one sub-sampled decoded frame

hevc_enc_swreg87_cmd_enc_addr_report_base
Address: Operational Base + offset (0x0015c)
addr report base

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr_report_base An address of report buffer An address of report buffer

hevc_enc_swreg87_cmd_enc_set_param_enable
Address: Operational Base + offset (0x0015c)
Encoder Parameter Change Setting

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	enable_set_nr_weight It enables to set NR_WEIGHT It enables to set NR_WEIGHT
25	RW	0x0	enable_set_nr_param It enables to set NR_PARAM It enables to set NR_PARAM
24:23	RO	0x0	reserved
22	RW	0x0	enable_set_rc_target_rate It enables to set RC_TARGET_RATE It enables to set RC_TARGET_RATE
21	RW	0x0	enable_set_rc_trans_rate It enables to set RC_TRANS_RATE It enables to set RC_TRANS_RATE
20	RW	0x0	enable_set_num_ticks_poc_diff_one It enables to set NUM_TICKS_POC_DIFF_ONE It enables to set NUM_TICKS_POC_DIFF_ONE

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Bit	Attr	Reset Value	Description
19	RW	0x0	enable_set_time_scale It enables to set TIME_SCALE It enables to set TIME_SCALE
18	RW	0x0	enable_set_num_units_in_tick It enables to set NUM_UNITS_IN_TICK It enables to set NUM_UNITS_IN_TICK
17:16	RO	0x0	reserved
15	RW	0x0	enable_set_rc_bit_ratio_layer_4_7 It enables to set RC_BIT_RATIO_LAYER_4_7 It enables to set RC_BIT_RATIO_LAYER_4_7
14	RW	0x0	enable_set_rc_bit_ratio_layer_0_3 It enables to set RC_BIT_RATIO_LAYER_0_3 It enables to set RC_BIT_RATIO_LAYER_0_3
13	RW	0x0	enable_set_rc_min_max_qp It enables to set RC_MIN_MAX_QP It enables to set RC_MIN_MAX_QP
12	RW	0x0	enable_set_rc_param It enables to set RC_PARAM It enables to set RC_PARAM
11	RO	0x0	reserved
10	RW	0x0	enable_set_enc_param It enables to set SEQ_ENC_PARAM It enables to set SEQ_ENC_PARAM
9	RW	0x0	enable_set_seq_intra_refresh It enables to set SEQ_INTRA_REFRESH It enables to set SEQ_INTRA_REFRESH.
8	RW	0x0	enable_set_seq_dependent_slice It enables to set SEQ_DEPENDENT_SLICE It enables to set SEQ_DEPENDENT_SLICE.
7	RW	0x0	enable_set_seq_independent_slice It enables to set SEQ_INDEPENDENT_SLICE It enables to set SEQ_INDEPENDENT_SLICE
6	RW	0x0	enable_set_seq_frame_rate It enables to set SEQ_FRAME_RATE It enables to set SEQ_FRAME_RATE
5	RW	0x0	enable_set_seq_conf_win_left_right It enables to set SEQ_CONF_WIN_LEFT_RIGHT It enables to set SEQ_CONF_WIN_LEFT_RIGHT
4	RW	0x0	enable_set_seq_conf_win_top_bot It enables to set SEQ_CONF_WIN_TOP It enables to set SEQ_CONF_WIN_TOP

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Bit	Attr	Reset Value	Description
3	RW	0x0	enable_set_seq_intra_param It enables to set SEQ_INTRA_PARAM It enables to set SEQ_INTRA_PARAM
2	RW	0x0	enable_set_seq_gop_param It enables to set SEQ_GOP_PARAM It enables to set SEQ_GOP_PARAM
1	RW	0x0	enable_set_seq_param It enables to set SEQ_PARAM It enables to set SEQ_PARAM.
0	RW	0x0	enable_set_seq_src_size It enables to set SEQ_SRC_SIZE. It enables to set SEQ_SRC_SIZE.

hevc_enc_swreg88_cmd_enc_report_size
Address: Operational Base + offset (0x00160)
addr report size

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	report_size A reporting size in byte A reporting size in byte

hevc_enc_swreg88_cmd_enc_vui_aspect_ratio_idc
Address: Operational Base + offset (0x00160)
enc vui aspect ratio idc

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	aspect_ratio_idc aspect_ratio_idc aspect_ratio_idc

hevc_enc_swreg88_addr_luma_base0
Address: Operational Base + offset (0x00160)
addr_luma_base0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base0 luma_base0 Luma base address of DPB index 0 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 0.

hevc_enc_swreg88_cmd_enc_seq_src_size
Address: Operational Base + offset (0x00160)
A size of Source Picture

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	src_heicht A height of source picture in pixel (128 - 8192) A height of source picture in pixel (128 - 8192)
15:0	RW	0x0000	src_windth A width of source picture in pixel(256 - 8192) A width of source picture in pixel (256 - 8192)

hevc_enc_swreg88_cmd_enc_custom_gop_param
Address: Operational Base + offset (0x00160)
custom gop param

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	derive_lambda_weight It derives a lambda_weight internally instead of using a lambda weight specified. It derives a lambda_weight internally instead of using a lambda weight specified.
3:0	RW	0x0	custom_gop_size The size of cyclic GOP (1 ~ 8) The size of cyclic GOP (1 ~ 8) This should be the same as the number of CUSTOM_GOP_PIC_PARAM set by host

hevc_enc_swreg89_cmd_enc_vui_sar_size
Address: Operational Base + offset (0x00164)
A size of vui sar

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sar_heicht sar_height (valid when aspect_ratio_idc is equal to 255) sar_height (valid when aspect_ratio_idc is equal to 255)
15:0	RW	0x0000	sar_windth sar_width (valid when aspect_ratio_idc is equal to 255) sar_width (valid when aspect_ratio_idc is equal to 255)

hevc_enc_swreg89_cmd_enc_custom_gop_pic_param_0
Address: Operational Base + offset (0x00164)
custom gop pic param0

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RW	0x0	temporal_id A temporal ID of 0th picture in the custom GOP A temporal ID of 0th picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 0th picture in the custom GOP A poc offset of reference L1 of 0th picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 0th picture int the custom GOP A poc offset of reference L0 of 0th picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 0th picture in the custom GOP A quantization parameter of 0th picture int the custom GOP
5:2	RW	0x0	poc_offset a poc offset of 0th in the custom GOP a poc offset of 0th in the custom GOP
1:0	RW	0x0	pic_type A picture type of 0th picture in the custom GOP A picture type of 0th picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg89_cmd_enc_report_param
Address: Operational Base + offset (0x00164)
report param

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	report_param report param report param

hevc_enc_swreg89_addr_cb_base0
Address: Operational Base + offset (0x00164)
addr_cb_base0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base0 cb_base0 Cb base address of DPB index 0 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 0.

hevc_enc_swreg90_cmd_enc_code_option
Address: Operational Base + offset (0x00168)
code option

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	VUI A flag to encode VUI nal unit explicitly A flag to encode VUI nal unit explicitly
8	RW	0x0	SEI A flag to encode SEI nal unit explicitly A flag to encode SEI nal unit explicitly
7	RW	0x0	EOB A flag to encode EOB nal unit explicitly A flag to encode EOB nal unit explicitly
6	RW	0x0	EOS A flag to encode EOS nal unit explicitly A flag to encode EOS nal unit explicitly
5	RW	0x0	aud A flag to encode AUD nal unit explicitly A flag to encode AUD nal unit explicitly
4	RW	0x0	pps A flag to encode PPS nal unit explicitly A flag to encode PPS nal unit explicitly
3	RW	0x0	sps A flag to encode SPS nal unit explicitly A flag to encode SPS nal unit explicitly
2	RW	0x0	vps A flag to encode VPS nal unit explicitly A flag to encode VPS nal unit explicitly
1	RW	0x0	vcl A flag to encode VCL nal unit explicitly A flag to encode VCL nal unit explicitly
0	RW	0x0	implicitly_header_encode A flag to enable to encode a header or headers A flag to enable to encode a header or headers (VPS, SPS, PPS) implicitly for generating bitstreams conforming to specification

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hevc_enc_swreg90_addr_cr_base0
Address: Operational Base + offset (0x00168)
addr_cr_base0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base0 cr_base0 Cr base address of DPB index 0 unless FBC is used.

hevc_enc_swreg90_addr_fbc_y_offset0
Address: Operational Base + offset (0x00168)
addr_fbc_y_offset0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_luma_offset_base0 fbc_luma_offset_base0 Compressed Luma offset table base address of DPB index 0 when FBC is used

hevc_enc_swreg90_cmd_enc_custom_gop_pic_param_1
Address: Operational Base + offset (0x00168)
enc_custom_gop_pic_param_1

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 1st picture in the custom GOP A temporal ID of 1st picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 1st picture in the custom GOP A poc offset of reference L1 of 1st picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 1st picture int the custom GOP A poc offset of reference L0 of 1st picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 1st picture in the custom GOP A quantization parameter of 1st picture int the custom GOP

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	poc_offset a poc offset of 1st in the custom GOP a poc offset of 1st in the custom GOP
1:0	RW	0x0	pic_type A picture type of 1st picture in the custom GOP A picture type of 1st picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg90_cmd_enc_vui_overscan_appropriate
Address: Operational Base + offset (0x00168)
vui_overscan_appropriate

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	overscan_appropriate_flag overscan_appropriate_flag overscan_appropriate_flag

hevc_enc_swreg91_cmd_enc_custom_gop_pic_param_2
Address: Operational Base + offset (0x0016c)
custom gop param2

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 2nd picture in the custom GOP A temporal ID of 2nd picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 2nd picture in the custom GOP A poc offset of reference L1 of 2nd picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 2nd picture int the custom GOP A poc offset of reference L0 of 2nd picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 2nd picture in the custom GOP A quantization parameter of 2nd picture int the custom GOP

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	poc_offset a poc offset of 2nd in the custom GOP a poc offset of 2nd in the custom GOP
1:0	RW	0x0	pic_type A picture type of 2nd picture in the custom GOP A profile indicator 0 - firmware determines a profile. 1 - main 2 - main10

hevc_enc_swreg91_cmd_enc_pic_parm
Address: Operational Base + offset (0x0016c)
enc_pic_parm

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	force_src_pic_type_flag A flag to use a force picture type by encoded/source index A flag to use a force picture type by encoded/source index
29	RW	0x0	force_src_pic_qp_flag A flag to use a force qp by encoded/source index A flag to use a force qp by encoded/source index
28:24	RO	0x0	reserved
23:21	RW	0x0	force_pic_type A force picture type (I, P, B, IDR, CRA) or anirap type A force picture type (I, P, B, IDR, CRA) or an irap type 0 : I picture 1 : P picture 2 : B picture 3 : IDR picture 4 : CRA picture
20	RW	0x0	use_force_pic_type A flag whether to use a force picture type or anirap type A flag whether to use a force picture type or anirap type

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Bit	Attr	Reset Value	Description
19:14	RW	0x00	force_pic_qp_b A force picture quantization parameter for B picture (0 ~ 51) A force picture quantization parameter for B picture (0 ~ 51)
13:8	RW	0x00	force_pic_qp_p A force picture quantization parameter for P picture (0 ~ 51) A force picture quantization parameter for P picture (0 ~ 51)
7:2	RW	0x00	force_pic_qp_i A force picture quantization parameter for I picture (0 ~ 51) A force picture quantization parameter for I picture (0 ~ 51)
1	RW	0x0	use_force_pic_qp A flag to use a force picture quantization parameter A flag to use a force picture quantization parameter
0	RW	0x0	pic_skip_flag A flag to skip the current picture A flag to skip the current picture

hevc_enc_swreg91_fbc_c_offset0
Address: Operational Base + offset (0x0016c)
fbc_c_offset0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base0 fbc_chroma_offset_base0 If FBC is used, this is chroma offset base address of DPB index 0. Otherwise, it is ignored.

hevc_enc_swreg91_cmd_enc_vui_video_signal
Address: Operational Base + offset (0x0016c)
vui video signal

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:20	RW	0x00	matrix_coeffs matrix_coeffs matrix_coeffs
19:12	RW	0x00	transfer_characteristics transfer_characteristics transfer_characteristics

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Bit	Attr	Reset Value	Description
11:4	RW	0x00	colour_primaries colour_primaries colour_primaries
3	RW	0x0	video_full_range_flag video_full_range_flag video_full_range_flag
2:0	RW	0x0	video_format video_format video_format

hevc_enc_swreg91_cmd_enc_seq_param
Address: Operational Base + offset (0x0016c)
It enables to set SEQ_PARAM

Bit	Attr	Reset Value	Description
31:27	RW	0x00	chroma_cr_qp_offset A value of chroma(Cr) QP offset (-10 ~ 10) A value of chroma(Cr) QP offset (-10 ~ 10)
26:22	RW	0x00	chroma_cb_qp_offset A value of chroma(Cb) QP offset (-10 ~ 10) A value of chroma(Cb) QP offset (-10 ~ 10)
21	RW	0x0	constrained_intra_pred It enables constrained intra prediction. It enables constrained intra prediction.
20	RW	0x0	use_lossless_coding It enables lossless coding It enables lossless coding
19:18	RW	0x0	chroma_format_idc A chroma format indicator (0 - 4:2:0) A chroma format indicator (0 - 4:2:0)
17:14	RW	0x0	bit_depth Bit depth (8 or 10) Bit depth (8 or 10)
13:12	RW	0x0	tier_idc A tier indicator A tier indicator 0 - main 1 - high
11:3	RW	0x000	level_idc A level indicator (level*10) A level indicator (level * 10) 0 - firmware determines a level.

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	profile_idc A profile indicator A profile indicator 0 - firmware determines a profile. 1 - main 2 - main10

hevc_enc_swreg92_cmd_enc_src_pic_idx
Address: Operational Base + offset (0x00170)
enc_src_pic_idx

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_idx A buffer index of a source picture (-2 : no more source picture) A buffer index of a source picture (-2 : no more source picture)

hevc_enc_swreg92_cmd_enc_seq_gop_param
Address: Operational Base + offset (0x00170)
cmd encode seq_gop_param

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	gop_preset_idx A GOP structur option A GOP structure option 0: Custom GOP 1 : I-I-I-I,..I (all intra, gop_size=1) 2 : I-P-P-P,…P (consecutive P, gop_size=1) 6 : I-P-P-P-P,…(consecutive P, gop_size=4)

hevc_enc_swreg92_addr_luma_base1
Address: Operational Base + offset (0x00170)
addr_luma_base1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base1 luma_base1 Luma base address of DPB index 1 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 1.

hevc_enc_swreg92_cmd_enc_custom_gop_pic_param_3
Address: Operational Base + offset (0x00170)
custom gop param3

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 3rd picture in the custom GOP A temporal ID of 3rd picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 3rd picture in the custom GOP A poc offset of reference L1 of 3rd picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 3rd picture int the custom GOP A poc offset of reference L0 of 3rd picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 3rd picture in the custom GOP A quantization parameter of 3rd picture int the custom GOP
5:2	RW	0x0	poc_offset a poc offset of 3rd in the custom GOP a poc offset of 3rd in the custom GOP
1:0	RW	0x0	pic_type A picture type of 3rd picture in the custom GOP A picture type of 3rd picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg92_cmd_enc_vui_chroma_sample_loc
Address: Operational Base + offset (0x00170)
vui_chroma_sample_loc

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	chroma_sample_loc_type_bottom_field chroma_sample_loc_type_bottom_field chroma_sample_loc_type_bottom_field
15:0	RW	0x0000	chroma_sample_loc_type_top_field chroma_sample_loc_type_top_field chroma_sample_loc_type_top_field

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hevc_enc_swreg93_cmd_enc_custom_gop_pic_param_4
Address: Operational Base + offset (0x00174)
custom gop param

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 4th picture in the custom GOP A temporal ID of 4th picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 4th picture in the custom GOP A poc offset of reference L1 of 4th picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 4th picture int the custom GOP A poc offset of reference L0 of 4th picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 4th picture in the custom GOP A quantization parameter of 4th picture int the custom GOP
5:2	RW	0x0	poc_offset a poc offset of 4th in the custom GOP a poc offset of 4th in the custom GOP
1:0	RW	0x0	pic_type A picture type of 4th picture in the custom GOP A picture type of 3rd picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg93_cmd_enc_vui_disp_win_left_right
Address: Operational Base + offset (0x00174)
vui_disp_win_left_right field

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	def_disp_win_right_offset def_disp_win_right_offset def_disp_win_right_offset

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	def_disp_win_left_offset def_disp_win_left_offset def_disp_win_left_offset

hevc_enc_swreg93_addr_cb_base1
Address: Operational Base + offset (0x00174)
addr_cb_base1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base1 cb_base1 Cb base address of DPB index 1 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 1.

hevc_enc_swreg93_cmd_enc_src_addr_y
Address: Operational Base + offset (0x00174)
enc_src_addr_y

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_addr_y An address of Y component of a source picture An address of Y component of a source picture

hevc_enc_swreg93_cmd_enc_seq_intra_param
Address: Operational Base + offset (0x00174)
enable seq_intra_param

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	intra_period A period of intra picture (0 - 1024) A period of intra picture (0 ~ 1024) 0 - implies an infinite period
15:9	RO	0x0	reserved
8:3	RW	0x00	intra_qp A quantization parameter of intra picture (0 - 51) A quantization parameter of intra picture (0 ~ 51)
2:0	RW	0x0	decoding_refresh_type A decoding refresh type of intra picture A decoding refresh type of intra picture 0 - Non-IRAP 1 - CRA 2 - IDR

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hevc_enc_swreg94_addr_cr_base1
Address: Operational Base + offset (0x00178)
addr_cr_base1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base1 cr_base1 Cr base address of DPB index 1 unless FBC is used.

hevc_enc_swreg94_cmd_enc_seq_conf_win_top_bot
Address: Operational Base + offset (0x00178)
It enables cmd enc_seq_config_win_top_bot

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	conformace_window_size_bottom A conformance top window size (0 - 8192) A conformance top window size (0 ~ 8192)
15:0	RW	0x0000	conformace_window_size_top A conformance top window size (0 - 8192) A conformance top window size (0 - 8192)

hevc_enc_swreg94_cmd_enc_custom_gop_pic_param_5
Address: Operational Base + offset (0x00178)
custom gop param5

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 5th picture in the custom GOP A temporal ID of 5th picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 5th picture in the custom GOP A poc offset of reference L1 of 5th picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 5th picture int the custom GOP A poc offset of reference L0 of 5th picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 5th picture in the custom GOP A quantization parameter of 5th picture int the custom GOP

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	poc_offset a poc offset of 5th in the custom GOP a poc offset of 5th in the custom GOP
1:0	RW	0x0	pic_type A picture type of 5th picture in the custom GOP A picture type of 3rd picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg94_cmd_enc_src_addr_u
Address: Operational Base + offset (0x00178)
enc_src_addr_u

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_addr_u An address of U component of a source picture An address of U component of a source picture

hevc_enc_swreg95_cmd_enc_custom_gop_pic_param_6
Address: Operational Base + offset (0x0017c)
custom gop param6

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 6th picture in the custom GOP A temporal ID of 6th picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 6th picture in the custom GOP A poc offset of reference L1 of 6th picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 6th picture int the custom GOP A poc offset of reference L0 of 6th picture int the custom GOP
13:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:6	RW	0x00	pic_qp A quantization parameter of 6th picture in the custom GOP A quantization parameter of 6th picture int the custom GOP
5:2	RW	0x0	poc_offset a poc offset of 6th in the custom GOP a poc offset of 6th in the custom GOP
1:0	RW	0x0	pic_type A picture type of 6th picture in the custom GOP A picture type of 3rd picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg95_cmd_seq_conf_win_left_right
Address: Operational Base + offset (0x0017c)
cmd SEQ_CONF_WIN_RIGHT

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	conformance_window_size_right A conformance rightwindow size (0 A conformance rightwindow size (0 ~ 8192)
15:0	RW	0x0000	conformance_window_size_left A conformance left window size A conformance left window size (0 ~ 8192)

hevc_enc_swreg95_cmd_enc_src_addr_v
Address: Operational Base + offset (0x0017c)
enc_src_addr_v

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_addr_v An address of V component of a source picture An address of V component of a source picture

hevc_enc_swreg95_addr_fbc_c_offset1
Address: Operational Base + offset (0x0017c)
FBC chroma offset base of index1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base1 fbc_chroma_offset_base1 If FBC is used, this is chroma offset base address of DPB index 1. Otherwise, it is ignored

hevc_enc_swreg96_cmd_enc_seq_frame_rate
Address: Operational Base + offset (0x00180)
ENC_SEQ_FRAME_RATE

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	frame_rate_idc A frame rate indicator (x 1024) A frame rate indicator (x 1024)

hevc_enc_swreg96_cmd_enc_custom_gop_pic_param_7
Address: Operational Base + offset (0x00180)
custom gop param7

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:24	RW	0x0	temporal_id A temporal ID of 7th picture in the custom GOP A temporal ID of 7th picture in the custom GOP
23:19	RW	0x00	ref_poc_l1 A poc offset of reference L1 of 7th picture in the custom GOP A poc offset of reference L1 of 7th picture in the custom GOP
18:14	RW	0x00	ref_poc_l0 A poc offset of reference L0 of 7th picture int the custom GOP A poc offset of reference L0 of 7th picture int the custom GOP
13:12	RO	0x0	reserved
11:6	RW	0x00	pic_qp A quantization parameter of 7th picture in the custom GOP A quantization parameter of 7th picture int the custom GOP
5:2	RW	0x0	poc_offset a poc offset of 7th in the custom GOP a poc offset of 7th in the custom GOP

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	pic_type A picture type of 7th picture in the custom GOP A picture type of 3rd picture in the custom GOP 0 : I picture 1 : P picture 2 : B picture

hevc_enc_swreg96_cmd_enc_src_stride
Address: Operational Base + offset (0x00180)
enc_src_stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	src_y_stride A luma stride of source picture in pixel A luma stride of source picture in pixel
15:0	RW	0x0000	src_c_stride A chroma stride of source picture in pixel A chroma stride of source picture in pixel

hevc_enc_swreg96_addr_luma_base2
Address: Operational Base + offset (0x00180)
luma base of index2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base2 luma base2 Luma base address of DPB index 2 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 2.

hevc_enc_swreg97_addr_cb_base2
Address: Operational Base + offset (0x00184)
Cb base of index2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base2 cb_base2 Cb base address of DPB index 2 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 2

hevc_enc_swreg97_cmd_enc_src_format
Address: Operational Base + offset (0x00184)
enc_src_format

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Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:6	RW	0x0	src_endian Endianess of source picture Endianess of source picture
5:3	RW	0x0	src_pixel_format src_pixel_format [2] left justified [1:0] 0 : 8bit 1 : 16bit (1 pixel / 2 byte) 2 : 32bit (3 pixel / 4 byte)
2:0	RW	0x0	src_frame_format src_frame_format 0 : Planar 1 : Tiled Sub-CTU frame map - within a sub-CTU(32x32), the first row of 16x32 is read in vertical direction and then the second row of 16x32 is read. 2 : CbCr interleaved (NV12) 3 : CrCb interleaved (NV21) 4 : Packed mode (YUYV) 5 : Packed mode (YVYU) 6 : Packed mode (UYVY) 7 : Packed mode (VYUY)

hevc_enc_swreg97_cmd_enc_seq_independent_slice
Address: Operational Base + offset (0x00184)
CMD_ENC_SEQ_INDEPENDENT_SLICE

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	argument A size of slice for the slice mode (2^16 - 1) A size of slice for the slice mode (2^16 - 1) The number of CTU for slice mode of 1
15:3	RO	0x0	reserved
2:0	RW	0x0	slice_mode An independent slice mode An independent slice mode 0 - No multi-slice 1 - CTU based slice

hevc_enc_swreg98_cmd_enc_custom_gop_pic_lambda_0
Address: Operational Base + offset (0x00188)
custom gop lambda0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 0th picture in the custom GOP A lamda weight of 0th picture in the custom GOP

hevc_enc_swreg98_addr_cr_base2
Address: Operational Base + offset (0x00188)
Cr base of index2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base2 cr_base2 Cr base address of DPB index 2 unless FBC is used.

hevc_enc_swreg98_cmd_enc_seq_dependent_slice
Address: Operational Base + offset (0x00188)
cmd_enc_seq_dependent_slice

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	argument A size of slice for the slice mode (2^16 - 1) A size of slice for the slice mode (2^16 - 1) The number of CTU for slice mode of 1 The number of byte for slice mode of 2
15:3	RO	0x0	reserved
2:0	RW	0x0	slice_mode A dependent slice mode A dependent slice mode 0 - No multi-slice 1 - CTU based slice 2 - Byte based slice

hevc_enc_swreg98_cmd_enc_prefix_sei_nal_addr
Address: Operational Base + offset (0x00188)
enc_prefix_sei_nal_addr

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	prefix_sei_nal_data_addr The start address of the total prefix SEI NALs to be encoded The start address of the total prefix SEI NALs to be encoded

hevc_enc_swreg99_addr_fbc_c_offset2
Address: Operational Base + offset (0x0018c)
FBC chroma offset base of index2

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base2 fbc_chroma_offset_base2 If FBC is used, this is chroma offset base address of DPB index 2. Otherwise, it is ignored.

hevc_enc_swreg99_cmd_enc_prefix_sei_info
Address: Operational Base + offset (0x0018c)
enc_prefix_sei_info

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	prefix_sei_nal_data_size prefix_sei_nal_data_size The total byte size of the prefix SEI
15:2	RO	0x0	reserved
1	RW	0x0	prefix_sei_timing_flag prefix-sei_timing_flag A flag whether to encode PREFIX_SEI_DATA with a picture of this command or with a source picture of the buffer at the moment 0 : encode PREFIX_SEI_DATA when a source picture is encoded. 1 : encode PREFIX_SEI_DATA at this command.
0	RW	0x0	prefix_sei_nal_data_enable_flag prefix_sei_nal_data_enable_flag It enables to encode the prefix SEI NAL which is given by host.

hevc_enc_swreg99_cmd_enc_custom_gop_pic_lambda_1
Address: Operational Base + offset (0x0018c)
custom gop lambda1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 1st picture in the custom GOP A lamda weight of 1st picture in the custom GOP

hevc_enc_swreg99_cmd_enc_seq_intra_refresh
Address: Operational Base + offset (0x0018c)
cmd_enc_seq_intra_refresh

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	intra_refresh_argument An argument of intra refresh mode (2^16 - 1) An argument of intra refresh mode (2^16 - 1) ?The number of consecutive CTU rows for intra refresh mode of 1 ?The number of consecutive CTU columns for intra refresh mode of 2 ?A step size in CTU for intra refresh mode of 3
15:3	RO	0x0	reserved
2:0	RW	0x0	intra_refresh_mode An intra refresh mode An intra refresh mode 0 - No intra refresh 1 - Row 2 - Column 3 - A step size in CTU

hevc_enc_swreg100_cmd_enc_custom_gop_pic_lambda_2
Address: Operational Base + offset (0x00190)
custom gop lambda2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 2nd picture in the custom GOP A lamda weight of 2nd picture in the custom GOP

hevc_enc_swreg100_addr_luma_base3
Address: Operational Base + offset (0x00190)
Luma base of index3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base3 luma_base3 Luma base address of DPB index 3 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 3.

hevc_enc_swreg100_cmd_enc_suffix_sei_nal_addr
Address: Operational Base + offset (0x00190)
suffix_sei_nal_addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	suffix_sei_nal_data_addr suffix_sei_nal_data_addr The start address of the total suffix SEI NALs to be encoded

hevc_enc_swreg100_cmd_enc_param
Address: Operational Base + offset (0x00190)
Display flag

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	use_intra_in_inter_slice It enables intra CU in inter slice It enables intra CU in inter slice
26:25	RO	0x0	reserved
24:21	RW	0x0	tc_offset_div2 TcOffsetDiv2 for deblocking filter TcOffsetDiv2 for deblocking filter
20:17	RW	0x0	beta_offset_div2 BetaOffsetDiv2 for deblocking filter BetaOffsetDiv2 for deblocking filter
16	RW	0x0	use_lf_cross_slice_boundary It enables the use of in-loop filtering across slice boundaries It enables the use of in-loop filtering across slice boundaries
15	RW	0x0	disable_dbk It disables the in-loop deblocking filter It disables the in-loop deblocking filter
14	RW	0x0	use_dynamic_merge_32x32 It enables 32x32 CUs to be dynamic merge candidates It enables 32x32 CUs to be dynamic merge candidates
13	RW	0x0	use_dynamic_merge_16x16 It enables 16x16 CUs to be dynamic merge candidates. It enables 16x16 CUs to be dynamic merge candidates.
12	RW	0x0	use_dynamic_merge_8x8 It enables 8x8 CUs for dynamic merge candidates It enables 8x8 CUs for dynamic merge candidates.

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Bit	Attr	Reset Value	Description
11:9	RW	0x0	max_num_merge Maximum number of merge candidates(0-2) Maximum number of merge candidates (0 ~ 2)
8	RO	0x0	reserved
7	RW	0x0	use_tmvp It enables temporal motion vectio prediction It enables temporal motion vectio prediction
6:4	RW	0x0	use_cu_size It enables CU size It enables CU size. [0] : 8x8 CU [1] : 16x16 CU [2] : fixed 0
3	RW	0x0	use_scaling_list It enables scaling list. It enables scaling list
2	RW	0x0	enable_ctu_qp_map It enables CTU QP map It enables CTU QP map.
1:0	RW	0x0	use_recommend_enc_param It uses recommended enc params. It uses recommended enc params. 0 : Custom 1 : Recommend enc params 2 ~ 3 : Reserved

hevc_enc_swreg100_cmd_enc_src_timestamp_low
Address: Operational Base + offset (0x00190)
src_timestamp_low

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_timestamp_low src_timestamp_low An MSB of timestamp of a source picture

hevc_enc_swreg101_addr_cb_base3
Address: Operational Base + offset (0x00194)
Cb base of index3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base_fbc_cbcr_base3 cb_base_fbc_cbcr_base3 Cb base address of DPB index 3 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 3

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hevc_enc_swreg101_cmd_enc_suffix_sei_info
Address: Operational Base + offset (0x00194)
enc_suffix_sei_info

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	suffix_sei_nal_data_size suffix_sei_nal_data_size The total byte size of the suffix SEI
15:2	RO	0x0	reserved
1	RW	0x0	suffix_sei_timing_flag suffix_sei_timing_flag A flag whether to encode suffix_SEI_DATA with a picture of this command or with a source picture of the buffer at the moment 0 : encode SUFFIX_SEI_DATA when a source picture is encoded. 1 : encode SUFFIX_SEI_DATA at this command.
0	RW	0x0	suffix_sei_nal_data_enable_flag suffix_sei_nal_data_enable_flag It enables to encode the suffix SEI NAL which is given by host.

hevc_enc_swreg101_cmd_enc_src_timestamp_high
Address: Operational Base + offset (0x00194)
src_timestamp_high

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_timestamp_high src_timestamp_high An MSB of timestamp of a source picture

hevc_enc_swreg101_cmd_enc_custom_gop_pic_lambda_3
Address: Operational Base + offset (0x00194)
custom gop lambda3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 3rd picture in the custom GOP A lamda weight of 3rd picture in the custom GOP

hevc_enc_swreg101_cmd_enc_reserved
Address: Operational Base + offset (0x00194)
reserved

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	reserved reserved reserved

hevc_enc_swreg102_cmd_enc_rc_param
Address: Operational Base + offset (0x00198)
cmd_enc_rc_param

Bit	Attr	Reset Value	Description
31:20	RW	0x000	initial_delay An inital cpb delay in msec (10 ~ 3000) An inital cpb delay in msec (10 ~ 3000)
19:14	RW	0x00	initial_rc_qp A value of initial RC QP by host. (0 ~ 51). A value of initial RC QP by host. (0 ~ 51). If host wants VPU to use a firmware-generated initial RC QP, set any value from 52 to 63
13	RW	0x0	en_seq_roi It enables ROI in sequence level. It enables ROI in sequence level.
12:9	RW	0x0	init_buf_levelx8 It specifies encoder initial delay It specifies encoder initial delay (encoder initial delay = InitialDelay * InitBufLevelx8 / 8).
8:7	RW	0x0	bit_alloc_mode It specifies picture bits allocation mode. It specifies picture bits allocation mode.
6:4	RW	0x0	hvs_qp_scale QP scaling factor for CU QP adjustment with en_hvs_qp = 1 QP scaling factor for CU QP adjustment with en_hvs_qp = 1
3	RW	0x0	en_hvs_qp_scale It enables QP scaling factor for CU QP adjustment with en_hvs_qp = 1 It enables QP scaling factor for CU QP adjustment with en_hvs_qp = 1.
2	RW	0x0	en_hvs_qp It enables CU QP adjustment for subjective quality enhancement It enables CU QP adjustment for subjective quality enhancement

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Bit	Attr	Reset Value	Description
1	RW	0x0	en_cu_level_rc It enables CU level rate control It enables CU level rate control
0	RW	0x0	en_rate_control It enables rate control. It enables rate control.

hevc_enc_swreg102_cmd_enc_custom_gop_pic_lambda_4
Address: Operational Base + offset (0x00198)
custom gop lambda4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 4th picture in the custom GOP A lamda weight of 4th picture in the custom GOP

hevc_enc_swreg102_addr_cr_base3
Address: Operational Base + offset (0x00198)
Cr base of index3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base3 cr_base_fbc_y_offset_base3 Cr base address of DPB index 3 unless FBC is used. If FBC is used, this is compressed Luma offset table base address of DPB index 3.

hevc_enc_swreg102_cmd_enc_longterm_pic
Address: Operational Base + offset (0x00198)
cmd_enc_longterm_pic

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	ref_longterm_pic_poc ref_longterm_pic_poc A POC of a longterm picture to be used as a reference picture
15:2	RO	0x0	reserved
1	RW	0x0	use_ref_longterm_pic use_ref_longterm_pic A flag to use a longterm picture in DPB to encode a picture

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Bit	Attr	Reset Value	Description
0	RW	0x0	use_src_longterm_pic use_src_longterm_pic A flag to use a source picture as a longterm reference picture later

hevc_enc_swreg102_addr_fbc_y_offset3
Address: Operational Base + offset (0x00198)
FBC luma offset base of index3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base3 cr_base_fbc_y_offset_base3 Cr base address of DPB index 3 unless FBC is used.

hevc_enc_swreg103_addr_fbc_c_offset3
Address: Operational Base + offset (0x0019c)
FBC chroma offset base of index3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base3 fbc_chroma_offset_base3 If FBC is used, this is chroma offset base address of DPB index 3. Otherwise, it is ignored.

hevc_enc_swreg103_cmd_enc_rc_min_max_qp
Address: Operational Base + offset (0x0019c)
It enables enc_rc_min_max_qp

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	intra_qp_offset A QP offset of intra picture A QP offset of intra picture
17:12	RW	0x00	max_delta_qp A maximum delta QP for rate control (0 ~ 51) A maximum delta QP for rate control (0 ~ 51)
11:6	RW	0x00	max_qp A maximum QP for rate control (0 ~ 51) A maximumQP for rate control (0 ~ 51)
5:0	RW	0x00	min_qp A minimum QP for rate control (0 ~ 51) A minimum QP for rate control (0 ~ 51)

hevc_enc_swreg103_cmd_enc_custom_gop_pic_lambda_5
Address: Operational Base + offset (0x0019c)
custom gop lambda5

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 5th picture in the custom GOP A lamda weight of 5th picture in the custom GOP

hevc_enc_swreg104_cmd_enc_custom_gop_pic_lambda_6
Address: Operational Base + offset (0x001a0)
custom gop lambda6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 6th picture in the custom GOP A lamda weight of 6th picture in the custom GOP

hevc_enc_swreg104_cmd_enc_rc_bit_ratio_layer_0_3
Address: Operational Base + offset (0x001a0)
enc_rc_bit_ratio_layer_0_3

Bit	Attr	Reset Value	Description
31:24	RW	0x00	fixed_bit_ratio_3 It specifies the ratio of 4th picture bits in encoding order in a GOP (1~255). It specifies the ratio of 4th picture bits in encoding order in a GOP (1~255).
23:16	RW	0x00	fixed_bit_ratio_2 It specifies the ratio of 3rd picture bits in encoding order in a GOP (1~255). It specifies the ratio of 3rd picture bits in encoding order in a GOP (1~255).
15:8	RW	0x00	fixed_bit_ratio_1 It specifies the ratio of 2nd picture bits in encoding order in a GOP (1~255). It specifies the ratio of 2nd picture bits in encoding order in a GOP (1~255).
7:0	RW	0x00	fixed_bit_ratio_0 It specifies the ratio of 1st picture bits in encoding order in a GOP (1~255) It specifies the ratio of 1st picture bits in encoding order in a GOP (1~255).

hevc_enc_swreg104_addr_luma_base4
Address: Operational Base + offset (0x001a0)
Luma base of index4

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base4 luma_base4 Luma base address of DPB index 4 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 4.

hevc_enc_swreg104_cmd_enc_roi_param
Address: Operational Base + offset (0x001a0)
enc_roi_param

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	ctu_map_stride ctu_map_stride Stride of CTU map Set this with (Width + CTB_SIZE - 1) / CTB_SIZE
15:12	RW	0x0	ctu_map_endian Endianess of CTU map Endianess of CTU map
11:10	RO	0x0	reserved
9	RW	0x0	ctu_qp_map_enable_flag ctu_qp_map_enable_flag It enables CTU QP map that allows CTUs to be encoded with the given QPs.
8	RW	0x0	ctu_mode_enable_flag ctu_mode_enable_flag It enables CTU mode map that allows CTUs to be encoded with force intra or to be skipped
7:1	RW	0x00	roi_delta_qp roi_delta_qp It specifies the delta QP for ROI important level
0	RW	0x0	roi_enable_flag roi_enable_flag It enables ROI map.

hevc_enc_swreg105_cmd_enc_custom_gop_pic_lambda_7
Address: Operational Base + offset (0x001a4)
custom gop lambda7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lambda_weith A lamda weight of 7th picture in the custom GOP A lamda weight of 7th picture in the custom GOP

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hevc_enc_swreg105_cmd_enc_roi_map_addr
Address: Operational Base + offset (0x001a4)
roi_map_addr

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	roi_map_addr roi_map_addr Start buffer address of ROI map The memory size is the number of CTUs of picture in bytes. For example if there are 64 CTUs within a picture, the size of ROI map is 64 bytes. All CTUs have their ROI importance level (0 ~ 8 ; 1 byte) in raster order. The higher an ROI level is, the more important the region is with a lower QP

hevc_enc_swreg105_addr_cb_base4
Address: Operational Base + offset (0x001a4)
Cb base of index4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base_fbc_cbcr_base4 cb_base_fbc_cbcr_base4 Cb base address of DPB index 4 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 4.

hevc_enc_swreg105_cmd_enc_rc_bit_ratio_layer_4_7
Address: Operational Base + offset (0x001a4)
cmd_enc_rc_bit_ratio_layer_4_7

Bit	Attr	Reset Value	Description
31:24	RW	0x00	fixed_bit_ratio_7 It specifies the ratio of 8th picture bits in encoding order in a Gop (1~255). It specifies the ratio of 8th picture bits in encoding order in a Gop (1~255).
23:16	RW	0x00	fixed_bit_ratio_6 It specifies the ratio of 7th picture bits in encoding order in a Gop (1~255). It specifies the ratio of 7th picture bits in encoding order in a GOP (1~255).

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	fixed_bit_ratio_5 It specifies the ratio of 6th picture bits in encoding order in a Gop (1~255). It specifies the ratio of 6th picture bits in encoding order in a GOP (1~255).
7:0	RW	0x00	fixed_bit_ratio_4 It specifies the ratio of 5th picture bits in encoding order in a Gop (1~255). It specifies the ratio of 5th picture bits in encoding order in a GOP (1~255).

hevc_enc_swreg106_addr_cr_base4
Address: Operational Base + offset (0x001a8)
Cr base of index4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base4 cr_base_fbc_y_offset_base4 Cr base address of DPB index 4 unless FBC is used. If FBC is used, this is compressed Luma offset table base address of DPB index 4.

hevc_enc_swreg106_ret_enc_pic_idx
Address: Operational Base + offset (0x001a8)
enc_pic_ind

Bit	Attr	Reset Value	Description
31:24	RW	0x00	pic_idx pic_idx A frame buffer index of the encoded picture -2 : encoding delay (default) -1 : encoding end
23:0	RO	0x0	reserved

hevc_enc_swreg106_cmd_enc_nr_param
Address: Operational Base + offset (0x001a8)
cmd enc nr_param

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:18	RW	0x00	nr_noise_sigma_cr It specifies Cr noise standard deviation if ENABLE_NR_EST_NOISE is 0. (0 ~ 25 It specifies Cr noise standard deviation if ENABLE_NR_EST_NOISE is 0. (0 ~ 255)

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Bit	Attr	Reset Value	Description
17:11	RW	0x00	nr_noise_sigma_cb It specifies Cb noise standard deviation if ENABLE_NR_EST_NOISE is 0. (0 ~ 25 It specifies Cb noise standard deviation if ENABLE_NR_EST_NOISE is 0. (0 ~ 255)
10:4	RW	0x00	nr_noise_sigma_y It specifies Y noise standard deviation if ENABLE_NR_EST_NOISE is 0. (0 ~ 25 It specifies Y noise standard deviation if ENABLE_NR_EST_NOISE is 0. (0 ~ 255)
3	RW	0x0	enable_nr_est_noise It enables VPU to conduct noise estimation for reduction. It enables VPU to conduct noise estimation for reduction. When this is disabled, noise estimation is carried out ouside VPU.
2	RW	0x0	enable_nr_cr It enables noise reduction algorithm to Cr component. It enables noise reduction algorithm to Cr component.
1	RW	0x0	enable_nr_cb It enables noise reduction algorithm to Cb component. It enables noise reduction algorithm to Cb component.
0	RW	0x0	enable_nr_y It enables noise reduction algorithm to Y component It enables noise reduction algorithm to Y component

hevc_enc_swreg106_cmd_enc_ctu_mode_map_addr
Address: Operational Base + offset (0x001a8)
ctu_mode_map_addr

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ctu_mode_map_addr ctu_mode_map_addr Start buffer address of CTU mode map The memory size is the number of CTUs of picture in bytes. For example if there are 64 CTUs within a picture, the size of CTU map is 64 bytes. It should be given when CTU_MODE_ENABLE_FLAG is 1.

hevc_enc_swreg106_addr_fbc_y_offset4
Address: Operational Base + offset (0x001a8)
Cr base of index4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base4 cr_base_fbc_y_offset_base4 Cr base address of DPB index 4 unless FBC is used. If FBC is used, this is compressed Luma offset table base address of DPB index 4.

hevc_enc_swreg107_cmd_enc_ctu_qp_map_addr
Address: Operational Base + offset (0x001ac)
enc_ctu_qp_map_addr

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ctu_qp_map_addr ctu_ap_map_addr Start buffer address of CTU qp map The memory size is the number of CTUs of picture in bytes. For example if there are 64 CTUs within a picture, the size of CTU map is 64 bytes. It should be given when CTU_QP_MAP_ENABLE_FLAG is 1.

hevc_enc_swreg107_addr_fbc_c_offset4
Address: Operational Base + offset (0x001ac)
FBC Chroma offset base of index4

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base4 fbc_chroma_offset_base4 If FBC is used, this is chroma offset base address of DPB index 4. Otherwise, it is ignored.

hevc_enc_swreg107_ret_enc_pic_slice_num
Address: Operational Base + offset (0x001ac)
pic_slice_num

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	pic_dep_slice_num pic_dep_slice_num The total dependent slice segment number in the encoded picture
15:0	RW	0x0000	pic_indep_slice_num pic_indep_slice_num The total independent slice segment number in the encoded picture

hevc_enc_swreg107_cmd_enc_nr_weight
Address: Operational Base + offset (0x001ac)
cmd_enc_nr_weight

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:25	RW	0x00	nr_inter_weith_cr Weight to CR noise level for inter picture (0 ~ 31) Weight to CR noise level for inter picture (0 ~ 31)
24:20	RW	0x00	nr_inter_weight_cb Weight to CB noise level for inter picture (0 ~ 31) Weight to CB noise level for inter picture (0 ~ 31)

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Bit	Attr	Reset Value	Description
19:15	RW	0x00	nr_inter_weight_y Weight to Y noise level for inter picture (0 ~ 31) Weight to Y noise level for inter picture (0 ~ 31) Note NR_INTER_WEIGHT/4 is multiplied to the noise level that has been estimated. This weight is put for inter frame to be filtered more strongly or more weakly than just with the estimated noise level.
14:10	RW	0x00	nr_intra_weight_cr Weight to CR noise level for intra picture (0 ~ 31) Weight to CR noise level for intra picture (0 ~ 31)
9:5	RW	0x00	nr_intra_weight_cb Weight to CB noise level for intra picture (0 ~ 31) Weight to CB noise level for intra picture (0 ~ 31)
4:0	RW	0x00	nr_intra_weight_y Weight to Y noise level for intra picture (0 ~ 31) Weight to Y noise level for intra picture (0 ~ 31) Note NR_INTRA_WEIGHT/4 is multiplied to the noise level that has been estimated. This weight is put for intra frame to be filtered more strongly or more weakly than just with the estimated noise level.

hevc_enc_swreg108_ret_enc_skip
Address: Operational Base + offset (0x001b0)
ret_enc_skip

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:1	RW	0x00	pic_skip pic_skip A picture skip flag
0	RO	0x0	reserved

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hevc_enc_swreg108_addr_luma_base5
Address: Operational Base + offset (0x001b0)
Luma base of index5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base5 luma_base5 Luma base address of DPB index 5 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 5.

hevc_enc_swreg108_cmd_enc_num_units_in_tick
Address: Operational Base + offset (0x001b0)
cmd_enc_num_units_in_tick

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	num_units_in_tick Num of units in tick It specifies the number of time units of a clock operating at the frequency time_scale Hz.

hevc_enc_swreg109_cmd_enc_time_scale
Address: Operational Base + offset (0x001b4)
cmd_enc_time_scale

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	time_scale It specifies the number of time units that pass in one second. It specifies the number of time units that pass in one second.

hevc_enc_swreg109_addr_cb_base5
Address: Operational Base + offset (0x001b4)
Cb base of index5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base_fbc_cbcr_base5 cb_base_fbc_cbcr_base5 Cb base address of DPB index 5 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 5.

hevc_enc_swreg109_ret_enc_pic_num_intra
Address: Operational Base + offset (0x001b4)
pic_num_intra

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_num_intra pic_num_intra The number of intra block in 8x8 ENC_PIC Command Parameter Registers These are command I/O registers where host processor can set arguments for the ENC_PIC command or get return values.

hevc_enc_swreg110_cmd_enc_num_ticks_poc_diff_one
Address: Operational Base + offset (0x001b8)
cmd_enc_num_ticks_poc_diff_one

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	num_ticks_poc_diff_one difference of pictur order count valuse equal to 1 It specifies the number of clock ticks corresponding to a difference of picture order count values equal to 1.

hevc_enc_swreg110_ret_enc_pic_num_merge
Address: Operational Base + offset (0x001b8)
pic_num_merge

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_num_merge pic_num_merge The number of merge block in 8x8

hevc_enc_swreg110_addr_cr_base5
Address: Operational Base + offset (0x001b8)
Cr base of index5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base5 cr_base_fbc_y_offset_base5 Cr base address of DPB index 5 unless FBC is used. If FBC is used, this is compressed Luma offset table base address of DPB index 5.

hevc_enc_swreg110_addr_fbc_y_offset5
Address: Operational Base + offset (0x001b8)
FBC luma offset base of index5

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base5 cr_base_fbc_y_offset_base5 Cr base address of DPB index 5 unless FBC is used.

hevc_enc_swreg111_addr_fbc_c_offset5
Address: Operational Base + offset (0x001bc)
FBC chroma offset base of index5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base5 fbc_chroma_offset_base5 If FBC is used, this is chroma offset base address of DPB index 5. Otherwise, it is ignored.

hevc_enc_swreg111_cmd_enc_rc_trans_rate
Address: Operational Base + offset (0x001bc)
cmd_enc_rc_trans_rate

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	trans_rate A peak transmission bitrate in bps (0 ~ 700000000) A peak transmission bitrate in bps (0 ~ 700000000)

hevc_enc_swreg112_addr_luma_base6
Address: Operational Base + offset (0x001c0)
Luma base of index6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base6 luma_base6 Luma base address of DPB index 6 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 6.

hevc_enc_swreg112_cmd_enc_rc_target_rate
Address: Operational Base + offset (0x001c0)
CMD_ENC_RC_TARGET_RATE

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	target_rate A target rate (EN_SEPARATE_BITRATE = 0) (0 ~ 700000000) A target rate (EN_SEPARATE_BITRATE = 0) (0 ~ 700000000)

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hevc_enc_swreg112_ret_enc_pic_num_skip
Address: Operational Base + offset (0x001c0)
pic_num_skip

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_num_skip pic_num_skip The number of skip block in 8x8

hevc_enc_swreg113_addr_cb_base6
Address: Operational Base + offset (0x001c4)
Cb base of index6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base_fbc_cbcr_base6 cb_base_fbc_cbcr_base6 Cb base address of DPB index 6 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 6.

hevc_enc_swreg113_ret_enc_pic_avg_cu_qp
Address: Operational Base + offset (0x001c4)
avg_cu_qp

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_avg_cu_qp pic_avg_cu_qp An average value of CU QPs

hevc_enc_swreg114_addr_fbc_y_offset6
Address: Operational Base + offset (0x001c8)
FBC luma offset base of index6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base6 cr_base_fbc_y_offset_base6 Compressed Luma offset table base address of DPB index 6 when FBC is used.

hevc_enc_swreg114_ret_enc_pic_byte
Address: Operational Base + offset (0x001c8)
pic_byte

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_byte pic_byte The size of encoded picture in byte

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hevc_enc_swreg114_addr_cr_base6
Address: Operational Base + offset (0x001c8)
Cr base of index6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_base6 cr_base_fbc_y_base6 Cr base address of DPB index 6 unless FBC is used.

hevc_enc_swreg115_ret_enc_pic_idx
Address: Operational Base + offset (0x001cc)
pic_idx

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gop_pic_idx gop_pic_idx A picture index in GOP

hevc_enc_swreg115_ret_enc_min_fb_num
Address: Operational Base + offset (0x001cc)
ret_enc_min_fb_num

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	min_fb_num The minimun number of frame buffer for encoding The minimun number of frame buffer for encoding

hevc_enc_swreg115_addr_fbc_c_offset6
Address: Operational Base + offset (0x001cc)
FBC chroma offset base of index6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base6 fbc_chroma_offset_base6 If FBC is used, this is chroma offset base address of DPB index 6. Otherwise, it is ignored.

hevc_enc_swreg116_ret_enc_pic_poc
Address: Operational Base + offset (0x001d0)
pic_poc

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_poc pic_poc A POC value of the current picture

hevc_enc_swreg116_ret_enc_nal_info_to_be_encoded

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Address: Operational Base + offset (0x001d0)
user data flag

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	nal_info_to_be_encoded NAL unit types that will be encoded while run the next ENC_PIC command NAL unit types that will be encoded while run the next ENC_PIC command

hevc_enc_swreg116_addr_luma_base7
Address: Operational Base + offset (0x001d0)
Luma base of index7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	luma_base7 luma_base7 Luma base address of DPB index 7 unless FBC is used. If FBC is used, this is compressed Luma base address of DPB index 7.

hevc_enc_swreg117_addr_cb_base7
Address: Operational Base + offset (0x001d4)
Cb base of index7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cb_base_fbc_cbcr_base7 cb_base_fbc_cbcr_base7 Cb base address of DPB index 7 unless FBC is used. If FBC is used, this is compressed Cb and Cr base address of DPB index 7.

hevc_enc_swreg117_ret_frame_cycle
Address: Operational Base + offset (0x001d4)
a frame processing cycle

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	frame_cycle The number of processing cycle for a frame (1 ~ 16) The number of processing cycle for a frame (1 ~ 16)

hevc_enc_swreg118_addr_fbc_y_offset7
Address: Operational Base + offset (0x001d8)
FBC luma offset base of index7

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base7 cr_base_fbc_y_offset_base7 Compressed Luma offset table base address of DPB index 7 when FBC is used.

hevc_enc_swreg118_ret_enc_used_src_idx
Address: Operational Base + offset (0x001d8)
enc_used_src_idx

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	used_src_pic_idx used_src_pic_idx A source buffer index of the encoded picture (-2 : encoding delay)

hevc_enc_swreg118_ret_min_src_buf_num
Address: Operational Base + offset (0x001d8)
DPB out flag

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ret_min_src_buf_num The mininum number of source frame buffer for encoding (1 ~ 16) The mininum number of source frame buffer for encoding (1 ~ 16)

hevc_enc_swreg118_addr_cr_base7
Address: Operational Base + offset (0x001d8)
Cr base of index7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cr_base_fbc_y_offset_base7 cr_base_fbc_y_offset_base7 Cr base address of DPB index 7 unless FBC is used. If FBC is used, this is compressed Luma offset table base address of DPB index 7.

hevc_enc_swreg119_ret_enc_pic_num
Address: Operational Base + offset (0x001dc)
ret_enc_pic_num

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pic_num The encoded picture number The encoded picture number

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hevc_enc_swreg119_addr_fbc_c_offset7
Address: Operational Base + offset (0x001dc)
FBC chroma offset base of index7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	fbc_chroma_offset_base7 fbc_chroma_offset_base7 If FBC is used, this is chroma offset base address of DPB index 7. Otherwise, it is ignored.

hevc_enc_swreg120_addr_mv_col0
Address: Operational Base + offset (0x001e0)
Colocated mv buffer base of index 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base0 col_mv_buf_base0 Base address of colocated motion vecotors buffer of DPB index 0

hevc_enc_swreg120_ret_enc_pic_type
Address: Operational Base + offset (0x001e0)
ret_enc_pic_type

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	pic_typ pic_type The encoded picture type 0 : I slice 1 : P slice 2 : B slice

hevc_enc_swreg121_addr_mv_col1
Address: Operational Base + offset (0x001e4)
Colocated mv buffer base of index 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base1 col_mv_buf_base1 Base address of colocated motion vecotors buffer of DPB index 1

hevc_enc_swreg122_addr_mv_col2
Address: Operational Base + offset (0x001e8)
Colocated mv buffer base of index 2

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base2 col_mv_buf_base2 Base address of colocated motion vecotors buffer of DPB index 2

hevc_enc_swreg123_addr_mv_col3
Address: Operational Base + offset (0x001ec)
Colocated mv buffer base of index 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base3 col_mv_buf_base3 Base address of colocated motion vecotors buffer of DPB index 3

hevc_enc_swreg124_addr_mv_col4
Address: Operational Base + offset (0x001f0)
Colocated mv buffer base of index 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base4 col_mv_buf_base4 Base address of colocated motion vecotors buffer of DPB index 0

hevc_enc_swreg125_addr_mv_col5
Address: Operational Base + offset (0x001f4)
Colocated mv buffer base of index 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base5 col_mv_buf_base5 Base address of colocated motion vecotors buffer of DPB index 5

hevc_enc_swreg126_addr_mv_col6
Address: Operational Base + offset (0x001f8)
mv buffer base of index 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base6 col_mv_buf_base6 Base address of colocated motion vecotors buffer of DPB index 6

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hevc_enc_swreg127_addr_mv_col7
Address: Operational Base + offset (0x001fc)
Colocated mv buffer base of index 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	col_mv_buf_base7 col_mv_buf_base7 Base address of colocated motion vecotors buffer of DPB index 7

hevc_enc_swreg128_mmu_DTE_ADDR
Address: Operational Base + offset (0x00200)
MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR mmu dte base addr mmu dte base addr , the address must be 4kb aligned

hevc_enc_swreg129_mmu_STATUS
Address: Operational Base + offset (0x00204)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Field0000 Abstract Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE Field0000 Abstract The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x0	REPLAY_BUFFER_EMPTY Field0000 Abstract The MMU replay buffer is empty
3	RO	0x0	MMU_IDLE Field0000 Abstract The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE Field0000 Abstract MMU stall mode currently enabled. The mode is enabled by command

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Bit	Attr	Reset Value	Description
1	RO	0x0	PAGE_FAULT_ACTIVE Field0000 Abstract MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Field0000 Abstract Paging is enabled

hevc_enc_swreg130_mmu_COMMAND
Address: Operational Base + offset (0x00208)
MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x0	MMU_CMD Field0000 Abstract MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

hevc_enc_swreg131_mmu_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x0020c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR Field0000 Abstract address of last page fault

hevc_enc_swreg132_mmu_ZAP_ONE_LINE
Address: Operational Base + offset (0x00210)
MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE Field0000 Abstract address to be invalidated from the page table cache

hevc_enc_swreg133_mmu_INT_RAWSTAT
Address: Operational Base + offset (0x00214)
MMU raw interrupt status register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault

hevc_enc_swreg134_mmu_INT_CLEAR
Address: Operational Base + offset (0x00218)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error
0	WO	0x0	PAGE_FAULT Field0000 Abstract page fault

hevc_enc_swreg135_mmu_INT_MASK
Address: Operational Base + offset (0x0021c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	INT_STATUS Field0001 Abstract read bus error enable an interrupt source if the corresponding mask bit is set to 1
0	RW	0x0	PAGE_FAULT Field0000 Abstract page fault enable an interrupt source if the corresponding mask bit is set to 1

hevc_enc_swreg136_mmu_INT_STATUS
Address: Operational Base + offset (0x00220)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR Field0000 Abstract read bus error

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Bit	Attr	Reset Value	Description
0	RO	0x0	PAGE_FAULT Field0000 Abstract page fault

hevc_enc_swreg137_mmu_AUTO_GATING
Address: Operational Base + offset (0x00224)
mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_auto_gating mmu_auto_gating when it is 1'b1, the mmu will auto gating it self

hevc_enc_swreg138_Reserved
Address: Operational Base + offset (0x00228)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	Field0000 Abstract Field0000 Description

hevc_enc_swreg139_Reserved
Address: Operational Base + offset (0x0022c)
Register0001 Abstract

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	Field0000 Abstract Field0000 Description

hevc_enc_swreg140_Reserved
Address: Operational Base + offset (0x00230)
Register0002 Abstract

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	Field0000 Abstract Reserved

hevc_enc_swreg141_Reserved
Address: Operational Base + offset (0x00234)
Register0003 Abstract

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	Field0000 Abstract Field0000 Description

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hevc_enc_swreg142_Reserved
Address: Operational Base + offset (0x00238)
Register0004 Abstract

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	Field0000 Abstract Field0000 Description

hevc_enc_swreg143_Reserved
Address: Operational Base + offset (0x0023c)
Register0005 Abstract

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved
-1:0	RW	RW	Field0000 Abstract Field0000 Description

hevc_enc_swreg144_AXI_CTRL
Address: Operational Base + offset (0x00240)
Register0006 Abstract

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	axi_mmu_disable Field0000 Abstract disable mmu and bypass if the bit is set to 1
3:1	RO	0x0	reserved
0	RW	0x0	axi_pref_enable Field0000 Abstract enable detect axi performance if the bit is set to 1

hevc_enc_swreg145_PERF_LATENCY_CTRL0
Address: Operational Base + offset (0x00244)
Axi performance latency module contrl register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:12	RW	0x000	sw_rd_latency_thr sw_rd_latency_thr sw_rd_latency_thr
11:4	RW	0x00	sw_rd_latency_id sw_rd_latency_id sw_rd_latency_id

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Bit	Attr	Reset Value	Description
3	RW	0x0	sw_axi_cnt_type sw_axi_cnt_type sw_axi_cnt_type
2	RW	0x0	sw_axi_perf_frm_type sw_axi_perf_frm_type 1'b0: clear by frame end 1'b1: clear by software configuration
1	RW	0x0	sw_axi_perf_clr_e sw_axi_perf_clr_e 1'b0: software clear disable 1'b1: software clear enalbe
0	RW	0x0	sw_axi_perf_work_e sw_axi_perf_work_e 1'b0: disable 1'b1: enable

hevc_enc_swreg146_PERF_LATENCY_CTRL1
Address: Operational Base + offset (0x00248)
PERF_LATENCY_CTRL1

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20	RW	0x0	sw_rd_total_bytes_mode sw_rd_total_bytes_mode sw_rd_total_bytes_mode
19:12	RW	0x00	sw_aw_count_id sw_aw_count_id sw_aw_count_id
11:4	RW	0x00	sw_ar_count_id sw_ar_count_id sw_ar_count_id
3	RW	0x0	sw_aw_cnt_id_type sw_aw_cnt_id_type sw_aw_cnt_id_type
2	RW	0x0	sw_ar_cnt_id_type sw_ar_cnt_id_type sw_ar_cnt_id_type
1:0	RW	0x0	sw_addr_align_type sw_addr_align_type sw_addr_align_type

hevc_enc_swreg147_PERF_RD_MAX_LATENCY_NUM0
Address: Operational Base + offset (0x0024c)
Read max latency number

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	rd_max_latency_num_ch0 read max latency value of channel 0 Field0000 Description

hevc_enc_swreg148_PERF_RD_LATENCY_SAMP_NUM
Address: Operational Base + offset (0x00250)
The number of bigger than configed threshold value

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_thr_num_ch0 read latency thr number channel 0

hevc_enc_swreg149_PERF_RD_LATENCY_ACC_SUM
Address: Operational Base + offset (0x00254)
Total sample number

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_acc_sum

hevc_enc_swreg150_PERF_RD_AXI_TOTAL_BYTE
Address: Operational Base + offset (0x00258)
perf_rd_axi_total_byte

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	perf_rd_axi_total_byte perf_rd_axi_total_byte perf_rd_axi_total_byte

hevc_enc_swreg151_PERF_WR_AXI_TOTAL_BYTE
Address: Operational Base + offset (0x0025c)
perf_wr_axi_total_byte

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	perf_wr_axi_total_byte perf_wr_axi_total_byte perf_wr_axi_total_byte

hevc_enc_swreg152_PERF_WORKING_CNT
Address: Operational Base + offset (0x00260)
perf_working_cnt

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	perf_working_cnt perf_working_cnt perf_working_cnt

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15.6 Interface Description

15.7 Application Notes

15.7.1 HEVC Configuration Flow

1. Prepare the data in the DDR.
2. Set the HEVC general system configuration in RKVDEC.swreg2, such as working mode, in/out endian.
3. Set the picture parameters with RKVDEC.swreg3.
4. Set the input and output data base address and HEVC reference configuration with RKVDEC.swreg4~RKVDEC.swreg43.
5. If CABAC error detection is desired, set the RKVDEC.swreg44 to enable the corresponding error detection.
6. Set the interrupt configuration and start the HEVC with HEVC.swreg1.
7. Wait for the frame interrupt, and then get the processed results in the target DDR
8. Clear all the interrupts, and repeat Process2~Process8 to start a new frame decoding if the decoding is not finished yet.

15.7.2 H264 Configuration Flow

1. Prepare the data in the DDR, for normal mode, we should prepare bitstream, tbl, pps and rps.
2. Set the h264 general system configuration in RKVDEC.swreg2, such as working mode, in/out endian.
3. Set the picture parameters with RKVDEC.swreg3.
4. Set the input and output data base address and H264 reference configuration with RKVDEC.swreg4~RKVDEC.swreg43.
5. If stream error detection is desired, set the swreg77_h264_error_e and swreg44_strmd_error_en to enable the corresponding error detection.
6. If prefech function is desired, set the prefetch common registers and clear its TLB. Pay attention, there contains two caches, which are for Y channel and UV channel.
7. If MMU function is desires, set the MMU common registers and clear its TLB. Pay attention, there contains two MMUs, which are for read channel and write channel.
8. Set the interrupt configuration and start the decoder with RKVDEC.swreg1.
9. Wait for the frame interrupt, and then get the processed results in the target DDR. There may be decoded frame, error_info, cur frame colmv output.
When the stream mode is not frame by frame mode,we also wait buf empty,and then send the next pack,repeat it until sw_dec_rdy_sta.
10. Clear all the interrupts, and repeat Process2~Process9 to start a new frame decoding if the decoding is not finished yet.

15.7.3 VP9 Configuration Flow

1. Prepare the data in the DDR, for normal mode, we should prepare bitstream, prob, and segid from last decoded frame.
2. Set the VP9 general system configuration in RKVDEC.swreg2, such as working mode, in/out endian.
3. Set the picture parameters with RKVDEC.swreg3.
4. Set the input and output data base address and HEVC reference configuration with RKVDEC.swreg4~RKVDEC.swreg43.
5. If stream error detection is desired, set the RKVDEC.swreg44 to enable the corresponding error detection.
6. If prefech function is desired, set the prefetch common registers and clear its TLB. Pay attention, there contains two caches, which are for Y channel and UV channel.
7. If MMU function is desires, set the MMU common registers and clear its TLB. Pay attention, there contains two MMUs, which are for read channel and write channel.

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8. Set the interrupt configuration and start the decoder with RKVDEC.swreg1.
9. Wait for the frame interrupt, and then get the processed results in the target DDR. There may be decoded frame, cur frame segid, cur frame count, cur frame colmv output.
10. Clear all the interrupts, and repeat Process2~Process9 to start a new frame decoding if the decoding is not finished yet.

15.7.4 Other formats and encoder Configuration flow

1. Prepare the decoder data in the DDR memory, and in decoder other than JPEG decoder, the input stream buffer should at least contain a slice or a frame data, otherwise the decoder will produce an interrupt and show error and then reset itself.
2. Config all the registers will be used. and please notice that which be list as follows:
In encoder----- We can configure the registers to control the input picture data format (such as endian and swap), but some input data format are fixed, such as cabac_table data. And the register VEPU_SWREG0~31 are JPEG quantization registers. They are write only registers. When you want to write these registers, you should first set VEPU_SWREG103[0] to 1'b0 and VEPU_SWREG103[5:4] to 2'b10(select JPEG mode).
In decoder---- The decoder can support ref buffer mode or cacheable mode, but they can't be both enabled. We can config the swreg57[28],swreg57[29] to enable cache and config the swreg65 to control the ref buffer.
4. You should config VDPU_SWREG57[0] as 1'b1 to enable video decoder. And config VDPU_SWREG41[0] as 1'b1 to enable pp. If pp performed in pipeline with decoder, you should config VDPU_SWREG41[4] as 1'b1 and then config VDPU_SWREG57[0] as 1'b1 to enable decoder and pp. VEPU_SWREG103[0] set to 1'b1 to enable encoder.
5. Wait for the frame interrupt,and then check if the frame decoder ready interrupt is right or not, after that,you can get the processed results in the target DDR
6. Clear all the interrupts,repeat step 2~5 to start a new frame decoder or encoder.
7.Please note that before you config h.264 encoder to work,you should need config GRF_SOC_CON3[11] to 1'b0.

15.7.5 Link table mode configuration flow

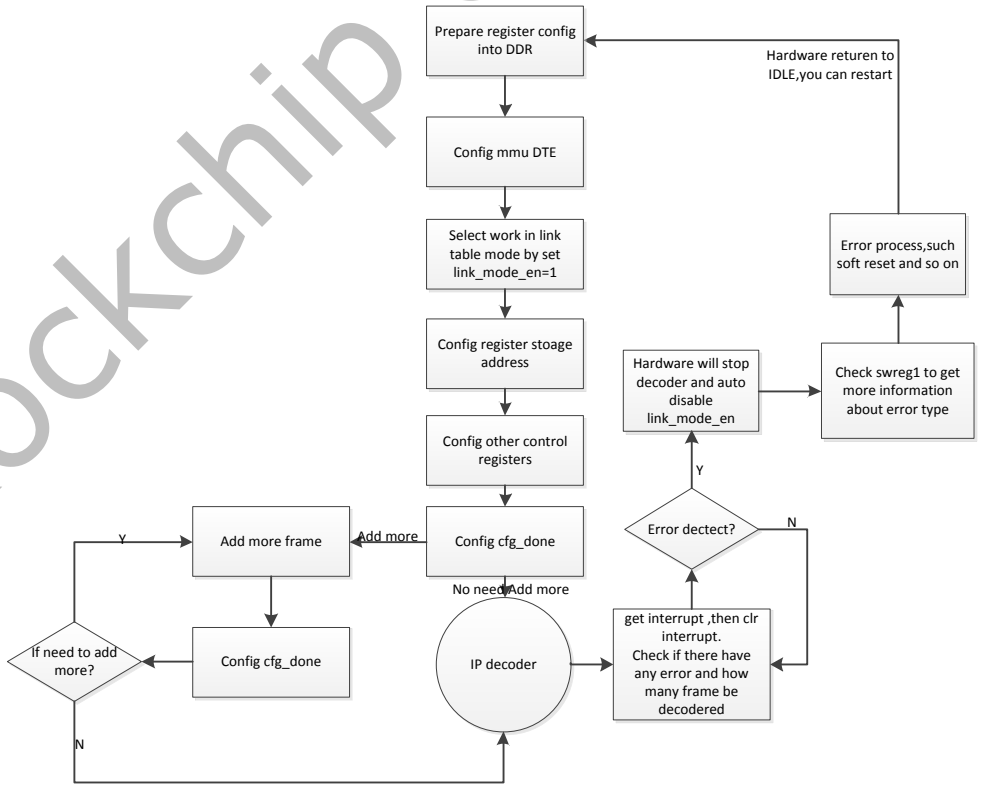

Fig. 16-29 Link table mode work flow

1.Prepare all the register config value into DDR.

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2.Config mmu DTE and enable mmu before config link table mode.
3.Config link_mode_en and other link table mode register.The register is from link_tale_swreg0~link_table_swreg2.
4.Config config_done(set link_table_swreg3[0]=1'b1) to make the link table register configuration effect,after that,hardware will begin to decoder frame by frame.
5.If you need to add more frame,you can use superaddition mode,you can add more frame when decoder at any status.
6.If all the frame be decodered,the hardware will hold the working status wait user to process,you can add more frame or set link_mode_en=0(need to set config_done=1'b1) to finish link table mode.
7.When you get interrupt,you will need to check if there have any error by check link_table_swreg4[31]=1'b1 or not.If any error found by hardware,the hardware will disable link table mode and stop decoder until user restart next link table mode.And then,you can get more information about error type from normal register swreg1.
8. When you get interrupt,if there doesn't have any error,you may be to check the frame number which have been decodered.
9.Please note that:Any link table register be config,you need to config swreg3[0]=1'b1 to make it effective.

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15.7.6 HEVC ENCODER Configuration Flow

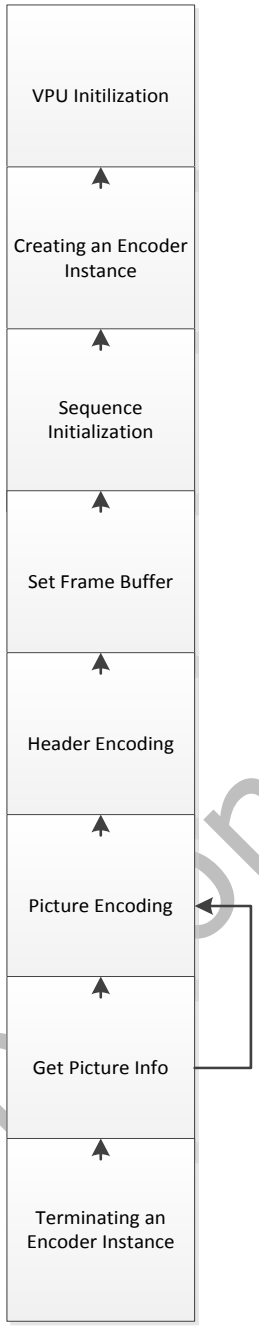

Fig. 16-30 HEVC ENCODER work flow

1. At first,you should need config GRF_SOC_CON3[11] to 1'b1 to select h.265 encoder to use share memory.
2.VPU initial:Firmware download to SDRAM,firmware download to Program Memory,set Working/Parameter/Code buffer,set interrupt for stream buffer and endianess.
3.Creating an Encoder Instance:Set video standard ,set stream buffer (address, size, mode, etc),set rotate/mirror option, picture size, GOP size, etc.
4.Sequence Initialization:Set all the encoder related information to Host Interface Registers of VPU and give the command ENC_SEQ_INI.
5.Set Frame Buffer:Set frame buffer base address, stride, etc.
6.Header Encoding:VPU make a header.
7.Picture Encoding: Set source frame address,post-option (rotate/mirror),secondary AXI options,frame skip mode, etc.And source yuv download to memory.
8.Get Picture Info:Wait pic end.And get encoding result such as MB Qstep, number of encoded bit, etc.If the sequence ends, jump to "Terminating an Encoder Instance"
, or jump to "Picture Encoding".
9.Terminating an Encoder Instance:Close current instance.

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Chapter 17 Video Input Processor (VIP)

16.1 Overview

The Video Input Processor, receives the data from Camera or CCIR656 encoder, and transfers the data into system main memory by AXI bus.
The Video Input Processor supports following features:

- Support YCbCr422 input
- Support Raw8bit input
- Support CCIR656(PAL/NTSC) input
- Support JPEG input
- Support YCbCr422/420 output
- Support UYVY/VYUY/YUYV/YVYU configurable
- Support up to 8192x8192 resolution source
- Support picture in picture
- Support arbitrary size window crop
- Support error/terminate interrupt and combined interrupt output
- Support clk/vsync/href polarity configurable
- Support one frame stop/ping-pong mode
- Support mmu

16.2 Block Diagram

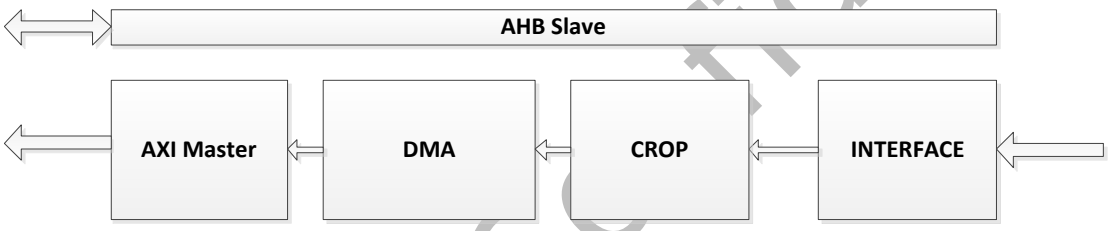

Fig. 18-1 VIP block diagram

The VIP comprises with:

- AHB Slave
- Host configure the registers via the AHB Slave
- AXI Master
- Transmit the data to chip memory via the AXI Master
- INTERFACE
- Translate the input video data into the requisite data format
- CROP
- Bypass or crop the source video data to a smaller size destination
- DMA
- Control the operation of AXI Master

16.3 Function Description

This chapter is used to illustrate the operational behavior of how VIP works. If YUV422 or ccir656 signal is received from external devices, VIP translate it into YUV422/420 data, and separate the data to Y and UV data, then store them to different memory via AXI bus separately. But if raw data is received, there are not any translations happened, the 8 data is considered as 16bit data and write directly to memory.

16.3.1 Support Vsync high active or low active

- Vsync Low active as below

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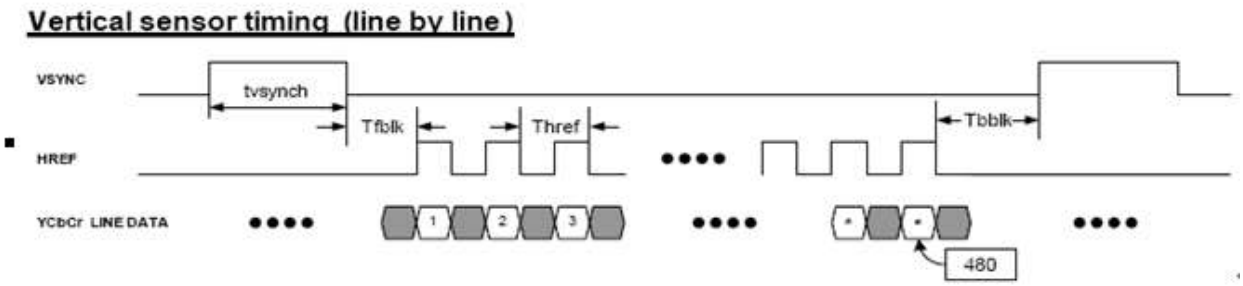

Fig. 18-2 Timing diagram for VIP when vsync low active

- Vsync High active

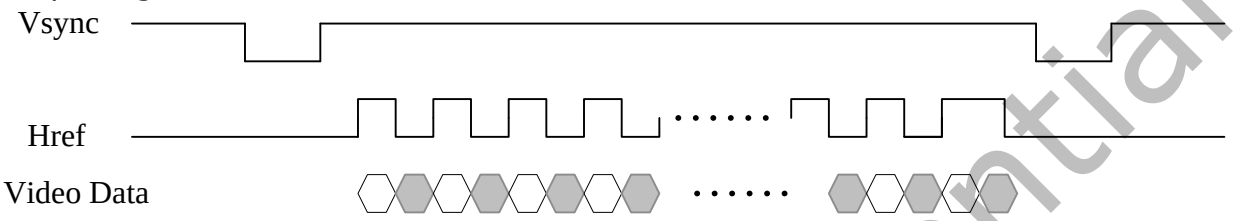

Fig. 17-3 Timing diagram for VIP when vsync high active

16.3.2 Support href high active or low active

- Href high active

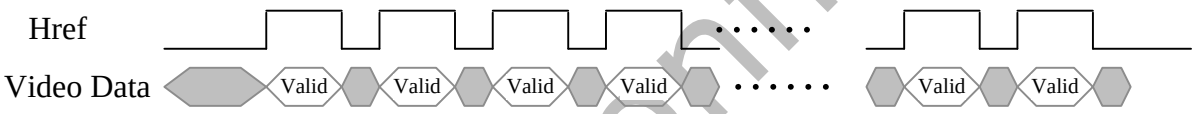

Fig. 18-4 Timing diagram for VIP when href high active

- Href Low active

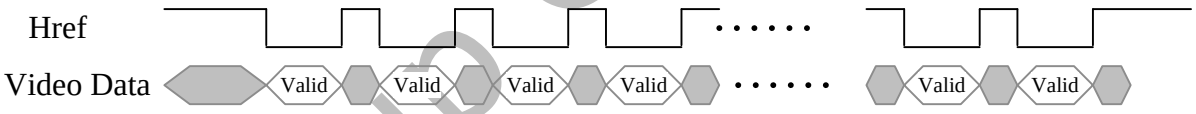

Fig. 18-5 Timing diagram for VIP when href low active

- Y first

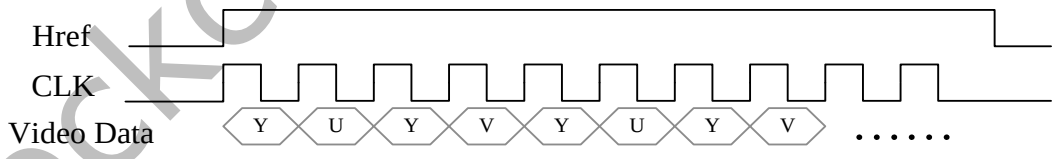

Fig. 18-6 Timing diagram for VIP when Y data first

- U first

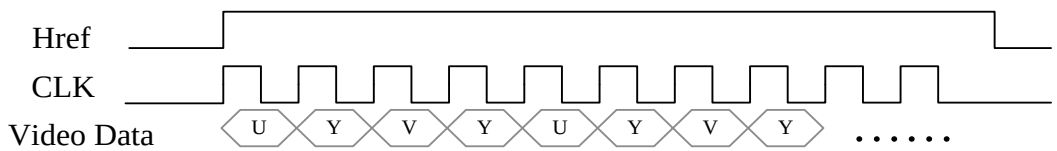

Fig. 18-7 Timing diagram for VIP when U data first

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16.3.3 Support CCIR656 (NTSC and PAL)

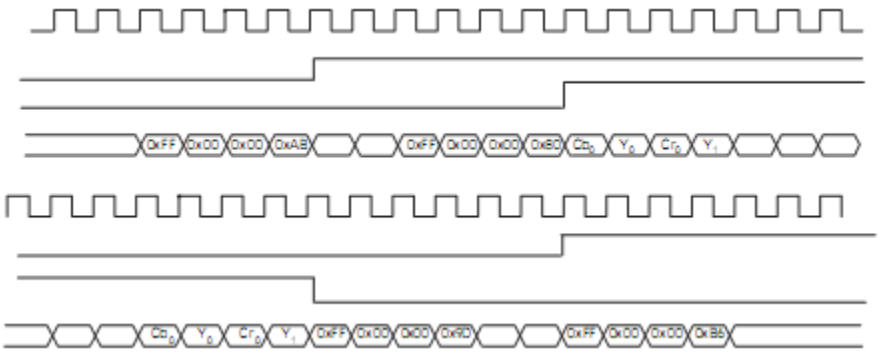

Fig. 18-8 CCIR656 timing

16.3.4 Support Raw data (8-bit) or JPEG

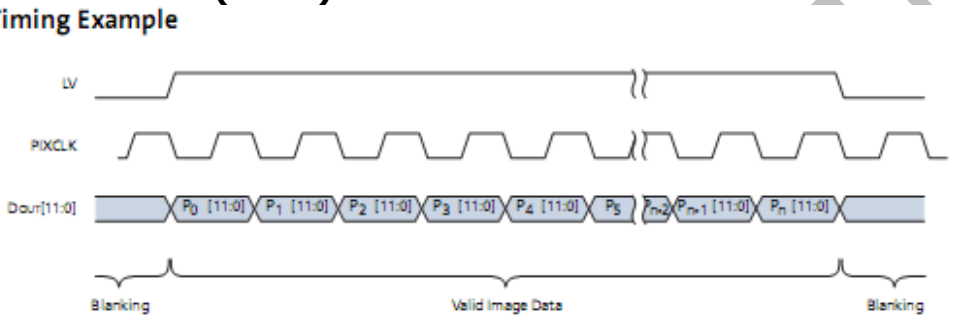

Fig. 17-9 Raw Data or JPEG Timing

VIP module can work in three modes: one frame stop mode, ping-pong mode.

One frame stop mode

In this mode, configure the parameter WORK_MODE to one frame stop mode. After one frame captured, VIP will automatic stop. After capturing, the image Y, UV data will be stored at main memory location defined by VIP_FRM0_ADDR_Y, FRM0_ADDR_UV separately.

Ping-Pong mode

After one frame(F1) captured, VIP will start to capture the next frame(F2) automatically, and host must assign new address pointer of frame1 and clear the frame1 status, thus VIP will capture the third frame automatically(by new F1 address) without any stop and so on for the following frames. But if host did not update the frame buffer address, the VIP will cover the pre-frame data stored in the memory with the following frame data.

Storage

Difference between the YUV mode and raw mode is that in the YUV mode or ccir656 mode, data will be storage in the Y data buffer and UV data buffer; but in the raw or jpeg mode, RGB data will be storage in the same buffer. In addition, in the yuv mode, the width of Y, U or V data is a byte in memory; in Raw or JPEGE mode, the width is a halfword no matter the data source is 8 bit.

CROP

The parameter START_Y and START_X defines the coordinate of crop start point. And the frame size after cropping is following the value of SET_WIDTH and SET_HEIGHT.

16.4 Register Description

16.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
VIP_CTRL	0x0000	W	0x00007000	VIP control
VIP_INTEN	0x0004	W	0x00000000	VIP interrupt status
VIP_INTSTAT	0x0008	W	0x00000000	VIP interrupt status
VIP_FOR	0x000c	W	0x00000000	VIP format

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Name	Offset	Size	Reset Value	Description
VIP_FRM0_ADDR_Y	0x0014	W	0x00000000	VIP frame0 y address
VIP_FRM0_ADDR_UV	0x0018	W	0x00000000	VIP frame0 uv address
VIP_FRM1_ADDR_Y	0x001c	W	0x00000000	VIP frame1 y address
VIP_FRM1_ADDR_UV	0x0020	W	0x00000000	VIP frame1 uv address
VIP_VIR_LINE_WIDTH	0x0024	W	0x00000000	VIP virtual line width
VIP_SET_SIZE	0x0028	W	0x01e002d0	VIP frame set size
VIP_CROP	0x0044	W	0x00000000	VIP crop start point
VIP_SCL_CTRL	0x0048	W	0x00000000	VIP scale control
VIP_FIFO_ENTRY	0x0054	W	0x00000000	VIP FIFO entry
VIP_FRAME_STATUS	0x0060	W	0x00000000	VIP frame status
VIP_CUR_DST	0x0064	W	0x00000000	VIP current destination address
VIP_LAST_LINE	0x0068	W	0x00000000	VIP last frame line number
VIP_LAST_PIX	0x006c	W	0x00000000	VIP last line pixel number
VIP_DTE_ADDR	0x0800	W	0x00000000	MMU current page Table address
VIP_STATUS	0x0804	W	0x00000000	MMU status register
VIP_COMMAND	0x0808	W	0x00000000	MMU command register
VIP_PAGE_FAULT_ADDR	0x080c	W	0x00000000	MMU logical address of last page fault
VIP_ZAP_ONE_LINE	0x0810	W	0x00000000	MMU Zap cache line register
VIP_INT_RAWSTAT	0x0814	W	0x00000000	MMU raw interrupt status register
VIP_INT_CLEAR	0x0818	W	0x00000000	MMU raw interrupt status register
VIP_INT_MASK	0x081c	W	0x00000000	MMU raw interrupt status register
VIP_INT_STATUS	0x0820	W	0x00000000	MMU raw interrupt status register
VIP_AUTO_GATING	0x0824	W	0x00000000	mmu auto gating

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.4.2 Detail Register Description

VIP_CTRL

Address: Operational Base + offset (0x0000)

VIP control

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:12	RW	0x7	AXI_BURST_TYPE axi master burst type 0-15 : burst1~16
11:3	RO	0x0	reserved
2:1	RW	0x0	WORK_MODE Working Mode 00 : one frame stop mode 01 : ping-pong mode 02 : line loop mode 03 : reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	CAP_EN capture enable 0 : disable 1 : enable

VIP_INTEN
Address: Operational Base + offset (0x0004)
VIP interrupt status

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	PST_INF_FRAME_END_EN frame end after interface FIFO interrupt enable 0 : disable 1 : enable
8	RW	0x0	PRE_INF_FRAME_END_EN frame end before interface FIFO interrupt enable 0 : disable 1 : enable
7	RO	0x0	reserved
6	W1C	0x0	BUS_ERR_EN axi master or ahb slave response error interrupt enable 0 : disable 1 : enable
5	RW	0x0	DFIFO_OF_EN DMA FIFO overflow interrupt enable 0 : disable 1 : enable
4	RW	0x0	IFIFO_OF_EN interface FIFO overflow interrupt enable 0 : disable 1 : enable
3	W1C	0x0	PIX_ERR_EN the pixel number of last line not equal to the set height interrupt enable 0 : disable 1 : enable
2	W1C	0x0	LINE_ERR_EN the line number of last frame not equal to the set height interrupt enable 0 : disable 1 : enable
1	W1C	0x0	LINE_END_EN line end interrupt enable 0 : disable 1 : enable

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Bit	Attr	Reset Value	Description
0	W1C	0x0	DMA_FRAME_END_EN dma frame end interrupt enable 0 : disable 1 : enable

VIP_INTSTAT
Address: Operational Base + offset (0x0008)
VIP interrupt status

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	PST_INF_FRAME_END frame end after interface FIFO interrupt 0 : no interrupt 1 : interrupt
8	RW	0x0	PRE_INF_FRAME_END frame end before interface FIFO interrupt 0 : no interrupt 1 : interrupt
7	RO	0x0	reserved
6	W1C	0x0	BUS_ERR axi master or ahb slave response error interrupt 0 : no interrupt 1 : interrupt
5	RW	0x0	DFIFO_OF DMA FIFO overflow interrupt 0 : no interrupt 1 : interrupt
4	RW	0x0	IFIFO_OF interface FIFO overflow interrupt 0 : no interrupt 1 : interrupt
3	W1C	0x0	PIX_ERR the pixel number of last line not equal to the set height interrupt 0 : no interrupt 1 : interrupt
2	W1C	0x0	LINE_ERR the line number of last frame not equal to the set height interrupt 0 : no interrupt 1 : interrupt
1	W1C	0x0	LINE_END line end interrupt 0 : no interrupt 1 : interrupt

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Bit	Attr	Reset Value	Description
0	W1C	0x0	DMA_FRAME_END dma frame end interrupt 0 : no interrupt 1 : interrupt

VIP_FOR
Address: Operational Base + offset (0x000c)
VIP format

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19	RW	0x0	UV_STORE_ORDER UV storage order 0 : UVUV 1 : VUVU
18	RW	0x0	RAW_END raw data endian 0 : little end 1 : big end
17	RW	0x0	OUT_420_ORDER output 420 order 0 : UV in the even line 1 : UV in the odd line Note: The first line is even line(line 0).
16	RW	0x0	OUTPUT_420 output 420 or 422 0 : output is 422 1 : output is 420
15	RO	0x0	reserved
14:13	RW	0x0	MIPI_MODE mipi mode 00 : 32 bit bypass mode; 01 : rgb mode; 10 : yuv mode; 11 : reserve;
12:11	RW	0x0	RAW_WIDTH raw data width 00 : 8bit raw data; 01 : 10bit raw data; 10 : 12bit raw data; 11 : reserve;
10	RW	0x0	JPEG_MODE JPEG mode 0 : other mode 1 : mode1

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Bit	Attr	Reset Value	Description
9	RW	0x0	FIELD_ORDER ccir input order 0 : odd field first 1 : even field first
8	RW	0x0	IN_420_ORDER 420 input order 0 : UV in the even line 1 : UV in the odd line Note: The first line is even line(line 0).
7	RW	0x0	INPUT_420 input 420 or 422 0 : 422 1 : 420
6:5	RW	0x0	YUV_IN_ORDER YUV input order 00 : UYVY 01 : YVYU 10 : VYUY 11 : YUYV
4:2	RW	0x0	INPUT_MODE input mode 000 : YUV 010 : PAL 011 : NTSC 100 : RAW 101 : JPEG 110 : MIPI Other : invalid
1	RW	0x0	HREF_POL href input polarity 0 : high active 1 : low active
0	RW	0x0	VSYNC_POL vsync input polarity 0 : low active 1 : high active

VIP_FRM0_ADDR_Y

Address: Operational Base + offset (0x0014)

VIP frame0 y address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM0_ADDR_Y frame0 y address

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VIP_FRM0_ADDR_UV
Address: Operational Base + offset (0x0018)
VIP frame0 uv address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM0_ADDR_UV frame0 uv address

VIP_FRM1_ADDR_Y
Address: Operational Base + offset (0x001c)
VIP frame1 y address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM1_ADDR_Y frame1 y address

VIP_FRM1_ADDR_UV
Address: Operational Base + offset (0x0020)
VIP frame1 uv address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	FRM1_ADDR_UV frame1 uv address

VIP_VIR_LINE_WIDTH
Address: Operational Base + offset (0x0024)
VIP virtual line width

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:0	RW	0x0000	VIR_LINE_WIDTH virtual line width

VIP_SET_SIZE
Address: Operational Base + offset (0x0028)
VIP frame set size

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x01e0	SET_HEIGHT set height
15:13	RO	0x0	reserved
12:0	RW	0x02d0	SET_WIDTH set width

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VIP_CROP
Address: Operational Base + offset (0x0044)
VIP crop start point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	START_Y start y point
15:13	RO	0x0	reserved
12:0	RW	0x0000	START_X start x point

VIP_SCL_CTRL
Address: Operational Base + offset (0x0048)
VIP scale control

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	MIPI_32B_BP mipi 32bit bypass 0 : no bypass 1 : bypass
5	RW	0x0	RAW_16B_BP raw 16 bit bypass 0 : no bypass 1 : bypass
4	RW	0x0	YUV_16B_BP YUV 16 bit bypass 0 : no bypass 1 : bypass
3:0	RO	0x0	reserved

VIP_FIFO_ENTRY
Address: Operational Base + offset (0x0054)
VIP FIFO entry

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17:9	RW	0x000	UV_FIFO_ENTRY valid UV double word in FIFO write 0 clear
8:0	RO	0x000	Y_FIFO_ENTRY valid Y double word in FIFO write 0 clear

VIP_FRAME_STATUS
Address: Operational Base + offset (0x0060)

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VIP frame status

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	FRAME_NUM complete frame number write 0 to clear
15:2	RO	0x0	reserved
1	RO	0x0	F1_STS frame 0 status 0 : frame 1 not ready 1 : frame 1 ready write 0 clear
0	RO	0x0	F0_STS frame 0 status 0 : frame 0 not ready 1 : frame 0 ready write 0 clear

VIP_CUR_DST
Address: Operational Base + offset (0x0064)
VIP current destination address

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CUR_DST current destination address maybe not the current, because the clock synchronization.

VIP_LAST_LINE
Address: Operational Base + offset (0x0068)
VIP last frame line number

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:0	RO	0x0000	LAST_LINE_NUM line number of last frame

VIP_LAST_PIX
Address: Operational Base + offset (0x006c)
VIP last line pixel number

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	LAST_UV_NUM y number of last line
15:13	RO	0x0	reserved
12:0	RO	0x0000	LAST_Y_NUM y number of last line

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VIP_DTE_ADDR
Address: Operational Base + offset (0x0800)
MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR MMU_DTE_ADDR MMU current page Table address

VIP_STATUS
Address: Operational Base + offset (0x0804)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID page fault bus id Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE page fault acess The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x0	REPLAY_BUFFER_EMPTY replay buffer empty status 1'b1:The MMU replay buffer is empty
3	RO	0x0	MMU_IDLE mmu idle status The MMU is idle when accesses are being translated and there are no unfinished translated accesses. 1'b1: MMU is idle
2	RO	0x0	STAIL_ACTIVE stall active status MMU stall mode currently enabled. The mode is enabled by command 1'b1: MMU is in stall active status
1	RO	0x0	PAGE_FAULT_ACTIVE page fault active status MMU page fault mode currently enabled . The mode is enabled by command. 1'b1: page fault is active
0	RO	0x0	PAGING_ENABLED Paging enabled status 1'b0: paging is disabled 1'b1: Paging is enabled

VIP_COMMAND

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Address: Operational Base + offset (0x0808)
MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD mmu cmd MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

VIP_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x080c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR page fault addr address of last page fault

VIP_ZAP_ONE_LINE
Address: Operational Base + offset (0x0810)
MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE zap one line address to be invalidated from the page table cache

VIP_INT_RAWSTAT
Address: Operational Base + offset (0x0814)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error read bus error status
0	RW	0x0	PAGE_FAULT page fault page fault status

VIP_INT_CLEAR
Address: Operational Base + offset (0x0818)
MMU raw interrupt status register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR read bus error write 1 to clear read bus error
0	WO	0x0	PAGE_FAULT page fault clear write 1 to page fault clear

VIP_INT_MASK
Address: Operational Base + offset (0x081c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error enable the read bus interrupt souce when this bit is set to 1'b1
0	RW	0x0	PAGE_FAULT page fault mask enable the page fault interrupt souce when this bit is set to 1'b1

VIP_INT_STATUS
Address: Operational Base + offset (0x0820)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR read bus error status 1'b1:read bus error status
0	RO	0x0	PAGE_FAULT page fault status 1'b1:page fault

VIP_AUTO_GATING
Address: Operational Base + offset (0x0824)
mmu auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_auto_clkgating mmu_auto_clkgating when it is 1'b1, the mmu will auto gating it self

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16.5 Interface Description

Module Pin	Dir	Pad Name	IOMUX Setting
vip_clkout	O	IO_TSPsyncm0_CIFclkoutm0SDMMC0EXTwp_GPIO3A3vccio6	GRF_GPIO3AL_IOMUX[11:9]=3'b010
	O	IOI2S1mclk_NOuse0_TSPsyncm1_CIFclkoutm1_GPIO2B7vccio5	GRF_GPIO2BH_IOMUX[2:0]=3'b100
vip_clkin	I	IO_TSLclk_CIFclkin_SDMMC0EXTclkout_SPIrxdm2_USB3PHYdebug3_I2S2sdim1_GPIO3A2vccio6	GRF_GPIO3AL_IOMUX[8:6]=3'b010
vip_href	I	IO_TSPfail_CIFhref_SDMMC0EXTdet_SPItxdm2_USB3PHYdebug2_I2S2sdom1_GPIO3A1vccio6	GRF_GPIO3AL_IOMUX[5:3]=3'b010
vip_vsync	I	IO_TSPvalid_CIFvsync_SDMMC0EXTcmd_SPIclkm2_USB3PHYdebug1_I2S2sclkm1_GPIO3A0vccio6	GRF_GPIO3AL_IOMUX[2:0]=3'b010
vip_data0	I	IO_TSPd0_CIFda0_SDMMC0EXTd0_UART1tx_USB3PHYdebug4_GPIO3A4vccio6	GRF_GPIO3AL_IOMUX[14:12]=3'b010
vip_data1	I	IO_TSPd1_CIFdata1_SDMMC0EXTd1_UART1rstn_USB3PHYdebug5_GPIO3A5vccio6	GRF_GPIO3AH_IOMUX[2:0]=3'b010
vip_data2	I	IO_TSPd2_CIFdata2_SDMMC0EXTd2_UART1rx_USB3PHYdebug6_GPIO3A6vccio6	GRF_GPIO3AH_IOMUX[5:3]=3'b010
vip_data3	I	IO_TSPd3_CIFdata3_SDMMC0EXTd3_UART1ctsn_USB3PHYdebug7_GPIO3A7vccio6	GRF_GPIO3AH_IOMUX[8:6]=3'b010
vip_data4	I	IO_TSPd4_CIFdata4_SPIcsn0m2_I2S2lrcktxm1_USB3PHYdebug8_I2S2lrckrxm1_GPIO3B0vccio6	GRF_GPIO3BL_IOMUX[2:0]=3'b010
vip_data5	I	IO_TSPd5m0_CIFdata5m0_GPIO3B1vccio6	GRF_GPIO3BH_IOMUX[3:2]=2'b10
	I	IO_I2S1lrckrx_NOuse1_TSPd5m1_CIFdata5m1_GPIO2C0vccio5	GRF_GPIO2CL_IOMUX[2:0]=3'b100
vip_data6	I	IO_TSPd6m0_CIFdata6m0_GPIO3B2vccio6	GRF_GPIO3BH_IOMUX[5:4]=2'b10
	I	IO_I2Slrcktx_SPDIFtxm1_TSPd6m1_GPIO2C1vccio5	GRF_GPIO2CL_IOMUX[5:3]=3'b100
vip_data7	I	IO_TSPd7m0_CIFdata7m0_GPIO3B3vccio6	GRF_GPIO3BH_IOMUX[7:6]=2'b10
	I	IO_I2S1sclk_PDMclkm0_TSPd7m1_CIFdata7m1_GPIO2C2vccio5	GRF_GPIO2CL_IOMUX[8:6]=3'b100

16.6 Application Notes

The biggest configuration requirement of all operations is the CAP_EN bit must be set after all the mode selection is ready. The configuration order of the input/output data format, YUV order, the address, frame size/width, AXI burst length and other options do not need to care. There are many debug registers to make it easy to read the internal operation information of VIP. The valid pixel number of scale result in FIFO can be known by read VIP_SCL_VALID_NUM. The line number of last frame and the pixel number of last line can be also known by read the VIP_LAST_LINE and VIP_LAST_PIX.

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Chapter 18 Video Output Processor (VOP)

16.7 Overview

VOP is the display interface from memory frame buffer to display device (HDMI and TV set). VOP is connected to an AHB bus through an AHB slave and AXI bus through an AXI master. The register setting is configured through the AHB slave interface and the display frame data is read through the AXI master interface.

16.7.1 Features

The features listed below which may or may not be present in actual product, may be subject to the third party licensing requirements. Please contact Rockchip for actual product feature configurations and licensing requirements.

- Display interface
  - TV Interface
    - 10bit TVE output
    - support PAL/NTSC
  - HDMI interface
    - Support 480p/576p/720p/1080p/2160p/1080i/
    - Support RGB/YUV420 format
  - Max resolution
    - Max input resolution：4096x2304
    - Max output resolution：4096x2160
  - Scaning timing 8192x4096
  - Support configurable polarity of DCLK/HSYNC/VSYNC/DEN
- Display process
  - BCSH,10bit
    - Brightness,Contrast,Saturation,Hue adjustment
    - YUV-10bit，RGB-10bit
  - Support display data swap
    - BG swap, RB swap, RG swap, dummy swap
  - Support three YUV2RGB transition modes:
    - 10bit-YUV: rec601-mpeg/rec601-jpeg/rec709-hd/bt2020
  - Support two RGB2YUV transition modes:
    - 10bit-RGB: rec601-full/rec601-jpeg/rec709-hd/bt2020
  - blank display
  - black display
  - standby mode
  - X-MIRROR,Y-MIRROR for win0/win1/hwc
  - scale down for TV overscan
    - after overlay
    - arbitrary non-integer scaling ratio
    - horizontal scale down using bilinear, 0.5~1.0
    - vertical scale down using bilinear, 0.5~1.0
- Layer process
  - Background layer
    - programmable 30-bit color
  - Win0/Win1/Win2 layer
    - Support data format
      - RGB888, ARGB888, RGB565,
      - YCbCr420SP, YCbCr422SP, YCbCr444SP
      - YUV-10bit
    - YUV clip
      - Y-8bit: 16~235;UV-10bit: 16~240
    - Support max input resolution 4096x8192
    - Support max output resolution 4096x2160

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- Support virtual display
- Support 1/8 to 8 scaling-down and scaling-up engine
  - scale up using bicubic and bilinear
  - scale down using bilinear and average
  - per-pix alpha + scale
- Support data swap
  - RGB/BPP: rb_swap
  - YUV: mid_swap,uv_swap
- transparency color key,prior to alpha blending and fading
- Support fading,prior to alpha blending
- Support alpha blending
- Support interlace and de-flicker for interlace output

- Hardware Cursor layer(HWC for short)
  - Support data format
    - RGB888, ARGB888, RGB565
    - little endian and big endian for BPP
  - Support four hwc size: 32x32,64x64,96x96,128x128
  - Support 2 color modes: normal and reversed color
  - Support fading,prior to alpha blending
  - Support alpha blending
  - Support displaying out of panel,right or bottom
  - Support NORMAL color and REVERSE color mode
  - Support interlace read and interlace output
- Overlay
  - support RGB and YUV domain overlay(display format is not hdr10 or bt2020)
  - Support 6 layers,background/win0/win1/win2/hwc
  - Win1/Win2 overlay position exchangeable
  - Alpha blending
    - Support 12 alpha blending modes
    - Support pre-multiplied alpha
    - Support global alpha and per_pix alpha
    - Support 256 level alpha
    - layer1/layer2/layer3/hwc support alpha
- HDR10
  - Win0 supports HDR2SDR and SDR2HDR
  - Other layers support SDR2HDR

- Bus interface
  - Support AMBA 2.0 AHB slave interface for accessing internal registers and LUT memories，32bit data bus width
  - Support AMBA 3.0 AXI master read interface for loading frame data
    - 128bit data bus width
  - Support MMU
  - Support two transfer modes
    - auto outstanding transfer
    - configuruable outstanding transfer(gather transfer)
  - DMA line mode for YUV
  - Support NOC hurry for higher bus prioirity for win0/win1
  - Support DMA stop mode
  - Win AXI read ID configurable
  - Max read outstanding number
    - 32 when MMU disable
    - 31 when MMU enable
- Interrupt
  - One combined interrupt
    - high active
    - raw status readable
    - combinational with interrupt sources

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16.8 Block Diagram

The architecture is shown in the following figure.

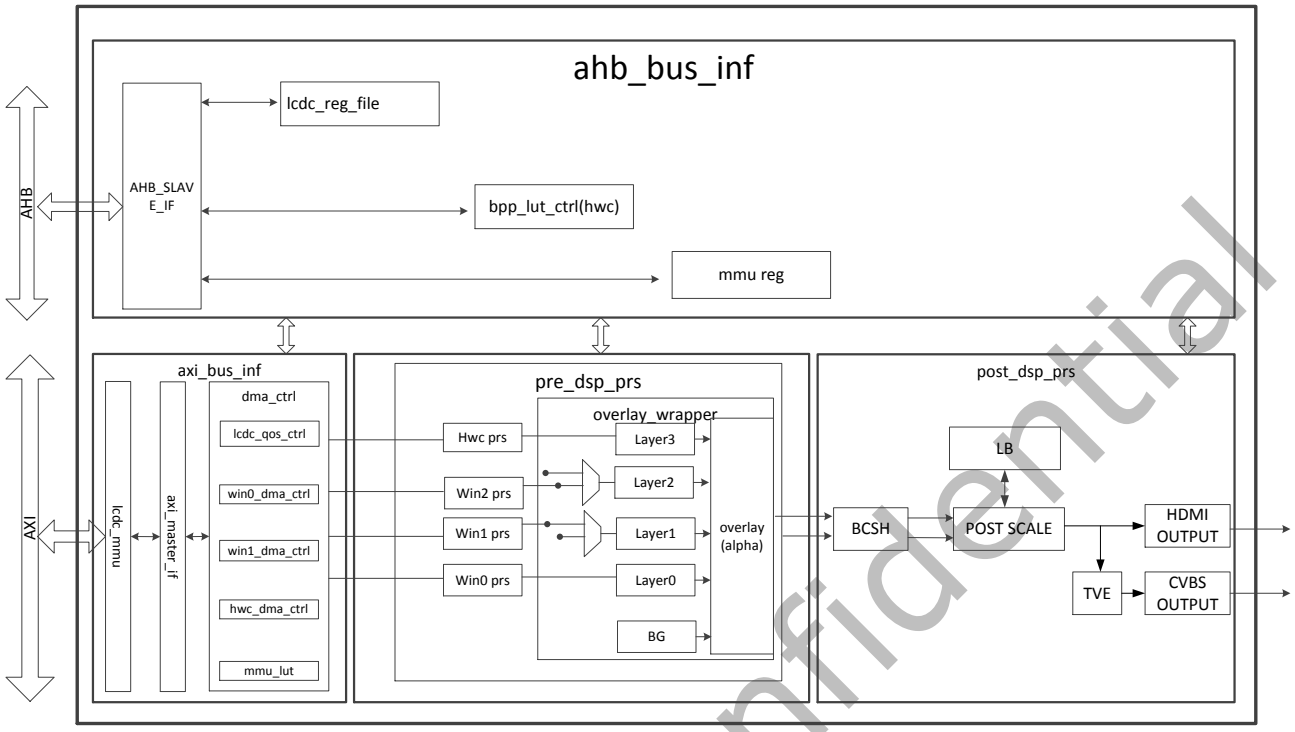

Fig. 18-1 VOP Block Diagram

16.9 Function Description

16.9.1 Pixel format

1.RGB

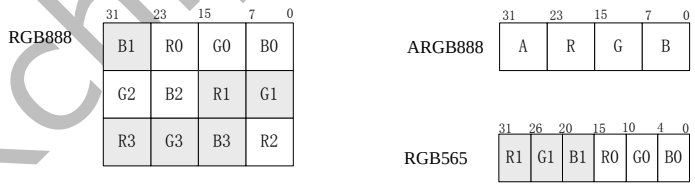

Fig. 18-2 RGB data format

2.YCbCr(8bit)

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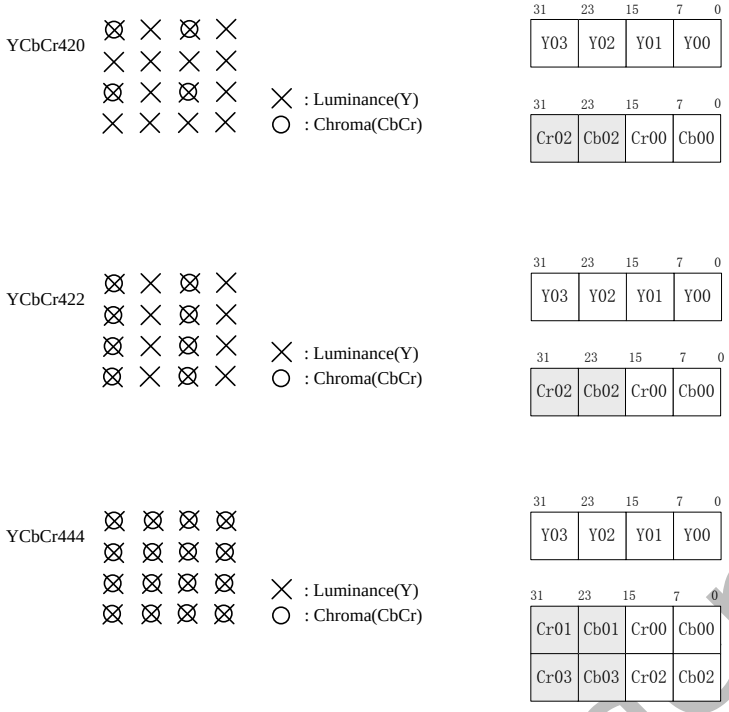

Fig. 18-3 YCbCr data format

YCbCr just support SP

YCbCr-8bit 32bit align

2.YCbCr(10bit)

YCbCr-10bit is the same as YCbCr-8bit except it is 640bit align(64 pixels alignn).

3.BPP

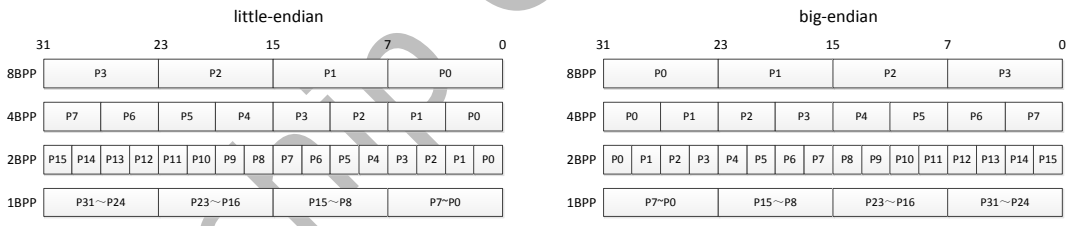

Fig. 18-4 BPP little/big endian data format

16.9.2 Pixel Data Path

There are only one data input path for VOP to get display layers' pixel data: internal DMA.

1.Internal DMA

Internal DMA can fetch the pixel data through AXI bus from system memory (DDR) for all the display layers. Data fetching is driven by display output requirement.

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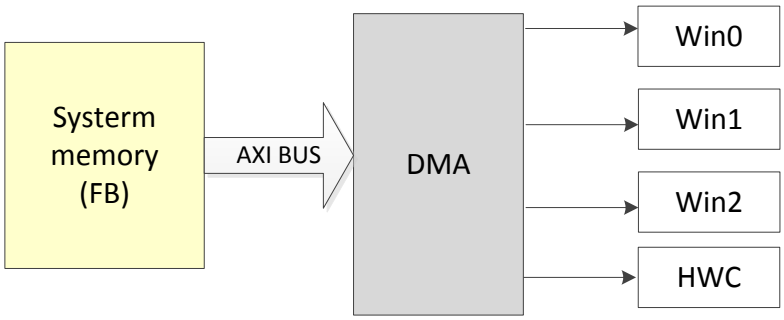

Fig. 18-5 VOP Internal DMA

16.9.3 Win Scaling

he scaling operation is the image resizing process by scaling-up or scaling-down the source image from active window size to display window size for displaying on LCD panel or TV set.

Horizontal scaling and vertical scaling are realized independently.

1.Scaling factor

（1）scale down or scale up using bilinear

Factor = ((src*2-3)<<11)/(dst-1);

（2）scale up using bicubic

Factor = ((src*2-3)<<15)/(dst-1);

（3）scale down using average

Factor = (dst<<17)/(src*2-1);

2.win scale line buffer mode

For YUV422/YUV420,

(1)LB_YUV_4096X5

If horizontal scale down and dsp_width is greater than 2560 or if horizontal scale up and act_width is greater than 2560,please config win_lb_mode as LB_YUV_4096X5;

(2)LB_YUV_2560X8

If horizontal scale down and dsp_width is less or equal than 2560 or if horizontal scale up and act_width is less or equal than 2560,please config win_lb_mode as LB_YUV_4096X5;

For YUV444/RGB,

(1)LB_RGB_4096X2

If horizontal scale down and dsp_width is greater than 2560 or if horizontal scale up and act_width is greater than 2560,please config win_lb_mode as LB_YUV_4096X2;

This mode does not support vertical scale.

(2)LB_RGB_2560X4

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horizontal scale down and dsp_width is greater than 1920 or if horizontal scale up and act_width is greater than 1920,please config win_lb_mode as LB_RGB_2560X4;

This mode only support bilinear for vertical scale up.

(3)LB_RGB_1920X5

horizontal scale down and dsp_width is greater than 1280 or if horizontal scale up and act_width is greater than 1280,please config win_lb_mode as LB_RGB_1920X5;

(4)LB_RGB_1280X8

horizontal scale down and dsp_width is less or equal than 2560 or if horizontal scale up and act_width is less or equal than 2560,please config win_lb_mode as LB_RGB_1280X8;

16.9.4 P2I

It is necessary to display a non-interlaced video signal on an interlaced display panel (such as TV set). Thus "non-interlaced-to-interlaced conversion" is required, we call it P2I.

When interlaced is required, no matter even field or odd field, we get the whole image from bus. After overlay, we discard the odd lines when even field(even lines when odd field). This method can be better.

16.9.5 Virtual display

When in virtual display, the active image is part of the virtual (original) image in frame buffer memory.

The virtual width is indicated by setting VIR_STRIDE for different data format. Note that RGB/BPP has one stride(yrgb_vir_stride)； YCbCr has two virtual stride(yrgb_vir_stride and cbcr_vir_stride).

For RGB-8bit and YUV-8bit，the stride should be multiples of word (32-bit), with dummy bytes in the end of virtual line if the original width is not 32-bit aligned.

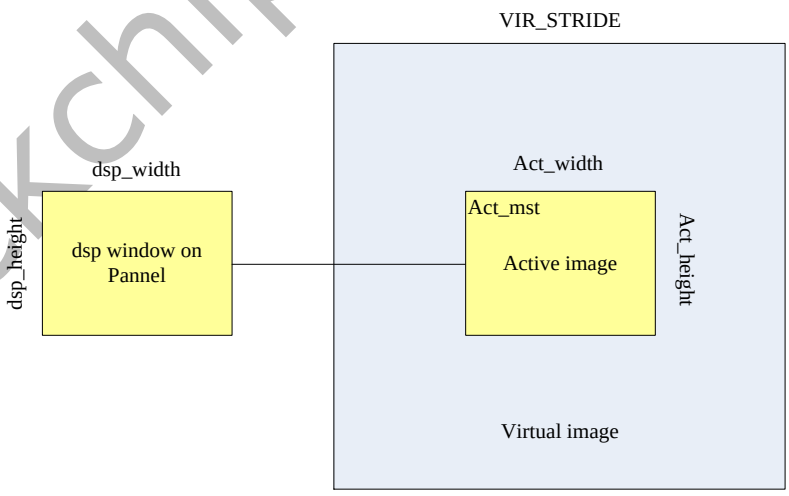

Fig. 08-6 Virtual display

16.9.6 MIRROR display

Mirror display is necessary for the panel with mirror timing interface. There are two types of mirror mode: horizontal mirror(X-mirror) and vertical mirror(Y-mirror).

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Win0/1 support X-mirror and Y-mirror;

VOP support X-mirror and Y-mirror after overlay.

The default display order is from left to right(L2R) in horizontal direction and from top to bottom(T2B) in vertical direction. However, when X-Mirror is enable, the horizontal display order is from right to left(R2L); when Y-MIRROR is enable, the vertical display order is from bottom to top(B2T).

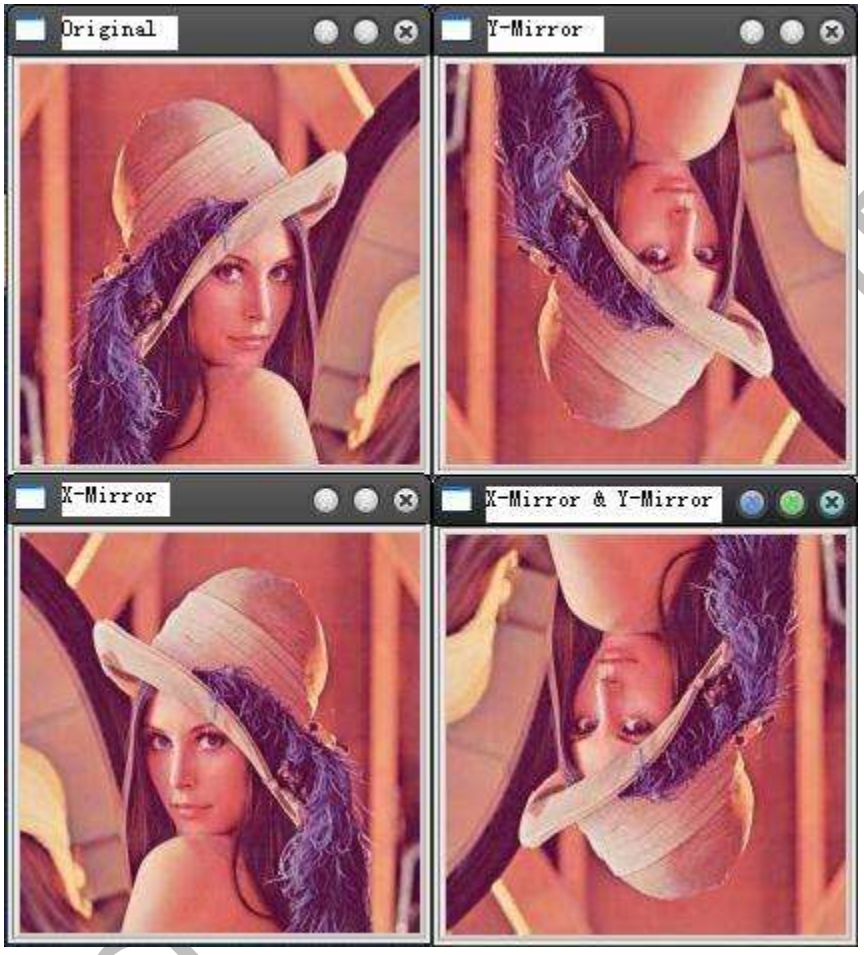

Fig. 08-7 X-Mirror and Y-Mirror

16.9.7 Display process

1. Overlay display

There are totally 4 layers for overlay display: Background, layer0, layer1, and hardware cursor layer(HWC).

Background is a programmable solid color layer, which is always in the bottom of the display screen.

HWC is always on the top of the display screen.

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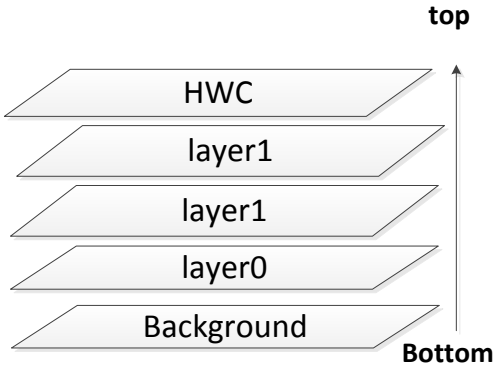

Fig. 08-8 overlay

Following figure is an example of overlay display for win0,win1 and hwc.

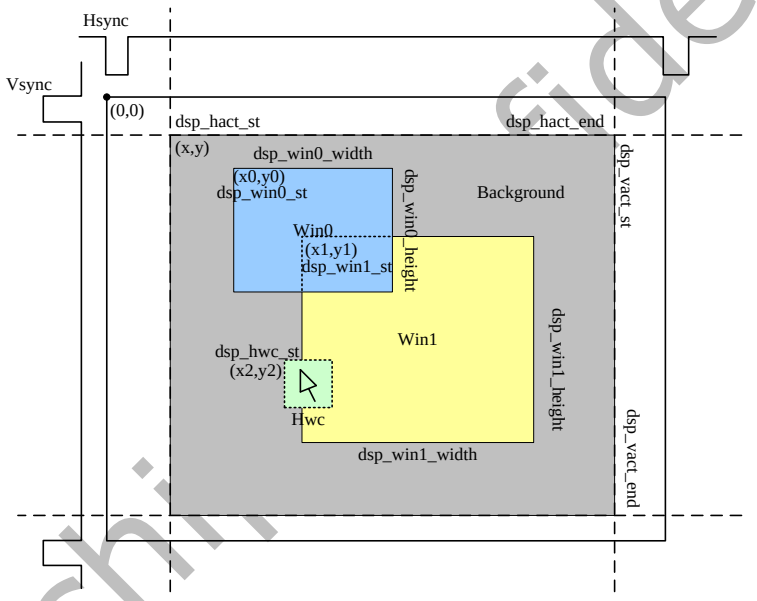

Fig. 08-9 overlay timing

2. Post scale down

Post scale down after overlay is supported to fix overscan,that draws the borders of the image beyond the normally visiable area on the screen.

The scale ratio of post scale down is 0.5~1.

Post timing setting

The post scale parameter ,such as,post_dsp_hact_st,post_dsp_hact_end, post_dsp_vact_st,post_dsp_vact_end can be configured.

When post scaling equal "1" ,the post scaler parameter are the same as dsp timing parameter.

eg:

post_dsp_hact_st = dsp_hact_st

post_dsp_hact_end = dap_hact_end

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post_dsp_vact_st = dsp_vact_st

post_dsp_vact_end = dsp_vact_end

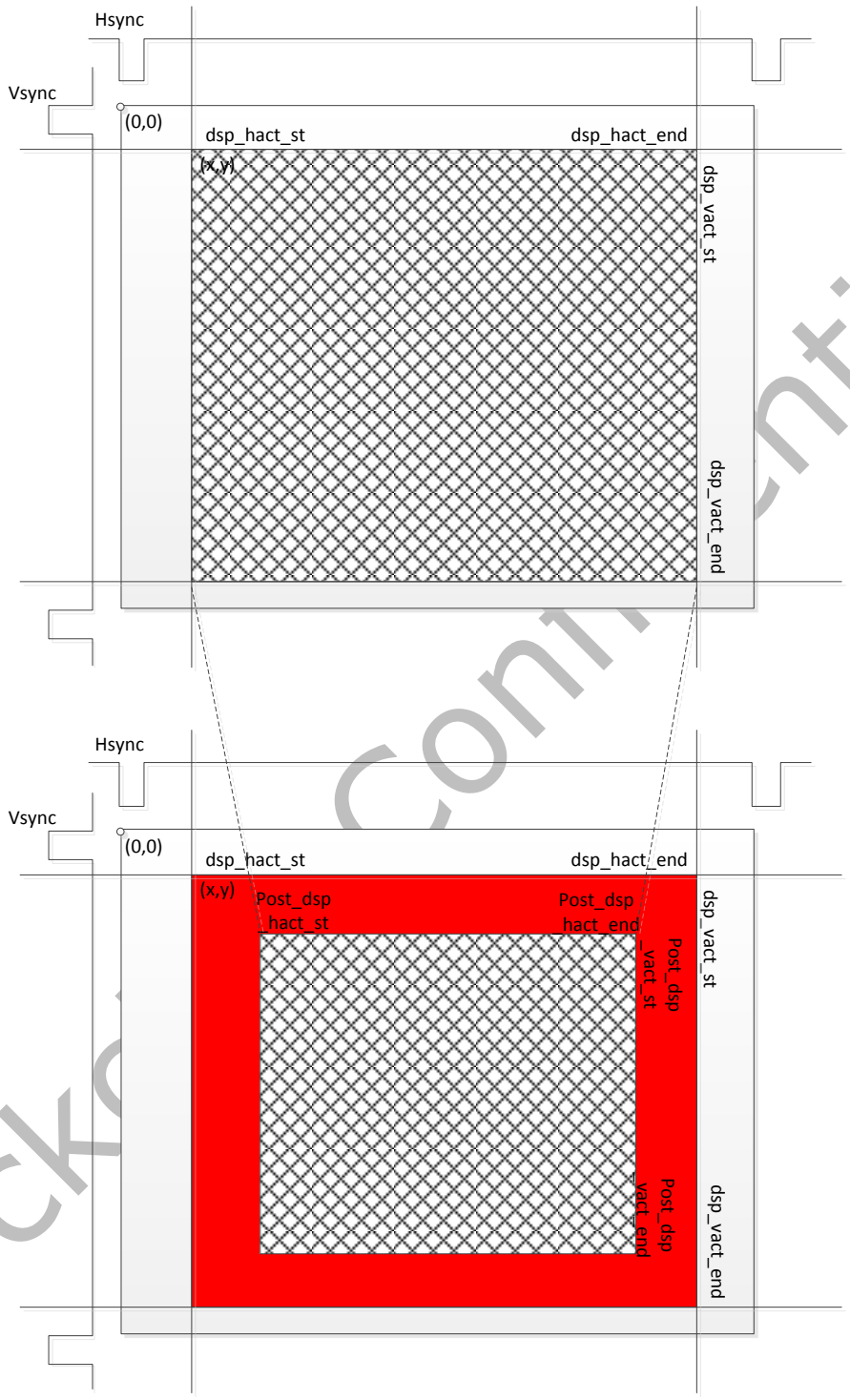

Fig. 08-10 post scaling timing

Post scale down factor

For horizontal scale down,factor = ((src_width*2-3)<<11)/(dst_width-1).

For vertical scale down,factor = ((src_width*2-3)<<11)/(dst_width-1).

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3. Transparency color key

The transparency color key value defines the pixel treated as transparent pixel. The pixel whose value is equal to the color key value could not be visible on the screen, instead of the pixel in the under layer or solid background color.

There are two transparency color key for win0 layer and win1 layer respectively. When color key is enable, the transparency process is done after scaling but before YUV2RGB color space converter.

Moreover, transparency color key is just available for non-scaling mode.

Following figure is an example of transparency color key for win0 and win1.

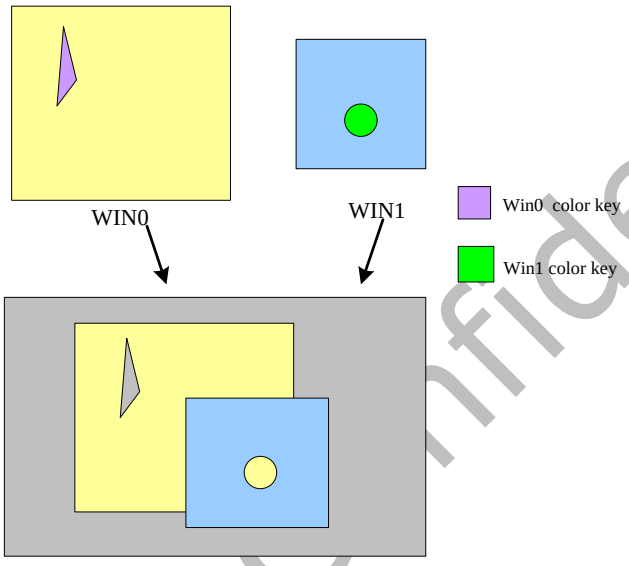

Fig. 08-11 Transparency Color Key

4. Replication(dither up)

If the size of panel data bus is lager than the size of source pixel data,i.e, the source input format is RGB565 and display output format is RGB888, you could do bit replication by replicating MSBs to LSBs if replication is enable (VOP_DSP_CTRL0[9]=1) or filling with "0" to LSBs if replication is disable (VOP_DSP_CTRL0[9]=0).

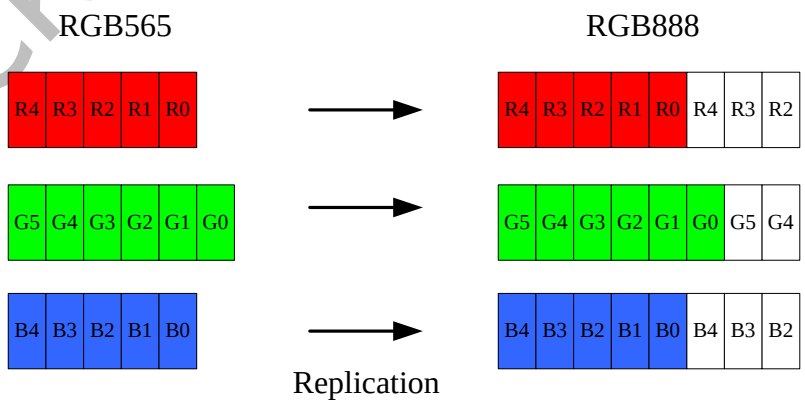

Fig. 08-12 Dither Up

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5. Alpha blending

There are 12 alpha blending mode between two overlay layers for layer1/layer2/hwc. Layer0 does not support alpha blending with background.VOP supports overlay on RGB or YUV domain.

When in per-pixel mode, the alpha value for every pixel is following with the pixel data. i.e, aRGB，and can be scaled like RGB data. Therefore it is just suitable for win0/win1/hwc layer with ARGB data format.

The alpha blending architechture is shown as follows.

Table 08-1 alpha blending mode settings

Blending Mode	Cs’	Fs	Cd’	Fd
AA_USER_DEFINED	X	User defined	Cd	User defined
AA_CLEAR	X	0	Cd	0
AA_SRC	X	0	Cd	1
AA_DST	X	1	Cd	1
AA_SRC_OVER	Cs	1	Cd	1-As’’
AA_DST_OVER	Cs	1-As’’	Cd	1
AA_SRC_IN	Cs	As’’	Cd	0
AA_DST_IN	X	0	Cd	As’’
AA_SRC_OUT	Cs	1-As’’	Cd	0
AA_DST_OUT	X	0	Cd	1-As’’
AA_SRC_ATOP	Cs	As’’	Cd	1-As’’
AA_DST_ATOP	Cs	1-As’’	Cd	As’’
AA_XOR	Cs	1-As’’	Cd	1-As’’
AA_SRC_OVER_GLOBAL	Cs*As’’	Ags’’	Cd	1-As’’

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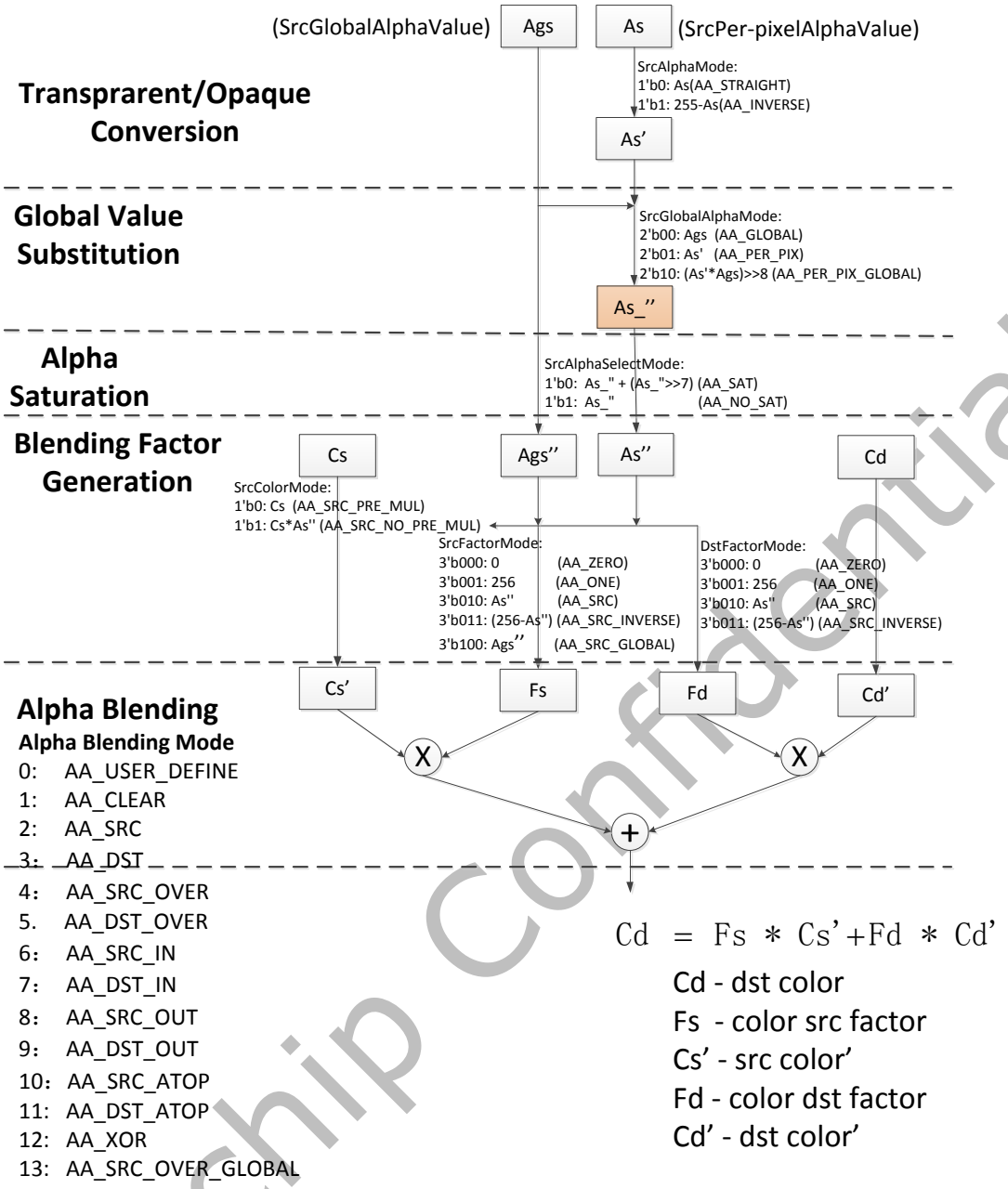

Fig. 08-13 alpha configuration flow

Pseudo Code:

```
switch(alpha_config->alpha_blending_mode)
{
    case AA_USER_DEFINE:
    break;
    case AA_CLEAR:
    alpha_config->src_factor_mode=AA_ZERO;
    alpha_config->dst_factor_mode=AA_ZERO;
    break;
    case AA_SRC:
    alpha_config->src_factor_mode=AA_ONE;
```

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```
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_DST:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_ONE;
break;
case AA_SRC_OVER:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_ONE;
alpha_config->dst_factor_mode=AA_SRC_INVERSE;
break;
case AA_DST_OVER:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC_INVERSE;
alpha_config->dst_factor_mode=AA_ONE;
break;
case AA_SRC_IN:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC;
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_DST_IN:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_SRC;
break;
case AA_SRC_OUT:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
alpha_config->src_factor_mode=AA_SRC_INVERSE;
alpha_config->dst_factor_mode=AA_ZERO;
break;
case AA_DST_OUT:
alpha_config->src_factor_mode=AA_ZERO;
alpha_config->dst_factor_mode=AA_SRC_INVERSE;
break;
case AA_SRC_ATOP:
alpha_config->src_color_mode=AA_SRC_PRE_MUL;
```

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```
        alpha_config->src_factor_mode=AA_SRC;
        alpha_config->dst_factor_mode=AA_SRC_INVERSE;
        break;
        case AA_DST_ATOP:
        alpha_config->src_color_mode=AA_SRC_PRE_MUL;
        alpha_config->src_factor_mode=AA_SRC_INVERSE;
        alpha_config->dst_factor_mode=AA_SRC;
        break;
        case AA_XOR:
        alpha_config->src_color_mode=AA_SRC_PRE_MUL;
        alpha_config->src_factor_mode=AA_SRC_INVERSE;
        alpha_config->dst_factor_mode=AA_SRC_INVERSE;
        break;
        case AA_SRC_OVER_GLOBAL:
        alpha_config->src_global_alpha_mode=AA_PER_PIX_GLOBAL;
        alpha_config->src_color_mode=AA_SRC_NO_PRE_MUL;
        alpha_config->src_factor_mode=AA_SRC_GLOBAL;
        alpha_config->dst_factor_mode=AA_SRC_INVERSE;
        break;
        default:
            printf("alpha mode error\n");
            break;
    }
```

6. Brightness,Contrast,Saturation,Hue(BCSH)

BCSH is used to adjust “Brightness,Contrast,Saturation,Hue”,like IEP BCSH-8bit. For details,please refer to IEP chapter. The brightness adjust support (-32,31).The yuv data of color bar are 8bits.

7. Color space conversion

There are 4 standards for YUV2RGB and RGB2YUV. For yuv2yuv ,we can use change among BT601L, BT601F, BT709L, BT2020 using coefficient registers.

YUV2RGB:

1. yuv to rgb (BT601L)

R = 1.164(Y-16) + 1.596(V-128)

G = 1.164(Y-16) - 0.391(U-128) - 0.813(V-128)

B = 1.164(Y-16) + 2.018(U-128)

2. yuv to rgb (BT601F)

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R = (Y-16) + 1.402(V-128)

G = (Y-16) - 0.344(U-128) - 0.714(V-128)

B = (Y-16) + 1.772(U-128)

3. yuv to rgb (BT709L)

R = 1.164(Y-16) + 1.793(V-128)

G = 1.164(Y-16) - 0.213(U-128) - 0.534(V-128)

B = 1.164(Y-16) + 2.115(U-128)

4. yuv to rgb(BT2020)

R = 1.1636(Y-64) + 1.6778(V-512)

G = 1.1636(Y-64) - 0.1872(U-512) - 0.6501(V-512)

B = 1.1636(Y-64) + 2.1406(U-512)

RGB2YUV:

RGB2YUV

1. rgb to yuv(BT601L)

Y = 0.257R + 0.504G + 0.098B + 16

Cb =-0.148R - 0.291G + 0.439B + 128

Cr = 0.439R - 0.368G - 0.071B + 128

2. rgb to yuv(BT601F)

Y = 0.299R + 0.5870G + 0.1140B + 0

Cb = -0.1687R - 0.3313G + 0.5000B + 512

Cr = 0.500R - 0.4187G - 0.0813B + 512

3. rgb to yuv(BT709L)

Y = 0.183R + 0.614G + 0.062B + 16

Cb =-0.101R - 0.338G + 0.439B + 128

Cr = 0.439R - 0.399G - 0.040B + 128

3. rgb to yuv(BT2020)

Y = 0.2250R + 0.5807G + 0.0508B + 64

Cb = -0.1223R - 0.3157G + 0.4380B + 512

Cr = 0.4380R - 0.4028G - 0.0352B + 512

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8. Pre-Dither Down

Dithering is an intentional applied form of noise, using to randomize quantization error, and thereby preventing large-scaling patterns such as "banding".

The pixel value is used by dithering process to display the data in a lower color depth on the LCD panel, i.e, the source input format is RGB101010 and display output format is RGB888. When dithering is enable(VOP_DSP_CTRL0[2]=1), the output data is generated by dithering algorithm based on the pixel position and the value of removed bits. Otherwise, the MSBs of the pixel color components are output as display data.

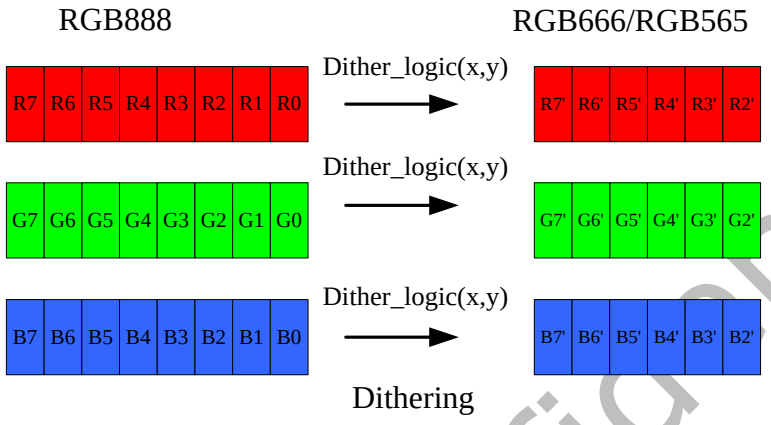

Fig. 08-14 Pre-Dither Down

9. Output format

Config dsp_out_mode register to adapt a variety of panel interface. As follow:

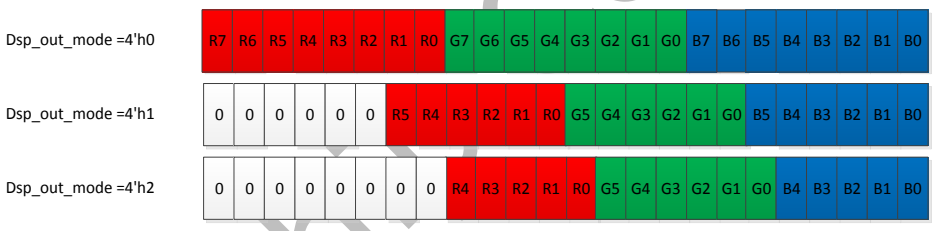

Fig. 08-15 dsp_out_mode description

10. HDR10

HDR10 consists of HDR2SDR and SDR2HDR as follow :

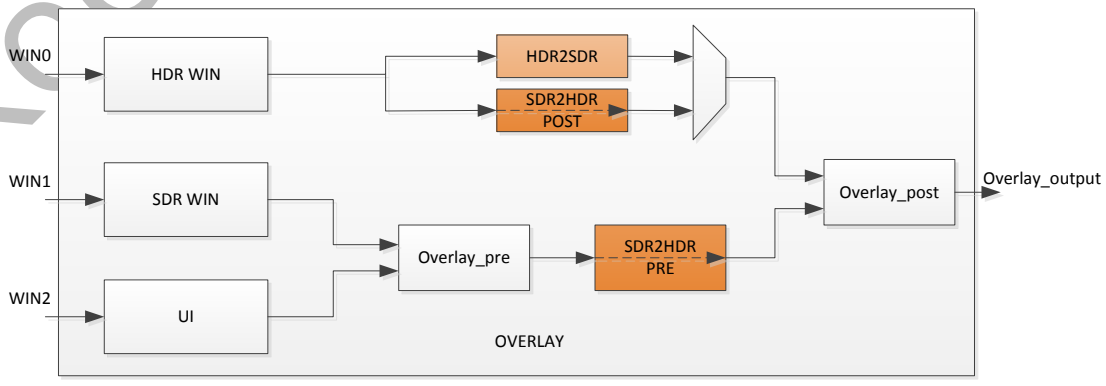

Fig. 08-19 HDR10 in overlay

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16.10 Register Description

16.10.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

16.10.2 VOP BASE Registers Summary

Name	Offset	Size	Reset Value	Description
VOP_REG_CFG_DONE	0x0000	W	0x00000000	Register config done flag
VOP_VERSION_INFO	0x0004	W	0x00000000	Version for vop
VOP_SYS_CTRL	0x0008	W	0x00821800	System control register0
VOP_SYS_CTRL1	0x000c	W	0x0003a000	System control register1
VOP_DSP_CTRL0	0x0010	W	0x00000000	Display control register0
VOP_DSP_CTRL1	0x0014	W	0x0000e400	Display control register1
VOP_DSP_BG	0x0018	W	0x00000000	Background color
VOP_AUTO_GATING_EN	0x003c	W	0x00000000	
VOP_LINE_FLAG	0x0040	W	0x00000000	Line flag config register
VOP_VOP_STATUS	0x0044	W	0x00000000	vop status register
VOP_BLANKING_VALUE	0x0048	W	0x00000000	Register0000 Abstract
VOP_WIN0_DSP_BG	0x0050	W	0x00000000	Win0 layer background color
VOP_WIN1_DSP_BG	0x0054	W	0x00000000	Win1 layer background color
VOP_DBG_PERF_LATENCY_CTRL0	0x00c0	W	0x00000000	Axi performance latency module contrl register0
VOP_DBG_PERF_RD_MAX_LATENCY_NUM0	0x00c4	W	0x00000000	Read max latency number
VOP_DBG_PERF_RD_LATENCY_THR_NUM0	0x00c8	W	0x00000000	The number of bigger than configed threshold value
VOP_DBG_PERF_RD_LATENCY_SAMP_NUM0	0x00cc	W	0x00000000	Total sample number
VOP_INTR_EN0	0x00e0	W	0x00000000	Interrupt enable register
VOP_INTR_CLEAR0	0x00e4	W	0x00000000	Interrupt clear register
VOP_INTR_STATUS0	0x00e8	W	0x00000000	interrupt status
VOP_INTR_RAW_STATUS0	0x00ec	W	0x00000000	raw interrupt status
VOP_INTR_EN1	0x00f0	W	0x00000000	Interrupt enable register
VOP_INTR_CLEAR1	0x00f4	W	0x00000000	Interrupt clear register
VOP_INTR_STATUS1	0x00f8	W	0x00000000	interrupt status
VOP_INTR_RAW_STATUS1	0x00fc	W	0x00000000	raw interrupt status
VOP_WIN0_CTRL0	0x0100	W	0x3a000040	Win0 ctrl register0
VOP_WIN0_CTRL1	0x0104	W	0x00000000	Win0 ctrl register1
VOP_WIN0_COLOR_KEY	0x0108	W	0x00000000	Win0 color key register
VOP_WIN0_VIR	0x010c	W	0x01400140	Win0 virtual stride
VOP_WIN0_YRGB_MST	0x0110	W	0x00000000	Win0 YRGB memory start address
VOP_WIN0_CBR_MST	0x0114	W	0x00000000	Win0 Cbr memory start address
VOP_WIN0_ACT_INFO	0x0118	W	0x00ef013f	Win0 active window width/height

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Name	Offset	Size	Reset Value	Description
VOP_WIN0_DSP_INFO	0x011c	W	0x00ef013f	Win0 display width/height on panel
VOP_WIN0_DSP_ST	0x0120	W	0x000a000a	Win0 display start point on panel
VOP_WIN0_SCL_FACTOR_YRGB	0x0124	W	0x10001000	Win0 YRGB scaling factor
VOP_WIN0_SCL_FACTOR_CBR	0x0128	W	0x10001000	Win0 Cbr scaling factor
VOP_WIN0_SCL_OFFSET	0x012c	W	0x00000000	Win0 scaling start point offset
VOP_WIN0_SRC_ALPHA_CTRL	0x0130	W	0x00000000	Win0 alpha source control register
VOP_WIN0_DST_ALPHA_CTRL	0x0134	W	0x00000000	Win0 alpha destination control register
VOP_WIN0_FADING_CTRL	0x0138	W	0x00000000	Win0 fading contrl register
VOP_WIN0_CTRL2	0x013c	W	0x00000021	Win0 ctrl register2
VOP_DBG_WIN0_REG0	0x01f0	W	0x00000000	Vop debug win0 register0
VOP_DBG_WIN0_REG1	0x01f4	W	0x00000000	Vop debug win0 register1
VOP_DBG_WIN0_REG2	0x01f8	W	0x00000000	Vop debug win0 register2
VOP_DBG_WIN0_RESERVED	0x01fc	W	0x00000000	Vop debug win0 register3 reserved
VOP_WIN1_CTRL0	0x0200	W	0x3a000040	Win1 ctrl register0
VOP_WIN1_CTRL1	0x0204	W	0x00000000	Win1 ctrl register1
VOP_WIN1_COLOR_KEY	0x0208	W	0x00000000	Win1 color key register
VOP_WIN1_VIR	0x020c	W	0x01400140	win1 virtual stride
VOP_WIN1_YRGB_MST	0x0210	W	0x00000000	Win1 YRGB memory start address
VOP_WIN1_CBR_MST	0x0214	W	0x00000000	Win1 Cbr memory start address
VOP_WIN1_ACT_INFO	0x0218	W	0x00ef013f	Win1 active window width/height
VOP_WIN1_DSP_INFO	0x021c	W	0x00ef013f	Win1 display width/height on panel
VOP_WIN1_DSP_ST	0x0220	W	0x000a000a	Win1 display start point on panel
VOP_WIN1_SCL_FACTOR_YRGB	0x0224	W	0x10001000	Win1 YRGB scaling factor
VOP_WIN1_SCL_FACTOR_CBR	0x0228	W	0x10001000	Win1 Cbr scaling factor
VOP_WIN1_SCL_OFFSET	0x022c	W	0x00000000	Win1 scaling start point offset
VOP_WIN1_SRC_ALPHA_CTRL	0x0230	W	0x00000000	Win1 alpha source control register
VOP_WIN1_DST_ALPHA_CTRL	0x0234	W	0x00000000	Win1 alpha destination control register
VOP_WIN1_FADING_CTRL	0x0238	W	0x00000000	Win1 fading contrl register
VOP_WIN1_CTRL2	0x023c	W	0x00000043	Win1 ctrl register2
VOP_DBG_WIN1_REG0	0x02f0	W	0x00000000	Vop debug win1 register0
VOP_DBG_WIN1_REG1	0x02f4	W	0x00000000	Vop debug win1 register1
VOP_DBG_WIN1_REG2	0x02f8	W	0x00000000	Vop debug win1 register2
VOP_DBG_WIN1_RESERVED	0x02fc	W	0x00000000	Vop debug win1 register3 reserved
VOP_WIN2_CTRL0	0x0300	W	0x3a000040	Win2 ctrl register0

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Name	Offset	Size	Reset Value	Description
VOP_WIN2_CTRL1	0x0304	W	0x00000000	Win2 ctrl register1
VOP_WIN2_COLOR_KEY	0x0308	W	0x00000000	Win2 color key register
VOP_WIN2_VIR	0x030c	W	0x01400140	win2 virtual stride
VOP_WIN2_YRGB_MST	0x0310	W	0x00000000	Win2 YRGB memory start address
VOP_WIN2_CBR_MST	0x0314	W	0x00000000	Win2 Cbr memory start address
VOP_WIN2_ACT_INFO	0x0318	W	0x00ef013f	Win2 active window width/height
VOP_WIN2_DSP_INFO	0x031c	W	0x00ef013f	Win2 display width/height on panel
VOP_WIN2_DSP_ST	0x0320	W	0x000a000a	Win2 display start point on panel
VOP_WIN2_SCL_FACTOR_YRGB	0x0324	W	0x10001000	Win2 YRGB scaling factor
VOP_WIN2_SCL_FACTOR_CBR	0x0328	W	0x10001000	Win2 Cbr scaling factor
VOP_WIN2_SCL_OFFSET	0x032c	W	0x00000000	Win2 scaling start point offset
VOP_WIN2_SRC_ALPHA_CTRL	0x0330	W	0x00000000	Win2 alpha source control register
VOP_WIN2_DST_ALPHA_CTRL	0x0334	W	0x00000000	Win2 alpha destination control register
VOP_WIN2_FADING_CTRL	0x0338	W	0x00000000	Win2 fading contrl register
VOP_WIN2_CTRL2	0x033c	W	0x00000043	Win2 ctrl register2
VOP_DBG_WIN2_REG0	0x03f0	W	0x00000000	Vop debug win2 register0
VOP_DBG_WIN2_REG1	0x03f4	W	0x00000000	Vop debug win2 register1
VOP_DBG_WIN2_REG2	0x03f8	W	0x00000000	Vop debug win2 register2
VOP_DBG_WIN2_RESERVED	0x03fc	W	0x00000000	Vop debug win2 register3 reserved
VOP_HWC_CTRL0	0x0500	W	0x00000000	Hwc ctrl register0
VOP_HWC_CTRL1	0x0504	W	0x00701d00	Hwc ctrl register1
VOP_HWC_MST	0x0508	W	0x00000000	Hwc memory start address
VOP_HWC_DSP_ST	0x050c	W	0x000a000a	Hwc display start point on panel
VOP_HWC_SRC_ALPHA_CTRL	0x0510	W	0x00000000	Hwc alpha source control register
VOP_HWC_DST_ALPHA_CTRL	0x0514	W	0x00000000	Hwc alpha destination control register
VOP_HWC_FADING_CTRL	0x0518	W	0x00000000	Hwc fading contrl register
VOP_HWC_RESERVED1	0x051c	W	0x00000000	Hwc reserved
VOP_POST_DSP_HACT_INFO	0x0600	W	0x000a014a	Post scaler down horizontal start and end
VOP_POST_DSP_VACT_INFO	0x0604	W	0x000a00fa	Panel active horizontal scanning start point and end point
VOP_POST_SCL_FACTOR_YRGB	0x0608	W	0x10001000	Post yrgb scaling factor
VOP_POST_RESERVED	0x060c	W	0x00000000	Post reserved
VOP_POST_SCL_CTRL	0x0610	W	0x00000000	Post scaling start point offset
VOP_POST_DSP_VACT_INFO_F1	0x0614	W	0x000a00fa	Panel active horizontal scanning start point and end point F1

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Name	Offset	Size	Reset Value	Description
VOP_DSP_HTOTAL_HS_END	0x0618	W	0x014a000a	Panel scanning horizontal width and hsync pulse end point
VOP_DSP_HACT_ST_END	0x061c	W	0x000a014a	Panel active horizontal scanning start point and end point
VOP_DSP_VTOTAL_VS_END	0x0620	W	0x00fa000a	Panel scanning vertical height and vsync pulse end point
VOP_DSP_VACT_ST_END	0x0624	W	0x000a00fa	Panel active vertical scanning start point and end point
VOP_DSP_VS_ST_END_F1	0x0628	W	0x00000000	Vertical scanning start point and vsync pulse end point of even filed in interlace mode
VOP_DSP_VACT_ST_END_F1	0x062c	W	0x00000000	Vertical scanning active start point and end point of even filed in interlace mode
VOP_BCSH_COLOR_BAR	0x0640	W	0x00000000	Color bar config register
VOP_BCSH_BCS	0x0644	W	0xd0010000	Brightness contrast saturation*contrast config register
VOP_BCSH_H	0x0648	W	0x01000000	Sin hue and cos hue config register
VOP_BCSH_CTRL	0x064c	W	0x00000000	BCSH contrl register
VOP_FRC_LOWER01_0	0x0678	W	0x12844821	FRC lookup table config register010
VOP_FRC_LOWER01_1	0x067c	W	0x21488412	FRC lookup table config register011
VOP_FRC_LOWER10_0	0x0680	W	0xa55a9696	FRC lookup table config register100
VOP_FRC_LOWER10_1	0x0684	W	0x5aa56969	FRC lookup table config register101
VOP_FRC_LOWER11_0	0x0688	W	0xdeb77bed	FRC lookup table config register110
VOP_FRC_LOWER11_1	0x068c	W	0xed7bb7de	FRC lookup table config register111
VOP_DBG_POST_REG0	0x06e8	W	0x00000000	Vop debug post register0
VOP_DBG_POST_RESERVED	0x06ec	W	0x00000000	Vop debug post register1 reserved
VOP_DBG_DATAO	0x06f0	W	0x00000000	debug data output path
VOP_DBG_DATAO_2	0x06f4	W	0x00000000	debug data output path 2
VOP_HWC_LUT_ADDR	0x0c00	W	0x00000000	Hwc lut base address
VOP_MMU_DTE_ADDR	0x3f00	W	0x00000000	MMU current page Table address
VOP_MMU_STATUS	0x3f04	W	0x00000000	MMU status register
VOP_MMU_COMMAND	0x3f08	W	0x00000000	MMU command register
VOP_MMU_PAGE_FAULT_ADDR	0x3f0c	W	0x00000000	MMU logical address of last page fault
VOP_MMU_ZAP_ONE_LINE	0x3f10	W	0x00000000	MMU Zap cache line register
VOP_MMU_INT_RAWSTAT	0x3f14	W	0x00000000	MMU raw interrupt status register

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Name	Offset	Size	Reset Value	Description
VOP_MMU_INT_CLEAR	0x3f18	W	0x00000000	MMU raw interrupt status register
VOP_MMU_INT_MASK	0x3f1c	W	0x00000000	MMU raw interrupt status register
VOP_MMU_INT_STATUS	0x3f20	W	0x00000000	MMU raw interrupt status register
VOP_MMU_AUTO_GATING	0x3f24	W	0x00000000	MMU auto gating

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.10.3 VOP BASE Detail Register Description

VOP_REG_CFG_DONE

Address: Operational Base + offset (0x0000)

Register config done flag

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15:9	RO	0x0	reserved
8	RW	0x0	reg_load_sys_en vop system register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the system register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
7	RW	0x0	reg_load_fbdc_en vop fbdc register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the fbdc register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
6	RW	0x0	reg_load_iep_en vop iep register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the iep register config finish(only 2 signals direct_path_en,direct_path_layer_sel), writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
5	RW	0x0	reg_load_hwc_en vop hwc register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the hwc register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.

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Bit	Attr	Reset Value	Description
4	RW	0x0	reg_load_win3_en vop win3 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win3 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
3	RW	0x0	reg_load_win2_en vop win2 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win2 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
2	RW	0x0	reg_load_win1_en vop win1 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win1 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
1	RW	0x0	reg_load_win0_en vop win0 register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the win0 register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.
0	WO	0x0	reg_load_en vop register config done flag In the first setting of the register, the new value was saved into the mirror register. When all the register config finish, writing this register to enable the copyright of the mirror register to real register. Then register would be updated at the start of every frame.

VOP_VERSION_INFO
Address: Operational Base + offset (0x0004)
Version for vop

Bit	Attr	Reset Value	Description
31:24	RO	0x00	major IP major vertion used for IP structure

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Bit	Attr	Reset Value	Description
23:16	RO	0x00	minor minor vertion big feature change under same structure
15:0	RO	0x0000	svnbuild rtl current svn number

VOP_SYS_CTRL
Address: Operational Base + offset (0x0008)
System control register0

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	sw_dac_sel dac output sel for tve in fpga dac output sel for tve in fpga 1'b0:dac 3 1'b1:dac 1
28	RW	0x0	sw_genlock genlock for tve genlock for tve in fpga 1'b0:master mode 1'b1:slave mode
27	RW	0x0	sw_uv_offset_en uv offset enable uv offset enable
26	RW	0x0	sw_tve_mode tve mode 1'b0:NTSC 1'b1:PAL
25	RW	0x0	sw_imd_tve_dclk_pol tve dclk pol tve dclk pol
24	RW	0x0	sw_imd_tve_dclk_en tve dclk enable tve dclk enable
23	RW	0x1	auto_gating_en LCDC layer axi-clk auto gating enable 1'b0 : disable auto gating 1'b1 : enable auto gating default auto gating enable

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Bit	Attr	Reset Value	Description
22	RW	0x0	vop_standby_en LCDC standby mode Writing "1" to turn LCDC into standby mode, All the layer would disable and the data transfer from frame buffer memory would stop at the end of current frame. The output would be blank. When writing "0" to this bit, standby mode would disable and the LCDC go back to work immediately. 1'b0 : disable 1'b1 : enable * Black display is recommended before setting standby mode enable.
21	RW	0x0	vop_dma_stop VOP DMA stop mode 1'b0 : disable 1'b1 : enable * If DMA is working, the stop mode would not be active until current bus transfer is finished.
20	RW	0x0	vop_mmu_en vop mmu enable signal 1'b0 : bypass mmu 1'b1 : enable mmu
19	RW	0x0	win23_pri_opt_mode 1'b0: win2 win3 dma priority enalbe 1'b1: win2 win3 dma priority disable
18	RW	0x0	post_lb_mode 1'b0 : 4x4096 1'b1 : 8x2048
17	RW	0x1	fs_same_addr_mask_en 1'b0 : disable 1'b1 : enable
16	RW	0x0	overlay_mode 1'b0: RGB overlay 1'b1: YUV overlay
15	RW	0x0	mipi_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : mipi interface enable
14	RW	0x0	edp_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : edp interface enable
13	RW	0x0	hdmi_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : hdmi interface enable

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Bit	Attr	Reset Value	Description
12	RW	0x1	rgb_out_en 1'b0 : gating output clk ,data and control signal 1'b1 : rgb/lvds interface enable
11	RW	0x1	global_regdone_en global config done enalbe 1'b0: DISABLE 1'b1: ENALBE
10	RW	0x0	edpi_wms_fs edpi wms mode ， frame st signal write "1": edpi_wms_mode frame start (when other register is config done) read : wms mode hold status
9	RW	0x0	edpi_wms_mode 1'b1: mipi command mode
8	RW	0x0	edpi_halt_en mipi flow ctrl enable
7:3	RO	0x0	reserved
2:1	RW	0x0	direct_path_layer_sel direct path layer select 2'b00 : select win0 2'b01 : select win1 2'b10 : select win2 2'b11 : select win3
0	RW	0x0	direct_path_en iep direct path enable signal 1'b0 : disable iep direct path 1'b1 : enable iep direct path

VOP_SYS_CTRL1
Address: Operational Base + offset (0x000c)
System control register1

Bit	Attr	Reset Value	Description
31	RW	0x0	fp_standby_en fp standby enable 1'b0 : disable 1'b1 : enable
30	RW	0x0	dcf_idle_en dcf idle enable 1'b0 : disable 1'b1 : enable
29	RW	0x0	post_bandwidth_ctrl deal with post empty 1'b0 : disable 1'b1 : enable

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Bit	Attr	Reset Value	Description
28	RW	0x0	level2_overlay_en 2-level overlay enable 1'b0 : disable 1'b1 : enable
27	RW	0x0	alpha_hard_calc calculate alpha 1'b0 : disable 1'b1 : enable
26:25	RO	0x0	reserved
24	RW	0x0	reg_done_frm reg done frame valid reg done framd valid 1'b0:reg done every field for interlace 1'b1:reg done every frame for interlace
23:22	RW	0x0	noc_hurry_w_value 2'b00: low priority 2'b11: high priority
21:20	RW	0x0	noc_hurry_w_mode 2'b00: noc_hurry_w disable 2'b01: left 1/4 fifo empty 2'b10: left 1/2 fifo empty 2'b11: left 3/4 fifo empty
19:18	RO	0x0	reserved
17:13	RW	0x1d	axi_outstanding_max_num axi bus max outstanding number
12	RW	0x0	axi_max_outstanding_en axi bus max outstanding enable
11:10	RW	0x0	noc_win_qos Noc win qos
9	RW	0x0	noc_qos_en Noc qos enable
8:3	RW	0x00	noc_hurry_threshold Noc hurry threshold value
2:1	RW	0x0	noc_hurry_value Noc hurry value
0	RW	0x0	noc_hurry_en Noc hurry enable

VOP_DSP_CTRL0
Address: Operational Base + offset (0x0010)
Display control register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
23	RW	0x0	dsp_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
22	RW	0x0	dsp_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror
21	RW	0x0	dsp_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
20	RW	0x0	dsp_ccir656_avg Cb-Cr filter in CCIR656 mode 1'b0 : drop mode 1'b1 : average mode
19	RW	0x0	dsp_black_en Black display mode When this bit enable, the pixel data output is all black (0x000000)
18	RW	0x0	dsp_blank_en Blank display mode When this bit enable, the Hsync/Vsync/Den output is blank
17	RW	0x0	dsp_out_zero Hsync/Vsync/Den output software ctrl 1'b0 : normal output 1'b1 : all output '0'
16	RW	0x0	dsp_dummy_swap Display dummy swap enable 1'b0 : B+G+R+dummy 1'b1 : dummy+B+G+R
15	RW	0x0	dsp_delta_swap Display delta swap enable 1'b0 : disable 1'b1 : enable *See detail description in Delta display charpter.
14	RW	0x0	dsp_rg_swap Display output red and green swap enable 1'b0 : RGB 1'b1 : GRB
13	RW	0x0	dsp_rb_swap Display output red and blue swap enable 1'b0 : RGB 1'b1 : BGR

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Bit	Attr	Reset Value	Description
12	RW	0x0	dsp_bg_swap Display output blue and green swap enable 1'b0 : RGB 1'b1 : RBG
11	RW	0x0	dsp_field_pol field polarity when interlace dsp 1'b0 : normal 1'b1 : invert
10	RW	0x0	dsp_interlace Interlace display enable 1'b0 : disable 1'b1 : enable *This mode is related to the ITU-R656 output, the display timing of odd field must be set correctly. (lcdc_dsp_vs_st_end_f1/lcdc_dsp_vact_end_f1)
9	RW	0x0	dsp_ddr_phase dclk phase lock 1'b0 : no lock 1'b1 : lock every line
8	RW	0x0	dsp_dclk_ddr dclk output mode 1'b0 : SDR 1'b1 : DDR
7:6	RO	0x0	reserved
5	RW	0x0	sw_p2i_en p2i enable 1'b0:disable p2i 1'b1:enable p2i
4	RW	0x0	sw_core_dclk_sel 1'b0: dclk_core sel dclk 1'b1: dclk_core sel dclk div2

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	dsp_out_mode Display output format 4'b0000: Parallel 24-bit RGB888 output R[7:0],G[7:0],B[7:0] 4'b0001: Parallel 18-bit RGB666 output 6'b0,R[5:0],G[5:0],B[5:0] 4'b0010: Parallel 16-bit RGB565 output 8'b0,R[4:0],G[5:0],B[4:0] 4'b0011: Parallel 24-bit RGB888 double pixel mix out phase0:G1[3:0],B1[7:0],G0[3:0],B0[7:0] phase1:R1[7:0],G1[7:4],R0[7:0],G0[7:4] 4'b0100: Serial 2x12-bit 12'b0,G[3:0],B[7:0] + 12'b0,R[7:0],G[7:4] 4'b0101: ITU-656 output mode0 16'b0,pixel_data[7:0] 4'b0110: ITU-656 output mode1 8'b0,pixel_data[7:0],8'b0 4'b0111: ITU-656 output mode2 9'b0,pixel_data[7:0],7'b0 4'b1000: Serial 3x8-bit RGB888 16'b0, B[7:0]+16'b0,G[7:0]+16'b0,R[7:0] 4'b1100: Serial 3x8-bit RGB888 + dummy 16'b0, B[7:0]+16'b0,G[7:0]+16'b0,R[7:0] + dummy 4'b1110: YUV420 output for HDMI 4'b1111: Parallel 30-bit RGBaaa output R[9:0],G[9:0],B[9:0] Others: Reserved.

VOP_DSP_CTRL1
Address: Operational Base + offset (0x0014)
Display control register1

Bit	Attr	Reset Value	Description
31	RW	0x0	mipi_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
30	RW	0x0	mipi_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
29	RW	0x0	mipi_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive

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Bit	Attr	Reset Value	Description
28	RW	0x0	mipi_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
27	RW	0x0	edp_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
26	RW	0x0	edp_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
25	RW	0x0	edp_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
24	RW	0x0	edp_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
23	RW	0x0	hdmi_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert
22	RW	0x0	hdmi_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
21	RW	0x0	hdmi_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
20	RW	0x0	hdmi_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
19	RW	0x0	rgb_lvds_dclk_pol DCLK invert enable 1'b0 : normal 1'b1 : invert default dclk invert

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Bit	Attr	Reset Value	Description
18	RW	0x0	rgb_lvds_den_pol DEN polarity 1'b0 : positive 1'b1 : negative
17	RW	0x0	rgb_lvds_vsync_pol VSYNC polarity 1'b0 : negative 1'b1 : positive
16	RW	0x0	rgb_lvds_hsync_pol HSYNC polarity 1'b0 : negative 1'b1 : positive
15:14	RW	0x3	dsp_layer3_sel layer3 selection
13:12	RW	0x2	dsp_layer2_sel layer2 selection
11:10	RW	0x1	dsp_layer1_sel layer1 selection
9:8	RW	0x0	dsp_layer0_sel layer0 selection
7	RO	0x0	reserved
6	RW	0x0	dither_up_en 1'b0 : no dither up 1'b1 : rgb565 dither up to rgb888
5	RO	0x0	reserved
4	RW	0x0	dither_down_sel dither down mode select 2'b0 : allegro 2'b1 : FRC
3	RW	0x0	dither_down_mode Dither-down mode 1'b0 : RGB888 to RGB565 1'b1 : RGB888 to RGB666
2	RW	0x0	dither_down_en Dither-down enable 1'b0 : disable 1'b1 : enable
1	RW	0x0	pre_dither_down_en 10bit -> 8bit (allegro)
0	RW	0x0	dsp_lut_en Display LUT ram enable 1'b0 : disable 1'b1 : enable *This bit should be "0" when CPU updates the LUT, and should be "1" when Display LUT mode enable.

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VOP_DSP_BG
Address: Operational Base + offset (0x0018)
Background color

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	dsp_bg_red Background Red color 8bit red color
15:8	RW	0x00	dsp_bg_green Background Green color 8bit green color
7:0	RW	0x00	dsp_bg_blue Background Blue color 8bit blue color

VOP_AUTO_GATING_EN
Address: Operational Base + offset (0x003c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	Field0000 Abstract Field0000 Description
6	RW	0x0	gamma_aclk_gating_en •
5	RW	0x0	overlay_aclk_gating_en
4	RW	0x0	hwc_aclk_gating_en
3:2	RO	0x0	reserved
1	RW	0x0	win1_aclk_gating_en
0	RW	0x0	win0_aclk_gating_en

VOP_LINE_FLAG
Address: Operational Base + offset (0x0040)
Line flag config register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dsp_line_flag_num_1 Line number of the Line flag interrupt 1 The display line number when the flag interrupt 1 occur, the range is (0~ DSP_VTOTAL-1).
15:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12:0	RW	0x0000	dsp_line_flag_num_0 Line number of the Line flag interrupt 0 The display line number when the flag interrupt occur, the range is (0~ DSP_VTOTAL-1).

VOP_VOP_STATUS
Address: Operational Base + offset (0x0044)
vop status register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	RW	0x0	dma_stop_valid dma stop valid
16	RW	0x0	mmu_idle mmu idle status
15:13	RO	0x0	reserved
12:0	RO	0x0000	dsp_vcnt read the dsp vertical counter

VOP_BLANKING_VALUE
Address: Operational Base + offset (0x0048)
Register0000 Abstract

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	blanking_value_config_en 1'b0 : disable blanking value when vop blank 1'b1 : enable blanking value when vop blank
23:0	RW	0x000000	blanking_value vop output data value when blanking,24bits

VOP_WIN0_DSP_BG
Address: Operational Base + offset (0x0050)
Win0 layer background color

Bit	Attr	Reset Value	Description
31	RW	0x0	win0_bg_en Win0 layer background enable 1'b0 : disable 1'b1 : enable
30:24	RO	0x0	reserved
23:16	RW	0x00	win0_dsp_bg_red Win0 layer Background Red color
15:8	RW	0x00	win0_dsp_bg_green Win0 layer Background Green color
7:0	RW	0x00	win0_dsp_bg_blue Win0 layer Background Blue color

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VOP_WIN1_DSP_BG
Address: Operational Base + offset (0x0054)
Win1 layer background color

Bit	Attr	Reset Value	Description
31	RW	0x0	win1_bg_en Win1 layer background enable 1'b0 : disable 1'b1 : enable
30:24	RO	0x0	reserved
23:16	RW	0x00	win1_dsp_bg_red Win1 layer Background Red color
15:8	RW	0x00	win1_dsp_bg_green Win1 layer Background Green color
7:0	RW	0x00	win1_dsp_bg_blue Win1 layer Background Blue color

VOP_DBG_PERF_LATENCY_CTRL0
Address: Operational Base + offset (0x00c0)
Axi performance latency module contrl register0

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:20	RW	0x00	rd_latency_st_num
19:8	RW	0x000	rd_latency_thr
7:4	RW	0x0	rd_latency_id0
3	RO	0x0	reserved
2	RW	0x0	rd_latency_mode read latency mode 1'b0: clear by frame end 1'b1: clear by software configuration
1	RW	0x0	hand_latency_clr read max latencyu software clear 1'b0: software clear disable 1'b1: software clear enalbe
0	RW	0x0	rd_latency_en vop_axi_rd_perf enable signal 1'b0: disable 1'b1: enable

VOP_DBG_PERF_RD_MAX_LATENCY_NUM0
Address: Operational Base + offset (0x00c4)

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Read max latency number

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RO	0x0	rd_latency_overflow_ch0 overflow flag(bigger than configed threshold value)
15:12	RO	0x0	reserved
11:0	RO	0x000	rd_max_latency_num_ch0 read max latency value of channel 0

VOP_DBG_PERF_RD_LATENCY_THR_NUM0
Address: Operational Base + offset (0x00c8)
The number of bigger than configed threshold value

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RO	0x000000	rd_latency_thr_num_ch0 read latency thr number channel 0

VOP_DBG_PERF_RD_LATENCY_SAMP_NUM0
Address: Operational Base + offset (0x00cc)
Total sample number

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RO	0x000000	rd_latency_samp_num_ch0

VOP_INTR_EN0
Address: Operational Base + offset (0x00e0)
Interrupt enable register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	intr_en_dma_finish 1'b0: disable 1'b1: enable
14	RW	0x0	intr_en_mmu 1'b0: disable 1'b1: enable
13	RW	0x0	intr_en_dsp_hold_valid display hold valid interrupt enable 1'b0: disable 1'b1: enable

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Bit	Attr	Reset Value	Description
12	RW	0x0	intr_en_pwm_gen pwm generated interrupt enable 1'b0: Channel 0 Interrupt not generated 1'b1: Channel 0 Interrupt generated
11	RW	0x0	intr_en_post_buf_empty post buffer empty interrupt enable 1'b0: disable 1'b1: enable
10	RW	0x0	intr_en_hwc_empty hwc data empty interrupt enable 1'b0: disable 1'b1: enable
9	RW	0x0	intr_en_win3_empty win3 data empty interrupt enable 1'b0: disable 1'b1: enable
8	RW	0x0	intr_en_win2_empty win2 data empty interrupt enable 1'b0: disable 1'b1: enable
7	RW	0x0	intr_en_win1_empty win1 data empty interrupt enable 1'b0: disable 1'b1: enable
6	RW	0x0	intr_en_win0_empty win0 data empty interrupt enable 1'b0: disable 1'b1: enable
5	RW	0x0	intr_en_bus_error Bus error Interrupt enable 1'b0: disable 1'b1: enable
4	RW	0x0	intr_en_line_flag1 Line flag 1 Interrupt enable 1'b0: disable 1'b1: enable
3	RW	0x0	intr_en_line_flag0 Line flag 0 Interrupt enable 1'b0: disable 1'b1: enable
2	RW	0x0	intr_en_addr_same memory start addr same interruption enable 1'b0: disable 1'b1: enable

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Bit	Attr	Reset Value	Description
1	RW	0x0	intr_en_fs_new Frame new start interrupt enable 1'b0: disable 1'b1: enable
0	RW	0x0	intr_en_fs Frame start interrupt enable 1'b0: disable 1'b1: enable

VOP_INTR_CLEAR0
Address: Operational Base + offset (0x00e4)
Interrupt clear register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask When every bit HIGH, enable the writing corresponding bit When every bit LOW, don't care the writing corresponding bit
15	RW	0x0	int_clr_dma_finish dma finish interrupt clear(Auto clear)
14	RW	0x0	int_clr_mmu mmu interrupt clear(Auto clear)
13	W1C	0x0	int_clr_dsp_hold_valid display hold valid interrupt clear(Auto clear)
12	W1C	0x0	int_clr_pwm_gen pwm generated interrupt clear(Auto clear) 1'b0: Channel 0 Interrupt not generated 1'b1: Channel 0 Interrupt generated
11	W1C	0x0	int_clr_post_buf_empty post buffer empty interrupt clear(Auto clear)
10	W1C	0x0	int_clr_hwc_empty hwc data empty interrupt clear(Auto clear)
9	W1C	0x0	int_clr_win3_empty win3 data empty interrupt clear(Auto clear)
8	W1C	0x0	int_clr_win2_empty win2 data empty interrupt clear(Auto clear)
7	W1C	0x0	int_clr_win1_empty win1 data empty interrupt clear(Auto clear)
6	W1C	0x0	int_clr_win0_empty win0 data empty interrupt clear(Auto clear)
5	W1C	0x0	int_clr_bus_error Bus error Interrupt clear(Auto clear)
4	W1C	0x0	int_clr_line_flag1 Line flag 1 Interrupt clear(Auto clear)
3	W1C	0x0	int_clr_line_flag0 Line flag 0 Interrupt clear(Auto clear)

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Bit	Attr	Reset Value	Description
2	W1C	0x0	int_clr_addr_same memory start addr same interruption clear(Auto clear)
1	W1C	0x0	int_clr_fs_new Frame new start interrupt clear (Auto clear)
0	W1C	0x0	int_clr_fs Frame start interrupt clear (Auto clear)

VOP_INTR_STATUS0
Address: Operational Base + offset (0x00e8)
interrupt status

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	int_status_dma_finish dma finish interrupt status
14	RW	0x0	int_status_mmu mmu interrupt status
13	RO	0x0	int_status_dsp_hold_valid display hold valid interrupt status
12	RO	0x0	int_status_pwm_gen pwm generated interrupt status 1'b0: Channel 0 Interrupt not generated 1'b1: Channel 0 Interrupt generated
11	RO	0x0	int_status_post_buf_empty post buffer empty interrupt status
10	RO	0x0	int_status_hwc_empty hwc data empty interrupt status
9	RO	0x0	int_status_win3_empty win3 data empty interrupt status
8	RO	0x0	int_status_win2_empty win2 data empty interrupt status
7	RO	0x0	int_status_win1_empty win1 data empty interrupt status
6	RO	0x0	int_status_win0_empty win0 data empty interrupt status
5	RO	0x0	int_status_bus_error Bus error Interrupt status
4	RO	0x0	int_status_line_flag1 Line flag 1 Interrupt status
3	RO	0x0	int_status_line_flag0 Line flag 0 Interrupt status
2	RW	0x0	int_status_addr_same memory start addr same interruption status
1	RO	0x0	int_status_fs_new Frame start interrupt status(when memory start addr are same,no interruption)

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Bit	Attr	Reset Value	Description
0	RO	0x0	int_status_fs Frame start interrupt status

VOP_INTR_RAW_STATUS0
Address: Operational Base + offset (0x00ec)
raw interrupt status

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	int_raw_status_dma_finish dma finish interrupt raw status
14	RO	0x0	int_raw_status_mmu mmu interrupt raw status
13	RO	0x0	int_raw_status_dsp_hold_valid display hold valid interrupt raw status
12	RO	0x0	int_raw_status_pwm_gen pwm generated interrupt raw status 1'b0: Channel 0 Interrupt not generated 1'b1: Channel 0 Interrupt generated
11	RO	0x0	int_raw_status_post_buf_empty post buffer empty interrupt raw status
10	RO	0x0	int_raw_status_hwc_empty hwc data empty interrupt raw status
9	RO	0x0	int_raw_status_win3_empty win3 data empty interrupt raw status
8	RO	0x0	int_raw_status_win2_empty win2 data empty interrupt raw status
7	RO	0x0	int_raw_status_win1_empty win1 data empty interrupt raw status
6	RO	0x0	int_raw_status_win0_empty win0 data empty interrupt raw status
5	RO	0x0	int_raw_status_bus_error Bus error Interrupt raw status
4	RO	0x0	int_raw_status_line_frag1 Line flag 1 Interrupt raw status
3	RO	0x0	int_raw_status_line_frag0 Line flag 0 Interrupt raw status
2	RO	0x0	int_raw_status_addr_same memory start addr same interruption raw status
1	RO	0x0	int_raw_status_fs_new Frame start interrupt raw status(when memory start addr are same)
0	RO	0x0	int_raw_status_fs Frame start raw interrupt status Frame start raw interrupt status

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VOP_INTR_EN1
Address: Operational Base + offset (0x00f0)
Interrupt enable register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask
15	RO	0x0	reserved
14	RW	0x0	int_en_wb_finish 1'b0: disable 1'b1: enable
13	RW	0x0	int_en_wb_uv_fifo_full 1'b0: disable 1'b1: enable
12	RW	0x0	int_en_wb_yrgb_fifo_full 1'b0: disable 1'b1: enable
11	RW	0x0	int_en_afbcd3_hreg_axi_rresp 1'b0: disable 1'b1: enable
10	RW	0x0	int_en_afbcd3_hreg_dec_resp 1'b0: disable 1'b1: enable
9	RW	0x0	int_en_afbcd2_hreg_axi_rresp 1'b0: disable 1'b1: enable
8	RW	0x0	int_en_afbcd2_hreg_dec_resp 1'b0: disable 1'b1: enable
7	RW	0x0	int_en_afbcd1_hreg_axi_rresp 1'b0: disable 1'b1: enable
6	RW	0x0	int_en_afbcd1_hreg_dec_resp 1'b0: disable 1'b1: enable
5	RW	0x0	int_en_afbcd0_hreg_axi_rresp 1'b0: disable 1'b1: enable
4	RW	0x0	int_en_afbcd0_hreg_dec_resp 1'b0: disable 1'b1: enable
3	RW	0x0	int_en_fbcd3 1'b0: disable 1'b1: enable
2	RW	0x0	int_en_fbcd2 1'b0: disable 1'b1: enable

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Bit	Attr	Reset Value	Description
1	RW	0x0	int_en_fbcd1 1'b0: disable 1'b1: enable
0	RW	0x0	int_en_fbcd0 1'b0: disable 1'b1: enable

VOP_INTR_CLEAR1
Address: Operational Base + offset (0x00f4)
Interrupt clear register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RW	0x0	int_clr_afbcd3_hreg_axi_rresp interrupt clear (Auto clear)
10	RW	0x0	int_clr_afbcd3_hreg_dec_resp interrupt clear (Auto clear)
9	RW	0x0	int_clr_afbcd2_hreg_axi_rresp interrupt clear (Auto clear)
8	RW	0x0	int_clr_afbcd2_hreg_dec_resp interrupt clear (Auto clear)
7	RW	0x0	int_clr_afbcd1_hreg_axi_rresp interrupt clear (Auto clear)
6	RW	0x0	int_clr_afbcd1_hreg_dec_resp interrupt clear (Auto clear)
5	RW	0x0	int_clr_afbcd0_hreg_axi_rresp interrupt clear (Auto clear)
4	RW	0x0	int_clr_afbcd0_hreg_dec_resp interrupt clear (Auto clear)
3	RW	0x0	int_clr_fbcd3 interrupt clear (Auto clear)
2	RW	0x0	int_clr_fbcd2 interrupt clear (Auto clear)
1	W1C	0x0	int_clr_fbcd1 interrupt clear (Auto clear)
0	W1C	0x0	int_clr_fbcd0 interrupt clear (Auto clear)

VOP_INTR_STATUS1
Address: Operational Base + offset (0x00f8)
interrupt status

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RW	0x0	int_status_afbcd4_hreg_dec_resp interrupt status

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Bit	Attr	Reset Value	Description
10	RW	0x0	int_status_afbcd3_hreg_dec_resp interrupt status
9	RW	0x0	int_status_afbcd2_hreg_axi_rresp interrupt status
8	RW	0x0	int_status_afbcd2_hreg_dec_resp interrupt status
7	RW	0x0	int_status_afbcd1_hreg_axi_rresp interrupt status
6	RW	0x0	int_status_afbcd1_hreg_dec_resp interrupt status
5	RW	0x0	int_status_afbcd0_hreg_axi_rresp interrupt status
4	RW	0x0	int_status_afbcd0_hreg_dec_resp interrupt status
3	RW	0x0	int_status_fbcd3 interrupt status
2	RW	0x0	int_status_fbcd2 interrupt status
1	RO	0x0	int_status_fbcd1 interrupt status
0	RO	0x0	int_status_fbcd0 interrupt status

VOP_INTR_RAW_STATUS1
Address: Operational Base + offset (0x00fc)
raw interrupt status

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RW	0x0	int_raw_status_afbcd3_hreg_axi_rresp interrupt raw status
10	RW	0x0	int_raw_status_afbcd3_hreg_dec_resp interrupt raw status
9	RW	0x0	int_raw_status_afbcd2_hreg_axi_rresp interrupt raw status
8	RW	0x0	int_raw_status_afbcd2_hreg_dec_resp interrupt raw status
7	RW	0x0	int_raw_status_afbcd1_hreg_axi_rresp interrupt raw status
6	RW	0x0	int_raw_status_afbcd1_hreg_dec_resp interrupt raw status
5	RW	0x0	int_raw_status_afbcd0_hreg_axi_rresp interrupt raw status
4	RW	0x0	int_raw_status_afbcd0_hreg_dec_resp interrupt raw status

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Bit	Attr	Reset Value	Description
3	RW	0x0	int_raw_status_fbcd3 interrupt raw status
2	RW	0x0	int_raw_status_fbcd2 interrupt raw status
1	RW	0x0	int_raw_status_fbcd1 interrupt raw status
0	RO	0x0	int_raw_status_fbcd0 interrupt raw status

VOP_WIN0_CTRL0
Address: Operational Base + offset (0x0100)
Win0 ctrl register0

Bit	Attr	Reset Value	Description
31:30	RW	0x0	win0_dma_burst_length WIN0 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
29:25	RW	0x1d	win0_axi_outstanding_max_num win0 out standing max number
24	RW	0x0	win0_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
23	RO	0x0	reserved
22	RW	0x0	win0_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
21	RW	0x0	win0_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror
20	RW	0x0	win0_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
19	RW	0x0	win0_cbr_deflick Win0 Cbr deflick mode 1'b0 : disable 1'b1 : enable
18	RW	0x0	win0_yrgb_deflick win0 YRGB deflick mode 1'b0 : disable 1'b1 : enable
17	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
16	RW	0x0	win0_hw_pre_mul_en 1'b0: no hardware pre multiply mode 1'b1: hardware pre multiply mode
15	RW	0x0	win0_uv_swap Win0 CbCr swap 1'b0 : CrCb 1'b1 : CbCr
14	RW	0x0	win0_mid_swap Win0 Y middle swap 1'b0 : Y3Y2Y1Y0 1'b1 : Y3Y1Y2Y0
13	RW	0x0	win0_alpha_swap win0 alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	win0_rb_swap win0 RGB RED and BLUE swap 1'b0 : RGB 1'b1 : BGR
11:10	RW	0x0	win0_csc_mode Win0 YUV2RGB or RGB2YUV Color space conversion(YUV2RGB): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020 Color space conversion(RGB2YUV): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020
9	RW	0x0	win0_no_outstanding win0 AXI master read outstanding 1'b0 : enable 1'b1 : disable
8	RW	0x0	win0_interlace_read Win0 interlace read mode 1'b0 : disable 1'b1 : enable
7:5	RW	0x2	win0_lb_mode win0 line buffer mode,calc by driver.
4	RW	0x0	win0_fmt_10 0: yuv 8bit fmt mode 1: yuv 10bit fmt mode

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Bit	Attr	Reset Value	Description
3:1	RW	0x0	win0_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100 : YcbCr420 3'b101 : YcbCr422 3'b110 : YcbCr444
0	RW	0x0	win0_en 1'b0 : disable 1'b1 : enable

VOP_WIN0_CTRL1
Address: Operational Base + offset (0x0104)
Win0 ctrl register1

Bit	Attr	Reset Value	Description
31	RW	0x0	win0_cbr_vsd_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average
30	RW	0x0	win0_cbr_vsu_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : bicubic
29:28	RW	0x0	win0_cbr_hsd_mode win0 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : bicubic 2'b10 : average
27:26	RW	0x0	win0_cbr_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
25:24	RW	0x0	win0_cbr_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
23	RW	0x0	win0_yrgb_vsd_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average

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Bit	Attr	Reset Value	Description
22	RW	0x0	win0_yrgb_vsu_mode win0 vertical scaler down mode select 1'b0 : bilinear 1'b1 : bicubic
21:20	RW	0x0	win0_yrgb_hsd_mode win0 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : average
19:18	RW	0x0	win0_yrgb_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
17:16	RW	0x0	win0_yrgb_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
15	RW	0x0	win0_line_load_mode when yuv fmt, 1'b0: load data by axi trans 1'b1: load data by lines
14:12	RW	0x0	win0_cbr_axi_gather_num win0 axi cbr data transfer gather number
11:8	RW	0x0	win0_yrgb_axi_gather_num win0 axi yrgb data transfer gather number
7	RW	0x0	win0_vsd_cbr_gt2 cbr_src/cbr_dst >= 2
6	RW	0x0	win0_vsd_cbr_gt4 cbr_src/cbr_dst >= 4
5	RW	0x0	win0_vsd_yrgb_gt2 yrgb_src/yrgb_dst >= 2
4	RW	0x0	win0_vsd_yrgb_gt4 yrgb_src/yrgb_dst >= 4
3:2	RW	0x0	win0_bic_coe_sel 2'b00 : PRECISE 2'b01 : SPLINE 2'b10 : CATROM 2'b11 : MITCHELL
1	RW	0x0	win0_cbr_axi_gather_en win0 axi bus cbr data gather transfer enable
0	RW	0x0	win0_yrgb_axi_gather_en win0 axi bus yrgb data gather transfer enable

VOP_WIN0_COLOR_KEY

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Address: Operational Base + offset (0x0108)
Win0 color key register

Bit	Attr	Reset Value	Description
31	RW	0x0	win0_key_en Win0 transparency color key enable 1'b0 : disable; 1'b1 : enable;
30:24	RO	0x0	reserved
23:0	RW	0x000000	win0_key_color Win0 key color 24 bit RGB888

VOP_WIN0_VIR
Address: Operational Base + offset (0x010c)
Win0 virtual stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win0_vir_stride_uv Number of words of Win0 uv Virtual width
15:0	RW	0x0140	win0_vir_stride Win0 Virtual stride Number of words of Win0 yrgb Virtual width ARGB888 : win0_vir_width RGB888 : (win0_vir_width*3/4) + (win0_vir_width%3) RGB565 : ceil(win0_vir_width/2) YUV : ceil(win0_vir_width/4)

VOP_WIN0_YRGB_MST
Address: Operational Base + offset (0x0110)
Win0 YRGB memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win0_yrgb_mst win0 YRGB frame buffer memory start address

VOP_WIN0_CBR_MST
Address: Operational Base + offset (0x0114)
Win0 Cbr memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win0_cbr_mst win0 CBR frame buffer memory start address

VOP_WIN0_ACT_INFO
Address: Operational Base + offset (0x0118)
Win0 active window width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
28:16	RW	0x00ef	win0_act_height Win0 active(original) window height win_act_height = (win0 vertical size -1)
15:13	RO	0x0	reserved
12:0	RW	0x013f	win0_act_width Win0 active(original) window width win_act_width = (win0 horizontal size -1)

VOP_WIN0_DSP_INFO
Address: Operational Base + offset (0x011c)
Win0 display width/height on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win0_dsp_height Win0 display window height win0_dsp_height = (win0 vertical size -1)
15:12	RO	0x0	reserved
11:0	RW	0x13f	win0_dsp_width Win0 display window width win0_dsp_width = (win0 horizontal size -1)

VOP_WIN0_DSP_ST
Address: Operational Base + offset (0x0120)
Win0 display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win0_dsp_yst Win0 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win0_dsp_xst Win0 horizontal start point(x) of the Panel scanning

VOP_WIN0_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x0124)
Win0 YRGB scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win0_vs_factor_yrgb Win0 YRGB vertical scaling factor: factor=((LCDC_WIN0_ACT_INFO[31:16]) /(LCDC_WIN0_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win0_hs_factor_yrgb Win0 YRGB horizontal scaling factor: factor=((LCDC_WIN0_ACT_INFO[15:0]) /(LCDC_WIN0_DSP_INFO[15:0]))*2^12

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VOP_WIN0_SCL_FACTOR_CBR
Address: Operational Base + offset (0x0128)
Win0 Cbr scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win0_vs_factor_cbr Win0 CBR vertical scaling factor: YCbCr420: factor=((LCDC_WIN0_ACT_INFO[31:16]/ 2) /(LCDC_WIN0_DSP_INFO[31:16]))*2^12 YCbCr422,YCbCr444: factor=((LCDC_WIN0_ACT_INFO[31:16]) /(LCDC_WIN0_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win0_hs_factor_cbr Win0 CBR horizontal scaling factor: YCbCr422,YCbCr420: factor=((LCDC_WIN0_ACT_INFO[15:0]/2) /(LCDC_WIN0_DSP_INFO[15:0]))*2^12 YCbCr444: factor=((LCDC_WIN0_ACT_INFO[15:0]) /(LCDC_WIN0_DSP_INFO[15:0]))*2^12

VOP_WIN0_SCL_OFFSET
Address: Operational Base + offset (0x012c)
Win0 scaling start point offset

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win0_vs_offset_cbr Cbr Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
23:16	RW	0x00	win0_vs_offset_yrgb Y Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
15:8	RW	0x00	win0_hs_offset_cbr Cbr Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99
7:0	RW	0x00	win0_hs_offset_yrgb Y Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99

VOP_WIN0_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x0130)
Win0 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win0_fading_value win0 fading value ,8bits
23:16	RW	0x00	win0_src_global_alpha src global alpha

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Bit	Attr	Reset Value	Description
15:9	RO	0x0	reserved
8:6	RW	0x0	win0_src_factor_mode src factor mode
5	RW	0x0	win0_src_alpha_cal_mode src alpha calc mode
4:3	RW	0x0	win0_src_blend_mode src blend mode
2	RW	0x0	win0_src_alpha_mode src alpha mode
1	RW	0x0	win0_src_color_mode src color mode
0	RW	0x0	win0_src_alpha_en src alpha en

VOP_WIN0_DST_ALPHA_CTRL
Address: Operational Base + offset (0x0134)
Win0 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win0_dst_factor_mode dst factor mode
5:0	RW	0x00	win0_dst_m0_reserved reserved

VOP_WIN0_FADING_CTRL
Address: Operational Base + offset (0x0138)
Win0 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	layer0_fading_en fading enable
23:16	RW	0x00	layer0_fading_offset_b fading offset blue value
15:8	RW	0x00	layer0_fading_offset_g fading offset green value
7:0	RW	0x00	layer0_fading_offset_r fading offset red value

VOP_WIN0_CTRL2
Address: Operational Base + offset (0x013c)
Win0 ctrl register2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x2	win_rid_win0_cbr axi read id of win0 cbr channel
3:0	RW	0x1	win_rid_win0_yrgb axi read id of win0 yrgb channel

VOP_DBG_WIN0_REG0
Address: Operational Base + offset (0x01f0)
Vop debug win0 register0

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_WIN0_REG1
Address: Operational Base + offset (0x01f4)
Vop debug win0 register1

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_WIN0_REG2
Address: Operational Base + offset (0x01f8)
Vop debug win0 register2

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dbg_win0_yrgb_cmd_line_cnt
15:0	RO	0x0	reserved

VOP_DBG_WIN0_RESERVED
Address: Operational Base + offset (0x01fc)
Vop debug win0 register3 reserved

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	reserved reserved

VOP_WIN1_CTRL0
Address: Operational Base + offset (0x0200)
Win1 ctrl register0

Bit	Attr	Reset Value	Description
31:30	RW	0x0	win1_dma_burst_length WIN1 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved

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Bit	Attr	Reset Value	Description
29:25	RW	0x1d	win1_axi_max_outstanding_num win1 out standing max number
24	RW	0x0	win1_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
23	RO	0x0	reserved
22	RW	0x0	win1_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
21	RW	0x0	win1_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror
20	RW	0x0	win1_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
19	RW	0x0	win1_cbr_deflick Win1 Cbr deflick mode 1'b0 : disable 1'b1 : enable
18	RW	0x0	win1_yrgb_deflick win1 YRGB deflick mode 1'b0 : disable 1'b1 : enable
17	RO	0x0	reserved
16	RW	0x0	win1_hw_pre_mul_en 1'b0: no hardware pre multiply mode 1'b1: hardware pre multiply mode
15	RW	0x0	win1_uv_swap Win1 CbCr swap 1'b0 : CrCb 1'b1 : CbCr
14	RW	0x0	win1_mid_swap Win1 Y middle 8-bit swap 1'b0 : Y3Y2Y1Y0 1'b1 : Y3Y1Y2Y0
13	RW	0x0	win1_alpha_swap win1 alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	win1_rb_swap win1 RGB RED and BLUE swap 1'b0 : RGB 1'b1 : BGR

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Bit	Attr	Reset Value	Description
11:10	RW	0x0	win1_csc_mode Win1 YUV2RGB or RGB2YUV Color space conversion(YUV2RGB): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020 Color space conversion(RGB2YUV): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020
9	RW	0x0	win1_no_outstanding win1 AXI master read outstanding 1'b0 : enable 1'b1 : disable
8	RW	0x0	win1_interlace_read Win1 interlace read mode 1'b0 : disable 1'b1 : enable
7:5	RW	0x2	win1_lb_mode win1 line buffer mode,calc by driver.
4	RW	0x0	win1_fmt_10 1'b0: yuv 8bit fmt mode 1'b1: yuv 10bit fmt mode
3:1	RW	0x0	win1_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100 : YcbCr420 3'b101 : YcbCr422 3'b110 : YcbCr444
0	RW	0x0	win1_en 1'b0 : disable 1'b1 : enable

VOP_WIN1_CTRL1

Address: Operational Base + offset (0x0204)

Win1 ctrl register1

Bit	Attr	Reset Value	Description
31	RW	0x0	win1_cbr_vsd_mode win1 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average

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Bit	Attr	Reset Value	Description
30	RW	0x0	win1_cbr_vsu_mode win1 vertical scaler up mode select 1'b0 : bilinear 1'b1 : bicubic
29:28	RW	0x0	win1_cbr_hsd_mode win1 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : bicubic 2'b10 : average
27:26	RW	0x0	win1_cbr_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
25:24	RW	0x0	win1_cbr_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
23	RW	0x0	win1_yrgb_vsd_mode win1 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average
22	RW	0x0	win1_yrgb_vsu_mode win1 vertical scaler up mode select 1'b0 : bilinear 1'b1 : bicubic
21:20	RW	0x0	win1_yrgb_hsd_mode win1 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : average
19:18	RW	0x0	win1_yrgb_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
17:16	RW	0x0	win1_yrgb_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
15	RW	0x0	win1_line_load_mode when yuv fmt, 1'b0: load data by pixels 1'b1: load data by lines

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Bit	Attr	Reset Value	Description
14:12	RW	0x0	win1_cbr_axi_gather_num win1 axi cbr data transfer gather number
11:8	RW	0x0	win1_yrgb_axi_gather_num win1 axi yrgb data transfer gather number
7	RW	0x0	win1_vsd_cbr_gt2 cbr_src/cbr_dst >= 2
6	RW	0x0	win1_vsd_cbr_gt4 cbr_src/cbr_dst >= 4
5	RW	0x0	win1_vsd_yrgb_gt2 yrgb_src/yrgb_dst >= 2
4	RW	0x0	win1_vsd_yrgb_gt4 yrgb_src/yrgb_dst >= 4
3:2	RW	0x0	win1_bic_coe_sel 2'b00 : PRECISE 2'b01 : SPLINE 2'b10 : CATROM 2'b11 : MITCHELL
1	RW	0x0	win1_cbr_axi_gather_en win1 cbr axi bus gather enable
0	RW	0x0	win1_yrgb_axi_gather_en win1 yrgb axi bus gather enable

VOP_WIN1_COLOR_KEY
Address: Operational Base + offset (0x0208)
Win1 color key register

Bit	Attr	Reset Value	Description
31	RW	0x0	win1_key_en Win1 transparency color key enable 1'b0 : disable; 1'b1 : enable;
30:24	RO	0x0	reserved
23:0	RW	0x000000	win1_key_color Win1 key color 24 bit RGB888

VOP_WIN1_VIR
Address: Operational Base + offset (0x020c)
win1 virtual stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win1_vir_stride_uv Number of words of Win1 uv Virtual width

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Bit	Attr	Reset Value	Description
15:0	RW	0x0140	win1_vir_stride Win1 Virtual stride Number of words of Win1 yrgb Virtual width ARGB888 : win1_vir_width RGB888 : (win1_vir_width*3/4) + (win1_vir_width%3) RGB565 : ceil(win1_vir_width/2) YUV : ceil(win1_vir_width/4)

VOP_WIN1_YRGB_MST
Address: Operational Base + offset (0x0210)
Win1 YRGB memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win1_yrgb_mst win1 YRGB frame buffer memory start address

VOP_WIN1_CBR_MST
Address: Operational Base + offset (0x0214)
Win1 Cbr memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win1_cbr_mst win1 CBR frame buffer memory start address

VOP_WIN1_ACT_INFO
Address: Operational Base + offset (0x0218)
Win1 active window width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00ef	win1_act_height Win1 active(original) window height win_act_height = (win1 vertical size -1)
15:13	RO	0x0	reserved
12:0	RW	0x013f	win1_act_width Win1 active(original) window width win_act_width = (win1 horizontial size -1)

VOP_WIN1_DSP_INFO
Address: Operational Base + offset (0x021c)
Win1 display width/height on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win1_dsp_height Win1 display window height win1_dsp_height = (win1 vertical size -1)
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x13f	win1_dsp_width Win1 display window width win1_dsp_width = (win1 horizontial size -1)

VOP_WIN1_DSP_ST
Address: Operational Base + offset (0x0220)
Win1 display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win1_dsp_yst Win1 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win1_dsp_xst Win1 horizontal start point(x) of the Panel scanning

VOP_WIN1_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x0224)
Win1 YRGB scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win1_vs_factor_yrgb Win1 YRGB vertical scaling factor: factor=((LCDC_WIN1_ACT_INFO[31:16]) /(LCDC_WIN1_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win1_hs_factor_yrgb Win1 YRGB horizontal scaling factor: factor=((LCDC_WIN1_ACT_INFO[15:0]) /(LCDC_WIN1_DSP_INFO[15:0]))*2^12

VOP_WIN1_SCL_FACTOR_CBR
Address: Operational Base + offset (0x0228)
Win1 Cbr scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win1_vs_factor_cbr Win1 CBR vertical scaling factor: YCbCr420: factor=((LCDC_WIN1_ACT_INFO[31:16]/ 2) /(LCDC_WIN1_DSP_INFO[31:16]))*2^12 YCbCr422,YCbCr444: factor=((LCDC_WIN1_ACT_INFO[31:16]) /(LCDC_WIN1_DSP_INFO[31:16]))*2^12

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Bit	Attr	Reset Value	Description
15:0	RW	0x1000	win1_hs_factor_cbr Win1 Cbr horizontal scaling factor: YCbCr422,YCbCr420: factor=((LCDC_WIN1_ACT_INFO[15:0]/2) /(LCDC_WIN1_DSP_INFO[15:0]))*2^12 YCbCr444: factor=((LCDC_WIN1_ACT_INFO[15:0]) /(LCDC_WIN1_DSP_INFO[15:0]))*2^12

VOP_WIN1_SCL_OFFSET
Address: Operational Base + offset (0x022c)
Win1 scaling start point offset

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win1_vs_offset_cbr Cbr Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
23:16	RW	0x00	win1_vs_offset_yrgb Y Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
15:8	RW	0x00	win1_hs_offset_cbr Cbr Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99
7:0	RW	0x00	win1_hs_offset_yrgb Y Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99

VOP_WIN1_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x0230)
Win1 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win1_fading_value fading value,8bit
23:16	RW	0x00	win1_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	win1_src_factor_mode src factor mode
5	RW	0x0	win1_src_alpha_cal_mode src alpha calc mode
4:3	RW	0x0	win1_src_blend_mode src blend mode
2	RW	0x0	win1_src_alpha_mode src alpha mode

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Bit	Attr	Reset Value	Description
1	RW	0x0	win1_src_color_mode src color mode
0	RW	0x0	win1_src_alpha_en src alpha en

VOP_WIN1_DST_ALPHA_CTRL
Address: Operational Base + offset (0x0234)
Win1 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win1_dst_factor_m0 dst factor mode
5:0	RW	0x00	win1_dsp_m0_reserved reserved

VOP_WIN1_FADING_CTRL
Address: Operational Base + offset (0x0238)
Win1 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win1_fading_en fading enable
23:16	RW	0x00	win1_fading_offset_b fading offset blue value
15:8	RW	0x00	win1_fading_offset_g fading offset green value
7:0	RW	0x00	win1_fading_offset_r fading offset red value

VOP_WIN1_CTRL2
Address: Operational Base + offset (0x023c)
Win1 ctrl register2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x4	win_rid_win1_cbr axi read id of win1 cbr channel
3:0	RW	0x3	win_rid_win1_yrgb axi read id of win1 yrgb channel

VOP_DBG_WIN1_REG0
Address: Operational Base + offset (0x02f0)
Vop debug win1 register0

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

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VOP_DBG_WIN1_REG1
Address: Operational Base + offset (0x02f4)
Vop debug win1 register1

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_WIN1_REG2
Address: Operational Base + offset (0x02f8)
Vop debug win1 register2

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_WIN1_RESERVED
Address: Operational Base + offset (0x02fc)
Vop debug win1 register3 reserved

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	reserved reserved

VOP_WIN2_CTRL0
Address: Operational Base + offset (0x0300)
Win2 ctrl register0

Bit	Attr	Reset Value	Description
31:30	RW	0x0	win2_dma_burst_length WIN1 DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
29:25	RW	0x1d	win2_axi_max_outstanding_num win1 out standing max number
24	RW	0x0	win2_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
23	RO	0x0	reserved
22	RW	0x0	win2_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
21	RW	0x0	win2_x_mir_en 1'b0 : no x_mirror 1'b1 : x_mirror

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Bit	Attr	Reset Value	Description
20	RW	0x0	win2_yuv_clip YCrCb clip 1'b0 : disable, YCbCr no clip 1'b1 : enable, YCbCr clip before YCbCr2RGB *Y clip: 16~235, CbCr clip: 16~239
19	RW	0x0	win2_cbr_deflick Win1 Cbr deflick mode 1'b0 : disable 1'b1 : enable
18	RW	0x0	win2_yrgb_deflick win1 YRGB deflick mode 1'b0 : disable 1'b1 : enable
17	RO	0x0	reserved
16	RW	0x0	win2_hw_pre_mul_en 1'b0: no hardware pre multiply mode 1'b1: hardware pre multiply mode
15	RW	0x0	win2_uv_swap Win1 CbCr swap 1'b0 : CrCb 1'b1 : CbCr
14	RW	0x0	win2_mid_swap Win1 Y middle 8-bit swap 1'b0 : Y3Y2Y1Y0 1'b1 : Y3Y1Y2Y0
13	RW	0x0	win2_alpha_swap win1 alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	win2_rb_swap win1 RGB RED and BLUE swap 1'b0 : RGB 1'b1 : BGR

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Bit	Attr	Reset Value	Description
11:10	RW	0x0	win2_csc_mode Win1 YUV2RGB or RGB2YUV Color space conversion(YUV2RGB): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020 Color space conversion(RGB2YUV): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020 2'bx1: BT709
9	RW	0x0	win2_no_outstanding win1 AXI master read outstanding 1'b0 : enable 1'b1 : disable
8	RW	0x0	win2_interlace_read Win1 interlace read mode 1'b0 : disable 1'b1 : enable
7:5	RW	0x2	win2_lb_mode win1 line buffer mode,calc by driver.
4	RW	0x0	win2_fmt_10 1'b0: yuv 8bit fmt mode 1'b1: yuv 10bit fmt mode
3:1	RW	0x0	win2_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100 : YcbCr420 3'b101 : YcbCr422 3'b110 : YcbCr444
0	RW	0x0	win2_en 1'b0 : disable 1'b1 : enable

VOP_WIN2_CTRL1

Address: Operational Base + offset (0x0304)

Win2 ctrl register1

Bit	Attr	Reset Value	Description
31	RW	0x0	win2_cbr_vsd_mode win2 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average

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Bit	Attr	Reset Value	Description
30	RW	0x0	win2_cbr_vsu_mode win2 vertical scaler up mode select 1'b0 : bilinear 1'b1 : bicubic
29:28	RW	0x0	win2_cbr_hsd_mode win2 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : bicubic 2'b10 : average
27:26	RW	0x0	win2_cbr_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
25:24	RW	0x0	win2_cbr_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
23	RW	0x0	win2_yrgb_vsd_mode win2 vertical scaler down mode select 1'b0 : bilinear 1'b1 : average
22	RW	0x0	win2_yrgb_vsu_mode win2 vertical scaler up mode select 1'b0 : bilinear 1'b1 : bicubic
21:20	RW	0x0	win2_yrgb_hsd_mode win2 horizontal scaler down mode select 2'b00 : bilinear 2'b01 : average
19:18	RW	0x0	win2_yrgb_ver_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
17:16	RW	0x0	win2_yrgb_hor_scl_mode 2'b00 : no scale 2'b01 : scale up 2'b10 : scale down 2'b11 : no scale
15	RW	0x0	win2_line_load_mode when yuv fmt, 1'b0: load data by pixels 1'b1: load data by lines

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Bit	Attr	Reset Value	Description
14:12	RW	0x0	win2_cbr_axi_gather_num win1 axi cbr data transfer gather number
11:8	RW	0x0	win2_yrgb_axi_gather_num win1 axi yrgb data transfer gather number
7	RW	0x0	win2_vsd_cbr_gt2 cbr_src/cbr_dst >= 2
6	RW	0x0	win2_vsd_cbr_gt4 cbr_src/cbr_dst >= 4
5	RW	0x0	win2_vsd_yrgb_gt2 yrgb_src/yrgb_dst >= 2
4	RW	0x0	win2_vsd_yrgb_gt4 yrgb_src/yrgb_dst >= 4
3:2	RW	0x0	win2_bic_coe_sel 2'b00 : PRECISE 2'b01 : SPLINE 2'b10 : CATROM 2'b11 : MITCHELL
1	RW	0x0	win2_cbr_axi_gather_en win2 cbr axi bus gather enable
0	RW	0x0	win2_yrgb_axi_gather_en win2 yrgb axi bus gather enable

VOP_WIN2_COLOR_KEY
Address: Operational Base + offset (0x0308)
Win2 color key register

Bit	Attr	Reset Value	Description
31	RW	0x0	win2_key_en Win2 transparency color key enable 1'b0 : disable; 1'b1 : enable;
30:24	RO	0x0	reserved
23:0	RW	0x000000	win2_key_color Win2 key color 24 bit RGB888

VOP_WIN2_VIR
Address: Operational Base + offset (0x030c)
win2 virtual stride

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	win1_vir_stride_uv Number of words of Win1 uv Virtual width

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Bit	Attr	Reset Value	Description
15:0	RW	0x0140	win1_vir_stride Win1 Virtual stride Number of words of Win1 yrgb Virtual width ARGB888 : win1_vir_width RGB888 : (win1_vir_width*3/4) + (win1_vir_width%3) RGB565 : ceil(win1_vir_width/2) YUV : ceil(win1_vir_width/4)

VOP_WIN2_YRGB_MST
Address: Operational Base + offset (0x0310)
Win2 YRGB memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_yrgb_mst win1 YRGB frame buffer memory start address

VOP_WIN2_CBR_MST
Address: Operational Base + offset (0x0314)
Win2 Cbr memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	win2_cbr_mst win1 CBR frame buffer memory start address

VOP_WIN2_ACT_INFO
Address: Operational Base + offset (0x0318)
Win2 active window width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00ef	win2_act_height Win2 active(original) window height win_act_height = (win1 vertical size -1)
15:13	RO	0x0	reserved
12:0	RW	0x013f	win2_act_width Win2 active(original) window width win_act_width = (win1 horizontial size -1)

VOP_WIN2_DSP_INFO
Address: Operational Base + offset (0x031c)
Win2 display width/height on panel

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:16	RW	0x0ef	win2_dsp_height Win2 display window height win2_dsp_height = (win2 vertical size -1)
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11:0	RW	0x13f	win2_dsp_width Win2 display window width win2_dsp_width = (win1 horizontial size -1)

VOP_WIN2_DSP_ST
Address: Operational Base + offset (0x0320)
Win2 display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	win2_dsp_yst Win2 vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	win2_dsp_xst Win2 horizontal start point(x) of the Panel scanning

VOP_WIN2_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x0324)
Win2 YRGB scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win2_vs_factor_yrgb Win2 YRGB vertical scaling factor: factor=((LCDC_WIN2_ACT_INFO[31:16]) /(LCDC_WIN2_DSP_INFO[31:16]))*2^12
15:0	RW	0x1000	win2_hs_factor_yrgb Win2 YRGB horizontal scaling factor: factor=((LCDC_WIN2_ACT_INFO[15:0]) /(LCDC_WIN2_DSP_INFO[15:0]))*2^12

VOP_WIN2_SCL_FACTOR_CBR
Address: Operational Base + offset (0x0328)
Win2 Cbr scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	win2_vs_factor_cbr Win2 CBR vertical scaling factor: YCbCr420: factor=((LCDC_WIN2_ACT_INFO[31:16]/ 2) /(LCDC_WIN2_DSP_INFO[31:16]))*2^12 YCbCr422,YCbCr444: factor=((LCDC_WIN2_ACT_INFO[31:16]) /(LCDC_WIN2_DSP_INFO[31:16]))*2^12

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Bit	Attr	Reset Value	Description
15:0	RW	0x1000	win2_hs_factor_cbr Win2 Cbr horizontal scaling factor: YCbCr422,YCbCr420: factor=((LCDC_WIN2_ACT_INFO[15:0]/2) /(LCDC_WIN2_DSP_INFO[15:0]))*2^12 YCbCr444: factor=((LCDC_WIN2_ACT_INFO[15:0]) /(LCDC_WIN2_DSP_INFO[15:0]))*2^12

VOP_WIN2_SCL_OFFSET
Address: Operational Base + offset (0x032c)
Win2 scaling start point offset

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win2_vs_offset_cbr Cbr Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
23:16	RW	0x00	win2_vs_offset_yrgb Y Vertical scaling start point offset (0x00~0xff)/0x100 = 0~0.99
15:8	RW	0x00	win2_hs_offset_cbr Cbr Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99
7:0	RW	0x00	win2_hs_offset_yrgb Y Horizontal scaling start point offset (0x00~0xff)/0x100 = 0~0.99

VOP_WIN2_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x0330)
Win2 alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	win2_fading_value fading value,8bit
23:16	RW	0x00	win2_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	win2_src_factor_mode src factor mode
5	RW	0x0	win2_src_alpha_cal_mode src alpha calc mode
4:3	RW	0x0	win2_src_blend_mode src blend mode
2	RW	0x0	win2_src_alpha_mode src alpha mode

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Bit	Attr	Reset Value	Description
1	RW	0x0	win2_src_color_mode src color mode
0	RW	0x0	win2_src_alpha_en src alpha en

VOP_WIN2_DST_ALPHA_CTRL
Address: Operational Base + offset (0x0334)
Win2 alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	win2_dst_factor_m0 dst factor mode
5:0	RW	0x00	win2_dsp_m0_reserved reserved

VOP_WIN2_FADING_CTRL
Address: Operational Base + offset (0x0338)
Win2 fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	win2_fading_en fading enable
23:16	RW	0x00	win2_fading_offset_b fading offset blue value
15:8	RW	0x00	win2_fading_offset_g fading offset green value
7:0	RW	0x00	win2_fading_offset_r fading offset red value

VOP_WIN2_CTRL2
Address: Operational Base + offset (0x033c)
Win2 ctrl register2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x4	win_rid_win2_cbr axi read id of win2 cbr channel
3:0	RW	0x3	win_rid_win2_yrgb axi read id of win2 yrgb channel

VOP_DBG_WIN2_REG0
Address: Operational Base + offset (0x03f0)
Vop debug win2 register0

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

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VOP_DBG_WIN2_REG1
Address: Operational Base + offset (0x03f4)
Vop debug win2 register1

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_WIN2_REG2
Address: Operational Base + offset (0x03f8)
Vop debug win2 register2

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_WIN2_RESERVED
Address: Operational Base + offset (0x03fc)
Vop debug win2 register3 reserved

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	reserved reserved

VOP_HWC_CTRL0
Address: Operational Base + offset (0x0500)
Hwc ctrl register0

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	hwc_endian_swap hwc 8pp palette data Big-endian/ Little-endian select 1'b0 : Big-endian 1'b1 : Little-endian
13	RW	0x0	hwc_alpha_swap hwc RGB alpha swap 1'b0 : ARGB 1'b1 : RGBA
12	RW	0x0	hwc_rb_swap hwc RGB Red and Blue swap 1'b0 : RGB 1'b1 : BGR
11	RO	0x0	reserved
10	RW	0x0	hwc_csc_mode hwc RGB2YUV Color space conversion: 2'bx0: BT601 2'bx1: BT709
9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8	RW	0x0	hwc_interlace_read hwc interlace read mode 1'b0 : disable 1'b1 : enable
7	RO	0x0	reserved
6:5	RW	0x0	hwc_size 2'b00 : 32x32 2'b01 : 64x64 2'b10 : 96x96 2'b11 : 128x128
4	RW	0x0	hwc_mode hwc color mode 1'b0 : normal color mode 1'b1 : reversed color mode
3:1	RW	0x0	hwc_data_fmt 3'b000 : ARGB888 3'b001 : RGB888 3'b010 : RGB565 3'b100: 8bpp 3'b101: 4bpp 3'b110: 2bpp 3'b111: 1bpp
0	RW	0x0	hwc_en 1'b0 : disable 1'b1 : enable

VOP_HWC_CTRL1
Address: Operational Base + offset (0x0504)
Hwc ctrl register1

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:20	RW	0x7	win_rid_hwc axi read id of hwc channel
19:17	RO	0x0	reserved
16	RW	0x0	hwc_lut_en 1'b0 : disable 1'b1 : enable
15	RW	0x0	hwc_y_mir_en 1'b0 : no y_mirror 1'b1 : y_mirror
14	RW	0x0	hwc_no_outstanding hwc AXI master read outstanding 1'b0 : enable 1'b1 : disable

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Bit	Attr	Reset Value	Description
13	RW	0x0	hwc_rgb2yuv_en 1'b0 : enable 1'b1 : disable
12:8	RW	0x1d	hwc_axi_max_outstanding_num hwc axi bus max outstanding number
7	RO	0x0	reserved
6:4	RW	0x0	hwc_axi_gather_num hwc axi gather transfer number
3:2	RW	0x0	hwc_dma_burst_length HWC DMA read Burst length 2'b00 : burst16 (burst 15 in rgb888 pack mode) 2'b01 : burst8 (burst 12 in rgb888 pack mode) 2'b10 : burst4 (burst 6 in rgb888 pack mode) 2'b11 : reserved
1	RW	0x0	hwc_axi_max_outstanding_en 1'b0 : disable 1'b1 : enable
0	RW	0x0	hwc_axi_gather_en 1'b0 : disable 1'b1 : enable

VOP_HWC_MST
Address: Operational Base + offset (0x0508)
Hwc memory start address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	hwc_mst HWC data memory start address

VOP_HWC_DSP_ST
Address: Operational Base + offset (0x050c)
Hwc display start point on panel

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	hwc_dsp_yst HWC vertical start point(y) of the Panel scanning
15:13	RO	0x0	reserved
12:0	RW	0x000a	hwc_dsp_xst HWC horizontal start point(x) of the Panel scanning

VOP_HWC_SRC_ALPHA_CTRL
Address: Operational Base + offset (0x0510)
Hwc alpha source control register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	hwc_fading_value fading value

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	hwc_src_global_alpha src global alpha
15:9	RO	0x0	reserved
8:6	RW	0x0	hwc_src_factor_mode src factor mode
5	RW	0x0	hwc_src_alpha_cal_mode src alpha calc mode
4:3	RW	0x0	hwc_src_blend_mode src blend mode
2	RW	0x0	hwc_src_alpha_mode src alpha mode
1	RW	0x0	hwc_src_color_mode src color mode
0	RW	0x0	hwc_src_alpha_en src alpha enable

VOP_HWC_DST_ALPHA_CTRL
Address: Operational Base + offset (0x0514)
Hwc alpha destination control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:6	RW	0x0	hwc_dst_factor_mode dst factor mode
5:0	RW	0x00	hwc_dst_m0_reserved reserved

VOP_HWC_FADING_CTRL
Address: Operational Base + offset (0x0518)
Hwc fading contrl register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	hwc_fading_en 1'b0 : disable 1'b1 : enable
23:16	RW	0x00	hwc_fading_offset_b fading offset blue
15:8	RW	0x00	hwc_fading_offset_g fading offset green
7:0	RW	0x00	hwc_fading_offset_r fading offset red

VOP_HWC_RESERVED1
Address: Operational Base + offset (0x051c)
Hwc reserved

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Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_POST_DSP_HACT_INFO
Address: Operational Base + offset (0x0600)
Post scaler down horizontal start and end

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_hact_st_post Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x014a	dsp_hact_end_post Panel display scanning horizontal active end point

VOP_POST_DSP_VACT_INFO
Address: Operational Base + offset (0x0604)
Panel active horizontal scanning start point and end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_vact_st_post Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x00fa	dsp_vact_end_post Panel display scanning horizontal active end point

VOP_POST_SCL_FACTOR_YRGB
Address: Operational Base + offset (0x0608)
Post yrgb scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x1000	post_vs_factor_yrgb post YRGB vertical scaling factor: factor=((src_height[31:16]) /(dst_height[31:16]))*2^12
15:0	RW	0x1000	post_hs_factor_yrgb Post YRGB horizontal scaling factor: factor=((src_width[15:0]) /(dst_width[15:0]))*2^12

VOP_POST_RESERVED
Address: Operational Base + offset (0x060c)
Post reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_POST_SCL_CTRL

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Address: Operational Base + offset (0x0610)
Post scaling start point offset

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	post_ver_sd_en 1'b0 : post ver scl down disable 1'b1 : post ver scl down enable
0	RW	0x0	post_hor_sd_en 1'b0 : post hor scl down disable 1'b1 : post hor scl down enable

VOP_POST_DSP_VACT_INFO_F1
Address: Operational Base + offset (0x0614)
Panel active horizontal scanning start point and end point F1

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_vact_st_post Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x00fa	dsp_vact_end_post Panel display scanning horizontal active end point

VOP_DSP_HTOTAL_HS_END
Address: Operational Base + offset (0x0618)
Panel scanning horizontal width and hsync pulse end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x014a	dsp_htotal Panel display scanning horizontal period
15:13	RO	0x0	reserved
12:0	RW	0x000a	dsp_hs_end Panel display scanning hsync pulse width

VOP_DSP_HACT_ST_END
Address: Operational Base + offset (0x061c)
Panel active horizontal scanning start point and end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_hact_st Panel display scanning horizontal active start point
15:13	RO	0x0	reserved
12:0	RW	0x014a	dsp_hact_end Panel display scanning horizontal active end point

VOP_DSP_VTOTAL_VS_END
Address: Operational Base + offset (0x0620)

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Panel scanning vertical height and vsync pulse end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00fa	dsp_vtotal Panel display scanning vertical period.
15	RW	0x0	sw_dsp_vtotal_imd dsp vtotal number valid immediately enable. 1'b0 : valid after frame start 1'b1 : valid immediately
14:13	RO	0x0	reserved
12:0	RW	0x000a	dsp_vs_end Panel display scanning vsync pulse width

VOP_DSP_VACT_ST_END

Address: Operational Base + offset (0x0624)

Panel active vertical scanning start point and end point

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x000a	dsp_vact_st Panel display scanning vertical active start point
15:13	RO	0x0	reserved
12:0	RW	0x00fa	dsp_vact_end Panel display scanning vertical active end point

VOP_DSP_VS_ST_END_F1

Address: Operational Base + offset (0x0628)

Vertical scanning start point and vsync pulse end point of even filed in interlace mode

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dsp_vs_st_f1 Panel display scanning vertical vsync start point of 2nd field (interlace display mode)
15:13	RO	0x0	reserved
12:0	RW	0x0000	dsp_vs_end_f1 Panel display scanning vertical vsync end point of 2nd field(interlace display mode)

VOP_DSP_VACT_ST_END_F1

Address: Operational Base + offset (0x062c)

Vertical scanning active start point and end point of even filed in interlace mode

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	dsp_vact_st_f1 Panel display scanning vertical active start point of 2nd field (interlace display mode)
15:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12:0	RW	0x0000	dsp_vact_end_f1 Panel display scanning vertical active end point of 2nd field (interlace display mode)

VOP_BCSH_COLOR_BAR
Address: Operational Base + offset (0x0640)
Color bar config register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	color_bar_v v color value
23:16	RW	0x00	color_bar_u u color value
15:8	RW	0x00	color_bar_y y color value
7:1	RO	0x0	reserved
0	RW	0x0	bcsh_en 1'b0 : bcsh bypass 1'b1 : bcsh enable

VOP_BCSH_BCS
Address: Operational Base + offset (0x0644)
Brightness contrast saturation*contrast config register

Bit	Attr	Reset Value	Description
31:30	RW	0x3	out_mode video out mode config register 2'b00 : black 2'b01 : blue 2'b10 : color bar 2'b11 : normal video
29:20	RW	0x100	sat_con Saturation*Contrast*256 : 0,1.992*1.992
19:17	RO	0x0	reserved
16:8	RW	0x100	contrast Contrast*256 : 0,1.992
7:0	RW	0x00	brightness Brightness : -32,31

VOP_BCSH_H
Address: Operational Base + offset (0x0648)
Sin hue and cos hue config register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RW	0x100	cos_hue cos hue value
15:9	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
8:0	RW	0x000	sin_hue sin hue value

VOP_BCSH_CTRL
Address: Operational Base + offset (0x064c)
BCSH contrl register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	bcsh_r2y_csc_mode Color space conversion(RGB2YUV): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020
5	RO	0x0	reserved
4	RW	0x0	bcsh_r2y_en 1'b0:bypass 1'b1:enable
3:2	RW	0x0	bcsh_y2r_csc_mode Color space conversion(YUV2RGB): 2'b00 : 601-l 2'b01 : 709-l 2'b10 : 601-f 2'b11 : bt2020
1	RO	0x0	reserved
0	RW	0x0	bcsh_y2r_en 1'b0:bypass 1'b1:enable

VOP_FRC_LOWER01_0
Address: Operational Base + offset (0x0678)
FRC lookup table config register010

Bit	Attr	Reset Value	Description
31:16	RW	0x1284	lower01_frm1 frc parameter lowerbit = 2'b01,frm1
15:0	RW	0x4821	lower01_frm0 frc parameter lowerbit = 2'b01,frm0

VOP_FRC_LOWER01_1
Address: Operational Base + offset (0x067c)
FRC lookup table config register011

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x2148	lower01_frm3 frc parameter lowerbit = 2'b01,frm3
15:0	RW	0x8412	lower01_frm2 frc parameter lowerbit = 2'b01,frm2

VOP_FRC_LOWER10_0
Address: Operational Base + offset (0x0680)
FRC lookup table config register100

Bit	Attr	Reset Value	Description
31:16	RW	0xa55a	lower10_frm1 frc parameter lowerbit = 2'b10,frm1
15:0	RW	0x9696	lower10_frm0 frc parameter lowerbit = 2'b10,frm0

VOP_FRC_LOWER10_1
Address: Operational Base + offset (0x0684)
FRC lookup table config register101

Bit	Attr	Reset Value	Description
31:16	RW	0x5aa5	lower10_frm3 frc parameter lowerbit = 2'b10,frm3
15:0	RW	0x6969	lower10_frm2 frc parameter lowerbit = 2'b10,frm2

VOP_FRC_LOWER11_0
Address: Operational Base + offset (0x0688)
FRC lookup table config register110

Bit	Attr	Reset Value	Description
31:16	RW	0xdeb7	lower11_frm1 frc parameter lowerbit = 2'b11,frm1
15:0	RW	0x7bed	lower11_frm0 frc parameter lowerbit = 2'b11,frm0

VOP_FRC_LOWER11_1
Address: Operational Base + offset (0x068c)
FRC lookup table config register111

Bit	Attr	Reset Value	Description
31:16	RW	0xed7b	lower11_frm3 frc parameter lowerbit = 2'b11,frm3
15:0	RW	0xb7de	lower11_frm2 frc parameter lowerbit = 2'b11,frm2

VOP_DBG_POST_REG0
Address: Operational Base + offset (0x06e8)
Vop debug post register0

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Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

VOP_DBG_POST_RESERVED
Address: Operational Base + offset (0x06ec)
Vop debug post register1 reserved

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	reserved reserved

VOP_DBG_DATAO
Address: Operational Base + offset (0x06f0)
debug data output path

Bit	Attr	Reset Value	Description
31:30	RW	0x0	sw_datao_sel debug data ouptut sel 2'b00:rgb output 2'bxx:reserved
29:0	RO	0x0	reserved

VOP_DBG_DATAO_2
Address: Operational Base + offset (0x06f4)
debug data output path 2

Bit	Attr	Reset Value	Description
31:30	RW	0x0	sw_datao_sel_2 debug data ouptut sel 2 2'b00:hdmi 2'b01:mipi 2'b10:lvds 2'b11:edp
29:0	RW	0x00000000	vop_data_o_2 debug vop data output 2 debug vop data output 2

VOP_HWC_LUT_ADDR
Address: Operational Base + offset (0x0c00)
Hwc lut base address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	hwc_lut_addr the head of hwc lut address

VOP_MMU_DTE_ADDR
Address: Operational Base + offset (0x3f00)
MMU current page Table address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR MMU DTE ADDR

VOP_MMU_STATUS
Address: Operational Base + offset (0x3f04)
MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE The direction of access for last page fault: 1'b0 = Read 1'b1 = Write
4	RO	0x0	REPLAY_BUFFER_EMPTY The MMU replay buffer is empty
3	RO	0x0	MMU_IDLE The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Paging is enabled

VOP_MMU_COMMAND
Address: Operational Base + offset (0x3f08)
MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

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VOP_MMU_PAGE_FAULT_ADDR
Address: Operational Base + offset (0x3f0c)
MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR address of last page fault

VOP_MMU_ZAP_ONE_LINE
Address: Operational Base + offset (0x3f10)
MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE address to be invalidated from the page table cache

VOP_MMU_INT_RAWSTAT
Address: Operational Base + offset (0x3f14)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error
0	RW	0x0	PAGE_FAULT page fault

VOP_MMU_INT_CLEAR
Address: Operational Base + offset (0x3f18)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR read bus error
0	WO	0x0	PAGE_FAULT page fault

VOP_MMU_INT_MASK
Address: Operational Base + offset (0x3f1c)
MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error
0	RW	0x0	PAGE_FAULT page fault

VOP_MMU_INT_STATUS
Address: Operational Base + offset (0x3f20)

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MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR read bus error
0	RO	0x0	PAGE_FAULT page fault

VOP_MMU_AUTO_GATING
Address: Operational Base + offset (0x3f24)
MMU auto gating

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_auto_gating mmu auto gating when it is 1'b1, the mmu will auto gating it self

16.10.4 SDR2HDR Registers Summary

Name	Offset	Size	Reset Value	Description
SDR2HDR_SDR2HDR_CTRL	0x0000	W	0x00000000	sdr2hdr control
SDR2HDR_eotf_oetf_y0	0x0004	W	0x00000000	y0 of eotf and oetf
SDR2HDR_eotf_oetf_y1	0x0010	W	0x00000000	y1 of eotf and oetf
SDR2HDR_eotf_oetf_y2	0x0014	W	0x00000000	y2 of eotf and oetf
SDR2HDR_eotf_oetf_y3	0x0018	W	0x00000000	y3 of eotf and oetf
SDR2HDR_eotf_oetf_y4	0x001c	W	0x00000000	y4 of eotf and oetf
SDR2HDR_eotf_oetf_y5	0x0020	W	0x00000000	y5 of eotf and oetf
SDR2HDR_eotf_oetf_y6	0x0024	W	0x00000000	y6 of eotf and oetf
SDR2HDR_eotf_oetf_y7	0x0028	W	0x00000000	y7 of eotf and oetf
SDR2HDR_eotf_oetf_y8	0x002c	W	0x00000000	y8 of eotf and oetf
SDR2HDR_eotf_oetf_y9	0x0030	W	0x00000000	9 of eotf and oetf
SDR2HDR_eotf_oetf_y10	0x0034	W	0x00000000	y10 of eotf and oetf
SDR2HDR_eotf_oetf_y11	0x0038	W	0x00000000	y11 of eotf and oetf
SDR2HDR_eotf_oetf_y12	0x003c	W	0x00000000	y12 of eotf and oetf
SDR2HDR_eotf_oetf_y13	0x0040	W	0x00000000	y13 of eotf and oetf
SDR2HDR_eotf_oetf_y14	0x0044	W	0x00000000	y13 of eotf and oetf
SDR2HDR_eotf_oetf_y15	0x0048	W	0x00000000	y15 of eotf and oetf
SDR2HDR_eotf_oetf_y16	0x004c	W	0x00000000	y16 of eotf and oetf
SDR2HDR_eotf_oetf_y17	0x0050	W	0x00000000	y17 of eotf and oetf
SDR2HDR_eotf_oetf_y18	0x0054	W	0x00000000	y18 of eotf and oetf
SDR2HDR_eotf_oetf_y19	0x0058	W	0x00000000	y19 of eotf and oetf
SDR2HDR_eotf_oetf_y20	0x005c	W	0x00000000	y20 of eotf and oetf
SDR2HDR_eotf_oetf_y21	0x0060	W	0x00000000	y21 of eotf and oetf
SDR2HDR_eotf_oetf_y22	0x0064	W	0x00000000	y22 of eotf and oetf
SDR2HDR_eotf_oetf_y23	0x0068	W	0x00000000	y23 of eotf and oetf

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Name	Offset	Size	Reset Value	Description
SDR2HDR_eotf_oetf_y24	0x006c	W	0x00000000	y24 of eotf and oetf
SDR2HDR_eotf_oetf_y25	0x0070	W	0x00000000	y25 of eotf and oetf
SDR2HDR_eotf_oetf_y26	0x0074	W	0x00000000	y26 of eotf and oetf
SDR2HDR_eotf_oetf_y27	0x0078	W	0x00000000	y27 of eotf and oetf
SDR2HDR_eotf_oetf_y28	0x007c	W	0x00000000	y28 of eotf and oetf
SDR2HDR_eotf_oetf_y29	0x0080	W	0x00000000	y29 of eotf and oetf
SDR2HDR_eotf_oetf_y30	0x0084	W	0x00000000	y30 of eotf and oetf
SDR2HDR_eotf_oetf_y31	0x0088	W	0x00000000	y31 of eotf and oetf
SDR2HDR_eotf_oetf_y32	0x008c	W	0x00000000	y32 of eotf and oetf
SDR2HDR_eotf_oetf_y33	0x0090	W	0x00000000	y33 of eotf and oetf
SDR2HDR_eotf_oetf_y34	0x0094	W	0x00000000	y34 of eotf and oetf
SDR2HDR_eotf_oetf_y35	0x0098	W	0x00000000	y35 of eotf and oetf
SDR2HDR_eotf_oetf_y36	0x009c	W	0x00000000	y36 of eotf and oetf
SDR2HDR_eotf_oetf_y37	0x00a0	W	0x00000000	y37 of eotf and oetf
SDR2HDR_eotf_oetf_y38	0x00a4	W	0x00000000	y38 of eotf and oetf
SDR2HDR_eotf_oetf_y39	0x00a8	W	0x00000000	y39 of eotf and oetf
SDR2HDR_eotf_oetf_y40	0x00ac	W	0x00000000	y40 of eotf and oetf
SDR2HDR_eotf_oetf_y41	0x00b0	W	0x00000000	y41 of eotf and oetf
SDR2HDR_eotf_oetf_y42	0x00b4	W	0x00000000	y42 of eotf and oetf
SDR2HDR_eotf_oetf_y43	0x00b8	W	0x00000000	y43 of eotf and oetf
SDR2HDR_eotf_oetf_y44	0x00bc	W	0x00000000	y44 of eotf and oetf
SDR2HDR_eotf_oetf_y45	0x00c0	W	0x00000000	y45 of eotf and oetf
SDR2HDR_eotf_oetf_y46	0x00c4	W	0x00000000	y46 of eotf and oetf
SDR2HDR_eotf_oetf_y47	0x00c8	W	0x00000000	y47 of eotf and oetf
SDR2HDR_eotf_oetf_y48	0x00cc	W	0x00000000	y48 of eotf and oetf
SDR2HDR_eotf_oetf_y49	0x00d0	W	0x00000000	y49 of eotf and oetf
SDR2HDR_eotf_oetf_y50	0x00d4	W	0x00000000	y50 of eotf and oetf
SDR2HDR_eotf_oetf_y51	0x00d8	W	0x00000000	y51 of eotf and oetf
SDR2HDR_eotf_oetf_y52	0x00dc	W	0x00000000	y52 of eotf and oetf
SDR2HDR_eotf_oetf_y53	0x00e0	W	0x00000000	y53 of eotf and oetf
SDR2HDR_eotf_oetf_y54	0x00e4	W	0x00000000	y54 of eotf and oetf
SDR2HDR_eotf_oetf_y55	0x00e8	W	0x00000000	y55 of eotf and oetf
SDR2HDR_eotf_oetf_y56	0x00ec	W	0x00000000	y56 of eotf and oetf
SDR2HDR_eotf_oetf_y57	0x00f0	W	0x00000000	y57 of eotf and oetf
SDR2HDR_eotf_oetf_y58	0x00f4	W	0x00000000	y58 of eotf and oetf
SDR2HDR_eotf_oetf_y59	0x00f8	W	0x00000000	y59 of eotf and oetf
SDR2HDR_eotf_oetf_y60	0x00fc	W	0x00000000	y60 of eotf and oetf
SDR2HDR_eotf_oetf_y61	0x0100	W	0x00000000	y61 of eotf and oetf
SDR2HDR_eotf_oetf_y62	0x0104	W	0x00000000	y62 of eotf and oetf
SDR2HDR_eotf_oetf_y63	0x0108	W	0x00000000	y63 of eotf and oetf
SDR2HDR_eotf_oetf_y64	0x010c	W	0x00000000	y64 of eotf and oetf
SDR2HDR_oetf_dx_dxpow1	0x0110	W	0x00000000	dx1 and dxpow1 of oetf

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Name	Offset	Size	Reset Value	Description
SDR2HDR_oetf_dx_dxpow2	0x0114	W	0x00000000	dx2 and dxpow2 of oetf
SDR2HDR_oetf_dx_dxpow3	0x0118	W	0x00000000	dx3 and dxpow3 of oetf
SDR2HDR_oetf_dx_dxpow4	0x011c	W	0x00000000	dx4 and dxpow4 of oetf
SDR2HDR_oetf_dx_dxpow5	0x0120	W	0x00000000	dx5 and dxpow5 of oetf
SDR2HDR_oetf_dx_dxpow6	0x0124	W	0x00000000	dx6 and dxpow6 of oetf
SDR2HDR_oetf_dx_dxpow7	0x0128	W	0x00000000	dx7 and dxpow7 of oetf
SDR2HDR_oetf_dx_dxpow8	0x012c	W	0x00000000	dx8 and dxpow8 of oetf
SDR2HDR_oetf_dx_dxpow9	0x0130	W	0x00000000	dx9 and dxpow9 of oetf
SDR2HDR_oetf_dx_dxpow10	0x0134	W	0x00000000	dx10 and dxpow10 of oetf
SDR2HDR_oetf_dx_dxpow11	0x0138	W	0x00000000	dx11 and dxpow11 of oetf
SDR2HDR_oetf_dx_dxpow12	0x013c	W	0x00000000	dx12 and dxpow12 of oetf
SDR2HDR_oetf_dx_dxpow13	0x0140	W	0x00000000	dx13 and dxpow13 of oetf
SDR2HDR_oetf_dx_dxpow14	0x0144	W	0x00000000	dx14 and dxpow14 of oetf
SDR2HDR_oetf_dx_dxpow15	0x0148	W	0x00000000	dx15 and dxpow15 of oetf
SDR2HDR_oetf_dx_dxpow16	0x014c	W	0x00000000	dx16 and dxpow16 of oetf
SDR2HDR_oetf_dx_dxpow17	0x0150	W	0x00000000	dx17 and dxpow17 of oetf
SDR2HDR_oetf_dx_dxpow18	0x0154	W	0x00000000	dx18 and dxpow18 of oetf
SDR2HDR_oetf_dx_dxpow19	0x0158	W	0x00000000	dx19 and dxpow19 of oetf
SDR2HDR_oetf_dx_dxpow20	0x015c	W	0x00000000	dx20 and dxpow20 of oetf
SDR2HDR_oetf_dx_dxpow21	0x0160	W	0x00000000	dx21 and dxpow21 of oetf
SDR2HDR_oetf_dx_dxpow22	0x0164	W	0x00000000	dx22 and dxpow22 of oetf
SDR2HDR_oetf_dx_dxpow23	0x0168	W	0x00000000	dx23 and dxpow23 of oetf

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Name	Offset	Size	Reset Value	Description
SDR2HDR_oetf_dx_dxpow24	0x016c	W	0x00000000	dx24 and dxpow24 of oetf
SDR2HDR_oetf_dx_dxpow25	0x0170	W	0x00000000	dx25 and dxpow25 of oetf
SDR2HDR_oetf_dx_dxpow26	0x0174	W	0x00000000	dx26 and dxpow26 of oetf
SDR2HDR_oetf_dx_dxpow27	0x0178	W	0x00000000	dx27 and dxpow27 of oetf
SDR2HDR_oetf_dx_dxpow28	0x017c	W	0x00000000	dx28 and dxpow28 of oetf
SDR2HDR_oetf_dx_dxpow29	0x0180	W	0x00000000	dx29 and dxpow29 of oetf
SDR2HDR_oetf_dx_dxpow30	0x0184	W	0x00000000	dx30 and dxpow31 of oetf
SDR2HDR_oetf_dx_dxpow31	0x0188	W	0x00000000	dx31 and dxpow31 of oetf
SDR2HDR_oetf_dx_dxpow32	0x018c	W	0x00000000	dx32 and dxpow32 of oetf
SDR2HDR_oetf_dx_dxpow33	0x0190	W	0x00000000	dx33 and dxpow33 of oetf
SDR2HDR_oetf_dx_dxpow34	0x0194	W	0x00000000	dx34 and dxpow34 of oetf
SDR2HDR_oetf_dx_dxpow35	0x0198	W	0x00000000	dx35 and dxpow35 of oetf
SDR2HDR_oetf_dx_dxpow36	0x019c	W	0x00000000	dx36 and dxpow36 of oetf
SDR2HDR_oetf_dx_dxpow37	0x01a0	W	0x00000000	dx37 and dxpow37 of oetf
SDR2HDR_oetf_dx_dxpow38	0x01a4	W	0x00000000	dx38 and dxpow38 of oetf
SDR2HDR_oetf_dx_dxpow39	0x01a8	W	0x00000000	dx39 and dxpow39 of oetf
SDR2HDR_oetf_dx_dxpow40	0x01ac	W	0x00000000	dx40 and dxpow40 of oetf
SDR2HDR_oetf_dx_dxpow41	0x01b0	W	0x00000000	dx41 and dxpow41 of oetf
SDR2HDR_oetf_dx_dxpow42	0x01b4	W	0x00000000	dx42 and dxpow42 of oetf
SDR2HDR_oetf_dx_dxpow43	0x01b8	W	0x00000000	dx43 and dxpow43 of oetf
SDR2HDR_oetf_dx_dxpow44	0x01bc	W	0x00000000	dx44 and dxpow44 of oetf
SDR2HDR_oetf_dx_dxpow45	0x01c0	W	0x00000000	dx45 and dxpow45 of oetf

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Name	Offset	Size	Reset Value	Description
SDR2HDR_oetf_dx_dxpow46	0x01c4	W	0x00000000	dx46 and dxpow46 of oetf
SDR2HDR_oetf_dx_dxpow47	0x01c8	W	0x00000000	dx47 and dxpow47 of oetf
SDR2HDR_oetf_dx_dxpow48	0x01cc	W	0x00000000	dx48 and dxpow48 of oetf
SDR2HDR_oetf_dx_dxpow49	0x01d0	W	0x00000000	dx49 and dxpow49 of oetf
SDR2HDR_oetf_dx_dxpow50	0x01d4	W	0x00000000	dx50 and dxpow50 of oetf
SDR2HDR_oetf_dx_dxpow51	0x01d8	W	0x00000000	dx51 and dxpow51 of oetf
SDR2HDR_oetf_dx_dxpow52	0x01dc	W	0x00000000	dx52 and dxpow52 of oetf
SDR2HDR_oetf_dx_dxpow53	0x01e0	W	0x00000000	dx53 and dxpow53 of oetf
SDR2HDR_oetf_dx_dxpow54	0x01e4	W	0x00000000	dx54 and dxpow54 of oetf
SDR2HDR_oetf_dx_dxpow55	0x01e8	W	0x00000000	dx55 and dxpow55 of oetf
SDR2HDR_oetf_dx_dxpow56	0x01ec	W	0x00000000	dx56 and dxpow56 of oetf
SDR2HDR_oetf_dx_dxpow57	0x01f0	W	0x00000000	dx57 and dxpow57 of oetf
SDR2HDR_oetf_dx_dxpow58	0x01f4	W	0x00000000	dx58 and dxpow58 of oetf
SDR2HDR_oetf_dx_dxpow59	0x01f8	W	0x00000000	dx59 and dxpow59 of oetf
SDR2HDR_oetf_dx_dxpow60	0x01fc	W	0x00000000	dx60 and dxpow60 of oetf
SDR2HDR_oetf_dx_dxpow61	0x0200	W	0x00000000	dx61 and dxpow61 of oetf
SDR2HDR_oetf_dx_dxpow62	0x0204	W	0x00000000	dx62 and dxpow62 of oetf
SDR2HDR_oetf_dx_dxpow63	0x0208	W	0x00000000	dx63 and dxpow63 of oetf
SDR2HDR_oetf_dx_dxpow64	0x020c	W	0x00000000	dx64 and dxpow64 of oetf
SDR2HDR_oetf_xn1	0x0210	W	0x00000000	xn1 of eotf
SDR2HDR_oetf_xn2	0x0214	W	0x00000000	xn2 of eotf
SDR2HDR_oetf_xn3	0x0218	W	0x00000000	xn3 of eotf
SDR2HDR_oetf_xn4	0x021c	W	0x00000000	xn4 of eotf
SDR2HDR_oetf_xn5	0x0220	W	0x00000000	xn5 of eotf
SDR2HDR_oetf_xn6	0x0224	W	0x00000000	Xn6 of eotf
SDR2HDR_oetf_xn7	0x0228	W	0x00000000	xn7 of eotf

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Name	Offset	Size	Reset Value	Description
SDR2HDR_oetf_xn8	0x022c	W	0x00000000	xn8 of eotf
SDR2HDR_oetf_xn9	0x0230	W	0x00000000	xn9 of eotf
SDR2HDR_oetf_xn10	0x0234	W	0x00000000	xn10 of eotf
SDR2HDR_oetf_xn11	0x0238	W	0x00000000	xn11 of eotf
SDR2HDR_oetf_xn12	0x023c	W	0x00000000	xn12 of eotf
SDR2HDR_oetf_xn13	0x0240	W	0x00000000	xn13 of eotf
SDR2HDR_oetf_xn14	0x0244	W	0x00000000	xn14 of eotf
SDR2HDR_oetf_xn15	0x0248	W	0x00000000	xn15 of eotf
SDR2HDR_oetf_xn16	0x024c	W	0x00000000	xn16 of eotf
SDR2HDR_oetf_xn17	0x0250	W	0x00000000	xn17 of eotf
SDR2HDR_oetf_xn18	0x0254	W	0x00000000	xn18 of eotf
SDR2HDR_oetf_xn19	0x0258	W	0x00000000	xn19 of eotf
SDR2HDR_oetf_xn20	0x025c	W	0x00000000	xn20 of eotf
SDR2HDR_oetf_xn21	0x0260	W	0x00000000	xn21 of eotf
SDR2HDR_oetf_xn22	0x0264	W	0x00000000	xn22 of eotf
SDR2HDR_oetf_xn23	0x0268	W	0x00000000	xn23 of eotf
SDR2HDR_oetf_xn24	0x026c	W	0x00000000	xn24 of eotf
SDR2HDR_oetf_xn25	0x0270	W	0x00000000	xn25 of eotf
SDR2HDR_oetf_xn26	0x0274	W	0x00000000	xn26 of eotf
SDR2HDR_oetf_xn27	0x0278	W	0x00000000	xn27 of eotf
SDR2HDR_oetf_xn28	0x027c	W	0x00000000	xn28 of eotf
SDR2HDR_oetf_xn29	0x0280	W	0x00000000	xn29 of eotf
SDR2HDR_oetf_xn30	0x0284	W	0x00000000	xn30 of eotf
SDR2HDR_oetf_xn31	0x0288	W	0x00000000	xn31 of eotf
SDR2HDR_oetf_xn32	0x028c	W	0x00000000	xn32 of eotf
SDR2HDR_oetf_xn33	0x0290	W	0x00000000	xn33 of eotf
SDR2HDR_oetf_xn34	0x0294	W	0x00000000	xn34 of eotf
SDR2HDR_oetf_xn35	0x0298	W	0x00000000	xn35 of eotf
SDR2HDR_oetf_xn36	0x029c	W	0x00000000	xn36 of eotf
SDR2HDR_oetf_xn37	0x02a0	W	0x00000000	xn37 of eotf
SDR2HDR_oetf_xn38	0x02a4	W	0x00000000	xn38 of eotf
SDR2HDR_oetf_xn39	0x02a8	W	0x00000000	xn39 of eotf
SDR2HDR_oetf_xn40	0x02ac	W	0x00000000	xn40 of eotf
SDR2HDR_oetf_xn41	0x02b0	W	0x00000000	xn41 of eotf
SDR2HDR_oetf_xn42	0x02b4	W	0x00000000	xn42 of eotf
SDR2HDR_oetf_xn43	0x02b8	W	0x00000000	xn43 of eotf
SDR2HDR_oetf_xn44	0x02bc	W	0x00000000	xn44 of eotf
SDR2HDR_oetf_xn45	0x02c0	W	0x00000000	xn45 of eotf
SDR2HDR_oetf_xn46	0x02c4	W	0x00000000	xn46 of eotf
SDR2HDR_oetf_xn47	0x02c8	W	0x00000000	xn47 of eotf
SDR2HDR_oetf_xn48	0x02cc	W	0x00000000	xn48 of eotf
SDR2HDR_oetf_xn49	0x02d0	W	0x00000000	xn49 of eotf
SDR2HDR_oetf_xn50	0x02d4	W	0x00000000	xn50 of eotf
SDR2HDR_oetf_xn51	0x02d8	W	0x00000000	xn51 of eotf

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Name	Offset	Size	Reset Value	Description
SDR2HDR_oetf_xn52	0x02dc	W	0x00000000	xn52 of eotf
SDR2HDR_oetf_xn53	0x02e0	W	0x00000000	xn53 of eotf
SDR2HDR_oetf_xn54	0x02e4	W	0x00000000	xn54 of eotf
SDR2HDR_oetf_xn55	0x02e8	W	0x00000000	xn55 of eotf
SDR2HDR_oetf_xn56	0x02ec	W	0x00000000	xn56 of eotf
SDR2HDR_oetf_xn57	0x02f0	W	0x00000000	xn57 of eotf
SDR2HDR_oetf_xn58	0x02f4	W	0x00000000	xn58 of eotf
SDR2HDR_oetf_xn59	0x02f8	W	0x00000000	xn59 of eotf
SDR2HDR_oetf_xn60	0x02fc	W	0x00000000	xn60 of eotf
SDR2HDR_oetf_xn61	0x0300	W	0x00000000	xn61 of eotf
SDR2HDR_oetf_xn62	0x0304	W	0x00000000	xn62 of eotf
SDR2HDR_oetf_xn63	0x0308	W	0x00000000	xn63 of eotf

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.10.5 SDR2HDR Detail Register Description

SDR2HDR_CTRL

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0000)
sdr2hdr control

Bit	Attr	Reset Value	Description
31	RW	0x0	win_csc_mode_sel 1'b0 : according to win datafmt and overlaymode , auto decide 1'b1 : configuable
30:18	RO	0x0	reserved
17	RW	0x0	hwc_y2r_en 1'b0 : disable 1'b1 : enable
16	RW	0x0	hwc_r2y_en 1'b0 : disable 1'b1 : enable
15	RW	0x0	win3_y2r_en 1'b0 : disable 1'b1 : enable
14	RW	0x0	win3_r2y_en 1'b0 : disable 1'b1 : enable
13	RW	0x0	win2_y2r_en 1'b0 : disable 1'b1 : enable
12	RW	0x0	win2_r2y_en 1'b0 : disable 1'b1 : enable
11	RW	0x0	win1_y2r_en 1'b0 : disable 1'b1 : enable

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Bit	Attr	Reset Value	Description
10	RW	0x0	win1_r2y_en 1'b0 : disable 1'b1 : enable
9	RW	0x0	win0_y2r_en 1'b0 : disable 1'b1 : enable
8	RW	0x0	win0_r2y_en 1'b0 : disable 1'b1 : enable
7	RW	0x0	st2084oetf_post_conv_en 1'b0 : disable 1'b1 : enable
6	RW	0x0	rgb2rgb_post_conv_mode 1'b0 : disable 1'b1 : enable
5	RW	0x0	rgb2rgb_post_conv_en 1'b0 : disable 1'b1 : enable
4	RW	0x0	bt1886eotf_post_conv_en 1'b0 : disable 1'b1 : enable
3	RW	0x0	st2084oetf_pre_conv_en 1'b0 : disable 1'b1 : enable
2	RW	0x0	rgb2rgb_pre_conv_mode 1'b0 : rgb709 to rgb2020 1'b1 : rgb2020 to rgb709
1	RW	0x0	rgb2rgb_pre_conv_en rgb2rgb conv enable 1'b0 : disable 1'b1 : enable
0	RW	0x0	bt1886eotf_pre_conv_en 1'b0 : disable 1'b1 : enable

SDR2HDR_eotf_oetf_y0

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0004)

y0 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y1

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0010)
y1 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y2
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0014)
y2 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y3
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0018)
y3 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y4
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x001c)
y4 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y5
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0020)
y5 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y6
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0024)

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y6 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y7

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0028)

y7 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y8

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x002c)

y8 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y9

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0030)

9 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y10

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0034)

y10 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y11

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0038)

y11 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y12
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x003c)
y12 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y13
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0040)
y13 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y14
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0044)
y13 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y15
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0048)
y15 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y16
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x004c)
y16 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y17
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0050)
y17 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y18
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0054)
y18 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y19
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0058)
y19 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y20
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x005c)
y20 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y21
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0060)
y21 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y22
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0064)
y22 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y23
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0068)
y23 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y24
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x006c)
y24 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y25
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0070)
y25 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y26
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0074)
y26 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y27
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0078)
y27 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y28
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x007c)
y28 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y29
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0080)
y29 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y30
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0084)
y30 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y31
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0088)
y31 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y32
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x008c)
y32 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y33
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0090)
y33 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y34
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0094)
y34 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y35
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0098)
y35 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y36
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x009c)
y36 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y37
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00a0)
y37 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y38
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00a4)
y38 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y39
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00a8)
y39 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y40
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00ac)
y40 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y41
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00b0)
y41 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y42
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00b4)
y42 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y43
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00b8)
y43 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y44
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00bc)
y44 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y45
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00c0)
y45 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y46
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00c4)
y46 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y47
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00c8)
y47 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y48
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00cc)
y48 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y49
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00d0)
y49 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y50
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00d4)
y50 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y51
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00d8)
y51 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y52
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00dc)
y52 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y53
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00e0)
y53 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y54
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00e4)
y54 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y55
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00e8)
y55 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y56
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00ec)
y56 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y57
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00f0)
y57 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y58
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00f4)
y58 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y59
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00f8)
y59 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y60
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x00fc)
y60 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y61
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0100)
y61 of eotf and oetf

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Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y62
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0104)
y62 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y63
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0108)
y63 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_eotf_oetf_y64
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x010c)
y64 of eotf and oetf

Bit	Attr	Reset Value	Description
31:18	RW	0x0000	st2084oetf_y y of st2084oetf
17:0	RW	0x00000	bt1886eotf_y bt1886eotf y

SDR2HDR_oetf_dx_dxpow1
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0110)
dx1 and dxpow1 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow2
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0114)

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dx2 and dxpow2 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow3

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0118)

dx3 and dxpow3 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow4

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x011c)

dx4 and dxpow4 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow5

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0120)

dx5 and dxpow5 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow6

Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0124)

dx6 and dxpow6 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow7
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0128)
dx7 and dxpow7 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow8
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x012c)
dx8 and dxpow8 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow9
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0130)
dx9 and dxpow9 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow10
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0134)
dx10 and dxpow10 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow11

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0138)
dx11 and dxpow11 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow12
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x013c)
dx12 and dxpow12 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow13
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0140)
dx13 and dxpow13 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow14
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0144)
dx14 and dxpow14 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow15
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0148)
dx15 and dxpow15 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow16
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x014c)
dx16 and dxpow16 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow17
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0150)
dx17 and dxpow17 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow18
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0154)
dx18 and dxpow18 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow19
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0158)
dx19 and dxpow19 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow20

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x015c)
dx20 and dxpow20 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow21
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0160)
dx21 and dxpow21 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow22
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0164)
dx22 and dxpow22 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow23
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0168)
dx23 and dxpow23 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow24
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x016c)
dx24 and dxpow24 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow25
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0170)
dx25 and dxpow25 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow26
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0174)
dx26 and dxpow26 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow27
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0178)
dx27 and dxpow27 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow28
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x017c)
dx28 and dxpow28 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow29

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0180)
dx29 and dxpow29 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow30
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0184)
dx30 and dxpow31 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow31
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0188)
dx31 and dxpow31 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow32
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x018c)
dx32 and dxpow32 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow33
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0190)
dx33 and dxpow33 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow34
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0194)
dx34 and dxpow34 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow35
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0198)
dx35 and dxpow35 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow36
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x019c)
dx36 and dxpow36 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow37
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01a0)
dx37 and dxpow37 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow38

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01a4)
dx38 and dxpow38 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow39
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01a8)
dx39 and dxpow39 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow40
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01ac)
dx40 and dxpow40 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow41
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01b0)
dx41 and dxpow41 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow42
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01b4)
dx42 and dxpow42 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow43
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01b8)
dx43 and dxpow43 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow44
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01bc)
dx44 and dxpow44 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow45
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01c0)
dx45 and dxpow45 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow46
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01c4)
dx46 and dxpow46 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow47

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01c8)
dx47 and dxpow47 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow48
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01cc)
dx48 and dxpow48 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow49
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01d0)
dx49 and dxpow49 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow50
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01d4)
dx50 and dxpow50 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow51
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01d8)
dx51 and dxpow51 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow52
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01dc)
dx52 and dxpow52 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow53
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01e0)
dx53 and dxpow53 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow54
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01e4)
dx54 and dxpow54 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow55
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01e8)
dx55 and dxpow55 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow56

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01ec)
dx56 and dxpow56 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow57
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01f0)
dx57 and dxpow57 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow58
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01f4)
dx58 and dxpow58 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow59
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01f8)
dx59 and dxpow59 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow60
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x01fc)
dx60 and dxpow60 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow61
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0200)
dx61 and dxpow61 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow62
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0204)
dx62 and dxpow62 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow63
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0208)
dx63 and dxpow63 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_dx_dxpow64
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x020c)
dx64 and dxpow64 of oetf

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	oetf_dxpow dxpow of oetf
15:0	RW	0x0000	oetf_dx dx of oetf

SDR2HDR_oetf_xn1

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Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0210)
xn1 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn2
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0214)
xn2 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn3
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0218)
xn3 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn4
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x021c)
xn4 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn5
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0220)
xn5 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn6
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0224)
xn of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn7
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0228)
xn7 of eotf

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn8
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x022c)
xn8 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn9
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0230)
xn9 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn10
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0234)
xn10 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn11
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0238)
xn11 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn12
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x023c)
xn12 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn13
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0240)
xn13 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

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SDR2HDR_oetf_xn14
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0244)
xn14 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn15
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0248)
xn15 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn16
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x024c)
xn16 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn17
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0250)
xn17 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn18
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0254)
xn18 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn19
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0258)
xn19 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn20
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x025c)

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xn20 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn21
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0260)
xn21 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn22
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0264)
xn22 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn23
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0268)
xn23 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn24
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x026c)
xn24 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn25
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0270)
xn25 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn26
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0274)
xn26 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn27
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0278)
xn27 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn28
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x027c)
xn28 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn29
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0280)
xn29 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn30
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0284)
xn30 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn31
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0288)
xn31 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn32
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x028c)
xn32 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

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SDR2HDR_oetf_xn33
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0290)
xn33 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn34
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0294)
xn34 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn35
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0298)
xn35 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn36
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x029c)
xn36 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn37
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02a0)
xn37 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn38
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02a4)
xn38 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn39
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02a8)
xn39 of eotf

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn40
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02ac)
xn40 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn41
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02b0)
xn41 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn42
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02b4)
xn42 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn43
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02b8)
xn43 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn44
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02bc)
xn44 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn45
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02c0)
xn45 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

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SDR2HDR_oetf_xn46
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02c4)
xn46 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn47
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02c8)
xn47 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn48
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02cc)
xn48 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn49
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02d0)
xn49 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn50
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02d4)
xn50 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn51
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02d8)
xn51 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn52
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02dc)

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xn52 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn53
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02e0)
xn53 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn54
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02e4)
xn54 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn55
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02e8)
xn55 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn56
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02ec)
xn56 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn57
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02f0)
xn57 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn58
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02f4)
xn58 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn59
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02f8)
xn59 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn60
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x02fc)
xn60 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn61
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0300)
xn61 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn62
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0304)
xn62 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

SDR2HDR_oetf_xn63
Address: Operational Base + SDR2HDR Base(0x0700) + offset (0x0308)
xn63 of eotf

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	eotf_xn

16.10.6 HDR2SDR Registers Summary

Name	Offset	Size	Reset Value	Description
HDR2SDR_hdr2sdr_ctrl	0x0000	W	0x00000000	hdr2sdr control
HDR2SDR_hdr2sdr_src_range	0x0004	W	0x00000000	source lumimance range
HDR2SDR_hdr2sdr_normfaceetf	0x0008	W	0x00000000	eetf normalization factor

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Name	Offset	Size	Reset Value	Description
HDR2SDR_hdr2sdr_dst_range	0x0010	W	0x00000000	destination lumimance range
HDR2SDR_hdr2sdr_normfaccgamma	0x0014	W	0x00000000	eetf normalization factor
HDR2SDR_eetf_oetf_y0	0x0018	W	0x00000000	y0 of eetf and oetf
HDR2SDR_sat_y0	0x001c	W	0x00000000	y0 of sat
HDR2SDR_eetf_oetf_y1	0x0020	W	0x00000000	y1 of eetf and oetf
HDR2SDR_eetf_oetf_y2	0x0024	W	0x00000000	y2 of eetf and oetf
HDR2SDR_eetf_oetf_y3	0x0028	W	0x00000000	y3 of eetf and oetf
HDR2SDR_eetf_oetf_y4	0x002c	W	0x00000000	y4 of eetf and oetf
HDR2SDR_eetf_oetf_y5	0x0030	W	0x00000000	y5 of eetf and oetf
HDR2SDR_eetf_oetf_y6	0x0034	W	0x00000000	y6 of eetf and oetf
HDR2SDR_eetf_oetf_y7	0x0038	W	0x00000000	y7 of eetf and oetf
HDR2SDR_eetf_oetf_y8	0x003c	W	0x00000000	y8 of eetf and oetf
HDR2SDR_eetf_oetf_y9	0x0040	W	0x00000000	y9 of eetf and oetf
HDR2SDR_eetf_oetf_y10	0x0044	W	0x00000000	y10 of eetf and oetf
HDR2SDR_eetf_oetf_y11	0x0048	W	0x00000000	y11 of eetf and oetf
HDR2SDR_eetf_oetf_y12	0x004c	W	0x00000000	y12 of eetf and oetf
HDR2SDR_eetf_oetf_y13	0x0050	W	0x00000000	y13 of eetf and oetf
HDR2SDR_eetf_oetf_y14	0x0054	W	0x00000000	y14 of eetf and oetf
HDR2SDR_eetf_oetf_y15	0x0058	W	0x00000000	y15 of eetf and oetf
HDR2SDR_eetf_oetf_y16	0x005c	W	0x00000000	y16 of eetf and oetf
HDR2SDR_eetf_oetf_y17	0x0060	W	0x00000000	y17 of eetf and oetf
HDR2SDR_eetf_oetf_y18	0x0064	W	0x00000000	y18 of eetf and oetf
HDR2SDR_eetf_oetf_y19	0x0068	W	0x00000000	y19 of eetf and oetf
HDR2SDR_eetf_oetf_y20	0x006c	W	0x00000000	y20 of eetf and oetf
HDR2SDR_eetf_oetf_y21	0x0070	W	0x00000000	y21 of eetf and oetf
HDR2SDR_eetf_oetf_y22	0x0074	W	0x00000000	y22 of eetf and oetf
HDR2SDR_eetf_oetf_y23	0x0078	W	0x00000000	y23 of eetf and oetf
HDR2SDR_eetf_oetf_y24	0x007c	W	0x00000000	y24 of eetf and oetf
HDR2SDR_eetf_oetf_y25	0x0080	W	0x00000000	y25 of eetf and oetf
HDR2SDR_eetf_oetf_y26	0x0084	W	0x00000000	y26 of eetf and oetf
HDR2SDR_eetf_oetf_y27	0x0088	W	0x00000000	y27 of eetf and oetf
HDR2SDR_eetf_oetf_y28	0x008c	W	0x00000000	y28 of eetf and oetf
HDR2SDR_eetf_oetf_y29	0x0090	W	0x00000000	y29 of eetf and oetf
HDR2SDR_eetf_oetf_y30	0x0094	W	0x00000000	y30 of eetf and oetf
HDR2SDR_eetf_oetf_y31	0x0098	W	0x00000000	y31 of eetf and oetf
HDR2SDR_eetf_oetf_y32	0x009c	W	0x00000000	y32 of eetf and oetf
HDR2SDR_sat_y1	0x00a0	W	0x00000000	y1 of sat
HDR2SDR_sat_y2	0x00a4	W	0x00000000	y2 of sat
HDR2SDR_sat_y3	0x00a8	W	0x00000000	y3 of sat
HDR2SDR_sat_y4	0x00ac	W	0x00000000	y4 of sat
HDR2SDR_sat_y5	0x00b0	W	0x00000000	y5 of sat
HDR2SDR_sat_y6	0x00b4	W	0x00000000	y6 of sat

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Name	Offset	Size	Reset Value	Description
HDR2SDR_sat_y7	0x00b8	W	0x00000000	y7 of sat
HDR2SDR_sat_y8	0x00bc	W	0x00000000	y8 of sat

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.10.7 HDR2SDR Detail Register Description

HDR2SDR_hdr2sdr_ctrl

Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0000)
hdr2sdr control

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	hdr2sdr_en 1'b0 : disable 1'b1 : enable

HDR2SDR_hdr2sdr_src_range

Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0004)
source lumimance range

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	src_max source max lumimance
15:14	RO	0x0	reserved
13:0	RW	0x0000	src_min source min lumimance

HDR2SDR_hdr2sdr_normfaceetf

Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0008)
eetf normalization factor

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x000	normfaceetf normalization factor

HDR2SDR_hdr2sdr_dst_range

Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0010)
destination lumimance range

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	dst_max destination max lumimance
15:0	RW	0x0000	dst_min destination min lumimance

HDR2SDR_hdr2sdr_normfaccgamma

Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0014)
eetf normalization factor

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	normfaccgamma gamma normalization factor

HDR2SDR_eetf_oetf_y0
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0018)
y0 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_sat_y0
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x001c)
y0 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_eetf_oetf_y1
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0020)
y1 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y2
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0024)
y2 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

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HDR2SDR_eetf_oetf_y3
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0028)
y3 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y4
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x002c)
y4 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y5
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0030)
y5 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y6
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0034)
y6 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

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HDR2SDR_eetf_oetf_y7
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0038)
y7 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y8
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x003c)
y8 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y9
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0040)
y9 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y10
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0044)
y10 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y11
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0048)

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y11 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y12
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x004c)
y12 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y13
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0050)
y13 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y14
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0054)
y14 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y15
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0058)
y15 of eetf and oetf

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y16
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x005c)
y16 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y17
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0060)
y17 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y18
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0064)
y18 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y19
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0068)
y19 of eetf and oetf

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y20
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x006c)
y20 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y21
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0070)
y21 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y22
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0074)
y22 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y23
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0078)
y23 of eetf and oetf

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y24
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x007c)
y24 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y25
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0080)
y25 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y26
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0084)
y26 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y27
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0088)
y27 of eetf and oetf

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y28
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x008c)
y28 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y29
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0090)
y29 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y30
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0094)
y30 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y31
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x0098)
y31 of eetf and oetf

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_eetf_oetf_y32
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x009c)
y32 of eetf and oetf

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	bt1886oetf_y y of bt1886oetf
15:14	RO	0x0	reserved
13:0	RW	0x0000	eetf_y y of eetf

HDR2SDR_sat_y1
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00a0)
y1 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y2
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00a4)
y2 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y3
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00a8)
y3 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y4
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00ac)

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y4 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y5
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00b0)
y5 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y6
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00b4)
y6 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y7
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00b8)
y7 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

HDR2SDR_sat_y8
Address: Operational Base + HDR2SDR Base(0x0a10) + offset (0x00bc)
y8 of sat

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:0	RW	0x0000	sat_y y of sat

16.10.8 TVE Registers Summary

Name	Offset	Size	Reset Value	Description
TVE_TVE_MODE_CTRL	0x0000	W	0x00000000	Mode Control
TVE_TVE_HOR_TIMING1	0x0004	W	0x00000000	Horizontal Timing 1
TVE_TVE_HOR_TIMING2	0x0008	W	0x00000000	Horizontal Timing 2
TVE_TVE_HOR_TIMING3	0x000c	W	0x00000000	Horizontal Timing 3
TVE_TVE_SUB_CAR_FRQ	0x0010	W	0x00000000	Sub-carrier Frequency
TVE_TVE_LUMA_FILTER1	0x0014	W	0x00000000	Luma Filter 1

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Name	Offset	Size	Reset Value	Description
TVE_TVE_LUMA_FILTER2	0x0018	W	0x00000000	Luma Filter 2
TVE_TVE_LUMA_FILTER3	0x001c	W	0x00000000	Luma Filter 3
TVE_TVE_MACROVISION_RESERVED0	0x0020	W	0x00000000	tve macrovision reserved0 register
TVE_TVE_MACROVISION_RESERVED1	0x0024	W	0x00000000	tve macrovision reserved1 register
TVE_TVE_MACROVISION_RESERVED2	0x0028	W	0x00000000	tve macrovision reserved2 register
TVE_TVE_MACROVISION_RESERVED3	0x002c	W	0x00000000	tve macrovision reserved3 register
TVE_TVE_MACROVISION_RESERVED4	0x0030	W	0x00000000	tve macrovision reserved3 register
TVE_TVE_IMAGE_POSITION	0x0034	W	0x00000000	tve image position
TVE_TVE_ROUTING	0x0038	W	0x00000000	Tve Routing
TVE_TVE_VBID_ST_TIMING	0x003c	W	0x00000000	tve vbid start timing
TVE_TVE_VBID_LINES	0x0040	W	0x00000000	tve vbid lines
TVE_TVE_VBID_CC_DATA	0x0044	W	0x00000000	tve vbid cc data
TVE_TVE_VBID_XDS_DATA	0x0048	W	0x00000000	tve vbid xds data
TVE_TVE_VBID_CGMS_DATA	0x004c	W	0x00000000	tve vbid cgms data
TVE_TVE_SYNC_ADJUST	0x0050	W	0x00000000	Sync Adjust
TVE_TVE_STATUS	0x0054	W	0x00000000	TVE Status (Read Only)
TVE_TVE_SURFACE_SIZE_RESERVED	0x0058	W	0x00000000	tve surface size reserved
TVE_TVE_STREAM_BASE_ADDR_CTRL_RESERVED	0x005c	W	0x00000000	tve stream base addr ctrl reserved
TVE_TVE_STRIDE_RESERVED	0x0060	W	0x00000000	tve stride reserved
TVE_TVE_MEM_INF_CTRL_RESERVED	0x0064	W	0x00000000	tve mem inf ctrl reserved
TVE_TVE_CTRL	0x0068	W	0x00000000	TVE Control
TVE_TVE_INTR_STATUS	0x006c	W	0x00000000	Interrupt Status
TVE_TVE_INTR_EN	0x0070	W	0x00000000	Interrupt Enable
TVE_TVE_INTR_CLR	0x0074	W	0x00000000	Interrupt Clear
TVE_TVE_COLOR_BUSRT_SAT	0x0078	W	0x00000000	Colour Burst and Saturation
TVE_TVE_TEXT_HOR_ACT_RESERVED	0x007c	W	0x00000000	Teletext Horizontal Active
TVE_TVE_TEXT_VER_ACT_RESERVED	0x0080	W	0x00000000	tve text ver act reserved

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Name	Offset	Size	Reset Value	Description
TVE_TVE_TEXT_MODE_RATE_RESERVED	0x0084	W	0x00000000	tve text mode rate reserved
TVE_TVE_TEXT_FETCH_RESERVED	0x0088	W	0x00000000	tve text fetch reserved
TVE_TVE_CHROMA_BANDWIDTH	0x008c	W	0x00000000	Chroma bandwidth
TVE_TVE_BRIGHTNESS_CONTRAST	0x0090	W	0x00000000	Brightness and Contrast
TVE_TVE_TEXT_DATA_ID_RESERVED	0x0094	W	0x00000000	tve text data id reserved
TVE_TVE_ID	0x0098	W	0x0a010000	TVE ID (Read Only)
TVE_TVE_REVISION	0x009c	W	0x00010008	TVE Revision (Read Only)
TVE_TVE_CLAMP	0x00a0	W	0x00000000	TVE Clamp

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.10.9 TVE Detail Register Description

TVE_TVE_MODE_CTRL

Address: Operational Base + TVE Base(0x3e00) + offset (0x0000)
Mode Control

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	pal 1'b0 Encoding is NTSC 1'b1 Encoding is PAL Note: 525/60 and 625/50 timing is selected separately
23:20	RO	0x0	reserved
19:16	RW	0x0	clock_ratios 4'bXX00: Reserved 4'bXX01: Timing enable is 1 * pix_clk (HDTV) 4'bXX10: Timing enable is 1 * pix_clk (SDTV) 4'bXX11: Reserved 4'b00XX: Reserved 4'b01XX: Upstream enable is 1 * pix_clk (HDTV) 4'b10XX: Upstream enable is 1/2* pix_clk (SDTV) 4'b11XX: Reserved
15:14	RW	0x0	function_select 2'b00: AV Off (TV displays broadcast signal, not SCART signal) 2'b01: Interface dependent effect 2'b10: Interface dependent effect 2'b11: Interface dependent effect Note: these bits are used to generate the SCART Function Select (pin 8, Slow Switching) signal via an optional external interface. They have no effect on the internal operation of the TVE, and could be re-assigned to another board level function.

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Bit	Attr	Reset Value	Description
13	RW	0x0	fast_blank 1'b0: TV uses composite SCART signal 1'b1: TV uses RGB SCART signal Note: This bit is used to generate the SCART Fast Blank signal via an optional external interface, which is used for static switching only. This bit has no effect on the internal operation of the TVE.
12:11	RO	0x0	reserved
10:9	RW	0x0	luma_filter_gain 2'b00: Luma Filter gain is 1.0 (default value) 2'b01: Luma Filter gain is 0.5 2'b10: Luma Filter gain is 2.0 2'b11: Reserved
8	RW	0x0	luma_upsample 1'b0: Luma Filter output is sampled at 13.5 MHz 1'b1: Luma Filter output is up-sampled to 27 MHz
7:3	RO	0x0	reserved
2:1	RW	0x0	csc_input_sel 2'b00: RGB input to 4:4:4+RGB output 2'b01: 4:2:2 input to 4:4:4 output 2'b10: Reserved 2'b11: Bypass (4:4:4 input to 4:4:4 output) The PAL/NTSC encoder requires 4:4:4 out from the CSC block, component out requires RGB or CrYCb for RGB or PbYPr respectively.
0	RW	0x0	color_space 1'b0: BT.601 (PAL/NTSC) 1'b1: BT.709 (ATSC/EIA-720.3) RGB input is converted to 4:4:4 in either colourspace. The PAL/NTSC encoder requires BT.601 colour space, component out requires BT.601 for 480i/576i or BT.709 for 720p/1080i.

TVE_TVE_HOR_TIMING1

Address: Operational Base + TVE Base(0x3e00) + offset (0x0004)

Horizontal Timing 1

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	h_burst_end Delay to the end of colour burst (from H Sync Start) Unsigned value, LSB = 37 ns when timing enable = 27 MHz.
15:8	RW	0x00	h_breeze_end Delay to the start of the colour burst (from H Sync Start) Unsigned value, LSB = 37 ns when timing enable = 27 MHz.

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	h_sync_end Width of horizontal sync from H Sync Start Unsigned value, LSB = 37 ns when timing enable = 27 MHz.

TVE_TVE_HOR_TIMING2
Address: Operational Base + TVE Base(0x3e00) + offset (0x0008)
Horizontal Timing 2

Bit	Attr	Reset Value	Description
31:28	RW	0x0	tv_syncs 4'b0000: Sync generator is stopped 4'b0001: 525/60 interlaced 4'b0010: 625/50 interlaced 4'b0011: PALM 4'b0100: PALN z 4'b0101: i480 (EIA-770.2) 4'b0110: p480 (EIA-770.2) 4'b0111: p720 (EIA-770.3) 4'b1001: i1080 (EIA-770.3) 4'b1101: 262 non-interlaced 4'b1110: 312 non-interlaced Note that PAL colour encoding is separately controlled
27:16	RW	0x000	h_active_end Delay to the end of active video (from H Sync Start) Unsigned value, LSB = 37 ns when timing enable = 27 MHz.
15:13	RO	0x0	reserved
12	RW	0x0	sc_reset_en SC Reset Enable Bit 1'b0: Subcarrier phase is not reset regularly 1'b1: Subcarrier phase is reset every four fields (NTSC) or eight fields (PAL)
11:0	RW	0x000	h_active_st Delay to the start of the active video (from H Sync Start) Unsigned value, LSB = 37 ns.

TVE_TVE_HOR_TIMING3
Address: Operational Base + TVE Base(0x3e00) + offset (0x000c)
Horizontal Timing 3

Bit	Attr	Reset Value	Description
31:28	RW	0x0	luma_delay_1 Programmable delay for Luma relative to Chroma which is applied to CVBS0 and Y/C outputs.

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Bit	Attr	Reset Value	Description
27:16	RW	0x000	h_total Total number of output pixels in a TV line in units of the timing enable. Applies to SD and HD formats. 525/60 = 1716 625/50 = 1728 720P/60 = 1650 1080IP/60 Hz = 2200 1080IP/50 Hz = 2640.
15:8	RO	0x0	reserved
7:0	RW	0x00	sc_horiz Phase offset applied if subcarrier phase is reset. Signed value adjusts subcarrier reset phase in units of 1.4 degrees per lsb

TVE_TVE_SUB_CAR_FRQ
Address: Operational Base + TVE Base(0x3e00) + offset (0x0010)
Sub-carrier Frequency

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dds_incr Phase increment used in the synthesis of the colour subcarrier Calculated as 2^32*Fsc/Fref, where Fref is the timing enable rate (normally 27 MHz). Note that the Fref must be accurate to 20 ppm for encoded formats. Recommended settings (for 27.00 MHz) are: NTSC/M, NTSC-J 0x21F07BD7 PAL (B, D, G, H, I, N) 0x2A098ACB PAL/Nc 0x21F69446 PAL/M 0x21E6EFA4

TVE_TVE_LUMA_FILTER1
Address: Operational Base + TVE Base(0x3e00) + offset (0x0014)
Luma Filter 1

Bit	Attr	Reset Value	Description
31:24	RW	0x00	luma_coeff3 8-bit signed
23:16	RW	0x00	luma_coeff2 8-bit signed
15:8	RW	0x00	luma_coeff1 8-bit signed
7:0	RW	0x00	luma_coeff0 8-bit signed

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TVE_TVE_LUMA_FILTER2
Address: Operational Base + TVE Base(0x3e00) + offset (0x0018)
Luma Filter 2

Bit	Attr	Reset Value	Description
31:24	RW	0x00	luma_coeff7 8-bit signed
23:16	RW	0x00	luma_coeff6 8-bit signed
15:8	RW	0x00	luma_coeff5 8-bit signed
7:0	RW	0x00	luma_coeff4 8-bit signed

TVE_TVE_LUMA_FILTER3
Address: Operational Base + TVE Base(0x3e00) + offset (0x001c)
Luma Filter 3

Bit	Attr	Reset Value	Description
31:24	RW	0x00	luma_coeff11 8-bit Unsigned
23:16	RW	0x00	luma_coeff10 8-bit signed
15:8	RW	0x00	luma_coeff9 8-bit signed
7:0	RW	0x00	luma_coeff8 8-bit signed

TVE_TVE_MACROVISION_RESERVED0
Address: Operational Base + TVE Base(0x3e00) + offset (0x0020)
tve macrovision reserved0 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_MACROVISION_RESERVED1
Address: Operational Base + TVE Base(0x3e00) + offset (0x0024)
tve macrovision reserved1 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_MACROVISION_RESERVED2
Address: Operational Base + TVE Base(0x3e00) + offset (0x0028)
tve macrovision reserved2 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_MACROVISION_RESERVED3

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Address: Operational Base + TVE Base(0x3e00) + offset (0x002c)
tve macrovision reserved3 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_MACROVISION_RESERVED4
Address: Operational Base + TVE Base(0x3e00) + offset (0x0030)
tve macrovision reserved3 register

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_IMAGE_POSITION
Address: Operational Base + TVE Base(0x3e00) + offset (0x0034)
tve image position

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26:16	RW	0x000	v_offset Vertical picture start offset (independent of picture blanking) The active area of the source image can be offset above or below the active area of the display. The value is the display line number at which the top line of the frame will be placed. The top of the frame cannot be earlier than line 1.
15:12	RO	0x0	reserved
11:0	RW	0x000	h_offset Horizontal picture start offset (independent of picture blanking) The active area of the source image can be offset to the left or right of the active area of the display. The value is the display pixel number at which the first pixel of the line will be placed. The left edge cannot be earlier than pixel 1.

TVE_TVE_ROUTING
Address: Operational Base + TVE Base(0x3e00) + offset (0x0038)
Tve Routing

Bit	Attr	Reset Value	Description
31	RW	0x0	RB_swap 1'b0: No swap 1'b1: Swap Blue/Pb and Red/Pr channels on DACs, relative to the routing shown in 30:28

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Bit	Attr	Reset Value	Description
30:28	RW	0x0	dac_fmt DAC3 DAC2 DAC1 DAC0 3'b000: CVBS Blue Green Red 3'b001: CVBS1 CVBS2 Y C 3'b011: Y CVBS2 CVBS1 C 3'b100: CVBS Pb Y Pr 3'b101: C Pb Y Pr
27:24	RW	0x0	sense_on When ON, outputs a steady mid level on the selected DAC(s) to allow load sensing via DAC or external comparators (application dependent). May be applied singly or together DAC3 DAC2 DAC1 DAC0 4'b0000: OFF OFF OFF OFF 4'b0001: OFF OFF OFF ON 4'b0010: OFF OFF ON OFF 4'b0100: OFF ON OFF OFF 4'b1000: ON OFF OFF OFF 4'b1111: ON ON ON ON
23:20	RW	0x0	vbid_on When ON, adds any enabled VBI Data to the specified channel Luma Red/Pr Green Blue/Pb 4'b0000: OFF OFF OFF OFF 4'bXXX1: ON 4'bXX1X: ON 4'bX1XX: ON 4'b1XXX: ON
19:16	RW	0x0	setup_on When ON, adds 7.5IRE setup to the specified channel Luma Red/Pr Green Blue/Pb 4'b0000: OFF OFF OFF OFF 4'bXXX1: ON 4'bXX1X: ON 4'bX1XX: ON 4'b1XXX: ON
15:12	RW	0x0	AGC_pluse_on When ON, adds AGC and EOF pulses to the specified channel. N.B. This field should not be used for disabling macrovision functionality. Y/C Red/Pr Green Blue/Pb 4'b0000: OFF OFF OFF OFF 4'bXXX1: ON 4'bXX1X: ON 4'bX1XX: ON 4'b1XXX: ON

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Bit	Attr	Reset Value	Description
11:8	RW	0x0	video_on When ON, adds appropriate video format to the specified channel Y/C Red/Pr Green Blue/Pb 4'b0000 : OFF OFF OFF OFF 4'bXXX1: ON 4'bXX1X: ON 4'bX1XX: ON 4'b1XXX: ON
7:4	RW	0x0	sync_on When ON, adds composite sync to the specified channel N.B. Macrovision pseudo-sync pulses (if enabled) will be added to each channel that has sync enabled. Luma Red/Pr Green Blue/Pb 4'b0000: OFF OFF OFF OFF 4'bXXX1: ON 4'bXX1X: ON 4'bX1XX: ON 4'b1XXX: ON
3	RW	0x0	YPP 1'b0: Component output is RGB 1'b1: Component output is YPbPr
2	RW	0x0	chroma_off 1'b0: Chroma is switched ON (normal colour display) 1'b1: Chroma is switched OFF (monochrome display)
1:0	RW	0x0	Picture_Sync_amplitudes 2'bX0: Component has picture/sync ratio of 714/286 (RS343A) 2'bX1: Component has picture/sync ratio of 700/300 (ATSC) 2'b0X : Composite has picture/sync ratio of 714/286 (NTSC) 2'b1X : Composite has picture/sync ratio of 700/300 (PAL)

TVE_TVE_VBID_ST_TIMING
Address: Operational Base + TVE Base(0x3e00) + offset (0x003c)
tve vbid start timing

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_VBID_LINES
Address: Operational Base + TVE Base(0x3e00) + offset (0x0040)
tve vbid lines

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_VBID_CC_DATA
Address: Operational Base + TVE Base(0x3e00) + offset (0x0044)

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tve vbid cc data

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_VBID_XDS_DATA
Address: Operational Base + TVE Base(0x3e00) + offset (0x0048)
tve vbid xds data

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_VBID_CGMS_DATA
Address: Operational Base + TVE Base(0x3e00) + offset (0x004c)
tve vbid cgms data

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_SYNC_ADJUST
Address: Operational Base + TVE Base(0x3e00) + offset (0x0050)
Sync Adjust

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	line_adj Signed number of lines by which to change frame length. Affects every frame whilst a non-zero value.
7:0	RW	0x00	pix_adj Signed number of (27MHz) pixels by which to adjust frame length. Affects every frame whilst a non-zero value.

TVE_TVE_STATUS
Address: Operational Base + TVE Base(0x3e00) + offset (0x0054)
TVE Status (Read Only)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RO	0x0	pixel_manager_vstt Pixel Manager Vertical Start State 1'b0: Vertical Start not active 1'b1: Vertical Start active
6:4	RO	0x0	pixel_manager_sta Pixel Manager Flow Controller State 3'b000: Idle 3'b001: Seeking 3'b010: Ready 3'b011: First pixel 3'b100: Active 3'b111: Other
3:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RO	0x0	sync_gen_VBI Sync Generator Vertical Blanking Interval 1'b0: In VBI 1'b1: Not in VBI
0	RO	0x0	sync_gen_FID Sync Generator Field Identity 1'b0: First Field 1'b1: Second Filed

TVE_TVE_SURFACE_SIZE_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0058)
tve surface size reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_STREAM_BASE_ADDR_CTRL_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x005c)
tve stream base addr ctrl reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_STRIDE_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0060)
tve stride reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_MEM_INF_CTRL_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0064)
tve mem inf ctrl reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_CTRL
Address: Operational Base + TVE Base(0x3e00) + offset (0x0068)
TVE Control

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	disp_rst Software reset for all other TVE functions: 1'b0: Normal operation. 1'b1: Software reset (please do NOT reset the TVE every field or frame).
0	RO	0x0	reserved

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TVE_TVE_INTR_STATUS
Address: Operational Base + TVE Base(0x3e00) + offset (0x006c)
Interrupt Status

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	ints_str1orun Indicates over-run for stream str1
0	RW	0x0	ints_str1urun Indicates under-run for stream str1

TVE_TVE_INTR_EN
Address: Operational Base + TVE Base(0x3e00) + offset (0x0070)
Interrupt Enable

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	inte_str1orun Enables over-run interrupt for stream str1
0	RW	0x0	inte_str1urun Enables under-run interrupt for stream str1

TVE_TVE_INTR_CLR
Address: Operational Base + TVE Base(0x3e00) + offset (0x0074)
Interrupt Clear

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	intclr_str1orun Writing '1' clears over-run interrupt for stream str1. Writing '0' has no effect
0	RW	0x0	intclr_str1urun Writing ‘礸’ clears under-run interrupt for stream str1. Writing ‘礫’ has no effect

TVE_TVE_COLOR_BUSRT_SAT
Address: Operational Base + TVE Base(0x3e00) + offset (0x0078)
Colour Burst and Saturation

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	burst_size Colour burst amplitude
15:8	RW	0x00	V_weight Conversion factor for Cr to V
7:0	RW	0x00	U_weight Conversion factor for Cb to U

TVE_TVE_TEXT_HOR_ACT_RESERVED

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Address: Operational Base + TVE Base(0x3e00) + offset (0x007c)
Teletext Horizontal Active

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_TEXT_VER_ACT_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0080)
tve text ver act reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_TEXT_MODE_RATE_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0084)
tve text mode rate reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_TEXT_FETCH_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0088)
tve text fetch reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_CHROMA_BANDWIDTH
Address: Operational Base + TVE Base(0x3e00) + offset (0x008c)
Chroma bandwidth

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	chroma_bandwidth 2'b00: Chroma bandpass filter is bypassed 2'b01: Chroma bandpass filter is centred on 3.58 MHz 2'b10: Chroma bandpass filter is centred on 4.43 MHz
3:0	RW	0x0	cdiff_bandwidth 4'b0000: Colour difference filters OFF (no colour) 4'b0001: Colour difference bandwidth is 0.6 MHz 4'b0010: Colour difference bandwidth is 1.3 MHz 4'b0011: Colour difference bandwidth is 2.0 MHz

TVE_TVE_BRIGHTNESS_CONTRAST
Address: Operational Base + TVE Base(0x3e00) + offset (0x0090)
Brightness and Contrast

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	Contrast This value determines the gain of the luma channel. It does not affect the chroma gain.

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	Brightness This value determines the black level on the luma channel during active video only. The setup (if applicable) is applied in addition to this value.

TVE_TVE_TEXT_DATA_ID_RESERVED
Address: Operational Base + TVE Base(0x3e00) + offset (0x0094)
tve text data id reserved

Bit	Attr	Reset Value	Description
31:0	RO	0x0	reserved

TVE_TVE_ID
Address: Operational Base + TVE Base(0x3e00) + offset (0x0098)
TVE ID (Read Only)

Bit	Attr	Reset Value	Description
31:24	RO	0x0a	group_id 0x0a
23:16	RO	0x01	core_id 0x01
15:0	RO	0x0	reserved

TVE_TVE_REVISION
Address: Operational Base + TVE Base(0x3e00) + offset (0x009c)
TVE Revision (Read Only)

Bit	Attr	Reset Value	Description
31:24	RO	0x00	oem OEM field = 鰬”
23:16	RO	0x01	major Major revision = 鯨”
15:8	RO	0x00	minor Minor revision= 鯨”
7:0	RO	0x08	maint Maintenance revision = □”

TVE_TVE_CLAMP
Address: Operational Base + TVE Base(0x3e00) + offset (0x00a0)
TVE Clamp

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	clamp_off Horizontal position in pix_clks/4 where clamp pulse goes low
7:0	RW	0x00	clamp_on Horizontal position in pix_clks/4 where clamp pulse goes high

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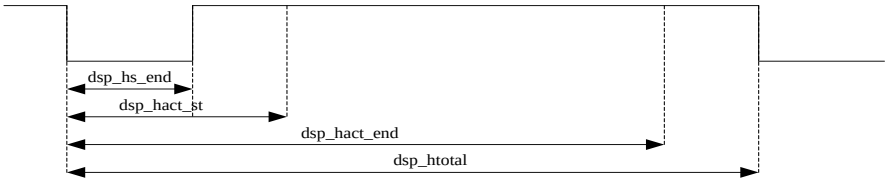
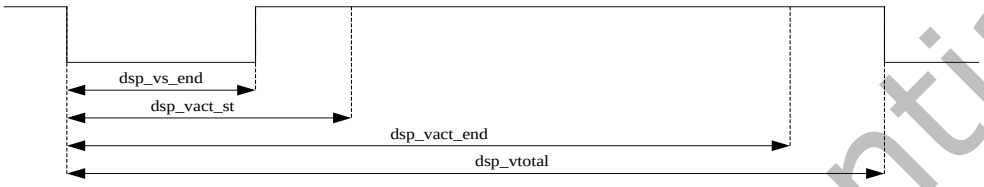
16.11 Timing Diagram

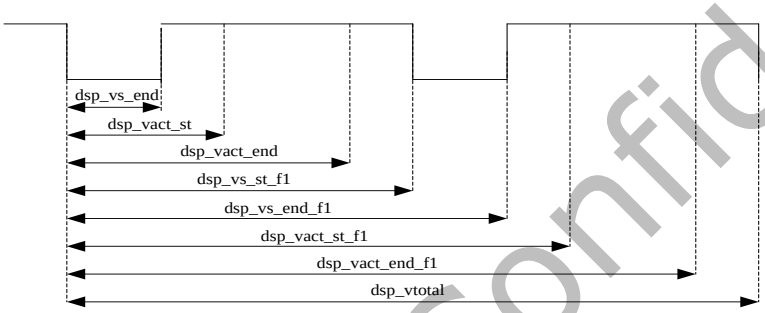
16.11.1 LCD interface timing



Horizontal timing



Vertical timing(Progressive mode)



Vertical timing(Interlace mode)

Fig. 08-16 VOP RGB interface timing (SDR)

16.12 Application Notes

16.12.1 DMA transfer mode

There are three DMA transfer modes for loading win0 or win1 frame data determined by following parameters(X=0,1,2,3):

dma_burst_length

winX_no_outstanding

winX_gather_en

winX_gather_thres

1. auto outstanding transfer mode(random transfer)

When winX_no_outstanding is 0, multi-bursts transfer command could be sent out to AXI master interface continuously if the internal memory has enough space to store new data. The continuous random burst number is in the range of 1 to 4, mainly depending on the empty level of internal memory, dma_burst_length, data format and active image width.

2. configured outstanding transfer mode(fixed transfer)

When winX_gather_en is 1, fixed-number of bursts transfer command should be sent out to AXI master interface continuously if the internal memory has enough space to store new data. The fixed-number is determined by winX_gather_thres. Since the internal memory size is limited, there is some restriction for the winX_gather_thres as follows.

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Table 08-2 Gather configuration for all format

Gather Threshold	dma_burst_length =2'b00(burst16)	dma_burst_length =2'b01(burst8)	dma_burst_length =2'b10(burst4)
YUV420 YUV422 YUV444	0	0,1,2	0,1,2,3
ARGB888 RGB888 RGB565	0,1,2,3	0,1,2,3	0,1,2,3
8BPP 4BPP 2BPP 1BPP	0,1,2,3	0,1,2,3	0,1,2,3

16.12.2 Win0/Win1/Win2 dma load mode

If you want to improve the efficiency of accessing external memory for loading winX frame data, you could assert winX_dma_load. When winX_dma_load is high, winX frame data is loaded in the unit of line composing with one or more burst transfers; otherwise, loaded in the unit of burst transfer. However, it is not suitable for data format YUV420, no-scaling and active width less than 256.

16.12.3 DMA QoS request

If you want to get higher priority for VOP to access external memory when the frame data is urgent, a QoS request can be generated and sent out basing on the configured values:

noc_hurry_en

noc_hurry_value

noc_qos_en

noc_win_qos

If noc_qos_en is enable, a win0/1_qos_req is asserted when the empty level of win0/1's linebuffer is greater than the threshold configured in noc_win_qos. And it will be disserted when the empty level is smaller than the threshold or noc_qos_en is disable.

If noc_qos_en is enable, a win0/1_hurry_req is asserted when the empty level of win2/3's fifo is greater than the threshold configured in noc_win_qos. And it will be disserted when the empty level is smaller than the threshold or noc_qos_en is disable.

Either win0/1_qos_req or win2/3_hurry_req is high, a QoS request will be sent out for VOP.

16.12.4 Mirror display

If Y-Mirror display is enable, the frame data is loaded from last line to first line, where the start address of first pixel in last line is defined in

WIN0/1_YRGB0_MST/WIN0/1_CBR0_MST/WIN0/1_YRGB1_MST/WIN0/1_CBR_MST for win0/1/2/3 respectively.

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Otherwise, the win's frame line data width and virtual stride should be 64bit-aligned for 8bit-RGB/YUV or 128bit-aligned for 10bit YUV if X-Mirror or Y-Mirror display is enable.

16.12.5 Interrupt

VOP interrupt is comprised of 10 interrupt sources:

2 frame start interrupt

same frame address interrupt

2 line flag interrupt

hold interrupt

bus error interrupt

win0 empty interrupt

win1 empty interrupt

hwc empty interrupt

post empty interrupt

irq_mmu

irq_tve

Every interrupt has independent interrupt enable (VOP_INT_EN,with bit write mask), interrupt clear (VOP_INT_CLR,with bit write mask), interrupt status (VOP_INT_STATUS).

16.12.6 RGB display mode

RGB display mode is used for RGB panel display and CCIR656 output. It is a continuous frames display mode.

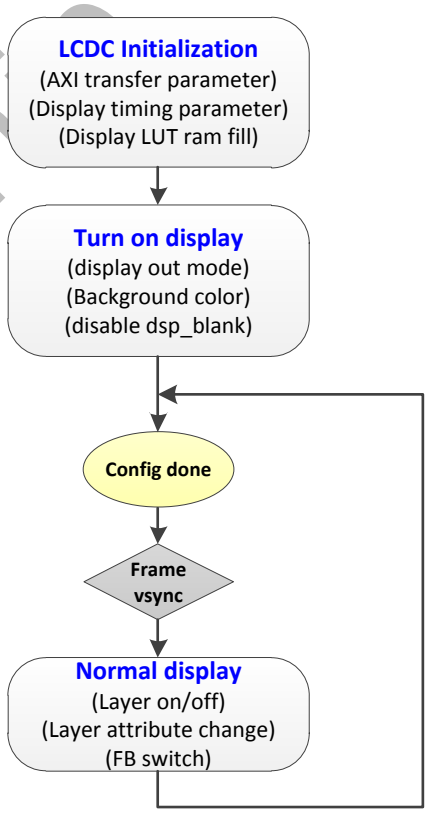

Fig. 08-17 VOP RGB mode Programming flow

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1.VOP initialization

VOP initialization should be done before turning display on.

First, AXI bus parameter (VOP_SYS_CTRL) should be set for DMA transfer.

Second, display panel/interface timing should be set for display output. The registers are: VOP_DSP_HTOTAL_HS_END/ VOP_DSP_HACT_ST_END/ VOP_DSP_VTOTAL_HS_END/ VOP_DSP_VACT_ST_END/ VOP_DSP_VS_ST_END_F1/ VOP_DSP_VACT_ST_END_F1

2.Background display

Before normal display, the background display could be turn on.

First, set display output mode (VOP_DSP_CTRL0/1) according to display device.

Second, disable dsp_blank mode, which would not be enable until frame synchronization.

Finally, writing '1' to "VOP_REG_CFG_DONE" register then all the frame-sync registers will be enable at the beginning of next frame.

3.Normal display

In normal display, all the display layers' attribute could be different according display scenario. So there is a programming loop in this mode.

First, configure all the display layers' attribute registers for the change of image format, location, size, scaling factor, alpha and overlay and so on. Those register would not be enable until frame synchronization.

Finally, write '1' to "VOP_REG_CFG_DONE" register then all the frame-sync registers will be enable at the beginning of next frame.

Immediately control register

There are two type register in VOP , one type is effective immediately,the other is effective by frame sync.

Effective immediately registers list as follows,other registers are all effective by frame sync.

Table 08-3 effective immediately register table

register address	Description
0x008[23:21],0x008[15:8]	some dsp ctrl function bit
0x00c	sys ctrl1 register
0x018	background color register
0x01c	mcu ctrl register
0x038	win0 color key register
0x078	win1 color key register
0x188~0x19c	dsp_timing ctrl registers

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0x1a0~0x1a8	pwm ctrl registers
0x1c8~0x1dc	cabc_gauss_parameter registers
0x1ec~0x1f4	frc pattern parameter registers

16.12.7 TV ENCODE

The TV ENCODE core feature :

Single 10-bit output to on-chip VDAC

PAL/NTSC encoding

Programmable luma filter coefficient

Programmable luma/chroma delay

Programmable brightness/contrast

There is a special register to config In TV mode.

tve_mode -> dsp_ctrl0[25]

0 : NTSC mode

1 : PAL mode

Other registers reference to register description.

There list NTSC and PAL typical register config as follow:

```
ntsc_config[][2] ={
    {0x00 , 0x000a0000} ,
    {0x04 , 0x00C07a81} ,
    {0x08 , 0x169800FC} ,
    {0x0C , 0x96B40000} ,
    {0x10 , 0x21F07BD7} ,
    {0x14 , 0x02ff0000} ,
    {0x18 , 0xF40202fd} ,
    {0x1C , 0xF332d919} ,
    {0x34 , 0x001500D6} ,
    {0x38 , 0x0100888C} ,
    {0x3C , 0x00000000} ,
    {0x50 , 0x00000000} ,
    {0x68 , 0x00000000} ,
    {0x78 , 0x0052543C} ,
    {0x8C , 0x00000002} ,
```

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```
    {0x90 , 0x00008300}};


pal_config[][2] ={
    {0x00 , 0x010a0006} ,
    {0x04 , 0x00C28381} ,
    {0x08 , 0x2674013D} ,
    {0x0C , 0x00000880} ,
    {0x10 , 0x2A098ACB} ,
    {0x14 , 0x02ff0000} ,
    {0x18 , 0xF40202fd} ,
    {0x1C , 0xF332d919} ,
    {0x34 , 0x001500F6} ,
    {0x38 , 0x4100088A} ,
    {0x3C , 0x00000000} ,
    {0x50 , 0x00000000} ,
    {0x68 , 0x00000000} ,
    {0x78 , 0x0035604c} ,
    {0x8C , 0x00000022} ,
    {0x90 , 0x00009800}};
```

When working in TV mode ,the dclk is 27Mhz,the sw_core_clk_sel should enable.

16.12.8 HDR10

1.Scenario one

WIN0 is HDR10 video or picture , WIN1 is SDR UI and display format is SDR , shown as follow .

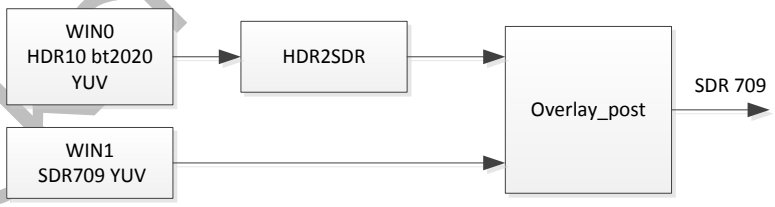

Fig. 08-18 HDR10 Scenario one

The reference register configuration is listed as follow .

1)Set SYS_CTRL[16] to 0 to set RGB overlay mode

2)Set SYS_CTRL1[28] to 1 to enable 2-level overlay mode

3)Configure hdr2sdr_eetf_yn

```
hdr2sdr_eetf_yn[33]=[647 ,945,1254 ,1573,
                     1903,2241,2586,2940,
                     3299,3664,4035,4410,
                     4789,5171,5556,5943,
```

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6333,6723,7100,7458,
7795,8110,8402,8670,
8913,9131,9323,9487,
9624,9731,9809,9856,
9872] ;

4)Configure hdr2sdr_sat_yn

hdr2sdr_sat_yn[0] =[0 ,1792 ,3584 ,3472 ,
2778 ,2083 ,1389 ,694 , 0];

5)Configure hdr2sdr_bt1886oetf[33]

hdr2sdr_bt1886oetf_yn[0] =[0 ,0 ,0 ,0 ,
0 ,0 ,57 ,295 ,
612 ,1036 ,1602 ,2358 ,
2912 ,3366 ,3758 ,4106 ,
4422 ,4712 ,5235 ,5700 ,
6122 ,6509 ,7207 ,7828 ,
8390 ,8908 ,9839 ,10668,
11419,12109,13353,14459,
16383];

6)Set HDR2SDR_CTRL = 1 to enable HDR2SDR

2.Scenario two

WIN0 is HDR10 video or picture , WIN1 is SDR UI and display format is HDR , shown as follow .

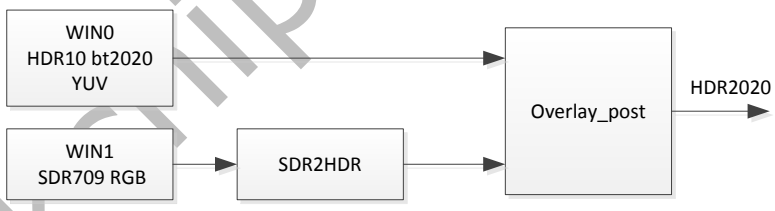

Fig. 08-22 HDR10 Scenario two

1)Set SYS_CTRL[16] to 0 to set RGB overlay mode

2)Set SYS_CTRL1[28] to 1 to enable 2-level overlay mode

3)Configure sdr2hdr_bt1886eotf_yn[65]

sdr2hdr_bt1886eotf_yn[65] =[0 ,11 ,45 ,107 ,
204 ,338 ,513 ,732 ,
998 ,1313 ,1679 ,2098 ,
2573 ,3106 ,3698 ,4351 ,
5066 ,5846 ,6692 ,7605 ,
8587 ,9105 ,9640 ,10193,
10764,11353,11960,12587,

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13232,13895,14578,15280,
16002,16369,16742,17120,
17503,17890,18283,18681,
19083,19491,19903,20321,
20744,21172,21604,22042,
22486,22709,22934,23160,
23388,23616,23846,24078,
24310,24544,24779,25016,
25254,25493,25733,25975,
26214];

4)Configure sdr2hdr_st2084oetf_yn

sdr2hdr_st2084oetf_yn[65] =[0 ,668 ,910 ,1217 ,
1600 ,2068 ,2384 ,2627 ,
3282 ,3710 ,4033 ,4879 ,
5416 ,5815 ,6135 ,6401 ,
6631 ,6833 ,7176 ,7462 ,
7707 ,7921 ,8113 ,8285 ,
8442 ,8586 ,8843 ,9068 ,
9268 ,9447 ,9760 ,10027,
10259,10465,10650,10817,
10971,11243,11480,11689,
11877,12047,12202,12345,
12477,12601,12716,12926,
13115,13285,13441,13583,
13716,13839,13953,14163,
14350,14519,14673,14945,
15180,15570,15887,16153,
16383];

5)Configure sdr2hdr_st2084oetf_xn

sdr2hdr_st2084oetf_xn[63] =[1 ,2 ,4 ,8 ,
16 ,24 ,32 ,64 ,
96 ,128 ,256 ,384 ,
512 ,640 ,768 ,896 ,
1024 ,1280 ,1536 ,1792 ,
2048 ,2304 ,2560 ,2816 ,
3072 ,3584 ,4096 ,4608 ,

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5120 ,6144 ,7168 ,8192 ,
9216 ,10240 ,11264 ,12288 ,
14336 ,16384 ,18432 ,20480 ,
22528 ,24576 ,26624 ,28672 ,
30720 ,32768 ,36864 ,40960 ,
45056 ,49152 ,53248 ,57344 ,
61440 ,65536 ,73728 ,81920 ,
90112 ,98304 ,114688,131072,
163840,196608,229376];

6)Configure sdr2hdr_st2084oetf_dxn

sdr2hdr_st2084oetf_dxn[64] =[1 ,1 ,2 ,4 ,
8 ,8 ,8 ,32 ,
32 ,32 ,128 ,128 ,
128 ,128 ,128 ,128 ,
128 ,256 ,256 ,256 ,
256 ,256 ,256 ,256 ,
256 ,512 ,512 ,512 ,
512 ,1024 ,1024 ,1024 ,
1024 ,1024 ,1024 ,1024 ,
2048 ,2048 ,2048 ,2048 ,
2048 ,2048 ,2048 ,2048 ,
2048 ,2048 ,4096 ,4096 ,
4096 ,4096 ,4096 ,4096 ,
4096 ,4096 ,8192 ,8192 ,
8192 ,8192 ,16384,16384,
32768,32768,32768,32768] ;

7)Configure sdr2hdr_st2084oetf_dxn_pow2

sdr2hdr_st2084oetf_dxn_pow2[0] =[0 ,0 ,1 ,2 ,
3 ,3 ,3 ,5 ,
5 ,5 ,7 ,7 ,
7 ,7 ,7 ,7 ,
7 ,8 ,8 ,8 ,
8 ,8 ,8 ,8 ,
8 ,9 ,9 ,9 ,
9 ,10 ,10 ,10 ,
10 ,10 ,10 ,10 ,

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```
                                    11 ,11 ,11 ,11 ,
                                    11 ,11 ,11 ,11 ,
                                    11 ,11 ,12 ,12 ,
                                    12 ,12 ,12 ,12 ,
                                    12 ,12 ,13 ,13 ,
                                    13 ,13 ,14 ,14 ,
                                    15 ,15 ,15 ,15 ];
```

8)Configure SDR2HDR_CTRL

Write 0x8000020b to SDR2HDR_CTRL

Chapter 18 Raster Graphic Acceleration(RGA)

16.13 Overview

RGAis a separate 2D raster graphic acceleration unit. It accelerates 2D graphics operations, such as point/line drawing, image scaling, rotation, BitBLT, alpha blending and image blur/sharpness.

RGA supports the following features:

- **Data format**
  - Input data:

    ARGB/RGB888/RGB565/RGB4444/RGB5551/YUV420/YUV422

    Support YUV422SP10bit/YUV420SP10bit

  - Output data:

    ARGB/RGB888/RGB565/RGB4444/RGB5551/YUV420/YUV422

  - Pixel Format conversion, BT.601/BT.709
  - Dither operation
  - Max resolution: 8192x8192 source, 4096x4096 destination
- **Scaling**
  - Down-scaling: Average filter
  - Up-scaling: Bi-cubic filter(source>2048 would use Bi-linear)
  - Arbitrary non-integer scaling ratio，from 1/8 to 8
- **Rotation**
  - 0, 90, 180, 270 degree rotation
  - x-mirror, y-mirror& rotation operation
- **BitBLT**
  - Block transfer
  - Color palette/Color fill, support with alpha
  - Transparency mode (color keying/stencil test, specified value/value range)
  - Two source BitBLT:
  - A+B=B only BitBLT, A support rotate&scale when B fixed

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- A+B=C second source (B) has same attribute with (C) plus rotation function

- **Alpha Blending**
  - New comprehensive per-pixel alpha(color/alpha channel separately)
  - Fading
  - SRC1(R2Y)&&SRC0(YUV)—alpha->DST(YUV)
- **MMU**
  - 4k/64k page size
  - Four channel: SRC/SRC1/DST/CMD, individual base address and enable control bit
  - TLB pre-fetch

16.14 Block Diagram

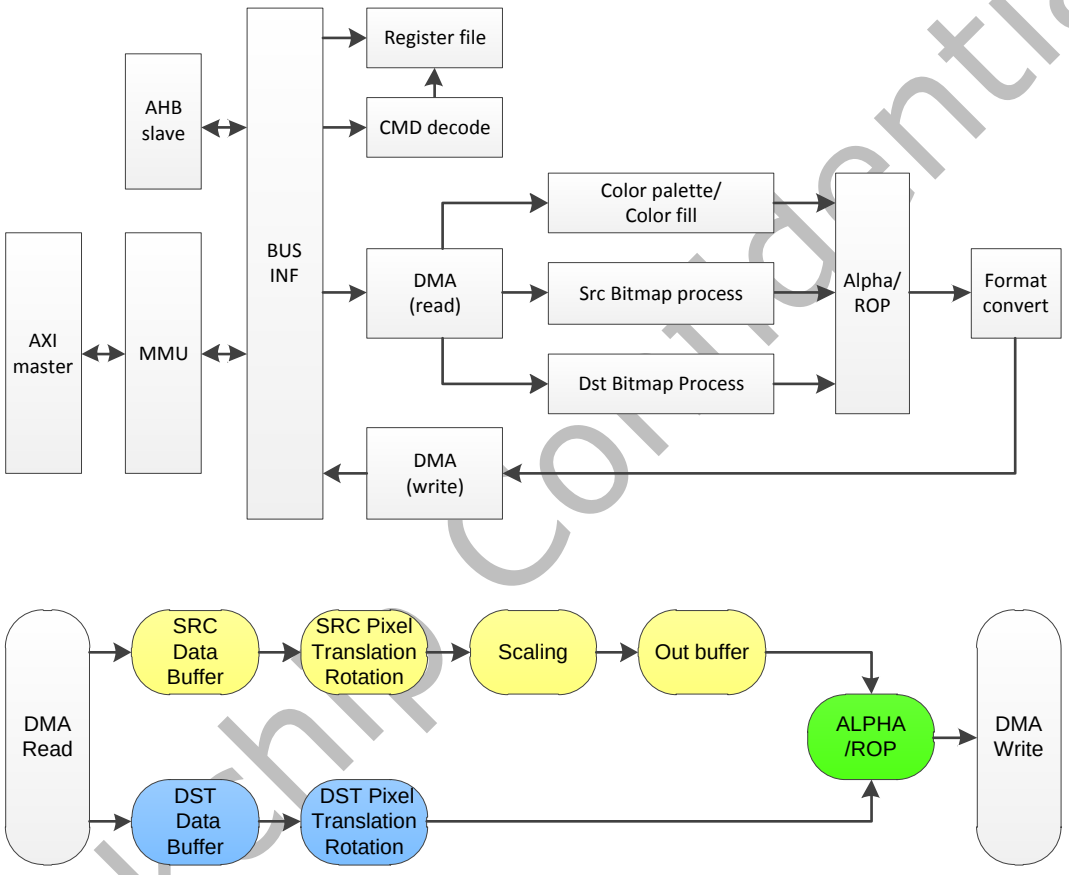

Fig.18-1RGA Block Diagram

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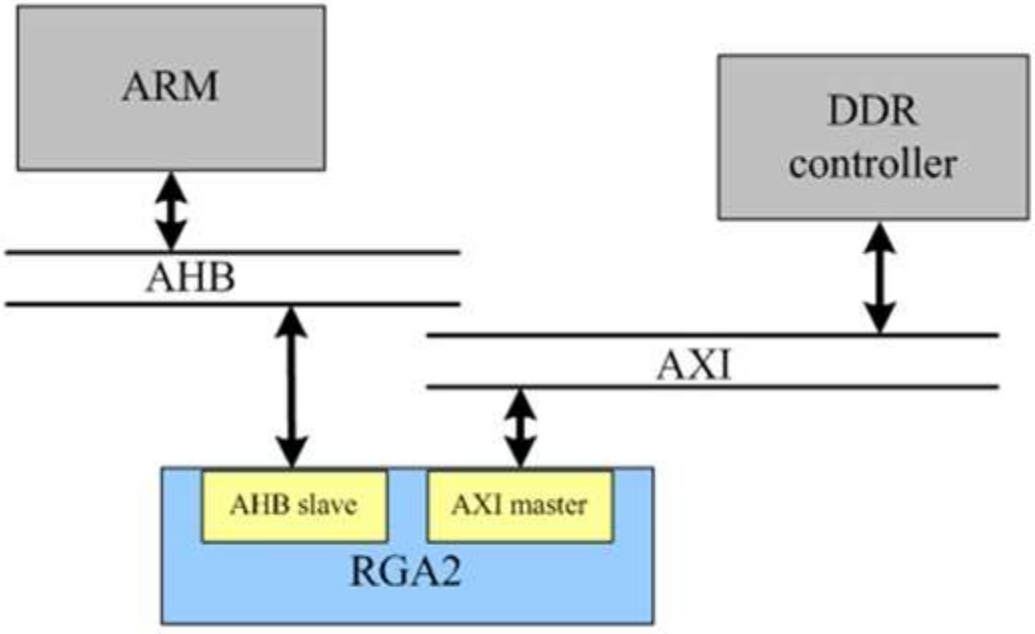

Fig.18-2 RGA2 in SOC

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16.15 Function Description

16.15.1 Data Format

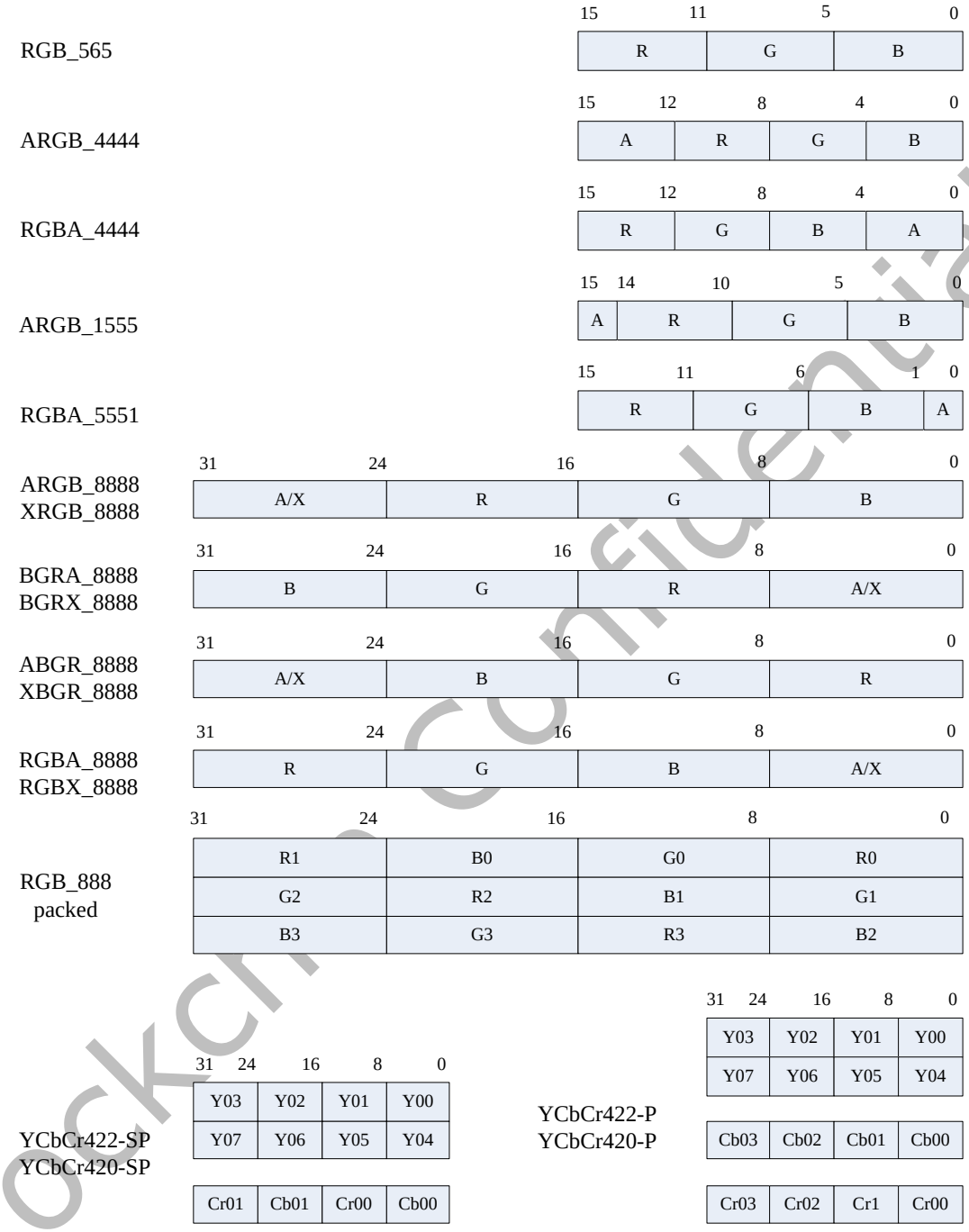

Fig.18-3 RGA Input Data Format

All input datas (defined by SRC_IN_FMT/DST_IN_FMT) are converted to ABGR8888. The results are converted to the output data format (defined by DST_OUT_FMT).

16.15.2 Dithering

There could have dithering operation for source image when the source image format is not RGB565 and the destination format is RGB565.

The down-dithering is done using Dither Allegro.

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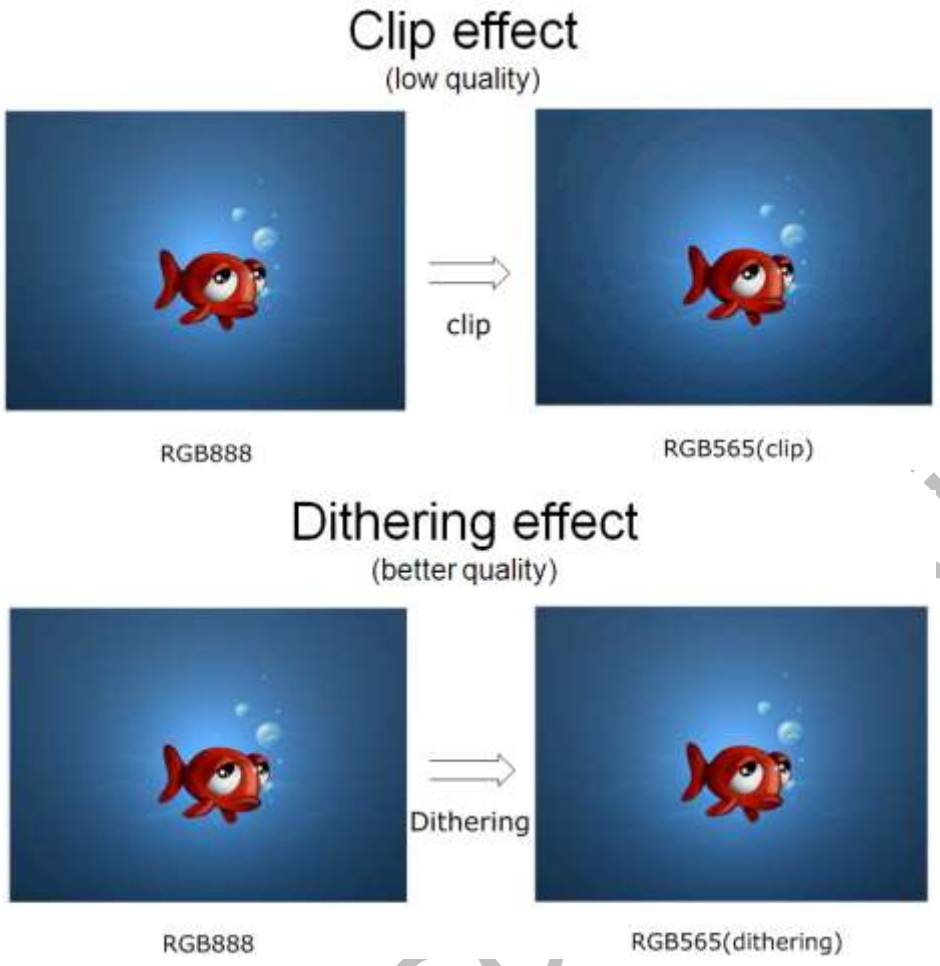

Fig.18-4 RGA Dither effect

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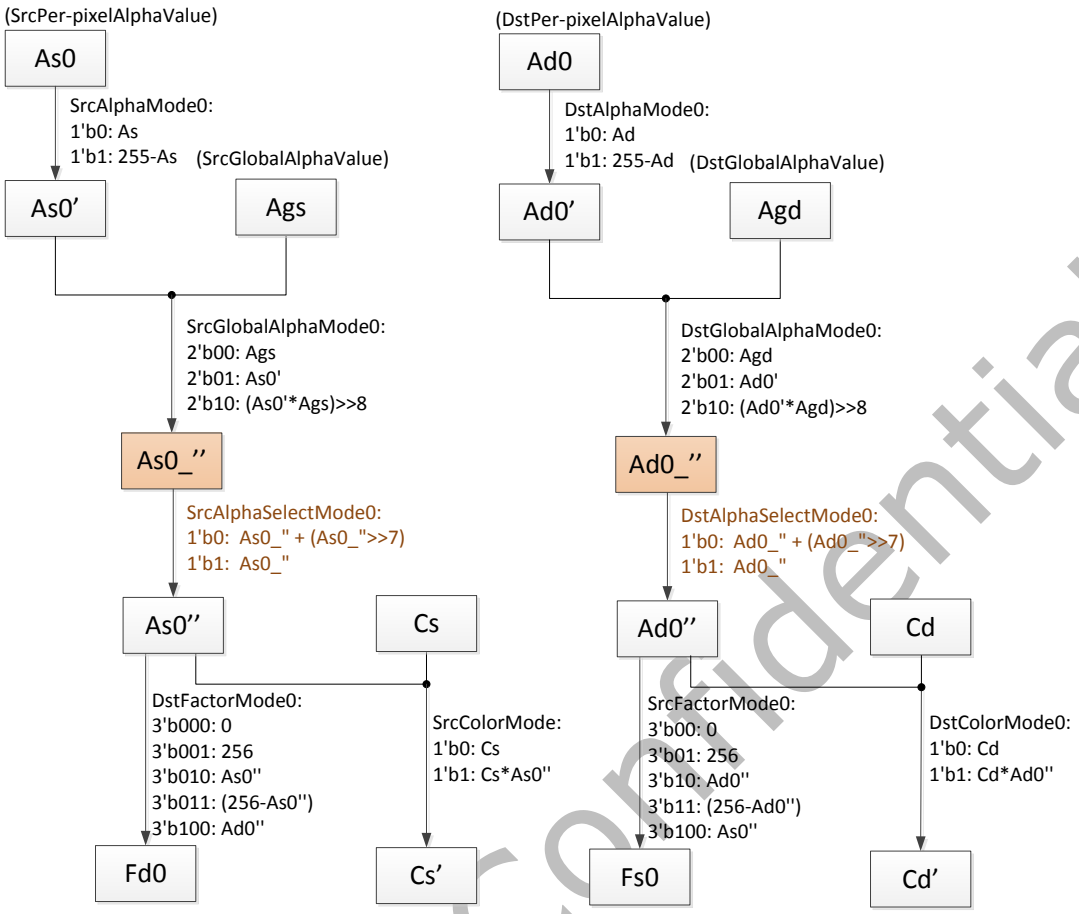
16.15.3 Alpha mode



$$Cd = Fs0 * Cs' + Fd0 * Cd' \quad (1)$$

(Cd – dst color, Fs0 – color src factor0, Cs' – src color', Fd0 – color dst factor0, Cd' – dst color')

Fig.18-5layer0 alpha blending calculate flow

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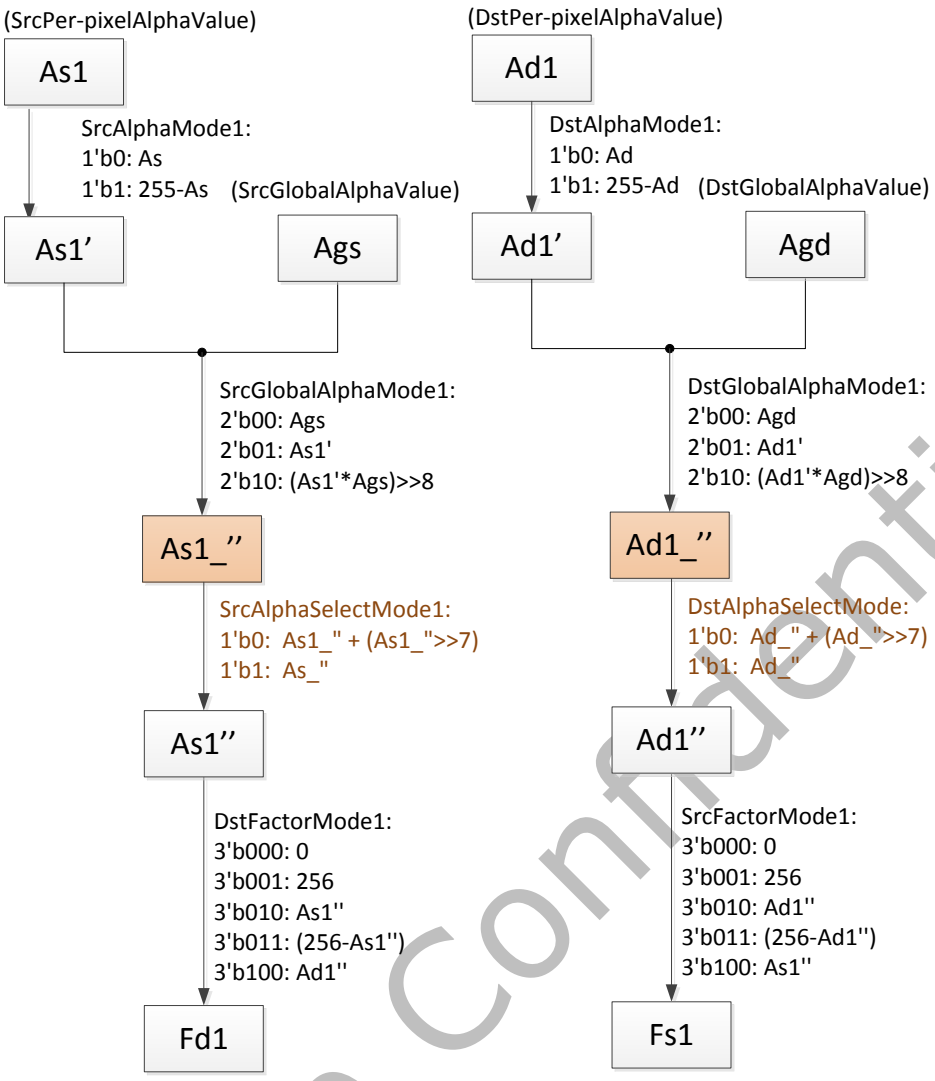

$$Ad = Fs1 * As1'' + Fd1 * Ad1'' \quad (2)$$

(Ad – dst alpha, Fs1 – alpha src factor1, As1'' – src alpha'', Fd1 – alpha dst factor1, Ad1' – dst alpha'')

Fig.18-6 layer1 alpha blending calculate flow

16.15.4 Color fill

Two modes of color fill can be done by RGA: solid fill and gradient fill.

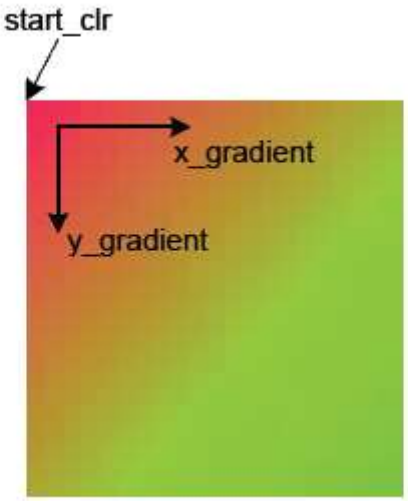

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Fig.18-7 RGA Gradient Fill

Gradient fill using following equations for ARGB calculation of every pixel in different coordinary.

A_cur = (A_start + x*x_A_gradient) +y*y_A_gradient;

R_cur = (R_start + x*x_R_gradient) +y*y_R_gradient;

G_cur = (G_start + x*x_G_gradient) +y*y_G_gradient;

B_cur = (B_start + x*x_B_gradient) +y*y_B_gradient;

A_start, R_start, G_start, B_start is the ARGB value of start point. There are four pairs of values for horizontal and vertical gradient. Saturation operation could be enabled or disabled if the color overflows 255 or underflows 0.

16.15.5 Raster Operation (ROP)

Raster operation (ROP) is a Booleanoperation between operands, which involve AND, OR, XOR, and NOT operations. For ROP2, operands are P (select pan) and D (Destination bitmap). For ROP3, operands are P (pattern), S (source bitmap) and D (Destination bitmap). For ROP4, operands are P (pattern), S (source bitmap), D (Destination bitmap) and MASK.

Table 18-1 RGAROP Boolean operations

Operator	Meaning
a	Bitwise AND
n	Bitwise NOT (inverse)
o	Bitwise OR
x	Bitwise exclusive OR (XOR)

16.15.6 Scaling

The scaling operation is the imageresizing processingof source image. Scaling is done base on ARGB8888 format.

There are three scale modes: scale down (bilinear, Average); scale up(bilinear, Bi-cubic);

16.16 Register Description

16.16.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

16.16.2 Registers Summary

Name	Offset	Size	Reset Value	Description
RGA2_SYS_CTRL	0x0000	W	0x00000044	RGA system control register
RGA2_CMD_CTRL	0x0004	W	0x00000000	RGA command control register

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Name	Offset	Size	Reset Value	Description
RGA2_CMD_BASE	0x0008	W	0x12345678	RGA command codes base address register
RGA2_STATUS1	0x000c	W	0x00000000	RGA status register
RGA2_INT	0x0010	W	0x00000000	RGA interrupt register
RGA2_MMU_CTRL0	0x0014	W	0x00000000	RGA MMU control 0 register
RGA2_MMU_CMD_BASE	0x0018	W	0x00000000	Register0000 Abstract
RGA2_STATUS2	0x001c	W	0x00000000	RGA status register
RGA2_WORK_CNT	0x0020	W	0x00000000	work counter
RGA2_VERSION_INFO	0x0028	W	0x00000000	Version number for rga
RGA2_PERF_LATENCY_CTRL0	0x0040	W	0x00000024	Axi performance latency module contrl register0
RGA2_PERF_LATENCY_CTRL1	0x0044	W	0x00000021	PERF_LATENCY_CTRL1
RGA2_PERF_RD_MAX_LATENCY_NUM0	0x0048	W	0x00000000	Read max latency number
RGA2_PERF_RD_LATENCY_SAMP_NUM	0x004c	W	0x00000000	The number of bigger than configed threshold value
RGA2_PERF_RD_LATENCY_ACC_SUM	0x0050	W	0x00000000	Total sample number
RGA2_PERF_RD_AXI_TOTAL_BYTE	0x0054	W	0x00000000	perf_rd_axi_total_byte
RGA2_PERF_WR_AXI_TOTAL_BYTE	0x0058	W	0x00000000	perf_wr_axi_total_byte
RGA2_PERF_WORKING_CNT	0x005c	W	0x00000000	perf_working_cnt
RGA2_MODE_CTRL	0x0100	W	0x00000000	RGA mode control register
RGA2_SRC_INFO	0x0104	W	0x00000000	RGA source information register
RGA2_SRC_BASE0	0x0108	W	0x00000000	source image Y/RGB base address
RGA2_SRC_BASE1	0x010c	W	0x00000000	RGA source image Cb/Cbr base address register
RGA2_SRC_BASE2	0x0110	W	0x00000000	RGA source image Cr base address register
RGA2_SRC_BASE3	0x0114	W	0x00000000	RGA source image 1 base address register
RGA2_SRC_VIR_INFO	0x0118	W	0x00000000	RGA source image virtual stride / RGA source image tile number
RGA2_SRC_ACT_INFO	0x011c	W	0x00000000	RGA source image active width/height register
RGA2_SRC_X_FACTOR	0x0120	W	0x00000000	RGA source image horizontal scaling factor
RGA2_SRC_Y_FACTOR	0x0124	W	0x00000000	RGA source image vertical scaling factor
RGA2_SRC_BG_COLOR	0x0128	W	0x00000000	RGA source image background color

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Name	Offset	Size	Reset Value	Description
RGA2_SRC_FG_COLOR	0x012c	W	0x00000000	RGA source image foreground color
RGA2_SRC_TR_COLOR0	0x0130	W	0x00000000	RGA source image transparency color min value
RGA2_CP_GR_A	0x0130	W	0x00000000	RGA color gradient fill step register
RGA2_SRC_TR_COLOR1	0x0134	W	0x00000000	RGA source image transparency color max value
RGA2_CP_GR_B	0x0134	W	0x00000000	RGA color gradient fill step register
RGA2_DST_INFO	0x0138	W	0x00000000	RGA destination format register
RGA2_DST_BASE0	0x013c	W	0x00000000	RGA destination image base address 0 register
RGA2_DST_BASE1	0x0140	W	0x00000000	RGA destination image base address 1 register
RGA2_DST_BASE2	0x0144	W	0x00000000	RGA destination image base address 2 register
RGA2_DST_VIR_INFO	0x0148	W	0x00000000	RGA destination image virtual width/height register
RGA2_DST_ACT_INFO	0x014c	W	0x00000000	RGA destination image active width/height register
RGA2_ALPHA_CTRL0	0x0150	W	0x00000000	Alpha control register 0
RGA2_ALPHA_CTRL1	0x0154	W	0x00000000	Register0000 Abstract
RGA2_FADING_CTRL	0x0158	W	0x00000000	Fading control register
RGA2_PAT_CON	0x015c	W	0x00000000	Pattern size/offset register
RGA2_ROP_CON0	0x0160	W	0x00000000	ROP code 0 control register
RGA2_CP_GR_G	0x0160	W	0x00000000	RGA color gradient fill step register (color fill mode)
RGA2_ROP_CON1	0x0164	W	0x00000000	ROP code 1 control register
RGA2_CP_GR_R	0x0164	W	0x00000000	RGA color gradient fill step register (color fill mode)
RGA2_MASK_BASE	0x0168	W	0x00000000	RGA mask base address register
RGA2_MMU_CTRL1	0x016c	W	0x00000000	RGA MMU control register 1
RGA2_MMU_SRC_BASE	0x0170	W	0x00000000	RGA source MMU TLB base address
RGA2_MMU_SRC1_BASE	0x0174	W	0x00000000	RGA source1 MMU TLB base address
RGA2_MMU_DST_BASE	0x0178	W	0x00000000	RGA destination MMU TLB base address
RGA2_MMU_ELS_BASE	0x017c	W	0x00000000	RGA ELSE MMU TLB base address

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.16.3 Detail Register Description

RGA2_SYS_CTRL

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Address: Operational Base + offset (0x0000)

RGA system control register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	sw_rst_handsave_p it would save protect-rstn into initial status if long time dead in protect-rstn status. (auto clear into '0')
6	RW	0x1	sw_rst_protect_e protect-rstn mode enable. it would be ensure all axi write/read operation into completion status when sw_cclk_sreset_p or sw_aclk_sreset_p valid.
5	RW	0x0	sw_auto_rst it would auto-resetn after one frame finish. 0: disable 1: enable
4	RW	0x0	sw_cclk_sreset_p RGA core clk domain Soft reset, write '1' to this would reset the RGA engine except config registers.
3	WO	0x0	sw_aclk_sreset_p RGA aclk domain Soft reset, write '1' to this would reset the RGA engine except config registers.
2	WO	0x1	sw_auto_ckg RGA auto clock gating enable bit 0：disable 1：enable
1	WO	0x0	sw_cmd_mode RGA command mode 0：slave mode 1：master mode
0	W1C	0x0	sw_cmd_op_st_p RGA operation start bit Only used in passive (slave) control mode

RGA2_CMD_CTRL

Address: Operational Base + offset (0x0004)

RGA command control register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:3	RW	0x000	sw_cmd_incr_num RGA command increment number

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Bit	Attr	Reset Value	Description
2	WO	0x0	sw_cmd_stop RGA command stop mode Command execution would stop after the current graphic operation finish if set this bit to 1
1	WO	0x0	sw_cmd_incr_valid_p RGA command increment valid (Auto cleared) When setting this bit, 1. The total cmd number would increase by the RGA_INCR_CMD_NUM. 2. RGA would continue running if idle.
0	RW	0x0	sw_cmd_line_st_p RGA command line fetch start (command line reset) (Auto cleared) When fetch start, the total cmd number would reset to RGA_INCR_CMD_NUM.

RGA2_CMD_BASE

Address: Operational Base + offset (0x0008)

RGA command codes base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x12345678	sw_cmd_base RGA command codes base address

RGA2_STATUS1

Address: Operational Base + offset (0x000c)

RGA status register

Bit	Attr	Reset Value	Description
31:20	RO	0x000	sw_cmd_total_num RGA command total number
19:8	RO	0x000	sw_cmd_cur_num RGA command current number
7:1	RO	0x00	Reserved Reserved
0	RO	0x0	sw_rga_sta RGA engine status 0: idle 1: working

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RGA2_INT

Address: Operational Base + offset (0x0010)

RGA interrupt register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10	RW	0x0	sw_intr_af_e All command finished interrupt enable
9	RW	0x0	sw_intr_mmu_e MMU interrupt enable
8	RW	0x0	sw_intr_err_e Error interrupt enable
7	WO	0x0	sw_intr_cf_clr Current command finished interrupt clear
6	WO	0x0	sw_intr_af_clr All command finished interrupt clear
5	WO	0x0	sw_intr_mmu_clr MMU interrupt clear
4	WO	0x0	sw_intr_err_clr Error interrupt clear
3	RO	0x0	sw_intr_cf Current command finished interrupt flag
2	RO	0x0	sw_intr_af All command finished interrupt flag
1	RO	0x0	sw_intr_mmu MMU interrupt
0	RO	0x0	sw_intr_err Error interrupt flag

RGA2_MMU_CTRL0

Address: Operational Base + offset (0x0014)

RGA MMU control 0 register

Bit	Attr	Reset Value	Description
31:11	RW	0x000000	Reserved Reserved
10:9	RW	0x0	sw_els_ch_priority sw_els_ch_priority

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Bit	Attr	Reset Value	Description
8:7	RW	0x0	sw_dst_ch_priority sw_dst_ch_priority
6:5	RW	0x0	sw_src1_ch_priority sw_src1_ch_priority
4:3	RW	0x0	sw_src_ch_priority sw_src_ch_priority
2	RW	0x0	sw_cmd_mmu_flush RGA CMD channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
1	RW	0x0	sw_cmd_mmu_en RGA CMD channel MMU enable 0: disable 1: enable
0	RW	0x0	sw_mmu_page_size RGA MMU Page table size 0: 4KB page 1: 64KB page

RGA2_MMU_CMD_BASE

Address: Operational Base + offset (0x0018)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_cmd_base RGA command MMU TLB base address (word)

RGA2_STATUS2

Address: Operational Base + offset (0x001c)

RGA status register

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	Reserved
12:11	RO	0x0	rpp_mkram_rready rpp_mkram_rready
10:6	RO	0x00	dstrpp_outbuf_rready dstrpp_outbuf_rready

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Bit	Attr	Reset Value	Description
5:2	RO	0x0	srcrpp_outbuf_rready dstrpp_outbuf_rready
1	RO	0x0	bus_error
0	RO	0x0	rpp_error

RGA2_WORK_CNT

Address: Operational Base + offset (0x0020)

work counter

Bit	Attr	Reset Value	Description
31:27	RW	0x00	Reserved
26:0	RO	0x0000000	sw_work_cnt working counter register RGA total working counter

RGA2_VERSION_INFO

Address: Operational Base + offset (0x0028)

Version number for rga

Bit	Attr	Reset Value	Description
31:24	RW	0x00	major IP major vertion used for IP structure version infomation
23:20	RW	0x0	minor minor vertion big feature change under same structure
19:0	RW	0x00000	svnbuild rtl current svn number

RGA2_PERF_LATENCY_CTRL0

Address: Operational Base + offset (0x0040)

Axi performance latency module contrl register0

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Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:8	RW	0x000	sw_rd_latency_thr sw_rd_latency_thr
7:4	RW	0x2	sw_rd_latency_id sw_rd_latency_id
3	RW	0x0	sw_axi_cnt_type sw_axi_cnt_type sw_axi_cnt_type
2	RW	0x1	sw_axi_perf_frm_type latency mode 1'b0: clear by software configuration 1'b1: clear by frame end
1	RW	0x0	sw_axi_perf_clr_e sw_axi_perf_clr_e 1'b0: software clear disable 1'b1: software clear enalbe
0	RW	0x0	sw_axi_perf_work_e sw_axi_perf_work_e 1'b0: disable 1'b1: enable

RGA2_PERF_LATENCY_CTRL1

Address: Operational Base + offset (0x0044)

PERF_LATENCY_CTRL1

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:8	RW	0x0	sw_aw_count_id sw_aw_count_id sw_aw_count_id
7:4	RW	0x2	sw_ar_count_id sw_ar_count_id sw_ar_count_id
3	RW	0x0	sw_aw_cnt_id_type sw_aw_cnt_id_type sw_aw_cnt_id_type

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Bit	Attr	Reset Value	Description
2	RW	0x0	sw_ar_cnt_id_type sw_ar_cnt_id_type sw_ar_cnt_id_type
1:0	RW	0x1	sw_addr_align_type sw_addr_align_type sw_addr_align_type

RGA2_PERF_RD_MAX_LATENCY_NUM0

Address: Operational Base + offset (0x0048)

Read max latency number

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	rd_max_latency_num_ch0 read max latency value of channel 0

RGA2_PERF_RD_LATENCY_SAMP_NUM

Address: Operational Base + offset (0x004c)

The number of bigger than configed threshold value

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_thr_num_ch0 read latency thr number channel 0

RGA2_PERF_RD_LATENCY_ACC_SUM

Address: Operational Base + offset (0x0050)

Total sample number

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rd_latency_acc_sum

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RGA2_PERF_RD_AXI_TOTAL_BYTE

Address: Operational Base + offset (0x0054)

perf_rd_axi_total_byte

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_rd_axi_total_byte perf_rd_axi_total_byte perf_rd_axi_total_byte

RGA2_PERF_WR_AXI_TOTAL_BYTE

Address: Operational Base + offset (0x0058)

perf_wr_axi_total_byte

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_wr_axi_total_byte perf_wr_axi_total_byte perf_wr_axi_total_byte

RGA2_PERF_WORKING_CNT

Address: Operational Base + offset (0x005c)

perf_working_cnt

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	perf_working_cnt perf_working_cnt perf_working_cnt

RGA2_MODE_CTRL

Address: Operational Base + offset (0x0100)

RGA mode control register

Bit	Attr	Reset Value	Description
31:8	RW	0x000000	Reserved Reserved

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Bit	Attr	Reset Value	Description
7	RW	0x0	sw_intr_cf_e Current command finished interrupt enable
6	RW	0x0	sw_gradient_sat Gradient saturation calculation mode 0:clip 1:not-clip
5	RW	0x0	sw_alpha_zero_key ARGB888 alpha zero key mode 0x000000 would be changed to 0x000100(RGB888)/0x0020(RGB565)for ARGB888 to RGBX/RGB565 color key 0: disable 1: enable
4	RW	0x0	sw_cf_rop4_pat Color fill/ROP4 pattern 0: solid color 1: pattern color
3	RW	0x0	sw_bb_mode Bitblt mode 0: SRC + DST => DST 1: SRC + SRC1 => DST
2:0	RW	0x0	sw_render_mode RGA 2D render mode 000: Bitblt 001: Color palette 010: Rectangle fill 011: Update palette LUT/pattern ram

RGA2_SRC_INFO

Address: Operational Base + offset (0x0104)

RGA source information register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RW	0x0	sw_src_yuv10_round_e this bit valid when RGA support yuv 10bit picture input 0: yuv 10bit to 8bit round disable 1: yuv 10bit to 8bit round enable

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Bit	Attr	Reset Value	Description
27	RW	0x0	sw_src_yuv10_e this bit valid when RGA support yuv 10bit picture input 0: yuv 10bit disable 1: yuv 10bit enalbe
26	RW	0x0	sw_vsp_mode 0:by-cubic 1:bi-linear
25:24	RW	0x0	sw_bic_coe_sel SRC bicubic scaling coefficient select 00: CATROM 01: MITCHELL 10: HERMITE 11: B-SPLINE
23	RW	0x0	sw_src_dither_up SRC dither up enable 0:disable 1:enable
22:19	RW	0x0	sw_src_trans_e Source transparency enable bits [3]: A value stencil test enable bit [2]: B value stencil test enable bit [1]: G value stencil test enable bit [0]: R value stencil test enable bit
18	RW	0x0	sw_src_trans_mode Source transparency mode 0: normal stencil test (color key) 1: inverted stencil test
17:16	RW	0x0	sw_src_vscl_mode SRC vertical scaling mode 00：no scaling 01: down-scaling 10: up-scaling
15:14	RW	0x0	sw_src_hscl_mode SRC horizontal scaling mode 00：no scaling 01: down-scaling 10: up-scaling
13:12	RW	0x0	sw_src_mir_mode SRC mirror mode 00: no mirror 01: x mirror 10: y mirror 11: x mirror + y mirror

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Bit	Attr	Reset Value	Description
11:10	RW	0x0	sw_src_rot_mode SRC rotation mode 00: 0 degree 01: 90 degree 10: 180 degree 11: 270 degree
9:8	RW	0x0	sw_src_csc_mode Source bitmap YUV2RGB conversion mode 00: bypass 01: BT.601-range0 10: BT.601-range1 11: BT.709-range0
7	RW	0x0	sw_cp_endian Source Color palette endian swap 0: big endian 1: little endian
6	RW	0x0	sw_src_uvswap Source Cb-Cr swap 0: CrCb 1: CbCr
5	RW	0x0	sw_src_alpha_swap Source bitmap data alpha swap 0: ABGR 1: BGRA
4	RW	0x0	sw_src_rbswap Source bitmap data RB swap 0: BGR 1: RGB
3:0	RW	0x0	sw_src_fmt Source bitmap data format 0000: ABGR888 0001: XBGR888 0010: BGR packed 0100: RGB565 0101: ARGB1555 0110: ARGB4444 1000: YUV422SP 1001: YUV422P 1010: YUV420SP 1011: YUV420P 1100: 1BPP (color palette) 1101: 2BPP (color palette) 1110: 4BPP (color palette) 1111: 8BPP (color palette)

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RGA2_SRC_BASE0

Address: Operational Base + offset (0x0108)

source image Y/RGB base address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_base0 source image Y/RGB base address

RGA2_SRC_BASE1

Address: Operational Base + offset (0x010c)

RGA source image Cb/Cbr base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_base1 source image Cb base address (YUV422/420-P) source image Cb/Cr base address (YU,V422/420-SP)

RGA2_SRC_BASE2

Address: Operational Base + offset (0x0110)

RGA source image Cr base address register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	sw_src_base2 source image Cr base address (YUV422/420-P)

RGA2_SRC_BASE3

Address: Operational Base + offset (0x0114)

RGA source image 1 base address register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	sw_src_base3 source image 1 RGB base address (source bitblt mode1)

RGA2_SRC_VIR_INFO

Address: Operational Base + offset (0x0118)

RGA source image virtual stride / RGA source image tile number

Bit	Attr	Reset Value	Description
31:26	RW	0x00	Reserved
25:16	RW	0x000	sw_mask_vir_stride mask image virtual stride (words)
15	RW	0x0	Reserved Reserved
14:0	RW	0x0000	sw_src_vir_stride src image virtual stride (words)

RGA2_SRC_ACT_INFO

Address: Operational Base + offset (0x011c)

RGA source image active width/height register

Bit	Attr	Reset Value	Description
31:29	RW	0x0	Reserved2 Reserved
28:16	RW	0x0000	sw_src_act_height source image active height
15:13	RW	0x0	Reserved1 Reserved
12:0	RW	0x0000	sw_src_act_width source image active width

RGA2_SRC_X_FACTOR

Address: Operational Base + offset (0x0120)

RGA source image horizontal scaling factor

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_src_hsp_factor Source image horizontal up-scaling factor =(DST_ACT_WIDTH/SRC_ACT_WIDTH) * 65536
15:0	RW	0x0000	sw_src_hsd_factor Source image horizontal down-scaling factor =(SRC_ACT_WIDTH/DST_ACT_WIDTH) * 65536

RGA2_SRC_Y_FACTOR

Address: Operational Base + offset (0x0124)

RGA source image vertical scaling factor

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_src_vsp_factor Source image vertical up-scaling factor (DST_ACT_HEIGHT/SRC_ACT_HEIGHT) * 65536
15:0	RW	0x0000	sw_src_vsd_factor Source image vertical down-scaling factor (SRC_ACT_HEIGHT/DST_ACT_HEIGHT) * 65536

RGA2_SRC_BG_COLOR

Address: Operational Base + offset (0x0128)

RGA source image background color

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_bg_color Source image background color ("0" bit color for mono expansion.)

RGA2_SRC_FG_COLOR

Address: Operational Base + offset (0x012c)

RGA source image foreground color

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_src_fg_color Source image foreground color Source image foreground color ("1" bit color for mono expansion.) Color fill color, Pan color

RGA2_SRC_TR_COLOR0

Address: Operational Base + offset (0x0130)

RGA source image transparency color min value

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_src_trans_amin source image transparency color A min value
23:16	RW	0x00	sw_src_trans_bmin source image transparency color B min value
15:8	RW	0x00	sw_src_trans_gmin source image transparency color G min value
7:0	RW	0x00	sw_src_trans_rmin source image transparency color R min value

RGA2_CP_GR_A

Address: Operational Base + offset (0x0130)

RGA color gradient fill step register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_a Y gradient value of Alpha (signed 8.8)
15:0	RW	0x0000	sw_gradient_x_a X gradient value of Alpha (signed 8.8)

RGA2_SRC_TR_COLOR1

Address: Operational Base + offset (0x0134)

RGA source image transparency color max value

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_src_trans_amax source image transparency color A max value
23:16	RW	0x00	sw_src_trans_bmax source image transparency color B max value
15:8	RW	0x00	sw_src_trans_gmax source image transparency color G max value
7:0	RW	0x00	sw_src_trans_rmax source image transparency color R max value

RGA2_CP_GR_B

Address: Operational Base + offset (0x0134)

RGA color gradient fill step register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_b Y gradient value of Blue (signed 8.8)
15:0	RW	0x0000	sw_gradient_x_b X gradient value of Blue (signed 8.8)

RGA2_DST_INFO

Address: Operational Base + offset (0x0138)

RGA destination format register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	sw_src1_csc_clip src1 read BGR2YUV Clip mode(from 0~255 clip to 36~235) 1: clip enable; 0: unclip
20:19	RW	0x0	sw_src1_csc_mode sw_dst_csc_mode SRC1 read bitmap RGB2YUV conversion mode 00: Bypass 01: BT.601-range0 10: BT.601-range1 11: BT.709-range0

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Bit	Attr	Reset Value	Description
18	RW	0x0	sw_dst_csc_clip dst write BGR2YUV Clip mode(from 0~255 clip to 36~235) 1: clip enable; 0: unclip
17:16	RW	0x0	sw_dst_csc_mode sw_dst_csc_mode sw_dst_csc_mode DST write bitmap RGB2YUV conversion mode 00: Bypass 01: BT.601-range0 10: BT.601-range1 11: BT.709-range0
15:14	RW	0x0	sw_dither_mode sw_dither_mode sw_dither_mode DST dither down bit mode 00: 888 to 666 01: 888 to 565 10: 888 to 555 11: 888 to 444
13	RW	0x0	sw_dither_down sw_dither_down sw_dither_down DST dither down enable 0:disable 1:enable
12	RW	0x0	sw_src1_dither_up sw_src1_dither_up sw_src1_dither_up DST/SRC1 dither up enable 0:disable 1:enable
11	RW	0x0	sw_src1_alpha_swap sw_src1_alpha_swap sw_src1_alpha_swap Source 1 bitmap data alpha swap 0: ABGR 1: BGRA
10	RW	0x0	sw_src1_rbswap sw_src1_rbswap sw_src1_rbswap Source 1 bitmap data RB swap 0: BGR 1: RGB

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Bit	Attr	Reset Value	Description
9:7	RW	0x0	sw_src1_fmt Source 1 bitmap data format sw_src1_fmt Source 1 bitmap data format 000: ABGR888 001: XBGR888 010: BGR packed 100: RGB565 101: ARGB1555 110: ARGB4444
6	RW	0x0	sw_dst_uvswap Destination Cb-Cr swap Destination Cb-Cr swap 0: CrCb 1: CbCr
5	RW	0x0	sw_dst_alpha_swap Destination bitmap data alpha swap Destination bitmap data alpha swap 0: ABGR 1: BGRA
4	RW	0x0	sw_dst_rbswap Destination bitmap data RB swap Destination bitmap data RB swap 0: BGR 1: RGB
3:0	RW	0x0	sw_dst_fmt Destination bitmap data format Destination bitmap data format 0000: ABGR888 0001: XBGR888 0010: BGR packed 0100: RGB565 0101: ARGB1555 0110: ARGB4444 1000: YUV422SP 1001: YUV422P 1010: YUV420SP 1011: YUV420P If RGA has yuyv output format feature: 1100: YVYU422(U, LSB) 1101: YVYU420(U, LSB) 1110: VYUY422(Y, LSB) 1111: VYUY420(Y, LSB)

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RGA2_DST_BASE0

Address: Operational Base + offset (0x013c)

RGA destination image base address 0 register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_dst_base0 destination image Y/RGB base address

RGA2_DST_BASE1

Address: Operational Base + offset (0x0140)

RGA destination image base address 1 register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_dst_base1 destination image Cb/CbCr base address

RGA2_DST_BASE2

Address: Operational Base + offset (0x0144)

RGA destination image base address 2 register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_dst_base2 destination image Cr base address

RGA2_DST_VIR_INFO

Address: Operational Base + offset (0x0148)

RGA destination image virtual width/height register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	Reserved2 Reserved
27:16	RW	0x000	sw_src1_vir_stride source image 1 virtual stride (words)

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Bit	Attr	Reset Value	Description
15:12	RW	0x0	Reserved1 Reserved
11:0	RW	0x000	sw_dst_vir_stride destination image virtual stride(words)

RGA2_DST_ACT_INFO

Address: Operational Base + offset (0x014c)

RGA destination image active width/height register

Bit	Attr	Reset Value	Description
31:28	RW	0x0	Reserved2 Reserved
27:16	RW	0x000	sw_dst_act_height Destination image active height
15:12	RW	0x0	Reserved1 Reserved
11:0	RW	0x000	sw_dst_act_width Destination image active width

RGA2_ALPHA_CTRL0

Address: Operational Base + offset (0x0150)

Alpha control register 0

Bit	Attr	Reset Value	Description
31:21	RW	0x000	Reserved Reserved
20	RW	0x0	sw_mask_endian sw_mask_endian ROP4 mask endian swap 0: big endian 1: little endian
19:12	RW	0x00	sw_dst_global_alpha global alpha value of DST(Agd)
11:4	RW	0x00	sw_src_global_alpha global alpha value of SRC(Ags) fading value in fading mod

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Bit	Attr	Reset Value	Description
3:2	RW	0x0	sw_rop_mode ROP mode select 00: ROP 2 01: ROP 3 10: ROP 4
1	RW	0x0	sw_alpha_rop_sel Alpha or ROP select 0: alpha 1: ROP
0	RW	0x0	sw_alpha_rop_e Alpha or ROP enable 0: disable 1: enable

RGA2_ALPHA_CTRL1

Address: Operational Base + offset (0x0154)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31:30	RW	0x0	Reserved Reserved
29	RW	0x0	sw_src_alpha_m1 Src Transparent/opaque of alpha channel (As1') 0: As 1: 255-As
28	RW	0x0	sw_dst_alpha_m1 Dst Transparent/opaque of alpha channel (Ad1') 0: Ad 1: 255-Ad
27:26	RW	0x0	sw_src_blend_m1 Alpha src blend mode select of alpha channel (As1_'') 00: Ags 01: As1' 10: (As1'*Ags)>>8 11: reserved
25:24	RW	0x0	sw_dst_blend_m1 Alpha dst blend mode select of alpha channel(Ad1_'') 00: Agd 01: Ad1' 10: (Ad1'*Agd)>>8 11: reserved

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Bit	Attr	Reset Value	Description
23	RW	0x0	sw_src_alpha_cal_m1 Alpha src calculate mode of alpha channel(As1'') 0: As1''= As1_''+ (As1_''>>7) 1: As1''= As1 _''
22	RW	0x0	sw_dst_alpha_cal_m1 Alpha dst calculate mode of alpha channel(Ad1'') 0: Ad1''= Ad1_'' + (Ad1_''>>7) 1: Ad1''= Ad1_''
21:19	RW	0x0	w_src_factor_m1 Src factore mode of alpha channel(Fs1) 000: 0 001: 256 010: Ad1'' 011: 256-Ad1'' 100: As1''
18:16	RW	0x0	sw_dst_factor_m1 Dst factore mode of alpha channel(Fd1) 000: 0 001: 256 010: As1'' 011: 256-As1'' 100: Ad1''
15	RW	0x0	sw_src_alpha_m0 Src Transparent/opaque of color channel (As0') 0: As 1: 255-As
14	RW	0x0	sw_dst_alpha_m0 Dst Transparent/opaque of color channel (Ad0') 0: Ad 1: 255-Ad
13:12	RW	0x0	sw_src_blend_m0 Alpha src blend mode select of color channel (As0_'') 00: Ags 01: As0' 10: (As0'*Ags)>>8 11: reserved
11:10	RW	0x0	sw_dst_blend_m0 Alpha dst blend mode select of color channel(Ad0_'') 00: Agd 01: Ad0' 10: (Ad0'*Agd)>>8 11: reserved

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Bit	Attr	Reset Value	Description
9	RW	0x0	sw_src_alpha_cal_m0 Alpha src calculate mode of color channel(As0'') 0: As0''= As0_''+ (As0_''>>7) 1: As0''= As0 _''
8	RW	0x0	sw_dst_alpha_cal_m0 Alpha dst calculate mode of color channel(Ad0'') 0: Ad0''= Ad0_'' + (Ad0_''>>7) 1: Ad0''= Ad0_''
7:5	RW	0x0	sw_src_factor_m0 Src factore mode of color channel(Fs0) 000: 0 001: 256 010: Ad0'' 011: 256-Ad0'' 100: As0''
4:2	RW	0x0	sw_dst_factor_m0 Dst factore mode of color channel(Fd0) 000: 0 001: 256 010: As0'' 011: 256-As0'' 100: Ad0''
1	RW	0x0	sw_src_color_m0 SRC color select(Cs') 0: Cs 1: Cs * As0''
0	RW	0x0	sw_dst_color_m0 SRC color select(Cd') 0: Cd 1: Cd * Ad0''

RGA2_FADING_CTRL

Address: Operational Base + offset (0x0158)

Fading control register

Bit	Attr	Reset Value	Description
31:25	RW	0x00	Reserved Reserved
24	RW	0x0	sw_fading_en Fading enable

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	sw_fading_offset_b Fading offset B value
15:8	RW	0x00	sw_fading_offset_g Fading offset G value (Pattern total number when pattern loading)
7:0	RW	0x00	sw_fading_offset_r Fading offset R value (Start point of pattern ram in pattern mode)

RGA2_PAT_CON

Address: Operational Base + offset (0x015c)

Pattern size/offset register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	sw_pat_offset_y Pattern y offset
23:16	RW	0x00	sw_pat_offset_x Pattern x offset
15:8	RW	0x00	sw_pat_height Pattern height
7:0	RW	0x00	sw_pat_width Pattern width

RGA2_ROP_CON0

Address: Operational Base + offset (0x0160)

ROP code 0 control register

Bit	Attr	Reset Value	Description
31:25	RW	0x00	Reserved Reserved
24:0	RW	0x0000000	sw_rop3_code0 Rop3 code 0 control bits

RGA2_CP_GR_G

Address: Operational Base + offset (0x0160)

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RGA color gradient fill step register (color fill mode)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_g Y gradient value of Green (signed 8.8)
15:0	RW	0x0000	sw_gradient_x_g X gradient value of Green (signed 8.8)

RGA2_ROP_CON1

Address: Operational Base + offset (0x0164)

ROP code 1 control register

Bit	Attr	Reset Value	Description
31:25	RW	0x00	Reserved Reserved
24:0	RW	0x0000000	sw_rop3_code1 Rop3 code 1 control bits

RGA2_CP_GR_R

Address: Operational Base + offset (0x0164)

RGA color gradient fill step register (color fill mode)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	sw_gradient_y_r Y gradient value of Red(signed 8.8)
15:0	RW	0x0000	sw_gradient_x_r X gradient value of Red(signed 8.8)

RGA2_MASK_BASE

Address: Operational Base + offset (0x0168)

RGA mask base address register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	sw_mask_base mask base address in ROP4 mode LUT/ pattern load base address

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RGA2_MMU_CTRL1

Address: Operational Base + offset (0x016c)

RGA MMU control register 1

Bit	Attr	Reset Value	Description
31:14	RW	0x00000	Reserved Reserved Reserved
13	RW	0x0	sw_els_mmu_flush RGA ELSE channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
12	RW	0x0	sw_els_mmu_en RGA ELSE channel MMU enable 0: disable 1: enable
11	RW	0x0	sw_dst_mmu_prefetch_dir sw_dst_mmu_prefetch_dir 0:forward 1:backward
10	RW	0x0	sw_dst_mmu_prefetch_en sw_dst_mmu_prefetch_en 0:disable 1:enable
9	RW	0x0	sw_dst_mmu_flush sw_dst_mmu_flush RGA DST channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
8	RW	0x0	sw_dst_mmu_en sw_dst_mmu_en RGA DST channel MMU enable 0: disable 1: enable
7	RW	0x0	sw_src1_mmu_prefetch_dir sw_src1_mmu_prefetch_dir 0:forward 1:backward
6	RW	0x0	sw_src1_mmu_prefetch_en sw_src1_mmu_prefetch_en 0:disable 1:enable

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Bit	Attr	Reset Value	Description
5	RW	0x0	sw_src1_mmu_flush sw_src1_mmu_flush RGA SRC1 channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
4	RW	0x0	sw_src1_mmu_en sw_src1_mmu_en RGA SRC1 channel MMU enable 0: disable 1: enable
3	RW	0x0	sw_src_mmu_prefetch_dir sw_src_mmu_prefetch_dir 0:forward 1:backward
2	RW	0x0	sw_src_mmu_prefetch_en sw_src_mmu_prefetch_en 0:disable 1:enable
1	RW	0x0	sw_src_mmu_flush RGA SRC channel MMU TLB flush: Set 1 to this bit to flush MMU TLB, auto clear
0	RW	0x0	sw_src_mmu_en RGA SRC channel MMU enable 0: disable 1: enable

RGA2_MMU_SRC_BASE

Address: Operational Base + offset (0x0170)

RGA source MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_src_base RGA source MMU TLB base address (128-bit)

RGA2_MMU_SRC1_BASE

Address: Operational Base + offset (0x0174)

RGA source1 MMU TLB base address

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Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_src1_base RGA source1 MMU TLB base address (128-bit)

RGA2_MMU_DST_BASE

Address: Operational Base + offset (0x0178)

RGA destination MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_dst_base RGA destination MMU TLB base address (128-bit)

RGA2_MMU_ELS_BASE

Address: Operational Base + offset (0x017c)

RGA ELSE MMU TLB base address

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	sw_mmu_els_base RGA destination MMU TLB base address (128-bit)

16.17 Application Notes

16.17.1 Register Partition

There are two types of register in RGA. The first 8 registers (0x0 - 0x1C) are general registers for system configuration including command mode, command parameter, RGA status, general interrupts. The other registers (from 0x100) are command registers for command codes.

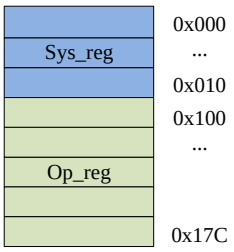

Fig. 18-8RGAsoftware main register-region

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16.17.2 Command Modes

RGA has two command modes: slave mode and master mode. In salve mode (RGA_SYS_CTRL[1] = 1'b0), 2D graphic command only could be run one by one. CPU set all the command registers in RGA and then start RGA running by setting RGA_SYS_CTRL[0] to '1'. In master mode (RGA_SYS_CTRL[1] = 1'b1), 2D graphic commands could be run sequentially. After setting command's number to RGA_CMD_CTRL[12:3], writing '1' to RGA_CMD_CTRL[0] will start the command fetch, then Internal command DMA fetch commands from external command line.

Command line is a collection of several command codes with continuous address. At the first start, the command start address (RGA_CMD_ADDR) and command number (RGA_CMD_CTRL[12:3]) should be set, then write '1' to cmd_line_st (RGA_CMD_CTRL[0]) to start the command line fetch. Incremental command is supported by setting cmd_incr_num (RGA_CMD_CTRL[12:3]) and cmd_incr_valid (RGA_CMD_CTRL[1]=1'b1)

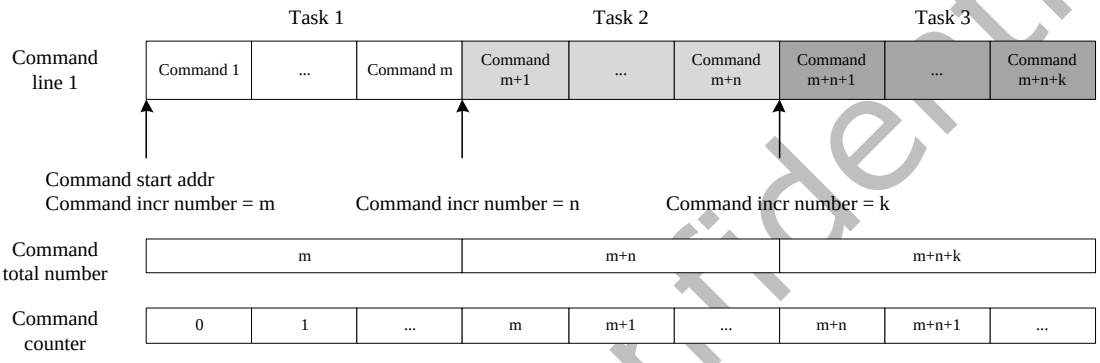

Fig. 18-9RGA command line and command counter

16.17.3 Command Sync

In slave command mode, command sync is controlled by CPU.

In master command mode, user can enable the current_cmd_int(sw_intr_cf), command by command to generate a interrupt at the end point of target command operation.

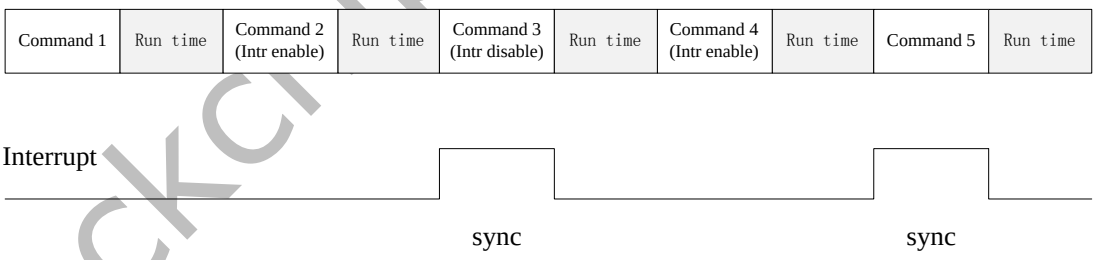

Fig.18-10RGA command sync generation

16.17.4 ColorPaletteApplication Notes

1. Palette/LUT Load into special RAM in ELS_BUF_CTRL;

2. ColorPalette/Pattern interval operations no need to initialLUT/pattern ram if LUT/patterncontentno update;

16.17.5 Some special application constraint

1. The algorithm of vertical scale up: must select bicubic algorithm when source picture is smaller or equal to 2k and must select bilinear when bigger than 2k

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2. The effects that The output's definition is near 2k or 4k may not very well when at the scenario that the vertical side is scale up and the horizontal is scale down within range of 2%(such as:2048x32→2008x64)

3. At the scenario A+B->C, the size among the A B C has some constraint :

   A's size must be equal to C. C's size must equal to B when A+C is no rotation. C's rotation (90degree)size must equal to B1 when A+C is rotation 90 degree .

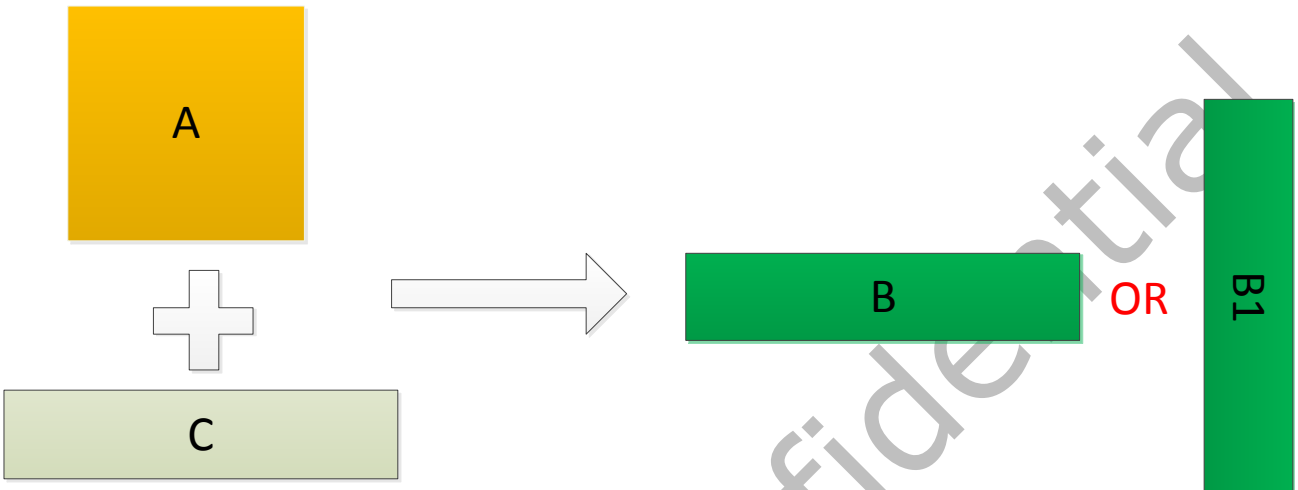

Fig.18-11the sizeconstraint among A B C

4. YUV420/422-8bit virtual stride need 8byte align，xoff/yoff need 2byte algin；

5. YUV420/422-10bit virtual stride need 16byte align，not support xoff/yoff；

6. Vertical scale down or not && Horizontal bi-cubic scale up src0 width<=2048；

   Vertical scale up && Horizontal bi-cubic scale up src0 width<=1928；

7. Vertical scale down or not && Horizontal bilinear scale up src0 width<=4096；

   Vertical scale up && Horizontal bilinear scale up src0 width<=3856；

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Chapter 19 Image Enhancement Processor（IEP）

16.18 Overview

The Image Enhancement Processor (IEP) receives data from and transmits data to system main memory by AXI bus.

The features of IEP are as follow:

- Image formatw
  - Input data: XRGB/RGB565/YUV420/YUV422
  - Output data: ARGB/RGB565/YUV420/YUV422
  - ARGB/XRGB/RGB565/YUV swap
  - YUV semi-planar/planar
  - BT601_l/BT601_f/BT709_l/BT709_f color space conversion
  - RGB dither up/down conversion
  - YUV up/down sampling conversion
  - Max resolution for static image up to 8192x8192
  - Max resolution for dynamic image up to 1920x1080
- Enhancement
  - Hue/Saturation/Brightness/Contrast enhancement
- De-interlace
  - Input 4 fields, output 2 frames mode
  - Input 4 fields, output 1 frames mode
  - Input 2 fields, output 1 frames mode
  - Programmable motion detection coefficient
  - Programmable high frequency factor
  - Programmable edge interpolation parameter
- Interface
  - 32bit AHB bus slave
  - 64bit AXI bus master
  - Combined interrupt output

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16.19 Block Diagram

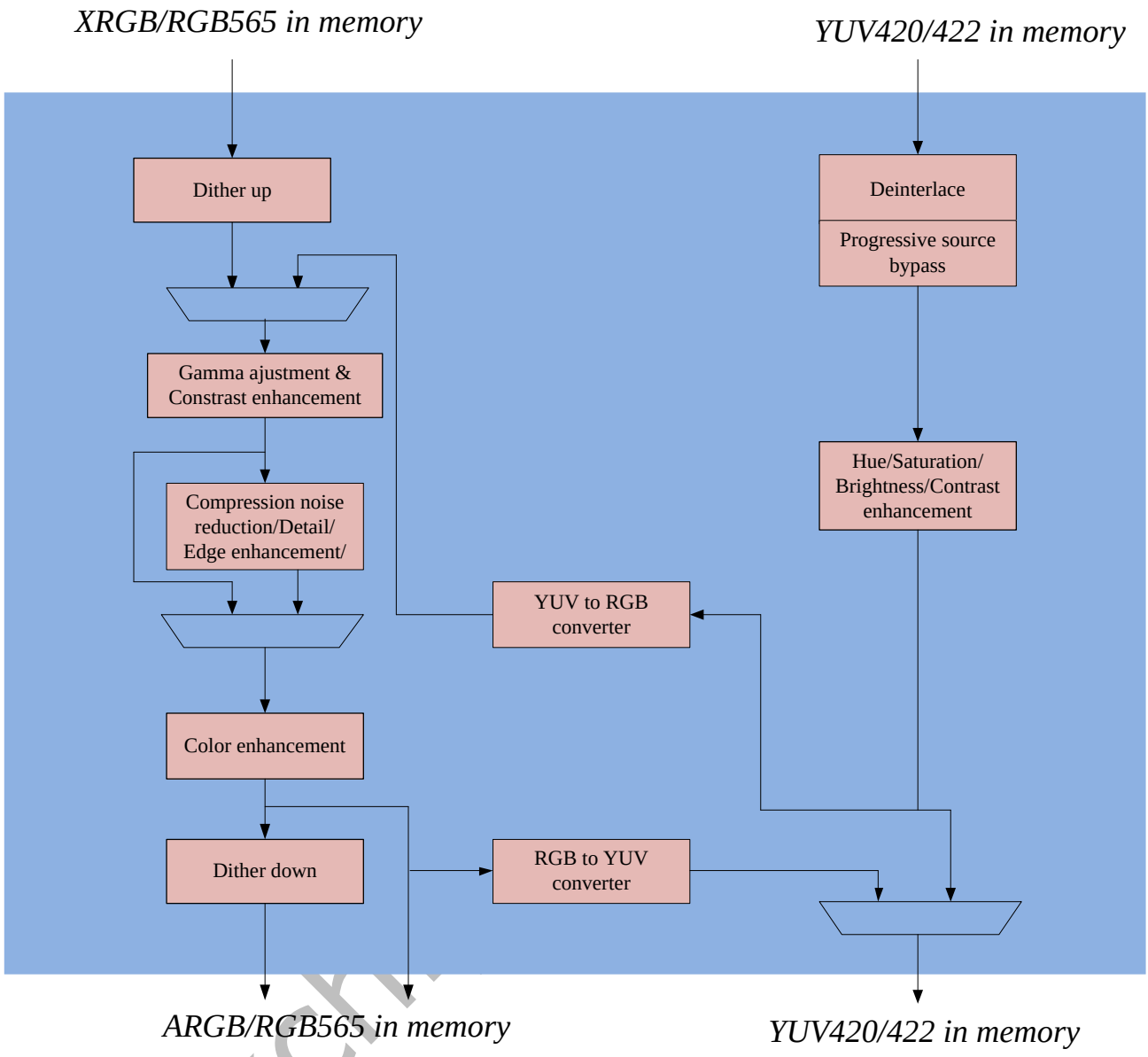

Fig.19-1IEP block diagram

The data path in IEP is in the previously diagram. The IEP comprises with:

- Deinterlace

There are five deinterlace mode including I4O2 (input 4 fields and output 2 frames once), I4O1B, I4O1T, I2O1B, I2O1T in the deinterlace block. YUV bypass is also supported. Pay attention if compression noise reduction, detail or edge enhancement work together with deinterlace is not allowed.

- Enhancement

Not only hue, saturation, brightness, contrast enhancement, but also blue screen, black screen and color bar are supported in YUV domain enhancement block. Gamma adjustment, edge enhancement, detail enhancement and color enhancement are supported in RGB domain enhancement block.

- Noise Reduction

Spatial and temporal sampling noise can be reduced in YUV domain noise reduction block.

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Compression noise can be reduced in RGB domain noise reduction block.

16.20 Function Description

16.20.1 Deinterlace

There are five deinterlace mode including I4O2, I4O1B, I4O1T, I2O1B and I2O1T in the deinterlace block. The I4O2 mode represents for 4 fields of input images and 2 frames of output images, so all of the two groups of source address registers and two groups of destination address registers need to be configured. For example, if source and destination format are both YUV420, the source address register IEP_SRC_ADDR_YRGB, IEP_SRC_ADDR_CBCR are used for source field0 and field 1, the source address register IEP_SRC_ADDR_Y1, IEP_SRC_ADDR_CBCR1 are used for source field2 and field3. The I4O1B and I4O1T mode have the same input images as the I4O2 mode, but only one frame output is generated once. The I2O1B and I2O1T mode have the same output as I4O1B and I4O1T mode, but only two fields input are needed. If bypass mode is selected, there are not any deinterlace operations. The parameter dil_ei_sel, dil_ei_radius, dil_ei_smooth, dil_ei_mode, dil_hf_en and dil_hf_fct in register IEP_CONFIG0 and registers IEP_DIL_MTN_TAB0~7 may have different influence in deinterlace effect depend on the type of the image source.

16.20.2 Noise reduction

Compression noise reduction is used for reducing the noise after the decompression of picture or video. Before the compression noise reduction is enabled, the IEP_ENH_DDE_COE0/1 from address 0x400 to 0x5FC for difference and distance coefficients must be written firstly. The filter matrix can be selected from 3x3/5x5 and the filter weight can be programmed by configuring IEP_ENH_RGB_CNFG.

16.20.3 Enhancement

Not only hue, saturation, brightness, contrast enhancement, but also blue screen, black screen and color bar are supported in this block. IEP_ENH_YUV_CNFG_0/1/2 registers can be configured to modify the YUV enhance parameters to satisfied with the requirement.

Before the gamma adjustment or contrast enhancement is enabled in RGB domain, the IEP_ENH_CG_TAB from address 0x100 to 0x3FC for B, G, R mapping must be written firstly. If the color enhancement is enabled, the IEP_ENH_C_COE must be written the required value.

Before the edge or detail enhancement is enabled, the IEP_ENH_DDE_COE0/1 from address 0x400 to 0x5FC for difference and distance coefficients must be written firstly. The filter matrix can be selected from 3x3/5x5 and the filter weight can be programmed by configuring IEP_ENH_RGB_CNFG.

16.20.4 Format conversion

The color space conversion either from RGB to YUV or from YUV to RGB has the selections including BT601/709_L/F mode, and the input can be clipped or not.

If the source format is RGB565, dither up must be enabled. In contrary to the destination format is RGB565, dither down must be enabled.

16.20.5 Shadow registers

The configuration registers can be configured at any time, but they cannot have any effect immediately unless config_done is available and a new frame_start is enabled. The registers IEP_RAW_CONFIG0/1, IEP_RAW_VIR_IMG_WIDTH, IEP_RAW_IMG_SCL_FCT, IEP_RAW_SRC_IMG_SIZE, IEP_RAW_ENH_YUV_CNFG_0/1/2 corresponding to the registers have the similar names but without letters _RAW. They are used for raw register value reading before the configurations really have effect on the new frame.

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16.20.6 VOP direct path

The IEP_DST_ADDR for DMA writing is useless if vop_path_en bit is set, because all RGB or YUV data is supplied for VOP directly from local bus via VOP and IEP.

16.21 Register Description

16.21.1 Registers Summary

Name	Offset	Size	Reset Value	Description
IEP_CONFIG0	0x0000	W	0x00000000	configuration register0
IEP_CONFIG1	0x0004	W	0x00000000	configuration register1
IEP_STATUS	0x0008	W	0x00000000	status register
IEP_INT	0x000c	W	0x00000000	interrupt register
IEP_FRM_START	0x0010	W	0x00000000	frame start
IEP_CONFIG_DONE	0x0018	W	0x00000000	configuration done
IEP_FRM_CNT	0x001c	W	0x00000000	frame counter
IEP_VIR_IMG_WIDTH	0x0020	W	0x01400140	Image virtual width
IEP_IMG_SCL_FCT	0x0024	W	0x20002000	scaling factor
IEP_SRC_IMG_SIZE	0x0028	W	0x00f00140	Source image width/height
IEP_DST_IMG_SIZE	0x002c	W	0x00f00140	Destination image width/height
IEP_DST_IMG_WIDTH_TILE0	0x0030	W	0x00000000	Destination image tile0 width
IEP_DST_IMG_WIDTH_TILE1	0x0034	W	0x00000000	Destination image tile1 width
IEP_DST_IMG_WIDTH_TILE2	0x0038	W	0x00000000	Destination image tile2 width
IEP_DST_IMG_WIDTH_TILE3	0x003c	W	0x00000000	Destination image tile3 width
IEP_ENH_YUV_CNFG_0	0x0040	W	0x00000000	brightness,contrast,saturation adjustment

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IEP_ENH_YUV_CNFG_1	0x0044	W	0x00000000	Hue configuration
IEP_ENH_YUV_CNFG_2	0x0048	W	0x00000000	color bar configuration
IEP_ENH_RGB_CNFG	0x004c	W	0x00000000	enhancement RGB configuration
IEP_ENH_C_COE	0x0050	W	0x00000000	rgb color enhancement coefficient
IEP_RAW_CONFIG0	0x0058	W	0x00000000	configuration register0
IEP_RAW_CONFIG1	0x005c	W	0x00000000	configuration register1
IEP_RAW_VIR_IMG_WIDTH	0x0060	W	0x01400140	Image virtual width
IEP_RAW_IMG_SCL_FCT	0x0064	W	0x20002000	scaling factor
IEP_RAW_SRC_IMG_SIZE	0x0068	W	0x00f00140	Source image width/height
IEP_RAW_DST_IMG_SIZE	0x006c	W	0x00f00140	Destination image width/height
IEP_RAW_ENH_YUV_CNFG_0	0x0070	W	0x00000000	brightness,contrast,saturation adjustment
IEP_RAW_ENH_YUV_CNFG_1	0x0074	W	0x00000000	Hue configuration
IEP_RAW_ENH_YUV_CNFG_2	0x0078	W	0x00000000	color bar configuration
IEP_RAW_ENH_RGB_CNFG	0x007c	W	0x00000000	enhancement RGB configuration
IEP_SRC_ADDR_YRGB	0x0080	W	0x00000000	Start address of source image(Y/RGB)
IEP_SRC_ADDR_CBCR	0x0084	W	0x00000000	Start address of source image(Cb/Cr)
IEP_SRC_ADDR_CR	0x0088	W	0x00000000	Start address of source image(Cr)
IEP_SRC_ADDR_Y1	0x008c	W	0x00000000	Start address of source image(Y)
IEP_SRC_ADDR_CBCR1	0x0090	W	0x00000000	Start address of source image(Cb/Cr)

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IEP_SRC_ADDR_CR1	0x0094	W	0x00000000	Start address of source image(Cr)
IEP_SRC_ADDR_Y_ITEMP	0x0098	W	0x00000000	Start address of source image(Y integer part)
IEP_SRC_ADDR_CBCR_ITEMP	0x009c	W	0x00000000	Start address of source image(CBCR integer part)
IEP_SRC_ADDR_CR_ITEMP	0x00a0	W	0x00000000	Start address of source image(CR integer part)
IEP_SRC_ADDR_Y_FTEMP	0x00a4	W	0x00000000	Start address of source image(Y fraction part)
IEP_SRC_ADDR_CBCR_FTEMP	0x00a8	W	0x00000000	Start address of source image(CBCR fraction part)
IEP_SRC_ADDR_CR_FTEMP	0x00ac	W	0x00000000	Start address of source image(CR fraction part)
IEP_DST_ADDR_YRGB	0x00b0	W	0x00000000	Start address of destination image(Y/RGB)
IEP_DST_ADDR_CBCR	0x00b4	W	0x00000000	Start address of destination image(Cb/Cr)
IEP_DST_ADDR_CR	0x00b8	W	0x00000000	Start address of destination image(Cr)
IEP_DST_ADDR_Y1	0x00bc	W	0x00000000	Start address of destination image(Y)
IEP_DST_ADDR_CBCR1	0x00c0	W	0x00000000	Start address of destination image(Cb/Cr)
IEP_DST_ADDR_CR1	0x00c4	W	0x00000000	Start address of destination image(Cr)
IEP_DST_ADDR_Y_ITEMP	0x00c8	W	0x00000000	Start address of destination image(Y integer part)
IEP_DST_ADDR_CBCR_ITEMP	0x00cc	W	0x00000000	Start address of destination image(CBCR integer part)

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IEP_DST_ADDR_CR_ITEMP	0x00d0	W	0x00000000	Start address of destination image(CR integer part)
IEP_DST_ADDR_Y_FTEMP	0x00d4	W	0x00000000	Start address of destination image(Y fraction part)
IEP_DST_ADDR_CBCR_FTEMP	0x00d8	W	0x00000000	Start address of destination image(CBCR fraction part)
IEP_DST_ADDR_CR_FTEMP	0x00dc	W	0x00000000	Start address of destination image(CR fraction part)
IEP_DIL_MTN_TAB0	0x00e0	W	0x00000000	Deinterlace motion table0
IEP_DIL_MTN_TAB1	0x00e4	W	0x00000000	Deinterlace motion table1
IEP_DIL_MTN_TAB2	0x00e8	W	0x00000000	Deinterlace motion table2
IEP_DIL_MTN_TAB3	0x00ec	W	0x00000000	Deinterlace motion table3
IEP_DIL_MTN_TAB4	0x00f0	W	0x00000000	Deinterlace motion table4
IEP_DIL_MTN_TAB5	0x00f4	W	0x00000000	Deinterlace motion table5
IEP_DIL_MTN_TAB6	0x00f8	W	0x00000000	Deinterlace motion table6
IEP_DIL_MTN_TAB7	0x00fc	W	0x00000000	Deinterlace motion table7
IEP_ENH_CG_TAB	0x0100	W	0x00000000	contrast and gamma enhancement table
IEP_ENH_DDE_COE0	0x0400	W	0x00000000	denoise,detail and edge enhancement coefficient
IEP_ENH_DDE_COE1	0x0500	W	0x00000000	denoise,detail and edge enhancement coefficient
IEP_MMU_DTE_ADDR	0x0800	W	0x00000000	MMU current page table address
IEP_MMU_STATUS	0x0804	W	0x00000018	MMU status register
IEP_MMU_CMD	0x0808	W	0x00000000	MMU command register
IEP_MMU_PAGE_FAULT_ADDR	0x080c	W	0x00000000	MMU logic address of last page fault

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IEP_MMU_ZAP_ONE_LINE	0x0810	W	0x00000000	MMU zap cache line register
IEP_MMU_INT_RAWSTAT	0x0814	W	0x00000000	MMU raw interrupt status register
IEP_MMU_INT_CLEAR	0x0818	W	0x00000000	MMU interrupt clear register
IEP_MMU_INT_MASK	0x081c	W	0x00000000	MMU interrupt mask register
IEP_MMU_INT_STATUS	0x0820	W	0x00000000	MMU interrupt status register
IEP_MMU_AUTO_GATING	0x0824	W	0x00000001	MMU clock auto gating register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

16.21.2 Detail Register Description

IEP_CONFIG0

Address: Operational Base + offset (0x0000)

configuration register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x0	dil_ei_sel deinterlace edge interpolation select
22:21	RW	0x0	dil_ei_radius deinterlace edge interpolation radius
20	RW	0x0	rgb_con_gam_order RGB contrast enhancement and gamma adjustment operation order select. 0:CG prior to DDE 1:DDE prior to CG (CG represent for contrast & gamma operation, and DDE represent for denoise, detail or edge enhancement operation)

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19:18	RW	0x0	rgb_enh_sel RGB enhancement select 00: no operation 01: denoise 10: detail enhancement 11: edge enhancement
17	RW	0x0	rgb_con_gam_en RGB contrast enhancement and gamma adjustment enable 0:disable 1:enable
16	RW	0x0	rgb_color_enh_en RGB color enhancement enable 0:disable 1:enable
15	RW	0x0	dil_ei_smooth deinterlace edge interpolation for smooth effect 0: disable 1: enable
14	RW	0x0	yuv_enh_en yuv enhancement enable 0:disable 1:enable
13	RW	0x0	yuv_dns_en YUV 3D denoise enable 0:disable 1:enable

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12	RW	0x0	dil_ei_mode deinterlace edge interpolation 0: disable 1: enable
11	RW	0x0	dil_hf_en deinterlace high frequency calculation enable 0: disable 1: enable
10:8	RW	0x0	dil_mode Deinterlace mode select: 000: YUV deinterlace and bypass path disable; 001: I4O2 mode 010: I4O1B mode 011: I4O1T mode 100: I2O1B mode 101: I2O1T mode 110: bypass mode
7:1	RW	0x00	dil_hf_fct deinterlace high frequency factor
0	RW	0x0	vop_path_en VOP direct path enable 0:disable 1:enable

IEP_CONFIG1

Address: Operational Base + offset (0x0004)

configuration register1

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	glb_alpha global alpha value only valid when destination format is ARGB
23	RW	0x0	rgb2yuv_input_clip RGB to YUV input range 0:R/G/B=[0,255] 1:R/G/B=[16,235]
22	RW	0x0	yuv2rgb_input_clip YUV to RGB input range 0:Y/U/V=[0,255] 1:Y=[16,235],U/V=[16,240]
21	RW	0x0	rgb_to_yuv_en RGB to YUV conversion enable 0: disable 1: enable
20	RW	0x0	yuv_to_rgb_en YUV to RGB conversion enable 0: disable 1: enable
19:18	RW	0x0	rgb2yuv_coe_sel rgb2yuv coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f

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17:16	RW	0x0	yuv2rgb_coe_sel yuv2rgb coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
15	RW	0x0	dthr_down_en dither down enable 0: disable 1: enable
14	RW	0x0	dthr_up_en dither up enable 0: disable 1: enable
13:12	RW	0x0	dst_yuv_swap destination YUV swap 00:SP UV 01:SP VU 10, 11:P

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11:10	RW	0x0	dst_rgb_swap destination RGB swap ARGB destination 00:ARGB 01:ABGR 10:RGBA 11:BGRA RGB565 destination 00,10:RGB 01,11:BGR
9:8	RW	0x0	dst_fmt Ouput image Format 00 : ARGB 01 : RGB565 10 : YUV422 11 : YUV420
7:6	RO	0x0	reserved
5:4	RW	0x0	src_yuv_swap source YUV swap 00:SP UV 01:SP VU 10, 11:P

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3:2	RW	0x0	src_rgb_swap source RGB swap XRGB source 00:XRGB 01:XBGR 10:RGBX 11:BGRX RGB565 source 00,10:RGB 01,11:BGR
1:0	RW	0x0	src_fmt Input image Format 00 : XRGB 01 : RGB565 10 : YUV422 11 : YUV420

IEP_STATUS

Address: Operational Base + offset (0x0008)

status register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19	RO	0x0	rrgb_idle_ack RGB read DMA idle acknowlege
18	RO	0x0	wrgb_idle_ack RGB write DMA idle acknowlege

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17	RO	0x0	ryuv_idle_ack YUV read DMA idle acknowlege
16	RO	0x0	wyuv_idle_ack YUV write DMA idle acknowlege
15:9	RO	0x0	reserved
8	RO	0x0	voi_sts vop direct path status 00:idle 01:working
7	RO	0x0	rrgb_sts RGB DMA read status 00:idle 01:working
6	RO	0x0	wrgb_sts RGB DMA write status 00:idle 01:working
5	RO	0x0	ryuv_sts YUV DMA read status 00:idle 01:working
4	RO	0x0	wyuv_sts YUV DMA write status 00:idle 01:working

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3	RO	0x0	dde_sts RGB denoise/enhancement status 00:idle 01:working
2	RO	0x0	dil_sts de-interlace or yuv bypass status 00:idle 01:working
1	RO	0x0	scl_sts scaling status 00:idle 01:working
0	RO	0x0	dns_sts YUV 3D denoise status 00:idle 01:working

IEP_INT

Address: Operational Base + offset (0x000c)

interrupt register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	W1C	0x0	frm_done_int_clr Frame process done interrupt clear After be set to 1, this bit will be clear automatically.
15:9	RO	0x0	reserved

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8	RW	0x0	frm_done_int_en Frame process done interrupt enable: 0: disable; 1: enable;
7:1	RO	0x0	reserved
0	RO	0x0	frm_done_int Frame process done interrupt 0: inactive; 1: active;

IEP_FRM_START

Address: Operational Base + offset (0x0010)

frame start

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	frm_start frame start Write 1, self clear.

IEP_CONFIG_DONE

Address: Operational Base + offset (0x0018)

configuration done

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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0	RW	0x0	config_done configuration done Wait for frame start to update raw register configuration to really used registers.

IEP_FRM_CNT

Address: Operational Base + offset (0x001c)

frame counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	frm_cnt frame counter Self increase one after a frame operation is finished. Write arbitrary value to clear to zero.

IEP_VIR_IMG_WIDTH

Address: Operational Base + offset (0x0020)

Image virtual width

Bit	Attr	Reset Value	Description
31:16	RW	0x0140	dst_vir_image_width Destination virtual image width
15:0	RW	0x0140	src_vir_image_width Source virtual image width

IEP_IMG_SCL_FCT

Address: Operational Base + offset (0x0024)

scaling factor

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Bit	Attr	Reset Value	Description

31:16	RW	0x2000	vrt_scl_fct
			Vertical scale factor
			up scaling:
			vrt_scl_fct=floor(src_image_height/dst_image_height);
			down scaling:
			vrt_scl_fct=ceiling((dst_image_height+1)/(src_image_height+1));

15:0	RW	0x2000	hrz_scl_fct
			Horizontal scale factor
			up scaling:
			hrz_scl_fct=floor(src_image_width/dst_image_width);
			down scaling:
			hrz_scl_fct=ceiling((dst_image_width+1)/(src_image_width+1));

IEP_SRC_IMG_SIZE

Address: Operational Base + offset (0x0028)

Source image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00f0	src_image_height source image height
15:13	RO	0x0	reserved
12:0	RW	0x0140	src_image_width source image width

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IEP_DST_IMG_SIZE

Address: Operational Base + offset (0x002c)

Destination image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x00f0	dst_image_height Destination image height
15:13	RO	0x0	reserved
12:0	RW	0x0140	dst_image_width Destination image width

IEP_DST_IMG_WIDTH_TILE0

Address: Operational Base + offset (0x0030)

Destination image tile0 width

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	dst_image_width_tile0 Destination image tile0 width

IEP_DST_IMG_WIDTH_TILE1

Address: Operational Base + offset (0x0034)

Destination image tile1 width

Bit	Attr	Reset Value	Description

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31:11	RO	0x0	reserved
10:0	RW	0x000	dst_image_width_tile1 Destination image tile1 width

IEP_DST_IMG_WIDTH_TILE2

Address: Operational Base + offset (0x0038)

Destination image tile2 width

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x000	dst_image_width_tile2 Destination image tile2 width

IEP_DST_IMG_WIDTH_TILE3

Address: Operational Base + offset (0x003c)

Destination image tile3 width

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:0	RW	0x000	dst_image_width_tile3 Destination image tile3 width

IEP_ENH_YUV_CNFG_0

Address: Operational Base + offset (0x0040)

brightness,contrast,saturation adjustment

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Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RW	0x000	sat_con YUV saturation and contrast adjustment saturation * contrast range from 0 to 1.992*1.992, and this value is saturation* contrast * 128
15:8	RW	0x00	contrast YUV contrast adjustment contrast value range from 0 to 1.992, and this value is contrast*128.
7:6	RO	0x0	reserved
5:0	RW	0x00	brightness YUV brightness adjustment range from -32 to 31 000000:0; 000001:1; …… 011111:31; 100000:-32; 100001:-31; …… 111110:-2; 111111:-1;

IEP_ENH_YUV_CNFG_1

Address: Operational Base + offset (0x0044)

Hue configuration

Bit	Attr	Reset Value	Description

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31:16	RO	0x0	reserved
15:8	RW	0x00	cos_hue the cos function value for hue adjustment sin function value range from 0.866 to 1 ,and this value is cos * 128 ,no sign bit
7:0	RW	0x00	sin_hue the sin function value for hue adjustment sin function value range from -0.5 to 0.5 ,and this value is sin * 128 ,and the high bit is sign bit

IEP_ENH_YUV_CNFG_2

Address: Operational Base + offset (0x0048)

color bar configuration

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:24	RW	0x0	video_mode video mode 00:black screen 01:blue screen 10:color bars 11:normal video
23:16	RW	0x00	color_bar_v color bar v value
15:8	RW	0x00	color_bar_u color bar u value

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7:0	RW	0x00	color_bar_y color bar y value

IEP_ENH_RGB_CNFG

Address: Operational Base + offset (0x004c)

enhancement RGB configuration

Bit	Attr	Reset Value	Description
31:30	RW	0x0	luma_spat_sel 3D denoiseluma spatial coefficient select
29:28	RW	0x0	luma_temp_sel 3D denoiseluma temporal coefficient select
27:26	RW	0x0	chroma_spat_sel 3D denoisechroma spatial coefficient select
25:24	RW	0x0	chroma_temp_sel 3D denoisechroma temporal coefficient select
23:16	RW	0x00	enh_threshold enhancement threshold In denoise and detail enhancement operation, more than the threshold, considering as detail; but if less than the threshold, considering as noise, need to to be filtered.
15	RO	0x0	reserved

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14:8	RW	0x00	enh_alpha enhancement alpha value 0000000:0 0000001:1/16 0000010:2/16 …… 0001111:15/16 0010000:1 0010001:1+1/16; 0010010:1+2/16; 0010011:1+3/16; …… 0100000:2; …… 0110000:3; …… 1000000:4; …… 1010000:5; …… 1100000:6; other : reserved
7:2	RO	0x0	reserved
1:0	RW	0x0	enh_radius enhancement radius 00:R=1 01:R=2 10:R=3 11:R=4

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IEP_ENH_C_COE

Address: Operational Base + offset (0x0050)

rgb color enhancement coefficient

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:5	RW	0x0	c_int_coe color enhancement integer coefficient
4:0	RW	0x00	c_frac_coe color enhancement fraction coefficient

IEP_RAW_CONFIG0

Address: Operational Base + offset (0x0058)

configuration register0

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RO	0x0	dil_ei_sel deinterlace edge interpolation select
22:21	RO	0x0	dil_ei_radius deinterlace edge interpolation radius

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20	RO	0x0	rgb_con_gam_order RGB contrast enhancement and gamma adjustment operation order select. 0:CG prior to DDE 1:DDE prior to CG (CG represent for contrast & gamma operation, and DDE represent for de-noise, detail or edge enhancement operation)
19:18	RO	0x0	rgb_enh_sel RGB enhancement select 00: no operation 01: denoise 10: detail enhancement 11: edge enhancement
17	RO	0x0	rgb_con_gam_en RGB contrast enhancement and gamma adjustment enable 0:disable 1:enable
16	RO	0x0	rgb_color_enh_en RGB color enhancement enable 0:disable 1:enable
15	RO	0x0	dil_ei_smooth deinterlace edge interpolation for smooth effect 0: disable 1: enable
14	RO	0x0	yuv_enh_en yuv enhancement enable 0:disable 1:enable

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13	RO	0x0	yuv_dns_en YUV 3D denoise enable 0:disable 1:enable
12	RO	0x0	dil_ei_mode deinterlace edge interpolation 0: disable 1: enable
11	RO	0x0	dil_hf_en deinterlace high frequency calculation enable 0: disable 1: enable
10:8	RO	0x0	dil_mode Deinterlace mode select: 000: YUV deinterlace and bypass path disable; 001: I4O2 mode 010: I4O1B mode 011: I4O1T mode 100: I2O1B mode 101: I2O1T mode 110: bypass mode
7:1	RO	0x00	dil_hf_fct deinterlace high frequency factor
0	WO	0x0	vop_path_en VOP direct path enable 0:disable 1:enable

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IEP_RAW_CONFIG1

Address: Operational Base + offset (0x005c)

configuration register1

Bit	Attr	Reset Value	Description
31:24	RO	0x00	glb_alpha global alpha value only valid when destination format is ARGB
23	RO	0x0	rgb2yuv_input_clip RGB to YUV input range 0:R/G/B=[0,255] 1:R/G/B=[16,235]
22	RO	0x0	yuv2rgb_input_clip YUV to RGB input range 0:Y/U/V=[0,255] 1:Y=[16,235],U/V=[16,240]
21	RO	0x0	rgb_to_yuv_en RGB to YUV conversion enable 0: disable 1: enable
20	RO	0x0	yuv_to_rgb_en YUV to RGB conversion enable 0: disable 1: enable

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19:18	RO	0x0	rgb2yuv_coe_sel rgb2yuv coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
17:16	RO	0x0	yuv2rgb_coe_sel yuv2rgb coefficient select 00:bt601_1 01:bt601_f 10:bt709_1 11:bt709_f
15	RO	0x0	dthr_down_en dither down enable 0: disable 1: enable
14	RO	0x0	dthr_up_en dither up enable 0: disable 1: enable
13:12	RO	0x0	dst_yuv_swap destination YUV swap 00:SP UV 01:SP VU 10, 11:P

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11:10	RO	0x0	dst_rgb_swap destination RGB swap ARGB destination 00:ARGB 01:ABGR 10:RGBA 11:BGRA RGB565 destination 00,10:RGB 01,11:BGR
9:8	RO	0x0	dst_fmt Ouput image Format 00 : ARGB 01 : RGB565 10 : YUV422 11 : YUV420
7:6	RO	0x0	reserved
5:4	RO	0x0	src_yuv_swap source YUV swap 00:SP UV 01:SP VU 10, 11:P

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3:2	RO	0x0	src_rgb_swap source RGB swap XRGB source 00:XRGB 01:XBGR 10:RGBX 11:BGRX RGB565 source 00,10:RGB 01,11:BGR
1:0	RO	0x0	src_fmt Input image Format 00 : XRGB 01 : RGB565 10 : YUV422 11 : YUV420

IEP_RAW_VIR_IMG_WIDTH

Address: Operational Base + offset (0x0060)

Image virtual width

Bit	Attr	Reset Value	Description
31:16	RO	0x0140	dst_vir_image_width Destination virtual image width
15:0	RO	0x0140	src_vir_image_width Source virtual image width

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IEP_RAW_IMG_SCL_FCT

Address: Operational Base + offset (0x0064)

scaling factor

Bit	Attr	Reset Value	Description

31:16	RO	0x2000	vrt_scl_fct
			Vertical scale factor
			up scaling:
			vrt_scl_fct=floor(src_image_height/dst_image_height);
			down scaling:
			vrt_scl_fct=ceiling((dst_image_height+1)/(src_image_height+1));
			;

15:0	RO	0x2000	hrz_scl_fct
			Horizontal scale factor
			up scaling:
			hrz_scl_fct=floor(src_image_width/dst_image_width);
			down scaling:
			hrz_scl_fct=ceiling((dst_image_width+1)/(src_image_width+1));

IEP_RAW_SRC_IMG_SIZE

Address: Operational Base + offset (0x0068)

Source image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RO	0x00f0	src_image_height source image height
15:13	RO	0x0	reserved

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12:0	RO	0x0140	src_image_width source image width

IEP_RAW_DST_IMG_SIZE

Address: Operational Base + offset (0x006c)

Destination image width/height

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RO	0x00f0	dst_image_height Destination image height
15:13	RO	0x0	reserved
12:0	RO	0x0140	dst_image_width Destination image width

IEP_RAW_ENH_YUV_CNFG_0

Address: Operational Base + offset (0x0070)

brightness,contrast,saturation adjustment

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:16	RO	0x000	sat_con YUV saturation and contrast adjustment saturation * contrast range from 0 to 1.992*1.992, and this value is saturation* contrast * 128

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15:8	RO	0x00	contrast YUV contrast adjustment contrast value range from 0 to 1.992, and this value is contrast*128.
7:6	RO	0x0	reserved
5:0	RO	0x00	brightness YUV brightness adjustment range from -32 to 31 000000:0; 000001:1; …… 011111:31; 100000:-32; 100001:-31; …… 111110:-2; 111111:-1;

IEP_RAW_ENH_YUV_CNFG_1

Address: Operational Base + offset (0x0074)

Hue configuration

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RO	0x00	cos_hue the cos function value for hue adjustment sin function value range from 0.866 to 1 ,and this value is cos * 128 ,no sign bit

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7:0	RO	0x00	sin_hue the sin function value for hue adjustment sin function value range from -0.5 to 0.5 ,and this value is sin * 128 ,and the high bit is sign bit

IEP_RAW_ENH_YUV_CNFG_2

Address: Operational Base + offset (0x0078)

color bar configuration

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:24	RO	0x0	video_mode video mode 00:black screen 01:blue screen 10:color bars 11:normal video
23:16	RO	0x00	color_bar_v color bar v value
15:8	RO	0x00	color_bar_u color bar u value
7:0	RO	0x00	color_bar_y color bar y value

IEP_RAW_ENH_RGB_CNFG

Address: Operational Base + offset (0x007c)

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enhancement RGB configuration

Bit	Attr	Reset Value	Description
31:30	RO	0x0	luma_spat_sel 3D denoiseluma spatial coefficient select
29:28	RO	0x0	luma_temp_sel 3D denoiseluma temporal coefficient select
27:26	RO	0x0	chroma_spat_sel 3D denoisechroma spatial coefficient select
25:24	RW	0x0	chroma_temp_sel 3D denoisechroma temporal coefficient select
23:16	RO	0x00	enh_threshold enhancement threshold In denoise and detail enhancement operation, more than the threshold, considering as detail; but if less than the threshold, considering as noise, need to to be filtered.
15	RO	0x0	reserved

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14:8	RO	0x00	enh_alpha enhancement alpha value 0000000:0 0000001:1/16 0000010:2/16 …… 0001111:15/16 0010000:1 0010001:1+1/16; 0010010:1+2/16; 0010011:1+3/16; …… 0100000:2; …… 0110000:3; …… 1000000:4; …… 1010000:5; …… 1100000:6; other : reserved
7:2	RO	0x0	reserved
1:0	RO	0x0	enh_radius enhancement radius 00:R=1 01:R=2 10:R=3 11:R=4

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IEP_SRC_ADDR_YRGB

Address: Operational Base + offset (0x0080)

Start address of source image(Y/RGB)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_yrgb_mst Source image data YRGB start address in Memory

IEP_SRC_ADDR_CBCR

Address: Operational Base + offset (0x0084)

Start address of source image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst Source image data CbCr start address in Memory

IEP_SRC_ADDR_CR

Address: Operational Base + offset (0x0088)

Start address of source image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst Source image data Cr start address in Memory

IEP_SRC_ADDR_Y1

Address: Operational Base + offset (0x008c)

Start address of source image(Y)

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_y_mst Source image data Y start address in Memory

IEP_SRC_ADDR_CBCR1

Address: Operational Base + offset (0x0090)

Start address of source image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst Source image data CbCr start address in Memory

IEP_SRC_ADDR_CR1

Address: Operational Base + offset (0x0094)

Start address of source image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst Source image data Cr start address in Memory

IEP_SRC_ADDR_Y_ITEMP

Address: Operational Base + offset (0x0098)

Start address of source image(Y integer part)

Bit	Attr	Reset Value	Description

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31:0	RW	0x00000000	src_image_y_mst_itemp Interger part source image data Y start address in Memory

IEP_SRC_ADDR_CBCR_ITEMP

Address: Operational Base + offset (0x009c)

Start address of source image(CBCR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst_cbcr_itemp Interger part source image data CBCR start address in Memory

IEP_SRC_ADDR_CR_ITEMP

Address: Operational Base + offset (0x00a0)

Start address of source image(CR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst_cr_itemp Interger part source image data CR start address in Memory

IEP_SRC_ADDR_Y_FTEMP

Address: Operational Base + offset (0x00a4)

Start address of source image(Y fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_y_mst_ftemp Fraction part source image data Y start address in Memory

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IEP_SRC_ADDR_CBCR_FTEMP

Address: Operational Base + offset (0x00a8)

Start address of source image(CBCR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cbcr_mst_ftemp Fraction part source image data CBCR start address in Memory

IEP_SRC_ADDR_CR_FTEMP

Address: Operational Base + offset (0x00ac)

Start address of source image(CR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_image_cr_mst_ftemp Fraction part source image data CR start address in Memory

IEP_DST_ADDR_YRGB

Address: Operational Base + offset (0x00b0)

Start address of destination image(Y/RGB)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_yrgb_mst Destination image data YRGB start address in Memory

IEP_DST_ADDR_CBCR

Address: Operational Base + offset (0x00b4)

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Start address of destination image(Cb/Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst Destination image data CBCR start address in Memory

IEP_DST_ADDR_CR

Address: Operational Base + offset (0x00b8)

Start address of destination image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst Destination image data CR start address in Memory

IEP_DST_ADDR_Y1

Address: Operational Base + offset (0x00bc)

Start address of destination image(Y)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_y_mst Destination image data Y start address in Memory

IEP_DST_ADDR_CBCR1

Address: Operational Base + offset (0x00c0)

Start address of destination image(Cb/Cr)

Bit	Attr	Reset Value	Description

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31:0	RW	0x00000000	dst_image_cbcr_mst Destination image data CbCr start address in Memory

IEP_DST_ADDR_CR1

Address: Operational Base + offset (0x00c4)

Start address of destination image(Cr)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst Destination image data Cr start address in Memory

IEP_DST_ADDR_Y_ITEMP

Address: Operational Base + offset (0x00c8)

Start address of destination image(Y integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_y_mst_itemp Interger part destination image data Y start address in Memory

IEP_DST_ADDR_CBCR_ITEMP

Address: Operational Base + offset (0x00cc)

Start address of destination image(CBCR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst_itemp Int part destination image data CBCR start address in Memory

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IEP_DST_ADDR_CR_ITEMP

Address: Operational Base + offset (0x00d0)

Start address of destination image(CR integer part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst_itemp Interger part destination image data CR start address in Memory

IEP_DST_ADDR_Y_FTEMP

Address: Operational Base + offset (0x00d4)

Start address of destination image(Y fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_y_mst_ftemp Fraction part destination image data Y start address in Memory

IEP_DST_ADDR_CBCR_FTEMP

Address: Operational Base + offset (0x00d8)

Start address of destination image(CBCR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cbcr_mst_ftemp Fraction part destination image data CBCR start address in Mem

IEP_DST_ADDR_CR_FTEMP

Address: Operational Base + offset (0x00dc)

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Start address of destination image(CR fraction part)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dst_image_cr_mst_ftemp Fraction part destination image data CR start address

IEP_DIL_MTN_TAB0

Address: Operational Base + offset (0x00e0)

Deinterlace motion table0

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

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IEP_DIL_MTN_TAB1

Address: Operational Base + offset (0x00e4)

Deinterlace motion table1

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

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IEP_DIL_MTN_TAB2

Address: Operational Base + offset (0x00e8)

Deinterlace motion table2

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB3

Address: Operational Base + offset (0x00ec)

Deinterlace motion table3

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved

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22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB4

Address: Operational Base + offset (0x00f0)

Deinterlace motion table4

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved

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14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB5

Address: Operational Base + offset (0x00f4)

Deinterlace motion table5

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved

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6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB6

Address: Operational Base + offset (0x00f8)

Deinterlace motion table6

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_DIL_MTN_TAB7

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Address: Operational Base + offset (0x00fc)

Deinterlace motion table7

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RW	0x00	mtn_sub_tab3 motion sub table3
23	RO	0x0	reserved
22:16	RW	0x00	mtn_sub_tab2 motion sub table2
15	RO	0x0	reserved
14:8	RW	0x00	mtn_sub_tab1 motion sub table1
7	RO	0x0	reserved
6:0	RW	0x00	mtn_sub_tab0 motion sub table0

IEP_ENH_CG_TAB

Address: Operational Base + offset (0x0100)

contrast and gamma enhancement table

Bit	Attr	Reset Value	Description
31:24	RW	0x00	cg_tab_3 cg table 3 pixel value 3,7,11,15,……mapping

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23:16	RW	0x00	cg_tab_2 cg table 2 pixel value 2,6,10,14,……mapping
15:8	RW	0x00	cg_tab_1 cg table 1 pixel value 1,5,9,13,……mapping
7:0	RW	0x00	cg_tab_0 cg table 0 256x8bit contrast & gamma mapping table pixel value 0,4,8,12,……mapping

IEP_ENH_DDE_COE0

Address: Operational Base + offset (0x0400)

denoise,detail and edge enhancement coefficient

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RW	0x00	dde_coe_3 dde coefficient 3 coefficient number 3,7,11,15,……
23:22	RO	0x0	reserved
21:16	RW	0x00	dde_coe_2 dde coefficient 2 coefficient number 2,6,10,14,……
15:14	RO	0x0	reserved
13:8	RW	0x00	dde_coe_1 dde coefficient 1 coefficient number 1,5,9,13,……

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7:6	RO	0x0	reserved
5:0	RW	0x00	dde_coe_0 dde coefficient 0 256x6bit cofficient for denoise and detail enhancement coefficient number 0,4,8,12,……

IEP_ENH_DDE_COE1

Address: Operational Base + offset (0x0500)

denoise,detail and edge enhancement coefficient

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:24	RW	0x00	dde_coe_3 dde coefficient 3 coefficient number 3,7,11,15,……
23:22	RO	0x0	reserved
21:16	RW	0x00	dde_coe_2 dde coefficient 3 coefficient number 2,6,10,14,……
15:14	RO	0x0	reserved
13:8	RW	0x00	dde_coe_1 dde coefficient 1 coefficient number 1,5,9,13,……
7:6	RO	0x0	reserved

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5:0	RW	0x00	dde_coe_0 dde coefficient 1 81x6bit cofficient for denoise and detail enhancement coefficient number 0,4,8,12,……

IEP_MMU_DTE_ADDR

Address: Operational Base + offset (0x0800)

MMU current page table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mmu_dte_addr page table address

IEP_MMU_STATUS

Address: Operational Base + offset (0x0804)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RW	0x00	mmu_page_fault_bus_id Index of master responsible for the last page fault
5	RW	0x0	mmu_page_fault_is_write The direction of access for last page fault: 0: read 1:write
4	RW	0x1	mmu_replay_buffer_empty The MMU replay buffer is empty.

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3	RW	0x1	mmu_idle the MMU is idle when accesses are being translated and there are no unfinished translated access. The MMU_IDLE signal only reports idle when the MMU processor is idle and accesses are active on the external bus. Note: the MMU can be idle in page fault mode.
2	RW	0x0	mmu_stall_active MMU stall mode currently enabled. The mode is enabled by command.
1	RW	0x0	mmu_page_fault_active MMU page fault mode currently enabled.The mode is enabled by command
0	RW	0x0	mmu_paging_enabled mmu paging is enabled

IEP_MMU_CMD

Address: Operational Base + offset (0x0808)

MMU command register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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2:0	RW	0x0	mmu_cmd 0: MMU_ENABLE_PAGING. enable paging. 1: MMU_DISABLE_PAGING. disable paging. 2: MMU_ENABLE_STALL. turn on stall mode. 3: MMU_DISABLE_STALL. turn off stall mode. 4: MMU_ZAP_CACHE. zap the entire page table cache. 5: MMU_PAGE_FAULT_DONE. leave page fault mode. 6: MMU_FORCE_RESET. reset the mmu. The MMU_ENABLE_STALL command can always be issued. Other commands are ignored unless the MMU is idle or stalled.

IEP_MMU_PAGE_FAULT_ADDR

Address: Operational Base + offset (0x080c)

MMU logic address of last page fault

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	mmu_page_fault_addr address of last page fault

IEP_MMU_ZAP_ONE_LINE

Address: Operational Base + offset (0x0810)

MMU zap cache line register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mmu_zap_one_line address to be invalidated from the page table cache.

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IEP_MMU_INT_RAWSTAT

Address: Operational Base + offset (0x0814)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error read bus error
0	RW	0x0	page_fault page fault

IEP_MMU_INT_CLEAR

Address: Operational Base + offset (0x0818)

MMU interrupt clear register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error_clear read bus error interrupt clear. write 1 to this register can clear read bus error interrupt.
0	RW	0x0	page_fault_clear page fault interrupt clear, write 1 to this register can clear page fault interrupt.

IEP_MMU_INT_MASK

Address: Operational Base + offset (0x081c)

MMU interrupt mask register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error_int_en read bus error interrupt enable
0	RW	0x0	page_fault_int_en page fault interrupt enable

IEP_MMU_INT_STATUS

Address: Operational Base + offset (0x0820)

MMU interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	read_bus_error read bus error interrupt
0	RW	0x0	page_fault page fault interrupt

IEP_MMU_AUTO_GATING

Address: Operational Base + offset (0x0824)

MMU clock auto gating register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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0	RW	0x1	mmu_auto_gating mmu clock auto gating when it is 1, the mmu will auto gating itself

16.22 Application Notes

16.22.1 VOP path disabled configure flow

1. Keep IEP direct path disabled.
2. Configure all registers which are needed at any time.
3. Configure IEP_CONFIG_DONE.
4. Configure IEP_FRM_START.

16.22.2 VOP path enabled configure flow

1. Keep IEP direct path enabled.
2. Configure all IEP registers which are needed.
3. Configure VOP related registers which are needed.
4. Configure CONFIG_DONE register in VOP only.
5. Wait for frame start from VOP and IEP direct path.

16.22.3 VOP path turn on flow

1. Configure all IEP registers which are needed.
2. Configure VOP related registers which are needed.
3. Enable IEP direct path.
4. Enable VOP direct path.
5. Configure CONFIG_DONE register in VOP only.
6. Wait for frame start from VOP and IEP direct path.

16.22.4 VOP path turn off flow

1. Disable VOP direct path.
2. Disable IEP direct path, so IEP do not receive any other CONFIG_DONE and frame start from VOP immediately.
3. Configure CONFIG_DONE register in VOP.
4. Wait for frame start from VOP and IEP direct path, so VOP quit direct path mode completely.
5. Configure IEP registers which are needed at any time.
6. Configure IEP_CONFIG_DONE.
7. Configure IEP_FRM_START, IEP is working at write back mode now.

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Chapter 1 Video Digital Analog Converter (VDAC)

16.23 Overview

Video DAC PHY is a small-sized, 27~300MHz, 1-channel, 10bit, high-speed D/A converter optimized for video or graphic applications. This IP designed to support Component(Pr,Y,Pb),Composite(CVBS), and S-Video(Y,C) signal standards for "consumer quality".

16.23.1 Features

- 10-bit resolution
- Single channel
- Up to 300Msps throughput rate
- Programmable current output: 14.7mA~ 34.8mA with 64 adjustable steps
- Current consumption: 1mA @Iout = 14.7mA, 39mA @Iout = 34.8mA
- 57dBc SFDR @Iout = 14.7, fclk = 300MHz and fout = 5MHz;45dBc SFDR @Iout = 34.8, fclk = 300MHz and fout = 5MHz;
- Clock frequency : 27MHz to 300MHz
- Cable connection detection
- Build-in bandgap reference
- 1.8V supply for analog and 1.0V supply for digital

16.24 Block Diagram

The architecture is shown in the following figure.

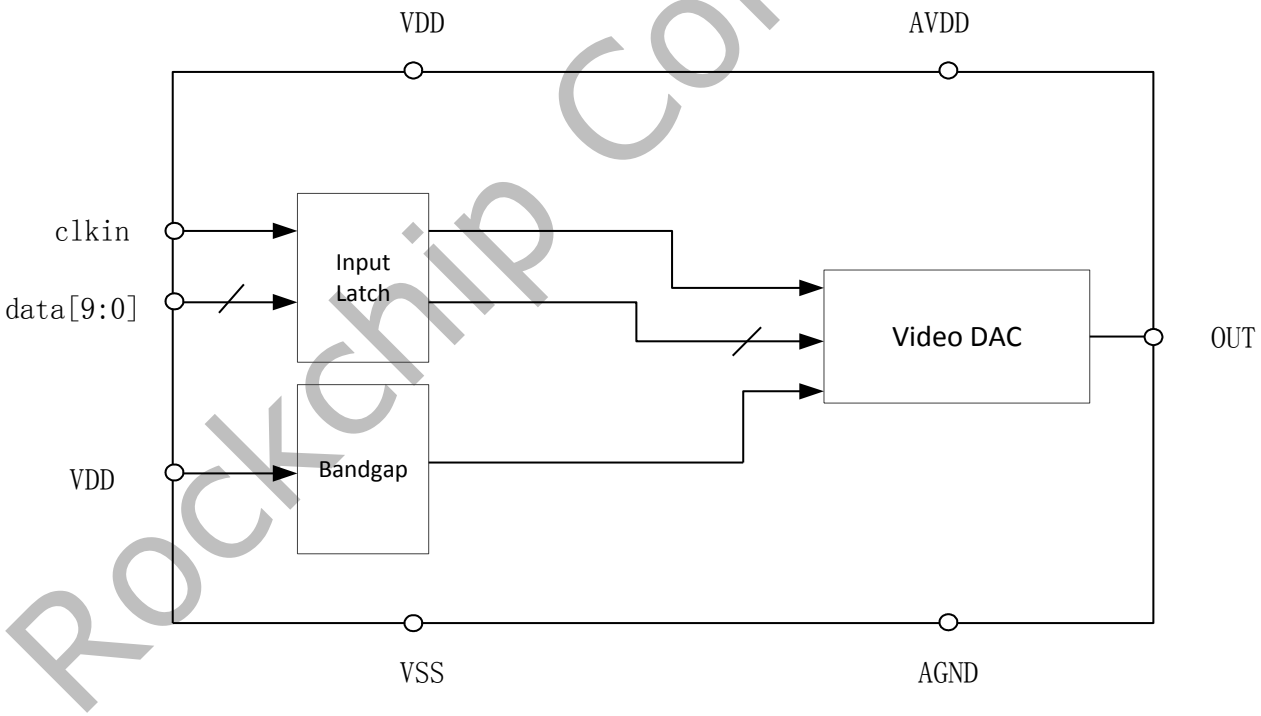

Fig. 0-1 VDAC Block Diagram

16.25 Function Description

16.25.1 System configure write timing for apb bus

The Write transfer starts with the address, write data, write signl all changing after the rising edge of the clock. The first clock cycle of the transfer is called the SETUP cycle. After the

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following clock edge the enable signal PENABLE is asserted and this indicates that ENABLE cycle is taking place. The address, data and control signals all remain valid throughout the ENABLE cycle. The transfer completes at the end of this cycle.

The enable signal, PENABLE, will be de-asserted at the end of the transfer. The select signal will also go LOW, unless the transfer is to be immediately follower by another transfer to the sample peripheral.

In order to reduce power consumption the address signal and the write signal will not change after a transfer until the next access occurs.

16.25.2 System configure read timing for apb bus

The timing of the address, write, select and strobe signals are all the same as for the write transfer. In the case of a read, the slave must provide the data during then ENABLE cycle. The data is sampled on the rising edge of clock at the end of the ENABLE cycle.

16.26 Register Description

16.26.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

16.26.2 Registers Summary

Name	Offset	Size	Reset Value	Description
VDAC_VDAC0	0x0000	W	0x000000c0	VDAC0
VDAC_VDAC1	0x0280	W	0x00000070	VDAC1
VDAC_VDAC2	0x0284	W	0x00000020	VDAC2
VDAC_VDAC3	0x0288	W	0x00000030	VDAC3

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

1.4.3 Detail Register Description

VDAC_VDAC0
Address: Operational Base + offset (0x0000)
VDAC0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	RST_ANA soft analog reset_n, low reset soft analog reset_n, low reset
6	RW	0x1	RST_DIG soft digital reset_n, low reset soft digital reset_n, low reset
5:0	RO	0x0	reserved

VDAC_VDAC1
Address: Operational Base + offset (0x0280)
VDAC1

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x7	CUR_REF select typical current reference select typical current reference
3:2	RO	0x0	reserved
1	RW	0x0	DR_PWR_DOWN vdac driver power down vdac driver power down 1: power down 0: power on
0	RW	0x0	BG_PWR_DOWN vdac band gap power down vdac band gap power down 1: power down 0: power on

VDAC_VDAC2
Address: Operational Base + offset (0x0284)
VDAC2

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x20	CUR_CTR output current control for DAC output current control for DAC tvdac_sw[5:0]

VDAC_VDAC3
Address: Operational Base + offset (0x0288)
VDAC3

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x1	CAB_EN Enable cable connection detection for DAC Enable cable connection detection for DAC 1: enable 0: disable
4	RW	0x1	CAB_REF reference voltage for cable disconnection detection of DAC reference voltage for cable disconnection detection of DAC 0: select 500mV 1: select 800mV
3:1	RO	0x0	reserved
0	RW	0x0	CAB_FLAG status output for DAC cable connection detection (1 means cable disconnection)

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16.27 Application Notes

16.27.1 CABLE DETECTION

The DAC channel contains a cable detection circuit to detect the cable plug condition. For typical application, cable with 75Ω characteristic impedance is used and DAC output is terminated by 75Ω double termination. In such case, a 75Ω source termination resistance is connected to ground at DAC output end. The 75Ω source termination resistance combined with 75Ω load termination resistance results in an equivalent load resistance of 37.5Ω.

Therefore, the equivalent load resistance for DAC output is 37.5Ω when cable is connected. It becomes 75Ω when cable is not connected. Compared to the case cable is connected, DAC output level will be twice in the case that cable is not connected with identical output current.

To start cable detection, controller should enable this function (controlled by register tvdac_dispdet_en) and set the 10-bit input data for a DAC channel to be middle level. Then controller should select a proper reference voltage(controlled by register tvdac_sw), which will be compared with DAC output level to judge whether cable is connected or not. The reference voltage selection is shown in following table.

Tvdac_sw	Tvdac_dispdet_sel	Reference voltage
6'b000000~6'b011111	1'b0	500mV
6'b100000~6'b111111	1'b1	800mV

If DAC output level is larger than the reference voltage, the cable detection flag signal(tvdac_dispdet) will be high and it means cable is disconnected. Otherwise, the cable detection flag signal will be low and it means cable is connected.

Tvdac_dispdet	1	Cable is connected
	0	Cable is disconnected

16.27.2 TYPICAL CONFIGURATION

The typical configuration is shown in following figure. DAC output is connected through 75Ω cable with 75Ω double termination.

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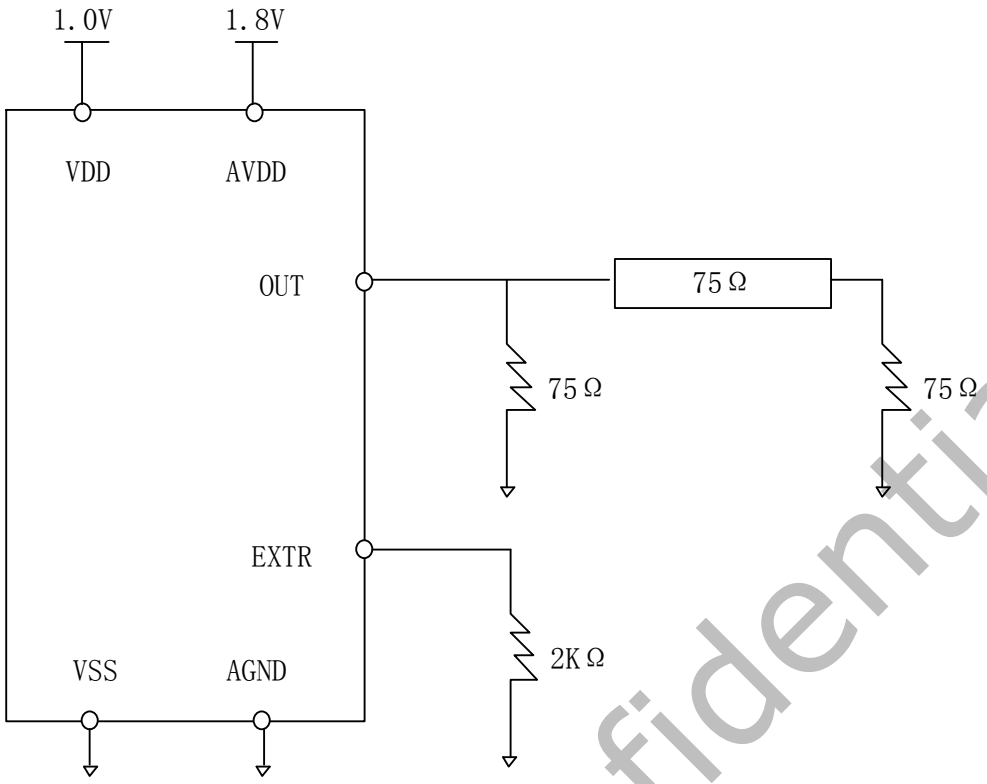

Fig. 0-2 VDAC Block Diagram

Analog supply AVDD should be connected to 1.8V power with decoupling. The digital supply VDD should be connected to digital core.

If external resistor is selected to produce reference current, EXTR should connect a 2KΩ resistor to ground.

Video DAC is suggested to placed close to the connector, in order to reduce signal noise and reflection due to impedance mismatch.

The DAC outputs are suggested to connect a 75Ω source termination resistance to ground. The termination resistors should be placed close to video DAC outputs to minimize reflection.

16.27.3 INSTRUCTION TO BRING UP VDAC

The following is a step by step instruction for bringing up the VDAC to your system, we use APB bus to configure VDAC.

Step1. Turn on entire system.

Step2. Configure 0xb3(data) to 0x280(address) to disable VDAC.

Step3. Configure 0x39(data) to 0x284(address) for current control.

Step4. Configure VOP.

Step5. Configure 0xb0(data) to 0x280(address) to enable VDAC and for typical current reference.

Step6. Now, TVDAC is ready to go. Start your test.

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Chapter 20 DDR Converser of Frequency (DCF)

17.1 Overview

DDR Converser of Frequency (DCF) is used to implement DDR frequency conversion without the participation of CPU. DCF is connected to SOC through an AXI Master and an APB Slave, with a DMAC and instruction buffer inside. Beforehand, software puts into SRAM formulary pieces of instructions which consist of all the register-related configurations in the process of frequency conversion. When started-up, DCF automatically reads instructions to the inside buffer by means of DMA, and implement register configuration after analysis. A done signal will be provided for CPU after all the instructions are analyzed.
DCF supports the following features:

- AMBA 32-bit AXI compliant
- Support DMA operation from SRAM to buffer
  - Support burst8 only
  - LLP not supported
- Support 2 internal instruction registers : r0 & r1
- Support following software instructions:
  - IDL (delay)
  - LDR (read register)
  - STR (write register)
  - ISB (write with flush)
  - Bitwise AND
  - Bitwise OR
  - Bitwise INV
  - LSR (left shift)
  - RSR (right shift)
  - ADD
  - SUB
  - POLLEQ (poll a register until the value match)
  - POLLNEQ (poll a register until the value dismatch)
  - CMPEQ (compare if equal)
  - CMPNEQ (compare if unequal)
  - BL (jump upwards or downwards, 8bytes align)
- Support following program functions through combination of basic instructions
  - If-else
  - while
  - loop
  - for

17.2 Block Diagram

DCF comprises with:

- An AXI Master Interface
- An APB Slave Interface
- A DMA controller
- An Instruction Buffer
- Instuction analyzer

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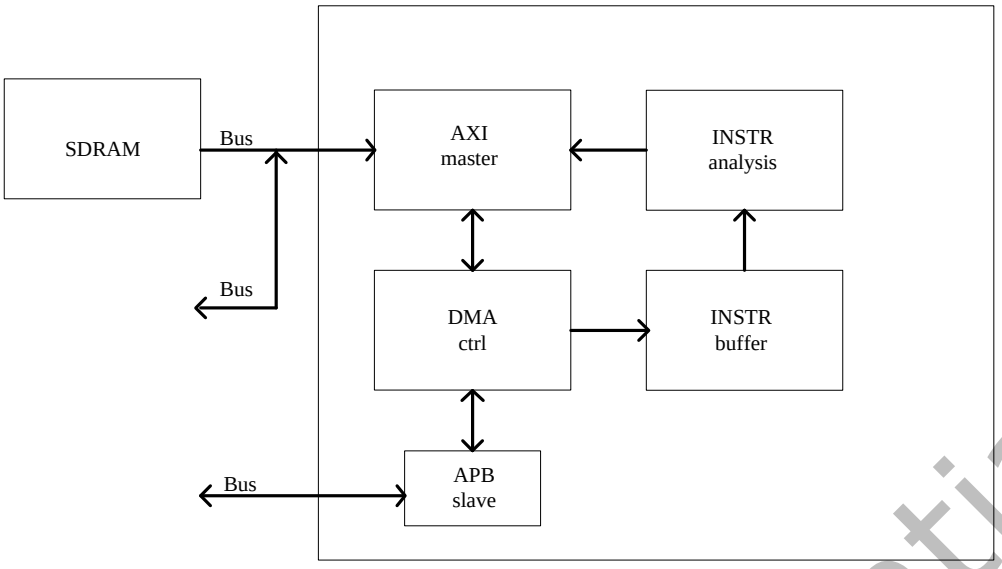

Fig.20-1DCF Block Diagram

17.3 Register Description

17.3.1 Registers Summary

Name	Offset	Size	Reset Value	Description
DCF_CTRL	0x0000	W	0x00000000	control Register
DCF_START_ADDR	0x0008	W	0x00000000	instruction start_addr
DCF_ISR	0x000c	W	0x00000000	interrupt Status Register
DCF_TIMEOUT_CYC	0x0014	W	0xffffffff	instruction timeout cycle
DCF_CURR_R0	0x0020	W	0x00000000	current internal r0 value
DCF_CURR_R1	0x0024	W	0x00000000	current internal r1 value
DCF_CMD_CNT	0x0028	W	0x00000000	current cmd_counter value
DCF_LAST_ADDR1	0x0030	W	0x00000000	last 1 instruction address
DCF_LAST_ADDR2	0x0034	W	0x00000000	last 2 instruction address
DCF_LAST_ADDR3	0x0038	W	0x00000000	last 3 instruction address
DCF_LAST_ADDR4	0x003c	W	0x00000000	last 4 instruction address

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

17.3.2 Detail Register Description

DCF_CTRL

Address: Operational Base + offset (0x0000)

control Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	vop_hw_en DCF hardware trigger enable 1'b1 : enable 1'b0 : disable
1	R/W SC	0x0	timeout_en DCF timeout enable 1: enable 0: disable

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Bit	Attr	Reset Value	Description
0	R/W SC	0x0	start DCF enable This bit will be auto cleared when DCF is done. 1: start DCF 0: stop DCF

DCF_START_ADDR
Address: Operational Base + offset (0x0008)
instruction start_addr

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr DMA transaction starting address

DCF_ISR
Address: Operational Base + offset (0x000c)
interrupt Status Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	timeout_error timeout_error 1'b1 : DCF timeout
1	W1 C	0x0	dcf_error DCF error 1: error response during DDR change frequency ; 0: no error response during DDR change frequency ;
0	W1 C	0x0	dcf_done DCF done 1: DDR change frequency completed; 0: DDR change frequency not completed;

DCF_TIMEOUT_CYC
Address: Operational Base + offset (0x0014)
instruction timeout cycle

Bit	Attr	Reset Value	Description
31:0	RW	0xffffffff	timeout dcf instruction timeout clock cycle number calculated by axi clock

DCF_CURR_R0
Address: Operational Base + offset (0x0020)
current internal r0 value

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	curr_r0 current r0 value r0 value is sampled when dcf_done or dcf_error occurs

DCF_CURR_R1
Address: Operational Base + offset (0x0024)
current internal r1 value

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	curr_r1 current r1 value r1 value is sampled when dcf_done or dcf_error occurs

DCF_CMD_CNT
Address: Operational Base + offset (0x0028)
current cmd_counter value

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	cmd_counter current cmd_counter value cmd_counter value is sampled when dcf_done or dcf_error occurs it records the number of pieces of instructions among DCF

DCF_LAST_ADDR1
Address: Operational Base + offset (0x0030)
last 1 instruction address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	last_addr1 last instruction address value is sampled when dcf_done or dcf_error occurs it records the last 1 instruction address

DCF_LAST_ADDR2
Address: Operational Base + offset (0x0034)
last 2 instruction address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	last_addr2 last 2 instruction address value is sampled when dcf_done or dcf_error occurs it records the last 2 instruction address

DCF_LAST_ADDR3
Address: Operational Base + offset (0x0038)
last 3 instruction address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	last_addr3 last 3 instruction address value is sampled when dcf_done or dcf_error occurs it records the last 3 instruction address

DCF_LAST_ADDR4
Address: Operational Base + offset (0x003c)
last 4 instruction address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	last_addr4 last 4 instruction address value is sampled when dcf_done or dcf_error occurs it records the last 4 instruction address

17.4 Application Notes

17.4.1 DCF work flow

1、 software needs to open up a separate space in SRAM and loads a series of instructions in advance. The instruction should consists of:
   a、 config msch and cpu idle
   b、 config Memory Controller to move DDR into Low-power State
   c、 reset DDRPHY if needed
   d、 config Clock frequency, wait for PLL lock
   e、 config all the Timing-relative registers
   f、 de-assert DDRPHY reset if needed
   g、 initialize DDRPHY, do calibration
   h、 config Memory Controller to move DDR into Access State
2、 software config DCF, notifying the start_addr, and then start DCF
3、 DCF will transfer instruction data through DMA from SRAM to an internal buffer
4、 In the meanwhile, An instruction analysis module will read these instructions from buffer, and transform them to corresponding bus-relative operations, including Write&Read registers, delay some clock cycles, arithmetic operation and so on.
5、 By Axi master, DCF will config uPCTL、DDRPHY、CRU module to implement the procedure of DDR frequency conversion
6、 DCF will recognize the last command, and generate a dcf_done interrupt for CPU

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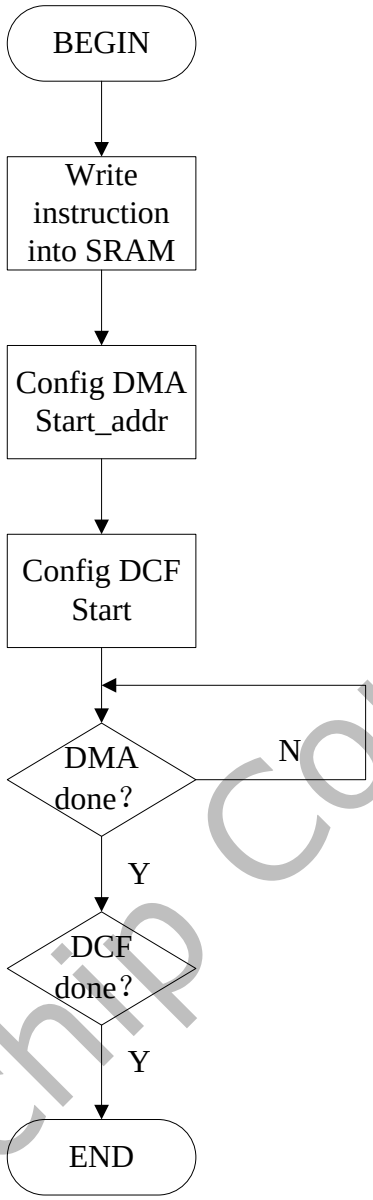

Fig.207-2DCF work flow

17.4.2 Instruction format

One piece of instruction, which is 64bit, should consist of the information of :

1、 Address
2、 Data
3、 Command

63:56	55:34	33:32	31:0
cmd[7:0]	addr[23:2]	(r1,r0)	data

The overall principle of instruction information is:

1、 addr[31:24] is reserved for 8bit command, which represents the corresponding operation
2、 addr[1:0] is used to indicate operation of r0 or r1

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3、 addr[23:2] is the real bus address. If 0, it means no bus operation ; if not 0, it means a combination of 2 instructions with a bus operation ahead and an arithmetic c operation followed in order to improve efficiency.

For example, let us analyze the instruction: 01620002_00000003

1、 command is 1,which represents an bitwise AND operation

2、 address is 0xff620000, represents a bus-read operation

3、 r1 is indicated, represents that the middle result is stored into internal register r1.

4、 Data is 0x00000003, represents that the operation value is 0x3

So, this instruction will do following operations:

1、 LDR #0xff620000, r1 ; //read register 0xff620000, and store value into r1

2、 AND r1, 0x00000003 ; //r1 is bitwise AND with 0x3, and re-store the result.

The following table lists all the supported command

INSTR	cmd[7:0]	addr[23:2]	R1	R0	Data[31:0]	
IDLE	8’h00	NA	NA	NA	#data	IDL #data
AND	8’h01	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1
			0	1	#data	ldr #addr r0 ; AND r0 #data
			1	0	#data	ldr #addr r1 ; AND r1 #data
			1	1	NA	ldr #addr r1 ; AND r1 r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	AND r0 #data
			1	0	#data	AND r1 #data
			1	1	NA	AND r1 r0
OR	8’h02	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1
			0	1	#data	ldr #addr r0 ; OR r0 #data
			1	0	#data	ldr #addr r1 ; OR r1 #data
			1	1	NA	ldr #addr r1 ; OR r1 r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	OR r0 #data
			1	0	#data	OR r1 #data
			1	1	NA	OR r1 r0
INV	8’h03	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1

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			0	1	#data	ldr #addr r0 ; XOR r0 ^#data
			1	0	#data	ldr #addr r1 ; XOR r1 ^#data
			1	1	#data	ldr #addr r1 ; XOR r1 ^r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	NA	INV r0
			1	0	NA	INV R1
			1	1	NA	SWP r0 r1
LSR	8’h04	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1
			0	1	#data	ldr #addr r0 ; LSR r0 #data
			1	0	#data	ldr #addr r1 ; LSR r1 #data
			1	1	NA	ldr #addr r1 ; LSR r1 r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	LSR r0 #data
			1	0	#data	LSR r1 #data
			1	1	NA	LSR r1 r0
RSR	8’h05	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1
			0	1	#data	ldr #addr r0 ; RSR r0 #data
			1	0	#data	ldr #addr r1 ; RSR r1 #data
			1	1	NA	ldr #addr r1 ; RSR r1 r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	RSR r0 #data
			1	0	#data	RSR r1 #data
			1	1	NA	RSR r1 r0
CMPEQ	8’h06	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1

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			0	1	#data	ldr #addr r0 ; CMPEQ r0 #data,flag
			1	0	#data	ldr #addr r0 ; CMPEQ r1 #data,flag
			1	1	NA	ldr #addr r0 ; CMPEQ r1 r0,flag
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	CMPEQ r0 #data, flag
			1	0	#data	CMPEQ r1 #data, flag
			1	1	NA	CMPEQ r1 r0, flag
CMPNE	8’h07	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1
			0	1	#data	ldr #addr r0 ; CMPNE r0 #data,flag
			1	0	#data	ldr #addr r1 ; CMPNE r1 #data,flag
			1	1	NA	ldr #addr r1 ; CMPNE r1 r0,flag
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	CMPNE r0 #data, flag
			1	0	#data	CMPNE r1 #data, flag
			1	1	NA	CMPNE r1 r0, flag
ADD	8’h08	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1
			0	1	#data	ldr #addr r0 ; ADD r0 #data
			1	0	#data	ldr #addr r0 ; ADD r1 #data
			1	1	NA	ldr #addr r0 ; ADD r1 r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	ADD r0 #data
			1	0	#data	ADD r1 #data
			1	1	NA	ADD r1 r0
SUB	8’h09	#addr	0	0	NA	ldr #addr r0 ; ldr #addr r1

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			0	1	#data	ldr #addr r0 ; SUB r0 #data
			1	0	#data	ldr #addr r0 ; SUB r1 #data
			1	1	NA	ldr #addr r0 ; SUB r1 r0
		All 0	0	0	NA	mov r0 r0 ; mov r1 r1
			0	1	#data	SUB r0 #data
			1	0	#data	SUB r1 #data
			1	1	NA	SUB r1 r0
STR	8’h0a	#addr	0	0	#data	STR #addr #data
			0	1	NA	STR #ADDR r0
			1	0	NA	STR #ADDR r1
			1	1	#data	STR #addr #data
ISB	8’h0b	#addr	0	0	#data	STR #addr #data
			0	1	NA	STR #ADDR r0
			1	0	NA	STR #ADDR r1
			1	1	#data	STR #addr #data
POLEQ	8’h0c	NA	0	1	#data	poll r0=#data,repeat last command
			1	0	#data	poll r1=#data,repeat last command
			1	1	NA	poll r1=r0,repeat last command
POLNEQ	8’h0d	NA	0	1	#data	poll r0!=#data,repeat last command
			1	0	#data	poll r1!=#data,repeat last command
			1	1	NA	poll r1!=r0,repeat last command
BL_U	8’h0e	NA	NA	NA	#data	brr #data,?flag (upwards)
		NA	0	1	NA	brr r0, ?flag
		NA	1	0	NA	brr r1, ?flag

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BL_D	8’h0f	NA	NA	NA	#data	brr #data,?flag (downwards)
		NA	0	1	NA	brr r0, ?flag
		NA	1	0	NA	brr r1, ?flag
DMA_S	8’h10	NA	NA	NA	#data	set dma_start_addr = #data
DMA_D	8’h11	NA	NA	NA	#data	set dma_end_addr = #data
DMA_DO	8’h12	NA	NA	NA	#data	set dma_length = #data (byte) dma_start
END	8’hed	NA	NA	NA	NA	End of instruction

17.4.3 Hardware trigger flow

When DCF_CTRL.vop_hw_en is enabled, DCF can be triggered by any of the followed three sources ： dma_finish 、vop_standby、vop_clkgate_en。

DCF is edge sensitive for dma_finish signal, and level sensitive for vop_standby and vop_clkgate_en signal.

When DCF is working， a dcf_idle is driven to low to indicate vop not to exit vop_standby status. And when DCF is not working, dcf_idle is driven to high for SOC and VOP to inquire.

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Chapter 2 Power Management Unit (PMU)

18.1 Overview

In order to meet low power requirements, a power management unit (PMU) is designed for controlling power resources in RK3228H. The RK3228H PMU is dedicated for managing the power of the whole chip.

18.1.1 Features

- Support DDR self-refresh
- Support DDR retention
- Support CPU2/CPU3 power down/up by software
- Support CPU2/CPU3 auto-power management
- Support L2 flush interface

18.2 Block Diagram

18.2.1 Voltage partition

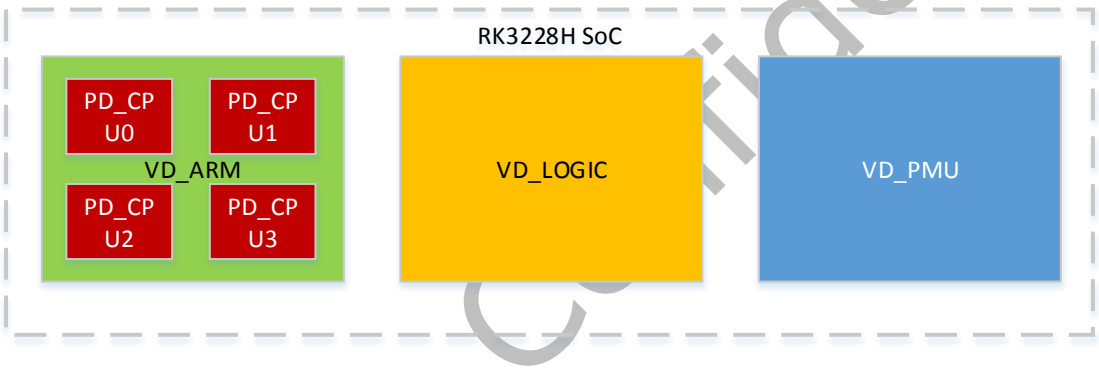

Fig. 4-1RK3228H Power Domain Partition

The above diagram describes voltage domain partition, notice that there are no power domains inside RK3228H except PD_CPU2 and PD_CPU3. PD_CPU2 and PD_CPU3 have MTCMOS inside, and for the blocks with name pd_xxx are not real power domains.

Table 4-1RK3228H Power Domain and Voltage Domain Summary

Voltage Domain	Blocks (not real power domain)	Description
VD_ARM	PD_CPU0	CPU Core 0 with NEON and FPU, DAP-lite
	PD_CPU1	CPU Core 1 with NEON and FPU, DAP-lite
	PD_CPU2	CPU Core 2 with NEON and FPU, DAP-lite
	PD_CPU3	CPU Core 3 with NEON and FPU, DAP-lite
	PD_SCU	DAP Lite, SCU and 256KB L2
VD_LOGIC	PD_GPU	Mali-450
	PD_RKVENC	Video encoder
	PD_RKVDEC	Video decoder, NANDC, EMMC, SDIO, SDMMC, GMAC2PHY, GMAC2IO
	PD_VIO	ISP, IEP, VOP, RGA, CIF0/1/2/3, TV decoder, HDMI host, DSI host
	PD_PERI	Peri NIU
	PD_DDR	UPCTL, MEM scheduler, DDR mon, DDR GRF

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	PD_BUS & TOP	CRYPTO, SPDIF, I2S0/1/2, PDM, TSP, SGRF, SEFUSE, SOTP, SRAM(36KB), ROM(20KB), DDRPHY, ACODEC, VDAC, HDMI PHY, PLLx4, GRF, I2Cx4, WDT, CRU, TIMERx6, EFUSE1024, SCR, TSADC, PMU, SARADC, SPI, PWMx4, GPIOx4, UARTx3, DFI monitor, TSADC CTL, Stimerx2, DCF, NSEFUSE, NSOTP
	PD_VPU	VPU
VD_PMU	PD_PMU	OSC, Pmux, and PAD ring

18.2.2 PMU block diagram

The following figure is the PMU block diagram. The PMU includes the 3 following sections:

- APB interface and register, which can accept the system configuration
- Low Power State Control, which generate low power control signals.
- Power Switch Control, which control all power domain switch

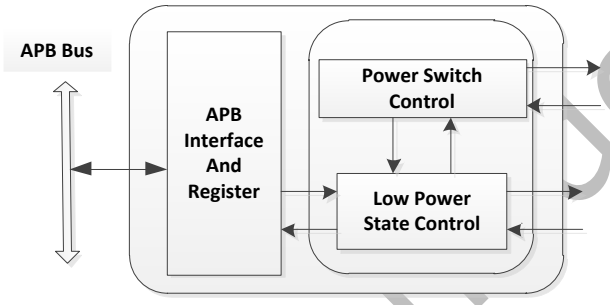

Fig. 4-2 PMU Bock Diagram

18.3 Function Description

First of all, we define two operation modes of PMU, normal mode and low power mode.
When operating at normal mode, that means software can manage power sources directly by accessing PMU register.
For example, Cortex-A53 CPU can write PMU_PWRDN_CON register to determine that power off/on which power domain independently.
When operating at low power mode, software manages power sources indirectly through FSM (Finite States Machine) in PMU and those settings always not take effect immediately.
That means software also can configure PMU registers to power down/up some power resources, but these setting will not be executed immediately after configuration. They will delay to execute after FSM running in particular phase.
To entering low power mode, after setting some power configurations, the PMU_POWER_MODE[0] bit must be set 1 to enable PMU FSM. Then Cortex-A53 CPU needs to execute a WFI command to perform ready signal. After PMU detects all Cortex-A53 CPUs in WFI status, then the FSM will be fetched. And the specific power sources will be controlled during specific status in FSM. So the low power mode is a "delay affect" way to handle power sources inside the RK3228H chip.

18.4 Register Description

18.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
PMU_PMU_WAKEUP_CFG0	0x0000	W	0x00000000	
PMU_PMU_PWRDN_CON	0x000c	W	0x00000000	

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Name	Offset	Size	Reset Value	Description
PMU_PMU_PWRDN_ST	0x0010	W	0x00000000	
PMU_PMU_PWRMODE_COMMON_CON	0x0018	W	0x00000000	
PMU_PMU_SFT_CON	0x001c	W	0x00000000	
PMU_PMU_INT_CON	0x0020	W	0x00000000	
PMU_PMU_INT_ST	0x0024	W	0x00000000	
PMU_PMU_POWER_ST	0x0044	W	0x00000000	
PMU_PMU_CPU0APM_CON	0x0080	W	0x00000000	
PMU_PMU_CPU1APM_CON	0x0084	W	0x00000000	
PMU_PMU_CPU2APM_CON	0x0088	W	0x00000000	
PMU_PMU_CPU3APM_CON	0x008c	W	0x00000000	
PMU_PMU_SYS_REG0	0x00a0	W	0x00000000	
PMU_PMU_SYS_REG1	0x00a4	W	0x00000000	
PMU_PMU_SYS_REG2	0x00a8	W	0x00000000	
PMU_PMU_SYS_REG3	0x00ac	W	0x00000000	

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

18.4.2 Detail Register Description

PMU_PMU_WAKEUP_CFG0
Address: Operational Base + offset (0x0000)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	wakeup_int_cluster_en interrupt wakeup enable 0: disable 1: enable

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PMU_PMU_PWRDN_CON
Address: Operational Base + offset (0x000c)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	pd_a53_3_pwrdwn_en a53 cpu3 power down enable 0: disable 1: enable
2	RW	0x0	pd_a53_2_pwrdwn_en a53 cpu2 power down enable 0: disable 1: enable
1	RW	0x0	pd_a53_1_pwrdwn_en a53 cpu1 power down enable 0: disable 1: enable
0	RW	0x0	pd_a53_0_pwrdwn_en a53 cpu0 power down enable 0: disable 1: enable

PMU_PMU_PWRDN_ST
Address: Operational Base + offset (0x0010)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	pd_a53_3_pwr_stat CPU3 power status 0: power up 1: power down

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Bit	Attr	Reset Value	Description
2	RW	0x0	pd_a53_2_pwr_stat CPU2 power status 0: power up 1: power down
1	RW	0x0	pd_a53_1_pwr_stat CPU1 power status 0: power up 1: power down
0	RW	0x0	pd_a53_0_pwr_stat CPU0 power status 0: power up 1: power down

PMU_PMU_PWRMODE_COMMON_CON
Address: Operational Base + offset (0x0018)

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	ddrio_ret_en ddrio retention enable 0: disable 1: enable
7	RW	0x0	ddrio_ret_de_req ddrio retention de request 0: disable 1: enable

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Bit	Attr	Reset Value	Description
6	RW	0x0	l2_idle_en wait for L2 idle enable 0: disable 1: enable
5	RW	0x0	l2_flush_en flush L2 during power mode 0: disable 1: enable
4	RW	0x0	wait_wakeup_begin_cfg pmu start to observe for wakeup signals 0: disable 1: enable
3	RW	0x0	cpu0_pd_en power down cpu0 enable 0: disable 1: enable
2	RW	0x0	global_int_disable_cfg global interrupt disable configure 0: enable interrupt 1: disable interrupt
1	RW	0x0	sref_enter_en DDR enter self-refresh enable when in power mode 0: disable 1: enable
0	RW	0x0	power_mode_en enable FSM 0: disable 1: enable

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PMU_PMU_SFT_CON
Address: Operational Base + offset (0x001c)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	ddr_io_ret_cfg software request ddr retention 0: disable 1: enable
1	RW	0x0	l2flushreq_req software request l2 flush 0: disable 1: enable
0	RW	0x0	upctl_c_sysreq_cfg software request ddr self-refresh 0: disable 1: enalbe

PMU_PMU_INT_CON
Address: Operational Base + offset (0x0020)

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18	RW	0x0	a53_l3_pwr_switch_int_en a53 CPU3 power switch interrupt enable 0: disable 1: enable
17	RW	0x0	a53_l2_pwr_switch_int_en a53 CPU2 power switch interrupt enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
16	RW	0x0	a53_l1_pwr_switch_int_en a53 CPU1 power switch interrupt enable 0: disable 1: enable
15	RW	0x0	a53_l0_pwr_switch_int_en a53 CPU0 power switch interrupt enable 0: disable 1: enable
14:5	RO	0x0	reserved
4	RW	0x0	wakeup_int_en interrupt wakeup interrupt enable 0: disable 1: enable
3:2	RO	0x0	reserved
1	RW	0x0	pwrmode_wakeup_int_en power mode wakeup interrupt enable 0: disable 1: enable
0	RW	0x0	pmu_int_en pmu interrupt global enable 0: disable 1: enable

PMU_PMU_INT_ST
Address: Operational Base + offset (0x0024)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	a53_l3_pwr_switch_status a53 cpu3 power switch interrupt status
4	RW	0x0	a53_l2_pwr_switch_status a53 cpu2 power switch interrupt status
3	RW	0x0	a53_l1_pwr_switch_status a53 cpu1 power switch interrupt status
2	RW	0x0	a53_l0_pwr_switch_status a53 cpu0 power switch interrupt status
1	RW	0x0	wakeup_int_status interrupt wakeup status
0	RW	0x0	pwrmode_wakeup_status power mode wakeup status

PMU_PMU_POWER_ST
Address: Operational Base + offset (0x0044)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	pwr_status pmu power FSM value

PMU_PMU_CPU0APM_CON
Address: Operational Base + offset (0x0080)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	cpu0_sft_wakeup cpu0 software wakeup enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
2	RW	0x0	global_int_disable_0_cfg disable interrupt to cpu0 0: enable interrupt 1: disable interrupt
1	RW	0x0	cpu0_int_wakeup_en cpu0 interrupt wakeup enable 0: disable 1: enable
0	RW	0x0	cpu0_wfi_pwrdn_en cpu0 WFI power down enable 0: disable 1: enable

PMU_PMU_CPU1APM_CON
Address: Operational Base + offset (0x0084)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	cpu1_sft_wakeup cpu1 software wakeup enable 0: disable 1: enable
2	RW	0x0	global_int_disable_1_cfg disable interrupt to cpu1 0: enable interrupt 1: disable interrupt

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Bit	Attr	Reset Value	Description
1	RW	0x0	cpu1_int_wakeup_en cpu1 interrupt wakeup enable 0: disable 1: enable
0	RW	0x0	cpu1_wfi_pwrdn_en cpu1 WFI power down enable 0: disable 1: enable

PMU_PMU_CPU2APM_CON
Address: Operational Base + offset (0x0088)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	cpu2_sft_wakeup cpu2 software wakeup enable 0: disable 1: enable
2	RW	0x0	global_int_disable_2_cfg disable interrupt to cpu2 0: enable interrupt 1: disable interrupt
1	RW	0x0	cpu2_int_wakeup_en cpu2 interrupt wakeup enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
0	RW	0x0	cpu2_wfi_pwrdn_en cpu2 WFI power down enable 0: disable 1: enable

PMU_PMU_CPU3APM_CON
Address: Operational Base + offset (0x008c)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	cpu3_sft_wakeup cpu3 software wakeup enable 0: disable 1: enable
2	RW	0x0	global_int_disable_3_cfg disable interrupt to cpu3 0: enable interrupt 1: disable interrupt
1	RW	0x0	cpu3_int_wakeup_en cpu3 interrupt wakeup enable 0: disable 1: enable
0	RW	0x0	cpu3_wfi_pwrdn_en cpu3 WFI power down enable 0: disable 1: enable

PMU_PMU_SYS_REG0
Address: Operational Base + offset (0x00a0)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg0 system register 0

PMU_PMU_SYS_REG1
Address: Operational Base + offset (0x00a4)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg1 system register 1

PMU_PMU_SYS_REG2
Address: Operational Base + offset (0x00a8)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg2 system register 2

PMU_PMU_SYS_REG3
Address: Operational Base + offset (0x00ac)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pmu_sys_reg3 system register 3

18.5 Timing Diagram

18.5.1 Each domain power switch timing

The following figure is the each domain power down and power up timing.

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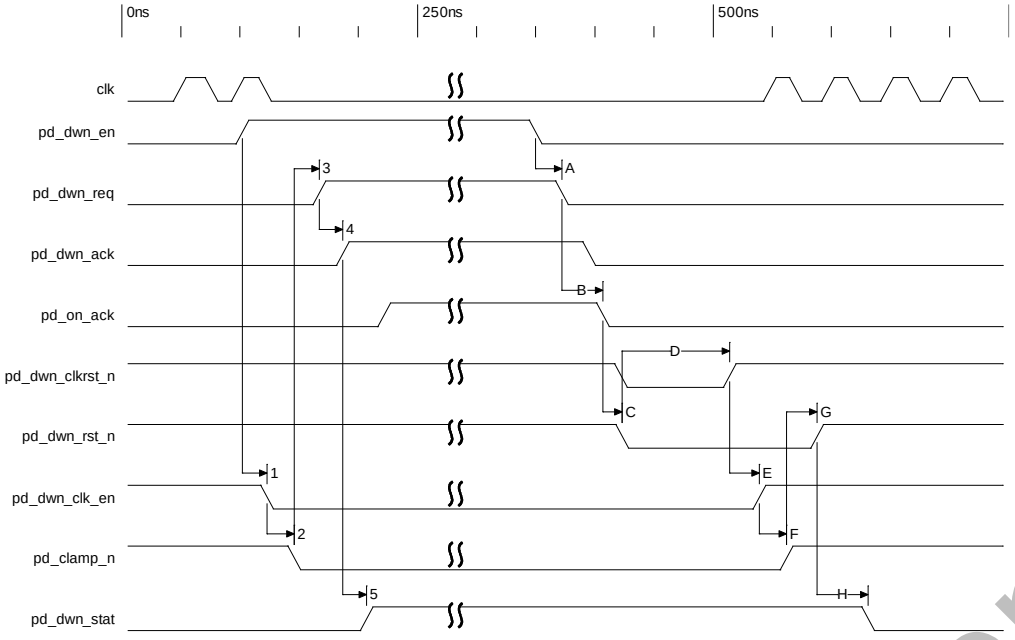

Fig. 4-5 Each Domain Power Switch Timing

18.5.2 External wakeup PAD timing

The PMU supports a lot of external wakeup sources, such as SD/MMDC, USBDEV, SIM detect wakeup, GPIO0 wakeup source and so on. All these external wakeup sources must meet the timing requirement (at least 200us) when the wakeup event is asserted. The following figure gives the timing information.

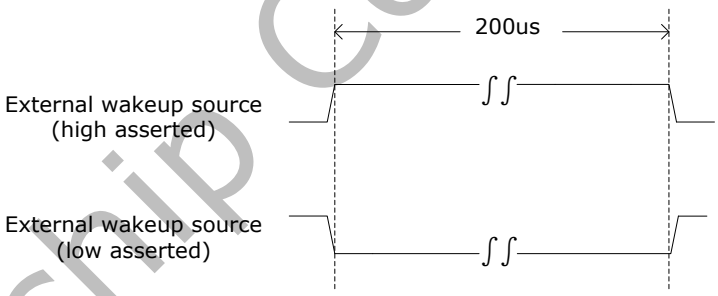

Fig. 4-6 External Wakeup Source PAD Timing

18.6 Application Note

18.6.1 Low power mode

PMU can work in the Low power mode by setting bit[0] of PMU_PWRMODE_CON register. After setting this bit and all CPU cores enters WFI states, PMU low power FSM will start to run. In the low power mode, PMU will manage power resources by hardware, such as power on/off the specified power domain, send idle request to specified power domain, shut down/up PLL and so on. All of above are configurable by setting corresponding registers.
ALL FSM power states could be monitored through IO. The following table describes all power states of PMU FSM.

Table 4-4 Low Power State

Num	STATES	Description
0	ST_NORMAL	Still in normal state
1	ST_CPU0_PWRDN	Hold CPU0 in reset status, not really

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		power down
2	ST_L2_FLUSH	Flush L2 by hardware
3	ST_L2_IDLE	Wait for L2 idle
4	ST_SREF_ENTER	Enter DDR self-refresh
5	ST_DDR_IO_RET	DDR IO retention
6	ST_WAIT_WAKEUP	Wait for wake up
7	ST_SREF_EXIT	Exit DDR self-refresh
8	ST_CPU0_PWRUP	De-assert reset for CPU0

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Chapter 21 TSP(Transport Stream Processing Module)

19.1 Overview

The Transport Stream Processing Module(TSP) is designed for processing Transport Stream Packets, including receiving TS packets, PID filtering, TS descrambling, De-multiplexing and TS outputting. Processed data are transferred to memory buffer which are continued to be processing by software.

TSP supports the following features:

- Supports 1 TS input channels
- Supports 4 TS Input Mode: sync/valid mode in the case of serial TS input; nosync/valid mode, sync/valid, sync/burst mode in the case of parallel TS input
- Supports 2 TS sources: demodulators and local memory
- Supports 1 Built-in PTIs(Programmable Transport Interface) to process TS simultaneously
- Supports 1 PVR(Personal Video Recording) output channel
- 1 built-in multi-channel DMA Controller
- Support DMA LLP transfer
- Each PTI supports
  - 64 PID filters
  - TS descrambling with 16 sets of Control Word under CSA v2.0 standard, up to 104Mbps
  - 16 PES/ES filters with PTS/DTS extraction and ES start code detection
  - 4/8 PCR extraction channels
  - 64 Section filters with CRC check, and three interrupt mode: stop per unit, full-stop, recycle mode with version number check
  - PID done and error interrupts for each channel
  - PCR/DTS/PTS extraction interrupt for each channel
- Support 32 bit AXI MMU.

19.2 Block Diagram

The TSP wrapper comprises of following components:

- TSP module (include: AHB slave, register block ,PTI ,DMAC, AHB master)
- AHB/AXI bridge
- 32bit AXI MMU

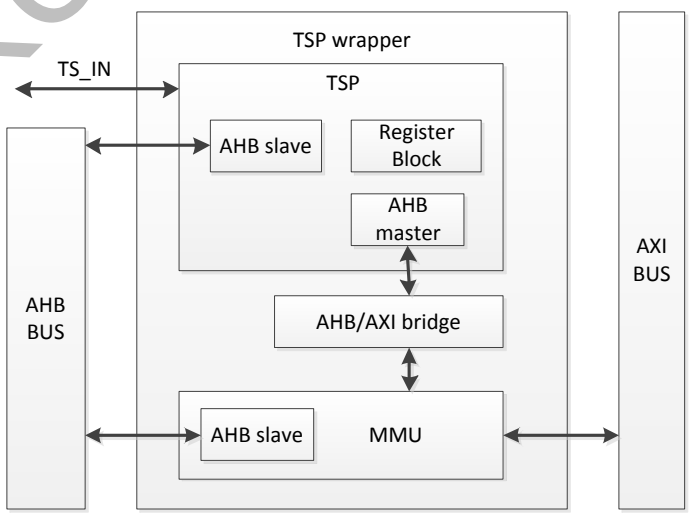

Fig. 1-1 TSP architecture

AHB Slave INTERFACE

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The host processor can get access to the TSP and MMU register block through AHB slave interface. The slave interface supports 32bit access.

Register block

All registers in the TSP are addressed at 32-bit boundaries to remain consistent with the AHB bus. Where the physical size of any register is less than 32-bits wide, the upper unused bits of the 32-bit boundary are reserved. Writing to these bits has no effect; reading from these bits returns 0.

PTI

Most of the TS processing are dealt with PTI. TS packets are re-synchronized, filtered, descrambled and demultiplexing, and the processed packets are transferred to memory buffer to be processed further by software. The embedded TS in interface can receive TS packets by connecting to a compliant TS demodulator. TS stream stored in the local memory is another source to fed into PTI through by using LLP DMA mode.

DMAC

The DMAC performs all DMA transfers which get access to memory.

AHB/AXI bridge

Convert AHB master to AXI master.

MMU

Support AXI interface,4K page size and TLB pre-fetch. Data bus width is 32 bit.

19.3 Function Description

19.3.1 TS Stream of TS_IN Interface

TS_IN interface supports 4 input TS stream mode: sync/valid serial mode, sync/valid parallel mode, sync/burst parallel mode, nosync/valid parallel mode.

A.Sync/Valid Serial Mode

In this mode, TS_IN interface takes use of TSI_SYNC and TSI_VALID clocked with TSI_CLK signal to sample input serial TS packet data.

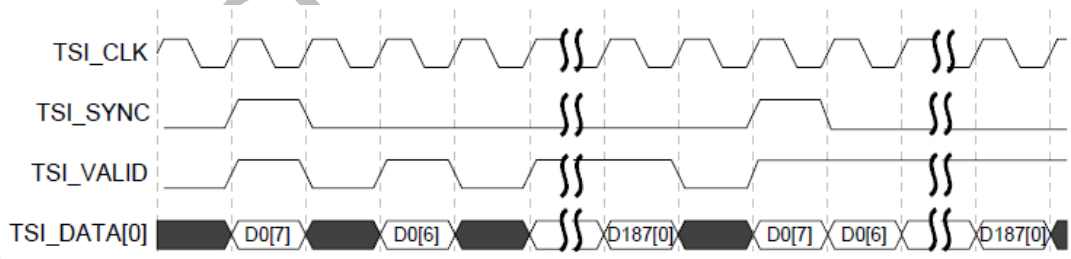

Fig. 1-2Sync/Valid Serial Mode with Msb-Lsb Bit Ordering

TSI_SYNC must be active high together with TSI_VALID when indicating the first valid bit of a TS packet, and TSI_VALID indicates the 188*8 valid bits of a TS packet. TSI supports both msb-lsb and lsb-msb bit ordering.

B. Sync/Valid Parallel Mode

In this mode, TS_IN interface takes use of TSI_SYNC and TSI_VALID clocked with TSI_CLK signal to sample input parallel TS packet data.

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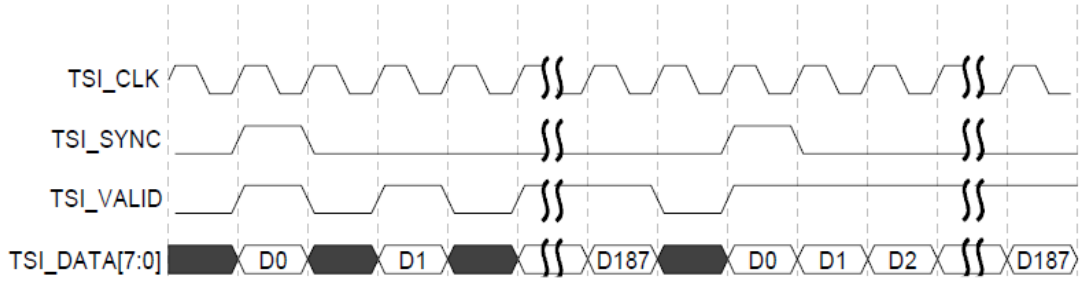

Fig. 1-3Sync/valid Parallel Mode

TSI_SYNC must be active high together with TSI_VALID when indicating the first valid byte of a TS packet, and TSI_VALID indicates the 188 valid byte of a TS packet.

C. Sync/Burst Parallel Mode

In this mode, TSI only takes use of TSI_SYNC to sample input parallel TS packet data.

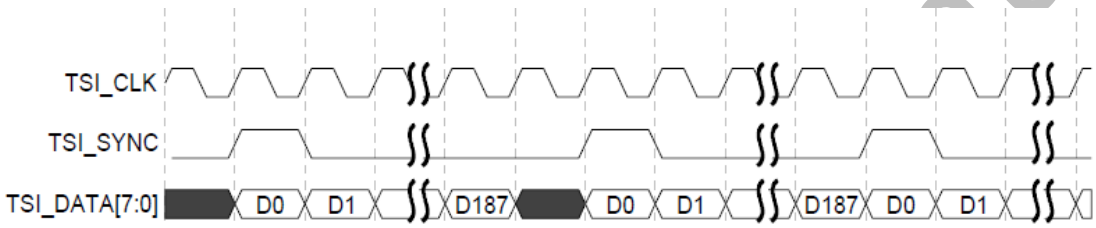

Fig. 1-4Sync/Burst Parallel Mode

When active high, TSI_SYNC implies the first valid byte of a TS packet and remaining 187 valid bytes of a TS packet are upcoming within the following successive 187 clock cycles.

D. Nosync/Valid Parallel Mode

In this mode, TSI only takes uses of TSI_VALID to sample input parallel TS packet data.

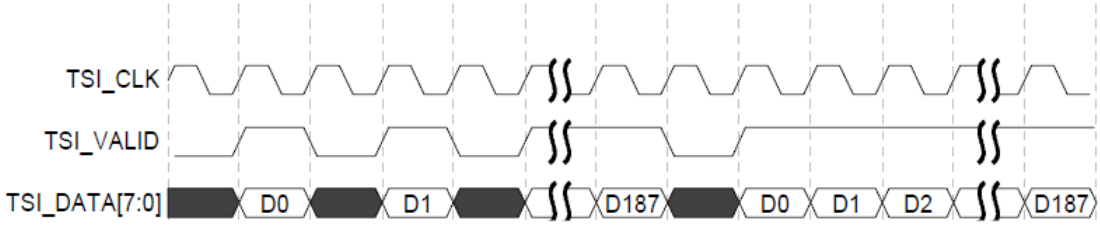

Fig. 1-5Nosync/Valid Parallel Mode

When active high, TSI_VALID implies a valid byte of a TS packet.

19.3.2 TS output of TS Out Interface

TS out interface transmit the TS data in two mode: serial mode and parallel mode. In the serial mode, the bit order can be lsb-msb or msb-lsb.

The TS_SYNC will be active high when indicating the header of the TS packets, and it only lasts for one cycle. TS_VALID will be active high when the output TS data is valid. The output data is 188 byte TS packet data.

TS out interface also stamp the TS output stream with new PCR value, making PCR adjustment. PCR is used to measure the transport rate.

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$$PCR(i) = PCR_base(i) \times 300 + PCR_ext(i)$$

where:

$$PCR_base(i) = ((system_clock_frequency \times t(i)) \, DIV \, 300) \, \% \, 2^{33}$$

$$PCR_ext(i) = ((system_clock_frequency \times t(i)) \, DIV \, 1) \, \% \, 300$$

$$transport_rate(i) = \frac{((i' - i'') \times system_clock_frequency)}{PCR(i') - PCR(i'')}$$

Where

i' is the index of the byte containing the last bit of the immediately following program_clock_reference_base field applicable to the program being decoded.

i is the is the index of any byte in the Transport Stream for i''< i < i'.

i''is the index of the byte containing the last bit of the most recent program_clock_reference_base fieldapplicable to the program being decoded.

System clock is 27Mhz.

19.3.3 Demux and descrambling

Each PTI has 64 PID channels to deal with demultiplexing and descrambling operation.

The PTI can descramble the TS Packets which are scrambled with CSA v2.0 standard. The TS packets can be scrambled either in TS level or PES level.

The demux module can do the section filtering, pes filtering and es filtering, or directly output TS packets.

19.4 Register Description

19.4.1 TSP Register Summary

Name	Offset	Size	Reset Value	Description
TSP_GCFG	0x0000	W	0x00000000	Global Configuration Register
TSP_PVR_CTRL	0x0004	W	0x00000000	PVR Control Register
TSP_PVR_LEN	0x0008	W	0x00000000	PVR DMA Transaction Length
TSP_PVR_BASE_ADDR	0x000c	W	0x00000000	PVR DMA transaction starting address
TSP_PVR_INT_STS	0x0010	W	0x00000000	PVR DMA Interrupt Status Register
TSP_PVR_INT_ENA	0x0014	W	0x00000000	DMA Interrupt Enable Register
TSP_TSOUT_CTRL	0x0018	W	0x00000000	TS Out Control Register
TSP_PVR_TOP_ADDR	0x001c	W	0x00000000	PVR buffer top address
TSP_PVR_WRITE_ADDR	0x0020	W	0x00000000	PVR buffer write point
TSP_PTIx_CTRL	0x0100	W	0x00000000	PTI Channel Control Register
TSP_PTIx_LLP_CFG	0x0104	W	0x00000000	LLP DMA Control Register

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Name	Offset	Size	Reset Value	Description
TSP_PTIx_LLP_BASE	0x0108	W	0x00000000	LLP Descriptor BASE Address
TSP_PTIx_LLP_WRITE	0x010c	W	0x00000000	LLP DMA Writing Software Descriptor Counter
TSP_PTIx_LLP_READ	0x0110	W	0x00000000	LLP DMA Reading Hardware Descriptor Counter
TSP_PTIx_PID_STS0	0x0114	W	0x00000000	PTI PID Channel Status 0 Register
TSP_PTIx_PID_STS1	0x0118	W	0x00000000	PTI PID Channel Status 1 Register
TSP_PTIx_PID_STS2	0x011c	W	0x00000000	PTI PID Channel Status 2 Register
TSP_PTIx_PID_STS3	0x0120	W	0x00000000	PTI PID Channel Status 3 Register
TSP_PTIx_PID_INT_ENA0	0x0124	W	0x00000000	PID Interrupt Enable Register 0
TSP_PTIx_PID_INT_ENA1	0x0128	W	0x00000000	PID Interrupt Enable Register 1
TSP_PTIx_PID_INT_ENA2	0x012c	W	0x00000000	PID Interrupt Enable Register 2
TSP_PTIx_PID_INT_ENA3	0x0130	W	0x00000000	PID Interrupt Enable Register 3
TSP_PTIx_PCR_INT_STS	0x0134	W	0x00000000	PTI PCR Interrupt Status Register
TSP_PTIx_PCR_INT_ENA	0x0138	W	0x00000000	PTI PCR Interrupt Enable Register
TSP_PTIx_PCRn_CTRL	0x013c	W	0x00000000	PID PCR Control Register
TSP_PTIx_PCRn_H	0x015c	W	0x00000000	High Order PCR value
TSP_PTIx_PCRn_L	0x0160	W	0x00000000	Low Order PCR value
TSP_PTIx_DMA_STS	0x019c	W	0x00000000	LLP DMA Interrupt Status Register
TSP_PTIx_DMA_ENA	0x01a0	W	0x00000000	DMA Interrupt Enable Register
TSP_PTIx_DATA_FLAG0	0x01a4	W	0x00000000	PTI_PID_WRITE Flag 0
TSP_PTIx_DATA_FLAG1	0x01a8	W	0x00000000	PTI_PID_WRITE Flag 1
TSP_PTIx_LIST_FLAG	0x01ac	W	0x00000000	PTIx_LIST_WRITE Flag
TSP_PTIx_DST_STS0	0x01b0	W	0x00000000	PTI Destination Status Register
TSP_PTIx_DST_STS1	0x01b4	W	0x00000000	PTI Destination Status Register
TSP_PTIx_DST_ENA0	0x01b8	W	0x00000000	PTI Destination Interrupt Enable Register
TSP_PTIx_DST_ENA1	0x01bc	W	0x00000000	PTI Destination Interrupt Enable Register
TSP_PTIx_ECWn_H	0x0200	W	0x00000000	The Even Control Word High Order
TSP_PTIx_ECWn_L	0x0204	W	0x00000000	The Even Control Word Low Order
TSP_PTIx_OCWn_H	0x0208	W	0x00000000	The Odd Control Word High Order
TSP_PTIx_OCWn_L	0x020c	W	0x00000000	The Odd Control Word Low Order
TSP_PTIx_PIDn_CTRL	0x0300	W	0x00000000	PID Channel Control Register
TSP_PTIx_PIDn_BASE	0x0400	W	0x00000000	PTI Data Memory Buffer Base Address
TSP_PTIx_PIDn_TOP	0x0404	W	0x00000000	PTI Data Memory Buffer Top Address
TSP_PTIx_PIDn_WRITE	0x0408	W	0x00000000	PTI Data Memory Buffer Hardware Writing Address
TSP_PTIx_PIDn_READ	0x040c	W	0x00000000	PTI Data Memory Buffer Software Reading Address

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Name	Offset	Size	Reset Value	Description
TSP_PTIx_LISTn_BASE	0x0800	W	0x00000000	PTI List Memory Buffer Base Address
TSP_PTIx_LISTn_TOP	0x0804	W	0x00000000	PTI List Memory Buffer Top Address
TSP_PTIx_LISTn_WRITE	0x0808	W	0x00000000	PTI List Memory Buffer Hardware Writing Address
TSP_PTIx_LISTn_READ	0x080c	W	0x00000000	PTI List Memory Buffer Software Reading Address
TSP_PTIx_PIDn_CFG	0x0900	W	0x00000008	PID Demux Configure Register
TSP_PTIx_PIDn_FILT_0	0x0904	W	0x00000000	Fliter Word 0
TSP_PTIx_PIDn_FILT_1	0x0908	W	0x00000000	Fliter Word 1
TSP_PTIx_PIDn_FILT_2	0x090c	W	0x00000000	Fliter Word 2
TSP_PTIx_PIDn_FILT_3	0x0910	W	0x00000000	Fliter Word 3

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

19.4.2 TSP Detail Register Description

TSP_GCFG

Address: Operational Base + offset (0x0000)

Global Configuration Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x0	arbit_cnt DMA channel arbiter counter This field is used to adjust the priority of DMA channels to prevent one channel holds the highest priority for a long time. The 3-bit field sets the largest times for a DMA channel to hold the highest priority to send the bus request. After requested times reach this limit, the highest priority is passed to next DMA channel in order.
3	RW	0x0	tsout_on TS Output Module Switch 1: TS output module switched on 0: TS output module switched off
2	RW	0x0	pvr_on PVR Module Switch 1: PVR function turned on ; 0: PVR function turned off ;
1	RW	0x0	pti1_on PTI0 channel switch 1: PTI1 channel switched on 0: PTI1 channel switched off

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Bit	Attr	Reset Value	Description
0	RW	0x0	pti0_on PTI0 channel switch 1: PTI0 channel switched on 0: PTI1 channel switched off

TSP_PVR_CTRL

Address: Operational Base + offset (0x0004)

PVR Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	fixaddr_en Fix Address Mode Select 1: fixed address mode; 0: incrementing address mode;
5:4	RW	0x0	burst_mode PVR burst mode PVR DMA burst mode 2'b00: INCR4 2'b01: INCR8 2'b10: INCR16 2'b11: Reserverd
3:2	RW	0x0	source PVR Source Select TS source for PVR output. 00: non-PID-filtered TS packets in PTI0; 01: PID filtered TS packets in PTI0; 10: non-PID-filtered TS packets in PTI1; 11: PID-filtered TS packets in PTI1;
1	R/WSC	0x0	stop PVR stop Write 1 to stop DMA channel. DMA will complete current burst transfer and then stop. It may takes several cycles. 1: PVR Stop ; 0: no effect ;
0	R/WSC	0x0	start PVR start Write 1 to start PVR. This bit will be cleared if PVR is stopped or PVR transaction is completed. 1: start PVR 0: no effect.

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TSP_PVR_LEN

Address: Operational Base + offset (0x0008)

PVR DMA Transaction Length

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	len Transaction Length Transaction Length

TSP_PVR_BASE_ADDR

Address: Operational Base + offset (0x000c)

PVR DMA transaction starting address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr PVR DMA transaction starting address PVR DMA transaction starting address

TSP_PVR_INT_STS

Address: Operational Base + offset (0x0010)

PVR DMA Interrupt Status Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	W1C	0x0	pvr_update_flag pvr address pageover flag When write_addr >= (base + top_addr/2), or write addr >= top_addr, the pvr_update_flag will assert HIGH. The application can write 1 to this bit to clear it.
1	W1C	0x0	pvr_error PVR DMA transaction error 1: error response during PVR DMA transaction; 0: no error response during PVR DMA transaction;
0	W1C	0x0	pvr_done PVR DMA transaction done 1: PVR DMA transaction completed; 0: PVR DMA transaction not completed;

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TSP_PVR_INT_ENA

Address: Operational Base + offset (0x0014)

DMA Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	pvr_update_ena 1: pvr_update interrupt enable 0: pvr_update interrupt disable
1	RW	0x0	pvr_error_ena PVR DMA Transcation Error Interrupt Enable 1: Error Interrupt Enabled 0: Error Interrupt Disabled
0	RW	0x0	pvr_done_ena PVR DMA Transaction Done Interrupt Enable 1: Done Interrupt Enabled 0: Done Interrupt Disabled

TSP_TSOUT_CTRL

Address: Operational Base + offset (0x0018)

TS Out Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	tso_sdo_sel TS serial data output 1: bit[0] use as serial data output ; 0: bit[7] use as serial data output ;
5	RW	0x0	tso_clk_phase TS output clock phase 0: ts output clock; 1: inverse of ts output clock.
4	RW	0x0	mode TS Output mode Selection Output mode select: 0: Serial Mode 1: Parallel Mode

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Bit	Attr	Reset Value	Description
3	RW	0x0	bit_order ts output serial data byte order Indicates that the output serial data byte order, ignored in the parallel: 0: MSB to LSB 1: LSB to MSB
2:1	RW	0x0	source TS Output Source Select TS source for TS out. 00: non-PID-filtered TS packets in PTI0; 01: PID filtered TS packets in PTI0; 10: non-PID-filtered TS packets in PTI1; 11: PID-filtered TS packets in PTI1;
0	RW	0x0	start TS out start 1: to start TS out function ; 0: to stop TS out function;

TSP_PVR_TOP_ADDR

Address: Operational Base + offset (0x001c)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pvr_top_addr top address in pvr mode

TSP_PVR_WRITE_ADDR

Address: Operational Base + offset (0x0020)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	pvr_write_addr The core will update this register to show the PVR write addr

TSP_PTIx_CTRL

Address: Operational Base + offset (0x0100)

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PTI Channel Control Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	tsi_sdi_sel TS Serial Data Input Select 1: bit[0] use as serial input data 0: bit[7] use as serial input data
20:19	RW	0x0	tsi_error_handle TS ERROR Handle 00: don't output 01: set the error indicator to 1 10: don't care
18	RW	0x0	clk_phase_sel ts input clock phase select 1'b0: ts input clock 1'b1: inverse of ts input clock
17:16	RW	0x0	demux_burst_mode Demux DMA Burst Mode Demux DMA Mode 2'b00: INCR4 2'b01: INCR8 2'b10: INCR16 2'b11: Reserved
15	RW	0x0	sync_bypass Bypass mode Selection 1'b1: Bypass mode, indicating that input TS packets will not be resynchronized and directly fed into the following modules; 1'b0: Synchronous mode, default, indicating that input TS packets will be resynchronized;
14	RW	0x0	cw_byteorder Control Word format Configuration 0: Default: first byte of the word is the highest byte 1: first byte of the word is the lowest byte
13	RW	0x0	cm_on CSA Conformance Mechanism Configuration CSA Conformance Mechanism 0: CM turned off 1: CM turned on
12:11	RW	0x0	tsi_mode TSI Input Mode Selection Input mode selection: 00: Serial Sync/valid Mode 01: Parallel Sync/valid Mode 10: Parallel Sync/burst Mode 11: Parallel Nosync/valid Mode

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Bit	Attr	Reset Value	Description
10	RW	0x0	tsi_bit_order input serial data order Indicates that the input serial data byte order, ignored in the parallel mode: 0: MSB to LSB 1: LSB to MSB
9	RW	0x0	tsi_sel TS Input Source Select Select input TS source 1'b1: HSADC ; 1'b0: internel memory ;
8	RW	0x0	out_byteswap Output byteswap function When enabled, the word to be transferred to memory buffer "B4B3B2B1" is performed byteswapping to "B1B2B3B4".
7	RW	0x0	in_byteswap Input TS Word Byteswap When enabled, the input TS word "B4B3B2B1" is perfomed byteswapping to "B1B2B3B4".
6:4	RW	0x0	unsync_times TS Header Unsynchronized Times If synchronous mode is selected. This field sets the successive times of TS packet header error to re-lock TS header when TS is in locked status;
3:1	RW	0x0	sync_times TS Header Synchronized Times If synchronous mode is selected. This field sets the successive times of finding TS packet header to lock the TS header when TS is in unlocked status;
0	R/W SC	0x0	clear Software clear signal It will reset the core register . It will table several cycles. After reset done, soft_reset will be low. 1. reset; 0. no effect.

TSP_PTIx_LLP_CFG

Address: Operational Base + offset (0x0104)

LLP DMA Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:8	RW	0x0	threshold LLP Transfer Threshold The depth for LLP descriptors is 64. An interrupt will be asserted when transfer reaches the threshold set if DMA transfer interrupt is enabled. 00: 1/1 depth 01: 1/2 depth 10: 1/4 depth 11: 1/8 depth
7:6	RW	0x0	burst_mode LLP DMA Burst Mode LLP DMA Burst Mode 2'b00: INCR4 2'b01: INCR8 2'b10: INCR16 2'b11: Reserverd
5	RW	0x0	hw_trigger Hardware Trigger Select 1. hardware trigger; 0. software trigger;
4	RW	0x0	fix_addr_en Fix Address Mode Select 1: fixed address mode; 0: incrementing address mode;
3	W1C	0x0	cfg_done LLP DMA Configuration Done When all descriptors of LLP are configured, write 1 to to this bit. The core will clear this bit when llp transction is finished ;
2	RW	0x0	pause LLP DMA Pause Write 1 to Pause DMA channel . DMA will complete current burst transfer and then pause. All register stay unchange. If software write 0 later , It will continue to work. It may take several cycles to pause. 1: pause; 0: continue to work ;
1	W1C	0x0	stop LLP DMA Stop Write 1 to stop DMA channel. DMA will complete current burst transter and then stop. It may takes several cycles. 1: stop ; 0: no effect ;

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Bit	Attr	Reset Value	Description
0	W1C	0x0	start LLP DMA start Write 1 to start DMA Channel , self clear after 1 cycle. 1: start ; 0: no effect

TSP_PTIx_LLP_BASE

Address: Operational Base + offset (0x0108)

LLP Descriptor BASE Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	addr LLP Descriptor BASE Address LLP Descriptor BASE address

TSP_PTIx_LLP_WRITE

Address: Operational Base + offset (0x010c)

LLP DMA Writing Software Descriptor Counter

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	counter LLP DMA Writing Software Descriptor Counter LLP DMA Writing Software Descriptor Counter

TSP_PTIx_LLP_READ

Address: Operational Base + offset (0x0110)

LLP DMA Reading Hardware Descriptor Counter

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	counter LLP DMA Reading Hardware Descriptor Counter LLP DMA Reading Hardware Descriptor Counter

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TSP_PTIx_PID_STS0

Address: Operational Base + offset (0x0114)

PTI PID Channel Status 0 Register

Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_done PID31 Channel Status 1 means done
30	W1C	0x0	pid30_done PID30 Channel Status 1 means done
29	W1C	0x0	pid29_done PID29 Channel Status 1 means done
28	W1C	0x0	pid28_done PID28 Channel Status 1 means done
27	W1C	0x0	pid27_done PID27 Channel Status 1 means done
26	W1C	0x0	pid26_done PID26 Channel Status 1 means done
25	W1C	0x0	pid25_done PID25 Channel Status 1 means done
24	W1C	0x0	pid24_done PID24 Channel Status 1 means done
23	W1C	0x0	pid23_done PID23 Channel Status 1 means done
22	W1C	0x0	pid22_done PID22 Channel Status 1 means done
21	W1C	0x0	pid21_done PID21 Channel Status 1 means done
20	W1C	0x0	pid20_done PID20 Channel Status 1 means done
19	W1C	0x0	pid19_done PID19 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
18	W1C	0x0	pid18_done PID18 Channel Status 1 means done
17	W1C	0x0	pid17_done PID17 Channel Status 1 means done
16	W1C	0x0	pid16_done PID16 Channel Status 1 means done
15	W1C	0x0	pid15_done PID15 Channel Status 1 means done
14	W1C	0x0	pid14_done PID14 Channel Status 1 means done
13	W1C	0x0	pid13_done PID13 Channel Status 1 means done
12	W1C	0x0	pid12_done PID12 Channel Status 1 means done
11	W1C	0x0	pid11_done PID11 Channel Status 1 means done
10	W1C	0x0	pid10_done PID10 Channel Status 1 means done
9	W1C	0x0	pid9_done PID9 Channel Status 1 means done
8	W1C	0x0	pid8_done PID8 Channel Status 1 means done
7	W1C	0x0	pid7_done PID7 Channel Status 1 means done
6	W1C	0x0	pid6_done PID6 Channel Status 1 means done
5	W1C	0x0	pid5_done PID5 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
4	W1C	0x0	pid4_done PID4 Channel Status 1 means done
3	W1C	0x0	pid3_done PID3 Channel Status 1 means done
2	RW	0x0	pid2_done PID2 Channel Status 1 means done
1	W1C	0x0	pid1_done PID1 Channel Status 1 means done
0	W1C	0x0	pid0_done PID0 Channel Status 1 means done

TSP_PTIx_PID_STS1

Address: Operational Base + offset (0x0118)

PTI PID Channel Status 1 Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	pid63_done PID63 Channel Status 1 means done
30	W1C	0x0	pid62_done PID62 Channel Status 1 means done
29	W1C	0x0	pid61_done PID61 Channel Status 1 means done
28	W1C	0x0	pid60_done PID60 Channel Status 1 means done
27	W1C	0x0	pid59_done PID59 Channel Status 1 means done
26	W1C	0x0	pid58_done PID58 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
25	W1C	0x0	pid57_done PID57 Channel Status 1 means done
24	W1C	0x0	pid56_done PID56 Channel Status 1 means done
23	W1C	0x0	pid55_done PID55 Channel Status 1 means done
22	W1C	0x0	pid54_done PID54 Channel Status 1 means done
21	W1C	0x0	pid53_done PID53 Channel Status 1 means done
20	W1C	0x0	pid52_done PID52 Channel Status 1 means done
19	W1C	0x0	pid51_done PID51 Channel Status 1 means done
18	W1C	0x0	pid50_done PID51 Channel Status 1 means done
17	W1C	0x0	pid49_done PID49 Channel Status 1 means done
16	W1C	0x0	pid48_done PID48 Channel Status 1 means done
15	W1C	0x0	pid47_done PID47 Channel Status 1 means done
14	W1C	0x0	pid46_done PID46 Channel Status 1 means done
13	W1C	0x0	pid45_done PID45 Channel Status 1 means done
12	W1C	0x0	pid44_done PID44 Channel Status 1 means done

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Bit	Attr	Reset Value	Description
11	W1C	0x0	pid43_done PID43 Channel Status 1 means done
10	W1C	0x0	pid42_done PID42 Channel Status 1 means done
9	W1C	0x0	pid41_done PID41 Channel Status 1 means done
8	W1C	0x0	pid40_done PID40 Channel Status 1 means done
7	W1C	0x0	pid39_done PID39 Channel Status 1 means done
6	W1C	0x0	pid38_done PID38 Channel Status 1 means done
5	W1C	0x0	pid37_done PID37 Channel Status 1 means done
4	W1C	0x0	pid36_done PID36 Channel Status 1 means done
3	RW	0x0	pid35_done PID35 Channel Status 1 means done
2	W1C	0x0	pid34_done PID34 Channel Status 1 means done
1	W1C	0x0	pid33_done PID33 Channel Status 1 means done
0	RW	0x0	pid32_done PID32 Channel Status 1 means done

TSP_PTIx_PID_STS2

Address: Operational Base + offset (0x011c)

PTI PID Channel Status 2 Register

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Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_error PID31 Error Interrupt Status 1 means error detected
30	W1C	0x0	pid30_error PID30 Error Interrupt Status 1 means error detected
29	W1C	0x0	pid29_error PID29 Error Interrupt Status 1 means error detected
28	W1C	0x0	pid28_error PID28 Error Interrupt Status 1 means error detected
27	W1C	0x0	pid27_error PID27 Error Interrupt Status 1 means error detected
26	W1C	0x0	pid26_error PID26 Error Interrupt Status 1 means error detected
25	W1C	0x0	pid25_error PID25 Error Interrupt Status 1 means error detected
24	W1C	0x0	pid24_error PID24 Error Interrupt Status 1 means error detected
23	W1C	0x0	pid23_error PID23 Error Interrupt Status 1 means error detected
22	W1C	0x0	pid22_error PID22 Error Interrupt Status 1 means error detected
21	W1C	0x0	pid21_error PID21 Error Interrupt Status 1 means error detected
20	W1C	0x0	pid20_error PID20 Error Interrupt Status 1 means error detected
19	W1C	0x0	pid19_error PID19 Error Interrupt Status 1 means error detected
18	W1C	0x0	pid18_error PID18 Error Interrupt Status 1 means error detected

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Bit	Attr	Reset Value	Description
17	W1C	0x0	pid17_error PID17 Error Interrupt Status 1 means error detected
16	W1C	0x0	pid16_error PID16 Error Interrupt Status 1 means error detected
15	W1C	0x0	pid15_error PID15 Error Interrupt Status 1 means error detected
14	W1C	0x0	pid14_error PID14 Error Interrupt Status 1 means error detected
13	W1C	0x0	pid13_error PID13 Error Interrupt Status 1 means error detected
12	W1C	0x0	pid12_error PID12 Error Interrupt Status 1 means error detected
11	W1C	0x0	pid11_error PID11 Error Interrupt Status 1 means error detected
10	W1C	0x0	pid10_error PID10 Error Interrupt Status 1 means error detected
9	W1C	0x0	pid9_error PID9 Error Interrupt Status 1 means error detected
8	W1C	0x0	pid8_error PID8 Error Interrupt Status 1 means error detected
7	W1C	0x0	pid7_error PID7 Error Interrupt Status 1 means error detected
6	W1C	0x0	pid6_error PID6 Error Interrupt Status 1 means error detected
5	W1C	0x0	pid5_error PID5 Error Interrupt Status 1 means error detected
4	W1C	0x0	pid4_error PID4 Error Interrupt Status 1 means error detected

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Bit	Attr	Reset Value	Description
3	W1C	0x0	pid3_error PID3 Error Interrupt Status 1 means error detected
2	W1C	0x0	pid2_error PID2 Error Interrupt Status 1 means error detected
1	W1C	0x0	pid1_error PID1 Error Interrupt Status 1 means error detected
0	W1C	0x0	pid0_error PID0 Error Interrupt Status 1 means error detected

TSP_PTIx_PID_STS3

Address: Operational Base + offset (0x0120)

PTI PID Channel Status 3 Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	pid63_error PID63 Error Interrupt Status 1 means error detected
30	W1C	0x0	pid62_error PID62 Error Interrupt Status 1 means error detected
29	W1C	0x0	pid61_error PID61 Error Interrupt Status 1 means error detected
28	W1C	0x0	pid60_error PID60 Error Interrupt Status 1 means error detected
27	W1C	0x0	pid59_error PID59 Error Interrupt Status 1 means error detected
26	W1C	0x0	pid58_error PID58 Error Interrupt Status 1 means error detected
25	W1C	0x0	pid57_error PID57 Error Interrupt Status 1 means error detected

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Bit	Attr	Reset Value	Description
24	W1C	0x0	pid56_error PID56 Error Interrupt Status 1 means error detected
23	W1C	0x0	pid55_error PID55 Error Interrupt Status 1 means error detected
22	W1C	0x0	pid54_error PID54 Error Interrupt Status 1 means error detected
21	W1C	0x0	pid53_error PID53 Error Interrupt Status 1 means error detected
20	W1C	0x0	pid52_error PID52 Error Interrupt Status 1 means error detected
19	W1C	0x0	pid51_error PID51 Error Interrupt Status 1 means error detected
18	W1C	0x0	pid50_error PID50 Error Interrupt Status 1 means error detected
17	W1C	0x0	pid49_error PID49 Error Interrupt Status 1 means error detected
16	W1C	0x0	pid48_error PID48 Error Interrupt Status 1 means error detected
15	W1C	0x0	pid47_error PID47 Error Interrupt Status 1 means error detected
14	W1C	0x0	pid46_error PID46 Error Interrupt Status 1 means error detected
13	W1C	0x0	pid45_error PID45 Error Interrupt Status 1 means error detected
12	W1C	0x0	pid44_error PID44 Error Interrupt Status 1 means error detected
11	W1C	0x0	pid43_error PID43 Error Interrupt Status 1 means error detected

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Bit	Attr	Reset Value	Description
10	W1C	0x0	pid42_error PID42 Error Interrupt Status 1 means error detected
9	W1C	0x0	pid41_error PID41 Error Interrupt Status 1 means error detected
8	W1C	0x0	pid40_error PID40 Error Interrupt Status 1 means error detected
7	W1C	0x0	pid39_error PID39 Error Interrupt Status 1 means error detected
6	W1C	0x0	pid38_error PID38 Error Interrupt Status 1 means error detected
5	W1C	0x0	pid37_error PID37 Error Interrupt Status 1 means error detected
4	W1C	0x0	pid36_error PID36 Error Interrupt Status 1 means error detected
3	W1C	0x0	pid35_error PID35 Error Interrupt Status 1 means error detected
2	W1C	0x0	pid34_error PID34 Error Interrupt Status 1 means error detected
1	W1C	0x0	pid33_error PID33 Error Interrupt Status 1 means error detected
0	W1C	0x0	pid32_error PID32 Error Interrupt Status 1 means error detected

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TSP_PTIx_PID_INT_ENA0

Address: Operational Base + offset (0x0124)

PID Interrupt Enable Register 0

Bit	Attr	Reset Value	Description

Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_done_ena PID31 Done Enable 1:enabled 0:disabled
30	RW	0x0	pid30_done_ena PID30 Done Enable 1:enabled 0:disabled
29	RW	0x0	pid29_done_ena PID29 Done Enable 1:enabled 0:disabled
28	RW	0x0	pid28_done_ena PID28 Done Enable 1:enabled 0:disabled
27	RW	0x0	pid27_done_ena PID27 Done Enable 1:enabled 0:disabled
26	RW	0x0	pid26_done_ena PID26 Done Enable 1:enabled 0:disabled
25	RW	0x0	pid25_done_ena PID25 Done Enable 1:enabled 0:disabled
24	RW	0x0	pid24_done_ena PID24 Done Enable 1:enabled 0:disabled
23	RW	0x0	pid23_done_ena PID23 Done Enable 1:enabled 0:disabled
22	RW	0x0	pid22_done_ena PID22 Done Enable 1:enabled 0:disabled
21	RW	0x0	pid21_done_ena PID21 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
20	RW	0x0	pid20_done_ena PID20 Done Enable 1:enabled 0:disabled
19	RW	0x0	pid19_done_ena PID19 Done Enable 1:enabled 0:disabled
18	RW	0x0	pid18_done_ena PID18 Done Enable 1:enabled 0:disabled
17	RW	0x0	pid17_done_ena PID17 Done Enable 1:enabled 0:disabled
16	RW	0x0	pid16_done_ena PID16 Done Enable 1:enabled 0:disabled
15	RW	0x0	pid15_done_ena PID15 Done Enable 1:enabled 0:disabled
14	RW	0x0	pid14_done_ena PID14 Done Enable 1:enabled 0:disabled
13	RW	0x0	pid13_done_ena PID13 Done Enable 1:enabled 0:disabled
12	RW	0x0	pid12_done_ena PID12 Done Enable 1:enabled 0:disabled
11	RW	0x0	pid11_done_ena PID11 Done Enable 1:enabled 0:disabled
10	RW	0x0	pid10_done_ena PID10 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
9	RW	0x0	pid9_done_ena PID9 Done Enable 1:enabled 0:disabled
8	RW	0x0	pid8_done_ena PID8 Done Enable 1:enabled 0:disabled
7	RW	0x0	pid7_done_ena PID7 Done Enable 1:enabled 0:disabled
6	RW	0x0	pid6_done_ena PID6 Done Enable 1:enabled 0:disabled
5	RW	0x0	pid5_done_ena PID5 Done Enable 1:enabled 0:disabled
4	RW	0x0	pid4_done_ena PID4 Done Enable 1:enabled 0:disabled
3	RW	0x0	pid3_done_ena PID3 Done Enable 1:enabled 0:disabled
2	RW	0x0	pid2_done_ena PID2 Done Enable 1:enabled 0:disabled
1	RW	0x0	pid1_done_ena PID1 Done Enable 1:enabled 0:disabled
0	RW	0x0	pid0_done_ena PID0 Done Enable 1:enabled 0:disabled

TSP_PTIx_PID_INT_ENA1

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Address: Operational Base + offset (0x0128)

PID Interrupt Enable Register 1

Bit	Attr	Reset Value	Description
31	RW	0x0	pid63_done PID63 Done Enable 1:enabled 0:disabled
30	RW	0x0	pid62_done PID62 Done Enable 1:enabled 0:disabled
29	RW	0x0	pid61_done PID61 Done Enable 1:enabled 0:disabled
28	RW	0x0	pid60_done PID60 Done Enable 1:enabled 0:disabled
27	RW	0x0	pid59_done PID59 Done Enable 1:enabled 0:disabled
26	RW	0x0	pid58_done PID58 Done Enable 1:enabled 0:disabled
25	RW	0x0	pid57_done PID57 Done Enable 1:enabled 0:disabled
24	RW	0x0	pid56_done PID56 Done Enable 1:enabled 0:disabled
23	RW	0x0	pid55_done PID55 Done Enable 1:enabled 0:disabled
22	RW	0x0	pid54_done PID54 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
21	RW	0x0	pid53_done PID53 Done Enable 1:enabled 0:disabled
20	RW	0x0	pid52_done PID52 Done Enable 1:enabled 0:disabled
19	RW	0x0	pid51_done PID51 Done Enable 1:enabled 0:disabled
18	RW	0x0	pid50_done PID50 Done Enable 1:enabled 0:disabled
17	RW	0x0	pid49_done PID49 Done Enable 1:enabled 0:disabled
16	RW	0x0	pid48_done PID48 Done Enable 1:enabled 0:disabled
15	RW	0x0	pid47_done PID47 Done Enable 1:enabled 0:disabled
14	RW	0x0	pid46_done PID46 Done Enable 1:enabled 0:disabled
13	RW	0x0	pid45_done PID45 Done Enable 1:enabled 0:disabled
12	RW	0x0	pid44_done PID44 Done Enable 1:enabled 0:disabled
11	RW	0x0	pid43_done PID43 Done Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
10	RW	0x0	pid42_done PID42 Done Enable 1:enabled 0:disabled
9	RW	0x0	pid41_done PID41 Done Enable 1:enabled 0:disabled
8	RW	0x0	pid40_done PID40 Done Enable 1:enabled 0:disabled
7	RW	0x0	pid39_done PID39 Done Enable 1:enabled 0:disabled
6	RW	0x0	pid38_done PID38 Done Enable 1:enabled 0:disabled
5	RW	0x0	pid37_done PID37 Done Enable 1:enabled 0:disabled
4	RW	0x0	pid36_done PID36 Done Enable 1:enabled 0:disabled
3	RW	0x0	pid35_done PID35 Done Enable 1:enabled 0:disabled
2	RW	0x0	pid34_done PID34 Done Enable 1:enabled 0:disabled
1	RW	0x0	pid33_done PID33 Done Enable 1:enabled 0:disabled
0	RW	0x0	pid32_done PID32 Done Enable 1:enabled 0:disabled

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TSP_PTIx_PID_INT_ENA2

Address: Operational Base + offset (0x012c)

PID Interrupt Enable Register 2

Bit	Attr	Reset Value	Description
31	RW	0x0	pid31_error PID31 Error Interrupt Enable 1:enabled 0:disabled
30	RW	0x0	pid30_error PID30 Error Interrupt Enable 1:enabled 0:disabled
29	RW	0x0	pid29_error PID29 Error Interrupt Enable 1:enabled 0:disabled
28	RW	0x0	pid28_error PID28 Error Interrupt Enable 1:enabled 0:disabled
27	RW	0x0	pid27_error PID27 Error Interrupt Enable 1:enabled 0:disabled
26	RW	0x0	pid26_error PID26 Error Interrupt Enable 1:enabled 0:disabled
25	RW	0x0	pid25_error PID25 Error Interrupt Enable 1:enabled 0:disabled
24	RW	0x0	pid24_error PID24 Error Interrupt Enable 1:enabled 0:disabled
23	RW	0x0	pid23_error PID23 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
22	RW	0x0	pid22_error PID22 Error Interrupt Enable 1:enabled 0:disabled
21	RW	0x0	pid21_error PID21 Error Interrupt Enable 1:enabled 0:disabled
20	RW	0x0	pid20_error PID20 Error Interrupt Enable 1:enabled 0:disabled
19	RW	0x0	pid19_error PID19 Error Interrupt Enable 1:enabled 0:disabled
18	RW	0x0	pid18_error PID18 Error Interrupt Enable 1:enabled 0:disabled
17	RW	0x0	pid17_error PID17 Error Interrupt Enable 1:enabled 0:disabled
16	RW	0x0	pid16_error PID16 Error Interrupt Enable 1:enabled 0:disabled
15	RW	0x0	pid15_error PID15 Error Interrupt Enable 1:enabled 0:disabled
14	RW	0x0	pid14_error PID14 Error Interrupt Enable 1:enabled 0:disabled
13	RW	0x0	pid13_error PID13 Error Interrupt Enable 1:enabled 0:disabled
12	RW	0x0	pid12_error PID12 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
11	RW	0x0	pid11_error PID11 Error Interrupt Enable 1:enabled 0:disabled
10	RW	0x0	pid10_error PID10 Error Interrupt Enable 1:enabled 0:disabled
9	RW	0x0	pid9_error PID9 Error Interrupt Enable 1:enabled 0:disabled
8	RW	0x0	pid8_error PID8 Error Interrupt Enable 1:enabled 0:disabled
7	RW	0x0	pid7_error PID7 Error Interrupt Enable 1:enabled 0:disabled
6	RW	0x0	pid6_error PID6 Error Interrupt Enable 1:enabled 0:disabled
5	RW	0x0	pid5_error PID5 Error Interrupt Enable 1:enabled 0:disabled
4	RW	0x0	pid4_error PID4 Error Interrupt Enable 1:enabled 0:disabled
3	RW	0x0	pid3_error PID3 Error Interrupt Enable 1:enabled 0:disabled
2	RW	0x0	pid2_error PID2 Error Interrupt Enable 1:enabled 0:disabled
1	RW	0x0	pid1_error PID1 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
0	RW	0x0	pid0_error PID0 Error Interrupt Enable 1:enabled 0:disabled

TSP_PTIx_PID_INT_ENA3

Address: Operational Base + offset (0x0130)

PID Interrupt Enable Register 3

Bit	Attr	Reset Value	Description
31	RW	0x0	pid63_error PID63 Error Interrupt Enable 1:enabled 0:disabled
30	RW	0x0	pid62_error PID62 Error Interrupt Enable 1:enabled 0:disabled
29	RW	0x0	pid61_error PID61 Error Interrupt Enable 1:enabled 0:disabled
28	RW	0x0	pid60_error PID60 Error Interrupt Enable 1:enabled 0:disabled
27	RW	0x0	pid59_error PID59 Error Interrupt Enable 1:enabled 0:disabled
26	RW	0x0	pid58_error PID58 Error Interrupt Enable 1:enabled 0:disabled
25	RW	0x0	pid57_error PID57 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
24	RW	0x0	pid56_error PID56 Error Interrupt Enable 1:enabled 0:disabled
23	RW	0x0	pid55_error PID55 Error Interrupt Enable 1:enabled 0:disabled
22	RW	0x0	pid54_error PID54 Error Interrupt Enable 1:enabled 0:disabled
21	RW	0x0	pid53_error PID53 Error Interrupt Enable 1:enabled 0:disabled
20	RW	0x0	pid52_error PID52 Error Interrupt Enable 1:enabled 0:disabled
19	RW	0x0	pid51_error PID51 Error Interrupt Enable 1:enabled 0:disabled
18	RW	0x0	pid50_error PID50 Error Interrupt Enable 1:enabled 0:disabled
17	RW	0x0	pid49_error PID49 Error Interrupt Enable 1:enabled 0:disabled
16	RW	0x0	pid48_error PID48 Error Interrupt Enable 1:enabled 0:disabled
15	RW	0x0	pid47_error PID47 Error Interrupt Enable 1:enabled 0:disabled
14	RW	0x0	pid46_error PID46 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
13	RW	0x0	pid45_error PID45 Error Interrupt Enable 1:enabled 0:disabled
12	RW	0x0	pid44_error PID44 Error Interrupt Enable 1:enabled 0:disabled
11	RW	0x0	pid43_error PID43 Error Interrupt Enable 1:enabled 0:disabled
10	RW	0x0	pid42_error PID42 Error Interrupt Enable 1:enabled 0:disabled
9	RW	0x0	pid41_error PID41 Error Interrupt Enable 1:enabled 0:disabled
8	RW	0x0	pid40_error PID40 Error Interrupt Enable 1:enabled 0:disabled
7	RW	0x0	pid39_error PID39 Error Interrupt Enable 1:enabled 0:disabled
6	RW	0x0	pid38_error PID38 Error Interrupt Enable 1:enabled 0:disabled
5	RW	0x0	pid37_error PID37 Error Interrupt Enable 1:enabled 0:disabled
4	RW	0x0	pid36_error PID36 Error Interrupt Enable 1:enabled 0:disabled
3	RW	0x0	pid35_error PID35 Error Interrupt Enable 1:enabled 0:disabled

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Bit	Attr	Reset Value	Description
2	RW	0x0	pid34_error PID34 Error Interrupt Enable 1:enabled 0:disabled
1	RW	0x0	pid33_error PID33 Error Interrupt Enable 1:enabled 0:disabled
0	RW	0x0	pid32_error PID32 Error Interrupt Enable 1:enabled 0:disabled

TSP_PTIx_PCR_INT_STS

Address: Operational Base + offset (0x0134)

PTI PCR Interrupt Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	W1C	0x0	pcr7_done PCR7 Status 1: done; 0: not done;
6	W1C	0x0	pcr6_done PCR6 Status 1: done; 0: not done;
5	W1C	0x0	pcr5_done PCR5 Status 1: done; 0: not done;
4	W1C	0x0	pcr4_done PCR4 Status 1: done; 0: not done;
3	W1C	0x0	pcr3_done PCR3 Status 1: done; 0: not done;

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Bit	Attr	Reset Value	Description
2	W1C	0x0	pcr2_done PCR2 Status 1: done; 0: not done;
1	W1C	0x0	pcr1_done PCR1 Status 1: done; 0: not done;
0	W1C	0x0	pcr0_done PCR0 Status 1: done; 0: not done;

TSP_PTIx_PCR_INT_ENA

Address: Operational Base + offset (0x0138)

PTI PCR Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	pcr7_done_ena pcr7 done interrupt enable 1: enabled; 0: disabled;
6	RW	0x0	pcr6_done_ena pcr6 done interrupt enable 1: enabled; 0: disabled;
5	RW	0x0	pcr5_done_ena pcr5 done interrupt enable 1: enabled; 0: disabled;
4	RW	0x0	pcr4_done_ena pcr4 done interrupt enable 1: enabled; 0: disabled;
3	RW	0x0	pcr3_done_ena pcr3 done interrupt enable 1: enabled; 0: disabled;

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Bit	Attr	Reset Value	Description
2	RW	0x0	pcr2_done_ena pcr2 done interrupt enable 1: enabled; 0: disabled;
1	RW	0x0	pcr1_done_ena pcr1 done interrupt enable 1: enabled; 0: disabled;
0	RW	0x0	pcr0_done_ena pcr0 done interrupt enable 1: enabled; 0: disabled;

TSP_PTIx_PCRn_CTRL

Address: Operational Base + offset (0x013c)

PID PCR Control Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:1	RW	0x0000	pid PCR Extraction PID number This 13-bit field sets the PID number that needs PCR extraction.
0	RW	0x0	on PCR Extraction Switch 1'b1: PCR extraction switched on ; 1'b0: PCR extraction switched off ;

TSP_PTIx_PCRn_H

Address: Operational Base + offset (0x015c)

High Order PCR value

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	pcr PCR[32] pcr[32]

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TSP_PTIx_PCRn_L

Address: Operational Base + offset (0x0160)

Low Order PCR value

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	pcr pcr[31:0] pcr[31:0]

TSP_PTIx_DMA_STS

Address: Operational Base + offset (0x019c)

LLP DMA Interrupt Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	W1C	0x0	llp_error LLP DMA Error Status 1: error response during DMA transaction; 0: no error response during DMA transaction;
0	W1C	0x0	llp_done LLP DMA Done Status 1: DMA transaction completed; 0: DMA transaction not completed;

TSP_PTIx_DMA_ENA

Address: Operational Base + offset (0x01a0)

DMA Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	llp_error_ena LLP DMA Error Interrupt Enable 1: enabled 0: disabled
0	RW	0x0	llp_done_ena LLP DMA Done Interrupt Enable 1: enabled 0: disabled

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TSP_PTIx_DATA_FLAG0

Address: Operational Base + offset (0x01a4)

PTI_PID_WRITE Flag 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data_write_flag_0 From PID0 TO PID31

TSP_PTIx_DATA_FLAG1

Address: Operational Base + offset (0x01a8)

PTI_PID_WRITE Flag 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	data_write_flag_1 From PID32 TO PID63

TSP_PTIx_LIST_FLAG

Address: Operational Base + offset (0x01ac)

PTIx_LIST_WRITE Flag

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	list_write_flag From PID0 TO PID15

TSP_PTIx_DST_STS0

Address: Operational Base + offset (0x01b0)

PTI Destination Status Register

Bit	Attr	Reset Value	Description
31:0	W1C	0x00000000	demux_dma_status_0 From 0 to 31 channel

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TSP_PTIx_DST_STS1

Address: Operational Base + offset (0x01b4)

PTI Destination Status Register

Bit	Attr	Reset Value	Description
31:0	W1C	0x00000000	demux_dma_status_0 From 32 to 63 channel

TSP_PTIx_DST_ENA0

Address: Operational Base + offset (0x01b8)

PTI Destination Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	demux_dma_enable_0 From 0 to 31 channel

TSP_PTIx_DST_ENA1

Address: Operational Base + offset (0x01bc)

PTI Destination Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	demux_dma_enable_1 From 32 to 63 channel

TSP_PTIx_ECWn_H

Address: Operational Base + offset (0x0200)

The Even Control Word High Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ecw_h The Even Control Word High Order ECW[63:32]

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TSP_PTIx_ECWn_L

Address: Operational Base + offset (0x0204)

The Even Control Word Low Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ecw_l The Even Control Word Low Order ECW[31:0]

TSP_PTIx_OCWn_H

Address: Operational Base + offset (0x0208)

The Odd Control Word High Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ocw_h The Odd Control Word High order OCW[63:32]

TSP_PTIx_OCWn_L

Address: Operational Base + offset (0x020c)

The Odd Control Word Low Order

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ocw_l The Odd Control Word Low Order OCW[31:0]

TSP_PTIx_PIDn_CTRL

Address: Operational Base + offset (0x0300)

PID Channel Control Register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
19:16	RW	0x0	cw_num Control Word Order Number This fields indicates the corresponding order number of control word to be used to descramble TS packets.
15:3	RW	0x0000	pid PID number This 13-bit sets the desired PID number to be processed by PTI channel.
2	RW	0x0	csa_on Descrambling Switch 1'b1: Descrambling function turned on; 1'b0: Descrambling function turned off;
1	R/W SC	0x0	clear PID Channel Clear Write 1 to clear PID channel. This bit will be set to 0 if the channel is clear.
0	R/W SC	0x0	en PID Channel Enable Write 1 to enable channel. Write 0 to this bit will not take any effect. This bit will be 0 when channel is cleared.

TSP_PTIx_PIDn_BASE

Address: Operational Base + offset (0x0400)

PTI Data Memory Buffer Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Base Address PTI Data Memory Buffer Base Address

TSP_PTIx_PIDn_TOP

Address: Operational Base + offset (0x0404)

PTI Data Memory Buffer Top Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Top Address PTI Data Memory Buffer Top Address

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TSP_PTIx_PIDn_WRITE

Address: Operational Base + offset (0x0408)

PTI Data Memory Buffer Hardware Writing Address

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	address PTI Data Memory Buffer Hardware Writing Address PTI Data Memory Buffer Hardware Writing Address

TSP_PTIx_PIDn_READ

Address: Operational Base + offset (0x040c)

PTI Data Memory Buffer Software Reading Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Software Reading Address PTI Data Memory Buffer Software Reading Address

TSP_PTIx_LISTn_BASE

Address: Operational Base + offset (0x0800)

PTI List Memory Buffer Base Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI Data Memory Buffer Software Reading Address PTI Data Memory Buffer Software Reading Address

TSP_PTIx_LISTn_TOP

Address: Operational Base + offset (0x0804)

PTI List Memory Buffer Top Address

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI List Memory Buffer Top Address PTI List Memory Buffer Top Address

TSP_PTIx_LISTn_WRITE

Address: Operational Base + offset (0x0808)

PTI List Memory Buffer Hardware Writing Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI List Memory Buffer Hardware Writing Address PTI List Memory Buffer Hardware Writing Address

TSP_PTIx_LISTn_READ

Address: Operational Base + offset (0x080c)

PTI List Memory Buffer Software Reading Address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	address PTI List Memory Buffer Software Reading Address PTI List Memory Buffer Software Reading Address

TSP_PTIx_PIDn_CFG

Address: Operational Base + offset (0x0900)

PID Demux Configure Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	filter_en Filter Byte Enable The proper position of filter byte Enable. For Section filter. the 1st,4th,5th,..18th byte of section header are used to be filtered; For PES filter, the 4th,7th,8th...21th byte of pes header are used to be filtered.
15:12	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
11	RW	0x0	scd_en Start Code Detection Switch Start code detection 1: enabled; 0: disabled; This bit is only valid when n < 16.
10	RW	0x0	cni_on Current Next Indicator Abort when current_next_indicator == 1'b1, 1'b1: abort ; 1'b0: do nothing ;
9:8	RW	0x0	filt_mode Section Filter Mode Filter Mode when the filter mode is configured as section filter. 2'b00: stop per unit; 2'b01: full stop; 2'b10: recycle, update when version number change 2'b11: reserverd
7:6	RW	0x0	video_type Video filtering Type 2'b00: MPEG2 2'b01: H264 2'b10: VC-1 2'b11: Reserved
5:4	RW	0x0	filt_type Filter Type 2'b00: section filtering; 2'b01: pes filtering; 2'b10: es filtering; 2'b11: ts filtering; if n>=16, it is reserved as only section filtering, other values are invalid.
3	RW	0x1	cc_abort Continue Counter Error Abort when continuity counter error happens: 1: abort; 0: do nothing;
2	RW	0x0	tei_abort Ts_error_indicator Abort when ts_error_indicator == 1: 1'b1: abort ; 1'b0: do nothing;

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Bit	Attr	Reset Value	Description
1	RW	0x0	crc_abort CRC Error Abort This bit is valid only when crc_on == 1'b1. When crc error happens, 1'b1: abort ; 1'b0: do nothing.
0	RW	0x0	crc_on CRC Check 1'b1: CRC check function turned on 1'b0: CRC check function turned off

TSP_PTIx_PIDn_FILT_0

Address: Operational Base + offset (0x0904)

Fliter Word 0

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 6th byte of section header or 9th byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 5th byte of section header or 8th byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 4th byte of section header or 7th byte of pes header
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 1st byte of section header or 4th byte of pes header

TSP_PTIx_PIDn_FILT_1

Address: Operational Base + offset (0x0908)

Fliter Word 1

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 10th byte of section header or 13rd byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 9th byte of section header or 12nd byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 8th byte of section header or 11st byte of pes header
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 7th byte of section header or 10th byte of pes header

TSP_PTIx_PIDn_FILT_2

Address: Operational Base + offset (0x090c)

Fliter Word 2

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 14th byte of section header or 17th byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 13rd byte of section header or 16th byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 12nd byte of section header or 15th byte of pes header
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 11st byte of section header or 14th byte of pes header

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TSP_PTIx_PIDn_FILT_3

Address: Operational Base + offset (0x0910)

Fliter Word 3

Bit	Attr	Reset Value	Description
31:24	RW	0x00	filt_byte_3 Fliter Byte 2 This byte refers to 18th byte of section header or 21st byte of pes header
23:16	RW	0x00	filt_byte_2 Fliter Byte 2 This byte refers to 17th byte of section header or 20th byte of pes header
15:8	RW	0x00	filt_byte_1 Fliter Byte 1 This byte refers to 16th byte of section header or 19th byte of pes header
7:0	RW	0x00	filt_byte_0 Fliter Byte 0 This byte refers to 15th byte of section header or 18th byte of pes header

19.4.3 MMU Register Summary

Name	Offset	Size	Reset Value	Description
TSP_MMU_DTE_ADDR	0x08800	W	0x00000000	MMU current page Table address
TSP_MMU_STATUS	0x08804	W	0x00000018	MMU status register
TSP_MMU_COMMAND	0x08808	W	0x00000000	MMU command register
TSP_MMU_PAGE_FAULT_ADDR	0x0880c	W	0x00000000	MMU logical address of last page fault
TSP_MMU_ZAP_ONE_LINE	0x08810	W	0x00000000	MMU Zap cache line register
TSP_MMU_INT_RAWSTAT	0x08814	W	0x00000000	MMU raw interrupt status register
TSP_MMU_INT_CLEAR	0x08818	W	0x00000000	MMU interrupt clear register
TSP_MMU_INT_MASK	0x0881c	W	0x00000000	MMU interrupt mask register
TSP_MMU_INT_STATUS	0x08820	W	0x00000000	MMU interrupt status register
TSP_MMU_AUTO_GATING	0x08824	W	0x00000001	MMU auto gating
TSP_MMU_MISS_CNT	0x08828	W	0x00000000	MMU miss counter
TSP_MMU_BURST_CNT	0x0882c	W	0x00000000	MMU burst counter

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

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19.4.4 MMU Detail Register Description

TSP_MMU_DTE_ADDR

Address: Operational Base + offset (0x08800)

MMU current page Table address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	MMU_DTE_ADDR MMU dte base addr MMU dte base addr , the address must be 4kb aligned

TSP_MMU_STATUS

Address: Operational Base + offset (0x08804)

MMU status register

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:6	RO	0x00	PAGE_FAULT_BUS_ID Index of master reponsible for last page fault
5	RO	0x0	PAGE_FAULT_IS_WRITE The direction of access for last page fault: 0 = Read 1 = Write
4	RO	0x1	REPLAY_BUFFER_EMPTY The MMU replay buffer is empty
3	RO	0x1	MMU_IDLE The MMU is idle when accesses are being translated and there are no unfinished translated accesses.
2	RO	0x0	STAIL_ACTIVE MMU stall mode currently enabled. The mode is enabled by command
1	RO	0x0	PAGE_FAULT_ACTIVE MMU page fault mode currently enabled . The mode is enabled by command.
0	RO	0x0	PAGING_ENABLED Paging is enabled

TSP_MMU_COMMAND

Address: Operational Base + offset (0x08808)

MMU command register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	WO	0x0	MMU_CMD MMU_CMD. This can be: 0: MMU_ENABLE_PAGING 1: MMU_DISABLE_PAGING 2: MMU_ENABLE_STALL 3: MMU_DISABLE_STALL 4: MMU_ZAP_CACHE 5: MMU_PAGE_FAULT_DONE 6: MMU_FORCE_RESET

TSP_MMU_PAGE_FAULT_ADDR

Address: Operational Base + offset (0x0880c)

MMU logical address of last page fault

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	PAGE_FAULT_ADDR address of last page fault

TSP_MMU_ZAP_ONE_LINE

Address: Operational Base + offset (0x08810)

MMU Zap cache line register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	MMU_ZAP_ONE_LINE address to be invalidated from the page table cache

TSP_MMU_INT_RAWSTAT

Address: Operational Base + offset (0x08814)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error
0	RW	0x0	PAGE_FAULT page fault

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TSP_MMU_INT_CLEAR

Address: Operational Base + offset (0x08818)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	WO	0x0	READ_BUS_ERROR read bus error
0	WO	0x0	PAGE_FAULT page fault

TSP_MMU_INT_MASK

Address: Operational Base + offset (0x0881c)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	READ_BUS_ERROR read bus error enable an interrupt source if the corresponding mask bit is set to 1
0	RW	0x0	PAGE_FAULT page fault enable an interrupt source if the corresponding mask bit is set to 1

TSP_MMU_INT_STATUS

Address: Operational Base + offset (0x08820)

MMU raw interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	READ_BUS_ERROR read bus error
0	RO	0x0	PAGE_FAULT page fault

TSP_MMU_AUTO_GATING

Address: Operational Base + offset (0x08824)

mmu auto gating

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x1	mmu_auto_gating when it is 1'b1, the mmu will auto gating it self

TSP_MMU_mmu_miss_cnt

Address: Operational Base + offset (0x08828)

Register0000 Abstract

Bit	Attr	Reset Value	Description
31	RW	0x0	cnt_ctrl_sel sel the counter for mmu_miss or mmu_real_miss 1'b0: mmu_real_miss 1'b1: mmu_miss When sel 1'b1, an axi command miss may count for several times; when sel 1'b0, an axi command only count for a time
30	RW	0x0	miss_cnt_overflow_flag miss cnt overflow flag
29:0	RW	0x00000000	miss_cnt count for miss AXI command

TSP_MMU_mmu_burst_cnt

Address: Operational Base + offset (0x0882c)

Register0001 Abstract

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	bust_cnt_overflow_flag
29:0	RW	0x00000000	burst_cnt The AXI input burst counter

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19.5 Interface Description

Table 1-1 TSP interface description

Module Pin	Dir	Pad Name	IOMUX Setting
IOMUX0			

Module Pin	Dir	Pad Name	IOMUX Setting
tsp_valid	I	IO_TSPvalidm0_CIFvsyncm0_SDMMC0EXTcmd_SPIclkm2_USB3PHYdebug1_I2S2sclkm1_GPIO3A0vccio6	GRF_GPIO3AL_iomux [2:0] = 3'b001
tsp_fail	I	IO_TSPfail_CIFhref_SDMMC0EXTdet_SPItxdm2_USB3PHYdebug2_I2S2sdom1_GPIO3A1vccio6	GRF_GPIO3AL_iomux [5:3] = 3'b001
tsp_clk	I	IO_TSPclk_CIFclkin_SDMMC0EXTclkout_SPIrxdm2_USB3PHYdebug3_I2S2sdim1_GPIO3A2vccio6	GRF_GPIO3AL_iomux [8:6] = 3'b001
tsp_syncm0	I	IO_TSPsync_CIFclkout_SDMMC0EXTwp_GPIO3A3vccio6	GRF_GPIO3AL_iomux [11:9] = 3'b001
tsp_d0	I	IO_TSPd0_CIFda0_SDMMC0EXTd0_UART1tx_USB3PHYdebug4_GPIO3A4vccio6	GRF_GPIO3AL_iomux [14:12] = 3'b001
tsp_d1	I	IO_TSPd1_CIFdata1_SDMMC0EXTd1_UART1rtsn_USB3PHYdebug5_GPIO3A5vccio6	GRF_GPIO3AH_iomux [2:0] = 3'b001
tsp_d2	I	IO_TSPd2_CIFdata2_SDMMC0EXTd2_UART1rx_USB3PHYdebug6_GPIO3A6vccio6	GRF_GPIO3AH_iomux [5:3] = 3'b001
tsp_d3	I	IO_TSPd3_CIFdata3_SDMMC0EXTd3_UART1ctsn_USB3PHYdebug7_GPIO3A7vccio6	GRF_GPIO3AH_iomux [8:6] = 3'b001
tsp_d4	I	IO_TSPd4_CIFdata4_SPIcsn0m2_I2S2lrcktxm1_USB3PHYdebug8_I2S2lrckrxm1_GPIO3B0vccio6	GRF_GPIO3BL_iomux [2:0] = 3'b001
tsp_d5m0	I	IO_TSPd5m0_CIFdata5m0_GPIO3B1vccio6	GRF_GPIO3BH_iomux [3:2] = 2'b01
tsp_d6m0	I	IO_TSPd6m0_CIFdata6m0_GPIO3B2vccio6	GRF_GPIO3BH_iomux [5:4] = 2'b01
tsp_d7m0	I	IO_TSPd7m0_CIFdata7m0_GPIO3B3vccio6	GRF_GPIO3BH_iomux [7:6] = 2'b01
IOMUX1			
tsp_syncm1	I	IO_I2S1mclk_NOuse0_TSPsyncm1_CIFclkoutm1_GPIO2B7vccio5	GRF_GPIO2BH_iomux [2:0] = 3'b011
tsp_d5m1	I	IO_I2S1lrckrx_NOuse1_TSPd5m1_CIFdata5m1_GPIO2C0vccio5	GRF_GPIO2CL_iomux [2:0] = 3'b011
tsp_d6m1	I	IO_I2S1lrcktx_SPDIFtxm1_TSPd6m1_CIFdata6m1_GPIO2C1vccio5	GRF_GPIO2CL_iomux [5:3] = 3'b011
tsp_d7m1	I	IO_I2S1sclk_PDMclkm0_TSPd7m1_CIFdata7m1_GPIO2C2vccio5	GRF_GPIO2CL_iomux [8:6] = 3'b011

There are two groups of IO for tsp_sync and tsp_data[7:5]. Which group of IO to be used is

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controlled by GRF_IOMUX_CON[8] , this bit has a default value 1'b0. If this bit is set to 1'b1, the second group of IO is selected.

19.6 Application Notes

19.6.1 Overall Operation Sequence

- Enable desired modules to work by writing correspond bit with '1' in TSP_GCFG. Note: it is important to do this step at first, otherwise writing the corresponding registers will not take effect.
- Set up TS configuration by writing corresponding registers.
- Wait for the interrupts to pick up the desired TS packets following the rules detailed in the following section.

19.6.2 TS Source

TS source can be chosen by writing the bit 9 of TSP_PTIx_CTRL(x=0,1), '1' for demodulator, '0' for local memory.

1.TS_IN Interface

Writing bit 10 of TSP_PTIx_CTRL to choose bit ordering, and writing bit [12:11] to choose input TS mode.

TS_IN interface supports 4 input TS stream mode: sync/valid serial mode, sync/valid parallel mode, sync/burst parallel mode, nosync/valid parallel mode.

2.Local Memory

PTI also can process the TS data read from local memory by using LLP DMA mode.

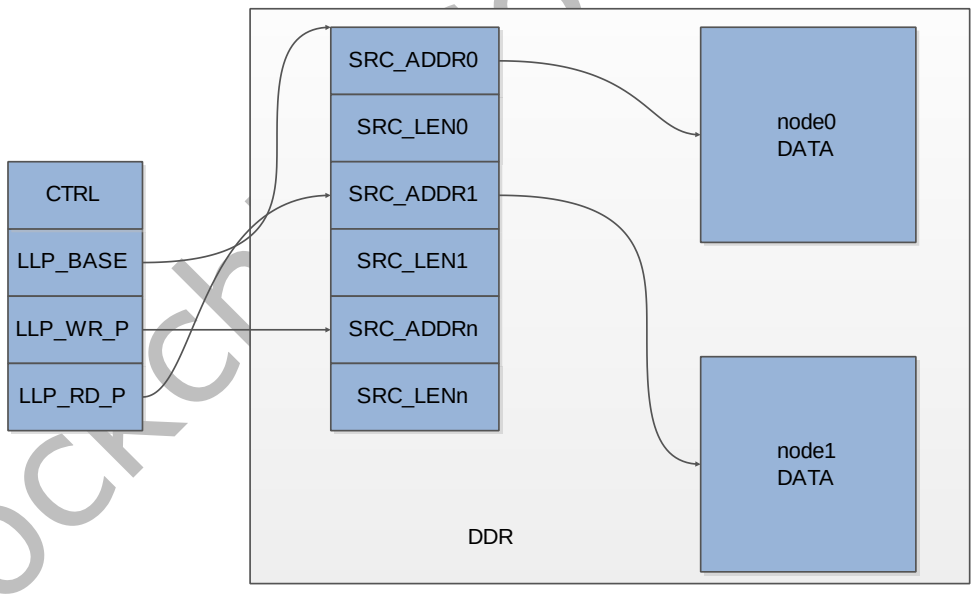

Fig. 1-6LLPaddress architecture

(1) Write PTIx_LLP_BASE with the list base address;
(2) Starting from the list base address, write the list nodes. One list node comprised of two words. The first word describes the TS data base address, the second one describes the length of TS data in unit of word.
(3) Write the PTIx_LLP_WRITE with the number of words that you have written in list memory. Note it is not the number of LLP nodes, so that the number you are writing should be an even one.
(4) Write PTIx_LLP_CFG with the configuration you want. Write the bit 0 with 1 to start LLP DMA. If all the list nodes are written, don't forget to write 1 to bit 3 to tell DMAC that the

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configuration is finished.

Note:

- *The MSB(bit7) of the 8-bit pointer in the PTIx_LLP_Write and PTIx_LLP_Read is used as the flag bit, and remaining 7 bits are used for addressing. Therefore the the pointer is referred to 7-bit space, not 8-bit space, and remember write the pointer with the correct flag bit. For example, if you have configured 63 LLP nodes and then you have to write the 64th LLP node starting from the list base address,*
- *PTIx_LLP_READ informs that how many words has been processed by LLP DMA. An interrupt may be generated when number of the processed words has reach to the threshold set in the PTIx_LLP_CFG.*
- *If you write the PTIx_LLP_Write several times in a complete DMA transaction, it is important to notice the flag bit of PTIx_LLP_Write, and never make the writing pointer catch up with the reading pointer.*

19.6.3 TS Synchronous Operation

Synchronous mode and Bypass mode can be switched by writing bit 15 of TSP_PTIx_CTRL.

In the synchronous mode, 188/192/204 byte TS packets are supported and self-adjusted. Set up locked times in TSP_PTIx_CTRL to inform the successive times of TS packet header detection needs to lock the header of TS packets when in the unlocked mode, and set up unlocked times to informs the successive times of TS packet header error needs to re-lock header of TS packets in the locked mode. It is recommended to use 2-3 as the locked times to quickly and correctly locked the header, and 2-3 as unlocked times to avoid unnecessarily entering into unlocked searching mode.

In the bypass mode, the input TS data will not be re-synchronized and directly fed into the PTI channel.

19.6.4 Descrambling Operation

Descrambler can achieve PES or TS level descrambling which conforms to the CSA v2.0.

Enable the channel you want by writing 1 to bit 0 of TSP_PTIx_PIDn_CTRL (x=0~1, n= 0~64);

Set the desired PID number

Turn on descrambling function by setting 1 to bit 2. If the corresponding CW is available or TS is required to be left undescrambled, CSA_ON bit is set to 0;

Choose corresponding Control Word by setting bit[19:16], and 16 set Control Word are available to be chosen. Don’t forget Control Word should be preprared before the descrambling function is enabled.

Note: If the enabled channel is needed to be disabled, write the CLEAR bit to disabled the channel rather than write ‘0’ to EN bit.

19.6.5 Demux Operation

Refer to TSP_PTIx_PIDn_CFG for Demux operation. The software users should be familiar with the demux knowledge.

Users should create a separate memory buffer to receive the processed data for each desired PID channel, and write the base and top address information of the memory buffer into TSP_PTIx_PIDn_BASE and TSP_PTIx_PIDn respectively. Also initial writing address and reading address, normally the same as base address, are also needed to be written into TSP_PTIx_PIDn_WRITE and TSP_PTIx_PIDn_READ respectively. For ES/PES filter, another

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separate memory needs to be created to store list data, which is used to assist obtaining PES/ES data. List base address, top address, initial writing address and reading address are also needed to write into corresponding registers.

Note:

For channel whose PID channel number larger than 15, the channels can only be used section filter. For others, there is no such limit. They can be configured as section filter, pes filter, es filter or ts filter.
Data memory address boundary should be aligned with word-size, and list memory address boundary should be aligned with word size. If the memory buffer is not larger to store processed data so that writing address reaches the top address, TSP will return to the base address to write data. So fetch the data in time, don't make the writing address catches up with reading address. The list memory buffer has the same issue.

Demux data obtain

- **TS filter**

To obtain TS data and section data, when an desired PID done interrupt is generated, read TSP_PTIx_PIDn_READ firstly to know the address that last reading stops, and then read TSP_PTIx_PIDn_WRITE to know the address that hardware has reached. For ts data, start from the TSP_PTIx_PIDn_READ address to get the TS packet data, and stop at the address you want. However, the ending address should not catch up with writing address. It is recommended to obtain the TS data in the unit of TS packet which is 47-word size. At last, don't forget to write the ending address into TSP_PTIx_PIDn_READ to leave a hint where current reading stops.

B. Section filter

Section filter can run three mode to meet different needs: stop-per-unit; full stop; recycle , update when version number change. The PID done interrupt will be generated after each part of a complete section is processed in the first mode, and the PID done will be generated only after the whole section is completed in the last two modes. In the frist two mode, the PID channel will be disabled after the whole section is completed. In the recycle mode, the channel will remain active and start a new section processing when the version number changes. Section filter also supports 16-byte filtering function, which can assign 1st , 4th to 18th byte to be filtered.

The process to obtain section data is similar to the process for TS data. After a PID done interrupt done is generated, refer to the corresponding PID error status register to check if the section data is correct. Read the frist word of the section start address to know the total length of the section according to the format of section data.

Section Length = {First Word[11:8], First Word[23:16]};

Total Length = Section Length;

Then start to fetch section data according to the total length. Again don't forget to write the stopped address.

C. PES/ES filter

PES filter supports 16-byte filtering function, which can assign 4th, 7th to 21st byte to be filtered.

ES filter supports start code detection, including MPEG2 start code 0x000001b3, 0x00000100, VC-1 start code 0x0000010d, 0x000010f, H264 start code 0x00001.

To obtain the pes/es data, the assistant of list descriptor is needed.

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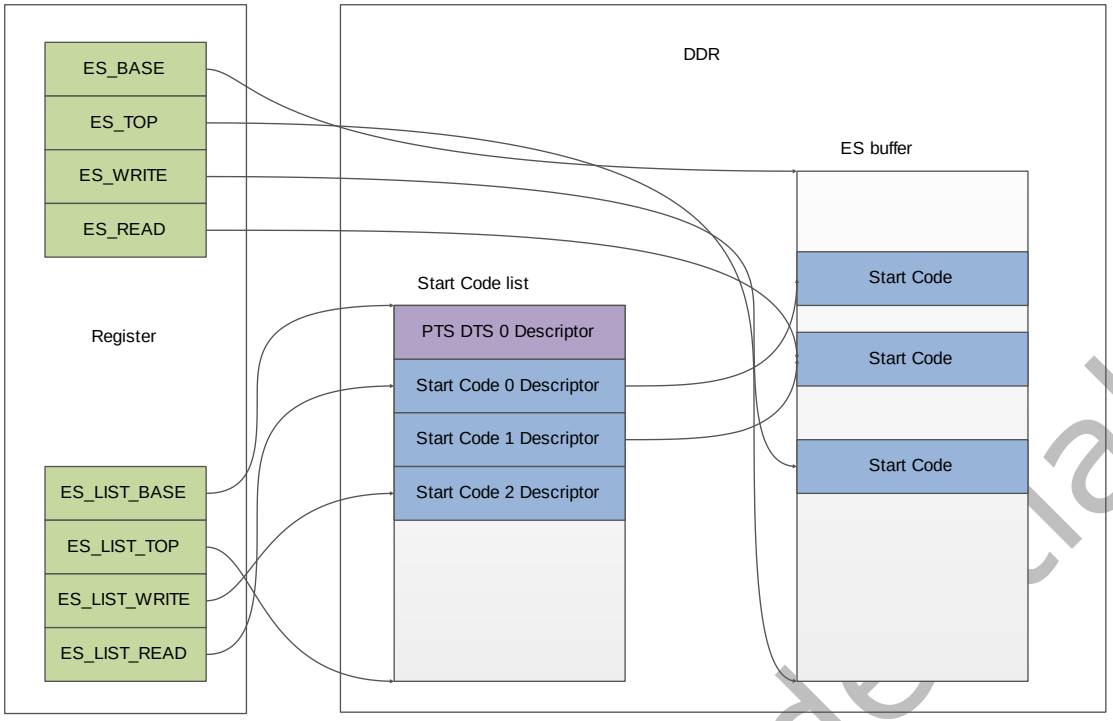

Fig. 1-7LLPmemory architecture

List memory buffer contains descriptors which contains information to obtain es/pes data which are stored in data memory buffer.

The descriptor stored in list memory buffer can be separated into two groups: PTS_DTS Descriptor and Start Code Descriptor. The descriptor is composed by 4 word content, word_0, word_1, word_2 and word_3. The word_x (x means the sequence number in a descriptor, and they are stored in the memory in sequence order). The format of the 4 words are listed as follows:

start code descriptor

Word_0:

Word_0[29:28] indicates the attributes of the bytes of the pointed word. 2'b00 means the whole word belongs to the new ES/PES packet; 2'b01 means that word[7:0] belongs to the previous packet, and the remaining bytes belong to the new packet; 2'b10 means means that word[15:0] belongs to the previous packet, and the remaining bytes belong to the new packet; 2'b11 means 'b10 means means that word[23:0] belongs to the previous packet, and the remaining bytes belong to the new packet. This pointed word is the word where start code starts, word_2 describes the location of start code.

Word_0[27:24] is equal to 0x0 in the start code descriptor. Users can used to tell two kinds of descriptor.

If the video type is H.264, word_0[23:8] means first_mb_in slice, and word_0 means nal_nuit_type.

Word_1:

the start code of stream.

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Word_2:

DDR offset address in the DDR of the word where the start code is located.

Word_3:

0x0

PTS_DTS Descriptor

Word_0:

Word_0[29:28]: the same as start code descriptor

Word_0[27:24]: 0x1 in PTS_DTS descriptor.

Word_0[3] : PTS[32];

Word_0[2] : DTS[32];

Word_0[1:0] : pts_dts_flag;

Word_1:

DDR offset address of the word that valid data starts.

Word_2:

PTS[31:0]

Word_3

DTS[31:0]

To obtain PES data or ES data when start code detection is disabled, use PTS_DTS descriptor. To obtain ES data when start code detection is enabled, use start code descriptor.

When a PID done interrupt is generated, make sure there is no corresponding PID error generated. Read the TSP_PTIx_LISTn_READ to know the list reading address in the last time. Start from here, read the 4-word descriptor one by one to know the offset of the packets. Refer to the offset in the DDR where in the data memory buffer to obtain data. Finally write TSP_PTIx_LISTn_READ and TSP_PTIx_PIDn_READ with corresponding reading address.

19.6.6 PVR

PVR module provide you with the function to record the programs you want. The 4 sources can be assigned with PVR, and they are the same as TS out interface.

Assign the PVR length and PVR address, and then configure TSP_PVR_CTRL to start PVR module. If you want to stop PVR function during recording, write '1' to STOP bit (bit 0) to to TSP_PVR_CTRL to stop it. Remember to take care of the status of PVR_ON bit of TSP_GFCG when programming the PVR-related registers.

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19.6.7 PCR extraction

PCR extraction can be enabled by configure PTIx_PCRn_CTRL. Then if the PID-matched TS data contain PCR field, the 33-bit PCR_base field will be written corresponding PTIx_PCRn_H and PTIx_PCRn_L registers. An interrupt will be asserted if PCR interrupt is enabled.

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Chapter 22 USB2.0 Host

20.1 Overview

USB2.0 host controller supports fully USB2.0 functions withone EHCI host controller and one OHCI host controller, and each host controller has one USB port. OHCI host controller only supports full-speed and low-speed mode and is used for full-speed devices and low-speed devices. EHCI only supports high-speed mode and is used for high-speed devices. OHCI host controller and EHCI host controller shares the same USB port, EHCI host controller will auto select the owner (OHCI or EHCI) of this USB port depending on the speed mode of attached devices, when selecting OHCI as owner, OHCI host controller will serve for the attached device; when selecting EHCI as owner, EHCI host controller will serve for the attached device.
USB2.0 Host Controller supports the following features:

- Compatible Specifications
  - Universal Serial Bus Specification, Revision 2.0
  - Enhanced Host Controller Interface Specification (EHCI), Revision 1.0
  - Open Host Controller Interface Specification (OHCI), Revision 1.0a
- Support High-speed (480Mbps), Full-speed (12Mbps) and Low-speed (1.5Mbps)

20.2 Block Diagram

USB2.0 Host Controller comprises with:

- EHCI Host Controller: Perform High-speed transactions
- OHCI Host Controller: Perform full/low-speed transactions
- Port Routing Control: Select EHCI Host Controller or OHCI Host Controller

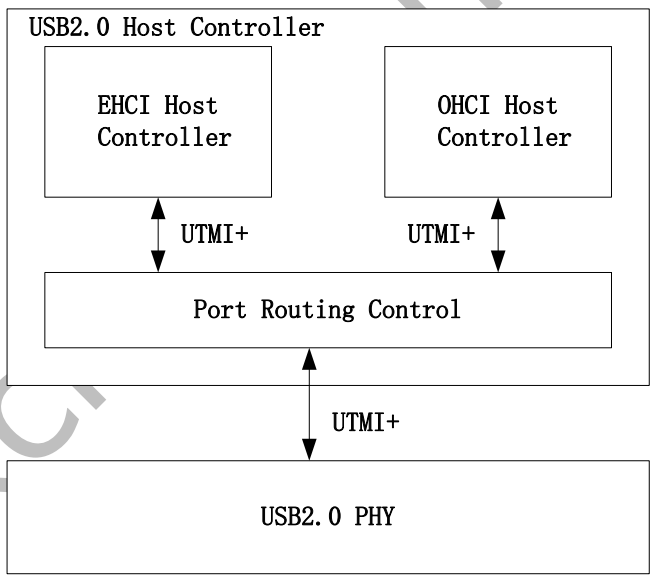

Fig.22-1USB2.0 Host Controller Block Diagram

20.3 Function Description

20.3.1 EHCI Host Controller

It performs descriptors and data read or write from or to system memory and packs or un-pack USB transactions from or to UTMI+ interface defined in EHCI specification for high-speed data transmission.

20.3.2 OHCI Host Controller

It performs descriptors and data read/write from/to system memory and packs or un-pack USB transactions from or to UTMI+ interface defined in OHCI specification for full-speed or low-speed data transmission.

20.3.3 Port Routing Control

As part of logic in the EHCI host controller, it is used to auto-select EHCI or OHCI host

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controller to serve the attached device depending on the speed of the attached device.

20.4 Register Description

20.4.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

Table 22-1USB2.0 Host Controller Address Mapping

Base Address[16]	Device	Address Length	Offset Address Range
1'b0	EHCI	64K BYTE	0x00000 ~ 0x0ffff
1'b1	OHCI	64K BYTE	0x10000 ~ 0x1ffff

EHCI and OHCI register definitions, please refer to Enhanced Host Controller Interface Specification (EHCI), Revision 1.0 and Open Host Controller Interface Specification (OHCI), Revision 1.0a.

20.5 Interface Description

Table 22-2 USB2.0 PHY Interface Description

Module Pin	Direction	Pad Name	Descriptions
USB0ID	I	IO_USB0_ID	USB2.0 PHY OTG Port ID, left unused for TypeC
USB0PN	I/O	IO_USB0_PN	USB2.0 PHY OTG Port PN
USB0PP	I/O	IO_USB0_PP	USB2.0 PHY OTG Port PP
VBUS	I	IO_USB0_VBUS	USB2.0 PHY OTG Port VBUS
USB1PN	I/O	IO_USB1_PN	USB2.0 PHY Host Port PN
USB1PP	I/O	IO_USB1_PP	USB2.0 PHY Host Port PP
USBRBIAS	I/O	IO_USB0_RBIAS	USB2.0 PHY Shared RBIAS

20.6 Application Notes

20.6.1 Special Setting

Set USB2.0 host controller master secure setting (see Chapter SGRF) before initialization.

20.6.2 Program flow

Please refer to Enhanced Host Controller Interface Specification (EHCI), Revision 1.0 and Open Host Controller Interface Specification (OHCI), Revision 1.0a.

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Chapter 23 USB OTG2.0

21.1 Overview

USB OTG 2.0 is a Dual-Role Device controller, which supports both device andhost functions and is fully compliant with OTG Supplement to USB2.0specificationandsupport high-speed(480Mbps),full-speed(12Mbps),low-speed(1.5Mbps) transfer.

USB OTG 2.0 isoptimized for portable electronic devices, point-to-point applications(no hub, directconnection to device) and multi-point applications to devices.

21.1.1 Features

- Compliant with the OTG Supplement to the USB2.0 Specification
- Operates in High-Speed and Full-Speed mode
- Support 9 channels in host mode
- 9 Device mode endpoints in addition to control endpoint 0, 4 in, 3 out and 2 IN/OUT
- Built-in one 1024x35 bits FIFO
- Internal DMA with scatter/gather function
- Supports packet-based, dynamic FIFO memory allocation for endpoints for flexible, efficient use of RAM
- Support dynamic FIFO sizing

21.2 Block Diagram

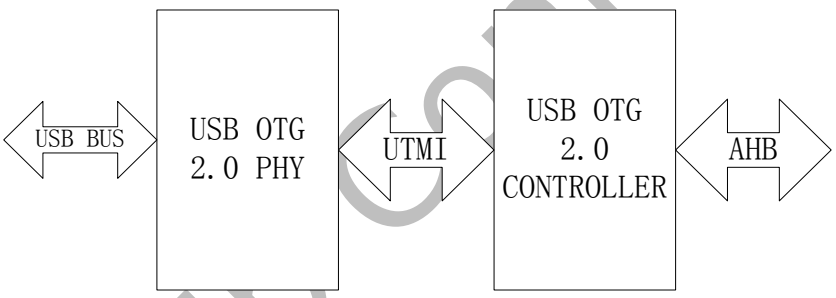

Fig.23-1 USB OTG 2.0 Architecture

The Fig shows the architecture of USB OTG 2.0. It is broken up into two separate units: USB OTG 2.0 controller and USB OTG 2.0 PHY. The two units are interconnected with UTMI interface.

21.2.1 USB OTG 2.0 Controller Function

The USB OTG 2.0 Controller controlsSIE(Serial Interface Engine) logic, the endpointlogic, the channel logic and the internal DMA logic.

The SIElogic contains the USB PID and address recognition logic, and other sequencingand state machine logic to handle USB packets and transactions. Generally the SIE Logicis required for any USB implementation while the number and types of endpoints will vary as function ofapplication and performance requirements.

The endpoint logic contains the endpointspecific logic: endpoint number recognition, FIFOs and FIFO control, etc.

The channel Logic contains the channel tasks schedule, FIFOs and FIFO control, etc.

The internal DMA logic controls data transaction between system memory and USB FIFOs.

21.2.2 USB OTG 2.0 PHY Function

The USB OTG 2.0 PHY handles the low level USB protocol and signaling. This includes features such as dataserialization and de-serialization, bit stuffing and clock recovery and

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synchronization. The primary focus ofthis block is to shift the clock domain of the data from the USB 2.0 rate to the frequency of UTMI clock which is 30MHz.

21.2.3 UTMI Interface

- Transmit

Transmit must be asserted to enable any transmissions.

The USB OTG2.0 CONTROLLER asserts TXValid to begin a transmission and negates TXValid to end a transmission.After the USB OTG2.0 CONTROLLER asserts TXValid it can assume that the transmission has started when it detects TXReady asserted.

The USB OTG2.0 CONTROLLER assumes that the USB OTG2.0 PHYhas consumed a data byte if TXReady and TXValid are asserted.

The USB OTG2.0 CONTROLLER must have valid packet information (PID) asserted on the DataIn bus coincident with theassertion of TXValid. Depending on the USB OTG2.0 PHYimplementation, TXReady may be asserted by theTransmit State Machine as soon as one CLK after the assertion of TXValid.TXValid and TXReady are sampled on the rising edge of CLK.

The Transmit State Machine does NOT automatically generate Packet ID's (PIDs) or CRC. Whentransmitting, the USB OTG2.0 CONTROLLER is always expected to present a PID as the first byte of the data stream and ifappropriate, CRC as the last bytes of the data stream.

The USB OTG2.0 CONTROLLER must use LineState to verify a Bus Idle condition before asserting TXValid in the TX Waitstate.

The state of TXReady in the TX Wait and Send SYNC states is undefined. An MTU implementationmay prepare for the next transmission immediately after the Send EOP state and assert TXReady inthe TX Wait state. An MTU implementation may also assert TXReady in the Send SYNC state. Thefirst assertion of TXReady is Macrocell implementation dependent. The USB OTG2.0 CONTROLLER must prepare DataIn forthe first byte to be transmitted before asserting TXValid.

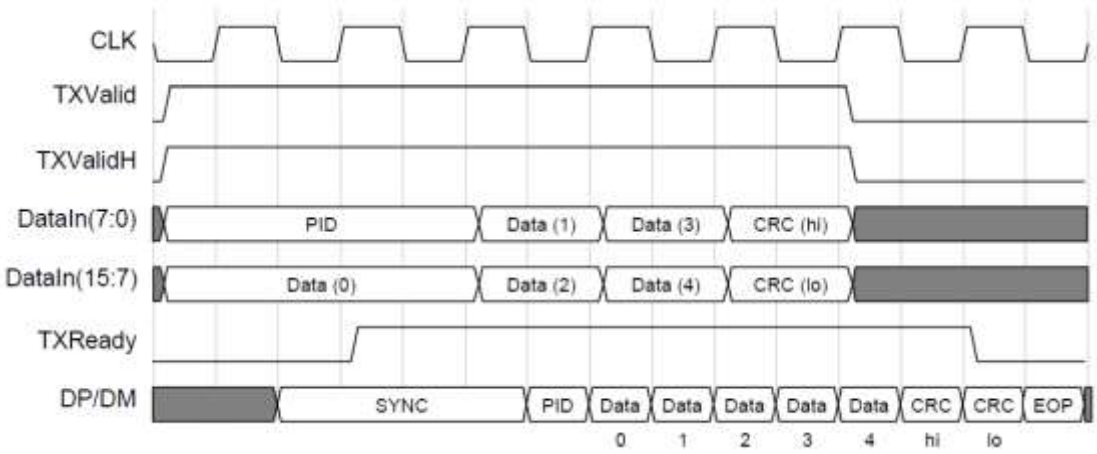

Fig.23-2 UTMI interface – Transmit timing for a data packe

- Receive

RXActive and RXValid are sampled on the rising edge of CLK.

In the RX Wait state the receiver is always looking for SYNC.

The USB OTG 2.0 PHY asserts RXActive when SYNC is detected (Strip SYNC state).

The USB OTG 2.0 PHY negates RXActive when an EOP is detected (Strip EOP state).

When RxActive is asserted, RXValid will be asserted if the RX Holding Register is full.

RXValid will be negated if the RX Holding Register was not loaded during the previous byte time.

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This will occur if 8 stuffed bits have been accumulated.

The USB OTG2.0 Controller must be ready to consume a data byte if RXActive and RXValid are asserted (RX Data state).

In FS mode, if a bit stuff error is detected then Receive State Machine will negate RXActive andRXValid, and return to the RXWait state.

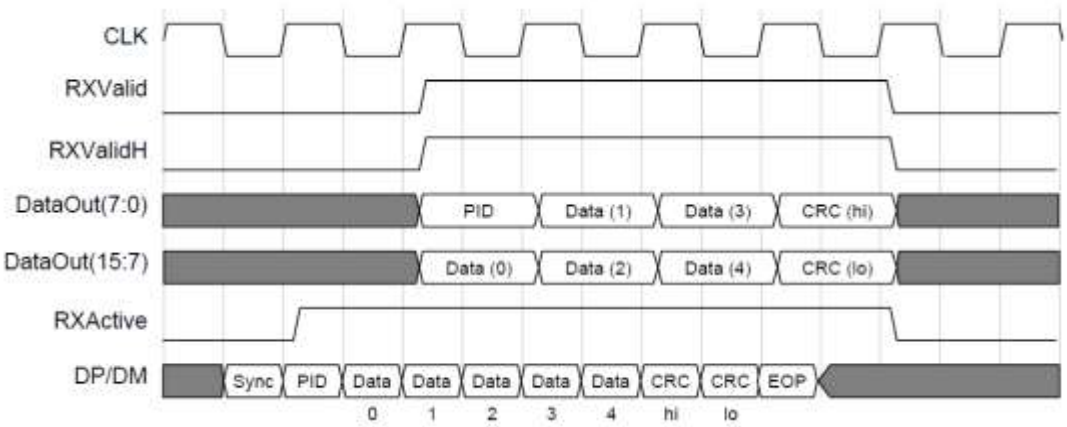

Fig.23-3 UTMI interface – Receive timing for a data packet

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21.3 USB OTG2.0 Controller

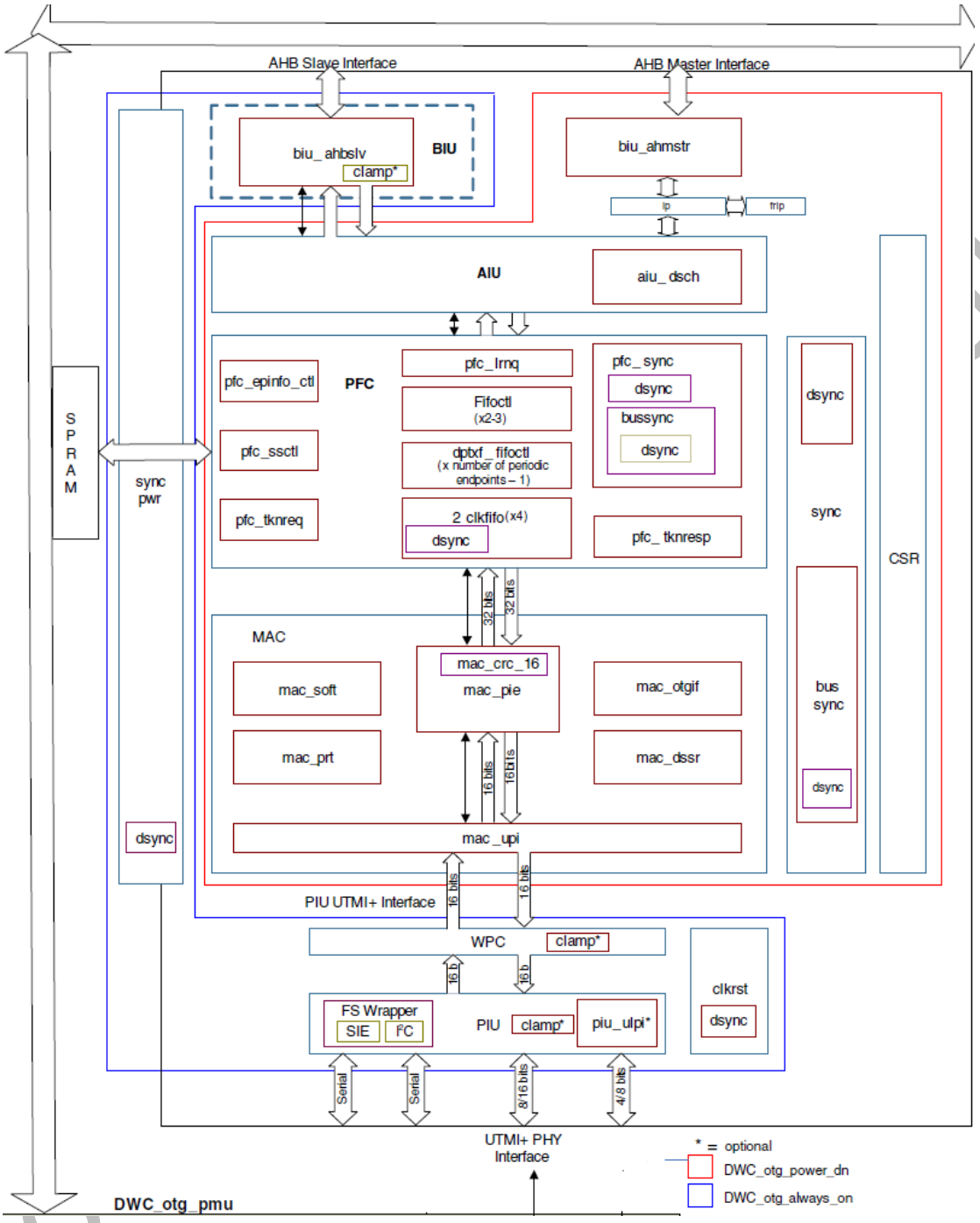

Fig.23-4 USB OTG2.0 Controller Architecture

1). AHB Slave Bus Interface Unit (BIUS)

The AHB Slave interface unit converts AHB cycles to CSR write/read, Data-FIFO read/write, and DFIFO push/pop signals.

2) Control and Status Registers (CSR)

The CSR block resides in the AHB clock domain, and contains all registers except the Power and Clock Gating Control Register (PCGCCTL) and bits 31:29 of the Core Interrupt register (GINTSTS).

3) Application Interface Unit (AIU)

The application Interface Unit (AIU) consists of the following interfaces:

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AHB Master

AHB Slave

Packet FIFO Controller

Control and Status registers

4). DMA Scheduler (DSCH)

This block is used only in DMA mode. It controls the transfer of data packets between the system memory and the USB OTG 2.0 Controller for both Internal and External DMA.

5).Packet FIFO Controller (PFC)

Several FIFOs are used in Device and Host modes to store data inside the core before transmitting it on either the AHB or the USB. PFC connects the Data FIFO interface to an industry-standard, single-port synchronous SRAM. Address, write data, and control outputs are driven late by the USB OTG 2.0 Controller, but in time to meet the SRAM setup requirements. Input read data is expected late from the SPRAM and registered inside the core before being used.

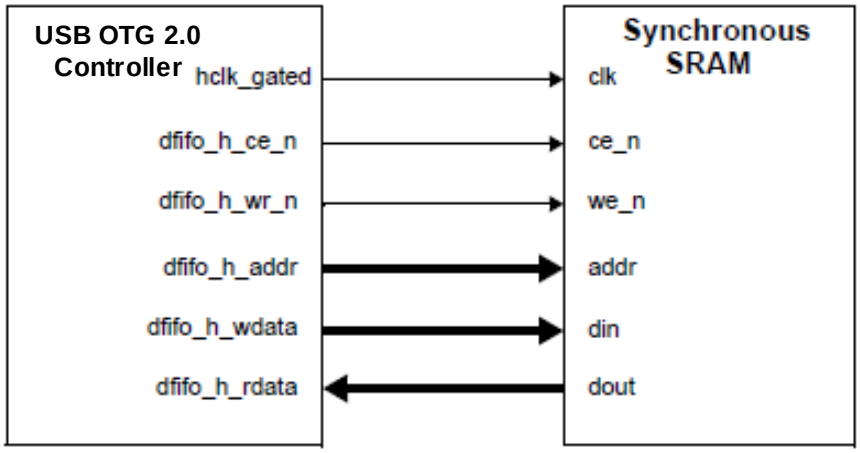

Fig.23-5 DFIFO single-port synchronous SRAM interface

6).Media Access Controller(MAC)

The Media Access Controller (MAC) module handles USB transactions, and device, host, and OTG protocols.

7) PHY Interface Unit (PIU)

The core uses 16-bit UTMI+ Interface.

8) Wakeup and Power Controller (WPC)

When the USB is suspended or the session is not valid, the PHY is driven into Suspend mode and the PHY clock is stopped to reduce PHY and the core power consumption. To reduce power consumption further, the core also supports AHB clock gating.

21.3.1 Host Architecture

The host uses one transmit FIFO for all non-periodic OUT transactions and one transmit FIFO for all periodic OUT transactions. These transmit FIFOs are used as transmit buffers to hold the data (payload of the transmit packet) to be transmitted over USB.

The host pipes the USB transactions through Request queues (one for periodic and one for non-periodic). Eachentryin the Request - queue holds the IN or OUT channel number along with other information to perform a transaction on the USB. The order in which the requests

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are written into the queue determines the sequence of transactions on the USB. The host processes the periodic Request queue first, followed by the non-periodic Request queue, at the beginning of each (micro)frame.

The host uses one Receive-FIFO for all periodic and non-periodic transactions. The FIFO is used as a Receive-buffer to hold the received data (payload of the received packet) from the USB until it is transferred to the system memory. The status of each packet received also goes into the FIFO. The status entry holds the IN channel number along with other information, such as received byte count and validity status, to perform a transaction on the AHB.

21.3.2 Device Architecture

The core uses Dedicated Transmit FIFO Operation. In this mode, there are individualtransmit FIFOs for each IN endpoint.

The OTG device uses a single receive FIFO to receive the data for all the OUT endpoints. The receive FIFO holds the status of the received data packet, such as byte count, data PID and the validity of the received data. The DMA or the application reads the data out of the receive FIFO as it is received.

21.3.3 FIFO Mapping

- FIFO mapping in Host mode.

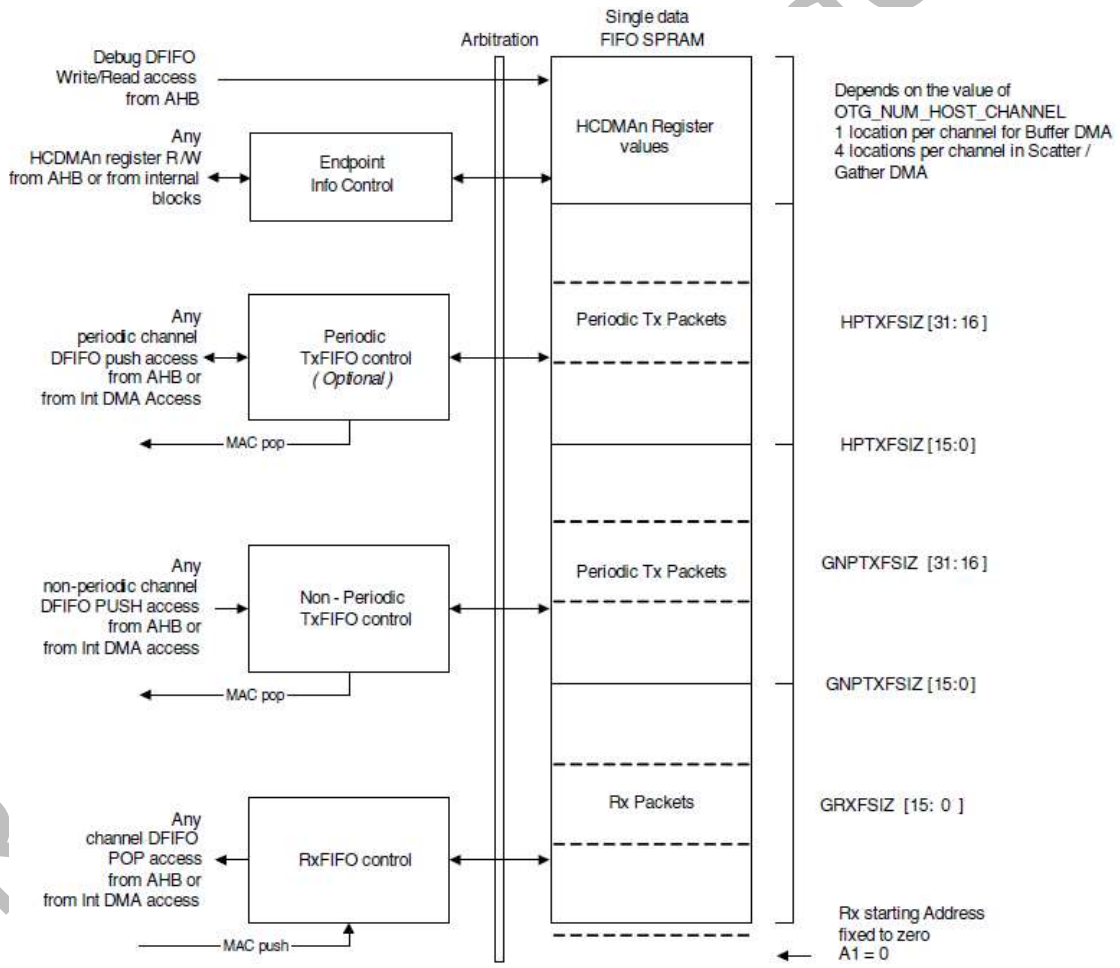

Fig 23-6 USB OTG 2.0 Controller host mode FIFO address mapping

Note: When the device is operating in Internal DMA mode, the last locations of the SPRAM are used to store the DMAADDR values for each channel.

- FIFO mapping in Device mode.

When the device is operating in non-Descriptor Internal DMA mode, the last locations of the SPRAM are used to store the DMAADDR values for each channel. When the device is operating

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in Descriptor mode, then the last locations of the SPRAM store the Base Descriptor address, Current Descriptor address, Current Buffer address, and status quadlet information for each endpoint direction.

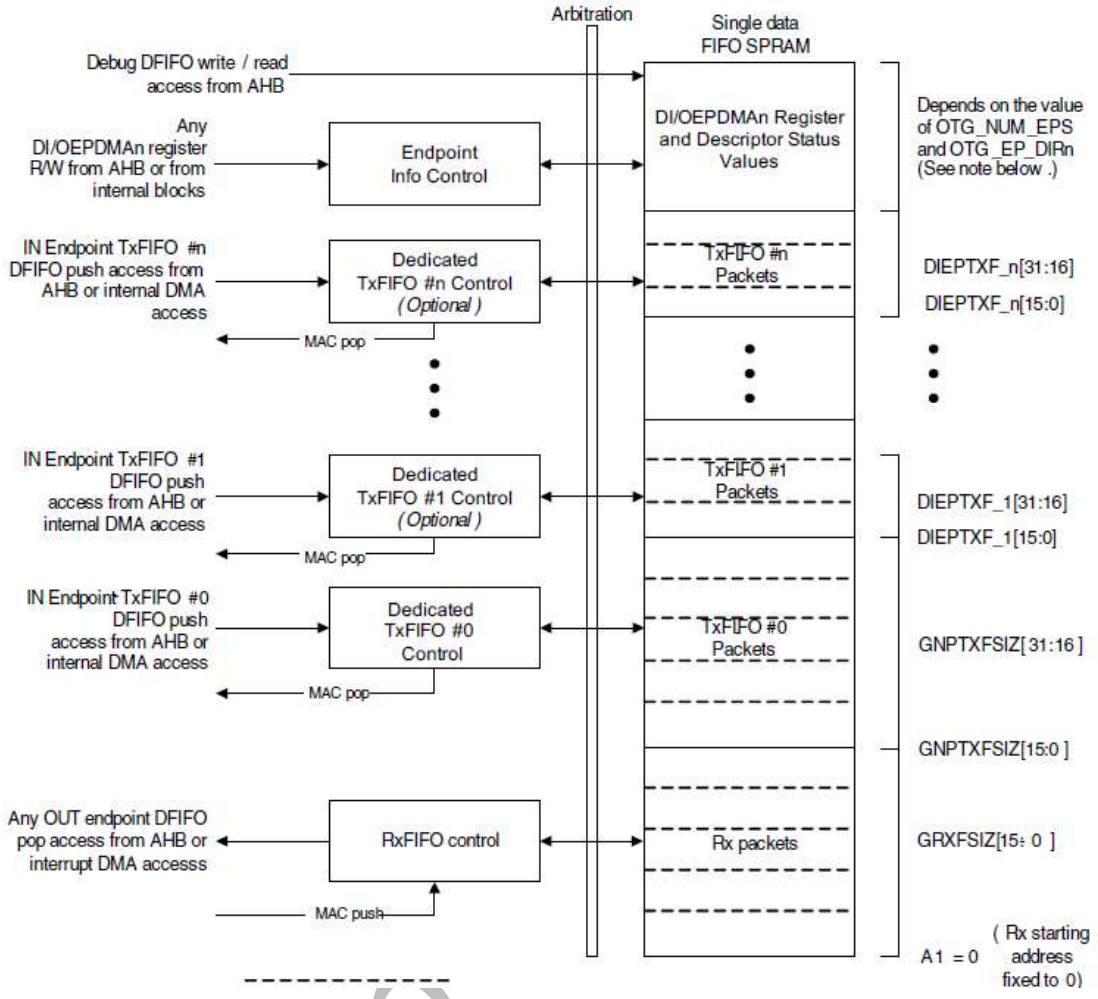

Fig23-7 USB OTG 2.0 Controller device mode FIFO address mapping

Note: When the device is operating in non-Scatter Gather Internal DMA mode, the last locations of the SPRAM are used to store the DMAADDR values for each Endpoint (1 location per endpoint).When the device is operating in Scatter Gather mode, then the last locations of the SPRAM store the Base Descriptor address, Current Descriptor address, Current Buffer address, and status quadlet information for each endpoint direction (4 locations per Endpoint). If an Endpoint is bidirectional,then 4 locations will be used for IN, and another 4 for OUT).

21.4 USB OTG2.0 PHY

USB PHY supports dual OTG ports'functions and is fully compliant with USB2.0 specification, andsupport High-speed(480Mbps),full-speed(12Mbps),low-speed(1.5Mbps) transfer. It provides a complete on-chip transceiver physical solution with ESD protection. A minimum number ofexternal components are needed, which include a 45 ohm resistor for resistance calibration purpose. Its feature contains:

- provide dual UTMI ports
- OTG0 Support UART Bypass Function
- Fully compliant with USB specifications Rev 2.0,1.1 HOST/Device and OTG V1.2.
- Supports 480Mbps (HS), 12Mbps (FS)&1.5Mbps(LS) serial data transmission
- Supports low latency hub mode with 40 bit timeround trip delay

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- 8 bit or 16 bit UTMI interface compliant withUTMI+ specification level 3 Rev 1.
- Loop back BIST mode supported
- Built-in I/O and ESD structure
- On-die self-calibrated HS/FS/LS termination
- 12MHz crystal oscillator with integratedphase-locked loop (PLL) oscillator
- Dual 3.3V / 1.2V supply

21.4.1 Block Diagram

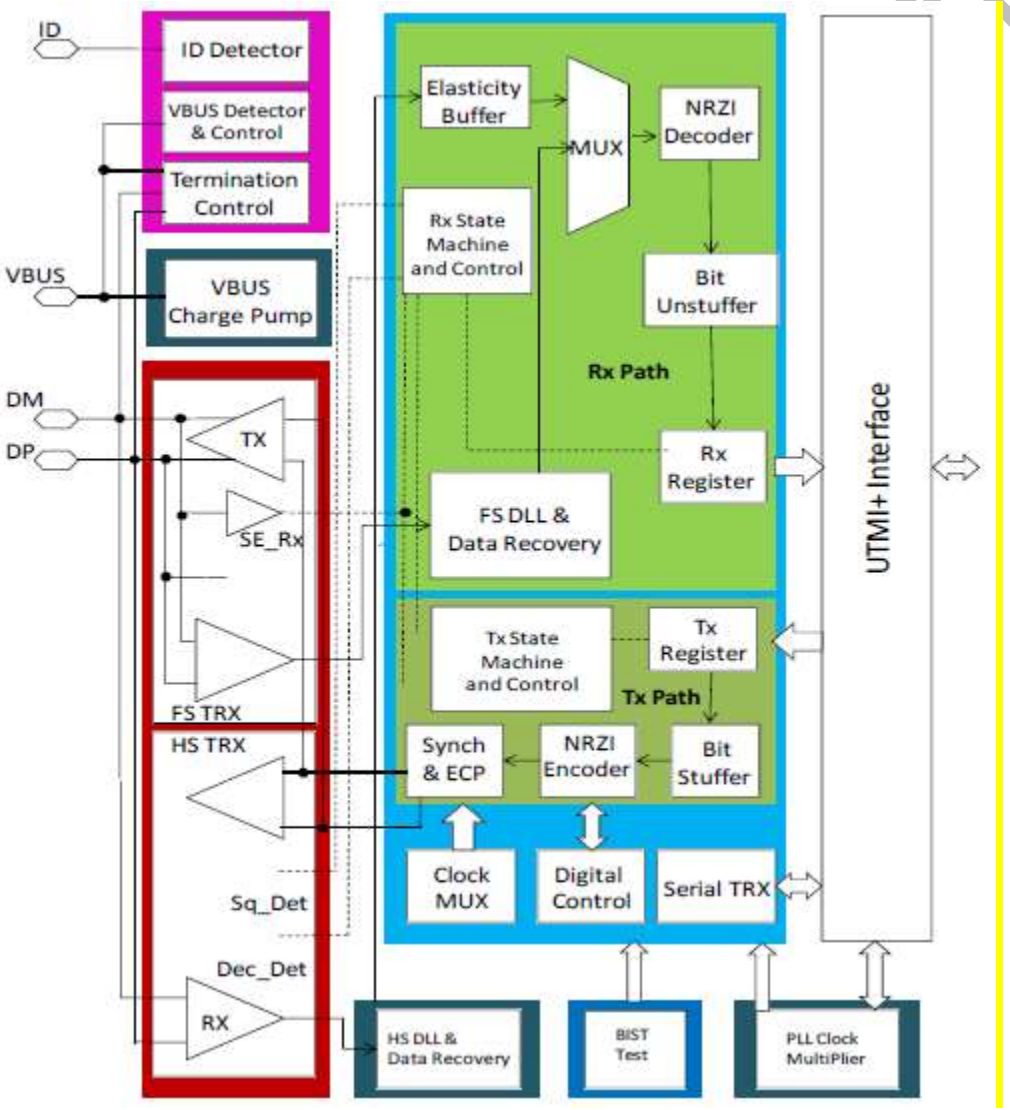

Fig.23-8usb phy architecture

HS AFE

The HS AFEcontains the low-level analog circuitry, and also the HS differential data transmitter and receiver, toperform HS transmission envelope detection and host disconnection detection. It works in HS mode only.

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HS Transmit driver

The HS transmit driver is active only when transmit is asserted. In HS transceiver enabled mode and thetransmit state machine has data to send, the XCVR selects input. Data from transmit data path will bedriven onto the DP/DM signal lines when enabled.

HS Differential Receiver

When enabled, received HS data will be multiplexed through the receive data path to the receive shift andhold registers. It is active only in HS mode.

Transmission envelope detector (Squelch detector)

When the amplitude of the differential signal at a receiver's inputs falls below the squelch threshold, theenvelope detector will indicate the invalid data. It must indicate squelch when the signal drops below 100mV differential amplitude, and also, it must indicate that the line is not in the squelch state when the signalexceeds 150mV differential amplitude.

Disconnection envelope detector

In host mode, this envelope detector is active to detect the high speed disconnect state on the line.Disconnection must be indicated when the amplitude of the differential signal at the downstream facingdriver's connector is more than 625 mV, and it must not be indicated when the signal amplitude is less than525 mV.

FS/LS AFE

In FS or LS mode, the FS/LS AFE is active to send and receive the FS or LS data on the USB bus. Also itsupports the reset, suspend and resume detection through the data line single ended receivers.

FS/LS Transmitter

The FS/LHS transmitter is active only when transmit is asserted. In FS or LS transceiver enabled mode andthe transmit state machine has data to send, the XCVR selects input. Data from transmit data path will bedriven onto the DP/DM signal lines when enabled.

FS/LS Differential Receiver

When enabled, received FS or LS data will be multiplexed through the receive data path to the receive shiftand hold registers. It is active only in FS or LS modes.

Single ended receivers

The single ended receivers are used for low-speed and full-speed signaling detection.

Digital Core TX Path

The digital core TX path has some blocks responsible for SYNC and EOP generation, data encoding, bitstuffing and data serialization. And meanwhile, also a TX state machine is involved to manage thecommunication with the controller.

TX Shift/Hold Register

The TX shift/Hold register module consists of an 8-bit primary shift register for parallel/serial conversionand 8-bit hold register used to buffer the next data to serialize. This module is responsible for readingparallel data from the parallel application bus interface upon command and serializing for transmissionover USB.

Bit stuffer

To ensure adequate signal transitions, when sending a packet on USB, a bit stuffer is employed by thetransmitter. A '0' has to be inserted after every six consecutive ones in the

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data stream before the data inNRZI encoded, to force a transition in the NRZI data stream.

NRZI Encoder

The High speed, Full speed or low speed serial transmitted data are encoded by The NRZI encoder. As astate transition, a '0' is encoded, and as no state transition, a '1' is encoded.

Transmit state machine

The communication between the controller and the PHY in TX path is controlled by the transmit statemachine, which synchronizes the Data with the Sync and the EOP, and also supports the LS, FS and HSModes.

Digital Core RX Path

The digital core RX path includes blocks responsible for SYNC and EOP detection and stripping, datadecoding, bit un-stuffing and data de-serialization. Also a RX state machine is involved to manage thecommunication with the controller. FS/LS data and clock is recovered in this section.

Elasticity buffer

To compensate for differences between transmitting and receiving clocks, the Elasticity Buffer is used tosynchronize the HS extracted data with the PLL internal clock.

Mux

The Mux block allows the data from the HS or FS/LS receivers to be routed to the shared receive logic.The state of the Mux is determined by the Xcvr Select input.

NRZI Decoder

The NRZI is responsible for decoding the High speed or Full speed received NRZI encoded data. Achange in level is decoded as '0' and no change in level is decoded as '1'.

Bit Un-stuffer

The Bit Un-stuffer not only recognizes the stuffed bits from the data stream, but also discards them. Alsoit detects bit stuff error, which is interpreted as HS EOP.

RX Shift/Hold Register

This module de-serializes received data and transmits 8-bit parallel data to the application bus interface.It consists of an 8-bit primary shit register for serial to parallel conversion and an 8-bit hold register forbuffering the last de-serialized data byte.

Receiver state machine

The receiver state machine controls the communication between the controller and the PHY in the RXpath, strips the SYNC and the EOP from the Data and supports the LS, FS and HS Modes.

PLL Clock Multiplier

This module is composed of the off-chip crystal and the on-chip clock multiplier. It generates theappropriate internal clocks for the UTM and the CLK output signal. All data transfer signals aresynchronized with the CLK signal.

External Crystal

The external crystal is composed of a precise resonance frequency crystal and a crystal oscillator. It isoptional to have this crystal oscillator integrated on-chip or have it off-chip. This crystal/crystal oscillatorprovides a very precise clock of 12 MHz with deviation of ±100 ppm. The oscillator is not a part of thePHY, but external.

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Clock Multiplier

The UTM interface is described as an un-directional/bi-directional 8-bit/16-bit parallel interface and theCLK signal is a 60/30 MHz signal. All data transfer signals should be synchronized with the CLK signal.CLK usable signal is internally implemented which blocks any transitions of CLK until it is usable.Meanwhile, the clock multiplier provides another three clocks in addition to the CLK signal. That is a480 MHz and 7.5 MHz clock signals.

Clock MUX

The Clock Multiplexer supplies both the transmitter and receiver paths with the adequate bit clockdepending on the XcvrSelect signal and to ensure smooth clock switching. It also includes clock gatingand power-down features.

Control Logic Block

This block is responsible for controlling, enabling and disabling the different blocks in the system.

OTG Circuitry （optional）

With the OTG circuitry, the system has the capability to dynamically switch between host and peripheral,enable dual role device behavior and point-to-point communication. The OTG circuitry functions asVBUS generation and detection. Both ID detection and terminations control are implemented in it.

ID Detector （optional）

To provide the ID signal that is used to indicate the state of the ID pin on the USB mini receptacle. Thispin makes it able to determine which kind of plug is connected and to confirm if the device default stateis A device or B device.

VBUS Detector and termination control

The VBUS detector is a set of comparators, functions to monitor and sense the voltage on USB buspower line. For VBUS signaling and discharging, VBUS pull up and pull-down resistors are alsoimplemented.

Automatic Test Functions

Loop-back test to address all IP components.

In loop-back test mode, all transmitted data packets are received back in an internal loop to check IPfunctional integrity. There are some digital components that cannot be tested with the scan techniquedue to the high-speed nature of the digital part. To be regarded as a good idea, Loop-back allows testingfull design paths at speed. It should complement the testing suite for digital core to achieve the highestcoverage possible. According to the UTMI specification Section 5.18, version 1.05 Page 34, the 8 bits un-directional data bus can be implemented as 8 bits bi-directional one. This implementation will hinderthe loop-back test functionality.

21.5 UART BYPASS FUNCITON

When in UART bypass mode,UART2 is connect to USB interface; Otherwise, UART2 use normal UART interface。

Signal	CONNECT	I/O	Description
BYPASSDMDATA0	uart2_sout	I	Data for DM0 Transmitter Digital Bypass

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BYPASSDMEN0	grf_uoc1_con4[11]	I	DM0 Transmitter Digital Bypass Enable
BYPASSSEL0	grf_uoc1_con4[13]	I	Transmitter Digital Bypass mode Enable
FSVPLUS0	uart2_sin	O	Single-Ended D- Indicator The controller signal indicates the state of the DP during normal operation or UART data reception
OTGDISABLE0	grf_uoc1_con4[10]	I	1'b1: OTG0 disable; 1'b0: OTG0 normal mode
COMMONONN	grf_uoc0_con5[11]	I	Common Block Power-Down Control This signal controls the power-down signals in PLL blocks when the USBPHY is in Suspend Mode. 1: PLL blocks are powered down. 0: PLL blocks remain powered This signal is a strapping option that must be set prior to a power-on reset and remain static during normal operation.

Note: USB OTG2.0 PHY support UART Bypass Function.

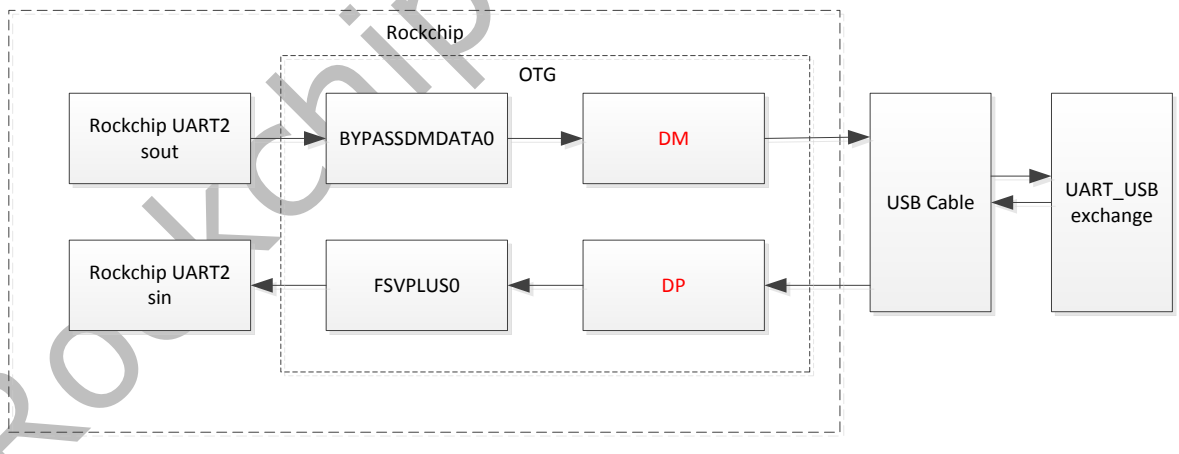

Fig 23-9UART Application

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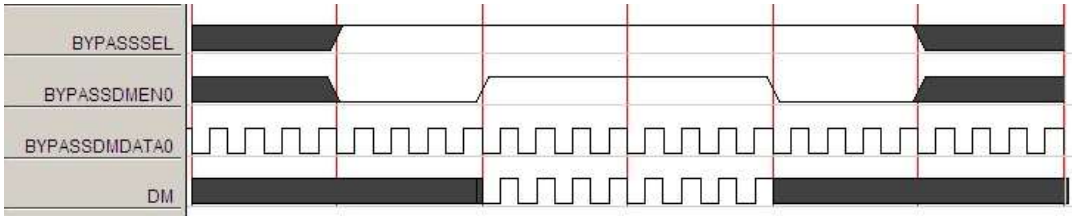

Fig.23-10UART Timing Sequence

To use UART and Autoresume functions:

1. Disable the OTG block by setting OTGDISABLE0 to 1’b1.

2. Disable the pull-up resistance on the DP line by setting OPMODE0[1:0] to 2’b01.

3. To ensure that the XO, Bias, and PLL blocks are powered down in Suspend mode, set COMMONONN to 1’b1.

4. Place the USB PHY in Suspend mode by setting SUSPENDM0 to 1’b0.

5. Set BYPASSSEL0 to 1’b1.

6. To transmit data, controls BYPASSDMEN0, and BYPASSDMDATA0.

To receive data, monitor FSVPLUS0.

To return to normal operating mode:

1. Ensure that there is no activity on the USB.

2. Set BYPASSSEL0 to 1'b0.

3. Set SUSPENDM0 to 1’b1.Resume the USB PHY.

4. Set COMMONONN to 1’b0.

5. set OTGDISABLE0 to 1'b0.

21.6 Register Description

21.6.1 Register Summary

Name	Offset	Size	Reset Value	Description
USBOTG_GOTGCTL	0x0000	W	0x00000000	Control and Status Register
USBOTG_GOTGINT	0x0004	W	0x00000000	Interrupt Register
USBOTG_GAHBCFG	0x0008	W	0x00000000	AHB Configuration Register
USBOTG_GUSBCFG	0x000c	W	0x00001400	USB Configuration Register
USBOTG_GRSTCTL	0x0010	W	0x80000000	Reset Register
USBOTG_GINTSTS	0x0014	W	0x00000000	Interrupt Register
USBOTG_GINTMSK	0x0018	W	0x00000000	Interrupt Mask Register
USBOTG_GRXSTSR	0x001c	W	0x00000000	Receive Status Debug Read Register
USBOTG_GRXSTSP	0x0020	W	0x00000000	Receive Status Read and Pop Register
USBOTG_GRXFSIZ	0x0024	W	0x00000000	Receive FIFO Size Register

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Name	Offset	Size	Reset Value	Description
USBOTG_GNPTXFSIZ	0x0028	W	0x00000000	Non-Periodic Transmit FIFO Size Register
USBOTG_GNPTXSTS	0x002c	W	0x00000000	Non-Periodic Transmit FIFO/Queue Status Register
USBOTG_GI2CCTL	0x0030	W	0x11000000	I2C Address Register
USBOTG_GPVNDCTL	0x0034	W	0x00000000	PHY Vendor Control Register
USBOTG_GGPIO	0x0038	W	0x00000000	General Purpost Input/Output Register
USBOTG_GUID	0x003c	W	0x00000000	User ID Register
USBOTG_GSNPSID	0x0040	W	0x00004f54	Core ID Register
USBOTG_GHWCFG1	0x0044	W	0x00000000	User HW Config1 Register
USBOTG_GHWCFG2	0x0048	W	0x00000000	User HW Config2 Register
USBOTG_GHWCFG3	0x004c	W	0x00000000	User HW Config3 Register
USBOTG_GHWCFG4	0x0050	W	0x00000000	User HW Config4 Register
USBOTG_GLPMCFG	0x0054	W	0x00000000	Core LPM Configuration Register
USBOTG_GPWRDN	0x0058	W	0x00000000	Global Power Down Register
USBOTG_GDFIFOCFG	0x005c	W	0x00000000	Global DFIFO Software Config Register
USBOTG_GADPCTL	0x0060	W	0x00000000	ADP Timer,Control and Status Register
USBOTG_HPTXFSIZ	0x0100	W	0x00000000	Host Periodic Transmit FIFO Size Register
USBOTG_DIEPTXFn	0x0104	W	0x00000000	Device Periodic Transmit FIFO-n Size Register
USBOTG_HCFG	0x0400	W	0x00000000	Host Configuration Register
USBOTG_HFIR	0x0404	W	0x00000000	Host Frame Interval Register
USBOTG_HFNUM	0x0408	W	0x0000ffff	Host Frame Number/Frame Time Remaining Register
USBOTG_HPTXSTS	0x0410	W	0x00000000	Host Periodic Transmit FIFO/Queue Status Register
USBOTG_HAINT	0x0414	W	0x00000000	Host All Channels Interrupt Register
USBOTG_HAINTMSK	0x0418	W	0x00000000	Host All Channels Interrupt Mask Register
USBOTG_HPRT	0x0440	W	0x00000000	Host Port Control and Status Register
USBOTG_HCCHARn	0x0500	W	0x00000000	Host Channel-n Characteristics Register
USBOTG_HCSPLTn	0x0504	W	0x00000000	Host Channel-n Split Control Register
USBOTG_HCINTn	0x0508	W	0x00000000	Host Channel-n Interrupt Register
USBOTG_HCINTMSKn	0x050c	W	0x00000000	Host Channel-n Interrupt Mask Register

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Name	Offset	Size	Reset Value	Description
USBOTG_HCTSIZn	0x0510	W	0x00000000	Host Channel-n Transfer Size Register
USBOTG_HCDMAn	0x0514	W	0x00000000	Host Channel-n DMA Address Register
USBOTG_HCDMABn	0x051c	W	0x00000000	Host Channel-n DMA Buffer Address Register
USBOTG_DCFG	0x0800	W	0x08200000	Device Configuration Register
USBOTG_DCTL	0x0804	W	0x00002000	Device Control Register
USBOTG_DSTS	0x0808	W	0x00000000	Device Status Register
USBOTG_DIEPMSK	0x0810	W	0x00000000	Device IN Endpoint common interrupt mask register
USBOTG_DOEPMSK	0x0814	W	0x00000000	Device OUT Endpoint common interrupt mask register
USBOTG_DAINT	0x0818	W	0x00000000	Device All Endpoints interrupt register
USBOTG_DAINTMSK	0x081c	W	0x00000000	Device All Endpoint interrupt mask register
USBOTG_DTKNQR1	0x0820	W	0x00000000	Device IN token sequence learning queue read register1
USBOTG_DTKNQR2	0x0824	W	0x00000000	Device IN token sequence learning queue read register2
USBOTG_DVBUSDIS	0x0828	W	0x00000b8f	Device VBUS discharge time register
USBOTG_DVBUSPULSE	0x082c	W	0x00000000	Device VBUS Pulsing Timer Register
USBOTG_DTHRCTL	0x0830	W	0x08100020	Device Threshold Control Register
USBOTG_DIEPEMPMSK	0x0834	W	0x00000000	Device IN endpoint FIFO empty interrupt mask register
USBOTG_DEACHINT	0x0838	W	0x00000000	Device each endpoint interrupt register
USBOTG_DEACHINTMSK	0x083c	W	0x00000000	Device each endpoint interrupt register mask
USBOTG_DIEPEACHMSKn	0x0840	W	0x00000000	Device each IN endpoint -n interrupt Register
USBOTG_DOEPEACHMSKn	0x0880	W	0x00000000	Device each out endpoint-n interrupt register
USBOTG_DIEPCTL0	0x0900	W	0x00008000	Device control IN endpoint 0 control register
USBOTG_DIEPINTn	0x0908	W	0x00000000	Device Endpoint-n Interrupt Register
USBOTG_DIEPTSIZn	0x0910	W	0x00000000	Device endpoint n transfer size register
USBOTG_DIEPDMAn	0x0914	W	0x00000000	Device endpoint-n DMA address register

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Name	Offset	Size	Reset Value	Description
USBOTG_DTXFSTSn	0x0918	W	0x00000000	Device IN endpoint transmit FIFO status register
USBOTG_DIEPDMABn	0x091c	W	0x00000000	Device endpoint-n DMA buffer address register
USBOTG_DIEPCTLn	0x0920	W	0x00000000	Device endpoint-n control register
USBOTG_DOEPCTL0	0x0b00	W	0x00000000	Device control OUT endpoint 0 control register
USBOTG_DOEPINTn	0x0b08	W	0x00000000	Device endpoint-n control register
USBOTG_DOEPTSIZn	0x0b10	W	0x00000000	Device endpoint n transfer size register
USBOTG_DOEPDMAn	0x0b14	W	0x00000000	Device Endpoint-n DMA Address Register
USBOTG_DOEPDMABn	0x0b1c	W	0x00000000	Device endpoint-n DMA buffer address register
USBOTG_DOEPCTLn	0x0b20	W	0x00000000	Device endpoint-n control register
USBOTG_PCGCR	0x0b24	W	0x200b8000	Power and clock gating control register
USBOTG_EPBUF0	0x1000	W	0x00000000	Device endpoint 0 / host out channel 0 address
USBOTG_EPBUF1	0x2000	W	0x00000000	Device endpoint 1 / host out channel 1 address
USBOTG_EPBUF2	0x3000	W	0x00000000	Device endpoint 2 / host out channel 2 address
USBOTG_EPBUF3	0x4000	W	0x00000000	Device endpoint 3 / host out channel 3 address
USBOTG_EPBUF4	0x5000	W	0x00000000	Device endpoint 4 / host out channel 4 address
USBOTG_EPBUF5	0x6000	W	0x00000000	Device endpoint 5 / host out channel 5 address
USBOTG_EPBUF6	0x7000	W	0x00000000	Device endpoint 6 / host out channel 6 address
USBOTG_EPBUF7	0x8000	W	0x00000000	Device endpoint 7 / host out channel 7 address

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

21.6.2 Detail Register Description

USBOTG_GOTGCTL

Address: Operational Base + offset (0x0000)

Control and Status Register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27	RW	0x0	ChirpEn Chirp on enable This bit when programmed to 1'b1 results in the core asserting chirp_on before sending an actual Chirp "K" signal on USB. This bit is present only if OTG_BC_SUPPORT = 1. If OTG_BC_SUPPORT != 1, this bit is a reserved bit.
26:22	RO	0x00	MultValidBc Multi Valued ID pin Battery Charger ACA inputs in the following order: Bit 26 - rid_float. Bit 25 - rid_gnd Bit 24 - rid_a Bit 23 - rid_b Bit 22 - rid_c These bits are present only if OTG_BC_SUPPORT = 1. Otherwise, these bits are reserved and will read 5'h0.
21	RO	0x0	reserved
20	RW	0x0	OTGVer OTG version Indicates the OTG revision. 0: OTG Version 1.3. In this version the core supports Data line pulsing and VBus pulsing for SRP. 1: OTG Version 2.0. In this version the core supports only Data line pulsing for SRP.
19	RO	0x0	BSesVld B-session valid Indicates the Device mode transceiver status. 0: B-session is not valid. 1: B-session is valid. In OTG mode, you can use this bit to determine if the device is connected or disconnected. Note: If you do not enabled OTG features (such as SRP and HNP), the read reset value will be 1.The vbus assigns the values internally for non-SRP or non-HNP configurations.
18	RO	0x0	ASesVld A-session valid Indicates the Host mode transceiver status. 0: A-session is not valid 1: A-session is valid Note: If you do not enabled OTG features (such as SRP and HNP), the read reset value will be 1.The vbus assigns the values internally for non-SRP or non-HNP configurations.

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Bit	Attr	Reset Value	Description
17	RO	0x0	DbnTime Long/short debounce time Indicates the debounce time of a detected connection. 0: Long debounce time, used for physical connections (100 ms + 2.5 us) 1: Short debounce time, used for soft connections (2.5 us)
16	RO	0x0	ConIDSts Connector ID Status Indicates the connector ID status on a connect event. 0: The core is in A-Device mode 1: The core is in B-Device mode
15:12	RO	0x0	reserved
11	RW	0x0	DevHNPEn Device HNP Enable The application sets this bit when it successfully receives a SetFeature. SetHNPEnable command from the connected USB host. 0: HNP is not enabled in the application 1: HNP is enabled in the application
10	RW	0x0	HstSetHNPEn Host set HNP enable The application sets this bit when it has successfully enabled HNP (using the SetFeature.SetHNPEnable command) on the connected device. 0: Host Set HNP is not enabled 1: Host Set HNP is enabled
9	RW	0x0	HNPReq HNP request The application sets this bit to initiate an HNP request to the connected USB host. The application can clear this bit by writing a 0 when the Host Negotiation Success Status Change bit in the OTG Interrupt register (GOTGINT.HstNegSucStsChng) is set. The core clears this bit when the HstNegSucStsChng bit is cleared. 0: No HNP request 1: HNP request
8	RO	0x0	HstNegScs Host Negotiation Success The core sets this bit when host negotiation is successful. The core clears this bit when the HNP Request (HNPReq) bit in this register is set. 0: Host negotiation failure 1: Host negotiation success
7:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	SesReq Session Request The application sets this bit to initiate a session request on the USB. The application can clear this bit by writing a 0 when the Host Negotiation Success Status Change bit in the OTG Interrupt register (GOTGINT.HstNegSucStsChng) is set. The core clears this bit when the HstNegSucStsChng bit is cleared. If you use the USB 1.1 Full-Speed Serial Transceiver interface to initiate the session request, the application must wait until the VBUS discharges to 0.2 V, after the B-Session Valid bit in this register (GOTGCTL.BSesVld) is cleared. This discharge time varies between different PHYs and can be obtained from the PHY vendor. 0: No session request 1: Session request
0	RO	0x0	SesReqScs Session Request Success The core sets this bit when a session request initiation is successful. 0: Session request failure 1: Session request success

USBOTG_GOTGINT

Address: Operational Base + offset (0x0004)

Interrupt Register

Bit	Attr	Reset Value	Description
31:21	RO	0x0	reserved
20	W1C	0x0	MultiValueChg Multi-Valued input changed This bit when set indicates that there is a change in the value of at least one ACA pin value. This bit is present only if OTG_BC_SUPPORT = 1, otherwise it is reserved.
19	W1C	0x0	DbnceDone Debounce Done The core sets this bit when the debounce is completed after the device connected. The application can start driving USB reset after seeing this interrupt. This bit is only valid when the HNP Capable or SRP Capable bit is set in the Core USB Configuration register (GUSBCFG.HNPCap or GUSBCFG.SRPCap, respectively).

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Bit	Attr	Reset Value	Description
18	W1C	0x0	ADevTOUTChg A-Device Timeout Change The core sets this bit to indicate that the A-device has timed out while waiting for the B-device to connect.
17	W1C	0x0	HstNegDet Host Negotiation Detected The core sets this bit when it detects a host negotiation request on the USB
16:10	RO	0x0	reserved
9	W1C	0x0	HstNegSucStsChng Host Negotiation Success Status Change The core sets this bit on the success or failure of a USB host negotiation request. The application must read the Host Negotiation Success bit of the OTG Control and Status register (GOTGCTL.HstNegScs) to check for success or failure
8	W1C	0x0	SesReqSucStsChng Session Request Success Status Change The core sets this bit on the success or failure of a session request.The application must read the Session Request Success bit in the OTG Control and Status register (GOTGCTL.SesReqScs) to check for success or failure.
7:3	RO	0x0	reserved
2	W1C	0x0	SesEndDet Session End Detected The core sets this bit when the utmisrp_bvalid signal is deasserted
1:0	RO	0x0	reserved

USBOTG_GAHBCFG

Address: Operational Base + offset (0x0008)

AHB Configuration Register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22	RW	0x0	NotiAllDmaWrit Notify All Dma Write Transactions This bit is programmed to enable the System DMA Done functionality for all the DMA write Transactions corresponding to the Channel/Endpoint. This bit is valid only when GAHBCFG.RemMemSupp is set to 1. GAHBCFG.NotiAllDmaWrit = 1. HSOTG core asserts int_dma_req for all the DMA write transactions on the AHB interface along with int_dma_done, chep_last_transact and chep_number signal informations. The core waits for sys_dma_done signal for all the DMA write transactions in order to complete the transfer of a particular Channel/Endpoint. GAHBCFG.NotiAllDmaWrit = 0. HSOTG core asserts int_dma_req signal only for the last transaction of DMA write transfer corresponding to a particular Channel/Endpoint. Similarly, the core waits for sys_dma_done signal only for that transaction of DMA write to complete the transfer of a particular Channel/Endpoint.
21	RW	0x0	RemMemSupp Remote Memory Support This bit is programmed to enable the functionality to wait for the system DMA Done Signal for the DMA Write Transfers. GAHBCFG.RemMemSupp=1. The int_dma_req output signal is asserted when HSOTG DMA starts write transfer to the external memory. When the core is done with the Transfers it asserts int_dma_done signal to flag the completion of DMA writes from HSOTG. The core then waits for sys_dma_done signal from the system to proceed further and complete the Data Transfer corresponding to a particular Channel/Endpoint. GAHBCFG.RemMemSupp=0. The int_dma_req and int_dma_done signals are not asserted and the core proceeds with the assertion of the XferComp interrupt as soon as the DMA write transfer is done at the HSOTG Core Boundary and it does not wait for the sys_dma_done signal to complete the DATA transfers.
20:9	RO	0x0	reserved
8	RW	0x0	PTxFEmpLvl Periodic TxFIFO Empty Level Indicates when the Periodic TxFIFO Empty Interrupt bit in the Core Interrupt register (GINTSTS.PTxFEmp) is triggered. This bit is used only in Slave mode. 0: GINTSTS.PTxFEmp interrupt indicates that the Periodic TxFIFO is half empty 1: GINTSTS.PTxFEmp interrupt indicates that the Periodic TxFIFO is completely empty

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Bit	Attr	Reset Value	Description
7	RW	0x0	NPTxFEmpLvl Non-Periodic TxFIFO Empty Level This bit is used only in Slave mode. In host mode and with Shared FIFO with device mode, this bit indicates when the Non-Periodic TxFIFO Empty Interrupt bit in the Core Interrupt register GINTSTS.NPTxFEmp) is triggered. With dedicated FIFO in device mode, this bit indicates when IN endpoint Transmit FIFO empty interrupt (DIEPINTn.TxFEmp) is triggered. Host mode and with Shared FIFO with device mode: 1'b0: GINTSTS.NPTxFEmp interrupt indicates that the Non-Periodic TxFIFO is half empty 1'b1: GINTSTS.NPTxFEmp interrupt indicates that the Non-Periodic TxFIFO is completely empty Dedicated FIFO in device mode: 1'b0: DIEPINTn.TxFEmp interrupt indicates that the IN Endpoint TxFIFO is half empty 1'b1: DIEPINTn.TxFEmp interrupt indicates that the IN Endpoint TxFIFO is completely empty
6	RO	0x0	reserved
5	RW	0x0	DMAEn DMA Enable 0: Core operates in Slave mode 1: Core operates in a DMA mode This bit is always 0 when Slave-Only mode has been selected.

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Bit	Attr	Reset Value	Description
4:1	RW	0x0	HBstLen Burst Length/Type This field is used in both External and Internal DMA modes. In External DMA mode, these bits appear on dma_burst[3:0] ports, External DMA Mode defines the DMA burst length in terms of 32-bit words: 4'b0000: 1 word 4'b0001: 4 words 4'b0010: 8 words 4'b0011: 16 words 4'b0100: 32 words 4'b0101: 64 words 4'b0110: 128 words 4'b0111: 256 words Others: Reserved Internal DMA Mode AHB Master burst type: 4'b0000: Single 4'b0001: INCR 4'b0011: INCR4 4'b0101: INCR8 4'b0111: INCR16 Others: Reserved
0	RW	0x0	GlblIntrMsk Global Interrupt Mask The application uses this bit to mask or unmask the interrupt line assertion by itself. Irrespective of this bit's setting, the interrupt status registers are updated by the core. 1'b0: Mask the interrupt assertion to the application. 1'b1: Unmask the interrupt assertion to the application.

USBOTG_GUSBCFG

Address: Operational Base + offset (0x000c)

USB Configuration Register

Bit	Attr	Reset Value	Description
31	RW	0x0	CorruptTxpacket Corrupt Tx packet This bit is for debug purposes only. Never set this bit to 1.

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Bit	Attr	Reset Value	Description
30	RW	0x0	ForceDevMode Force Device Mode Writing a 1 to this bit forces the core to device mode irrespective of utmiotg_iddig input pin. 1'b0: Normal Mode 1'b1: Force Device Mode After setting the force bit, the application must wait at least 25 ms before the change to take effect. When the simulation is in scale down mode, waiting for 500 us is sufficient. This bit is valid only when OTG_MODE = 0, 1 or 2. In all other cases, this bit reads 0.
29	RW	0x0	ForceHstMode Force Host Mode Writing a 1 to this bit forces the core to host mode irrespective of utmiotg_iddig input pin. 1'b0: Normal Mode 1'b1: Force Host Mode After setting the force bit, the application must wait at least 25 ms before the change to take effect. When the simulation is in scale down mode, waiting for 500 us is sufficient. This bit is valid only when OTG_MODE =0, 1 or 2. In allother cases, this bit reads 0.
28	RW	0x0	TxEndDelay Tx End Delay Writing a 1 to this bit enables the TxEndDelay timers in the core. 1'b0: Normal mode 1'b1: Introduce Tx end delay timers
27	RW	0x0	IC_USBTrafCtl IC_USB TrafficPullRemove Control When this bit is set, pullup/pulldown resistors are detached from the USB during traffic signaling.This bit is valid only when configuration parameter OTG_ENABLE_IC_USB = 1 and register field GUSBCFG.IC_USBCap is set to 1.
26	RW	0x0	IC_USBCap IC_USB-Capable The application uses this bit to control the IC_USB capabilities. 1'b0: IC_USB PHY Interface is not selected. 1'b1: IC_USB PHY Interface is selected. This bit is writable only if OTG_ENABLE_IC_USB=1 and OTG_FSPHY_INTERFACE!=0.The reset value depends on the configuration parameter OTG_SELECT_IC_USB when OTG_ENABLE_IC_USB = 1. In all other cases, this bit is set to 1'b0 and the bit is read only.

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Bit	Attr	Reset Value	Description
25	RW	0x0	ULPIIfDis ULPI Interface Protect Disable Controls circuitry built into the PHY for protecting the ULPI interface when the link tri-states STP and data. Any pull-ups or pull-downs employed by this feature can be disabled. Please refer to the ULPI Specification for more detail. 1'b0: Enables the interface protect circuit 1'b1: Disables the interface protect circuit
24	RW	0x0	IndPassThrough Indicator Pass Through Controls whether the Complement Output is qualified with the Internal Vbus Valid comparator before being used in the Vbus State in the RX CMD. Please refer to the ULPI Specification for more detail. 1'b0: Complement Output signal is qualified with the Internal VbusValid comparator. 1'b1: Complement Output signal is not qualified with the Internal VbusValid comparator.
23	RW	0x0	IndComple Indicator Complement Controls the PHY to invert the ExternalVbusIndicator input signal,generating the Complement Output. Please refer to the ULPI Specification for more detail 1'b0: PHY does not invert ExternalVbusIndicator signal 1'b1: PHY does invert ExternalVbusIndicator signal
22	RW	0x0	TermSelDLPulse TermSel DLine Pulsing Selection This bit selects utmi_termselect to drive data line pulse during SRP. 1'b0: Data line pulsing using utmi_txvalid (default). 1'b1: Data line pulsing using utmi_termsel.
21	RW	0x0	ULPIExtVbusIndicator ULPI External VBUS Indicator This bit indicates to the ULPI PHY to use an external VBUS over-current indicator. 1'b0: PHY uses internal VBUS valid comparator. 1'b1: PHY uses external VBUS valid comparator. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
20	RW	0x0	ULPIExtVbusDrv ULPI External VBUS Drive This bit selects between internal or external supply to drive 5V on VBUS,in ULPI PHY. 1'b0: PHY drives VBUS using internal charge pump (default). 1'b1: PHY drives VBUS using external supply. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)

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Bit	Attr	Reset Value	Description
19	RW	0x0	ULPIClkSusM ULPI Clock SuspendM This bit sets the ClockSuspendM bit in the Interface Control register on the ULPI PHY. This bit applies only in serial or carkit modes. 1'b0: PHY powers down internal clock during suspend. 1'b1: PHY does not power down internal clock. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
18	RW	0x0	ULPIAutoRes ULPI Auto Resume This bit sets the AutoResume bit in the Interface Control register on the ULPI PHY. 1'b0: PHY does not use AutoResume feature. 1'b1: PHY uses AutoResume feature. (Valid only when RTL parameter OTG_HSPHY_INTERFACE = 2 or 3)
17	RW	0x0	ULPIFsLs ULPI FS/LS Select The application uses this bit to select the FS/LS serial interface for the ULPI PHY. This bit is valid only when the FS serial transceiver is selected on the ULPI PHY. 1'b0: ULPI interface 1'b1: ULPI FS/LS serial interface (Valid only when RTL parameters OTG_HSPHY_INTERFACE = 2 or 3 and OTG_FSPHY_INTERFACE = 1, 2, or 3)
16	RW	0x0	OtgI2CSel UTMIFS or I2C Interface Select The application uses this bit to select the I2C interface. 1'b0: UTMI USB 1.1 Full-Speed interface for OTG signals 1'b1: I2C interface for OTG signals This bit is writable only if I2C and UTMIFS were specified for Enable I2C Interface (parameter OTG_I2C_INTERFACE = 2). Otherwise, reads return 0.
15	RW	0x0	PhyLPwrClkSel PHY Low-Power Clock Select Selects either 480-MHz or 48-MHz (low-power) PHY mode. In FS and LS modes, the PHY can usually operate on a 48-MHz clock to save power. 1'b0: 480-MHz Internal PLL clock 1'b1: 48-MHz External Clock In 480 MHz mode, the UTMI interface operates at either 60 or 30-MHz, depending upon whether 8- or 16-bit data width is selected. In 48-MHz mode, the UTMI interface operates at 48 MHz in FS and LS modes. This bit drives the utmi_fsls_low_power core output signal, and is valid only for UTMI+ PHYs.
14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13:10	RW	0x5	USBTrdTim USB Turnaround Time Sets the turnaround time in PHY clocks. Specifies the response time for a MAC request to the Packet FIFO Controller (PFC) to fetch data from the DFIF(SPRAM). This must be programmed to 4'h5: When the MAC interface is 16-bit UTMI+. 4'h9: When the MAC interface is 8-bit UTMI+. Note: The values above are calculated for the minimum AHB frequency of 30MHz. USB turnaround time is critical for certification where long cables and 5-Hubs are used, so if you need the AHB to run at less than 30 MHz, and if USB turnaround time is not critical, these bits can be programmed to a larger value.
9	RW	0x0	HNPCap HNP-Capable The application uses this bit to control the DWC_otg core's HNP capabilities. 0: HNP capability is not enabled. 1: HNP capability is enabled. This bit is writable only if an HNP mode was specified for Mode of Operation (parameter OTG_MODE). Otherwise, reads return 0.
8	RW	0x0	SRPCap SRP-Capable The application uses this bit to control the DWC_otg core SRP capabilities. If the core operates as a non-SRP-capable B-device, it cannot request the connected A-device (host) to activate VBUS and start a session. 0: SRP capability is not enabled. 1: SRP capability is enabled. This bit is writable only if an SRP mode was specified for Mode of Operation in coreConsultant (parameter OTG_MODE). Otherwise, reads return 0.
7	RW	0x0	DDRSel ULPI DDR Select The application uses this bit to select a Single Data Rate (SDR) or Double Data Rate (DDR) or ULPI interface. 0: Single Data Rate ULPI Interface, with 8-bit-wide data bus 1: Double Data Rate ULPI Interface, with 4-bit-wide data bus

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Bit	Attr	Reset Value	Description
6	RW	0x0	PHYSel USB 2.0 High-Speed PHY or USB 1.1 Full-Speed Serial Transcei The application uses this bit to select either a high-speed UTMI+ or ULPI PHY, or a full-speed transceiver. 0: USB 2.0 high-speed UTMI+ or ULPI PHY 1: USB 1.1 full-speed serial transceiver If a USB 1.1 Full-Speed Serial Transceiver interface was not selected (parameter OTG_FSPHY_INTERFACE = 0), this bit is always 0, with Write Only access. If a high-speed PHY interface was not selected (parameter OTG_HSPHY_INTERFACE = 0), this bit is always 1, with Write Only access. If both interface types were selected in coreConsultant (parameters have non-zero values), the application uses this bit to select which interface is active, and access is Read and Write.
5	RW	0x0	FSIntf Full-Speed Serial Interface Select The application uses this bit to select either a unidirectional or bidirectional USB 1.1 full-speed serial transceiver interface. 0: 6-pin unidirectional full-speed serial interface 1: 3-pin bidirectional full-speed serial interface If a USB 1.1 Full-Speed Serial Transceiver interface was not selected (parameter OTG_FSPHY_INTERFACE = 0), this bit is always 0, with Write Only access. If a USB 1.1 FS interface was selected (parameter OTG_FSPHY_INTERFACE! = 0), then the application can set this bit to select between the 3- and 6-pin interfaces, and access is Read and Write.
4	RW	0x0	ULPI_UTMI_Sel ULPI or UTMI+ Select The application uses this bit to select either a UTMI+ interface or ULPI Interface. 0: UTMI+ Interface 1: ULPI Interface This bit is writable only if UTMI+ and ULPI was specified for High-Speed PHY Interface(s) in coreConsultant configuration (parameter OTG_HSPHY_INTERFACE = 3). Otherwise, reads return either 0 or 1, depending on the interface selected using the OTG_HSPHY_INTERFACE parameter.

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Bit	Attr	Reset Value	Description
3	RW	0x0	PHYIf PHY Interface The application uses this bit to configure the core to support a UTMI+ PHY with an 8- or 16-bit interface. When a ULPI PHY is chosen, this must be set to 8-bit mode. 0: 8 bits 1: 16 bits This bit is writable only if UTMI+ and ULPI were selected (parameter OTG_HSPHY_DWIDTH = 3). Otherwise, this bit returns the value for the power-on interface selected during configuration.
2:0	RW	0x0	TOutCal HS/FS Timeout Calibration The number of PHY clocks that the application programs in this field is added to the high-speed/full-speed inter-packet timeout duration in the core to account for any additional delays introduced by the PHY. This can be required, because the delay introduced by the PHY in generating the line state condition can vary from one PHY to another. The USB standard timeout value for high-speed operation is 736 to 816 (inclusive) bit times. The USB standard timeout value for full-speedoperation is 16 to 18 (inclusive) bit times. The application must program this field based on the speed of enumeration. The number of bit times added per PHY clock are: High-speed operation: One 30-MHz PHY clock = 16 bit times One 60-MHz PHY clock = 8 bit times Full-speed operation: One 30-MHz PHY clock = 0.4 bit times One 60-MHz PHY clock = 0.2 bit times One 48-MHz PHY clock = 0.25 bit times

USBOTG_GRSTCTL

Address: Operational Base + offset (0x0010)

Reset Register

Bit	Attr	Reset Value	Description
31	RO	0x1	AHBIdle AHB Master Idle Indicates that the AHB Master State Machine is in the IDLE condition.

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Bit	Attr	Reset Value	Description
30	RO	0x0	DMAReq DMA Request Signal Indicates that the DMA request is in progress. Used for debug.
29:11	RO	0x0	reserved
10:6	RW	0x00	TxFNum TxFIFO Number This is the FIFO number that must be flushed using the TxFIFO Flush bit. This field must not be changed until the core clears the TxFIFO Flush bit. 5'h0: Non-periodic TxFIFO flush in Host mode; Non-periodic TxFIFO flush in device mode when in shared FIFO operation. Tx FIFO 0 flush in device mode when in dedicated FIFO mode. 5'h1: Periodic TxFIFO flush in Host mode: Periodic TxFIFO 1 flush in Device mode when in shared FIFO operation; TXFIFO 1 flush in device mode when in dedicated FIFO mode. 5'h2: Periodic TxFIFO 2 flush in Device mode when in shared FIFO operation: TXFIFO 2 flush in device mode when in dedicated FIFO mode. ... 5'hF: Periodic TxFIFO 15 flush in Device mode when in shared FIFO operation: TXFIFO 15 flush in device mode when in dedicated FIFO mode. 5'h10: Flush all the transmit FIFOs in device or host mode.
5	R/W SC	0x0	TxFFlsh TxFIFO Flush This bit selectively flushes a single or all transmit FIFOs, but cannot do so if the core is in the midst of a transaction. The application must write this bit only after checking that the core is neither writing to the TxFIFO nor reading from the TxFIFO. Verify using these registers: Read NAK Effective Interrupt ensures the core is not reading from the FIFO. Write GRSTCTL.AHBIdle ensures the core is not writing anything to the FIFO. Flushing is normally recommended when FIFOs are re-configured or when switching between Shared FIFO and Dedicated Transmit FIFO operation. FIFO flushing is also recommended during device endpoint disable. The application must wait until the core clears this bit before performing any operations. This bit takes eight clocks to clear, using the slower clock of phy_clk or hclk.

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Bit	Attr	Reset Value	Description
4	R/W SC	0x0	RxFFlsh RxFIFO Flush The application can flush the entire RxFIFO using this bit, but must first ensure that the core is not in the middle of a transaction. The application must only write to this bit after checking that the core is neither reading from the RxFIFO nor writing to the RxFIFO. The application must wait until the bit is cleared before performing any other operations. This bit requires 8 clocks (slowest of PHY or AHB clock) to clear.
3	R/W SC	0x0	INTknQFlsh IN Token Sequence Learning Queue Flush This bit is valid only if OTG_EN_DED_TX_FIFO = 0. The application writes this bit to flush the IN Token Sequence Learning Queue.
2	W1 C	0x0	FrmCntrRst Host Frame Counter Reset The application writes this bit to reset the (micro)frame number counter inside the core. When the (micro)frame counter is reset, the subsequent SOF sent out by the core has a (micro)frame number of 0.
1	R/W SC	0x0	Reset A write to this bit issues a soft reset to the DWC_otg_power_dn module of the core.

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Bit	Attr	Reset Value	Description
0	R/W SC	0x0	CSftRst Core Soft Reset Resets the hclk and phy_clock domains as follows: Clears the interrupts and all the CSR registers except the following register bits: CGCCTL.RstPdwnModule PCGCCTL.GateHclk PCGCCTL.PwrClmp PCGCCTL.StopPPhyLPwrClkSelclk GUSBCFG.PhyLPwrClkSel GUSBCFG.DDRSel GUSBCFG.PHYSel GUSBCFG.FSIntf GUSBCFG.ULPI_UTMI_Sel GUSBCFG.PHYIf HCFG.FSLSPclkSel DCFG.DevSpd GGPIO GPWRDN GADPCTL All module state machines (except the AHB Slave Unit) are reset to the IDLE state, and all the transmit FIFOs and the receive FIFO are flushed. Any transactions on the AHB Master are terminated as soon as possible, after gracefully completing the last data phase of an AHB transfer. Any transactions on the USB are terminated immediately. When Hibernation or ADP feature is enabled, the PMU module is not reset by the Core Soft Reset. The application can write to this bit any time it wants to reset the core. This is a self-clearing bit and the core clears this bit after all the necessary logic is reset in the core, which can take several clocks, depending on the current state of the core. Once this bit is cleared software must wait at least 3 PHY clocks before doing any access to the PHY domain (synchronization delay). Software must also must check that bit 31 of this register is 1 (AHB Master is IDLE) before starting any operation. Typically software reset is used during software development and also when you dynamically change the PHY selection bits in the USB configuration registers listed above. When you change the PHY, the corresponding clock for the PHY is selected and used in the PHY domain. Once a new clock is selected, the PHY domain has to be reset for proper operation.

USBOTG_GINTSTS

Address: Operational Base + offset (0x0014)

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Interrupt Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	WkUpInt Resume/Remote Wakeup Detected Interrupt Wakeup Interrupt during Suspend(L2) or LPM(L1) state. During Suspend(L2): Device Mode:This interrupt is asserted only when Host Initiated Resume is detected on USB. Host Mode:This interrupt is asserted only when Device Initiated Remote Wakeup is detected on USB. During LPM(L1): Device Mode:This interrupt is asserted for either Host Initiated Resume or Device Initiated Remote Wakeup on USB. Host Mode:This interrupt is asserted for either Host Initiated Resume or Device Initiated Remote Wakeup on USB.
30	W1C	0x0	SessReqInt Session Request/New Session Detected Interrupt In Host mode, this interrupt is asserted when a session request is detected from the device. In Host mode, this interrupt is asserted when a session request is detected from the device. In Device mode, this interrupt is asserted when the utmisrp_bvalid signal goes high.
29	W1C	0x0	DisconnInt Disconnect Detected Interrupt This interrupt is asserted when a device disconnect is detected.
28	W1C	0x0	ConIDStsChng Connector ID Status Change This interrupt is asserted when there is a change in connector ID status.
27	W1C	0x0	LPM_Int LPM Transaction Received Interrupt Device Mode: This interrupt is asserted when the device receives an LPM transaction and responds with a non-ERRORed response. Host Mode: This interrupt is asserted when the device responds to an LPM transaction with a non-ERRORed response or when the host core has completed LPM transactions for the programmed number of times (GLPMCFG.RetryCnt). This field is valid only if the Core LPM Configuration register's LPMCapable (LPMCap) field is set to 1.
26	RO	0x0	PTxFEmp Periodic TxFIFO Empty This interrupt is asserted when the Periodic Transmit FIFO is either half or completely empty and there is space for at least one entry to be written in the Periodic Request Queue. The half or completely empty status is determined by the Periodic TxFIFO Empty Level bit in the Core AHB Configuration register (GAHBCFG.PTxFEmpLvl).

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Bit	Attr	Reset Value	Description
25	RO	0x0	HChInt Host Channels Interrupt The core sets this bit to indicate that an interrupt is pending on one of the channels of the core (in Host mode). The application must read the Host All Channels Interrupt (HAINT) register to determine the exact number of the channel on which the interrupt occurred, and then read the corresponding Host Channel-n Interrupt (HCINTn) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the HCINTn register to clear this bit.
24	RO	0x0	PrtInt Host Port Interrupt The core sets this bit to indicate a change in port status of one of the DWC_otg core ports in Host mode. The application must read the Host Port Control and Status (HPRT) register to determine the exact event that caused this interrupt. The application must clear the appropriate status bit in the Host Port Control and Status register to clear this bit.
23	RW	0x0	ResetDet Reset Detected Interrupt The core asserts this interrupt in Device mode when it detects a reset on the USB in Partial Power-Down mode when the device is in Suspend. This interrupt is not asserted in Host mode.
22	W1C	0x0	FetSusp Data Fetch Suspended This interrupt is valid only in DMA mode. This interrupt indicates that the core has stopped fetching data for IN endpoints due to the unavailability of TxFIFO space or Request Queue space. This interrupt is used by the application for an endpoint mismatch algorithm. For example, after detecting an endpoint mismatch, the application: Sets a global non-periodic IN NAK handshake, Disables In endpoints, Flushes the FIFO, Determines the token sequence from the IN Token Sequence Learning Queue, Re-enables the endpoints, Clear the global non-periodic IN NAK handshake. If the global non-periodic IN NAK is cleared, the core has not yet fetched data for the IN endpoint, and the IN token is received: the core generates an "IN token received when FIFO empty" interrupt. The OTG then sends the host a NAK response. To avoid this scenario, the application can check the GINTSTS.FetSusp interrupt, which ensures that the FIFO is full before clearing a global NAK handshake. Alternatively, the application can mask the IN token received when FIFO empty interrupt when clearing a global IN NAK handshake.

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Bit	Attr	Reset Value	Description
21	W1C	0x0	incomplP Incomplete Periodic Transfer In Host mode, the core sets this interrupt bit when there are incomplete periodic transactions still pending which are scheduled for the current microframe. Incomplete Isochronous OUT Transfer (incompISOOUT) The Device mode, the core sets this interrupt to indicate that there is at least one isochronous OUT endpoint on which the transfer is not completed in the current microframe. This interrupt is asserted along with the End of Periodic Frame Interrupt (EOPF) bit in this register.
20	W1C	0x0	incompISOIN Incomplete Isochronous IN Transfer The core sets this interrupt to indicate that there is at least one isochronous IN endpoint on which the transfer is not completed in the current microframe. This interrupt is asserted along with the End of Periodic Frame Interrupt (EOPF) bit in this register. Note: This interrupt is not asserted in Scatter/Gather DMA mode.
19	RO	0x0	OEPInt OUT Endpoints Interrupt The core sets this bit to indicate that an interrupt is pending on one of the OUT endpoints of the core (in Device mode). The application must read the Device All Endpoints Interrupt (DAINT) register to determine the exact number of the OUT endpoint on which the interrupt occurred, and then read the corresponding Device OUT Endpoint-n Interrupt (DOEPINTn) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the corresponding DOEPINTn register to clear this bit.
18	RO	0x0	IEPInt IN Endpoints Interrupt The core sets this bit to indicate that an interrupt is pending on one of the IN endpoints of the core (in Device mode). The application must read the Device All Endpoints Interrupt (DAINT) register to determine the exact number of the IN endpoint on which the interrupt occurred, and then read the corresponding Device IN Endpoint-n Interrupt (DIEPINTn) register to determine the exact cause of the interrupt. The application must clear the appropriate status bit in the corresponding DIEPINTn register to clear this bit.
17	W1C	0x0	EPMis Endpoint Mismatch Interrupt Note: This interrupt is valid only in shared FIFO operation. Indicates that an IN token has been received for a non-periodic endpoint, but the data for another endpoint is present in the top of the Non-periodic Transmit FIFO and the IN endpoint mismatch count programmed by the application has expired.

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Bit	Attr	Reset Value	Description
16	W1C	0x0	RstrDoneInt Restore Done Interrupt The core sets this bit to indicate that the restore command after Hibernation was completed by the core. The core continues from Suspended state into the mode dictated by PCGCCTL.RestoreMode field.This bit is valid only when Hibernation feature is enabled.
15	W1C	0x0	EOPF End of Periodic Frame Interrupt Indicates that the period specified in the Periodic Frame Interval field of the Device Configuration register (DCFG.PerFrInt) has been reached in the current microframe.
14	W1C	0x0	ISOOutDrop Isochronous OUT Packet Dropped Interrupt The core sets this bit when it fails to write an isochronous OUT packet into the RxFIFO because the RxFIFO does not have enough space to accommodate a maximum packet size packet for the isochronous OUT endpoint.
13	W1C	0x0	EnumDone Enumeration Done The core sets this bit to indicate that speed enumeration is complete. The application must read the Device Status (DSTS) register to obtain the enumerated speed.
12	W1C	0x0	USBRst USB Reset The core sets this bit to indicate that a reset is detected on the USB.
11	W1C	0x0	USBSusp USB Suspend The core sets this bit to indicate that a suspended was detected on the USB. The core enters the Suspended state when there is no activity on the utmi_linestate signal for an extended period of time.
10	W1C	0x0	ErlySusp Early Suspend The core sets this bit to indicate that an Idle state has been detected on the USB for 3ms.
9	W1C	0x0	I2CINT I2C Interrupt The core sets this interrupt when I2C access is completed on the I2C interface. This field is used only if the I2C interface was enabled. Otherwise, reads return 0.

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Bit	Attr	Reset Value	Description
8	W1C	0x0	ULPICKINT ULPI Carkit Interrupt This field is used only if the Carkit interface was enabled. Otherwise, reads return 0. The core sets this interrupt when a ULPI Carkit interrupt is received. The core's PHY sets ULPI Carkit interrupt in UART or Audio mode. I2C Carkit Interrupt (I2CCKINT) This field is used only if the I2C interface was enabled. Otherwise, reads return 0.The core sets this interrupt when a Carkit interrupt is received. The core's PHY sets the I2C Carkit interrupt in Audio mode.
7	RO	0x0	GOUTNakEff Global OUT NAK Effective Indicates that the Set Global OUT NAK bit in the Device Control register DCTL.SGOUTNak), set by the application, has taken effect in the core. This bit can be cleared by writing the Clear Global OUT NAK bit in the Device Control register (DCTL.CGOUTNak).
6	RO	0x0	GINNakEff Global IN Non-Periodic NAK Effective Indicates that the Set Global Non-periodic IN NAK bit in the Device Control register (DCTL.SGNPInNak), set by the application, have taken effect in the core. That is, the core has sampled the Global IN NAK bit set by the application. This bit can be cleared by clearing the Clear Global Nonperiodic IN NAK bit in the Device Control register (DCTL.CGNPInNak). This interrupt does not necessarily mean that a NAK handshake is sent out on the USB. The STALL bit takes precedence over the NAK bit.
5	RO	0x0	NPTxFEmp Non-Periodic TxFIFO Empty This interrupt is valid only when OTG_EN_DED_TX_FIFO = 0. This interrupt is asserted when the Non-periodic TxFIFO is either half or completely empty, and there is space for at least one entry to be written to the Non-periodic Transmit Request Queue. The half or completely empty status is determined by the Non-periodic TxFIFO Empty Level bit in the Core AHB Configuration register(GAHBCFG.NPTxFEmpLvl).
4	RO	0x0	RxFLvl RxFIFO Non-Empty Indicates that there is at least one packet pending to be read from the RxFIFO.

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Bit	Attr	Reset Value	Description
3	W1C	0x0	Sof Start of (micro)Frame In Host mode, the core sets this bit to indicate that an SOF (FS), micro-SOF(HS), or Keep-Alive (LS) is transmitted on the USB. The application must write a 1 to this bit to clear the interrupt. In Device mode, in the core sets this bit to indicate that an SOF token has been received on the USB. The application can read the Device Status register to get the current (micro)frame number. This interrupt is seen only when the core is operating at either HS or FS.
2	RO	0x0	OTGInt OTG Interrupt The core sets this bit to indicate an OTG protocol event. The application must read the OTG Interrupt Status (GOTGINT) register to determine the exact event that caused this interrupt. The application must clear the appropriate status bit in the GOTGINT register to clear this bit.
1	W1C	0x0	ModeMis Mode Mismatch Interrupt The core sets this bit when the application is trying to access: A Host mode register, when the core is operating in Device mode; A Device mode register, when the core is operating in Host mode. The register access is completed on the AHB with an OKAY response, but is ignored by the core internally and does not affect the operation of the core.
0	RO	0x0	CurMod Current Mode of Operation Indicates the current mode. 1'b0: Device mode 1'b1: Host mode

USBOTG_GINTMSK

Address: Operational Base + offset (0x0018)

Interrupt Mask Register

Bit	Attr	Reset Value	Description
31	RW	0x0	WkUpIntMsk Resume/Remote Wakeup Detected Interrupt Mask
30	RW	0x0	SessReqIntMsk Session Request/New Session Detected Interrupt Mask
29	RW	0x0	DisconnIntMsk Disconnect Detected Interrupt Mask

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Bit	Attr	Reset Value	Description
28	RW	0x0	ConIDStsChngMsk Connector ID Status Change Mask
27	RW	0x0	LPM_IntMsk LPM Transaction Received Interrupt Mask
26	RW	0x0	PTxFEmpMsk Periodic TxFIFO Empty Mask
25	RW	0x0	HChIntMsk Host Channels Interrupt Mask
24	RW	0x0	PrtIntMsk Host Port Interrupt Mask
23	RW	0x0	ResetDetMsk Reset Detected Interrupt Mask
22	RW	0x0	FetSuspMsk Data Fetch Suspended Mask
21	RW	0x0	incomplPMsk_incompISOOUTMsk Incomplete Periodic Transfer Mask(Host only) Incomplete Isochronous OUT Transfer Mask(Device only)
20	RW	0x0	incompISOINMsk Incomplete Isochronous IN Transfer Mask
19	RW	0x0	OEPIntMsk OUT Endpoints Interrupt Mask
18	RW	0x0	IEPIntMsk IN Endpoints Interrupt Mask
17	RW	0x0	EPMisMsk Endpoint Mismatch Interrupt Mask
16	RW	0x0	RstrDoneIntMsk Restore Done Interrupt Mask This field is valid only when Hibernation feature is enabled.
15	RW	0x0	EOPFMsk End of Periodic Frame Interrupt Mask
14	RW	0x0	ISOOutDropMsk Isochronous OUT Packet Dropped Interrupt Mask
13	RW	0x0	EnumDoneMsk Enumeration Done Mask
12	RW	0x0	USBRstMsk USB Reset Mask
11	RW	0x0	USBSuspMsk USB Suspend Mask
10	RW	0x0	ErlySuspMsk Early Suspend Mask
9	RW	0x0	I2CIntMsk I2C Interrupt Mask

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Bit	Attr	Reset Value	Description
8	RW	0x0	ULPICKINTMsk_I2CCKINTMsk ULPI Carkit Interrupt Mask (ULPICKINTMsk) I2C Carkit Interrupt Mask (I2CCKINTMsk)
7	RW	0x0	GOUTNakEffMsk Global OUT NAK Effective Mask
6	RW	0x0	GINNakEffMsk Global Non-periodic IN NAK Effective Mask
5	RW	0x0	NPTxFEmpMsk Non-periodic TxFIFO Empty Mask
4	RW	0x0	RxFLvlMsk Receive FIFO Non-Empty Mask
3	RW	0x0	SofMsk Start of (micro)Frame Mask
2	RW	0x0	OTGIntMsk OTG Interrupt Mask
1	RW	0x0	ModeMisMsk Mode Mismatch Interrupt Mask
0	RO	0x0	reserved

USBOTG_GRXSTSR

Address: Operational Base + offset (0x001c)

Receive Status Debug Read Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:21	RO	0x0	FN Frame Number (Device Only)This is the least significant 4 bits of the (micro)frame number in which the packet is received on the USB. This field is supported only when isochronous OUT endpoints are supported.

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Bit	Attr	Reset Value	Description
20:17	RO	0x0	PktSts Packet Status Indicates the status of the received packet(Host Only) 4'b0010: IN data packet received 4'b0011: IN transfer completed (triggers an interrupt) 4'b0101: Data toggle error (triggers an interrupt) 4'b0111: Channel halted (triggers an interrupt) Others: Reserved Indicates the status of the received packet(Device only) 4'b0001: Global OUT NAK (triggers an interrupt) 4'b0010: OUT data packet received 4'b0011: OUT transfer completed (triggers an interrupt) 4'b0100: SETUP transaction completed (triggers an interrupt) 4'b0110: SETUP data packet received Others: Reserved
16:15	RO	0x0	DPID Data PID Indicates the Data PID of the received packet 2'b00: DATA0 2'b10: DATA1 2'b01: DATA2 2'b11: MDATA
14:4	RW	0x000	BCnt Byte Count Indicates the byte count of the received data packet.
3:0	RO	0x0	ChNum_EPNum Channel Number(Host) Endpoint Number(Device) (Host Only) Indicates the channel number to which the current received packet belongs. (Device Only) Indicates the endpoint number to which the current received packet belongs.

USBOTG_GRXSTSP

Address: Operational Base + offset (0x0020)

Receive Status Read and Pop Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
24:21	RO	0x0	FN Frame Number (Device Only) This is the least significant 4 bits of the (micro)frame number in which the packet is received on the USB. This field is supported only when isochronous OUT endpoints are supported.
20:17	RO	0x0	PktSts Packet Status Indicates the status of the received packet(Host Only) 4'b0010: IN data packet received 4'b0011: IN transfer completed (triggers an interrupt) 4'b0101: Data toggle error (triggers an interrupt) 4'b0111: Channel halted (triggers an interrupt) Others: Reserved Indicates the status of the received packet(Device only) 4'b0001: Global OUT NAK (triggers an interrupt) 4'b0010: OUT data packet received 4'b0011: OUT transfer completed (triggers an interrupt) 4'b0100: SETUP transaction completed (triggers an interrupt) 4'b0110: SETUP data packet received Others: Reserved
16:15	RO	0x0	DPID Data PID Indicates the Data PID of the received OUT data packet 2'b00: DATA0 2'b10: DATA1 2'b01: DATA2 2'b11: MDATA
14:4	RO	0x000	BCnt Byte Count Indicates the byte count of the received data packet.
3:0	RO	0x0	ChNum_EPNum Channel Number(Host) Endpoint Number(Device) (Host Only) Indicates the channel number to which the current received packet belongs. (Device Only) Indicates the endpoint number to which the current received packet belongs.

USBOTG_GRXFSIZ

Address: Operational Base + offset (0x0024)

Receive FIFO Size Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	RxFDep RxFIFO Depth This value is in terms of 32-bit words. Minimum value is 16. Maximum value is 32,768. The power-on reset value of this register is specified as the Largest Rx Data FIFO Depth. If Enable Dynamic FIFO Sizing was deselected, these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing wasselected, you can write a new value in this field. You can write a new value in this field. Programmed values must not exceed the power-on value.

USBOTG_GNPTXFSIZ

Address: Operational Base + offset (0x0028)

Non-Periodic Transmit FIFO Size Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	NPTxFDep Non-periodic TxFIFO For host mode, this field is always valid. For Device mode, this field is valid only when OTG_EN_DED_TX_FIFO==0. This value is in terms of 32-bit words. Minimum value is 16. Maximum value is 32,768 This field is determined by Enable Dynamic FIFO Sizing. OTG_DFIFO_DYNAMIC = 0: These flops are optimized, and reads return the power-on value. OTG_DFIFO_DYNAMIC = 1: The application can write a new value in this field. Programmed values must not exceed the power-on value. The power-on reset value of this field is specified by OTG_EN_DED_TX_FIFO: OTG_EN_DED_TX_FIFO = 0:The reset value is the Largest Non-periodic Tx Data FIFO Depth parameter, OTG_TX_NPERIO_DFIFO_DEPTH. OTG_EN_DED_TX_FIFO = 1:The reset value is parameter OTG_TX_HNPERIO_DFIFO_DEPTH.

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	NPTxFStAddr Non-periodic Transmit RAM For host mode, this field is always valid. This field contains the memory start address for Non-periodic Transmit FIFO RAM. This field is determined by Enable Dynamic FIFO Sizing(OTG_DFIFO_DYNAMIC): OTG_DFIFO_DYNAMIC = 0:These flops are optimized, and reads return the power-on value. OTG_DFIFO_DYNAMIC = 1:The application can write a new value in this field. Programmed values must not exceed the power-on value. The power-on reset value of this field is specified by Largest Rx Data FIFO Depth (parameter OTG_RX_DFIFO_DEPTH).

USBOTG_GNPTXSTS

Address: Operational Base + offset (0x002c)

Non-Periodic Transmit FIFO/Queue Status Register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:24	RO	0x00	NPTxQTop Top of the Non-periodic Transmit Request Queue Entry in the Non-periodic Tx Request Queue that is currently being processed by the MAC. Bits [30:27]: Channel/endpoint number Bits [26:25]: 2'b00: IN/OUT token 2'b01: Zero-length transmit packet (device IN/host OUT) 2'b10: PING/CSPLIT token 2'b11: Channel halt command Bit [24]: Terminate (last entry for selected channel/endpoint)
23:16	RO	0x00	NPTxQSpcAvail Non-periodic Transmit Request Queue Space Available Indicates the amount of free space available in the Non-periodic Transmit Request Queue. This queue holds both IN and OUT requests in Host mode. Device mode has only IN requests. 8'h0: Non-periodic Transmit Request Queue is full 8'h1: 1 location available 8'h2: 2 locations available n: n locations available (0 <=n <= 8) Others: Reserved

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	NPTxFSpcAvail Non-periodic TxFIFO Space Avail Indicates the amount of free space available in the Non-periodic TxFIFO. Values are in terms of 32-bit words. 16'h0: Non-periodic TxFIFO is full 16'h1: 1 word available 16'h2: 2 words available 16'hn: n words available (where 0 <=n <=32,768) 16'h8000: 32,768 words available Others: Reserved

USBOTG_GI2CCTL

Address: Operational Base + offset (0x0030)

I2C Address Register

Bit	Attr	Reset Value	Description
31	R/W SC	0x0	BsyDne I2C Busy/Done The application sets this bit to 1'b1 to start a request on the I2C interface. When the transfer is complete, the core de-asserts this bit to 1'b0. As long as the bit is set, indicating that the I2C interface is busy, the application cannot start another request on the interface.
30	RW	0x0	RW Read/Write Indicator Indicates whether a read or write register transfer must be performed on the interface.Read/write bursting is not supported for registers. 1'b1: Read 1'b0: Write
29	RO	0x0	reserved
28	RW	0x1	I2CDatSe0 I2C DatSe0 USB Mode Selects the FS interface USB mode. 1'b1: VP_VM USB mode 1'b0: DAT_SE0 USB mode

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Bit	Attr	Reset Value	Description
27:26	RW	0x0	I2CDevAdr I2C Device Address Selects the address of the I2C Slave on the USB 1.1 full-speed serial transceiver that the core uses for OTG signaling. 2'b00: 7'h2C 2'b01: 7'h2D 2'b10: 7'h2E 2'b11: 7'h2F
25	RW	0x0	I2CSuspCtl I2C Suspend Control Selects how Suspend is connected to a full-speed transceiver in I2C mode. 1'b0: Use the dedicated utmi_suspend_n pin 1'b1: Use an I2C write to program the Suspend bit in the PHY register
24	RO	0x1	Ack I2C ACK Indicates whether an ACK response was received from the I2C Slave. This bit is valid when BsyDne is cleared by the core, after application has initiated an I2C access. 1'b0: NAK 1'b1: ACK
23	RW	0x0	I2CEn I2C Enable Enables the I2C Master to initiate I2C transactions on the I2C interface
22:16	RW	0x00	Addr I2C Address This is the 7-bit I2C device address used by software to access any external I2C Slave,including the I2C Slave on a USB 1.1 OTG full-speed serial transceiver. Software can change this address to access different I2C Slaves.
15:8	RW	0x00	RegAddr I2C Register Addr This field programs the address of the register to be read from or written to.
7:0	RW	0x00	RWData I2C Read/Write Data After a register read operation, this field holds the read data for the application. During a write operation, the application can use this register to program the write data to be written to a register. During writes, this field holds the write data.

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USBOTG_GPVNDCTL

Address: Operational Base + offset (0x0034)

PHY Vendor Control Register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	DisUlpiDrvr Disable ULPI Drivers This field is used only if the Carkit interface was enabled in coreConsultant (parameter OTG_ULPI_CARKIT = 1). Otherwise, reads return 0. The application sets this bit when it has finished processing the ULPI Carkit Interrupt (GINTSTS.ULPICKINT). When set, the DWC_otg core disables drivers for output signals and masks input signal for the ULPI interface. DWC_otg clears this bit before enabling the ULPI interface.
30:28	RO	0x0	reserved
27	R/WSC	0x0	VStsDone VStatus Done The core sets this bit when the vendor control access is done. This bit is cleared by the core when the application sets the New Register Request bit (bit 25).
26	RO	0x0	VStsBsy VStatus Busy The core sets this bit when the vendor control access is in progress and clears this bit when done.
25	R/WSC	0x0	NewRegReq New Register Request The application sets this bit for a new vendor control access.
24:23	RO	0x0	reserved
22	RW	0x0	RegWr Register Write Set this bit for register writes, and clear it for register reads.
21:16	RW	0x00	RegAddr Register Address The 6-bit PHY register address for immediate PHY Register Set access. Set to 6'h2F for Extended PHY Register Set access.
15:8	RW	0x00	VCtrl UTMI+ Vendor Control Register Address The 4-bit register address a vendor defined 4-bit parallel output bus. Bits 11:8 of this field are placed on utmi_vcontrol[3:0]. ULPI Extended Register Address (ExtRegAddr) The 6-bit PHY extended register address.
7:0	RW	0x00	RegData Register Data Contains the write data for register write. Read data for register read, valid when VStatus Done is set.

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USBOTG_GGPIO

Address: Operational Base + offset (0x0038)

General Purpost Input/Output Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	GPO General Purpose Output This field is driven as an output from the core, gp_o[15:0]. The application can program this field to determine the corresponding value on the gp_o[15:0] output.
15:0	RO	0x0000	GPI General Purpose Input This field's read value reflects the gp_i[15:0] core input value.

USBOTG_GUID

Address: Operational Base + offset (0x003c)

User ID Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	UserID Application-programmable ID field.

USBOTG_GSNPSID

Address: Operational Base + offset (0x0040)

Core ID Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00004f54	CoreID Release number of the core being used

USBOTG_GHWCFG1

Address: Operational Base + offset (0x0044)

User HW Config1 Register

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	epdir Endpoint Direction This 32-bit field uses two bits per endpoint to determine the endpoint direction. Endpoint Bits [31:30]: Endpoint 15 direction Bits [29:28]: Endpoint 14 direction ... Bits [3:2]: Endpoint 1 direction Bits[1:0]: Endpoint 0 direction (always BIDIR) Direction 2'b00: BIDIR (IN and OUT) endpoint 2'b01: IN endpoint 2'b10: OUT endpoint 2'b11: Reserved

USBOTG_GHWCFG2

Address: Operational Base + offset (0x0048)

User HW Config2 Register

Bit	Attr	Reset Value	Description
31	RO	0x0	OTG_ENABLE_IC_USB IC_USB mode specified for mode of operation (parameter OTG_ENABLE_IC_USB). To choose IC_USB_MODE, both OTG_FSPHY_INTERFACE and OTG_ENABLE_IC_USB must be 1.
30:26	RO	0x00	TknQDepth Device Mode IN Token Sequence Learning Queue Depth Range: 0-30
25:24	RO	0x0	PTxQDepth Host Mode Periodic Request Queue Depth 2'b00: 2 2'b01: 4 2'b10: 8 Others: Reserved
23:22	RO	0x0	NPTxQDepth Non-periodic Request Queue Depth 2'b00: 2 2'b01: 4 2'b10: 8 Others: Reserved
21	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
20	RO	0x0	MultiProcIntrpt Multi Processor Interrupt Enabled 1'b0: No 1'b1: Yes
19	RO	0x0	DynFifoSizing Dynamic FIFO Sizing Enabled 1'b0: No 1'b1: Yes
18	RO	0x0	PerioSupport Periodic OUT Channels Supported in Host Mode 1'b0: No 1'b1: Yes
17:14	RO	0x0	NumHstChnl Number of Host Channels Indicates the number of host channels supported by the core in Host mode. The range of this field is 0-15: 0 specifies 1 channel, 15 specifies 16 channels.
13:10	RO	0x0	NumDevEps Number of Device Endpoints Indicates the number of device endpoints supported by the core in Device mode in addition to control endpoint 0. The range of this field is 1-15.
9:8	RO	0x0	FSPhyType Full-Speed PHY Interface Type 2'b00: Full-speed interface not supported 2'b01: Dedicated full-speed interface 2'b10: FS pins shared with UTMI+ pins 2'b11: FS pins shared with ULPI pins
7:6	RO	0x0	HSPhyType High-Speed PHY Interface Type 2'b00: High-Speed interface not supported 2'b01: UTMI+ 2'b10: ULPI 2'b11: UTMI+ and ULPI
5	RO	0x0	SingPnt Point-to-Point 1'b0: Multi-point application (hub and split support) 1'b1: Single-point application (no hub and no split support)
4:3	RO	0x0	OtgArch Architecture 2'b00: Slave-Only 2'b01: External DMA 2'b10: Internal DMA Others: Reserved

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Bit	Attr	Reset Value	Description
2:0	RO	0x0	OtgMode Mode of Operation 3'b000: HNP- and SRP-Capable OTG (Host and Device) 3'b001: SRP-Capable OTG (Host and Device) 3'b010: Non-HNP and Non-SRP Capable OTG (Host and Device) 3'b011: SRP-Capable Device 3'b100: Non-OTG Device 3'b101: SRP-Capable Host 3'b110: Non-OTG Host Others: Reserved

USBOTG_GHWCFG3

Address: Operational Base + offset (0x004c)

User HW Config3 Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	DfifoDepth DFIFO Depth This value is in terms of 32-bit words. Minimum value is 32 Maximum value is 32,768
15	RO	0x0	OTG_ENABLE_LPM LPM mode specified for Mode of Operation (parameter OTG_ENABLE_LPM).
14	RO	0x0	OTG_BC_SUPPORT This bit indicates the HS OTG controller support for Battery Charger. 0 : No Battery Charger Support 1 : Battery Charger support present.
13	RO	0x0	OTG_ENABLE_HSIC HSIC mode specified for Mode of Operation (parameter OTG_ENABLE_HSIC). Value Range: 0-1 1: HSIC-capable with shared UTMI PHY interface 0: Non-HSIC-capable
12	RO	0x0	OTG_ADP_SUPPORT This bit indicates whether ADP logic is present within or external to the HS OTG controller 0: No ADP logic present with HSOTG controller 1: ADP logic is present along with HSOTG controller.

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Bit	Attr	Reset Value	Description
11	RO	0x0	RstType Reset Style for Clocked always Blocks in RTL 1'b0: Asynchronous reset is used in the core 1'b1: Synchronous reset is used in the core
10	RO	0x0	OptFeature Optional Features Removed Indicates whether the User ID register, GPIO interface ports, and SOF toggle and counter ports were removed for gate count optimization by enabling Remove Optional Features? 1'b0: No 1'b1: Yes
9	RO	0x0	VndctlSupt Vendor Control Interface Support 1'b0: Vendor Control Interface is not available on the core. 1'b1: Vendor Control Interface is available.
8	RO	0x0	I2CIntSel I2C Selection 1'b0: I2C Interface is not available on the core. 1'b1: I2C Interface is available on the core.
7	RO	0x0	OtgEn OTG Function Enabled The application uses this bit to indicate the DWC_otg core's OTG capabilities. 1'b0: Not OTG capable 1'b1: OTG Capable
6:4	RO	0x0	PktSizeWidth Width of Packet Size Counters 3'b000: 4 bits 3'b001: 5 bits 3'b010: 6 bits 3'b011: 7 bits 3'b100: 8 bits 3'b101: 9 bits 3'b110: 10 bits Others: Reserved
3:0	RO	0x0	XferSizeWidth Width of Transfer Size Counters 4'b0000: 11 bits 4'b0001: 12 bits ... 4'b1000: 19 bits Others: Reserved

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USBOTG_GHWCFG4

Address: Operational Base + offset (0x0050)

User HW Config4 Register

Bit	Attr	Reset Value	Description
31	RO	0x0	SGDMA Scatter/Gather DMA 1'b1: Dynamic configuration
30	RO	0x0	SGDMACon Scatter/Gather DMA configuration 1'b0: Non-Scatter/Gather DMA configuration 1'b1: Scatter/Gather DMA configuration
29:26	RO	0x0	INEps Number of Device Mode IN Endpoints Including Control Endpoint Range 0 -15 0:1 IN Endpoint 1:2 IN Endpoints .... 15:16 IN Endpoints
25	RW	0x0	DedFifoMode Enable Dedicated Transmit FIFO for device IN Endpoints 1'b0: Dedicated Transmit FIFO Operation not enabled. 1'b1: Dedicated Transmit FIFO Operation enabled.
24	RW	0x0	SessEndFltr session_end Filter Enabled 1'b0: No filter 1'b1: Filter
23	RW	0x0	BValidFltr "b_valid" Filter Enabled 1'b0: No filter 1'b1: Filter
22	RO	0x0	AValidFltr "a_valid" Filter Enabled 1'b0: No filter 1'b1: Filter
21	RO	0x0	VBusValidFltr "vbus_valid" Filter Enabled 1'b0: No filter 1'b1: Filter
20	RO	0x0	IddgFltr "iddig" Filter Enable 1'b0: No filter 1'b1: Filter

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Bit	Attr	Reset Value	Description
19:16	RO	0x0	NumCtlEps Number of Device Mode Control Endpoints in Addition to Endpoin Range: 0-15
15:14	RO	0x0	PhyDataWidth UTMI+ PHY/ULPI-to-Internal UTMI+ Wrapper Data Width When a ULPI PHY is used, an internal wrapper converts ULPI to UTMI+. 2'b00: 8 bits 2'b01: 16 bits 2'b10: 8/16 bits, software selectable Others: Reserved
13:7	RO	0x0	reserved
6	RO	0x0	EnHiber Enable Hibernation 1'b0: Hibernation feature not enabled 1'b1: Hibernation feature enabled
5	RO	0x0	AhbFreq Minimum AHB Frequency Less Than 60 MHz 1'b0: No 1'b1: Yes
4	RO	0x0	EnParPwrDown Enable Partial Power Down 1'b0: Partial Power Down Not Enabled 1'b1: Partial Power Down Enabled
3:0	RO	0x0	NumDevPerioEps Number of Device Mode Periodic IN Endpoints Range: 0-15

USBOTG_GLPMCFG

Address: Operational Base + offset (0x0054)

Core LPM Configuration Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	InvSelHsic HSIC-Invert Select HSIC The application uses this bit to control the DWC_otg core HSIC enable/disable. This bit overrides and functionally inverts the if_sel_hsic input port signal. If the core operates as non-HSIC-capable, it can only connect to non-HSIC-capable PHYs. If the core operates as HSIC-capable, it can only connect to HSICcapable PHYs. If the if_sel_hsic input signal is1: 1'b1: HSIC capability is not enabled. 1'b0: HSIC capability is enabled, If InvSelHsic = 1'b0: HSIC capability is enabled. If the if_sel_hsic input signal is 0: 1'b1: HSIC capability is enabled, 1'b0: HSIC capability is not enabled. This bit is writable only if an HSIC mode was specified for Mode of Operation (parameter OTG_ENABLE_HSIC). This bit is valid only if OTG_ENABLE_HSIC is enabled.
30	RW	0x0	HSICCon HSIC-Connect The application must use this bit to initiate the HSIC Attach sequence. Host Mode: Once this bit is set, the host core configures to drive the HSIC Idle state (STROBE = 1 & DATA = 0) on the bus. It then waits for the device to initiate the Connect sequence. Device Mode: Once this bit is set, the device core waits for the HSIC Idle line state on the bus. Upon receiving the Idle line state, it initiates the HSIC Connect sequence. This bit is valid only if OTG_ENABLE_HSIC is 1, if_sel_hsic = 1 and InvSelHSIC is 0. Otherwise, it is read-only.
29:28	RO	0x0	reserved
27:25	RO	0x0	LPM_RetryCnt_Sts LPM Retry Count Status Number of LPM host retries remaining to be transmitted for the current LPM sequence.
24	RW	0x0	SndLPM Send LPM Transaction Host Mode: When the application software sets this bit, an LPM transaction containing two tokens, EXT and LPM, is sent. The hardware clears this bit once a valid response (STALL, NYET, or ACK) is received from the device or the core has finished transmitting the programmed number of LPM retries. Note: This bit must only be set when the host is connected to a local port.

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Bit	Attr	Reset Value	Description
23:21	R/WSC	0x0	LPM_Retry_Cnt LPM Retry Count When the device gives an ERROR response, this is the number of additional LPM retries that the host performs until a valid device response (STALL, NYET, or ACK) is received.
20:17	RW	0x0	LPM_Chnl_Indx LPM Channel Index The channel number on which the LPM transaction must be applied while sending an LPM transaction to the local device. Based on the LPM channel index, the core automatically inserts the device address and endpoint number programmed in the corresponding channel into the LPM transaction.
16	RO	0x0	L1ResumeOK Sleep State Resume OK Indicates that the application or host can start a resume from the Sleep state. This bit is valid in the LPM Sleep (L1) state. It is set in Sleep mode after a delay of 50 us (TL1Residency). The bit is reset when SlpSts is 0 1'b1: The application/core can start resume from the Sleep state 1'b0: The application/core cannot start resume from the Sleep state

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Bit	Attr	Reset Value	Description
15	RO	0x0	SlpSts Port Sleep Status Device Mode: This bit is set as long as a Sleep condition is present on the USB bus. The core enters the Sleep state when an ACK response is sent to an LPM transaction and the timer TL1TokenRetry has expired. To stop the PHY clock, the application must set the Port Clock Stop bit, which asserts the PHY Suspend input pin. The application must rely on SlpSts and not ACK in CoreL1Res to confirm transition into sleep. The core comes out of sleep: When there is any activity on the USB line_state, When the application writes to the Remote Wakeup Signaling bit in the Device Control register (DCTL.RmtWkUpSig) or when the application resets or soft-disconnects the device. Host Mode: The host transitions to the Sleep (L1) state as a side-effect of a successful LPM transaction by the core to the local port with an ACK response from the device. The read value of this bit reflects the port's current sleep status. The core clears this bit after: The core detects a remote L1 Wakeup signal, The application sets the Port Reset bit or the Port L1Resume bit in the HPRT register or The application sets the L1Resume/ Remote Wakeup Detected Interrupt bit or Disconnect Detected Interrupt bit in the Core Interrupt register (GINTSTS.L1WkUpInt or GINTSTS.DisconnInt, respectively). Values: 1'b0: Core not in L1 1'b1: Core in L1
14:13	RO	0x0	CoreL1Res LPM Response Device Mode: The core's response to the received LPM transaction is reflected in these two bits. Host Mode: The handshake response received from the local device for LPM transaction. 11: ACK 10: NYET 01: STALL 00: ERROR (No handshake response)

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Bit	Attr	Reset Value	Description
12:8	RW	0x00	HIRD_Thres HIRD Threshold Device Mode: The core asserts L1SuspendM to put the PHY into Deep Low-Power mode in L1 when the HIRD value is greater than or equal to the value defined in this field (GLPMCFG.HIRD_Thres[3:0]), and HIRD_Thres[4] is set to 1'b1. Host Mode: The core asserts L1SuspendM to put the PHY into Deep Low-Power mode in L1 when HIRD_Thres[4] is set to 1'b1. HIRD_Thres[3:0] specifies the time for which resume signaling is to be reflected by the host TL1HubDrvResume2) on the USB when it detects device-initiated resume. HIRD_Thres must not be programmed with a value greater than 4'b1100 in Host mode, because this exceeds maximum TL1HubDrvResume2. Sl. No HIRD_Thres[3:0] Host mode resume time(us) 1 4'b0000 60 2 4'b0001 135 3 4'b0010 210 4 4'b0011 285 5 4'b0100 360 6 4'b0101 435 7 4'b0110 510 8 4'b0111 585 9 4'b1000 660 10 4'b1001 735 11 4'b1010 810 12 4'b1011 885 13 4'b1100 960 14 4'b1101 invalid 15 4'b1110 invalid 16 4'b1111 invalid

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Bit	Attr	Reset Value	Description
7	RW	0x0	EnblSlpM Enable utmi_sleep_n For ULPI interface: The application uses this bit to write to the function control [7] in the L1 state, to enable the PHY to go into Low Power mode. For the host, this bit is valid only in Local Device mode. 1'b0: Writes to the ULPI Function Control Bit[7] are disabled. 1'b1: The core is enabled to write to the ULPI Function Control Bit[7], which enables the PHY to enter Low-Power mode. Note: When a ULPI interface is configured, enabling this bit results in a write to Bit 7 of the ULPI Function Control register. The ULPI PHY supports writing to this bit, and in the L1 state asserts SleepM when utmi_l1_suspend_n cannot be asserted. When a ULPI interface is configured, this bit must always be set if you are using the ULPI PHY. Note: For ULPI interface, do not clear this bit during the resume. For all other interfaces: The application uses this bit to control utmi_sleep_n assertion to the PHY in the L1 state. For the host, this bit is valid only in Local Device mode. 1'b0: utmi_sleep_n assertion from the core is not transferred to the external PHY. 1'b1: utmi_sleep_n assertion from the core is transferred to the external PHY when utmi_l1_suspend_n cannot be asserted.
6	RW	0x0	bRemoteWake RemoteWakeEnable Host Mode: The remote wakeup value to be sent in the LPM transaction's wIndex field. Device Mode: This field is updated with the received bRemoteWake LPM token's bmAttribute when an ACK/NYET/STALL response is sent to an LPM transaction.

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	HIRD Host-Initiated Resume Duration Host Mode: The value of HIRD to be sent in an LPM transaction. This value is also used to initiate resume for a durationL1HubDrvResume1 for host initiated resume. Device Mode: This field is updated with the Received LPM Token HIRD bmAttribute when an ACK/NYET/STALL response is sent to an LPM transaction Sl. No HIRD[3:0] THIRD (us) 1 4'b0000 50 2 4'b0001 125 3 4'b0010 200 4 4'b0011 275 5 4'b0100 350 6 4'b0101 425 7 4'b0110 500 8 4'b0111 575 9 4'b1000 650 10 4'b1001 725 11 4'b1010 800 12 4'b1011 875 13 4'b1100 950 14 4'b1101 1025 15 4'b1110 1100 16 4'b1111 1175
1	RW	0x0	AppL1Res LPM response programmed by application Handshake response to LPM token pre-programmed by device application software. The response depends on GLPMCFG.LPMCap. If GLPMCFG.LPMCap is 1'b0, the core always responds with a NYET. If GLPMCFG.LPMCap is 1'b1, the core responds as follows: 1: ACK. Even though an ACK is pre-programmed, the core responds with an ACK only on a successful LPM transaction. The LPM transaction is successful if: There are no PID/CRC5 errors in both the EXT token and the LPM token (else ERROR); A valid bLinkState = 0001B (L1) is received in the LPM transaction (else STALL); No data is pending in the Transmit queue (else NYET) 0: NYET. The pre-programmed software bit is overridden for response to LPM token when:(1)The received bLinkState is not L1 (STALL response); (2)An error is detected in either of the LPM token packets due to corruption (ERROR response).

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Bit	Attr	Reset Value	Description
0	RW	0x0	LPMCap LPM-Capable The application uses this bit to control the DWC_otg core LPM capabilities. If the core operates as a non-LPM-capable host, it cannot request the connected device/hub to activate LPM mode. If the core operates as a non-LPM-capable device, it cannot respond to any LPM transactions. 1'b0: LPM capability is not enabled. 1'b1: LPM capability is enabled. This bit is writable only if an LPM mode was specified for Mode of Operation (parameter OTG_ENABLE_LPM). Otherwise, reads return 0.

USBOTG_GPWRDN

Address: Operational Base + offset (0x0058)

Global Power Down Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RO	0x00	MultValIdBC Multi Valued ID pin Battery Charger ACA inputs in the following order: Bit 26 - rid_float. Bit 25 - rid_gnd Bit 24 - rid_a Bit 23 - rid_b Bit 22 - rid_c These bits are present only if OTG_BC_SUPPORT = 1. Otherwise, these bits are reserved and will read 5'h0.
23	W1C	0x0	ADPInt ADP Interupt This bit is set whenever there is a ADP event.
22	RO	0x0	BSessVld B Session Valid This field reflects the B session valid status signal from the PHY. 1'b0: B-Valid is 0. 1'b1: B-Valid is 1. This bit is valid only when GPWRDN.PMUActv is 1.

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Bit	Attr	Reset Value	Description
21	RO	0x0	IDDIG This bit indicates the status of the signal IDDIG. The application must read this bit after receiving GPWRDN.StsChngInt and decode based on the previous value stored by the application. Indicates the current mode. 1'b1: Device mode 1'b0: Host mode This bit is valid only when GPWRDN.PMUActv is 1.
20:19	RO	0x0	LineState This field indicates the current linestate on USB as seen by the PMU module. 2'b00: DM = 0, DP = 0. 2'b01: DM = 0, DP = 1. 2'b10: DM = 1, DP = 0. 2'b11: Not-defined. This bit is valid only when GPWRDN.PMUActv is 1.
18	RW	0x0	StsChngIntMsk Mask For StsChng Interrupt
17	W1C	0x0	StsChngInt This field indicates a status change in either the IDDIG or BSessVld signal. 1'b0: No Status change 1'b1: status change detected After receiving this interrupt the application should read the GPWRDN register and interpret the change in IDDIG or BSesVld with respect to the previous value stored by the application.
16	RW	0x0	SRPDetectMsk Mask For SRPDetect Interrupt
15	W1C	0x0	SRPDetect This field indicates that SRP has been detected by the PMU.This field generates an interrupt. After detecting SRP during hibernation the application should not restore the core. The application should get into the initialization process. 1'b0: SRP not detected 1'b1: SRP detected
14	RW	0x0	ConnDetMsk Mask for ConnectDet interrupt This bit is valid only when OTG_EN_PWROPT = 2.
13	W1C	0x0	ConnectDet This field indicates that a new connect has been detected 1'b0: Connect not detected 1'b1: Connect detected This bit is valid only when OTG_EN_PWROPT = 2.

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Bit	Attr	Reset Value	Description
12	RW	0x0	DisconnectDetectMsk Mask For DisconnectDetect Interrupt This bit is valid only when OTG_EN_PWROPT = 2.
11	W1C	0x0	DisconnectDetect This field indicates that Disconnect has been detected by the PMU. This field generates an interrupt. After detecting disconnect during hibernation the application must not restore the core, but instead start the initialization process. 1'b0: Disconnect not detected 1'b1: Disconnect detected This bit is valid only when OTG_EN_PWROPT = 2.
10	RW	0x0	ResetDetMsk Mask For ResetDetected interrupt.This bit is valid only when OTG_EN_PWROPT = 2.
9	W1C	0x0	ResetDetected This field indicates that Reset has been detected by the PMU module. This field generates an interrupt. 1'b0: Reset Not Detected 1'b1: Reset Detected This bit is valid only when OTG_EN_PWROPT = 2.
8	RW	0x0	LineStageChangeMsk Mask For LineStateChange interrupt This bit is valid only when OTG_EN_PWROPT = 2.
7	W1C	0x0	LnStsChng Line State Change This interrupt is asserted when there is a Linestate Change detected by the PMU. The application should read GPWRDN.Linestate to determine the current linestate on USB. 1'b0: No LineState change on USB 1'b1: LineState change on USB This bit is valid only when GPWRDN.PMUActv is 1.This bit is valid only when OTG_EN_PWROPT = 2.
6	RW	0x0	DisableVBUS The application should program this bit if HPRT0.PrtPwr was programmed to 0 before entering Hibernation. This is to indicate PMU whether session was ended before entering Hibernation. 1'b0: HPRT0.PrtPwr was not programmed to 0. 1'b1: HPRT0.PrtPwr was programmed to 0.

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Bit	Attr	Reset Value	Description
5	RW	0x0	PwrDnSwtch Power Down Switch This bit indicates to the DWC_otg core VDD switch is in ON/OFF state 1'b0: DWC_otg is in ON state 1'b1: DWC_otg is in OFF state *Note: This bit must not be written to during normal mode of operation.*
4	RW	0x0	PwrDnRst_n Power Down ResetN The application must program this bit to reset the DWC OTG core during the Hibernation exit process or during ADP when powering up the core (in case the DWC OTG core was powered off during ADP process). 1'b1: DWC_otg is in normal operation 1'b0: reset DWC_otg *Note: This bit must not be written to during normal mode of operation.*
3	RW	0x0	PwrDnClmp Power Down Clamp The application must program this bit to enable or disable the clamps to all the outputs of the DWC OTG core module to prevent the corruption of other active logic. 1'b0: Disable PMU power clamp 1'b1: Enable PMU power clamp
2	RW	0x0	Restore The application should program this bit to enable or disable restore mode from the PMU module. 1'b0: DWC_otg in normal mode of operation 1'b1: DWC_otg in restore mode *Note: This bit must not be written to during normal mode of operation. This bit is valid only when OTG_EN_PWROPT = 2*.
1	RW	0x0	PMUActv PMU Active This is bit is to enable or disable the PMU logic. 1'b0: Disable PMU module 1'b1: Enable PMU module *Note: This bit must not be written to during normal mode of operation.*

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Bit	Attr	Reset Value	Description
0	RW	0x0	PMUIntSel PMU Interrupt Select When the hibernation functionality is selected using the configuration option OTG_EN_PWR_OPT = 2, a write to this bit with 1'b1 enables the PMU to generate interrupts to the application. During this state all interrupts from the core module are blocked to the application. Note: This bit must be set to 1'b1 before the core is put into hibernation 1'b0: Internal DWC_otg_core interrupt is selected 1'b1: the external DWC_otg_pmu interrupt is selected Note: This bit must not be written to during normal mode of operation.

USBOTG_GDFIFOCFG

Address: Operational Base + offset (0x005c)

Global DFIFO Software Config Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	EPInfoBaseAddr This field provides the start address of the EP info controller.
15:0	RW	0x0000	GDFIFOCfg This field is for dynamic programming of the DFIFO Size. This value takes effect only when the application programs a non-zero value to this register.The core does not have any corrective logic if the FIFO sizes are programmed incorrectly.

USBOTG_GADPCTL

Address: Operational Base + offset (0x0060)

ADP Timer,Control and Status Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:27	R/W SC	0x0	AR Access Request 2'b00 Read/Write Valid (updated by the core) 2'b01 Read 2'b10 Write 2'b11 Reserved

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Bit	Attr	Reset Value	Description
26	RW	0x0	AdpTmoutMsk ADP Timeout Interrupt Mask When this bit is set, it unmasks the interrupt because of AdpTmoutInt. This bit is valid only if OTG_Ver = 1'b1(GOTGCTL[20]).
25	RW	0x0	AdpSnsIntMsk ADP Sense Interrupt Mask When this bit is set, it unmasks the interrupt due to AdpSnsInt. This bit is valid only if OTG_Ver = 1'b1(GOTGCTL[20]).
24	RW	0x0	AdpPrbIntMsk ADP Probe Interrupt Mask When this bit is set, it unmasks the interrupt due to AdpPrbInt.This bit is valid only if OTG_Ver = 1'b1(GOTGCTL[20]).
23	W1C	0x0	AdpTmoutInt ADP Timeout Interrupt This bit is relevant only for an ADP probe. When this bit is set, it means that the ramp time has completed (GADPCTL.RTIM has reached its terminal value of 0x7FF). This is a debug feature that allows software to read the ramp time after each cycle. This bit is valid only if OTG_Ver = 1'b1.
22	W1C	0x0	AdpSnsInt ADP Sense Interrupt When this bit is set, it means that the VBUS voltage is greater than VadpSns value or VadpSns is reached. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
21	W1C	0x0	AdpPrbInt ADP Probe Interrupt When this bit is set, it means that the VBUS voltage is greater than VadpPrb or VadpPrb is reached. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
20	RW	0x0	ADPEn ADP Enable When set, the core performs either ADP probing or sensing based on EnaPrb or EnaSns. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
19	R/W SC	0x0	ADPRes ADP Reset When set, ADP controller is reset. This bit is auto-cleared after the reset procedure is complete in ADP controller.This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
18	RW	0x0	EnaSns Enable Sense When programmed to 1'b1, the core performs a sense operation. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).

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Bit	Attr	Reset Value	Description
17	RW	0x0	EnaPrb Enable Probe When programmed to 1'b1, the core performs a probe operation. This bit is valid only if OTG_Ver = 1'b1 (GOTGCTL[20]).
16:6	RO	0x000	RTIM RAMP TIME These bits capture the latest time it took for VBUS to ramp from VADP_SINK to VADP_PRB. The bits are defined in units of 32 kHz clock cycles as follows: 0x000 - 1 cycles 0x001 - 2 cycles 0x002 - 3 cycles and so on till 0x7FF - 2048 cycles A time of 1024 cycles at 32 kHz corresponds to a time of 32 msec. (Note for scaledown ramp_timeout = prb_delta == 2'b00 => 200 cycles prb_delta == 2'b01 => 100 cycles prb_delta == 2'b01 => 50 cycles prb_delta == 2'b01 => 25 cycles.)
5:4	RW	0x0	PrbPer Probe Period These bits sets the TadpPrd as follows: 2'b00 - 0.625 to 0.925 sec (typical 0.775 sec) 2'b01 - 1.25 to 1.85 sec (typical 1.55 sec) 2'b10 - 1.9 to 2.6 sec (typical 2.275 sec) 2'b11 - Reserved (PRB_PER is also scaledown prb_per== 2'b00 => 400 ADP clocks prb_per== 2'b01 => 600 ADP clocks prb_per== 2'b10 => 800 ADP clocks prb_per==2'b11 => 1000 ADP clocks)
3:2	RW	0x0	PrbDelta Probe Delta These bits set the resolution for RTIM value. The bits are defined in units of 32 kHz clock cycles as follows: 2'b00 - 1 cycles 2'b01 - 2 cycles 2'b10 - 3 cycles 2'b11 - 4 cycles For example if this value is chosen to 2'b01, it means that RTIM increments forevery three 32Khz clock cycles.

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	PrbDschg Probe Discharge These bits set the times for TadpDschg. These bits are defined as follows: 2'b00 4 msec (Scaledown 2 32Khz clock cycles) 2'b01 8 msec (Scaledown 4 32Khz clock cycles) 2'b10 16 msec (Scaledown 8 32Khz clock cycles) 2'b11 32 msec (Scaledown 16 32Khz clock cycles)

USBOTG_HPTXFSIZ

Address: Operational Base + offset (0x0100)

Host Periodic Transmit FIFO Size Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	PTxFSize Host Periodic TxFIFO Depth This value is in terms of 32-bit words. Minimum value is 16 Maximum value is 32,768 The power-on reset value of this register is specified as the Largest Host Mode Periodic Tx Data FIFO Depth (parameter OTG_TX_HPERIO_DFIFO_DEPTH). If Enable Dynamic FIFO Sizing was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads returnthe power-on value. If Enable Dynamic FIFO Sizing was selected (parameter OTG_DFIFO_DYNAMIC = 1), you can write a new value in this field. Programmed values must not exceed the power-on value set.

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	PTxFStAddr Host Periodic TxFIFO Start Address The power-on reset value of this register is the sum of the Largest Rx Data FIFO Depth and Largest Non-periodic Tx Data FIFO Depth specified. These parameters are: In shared FIFO operation: OTG_RX_DFIFO_DEPTH + OTG_TX_NPERIO_DFIFO_DEPTH. In dedicated FIFO mode: OTG_RX_DFIFO_DEPTH + OTG_TX_HNPERIO_DFIFO_DEPTH. If Enable Dynamic FIFO Sizing was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing was selected(parameter OTG_DFIFO_DYNAMIC = 1), you can write a new value in this field.Programmed values must not exceed the power-on value.

USBOTG_DIEPTXFn

Address: Operational Base + offset (0x0104)

Device Periodic Transmit FIFO-n Size Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	INEP1TxFDep IN Endpoint TxFIFO Depth This value is in terms of 32-bit words. Minimum value is 16 Maximum value is 32,768 The power-on reset value of this register is specified as the Largest IN Endpoint FIFO number Depth (parameter OTG_TX_DINEP_DFIFO_DEPTH_n)(0 < n <= 15). If Enable Dynamic FIFO Sizing was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing was selected (parameter OTG_DFIFO_DYNAMIC = 1), you can write a new value in this field.Programmed values must not exceed the power-on value.

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	INEP1TxFStAddr IN Endpoint FIFO1 Transmit RAM Start Address This field contains the memory start address for IN endpoint Transmit FIFOn (0 < n <= 15). The power-on reset value of this register is specified as the Largest Rx Data FIFO Depth (parameter OTG_RX_DFIFO_DEPTH). OTG_RX_DFIFO_DEPTH + SUM 0 to n-1 (OTG_DINEP_TXFIFO_DEPTH_n) For example start address of IN endpoint FIFO 1 is OTG_RX_DFIFO_DEPTH + OTG_DINEP_TXFIFO_DEPTH_0. The start address of IN endpoint FIFO 2 is OTG_RX_DFIFO_DEPTH + OTG_DINEP_TXFIFO_DEPTH_0 + OTG_DINEP_TXFIFO_DEPTH_1. If Enable Dynamic FIFO Sizing? was deselected (parameter OTG_DFIFO_DYNAMIC = 0), these flops are optimized, and reads return the power-on value. If Enable Dynamic FIFO Sizing was selected (parameter OTG_DFIFO_DYNAMIC = 1), and you have programmed a new value for RxFIFO depth, you can write that value in this field. Programmed values must not exceed the power-on value set.

USBOTG_HCFG

Address: Operational Base + offset (0x0400)

Host Configuration Register

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	PerSchedEna Enable Periodic Scheduling Applicable in Scatter/Gather DMA mode only. Enables periodic scheduling within the core. Initially, the bit is reset. The core will not process any periodic channels. As soon as this bit is set, the core will get ready to start scheduling periodic channels and sets HCFG.PerSchedStat. The setting of HCFG.PerSchedStat indicates the core has enabled periodic scheduling. Once HCFG.PerSchedEna is set, the application is not supposed to again reset the bit unless HCFG.PerSchedStat is set. As soon as this bit is reset, the core will get ready to stop scheduling periodic channels and resets HCFG.PerSchedStat. In non Scatter/Gather DMA mode, this bit is reserved.

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Bit	Attr	Reset Value	Description
25:24	RW	0x0	FrListEn Frame List Entries The value in the register specifies the number of entries in the Frame list. This field is valid only in Scatter/Gather DMA mode.
23	RW	0x0	DescDMA Enable Scatter/gather DMA in Host mode When the Scatter/Gather DMA option selected during configuration of the RTL, the application can set this bit during initialization to enable the Scatter/Gather DMA operation. NOTE: This bit must be modified only once after a reset. The following combinations are available for programming: GAHBCFG.DMAEn=0, HCFG.DescDMA=0 => Slave mode GAHBCFG.DMAEn=0, HCFG.DescDMA=1 => Invalid GAHBCFG.DMAEn=1, HCFG.DescDMA=0 => Buffered DMA mode GAHBCFG.DMAEn=1, HCFG.DescDMA=1 => Scatter/Gather DMA mode In non Scatter/Gather DMA mode, this bit is reserved.
22:16	RO	0x0	reserved
15:8	RW	0x00	ResValid Resume Validation Period This field is effective only when HCFG.Ena32KHzS is set. It controls the resume period when the core resumes from suspend. The core counts the ResValid number of clock cycles to detect a valid resume when this is set.
7	RW	0x0	Ena32KHzS Enable 32-KHz Suspend Mode This bit can only be set if the USB 1.1 Full-Speed Serial Transceiver Interface has been selected. If USB 1.1 Full-Speed Serial Transceiver Interface has not been selected, this bit must be zero. When the USB 1.1 Full-Speed Serial Transceiver Interface is chosen and this bit is set, the core expects the 48-MHz PHY clock to be switched to 32 KHz during a suspend.
6:3	RO	0x0	reserved
2	RW	0x0	FSLSSupp FS- and LS-Only Support The application uses this bit to control the core enumeration speed. Using this bit, the application can make the core enumerate as a FS host, even if the connected device supports HS traffic. Do not make changes to this field after initial programming. 1'b0: HS/FS/LS, based on the maximum speed supported by the connected device 1'b1: FS/LS-only, even if the connected device can support HS

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	FSLSPclkSel FS/LS PHY Clock Select 2'b00: PHY clock is running at 30/60 MHz 2'b01: PHY clock is running at 48 MHz Others: Reserved

USBOTG_HFIR

Address: Operational Base + offset (0x0404)

Host Frame Interval Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	FrInt Frame Interval The value that the application programs to this field specifies the interval between two consecutive SOFs (FS) or micro-SOFs (HS) or Keep-Alive tokens (HS). This field contains the number of PHY clocks that constitute the required frame interval. The default value set in this field for a FS operation when the PHY clock frequency is 60MHz. The application can write a value to this register only after the Port Enable bit of the Host Port Control and Status register (HPRT.PrtEnaPort) has been set. If no value is programmed, the core calculates the value based on the PHY clock specified in the FS/LS PHY Clock Select field of the Host Configuration register (HCFG.FSLSPclkSel). Do not change the value of this field after the initial configuration. 125 us * (PHY clock frequency for HS) 1 ms * (PHY clock frequency for FS/LS)

USBOTG_HFNUM

Address: Operational Base + offset (0x0408)

Host Frame Number/Frame Time Remaining Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0000	FrRem Frame Time Remaining Indicates the amount of time remaining in the current microframe (HS) or frame (FS/LS), in terms of PHY clocks. This field decrement on each PHY clock. When it reaches zero, this field is reloaded with the value in the Frame Interval register and a new SOF is transmitted on the USB.
15:0	RO	0xffff	FrNum Frame Number This field increment when a new SOF is transmitted on the USB, and is reset to 0 when it reaches 16'h3FFF. This field is writable only if Remove Optional Features was not selected (OTG_RM_OTG_FEATURES = 0). Otherwise, reads return the frame number value.

USBOTG_HPTXSTS

Address: Operational Base + offset (0x0410)

Host Periodic Transmit FIFO/Queue Status Register

Bit	Attr	Reset Value	Description
31:24	RO	0x00	PTxQTop Top of the Periodic Transmit Request Queue This indicates the entry in the Periodic Tx Request Queue that is currently being processes by the MAC. This register is used for debugging. Bit [31]: Odd/Even (micro)frame 1'b0: send in even (micro)frame 1'b1: send in odd (micro)frame Bits [30:27]: Channel/endpoint number Bits [26:25]: Type 2'b00: IN/OUT 2'b01: Zero-length packet 2'b10: CSPLIT 2'b11: Disable channel command Bit [24]: Terminate (last entry for the selected channel/endpoint)

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Bit	Attr	Reset Value	Description
23:16	RO	0x00	PTxQSpcAvail Periodic Transmit Request Queue Space Available Indicates the number of free locations available to be written in the Periodic Transmit Request Queue. This queue holds both IN and OUT requests. 8'h0: Periodic Transmit Request Queue is full 8'h1: 1 location available 8'h2: 2 locations available n: n locations available (0 <=n <= 16) Others: Reserved
15:0	RW	0x0000	PTxFSpcAvail Periodic Transmit Data FIFO Space Available Indicates the number of free locations available to be written to in the Periodic TxFIFO. Values are in terms of 32-bit words 16'h0: Periodic TxFIFO is full 16'h1: 1 word available 16'h2: 2 words available 16'hn: n words available (where 0 . n . 32,768) 16'h8000: 32,768 words available Others: Reserved

USBOTG_HAINT

Address: Operational Base + offset (0x0414)

Host All Channels Interrupt Reigster

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	HAINT Channel Interrupts One bit per channel: Bit 0 for Channel 0, bit 15 for Channel 15

USBOTG_HAINTMSK

Address: Operational Base + offset (0x0418)

Host All Channels Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	HAINTMsk Channel Interrupt Mask One bit per channel: Bit 0 for channel 0, bit 15 for channel 15

USBOTG_HPRT

Address: Operational Base + offset (0x0440)

Host Port Control and Status Register

Bit	Attr	Reset Value	Description
31:19	RO	0x0	reserved
18:17	RO	0x0	PrtSpd Port Speed Indicates the speed of the device attached to this port. 2'b00: High speed 2'b01: Full speed 2'b10: Low speed 2'b11: Reserved
16:13	RW	0x0	PrtTstCtl Port Test Control The application writes a nonzero value to this field to put the port into a Test mode,and the corresponding pattern is signaled on the port. 4'b0000: Test mode disabled 4'b0001: Test_J mode 4'b0010: Test_K mode 4'b0011: Test_SE0_NAK mode 4'b0100: Test_Packet mode 4'b0101: Test_Force_Enable Others: Reserved
12	R/W SC	0x0	PrtPwr Port Power The application uses this field to control power to this port (write 1'b1 to set to 1'b1and write 1'b0 to set to 1'b0), and the core can clear this bit on an over current condition. 1'b0: Power off 1'b1: Power on
11:10	RO	0x0	PrtLnSts Port Line Status Indicates the current logic level USB data lines Bit [10]: Logic level of D+ Bit [11]: Logic level of D

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Bit	Attr	Reset Value	Description
9	RO	0x0	reserved
8	RW	0x0	PrtRst Port Reset When the application sets this bit, a reset sequence is started on this port. The application must time the reset period and clear this bit after the reset sequence is complete. 1'b0: Port not in reset 1'b1: Port in reset To start a reset on the port, the application must leave this bit set for at least the minimum duration mentioned below, as specified in the USB 2.0 specification, Section 7.1.7.5. The application can leave it set for another 10 ms in addition to the required minimum duration, before clearing the bit, even though there is no maximum limit set by the USB standard. High speed: 50 ms Full speed/Low speed: 10 ms
7	R/W SC	0x0	PrtSusp Port Suspend The application sets this bit to put this port in Suspend mode. The core only stops sending SOFs when this is set. To stop the PHY clock, the application must set the Port Clock Stop bit, which asserts suspend input pin of the PHY. The read value of this bit reflects the current suspend status of the port. This bit is cleared by the core after a remote wakeup signal is detected or the application sets the Port Reset bit or Port Resume bit in this register or the Resume/Remote Wakeup Detected Interrupt bit or Disconnect Detected Interrupt bit in the Core Interrupt register (GINTSTS.WkUpInt or GINTSTS.DisconnInt, respectively). 1'b0: Port not in Suspend mode 1'b1: Port in Suspend mode

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Bit	Attr	Reset Value	Description
6	R/W SC	0x0	PrtRes Port Resume The application sets this bit to drive resume signaling on the port. The core continues to drive the resume signal until the application clears this bit. If the core detects a USB remote wakeup sequence, as indicated by the Port Resume/Remote Wakeup Detected Interrupt bit of the Core Interrupt register (GINTSTS.WkUpInt), the core starts driving resume signaling without application intervention and clears this bit when it detects a disconnect condition. The read value of this bit indicates whether the core is currently driving resume signaling. 1'b0: No resume driven 1'b1: Resume driven When LPM is enabled and the core is in the L1 (Sleep) state, setting this bit results in the following behavior: The core continues to drive the resume signal until a pre-determined time specified in the GLPMCFG.HIRD_Thres[3:0] field. If the core detects a USB remote wakeup sequence, as indicated by the Port L1 Resume/Remote L1 Wakeup Detected Interrupt bit of the Core Interrupt register (GINTSTS.L1WkUpInt), the core starts driving resume signaling without application intervention and clears this bit at the end of the resume. The read value of this bit indicates whether the core is currently driving resume signaling. 1'b0: No resume driven 1'b1: Resume driven
5	W1C	0x0	PrtOvrCurrChng Port Overcurrent Change The core sets this bit when the status of the Port Overcurrent Active bit (bit 4) in this register changes.
4	RO	0x0	PrtOvrCurrAct Port Overcurrent Active Indicates the overcurrent condition of the port. 1'b0: No overcurrent condition 1'b1: Overcurrent condition
3	W1C	0x0	PrtEnChng Port Enable/Disable Change The core sets this bit when the status of the Port Enable bit [2] of this register changes.

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Bit	Attr	Reset Value	Description
2	W1C	0x0	PrtEna Port Enable A port is enabled only by the core after a reset sequence, and is disabled by an over-current condition, a disconnect condition, or by the application clearing this bit.The application cannot set this bit by a register write. It can only clear it to disable the port. This bit does not trigger any interrupt to the application. 1'b0: Port disabled 1'b1: Port enabled
1	W1C	0x0	PrtConnDet Port Connect Detected The core sets this bit when a device connection is detected to trigger an interrupt to the application using the Host Port Interrupt bit of the Core Interrupt register (GINTSTS.PrtInt). The application must write a 1 to this bit to clear the interrupt.
0	RO	0x0	PrtConnSts Port Connect Status 0: No device is attached to the port. 1: A device is attached to the port.

USBOTG_HCCHARn

Address: Operational Base + offset (0x0500)

Host Channel-n Characteristics Register

Bit	Attr	Reset Value	Description
31	R/W SC	0x0	ChEna Channel Enable When Scatter/Gather mode is enabled 1'b0: Indicates that the descriptor structure is not yet ready. 1'b1: Indicates that the descriptor structure and data buffer with data is setup and this channel can access the descriptor. When Scatter/Gather mode is disabled, This field is set by the application and cleared by the OTG host. 1'b0: Channel disabled 1'b1: Channel enabled
30	R/W SC	0x0	ChDis Channel Disable The application sets this bit to stop transmitting/receiving data on a channel, even before the transfer for that channel is complete. The application must wait for the Channel Disabled interrupt before treating the channel as disabled.

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Bit	Attr	Reset Value	Description
29	RW	0x0	OddFrm Odd Frame This field is set (reset) by the application to indicate that the OTG host must perform a transfer in an odd (micro)frame. This field is applicable for only periodic (isochronous and interrupt) transactions. 1'b0: Even (micro)frame 1'b1: Odd (micro)frame This field is not applicable for Scatter/Gather DMA mode and need not be programmed by the application and is ignored by the core.
28:22	RW	0x00	DevAddr Device Address This field selects the specific device serving as the data source or sink.
21:20	RW	0x0	MC_EC Multi Count (MC) / Error Count (EC) When the Split Enable bit of the Host Channel-n Split Control register (HCSPLTn.SpltEna) is reset (1'b0), this field indicates to the host the number of transactions that must be executed per microframe for this periodic endpoint. For non periodic transfers, this field is used only in DMA mode, and specifies the number packets to be fetched for this channel before the internal DMA engine changes arbitration. 2'b00: Reserved This field yields undefined results. 2'b01: 1 transaction 2'b10: 2 transactions to be issued for this endpoint per microframe 2'b11: 3 transactions to be issued for this endpoint per microframe When HCSPLTn.SpltEna is set (1'b1), this field indicates the number of immediate retries to be performed for a periodic split transactions on transaction errors. This field must be set to at least 2'b01.
19:18	RW	0x0	EPType Endpoint Type Indicates the transfer type selected. 2'b00: Control 2'b01: Isochronous 2'b10: Bulk 2'b11: Interrupt
17	RW	0x0	LSpdDev Low-Speed Device This field is set by the application to indicate that this channel is communicating to a low-speed device.
16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15	RW	0x0	EPDir Endpoint Direction Indicates whether the transaction is IN or OUT. 1'b0: OUT 1'b1: IN
14:11	RW	0x0	EPNum Endpoint Number Indicates the endpoint number on the device serving as the data source or sink.
10:0	RW	0x000	MPS Maximum Packet Size Indicates the maximum packet size of the associated endpoint.

USBOTG_HCSPLTn

Address: Operational Base + offset (0x0504)

Host Channel-n Split Control Register

Bit	Attr	Reset Value	Description
31	RW	0x0	SpltEna Split Enable The application sets this field to indicate that this channel is enabled to perform split transactions.
30:17	RO	0x0	reserved
16	RW	0x0	CompSplt Do Complete Split The application sets this field to request the OTG host to perform a complete split transaction.
15:14	RW	0x0	XactPos Transaction Position This field is used to determine whether to send all, first, middle, or last payloads with each OUT transaction. 2'b11: All. This is the entire data payload is of this transaction (which is less than or equal to 188 bytes). 2'b10: Begin. This is the first data payload of this transaction (which is larger than 188 bytes). 2'b00: Mid. This is the middle payload of this transaction (which is larger than 188bytes). 2'b01: End. This is the last payload of this transaction (which is larger than 188 bytes).

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Bit	Attr	Reset Value	Description
13:7	RW	0x00	HubAddr Hub Address This field holds the device address of the transaction translator's hub.
6:1	RO	0x0	reserved
0	RW	0x0	PrtAddr Port Address This field is the port number of the recipient transaction translator.

USBOTG_HCINTn

Address: Operational Base + offset (0x0508)

Host Channel-n Interrupt Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	W1C	0x0	DESC_LST_ROLLIntr Descriptor rollover interrupt This bit is valid only when Scatter/Gather DMA mode is enabled. The core sets this bit when the corresponding channel's descriptor list rolls over. For non Scatter/Gather DMA mode, this bit is reserved.
12	W1C	0x0	XCS_XACT_ERR Excessive Transaction Error This bit is valid only when Scatter/Gather DMA mode is enabled. The core sets this bit when 3 consecutive transaction errors occurred on the USB bus. XCS_XACT_ERR will not be generated for Isochronous channels. For non Scatter/Gather DMA mode, this bit is reserved.
11	W1C	0x0	BNAIntr BNA (Buffer Not Available) Interrupt This bit is valid only when Scatter/Gather DMA mode is enabled. The core generates this interrupt when the descriptor accessed is not ready for the Core to process. BNA will not be generated for Isochronous channels. For non Scatter/Gather DMA mode, this bit is reserved.
10	W1C	0x0	DataTglErr Data Toggle Error In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.

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Bit	Attr	Reset Value	Description
9	W1C	0x0	FrmOvrun Frame Overrun In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core
8	W1C	0x0	BblErr Babble Error In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
7	W1C	0x0	XactErr Transaction Error Indicates one of the following errors occurred on the USB: CRC check failure, Timeout, Bit stuff error, False EOP. In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
6	WO	0x0	NYET NYET Response Received Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
5	W1C	0x0	ACK ACK Response Received/Transmitted Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
4	W1C	0x0	NAK NAK Response Received Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
3	W1C	0x0	STALL STALL Response Received Interrupt In Scatter/Gather DMA mode, the interrupt due to this bit is masked in the core.
2	W1C	0x0	AHBErr AHB Error This is generated only in DMA mode when there is an AHB error during AHB read/write.The application can read the corresponding channel's DMA address register to get the error address.

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Bit	Attr	Reset Value	Description
1	W1C	0x0	ChHltd Channel Halted In non Scatter/Gather DMA mode, it indicates the transfer completed abnormally either because of any USB transaction error or in response to disable request by the application or because of a completed transfer.In Scatter/Gather DMA mode, this indicates that transfer completed due to any of the following: EOL being set in descriptor, AHB error, Excessive transaction errors, In response to disable request by the application,Babble, Stall, Buffer Not Available (BNA)
0	W1C	0x0	XferCompl Transfer Completed For Scatter/Gather DMA mode, it indicates that current descriptor processing got completed with IOC bit set in its descriptor. In non Scatter/Gather DMA mode, it indicates that Transfer completed normally without any errors.

USBOTG_HCINTMSKn

Address: Operational Base + offset (0x050c)

Host Channel-n Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	DESC_LST_ROLLIntrMsk Descriptor rollover interrupt Mask register This bit is valid only when Scatter/Gather DMA mode is enabled. In non Scatter/Gather DMA mode, this bit is reserved.
12	RO	0x0	reserved
11	RW	0x0	BNAIntrMsk BNA (Buffer Not Available) Interrupt mask register This bit is valid only when Scatter/Gather DMA mode is enabled. In non Scatter/Gather DMA mode, this bit is reserved.
10	RW	0x0	DataTglErrMsk Data Toggle Error Mask This bit is not applicable in Scatter/Gather DMA mode.
9	RW	0x0	FrmOvrunMsk Frame Overrun Mask This bit is not applicable in Scatter/Gather DMA mode.
8	RW	0x0	BblErrMsk Babble Error Mask This bit is not applicable in Scatter/Gather DMA mode.

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Bit	Attr	Reset Value	Description
7	RW	0x0	XactErrMsk Transaction Error Mask This bit is not applicable in Scatter/Gather DMA mode
6	RW	0x0	NyetMsk NYET Response Received Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
5	RW	0x0	AckMsk ACK Response Received/Transmitted Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
4	RW	0x0	NakMsk NAK Response Received Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
3	RW	0x0	StallMsk STALL Response Received Interrupt Mask This bit is not applicable in Scatter/Gather DMA mode.
2	RW	0x0	AHBErrMsk AHB Error Mask Note: This bit is only accessible when OTG_ARCHITECTURE = 2
1	RW	0x0	ChHltdMsk Channel Halted Mask
0	RW	0x0	XferComplMsk Transfer Completed Mask This bit is valid only when Scatter/Gather DMA mode is enabled. In non Scatter/Gather DMA mode, this bit is reserved.

USBOTG_HCTSIZn

Address: Operational Base + offset (0x0510)

Host Channel-n Transfer Size Register

Bit	Attr	Reset Value	Description
31	RW	0x0	DoPng Do Ping This bit is used only for OUT transfers. Setting this field to 1 directs the host to do PING protocol. Note: Do not set this bit for IN transfers. If this bit is set for IN transfers it disables the channel.

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	Pid PID The application programs this field with the type of PID to use for the initial transaction. The host maintains this field for the rest of the transfer. 2'b00: DATA0 2'b01: DATA2 2'b10: DATA1 2'b11: MDATA (non-control)/SETUP (control)
28:19	RW	0x000	PktCnt Packet Count This field is programmed by the application with the expected number of packets to be transmitted (OUT) or received (IN).The host decrements this count on every successful transmission or reception of an OUT/IN packet. Once this count reaches zero, the application is interrupted to indicate normal completion.The width of this counter is specified as Width of Packet Counters (parameter OTG_PACKET_COUNT_WIDTH).
18:0	RW	0x00000	XferSize Transfer Size For an OUT, this field is the number of data bytes the host sends during the transfer. For an IN, this field is the buffer size that the application has Reserved for the transfer. The application is expected to program this field as an integer multiple of the maximum packet size for IN transactions (periodic and non-periodic). The width of this counter is specified as Width of Transfer Size Counters (parameter OTG_TRANS_COUNT_WIDTH).

USBOTG_HCDMAn

Address: Operational Base + offset (0x0514)

Host Channel-n DMA Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAAddr DMA Address This field holds the start address in the external memory from which the data for the endpoint must be fetched or to which it must be stored. This register is incremented on every AHB transaction.

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USBOTG_HCDMABn

Address: Operational Base + offset (0x051c)

Host Channel-n DMA Buffer Address Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HCDMABn Holds the current buffer address This register is updated as and when the data transfer for the corresponding end point is in progress. This register is present only in Scatter/Gather DMA mode. Otherwise this field is reserved.

USBOTG_DCFG

Address: Operational Base + offset (0x0800)

Device Cconfiguration Register

Bit	Attr	Reset Value	Description
31:26	RW	0x02	ResValid Resume Validation Period This field controls the period when the core resumes from asuspended. When this bit is set, the core counts for the ResValid number of clock cycles to detect a valid resume. This field is effective only when DCFG.Ena32KHzSusp is set.
25:24	RW	0x0	PerSchIntvl Periodic Scheduling Interval PerSchIntvl must be programmed only for Scatter/Gather DMA mode. Description: This field specifies the amount of time the Internal DMA engine must allocate for fetching periodic IN endpoint data. Based on the number of periodic endpoints, this value must be specified as 25,50 or 75% of (micro)frame. When any periodic endpoints are active, the internal DMA engine allocates the specified amount of time in fetching periodic IN endpoint data. When no periodic endpoints are active, then the internal DMA engine services nonperiodic endpoints, ignoring this field. After the specified time within a (micro)frame, the DMA switches to fetching for nonperiodic endpoints. 2'b00: 25% of (micro)frame. 2'b01: 50% of (micro)frame. 2'b10: 75% of (micro)frame. 2'b11: Reserved.

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Bit	Attr	Reset Value	Description
23	RW	0x0	DescDMA Enable Scatter/Gather DMA in Device mode When the Scatter/Gather DMA option selected during configuration of the RTL, the application can set this bit during initialization to enable the Scatter/Gather DMA operation. NOTE: This bit must be modified only once after a reset. The following combinations are available for programming: GAHBCFG.DMAEn=0,DCFG.DescDMA=0 => Slave mode GAHBCFG.DMAEn=0,DCFG.DescDMA=1 => Invalid GAHBCFG.DMAEn=1,DCFG.DescDMA=0 => Buffered DMA mode GAHBCFG.DMAEn=1,DCFG.DescDMA=1 => Scatter/Gather DMA mode
22:18	RW	0x08	EPMisCnt IN Endpoint Mismatch Count This field is valid only in shared FIFO operation.The application programs this filed with a count that determines when the core generates an Endpoint Mismatch interrupt (GINTSTS.EPMis). The core loads this value into an internal counter and decrements it. The counter is reloaded whenever there is a match or when the counter expires. The width of this counter depends on the depth of the Token Queue.
17:13	RO	0x0	reserved
12:11	RW	0x0	PerFrInt Periodic Frame Interval Indicates the time within a (micro)frame at which the application must be notified using the End Of Periodic Frame Interrupt. This can be used to determine if all the isochronous traffic for that (micro)frame is complete. 2'b00: 80% of the (micro)frame interval 2'b01: 85% 2'b10: 90% 2'b11: 95%
10:4	RW	0x00	DevAddr Device Address The application must program this field after every SetAddress control command.
3	RW	0x0	Ena32KHzS Enable 32-KHz Suspend Mode When the USB 1.1 Full-Speed Serial Transceiver Interface is chosen and this bit is set, the core expects the 48-MHz PHY clock to be switched to 32 KHz during a suspend. This bit can only be set if USB 1.1 Full-Speed Serial Transceiver Interface has been selected. If USB 1.1 Full-Speed Serial Transceiver Interface has not been selected, this bit must be zero.

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Bit	Attr	Reset Value	Description
2	RW	0x0	NZStsOUTHShk Non-Zero-Length Status OUT Handshake The application can use this field to select the handshake the core sends on receiving a nonzero-length data packet during the OUT transaction of a control transfer's Status stage. 1'b1: Send a STALL handshake on a nonzero-length status OUT transaction and do not send the received OUT packet to the application. 1'b0: Send the received OUT packet to the application (zero-length or nonzerolength) and send a handshake based on the NAK and STALL bits for the endpoint in the Device Endpoint Control register.
1:0	RW	0x0	DevSpd Device Speed Indicates the speed at which the application requires the core to enumerate, or the maximum speed the application can support. However, the actual bus speed is determined only after the chirp sequence is completed, and is based on the speed of the USB host to which the core is connected. 2'b00: High speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) 2'b01: Full speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) 2'b10: Reserved 2'b11: Full speed (USB 1.1 transceiver clock is 48 MHz)

USBOTG_DCTL

Address: Operational Base + offset (0x0804)

Device Control Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	NakOnBble Set NAK automatically on babble The core sets NAK automatically for the endpoint on which babble is received.

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Bit	Attr	Reset Value	Description
15	RW	0x0	IgnrFrmNum Ignore frame number for isochronous endpoints in case of Scatte Do NOT program IgnrFrmNum bit to 1'b1 when the core is operating in Threshold mode. Note: When Scatter/Gather DMA mode is enabled this feature is not applicable to highspeed, high-bandwidth transfers. When this bit is enabled, there must be only one packet per descriptor. 0: The core transmits the packets only in the frame number in which they are intended to be transmitted. 1: The core ignores the frame number, sending packets immediately as the packets are ready. Scatter/Gather: In Scatter/Gather DMA mode, when this bit is enabled, the packets are not flushed when an ISOC IN token is received for an elapsed frame. When Scatter/Gather DMA mode is disabled, this field is used by the application to enable periodic transfer interrupt. The application can program periodic endpoint transfers for multiple micro)frames. 0: Periodic transfer interrupt feature is disabled; the application must program transfers for periodic endpoints every (micro)frame 1: Periodic transfer interrupt feature is enabled; the application can program transfers for multiple (micro)frames for periodic endpoints. In non-Scatter/Gather DMA mode, the application receives transfer complete interrupt after transfers for multiple (micro)frames are completed.
14:13	RW	0x1	GMC Global Multi Count GMC must be programmed only once after initialization. Applicable only for Scatter/Gather DMA mode. This indicates the number of packets to be serviced for that end point before moving to the next end point. It is only for nonperiodic endpoints. 2'b00: Invalid. 2'b01: 1 packet. 2'b10: 2 packets. 2'b11: 3 packets. When Scatter/Gather DMA mode is disabled, this field is reserved, and reads 2'b00.
12	RO	0x0	reserved
11	RW	0x0	PWROnPrgDone Power-On Programming Done The application uses this bit to indicate that register programming is completed after a wake-up from Power Down mode.

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Bit	Attr	Reset Value	Description
10	WO	0x0	CGOUTNak Clear Global OUT NAK A write to this field clears the Global OUT NAK.
9	WO	0x0	SGOUTNak Set Global OUT NAK A write to this field sets the Global OUT NAK. The application uses this bit to send a NAK handshake on all OUT endpoints. The application must set this bit only after making sure that the Global OUT NAK Effective bit in the Core Interrupt Register (GINTSTS.GOUTNakEff) is cleared.
8	WO	0x0	CGNPInNak Clear Global Non-periodic IN NAK A write to this field clears the Global Non-periodic IN NAK.
7	WO	0x0	SGNPInNak Set Global Non-periodic IN NAK A write to this field sets the Global Non-periodic IN NAK.The application uses this bit to send a NAK handshake on all non-periodic IN endpoints. The core can also set this bit when a timeout condition is detected on a non-periodic endpoint in shared FIFO operation. The application must set this bit only after making sure that the Global IN NAK Effective bit in the Core Interrupt Register (GINTSTS.GINNakEff) is cleared.
6:4	RW	0x0	TstCtl Test Control 3'b000: Test mode disabled 3'b001: Test_J mode 3'b010: Test_K mode 3'b011: Test_SE0_NAK mode 3'b100: Test_Packet mode 3'b101: Test_Force_Enable Others: Reserved
3	RO	0x0	GOUTNakSts Global OUT NAK Status 1'b0: A handshake is sent based on the FIFO Status and the NAK and STALL bit settings. 1'b1: No data is written to the RxFIFO, irrespective of space availability. Sends a NAK handshake on all packets, except on SETUP transactions. All isochronous OUT packets are dropped
2	RO	0x0	GNPINNakSts Global Non-periodic IN NAK Status 1'b0: A handshake is sent out based on the data availability in the transmit FIFO. 1'b1: A NAK handshake is sent out on all non-periodic IN endpoints, irrespective of the data availability in the transmit FIFO.

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Bit	Attr	Reset Value	Description
1	RW	0x0	SftDiscon Soft Disconnect The application uses this bit to signal the DWC_otg core to do a soft disconnect. As long as this bit is set, the host does not see that the device is connected, and the device does not receive signals on the USB. The core stays in the disconnected state until the application clears this bit. 1'b0: Normal operation. When this bit is cleared after a soft disconnect, the core drives the phy_opmode_o signal on the UTMI+ to 2'b00, which generates a device connect event to the USB host. When the device is reconnected, the USB host restarts device enumeration. 1'b1: The core drives the phy_opmode_o signal on the UTMI+ to 2'b01, which generates a device disconnect event to the USB host.
0	RW	0x0	RmtWkUpSig Remote Wakeup Signaling When the application sets this bit, the core initiates remote signaling to wake the USB host. The application must set this bit to instruct the core to exit the Suspend state. As specified in the USB 2.0 specification, the application must clear this bit 1蝡ms after setting it.If LPM is enabled and the core is in the L1 (Sleep) state, when the application sets this bit, the core initiates L1 remote signaling to wake up the USB host. The application must set this bit to instruct the core to exit the Sleep state. As specified in the LPM specification, the hardware automatically clears this bit 50 us (TL1DevDrvResume) after being set by the application. The application must not set this bit when GLPMCFG bRemoteWake from the previous LPM transaction is zero.

USBOTG_DSTS

Address: Operational Base + offset (0x0808)

Device Status Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21:8	RW	0x0000	SOFFN Frame or Microframe Number of the Received SOF When the core is operating at high speed, this field contains a microframe number. When the core is operating at full or low speed, this field contains a frame number.
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x0	ErrticErr Erratic Error The core sets this bit to report any erratic errors (phy_rxvalid_i/phy_rxvldh_i or phy_rxactive_i is asserted for at least 2 ms, due to PHY error) seen on the UTMI+. Due to erratic errors, the DWC_otg core goes into Suspended state and an interrupt is generated to the application with Early Suspend bit of the Core Interrupt register (GINTSTS.ErlySusp). If the early suspend is asserted due to an erratic error, the application can only perform a soft disconnect recover.
2:1	RW	0x0	EnumSpd Enumerated Speed Indicates the speed at which the DWC_otg core has come up after speed detection through a chirp sequence. 2'b00: High speed (PHY clock is running at 30 or 60 MHz) 2'b01: Full speed (PHY clock is running at 30 or 60 MHz) 2'b10: Low speed (PHY clock is running at 48 MHz, internal phy_clk at 6 MHz) 2'b11: Full speed (PHY clock is running at 48 MHz) Low speed is not supported for devices using a UTMI+ PHY.
0	RW	0x0	SuspSts Suspend Status In Device mode, this bit is set as long as a Suspend condition is detected on the USB. The core enters the Suspended state when there is no activity on the utmi_linestate signal for an extended period of time. The core comes out of the suspend: When there is any activity on the utmi_linestate signal, When the application writes to the Remote Wakeup Signaling bit in the Device Control register (DCTL.RmtWkUpSig).

USBOTG_DIEPMSK

Address: Operational Base + offset (0x0810)

Device IN Endpoint common interrupt mask register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	NAKMsk NAK interrupt Mask
12:10	RO	0x0	reserved
9	RW	0x0	BNAInIntrMsk BNA Interrupt Mask

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Bit	Attr	Reset Value	Description
8	RW	0x0	TxfifoUndrnMsk Fifo Underrun Mask
7	RO	0x0	reserved
6	RW	0x0	INEPNakEffMsk IN Endpoint NAK Effective Mask
5	RW	0x0	INTknEPMisMsk IN Token received with EP Mismatch Mask
4	RW	0x0	INTknTXFEmpMsk IN Token Received When TxFIFO Empty Mask
3	RW	0x0	TimeOUTMsk Timeout Condition Mask
2	RW	0x0	AHBErrMsk AHB Error Mask
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DOEPMSK

Address: Operational Base + offset (0x0814)

Device OUT Endpoint common interrupt mask register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	NYETMsk NYET Interrupt Mask
13	RW	0x0	NAKMsk NAK Interrupt Mask
12	RW	0x0	BbleErrMsk Babble Interrupt Mask
11:10	RO	0x0	reserved
9	RW	0x0	BnaOutIntrMsk BNA interrupt Mask
8	RW	0x0	OutPktErrMsk OUT Packet Error Mask
7	RO	0x0	reserved
6	RW	0x0	Back2BackSETup Back-to-Back SETUP Packets Received Mask Applies to control OUT endpoints only.
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	OUTTknEPdisMsk OUT Token Received when Endpoint Disabled Mask Applies to control OUT endpoints only.
3	RW	0x0	SetUPMsk SETUP Phase Done Mask Applies to control endpoints only.
2	RW	0x0	AHBErrMsk AHB Error
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DAINT

Address: Operational Base + offset (0x0818)

Device All Endpoints interrupt register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	OutEPInt OUT Endpoint Interrupt Bits One bit per OUT endpoint: Bit 16 for OUT endpoint 0, bit 31 for OUT endpoint 15
15:0	RO	0x0000	InEpInt IN Endpoint Interrupt Bits One bit per IN Endpoint: Bit 0 for IN endpoint 0, bit 15 for endpoint 15

USBOTG_DAINTMSK

Address: Operational Base + offset (0x081c)

Device All Endpoint interrupt mask register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	OutEpMsk OUT EP Interrupt Mask Bits One per OUT Endpoint: Bit 16 for OUT EP 0, bit 31 for OUT EP 15
15:0	RW	0x0000	InEpMsk IN EP Interrupt Mask Bits One bit per IN Endpoint: Bit 0 for IN EP 0, bit 15 for IN EP 15

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USBOTG_DTKNQR1

Address: Operational Base + offset (0x0820)

Device IN token sequence learning queue read register1

Bit	Attr	Reset Value	Description
31:8	RO	0x000000	EPTkn Endpoint Token Four bits per token represent the endpoint number of the token: Bits [31:28]: Endpoint number of Token 5 Bits [27:24]: Endpoint number of Token 4 ..... Bits [15:12]: Endpoint number of Token 1 Bits [11:8]: Endpoint number of Token 0
7	RO	0x0	WrapBit Wrap Bit This bit is set when the write pointer wraps. It is cleared when the learning queue is cleared.
6:5	RO	0x0	reserved
4:0	RO	0x00	INTknWPtr IN Token Queue Write Pointer

USBOTG_DTKNQR2

Address: Operational Base + offset (0x0824)

Device IN token sequence learning queue read register2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	EPTkn Endpoint Token Four bits per token represent the endpoint number of the token: Bits [31:28]: Endpoint number of Token 13 Bits [27:24]: Endpoint number of Token 12 ..... Bits [7:4]: Endpoint number of Token 7 Bits [3:0]: Endpoint number of Token 6

USBOTG_DVBUSDIS

Address: Operational Base + offset (0x0828)

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Device VBUS discharge time register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0b8f	DVBUSDis Device VBUS Discharge Time Specifies the VBUS discharge time after VBUS pulsing during SRP. This value equals:VBUS discharge time in PHY clocks / 1,024.The value you use depends whether the PHY is operating at 30 MHz (16-bit data width) or 60 MHz (8-bit data width). Depending on your VBUS load, this value can need adjustment.

USBOTG_DVBUSPULSE

Address: Operational Base + offset (0x082c)

Device VBUS Pulsing Timer Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:0	RW	0x000	DVBUSPulse Device VBUS Pulsing Time Specifies the VBUS pulsing time during SRP. This value equals:VBUS pulsing time in PHY clocks / 1,024.The value you use depends whether the PHY is operating at 30 MHz (16-bit data width) or 60 MHz (8-bit data width).

USBOTG_DTHRCTL

Address: Operational Base + offset (0x0830)

Device Threshold Control Register

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x1	ArbPrkEn Arbiter Parking Enable This bit controls internal DMA arbiter parking for IN endpoints. When thresholding is enabled and this bit is set to one, then the arbiter parks on the IN endpoint for which there is a token received on the USB. This is done to avoid getting into underrun conditions. By default the parking is enabled.
26	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
25:17	RW	0x008	RxThrLen Receive Threshold Length This field specifies Receive thresholding size in DWORDS. This field also specifies the amount of data received on the USB before the core can start transmitting on the AHB. The threshold length has to be at least eight DWORDS. The recommended value for ThrLen is to be the same as the programmed AHB Burst Length (GAHBCFG.HBstLen).
16	RW	0x0	RxThrEn Receive Threshold Enable When this bit is set, the core enables thresholding in the receive direction.
15:13	RO	0x0	reserved
12:11	RW	0x0	AHBThrRatio AHB Threshold Ratio These bits define the ratio between the AHB threshold and the MAC threshold for the transmit path only. The AHB threshold always remains less than or equal to the USB threshold, because this does not increase overhead. Both the AHB and the MAC threshold must be DWORD-aligned. The application needs to program TxThrLen and the AHBThrRatio to make the AHB Threshold value DWORD aligned. If the AHB threshold value is not DWORD aligned, the core might not behave correctly. When programming the TxThrLen and AHBThrRatio, the application must ensure that the minimum AHB threshold value does not go below 8 DWORDS to meet the USB turnaround time requirements. 2'b00: AHB threshold = MAC threshold 2'b01: AHB threshold = MAC threshold / 2 2'b10: AHB threshold = MAC threshold / 4 2'b11: AHB threshold = MAC threshold / 8
10:2	RW	0x008	TxThrLen Transmit Threshold Length This field specifies Transmit thresholding size in DWORDS. This field also forms the MAC threshold and specifies the amount of data, in bytes, to be in the corresponding endpoint transmit FIFO before the core can start a transmit on the USB. When the value of AHBThrRatio is 2'h00, the threshold length must be at least 8 DWORDS. If the AHBThrRatio is nonzero, the application must ensure that the AHB threshold value does not go below the recommended 8 DWORDs. This field controls both isochronous and non-isochronous IN endpoint thresholds. The recommended value for ThrLen is to be the same as the programmed AHB Burst Length (GAHBCFG.HBstLen).

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Bit	Attr	Reset Value	Description
1	RW	0x0	ISOThrEn ISO IN Endpoints Threshold Enable When this bit is set, the core enables thresholding for isochronous IN endpoints.
0	RW	0x0	NonISOThrEn Non-ISO IN Endpoints Threshold Enable When this bit is set, the core enables thresholding for Non Isochronous IN endpoints.

USBOTG_DIEPEMPMSK

Address: Operational Base + offset (0x0834)

Device IN endpoint FIFO empty interrupt mask register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	InEpTxfEmpMsk IN EP Tx FIFO Empty Interrupt Mask Bits These bits act as mask bits for DIEPINTn. TxFEmp interrupt One bit per IN Endpoint: Bit 0 for IN endpoint 0 ... Bit 15 for endpoint 15

USBOTG_DEACHINT

Address: Operational Base + offset (0x0838)

Device each endpoint interrupt register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	EchOutEPInt OUT Endpoint Interrupt Bits One bit per OUT endpoint: Bit 16 for OUT endpoint 0 ... Bit 31 for OUT endpoint 15

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	EchInEpInt IN Endpoint Interrupt Bits One bit per IN Endpoint: Bit 0 for IN endpoint 0 ... Bit 15 for endpoint 15

USBOTG_DEACHINTMSK

Address: Operational Base + offset (0x083c)

Device each endpoint interrupt register mask

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	EchOutEpMsk OUT EP Interrupt Mask Bits One per OUT Endpoint: Bit 16 for IN endpoint 0 ... Bit 31 for endpoint 15
15:0	RW	0x0000	EchInEpMsk IN EP Interrupt Mask Bits One bit per IN Endpoint: Bit 0 for IN endpoint 0 ... Bit 15 for endpoint 15

USBOTG_DIEPEACHMSKn

Address: Operational Base + offset (0x0840)

Device each IN endpoint -n interrupt Register

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	NAKMsk NAK interrupt Mask
12:10	RO	0x0	reserved
9	RW	0x0	BNAInIntrMsk BNA interrupt Mask
8	RW	0x0	TxfifoUndrnMsk Fifo Under run Mask

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Bit	Attr	Reset Value	Description
7	RO	0x0	reserved
6	RW	0x0	INEPNakEffMsk IN Endpoint NAK Effective Mask
5	RW	0x0	INTknEPMisMsk IN Token received with EP Mismatch Mask
4	RW	0x0	INTknTXFEmpMsk IN Token Received When TxFIFO Empty Mask
3	RW	0x0	TimeOUTMsk Timeout Condition Mask(Non-isochronous endpoints)
2	RW	0x0	AHBErrMsk AHB Error Mask
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DOEPEACHMSKn

Address: Operational Base + offset (0x0880)

Device each out endpoint-n interrupt register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	RW	0x0	NYETMsk NYET interrupt Mask
13	RW	0x0	NAKMsk NAK interrupt Mask
12	RW	0x0	BbleErrMsk Babble interrupt Mask
11:10	RO	0x0	reserved
9	RW	0x0	BnaOutIntrMsk BNA interrupt Mask
8	RW	0x0	OutPktErrMsk OUT Packet Error Mask
7	RO	0x0	reserved
6	RW	0x0	Back2BackSETup Back-to-Back SETUP Packets Received Mask Applies to control OUT endpoints only.
5	RO	0x0	reserved
4	RW	0x0	OUTTknEPdisMsk OUT Token Received when Endpoint Disabled Mask Applies to control OUT endpoints only.

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Bit	Attr	Reset Value	Description
3	RW	0x0	SetUPMsk SETUP Phase Done Mask Applies to control endpoints only.
2	RW	0x0	AHBErrMsk AHB Error
1	RW	0x0	EPDisbldMsk Endpoint Disabled Interrupt Mask
0	RW	0x0	XferComplMsk Transfer Completed Interrupt Mask

USBOTG_DIEPCTL0

Address: Operational Base + offset (0x0900)

Device control IN endpoint 0 control register

Bit	Attr	Reset Value	Description
31	R/WSC	0x0	EPEna Endpoint Enable When Scatter/Gather DMA mode is enabled, for IN endpoints this bit indicates that the descriptor structure and data buffer with data ready to transmit is setup. When Scatter/Gather DMA mode is disabled-such as in buffer-pointer based DMA mode-this bit indicates that data is ready to be transmitted on the endpoint. The core clears this bit before setting the following interrupts on this endpoint: Endpoint Disabled; Transfer Completed.
30	R/WSC	0x0	EPDis Endpoint Disable The application sets this bit to stop transmitting data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint Disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint Disabled Interrupt. The application must set this bit only if Endpoint Enable is already set for this endpoint.
29:28	RO	0x0	reserved
27	WO	0x0	SNAK Set NAK A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for an endpoint after a SETUP packet is received on that endpoint.

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Bit	Attr	Reset Value	Description
26	WO	0x0	CNAK Clear NAK A write to this bit clears the NAK bit for the endpoint.
25:23	RO	0x0	reserved
22	RW	0x0	TxFNum TxFIFO Number For Shared FIFO operation, this value is always set to 0, indicating that control IN endpoint 0 data is always written in the Non-Periodic Transmit FIFO. For Dedicated FIFO operation, this value is set to the FIFO number that is assigned to IN Endpoint 0.
21	R/WSC	0x0	Stall STALL Handshake The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority.
20	RO	0x0	reserved
19:18	RO	0x0	EPType Endpoint Type Hardcoded to 00 for control
17	RO	0x0	NAKSts NAK Status Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status 1'b1: The core is transmitting NAK handshakes on this endpoint. When this bit is set, either by the application or core, the core stops transmitting data, even if there is data available in the TxFIFO. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
16	RO	0x0	reserved
15	RO	0x1	USBActEP USB Active Endpoint This bit is always set to 1, indicating that control endpoint 0 is always active in all configurations and interfaces.
14:11	RW	0x0	NextEp Next Endpoint Applies to non-periodic IN endpoints only. Indicates the endpoint number to be fetched after the data for the current endpoint is fetched. The core can access this field, even when the Endpoint Enable (EPEna) bit is not set. This field is not valid in Slave mode. Note: This field is valid only for Shared FIFO operations.
10:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	MPS Maximum Packet Size Applies to IN and OUT endpoints. The application must program this field with the maximum packet size for the current logical endpoint. 2'b00: 64 bytes 2'b01: 32 bytes 2'b10: 16 bytes 2'b11: 8 bytes

USBOTG_DIEPINTn

Address: Operational Base + offset (0x0908)

Device Endpoint-n Interrupt Register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	W1C	0x0	NYETIntrpt NYET interrupt The core generates this interrupt when a NYET response is transmitted for a non isochronous OUT endpoint.
13	W1C	0x0	NAKIntrpt NAK interrupt The core generates this interrupt when a NAK is transmitted or received by the device.In case of isochronous IN endpoints the interrupt gets generated when a zero length packet is transmitted due to un-availability of data in the TXFifo.
12	W1C	0x0	BbleErrIntrpt BbleErr (Babble Error) interrupt The core generates this interrupt when babble is received for the endpoint.
11	W1C	0x0	PktDrpSts Packet Dropped Status This bit indicates to the application that an ISOC OUT packet has been dropped. This bit does not have an associated mask bit and does not generate an interrupt. Dependency: This bit is valid in non Scatter/Gather DMA mode when periodic transfer interrupt feature is selected.
10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	W1C	0x0	BNAIntr BNA (Buffer Not Available) Interrupt The core generates this interrupt when the descriptor accessed is not ready for the Core to process, such as Host busy or DMA done Dependency: This bit is valid only when Scatter/Gather DMA mode is enabled.
8	W1C	0x0	TxfifoUndrn FIFO Underrun Applies to IN endpoints only. The core generates this interrupt when it detects a transmit FIFO underrun condition for this endpoint. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1; Thresholding is enabled; OUT Packet Error(OutPktErr). Applies to OUT endpoints only. This interrupt is asserted when the core detects an overflow or a CRC error for an OUT packet. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1; Thresholding is enabled.
7	W1C	0x0	TxFEmp Transmit FIFO Empty This bit is valid only for IN Endpoints.This interrupt is asserted when the TxFIFO for this endpoint is either half or completely empty. The half or completely empty status is determined by the TxFIFO Empty Level bit in the Core AHB Configuration register (GAHBCFG.NPTxFEmpLvl)).
6	W1C	0x0	INEPNakEff IN Endpoint NAK Effective Applies to periodic IN endpoints only. This bit can be cleared when the application clears the IN endpoint NAK by writing to DIEPCTLn.CNAK. This interrupt indicates that the core has sampled the NAK bit set (either by the application or by the core). The interrupt indicates that the IN endpoint NAK bit set by the application has taken effect in the core. This interrupt does not guarantee that a NAK handshake is sent on the USB. A STALL bit takes priority over a NAK bit. This bit is applicable only when the endpoint is enabled. Back-to-Back SETUP Packets Received (Back2BackSETup) Applies to Control OUT endpoints only. This bit indicates that the core has received more than three back-to-back SETUP packets for this particular endpoint.

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Bit	Attr	Reset Value	Description
5	W1C	0x0	INTknEPMis IN Token Received with EP Mismatch Applies to non-periodic IN endpoints only. Indicates that the data in the top of the non-periodic TxFIFO belongs to an endpoint other than the one for which the IN token was received. This interrupt is asserted on the endpoint for which the IN token was received. Status Phase Received For Control Write (StsPhseRcvd) This interrupt is valid only for Control OUT endpoints and only in Scatter Gather DMA mode. This interrupt is generated only after the core has transferred all the data that the host has sent during the data phase of a control write transfer, to the system memory buffer. The interrupt indicates to the application that the host has switched from data phase to the status phase of a Control Write transfer. The application can use this interrupt to ACK or STALL the Status phase, after it has decoded the data phase. This is applicable only in case of Scatter Gather DMA mode.
4	W1C	0x0	INTknTXFEmp IN Token Received When TxFIFO is Empty Indicates that an IN token was received when the associated TxFIFO periodic/nonperiodic) was empty. This interrupt is asserted on the endpoint for which the IN token was received. OUT Token Received When Endpoint Disabled (OUTTknEPdis) Indicates that an OUT token was received when the endpoint was not yet enabled. This interrupt is asserted on the endpoint for which the OUT token was received.
3	W1C	0x0	TimeOUT Timeout Condition In shared TX FIFO mode, applies to non-isochronous IN endpoints only. In dedicated FIFO mode, applies only to Control IN endpoints. In Scatter/Gather DMA mode, the TimeOUT interrupt is not asserted. Indicates that the core has detected a timeout condition on the USB for the last IN token on this endpoint. SETUP Phase Done (SetUp) Applies to control OUT endpoints only. Indicates that the SETUP phase for the control endpoint is complete and no more back-to-back SETUP packets were received for the current control transfer. On this interrupt, the application can decode the received SETUP data packet.
2	W1C	0x0	AHBErr AHB Error Applies to IN and OUT endpoints. This is generated only in Internal DMA mode when there is an AHB error during an AHB read/write. The application can read the corresponding endpoint DMA address register to get the error address.

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Bit	Attr	Reset Value	Description
1	W1C	0x0	EPDisbld Endpoint Disabled Interrupt Applies to IN and OUT endpoints. This bit indicates that the endpoint is disabled per the application's request.
0	W1C	0x0	XferCompl Transfer Completed Interrupt Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled: For IN endpoint this field indicates that the requested data from the descriptor is moved from external system memory to internal FIFO. For OUT endpoint this field indicates that the requested data from the internal FIFO is moved to external system memory. This interrupt is generated only when the corresponding endpoint descriptor is closed, and the IOC bit for the corresponding descriptor is set. When Scatter/Gather DMA mode is disabled, this field indicates that the programmed transfer is complete on the AHB as well as on the USB, for this endpoint.

USBOTG_DIEPTSIZn

Address: Operational Base + offset (0x0910)

Device endpoint n transfer size register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	MC Multi Count Applies to IN endpoints only. For periodic IN endpoints, this field indicates the number of packets that must be transmitted per microframe on the USB. The core uses this field to calculate the data PID for isochronous IN endpoints. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets For non-periodic IN endpoints, this field is valid only in Internal DMA mode. It specifies the number of packets the core must fetch for an IN endpoint before it switches to the endpoint pointed to by the Next Endpoint field of the Device Endpoint-n Control register (DIEPCTLn.NextEp). Received Data PID (RxDPID) Applies to isochronous OUT endpoints only. This is the data PID received in the last packet for this endpoint. 2'b00: DATA0 2'b01: DATA2 2'b10: DATA1 2'b11: MDATA SETUP Packet Count (SUPCnt).Applies to control OUT Endpoints only.This field specifies the number of back-to-back SETUP data packets the endpoint can receive. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets
28:19	RW	0x000	PktCnt Packet Count Indicates the total number of USB packets that constitute the Transfer Size amount of data for this endpoint. The power-on value is specified for Width of Packet Counters during coreConsultant configuration (parameter OTG_PACKET_COUNT_WIDTH). IN Endpoints: This field is decremented every time a packet (maximum size or short packet) is read from the TxFIFO. OUT Endpoints: This field is decremented every time a packet (maximum size or short packet) is written to the RxFIFO.

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Bit	Attr	Reset Value	Description
18:0	RW	0x00000	XferSize Transfer Size This field contains the transfer size in bytes for the current endpoint. The power-on value is specified for Width of Transfer Size Counters during coreConsultant configuration (parameter OTG_TRANS_COUNT_WIDTH). The core only interrupts the application after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet. IN Endpoints: The core decrements this field every time a packet from the external memory is written to the TxFIFO.OUT Endpoints: The core decrements this field every time a packet is read from the RxFIFO and written to the external memory.

USBOTG_DIEPDMAn

Address: Operational Base + offset (0x0914)

Device endpoint-n DMA Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAAddr DMA Address Holds the start address of the external memory for storing or fetching endpoint data. Note: For control endpoints, this field stores control OUT data packets as well as SETUP transaction data packets. When more than three SETUP packets are received back-to-back, the SETUP data packet in the memory is overwritten. This register is incremented on every AHB transaction. The application can give only a DWORD-aligned address. When Scatter/Gather DMA mode is not enabled, the application programs the start address value in this field. When Scatter/Gather DMA mode is enabled, this field indicates the base pointer for the descriptor list.

USBOTG_DTXFSTSn

Address: Operational Base + offset (0x0918)

Device IN endpoint transmit FIFO status register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	INEPTxFSpcAvail IN Endpoint TxFIFO Space Avail Indicates the amount of free space available in the Endpoint TxFIFO.Values are in terms of 32-bit words. 16'h0: Endpoint TxFIFO is full 16'h1: 1 word available 16'h2: 2 words available 16'hn: n words available (where 0 . n . 32,768) 16'h8000: 32,768 words available Others: Reserved

USBOTG_DIEPDMABn

Address: Operational Base + offset (0x091c)

Device endpoint-n DMA buffer address register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	DMABufferAddr DMA Buffer Address Holds the current buffer address.This register is updated as and when the data transfer for the corresponding end point is in progress. This register is present only in Scatter/Gather DMA mode. Otherwise this field is reserved.

USBOTG_DIEPCTLn

Address: Operational Base + offset (0x0920)

Device endpoint-n control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	R/W SC	0x0	EPEna Endpoint Enable Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled, For IN endpoints this bit indicates that the descriptor structure and data buffer with data ready to transmit is setup. For OUT endpoint it indicates that the descriptor structure and data buffer to receive data is setup. When Scatter/Gather DMA mode is enabled-such as for buffer-pointer based DMA mode: For IN endpoints, this bit indicates that data is ready to be transmitted on the endpoint ; For OUT endpoints, this bit indicates that the application has allocated the memory to start receiving data from the USB. The core clears this bit before setting any of the following interrupts on this endpoint: SETUP Phase Done, Endpoint Disabled, Transfer Completed. Note: For control endpoints in DMA mode, this bit must be set to be able to transfer SETUP data packets in memory.
30	R/W SC	0x0	EPDis Endpoint Disable Applies to IN and OUT endpoints. The application sets this bit to stop transmitting/receiving data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint Disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint Disabled interrupt. The application must set this bit only if Endpoint Enable is already set for this endpoint.
29	WO	0x0	SetD1PID Set DATA1 PID Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA1.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. Set Odd (micro)frame (SetOddFr). Applies to isochronous IN and OUTendpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to odd (micro)frame. This field is not applicable for Scatter/Gather DMA mode.

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Bit	Attr	Reset Value	Description
28	WO	0x0	SetD0PID Set DATA0 PID Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA0.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. In non-Scatter/Gather DMA mode: Set Even (micro)frame (SetEvenFr) Applies to isochronous IN and OUT endpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to even (micro)frame. When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is in the transmit descriptor structure. The frame in which to receive data is updated in received descriptor structure.
27	WO	0x0	SNAK Set NAK Applies to IN and OUT endpoints. A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for OUT endpoints on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
26	WO	0x0	CNAK Clear NAK Applies to IN and OUT endpoints. A write to this bit clears the NAK bit for the endpoint.
25:22	RW	0x0	TxFNum TxFIFO Number Shared FIFO Operation:non-periodic endpoints must set this bit to zero. Periodic endpoints must map this to the corresponding Periodic TxFIFO number. 4'h0: Non-Periodic TxFIFO; Others: Specified Periodic TxFIFO number. Note: An interrupt IN endpoint can be configured as a non-periodic endpoint for applications such as mass storage. The core treats an IN endpoint as a non-periodic endpoint if the TxFNum field is set to 0. Otherwise, a separate periodic FIFO must be allocated for an interrupt IN endpoint using coreConsultant, and the number of this FIFO must be programmed into the TxFNum field. Configuring an interrupt IN endpoint as a non-periodic endpoint saves the extra periodic FIFO area. Dedicated FIFO Operation:these bits specify the FIFO number associated with this endpoint. Each active IN endpoint must be programmed to a separate FIFO number. This field is valid only for IN endpoints.

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Bit	Attr	Reset Value	Description
21	RW	0x0	Stall STALL Handshake Applies to non-control, non-isochronous IN and OUT endpoints only. The application sets this bit to stall all tokens from the USB host to this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Only the application can clear this bit, never the core. Applies to control endpoints only. The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
20	RW	0x0	Snp Snoop Mode Applies to OUT endpoints only. This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.
19:18	RW	0x0	EPType Endpoint Type Applies to IN and OUT endpoints. This is the transfer type supported by this logical endpoint. 2'b00: Control 2'b01: Isochronous 2'b10: Bulk 2'b11: Interrupt
17	RO	0x0	NAKSts NAK Status Applies to IN and OUT endpoints. Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status. 1'b1: The core is transmitting NAK handshakes on this endpoint. When either the application or the core sets this bit: The core stops receiving any data on an OUT endpoint, even if there is space in the RxFIFO to accommodate the incoming packet. For non-isochronous IN endpoints: The core stops transmitting any data on an IN endpoint, even if there data is available in the TxFIFO. For isochronous IN endpoints: The core sends out a zero-length data packet, even if there data is available in the TxFIFO. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.

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Bit	Attr	Reset Value	Description
16	RO	0x0	DPID Endpoint Data PID Applies to interrupt/bulk IN and OUT endpoints only. Contains the PID of the packet to be received or transmitted on this endpoint. The application must program the PID of the first packet to be received or transmitted on this endpoint, after the endpoint is activated. The applications use the SetD1PID and SetD0PID fields of this register to program either DATA0 or DATA1 PID. 1'b0: DATA0 1'b1: DATA1 This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode.Even/Odd (Micro)Frame (EO_FrNum) In non-Scatter/Gather DMA mode: Applies to isochronous IN and OUT endpoints only.Indicates the (micro) rame number in which the core transmits/receives isochronous data for this endpoint. The application must program the even/odd (micro) frame number in which it intends to transmit/receive isochronous data for this endpoint using the SetEvnFr and SetOddFr fields in this register. 1'b0: Even (micro)frame 1'b1: Odd (micro)frame When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is provided in the transmit descriptor structure. The frame in which data is received is updated in receive descriptor structure.
15	R/W SC	0x0	USBActEP USB Active Endpoint Applies to IN and OUT endpoints.Indicates whether this endpoint is active in the current configuration and interface. The core clears this bit for all endpoints (other than EP 0) after detecting a USB reset. After receiving the SetConfiguration and SetInterface commands, the application must program endpoint registers accordingly and set this bit.
14:11	RW	0x0	NextEp Next Endpoint Applies to non-periodic IN endpoints only.Indicates the endpoint number to be fetched after the data for the current endpoint is fetched. The core can access this field, even when the Endpoint Enable (EPEna) bit is low. This field is not valid in Slave mode operation.Note: This field is valid only for Shared FIFO operations.
10:0	RW	0x000	MPS Maximum Packet Size Applies to IN and OUT endpoints.The application must program this field with the maximum packet size for the current logical endpoint. This value is in bytes.

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USBOTG_DOEPCTL0

Address: Operational Base + offset (0x0b00)

Device control OUT endpoint 0 control register

Bit	Attr	Reset Value	Description
31	R/W SC	0x0	EPEna Endpoint Enable When Scatter/Gather DMA mode is enabled, for OUT endpoints this bit indicates that the descriptor structure and data buffer to receive data is setup. When Scatter/Gather DMA mode is disabled?such as for buffer-pointer based DMA mode)-this bit indicates that the application has allocated the memory to start receiving data from the USB. The core clears this bit before setting any of the following interrupts on this endpoint: SETUP Phase Done, Endpoint Disabled, Transfer Completed. *Note: In DMA mode, this bit must be set for the core to transfer SETUP data packets into memory.*
30	WO	0x0	EPDis Endpoint Disable The application cannot disable control OUT endpoint 0.
29:28	RO	0x0	reserved
27	WO	0x0	SNAK Set NAK A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set bit on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
26	WO	0x0	CNAK Clear NAK A write to this bit clears the NAK bit for the endpoint.
25:22	RO	0x0	reserved
21	R/W SC	0x0	Stall STALL Handshake The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
20	RW	0x0	Snp Snoop Mode This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.

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Bit	Attr	Reset Value	Description
19:18	RO	0x0	EPType Endpoint Type Hardcoded to 2'b00 for control.
17	RO	0x0	NAKSts NAK Status Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status. 1'b1: The core is transmitting NAK handshakes on this endpoint. When either the application or the core sets this bit, the core stops receiving data, even if there is space in the RxFIFO to accommodate the incoming packet. Irrespective of this bit setting, the core always responds to SETUP data packets with an ACK handshake.
16	RO	0x0	reserved
15	RO	0x0	USBActEP USB Active Endpoint This bit is always set to 1, indicating that a control endpoint 0 is always active in all configurations and interfaces.
14:2	RO	0x0	reserved
1:0	RO	0x0	MPS Maximum Packet Size The maximum packet size for control OUT endpoint 0 is the same as what is programmed in control IN Endpoint 0. 2'b00: 64 bytes 2'b01: 32 bytes 2'b10: 16 bytes 2'b11: 8 bytes

USBOTG_DOEPINTn

Address: Operational Base + offset (0x0b08)

Device endpoint-n control register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14	W1C	0x0	NYETIntrpt NYET interrupt The core generates this interrupt when a NYET response is transmitted for a non isochronous OUT endpoint.

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Bit	Attr	Reset Value	Description
13	W1C	0x0	NAKIntrpt NAK interrupt The core generates this interrupt when a NAK is transmitted or received by the device.In case of isochronous IN endpoints the interrupt gets generated when a zero length packet is transmitted due to un-availability of data in the TXFifo.
12	W1C	0x0	BbleErrIntrpt BbleErr (Babble Error) interrupt The core generates this interrupt when babble is received for the endpoint.
11	W1C	0x0	PktDrpSts Packet Dropped Status This bit indicates to the application that an ISOC OUT packet has been dropped. This bit does not have an associated mask bit and does not generate an interrupt. Dependency: This bit is valid in non Scatter/Gather DMA mode when periodic transfer interrupt feature is selected.
10	RO	0x0	reserved
9	W1C	0x0	BNAIntr BNA (Buffer Not Available) Interrupt The core generates this interrupt when the descriptor accessed is not ready for the Core to process, such as Host busy or DMA done. Dependency: This bit is valid only when Scatter/Gather DMA mode is enabled.
8	W1C	0x0	TxfifoUndrn FIFO Underrun Applies to IN endpoints only. The core generates this interrupt when it detects a transmit FIFO underrun condition for this endpoint. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1, Thresholding is enabled, OUT Packet Error (OutPktErr). Applies to OUT endpoints only. This interrupt is asserted when the core detects an overflow or a CRC error for an OUT packet. Dependency: This interrupt is valid only when both of the following conditions are true: Parameter OTG_EN_DED_TX_FIFO==1, Thresholding is enabled.
7	W1C	0x0	TxFEmp Transmit FIFO Empty This bit is valid only for IN Endpoints. This interrupt is asserted when the TxFIFO for this endpoint is either half or completely empty. The half or completely empty status is determined by the TxFIFO Empty Level bit in the Core AHB Configuration register(GAHBCFG.NPTxFEmpLvl)).

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Bit	Attr	Reset Value	Description
6	W1C	0x0	INEPNakEff IN Endpoint NAK Effective Applies to periodic IN endpoints only. This bit can be cleared when the application clears the IN endpoint NAK by writing to DIEPCTLn.CNAK. This interrupt indicates that the core has sampled the NAK bit set (either by the application or by the core). The interrupt indicates that the IN endpoint NAK bit set by the application has taken effect in the core. This interrupt does not guarantee that a NAK handshake is sent on the USB. A STALL bit takes priority over a NAK bit. This bit is applicable only when the endpoint is enabled. Back-to-Back SETUP Packets Received (Back2BackSETup) Applies to Control OUT endpoints only. This bit indicates that the core has received more than three back-to-back SETUP packets for this particular endpoint.
5	W1C	0x0	INTknEPMis IN Token Received with EP Mismatch Applies to non-periodic IN endpoints only. Indicates that the data in the top of the non-periodic TxFIFO belongs to an endpoint other than the one for which the IN token was received. This interrupt is asserted on the endpoint for which the IN token was received. Status Phase Received For Control Write (StsPhseRcvd) This interrupt is valid only for Control OUT endpoints and only in Scatter Gather DMA mode.
4	W1C	0x0	INTknTXFEmp IN Token Received When TxFIFO is Empty Indicates that an IN token was received when the associated TxFIFO periodic/nonperiodic) was empty. This interrupt is asserted on the endpoint for which the IN token was received. OUT Token Received When Endpoint Disabled (OUTTknEPdis) Indicates that an OUT token was received when the endpoint was not yet enabled. This interrupt is asserted on the endpoint for which the OUT token was received.
3	W1C	0x0	TimeOUT Timeout Condition In shared TX FIFO mode, applies to non-isochronous IN endpoints only. In dedicated FIFO mode, applies only to Control IN endpoints. In Scatter/Gather DMA mode, the TimeOUT interrupt is not asserted.Indicates that the core has detected a timeout condition on the USB for the last IN token on this endpoint. SETUP Phase Done (SetUp). Applies to control OUT endpoints only. Indicates that the SETUP phase for the control endpoint is complete and no more back-to-back SETUP packets were received for the current control transfer. On this interrupt, the application can decode the received SETUP data packet.

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Bit	Attr	Reset Value	Description
2	W1C	0x0	AHBErr AHB Error Applies to IN and OUT endpoints. This is generated only in Internal DMA mode when there is an AHB error during an AHB read/write. The application can read the corresponding endpoint DMA address register to get the error address.
1	W1C	0x0	EPDisbld Endpoint Disabled Interrupt Applies to IN and OUT endpoints. This bit indicates that the endpoint is disabled per the application's request.
0	W1C	0x0	XferCompl Transfer Completed Interrupt Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled. For IN endpoint this field indicates that the requested data from the descriptor is moved from external system memory to internal FIFO. For OUT endpoint this field indicates that the requested data from the internal FIFO is moved to external system memory. This interrupt is generated only when the corresponding endpoint descriptor is closed, and the IOC bit for the corresponding descriptor is set. When Scatter/Gather DMA mode is disabled, this field indicates that the programmed transfer is complete on the AHB as well as on the USB, for this endpoint.

USBOTG_DOEPTSIZn

Address: Operational Base + offset (0x0b10)

Device endpoint n transfer size register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
30:29	RW	0x0	MC Multi Count Applies to IN endpoints only. For periodic IN endpoints, this field indicates the number of packets that must be transmitted per microframe on the USB. The core uses this field to calculate the data PID for isochronous IN endpoints. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets For non-periodic IN endpoints, this field is valid only in Internal DMA mode. It specifies the number of packets the core must fetch for an IN endpoint before it switches to the endpoint pointed to by the Next Endpoint field of the Device Endpoint-n Control register (DIEPCTLn.NextEp). Received Data PID (RxDPID) Applies to isochronous OUT endpoints only. This is the data PID received in the last packet for this endpoint. 2'b00: DATA0 2'b01: DATA2 2'b10: DATA1 2'b11: MDATA SETUP Packet Count (SUPCnt).Applies to control OUT Endpoints only. This field specifies the number of back-to-back SETUP data packets the endpoint can receive. 2'b01: 1 packet 2'b10: 2 packets 2'b11: 3 packets
28:19	RW	0x000	PktCnt Packet Count Indicates the total number of USB packets that constitute the Transfer Size amount of data for this endpoint. The power-on value is specified for Width of Packet Counters during coreConsultant configuration (parameter OTG_PACKET_COUNT_WIDTH). IN Endpoints: This field is decremented every time a packet (maximum size or short packet) is read from the TxFIFO. OUT Endpoints: This field is decremented every time a packet (maximum size or short packet) is written to the RxFIFO.

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Bit	Attr	Reset Value	Description
18:0	RW	0x00000	XferSize Transfer Size This field contains the transfer size in bytes for the current endpoint. The power-on value is specified for Width of Transfer Size Counters during coreConsultant configuration (parameter OTG_TRANS_COUNT_WIDTH). The core only interrupts the application after it has exhausted the transfer size amount of data. The transfer size can be set to the maximum packet size of the endpoint, to be interrupted at the end of each packet. IN Endpoints: The core decrements this field every time a packet from the external memory is written to the TxFIFO. OUT Endpoints: The core decrements this field every time a packet is read from the RxFIFO and written to the external memory.

USBOTG_DOEPDMAn

Address: Operational Base + offset (0x0b14)

Device Endpoint-n DMA Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DMAAddr DMA Address Holds the start address of the external memory for storing or fetching endpoint data. Note: For control endpoints, this field stores control OUT data packets as well as SETUP transaction data packets. When more than three SETUP packets are received back-to-back, the SETUP data packet in the memory is overwritten. This register is incremented on every AHB transaction. The application can give only a DWORD-aligned address. When Scatter/Gather DMA mode is not enabled, the application programs the start address value in this field. When Scatter/Gather DMA mode is enabled, this field indicates the base pointer for the descriptor list.

USBOTG_DOEPDMABn

Address: Operational Base + offset (0x0b1c)

Device endpoint-n DMA buffer address register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	DMABufferAddr DMA Buffer Address Holds the current buffer address.This register is updated as and when the data transfer for the corresponding end point is in progress. This register is present only in Scatter/Gather DMA mode. Otherwise this field is reserved.

USBOTG_DOEPCTLn

Address: Operational Base + offset (0x0b20)

Device endpoint-n control register

Bit	Attr	Reset Value	Description
31	R/W SC	0x0	EPEna Endpoint Enable Applies to IN and OUT endpoints. When Scatter/Gather DMA mode is enabled, For IN endpoints this bit indicates that the descriptor structure and data buffer with data ready to transmit is setup. For OUT endpoint it indicates that the descriptor structure and data buffer to receive data is setup. When Scatter/Gather DMA mode is enabled-such as for buffer-pointer based DMA mode: For IN endpoints, this bit indicates that data is ready to be transmitted on the endpoint; For OUT endpoints, this bit indicates that the application has allocated the memory to start receiving data from the USB. The core clears this bit before setting any of the following interrupts on this endpoint: SETUP Phase Done, Endpoint Disabled, Transfer Completed. Note: For control endpoints in DMA mode, this bit must be set to be able to transfer SETUP data packets in memory.
30	R/W SC	0x0	EPDis Endpoint Disable Applies to IN and OUT endpoints. The application sets this bit to stop transmitting/receiving data on an endpoint, even before the transfer for that endpoint is complete. The application must wait for the Endpoint Disabled interrupt before treating the endpoint as disabled. The core clears this bit before setting the Endpoint Disabled interrupt. The application must set this bit only if Endpoint Enable is already set for this endpoint.

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Bit	Attr	Reset Value	Description
29	RO	0x0	SetD1PID Field0001 Abstract Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA1.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. Set Odd (micro)frame (SetOddFr). Applies to isochronous IN and OUT endpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to odd (micro)frame. This field is not applicable for Scatter/Gather DMA mode.
28	WO	0x0	SetD0PID Set DATA0 PID Applies to interrupt/bulk IN and OUT endpoints only.Writing to this field sets the Endpoint Data PID (DPID) field in this register to DATA0.This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. In non-Scatter/Gather DMA mode: Set Even (micro)frame (SetEvenFr) Applies to isochronous IN and OUT endpoints only. Writing to this field sets the Even/Odd (micro)frame (EO_FrNum) field to even (micro)frame. When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is in the transmit descriptor structure. The frame in which to receive data is updated in receive descriptor structure.
27	WO	0x0	SNAK Set NAK Applies to IN and OUT endpoints. A write to this bit sets the NAK bit for the endpoint. Using this bit, the application can control the transmission of NAK handshakes on an endpoint. The core can also set this bit for OUT endpoints on a Transfer Completed interrupt, or after a SETUP is received on the endpoint.
26	WO	0x0	CNAK Clear NAK Applies to IN and OUT endpoints. A write to this bit clears the NAK bit for the endpoint.

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Bit	Attr	Reset Value	Description
25:22	RW	0x0	TxFNum TxFIFO Number Shared FIFO Operation:non-periodic endpoints must set this bit to zero. Periodic endpoints must map this to the corresponding Periodic TxFIFO number. 4'h0: Non-Periodic TxFIFO; Others: Specified Periodic TxFIFO number. Note: An interrupt IN endpoint can be configured as a non-periodic endpoint for applications such as mass storage. The core treats an IN endpoint as a non-periodic endpoint if the TxFNum field is set to 0. Otherwise, a separate periodic FIFO must be allocated for an interrupt IN endpoint using coreConsultant, and the number of this FIFO must be programmed into the TxFNum field. Configuring an interrupt IN endpoint as a non-periodic endpoint saves the extra periodic FIFO area. Dedicated FIFO Operation:these bits specify the FIFO number associated with this endpoint. Each active IN endpoint must be programmed to a separate FIFO number. This field is valid only for IN endpoints.
21	RW	0x0	Stall STALL Handshake Applies to non-control, non-isochronous IN and OUT endpoints only. The application sets this bit to stall all tokens from the USB host to this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Only the application can clear this bit, never the core. Applies to control endpoints only. The application can only set this bit, and the core clears it, when a SETUP token is received for this endpoint. If a NAK bit, Global Non-periodic IN NAK, or Global OUT NAK is set along with this bit, the STALL bit takes priority. Irrespective of this bit's setting, the core always responds to SETUP data packets with an ACK handshake.
20	RW	0x0	Snp Snoop Mode Applies to OUT endpoints only. This bit configures the endpoint to Snoop mode. In Snoop mode, the core does not check the correctness of OUT packets before transferring them to application memory.
19:18	RW	0x0	EPType Endpoint Type Applies to IN and OUT endpoints. This is the transfer type supported by this logical endpoint. 2'b00: Control 2'b01: Isochronous 2'b10: Bulk 2'b11: Interrupt

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Bit	Attr	Reset Value	Description
17	RO	0x0	NAKSts NAK Status Applies to IN and OUT endpoints. Indicates the following: 1'b0: The core is transmitting non-NAK handshakes based on the FIFO status. 1'b1: The core is transmitting NAK handshakes on this endpoint. When either the application or the core sets this bit: The core stops receiving any data on an OUT endpoint, even if there is space in the RxFIFO to accommodate the incoming packet.
16	RO	0x0	DPID Endpoint Data PID Applies to interrupt/bulk IN and OUT endpoints only. Contains the PID of the packet to be received or transmitted on this endpoint. The application must program the PID of the first packet to be received or transmitted on this endpoint, after the endpoint is activated. The applications use the SetD1PID and SetD0PID fields of this register to program either DATA0 or DATA1 PID. 1'b0: DATA0 1'b1: DATA1 This field is applicable both for Scatter/Gather DMA mode and non-Scatter/Gather DMA mode. Even/Odd (Micro)Frame (EO_FrNum). In non-Scatter/Gather DMA mode: Applies to isochronous IN and OUT endpoints only. Indicates the (micro)frame number in which the core transmits/receives isochronous data for this endpoint. The application must program the even/odd (micro) frame number in which it intends to transmit/receive isochronous data for this endpoint using the SetEvnFr and SetOddFr fields in this register. 1'b0: Even (micro)frame 1'b1: Odd (micro)frame When Scatter/Gather DMA mode is enabled, this field is reserved. The frame number in which to send data is provided in the transmit descriptor structure. The frame in which data is received is updated in receive descriptor structure.
15	R/WSC	0x0	USBActEP USB Active Endpoint Applies to IN and OUT endpoints. Indicates whether this endpoint is active in the current configuration and interface. The core clears this bit for all endpoints (other than EP 0) after detecting a USB reset. After receiving the SetConfiguration and SetInterface commands, the application must program endpoint registers accordingly and set this bit.

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Bit	Attr	Reset Value	Description
14:11	RW	0x0	NextEp Next Endpoint Applies to non-periodic IN endpoints only. Indicates the endpoint number to be fetched after the data for the current endpoint is fetched. The core can access this field, even when the Endpoint Enable (EPEna) bit is low. This field is not valid in Slave mode operation. Note: This field is valid only for Shared FIFO operations.
10:0	RW	0x000	MPS Maximum Packet Size Applies to IN and OUT endpoints. The application must program this field with the maximum packet size for the current logical endpoint. This value is in bytes.

USBOTG_PCGCR

Address: Operational Base + offset (0x0b24)

Power and clock gating control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:14	RW	0x0802e	RestoreValue Restore Value (Applicable only when Hibernation is enabled (OTG_EN_PWROPT=2). Defines port clock select for different speeds. [31] if_dev_mode - 1: Device mode, core restored as device - 0: Host mode, core restored as host [30:29] p2hd_prt_spd (PRT speed) - 00: HS - 01: FS - 10: LS - 11: Reserved [28:27] p2hd_dev_enum_spd (Device enumerated speed) - 00: HS - 01: FS (30/60 MHz clk) - 10: LS - 11: FS (48 MHz clk) [26:20] mac_dev_addr (MAC device address) Device address [19] mac_termselect (Termination selection) - 0: HS_TERM (Program for High Speed) - 1: FS_TERM (Program for Full Speed) [18:17] mac_xcvrselect (Transceiver select) - 00: HS_XCVR (High Speed) - 01: FS_XCVR (Full Speed) - 10: LS_XCVR (Low Speed) - 11: LFS_XCVR (Reserved) [16] sh2pl_prt_ctl[0] - 1: prt_power enabled - 0: prt_power disabled [15:14] prt_clk_sel (Refer prt_clk_sel table)
13	RW	0x0	EssRegRestored Essential Register Values Restored (Applicable only when Hibernation is enabled (OTG_EN_PWROPT=2). When a value of 1 is written to this field, it indicates that register values of essential registers have been restored.
12:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9	RO	0x0	RestoreMode Restore Mode (Applicable only when Hibernation is enabled (OTG_EN_PWROPT=2). The application should program this bit to specify the restore mode during RESTORE POINT before programming PCGCCTL.EssRegRest bit is set. Host Mode: 1'b0: Host Initiated Resume, Host Initiated Reset 1'b1: Device Initiated Remote Wake up Device Mode: 1'b0: Device Initiated Remote Wake up 1'b1: Host Initiated Resume, Host Initiated Reset
8	RW	0x0	ResetAfterSusp Reset After Suspend Applicable in Partial power-down mode. In partial power-down mode of operation, this bit needs to be set in host mode before clamp is removed if the host needs to issue reset after suspend. If this bit is not set,then the host issues resume after suspend. This bit is not applicable in device mode and non-partial power-down mode.In Hibernation mode, this bit needs to be set at RESTORE_POINT before PCGCCTL.EssRegRestored is set. In this case, PCGCCTL.restore_mode needs to be set to wait_restore.
7	RO	0x0	L1Suspended Deep Sleep This bit indicates that the PHY is in deep sleep when in L1 state.
6	RO	0x0	PhySleep PHY in Sleep This bit indicates that the PHY is in the Sleep state.
5	RW	0x0	Enbl_L1Gating Enable Sleep Clock Gating When this bit is set, core internal clock gating is enabled in Sleep state if the core cannot assert utmi_l1_suspend_n. When this bit is not set, the PHY clock is not gated in Sleep state.
4	RO	0x0	reserved
3	RW	0x0	RstPdwnModule Reset Power-Down Modules This bit is valid only in Partial Power-Down mode. The application sets this bit when the power is turned off. The application clears this bit after the power is turned on and the PHY clock is up.

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Bit	Attr	Reset Value	Description
2	RW	0x0	PwrClmp Power Clamp This bit is valid only in Partial Power-Down mode (OTG_EN_PWROPT = 1). The application sets this bit before the power is turned off to clamp the signals between the power-on modules and the power-off modules. The application clears the bit to disable the clamping before the power is turned on.
1	RW	0x0	GateHclk Gate Hclk The application sets this bit to gate hclk to modules other than the AHB Slave and Master and wakeup logic when the USB is uspended or the session is not valid. The application clears this bit when the USB is resumed or a new session starts.
0	RW	0x0	StopPclk Stop Pclk The application sets this bit to stop the PHY clock (phy_clk) when the USB is suspended, the session is not valid, or the device is disconnected. The application clears this bit when the USB is resumed or a new session starts.

USBOTG_EPBUF0

Address: Operational Base + offset (0x1000)

Device endpoint 0 / host out channel 0 address

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	EPBUF0 From0x1000 to 0x2000, EPBUF for endport0

USBOTG_EPBUF1

Address: Operational Base + offset (0x2000)

Device endpoint 1 / host out channel 1 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF1 From0x2000 to 0x3000, EPBUF for endport1

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USBOTG_EPBUF2

Address: Operational Base + offset (0x3000)

Device endpoint 2 / host out channel 2 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF2 From0x3000 to 0x4000, EPBUF for endport2

USBOTG_EPBUF3

Address: Operational Base + offset (0x4000)

Device endpoint 3 / host out channel 3 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF3 From0x4000 to 0x5000, EPBUF for endport3

USBOTG_EPBUF4

Address: Operational Base + offset (0x5000)

Device endpoint 4 / host out channel 4 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF4 From0x5000 to 0x6000, EPBUF for endport4

USBOTG_EPBUF5

Address: Operational Base + offset (0x6000)

Device endpoint 5 / host out channel 5 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF5 From0x6000 to 0x7000, EPBUF for endport5

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USBOTG_EPBUF6

Address: Operational Base + offset (0x7000)

Device endpoint 6 / host out channel 6 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF6 From0x7000 to 0x8000, EPBUF for endport6

USBOTG_EPBUF7

Address: Operational Base + offset (0x8000)

Device endpoint 7 / host out channel 7 address

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	EPBUF7 From0x8000 to 0x9000, EPBUF for endport7

21.7 Interface description

Table 23-1USB OTG 2.0 Interface Description

Module Pin	Direction	Pad Name	pinmux
USB0PN	A	IO_USB0_PN	-
USBRBIAS	A	IO_USB_RBIAS_0	-
USB0PP	A	IO_USB0_PP	-
VBUS	A	IO_USB_VBUS_0	-
USB0ID	A	IO_USB0_ID	-

***Note**: **A**—Analog pad ;**AP**—Analog power; **AG**—Analog ground ;**DP**—Digital power ;**DG**–Digital ground;*

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21.8 Application Note

21.8.1 Suspend Mode

When PHY is in suspend state

- COMMONONN = 1'b1, 480M clock invalid
- COMMONONN = 1'b0, 480M clock output available.

Please refer to "Chapter GRF" for configuration details

21.8.2 Reset a port

Please refer to "Chapter CRU" for more details.

21.8.3 Relative GRF Registers

USBPHY contains 284 bit registers to configure USB PHY. These bits are used to adjust DP/DM SI. Please refer to "Chapter GRF" for more details.

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Chapter 24 USB3.0 OTG Controller

22.1 Overview

USB3.0OTG Controller can act as static host, static device, USB2.0/3.0 OTG A device or B device basing on the status of input ID from USB3.0 PHY. It can perform data transmission between host and device as host or device for Super-Speed / High-Speed / Full-Speed / Low-Speed.

USB3.0 OTG controller supports the following features:

- General Features
  1. Compatible Specification
     - Universal Serial Bus 3.0 Specification, Revision 1.0
     - Universal Serial Bus Specification, Revision 2.0
     - eXtensible Host Controller Interface for Universal Serial Bus (xHCI), Revision 1.1
  2. Support Control/Bulk(including stream)/Interrupt/Isochronous Transfer
  3. Simultaneous IN and OUT transfer for USB3.0, up to 8Gbps bandwidth
  4. Descriptor caching and data pre-fetching used to improve system performance in high-latency systems
  5. LPM protocol in USB 2.0 and U0, U1, U2, and U3 states for USB 3.0
  6. Dynamic FIFO memory allocation for endpoints
  7. Keep-Alive feature in LS mode and (micro-)SOFs in HS/FS modes
  8. Low MIPS requirement
     - Driver involved only in setting up transfers and high-level error recovery
     - Hardware handles data packing and routing to a specific pipe
- Application Interface Features
  1. AHB Slave interface
  2. AXI Master interface
     - Programmable burst lengths up to 16
     - Handle fixed burst address alignment
     - Programmable number of outstanding read/write requests up to 16
     - Concurrent read/write to get best performance of USB3.0 duplex operation
- USB3.0 Device Features
  1. Up to 7 IN endpoints, including control endpoint 0
  2. Up to 6 OUT endpoints, including control endpoint 0
  3. Up to 13 endpoint transfer resources, each one for each endpoint
  4. Flexible endpoint configuration for multiple applications/USB set-configuration modes
  5. Hardware handles ERDY and burst
  6. Stream-based bulk endpoints with controller automatically initiating data movement
  7. Isochronous endpoints with isochronous data in data buffers
  8. Flexible Descriptor with rich set of features to support buffer interrupt moderation, multiple transfers, isochronous, control, and scattered buffering support
- USB Class-Specific Device Features
  1. Stream support for UASP application
  2. Gathering of scattered packet to support Ethernet Over USB
  3. Scheduling of multiple Ethernet packets without interrupt
  4. Variable FIFO buffer allocation for each endpoint
  5. For isochronous applications, scheduling of variable-length payloads for each microframe
  6. Microframe precise scheduling for isochronous applications
  7. Configurable endpoint type selection and dynamic FIFO allocation to facilitate multi-function/composite device implementation. During set-config or alternate-setting, device resources are reconfigured to meet the configuration or alternate setting requirements.
- USB 3.0 xHCI Host Features
  1. Support up to 64 devices
  2. Support 1 interrupter

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3. Support 1 USB2.0 port and 1 Super-Speed port
4. Support xHCI Debug Capability
5. Concurrent USB3.0/USB2.0 traffic, up to 8.48Gbps bandwidth
6. Support standard or open-source xHCI and class driver

- USB 3.0 Dual-Role Device (DRD) Features
  1. Static Device Operation
  2. Static Host Operation
  3. USB3.0/USB2.0 OTG A device and B device basing on ID
  4. Not support USB3.0/USB2.0 OTG session request protocol(SRP), host negotiation protocol(HNP) and Role Swap Protocol(RSP)

22.2 Block Diagram

USB3.0 OTG Controller comprises with:

- Bus Interface/List Management: Register Interface/Data and Descriptors DMA management
- HS/FS/LS MAC : USB2.0 part logic
- SS MAC : SS part logic
- USB2.0 PHY: UTMI+ interface USB2.0 PHY, see Chapter USB2.0 PHY
- TypeC PHY: Pipe Interface Super-Speed PHY, see Chapter TypeC PHY

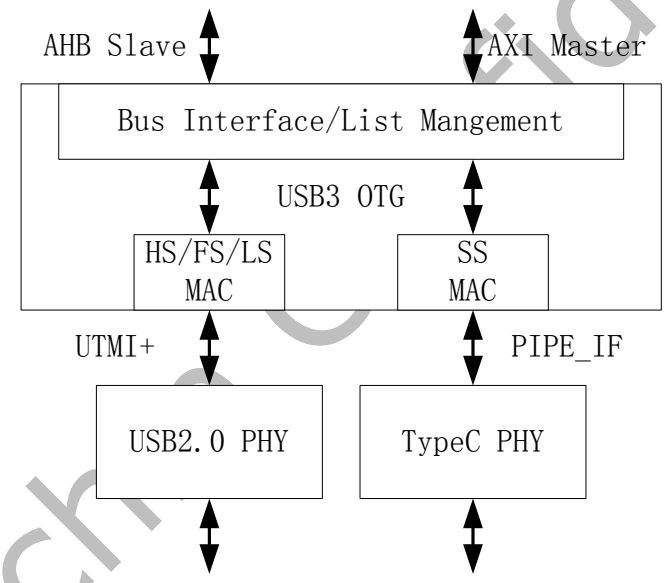

Fig.24-1USB3.0 OTG Block Diagram

22.3 Function Description

As a USB3.0 OTG controller, it can act as static xHCI host controller, static device controller, USB3.0/2.0 OTG A device or B device basing on ID of USB3.0 PHY.
As device controller, it can work on either USB2.0 speed or Super-Speed basing on speed of host attached to, and process USB tractions described in the descriptors (read back from external memory by AXI master) to/from UTMI+ interface and Pipe Interface of USB3.0 PHY.
As host controller, it can work on USB2.0 speed, Super-Speed or both basing on speed or type of attached device, and process USB tractions described in the descriptors (read back from external memory by AXI master) to/from UTMI+ interface and Pipe Interface of USB3.0 PHY.

22.4 Register Description

22.4.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

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Table 24-1USB3 Address Mapping

Offset Address Range	Register Type
0x00000 ~ 0x07FFF	xHCI Registers, see xHCI spec.
0x0C100 ~ 0x0C6FF	Global Registers
0x0C700 ~ 0x0CBFF	Device Controller Registers
0x0CC00 ~ 0x0CFFF	Unused/Reserved
0x40000 ~ 0x7FFFF	Internal RAM0 – Debug Access (256KB)
0x80000 ~ 0xBFFFF	Internal RAM1 – Debug Access (256KB)
0xC0000 ~ 0xFFFFF	Internal RAM2 – Debug Access (256KB)

22.4.2 Registers Summary

Name	Offset	Size	Reset Value	Description
USB3_GSBUSCFG0	0xc100	W	0x00000001	Global SoC Bus Configuration Register 0
USB3_GSBUSCFG1	0xc104	W	0x00000300	Global SoC Bus Configuration Register 1
USB3_GTXTHRCFG	0xc108	W	0x00000000	Global Tx Threshold Control Register
USB3_GRXTHRCFG	0xc10c	W	0x00000000	Global Rx Threshold Control Register
USB3_GCTL	0xc110	W	0x30c12004	Global Core Control Register
USB3_GPMSTS	0xc114	W	0x00000000	Global Power Management Status Register
USB3_GSTS	0xc118	W	0x7e800000	Global Status Register
USB3_GUCTL1	0xc11c	W	0x0004018a	Global User Control Register 1
USB3_GSNPSID	0xc120	W	0x5533290a	Global SNPS ID Register
USB3_GGPIO	0xc124	W	0x00000000	Global General Purpose Input/Output Register
USB3_GUID	0xc128	W	0x12345678	Global User ID Register
USB3_GUCTL	0xc12c	W	0x02008010	Global User Control Register
USB3_GBUSERRADDRLO	0xc130	W	0x00000000	Global SoC Bus Error Address Register - Low
USB3_GBUSERRADDRHI	0xc134	W	0x00000000	Global SoC Bus Error Address Register - High
USB3_GPRTBIMAPLO	0xc138	W	0x00000000	Global SS Port to Bus Instance Mapping Register - Low
USB3_GHWPARAMS0	0xc140	W	0x2020400a	Global Hardware Parameters Register 0
USB3_GHWPARAMS1	0xc144	W	0x0160c93b	Global Hardware Parameters Register 1
USB3_GHWPARAMS2	0xc148	W	0x12345678	Global Hardware Parameters Register 2
USB3_GHWPARAMS3	0xc14c	W	0x069cd085	Global Hardware Parameters Register 3
USB3_GHWPARAMS4	0xc150	W	0x47822008	Global Hardware Parameters Register 4
USB3_GHWPARAMS5	0xc154	W	0x04202088	Global Hardware Parameters Register 5

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Name	Offset	Size	Reset Value	Description
USB3_GHWPARAMS6	0xc158	W	0x077c8020	Global Hardware Parameters Register 6
USB3_GHWPARAMS7	0xc15c	W	0x03080756	Global Hardware Parameters Register 7
USB3_GDBGFIFOSPACE	0xc160	W	0x00420000	Global Debug Queue/FIFO Space Available Register
USB3_GDBGLTSSM	0xc164	W	0x41010440	Global Debug LTSSM Register
USB3_GDBGLNMCC	0xc168	W	0x00000000	Global Debug LNMCC Register
USB3_GDBGBMU	0xc16c	W	0x00000000	Global Debug BMU Register
USB3_GDBGLSPMUX	0xc170	W	0x003f0000	Global Debug LSP MUX Register - Device
USB3_GDBGLSP	0xc174	W	0x00000000	Global Debug LSP Register
USB3_GDBGEPINFO0	0xc178	W	0x00000000	Global Debug Endpoint Information Register 0
USB3_GDBGEPINFO1	0xc17c	W	0x00800000	Global Debug Endpoint Information Register 1
USB3_GPRTBIMAP_HSLO	0xc180	W	0x00000000	Global High-Speed Port to Bus Instance Mapping Register - Low
USB3_GPRTBIMAP_FSLO	0xc188	W	0x00000000	Global Full-Speed Port to Bus Instance Mapping Register - Low
USB3_GUSB2PHYCFG0	0xc200	W	0x40102400	Global USB2 PHY Configuration Register 0
USB3_GUSB3PIPECTL0	0xc2c0	W	0x010c0002	Global USB3 PIPE Control Register 0
USB3_GTXFIFOSIZ0~6	0xc300~ 0xc318	W	0x00000042	Global Transmit FIFO Size Register n
USB3_GRXFIFOSIZ0~2	0xc380~ 0xC388	W	0x00000285	Global Receive FIFO Size Register n
USB3_GEVNTADRLO0	0xc400	W	0x00000000	Global Event Buffer Address (Low) Register 0
USB3_GEVNTADRHI0	0xc404	W	0x00000000	Global Event Buffer Address (High) Register 0
USB3_GEVNTSIZ0	0xc408	W	0x00000000	Global Event Buffer Size Register 0
USB3_GEVNTCOUNT0	0xc40c	W	0x00000000	Global Event Buffer Count Register 0
USB3_GHWPARAMS8	0xc600	W	0x0000077c	Global Hardware Parameters Register 8
USB3_GTXFIFOPRIDEV	0xc610	W	0x00000000	Global Device TX FIFO DMA Priority Register
USB3_GTXFIFOPRIHST	0xc618	W	0x00000000	Global Host TX FIFO DMA Priority Register
USB3_GRXFIFOPRIHST	0xc61c	W	0x00000000	Global Host RX FIFO DMA Priority Register
USB3_GFIFOPRIDBC	0xc620	W	0x00000000	Global Host Debug Capability DMA Priority Register

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Name	Offset	Size	Reset Value	Description
USB3_GDMAHLRATIO	0xc624	W	0x00000008	Global Host FIFO DMA High-Low Priority Ratio Register
USB3_GFLADJ	0xc630	W	0x00000000	Global Frame Length Adjustment Register
USB3_DCFG	0xc700	W	0x00080004	Device Configuration Register
USB3_DCTL	0xc704	W	0x00f00000	Device Control Register
USB3_DEVTEN	0xc708	W	0x00000000	Device Event Enable Register
USB3_DSTS	0xc70c	W	0x00500004	Device Status Register
USB3_DGCMDPAR	0xc710	W	0x00000000	Device Generic Command Parameter Register
USB3_DGCMD	0xc714	W	0x00000000	Device Generic Command Register
USB3_DALEPENA	0xc720	W	0x00000000	Device Active USB Endpoint Enable Register
USB3_DEPnCMDPAR2	0xc800~ 0xc8c0	W	0x00000000	Device Physical Endpoint-n Command Parameter 2 Register
USB3_DEPnCMDPAR1	0xc804~ 0xc8c4	W	0x00000000	Device Physical Endpoint-n Command Parameter 1 Register
USB3_DEPnCMDPAR0	0xc808~ 0xc8c8	W	0x00000000	Device Physical Endpoint-n Command Parameter 0 Register
USB3_DEPnCMD	0xc80c~ 0xc8cc	W	0x00000000	Device Physical Endpoint-n Command Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

Name	Offset	Size	Reset Value	Description
USB3PHYUTMI_REG00	0x0000	W	0x00000000	PHYREG00
USB3PHYUTMI_REG04	0x0004	W	0x00000040	PHYREG04
USB3PHYUTMI_REG08	0x0008	W	0x00000000	PHYREG08
USB3PHYUTMI_REG0C	0x000c	W	0x00000000	PHYREG0C
USB3PHYUTMI_REG10	0x0010	W	0x00000000	PHYREG10
USB3PHYUTMI_REG14	0x0014	W	0x00000000	PHYREG14
USB3PHYUTMI_REG18	0x0018	W	0x00000000	PHYREG18
USB3PHYUTMI_REG1C	0x001c	W	0x00000010	PHYREG1C
USB3PHYUTMI_REG20	0x0020	W	0x00000000	PHYREG20
USB3PHYUTMI_REG24	0x0024	W	0x00000000	PHYREG24
USB3PHYUTMI_REG28	0x0028	W	0x00000000	PHYREG28
USB3PHYUTMI_REG2C	0x002c	W	0x00000020	PHYREG2C
USB3PHYUTMI_REG30	0x0030	W	0x00000005	PHYREG30
USB3PHYUTMI_REG34	0x0034	W	0x00000000	PHYREG34
USB3PHYUTMI_REG38	0x0038	W	0x00000000	PHYREG38
USB3PHYUTMI_REG40	0x0040	W	0x00000061	PHYREG40
USB3PHYUTMI_REG44	0x0044	W	0x00000008	PHYREG44
USB3PHYUTMI_REG48	0x0048	W	0x0000000e	PHYREG48
USB3PHYUTMI_REG4C	0x004c	W	0x00000040	PHYREG4C
USB3PHYUTMI_REG50	0x0050	W	0x00000003	PHYREG50
USB3PHYUTMI_REG54	0x0054	W	0x00000070	PHYREG54

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Name	Offset	Size	Reset Value	Description
USB3PHYUTMI_REG58	0x0058	W	0x00000055	PHYREG58
USB3PHYUTMI_REG5C	0x005c	W	0x00000001	PHYREG5C
USB3PHYUTMI_REG60	0x0060	W	0x00000003	PHYREG60
USB3PHYUTMI_REG64	0x0064	W	0x000000e0	PHYREG64
USB3PHYUTMI_REG68	0x0068	W	0x000000f8	PHYREG68
USB3PHYUTMI_REG6C	0x006c	W	0x00000012	PHYREG6C
USB3PHYUTMI_REG70	0x0070	W	0x000000b0	PHYREG70
USB3PHYUTMI_REG74	0x0074	W	0x00000070	PHYREG74
USB3PHYUTMI_REG78	0x0078	W	0x00000000	PHYREG78
USB3PHYUTMI_REG7C	0x007c	W	0x00000000	PHYREG7C
USB3PHYUTMI_REG80	0x0080	W	0x000000f8	PHYREG80
USB3PHYUTMI_REG84	0x0084	W	0x00000001	PHYREG84
USB3PHYUTMI_REG88	0x0088	W	0x00000000	PHYREG88
USB3PHYUTMI_REG8C	0x008c	W	0x00000000	PHYREG8C
USB3PHYUTMI_REG90	0x0090	W	0x00000055	PHYREG90
USB3PHYUTMI_REG94	0x0094	W	0x00000055	PHYREG94
USB3PHYUTMI_REG98	0x0098	W	0x00000055	PHYREG98
USB3PHYUTMI_REG9C	0x009c	W	0x00000055	PHYREG9C
USB3PHYUTMI_REGA0	0x00a0	W	0x000000f0	PHYREGA0
USB3PHYUTMI_REGA4	0x00a4	W	0x0000008d	PHYREGA4
USB3PHYUTMI_REGA8	0x00a8	W	0x00000000	PHYREGA8

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

Name	Offset	Size	Reset Value	Description
USB3PHYPIPE_REG20	0x0020	W	0x00000086	PHYREG20
USB3PHYPIPE_REG28	0x0028	W	0x00000071	PHYREG28
USB3PHYPIPE_REG30	0x0030	W	0x00000006	PHYREG30
USB3PHYPIPE_REG44	0x0044	W	0x00000000	PHYREG44
USB3PHYPIPE_REG58	0x0058	W	0x00000000	PHYREG58
USB3PHYPIPE_REG64	0x0064	W	0x00000040	PHYREG64
USB3PHYPIPE_REG68	0x0068	W	0x00000070	PHYREG68
USB3PHYPIPE_REG88	0x0088	W	0x00000000	PHYREG88
USB3PHYPIPE_REGD0	0x00d0	W	0x00000000	PHYREGD0
USB3PHYPIPE_REGE0	0x00e0	W	0x00000000	PHYREGE0
USB3PHYPIPE_REG118	0x0118	W	0x00000002	PHYREG118
USB3PHYPIPE_REG11C	0x011c	W	0x00000071	PHYREG11C
USB3PHYPIPE_REG12C	0x012c	W	0x00000000	PHYREG12C
USB3PHYPIPE_REG180	0x0180	W	0x00000000	PHYREG180
USB3PHYPIPE_REG188	0x0188	W	0x00000001	PHYREG188
USB3PHYPIPE_REG18C	0x018c	W	0x00000000	PHYREG18C
USB3PHYPIPE_REG190	0x0190	W	0x00000000	PHYREG190
USB3PHYPIPE_REG1A8	0x01a8	W	0x00000004	PHYREG1A8
USB3PHYPIPE_REG1AC	0x01ac	W	0x00000000	PHYREG1AC
USB3PHYPIPE_REG1B8	0x01b8	W	0x00000001	PHYREG1B8

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

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22.4.3 Detail Register Description

USB3_GSBUSCFG0
Address: Operational Base + offset (0xc100)
Global SoC Bus Configuration Register 0

Bit	Attr	Reset Value	Description
31:28	RW	0x0	DATRDREQINFO DATRDREQINFO AXI-cache for Data Read (DatRdReqInfo)
27:24	RW	0x0	DESRDREQINFO DESRDREQINFO AXI-cache for Descriptor Read (DesRdReqInfo).
23:20	RW	0x0	DATWRREQINFO DATWRREQINFO AXI-cache for Data Write (DatWrReqInfo).
19:16	RW	0x0	DESWRREQINFO DESWRREQINFO AXI-cache for Descriptor Write (DesWrReqInfo)
15:12	RO	0x0	reserved
11	RW	0x0	DATBIGEND Data Access is Big Endian This bit controls the endian mode for data accesses.0, Little-endian (default); 1, Big-endian;
10	RW	0x0	DESBIGEND Descriptor Access is Big Endian This bit controls the endian mode for descriptor accesses. 0, Little-endian (default); 1, Big-endian.
9:8	RO	0x0	reserved
7	RW	0x0	INCR256BRSTENA INCR256 Burst Type Enable If software set this bit to 1, the AXI master uses INCR to do the 256-beat burst.
6	RW	0x0	INCR128BRSTENA INCR128 Burst Type Enable If software set this bit to 1, the AXI master uses INCR to do the 128-beat burst.
5	RW	0x0	INCR64BRSTENA INCR64 Burst Type Enable If software set this bit to 1, AXI master uses INCR to do the 64-beat burst.
4	RW	0x0	INCR32BRSTENA INCR32 Burst Type Enable If software set this bit to 1, the AXI master uses INCR to do the 32-beat burst.

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Bit	Attr	Reset Value	Description
3	RW	0x0	INCR16BRSTENA INCR16 Burst Type Enable If software set this bit to 1, the AXI master uses INCR to do the 16-beat burst.
2	RW	0x0	INCR8BRSTENA INCR8 Burst Type Enable If software set this bit to 1, the AXI master uses INCR to do the 8-beat burst.
1	RW	0x0	INCR4BRSTENA INCR4 Burst Type Enable When this bit is enabled the controller is allowed to do bursts of beat length 1, 2, 3, and 4. It is highly recommended that this bit is enabled to prevent descriptor reads and writes from being broken up into separate transfers.
0	RW	0x1	INCRBRSTENA Undefined Length INCR Burst Type Enable This bit determines the set of burst lengths the master interface uses. It works in conjunction with the GSBUSCFG0[7:1] enables (INCR256/128/64/32/16/8/4). 0: INCRX burst mode ARLEN/AWLEN do not use INCR. They use only the following burst lengths: 1; 4 (if GSBUSCFG0.INCR4BrstEna = 1); 8 (if GSBUSCFG0.INCR8BrstEna = 1); 16 (if GSBUSCFG0.INCR16BrstEna = 1); 32 (if GSBUSCFG0.INCR32BrstEna = 1); 64 (if GSBUSCFG0.INCR64BrstEna = 1); 128 (if GSBUSCFG0.INCR128BrstEna = 1); 256 (if GSBUSCFG0.INCR256BrstEna = 1); 1: INCR (undefined length) burst mode ARLEN/AWLEN uses any length less than or equal to the largest-enabled burst length of INCR4/8/16/32/64/128/256. For cache line-aligned applications, this bit is typically set to 0 to ensure that the master interface uses only power-of-2 burst lengths (as enabled via GSBUSCFG0[7:0]).

USB3_GSBUSCFG1

Address: Operational Base + offset (0xc104)

Global SoC Bus Configuration Register 1

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
12	RW	0x0	EN1KPAGE 1K Page Boundary Enable By default (this bit is disabled) the AXI breaks transfers at the 4k page boundary. When this bit is enabled, the AXI master (DMA data) breaks transfers at the 1k page boundary.
11:8	RW	0x3	PipeTransLimit AXI Pipelined Transfers Burst Request Limit The field controls the number of outstanding pipelined transfer requests the AXI master pushes to the AXI slave. When the AXI master reaches this limit, it does not make any more requests on the AXI ARADDR and AWADDR buses until the associated data phases complete. This field is encoded as follows: 0: 1 request 1: 2 requests 2: 3 requests 3: 4 requests ... F: 16 requests
7:0	RO	0x0	reserved

USB3_GTXTHRCFG

Address: Operational Base + offset (0xc108)

Global Tx Threshold Control Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RO	0x0	USBTxPktCntSel USB Transmit Packet Count Enable This field enables/disables the USB transmission multi-packet thresholding: 0: USB transmission multi-packet thresholding is disabled; the core can only start transmission on the USB after the entire packet has been fetched into the corresponding TXFIFO. 1: USB transmission multi-packet thresholding is enabled. The core can only start transmission on the USB after USB Transmit Packet Count amount of packets for the USB transaction (burst) are already in the corresponding TXFIFO. This mode is only valid in the host mode. It is only used for SuperSpeed.
28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RW	0x0	USBTxPktCnt USB Transmit Packet Count This field specifies the number of packets that must be in the TXFIFO before the core can start transmission for the corresponding USB transaction (burst). This field is only valid when the USB Transmit Packet Count Enable field is set to one. Valid values are from 1 to 15. Note: This field must be less than or equal to the USB Maximum TX Burst Size field.
23:16	RW	0x00	USBMaxTxBurstSize USB Maximum TX Burst Size When USBTxPktCntSel is 1, this field specifies the Maximum Bulk OUT burst the core can execute. When the system bus is slower than the USB, TX FIFO can underrun during a long burst. You can program a smaller value to this field to limit the TX burst size that the core can execute. It only applies to SS Bulk, Isochronous, and Interrupt OUT endpoints in the host mode. Valid values are from 1 to 16.
15:0	RO	0x0	reserved

USB3_GRXTHRCFG
Address: Operational Base + offset (0xc10c)
Global Rx Threshold Control Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	USBRxPktCntSel USB ReceivePacket Count Enable This field enables/disables the USB reception multi-packet thresholding: 0: The core can only start reception on the USB when the RX FIFO has space for at least one packet. 1: The core can only start reception on the USB when the RX FIFO has space for at least USBRxPktCnt amount of packets. This mode is valid in both host and device mode. It is only used for SuperSpeed. In device mode, Setting this bit to 1 also enables the functionality of reporting NUMP in the ACK TP based on the RX FIFO space instead of reporting a fixed NUMP derived from DCFG.NUMP If you are using external buffer control (EBC) feature, disable this mode by setting USBRxPktCntSel to 0.
28	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
27:24	RW	0x0	USBRxPktCnt USB Receive Packet Count In host mode, this field specifies the space (in terms of the number of packets) that must be available in the RX FIFO before the core can start the corresponding USB RX transaction (burst). In device mode, this field specifies the space (in terms of the number of packets) that must be available in the RX FIFO before the core can send ERDY for a flow-controlled endpoint. This field is valid only when the USB Receive Packet Count Enable field is set to 1. The valid values for this field are from 1 to 15. Note: This field must be less than or equal to the USB Maximum Receive Burst Size field.
23:19	RW	0x00	USBMaxRxBurstSize USB Maximum Receive Burst Size In host mode, this field specifies the Maximum Bulk IN burst the usb3 controller can perform. When the system bus is slower than the USB, RX FIFO can overrun during a long burst. You can program a smaller value to this field to limit the RX burst size that the core can perform. It only applies to SS Bulk, Isochronous, and Interrupt IN endpoints in the host mode. In device mode, this field specifies the NUMP value that is sent in ERDY for an OUT endpoint. This field is valid only when USBRxPktCntSel is one. The valid values for this field are from 1 to 16.
18:0	RO	0x0	reserved

USB3_GCTL
Address: Operational Base + offset (0xc110)
Global Core Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:19	RW	0x0618	PWRDNSCALE Power Down Scale (PwrDnScale) The USB3 suspend_clk input replaces pipe3_rx_pclk as a clock source to a small part of the USB3 core that operates when the SS PHY is in its lowest power (P3) state, and therefore does not provide a clock. The Power Down Scale field specifies how many suspend_clk periods fit into a 16 kHz clock period. When performing the division, round up the remainder. For example, when using an 8-bit/16-bit/32-bit PHY and 25-MHz Suspend clock, Power Down Scale = 25000 kHz/16 kHz = 13'd1563 (rounder up) Note: Minimum Suspend clock frequency is 32 kHz Maximum Suspend clock frequency is 125 MHz The LTSSM uses Suspend clock for 12-ms and 100-ms timers during suspend mode. According to the USB 3.0 specification, the accuracy on these timers is 0% to +50%. 12 ms + 0~+50% accuracy = 18 ms (Range is 12 ms - 18 ms) 100 ms + 0~+50% accuracy = 150 ms (Range is 100 ms - 150ms). The suspend clock accuracy requirement is: (12,000/62.5) * (GCTL[31:19]) * actual suspend_clk_period must be between 12,000 and 18,000 (100,0000/62.5) * (GCTL[31:19]) * actual suspend_clk_period must be between 100,000 and 150,000 For example, if your suspend_clk frequency varies from 7.5 MHz to 10.5MHz, then the value needs to programmed is: Power Down Scale = 10500/16 = 657 (rounded up; and fastest frequency used).
18	RW	0x0	MASTERFILTBYPASS Master Filter Bypass When this bit is set to 1'b1, all the filters are bypassed. The double synchronizers to mac_clk preceding the filters are also bypassed. For enabling the filters, this bit must be 1'b0.

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Bit	Attr	Reset Value	Description
17	RW	0x0	BYPSSETADDR Bypass SetAddress in Device Mode. When BYPSSETADDR bit is set, the device core uses the value in the DCFG[DevAddr] bits directly for comparing the device address in the tokens. For simulation, you can use this feature to avoid sending an actual SET ADDRESS control transfer on the USB, and make the device core respond to a new address. When the xHCI Debug capability is enabled and this bit is set, the Debug Target immediately enters the configured state without requiring the Debug Host to send a SetAddress or SetConfig request. Note: You can set this bit for simulation purposes only. In the actual hardware, this bit must be set to 1'b0.
16	RW	0x1	U2RSTECN U2RSTECN If the SuperSpeed connection fails during POLL or LMP exchange, the device connects at non-SS mode. If this bit is set, then device attempts three more times to connect at SS, even if it previously failed to operate in SS mode. For each attempt, the device checks receiver termination eight times. From 2.60a release, this bit controls whether to check for Rx.Detect eight times or one time for every attempt. Device controller on USB 2.0 reset checks for receiver termination eight times per attempt if this bit is set to zero, or only once per attempt if the bit is set to one. Note: This bit is applicable only in device mode.
15:14	RW	0x0	FRMSCLDWN FRMSCLDWN This field scales down device view of a SOF/USOF/ITP duration. For SS/HS mode: Value of 2'h3 implements interval to be 15.625 us Value of 2'h2 implements interval to be 31.25 us Value of 2'h1 implements interval to be 62.5 us Value of 2'h0 implements interval to be 125us For FS mode, the scale-down value is multiplied by 8. When xHCI Debug Capability is enabled, this field also scales down the MaxPacketSize of the IN and OUT bulk endpoint to allow more traffic during simulation. It can only be changed from a non-zero value during simulation. 2'h0: 1024 bytes 2'h1: 512 bytes 2'h2: 256 bytes 2'h3: 128 bytes

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Bit	Attr	Reset Value	Description
13:12	RW	0x2	PRTCAPDIR PRTCAPDIR: Port Capability Direction (PrtCapDir) 2'b01: for Host configurations 2'b10: for Device configurations SW should base on IDDIG input to set usb3 controller as an OTG 2.0/3.0 device with A-device or B-device.
11	RW	0x0	CORESOFTRESET Core Soft Reset (CoreSoftReset) 1'b0 - No soft reset; 1'b1 - Soft reset to core Clears the interrupts and all the CSRs except the following registers: GCTL; GUCTL; GSTS; GSNPSID; GGPIO; GUID; GUSB2PHYCFGn registers; GUSB3PIPECTLn registers; DCFG; DCTL; DEVTEN; DSTS. When you reset PHYs (using GUBS3PHYCFG or GUSB3PIPECTL registers), you must keep the core in reset state until PHY clocks are stable. This controls the bus, ram, and mac domain resets. Note: This bit is for debug purposes only. Use USBCMD.HCRESET in xHCI Mode and DCTL.SoftReset in device mode for soft reset.

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Bit	Attr	Reset Value	Description
10	RW	0x0	SOFITPSYNC SOFITPSYNC If this bit is set to 0 operating in host mode, the core keeps the UTMI/ULPI PHY on the first port in a non-suspended state whenever there is a SuperSpeed port that is not in Rx.Detect, SS.Disable and U3. If this bit is set to 1 operating in host mode, the core keeps the UTMI/ULPI PHY on the first port in a non-suspended state whenever the other non-SuperSpeed ports are not in a suspended state. This feature is useful because it saves power by suspending UTMI/ULPI when SuperSpeed only is active, and it helps resolve when the PHY does not transmit a host resume unless it is placed in suspend state. This bit must be programmed as a part of initialization at power-on reset, and must not be dynamically changed afterwards. Note: USB2PHYCFGn[6].PhySusp eventually decides to put the UTMI/ULPI PHY in to suspend state. In addition, when this bit is set to 1, the core generates ITP from the ref_clk based counter. Otherwise, ITP and SOF are generated from utmi/ulpi_clk[0] based counter. To program the reference clock period inside the core, refer to GUCTL[31:22].REFCLKPER. This feature is valid in Host and DRD/OTG configurations and used only in Host mode operation. If you never use this feature or the GFLADJ.GFLADJ_REFCLK_LPM_SEL, the minimum frequency for the ref_clk can be as low as 32KHz. You can connect the suspend_clk (as low as 32 KHz) to the ref_clk. If you plan to enable hardware-based LPM or software-based LPM (PORTPMSC. HLE=1), then you cannot use this feature. Turn off this feature by setting this bit to 0 and use the GFLADJ.GFLADJ_REFCLK_LPM_SEL feature. If you set this bit to 1, the GUSB2PHYCFG.U2_FREECLK_EXISTS bit and the DWC_USB3_FREECLK_USB2_EXIST parameter must be set to 0. Program this bit to 0 if the core is intended to be operated in USB 3.0 mode.
9	RW	0x0	U1U2TimerScale Disable U1/U2 timer Scaledown (U1U2TimerScale). If set to 1 along with GCTL[5:4] (ScaleDown) = 2'bX1, disables the scale down of U1/U2 inactive timer values. This is for simulation mode only.

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Bit	Attr	Reset Value	Description
8	RW	0x0	DEBUGATTACH Debug Attach When this bit is set: 1. SS Link proceeds directly to the Polling link state (after RUN/STOP in the DCTL register is asserted) without checking remote termination; 2. Link LFPS polling timeout is infinite; 3. Polling timeout during TS1 is infinite (in case link is waiting for TXEQ to finish).
7:6	RW	0x0	RAMCLKSEL RAM Clock Select (RAMClkSel) 2'b00: bus clock 2'b01: pipe clock (Only used in device mode) 2'b10: In device mode, pipe/2 clock. In Host mode, controller switches ram_clk between pipe/2 clock, mac2_clk and bus_clk based on the status of the U2/U3 ports 2'b11: In device mode, selects mac2_clk as ram_clk (when 8-bit UTMI or ULPI used. Not supported in 16-bit UTMI mode); In Host mode, controller switches ram_clk between pipe_clk, mac2_clk and bus_clk based on the status of the U2/U3 ports. In device mode, upon a USB reset and USB disconnect, the hardware clears these bits to 2'b00.
5:4	RW	0x0	SCALEDOWN Scale-Down Mode (ScaleDown) When Scale-Down mode is enabled for simulation, the core uses scaled-down timing values, resulting in faster simulations. When Scale-Down mode is disabled, actual timing values are used. This is required for hardware operation. HS/FS/LS Modes: 2'b00: Disables all scale-downs. Actual timing values are used. 2'b01: Enables scale-down of all timing values except Device mode suspend and resume. These include Speed enumeration, HNP/SRP, and Host mode suspend and resume 2'b10: Enables scale-down of Device mode suspend and resume timing values only. 2'b11: Enables bit 0 and bit 1 scale-down timing values. SS Mode: 2'b00: Disables all scale-downs. Actual timing values are used. 2'b01: Enables scaled down SS timing and repeat values including: (1) Number of TxEq training sequences reduce to 8; (2) LFPS polling burst time reduce to 256 nS; (3) LFPS warm reset receive reduce to 30 uS. 2'b10: No TxEq training sequences are sent. Overrides Bit 4. 2'b11: Enables bit 0 and bit 1 scale-down timing values.

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Bit	Attr	Reset Value	Description
3	RW	0x0	DISSCRAMBLE Disable Scrambling (DisScramble) Transmit request to Link Partner on next transition to Recovery or Polling.
2	RW	0x1	U2EXIT_LFPS U2EXIT_LFPS If this bit is: 0: the link treats 248ns LFPS as a valid U2 exit. 1: the link waits for 8us of LFPS before it detects a valid U2 exit. This bit is added to improve interoperability with a third party host controller. This host controller in U2 state while performing receiver detection generates an LFPS glitch of about 4ms duration. This causes the device to exit from U2 state because the LFPS filter value is 248ns. With the new functionality enabled, the device can stay in U2 while ignoring this glitch from the host controller.
1	RO	0x0	GblHibernationEn GblHibernationEn This bit enables hibernation at the global level. If hibernation is not enabled through this bit, the PMU immediately accepts the D0->D3 and D3->D0 power state change requests, but does not save or restore any core state. In addition, the PMUs never drive the PHY interfaces and let the core continue to drive the PHY interfaces.
0	RW	0x0	DSBLCLKGTNG Disable Clock Gating (DsblClkGtng) This bit is set to 1 and the core is in Low Power mode, internal clock gating is disabled. You can set this bit to 1'b1 after Power On Reset.

USB3_GPMSTS
Address: Operational Base + offset (0xc114)
Global Power Management Status Register

Bit	Attr	Reset Value	Description
31:28	WO	0x0	PortSel Global Power Management Status Register This field selects the port number.
27:17	RO	0x0	reserved
16:12	RO	0x00	U3Wakeup U3Wakeup This field gives the following USB 3.0 port wakeup conditions: Bit [12]: Overcurrent Detected Bit [13]: Resume Detected Bit [14]: Connect Detected Bit [15]: Disconnect Detected Bit [16]: Last Connection State
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:0	RO	0x000	U2Wakeup U2Wakeup This field indicates the following USB 2.0 port wakeup conditions: Bit [0]: Overcurrent Detected Bit [1]: Resume Detected Bit [2]: Connect Detected Bit [3]: Disconnect Detected Bit [4]: Last Connection State Bit [5]: ID Change Detected Bit [6]: SRP Request Detected Bit [7]: ULPI Interrupt Detected Bit [8]: USB Reset Detected Bit [9]: Resume Detected Changed

USB3_GSTS
Address: Operational Base + offset (0xc118)
Global Status Register

Bit	Attr	Reset Value	Description
31:20	RO	0x7e8	CBELT Current BELT Value In Host mode, this field indicates the minimum value of all received device BELT values and the BELT value that is set by the Set Latency Tolerance Value command.
19:12	RO	0x0	reserved
11	RO	0x0	SSIC_IP SSIC interrupt pending This field indicates that there is a pending interrupt related to SSIC in the SEVT register. Note: When the DWC_USB3_NUM_SSIC_PORTS parameter is set to zero, this bit is reserved.
10	RO	0x0	OTG_IP OTG Interrupt Pending This field indicates that there is a pending interrupt pertaining to OTG in OEVT register.
9	RO	0x0	BC_IP Battery Charger Interrupt Pending This field indicates that there is a pending interrupt pertaining to BC in BCEVT register.
8	RO	0x0	ADP_IP ADP Interrupt Pending his field indicates that there is a pending interrupt pertaining to ADP in ADPEVT register.

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Bit	Attr	Reset Value	Description
7	RO	0x0	Host_IP Host Interrupt Pending This field indicates that there is a pending interrupt pertaining to xHC in the Host event queue.
6	RO	0x0	Device_IP Device Interrupt Pending This field indicates that there is a pending interrupt pertaining to peripheral (device) operation in the Device event queue.
5	W1C	0x0	CSRTimeout CSR Timeout When this bit is 1'b1, it indicates that the software performed a write or read to a core register that could not be completed within DWC_USB3_CSR_ACCESS_TIMEOUT bus clock cycles (default: h1FFFF).
4	W1C	0x0	BUSERRADDRVLD Bus Error Address Valid Indicates that the GBUSERRADDR register is valid and reports the first bus address that encounters a bus error.
3:2	RO	0x0	reserved
1:0	RO	0x0	CURMOD Current Mode of Operation Current Mode of Operation

USB3_GUCTL1

Address: Operational Base + offset (0xc11c)

Global User Control Register 1

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29	RW	0x0	FILTER_SE0_FSLS_EOP FILTER_SE0_FSLS_EOP 0: Default behaviour, no change in Linestate check for SE0 detection in FS/LS 1: Feature enabled, FS/LS SE0 is filtered for 2 clocks for detecting EOP This bit is applicable for FS/LS operation. If this feature is enabled, then SE0 on the linestate is validated for 2 consecutive utmi/ulpi clock edges for EOP detection. This feature is applicable only in FS in device mode and FS/LS mode of operation in host mode. Device mode: FS - If GUCTL1.FILTER_SE0_FSLS_EOP is set, then for device LPM hanshake, the core will ignore single SE0 glitch on the linestate during transmit. Only 2 or more SE0 is considered as a valid EOP on FS. Host mode: FS/LS - If GUCTL1.FILTER_SE0_FSLS_EOP is set, then the core will ignore single SE0 glitch on the linestate during transmit. Only 2 or more SE0 is considered as a valid EOP on FS/LS port. Enable this feature if the LineState has SE0 glitches during transmission. This bit is quasi-static, i.e., should not be changed during device operation.
28	RW	0x0	TX_IPGAP_LINECHECK_DIS TX_IPGAP_LINECHECK_DIS 0: Default behaviour, no change in Linestate check 1: Feature enabled, 2.0 MAC disables Linestate check during HS transmit This bit is applicable for HS operation of u2mac. If this feature is enabled, then the 2.0 mac operating in HS ignores the UTMI/ULPI Linestate during the transmit of a token (during token-to-token and token-to-data IPGAP). When enabled, the controller implements a fixed 40-bit TxEndDelay after the packet is given on UTMI and ignores the Linestate during this time. This fetaure is applicable only in HS mode of operation. Device mode: If GUCTL1.TX_IPGAP_LINECHECK_DIS is set, then for device LPM handshake, the core will ignore the linestate after TX and wait for a fixed clocks (40 bit times equivalent) after transmiting ACK on utmi. Host mode: If GUCTL1.TX_IPGAP_LINECHECK_DIS is set, then the ipgap between (tkn to tkn/data) is added by 40 bit times of TXENDDELAY, and linestate is ignored during this 40 bit times delay. Enable this bit if the LineState will not reflect the expected line state (J) during transmission. This bit is quasi-static, i.e., should not be changed during device operation.

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Bit	Attr	Reset Value	Description
27	RW	0x0	DEV_TRB_OUT_SPR_IND DEV_TRB_OUT_SPR_IND 0: Default behaviour, no change in TRB status dword 1: Feature enabled, OUT TRB status indicates Short Packet This bit is applicable for device mode only (and ignored in host mode). If the device application (SW/HW) wants to know if a short packet was received for an OUT in the TRB status itself, then this feature can be enabled, so that a bit is set in the TRB writeback in the buf_size dword. Bit[26] - SPR of the trbstatus, RSVD, SPR,PCM1, bufsize dword will be set during an OUT transfer TRB write back if this is the last TRB used for that transfer descriptor. This bit is quasi-static, i.e., should not be changed during device operation.
26	RW	0x0	DEV_FORCE_20_CLK_FOR_30_CLK DEV_FORCE_20_CLK_FOR_30_CLK 0: Default behaviour, Uses 3.0 clock when operating in 2.0 mode 1: Feature enabled This bit is applicable (and to be set) for device mode (DCFG.Speed!= SS) only. In the 3.0 device core, if the core is programmed to operate in 2.0 only (i.e., Device Speed is programmed to 2.0 speeds in DCFG[Speed]), then setting this bit makes the internal 2.0 (utmi/ulpi) clock to be routed as the 3.0 (pipe) clock. Enabling this feature allows the pipe3 clock to be not-running when forcibily operating in 2.0 device mode. Note: When using this feature, all pipe3 inputs must be in inactive mode, esp. pipe3 clocks not running and pipe3_phystatus_async must be tied to 0. This bit should not be set if the core is programmed to operate in SuperSpeed mode (even when it falls back to 2.0). This bit is quasi-static, i.e., should not be changed during operation.
25	RW	0x0	P3_IN_U2 P3_IN_U2 0: Default behaviour, When SuperSpeed link is in U2 , PowerState P2 is attempted on the PIPE Interface. 1: When SuperSpeed link is in U2, PowerState P3 is attempted if GUSB3PIPECTL[17] is set. Setting this bit enables P3 Power State when the SuperSpeed link is in U2. Another Power Saving option. Check with your PHY vendor before enabling this option.When setting this bit to 1 to enable P3 in P2, GUSB3PIPECTL[27] should be set to 0 to make sure that the U2 exit is attempted in P0.

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Bit	Attr	Reset Value	Description
24	RW	0x0	DEV_L1_EXIT_BY_HW DEV_L1_EXIT_BY_HW 0: Default behaviour, disables device L1 hardware exit logic 1: feature enabled This bit is applicable for device mode (2.0) only. This field enables device controller sending remote wakeup for L1 if the device becomes ready for sending/accepting data when in L1 state. If the host expects the device to send remote wkp signalling to resume after going into L1 in flow controlled state, then this bit can be set to send the remote wake signal automatically when the device controller becomes ready. This HW remote wake feature is applicable only to bulk and interrupt transfers, and not for Isoch/Control When control transfers are in progress, the LPM will be rejected (NYET response). Only after control transfers are completed (either with ACK/STALL), LPM will be accepted For Isoch transfers, the host needs to do the wake-up and start the transfer. Device controller will not do remote-wakeup when Isoch endpoints get ready. The device SW needs to keep the GUSB2PHYCFG[EnblSlpM] reset in order to keep the PHY clock to be running for keeping track of SOF intervals. When L1 hibernation is enabled, the controller will not do automatic exit for hibernation requests thru L1. This bit is quasi-static, i.e., should not be changed during device operation.
23:21	RW	0x0	IP_GAP_ADD_ON IP_GAP_ADD_ON This register field is used to add on to the default inter packet gap setting in the USB 2.0 MAC. This should be programmed to a non zero value only in case where you need to increase the default inter packet delay calculations in the USB 2.0 MAC module DWC_usb3_u2mac.v
20	RW	0x0	DEV_LSP_TAIL_LOCK_DIS DEV_LSP_TAIL_LOCK_DIS 0: Default behaviour, enables device lsp lock logic for tail TRB update 1: Fix disabled This is a bug fix for STAR 9000716195 that affects the CSP mode for OUT endpoints in device mode. The issue is that tail TRB index is not synchronized with the cache Scratchpad bytecount update. If the fast-forward request comes in-between the bytecount update ona newly fetched TRB and the tail-index write update in TPF, the RDP works on an incorrect tail index and misses the byte count decrement for the newly fetched TRB in the fast-forwarding process. This fix needs to be present all the times.

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Bit	Attr	Reset Value	Description
19	RW	0x0	NAK_PER_ENH_FS NAK_PER_ENH_FS 1: Enables performance enhancement for FS async endpoints in the presence of NAKs 0: Enhancement not applied If a periodic endpoint is present , and if a bulk endpoint which is also active is being NAKed by the device, then this could result in a decrease in performance of other Full Speed bulk endpoint which is ACked by the device. Setting this bit to 1, will enable the host controller to schedule more transactions to the async endpoints (bulk/ control) and hence will improve the performance of the bulk endpoint. This control bit should be enabled only if the existing performance with the default setting is not sufficient for your FullSpeed application. Setting this bit will only control, and is only required for Full Speed transfers.
18	RW	0x1	NAK_PER_ENH_HS NAK_PER_ENH_HS 1: Enables performance enhancement for HS async endpoints in the presence of NAKs 0: Enhancement not applied If a periodic endpoint is present , and if a bulk endpoint which is also active is being NAKed by the device, then this could result in a decrease in performance of other High Speed bulk endpoint which is ACked by the device. Setting this bit to 1, will enable the host controller to schedule more transactions to the async endpoints (bulk/ control) and hence will improve the performance of the bulk endpoint. This control bit should be enabled only if the existing performance with the default setting is not sufficient for your HighSpeed application. Setting this bit will only control, and is only required for High Speed transfers.
17	RW	0x0	PARKMODE_DISABLE_SS PARKMODE_DISABLE_SS This bit is used only in host mode, and is for debug purpose only. When this bit is set to 1 all SS bus instances in park mode are disabled.

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Bit	Attr	Reset Value	Description
16	RW	0x0	PARKMODE_DISABLE_HS PARKMODE_DISABLE_HS This bit is used only in host mode. When this bit is set to 1 all HS bus instances park mode are disabled. To improve performance in park mode, the xHCI scheduler queues in three requests of 4 packets each for High Speed asynchronous endpoints in a micro-frame. But if a device is slow and if it NAKs more than 3 times, then it is rescheduled only in the next micro-frame. This could decrease the performance of a slow device even further. In a few high speed devices (such as Sandisk Cruzer Blade 4GB VID:1921, PID:21863 and Flex Drive VID:3744, PID:8552) when an IN request is sent within 900ns of the ACK of the previous packet, these devices send a NAK. When connected to these devices, if required, the software can disable the park mode if you see performance drop in your system. When park mode is disabled, pipelining of multiple packet is disabled and instead one packet at a time is requested by the scheduler. This allows up to 12 NAKs in a micro-frame and improves performance of these slow devices.
15	RW	0x0	PARKMODE_DISABLE_FSLS PARKMODE_DISABLE_FSLS This bit is used only in host mode, and is for debug purpose only. When this bit is set to 1 all FS/LS bus instances in park mode disabled.
14:9	RO	0x0	reserved
8	RW	0x1	L1_SUSP_THRLD_EN_FOR_HOST L1_SUSP_THRLD_EN_FOR_HOST This bit is used only in host mode. The host controller asserts the utmi_l1_suspend_n and utmi_sleep_n output signals (see LPM Interface Signals table in the Databook) as follows: The controller asserts the utmi_l1_suspend_n signal to put the PHY into deep low-power mode in L1 when both of the following are true: The HIRD/BESL value used is greater than or equal to the value in L1_SUSP_THRLD_FOR_HOST field. The L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1'b1. The controller asserts utmi_sleep_n on L1 when one of the following is true: The HIRD/BESL value used is less than the value in L1_SUSP_THRLD_FOR_HOST field. The L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1'b0.

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Bit	Attr	Reset Value	Description
7:4	RW	0x8	L1_SUSP_THRLD_FOR_HOST L1_SUSP_THRLD_FOR_HOST This field is effective only when the L1_SUSP_THRLD_EN_FOR_HOST bit is set to 1. For more details, refer to the description of the L1_SUSP_THRLD_EN_FOR_HOST bit.
3	RW	0x1	HC_ERRATA_ENABLE Host ELD Enable When this bit is set to 1, it enables the Exit Latency Delta (ELD) support defined in the xHCI 1.0 Errata. This bit is used only in the host mode. This bit has to be set to 1 in Host mode.
2	RW	0x0	HC_PARCHK_DISABLE Host Parameter Check Disable When this bit is set to 0 (by default), the xHC checks that the input slot/EP context fields comply to the xHCI Specification. Upon detection of a parameter error during command execution, the xHC generates an event TRB with completion code indicating PARAMETER ERROR. When the bit is set to 1, the xHC does not perform parameter checks and does not generate PARAMETER ERROR completion code.
1	RW	0x1	OVRLD_L1_SUSP_COM OVRLD_L1_SUSP_COM If this bit is set, the utmi_l1_suspend_com_n is overloaded with the utmi_sleep_n signal. This bit is usually set if the PHY stops the port clock during L1 sleep condition.
0	RW	0x0	LOA_FILTER_EN LOA_FILTER_EN If this bit is set, the USB 2.0 port babble is checked at least three consecutive times before the port is disabled. This prevents false triggering of the babble condition when using low quality cables. Note: This bit is valid only in host mode.

USB3_GSNPSID

Address: Operational Base + offset (0xc120)

Global SNPS ID Register

Bit	Attr	Reset Value	Description
31:0	RO	0x5533300a	SNPSID SNPSID SNPSID[31:16] indicates Core Identification Number. 0x5533 is ASCII for U3 (DWC_usb3). SNPSID[15:0] indicates the release number. Current Release is 3.00a. Software uses this register to configure release-specific features in the driver.

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USB3_GGPIO
Address: Operational Base + offset (0xc124)
Global General Purpose Input/Output Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	GPO General Purpose Output This field's value is driven out on the gp_out[15:0] core output port.
15:0	RO	0x0000	GPI General Purpose Input This field's read value reflects the gp_in[15:0] core input value.

USB3_GUID
Address: Operational Base + offset (0xc128)
Global User ID Register

Bit	Attr	Reset Value	Description
31:0	RW	0x12345678	USERID USERID Application-programmable ID field.

USB3_GUCTL
Address: Operational Base + offset (0xc12c)
Global User Control Register

Bit	Attr	Reset Value	Description
31:22	RW	0x008	REFCLKPER REFCLKPER This field indicates in terms of nano seconds the period of ref_clk. The default value of this register is set to 'h8 (8ns/125 MHz). This field needs to be updated during power-on initialization, if GCTL.SOFITPSYNC or GFLADJ.GFLADJ_REFCLK_LPM_SEL is set to 1. The programmable maximum value is 62ns, and the minimum value is 8ns. You must use a reference clock with a period that is an integer multiple, so that ITP can meet the jitter margin of 32ns. The allowable ref_clk frequencies whose period is not integer multiples are 16/17/19.2/24/39.7MHz. This field must not be set to 0 at any time. If you never plan to use this feature, then set this field to 'h8, the default value.

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Bit	Attr	Reset Value	Description
21	RW	0x0	NoExtrDl No Extra Delay Between SOF and the First Some HS devices misbehave when the host sends a packet immediately after a SOF. However, adding an extra delay between a SOF and the first packet can reduce the USB data rate and performance. This bit is used to control whether the host must wait for 2 microseconds before it sends the first packet after a SOF, or not. User can set this bit to one to improve the performance if those problematic devices are not a concern in the user's host environment. 1'b0: Host waits for 2 microseconds after a SOF before it sends the first USB packet. 1'b1: Host doesn't wait after a SOF before it sends the first USB packet.
20:18	RO	0x0	reserved
17	RW	0x0	SprsCtrlTransEn Sparse Control Transaction Enable Some devices are slow in responding to Control transfers. Scheduling multiple transactions in one microframe/frame can cause these devices to misbehave. If this bit is set to 1'b1, the host controller schedules transactions for a Control transfer in different microframes/frames.
16	RW	0x0	ResBwHSEPS Reserving 85% Bandwidth for HS Periodic EPs By default, HC reserves 80% of the bandwidth for periodic EPs. If this bit is set, the bandwidth is relaxed to 85% to accommodate two high speed, high bandwidth ISOC EPs. USB 2.0 required 80% bandwidth allocated for ISOC traffic. If two High-bandwidth ISOC devices (HD Webcams) are connected, and if each requires 1024-bytes X 3 packets per Micro-Frame, then the bandwidth required is around 82%. If this bit is set, then it is possible to connect two Webcams of 1024bytes X 3 paylod per Micro-Frame each. Otherwise, you may have to reduce the resolution of the Webcams. This bit is valid in Host and DRD configuration and is used in host mode operation only. Ignore this bit in device mode.

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Bit	Attr	Reset Value	Description
15	RW	0x1	CMdevAddr Compliance Mode for Device Address When this bit is 1'b1, Slot ID may have different value than Device Address if max_slot_enabled < 128. 1'b1: Increment Device Address on each Address Device command. 1'b0: Device Address is equal to Slot ID. The xHCI compliance requires this bit to be set to 1. The 0 mode is for debug purpose only. This allows you to easily identify a device connected to a port in the Lecroy or Eliisys trace during hardware debug. This bit is valid in Host and DRD configuration and is used in host mode operation only. Ignore this bit in device mode.
14	RW	0x0	USBHstInAutoRetryEn Host IN Auto Retry When set, this field enables the Auto Retry feature. For IN transfers (non-isochronous) that encounter data packets with CRC errors or internal overrun scenarios, the auto retry feature causes the Host core to reply to the device with a non-terminating retry ACK (that is, an ACK transaction packet with Retry = 1 and NumP != 0). If the Auto Retry feature is disabled (default), the core will respond with a terminating retry ACK (that is, an ACK transaction packet with Retry = 1 and NumP = 0). 1'b0: Auto Retry Disabled 1'b1: Auto Retry Enabled Note: This bit is also applicable to the device mode.
13	RW	0x0	EnOverlapChk Enable Check for LFPS Overlap During Remote Ux 1'b1: The SuperSpeed link when exiting U1/U2/U3 waits for either the remote link LFPS or TS1/TS2 training symbols before it confirms that the LFPS handshake is complete. This is done to handle the case where the LFPS glitch causes the link to start exiting from the low power state. Looking for the LFPS overlap makes sure that the link partner also sees the LFPS. 1'b0: When the link exists U1/U2/U3 because of a remote exit, it does not look for an LFPS overlap.

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Bit	Attr	Reset Value	Description
12	RW	0x0	ExtCapSupptEN External Extended Capability Support Enable When set, this field enables extended capabilities to be implemented outside the core. When the ExtCapSupEN is set and the Debug Capability is enabled, the Next Capability pointer in Debug Capability returns 16. A read to the first DWORD of the last internal extended capability (the "xHCI Supported Protocol Capability for USB 3.0" when the Debug Capability is not enabled) returns a value of 4 in the Next Capability Pointer field. This indicates to software that there is another capability four DWORDs after this capability (for example, at address N+16 where N is the address of this DWORD). If enabled, an external address decoder that snoops the xHC slave interface must be implemented. If it sees an access to N+16 or greater, the slave access is re-routed to a piece of hardware which returns the external capability pointer register of the new capability and also handles reads/writes to this new capability and the side effects. If disabled, a read to the first DWORD of the last internal extended capability returns 0 in the 'Next Capability Pointer' field. This indicates there are no more capabilities.
11	RW	0x0	InsrtExtrFSBODI Insert Extra Delay Between FS Bulk OUT Some FS devices are slow to receive Bulk OUT data and can get stuck when there are consecutive Bulk OUT transactions with short inter-transaction delays. This bit is used to control whether the host inserts extra delay between consecutive Bulk OUT transactions to a FS Endpoint. 1'b0: Host doesn't insert extra delay between consecutive Bulk OUT transactions to a FS Endpoint. 1'b1: Host inserts about 12us extra delay between consecutive Bulk OUT transactions to a FS Endpoint to work around the device issue. Note: Setting this bit to one will reduce the Bulk OUT transfer performance for most of the FS devices.
10:9	RW	0x0	DTCT Device Timeout Coarse Tuning This field is a Host mode parameter which determines how long the host waits for a response from device before considering a timeout. The core first checks the DTCT value. If it is 0, then the timeout value is defined by the DTFT. If it is non-zero, then it uses the following timeout values: 2'b00: 0 usec -> use DTFT value instead 2'b01: 500 usec 2'b10: 1.5 msec 2'b11: 6.5 msec

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Bit	Attr	Reset Value	Description
8:0	RW	0x010	DTFT Device Timeout Fine Tuning This field is a Host mode parameter which determines how long the host waits for a response from device before considering a timeout. For the DTFT field to take effect, DTCT must be set to 2'b00. The DTFT value is the number of 125 MHz clocks * 256 to count before considering a device timeout. The minimum value of DTFT is 2. For example, if the mac3_clk is 125 MHz clk (8 ns period), this is calculated as follows: (DTFT value) * 256 * (8 ns) Quick Reference: if DTFT = 0x2, 2*256*8 = 4usec timeout if DTFT = 0x5, 5*256*8 = 10usec timeout if DTFT = 0xA, 10*256*8 = 20usec timeout if DTFT = 0x10, 16*256*8 = 32usec timeout if DTFT = 0x19, 25*256*8 = 51usec timeout if DTFT = 0x31, 49*256*8 = 100usec timeout if DTFT = 0x62, 98*256*8 = 200usec timeout

USB3_GBUSERRADDRLO

Address: Operational Base + offset (0xc130)

Gobal SoC Bus Error Address Register - Low

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	BUSERRADDR Bus Address - Low This register contains the lower 32 bits of the first bus address that encountered a SoC bus error. It is valid when the GSTS.BusErrAddrVld field is 1. It can only be cleared by resetting the core. Note: Only supported in AHB and AXI configurations.

USB3_GBUSERRADDRHI

Address: Operational Base + offset (0xc134)

Gobal SoC Bus Error Address Register - High

Bit	Attr	Reset Value	Description
31:0	RU	0x00000000	BUSERRADDR Bus Address - High his register contains the higher 32 bits of the first bus address that encountered a SoC bus error. It is valid when the GSTS.BusErrAddrVld field is 1. It can only be cleared by resetting the core. Note: Only supported in AHB and AXI configurations.

USB3_GPRTBIMAPLO

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Address: Operational Base + offset (0xc138)
Global SS Port to Bus Instance Mapping Register - Low

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	BINUM1 SS USB Instance Number for Port 1 Application-programmable ID field.

USB3_GHWPARAMS0
Address: Operational Base + offset (0xc140)
Global Hardware Parameters Register 0

Bit	Attr	Reset Value	Description
31:0	RO	0x2020400a	GHWPARAMS0 Global Hardware Parameters Register 0 Global Hardware Parameters Register 0

USB3_GHWPARAMS1
Address: Operational Base + offset (0xc144)
Global Hardware Parameters Register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x0160c93b	GHWPARAMS1 Global Hardware Parameters Register 1 Global Hardware Parameters Register 1

USB3_GHWPARAMS2
Address: Operational Base + offset (0xc148)
Global Hardware Parameters Register 2

Bit	Attr	Reset Value	Description
31:0	RO	0x12345678	GHWPARAMS2 Global Hardware Parameters Register 2 Global Hardware Parameters Register 2

USB3_GHWPARAMS3
Address: Operational Base + offset (0xc14c)
Global Hardware Parameters Register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x069cd085	GHWPARAMS3 Global Hardware Parameters Register 3 Global Hardware Parameters Register 3

USB3_GHWPARAMS4
Address: Operational Base + offset (0xc150)
Global Hardware Parameters Register 4

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x47822008	GHWPARAMS4 Global Hardware Parameters Register 4 Global Hardware Parameters Register 4

USB3_GHWPARAMS5
Address: Operational Base + offset (0xc154)
Global Hardware Parameters Register 5

Bit	Attr	Reset Value	Description
31:0	RO	0x04202088	GHWPARAMS5 Global Hardware Parameters Register 5 Global Hardware Parameters Register 5

USB3_GHWPARAMS6
Address: Operational Base + offset (0xc158)
Global Hardware Parameters Register 6

Bit	Attr	Reset Value	Description
31:0	RO	0x077c8020	GHWPARAMS6 Global Hardware Parameters Register 6 Global Hardware Parameters Register 6

USB3_GHWPARAMS7
Address: Operational Base + offset (0xc15c)
Global Hardware Parameters Register 7

Bit	Attr	Reset Value	Description
31:0	RO	0x03080756	GHWPARAMS7 Global Hardware Parameters Register 7 Global Hardware Parameters Register 7

USB3_GDBGFIFOSPACE
Address: Operational Base + offset (0xc160)
Global Debug Queue/FIFO Space Available Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0042	SPACE_AVAILABLE Space Avalible Space Avalible
15:9	RO	0x0	reserved
8:0	RW	0x000	FIFO_QUEUE_SELECT FIFO/Queue Select (or) Port-Select FIFO/Queue Select[8:5] indicates the FIFO/Queue Type FIFO/Queue Select[4:0] indicates the FIFO/Queue Number Port-Select[3:0] selects the port-number when accessing GDBGLTSSM register.

USB3_GDBGLTSSM

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Address: Operational Base + offset (0xc164)
Global Debug LTSSM Register

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RO	0x1	RxElecidle RxElecidle Reflect status of Pipe interface.
29:27	RO	0x0	reserved
26	RW	0x0	LTDBTIMEOUT LTDB Timeout LTDB Timeout
25:22	RO	0x4	LTDBLINKSTATE LTDB Link State LTDB Link State
21:18	RO	0x0	LTDBSUBSTATE LTDB Sub-State LTDB Sub-State
17	RO	0x0	ELASTICBUFFERMODE ELASTICBUFFERMODE Reflect status of Pipe interface.
16	RO	0x1	TXELECLDLE TXELECLDLE Reflect status of Pipe interface.
15	RO	0x0	RXPOLARITY RXPOLARITY Reflect status of Pipe interface.
14	RO	0x0	TxDetRxLoopback Tx Detect Rx/Loopback Reflect status of Pipe interface.
13:11	RO	0x0	LTDBPhyCmdState LTSSM PHY command State 000: PHY_IDLE (PHY command state is in IDLE. No PHY request pending) 001: PHY_DET (Request to start Receiver detection) 010: PHY_DET_3 (Wait for Phy_Status (Receiver detection)) 011: PHY_PWR_DLY (Delay Pipe3_PowerDown P0 -> P1/P2/P3 request) 100: PHY_PWR_A (Delay for internal logic) 101: PHY_PWR_B (Wait for Phy_Status(Power state change request))
10:9	RO	0x2	POWERDOWN POWERDOWN Reflect status of Pipe interface.
8	RO	0x0	RXEQTRAIN RXEQTRAIN Reflect status of Pipe interface.

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Bit	Attr	Reset Value	Description
7:6	RO	0x1	TXDEEMPHASIS TXDEEMPHASIS Reflect status of Pipe interface.
5:3	RO	0x0	LTDBClkState LTSSM Clock State In multi-port host configuration, the port number is defined by Port-Select[3:0] field in the GDBGFIFOSPACE register. Note:GDBGLTSSM register is not applicable for USB 2.0-only mode. 000: CLK_NORM (PHY is in non-P3 state and PCLK is running) 001: CLK_TO_P3 (P3 entry request to PHY); 010: CLK_WAIT1 (Wait for Phy_Status (P3 request)); 011: CLK_P3 (PHY is in P3 and PCLK is not running); 100: CLK_TO_P0 (P3 exit request to PHY); 101: CLK_WAIT2 (Wait for Phy_Status (P3 exit request))
2	RO	0x0	TXSWING TXSWING Reflect status of Pipe interface.
1	RO	0x0	RXTERMINATION RXTERMINATION Reflect status of Pipe interface.
0	RO	0x0	TXONESZEROS TXONESZEROS Reflect status of Pipe interface.

USB3_GDBGLNMCC
Address: Operational Base + offset (0xc168)
Global Debug LNMCC Register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RO	0x000	LNMCC_BERC LNMCC_BERC This field indicates the bit error rate information for the port selected in the GDBGFIFOSPACE.PortSelect field.This field is for debug purposes only.

USB3_GDBGBMU
Address: Operational Base + offset (0xc16c)
Global Debug BMU Register

Bit	Attr	Reset Value	Description
31:8	RW	0x000000	BMU_BCU BMU_BCU Debug information BMU_BCU Debug information
7:4	RO	0x0	BMU_DCU BMU_DCU Debug information BMU_DCU Debug information

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Bit	Attr	Reset Value	Description
3:0	RO	0x0	BMU_CCU BMU_CCU Debug information BMU_CCU Debug information

USB3_GDBGLSPMUX
Address: Operational Base + offset (0xc170)
Global Debug LSP MUX Register - Device

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x3f	logic_analyzer_trace Logic Analyzer Trace Port MUX Select Currently only bits[21:16] are used. A value of 6'h3F drives "0"s on the logic_analyzer_trace signal. If you plan to OR (instead using a mux) this signal with other trace signals in your system to generate a common trace signal, you can use this feature.
15	RW	0x0	EnDbc EnDbc Enable debugging of Debug capablity LSP in Host mode. Use HostSelect to select DbC LSP debug information presented in the GDBGLSP register.
14	RO	0x0	reserved
13:8	RW	0x00	HOSTSELECT Host LSP Select Selects the LSP debug information presented in the GDBGLSP register in host mode.
7:4	RW	0x0	DEVSELECT Device LSP Select Selects the LSP debug information presented in the GDBGLSP register in device mode. Or bit[7:4] of HOSTSELECT, Selects the LSP debug information presented in the GDBGLSP register in host mode.
3:0	RW	0x0	EPSELECT Device Endpoint Select Selects the Endpoint debug information presented in the GDBGEPINFO registers in device mode. Or bit[3:0] of HOSTSELECT, Selects the LSP debug information presented in the GDBGLSP register in host mode.

USB3_GDBGLSP
Address: Operational Base + offset (0xc174)
Global Debug LSP Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	LSPDEBUG LSP Debug Information LSP Debug Information

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USB3_GDBGEPINFO0
Address: Operational Base + offset (0xc178)
Global Debug Endpoint Information Register 0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	EPDEBUG Endpoint Debug Information Low 32-bit Endpoint Debug Information Low 32-bit

USB3_GDBGEPINFO1
Address: Operational Base + offset (0xc17c)
Global Debug Endpoint Information Register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00800000	EPDEBUG Endpoint Debug Information High 32-bit Endpoint Debug Information High 32-bit

USB3_GPRTBIMAP_HSLO
Address: Operational Base + offset (0xc180)
Global High-Speed Port to Bus Instance Mapping Register - Low

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	BINUM1 HS USB Instance Number for Port 1 Application-programmable ID field.

USB3_GPRTBIMAP_FSLO
Address: Operational Base + offset (0xc188)
Global Full-Speed Port to Bus Instance Mapping Register - Low

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	BINUM1 FS USB Instance Number for Port 1 Application-programmable ID field.

USB3_GUSB2PHYCFG0
Address: Operational Base + offset (0xc200)
Global USB2 PHY Configuration Register 0

Bit	Attr	Reset Value	Description
31	RW	0x0	PHYSOFTRST UTMI PHY Soft Reset Causes the usb2phy_reset signal to be asserted to reset a UTMI PHY. Not applicable to ULPI because ULPI PHYs are reset via their FunctionControl.Reset register, and the core automatically writes to this register when the core is reset (vcc_reset_n, USBCMD.HCRST, DCTL.SoftReset, or GCTL.SoftReset)

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Bit	Attr	Reset Value	Description
30	RW	0x1	U2_FREECLK_EXISTS U2_FREECLK_EXISTS Specifies whether your USB 2.0 PHY provides a free-running PHY clock, which is active when the clock control input is active. If your USB 2.0 PHY provides a free-running PHY clock, it must be connected to the utmi_clk[0] input. The remaining utmi_clk[n] must be connected to the respective port clocks. The core uses the Port-0 clock for generating the internal mac2 clock. 1'b0: USB 2.0 free clock does not exist 1'b1: USB 2.0 free clock exists Note: When the core is configured as device-only, do not set this bit to 1.
29:25	RO	0x0	reserved
24:22	RW	0x0	LSTRD LS Turnaround Time This field indicates the value of the Rx-to-Tx packet gap for LS devices. The encoding is as follows: 0: 2 bit times 1: 2.5 bit times 2: 3 bit times 3: 3.5 bit times 4: 4 bit times 5: 4.5 bit times 6: 5 bit times 7: 5.5 bit times Note: This field is applicable only in Host mode. For normal operation (to work with most LS devices), set the default value of this field to 3'h0 (2 bit times). The programmable LS device inter-packet gap and turnaround delays are provided to support some legacy LS devices that might require different delays than the default/fixed ones. For instance, the Open LS mouse requires 3 bit times of inter-packet gap to work correctly. Include your PHY delays when programming the LSIPD/LSTRDTIM values. For example, if your PHY's TxEndDelay in LS mode is 30 UTMI/ULPI CLKs, then subtract this delay (~1 LS bit time) from the device's delay requirement.

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Bit	Attr	Reset Value	Description
21:19	RW	0x2	LSIPD LS Inter-Packet Time This field indicates the value of Tx-to-Tx packet gap for LS devices. The encoding is as follows: 0: 2 bit times 1: 2.5 bit times 2: 3 bit times 3: 3.5 bit times 4: 4 bit times 5: 4.5 bit times 6: 5 bit times 7: 5.5 bit times Note: This field is applicable only in Host mode. For normal operation (to work with most LS devices), set the default value of this field to 3'h2 (3 bit times). The programmable LS device inter-packet gap and turnaround delays are provided to support some legacy LS devices that might require different delays than the default/fixed ones. For instance, the AOpen LS mouse requires 3 bit times of inter-packet gap to work correctly. Include your PHY delays when programming the LSIPD/LSTRDTIM values. For example, if your PHY's TxEndDelay in LS mode is 30 UTMI/ULPI CLKs, then subtract this delay (~1 LS bit time) from the device's delay requirement.
18:14	RO	0x0	reserved
13:10	RW	0x9	USBTRDTIM USB 2.0 Turnaround Time Sets the turnaround time in PHY clocks. Specifies the response time for a MAC request to the Packet FIFO Controller (PFC) to fetch data from the DFIFO (SPRAM). The following are the required values for the minimum SoC bus frequency of 60 MHz. USB turnaround time is a critical certification criteria when using long cables and five hub levels. The required values for this field: 4'h5: When the MAC interface is 16-bit UTMI+. 4'h9: When the MAC interface is 8-bit UTMI+/ULPI. If SoC bus clock is less than 60 MHz, and USB turnaround time is not critical, this field can be set to a larger value. Note: This field is valid only in device mode.

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Bit	Attr	Reset Value	Description
9	RW	0x0	XCVRDLY Transceiver Delay Enables a delay between the assertion of the UTMI/ULPI Transceiver Select signal (for HS) and the assertion of the TxValid signal during a HS Chirp. When this bit is set to 1, a delay (of approximately 2.5 us) is introduced from the time when the Transceiver Select is set to 2'b00 (HS) to the time the TxValid is driven to 0 for sending the chirp-K. This delay is required for some UTMI/ULPI PHYs. Note: If you enable the hibernation feature when the device core comes out of power-off, you must re-initialize this bit with the appropriate value because the core does not save and restore this bit value during hibernation. This bit is valid only in device mode.
8	RW	0x0	ENBLSLPM Enable utmi_sleep_n and utmi_l1_suspend_n The application uses this bit to control utmi_sleep_n and utmi_l1_suspend_n assertion to the PHY in the L1 state. 1'b0: utmi_sleep_n and utmi_l1_suspend_n assertion from the core is not transferred to the external PHY. 1'b1: utmi_sleep_n and utmi_l1_suspend_n assertion from the core is transferred to the external PHY. Note: This bit must be set high for Port0 if SNPS PHY is used. In Device mode - Before issuing any device endpoint command when operating in 2.0 speeds, disable this bit and enable it after the command completes. Without disabling this bit, if a command is issued when the device is in L1 state and if mac2_clk (utmi_clk/ulpi_clk) is gated off, the command will not get completed.
7	RO	0x0	PHYSEL USB 2.0 High-Speed PHY or USB 1.1 Full-Speed 1'b0: USB 2.0 high-speed UTMI+ or ULPI PHY. 1'b1: USB 1.1 full-speed serial transceiver.

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Bit	Attr	Reset Value	Description
6	RW	0x0	SUSPENDUSB20 Suspend USB2.0 HS/FS/LS PHY When set, USB2.0 PHY enters Suspend mode if Suspend conditions are valid. For DRD/OTG configurations, it is recommended that this bit is set to 0 during coreConsultant configuration. If it is set to 1, then the application must clear this bit after power-on reset. Application needs to set it to 1 after the core initialization completes. For all other configurations, this bit can be set to 1 during core configuration. Note: In host mode, on reset, this bit is set to 1. Software can override this bit after reset. In device mode, before issuing any device endpoint command when operating in 2.0 speeds, disable this bit and enable it after the command completes. If you issue a command without disabling this bit when the device is in L2 state and if mac2_clk (utmi_clk/ulpi_clk) is gated off, the command will not get completed.
5	RO	0x0	reserved
4	RO	0x0	ULPI_UTMI_Sel ULPI or UTMI+ Select 1'b0: UTMI+ Interface 1'b1: ULPI Interface
3	RW	0x0	PHYIF PHY Interface If UTMI+ is selected, the application uses this bit to configure the core to support a UTMI+ PHY with an 8- or 16-bit interface. 1'b0: 8 bits 1'b1: 16 bits

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	TOutCal HS/FS Timeout Calibration The number of PHY clocks, as indicated by the application in this field, is multiplied by a bit-time factor; this factor is added to the high-speed/full-speed interpacket timeout duration in the core to account for additional delays introduced by the PHY. This may be required, since the delay introduced by the PHY in generating the linestate condition may vary among PHYs. The USB standard timeout value for high-speed operation is 736 to 816 (inclusive) bit times. The USB standard timeout value for full-speed operation is 16 to 18 (inclusive) bit times. The application must program this field based on the speed of connection. The number of bit times added per PHY clock are: High-speed operation: One 30-MHz PHY clock = 16 bit times One 60-MHz PHY clock = 8 bit times Full-speed operation: One 30-MHz PHY clock = 0.4 bit times One 60-MHz PHY clock = 0.2 bit times

USB3_GUSB3PIPECTL0
Address: Operational Base + offset (0xc2c0)
Global USB3 PIPE Control Register 0

Bit	Attr	Reset Value	Description
31	RW	0x0	PHYSoftRst USB3 PHY Soft Reset After setting this bit to 1, the software needs to clear this bit.

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Bit	Attr	Reset Value	Description
30	RW	0x0	HstPrtCmpl HstPrtCmpl This feature tests the PIPE PHY compliance patterns without having to have a test fixture on the USB 3.0 cable. This bit enables placing the SS port link into a compliance state. By default, this bit must be set to 1'b0. In compliance lab testing, the SS port link enters compliance after failing the first polling sequence after power on. Set this bit to 0, when you run compliance tests. The sequence for using this functionality is as follows: 1. Disconnect any plugged in devices. 2. Perform USBCMD.HCRST or power-on-chip reset. 3. Set PORTSC.PP=0. 4. Set GUSB3PIPECTL. HstPrtCmpl=1. This places the link into compliance state. To advance the compliance pattern, follow this sequence (toggle the set GUSB3PIPECTL. HstPrtCmpl): 1. Set GUSB3PIPECTL.HstPrtCmpl=0. 2. Set GUSB3PIPECTL.HstPrtCmpl=1. This advances the link to the next compliance pattern. To exit from the compliance state perform USBCMD.HCRST or power-on-chip reset.
29	RW	0x0	U2SSInactP3ok P3 OK for U2/SSInactive 0: During link state U2/SS.Inactive, put PHY in P2 (Default) 1: During link state U2/SS.Inactive, put PHY in P3. Note: For a port, if GUSB3PIPECTL[7]=1 and GUSB3PIPECTL[29]=1, set GUSB3PIPECTL[11] to 1.
28	RW	0x0	DisRxDetP3 Disabled receiver detection in P3 0: If PHY is in P3 and Core needs to perform receiver detection, The core performs receiver detection in P3. (Default) 1: If PHY is in P3 and Core needs to perform receiver detection, The core changes the PHY power state to P2 and then performs receiver detection. After receiver detection, the cores changes PHY power state to P3.

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Bit	Attr	Reset Value	Description
27	RW	0x0	Ux_exit_in_Px Ux Exit in Px 0: The core does U1/U2/U3 exit in PHY power state P0 (default behavior). 1: The core does U1/U2/U3 exit in PHY power state P1/P2/P3 respectively. This bit is added for SS PHY workaround where SS PHY injects a glitch on pipe3_RxElecIdle while receiving Ux exit LFPS, and pipe3_PowerDown change is in progress. Note: This bit is used by third-party SS PHY. It must be set to 0 for SNPS PHY.
26	RW	0x0	ping_enhancement_en Ping Enhancement Enable When set, the Downstream port U1 ping receive timeout becomes 500 ms instead of 300 ms. Minimum Ping.LFPS receive duration is 8 ns (one mac3_clk). This field is valid for the downstream port only. Note: This bit is used by third-party SS PHY. It must be set to 0 for SNPS PHY.
25	RW	0x0	u1u2exitfail_to_recov U1U2exitfail to Recovery When set, and U1/U2 LFPS handshake fails, the LTSSM transitions from U1/U2 to Recovery instead of SS Inactive. If Recovery fails, then the LTSSM can enter SS.Inactive. This is an enhancement only. It prevents interoperability issue if the remote link does not do proper handshake.
24	RW	0x1	request_p1p2p3 Always Request P1/P2/P3 for U1/U2/U3 When set, the core always requests PHY power change from P0 to P1/P2/P3 during U0 to U1/U2/U3 transition. If this bit is 0, and immediate Ux exit (remotely initiated, or locally initiated) happens, the core does not request P1/P2/P3 power state change. Note: This bit must be set to 1 for SNPS PHY. For third-party SS PHY, check with your PHY vendor.
23	RW	0x0	StartRxDetU3RxDet Start Receiver Detection in U3/Rx.Detect If DWC_USB3_GUSB3PIPECTL_INIT[22] is set, and the link is in either U3 or Rx.Detect state, the core starts receiver detection on the rising edge of this bit. This can only be used for Downstream ports. This bit must be set to 0 for Upstream ports. This feature must not be enabled for normal operation. If have to use this feature, contact SNPS.

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Bit	Attr	Reset Value	Description
22	RW	0x0	DisRxDetU3RxDet Disable Receiver Detection in U3/Rx.Det When set, the core does not handle receiver detection in either U3 or Rx.Detect states. DWC_USB3_GUSB3PIPECTL_INIT[23] must be used to start receiver detection manually. This bit can only be used for the downstream port. This bit must be set to 0 for Upstream ports. This feature must not be enabled for normal operation. If you have to use this feature, contact SNPS.
21:19	RW	0x1	DelayP1P2P3 Delay P1P2P3 Delay P0 to P1/P2/P3 request when entering U1/U2/U3 until (DWC_USB3_GUSB3PIPECTL_INIT[21:19]*8) 8B10B error occurs, or Pipe3_RxValid drops to 0. DWC_USB3_GUSB3PIPECTL_INIT[18] must be 1 to enable this functionality.
18	RW	0x1	DELAYP1TRANS DELAYP1TRANS Delay PHY power change from P0 to P1/P2/P3 when link state changing from U0 to U1/U2/U3 respectively. 1'b1: When entering U1/U2/U3, delay the transition to P1/P2/P3 until the pipe3 signals, Pipe3_RxElecIlde is 1 and pipe3_RxValid is 0 1'b0: When entering U1/U2/U3, transition to P1/P2/P3 without checking for Pipe3_RxElecIlde and pipe3_RxValid. Note: This bit must be set to '1' for SNPS PHY. It is also used by third-party SS PHY.
17	RW	0x0	SUSPENDENABLE Suspend USB3.0 SS PHY When set, and if Suspend conditions are valid, the USB 3.0 PHY enters Suspend mode. For DRD/OTG configurations, it is recommended that this bit is set to '0' during coreConsultant configuration. If it is set to '1', then the application must clear this bit after power-on reset. Application needs to set it to '1' after the core initialization is completed. For all other configurations, this bit can be set to '1' during core configuration.
16:15	RW	0x0	DATWIDTH PIPE Data Width 2'b00: 32 bits 2'b01: 16 bits 2'b10: 8 bits Note: USB3 controller only support 32-bit width pipe interface.

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Bit	Attr	Reset Value	Description
14	RW	0x0	AbortRxDetInU2 Abort Rx Detect in U2 When set and the link state is U2, then the core will abort receiver detection if it receives U2 exit LFPS from the remote link partner. This bit is for the downstream port only. Note: This bit is used by third-party SS PHY. It must be set to '0' for SNPS PHY.
13	RW	0x0	SkipRxDet Skip Rx Detect When set, the core skips Rx Detection if pipe3_RxElecIdle is low. Skip is defined as waiting for the appropriate timeout, then repeating the operation.
12	RW	0x0	LFPSP0Algn LFPS P0 Align When set: 1. The core deasserts LFPS transmission on the clock edge that it requests Phy power state 0 when exiting U1, U2, or U3 low power states. Otherwise, LFPS transmission is asserted one clock earlier. 2. The core requests symbol transmission two pipe3_rx_pclks periods after the PHY asserts PhyStatus as a result of the PHY switching from P1 or P2 state to P0 state. Currently, this bit is only used in USB 3.0 HUB with SNPS PHY. For other USB 3.0 Host, Device, and DRD cores, this bit is not required.
11	RW	0x0	P3P2TranOK P3 P2 Transitions OK When set, the core transitions directly from Phy power state P2 to P3 or from state P3 to P2. When not set, P0 is always entered as an intermediate state during transitions between P2 and P3, as defined in the PIPE3 Specification. According to the PIPE3 Specification, any direct transition between P3 and P2 is illegal. Note: This bit is used by third-party SS PHY. It must be set to '0' for SNPS PHY.
10	RW	0x0	P3ExSigP2 P3 Exit Signal in P2 When this bit is set, the core always changes the PHY power state to P2, before attempting a U3 exit handshake. This bit is used only for some non-SNPS PHYs that cannot do LFPS in P3. Note: This bit is used by third-party SS PHY. It must be set to '0' for SNPS PHY.
9	RW	0x0	LFPSFILTER LFPS Filter When set, filter LFPS reception with pipe3_RxValid in PHY power state P0, that is, ignore LFPS reception from the PHY unless both pipe3_Rxelecidle and pipe3_RxValid are deasserted.

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Bit	Attr	Reset Value	Description
8	RW	0x0	RX_DETECT_to_Polling_L RX_DETECT to Polling.LFPS Control 1'b0 (Default): Enables a 400us delay to start Polling LFPS after RX_DETECT. This allows VCM offset to settle to a proper level. 1'b1: Disables the 400us delay to start Polling LFPS after RX_DETECT. During controller certification with third party PHY it is observed that the PHY is not able to meet the Tx AC common mode voltage active (VTX-CM-ACPP_ACTIVE <100mv) if the link starts polling within 80us from the time rx.detect is performed. To meet this VTX-CM-ACPP_ACTIVE specification, the polling must be delayed further. If the PHY does not have issue then they can set this bit to 1 which allows polling to start within 80us.
7	RO	0x0	reserved
6	RW	0x0	TX_SWING Tx Swing Drive the setting value to the pipe interface of PHY.
5:3	RW	0x0	TX_MARGIN Tx Margin[2:0] Drive the setting value to the pipe interface of PHY.
2:1	RW	0x1	TX_DE_EPPHASIS Tx Deemphasis The value driven to the PHY is controlled by the LTSSM during USB3 Compliance mode.
0	RW	0x0	ELASTIC_BUFFER_MODE Elastic Buffer Mode Drive the setting value to the pipe interface of PHY.

USB3_GTXFIFOSIZn

Address: Operational Base + offset (0xc300 + 4*n), n=0~6

Global Transmit FIFO Size Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	TXFSTADDR_N Transmit FIFOn RAM Start Address This field contains the memory start address for TxFIFOn in 64-bit words.
15:0	RW	0x0042	TXFDEP_N TxFIFO Depth This field contains the depth of TxFIFOn in 64-bit words. Minimum value: 32; Maximum value: 32,768

USB3_GRXFIFOSIZn

Address: Operational Base + offset (0xc380 + 4*n), n=0~2

Global Receive FIFO Size Register n

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	RXFSTADDR_N RxFIFOn RAM Start Address This field contains the memory start address for RxFIFOn in 64-bit words.
15:0	RW	0x0285	RXFDEP_N RxFIFO Depth This field contains the depth of RxFIFOn in 64-bit words. Minimum value: 32; Maximum value: 16,384

USB3_GEVNTADRLO0
Address: Operational Base + offset (0xc400)
Global Event Buffer Address (Low) Register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	EVNTADRLO Event Buffer Address Holds the lower 32 bits of start address of the external memory for the Event Buffer. During operation, hardware does not update this address.

USB3_GEVNTADRHI0
Address: Operational Base + offset (0xc404)
Global Event Buffer Address (High) Register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	EVNTADRHI Event Buffer Address Holds the higher 32 bits of start address of the external memory for the Event Buffer. During operation, hardware does not update this address.

USB3_GEVNTSIZ0
Address: Operational Base + offset (0xc408)
Global Event Buffer Size Register 0

Bit	Attr	Reset Value	Description
31	RW	0x0	EVNTINTRPTMASK Event Interrupt Mask When set to '1', this prevents the interrupt from being generated. However, even when the mask is set, the events are queued.
30:16	RO	0x0	reserved
15:0	RW	0x0000	EVENTSIZ Event Buffer Size in bytes Holds the size of the Event Buffer in bytes; must be a multiple of four. This is programmed by software once during initialization. The minimum size of the event buffer is 32 bytes.

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USB3_GEVNTCOUNT0
Address: Operational Base + offset (0xc40c)
Global Event Buffer Count Register 0

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	EVNTCOUNT Event Count When read, returns the number of valid events in the Event Buffer (in bytes). When written, hardware decrements the count by the value written. The interrupt line remains high when count is not 0.

USB3_GHWPARAMS8
Address: Operational Base + offset (0xc600)
Global Hardware Parameters Register 8

Bit	Attr	Reset Value	Description
31:0	RO	0x0000077c	ghwparams8_32_0 ghwparams8_32_0 ghwparams8_32_0

USB3_GTXFIFOPRIDEV
Address: Operational Base + offset (0xc610)
Global Device TX FIFO DMA Priority Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:0	RW	0x00	gtxfifopridev Device TxFIFO priority This register specifies the relative DMA priority level among the Device TXFIFOs (one per IN endpoint). Each register bit[n] controls the priority (1: high, 0: low) of each TXFIFO[n]. When multiple TXFIFOs compete for DMA service at a given time (that is, multiple TXQs contain TX DMA requests and their corresponding TXFIFOs have space available), the TX DMA arbiter grants access on a packet-basis in the following manner: 1. High-priority TXFIFOs are granted access using round-robin arbitration 2. Low-priority TXFIFOs are granted access using round-robin arbitration only after the high-priority TXFIFOs have no further processing to do (that is, either the TXQs are empty or the corresponding TXFIFOs are full). For scatter-gather packets, the arbiter grants successive DMA requests to the same FIFO until the entire packet is completed. When configuring periodic IN endpoints, software must set register bit[n]=1, where n is the TXFIFO assignment. This ensures that the DMA for isochronous or interrupt IN endpoints are prioritized over bulk or control IN endpoints. This register is present only when the core is configured to operate in the device mode. The register size corresponds to the number of Device IN endpoints.

USB3_GTXFIFOPRIHST
Address: Operational Base + offset (0xc618)
Global Host TX FIFO DMA Priority Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	gtxfifoprihst Host TxFIFO priority This register specifies the relative DMA priority level among the Host TXFIFOs (one per USB bus instance) within the associated speed group (SS or HS/FSLS). Each register bit[n] controls the priority (1: high, 0: low) of TXFIFO[n] within a speed group. When multiple TXFIFOs compete for DMA service at a given time (i.e., multiple TXQs contain TX DMA requests and their corresponding TXFIFOs have space available), the TX DMA arbiter grants access on a packet-basis in the following manner: 1. Among the FIFOs in the same speed group (SS or HS/FSLS): a. High-priority TXFIFOs are granted access using round-robin arbitration b. Low-priority TXFIFOs are granted access using round-robin arbitration only after the high-priority TXFIFOs have no further processing to do (that is, either the TXQs are empty or the corresponding TXFIFOs are full). 2. The TX DMA arbiter prioritizes the SS speed group or HS/FSLS speed group according to the ratio programmed in the GDMAHLRATIO register. For scatter-gather packets, the arbiter grants successive DMA requests to the same FIFO until the entire packet is completed. This register is present only when the core is configured to operate in the host mode (includes DRD and OTG modes). The register size corresponds to the number of configured USB bus instances; for example, in the default configuration, there are 3 USB bus instances (1 SS, 1 HS, and 1 FSLS).

USB3_GRXFIFOPRIHST
Address: Operational Base + offset (0xc61c)
Global Host RX FIFO DMA Priority Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	grxfifoprihst Host RxFIFO priority This register specifies the relative DMA priority level among the Host RXFIFOs (one per USB bus instance) within the associated speed group (SS or HS/FSLS). Each register bit[n] controls the priority (1: high, 0: low) of RXFIFO[n] within a speed group. When multiple RXFIFOs compete for DMA service at a given time (i.e., multiple RXQs contain RX DMA requests and their corresponding RXFIFOs have data available), the RX DMA arbiter grants access on a packet-basis in the following manner: 1. Among the FIFOs in the same speed group (SS or HS/FSLS): a. High-priority RXFIFOs are granted access using round-robin arbitration b. Low-priority RXFIFOs are granted access using round-robin arbitration only after high-priority RXFIFOs have no further processing to do (that is, either the RXQs are empty or the corresponding RXFIFOs do not have the required data). 2. The RX DMA arbiter prioritizes the SS speed group or HS/FSLS speed group according to the ratio programmed in the GDMAHLRATIO register. For scatter-gather packets, the arbiter grants successive DMA requests to the same FIFO until the entire packet is completed. This register is present only when the core is configured to operate in the host mode (includes DRD and OTG modes). The register size corresponds to the number of configured USB bus instances; for example, in the default configuration, there are 3 USB bus instances (1 SS, 1 HS, and 1 FSLS).

USB3_GFIFOPRIDBC
Address: Operational Base + offset (0xc620)
Global Host Debug Capability DMA Priority Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	gfifopridbc Host DbC DMA priority This register specifies the relative priority of the RXFIFOs and TXFIFOs associated with the DbC mode. It overrides the priority assigned in the corresponding indexes of the Host RXFIFO and TXFIFO DMA priority registers, when the DbC mode is enabled. Priority settings are specified in relation to the low-priority SS speed group: 1. Normal priority indicates that the DbC FIFOs are considered identical to the Host SS low-priority FIFOs. 2. Low priority indicates that the DbC FIFOs are considered to have lower priority than all Host SS FIFOs. 3. High priority indicates that the DbC FIFOs are considered higher priority than the Host SS low-priority FIFOs but lower priority than the Host SS high-priority FIFOs. This register is present only when the core is configured to operate in Host Debug Capability (DbC) mode.

USB3_GDMAHLRATIO
Address: Operational Base + offset (0xc624)
Global Host FIFO DMA High-Low Priority Ratio Register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	hstrxfifo Host RXFIFO DMA High-Low Priority This register specifies the relative priority of the SS FIFOs with respect to the HS/FSLS FIFOs. The DMA arbiter prioritizes the HS/FSLS round-robin arbiter group every DMA High-Low Priority Ratio grants as indicated in the register separately for TX and RX. To illustrate, consider that all FIFOs are requesting access simultaneously, and the ratio is 4. SS gets priority for 4 packets, HS/FSLS gets priority for 1 packet, SS gets priority for 4 packets, HS/FSLS gets priority for 1 packet, and so on. If FIFOs from both speed groups are not requesting access simultaneously then, 1. if SS got grants 4 out of the last 4 times, then HS/FSLS get the priority on any future request. 2. if HS/FSLS got the grant last time, SS gets the priority on the next request. 3. if there is a valid request on either SS or HS/FSLS, a grant is always awarded; there is no idle. This register is present if the core is configured to operate in host mode.
7:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x08	hsttxfifo Host TXFIFO DMA High-Low Priority This register specifies the relative priority of the SS FIFOs with respect to the HS/FSLS FIFOs. The DMA arbiter prioritizes the HS/FSLS round-robin arbiter group every DMA High-Low Priority Ratio grants as indicated in the register separately for TX and RX. To illustrate, consider that all FIFOs are requesting access simultaneously, and the ratio is 4. SS gets priority for 4 packets, HS/FSLS gets priority for 1 packet, SS gets priority for 4 packets, HS/FSLS gets priority for 1 packet, and so on. If FIFOs from both speed groups are not requesting access simultaneously then, 1. if SS got grants 4 out of the last 4 times, then HS/FSLS get the priority on any future request. 2. if HS/FSLS got the grant last time, SS gets the priority on the next request. 3. if there is a valid request on either SS or HS/FSLS, a grant is always awarded; there is no idle. This register is present if the core is configured to operate in host mode.

USB3_GFLADJ
Address: Operational Base + offset (0xc630)
Global Frame Length Adjustment Register

Bit	Attr	Reset Value	Description
31	RW	0x0	GFLADJ_REFCLK_240MHZDECR_PLS1 GFLADJ_REFCLK_240MHZDECR_PLS1 This field indicates that the decrement value that the controller applies for each ref_clk must be GFLADJ_REFCLK_240MHZ_DECR and GFLADJ_REFCLK_240MHZ_DECR +1 alternatively on each ref_clk. Set this bit to a 1 only if GFLADJ_REFCLK_LPM_SEL is set to 1 and the fractional component of 240/ref_frequency is greater than or equal to 0.5. Examples: If the ref_clk is 24 MHz then 1. GUCTL.REF_CLK_PERIOD = 41 2. GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = (240/24) = 10 3. GFLADJ.GFLADJ_REFCLK_240MHZDECR_PLS1 = 0

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Bit	Attr	Reset Value	Description
30:24	RW	0x00	GFLADJ_REFCLK_240MHZ_DECR GFLADJ_REFCLK_240MHZ_DECR This field indicates the decrement value that the controller applies for each ref_clk in order to derive a frame timer in terms of a 240-MHz clock. This field must be programmed to a non-zero value only if GFLADJ_REFCLK_LPM_SEL is set to 1. The value is derived as follows: GFLADJ_REFCLK_240MHZ_DECR = 240/ref_clk_frequency Examples: If the ref_clk is 24 MHz then 1. GUCTL.REF_CLK_PERIOD = 41 2. GFLADJ.GFLADJ_REFCLK_240MHZ_DECR = 240/24 = 10
23	RW	0x0	GFLADJ_REFCLK_LPM_SEL GFLADJ_REFCLK_LPM_SEL This bit enables the functionality of running SOF/ITP counters on the ref_clk. This bit must not be set to 1 if GCTL.SOFITPSYNC bit is set to 1. Similarly, if GFLADJ_REFCLK_LPM_SEL set to 1, GCTL.SOFITPSYNC must not be set to 1. When GFLADJ_REFCLK_LPM_SEL is set to 1 the overloading of the suspend control of the USB 2.0 first port PHY (UTMI/ULPI) with USB 3.0 port states is removed. For example, for SNPS PHY, the COMMONONN signal can be tied to 1. Note that the ref_clk frequencies supported in this mode are 16/17/19.2/20/24/39.7/40 MHz. The utmi_clk[0] signal of the core must be connected to the FREECLK of the PHY. Note: If you set this bit to 1, the GUSB2PHYCFG.U2_FREECLK_EXISTS bit must be set to 0.
22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21:8	RW	0x0000	GFLADJ_REFCLK_FLADJ GFLADJ_REFCLK_FLADJ This field indicates the frame length adjustment to be applied when SOF/ITP counter is running on the ref_clk. This register value is used to adjust the ITP interval when GCTL[SOFITPSYNC] is set to 1; SOF and ITP interval when GLADJ.GFLADJ_REFCLK_LPM_SEL is set to 1. This field must be programmed to a non-zero value only if GFLADJ_REFCLK_LPM_SEL is set to 1 or GCTL.SOFITPSYNC is set to 1. The value is derived as follows: FLADJ_REF_CLK_FLADJ=((125000/ref_clk_period_integer)-(125000/ref_clk_period)) * ref_clk_period where: 1. The ref_clk_period_integer is the integer value of the ref_clk period got by truncating the decimal (fractional) value that is programmed in the GUCTL.REF_CLK_PERIOD field. 2. The ref_clk_period is the ref_clk period including the fractional value. Examples: If the ref_clk is 24 MHz then 1. GUCTL.REF_CLK_PERIOD = 41 2. GFLADJ.GLADJ_REFCLK_FLADJ = ((125000/41)-(125000/41.6666))*41.6666 = 2032 (ignoring the fractional value)
7	RW	0x0	GFLADJ_30MHZ_SDBND_SEL GFLADJ_30MHZ_SDBND_SEL This field selects whether to use the input signal fladj_30mhz_reg or the GFLADJ.GFLADJ_30MHZ to adjust the frame length for the SOF/ITP. When this bit is set to: 1, the controller uses the register field GFLADJ.GFLADJ_30MHZ value 0, the controller uses the input signal fladj_30mhz_reg value
6	RO	0x0	reserved
5:0	RW	0x00	GFLADJ_30MHZ GFLADJ_30MHZ This field indicates the value that is used for frame length adjustment instead of considering from the sideband input signal fladj_30mhz_reg. This enables post-silicon frame length adjustment in case the input signal fladj_30mhz_reg is connected to a wrong value or is not valid. For details on how to set this value, refer to section 5.2.4, "Frame Length Adjustment Register (FLADJ)," of the xHCI Specification.

USB3_DCFG
Address: Operational Base + offset (0xc700)
Device Configuration Register

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Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23	RW	0x0	IgnStrmPP IgnoreStreamPP This bit only affects stream-capable bulk endpoints. When this bit is set to 0 and the controller receives a Data Packet with the Packet Pending (PP) bit set to 0 for OUT endpoints, or it receives an ACK with the NumP field set to 0 and PP set to 0 for IN endpoints, the core attempts to search for another stream (CStream) to initiate to the host. However, there are two situations where this behavior is not optimal: 1. When the host is setting PP=0 even though it has not finished the stream, or 2. When the endpoint on the device is configured with one transfer resource and therefore does not have any other streams to initiate to the host. When this bit is set to 1, the core ignores the Packet Pending bit for the purposes of stream selection and does not search for another stream when it receives DP(PP=0) or ACK(NumP=0, PP=0). This can enhance the performance when the device system bus bandwidth is low or the host responds to the core's ERDY transmission very quickly.
22	RW	0x0	LPMCAP LPM Capable The application uses this bit to control the DWC_usb3 core LPM capabilities. If the core operates as a non-LPM-capable device, it cannot respond to LPM transactions. 1'b0: LPM capability is not enabled. 1'b1: LPM capability is enabled.
21:17	RW	0x04	NUMP Number of Receive Buffers. This bit indicates the number of receive buffers to be reported in the ACK TP. The DWC_usb3 controller uses this field if GRXTHRCFG.USBRxPktCntSel is set to 0. The application can program this value based on RxFIFO size, buffer sizes programmed in descriptors, and system latency. For an OUT endpoint, this field controls the number of receive buffers reported in the NumP field of the ACK TP transmitted by the core. Note: This bit is used in host mode when Debug Capability is enabled.
16:12	RW	0x00	INTRNUM Interrupt number Indicates interrupt/EventQ number on which non-endpoint-specific device-related interrupts (see DEVT) are generated.
11:10	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
9:3	RW	0x00	DEVADDR Device Address. The application must perform the following: 1. Program this field after every SetAddress request. 2. Reset this field to zero after USB reset.
2:0	RW	0x4	DEVSPD Device Speed Indicates the speed at which the application requires the core to connect, or the maximum speed the application can support. However, the actual bus speed is determined only after the chirp sequence is completed, and is based on the speed of the USB host to which the core is connected. 3'b100: SuperSpeed (USB 3.0 PHY clock is 125 MHz or 250 MHz) 3'b000: High-speed (USB 2.0 PHY clock is 30 MHz or 60 MHz) 3'b001: Full-speed (USB 2.0 PHY clock is 30 MHz or 60 MHz)

USB3_DCTL
Address: Operational Base + offset (0xc704)
Device Control Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	RUN_STOP Run/Stop The software writes 1 to this bit to start the device controller operation. To stop the device controller operation, the software must remove any active transfers and write 0 to this bit. When the controller is stopped, it sets the DSTS.DevCtrlHlt bit when the core is idle and the lower layer finishes the disconnect process. The Run/Stop bit must be used in following cases as specified: 1. After power-on reset and CSR initialization, the software must write 1 to this bit to start the device controller. The controller does not signal connect to the host until this bit is set. 2. The software uses this bit to control the device controller to perform a soft disconnect. When the software writes 0 to this bit, the host does not see that the device is connected. The device controller stays in the disconnected state until the software writes 1 to this bit. The minimum duration of keeping this bit cleared is specified in the Note below. If the software attempts a connect after the soft disconnect or detects a disconnect event, it must set DCTL[8:5] to 5 before reasserting the Run/Stop bit. 3. When the USB or Link is in a lower power state and the Two Power Rails configuration is selected, software writes 0 to this bit to indicate that it is going to turn off the Core Power Rail. After the software turns on the Core Power Rail again and re-initializes the device controller, it must set this bit to start the device controller. Note: The following is the minimum duration under various conditions for which the soft disconnect (SftDiscon) bit must be set for the USB host to detect a device disconnect: 30ms: For SuperSpeed, when the device state is Suspended, Idle, Transmit, or Receive. 10ms: For high-speed, when the device state is Suspended, Idle, or not Idle/Suspended (performing transactions); For full-speed/low-speed, when the device state is Suspended, Idle, or not Idle/Supended (performing transactions) To accommodate clock jitter, it is recommended that the application add extra delay to the specified minimum duration.

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Bit	Attr	Reset Value	Description
30	R/W SC	0x0	CSFTRST Core Soft Reset Reset all clock domains as follows: 1. This bit clears the interrupts and all the CSRs except GSTS, GSNPSID, GGPIO, GUID, GUSB2PHYCFGn registers, GUSB3PIPECTLn registers, DCFG, DCTL, DEVTEN, and DSTS registers. 2. All module state machines (except the SoC Bus Slave Unit) are reset to the IDLE state, and all the TxFIFOs and the RxFIFO are flushed. 3. Any transactions on the SoC bus Master are terminated as soon as possible, after gracefully completing the last data phase of a SoC bus transfer. Any transactions on the USB are terminated immediately. The application can write this bit at any time to reset the core. This is a self-clearing bit; the core clears this bit after all necessary logic is reset in the core, which may take several clocks depending on the core's current state. Once this bit is cleared, the software must wait at least 3 PHY clocks before accessing the PHY domain (synchronization delay). Typically, software reset is used during software development and also when you dynamically change the PHY selection bits in the USB configuration registers listed above. When you change the PHY, the corresponding clock for the PHY is selected and used in the PHY domain. Once a new clock is selected, the PHY domain must be reset for proper operation.
29	RO	0x0	reserved
28:24	RW	0x00	HIRDTHRES HIRD Threshold The core asserts output signals utmi_l1_suspend_n and utmi_sleep_n on the basis of this signal: The core asserts utmi_l1_suspend_n to put the PHY into Deep Low-Power mode in L1 when both of the following are true: 1. HIRD value is greater than or equal to the value in DCTL.HIRD_Thres[3:0] 2. HIRD_Thres[4] is set to 1'b1. The core asserts utmi_sleep_n on L1 when one of the following is true: 1. If the HIRD value is less than HIRD_Thres[3:0] or 2. HIRD_Thres[4] is set to 1'b0. Note: This field must be set to '0' during SuperSpeed mode of operation.

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Bit	Attr	Reset Value	Description
23:20	RW	0xf	LPM_NYET_thres LPM NYET Threshold Handshake response to LPM token specified by device application. Response depends on DCFG.LPMCap. DCFG.LPMCap is 1'b0 - The core always responds with Timeout (that is, no response). DCFG.LPMCap is 1'b1 - The core responds with an ACK on successful LPM transaction, which requires that all of the following are satisfied: 1. There are no PID or CRC5 errors in both the EXT token and the LPM token (if not true, inactivity results in a timeout ERROR). 2. No data is pending in the Transmit FIFO and OUT endpoints not in flow controlled state (else NYET). 3. The BESL value in the LPM token is less than or equal to LPM_NYET_thres[3:0]
19	RW	0x0	KeepConnect KeepConnect When 1, this bit enables the save and restore programming model by preventing the core from disconnecting from the host when DCTL.RunStop is set to 0. It also enables the Hibernation Request Event to be generated when the link goes to U3 or L2. The device core disconnects from the host when DCTL.RunStop is set to 0. This bit indicates whether to preserve this behavior (0), or if the core must not disconnect when RunStop is set to 0 (1). This bit also prevents the LTSSM from automatically going to U0/L0 when the host requests resume from U3/L2.
18	RW	0x0	L1HibernationEn L1HibernationEn When this bit is set along with KeepConnect, the device core generates a Hibernation Request Event if L1 is enabled and the HIRD value in the LPM token is larger than the threshold programmed in DCTL.HIRD_Thres. The core does not exit the LPM L1 state until software writes Recovery into the DCTL.ULStChngReq field. This prevents corner cases where the device is entering hibernation at the same time the host is attempting to exit L1.
17	RW	0x0	CRS Controller Restore State This command is similar to the USBCMD.CRS bit in host mode and initiates the restore process. When software sets this bit to 1, the controller immediately sets DSTS.RSS to 1. When the controller has finished the restore process, it sets DSTS.RSS to 0. Note: When read, this field always returns 0.

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Bit	Attr	Reset Value	Description
16	RW	0x0	CSS Controller Save State This command is similar to the USBCMD.CSS bit in host mode and initiates the save process. When software sets this bit to 1, the controller immediately sets DSTS.SSS to 1. When the controller has finished the save process, it sets DSTS.SSS to 0. Note: When read, this field always returns 0.
15:13	RO	0x0	reserved
12	RW	0x0	INITU2ENA Initiate U2 Enable 1'b0: May not initiate U2 (default) 1'b1: May initiate U2 On USB reset, hardware clears this bit to 0. Software sets this bit after receiving SetFeature(U2_ENABLE), and clears this bit when ClearFeature(U2_ENABLE) is received. If DCTL[11] (AcceptU2Ena) is 0, the link immediately exits U2 state.
11	RW	0x0	ACCEPTU2ENA Accept U2 Enable 1'b0: Reject U2 except when Force_LinkPM_Accept bit is set (default) 1'b1: Core accepts transition to U2 state if nothing is pending on the application side. On USB reset, hardware clears this bit to 0. Software sets this bit after receiving a SetConfiguration command.
10	RW	0x0	INITU1ENA Initiate U1 Enable 1'b0: May not initiate U1 (default); 1'b1: May initiate U1. On USB reset, hardware clears this bit to 0. Software sets this bit after receiving SetFeature(U1_ENABLE), and clears this bit when ClearFeature(U1_ENABLE) is received. If DCTL[9] (AcceptU1Ena) is 0, the link immediately exits U1 state.
9	RW	0x0	ACCEPTU1ENA Accept U1 Enable 1'b0: Core rejects U1 except when Force_LinkPM_Accept bit is set (default) 1'b1: Core accepts transition to U1 state if nothing is pending on the application side. On USB reset, hardware clears this bit to 0. Software sets this bit after receiving a SetConfiguration command.

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Bit	Attr	Reset Value	Description
8:5	RW	0x0	ULSTCHNGREQ ULSTCHNGREQ Software writes this field to issue a USB/Link state change request. A change in this field indicates a new request to the core. If software wants to issue the same request back-to-back, it must write a 0 to this field between the two requests. The result of the state change request is reflected in the USB/Link State in DSTS. These bits are self-cleared on the MAC Layer exiting suspended state. If software is updating other fields of the DCTL register and not intending to force any link state change, then it must write a 0 to this field. SS Compliance mode is normally entered and controlled by the remote link partner. Refer to the USB 3.0 specification. Alternatively, you can force the local link directly into compliance mode, by resetting the SS link with the RUN/STOP bit set to zero. If you then write 10 to the USB/Link State Change field and 1 to RUN/STOP, the link goes to compliance mode. Once you are in compliance, you may alternately write zero and 10 to this field to advance the compliance pattern. In SS mode: Value: Requested Link State Transition/Action 0: No Action 4: SS.Disabled 5: Rx.Detect 6: SS.Inactive 8: Recovery 10: Compliance Others: Reserved In HS/FS/LS mode: Value:Requested USB state transition 8: Remote wakeup request Others: Reserved The Remote wakeup request must be issued 2us after the device goes into suspend state (DSTS[21:18] is 3). Note: After coming out of hibernation, software must write 8 (Recovery) into this field to confirm exit from the suspended state.
4:1	RW	0x0	TSTCTL Test Control 4'b000: Test mode disabled 4'b001: Test_J mode 4'b010: Test_K mode 4'b011: Test_SE0_NAK mode 4'b100: Test_Packet mode 4'b101: Test_Force_Enable Others: Reserved

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Bit	Attr	Reset Value	Description
0	RO	0x0	reserved

USB3_DEVTEN
Address: Operational Base + offset (0xc708)
Device Event Enable Register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	VENDEVTSTRCVDEN Vendor Device Test LMP Received Event 1: Enable this event 0: Disable this event
11:10	RO	0x0	reserved
9	RW	0x0	ERRTICERREVTEN Erratic Error Event Enable 1: Enable this event 0: Disable this event
8	RO	0x0	reserved
7	RW	0x0	SOFTEVTEN Start of (u)frame Event Enable 1: Enable this event 0: Disable this event
6	RW	0x0	U3L2L1SuspEn U3/L2-L1 Suspend Event Enable 1: Enable this event 0: Disable this event
5	RW	0x0	HibernationReqEvtEn Hibernation Request Event Enable 1: Enable this event 0: Disable this event
4	RW	0x0	WKUPEVTEN Resume/Remote Wakeup Detected Event Enable 1: Enable this event 0: Disable this event
3	RW	0x0	ULSTCNGEN USB/Link State Change Event Enable 1: Enable this event 0: Disable this event
2	RW	0x0	CONNECTDONEEVTEN Connection Done Event Enable 1: Enable this event 0: Disable this event
1	RW	0x0	USBRSTEVTEN USB Reset Event Enable 1: Enable this event 0: Disable this event

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Bit	Attr	Reset Value	Description
0	RW	0x0	DISSCONNEVTEN Disconnect Detected Event Enable 1: Enable this event 0: Disable this event

USB3_DSTS
Address: Operational Base + offset (0xc70c)
Device Status Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RO	0x0	DCNRD Device Controller Not Ready The bit indicates that the core is in the process of completing the state transitions after exiting from hibernation. To complete the state transitions, it takes 256 bus clock cycles from the time DCTL[31].Run/Stop is set. During hibernation, if the UTMI/ULPI PHY is in suspended state, then the 256-bus clock cycle delay starts after the PHY exited suspended state. Software must set DCTL[31].Run/Stop to 1 and wait for this bit to be de-asserted to zero before processing DSTS.USBLnkSt.
28:26	RO	0x0	reserved
25	RO	0x0	RSS RSS Restore State Status This bit is similar to the USBSTS.RSS in host mode. When the controller finishes the restore process, it completes the command by setting DSTS.RSS to 0.
24	RO	0x0	SSS SSS Save State Status This bit is similar to the USBSTS.SSS in host mode. When the controller has finished the save process, it completes the command by setting DSTS.SSS to 0.
23	RO	0x0	COREIDLE Core Idle The bit indicates that the core finished transferring all RxFIFO data to system memory, writing out all completed descriptors, and all Event Counts are zero. Note: While testing for Reset values, mask out the read value. This bit represents the changing state of the core and does not hold a static value.

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Bit	Attr	Reset Value	Description
22	RO	0x1	DEVCTRLHLT Device Controller Halted This bit is set to 0 when the Run/Stop bit in the DCTL register is set to 1. The core sets this bit to 1 when, after SW sets Run/Stop to 0, the core is idle and the lower layer finishes the disconnect process. When Halted=1, the core does not generate Device events. Note: The core does not set this bit to 1 if GEVNTCOUNTn has some valid value. Software needs to acknowledge the events that are generated (by writing to GEVNTCOUNTn) while it is waiting for this bit to be set to 1.
21:18	RO	0x4	USBLNKST USBLNKST SS mode: LTSSM State 4'h0: U0 4'h1: U1 4'h2: U2 4'h3: U3 4'h5: RX_DET 4'h6: SS_INACT 4'h7: POLL 4'h8: RECOV 4'h9: HRESET 4'ha: CMPLY 4'hb: LPBK 4'hf: Resume/Reset In HS/FS/LS mode: 4'h0: On state 4'h2: Sleep (L1) state 4'h3: Suspend (L2) state 4'h4: Disconnected state (Default state) 4'h5: Early Suspend state (valid only when Hibernation is disabled, GCTL[1].GblHibernationEn = 0) 4'he: Reset (valid only when Hibernation is enabled, GCTL[1].GblHibernationEn = 1) 4'hf: Resume (valid only when Hibernation is enabled, GCTL[1].GblHibernationEn = 1) The link state Resume/Reset indicates that the core received a resume or USB reset request from the host while the link was in hibernation. Software must write 8 (Recovery) to the DCTL.ULStChngReq field to acknowledge the resume/reset request. When Hibernation is enabled, GCTL[1].GblHibernationEn = 1, this field USBLnkSt is valid only when DCTL[31].Run/Stop set to 1 and DSTS[29].DCNRD = 0.
17:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2:0	RU	0x4	CONNECTSPD Connected Speed Indicates the speed at which the DWC_usb3 core has come up after speed detection through a chirp sequence. 3'b100: SuperSpeed (PHY clock is running at 125 or 250 MHz) 3'b000: High-speed (PHY clock is running at 30 or 60 MHz) 3'b001: Full-speed (PHY clock is running at 30 or 60 MHz)

USB3_DGCMDPAR
Address: Operational Base + offset (0xc710)
Device Generic Command Parameter Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PARAMETER PARAMETER This register indicates the device command parameter. This must be programmed before or along with the device command. The available device commands are listed in DGCMD register.

USB3_DGCMD
Address: Operational Base + offset (0xc714)
Device Generic Command Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:12	RO	0x0	CMDSTATUS Command Status 1: CmdErr: Indicates that the device controller encountered an error while processing the command. 0: Indicates command success
11	RO	0x0	reserved
10	R/WSC	0x0	CMDACT Command Active The software sets this bit to 1 to enable the device controller to execute the generic command. The device controller sets this bit to 0 after executing the command.
9	RO	0x0	reserved
8	RW	0x0	CMDIOC Command Interrupt on Complete When this bit is set, the device controller issues a Generic Command Completion event after executing the command. Note that this interrupt is mapped to DCFG.IntrNum. Note: This field must not set to 1 if the DCTL.RunStop field is 0.

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	CMDTYP Command Type Specifies the type of command the software driver is requesting the core to perform. 00h: Reserved 01h: Set Endpoint Configuration - 64 or 96-bit Parameter 02h: Set Endpoint Transfer Resource Configuration - 32-bit Parameter 03h: Get Endpoint State - No Parameter Needed 05h: Clear Stall (see Set Stall) - No Parameter Needed 06h: Start Transfer - 64-bit Parameter 07h: Update Transfer - No Parameter Needed 08h: End Transfer - No Parameter Needed 09h: Start New Configuration - No Parameter Needed

USB3_DALEPENA
Address: Operational Base + offset (0xc720)
Device Active USB Endpoint Enable Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	USBACTEP USB Active Endpoints This field indicates if a USB endpoint is active in the current configuration and interface. It applies to USB IN endpoints 0~15 and OUT endpoints 0~15, with one bit for each of the 32 possible endpoints. Even numbers are for USB OUT endpoints, and odd numbers are for USB IN endpoints, as follows: Bit[0]: USB EP0-OUT Bit[1]: USB EP0-IN Bit[2]: USB EP1-OUT Bit[3]: USB EP1-IN ... The entity programming this register must set bits 0 and 1 because they enable control endpoints that map to physical endpoints (resources) after USBReset. Hardware clears these bits for all endpoints (other than EP0-OUT and EP0-IN) after detecting a USB reset event. After receiving SetConfiguration and SetInterface requests, the application must program endpoint registers accordingly and set these bits. For more information, Pls see "Flexible Endpoint Mapping" section.

USB3_DEPnCMDPAR2
Address: Operational Base + offset (0xc800 + 4*n), n=0~12
Device Physical Endpoint-n Command Parameter 2 Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PARAMETER PARAMETER This register indicates the physical endpoint command Parameter 2. It must be programmed before issuing the command.

USB3_DEPnCMDPAR1
Address: Operational Base + offset (0xc804 + 4*n), n=0~12
Device Physical Endpoint-n Command Parameter 1 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PARAMETER PARAMETER This register indicates the physical endpoint command Parameter 1. It must be programmed before issuing the command.

USB3_DEPnCMDPAR0
Address: Operational Base + offset (0xc808 + 4*n), n=0~12
Device Physical Endpoint-n Command Parameter 0 Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PARAMETER PARAMETER This register indicates the physical endpoint command Parameter 0. It must be programmed before issuing the command.

USB3_DEPnCMD
Address: Operational Base + offset (0xc80c + 4*n), n=0~12
Device Physical Endpoint-n Command Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	COMMANDPARAM Command Parameters or Event Parameters when this register is written: For Start Transfer command: The 16-bit StreamID assigned to this transfer For Start Transfer command applied to an isochronous endpoint: StartMicroFramNum, Indicates the (micro)frame number to which the first TRB applies. For Update Transfer, End Transfer, and Start New Configuration commands: [22:16]: Transfer Resource Index (XferRscIdx). The hardware-assigned transfer resource index for the transfer, which was returned in response to the Start Transfer command. The application software-assigned transfer resource index for a Start New Configuration command. Event Parameters (EventParam), when this register is read. Please see bits [31:16] in the "Device Endpoint-n Events: DEPEVT" of the Databook.

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Bit	Attr	Reset Value	Description
15:12	RW	0x0	CMDSTATUS Command Completion Status The information is in the same format as bits 15:12 of the Endpoint Command Complete event, Please see "Device Endpoint-n Events: DEPEVT" in the Databook.
11	RW	0x0	HIPRI_FORCERM HighPriority/ForceRM HighPriority: Only valid for Start Transfer command ForceRM: Only valid for End Transfer command ClearPendIN: Only valid for Clear Stall command . Software sets this bit to clear any pending IN transaction (on that endpoint) stuck at the lower layers when a Clear Stall command is issued.
10	RW	0x0	CMDACT Command Active Software sets this bit to 1 to enable the device endpoint controller to execute the generic command. The device controller sets this bit to 0 when the CmdStatus field is valid and the endpoint is ready to accept another command. This does not imply that all the effects of the previously-issued command have taken place.
9	RO	0x0	reserved
8	RW	0x0	CMDIOC Command Interrupt on Complete When this bit is set, the device controller issues a generic Endpoint Command Complete event after executing the command. Note that this interrupt is mapped to DEPCFG.IntrNum. When the DEPCFG command is executed, the command interrupt on completion goes to the interrupt pointed by the DEPCFG.IntrNum in the current command. Note: This field must not set to 1 if the DCTL.RunStop field is 0.
7:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	CMDTYP Command Type Specifies the type of command the software driver is requesting the core to perform. 00h: Reserved 01h: Set Endpoint Configuration - -64 or 96-bit Parameter 02h: Set Endpoint Transfer Resource Configuration - 32-bitparameter 03h: Get Endpoint State - No Parameter Needed 04h: Set Stall - No Parameter Needed 05h: Clear Stall (see Set Stall) - No Parameter Needed 06h: Start Transfer - 64-bit Parameter 07h: Update Transfer - No Parameter Needed 08h: End Transfer - No Parameter Needed 09h: Start New Configuration - No Parameter Needed

USB3PHYUTMI_REG00
Address: Operational Base + offset (0x0000)
PHYREG00

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	Digital Debug

USB3PHYUTMI_REG04
Address: Operational Base + offset (0x0004)
PHYREG04

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x40	Digital Debug

USB3PHYUTMI_REG08
Address: Operational Base + offset (0x0008)
PHYREG08

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	Digital Debug

USB3PHYUTMI_REG0C
Address: Operational Base + offset (0x000c)
PHYREG0C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	Digital Debug

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USB3PHYUTMI_REG10
Address: Operational Base + offset (0x0010)
PHYREG10

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	Digital Debug

USB3PHYUTMI_REG14
Address: Operational Base + offset (0x0014)
PHYREG14

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	Digital Debug

USB3PHYUTMI_REG18
Address: Operational Base + offset (0x0018)
PHYREG18

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x0	Digital Debug

USB3PHYUTMI_REG1C
Address: Operational Base + offset (0x001c)
PHYREG1C

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x10	Digital Debug

USB3PHYUTMI_REG20
Address: Operational Base + offset (0x0020)
PHYREG20

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x0	Digital Debug

USB3PHYUTMI_REG24
Address: Operational Base + offset (0x0024)
PHYREG24

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:0	RW	0x00	Digital Debug

USB3PHYUTMI_REG28
Address: Operational Base + offset (0x0028)
PHYREG28

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	Digital Debug

USB3PHYUTMI_REG2C
Address: Operational Base + offset (0x002c)
PHYREG2C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x20	Digital Debug

USB3PHYUTMI_REG30
Address: Operational Base + offset (0x0030)
PHYREG30

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x5	Digital Debug

USB3PHYUTMI_REG34
Address: Operational Base + offset (0x0034)
PHYREG34

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x0	Digital Debug

USB3PHYUTMI_REG38
Address: Operational Base + offset (0x0038)
PHYREG38

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Digital Debug

USB3PHYUTMI_REG40
Address: Operational Base + offset (0x0040)
PHYREG40

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	Reserved, keep default value.
6	RW	0x1	half bit pre_emphasize enable, active high.
5:3	RW	0x4	TX HS pre-emphasize strength configure, 3'b111 represents the strongest, 3'b000 the weakest.
2:0	RW	0x1	TX HS driver strength configure, more "1" represents larger slew rate and eye height.

USB3PHYUTMI_REG44

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Address: Operational Base + offset (0x0044)
PHYREG44

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x0	USB RX Clock phase configure, 3'b000 represent the earliest phase, and 11 the latest, single step delay is 256ps.
4:2	RW	0x2	
1:0	RW	0x0	Reserved, keep default value.

USB3PHYUTMI_REG48
Address: Operational Base + offset (0x0048)
PHYREG48

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	
3	RW	0x1	
2:0	RW	0x6	USB TX Clock phase configure, 3'b000 represent the earliest phase, and 3'b111 the latest, single step delay is 256ps.

USB3PHYUTMI_REG4C
Address: Operational Base + offset (0x004c)
PHYREG4C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x40	

USB3PHYUTMI_REG50
Address: Operational Base + offset (0x0050)
PHYREG50

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x03	

USB3PHYUTMI_REG54
Address: Operational Base + offset (0x0054)
PHYREG54

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x70	

USB3PHYUTMI_REG58
Address: Operational Base + offset (0x0058)
PHYREG58

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x55	Field0000 Abstract Field0000 Description

USB3PHYUTMI_REG5C
Address: Operational Base + offset (0x005c)
PHYREG5C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x01	

USB3PHYUTMI_REG60
Address: Operational Base + offset (0x0060)
PHYREG60

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x0	Hs driver current compensation voltage reference. This is used to adjust the HS current by adjusting the reference voltage that tunes it. Useful to compensate for package and board level parasitic. Setting the compensation voltage lower, sets the current lower, the opposite , sets the current larger 3'b000: 200mV (default) 3'b001: 187.5mV 3'b010: 225mV 3'b011: 237.5mV 3'b100: 250mV 3'b101: 212.5mV 3'b110: 162.5mV 3'b111: 162.5mV
4:2	RW	0x0	This is used to adjust the 45 O HS termination by adjusting the reference voltage that tunes it. Useful to compensate for package and board level parasitic. Setting the compensation voltage lower, sets the ODT impedance lower. 3'b000: 400mV (default) 3'b001: 362.5mV 3'b010: 350mV 3'b011: 387.5mV 3'b100: 412.5mV 3'b101: 425mV 3'b110: 437.5mV 3'b111: 450mV
1:0	RW	0x3	Host disconnect detection trigger point configure, only used in host mode

USB3PHYUTMI_REG64
Address: Operational Base + offset (0x0064)

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PHYREG64

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	With bit0 of REG68 as host disconnect trigger point configure, only used in host mode.
6:5	RW	0x3	
4:3	RW	0x0	RX Squelch trigger point configure 0000:112.5 mV 1001:162.5 mV 1011:175 mV 1100:150 mV 1110:125 mV 1111:200 mV These register combinations can adjust the RX squelch trigger point under different signal amplitude conditions, usually the default value is suitable for all the condition. lower trigger point will reduced the receiving trigger point.
2:0	RW	0x0	Compensation voltage reference tuning 000:400mV 001:362 mV 110:437 mV 111:450 mV Bias current generator reference, higher reference voltage will increase the PHY bias current

USB3PHYUTMI_REG68

Address: Operational Base + offset (0x0068)

PHYREG68

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:1	RW	0x7c	Debug.
0	RW	0x0	See REG64 bit 7.

USB3PHYUTMI_REG6C

Address: Operational Base + offset (0x006c)

PHYREG6C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x12	

USB3PHYUTMI_REG70

Address: Operational Base + offset (0x0070)

PHYREG70

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	Enable Bias current generator, active high

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Bit	Attr	Reset Value	Description
6:3	RW	0x6	Debug
2	RW	0x0	Enable single-ended disconnection detection, active high, only used in HOST
1:0	RW	0x0	

USB3PHYUTMI_REG74
Address: Operational Base + offset (0x0074)
PHYREG74

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x70	

USB3PHYUTMI_REG78
Address: Operational Base + offset (0x0078)
PHYREG78

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:5	RW	0x0	
4:3	RW	0x0	OTG session reference tuning
2:0	RW	0x0	

USB3PHYUTMI_REG7C
Address: Operational Base + offset (0x007c)
PHYREG7C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	

USB3PHYUTMI_REG80
Address: Operational Base + offset (0x0080)
PHYREG80

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xf8	

USB3PHYUTMI_REG84
Address: Operational Base + offset (0x0084)
PHYREG84

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x01	

USB3PHYUTMI_REG88
Address: Operational Base + offset (0x0088)

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PHYREG88

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	

USB3PHYUTMI_REG8C
Address: Operational Base + offset (0x008c)
PHYREG8C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	

USB3PHYUTMI_REG90
Address: Operational Base + offset (0x0090)
PHYREG90

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x55	

USB3PHYUTMI_REG94
Address: Operational Base + offset (0x0094)
PHYREG94

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x55	

USB3PHYUTMI_REG98
Address: Operational Base + offset (0x0098)
PHYREG98

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x55	

USB3PHYUTMI_REG9C
Address: Operational Base + offset (0x009c)
PHYREG9C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x55	

USB3PHYUTMI_REGA0
Address: Operational Base + offset (0x00a0)
PHYREGA0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RW	0x1	core clock phase selection
6:4	RW	0x7	Debug
3:2	RW	0x0	BG output voltage reference adjust, debug interface, nothing to do with the working functions. Keeping the default value is strongly recommended.
1:0	RW	0x0	

USB3PHYUTMI_REGA4
Address: Operational Base + offset (0x00a4)
PHYREGA4

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	bypass chirp
6:3	RW	0x1	
2	RW	0x1	bypass ODT compensation refresh function, keeping the default value is greatly appreciate
1:0	RW	0x1	Debug

USB3PHYUTMI_REGA8
Address: Operational Base + offset (0x00a8)
PHYREGA8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	Debug

USB3PHYPIPE_REG20
Address: Operational Base + offset (0x0020)
PHYREG20

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x2	See REG28, bit9~8 of RX CDR frequency multiplier factor, denoted as P. Receiver data rate is calculated as (2*Fref*P)/Q, where Q is no less than 1.
5	RO	0x0	reserved
4:0	RW	0x06	TX PLL Div, denoted as N. Transmitter data rate is calculated as (2*Fref*M)/N, where N is no less than 1.

USB3PHYPIPE_REG28
Address: Operational Base + offset (0x0028)
PHYREG28

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x71	Field0000 Abstract See REG20, bit7~0 of RX CDR frequency multiplier factor, denoted as P.

USB3PHYPIPE_REG30
Address: Operational Base + offset (0x0030)
PHYREG30

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x06	RX CDR Div, denoted as Q.

USB3PHYPIPE_REG44
Address: Operational Base + offset (0x0044)
PHYREG44

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Band-gap power down control, only will be asserted in whole chip totally power down mode to save power, in this mode phy will not respond properly, active high 1'b0: normal operation mode 1'b1: power down mode
3:0	RO	0x0	reserved

USB3PHYPIPE_REG58
Address: Operational Base + offset (0x0058)
PHYREG58

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	RX CTLE enable, active low 1'b1: turn off RX equalizer 1'b0: normal operation mode with CTLE on
1:0	RO	0x0	reserved

USB3PHYPIPE_REG64
Address: Operational Base + offset (0x0064)
PHYREG64

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x4	RX equalizer tail current control.
3:0	RO	0x0	reserved

USB3PHYPIPE_REG68
Address: Operational Base + offset (0x0068)
PHYREG68

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x7	RX equalizer low frequency gain control.
3:0	RO	0x0	reserved

USB3PHYPIPE_REG88
Address: Operational Base + offset (0x0088)
PHYREG88

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	Tx driver pre-emphasize enable, active high 1'b0: pre-emphasize off 1'b1: pre-emphasize on

USB3PHYPIPE_REGD0
Address: Operational Base + offset (0x00d0)
PHYREGD0

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	TX detect RX bypass bit, debug usage only, active high. 1'b1: force assertion of TX detect RX, ignoring the result of detection 1'b0: normal mode, TX detecting RX results will be changed according to detection module.
3:0	RO	0x0	reserved

USB3PHYPIPE_REGE0
Address: Operational Base + offset (0x00e0)
PHYREGE0

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Serdes life clock power down control, active high 1'b0: normal operation mode 1'b1: power down mode, internal life clock stop toggling
3:0	RO	0x0	reserved

USB3PHYPIPE_REG118
Address: Operational Base + offset (0x0118)
PHYREG118

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x2	See REG11C. Bit11~8 of TX PLL frequency multiplier factor, denoted as M.

USB3PHYPIPE_REG11C

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Address: Operational Base + offset (0x011c)
PHYREG11C

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x71	See REG118. Bit7~0 of TX PLL frequency multiplier factor, denoted as M.

USB3PHYPIPE_REG12C
Address: Operational Base + offset (0x012c)
PHYREG12C

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	1'b0: PLL control by pipe_pd 1'b1: PLL always running

USB3PHYPIPE_REG180
Address: Operational Base + offset (0x0180)
PHYREG180

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:2	RW	0x0	Input reference voltage of unit gain buffer to create the bias current used in TX driver and RX EQ. Higher voltage reference will increase the bias current for the PHY. 2'b00: 200mv; 2'b01: 175mv; 2'b10: 225mv; 2'b11:250mv
1:0	RO	0x0	reserved

USB3PHYPIPE_REG188
Address: Operational Base + offset (0x0188)
PHYREG188

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	RX electrical idle detection threshold tuning reference, positive input selection between vref1 and vref2 defined in REG18C, output was define as vrefp. The final RX electrical idle detection threshold will be defined as |vrefp-vrefn|. 1'b0: vref2; 1'b1: vref1.
0	RW	0x1	RX electrical idle detection threshold tuning reference, negative input selection between vref3 and vref4 defined in REG18C, output was define as vrefn. The final RX electrical idle detection threshold will be defined as |vrefp-vrefn|. 1'b0: vref4; 1'b1: vref3.

USB3PHYPIPE_REG18C
Address: Operational Base + offset (0x018c)
PHYREG18C

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	Field0003 Abstract RX electrical idle detection threshold tuning reference, negative input of the threshold comparator, define as vref4. 2'b00: 600mv; 2'b01: 575mv; 2'b10: 625mv; 2'b11: 650mv
5:4	RW	0x0	RX electrical idle detection threshold tuning reference, negative input of the threshold comparator, define as vref3. 2'b00: 725mv; 2'b01: 675mv; 2'b10: 700mv; 2'b11: 750mv
3:2	RW	0x0	RX electrical idle detection threshold tuning reference, positive input of the threshold comparator, define as vref2. 2'b00: 750mv; 2'b01: 700mv; 2'b10: 725mv; 2'b11: 775mv
1:0	RW	0x0	RX electrical idle detection threshold tuning reference, positive input of the threshold comparator, define as vref1. 2'b00: 850mv; 2'b01: 800mv; 2'b10: 825mv; 2'b11: 875mv

USB3PHYPIPE_REG190
Address: Operational Base + offset (0x0190)
PHYREG190

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:2	RW	0x0	TX detect RX reference tuning bits, low reference voltage will increase the detection sensitivity. 2'b00: 800mv 2'b01: 750mv 2'b10: 775mv 2'b11: 825mv
1:0	RW	0x0	TX detect RX shutting reference tuning bits, higher reference voltage will induce high voltage puls on TXP/TXN to discover the RX. 2'b00: 850mv 2'b01: 800mv 2'b10: 825mv 2'b11: 875mv

USB3PHYPIPE_REG1A8
Address: Operational Base + offset (0x01a8)
PHYREG1A8

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	TX LDO reference voltage selection 1'b0: reference from analog power rail 1'b1: reference coming from band-gap

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Bit	Attr	Reset Value	Description
6:5	RW	0x0	TX DLO voltage tuning bits, only available when REG1A8[7] is set to 1. 2'b00: 1.0v from bang-gap 2'b01: 0.95v from bang-gap 2'b10: 1.05v from bang-gap 2'b11: 1.1v from bang-gap
4:3	RO	0x0	reserved
2	RW	0x1	PCIe device mode enable, active high
1	RW	0x0	PCIe PLL and CDR reference clock source select. 1'b0: reference clock from internal single ended in stage, sinking clock from pllrefclk 1'b1: reference clock from external differential pair.
0	RW	0x0	PCIe Host 100MHz differential clock output at REFCLKP/REFCLKN pins enable, active high, only available when PCIe was in Host mode and REG1A8[2] is set to 0. 1'b0: disable 100MHz clock output 1'b1: enable 100MHz clock output

USB3PHYPIPE_REG1AC
Address: Operational Base + offset (0x01ac)
PHYREG1AC

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	Serdes TX driver output common mode voltage tuning bits, these bit will tune the common mode voltage of TXP/TXN differential signals. 2'b00: 800mv 2'b01: 750mv 2'b10: 950mv 2'b11: 1100mv
3:0	RO	0x0	reserved

USB3PHYPIPE_REG1B8
Address: Operational Base + offset (0x01b8)
PHYREG1B8

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	TX driver pre-emphasize strength tuning bits with weight, 2'b11 represents the strongest and 2'b00 the weakest. Only available when REG88[4] was set to 1.

22.5 Interface Description

Table 24-1 USB3.0 PHY Interface Description

Module Pin	Direction	Pad Name	Descriptions

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Module Pin	Direction	Pad Name	Descriptions
USB0ID	I/O	IO_USB3_ID	
USB0PN	I/O	IO_USB3_PN	
USB0PP	I/O	IO_USB3_PP	
VBUS	I/O	IO_USB3_VBUS	
TXP	I/O	IO_USB3_TXP	
TXN	I/O	IO_USB3_TXN	
RXP	I/O	IO_USB3_RXP	
RXN	I/O	IO_USB3_RXN	
RBIAS	I/O	IO_USB3_RBIAS	

22.6 Application Notes

22.6.1 Some Special Settings before Initialization

Set USB3.0 OTG controller AXI master secure setting.

Clear USB2.0 only mode setting (bit 3 of register GRF_USB3PHY0/1_CON0 in Chapter GRF)

Set PHYIF to 1 to use 16-bit UTMI+ interface (see register GUSB2PHYCFG0)

Clear ENBLSLPM to 0 to disable sleep and l1 suspend (see register GUSB2PHYCFG0)

Clear U2_FREECLK_EXITSTS to 0 (see register GUSB2PHYCFG0)

Clear DEV_FORCE_20_CLK_FOR_30_CLK to 0 (see register GUCTL1)

Clear DELAYP1TRANS to 0 (see register GUSB3PIPECTL0)

Hold USB3.0 controller in resetting during USB3PHY initialization

22.6.2 OTG Programming Model

When detect ID change event (see USB3PHY_GRF) or VBUS change event (see USB3PHY_GRF) after disconnect, it means a USB3/2 OTG A device or B device may connect, then check status of ID(see USB3PHY_GRF). If ID==0, it will work as A device, then follow host programming flow; if ID==1, it will work as B device, then it follow device programming flow.
Note: USB3.0 OTG doesn't support host/device mode swapping through HNP and RSP.

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Chapter 25 SPDIF transmitter

23.1 Overview

The SPDIF transmitter is a self-clocking, serial and unidirectional interface for the interconnection of digital audio equipment in consumer and professional applications which uses linear PCM coded audio samples.
When used in professional application, the interface is primarily intended to carry monophonic or stereophonic programmes at a 48 kHz sampling frequency with a resolution of up to 24bits per sample. It may alternatively be used to carry signals sampled at 32 kHz or 44.1 kHz.
When used in consumer application, the interface is primarily intended to carry stereophonic programmes with a resolution of up to 20 bits per sample, an extension to 24 bits per sample being possible.
When used for other purposes, the interface is primarily intended to carry audio data coded other than linear PCM coded audio samples. Provision is also made to allow the interface to carry data related to computer software or signals coded using non-linear PCM. The maximum sample frequency can be up to 192 kHz for the non-linear PCM mode.
In all cases, the clock references and auxiliary information are transmitted along with the programme.

- Supports one internal 32-bit wide and 32-location deep sample data buffer
- Supports two 16-bit audio data store together in one 32-bit wide location
- Supports AHB bus interface
- Supports biphase format stereo audio data output
- Supports DMA handshake interface and configurable DMA water level
- Supports sample data buffer empty, block terminate and user data interrupt
- Supports combine interrupt output
- Supports 16 to 31 bit audio data left or right justified in 32-bit wide sample data buffer
- Support 16, 20, 24 bits audio data transfer in linear PCM mode
- Support non-linear PCM transfer

23.2 Block Diagram

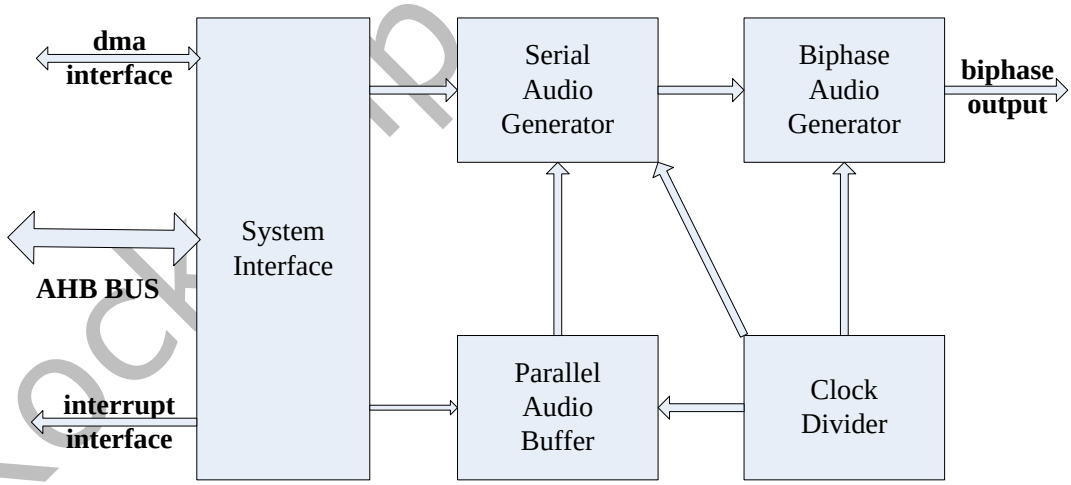

Fig.25-1 SPDIF transmitter Block Diagram

System Interface
The system interface implements the AHB slave operation. It contains not only control registers of transmitters and receiver inside but also interrupt and DMA handshake interface.
Clock Divider
The Clock Divider implements clock generation function. The input source clock to the module is MCLK. By the divider of the module, the clock divider generates work clock for digital audio data transformation.
Parallel Audio Buffer
The Parallel Audio Buffer is the buffer to store transmitted audio data. The size of the FIFO is 32bits x 32.

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Serial Audio Converter
The Serial Audio Converter reads parallel audio data from the Parallel Audio Buffer and converts it to serial audio data.
Biphase Audio Generator
The Biphase Audio Generator reads serial audio data from the Serial Audio Converter and generates biphase audio data based on IEC-60958 standard.

23.3 Function description

23.3.1 Frame Format

A frame is uniquely composed of two sub-frames. For linear coded audio applications, the rate of transmission of frames corresponds exactly to the source sampling frequency.
In the 2-channel operation mode, the samples taken from both channels are transmitted by time multiplexing in consecutive sub-frames. The first sub-frame(left channel in stereophonic operation and primary channel in monophonic operation) normally use preamble M. However, the preamble is changed to preamble B once every 192 frame to identify the start of the block structure used to organize the channel status information. The second sub-frame (right in stereophonic operation and secondary channel in monophonic operation) always use preamble W.

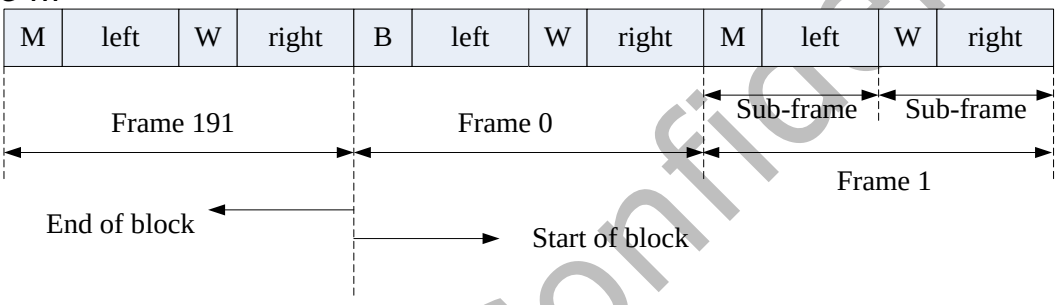

Fig.25-2 SPDIF Frame Format

In the single channel operation mode in a professional application, the frame format is the same as in the 2-channel mode. Data is carried only in the first sub-frame and may be duplicated in the second sub-frame. If the second sub-frame is not carrying duplicate data, then time slot 28 (validity flag) shall be set to logical '1' (not valid).

23.3.2 Sub-frame Format

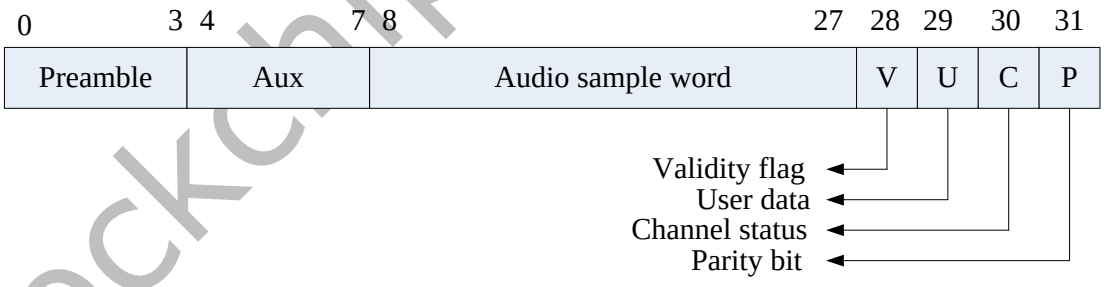

Fig.25-3 SPDIF Sub-frame Format

Each sub-frame is divided into 32 time slots, numbered from 0 to 31. Time slot 0 to 3 carries one of the three permitted preambles. Time slot 4 to 27 carry the audio sample word in linear 2's complement representation. The MSB is carried by time slot 27. When a 24-bit coding range is used, the LSB is in time slot 4. When a 20-bit coding range is used, time slot 8 to 27 carry the audio sample word with the LSB in time slot 8.Time slot 4 to 7 may be used for other application. Under these circumstances, the bits in the time slot 4 to 7 are designated auxiliary sample bits.
If the source provides fewer bits than the interface allows (either 24 or 20), the unused LSBs are set to a logical '0'. For a non-linear PCM audio application or a data application the main data field may carry any other information. Time slot 28 carries the validity flag associated with the main data field. Time slot 29 carries 1 bit of the user data associated with the audio channel transmitted in the same sub-frame. Time slot 30 carries one bit of the channel status words associated with the main data field channel transmitted in the same sub-frame. Time

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slot 31 carries a parity bit such that time slots 4 to 31 inclusive carries an even number of ones and an even number of zeros.

23.3.3 Channel Coding

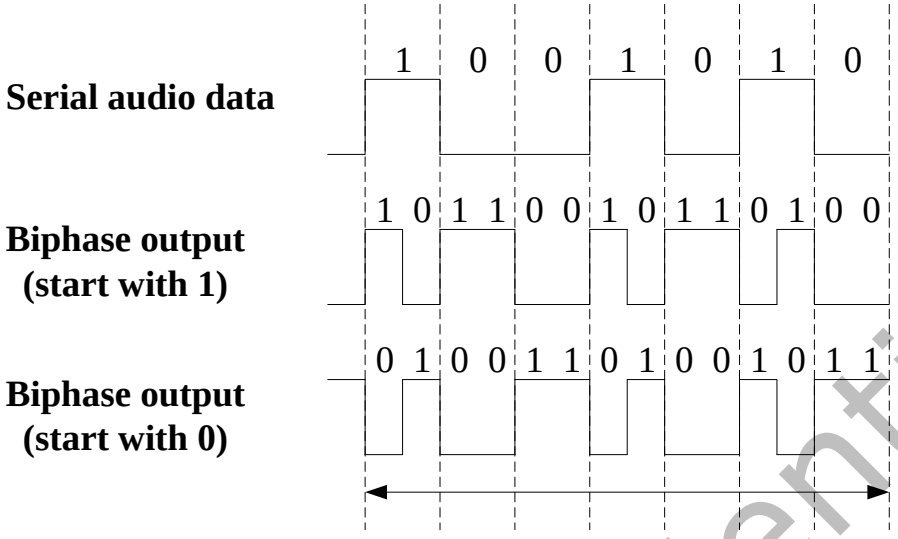

Fig.25-4 SPDIF Channel Coding

To minimize the direct current component on the transmission line, to facilitate clock recovery from the data stream and to make the interface insensitive to the polarity of connections, time slots 4 to 31 are encoded in biphase-mark.

Each bit to be transmitted is represented by a symbol comprising two consecutive binary states. The first state of a symbol is always different from the second state of the previous symbol. The second state of the symbol is identical to the first if the bit to be transmitted is logical '0'.However, it is different from the first if the bit is logical '1'.

23.3.4 Preamble

Preambles are specific patterns providing synchronization and identification of the sub-frames and blocks.

To achieve synchronization within one sampling period and to make this process completely reliable, these patterns violate the biphase-mark code rules, thereby avoiding the possibility of data imitating the preambles.

A set of three preambles is used. These preambles are transmitted in the time allocated to four time slots (time slots 0 to 3) and are represented by eight successive states. The first state of the preamble is always different from the second state of the previous symbol.

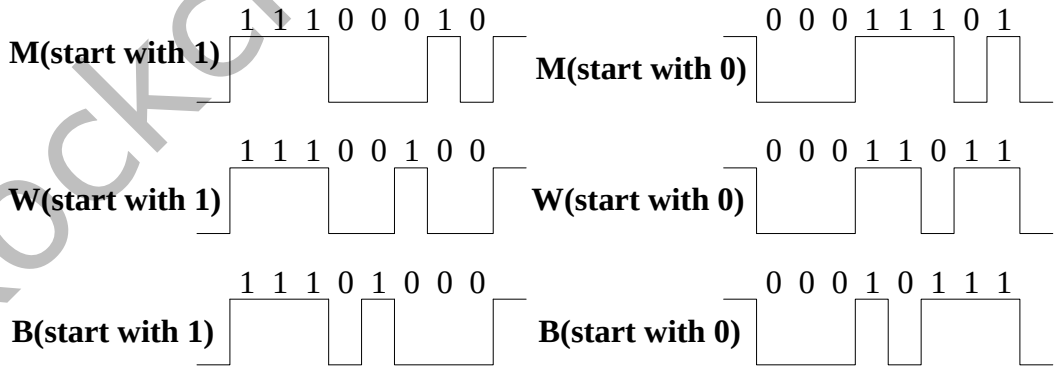

Fig.25-5 SPDIF Preamble

Like biphase code, these preambles are dc free and provide clock recovery. They differ in at least two states from any valid biphase sequence.

23.3.5 NON-LINEAR PCM ENCODED SOURCE(IEC 61937)

The non-linear PCM encoded audio bitstream is transferred using the basic 16-bit data area of the IEC 60958subframes, i.e. in time slots 12 to 27. Each IEC 60958 frame transfers 32-bit of the non-PCM data in consumer application mode.

If the SPDIF bitstream conveys linear PCM audio, the symbol frequency is 64 times the PCM sampling frequency(32 time slots per PCM sample times two channels). If a non-linear PCM

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encoded audio bitstream is conveyed by the interface, the symbol frequency is 64 times the sampling rate of the encoded audio within that bitstream. But in the case where a non-linear PCM encoded audio bitstream is conveyed by the interface containing audio with low sampling frequency, the symbol frequency is 128 times the sampling rate of the encoded audio within that bitstream.
Each data burst contains a burst-preamble consisting of four 16-bit words (Pa, Pb, Pc, Pd), followed by the burst payload which contains data of an encoded audio frame.
The burst-preamble consists of four mandatory fields. Pa and Pb represent a synchronization word. Pc gives information about the type of data and some information/control for the receiver. Pd gives the length of the burst payload, the number of bits or number of bytes according to data-type.
The four preamble words are contained in two sequential SPDIF frames. The frame beginning the data-burst contains preamble word Pa in subframe 0 and Pb in subframe 1. The next frame contains Pc in subframe 0 and Pd in subframe 1. When placed into a SPDIF subframe, the MSB of a 16-bit burst-preamble is placed into timeslot 27 and the LSB is placed into time slot 12.

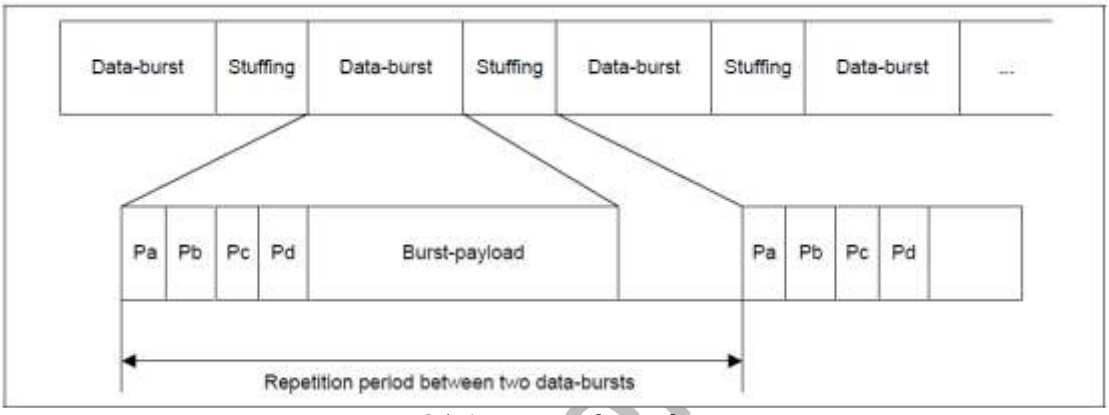

Fig.25-6 Format of Data-burst

23.4 Register description

23.4.1 Register Summary

Name	Offset	Size	Reset Value	Description
SPDIF_CFGR	0x0000	W	0x00000000	Transfer Configuration Register
SPDIF_SDBLR	0x0004	W	0x00000000	Sample Date Buffer Level Register
SPDIF_DMACR	0x0008	W	0x00000000	DMA Control Register
SPDIF_INTCR	0x000c	W	0x00000000	Interrupt Control Register
SPDIF_INTSR	0x0010	W	0x00000000	Interrupt Status Register
SPDIF_XFER	0x0018	W	0x00000000	Transfer Start Register
SPDIF_SMPDR	0x0020	W	0x00000000	Sample Data Register
SPDIF_VLDFRn	0x0060	W	0x00000000	Validity Flag Register n
SPDIF_USRDRn	0x0090	W	0x00000000	User Data Register n
SPDIF_CHNSRn	0x00c0	W	0x00000000	Channel Status Register n
SPDIF_BURTSINFO	0x0100	W	0x00000000	Channel Burst Info Register
SPDIF_REPETTION	0x0104	W	0x00000000	Channel Repetition Register
SPDIF_BURTSINFO_SHD	0x0108	W	0x00000000	Shadow Channel Burst Info Register
SPDIF_REPETTION_SHD	0x010c	W	0x00000000	Shadow Channel Repetition Register
SPDIF_USRDR_SHDn	0x0190	W	0x00000000	Shadow User Data Register n

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

23.4.2 Detail Register Description

SPDIF_CFGR
Address: Operational Base + offset (0x0000)

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Transfer Configuration Register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RW	0x00	MCD mclk divider Fmclk/Fsdo This parameter can be calculated by Fmclk/(Fs*128). Fs=the sample frequency be wanted
15:9	RO	0x0	reserved
8	RW	0x0	PCMTYPE PCM type 0: linear PCM 1: non-linear PCM
7	WO	0x0	CLR mclk domain logic clear Write 1 to clear mclk domain logic. Read return zero.
6	RW	0x0	CSE Channel status enable 0: disable 1: enable The bit should be set to 1 when the channel conveys non-linear PCM
5	RW	0x0	UDE User data enable 0: disable 1: enable
4	RW	0x0	VFE Validity flag enable 0: disable 1: enable
3	RW	0x0	ADJ audio data justified 0: Right justified 1: Left justified
2	RW	0x0	HWT Halfword word transform enable 0: disable 1: enable It is valid only when the valid data width is 16bit.
1:0	RW	0x0	VDW Valid data width 00: 16bit 01: 20bit 10: 24bit 11: reserved The valid data width is 16bit only for non-linear PCM

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SPDIF_SDBLR
Address: Operational Base + offset (0x0004)
Sample Date Buffer Level Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	SDBLR Sample Date Buffer Level Register Contains the number of valid data entries in the sample data buffer.

SPDIF_DMACR
Address: Operational Base + offset (0x0008)
DMA Control Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	TDE Transmit DMA Enable 0: Transmit DMA disabled 1: Transmit DMA enabled
4:0	RW	0x00	TDL Transmit Data Level This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the Sample Date Buffer is equal to or below this field value

SPDIF_INTCR
Address: Operational Base + offset (0x000c)
Interrupt Control Register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	W1C	0x0	UDTIC User Data Interrupt Clear Write '1' to clear the user data interrupt.
16	W1C	0x0	BTTIC Block/Data burst transfer finish interrupt clear Write 1 to clear the interrupt.
15:10	RO	0x0	reserved
9:5	RW	0x00	SDBT Sample Date Buffer Threshold Sample Date Buffer Threshold for empty interrupt
4	RW	0x0	SDBEIE Sample Date Buffer empty interrupt enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
3	RW	0x0	BTTIE Block transfer/repetition period end interrupt enable When enabled, an interrupt will be asserted when the block transfer is finished if the channel conveys linear PCM or when the repetition period is reached if the channel conveys non-linear PCM. 0: disable 1: enable
2	RW	0x0	UDTIE User Data Interrupt 0: disable 1: enable If enabled, an interrupt will be asserted when the content of the user data register is fed into the corresponding shadow register
1:0	RO	0x0	reserved

SPDIF_INTSR
Address: Operational Base + offset (0x0010)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	SDBEIS Sample Date Buffer empty interrupt status 0: inactive 1: active
3	RW	0x0	BTTIS Block/Data burst transfer interrupt status 0: inactive 1: active
2	RW	0x0	UDTIS User Data Interrupt Status 0: inactive 1: active
1:0	RO	0x0	reserved

SPDIF_XFER
Address: Operational Base + offset (0x0018)
Transfer Start Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	XFER Transfer Start Register Transfer Start Register

SPDIF_SMPDR
Address: Operational Base + offset (0x0020)

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Sample Data Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	SMPDR Sample Data Register Sample Data Register

SPDIF_VLDFRn

Address: Operational Base + offset (0x0060)

Validity Flag Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	VLDFR_SUB_1 Validity Flag Subframe 1 Validity Flag Register 0
15:0	RW	0x0000	VLDFR_SUB_0 Validity Flag Subframe 0 Validity Flag for Subframe 0

SPDIF_USRDRn

Address: Operational Base + offset (0x0090)

User Data Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	USR_SUB_1 User Data Subframe 1 User Data Bit for Subframe 1
15:0	RW	0x0000	USR_SUB_0 User Data Subframe 0 User Data Bit for Subframe 0

SPDIF_CHNSRn

Address: Operational Base + offset (0x00c0)

Channel Status Register n

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	CHNSR_SUB_1 Channel Status Subframe 1 Channel Status Bit for Subframe 1
15:0	RW	0x0000	CHNSR_SUB_0 Channel Status Subframe 0 Channel Status Bit for Subframe 0

SPDIF_BURTSINFO

Address: Operational Base + offset (0x00d0)

Channel Burst Info Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	PD pd Preamble Pd for non-linear pcm, indicating the length of burst payload in unit of bytes or bits.
15:13	RW	0x0	BSNUM Bitstream Number This field indicates the bitstream number. Usually the bitstream number is 0.
12:8	RW	0x00	DATAINFO Data-type-dependent info This field gives the data-type-dependent info
7	RW	0x0	ERRFLAG Error Flag 0: indicates a valid burst-payload 1: indicates that the burst-payload may contain errors

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Bit	Attr	Reset Value	Description
6:0	RW	0x00	DATATYPE Data type 0000000: null data 0000001: AC-3 data 0000011: Pause data 0000100: MPEG-1 layer 1 data 0000101: MPEG-1 layer 2 or 3 data or MPEG-2 without extension 0000110: MPEG-2 data with extension 0000111: MPEG-2 AAC 0001000: MPEG-2, layer-1 low sampling frequency 0001001: MPEG-2, layer-2 low sampling frequency 0001010: MPEG-2, layer-3 low sampling frequency 0001011: DTS type I 0001100: DTS type II 0001101: DTS type III 0001110: ATRAC 0001111: ATRAC 2/3 0010000: ATRAC-X 0010001: DTS type IV 0010010: WMA professional type I 0110010: WMA professional type II 1010010: WMA professional type III 1110010: WMA professional type IV 0010011: MPEG-2 AAC low sampling frequency 0110011: MPEG-2 AAC low sampling frequency 1010011: MPEG-2 AAC low sampling frequency 1110011: MPEG-2 AAC low sampling frequency 0010100: MPEG-4 AAC 0110100: MPEG-4 AAC 1010100: MPEG-4 AAC 1110100: MPEG-4 AAC 0010101: Enhanced AC-3 0010110: MAT others: reserved

SPDIF_REPETTION

Address: Operational Base + offset (0x0104)

Channel Repetition Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	REPETTION Repetition This define the repetition period when the channel conveys non-linear PCM

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SPDIF_BURTSINFO_SHD
Address: Operational Base + offset (0x0108)
Shadow Channel Burst Info Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	PD pd Preamble Pd for non-linear pcm, indicating the length of burst payload in unit of bytes or bits.
15:13	RO	0x0	BSNUM Bitstream Number This field indicates the bitstream number. Usually the birstream number is 0.
12:8	RO	0x00	DATAINFO Data-type-dependent info This field gives the data-type-dependent info
7	RO	0x0	ERRFLAG Error Flag 0: indicates a valid burst-payload 1: indicates that the burst-payload may contain errors

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Bit	Attr	Reset Value	Description
6:0	RO	0x00	DATATYPE Data type 0000000: null data 0000001: AC-3 data 0000011: Pause data 0000100: MPEG-1 layer 1 data 0000101: MPEG-1 layer 2 or 3 data or MPEG-2 without extension 0000110: MPEG-2 data with extension 0000111: MPEG-2 AAC 0001000: MPEG-2, layer-1 low sampling frequency 0001001: MPEG-2, layer-2 low sampling frequency 0001010: MPEG-2, layer-3 low sampling frequency 0001011: DTS type I 0001100: DTS type II 0001101: DTS type III 0001110: ATRAC 0001111: ATRAC 2/3 0010000: ATRAC-X 0010001: DTS type IV 0010010: WMA professional type I 0110010: WMA professional type II 1010010: WMA professional type III 1110010: WMA professional type IV 0010011: MPEG-2 AAC low sampling frequency 0110011: MPEG-2 AAC low sampling frequency 1010011: MPEG-2 AAC low sampling frequency 1110011: MPEG-2 AAC low sampling frequency 0010100: MPEG-4 AAC 0110100: MPEG-4 AAC 1010100: MPEG-4 AAC 1110100: MPEG-4 AAC 0010101: Enhanced AC-3 0010110: MAT others: reserved

SPDIF_REPETTION_SHD
Address: Operational Base + offset (0x010c)
Shadow Channel Repetition Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	REPETTION Repetition This register provides the repetition of the bitstream when channel conveys non-linear PCM. In the design, it is define the length bwtween Pa of the two consecutive data-burst. For the same audio format, the definition is different. Please convert the actual repetition in order to comply with the design.

SPDIF_USRDR_SHDn
Address: Operational Base + offset (0x0190)
Shadow User Data Register n

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	USR_SUB_1 User Data Subframe 1 User Data Bit for Subframe 1
15:0	RO	0x0000	USR_SUB_0 User Data Subframe 0 User Data Bit for Subframe 0

23.5 Interface description

Table 25-1 SPDIF Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
spdif_8ch_sdo	O	IO_SPDIFtx_GPIO3d3	GRF_GPIO3D_IOMUX[7:6]=2'b01
spdif_8ch_sdo	O	IO_TESTCLKout1_SPDIF1tx_GPIO3d7	GRF_GPIO3D_IOMUX[15:14]=2'b10

The output of SPDIF module which signals as spdif_8ch_sdo is also connected to the audio interface of HDMI.

Table 25-2 Interface Between SPDIF And HDMI

Module Pin	Direction	Module Pin	Direction
mclk_spdif_8ch	O	ispdifclk	I
spdif_8ch_sdo	O	ispdifdata	I

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23.6 Application Notes

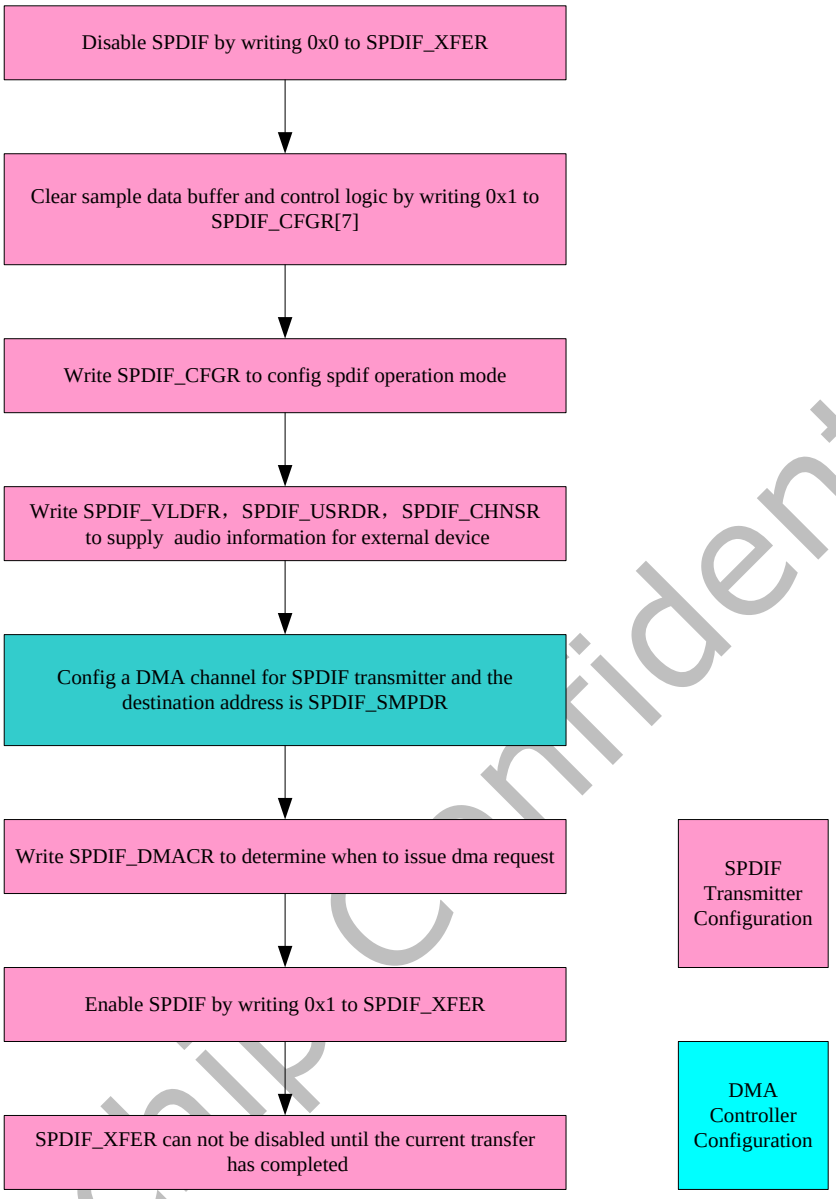

Fig.25-7 SPDIF transmitter operation flow chart

23.6.1 Channel Status Bit and Validity Flag Bit

Normally the channel status bits and validity flag bits are not necessarily updated frequently. If it is desired to change the channel status bits or validity flag, please write to the corresponding register after a block termination interrupt is asserted. The new value will take effect immediately.

23.6.2 User Data Bit

As the user data bits are updated frequently, the design takes use of the shadow register mechanism to store and convey the user data bit. When the SPDIF interface is disabled, the values of the shadow user data registers keeps the same with the corresponding user data registers. After the SPDIF starts, any change of the user data register will not go to the corresponding shadow user data registers until an user data interrupt is asserted. Therefore before the SPDIF transfer starts, prepare the first 384 user data bits by writing them to the SPDIF_USRDR registers. After the SPDIF transfer starts, writing the second 384 user data bits to the SPDIF_USRDR registers. Then wait for the assertion of user data interrupt. The second 384 user data bits goes to the shadow registers, and then third 384 user bits are written to SPDIF_USRDR.

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23.6.3 Burst Info and Repetition

The shadow register mechanism is also applied to the data of burst info and repetition as the user data. The difference is that the update of shadow register will be taken after assertion of the block termination interrupt.
It is important to note that the repetition defined in the design is a little different from the repetition defined in IEC-61957. The repetition is always defined as the length (measured in IEC-60958 frame) between Pa of two consecutive data-bursts. Therefore the user needs to calculate the new repetition value if the definition of the repetition is different for some audio formats such as AC-3.

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Chapter 26 Pulse Density Modulation Interface Controller

24.1 Overview

The Pulse Density Modulation Interface Controller (PDMC) is a PDM interface controller and decoder that support PDM format. It integrates a clock generator driving the PDM microphone and embeds filters which decimate the incoming bit stream to obtain most common audio rates.
PDMC supports the following features:

- Support one internal 32-bit wide and 128-location deep FIFOs for receiving audio data
- Support receive FIFO full, overflow interrupt and all interrupts can be masked
- Support configurable water level of receive FIFO full interrupt
- Support combined interrupt output
- Support AHB bus slave interface
- Support DMA handshaking interface and configurable DMA water level
- Support PDM master receive mode
- Support 4 paths. Each path is composed of two digital microphone channels, the PDMC can be used with four stereo or eight mono microphones. Each path is enabled or disabled independently
- Support 16 ~24 bit sample resolution
- Support sample rate:

  8khz,16khz,32kHz,64kHz,128khz,11.025khz,22.05khz,44.1khz,88.2khz,176.4khz,12khz,24khz,48khz,96khz,192khz
- Support two 16-bit audio data store together in one 32-bit wide location
- Support programmable data sampling sensibility (rising or falling edge)

24.2 Block Diagram

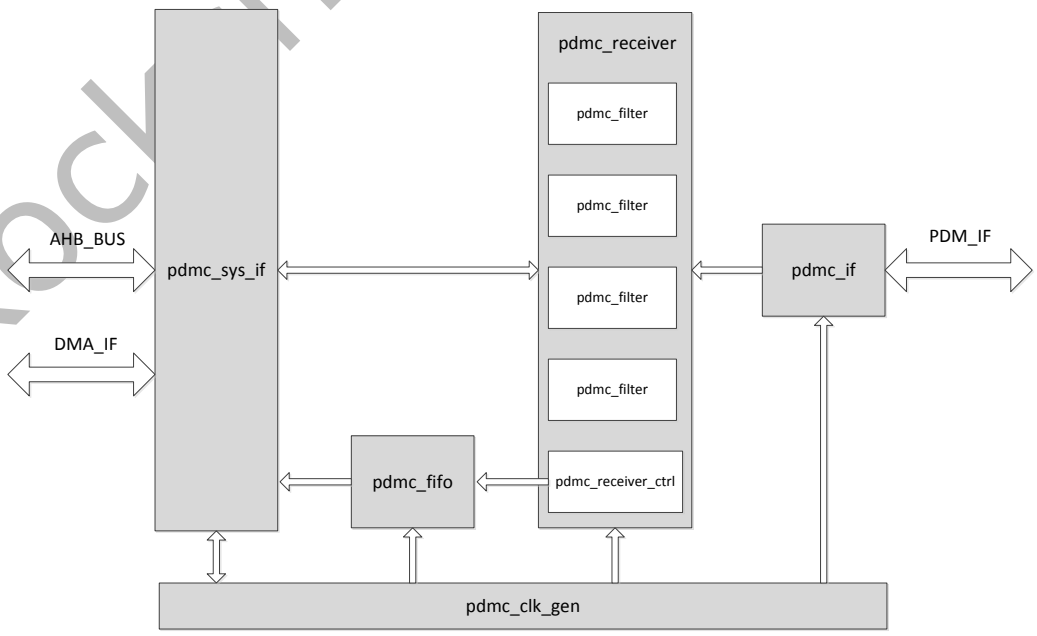

Fig.1- 1 PDMC Block Diagram

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System Interface

The system interface implements the APB slave operation. It contains not only control registers of receiver inside but also interrupt and DMA handshaking interface.

Clock Generator

The Clock Generator implements clock generation function. The input source clock to the module is MCLK, and by the divider of the module, the clock generator generates CLK_PDM to receiver.

Receiver

The receiver can act as a decimation filter of PDM. And export PCM format data.

Receive FIFO

The Receive FIFO is the buffer to store received audio data. The size of the FIFO is 32bits x 128.

PDM interface

The PDM interface implements PDM bit streams receive operation.

24.3 Function Description

24.3.1 AHB Interface

There is an AHB slave interface in PDMC. It is responsible for accessing registers and internal memories. The addresses of these registers and memories are listed in 29.4.1.

24.3.2 PDM Interface

The PDM interface is a 5-wire interface. The PDMC module can support up to four external stereo and eight digital microphones.

Fig.1-2 and Fig.1-3 show two cases of use of the PDMC, but all configurations are possible with stereo and mono digital microphones.

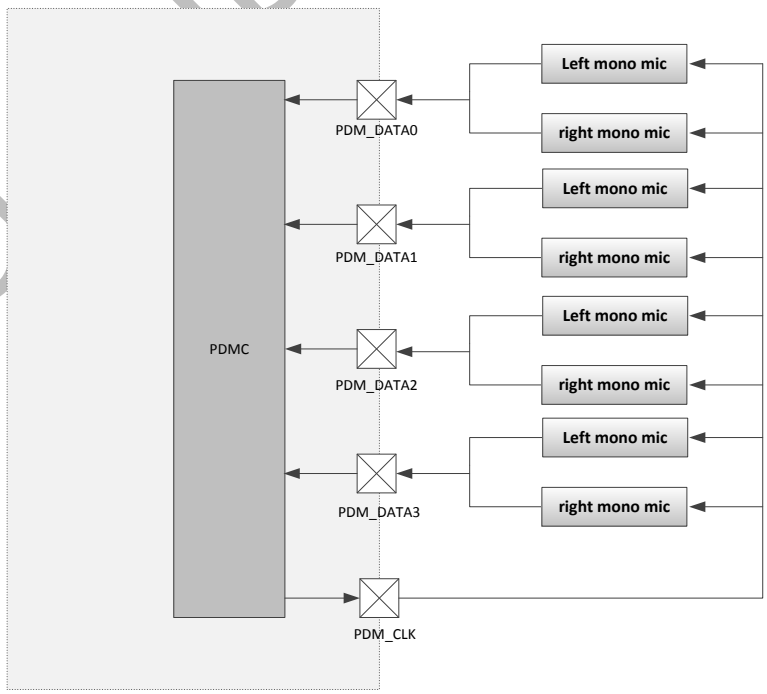

Fig.1- 2 PDMC with Eight Mono MIC

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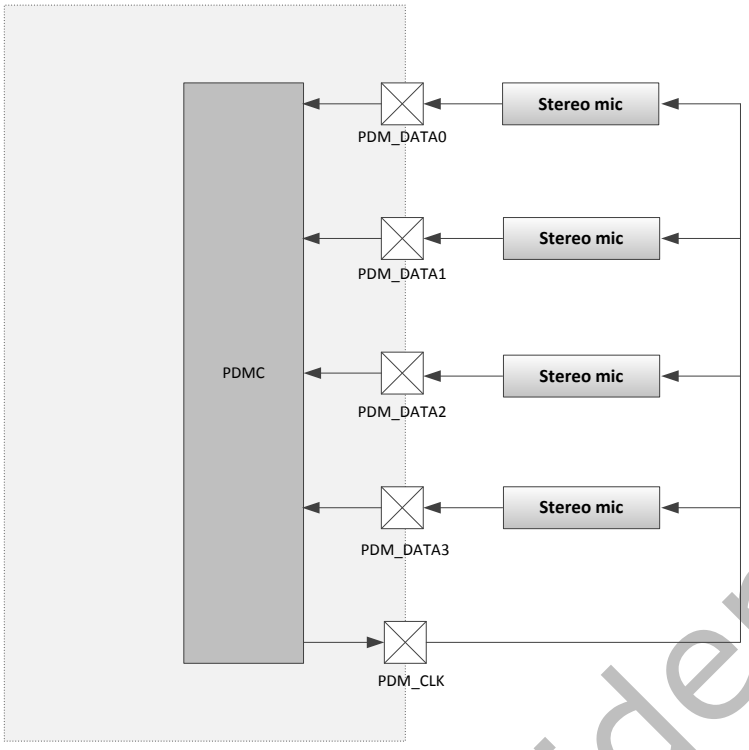

Fig.1- 3 PDMC with Four Stereo MIC

The PDM interface consists of a serial-data shift clock output (PDM_CLK) and a serial data input (PDM_DATA). The clock is fanned out to both digital mics, and both digital mics' data (left channel and right channel) outputs share a single signal line. To share a single line, the digital mics tristate their output during one phase of the clock(high or low part of cycle, depending on how they are configured via their L/R input).

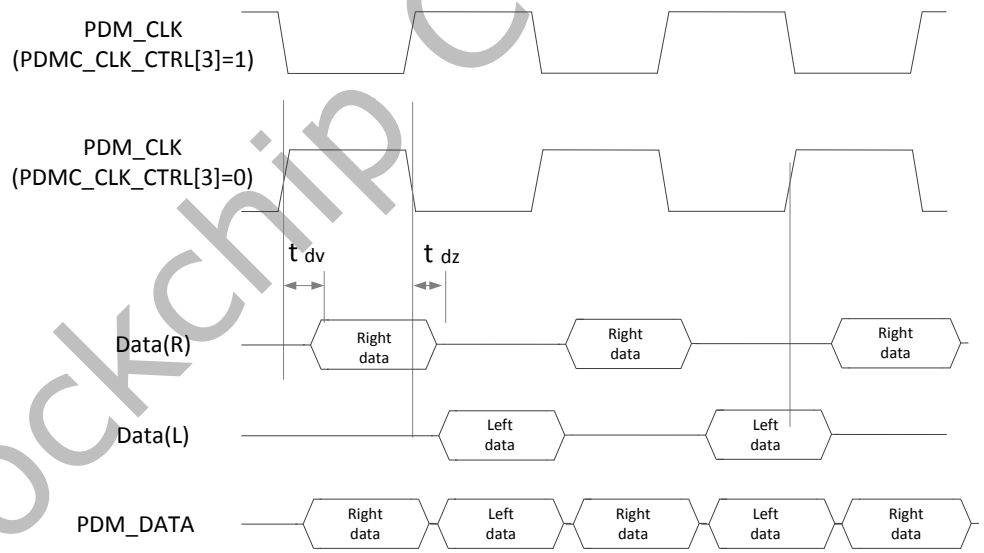

Fig.1- 4 PDMC interface diagram with external MIC

24.3.3 Digital Filter

The external PDMIC generates a PDM stream of bits and transfers it in one period or one half-period of the clock provided by the PDMC. The aim of the PDMC is to process data from the PDM interface, decimate and filter the data, and store the processed data in the FIFO.

The four paths are identical. Each path is composed of a left and a right channel. The PDM interface delivers eight parallel data of 1bit. Each bit goes to a filter. The aim of the filter is to limit the noise and export PCM format audio data.

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24.3.4 Clock Configuration

MCLK is the source clock signal. PDM_CLK is the output clocks generated in the PDMC and is fed to the external microphones. They are also the internal clock of the external microphones. User must take care about the value of PDM_CLK when selecting the source clock (MCLK).

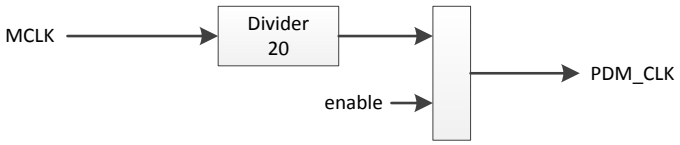

Fig.1- 5 PDMC Clock Structure

Table 3-1 Relation between MCLK, ASP_CLK and sample rate

MCLK	PDM_CLK	Sample rate
61.44Mhz	3.072Mhz	12khz,24khz,48khz,96khz,192khz
56.448Mhz	2.8224Mhz	11.025khz,22.05khz,44.1khz,88.2khz,176.4khz
40.96Mhz	2.048Mhz	8khz,16khz,32kHz,64kHz,128khz

24.4 Register Description

24.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
PDMC_SYSCONFIG	0x00000	W	0x00000000	PDMC system config register
PDMC_CTRL0	0x00004	W	0x780003f7	PDMC control register 0
PDMC_CTRL1	0x00008	W	0x000000ff	PDMC control register 1
PDMC_CLK_CTRL	0x0000c	W	0x00000000	PDMC clock control register
PDMC_HPF_CTRL	0x00010	W	0x00000000	PDMC high pass filter control register
PDMC_FIFO_CTRL	0x00014	W	0x00000000	PDMC FIFO control register
PDMC_DMA_CTRL	0x00018	W	0x0000001f	PDMC DMA control register

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Name	Offset	Size	Reset Value	Description
PDMC_INT_EN	0x0001c	W	0x00000000	PDMC interrupt enable register
PDMC_INT_CLR	0x00020	W	0x00000000	PDMC interrupt clear register
PDMC_INT_ST	0x00024	W	0x00000000	PDMC interrupt status register
PDMC_RXFIFO_DATA_REG	0x00030	W	0x00000000	PDMC receive FIFO data register
PDMC_DATA0R_REG	0x00034	W	0x00000000	PDMC path0 right channel data register
PDMC_DATA0L_REG	0x00038	W	0x00000000	PDMC path0 left channel data register
PDMC_DATA1R_REG	0x0003c	W	0x00000000	PDMC path1 right channel data register
PDMC_DATA1L_REG	0x00040	W	0x00000000	PDMC path1 left channel data register
PDMC_DATA2R_REG	0x00044	W	0x00000000	PDMC path2 right channel data register
PDMC_DATA2L_REG	0x00048	W	0x00000000	PDMC path2 left channel data register
PDMC_DATA3R_REG	0x0004c	W	0x00000000	PDMC path3 right channel data register
PDMC_DATA3L_REG	0x00050	W	0x00000000	PDMC path3 left channel data register
PDMC_DATA_VALID	0x00054	W	0x00000000	path data valid register
PDMC_VERSION	0x00058	W	0x59313030	PDMC version register
PDMC_RXDR	0x400~0x7fc	W	0x00000000	Receive FIFO data register

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*Notes: Size : **B** - Byte (8 bits) access, **HW** - Half WORD (16 bits) access, **W** -WORD (32 bits) access*

24.4.2 Detail Register Description

PDMC_SYSCONFIG

Address: Operational Base + offset (0x00000)

PDMC system config register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	rx_start RX transfer start bit RX Transfer start bit 0:stop RX transfer. 1:start RX transfer
1	RO	0x0	reserved
0	RW	0x0	rx_clr PDMC RX logic clear PDMC RX logic clear; This is a self cleard bit. High active. Write 0x1: clear RX logic Write 0x0: no action Read 0x1: clear ongoing Read 0x0: clear done

PDMC_CTRL0

Address: Operational Base + offset (0x00004)

PDMC control register 0

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31	RW	0x0	mode_sel Working mode selection: 0: PDM mode; 1: reserved;
30	RW	0x1	path3_en Path 3 enable; 1'b1: enable 1'b0: disable
29	RW	0x1	path2_en Path 2 enable; 1'b1: enable 1'b0: disable
28	RW	0x1	path1_en Path 1 enable; 1'b1: enable 1'b0: disable
27	RW	0x1	path0_en Path 0 enable; 1'b1: enable 1'b0: disable

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Bit	Attr	Reset Value	Description
26	RW	0x0	hwt_en HWT Halfword word transform Only valid when VDW select 16bit data. 0:32 bit data valid to AHB/APB bus. Low 16 bit for left channel and high 16 bit for right channel. 1:low 16bit data valid to AHB/APB bus, high 16 bit data invalid.
25	RW	0x0	Reserved
24	RW	0x0	Reserved
23	RW	0x0	Reserved
22	RW	0x0	Reserved
21:19	RW	0x0	Reserved
18	RW	0x0	Reserved
17	RW	0x0	Reserved
16	RW	0x0	Reserved
15:13	RO	0x0	reserved
12:10	RW	0x0	Reserved
9:5	RW	0x1f	Reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x17	data_vld_width (Can be written only when SYSCONFIG[2] is 0.) Valid Data width 0~14:reserved 15:16bit 16:17bit 17:18bit 18:19bit …… n:(n+1)bit …… 23:24bit

PDMC_CTRL1

Address: Operational Base + offset (0x00008)

PDMC control register 1

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8:0	RW	0x0ff	Reserved

PDMC_CLK_CTRL

Address: Operational Base + offset (0x0000c)

PDMC clock control register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	Reserved
5	RW	0x0	pdm_clk_en Pdm clk enable.working at PDM mode (Can be written only when SYSCONFIG[2] is 0.) 0:pdm clk disable 1:pdm clk enable
4	RO	0x0	reserved
3	RW	0x0	clk_polar PDM_CLK polarity selection (Can be written only when SYSCONFIG[2] is 0.) 0: no inverted 1: inverted
2:0	RW	0x0	pdm_ds_ratio DS_RATIO,working at PDM mode (Can be written only when SYSCONFIG[2] is 0.) 3'b000: sample rate 192k/176.5k/128k 3'b001: sample rate 96kk/88.2k/64k 3'b010: sample rate 48kk/44.1k/32k 3'b011: sample rate 24kk/22.05k/16k 3'b100: sample rate 12kk/11.025k/8k

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PDMC_HPF_CTRL

Address: Operational Base + offset (0x00010)

PDMC high pass filter control register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	hpfle HPFLE high pass filter enable for left channel 1'b0: high pass filter for right channel is diabled. 1'b1: high pass filter for right channel is enabled.
2	RW	0x0	hpfre HPFRE high pass filter enable for right channel 1'b0: high pass filter for right channel is diabled. 1'b1: high pass filter for right channel is enabled.
1:0	RW	0x0	hpf_cf HPF_CF high pass filter configure register high pass filter configure register 2'b00: 3.79Hz 2'b01: 60Hz 2'b10: 243Hz 2'b11: 493Hz

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PDMC_FIFO_CTRL

Address: Operational Base + offset (0x00014)

PDMC fifo control register

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved
14:8	RW	0x00	rft Receive FIFO Threshold When the number of receive FIFO entries is more than or equal to this threshold plus 1, the receive FIFO threshold interrupt is triggered.
7:0	RO	0x00	rfl RFL Receive FIFO Level Contains the number of valid data entries in the receive FIFO.

PDMC_DMA_CTRL

Address: Operational Base + offset (0x00018)

PDMC dma control register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	rde Receive DMA Enable 0 : Receive DMA disabled 1 : Receive DMA enabled

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Bit	Attr	Reset Value	Description
7	RO	0x0	reserved
6:0	RW	0x1f	rdl Receive Data Level This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1.

PDMC_INT_EN

Address: Operational Base + offset (0x0001c)

PDMC interrupt enable register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	rxoie RX overflow interrupt enable 0:disable 1:enable
0	RW	0x0	rxtie RX threshold interrupt enable 0:disable 1:enable

PDMC_INT_CLR

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Address: Operational Base + offset (0x00020)

PDMC interrupt clear register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	W1C	0x0	rxoic RX overflow interrupt clear, high active, auto clear.
0	RO	0x0	reserved

PDMC_INT_ST

Address: Operational Base + offset (0x00024)

PDMC interrupt status register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	rxoi RX overflow interrupt 0:inactive 1:active
0	RO	0x0	rxfi RX full interrupt 0:inactive 1:active

PDMC_RXFIFO_DATA_REG

Address: Operational Base + offset (0x00030)

PDMC receive fifo data register

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdr Receive FIFO shadow Register When the register is read, data in the receive FIFO is accessed.

PDMC_DATA0R_REG

Address: Operational Base + offset (0x00034)

PDMC path0 right channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data0r Data of the path 0 right channel

PDMC_DATA0L_REG

Address: Operational Base + offset (0x00038)

PDMC path0 leght channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data0l Data of the path 0 left channel

PDMC_DATA1R_REG

Address: Operational Base + offset (0x0003c)

PDMC path1 right channel data register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RO	0x0	data1r Data of the path 1 right channel

PDMC_DATA1L_REG

Address: Operational Base + offset (0x00040)

PDMC path1 left channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data1l Data of the path 1 left channel

PDMC_DATA2R_REG

Address: Operational Base + offset (0x00044)

PDMC path2 right channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data2r Data of the path 2 right channel

PDMC_DATA2L_REG

Address: Operational Base + offset (0x00048)

PDMC path2 left channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data2l Data of the path 2 left channel

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PDMC_DATA3R_REG

Address: Operational Base + offset (0x0004c)

PDMC path3 right channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data3r Data of the path 3 right channel

PDMC_DATA3L_REG

Address: Operational Base + offset (0x00050)

PDMC path3 left channel data register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	data3l Data of the path 3 left channel

PDMC_DATA_VALID

Address: Operational Base + offset (0x00054)

path data valid register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RC	0x0	path0_vld 0: DATA0R_REG, DATA0L_REG value is invalid; 1: DATA0R_REG, DATA0L_REG value is valid;

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Bit	Attr	Reset Value	Description
2	RC	0x0	path1_vld 0: DATA1R_REG, DATA1L_REG value is invalid; 1: DAT1R_REG, DATA1L_REG value is valid;
1	RC	0x0	path2_vld 0: DATA2R_REG, DATA2L_REG value is invalid; 1: DATA2R_REG, DATA2L_REG value is valid;
0	RC	0x0	path3_vld 0: DATA3R_REG, DATA3L_REG value is invalid; 1: DATA3R_REG, DATA3L_REG value is valid;

PDMC_VERSION

Address: Operational Base + offset (0x00058)

PDMC version register

Bit	Attr	Reset Value	Description
31:0	RO	0x59313030	version PDMC version

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24.5 Interface Description

Table 26-4 PDMC Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
O_pdm_clk	O	IO_I2S1sclk_PDMclkm0_TSPd7	PDMclkm0: GPIO2CL_IO[7:6]=2

		m1_CIFdata7m1_GPIO2C2vccio5/IO_I2Ssclkm0_GMACrxdvm1_PDMclkm1_GPIO1C6vccio4	PDMclkm1: GPIO1C_IO[13:12]=3
O_pdm_fsync	O	IO_I2S1sdo_PDMfsyncm0_GPIO2C7vccio5/IO_SDMMC1detn_GMACmdiom1_PDMfsyncm1_GPIO1C3vccio4	PDMfsyncm0: GPIO2CH_IO[15:14]=2 PDMfsyncm1: GPIO1C_IO[7:6]=3
I_pdm_data0	I	IO_I2S1sdi_PDMsdi0m0_CARDclkm1_GPIO2C3vccio5/IO_I2S2lrcktxm0_GMACmdcm1_PDMsdi0m1_GPIO1C7vccio4	PDMsdi0m0: GPIO2CL_IO[10:9]=2 PDMsdi0m1: GPIO1C_IO[1:0]=3
I_pdm_data1	I	IO_I2S1sdio1_PDMsdi1m0_CARDrstm1_GPIO2C4vccio5/IO_I2S2sdim0_GMACrxerm1_PDMsdi1m1_GPIO1D0vccio4	PDMsdi1m0: GPIO2CL_IO[13:12]=2 PDMsdi1m1: GPIO1D_IO[1:0]=3
I_pdm_data2	I	IO_I2S1sdio2_PDMsdi2m0_CARDdetm1_GPIO2C5vccio5/IO_I2S2sdom0_GMACtxenm1_PDMsdi2m1_GPIO1D1vccio4	PDMsdi2m0: GPIO2CH_IO[1:0]=2 PDMsdi2m1: GPIO1D_IO[3:2]=3
I_pdm_data3	I	IO_I2S1sdio3_PDMsdi3m0_CARDiom1_GPIO2C6vccio5/IO_I2S2lrckrxm0_CLKout_gmacm2_PDMsdi3m1_GPIO1D2vccio4	PDMsdi3m0: GPIO2CH_IO[4:3]=2 PDMsdi3m1: GPIO1D_IO[5:4]=3

Notes: I=input, O=output, I/O=input/output, bidirectional

Furthermore, different IOs are selected and connected to different flash interface, which is shown as follows.

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24.6 Application Notes

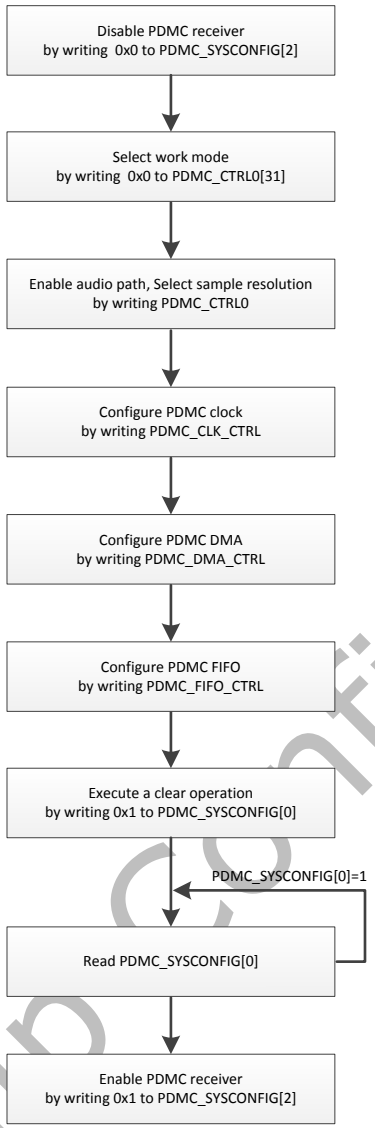

Fig.1- 6 PDMC operation flow

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Chapter 3 I2S/PCM Controller

1.1 Overview

The I2S/PCM controller is designed for interfacing between the AHB bus and the I2S bus.

The I2S bus (Inter-IC sound bus) is a serial link for digital audio data transfer between devices in the system and be invented by Philips Semiconductor. Now it is widely used by many semiconductor manufacturers.

Devices often use the I2S bus are ADC, DAC, DSP, CPU, etc. With the I2S interface, we can connect audio devices and the embedded SoC platform together and provide an audio interface solution for the system.

Not only I2S but also PCM mode surround audio output and stereo input are supported in I2S/PCM controller.

There are three I2S/PCM controllers embedded in the design, I2S0, I2S1 and I2S2. Different features between I2S/PCM controllers are as follows.

- Support four internal 32-bit wide and 32-location deep FIFOs for transmitting audio data for I2S0
- Support eight internal 32-bit wide and 32-location deep FIFOs, four for transmitting and four for receiving audio data for I2S1
- Support two internal 32-bit wide and 32-location deep FIFOs, one for transmitting and one for receiving audio data for I2S2
- Support 8 channels audio data transmitting in I2S mode for I2S0, 8 channels audio data transmitting or 8 channels audio data receiving for I2S1, 2 channels audio data transmitting and 2 channels audio data receiving for I2S2.

Common features for I2S0, I2S1 and I2S2 are as follows.

- Support AHB bus interface
- Support 16 ~ 32 bits audio data transfer
- Support master and slave mode
- Support DMA handshake interface and configurable DMA water level
- Support transmit FIFO empty, underflow, receive FIFO full, overflow interrupt and all interrupts can be masked
- Support configurable water level of transmit FIFO empty and receive FIFO full interrupt
- Support combine interrupt output
- Support 2 channels audio receiving in PCM mode
- Support I2S normal, left and right justified mode serial audio data transfer
- Support PCM early, late1, late2, late3 mode serial audio data transfer
- Support MSB or LSB first serial audio data transfer
- Support 16 to 31 bit audio data left or right justified in 32-bit wide FIFO
- Support two 16-bit audio data store together in one 32-bit wide location
- Support 2 independent LRCK signals, one for receiving and one for transmitting audio data. Single LRCK can be used for transmitting and receiving data if the sample rate are the same
- Support configurable SCLK and LRCK polarity

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1.2 Block Diagram

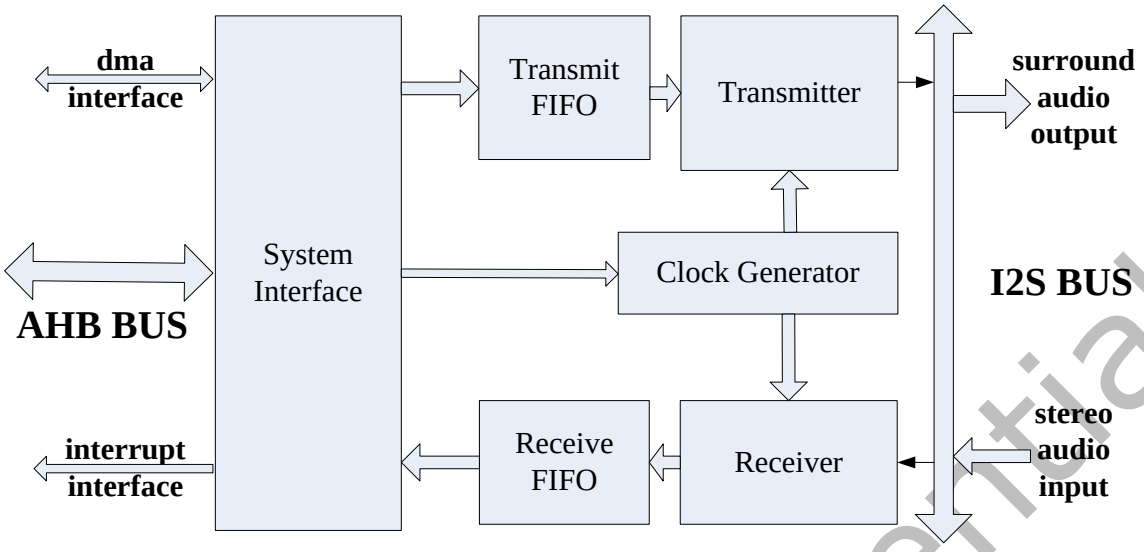

Fig.3-1 I2S/PCM controller (8 channel) Block Diagram

System Interface

The system interface implements the AHB slave operation. It contains not only control registers of transmitter and receiver inside but also interrupt and DMA handshake interface.

Clock Generator

The Clock Generator implements clock generation function. The input source clock to the module is MCLK_I2S, and by the divider of the module, the clock generator generates SCLK and LRCK to transmitter and receiver.

Transmitter

The Transmitter implements transmission operation. The transmitter can act as either master or slave, with I2S or PCM mode surround serial audio interface.

Receiver

The Receiver implements receive operation. The receiver can act as either master or slave, with I2S or PCM mode stereo serial audio interface.

Transmit FIFO

The Transmit FIFO is the buffer to store transmitted audio data. The size of the FIFO is 32bits x 32.

Receive FIFO

The Receive FIFO is the buffer to store received audio data. The size of the FIFO is 32bits x 32.

1.3 Function description

In the I2S/PCM controller, there are four conditions: transmitter-master & receiver-master; transmitter-master & receiver-slave; transmitter-slave & receiver-master; transmitter-slave & receiver-slave.

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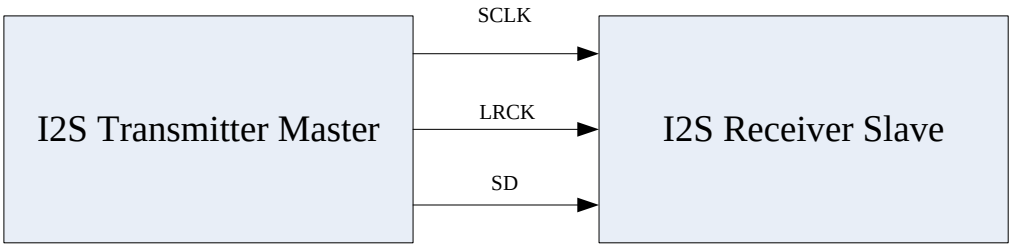

Fig.3-2 I2S transmitter-master & receiver-slave condition

When transmitter acts as a master, it sends all signals to receiver (slave), and CPU control when to send clock and data to the receiver. When acting as a slave, SD signal still goes from transmitter to receiver, but SCLK and LRCK signals are from receiver (master) to transmitter. Based on three interface specifications, transmitting data should be ready before transmitter receives SCLK and LRCK signals. CPU should know when the receiver to initialize a transaction and when to send data.

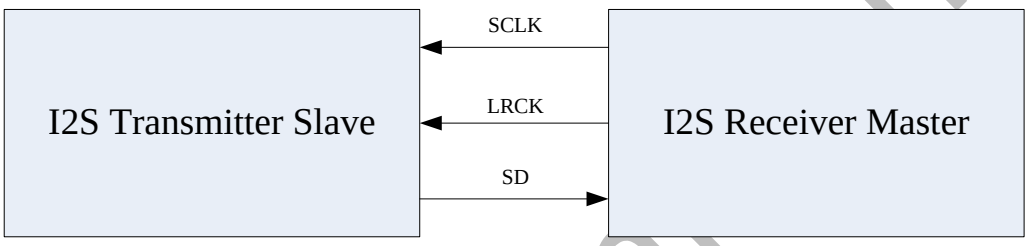

Fig.3-3 I2S transmitter-slave& receiver-master condition

When the receiver acts as a master, it sends SCLK and LRCK signals to the transmitter (slave) and receives serial data. So CPU must tell the transmitter when to start a transaction for it to prepare transmitting data then the receiver start a transfer and send clock and channel-select signals. When the receiver acts as a slave, CPU should only do initial setting and wait for all signals and then start reading data.

Before transmitting or receiving data, CPU need do initial setting to the I2S register. These includes CPU settings, I2S interface registers settings, and maybe the embedded SoC platform settings. These registers must be set before starting data transfer.

1.3.1 i2s normal mode

This is the waveform of I2S normal mode. For LRCK (i2s_lrck_rx/i2s_lrck_tx) signal, it goes low to indicate left channel and high to right channel. For SD (i2s_sdo,i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit one SCLK clock cycle after LRCK changes. The range of SD signal width is from 16 to 32bits.

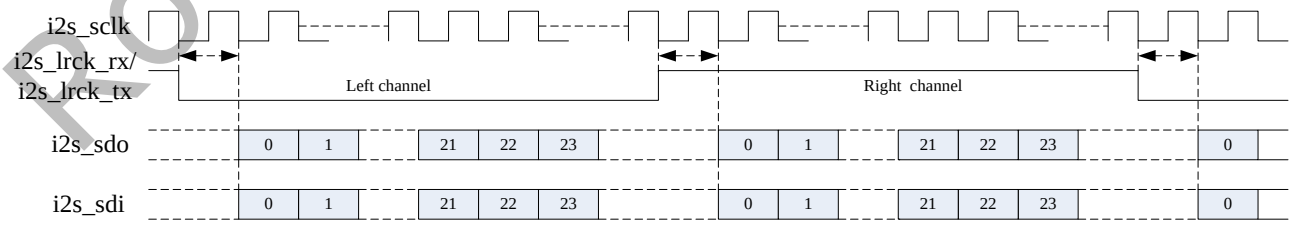

Fig.3-4 I2S normal mode timing format

1.3.2 i2s left justified mode

This is the waveform of I2S left justified mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx) signal, it goes high to indicate left channel and low to right channel. For SD (i2s_sdo, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit at the same time when LRCK changes. The

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range of SD signal width is from 16 to 32bits.

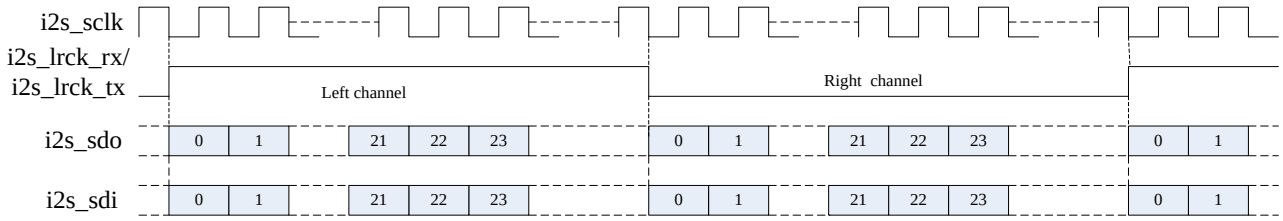

Fig.3-5 I2S left justified mode timing format

1.3.3 i2s right justified mode

This is the waveform of I2S right justified mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx) signal, it goes high to indicate left channel and low to right channel. For SD (i2s_sdo, i2s_sdi) signal, it transfers MSB or LSB first; but different from I2S normal or left justified mode, its data is aligned to last bit at the edge of the LRCK signal. The range of SD signal width is from 16 to 32bits.

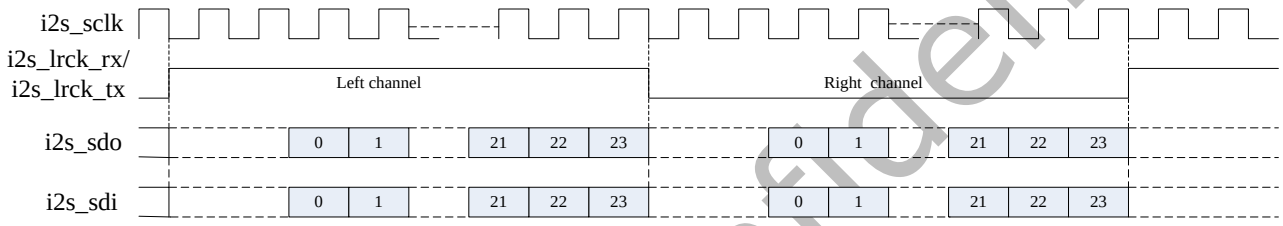

Fig.3-6 I2S right justified mode timing format

1.3.4 PCM early mode

This is the waveform of PCM early mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit at the same time when LRCK goes high. The range of SD signal width is from 16 to 32bits.

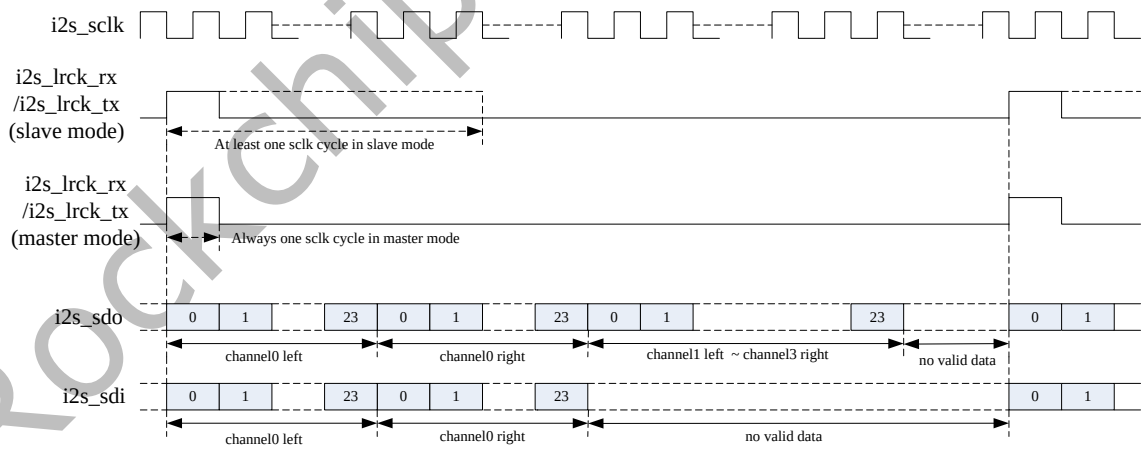

Fig.3-7 PCM early mode timing format

1.3.5 PCM late1 mode

This is the waveform of PCM late1 mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit one SCLK clock cycle after LRCK goes high. The range of SD signal width is from 16 to 32bits.

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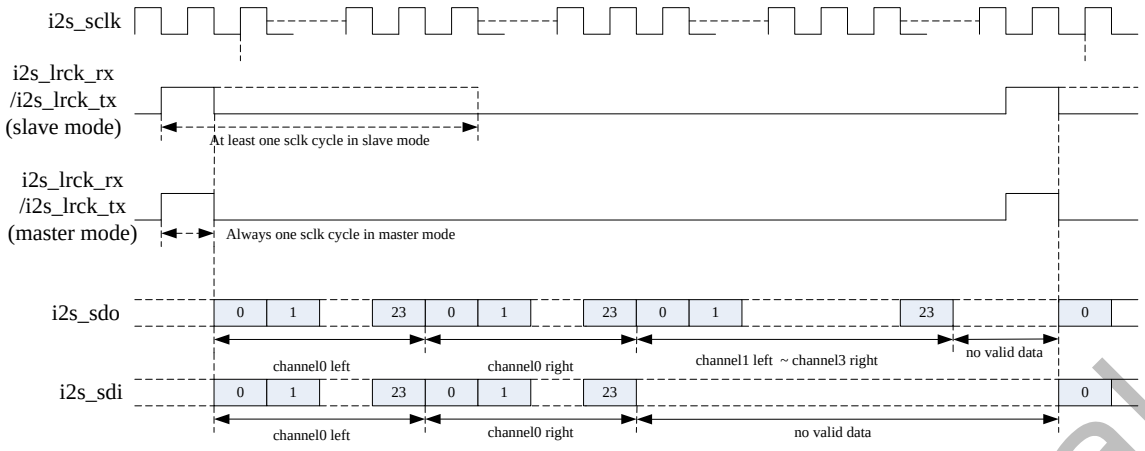

Fig.3-8 PCM late1 mode timing format

1.3.6 PCM late2 mode

This is the waveform of PCM late2 mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit two SCLK clock cycles after LRCK goes high. The range of SD signal width is from 16 to 32bits.

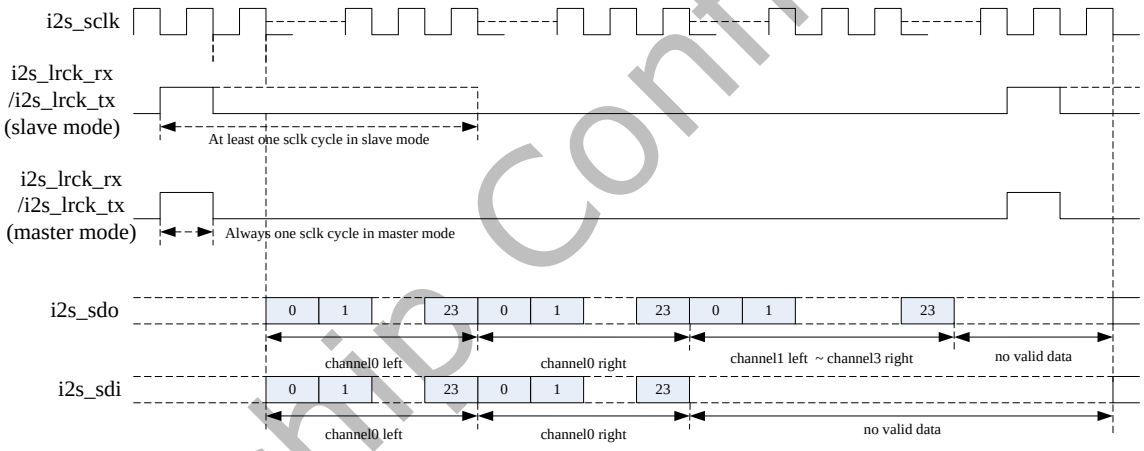

Fig.3-9 PCM late2 mode timing format

1.3.7 PCM late3 mode

This is the waveform of PCM late3 mode. For LRCK (i2s_lrck_rx / i2s_lrck_tx) signal, it goes high to indicate the start of a group of audio channels. For SD (i2s_sdo, i2s_sdi) signal, it transfers MSB or LSB first and sends the first bit three SCLK clock cycles after LRCK goes high. The range of SD signal width is from 16 to 32bits.

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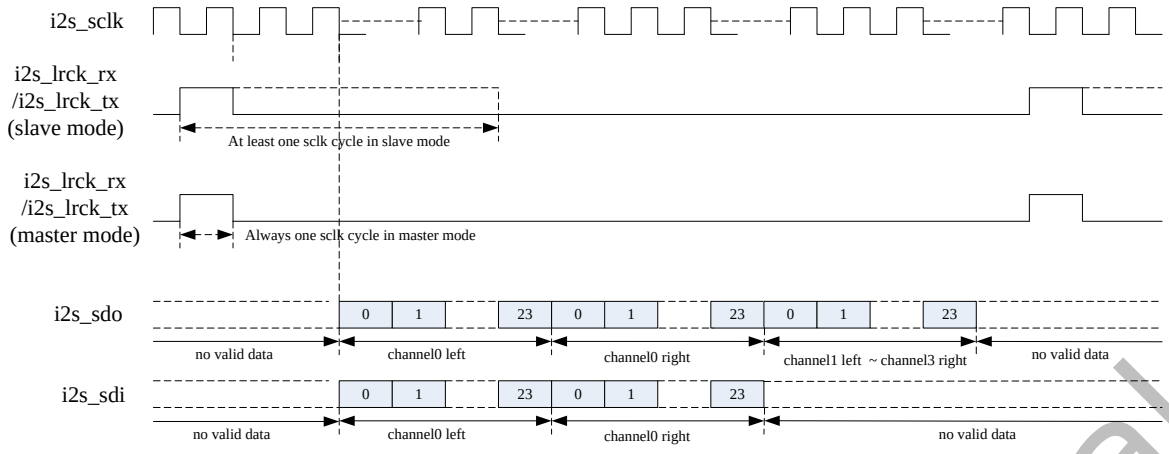

Fig.3-10 PCM late3 mode timing format

1.4 Register Description

This section describes the control/status registers of the design.

1.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
I2S_TXCR	0x0000	W	0x0000000f	transmit operation control register
I2S_RXCR	0x0004	W	0x0000000f	receive operation control register
I2S_CKR	0x0008	W	0x00071f1f	clock generation register
I2S_TXFIFOLR	0x000c	W	0x00000000	TX FIFO level register
I2S_DMACR	0x0010	W	0x001f0000	DMA control register
I2S_INTCR	0x0014	W	0x00000000	interrupt control register
I2S_INTSR	0x0018	W	0x00000000	interrupt status register
I2S_XFER	0x001c	W	0x00000000	Transfer Start Register
I2S_CLR	0x0020	W	0x00000000	SCLK domain logic clear Register
I2S_TXDR	0x0024	W	0x00000000	Transmit FIFO Data Register
I2S_RXDR	0x0028	W	0x00000000	Receive FIFO Data Register
I2S_RXFIFOLR	0x002c	W	0x00000000	RX FIFO level register
I2S_VERSION	0x0030	W	0x20150001	I2s version

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

1.4.2 Detail Register Description

I2S_TXCR

Address: Operational Base + offset (0x0000)

transmit operation control register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
22:17	RW	0x00	RCNT right justified counter (Can be written only when XFER[0] bit is 0.) Only valid in I2S Right justified format and slave tx mode is selected. Start to transmit data RCNT sclk cycles after left channel valid.
16:15	RW	0x0	TCSR TX Channel select register 2'b00:two channel 2'b01:four channel 2'b10:six channel 2'b11:eight channel
14	RW	0x0	HWT Halfword word transform (Can be written only when XFER[0] bit is 0.) Only valid when VDW select 16bit data. 0:32 bit data valid from AHB/APB bus. Low 16 bit for left channel and high 16 bit for right channel. 1:low 16bit data valid from AHB/APB bus, high 16 bit data invalid.
13	RO	0x0	reserved
12	RW	0x0	SJM Store justified mode SJM Store justified mode (Can be written only when XFER[1] bit is 0.) 16bit~31bit DATA stored in 32 bits width fifo. This bit is invalid if VDW select 16bit data and HWT select 0, Because every fifo unit contain two 16bit data and 32 bit space is full, it is impossible to choose justified mode. 0:right justified 1:left justified
11	RW	0x0	FBM First Bit Mode (Can be written only when XFER[0] bit is 0.) 0:MSB 1:LSB
10:9	RW	0x0	IBM I2S bus mode (Can be written only when XFER[0] bit is 0.) 0:I2S normal 1:I2S Left justified 2:I2S Right justified 3:reserved

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Bit	Attr	Reset Value	Description
8:7	RW	0x0	PBM PCM bus mode (Can be written only when XFER[0] bit is 0.) 0:PCM no delay mode 1:PCM delay 1 mode 2:PCM delay 2 mode 3:PCM delay 3 mode
6	RO	0x0	reserved
5	RW	0x0	TFS Transfer format select (Can be written only when XFER[0] bit is 0.) 0: I2S format 1: PCM format
4:0	RW	0x0f	VDW Valid Data width (Can be written only when XFER[0] bit is 0.) 0~14:reserved 15:16bit 16:17bit 17:18bit 18:19bit …… n:(n+1)bit …… 28:29bit 29:30bit 30:31bit 31:32bit

I2S_RXCR

Address: Operational Base + offset (0x0004)

receive operation control register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16:15	RW	0x0	RCSR RX Channel select register 2'b00:two channel 2'b01:four channel 2'b10:six channel 2'b11:eight channel

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Bit	Attr	Reset Value	Description
14	RW	0x0	HWT Halfword word transform (Can be written only when XFER[1] bit is 0.) Only valid when VDW select 16bit data. 0:32 bit data valid to AHB/APB bus. Low 16 bit for left channel and high 16 bit for right channel. 1:low 16bit data valid to AHB/APB bus, high 16 bit data invalid.
13	RO	0x0	reserved
12	RW	0x0	SJM Store justified mode (Can be written only when XFER[1] bit is 0.) 16bit~31bit DATA stored in 32 bits width fifo. If VDW select 16bit data, this bit is valid only when HWT select 0.Because if HWT is 1, every fifo unit contain two 16bit data and 32 bit space is full, it is impossible to choose justified mode. 0:right justified 1:left justified
11	RW	0x0	FBM First Bit Mode (Can be written only when XFER[1] bit is 0.) 0:MSB 1:LSB
10:9	RW	0x0	IBM I2S bus mode (Can be written only when XFER[1] bit is 0.) 0:I2S normal 1:I2S Left justified 2:I2S Right justified 3:reserved
8:7	RW	0x0	PBM PCM bus mode (Can be written only when XFER[1] bit is 0.) 0:PCM no delay mode 1:PCM delay 1 mode 2:PCM delay 2 mode 3:PCM delay 3 mode
6	RO	0x0	reserved
5	RW	0x0	TFS Transfer format select (Can be written only when XFER[1] bit is 0.) 0:i2s 1:pcm

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Bit	Attr	Reset Value	Description
4:0	RW	0x0f	VDW Valid Data width (Can be written only when XFER[1] bit is 0.) 0~14:reserved 15:16bit 16:17bit 17:18bit 18:19bit …… n:(n+1)bit …… 28:29bit 29:30bit 30:31bit 31:32bit

I2S_CKR

Address: Operational Base + offset (0x0008)

clock generation register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:28	RW	0x0	TRCM Tx and Rx Common Use 2'b00/2'b11:tx_lrck/rx_lrck are used as synchronous signal for TX /RX respectively. 2'b01:only tx_lrck is used as synchronous signal for TX and RX. 2'b10:only rx_lrck is used as synchronous signal for TX and RX.
27	RW	0x0	MSS Master/slave mode select (Can be written only when XFER[1] or XFER[0] bit is 0.) 0:master mode(sclk output) 1:slave mode(sclk input)
26	RW	0x0	CKP Sclk polarity (Can be written only when XFER[1] or XFER[0] bit is 0.) 0: sample data at posedge sclk and drive data at negedge sclk 1: sample data at negedge sclk and drive data at posedge sclk

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Bit	Attr	Reset Value	Description
25	RW	0x0	RLP Receive lrck polarity (Can be written only when XFER[1] or XFER[0] bit is 0.) 0:normal polarity (I2S normal: low for left channel, high for right channel I2S left/right just: high for left channel, low for right channel PCM start signal: high valid) 1:oppsite polarity (I2S normal: high for left channel, low for right channel I2S left/right just: low for left channel, high for right channel PCM start signal: low valid)
24	RW	0x0	TLP Transmit lrck polarity (Can be written only when XFER[1] or XFER[0] bit is 0.) 0:normal polarity (I2S normal: low for left channel, high for right channel I2S left/right just: high for left channel, low for right channel PCM start signal: high valid) 1:oppsite polarity (I2S normal: high for left channel, low for right channel I2S left/right just: low for left channel, high for right channel PCM start signal: low valid)
23:16	RW	0x07	MDIV mclk divider (Can be written only when XFER[1] or XFER[0] bit is 0.) Serial Clock Divider = Fmclk / Ftxsclk-1.(mclk frequecy / txsclk frequecy-1) 0 :Fmclk=Ftxsclk; 1 :Fmclk=2*Ftxsclk; 2,3 :Fmclk=4*Ftxsclk; 4,5 :Fmclk=6*Ftxsclk; …… 2n,2n+1:Fmclk=(2n+2)*Ftxsclk; …… 60,61:Fmclk=62*Ftxsclk; 62,63:Fmclk=64*Ftxsclk; …… 252,253:Fmclk=254*Ftxsclk; 254,255:Fmclk=256*Ftxsclk;

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Bit	Attr	Reset Value	Description
15:8	RW	0x1f	RSD Receive sclk divider (Can be written only when XFER[1] or XFER[0] bit is 0.) Receive sclk divider= Fsclk/Frxlrck 0~30:reserved 31: 32fs 32: 33fs 33: 34fs 34: 35fs …… n: (n+1)fs …… 253: 254fs 254: 255fs 255: 256fs
7:0	RW	0x1f	TSD Transmit sclk divider (Can be written only when XFER[1] or XFER[0] bit is 0.) Transmit sclk divider=Ftxsclk/Ftxlrck 0~30:reserved 31: 32fs 32: 33fs 33: 34fs 34: 35fs …… n: (n+1)fs …… 253: 254fs 254: 255fs 255: 256fs

I2S_TXFIFOLR

Address: Operational Base + offset (0x000c)

TX FIFO level register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:18	RO	0x00	TFL3 Transmit FIFO3 Level Contains the number of valid data entries in the transmit FIFO3.

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Bit	Attr	Reset Value	Description
17:12	RO	0x00	TFL2 Transmit FIFO2 Level Contains the number of valid data entries in the transmit FIFO2.
11:6	RO	0x00	TFL1 Transmit FIFO1 Level Contains the number of valid data entries in the transmit FIFO1.
5:0	RO	0x00	TFL0 Transmit FIFO0 Level Contains the number of valid data entries in the transmit FIFO0.

I2S_DMACR

Address: Operational Base + offset (0x0010)

DMA control register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	RDE Receive DMA Enable 0 : Receive DMA disabled 1 : Receive DMA enabled
23:21	RO	0x0	reserved
20:16	RW	0x1f	RDL Receive Data Level This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO (RXFIFO0 if RCSR=00;RXFIFO1 if RCSR=01,RXFIFO2 if RCSR=10,RXFIFO3 if RCSR=11)is equal to or above this field value + 1.
15:9	RO	0x0	reserved
8	RW	0x0	TDE Transmit DMA Enable 0 : Transmit DMA disabled 1 : Transmit DMA enabled
7:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	TDL Transmit Data Level This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the TXFIFO(TXFIFO0 if TCSR=00;TXFIFO1 if TCSR=01,TXFIFO2 if TCSR=10,TXFIFO3 if TCSR=11)is equal to or below this field value.

I2S_INTCR

Address: Operational Base + offset (0x0014)

interrupt control register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24:20	RW	0x00	RFT Receive FIFO Threshold When the number of receive FIFO entries (RXFIFO0 if RCSR=00; RXFIFO1 if RCSR=01, RXFIFO2 if RCSR=10, RXFIFO3 if RCSR=11) is more than or equal to this threshold plus 1, the receive FIFO full interrupt is triggered.
19	RO	0x0	reserved
18	WO	0x0	RXOIC RX overrun interrupt clear Write 1 to clear RX overrun interrupt.
17	RW	0x0	RXOIE RX overrun interrupt enable 0:disable 1:enable
16	RW	0x0	RXFIE RX full interrupt enable 0:disable 1:enable
15:9	RO	0x0	reserved
8:4	RW	0x00	TFT Transmit FIFO Threshold When the number of transmit FIFO (TXFIFO0 if TCSR=00; TXFIFO1 if TCSR=01, TXFIFO2 if TCSR=10, TXFIFO3 if TCSR=11) entries is less than or equal to this threshold, the transmit FIFO empty interrupt is triggered.
3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	WO	0x0	TXUIC TX underrun interrupt clear Write 1 to clear TX underrun interrupt.
1	RW	0x0	TXUIE TX underrun interrupt enable 0:disable 1:enable
0	RW	0x0	TXEIE TX empty interrupt enable 0:disable 1:enable

I2S_INTSR

Address: Operational Base + offset (0x0018)

interrupt status register

Bit	Attr	Reset Value	Description
31:18	RO	0x0	reserved
17	RO	0x0	RXOI RX overrun interrupt 0:inactive 1:active
16	RO	0x0	RXFI RX full interrupt 0:inactive 1:active
15:2	RO	0x0	reserved
1	RO	0x0	TXUI TX underrun interrupt 0:inactive 1:active
0	RO	0x0	TXEI TX empty interrupt 0:inactive 1:active

I2S_XFER

Address: Operational Base + offset (0x001c)

Transfer Start Register

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Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	RXS RX Transfer start bit 0:stop RX transfer. 1:start RX transfer
0	RW	0x0	TXS TX Transfer start bit 0:stop TX transfer. 1:start TX transfer

I2S_CLR

Address: Operational Base + offset (0x0020)

SCLK domain logic clear Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	RXC RX logic clear This is a self cleared bit. Write 1 to clear all receive logic.
0	RW	0x0	TXC TX logic clear This is a self cleared bit. Write 1 to clear all transmit logic.

I2S_TXDR

Address: Operational Base + offset (0x0024)

Transmit FIFO Data Register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	TXDR Transmit FIFO Data Register When it is written to, data are moved into the transmit FIFO.

I2S_RXDR

Address: Operational Base + offset (0x0028)

Receive FIFO Data Register

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	RXDR Receive FIFO Data Register When the register is read, data in the receive FIFO is accessed.

I2S_RXFIFOLR

Address: Operational Base + offset (0x002c)

RX FIFO level register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:18	RO	0x00	RFL3 Receive FIFO3 Level Contains the number of valid data entries in the receive FIFO3.
17:12	RO	0x00	RFL2 Receive FIFO2 Level Contains the number of valid data entries in the receive FIFO2.
11:6	RU	0x00	RFL1 Receive FIFO1 Level Contains the number of valid data entries in the receive FIFO1.
5:0	RO	0x00	RFL0 Receive FIFO0 Level Contains the number of valid data entries in the receive FIFO0.

I2S_VERSION

Address: Operational Base + offset (0x0030)

I2S version

Bit	Attr	Reset Value	Description
31:0	RO	0x20150001	I2S version

1.5 Interface description

Table 3-1 I2S Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
Interface for i2s1			
i2s1_mclk	I/O	IO_I2S1mclk_Nouse0_TSPd0m1_CIFdata7m1_GPIO2C2vccio5	GRF_GPIO2BH_IOMUX[8:6]=3'b001

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Module Pin	Direction	Pad Name	IOMUX Setting
i2s1_sclk	I/O	IO_I2S1sclk_PDMclkm0_TSPd7m1_CIFdata7m1_GPIO2C2vccio5	GRF_GPIO2CL_IOMUX[8:6]=3'b001
i2s1_lrck_rx	I/O	IO_I2S1lrckrx_NOuse1_TSPd5m1_CIFdata5m1_GPIO2C0vccio5	GRF_GPIO2CL_IOMUX[2:0]=3'b001
i2s1_lrck_tx	I/O	IO_I2S1lrcktx_SPDIFtxm1_TSPd6m1_CIFdata6m1_GPIO2C1vccio5	GRF_GPIO2CL_IOMUX[5:3]=3'b001
i2s1_sdo0	O	IO_I2S1sdo_PDMfsyncm0_GPIO2C7vccio5	GRF_GPIO2CH_IOMUX[7:6]=2'b01
i2s1_sdo1	O	IO_I2S1sdio1_PDMsdi1m0_CARDrstm1_GPIO2C4vccio5	GRF_GPIO2CL_IOMUX[14:12]=3'b001
i2s1_sdo2	O	IO_I2S1sdio2_PDMsdi2m0_CARDdetm1_GPIO2C5vccio5	GRF_GPIO2CH_IOMUX[2:0]=3'b001
i2s1_sdo3	O	IO_I2S1sdio3_PDMsdi3m0_CARDiom1_GPIO2C6vccio5	GRF_GPIO2CH_IOMUX[5:3]=3'b001
i2s1_sdi0	I	IO_I2S1sdi_PDMsdi0m0_CARDclkm1_GPIO2C3vccio5	GRF_GPIO2CL_IOMUX[11:9]=3'b001
i2s1_sdi1	I	IO_I2S1sdio1_PDMsdi1m0_CARDrstm1_GPIO2C4vccio5	GRF_GPIO2CL_IOMUX[14:12]=3'b001
i2s1_sdi2	I	IO_I2S1sdio2_PDMsdi2m0_CARDdetm1_GPIO2C5vccio5	GRF_GPIO2CH_IOMUX[2:0]=3'b001
i2s1_sdi3	I	IO_I2S1sdio3_PDMsdi3m0_CARDiom1_GPIO2C6vccio5	GRF_GPIO2CH_IOMUX[5:3]=3'b001
Interface for i2s2 M0 IO			
i2s2_mclk	I/O	IO_I2S2mclk_GMACclkm1_GPIO1C5vccio4	GRF_GPIO1C_IOMUX[11:10]=2'b01
i2s2_sclk	I/O	IO_I2S2sclkm0_GMACrxdvm1_PDMclkm1_GPIO1C6vccio4	GRF_GPIO1C_IOMUX[13:12]=2'b01
i2s2_lrck_tx	I/O	IO_I2S2lrcktxm0_GMACmdcm1_PDMsdi0m1_GPIO1C7vccio4	GRF_GPIO1C_IOMUX[15:14]=2'b01
i2s2_lrck_rx	I/O	IO_I2S2lrckrxm0_CLKout_gmacm2_PDMsdi3m1_GPIO1D2vccio4	GRF_GPIO1D_IOMUX[5:4]=2'b01
i2s2_sdi	I	IO_I2S2sdim0_GMACrxerm1_PDMsdi1m1_GPIO1D0vccio4	GRF_GPIO1D_IOMUX[1:0]=2'b01
i2s2_sdo	O	IO_I2S2sdom0_GMACtxenm1_PDMsdi2m1_GPIO1D1vccio4	GRF_GPIO1D_IOMUX[3:2]=2'b01
Interface for i2s2 M1 IO			
i2s2_sclk	I/O	IO_TSPvalid_CIFvsync_SDMMC0EXTcmd_SPIclkm2_USB3PHYdebug1_I2S2sclkm1_GPIO3A0vccio6	GRF_GPIO3AL_IOMUX[2:0]=3'b110

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Module Pin	Direction	Pad Name	IOMUX Setting
i2s2_lrck_tx	I/O	IO_TSPd4_CIFdata4_SPIcsn0m2_I2S2lrcktxm1_USB3PHYdebug8_I2S2lrckrxm1_GPIO3B0vccio6	GRF_GPIO3BL_IOMUX[2:0]=3'b100
i2s2_lrck_rx	I/O	IO_TSPd4_CIFdata4_SPIcsn0m2_I2S2lrcktxm1_USB3PHYdebug8_I2S2lrckrxm1_GPIO3B0vccio6	GRF_GPIO3BL_IOMUX[2:0]=3'b110
i2s2_sdi	I	IO_TSPclk_CIFclkin_SDMMC0EXTclkout_SPIrxdm2_USB3PHYdebug3_I2S2sdim1_GPIO3A2vccio6	GRF_GPIO3AL_IOMUX[8:6]=3'b110
i2s2_sdo	O	IO_TSPfail_CIFhref_SDMMC0EXTdet_SPItxdm2_USB3PHYdebug2_I2S2sdom1_GPIO3A1vccio6	GRF_GPIO3AL_IOMUX[5:3]=3'b110

Notes: I=input, O=output, I/O=input/output, bidirectional

The i2s1_sdix(x=1,2,3) and i2s1_sdox(x=1,2,3) signals shares the same IO, the direction is configured by setting GRF_CON_CON10 [4:2]. Each bit controls the direction of IO_I2S1sdio1_PDMsdi1m0_CARDrstm1_GPIO2C4vccio5, IO_I2S1sdio2_PDMsdi2m0_CARDdetm1_GPIO2C5vccio5 and IO_I2S1sdio3_PDMsdi3m0_CARDiom1_GPIO2C6vccio5 respectively with high level meaning output.

When M0 IO is used, I2S2 can used as transmitter and receiver and the same time.

When M1 IO is used, IO_TSPd4_CIFdata4_SPIcsn0m2_I2S2lrcktxm1_USB3PHYdebug8_I2S2lrckrxm1_GPIO3B0vccio6 is connected to either of i2s2_lrck_rx and i2s2_lrck_tx at the same time, so I2S2 cannot be used as transmitter and receiver and the same time.

The I2S1 is also connected to the ACODEC which supports master and slave mode. When the ACODEC acts as a master, the signal i2s1_lrck_tx_in which connected to I2S1 can be selected from ACODEC or external IO by setting GRF_SOC_CON2[15].

Table 3-2 Interface Between I2S1 and ACODEC

Module Pin	Direction	Module Pin	Direction
i2s1_mclk	O	pin_mclk	I
i2s1_sclk_out	O	pin_sck_i	I
i2s1_sclk_in	I	pin_sck_o	O
i2s1_lrck_tx_out	O	pin_dac_ws_i	I
i2s1_lrck_tx_in	I	pin_dac_ws_o	O
i2s1_sdo0	O	pin_dac_sd_i	I

The I2S0 module is connected to the audio interface of HDMI, which supports 8 channels audio data transmitting.

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Table 3-3 I2S Interface Between I2S2 and HDMI

Module Pin	Direction	Module Pin	Direction
i2s0_sclk_out	O	ii2sclk	I
i2s0_tx_lrck_out	O	ii2slrck	I
i2s0_sdo[3:0]	O	ii2sdata[3:0]	I

1.6 Application Notes

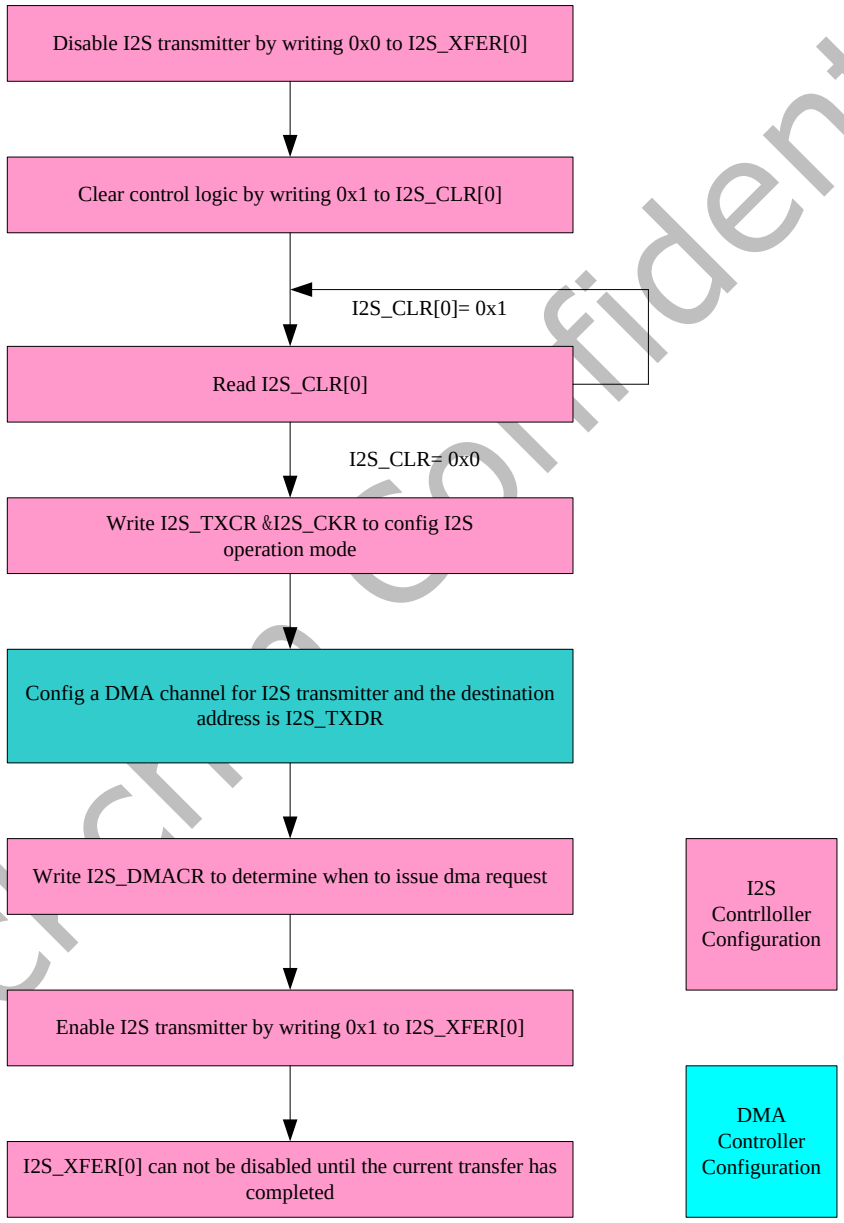

Fig.3-11 I2S/PCM controller transmit operation flow chart

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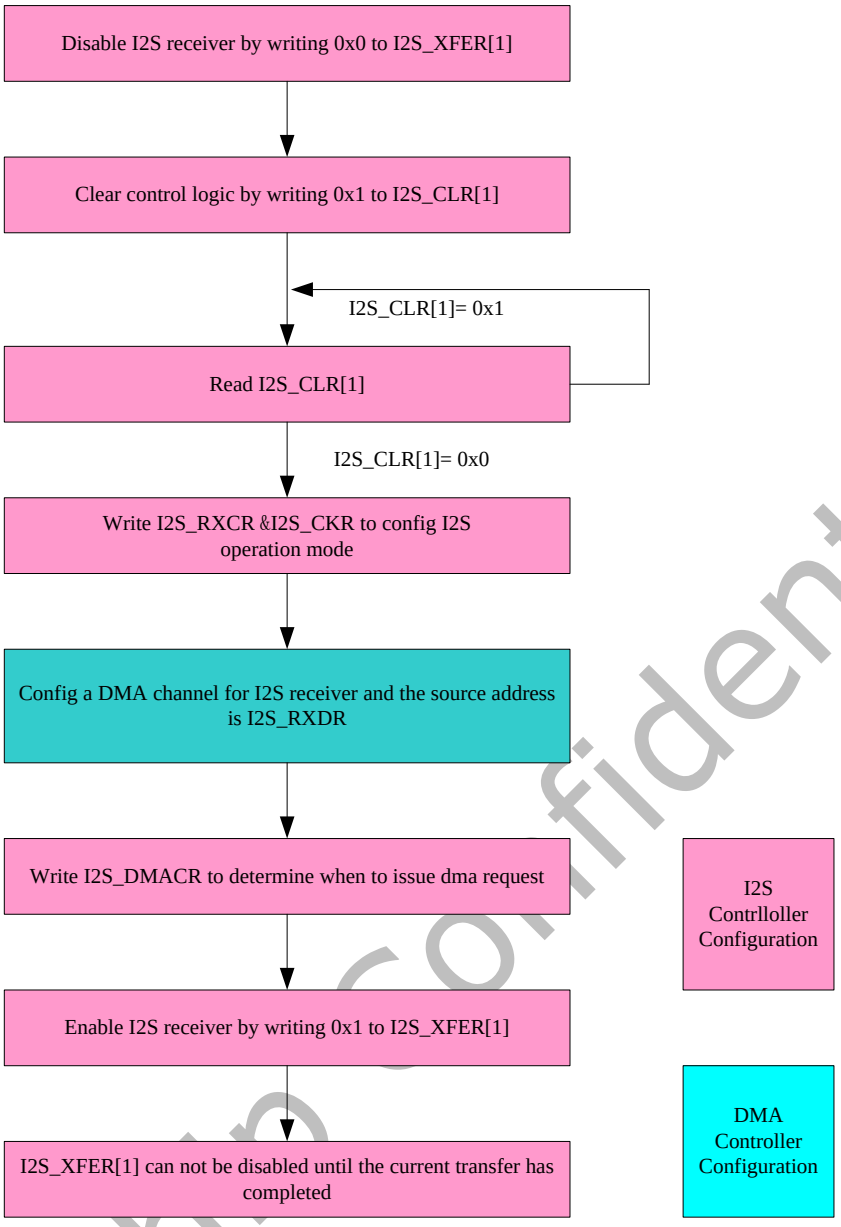

Fig.3-12 I2S/PCM controller receive operation flow chart

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Chapter 27 Audio Codec

2.1 Overview

The Audio Codec is a low power, high resolution, stereo CODEC solution that employs Sigma-Delta noise-shaping technique. The DAC with 24bit resolution and power amplifier are integrated.

It supports the following features:

- 24bit DAC
- Support 16Ω to 32Ω headphone output and speaker output
- Support Mono, Stereo channel
- Digital interpolation and decimation filter integrated
- Sampling rate of 8/12/16/24/32/44.1/48/96KHz
- Optional fractional PLL available that support 6MHz to 20MHz clock input to any clock

2.2 Block Diagram

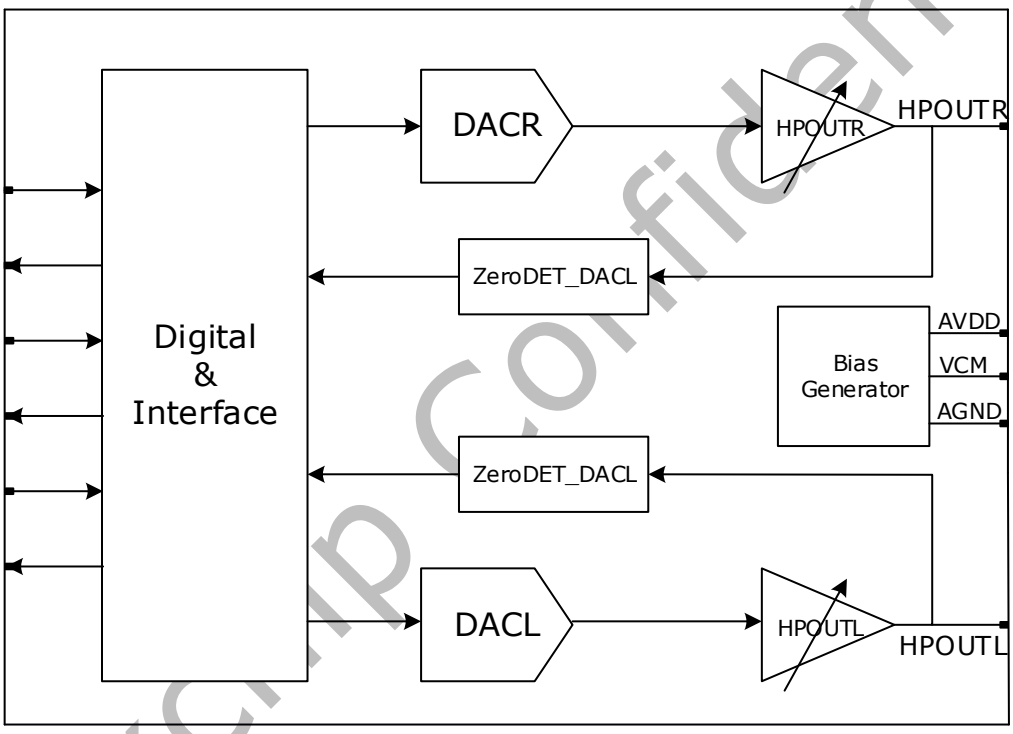

Fig. 27-1 Audio Codec Block Diagram

2.3 Function Description

2.3.1 Digital Interface

The Codec has the I2S PCM interface of audio data stream in for DAC, which can be configured in master or slave mode. Different audio data formats are available for different operating modes, which are demonstrated in below table.

Table 27-1 Supported Data Formats in Different Modes

Data Formats	DAC	
	Master	Slave
Left Justified	√	√
Right Justified	√	√
I²S	√	√
DSP/PCM mode A	√	√
DSP/PCM mode B	√	√

I2S_PCM interface supports five audio data formats: Left Justified mode, Right Justified mode, I2S mode, DSP/PCM mode A and mode B. They are valid when the device operates as a

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master or slave.
For Left Justified mode, the data format is illustrated in Fig. 24-2.The MSB is valid at the first rising edge of sck after ws transition is done. The other valid bits up to the LSB are transmitted sequentially. Due to varied word length, different sck frequency and sample rate, some unused sck cycles may appear before every ws transition, which means the data in this period is invalid.

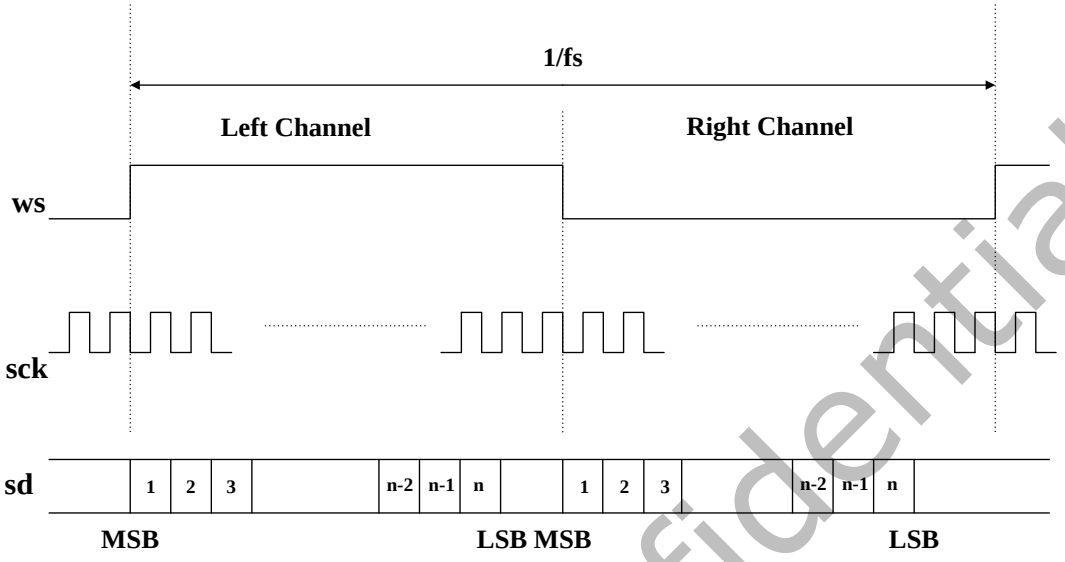

Fig.27-2 Left Justified Mode (assuming n-bit word length)

For Right Justified mode, the data format is shown in Fig. 24-3. The LSB becomes valid at the last rising edge of sck before ws transition is done. As the MSB is transmitted first, the other valid bits up to the MSB are followed in order. Due to varied word length, different sck frequency and sample rate, some unused sck cycles may exist after every ws transition, which means the data in this period is invalid.

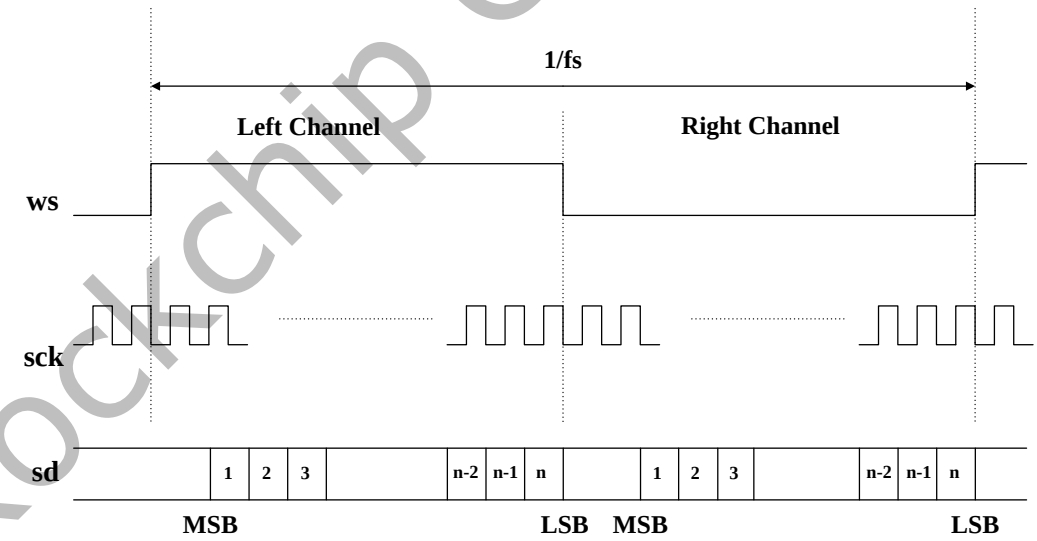

Fig.27-3 Right Justified Mode (assuming n-bit word length)

For I2S mode, the data format is depicted in Fig. 24-4. The MSB becomes available at the second rising edge of sck when ws transition is done. The other valid bits up to the LSB are transmitted in order. Due to varied word length, different sck frequency and sample rate, some unused sck cycles may appear between the LSB of the current sample and the MSB of the next one, which means the data in this period can be ignored.

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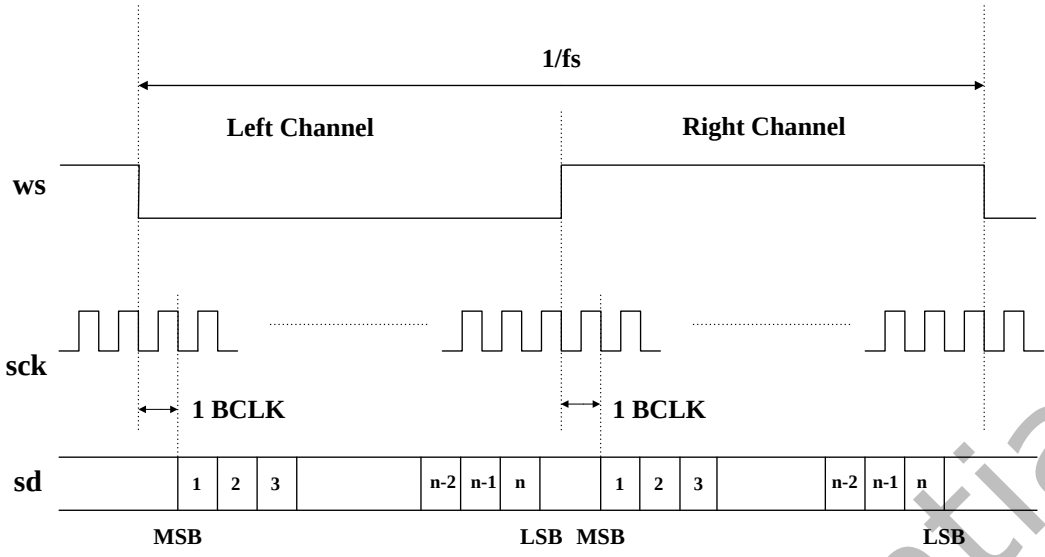

Fig.27-4 I2S Mode (assuming n-bit word length)

For DSP/PCM mode, the left channel data is transmitted first, followed by right channel data. For DSP/PCM mode A/B, the MSB is available at the second and first rising edge of sck after the rising edge of ws respectively, as shown in Fig. 24-5 and Fig.24-6. Based on word length, sck frequency and sample rate, there may be some invalid data between the LSB of the right channel data and the next sample.

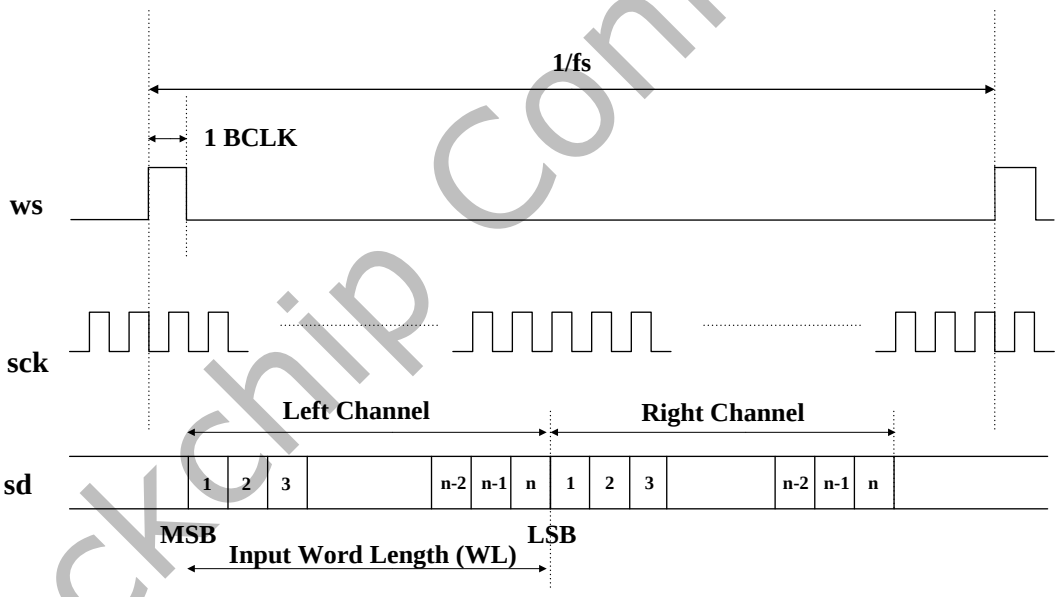

Fig.27-5 DSP/PCM Mode A (assuming n-bit word length)

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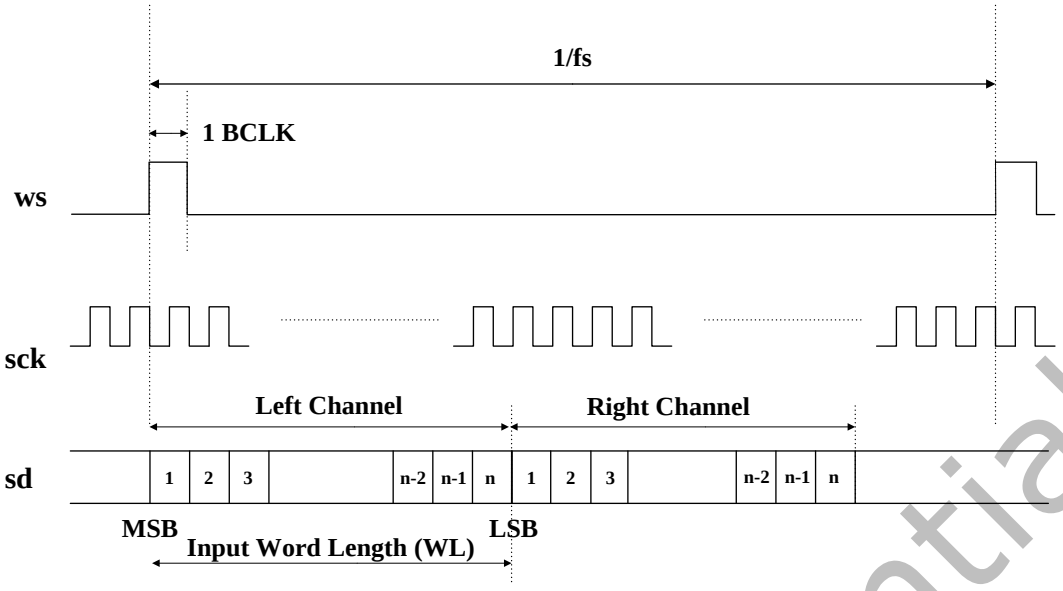

Fig.27-6 DSP/PCM Mode B (assuming n-bit word length)

2.3.2 Analog Interface

The Codec DAC output can drive 16Ω or 32Ω headphone load either through DC-blocking capacitor.

In the configuration using DC-blocking capacitor, shown in following figure, the headphone ground is connected to the real ground. The capacitance and the load resistance determine the lower cut-off frequency. For instance, if 16Ω headphone and 100uF DC-blocking capacitor are used, the lower cut-off frequency is:

$$f = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 16 \times 100 \times 10^{-6}} = 99.5Hz$$

The DC-blocking capacitor can be increased to lower the cut-off frequency for better bass response.

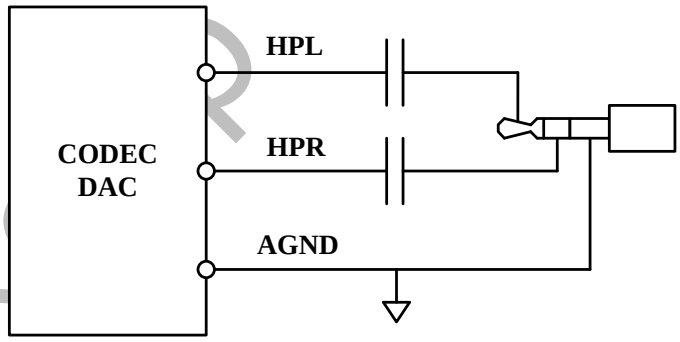

Fig.27-7 DC-blocking capacitor

Microphone bias output is used to bias external microphones. The bias voltage can varies from 0.5*AVDD to 0.85* AVDD with a step of 0.05*AVDD.

2.3.3 Interface Relationship

The relationship between I2S1 interface and the parallel audio data for DAC channels are shown in the following figure.

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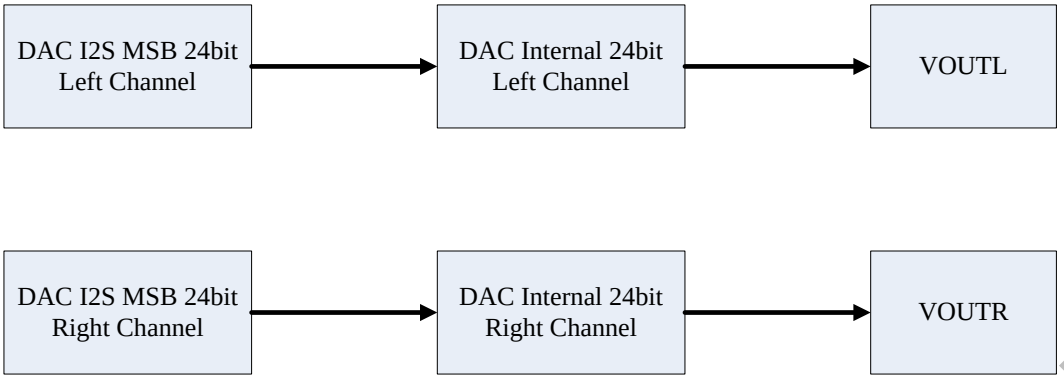

Fig.27-8 DAC Channel Relationship

Following table lists the corresponding signals between I2S1 and ACODEC.

Table 27-2 Signals Between I2S1 and ACODEC

Module Pin	Direction	Module Pin	Direction
i2s1_mclk	O	pin_mclk	I
i2s1_sclk_out	O	pin_sck_i	I
i2s1_sclk_in	I	pin_sck_o	O
i2s1_lrck_tx_out	O	pin_dac_ws_i	I
i2s1_lrck_tx_in	I	pin_dac_ws_o	O
i2s1_sdo0	O	pin_dac_sd_i	I

2.4 Register Description

2.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
Codec_REG0	0x0000	B	0x03	Audio Codec register 0
Codec_REG1	0x000c	B	0x00	Audio Codec register 1
Codec_REG2	0x0010	B	0x50	Audio Codec register 2
Codec_REG3	0x0014	B	0x0e	Audio Codec register 3
Codec_REG4	0x0034	B	0x00	Audio Codec register 4
Codec_REG5	0x0088	B	0x01	Audio Codec register 5
Codec_REG6	0x008c	B	0x00	Audio Codec register 6
Codec_REG7	0x0090	B	0x00	Audio Codec register 7
Codec_REG8	0x0094	B	0x00	Audio Codec register 8
Codec_REG9	0x0098	B	0x00	Audio Codec register 9
Codec_REG10	0x009c	B	0x00	Audio Codec register 10
Codec_REG11	0x00a0	B	0x00	Audio Codec register 11
Codec_REG12	0x00a4	B	0x00	Audio Codec register 12
Codec_REG13	0x00a8	B	0x11	Audio Codec register 13

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

2.4.2 Detail Register Description

Codec_REG0

Address: Operational Base + offset (0x0000)

Codec register 0

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RW	0x0	Inner-Bist Wave Generated model enable The Inner-Bist Wave Generated model enable signal. 1:Enable 0:Disable
6	RW	0x0	Power reset bypass 0:not 1:bypass
5:2	RO	0x0	reserved
1	RW	0x1	Codec digital core reset This reset only reset the codec data path. 0:reset 1:work
0	RW	0x1	Codec system reset This signal will reset the registers which control all the digital and analog part. 0:reset 1:work

Codec_REG1
Address: Operational Base + offset (0x000c)
Codec register 1

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	PINDIR Choose the input or output direction of the pin which is related to the I2S mode. 1'b0: Input(Slave Mode) 1'b1: Output(Master Mode)
4	RW	0x0	I2SMODE DAC I2S Mode Select 1'b0: Slave mode• 1'b1: Master mode
3:0	RO	0x0	reserved

Codec_REG2
Address: Operational Base + offset (0x0010)
Codec register 2

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	LRCP DAC LRC Polarity 1'b0: Normal 1'b1: Reversal

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Bit	Attr	Reset Value	Description
6:5	RW	0x2	VWL DAC Valid Word Length in one 1/2 Frame 2'b11: 32 bits 2'b10: 24 bits 2'b01: 20 bits 2'b00: 16 bits
4:3	RW	0x2	DACM DAC mode 2'b11: PCM Mode 2'b10: I2S Mode 2'b01: Left Justified Mode 2'b00: Right Justified Mode Note. Same word length in 1/2frame and valid data is not supported in Right Justified Mode. For example, 32/24 or 24/20 is supported, but 32/32 or 24/24 is not supported. (1/2frame length/valid data length)
2	RW	0x0	LRS DAC Left-Right Swap 1'b0: Normal 1'b1: Swap
1:0	RO	0x0	reserved

Codec_REG3
Address: Operational Base + offset (0x0014)
Codec register 3

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	CSRC Choose the digital source of the right channel 00:Choose the I2S right channel as the digital audio source. 01:Choose the sine wave generated by the inner-bist module. 10:Choose the square wave generated by the inner-bist module. 11:Choose the I2S left channel as the digital audio source.
3:2	RW	0x3	FWL Choose the word length in 1/2 Frame of the DAC I2S interface. 2'b11: 32 bits 2'b10: 24 bits 2'b01: 20 bits 2'b00: 16 bits
1	RW	0x1	DACR The reset signal of the DAC I2S module. 1'b0: Keep the current state 1'b1: Reset

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Bit	Attr	Reset Value	Description
0	RW	0x0	BCP DAC Bit Clock polarity 1'b0: Keep the current polarity 1'b1: Invert the polarity

Codec_REG4
Address: Operational Base + offset (0x0034)
Codec register 4

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	Choose the digital audio source of the left channel. 00:Choose the I2S right channel as the digital audio source. 01:Choose the sine wave generated by the inner-bist module. 10:Choose the square wave generated by the inner-bist module. 11:Choose the I2S right channel as the digital audio source.
5:4	RW	0x0	Choose the digital audio source of the right channel. 00:Choose the I2S right channel as the digital audio source. 01:Choose the sine wave generated by the inner-bist module. 10:Choose the square wave generated by the inner-bist module. 11:Choose the I2S left channel as the digital audio source.
3:0	RO	0x0	reserved

Codec_REG5
Address: Operational Base + offset (0x0088)
Codec register 5

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	The control signal of the PreCharge module for the DAC. 0:Discharge 1:Precharge
6	RW	0x0	CPC6 Choose the Precharge Current. 0:Choose the current 64*I0 1:Don't choose the current 64*I0
5	RW	0x0	CPC5 Choose the Precharge Current. 0:Choose the current 32*I0 1:Don't choose the current 32*I0
4	RW	0x0	CPC4 Choose the Precharge Current. 0:Choose the current 16*I0 1:Don't choose the current 16*I0

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Bit	Attr	Reset Value	Description
3	RW	0x0	CPC3 Choose the Precharge Current. 0:Choose the current 8*I0 1:Don't choose the current 8*I0
2	RW	0x0	CPC2 Choose the Precharge Current. 0:Choose the current 4*I0 1:Don't choose the current 4*I0
1	RW	0x0	CPC1 Choose the Precharge Current. 0:Choose the current 2*I0 1:Don't choose the current 2*I0
0	RW	0x1	CPC0 Choose the Precharge Current. 0:Choose the current I0 1:Don't choose the current I0

Codec_REG6
Address: Operational Base + offset (0x008c)
Codec register 6

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	The enable signal of current source for CODEC DAC. 0:Disable 1:Enable
5	RW	0x0	The enable signal of reference voltage buffer for left DAC path. 0:Stop working 1:Work
4	RW	0x0	The enable signal of zero-crossing detection module for HPOUTL. 0:Stop Working, output 0 electrical level 1:Work
3:2	RO	0x0	reserved
1	RW	0x0	The enable signal of reference voltage buffer for right DAC path. 0:Stop working 1:Work
0	RW	0x0	The enable signal of zero-crossing detection module for HOUTR. 0:Stop Working, output 0 electrical level 1:Work

Codec_REG7
Address: Operational Base + offset (0x0090)
Codec register 7

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RW	0x0	The enable signal of reference voltage buffer for DACL module. 0:Stop working 1:Work
6	RW	0x0	The enable signal of CLOCK module for DACL. 0:Set CLOCK to logic "1" 1:Work
5	RW	0x0	The enable signal of DACL module. 0:Stop work 1:Work
4	RW	0x0	The Initial signal of DACL module. 0:Initialization 1:Work
3	RW	0x0	The enable signal of reference voltage buffer for DACR module. 0:Stop working 1:Work
2	RW	0x0	The enable signal of CLOCK module for DACR. 0:Set CLOCK to logic "1" 1:Work
1	RW	0x0	The enable signal of DACR module. 0:Stop work 1:Work
0	RW	0x0	The Initial signal of DACR module. 0:Initialization 1:Work

Codec_REG8
Address: Operational Base + offset (0x0094)
Codec register 8

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	The enable signal of HPMIXL module. 0:Stop work 1:Work
5	RW	0x0	The initial1 signal of HPMIXL module. 0:Initialization 1:Work
4	RW	0x0	The initial2 signal of HPMIXL module. 0:Initialization 1:Work
3	RO	0x0	reserved
2	RW	0x0	The enable signal of HPMIXR module. 0:Stop work 1:Work

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Bit	Attr	Reset Value	Description
1	RW	0x0	The initial1 signal of HPMIXR module. 0:Initialization 1:Work
0	RW	0x0	The initial2 signal of HPMIXR module. 0:Initialization 1:Work

Codec_REG9
Address: Operational Base + offset (0x0098)
Codec register 9

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	Select output of DACL module. 0:Don't select 1:Select
3:1	RO	0x0	reserved
0	RW	0x0	The signal to select input of HPMIXR module. 0:Don't select 1:Select

Codec_REG10
Address: Operational Base + offset (0x009c)
Codec register 10

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	The enable signal of HPOUTL module. 0:Stop working 1:Work
6	RW	0x0	The Initial signal of HPOUTL module. 0:Initialization 1:Work
5	RW	0x0	The mute signal of HPOUTL module. 0:MUTE 1:Work
4	RW	0x0	The enable signal of HPOUTR module. 0:Stop working 1:Work
3	RW	0x0	The Initial signal of HPOUTR module. 0:Initialization 1:Work
2	RW	0x0	The mute signal of HPOUTR module. 0:MUTE 1:Work
1:0	RO	0x0	reserved

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Codec_REG11
Address: Operational Base + offset (0x00a0)
Codec register 11

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	HPOUTLGAIN The signal to select gain of HPOUTL module: 5'b00000: -39dB 5'b11010: 6dB 5'b11111: 0dB Step: 1.5dB

Codec_REG12
Address: Operational Base + offset (0x00a4)
Codec register 12

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	HPOUTRGAIN The signal to select gain of HPOUTR module: 5'b00000: -39dB 5'b11010: 6dB 5'b11111: 0dB Step: 1.5dB

Codec_REG13
Address: Operational Base + offset (0x00a8)
Codec register 13

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	HPOUTLPOPCTRL The POP Sound Control signal of the HPOUTR module. 2'b01: Precharge(or Discharge) 2'b10: Work
3:2	RO	0x0	reserved
1:0	RW	0x1	HPOUTRPOPCTRL The POP Sound Control signal of the HPOUTL module. 2'b01: Precharge(or Discharge) 2'b10: Work

2.5 Application Notes

This chapter will give some reference register configurations about how to power up/off Codec and make the codec in the stand by state.
There are some GRF register bits to control the connection between codec and i2s as below. GRF_SOC_CON2 [14] gates the input signal pin_dac_ws_i and pin_dac_sd_i of codec, please set GRF_SOC_CON2 [14] = 1 if acodec is used.
When the ACODEC acts as a master, the signal i2s1_lrck_tx_in which connected to I2S1 can be selected from ACODEC or external IO by setting GRF_SOC_CON2[15]. Please set

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GRF_SOC_CON2 [15] = 1 if acodec is used.

2.5.1 POWER UP

This section will give the suggest configuration steps to power up the Codec.
Step1: Supply the power of the digital part and reset the audio codec.
Step2: Configure the register Codec_REG5[6:0] to 7'h1.
Step3: Configure the register Codec_REG5[7] to 0 to enable the precharge function.
Step4: Configure the register Codec_REG5[6:0] to 7'b111_1111. The suggestion slot time of the step is 20ms.
Step5: Wait until the voltage of VCM keeps stable at the AVDD/2.
Step6: Configure the register Codec_REG5[6:0] to choose a small current to decrease the power consumption.
Notes:The suggestion slot of these steps is 10ms.

2.5.2 POWER OFF

This section will give the suggest configuration steps to power off the Codec.
Step1: Keep the power on and disable DAC according to the description of section DISABLE DAC.
Step2: Configure the register Codec_REG5[6:0] to 7'h1.
Step3: Configure the register Codec_REG5[7] to 1 to begin to discharge.
Step4: Configure the Precharge current through the register Codec_REG5[6:0]. Change the register Codec_REG5[6:0] to 7'h3f. The suggestion slot time of the step is 20ms.
Step5: Wait until the voltage of VCM keeps stable at AGND.
Step6: Configure the register Codec_REG5[6:0] to 7'h1.

2.5.3 STAND-BY

Step1: Keep the power on and disable DAC according to the description of section DISABLE DAC.
Step2: Configure the register Codec_REG5[6:0] to 7'h1.

2.5.4 ENABLE DAC

This section will give the suggestion configuration steps to enable the DAC.

Table 27-3 The suggestion configuration steps to enable the DAC

step	description
0	Power up the Codec and input the mute signal.
1	Set the register Codec_REG6[6] to 1 to enable the current source of Codec DAC.
2	Set the register Codec_REG6[5] to 1 to enable the reference voltage buffer of left DACL path. Set the register Codec_REG6[1] to 1 to enable the reference voltage buffer of left DACR path.
3	Set the register Codec_REG13[5:4] to 2'b10 to enable the POP sound control signal of HPOUTR module. Set the register Codec_REG13[1:0] to 2'b10 to enable the POP sound control signal of HPOUTL module.
4	Set the register Codec_REG8[6] to 1 to enable the HPMIXL module. Set the register Codec_REG8[2] to 1 to enable the HPMIXR module.
5	Set the register Codec_REG8[5] to 1 to enable the INITIAL1 signal of the HPMIXL module. Set the register Codec_REG8[1] to 1 to enable the INITIAL1 signal of the HPMIXR module.
6	Set the register Codec_REG10[7] to 1 to enable the HPOUTL module. Set the register Codec_REG10[4] to 1 to enable the HPOUTR module.
7	Set the register Codec_REG10[6] to 1 to initial the HPOUTL module. Set the register Codec_REG10[3] to 1 to initial the HPOUTR module.
8	Set the register Codec_REG7[7] to 1 to enable the voltage reference module of DACL. Set the register Codec_REG7[3] to 1 to enable the voltage reference module of DACR.

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9	Set the register Codec_REG7[6] to 1 to enable the clock module of DACL module. Set the register Codec_REG7[2] to 1 to enable the clock module of DACR module.
10	Set the register Codec_REG7[5] to 1 to enable the DACL module. Set the register Codec_REG7[1] to 1 to enable the DACR module.
11	Set the register Codec_REG7[4] to 1 to initial the DACL module. Set the register Codec_REG7[0] to 1 to initial the DACR module.
12	Set the register Codec_REG9[4] to 1 to enable the input of DACL. Set the register Codec_REG9[0] to 1 to enable the input of DACR.
13	Set the register Codec_REG8[4] to 1 to enable the INITIAL2 signal of the HPMIXL module. Set the register Codec_REG8[0] to 1 to enable the INITIAL2 signal of the HPMIXR module.
14	Set the register Codec_REG10[5] to 1 to disable the mute signal of HPOUTL module. Set the register Codec_REG10[2] to 1 to disable the mute signal of HPOUTR module.
15	Set the register Codec_REG11[4:0] to select the gain of HPOUTL module. Set the register Codec_REG12[4:0] to select the gain of HPOUTR module.
16	Input the music

Notes:The suggestion slot of these steps is 10ms.

2.5.5 DISABLE DAC

This section will give the suggestion configuration steps to disable the DAC.

Table 27-4 The suggestion configuration steps to disable the DAC

step	description
0	Keep the DAC work and input the mute signal.
1	Set the register Codec_REG11[4:0] to 0. Set the register Codec_REG12[4:0] to 0.
2	Set the register Codec_REG8[4] to 0. Set the register Codec_REG8[0] to 0.
3	Set the register Codec_REG9[4] to 0. Set the register Codec_REG9[0] to 0.
4	Set the register Codec_REG10[5] to 0. Set the register Codec_REG10[2] to 0.
5	Set the register Codec_REG10[6] to 0. Set the register Codec_REG10[3] to 0.
6	Set the register Codec_REG10[7] to 0. Set the register Codec_REG10[4] to 0.
7	Set the register Codec_REG8[6] to 0. Set the register Codec_REG8[2] to 0.
8	Set the register Codec_REG7[5] to 0. Set the register Codec_REG7[1] to 0.
9	Set the register Codec_REG7[6] to 0. Set the register Codec_REG7[2] to 0.
10	Set the register Codec_REG7[7] to 0. Set the register Codec_REG7[3] to 0.
11	Set the register Codec_REG13[5:4] to 00. Set the register Codec_REG13[1:0] to 00.
12	Set the register Codec_REG6[5] to 0. Set the register Codec_REG6[1] to 0.
13	Set the register Codec_REG6[6] to 0.
14	Set the register Codec_REG8[5] to 0. Set the register Codec_REG8[1] to 0.
15	Set the register Codec_REG7[4] to 0. Set the register Codec_REG7[0] to 0.

Notes:The suggestion slot of these steps is 10ms.

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Chapter 28 Smart Card Reader (SCR)

3.1 Overview

The Smart Card Reader (SCR) is a communication controller that transmits data between the superior system and the Smart Card. The controller can perform a complete smart card session, including card activation, card deactivation, cold/warm reset, Answer to Reset (ATR) response reception, data transfers, etc.
SCR supports the following features:

- Supports the ISO/IEC 7816-3:1997(E) and EMV2000 (4.0) specifications
- Performs functions needed for complete smart card sessions, including:
  - Card activation and deactivation
  - Cold/warm reset
  - Answer to Reset (ATR) response reception
  - Data transfers to and from the card
- Extensive interrupt support system
- Adjustable clock rate and bit (baud) rate
- Configurable automatic byte repetition
- Handles commonly used communication protocols:
  - T=0 for asynchronous half-duplex character transmission
  - T=1 for asynchronous half-duplex block transmission
- Automatic convention detection
- Configurable timing functions:
  - Smart card activation time
  - Smart card reset time
  - Guard time
  - Timeout timers
- Automatic operating voltage class selection
- Supports synchronous and any other non-ISO 7816 and non-EMV cards
- Advanced Peripheral Bus (APB) slave interface for easy integration with AMBA-based host systems Block Diagram

3.2 Block Diagram

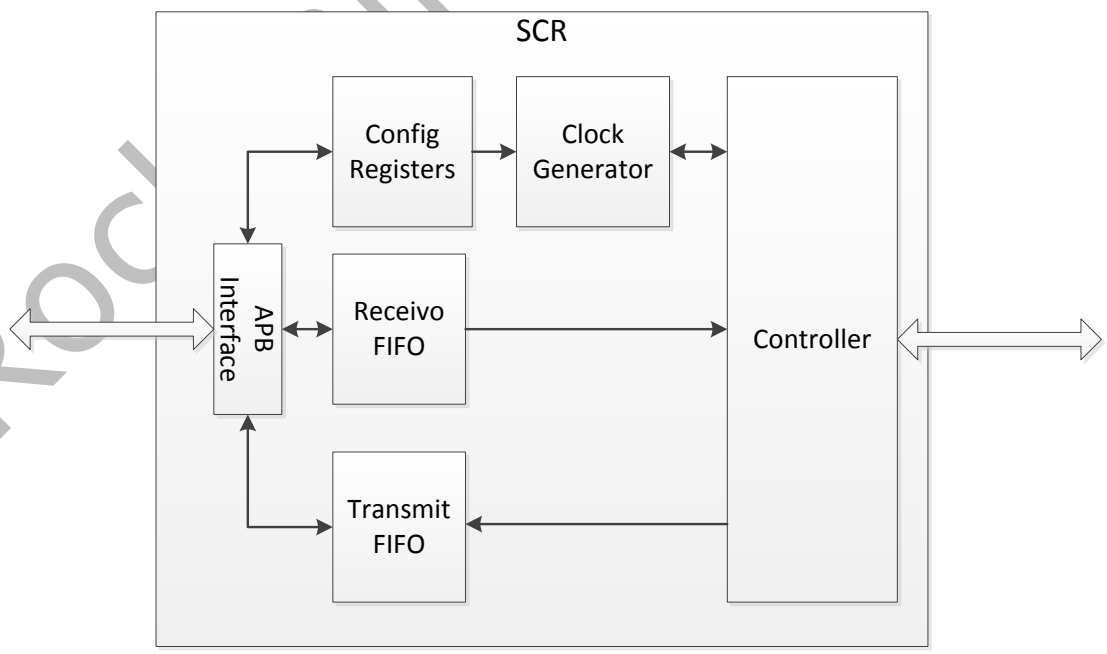

Fig.28-1 SCR Block Diagram

The host processor gets access to PWM Register Block through the APB slave interface.

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3.2.1 APB Interface

The host processor gets access to PWM Register Block through the APB slave interface.

3.2.2 Configuration Registers

The Configuration Registers block provides control over all functions of the Smart Card Reader

3.2.3 Controller

The Controller is the main block in the SCR core. This block controls receiving characters transmitted by the Smart Card, storing them in the RX FIFO, and transmitting them to the Smart Card. This block also performs card activation, deactivation, and cold and warm reset. After the card is reset, the Answer To Reset (ATR) sequence is received by the controller and stored in RX FIFO.
The parallel to serial conversion needed to transmit data from a Smart Card Reader to a Smart Card and the serial to parallel conversion needed to transmit data in the opposite direction is performed by the UART. The UART also performs the guard time, parity checking and character repeating functions.

3.2.4 Receive FIFO

The Receive FIFO is used to store the data received from the Smart Card until the data is read out by the superior system.

3.2.5 Transmit FIFO

The Transmit FIFO is used to store the data to be transmitted to the Smart Card.

3.2.6 Clock Generator

The Clock Generator generates the Smart Card Clock signal and the Baud Clock Impulse signal, used in timing the Smart Card Reader.)

3.3 Function Description

A Smart Card session consists of following stages:
1. Smart Card insertion
2. Activation of contacts and cold reset sequence
3. Answer To Reset sequence (ATR)
4. Execution of transaction
5. Deactivation of contacts
6. Smart Card removal

3.3.1 Smart Card Insertion

A Smart Card session starts with the insertion of the Smart Card. This event is signaled to the SCR using the SCDETECT input. The SCPRESENT bit is set and also the SCINS interrupt is asserted (if enabled).
When the external card detect switch is not used, the input pin SCDETECT must be tied to inactive state.

3.3.2 Automatic operating voltage class selection

There are three operating classes (1.8V - class C, 3V - class B and 5V - class A) defined in ISO/IEC 7816-3(2006) specification. Only 1.8V and 3.3V are supported by the SCR.
Before the activation of contacts, operating classes have to be enabled via bits VCC18, VCC33 in CTRL2 register. In case that no operating class is enabled, the controller performs activation for all two voltage classes (1.8V, 3V) in sequence.
When Smart Card Reader performs activation of contacts the lowest enabled voltage class is automatically applied first. When the first character start bit of ATR sequence is received, the selected voltage class is correct (even if the ATR is then received with errors). When the ATR sequence reception does not start, ATRFAIL interrupt is not activated, deactivation is performed and next higher enabled voltage class is applied. If the ATR sequence reception does not start and no other higher class is enabled was already applied the ATRFAIL interrupt is activated and the last applied voltage class remains active.
After the automatic voltage class selection is finished the selected class can be read from bits

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VCC18, VCC33 in CTRL2 register. If the automatic voltage class selection fails, these bits remain untouched.
There is a delay applied between deactivation of contacts with lower voltage class and activation of contacts with higher voltage class. This delay should be at least 10 ms according to the ISO/IEC 7816-3 specification.

3.3.3 Activation of Contacts and Cold Reset Sequence

When the Smart Card is properly inserted and the ACT bit in CTRL2 register is asserted, the activation of contacts can be started. The duration of each part of the activation is the time Ta, which is equal to the ADEATIME register value. If no Vpp is necessary, the activation and deactivation part of Vpp can be omitted by clearing the AUTOADEAVPP bit in SCPADS register. The Cold Reset sequence follows immediately after the activation. Time (Tc) is the duration of the Reset. The EMV specification recommends that this value should be between 40000 and 45000. The activation of contacts and cold reset sequence is shown inFig.28-2.

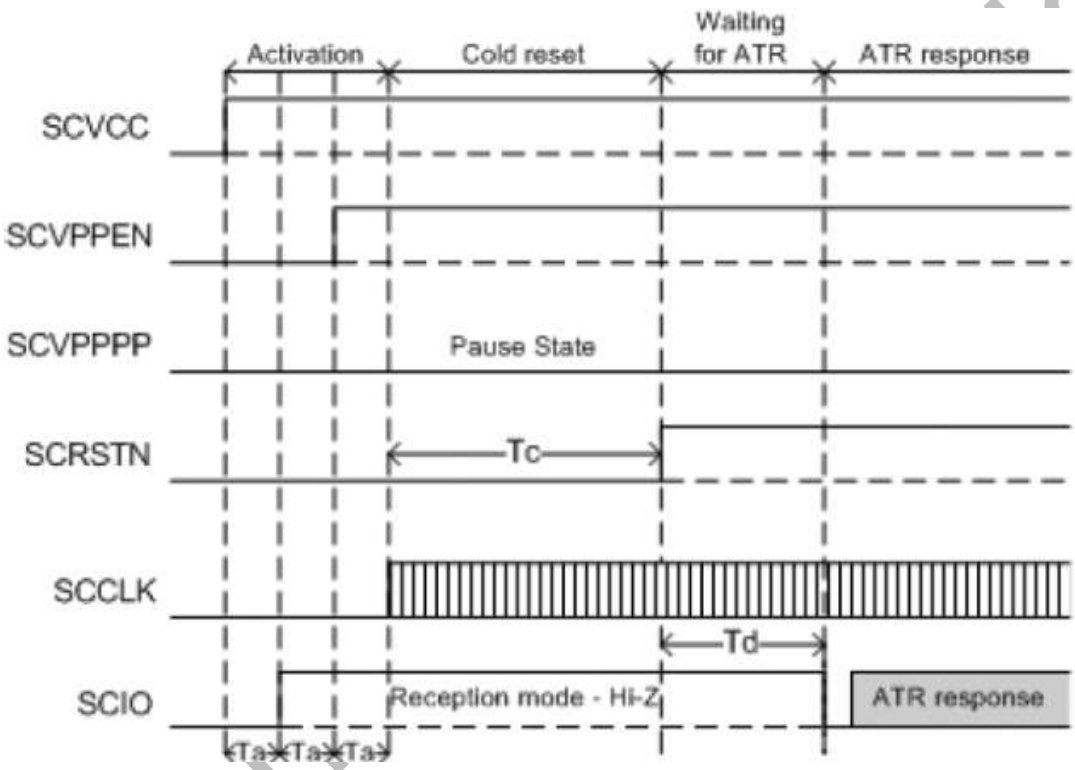

Fig.28-2 Activation, Cold Reset and ATR

3.3.4 Execution of Transaction

All transfers between the Smart Card Reader and a Smart Card are under the control of the superior system. It controls the number of characters sent to the Smart Card and it knows the number of characters expected to be returned from the Smart Card.

3.3.5 Warm Reset

The Warm Reset sequence is initialized by setting the WRST bit in the CTRL2 register to '1'.Smart Card Reader drives the SCRSTN signal to '0' to perform the Warm Reset as shown in Fig.28-3. After the SCRSTN assertion, the Warm Reset sequence then continues the same way as the Cold Reset sequence.

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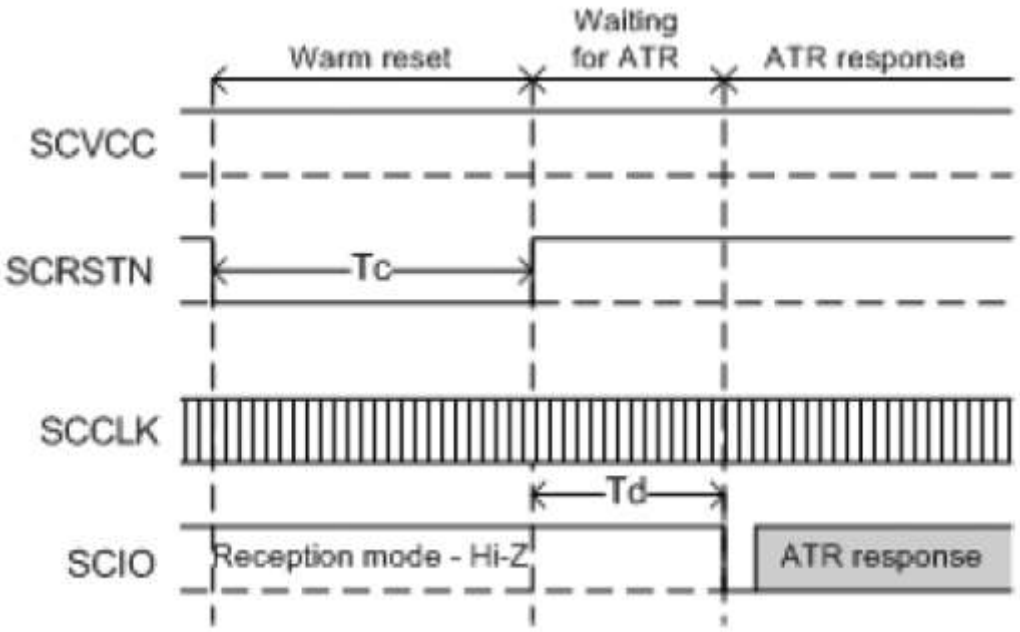

Fig.28-3 Warm Reset and ATR

3.3.6 Deactivation of Contacts

After the smart card reader detects the removal of the smart card (SCREM interrupt) or the superior system initiates deactivation by setting the DEACT bit in the CTRL2 register to '1', the deactivation is performed immediately as shown in . The duration time (Ta), of each part of the deactivation sequence time is defined in the ADEATIME register.

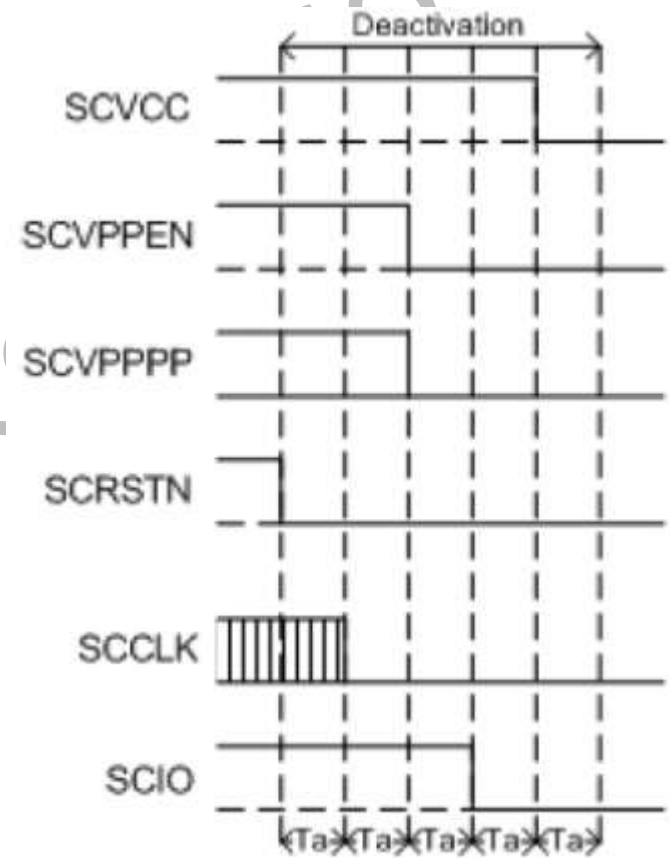

Fig.28-4 Deactivation Sequence

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3.4 Register Description

3.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
SCR_CTRL1	0x0000	HW	0x0000	Control Register 1
SCR_CTRL2	0x0004	HW	0x0000	Control Register 2
SCR_SCPADS	0x0008	HW	0x0000	Smart Card Pads Register
SCR_INTEN1	0x000c	HW	0x0000	Interrupt Enable Register 1
SCR_INTSTAT1	0x0010	HW	0x0000	Interrupt Status Register 1
SCR_FIFOCTRL	0x0014	HW	0x0000	FIFO Control Register
SCR_LEGTXFICNT	0x0018	B	0x00	Legacy TX FIFO Counter
SCR_LEGRXFICNT	0x0019	B	0x00	Legacy RX FIFO Counter
SCR_RXFITH	0x001c	HW	0x0000	RX FIFO Threshold
SCR_REP	0x0020	B	0x00	Repeat
SCR_SCCDDIV	0x0024	HW	0x0000	Smart Card Clock Divisor
SCR_BAUDDIV	0x0028	HW	0x0000	Baud Clock Divisor
SCR_SCGUTIME	0x002c	B	0x00	Smart Card Guard-time
SCR_ADEATIME	0x0030	HW	0x0000	Activation / Deactivation Time
SCR_LOWRSTTIME	0x0034	HW	0x0000	Reset Duration
SCR_ATRSTARTLIMIT	0x0038	HW	0x0000	ATR Start Limit
SCR_C2CLIM	0x003c	HW	0x0000	Two Characters Delay Limit
SCR_INTEN2	0x0040	HW	0x0000	Interrupt Enable Register 2
SCR_INTSTAT2	0x0044	HW	0x0000	Interrupt Status Register 2
SCR_TXFITH	0x0048	HW	0x0000	TX FIFO Threshold
SCR_TXFIFOCNT	0x004c	HW	0x0000	TX FIFO Counter
SCR_RXFIFOCNT	0x0050	HW	0x0000	RX FIFO Counter
SCR_BAUDTUNE	0x0054	B	0x00	Baud Tune Register

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Name	Offset	Size	Reset Value	Description
SCR_FIFO	0x0200	B	0x00	FIFO

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

3.4.2 Detail Register Description

SCR_CTRL1
Address: Operational Base + offset (0x0000)
Control Register 1

Bit	Attr	Reset Value	Description
15	RW	0x0	GINTEN Global Interrupt Enable When high, INTERRUPT output assertion is enabled.
14	RO	0x0	reserved
13	RW	0x0	TCKEN TCK enable When enabled all ATR bytes beginning from T0 are being XOR-ed. The result must be equal to TCK byte (when present). If the TCK byte does not match the computed value the ATR is considered to be malformed.
12	RW	0x0	ATRSTFLUSH ATR Start Flush FIFO When enabled, both FIFOs are flushed before the ATR is started.
11	RW	0x0	T0T1 T0/T1 Protocol Controls the using of T=0 or T=1 protocol. No character repeating is used when T=1 protocol is selected. The Character Guard-time (minimum delay between the leading edges of two consecutive characters) is reduced to 11 ETU when T=1 protocol is used and Guard-time value N = 255. The delay between the leading edge of the last received character and the leading edge of the first character transmitted is 16 ETU when T=0 protocol is used and 22 ETU when T=1 protocol is used.
10	RW	0x0	TS2FIFO TS to FIFO Enables to store the first ATR character TS in RX FIFO. During ideal card session there is no necessity to store TS character, so it can be disabled
9	RW	0x0	RXEN Receiving enable When enabled the characters sent by the Smart Card are received by the UART and stored in RX FIFO. Receiving is internally disabled while a transmission is in progress.
8	RW	0x0	TXEN Transmission enable When enabled the characters are read from TX FIFO and transmitted through UART to the Smart Card

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Bit	Attr	Reset Value	Description
7	RW	0x0	CLKSTOPVAL Clock Stop Value The value of the scclk output during the clock stop state.
6	RW	0x0	CLKSTOP Clock Stop Clock Stop. When this bit is asserted and the smart card I/O line is in 'Z' state, the SCR core stops driving of the smart card clock signal after the CLKSTOPDELAY time expires. The smart card clock is restarted immediately after the CLKSTOP signal is de-asserted. New character transmission can be started by superior system after the CLKSTARTDELAY time expires. The expiration of both times is signaled by the CLKSTOPRUN bit in the Interrupt registers. Reading '1' from this bit signals that the clock is stopped or CLKSTARTDELAY time not expired yet. Reading '0' from this bit signals that the clock is not stopped.
5:3	RO	0x0	reserved
2	RW	0x0	PECH2FIFO Character With Wrong Parity to FIFO Enables storage of the characters received with wrong parity in RX FIFO.
1	RW	0x0	INVORD Inverse Bit Ordering When High, inverse bit ordering convention(MSB-LSB) is used.
0	RW	0x0	INVLEV Inverse Bit Level When high, inverse level convention is used(A= '1', Z='0');

SCR_CTRL2

Address: Operational Base + offset (0x0004)

Control Register 2

Bit	Attr	Reset Value	Description
15:8	RO	0x00	Reserved3 Reserved Reserved bits are hard-wired to zero
7	RW	0x0	VCC50 Control 5V Smart Card Vcc Control 5V Smart Card Vcc. Setting of this bit allows selection of 5V Vcc for Smart Card session (Class A). After the selection of operating class is completed, this bit is in '1' if this class was selected. Default value after reset is '0'.

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Bit	Attr	Reset Value	Description
6	RW	0x0	VCC33 Control 3V Smart Card Vcc Setting of this bit allows selection of 3V Vcc for Smart Card session (Class B). After the selection of operating class is completed, this bit is in '1' if this class was selected. Default value after reset is '0'.
5	RW	0x0	VCC18 Control 1.8V Smart Card Vcc Control 1.8V Smart Card Vcc. Setting of this bit allows selection of 1.8V Vcc for Smart Card session (Class C). After the selection of operating class is completed, this bit is in '1' if this class was selected. Default value after reset is '0'.
4	RW	0x0	DEACT Deactivation Setting of this bit initializes the deactivation sequence. When the deactivation is finished, the DEACT bit is automatically cleared.
3	RW	0x0	ACT Activation Setting of this bit initializes the activation sequence. When the activation is finished, the ACT bit is automatically cleared.
2	WO	0x0	WARMRST Warm Reset Command Writing '1' to this bit initializes Warm Reset of the Smart Card. This bit is always read as '0'.
1:0	RO	0x0	reserved

SCR_SCPADS
Address: Operational Base + offset (0x0008)
Smart Card Pads Register

Bit	Attr	Reset Value	Description
15:10	RO	0x0	reserved
9	RO	0x0	SCPRESENT Smart Card presented This bit is set to '1' when the SCDETECT input is active at least for SCDETECTTIME
8	RW	0x0	DSCFCB Direct Smart Card Function Code Bit It provides direct access to SCFCB output
7	RW	0x0	DSCVPPPP Direct Smart Card Vpp Pause/Prog It provides direct access to SCVPPPP output
6	RW	0x0	DSCVPPEN Direct Smart Card Vpp Enable It provides direct access to SCVPPEN output

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Bit	Attr	Reset Value	Description
5	RW	0x0	AUTOADEAVPP Automatic Vpp Handling. When high, it enables automatic handling of DSCVPPEN and DSCVPPPP signals during activation and deactivation sequence.
4	RW	0x0	DSCVCC Direct Smart Card Vcc Direct Smart Card Vcc. When DIRACCPADS = '1', the DSCVCC bit provides direct access to SCVCCx outputs. The appropriate SCVCC18, SCVCC33 and SCVCC50 outputs are driven according to state of bits VCC18, VCC33 and VCC50 in CTRL2 register.
3	RW	0x0	DSCRST Direct Smart Card Reset When DIRACCPADS = '1', the DSCRST bit provides direct access to SCRST output
2	RW	0x0	DSCCLK Direct Smart Card Clock When DIRACCPADS = '1', the DSCCLK bit provides direct access to SCCLK output
1	RW	0x0	DSCIO Direct Smart Card Input/Output When DIRACCPADS = '1', the DSCIO bit provides direct access to SCIO pad.
0	RW	0x0	DIRACCPADS Direct Access To Smart Card Pads When high, it disables a serial interface functionality and enables direct control of the smart card pads using following 4 bits.

SCR_INTEN1
Address: Operational Base + offset (0x000c)
Interrupt Enable Register 1

Bit	Attr	Reset Value	Description
15	RW	0x0	SCDEACT Smart Card Deactivation Interrupt When enabled, this interrupt is asserted after the Smart Card deactivation sequence is complete.
14	RW	0x0	SCACT Smart Card Activation Interrupt. When enabled, this interrupt is asserted after the Smart Card activation sequence is complete.
13	RW	0x0	SCINS Smart Card Inserted Interrupt When enabled, this interrupt is asserted after the smart card insertion

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Bit	Attr	Reset Value	Description
12	RW	0x0	SCREM Smart Card Removed Interrupt. When enabled, this interrupt is asserted after the smart card removal.
11	RW	0x0	ATRDONE ATR Done Interrupt When enabled, this interrupt is asserted after the ATR sequence is successfully completed.
10	RW	0x0	ATRFAIL ATR Fail Interrupt When enabled, this interrupt is asserted if the ATR sequence fails.
9	RW	0x0	RXTHRESHOLD RX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in RX FIFO is equal or exceeds the RX FIFO threshold.
8	RW	0x0	C2CFULL Two Consecutive Characters Limit Interrupt When enabled, this interrupt is asserted if the time between two consecutive characters, transmitted between the Smart Card and the Reader in both directions, is equal the Two Characters Delay Limit described below. The C2CFULL interrupt is internally enabled from the ATR start to the deactivation or ATR restart initialization. It is recommended to use this counter to detect unresponsive Smart Cards.
7	RW	0x0	RXPERR Reception Parity Error Interrupt When enabled, this interrupt is asserted after the character with wrong parity was received when the number of repeated receptions exceeds RXREPEAT value or T=1 protocol is used
6	RW	0x0	TXPERR Transmission Parity Error Interrupt. When enabled, this interrupt is asserted if the Smart Card signals wrong character parity during the guard-time after the character transmission was repeated TXREPEAT-times
5	RW	0x0	RXDONE Reception Done Interrupt When enabled, this interrupt is asserted after a character was received from the Smart Card.
4	RW	0x0	TXDONE Transmission Done Interrupt When enabled, this interrupt is asserted after one character was transmitted to the Smart Card.

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Bit	Attr	Reset Value	Description
3	RW	0x0	CLKSTOPRUN Smart Card Clock Stop Interrupt When enabled, this interrupt is asserted in two cases: 1. When the smart card clock is stopped (after CLOCKSTOP assertion). 2. When the new character transfer can be started (the smart card clock is fully running after CLOCKSTOP de-assertion).
2	RW	0x0	RXFIFULL RX FIFO Full Interrupt When enabled, this interrupt is asserted if the RX FIFO is filled up.
1	RW	0x0	TXFIEMPTY TX FIFO Empty Interrupt. When enabled, this interrupt is asserted if the TX FIFO is emptied out.
0	RW	0x0	TXFIDONE TX FIFO Done Interrupt When enabled, this interrupt is asserted after all bytes from TX FIFO were transferred to the Smart Card

SCR_INTSTAT1
Address: Operational Base + offset (0x0010)
Interrupt Status Register 1

Bit	Attr	Reset Value	Description
15	RW	0x0	SCDEACT Smart Card Deactivation Interrupt When enabled, this interrupt is asserted after the Smart Card deactivation sequence is complete.
14	RW	0x0	SCACT Smart Card Activation Interrupt. When enabled, this interrupt is asserted after the Smart Card activation sequence is complete.
13	RW	0x0	SCINS Smart Card Inserted Interrupt When enabled, this interrupt is asserted after the smart card insertion
12	RW	0x0	SCREM Smart Card Removed Interrupt. When enabled, this interrupt is asserted after the smart card removal.
11	RW	0x0	ATRDONE ATR Done Interrupt When enabled, this interrupt is asserted after the ATR sequence is successfully completed.

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Bit	Attr	Reset Value	Description
10	RW	0x0	ATRFAIL ATR Fail Interrupt When enabled, this interrupt is asserted if the ATR sequence fails.
9	RW	0x0	RXTHRESHOLD RX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in RX FIFO is equal or exceeds the RX FIFO threshold.
8	RW	0x0	C2CFULL Two Consecutive Characters Limit Interrupt When enabled, this interrupt is asserted if the time between two consecutive characters, transmitted between the Smart Card and the Reader in both directions, is equal the Two Characters Delay Limit described below. The C2CFULL interrupt is internally enabled from the ATR start to the deactivation or ATR restart initialization. It is recommended to use this counter to detect unresponsive Smart Cards.
7	RW	0x0	RXPERR Reception Parity Error Interrupt When enabled, this interrupt is asserted after the character with wrong parity was received when the number of repeated receptions exceeds RXREPEAT value or T=1 protocol is used
6	RW	0x0	TXPERR Transmission Parity Error Interrupt. When enabled, this interrupt is asserted if the Smart Card signals wrong character parity during the guard-time after the character transmission was repeated TXREPEAT-times
5	RW	0x0	RXDONE Reception Done Interrupt When enabled, this interrupt is asserted after a character was received from the Smart Card.
4	RW	0x0	TXDONE Transmission Done Interrupt When enabled, this interrupt is asserted after one character was transmitted to the Smart Card.
3	RW	0x0	CLKSTOPRUN Smart Card Clock Stop Interrupt When enabled, this interrupt is asserted in two cases: 1. When the smart card clock is stopped (after CLOCKSTOP assertion). 2. When the new character transfer can be started (the smart card clock is fully running after CLOCKSTOP de-assertion).
2	RW	0x0	RXFIFULL RX FIFO Full Interrupt When enabled, this interrupt is asserted if the RX FIFO is filled up.

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Bit	Attr	Reset Value	Description
1	RW	0x0	TXFIEMPTY TX FIFO Empty Interrupt. When enabled, this interrupt is asserted if the TX FIFO is emptied out.
0	RW	0x0	TXFIDONE TX FIFO Done Interrupt When enabled, this interrupt is asserted after all bytes from TX FIFO were transferred to the Smart Card

SCR_FIFOCTRL
Address: Operational Base + offset (0x0014)
FIFO Control Register

Bit	Attr	Reset Value	Description
15:11	RO	0x0	reserved
10	WO	0x0	RXFIFLUSH Flush RX FIFO RX FIFO is flushed, when '1' is written to this bit.
9	RO	0x0	RXFIFULL RX FIFO Full RX FIFO Full
8	RO	0x0	RXFIEMPTY RX FIFO Empty RX FIFO Empty
7:3	RO	0x0	reserved
2	WO	0x0	TXFIFLUSH Flush TX FIFO. TX FIFO is flushed, when '1' is written to this bit.
1	RO	0x0	TXFIFULL TX FIFO Full TX FIFO Full
0	RO	0x0	TXFIEMPTY TX FIFO Empty. TX FIFO Empty.

SCR_LEGTXFICNT
Address: Operational Base + offset (0x0018)
Legacy TX FIFO Counter

Bit	Attr	Reset Value	Description
7:0	RO	0x00	LEGTXFICNT Legacy TX FIFO Counter It is equal to TX FIFO Counter up to value 255. All values above 255 are read as 255. It is recommended to use the 16-bit TX FIFO Counter instead of this register.

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SCR_LEGRXFICNT
Address: Operational Base + offset (0x0019)
Legacy RX FIFO Counter

Bit	Attr	Reset Value	Description
7:0	RO	0x00	LEGRXFICNT Legacy RX FIFO Counter It is equal to RX FIFO Counter up to value 255. All values above 255 are read as 255. It is recommended to use the 16-bit RX FIFO Counter instead of this register.

SCR_RXFITH
Address: Operational Base + offset (0x001c)
RX FIFO Threshold

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	RXFITH RX FIFO Threshold The interrupt is asserted when the number of bytes it receives is equal to, or exceeds the threshold

SCR_REP
Address: Operational Base + offset (0x0020)
Repeat

Bit	Attr	Reset Value	Description
7:4	RW	0x0	RXREP RX Repeat This is a 4-bit, read/write register that specifies the number of attempts to request character re-transmission after wrong parity was detected. The re-transmission of the character is requested using the 1 ETU long error signal during the guard-time
3:0	RW	0x0	TXREP TX Repeat This is a 4-bit, read/write register that specifies the number of attempts to re-transmit the character after the Smart Card signals the wrong parity during the guard-time.

SCR_SCCDDIV
Address: Operational Base + offset (0x0024)
Smart Card Clock Divisor

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	SCCDDIV Smart Card Clock Divisor This is a 16-bit, read/write register that defines the divisor value used to generate the Smart Card Clock from the system clock.

SCR_BAUDDIV
Address: Operational Base + offset (0x0028)

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Baud Clock Divisor

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	BAUDDIV Baud Clock Divisor This is a 16-bit, read/write register that defines a divisor value used to generate the Baud Clock impulses from the system clock

SCR_SCGUTIME
Address: Operational Base + offset (0x002c)
Smart Card Guard-time

Bit	Attr	Reset Value	Description
7:0	RW	0x00	SCGUTI Smart Card Guard-time This is an 8-bit, read/write register that sets a delay at the end of each character transmitted from the Smart Card Reader to the Smart Card. The value is in Elementary Time Units (ETU). The parity error is besides signaled during the guardtime

SCR_ADEATIME
Address: Operational Base + offset (0x0030)
Activation / Deactivation Time

Bit	Attr	Reset Value	Description
15:8	RW	0x00	ADEATIME Activation / Deactivation Time Sets the duration of each part of the activation and deactivation sequence. The value is in Smart Card Clock Cycles.
7:0	RW	0x00	Reserved Reserved Reserved bits are hard-wired to zero.

SCR_LOWRSTTIME
Address: Operational Base + offset (0x0034)
Reset Duration

Bit	Attr	Reset Value	Description
15:8	RW	0x00	LOWRSTTIME Reset Duration Sets the duration of the smart card reset sequence. This value is same for the cold and warm reset. The value is in terms of smart card clock cycles.
7:0	RW	0x00	Reserved Reserved Bits (7:0) of this register are hard-wired to zero.

SCR_ATRSTARTLIMIT
Address: Operational Base + offset (0x0038)
ATR Start Limit

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Bit	Attr	Reset Value	Description
15:8	RW	0x00	ATRSTARTLIMIT ATR Start Limit Defines the maximum time between the rising edge of the SCRSTN signal and the start of ATR response. The value is in terms of smart card clock cycles
7:0	RW	0x00	Reserved Reserved Bits (7:0) of this register are hard-wired to zero

SCR_C2CLIM
Address: Operational Base + offset (0x003c)
Two Characters Delay Limit

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	C2CLIM Two Characters Delay Limit This is a 16-bit, read/write register that sets the maximum time between the leading edges of two, consecutive characters. The value is in ETUs.

SCR_INTEN2
Address: Operational Base + offset (0x0040)
Interrupt Enable Register 2

Bit	Attr	Reset Value	Description
15:2	RO	0x0	reserved
1	RW	0x0	TCKERR TCK Error Interrupt. When enabled, this interrupt is asserted if the TCK byte does not match computed value.
0	RW	0x0	TXTHRESHOLD TX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in TX FIFO is equal or less than the TX FIFO threshold.

SCR_INTSTAT2
Address: Operational Base + offset (0x0044)
Interrupt Status Register 2

Bit	Attr	Reset Value	Description
15:2	RO	0x0	reserved
1	RW	0x0	TCKERR TCK Error Interrupt When enabled, this interrupt is asserted if the TCK byte does not match computed value.

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Bit	Attr	Reset Value	Description
0	RW	0x0	TXTHRESHOLD TX FIFO Threshold Interrupt When enabled, this interrupt is asserted if the number of bytes in TX FIFO is equal or less than the TX FIFO threshold.

SCR_TXFITH
Address: Operational Base + offset (0x0048)
TX FIFO Threshold

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	TXFITH TX FIFO Threshold The interrupt is asserted when the number of bytes in TX FIFO is equal or less than the threshold

SCR_TXFIFOCNT
Address: Operational Base + offset (0x004c)
TX FIFO Counter

Bit	Attr	Reset Value	Description
15:0	RO	0x0000	TXFIFOCNT TX FIFO Counter This is a 16-bit, read-only register that provides the number of bytes stored in the RX FIFO

SCR_RXFIFOCNT
Address: Operational Base + offset (0x0050)
RX FIFO Counter

Bit	Attr	Reset Value	Description
15:0	RO	0x0000	RXFIFOCNT RX FIFO Counter This is a 16-bit, read-only register that provides the number of bytes stored in the RX FIFO.

SCR_BAUDTUNE
Address: Operational Base + offset (0x0054)
Baud Tune Register

Bit	Attr	Reset Value	Description
7:4	RO	0x0	reserved
3:0	RW	0x0	BAUDTUNE Baud Tune Register This is a 3-bit, read/write register that defines an additional value used to increase the accuracy of the Baud Clock impulses

SCR_FIFO
Address: Operational Base + offset (0x0200)
FIFO

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	FIFO FIFO This is an 8-bit, read/write register that provides access to the receive and transmit FIFO buffers. The TX FIFO is accessed during the APB write transfer. The RX FIFO is accessed during the APB read transfer. All read/write accesses at address range 200h-3ffh are redirected to the FIFO.

3.5 Interface Description

Table 28-1SCR Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
sc_clk	O	IO_CARDclkm0_GPIO3B4vccio6	GPIO3B_IOMUX[9:8]=01 GRF_CON_IOMUX[7]=0
		IO_I2S1sdi_PWMsdi0m0_CARDclkm1_GPIO2C3vccio5	GPIO2CL_IOMUX[11:9]=011 GRF_CON_IOMUX[7]=1
sc_rst	O	IO_CARDrstm0_GPIO3B5vccio6	GPIO3B_IOMUX[11:10]=01 GRF_CON_IOMUX[7]=0
		IO_I2S1sdio1_PDMsdi1m0_CARDrstm1_GPIO2C4vccio5	GPIO2CL_IOMUX[14:12]=011 GRF_CON_IOMUX[7]=1
sc_detect	I	IO_CARDdetm0_GPIO3B6vccio6	GPIO3B_IOMUX[13:12]=01 GRF_CON_IOMUX[7]=0
		IO_I2S1sdio2_PDMsdi2m0_CARDdetm1_GPIO2C5vccio5	GPIO2CH_IOMUX[2:0]=011 GRF_CON_IOMUX[7]=1
sc_io	I	IO_CARDiom0_GPIO3B7vccio6	GPIO3B_IOMUX[15:14]=01 GRF_CON_IOMUX[7]=0
		IO_I2S1sdio3_PDMsdi3m0_CARDiom1_GPIO2C6vccio5	GPIO2CH_IOMUX[5:3]=011 GRF_CON_IOMUX[7]=1

Notes: I=input, O=output, I/O=input/output, bidirectional

3.6 Application Notes

3.6.1 BCHST/BCHLOC/BCHDE/SPARE Application

The Smart Card Clock signal is used as the main clock for the smart card. Its frequency canbeadjusted using the Smart Card Clock Divisor (SCCDIV). This value is used to divide the system clock.
The SCCLK frequency is given by the following equation:

$$\mathrm{SCCLK_{freq}} = \frac{\mathrm{CLK_freq}}{2 * (\mathrm{SCCDIV} + 1)}, \qquad \mathrm{SCCDIV} \cong \frac{\mathrm{CLK_{freq}}}{2 * \mathrm{SCCLK_{freq}}} - 1$$

SCCLK_freq– Smart Card Clock Frequency
CLK_freq- System Clock Frequency
The Baud Clock Impulse signal is used to transmit and receive serial data between the Smart CardReader and the Smart Card. The baud rate can be modified using the Baud Clock Divisor (BAUDDIV)which is used to divide the system clock. The BAUDDIV value must be >= 4. The BAUD rate is given by
the following equation:

$$\mathrm{BAUD_{rate}} = \frac{\mathrm{CLK_freq}}{2 * (\mathrm{BAUDDIV} + 1)}$$

The duration of one bit, Elementary Time Unit (ETU) and parameters F and D are defined in the ISO/IEC7816-3 specification.

$$\frac{1}{\mathrm{BAUD_rate}} \cong \mathrm{ETU} = \frac{\mathrm{F}}{\mathrm{D}} * \frac{1}{\mathrm{SCCLK_{freq}}}, \frac{\mathrm{F}}{\mathrm{D}} \cong \frac{\mathrm{BAUDDIV} + 1}{\mathrm{SCCDIV} + 1}$$

BAUDDIV equation based on SCCDIV value and Smart Card parameters F and D is following:

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$$\text{BAUDDIV} \cong (\text{SCCDIV} + 1) * \frac{\text{F}}{\text{D}} - 1$$

During the first answer to reset response after the cold reset, the initial ETU must be equal to 372 SmartCard Clock Cycles (given by parameters F=372 and D=1). In this case, the BAUDDIV should be:

$$\text{BAUDDIV} \cong (\text{SCCDIV} + 1) * \frac{372}{1} - 1$$

After the ATR is completed, the BAUDDIV register value can be changed according to Smart Cardparameters F and D.

Baud Tune Register (BAUDTUNE) 3-bit value that can be used to increase the accuracy of the BaudClock impulses timing by using the BAUDTUNE Increment from Table listed below in combination with BAUDDIVregister value.

Table 28-2BAUDTUNE register

BAUDTUNE	000	001	010	011	100	101	110	111
$\text{BAUDTUNE}_{\text{INCR}}$	+0	+0.125	0.25	+0.375	+0.5	+0.625	+0.75	+0.875

$$\text{BAUDDIV} + \text{BAUDTUNE}_{\text{INCR}} \cong (\text{SCCDIV} + 1) * \frac{\text{F}}{\text{D}} - 1$$

The BAUDDIV register value (nearest integer) can be computed using following equation:

$$\text{BAUDDIV} \cong (\text{SCCDIV} + 1) * \frac{\text{F}}{\text{D}} - 1 - \text{BAUDTUNE}_{\text{INCR}}$$

3.6.2 Smart Card Detect Application

It is configurable for SCR's detect pin when Smart Card is inserted.
When config GRF_SOC_CON7[0]=0, SCDETECT`s active state is 0.
When config GRF_SOC_CON7[0]=1, SCDETECT`s active state is 1.

Chapter 29 Chapter 1 Serial Peripheral Interface (SPI)

4.1 Overview

The serial peripheral interface is an APB slave device. A four wire full duplex serial protocol from Motorola. There are four possible combinations for the serial clock phase and polarity. The clock phase (SCPH) determines whether the serial transfer begins with the falling edge of slave select signals or the first edge of the serial clock. The slave select line is held high when the SPI is idle or disabled. This SPI controller can work as either master or slave mode.
SPI Controller supports the following features:

- Support Motorola SPI,TI Synchronous Serial Protocol and National Semiconductor Micro wire interface
- Support 32-bit APB bus
- Support two internal 16-bit wide and 32-location deep FIFOs, one for transmitting and the other for receiving serial data
- Support two chip select signals in master mode
- Support 4,8,16 bit serial data transfer
- Support configurable interrupt polarity
- Support asynchronous APB bus and SPI clock
- Support master and slave mode
- Support DMA handshake interface and configurable DMA water level
- Support transmit FIFO empty, underflow, receive FIFO full, overflow, interrupt and all interrupts can be masked
- Support configurable water level of transmit FIFO empty and receive FIFO full interrupt
- Support combine interrupt output
- Support up to half of SPI clock frequency transfer in master mode and one sixth of SPI clock frequency transfer in slave mode
- Support full and half duplex mode transfer

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- Stop transmitting SCLK if transmit FIFO is empty or receive FIFO is full in master mode
- Support configurable delay from chip select active to SCLK active in master mode
- Support configurable period of chip select inactive between two parallel data in master mode
- Support big and little endian, MSB and LSB first transfer
- Support two 8-bit audio data store together in one 16-bit wide location
- Support sample RXD 0~3 SPI clock cycles later
- Support configurable SCLK polarity and phase
- Support fix and incremental address access to transmit and receive FIFO

4.2 Block Diagram

The SPI Controller comprises with:

- AMBA APB interface and DMA Controller Interface
- Transmit and receive FIFO controllers and an FSM controller
- Register block
- Shift control and interrupt

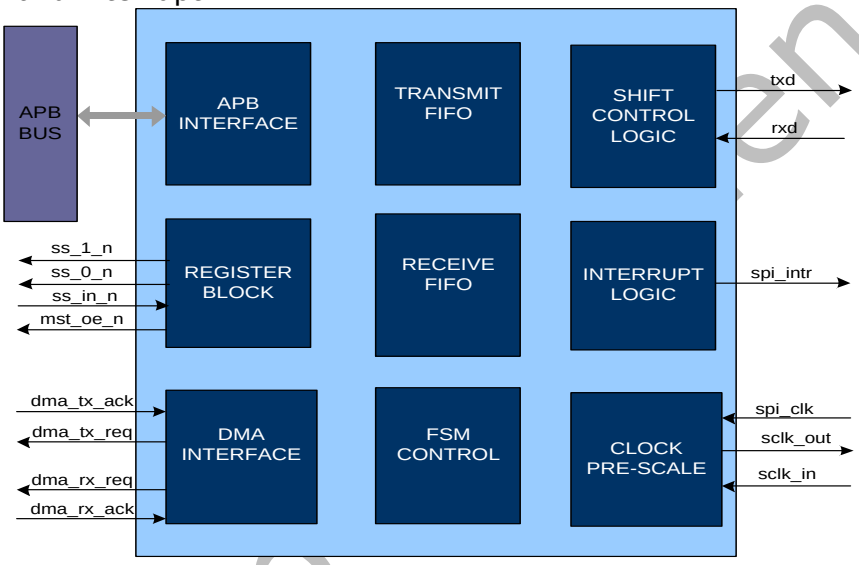

Fig. 29-1 SPI Controller Block diagram

APB INTERFACE

The host processor accesses data, control, and status information on the SPI through the APB interface. The SPI supports APB data bus widths of 32 bits and 8 or 16 bits when reading or writing internal FIFO if data frame size(SPI_CTRL0[1:0]) is set to 8 bits.

DMA INTERFACE

This block has a handshaking interface to a DMA Controller to request and control transfers. The APB bus is used to perform the data transfer to or from the DMA Controller.

FIFO LOGIC

For transmit and receive transfers, data transmitted from the SPI to the external serial device is written into the transmit FIFO. Data received from the external serial device into the SPI is pushed into the receive FIFO. Both fifos are 32x16bits.

FSM CONTROL

Control the state's transformation of the design.

REGISTER BLOCK

All registers in the SPI are addressed at 32-bit boundaries to remain consistent with the APB bus. Where the physical size of any register is less than 32-bits wide, the upper unused bits of the 32-bit boundary are reserved. Writing to these bits has no effect; reading from these bits returns 0.

SHIFT CONTROL

Shift control logic shift the data from the transmit fifo or to the receive fifo. This logic automatically right-justifies receive data in the receive FIFO buffer.

INTERRUPT CONTROL

The SPI supports combined and individual interrupt requests, each of which can be masked. The combined interrupt request is the ORed result of all other SPI interrupts after masking.

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4.3 Function Description

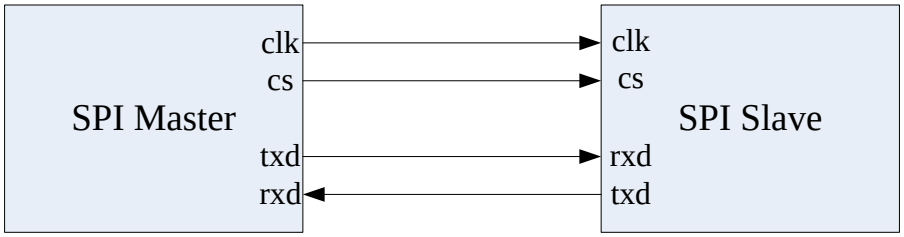

Fig. 29-2 SPI Master and Slave Interconnection

The SPI controller support dynamic switching between master and slave in a system. The diagram show how the SPI controller connects with other SPI devices.

Operation Modes

The SPI can be configured in the following two fundamental modes of operation: Master Mode when SPI_CTRLR0 [20] is 1'b0, Slave Mode when SPI_CTRLR0 [20] is 1'b1.

Transfer Modes

The SPI operates in the following three modes when transferring data on the serial bus.

1). Transmit and Receive

When SPI_CTRLR0 [19:18]== 2'b00, both transmit and receive logic are valid.

2).Transmit Only

When SPI_CTRLR0 [19:18] == 2'b01, the receive data are invalid and should not be stored in the receive FIFO.

3).Receive Only

When SPI_CTRLR0 [19:18]== 2'b10, the transmit data are invalid.

Clock Ratios

A summary of the frequency ratio restrictions between the bit-rate clock (sclk_out/sclk_in) and the SPI peripheral clock (spi_clk) are described as,

When SPI Controller works as master, the F_{spi_clk}>= 2 × (maximum F_{sclk_out})

When SPI Controller works as slave, the F_{spi_clk}>= 6 × (maximum F_{sclk_in})

With the SPI, the clock polarity (SCPOL) configuration parameter determines whether the inactive state of the serial clock is high or low. To transmit data, both SPI peripherals must have identical serial clock phase (SCPH) and clock polarity (SCPOL) values. The data frame can be 4/8/16 bits in length.

When the configuration parameter SCPH = 0, data transmission begins on the falling edge of the slave select signal. The first data bit is captured by the master and slave peripherals on the first edge of the serial clock; therefore, valid data must be present on the txd and rxd lines prior to the first serial clock edge. The following two figures show a timing diagram for a single SPI data transfer with SCPH = 0. The serial clock is shown for configuration parameters SCPOL = 0 and SCPOL = 1.

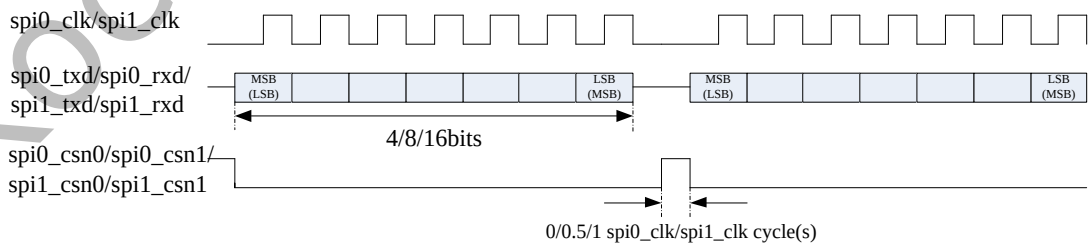

Fig. 29-3 SPI Format (SCPH=0 SCPOL=0)

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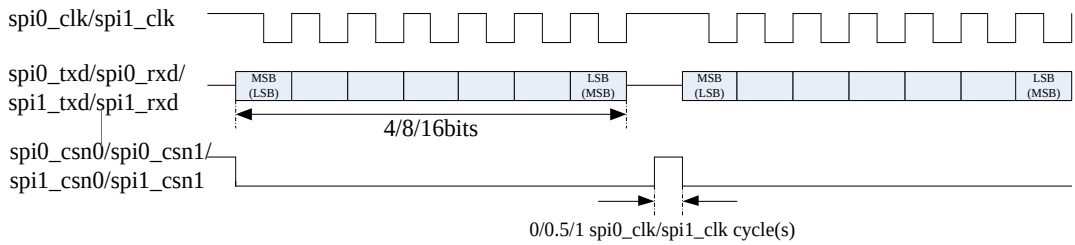

Fig. 29-4 SPI Format (SCPH=0 SCPOL=1)

When the configuration parameter SCPH = 1, both master and slave peripherals begin transmitting data on the first serial clock edge after the slave select line is activated. The first data bit is captured on the second (trailing) serial clock edge. Data are propagated by the master and slave peripherals on the leading edge of the serial clock. During continuous data frame transfers, the slave select line may be held active-low until the last bit of the last frame has been captured. The following two figures show the timing diagram for the SPI format when the configuration parameter SCPH = 1.

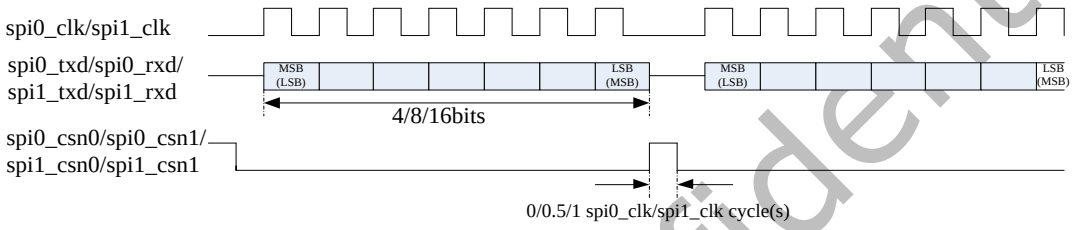

Fig. 29-5 SPI Format (SCPH=1 SCPOL=0)

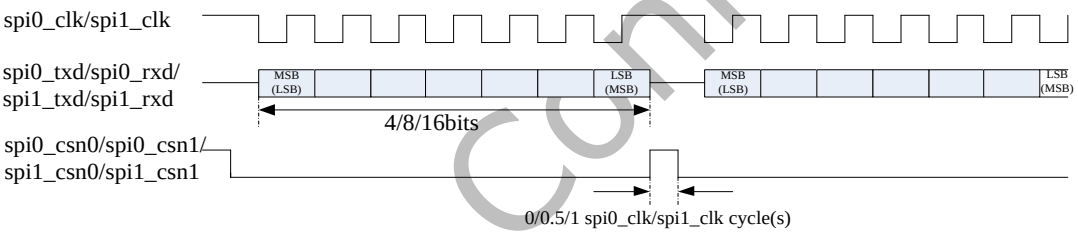

Fig. 29-6 SPI Format (SCPH=1 SCPOL=1)

4.4 Register Description

4.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
SPI_CTRLR0	0x0000	W	0x00000002	Control Register 0
SPI_CTRLR1	0x0004	W	0x00000000	Control Register 1
SPI_ENR	0x0008	W	0x00000000	SPI Enable
SPI_SER	0x000c	W	0x00000000	Slave Enable Register
SPI_BAUDR	0x0010	W	0x00000000	Baud Rate Select
SPI_TXFTLR	0x0014	W	0x00000000	Transmit FIFO Threshold Level
SPI_RXFTLR	0x0018	W	0x00000000	Receive FIFO Threshold Level
SPI_TXFLR	0x001c	W	0x00000000	Transmit FIFO Level
SPI_RXFLR	0x0020	W	0x00000000	Receive FIFO Level
SPI_SR	0x0024	W	0x0000000c	SPI Status
SPI_IPR	0x0028	W	0x00000000	Interrupt Polarity
SPI_IMR	0x002c	W	0x00000000	Interrupt Mask
SPI_ISR	0x0030	W	0x00000000	Interrupt Status
SPI_RISR	0x0034	W	0x00000001	Raw Interrupt Status

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Name	Offset	Size	Reset Value	Description
SPI_ICR	0x0038	W	0x00000000	Interrupt Clear
SPI_DMACR	0x003c	W	0x00000000	DMA Control
SPI_DMATDLR	0x0040	W	0x00000000	DMA Transmit Data Level
SPI_DMARDLR	0x0044	W	0x00000000	DMA Receive Data Level
SPI_TXDR	0x0048	W	0x00000000	Transmit FIFO Data
SPI_RXDR	0x004c	W	0x00000000	Receive FIFO Data

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

4.4.2 Detail Register Description

SPI_CTRLR0

Address: Operational Base + offset (0x0000)

Control Register 0

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	MTM Microwire Transfer Mode Valid when frame format is set to National Semiconductors Microwire. 1'b0: non-sequential transfer 1'b1: sequential transfer
20	RW	0x0	OPM Operation Mode 1'b0: Master Mode 1'b1: Slave Mode
19:18	RW	0x0	XFM Transfer Mode 2'b00 :Transmit & Receive 2'b01 : Transmit Only 2'b10 : Receive Only 2'b11 :reserved
17:16	RW	0x0	FRF Frame Format 2'b00: Motorola SPI 2'b01: Texas Instruments SSP 2'b10: National Semiconductors Microwire 2'b11 : Reserved

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Bit	Attr	Reset Value	Description
15:14	RW	0x0	RSD Rxd Sample Delay When SPI is configured as a master, if the rxd data cannot be sampled by the sclk_out edge at the right time, this register should be configured to define the number of the spi_clk cycles after the active sclk_out edge to sample rxd data later when SPI works at high frequency. 2'b00:do not delay 2'b01:1 cycle delay 2'b10:2 cycles delay 2'b11:3 cycles delay
13	RW	0x0	BHT Byte and Halfword Transform Valid when data frame size is 8bit. 1'b0:apb 16bit write/read, spi 8bit write/read 1'b1: apb 8bit write/read, spi 8bit write/read
12	RW	0x0	FBM First Bit Mode 1'b0:first bit is MSB 1'b1:first bit is LSB
11	RW	0x0	EM Endian Mode Serial endian mode can be configured by this bit. Apb endian mode is always little endian. 1'b0:little endian 1'b1:big endian
10	RW	0x0	SSD ss_n to sclk_out delay Valid when the frame format is set to Motorola SPI and SPI used as a master. 1'b0: the period between ss_n active and sclk_out active is half sclk_out cycles. 1'b1: the period between ss_n active and sclk_out active is one sclk_out cycle.
9:8	RW	0x0	CSM Chip Select Mode Valid when the frame format is set to Motorola SPI and SPI used as a master. 2'b00: ss_n keep low after every frame data is transferred. 2'b01:ss_n be high for half sclk_out cycles after every frame data is transferred. 2'b10: ss_n be high for one sclk_out cycle after every frame data is transferred. 2'b11:reserved

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Bit	Attr	Reset Value	Description
7	RW	0x0	SCPOL Serial Clock Polarity Valid when the frame format is set to Motorola SPI. 1'b0: Inactive state of serial clock is low 1'b1: Inactive state of serial clock is high
6	RW	0x0	SCPH Serial Clock Phase Valid when the frame format is set to Motorola SPI. 1'b0: Serial clock toggles in middle of first data bit 1'b1: Serial clock toggles at start of first data bit
5:2	RW	0x0	CFS Control Frame Size Selects the length of the control word for the Microwire frame format. 4'b0000~0010:reserved 4'b0011:4-bit serial data transfer 4'b0100:5-bit serial data transfer 4'b0101:6-bit serial data transfer 4'b0110:7-bit serial data transfer 4'b0111:8-bit serial data transfer 4'b1000:9-bit serial data transfer 4'b1001:10-bit serial data transfer 4'b1010:11-bit serial data transfer 4'b1011:12-bit serial data transfer 4'b1100:13-bit serial data transfer 4'b1101:14-bit serial data transfer 4'b1110:15-bit serial data transfer 4'b1111:16-bit serial data transfer
1:0	RW	0x2	DFS Data Frame Size Selects the data frame length. 2'b00:4bit data 2'b01:8bit data 2'b10:16bit data 2'b11:reserved

SPI_CTRLR1

Address: Operational Base + offset (0x0004)

Control Register 1

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	NDM Number of Data Frames When Transfer Mode is receive only, this register field sets the number of data frames to be continuously received by the SPI. The SPI continues to receive serial data until the number of data frames received is equal to this register value plus 1, which enables you to receive up to 64 KB of data in a continuous transfer.

SPI_ENR
Address: Operational Base + offset (0x0008)
SPI Enable

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	ENR SPI Enable 1'b1: Enable all SPI operations. 1'b0: Disable all SPI operations Transmit and receive FIFO buffers are cleared when the device is disabled.

SPI_SER
Address: Operational Base + offset (0x000c)
Slave Enable Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	SER1 Slave 1 Select Enable 1'b1: Enable chip select 1 1'b0: Disable chip select 1 This register is valid only when SPI is configured as a master device.
0	RW	0x0	SER0 Slave Select Enable 1'b1: Enable chip select 0 1'b0: Disable chip select 0 This register is valid only when SPI is configured as a master device.

SPI_BAUDR
Address: Operational Base + offset (0x0010)
Baud Rate Select

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	BAUDR Baud Rate Select SPI Clock Divider. This register is valid only when the SPI is configured as a master device. The LSB for this field is always set to 0 and is unaffected by a write operation, which ensures an even value is held in this register. If the value is 0, the serial output clock (sclk_out) is disabled. The frequency of the sclk_out is derived from the following equation: Fsclk_out = Fspi_clk/ SCKDV Where SCKDV is any even value between 2 and 65534. For example: for Fspi_clk = 3.6864MHz and SCKDV =2 Fsclk_out = 3.6864/2= 1.8432MHz

SPI_TXFTLR
Address: Operational Base + offset (0x0014)
Transmit FIFO Threshold Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	TXFTLR Transmit FIFO Threshold Level When the number of transmit FIFO entries is less than or equal to this value, the transmit FIFO empty interrupt is triggered.

SPI_RXFTLR
Address: Operational Base + offset (0x0018)
Receive FIFO Threshold Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	RXFTLR Receive FIFO Threshold Level When the number of receive FIFO entries is greater than or equal to this value + 1, the receive FIFO full interrupt is triggered.

SPI_TXFLR
Address: Operational Base + offset (0x001c)
Transmit FIFO Level

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RO	0x00	TXFLR Transmit FIFO Level Contains the number of valid data entries in the transmit FIFO.

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SPI_RXFLR
Address: Operational Base + offset (0x0020)
Receive FIFO Level

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RO	0x00	RXFLR Receive FIFO Level Contains the number of valid data entries in the receive FIFO.

SPI_SR
Address: Operational Base + offset (0x0024)
SPI Status

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	RFF Receive FIFO Full 1'b0: Receive FIFO is not full 1'b1: Receive FIFO is full
3	RO	0x1	RFE Receive FIFO Empty 1'b0: Receive FIFO is not empty 1'b1: Receive FIFO is empty
2	RO	0x1	TFE Transmit FIFO Empty 1'b0: Transmit FIFO is not empty 1'b1: Transmit FIFO is empty
1	RO	0x0	TFF Transmit FIFO Full 1'b0: Transmit FIFO is not full 1'b1: Transmit FIFO is full
0	RO	0x0	BSF SPI Busy Flag When set, indicates that a serial transfer is in progress; when cleared indicates that the SPI is idle or disabled. 1'b0: SPI is idle or disabled 1'b1: SPI is actively transferring data

SPI_IPR
Address: Operational Base + offset (0x0028)
Interrupt Polarity

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	IPR Interrupt Polarity Interrupt Polarity Register 1'b0:Active Interrupt Polarity Level is HIGH 1'b1: Active Interrupt Polarity Level is LOW

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SPI_IMR
Address: Operational Base + offset (0x002c)
Interrupt Mask

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	RFFIM Receive FIFO Full Interrupt Mask 1'b0: spi_rxf_intr interrupt is masked 1'b1: spi_rxf_intr interrupt is not masked
3	RW	0x0	RFOIM Receive FIFO Overflow Interrupt Mask 1'b0: spi_rxo_intr interrupt is masked 1'b1: spi_rxo_intr interrupt is not masked
2	RW	0x0	RFUIM Receive FIFO Underflow Interrupt Mask 1'b0: spi_rxu_intr interrupt is masked 1'b1: spi_rxu_intr interrupt is not masked
1	RW	0x0	TFOIM Transmit FIFO Overflow Interrupt Mask 1'b0: spi_txo_intr interrupt is masked 1'b1: spi_txo_intr interrupt is not masked
0	RW	0x0	TFEIM Transmit FIFO Empty Interrupt Mask 1'b0: spi_txe_intr interrupt is masked 1'b1: spi_txe_intr interrupt is not masked

SPI_ISR
Address: Operational Base + offset (0x0030)
Interrupt Status

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	RFFIS Receive FIFO Full Interrupt Status 1'b0: spi_rxf_intr interrupt is not active after masking 1'b1: spi_rxf_intr interrupt is full after masking
3	RO	0x0	RFOIS Receive FIFO Overflow Interrupt Status 1'b0: spi_rxo_intr interrupt is not active after masking 1'b1: spi_rxo_intr interrupt is active after masking
2	RO	0x0	RFUIS Receive FIFO Underflow Interrupt Status 1'b0: spi_rxu_intr interrupt is not active after masking 1'b1: spi_rxu_intr interrupt is active after masking

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Bit	Attr	Reset Value	Description
1	RO	0x0	TFOIS Transmit FIFO Overflow Interrupt Status 1'b0: spi_txo_intr interrupt is not active after masking 1'b1: spi_txo_intr interrupt is active after masking
0	RO	0x0	TFEIS Transmit FIFO Empty Interrupt Status 1'b0: spi_txe_intr interrupt is not active after masking 1'b1: spi_txe_intr interrupt is active after masking

SPI_RISR
Address: Operational Base + offset (0x0034)
Raw Interrupt Status

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	RFFRIS Receive FIFO Full Raw Interrupt Status 1'b0: spi_rxf_intr interrupt is not active prior to masking 1'b1: spi_rxf_intr interrupt is full prior to masking
3	RO	0x0	RFORIS Receive FIFO Overflow Raw Interrupt Status 1'b0 = spi_rxo_intr interrupt is not active prior to masking 1'b1 = spi_rxo_intr interrupt is active prior to masking
2	RO	0x0	RFURIS Receive FIFO Underflow Raw Interrupt Status 1'b0: spi_rxu_intr interrupt is not active prior to masking 1'b1: spi_rxu_intr interrupt is active prior to masking
1	RO	0x0	TFORIS Transmit FIFO Overflow Raw Interrupt Status 1'b0: spi_txo_intr interrupt is not active prior to masking 1'b1: spi_txo_intr interrupt is active prior to masking
0	RO	0x1	TFERIS Transmit FIFO Empty Raw Interrupt Status 1'b0: spi_txe_intr interrupt is not active prior to masking 1'b1: spi_txe_intr interrupt is active prior to masking

SPI_ICR
Address: Operational Base + offset (0x0038)
Interrupt Clear

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	WO	0x0	CTFOI Clear Transmit FIFO Overflow Interrupt Write 1 to Clear Transmit FIFO Overflow Interrupt

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Bit	Attr	Reset Value	Description
2	WO	0x0	CRFOI Clear Receive FIFO Overflow Interrupt Write 1 to Clear Receive FIFO Overflow Interrupt
1	WO	0x0	CRFUI Clear Receive FIFO Underflow Interrupt Write 1 to Clear Receive FIFO Underflow Interrupt
0	WO	0x0	CCI Clear Combined Interrupt Write 1 to Clear Combined Interrupt

SPI_DMACR
Address: Operational Base + offset (0x003c)
DMA Control

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	TDE Transmit DMA Enable 1'b0: Transmit DMA disabled 1'b1: Transmit DMA enabled
0	RW	0x0	RDE Receive DMA Enable 1'b0: Receive DMA disabled 1'b1: Receive DMA enabled

SPI_DMATDLR
Address: Operational Base + offset (0x0040)
DMA Transmit Data Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	TDL Transmit Data Level This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the transmit FIFO is equal to or below this field value, and Transmit DMA Enable (DMACR[1]) = 1.

SPI_DMARDLR
Address: Operational Base + offset (0x0044)
DMA Receive Data Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	RDL Receive Data Level This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1, and Receive DMA Enable(DMACR[0])=1.

SPI_TXDR
Address: Operational Base + offset (0x0048)
Transmit FIFO Data

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	WO	0x0000	TXDR Transimt FIFO Data Register. When it is written to, data are moved into the transmit FIFO.

SPI_RXDR
Address: Operational Base + offset (0x004c)
Receive FIFO Data

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	RXDR Receive FIFO Data Register. When the register is read, data in the receive FIFO is accessed.

4.5 Interface Description

Table 29-1 1SPI interface description

Module Pin	Direction	Pad Name	IOMUX Setting
spi0_clk	I/O	IO_SPIclkm0_GPIO2B0vccio5	GRF_GPIO2B_IOMUX[1:0]=2'b01
spi0_rxd	I	IO_SPIrxdm0_GPIO2B2vccio5	GRF_GPIO2B_IOMUX[5:4]=2'b01
spi0_txd	O	IO_SPItxdm0_GPIO2B1vccio5	GRF_GPIO2B_IOMUX[3:2]=2'b01
spi0_csn0	I/O	IO_SPIcsn0m0_GPIO2B3vccio5	GRF_GPIO2B_IOMUX[7:6]=2'b01
spi0_csn1	O	IO_SPIcsn1m0_FLASHvol_sel_GPIO2B4vccio5	GRF_GPIO2B_IOMUX[9:8]=2'b01
spi1_clk	I/O	IO_FLASHcs1_SPIclkm1_GPIO3C7vccio2	GRF_GPIO3C_IOMUX[15:14]=2'b10
spi1_rxd	I	IO_FLASHale_SPIrxdm1_GPIO3D0vccio2	GRF_GPIO3D_IOMUX[1:0]=2'b10
spi1_txd	O	IO_FLASHcle_SPItxdm1_GPIO3D1vccio2	GRF_GPIO3D_IOMUX[3:2]=2'b10
spi1_csn0	I/O	IO_FLASHwrn_SPIcsn0m1_GPIO	GRF_GPIO3D_IOMUX[5:4]=2'b1

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Module Pin	Direction	Pad Name	IOMUX Setting
		3D2vccio2	0
spi1_csn1	O	IO_FLASHcs0_SPIcsn1m1_GPIO 3Dvccio2	GRF_GPIO3D_IOMUX[7:6]=2'b10
spi2_clk	I/O	IO_TSPvalid_CIFvsync_SDMMC0EXTcmd_SPIclkm2_USB3PHYdebug1_I2S2sclkm1_GPIO3A0vccio6	GRF_GPIO3AL_IOMUX[2:0]=3'b100
spi2_rxd	I	IO_TSLclk_CIFclkin_SDMMC0EXTclkout_SPIrxdm2_USB3PHYdebug3_I2S2sdim1_GPIO3A2vccio6	GRF_GPIO3AL_IOMUX[5:3]=3'b100
spi2_txd	O	IO_TSPfail_CIFhref_SDMMC0EXTdet_SPItxdm2_USB3PHYdebug2_I2S2sdom1_GPIO3A1vccio6	GRF_GPIO3AL_IOMUX[8:6]=3'b100
spi2_csn0	I/O	IO_TSPd4_CIFdata4_SPIcsn0m2_I2S2lrcktxm1_USB3PHYdebug8_I2S2lrckrxm1_GPIO3B0vccio6	GRF_GPIO3BL_IOMUX[2:0]=3'b011

Notes: I=input, O=output, I/O=input/output, bidirectional. spi_csn1 can only be used in master mode

4.6 Application Notes

Clock Ratios

A summary of the frequency ratio restrictions between the bit-rate clock (sclk_out/sclk_in) and the SPI peripheral clock (spi_clk) are described as,

When SPI Controller works as master, the Fspi_clk>= 2 × (maximum Fsclk_out)

When SPI Controller works as slave, the Fspi_clk>= 6 × (maximum Fsclk_in)

Master Transfer Flow

When configured as a serial-master device, the SPI initiates and controls all serial transfers. The serial bit-rate clock, generated and controlled by the SPI, is driven out on the sclk_out line. When the SPI is disabled (SPI_ENR = 0), no serial transfers can occur and sclk_out is held in "inactive" state, as defined by the serial protocol under which it operates.

Slave Transfer Flow

When the SPI is configured as a slave device, all serial transfers are initiated and controlled by the serial bus master.

When the SPI serial slave is selected during configuration, it enables its txd data onto the serial bus. All data transfers to and from the serial slave are regulated on the serial clock line (sclk_in), driven from the serial-master device. Data are propagated from the serial slave on one edge of the serial clock line and sampled on the opposite edge.

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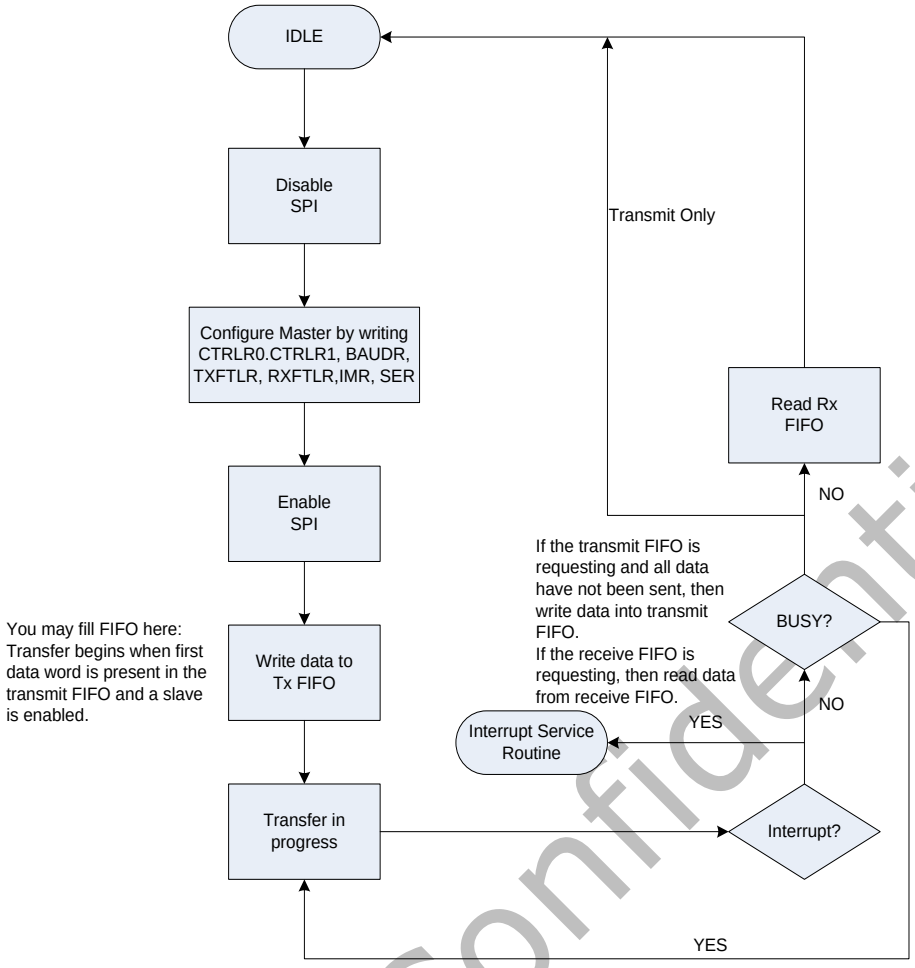

Fig. 29-7 SPI Master transfer flow diagram

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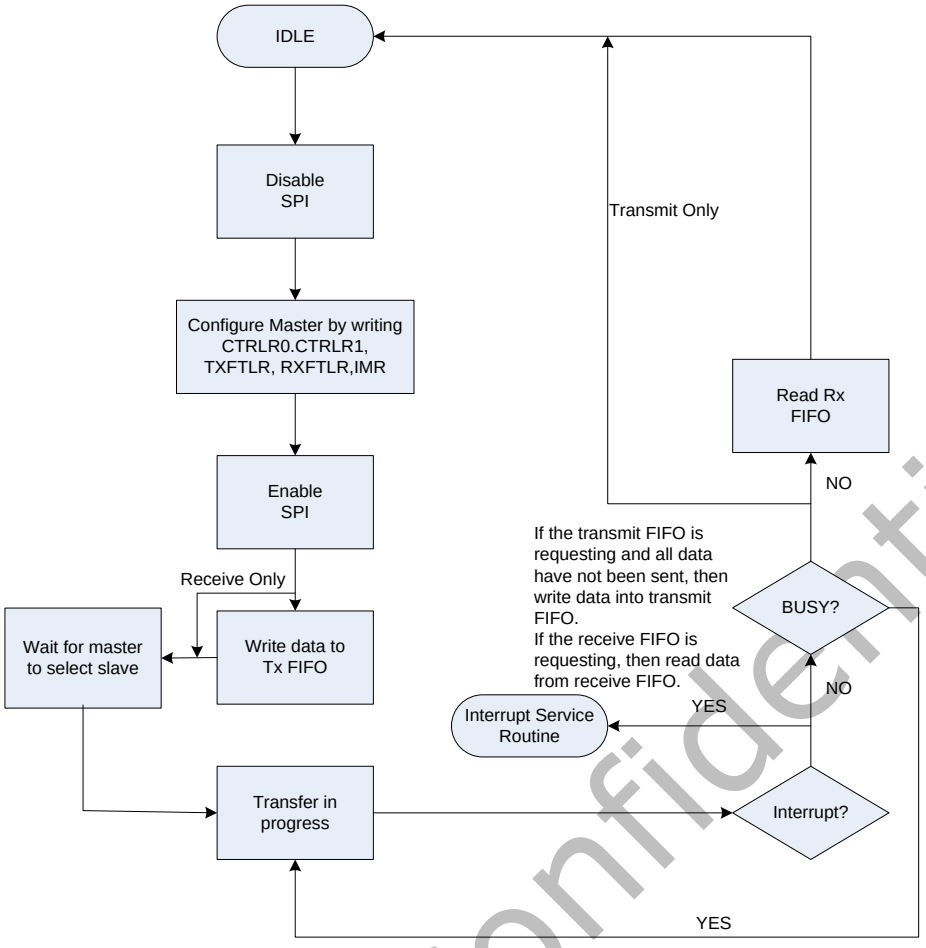

Fig. 29-8 SPI Slave transfer flow diagram

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Chapter 30 UART

1.1 Overview

The Universal Asynchronous Receiver/Transmitter (UART) is used for serial communication with a peripheral, modem (data carrier equipment, DCE) or data set. Data is written from a master (CPU) over the APB bus to the UART and it is converted to serial form and transmitted to the destination device. Serial data is also received by the UART and stored for the master (CPU) to read back.

UART Controller supports the following features:

- Support 3 independent UART controller: UART0, UART1, UART2
- UART0/UART1/UART2 all contain two 64Bytes FIFOs for data receive and transmit
- UART0/UART1/UART2 all support auto flow-control
- Support bit rates 115.2Kbps,460.8Kbps,921.6Kbps,1.5Mbps,3Mbps, 4Mbps
- Support programmable baud rates, even with non-integer clock divider
- Standard asynchronous communication bits (start, stop and parity)
- Support interrupt-based or DMA-based mode
- Support 5-8 bits width transfer

1.2 Block Diagram

This section provides a description about the functions and behavior under various conditions. The UART Controller comprises with:

- AMBA APB interface
- FIFO controllers
- Register block
- Modem synchronization block and baud clock generation block
- Serial receiver and serial transmitter

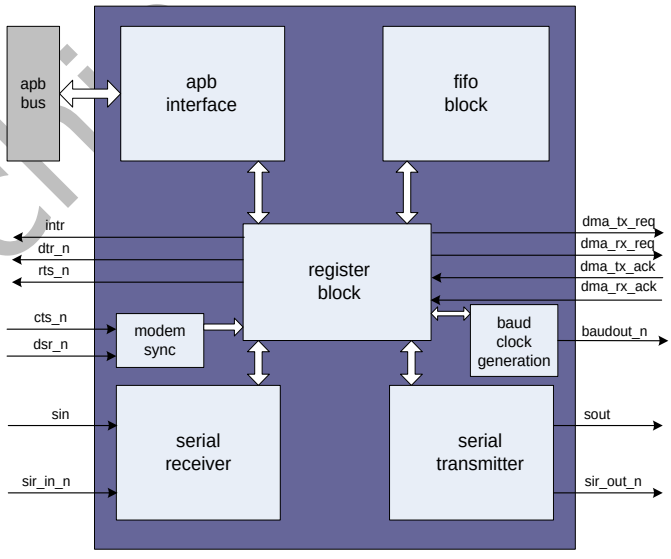

Fig. 29-9UART Architecture

APB INTERFACE

The host processor accesses data, control, and status information on the UART through the APB interface. The UART supports APB data bus widths of 8, 16, and 32 bits.

Register block

Be responsible for the main UART functionality including control, status and interrupt

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generation.

Modem Synchronization block

Synchronizes the modem input signal.

FIFO block

Be responsible for FIFO control and storage (when using internal RAM) or signaling to control external RAM (when used).

Baud Clock Generator

Generates the transmitter and receiver baud clockalong with the output reference clock signal (baudout_n).

Serial Transmitter

Converts the parallel data, written to the UART, into serial form and adds all additional bits, as specified by the control register, for transmission. This makeup of serial data, referred to as a character can exit the block in two forms, either serial UART format or IrDA 1.0 SIR format.

Serial Receiver

Converts the serial data character (as specified by the control register) received in either the UART or IrDA 1.0 SIR format to parallel form. Parity error detection, framing error detection and line break detection is carried out in this block.

1.3 Function Description

UART (RS232) Serial Protocol

Because the serial communication is asynchronous, additional bits (start and stop) are added to the serial data to indicate the beginning and end. An additional parity bit may be added to the serial character. This bit appears after the last data bit and before the stop bit(s) in the character structure to perform simple error checking on the received data, as shown in Figure.

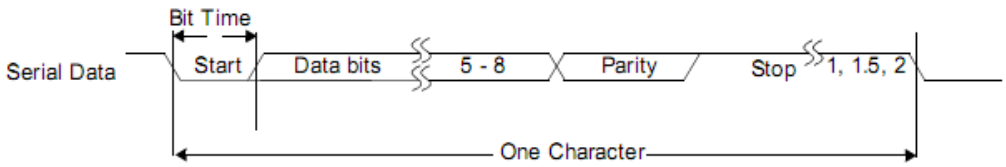

Fig. 29-10UART Serial protocol

IrDA 1.0 SIR Protocol

The Infrared Data Association (IrDA) 1.0 Serial Infrared (SIR) mode supports bi-directional datacommunications with remote devices using infrared radiation as the transmission medium. IrDA 1.0 SIR mode specifies a maximum baud rate of 115.2 Kbaud.

Transmitting a single infrared pulse signals a logic zero, while a logic one is represented by not sending a pulse. The width of each pulse is 3/16ths of a normal serial bit time. Data transfers can only occur in half-duplex fashion when IrDA SIR mode is enabled.

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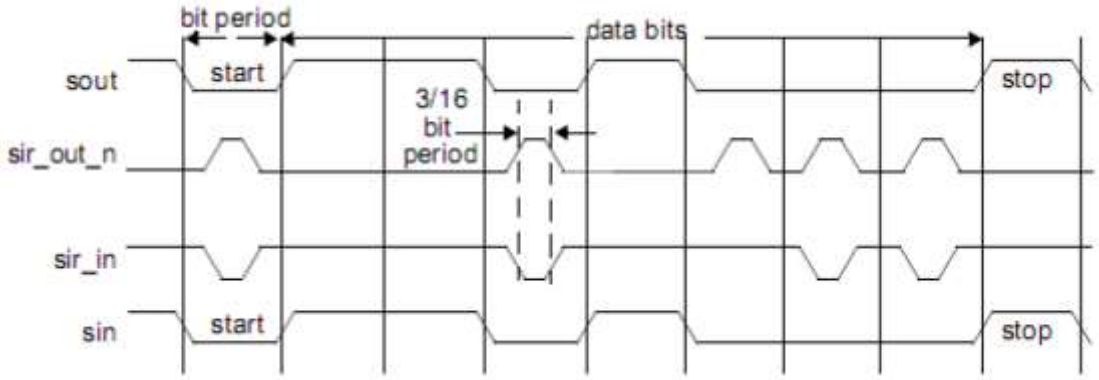

Fig. 29-11IrDA 1.0

Baud Clock

The baud rate is controlled by the serial clock (sclk or pclk in a single clock implementation) and the Divisor Latch Register (DLH and DLL). As the exact number of baud clocks that each bit was transmitted for is known, calculating the mid-point for sampling is not difficult, that is every 16 baud clocks after the mid-point sample of the start bit.

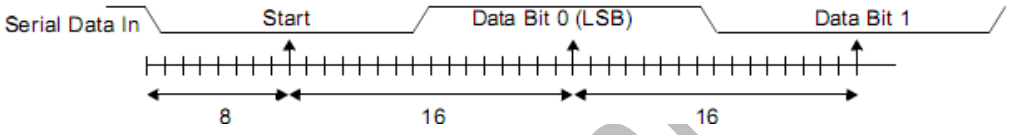

Fig. 29-12UART baud rate

FIFO Support

1. NONE FIFO MODE

If FIFO support is not selected, then no FIFOs are implemented and only a single receive data byte and transmit data byte can be stored at a time in the RBR and THR.

2. FIFO MODE

The FIFO depth of UART0/UART1/UART2is 64bytes. The FIFO mode of all the UART is enabled by register FCR[0].

Interrupts

The following interrupt types can be enabled with the IER register.

- Receiver Error
- Receiver Data Available
- Character Timeout (in FIFO mode only)
- Transmitter Holding Register Empty at/below threshold (in Programmable THRE Interrupt mode)
- Modem Status

DMA Support

The UART supports DMA signaling with the use of two output signals (dma_tx_req_n and dma_rx_req_n) to indicate when data is ready to be read or when the transmit FIFO is empty.

The dma_tx_req_n signal is asserted under the following conditions:

- When the Transmitter Holding Register is empty in non-FIFO mode.
- When the transmitter FIFO is empty in FIFO mode with Programmable THRE interrupt mode disabled.
- When the transmitter FIFO is at, or below the programmed threshold with Programmable THRE interrupt mode enabled.

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The dma_rx_req_n signal is asserted under the following conditions:

- When there is a single character available in the Receive Buffer Register in non-FIFO mode.
- When the Receiver FIFO is at or above the programmed trigger level in FIFO mode.

Auto Flow Control

The UART can be configured to have a 16750-compatible Auto RTS and Auto CTS serial data flow control mode available. If FIFOs are not implemented, then this mode cannot be selected. When Auto Flow Control mode has been selected, it can be enabled with the Modem Control Register (MCR[5]). Following figure shows a block diagram of the Auto Flow Control functionality.

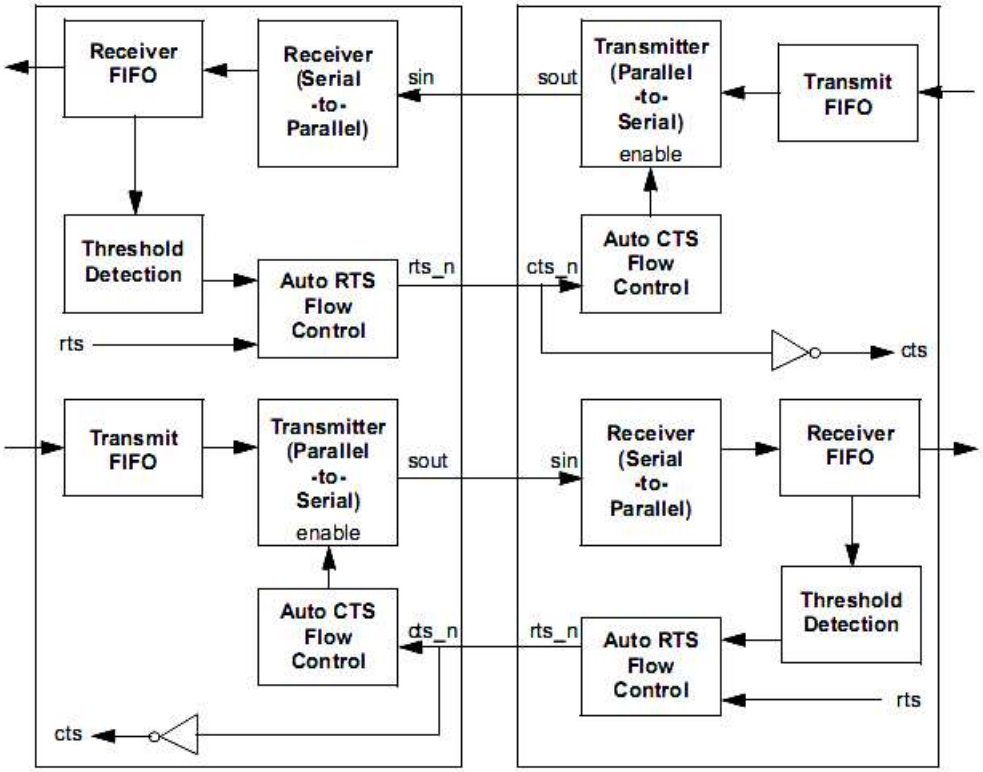

Fig. 29-13UART Auto flow control block diagram

Auto RTS – Becomes active when the following occurs:

- Auto Flow Control is selected during configuration
- FIFOs are implemented
- RTS (MCR[1] bit and MCR[5]bit are both set)
- FIFOs are enabled (FCR[0]) bit is set)
- SIR mode is disabled (MCR[6] bit is not set)

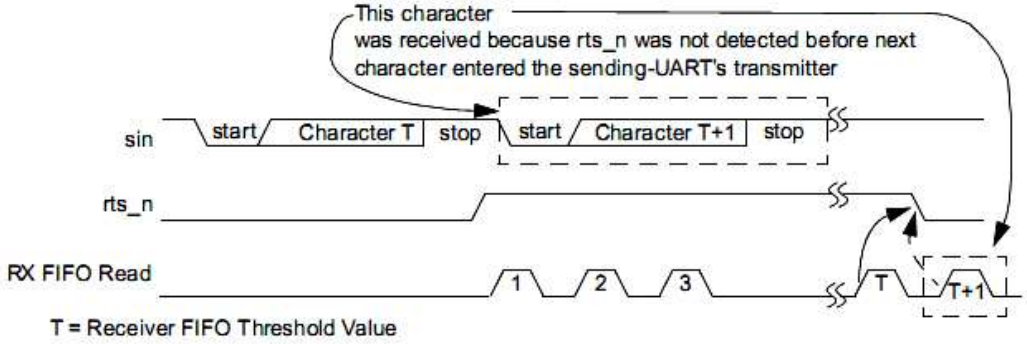

Fig. 29-14UART AUTO RTS TIMING

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Auto CTS – becomes active when the following occurs:

- Auto Flow Control is selected during configuration
- FIFOs are implemented
- AFCE (MCR[5] bit is set)
- FIFOs are enabled through FIFO Control Register FCR[0] bit
- SIR mode is disabled (MCR[6] bit is not set)

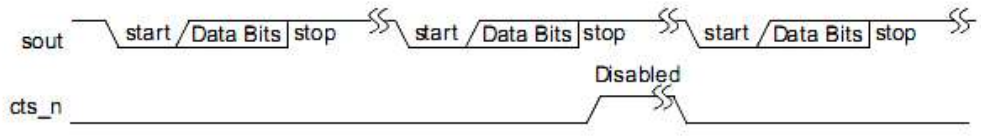

Fig. 29-15UART AUTO CTS TIMING

1.4 Register Description

This section describes the control/status registers of the design. There are 3 UARTs in RK3228, and each one has its own base address.

5.1.1 1.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
UART_RBR	0x0000	W	0x00000000	Receive Buffer Register
UART_THR	0x0000	W	0x00000000	Transmit Holding Register
UART_DLL	0x0000	W	0x00000000	Divisor Latch (Low)
UART_DLH	0x0004	W	0x00000000	Divisor Latch (High)
UART_IER	0x0004	W	0x00000000	Interrupt Enable Register
UART_IIR	0x0008	W	0x00000000	Interrupt Identification Register
UART_FCR	0x0008	W	0x00000000	FIFO Control Register
UART_LCR	0x000c	W	0x00000000	Line Control Register
UART_MCR	0x0010	W	0x00000000	Modem Control Register
UART_LSR	0x0014	W	0x00000000	Line Status Register
UART_MSR	0x0018	W	0x00000000	Modem Status Register
UART_SCR	0x001c	W	0x00000000	Scratchpad Register
UART_SRBR	0x0030	W	0x00000000	Shadow Receive Buffer Register
UART_STHR	0x006c	W	0x00000000	Shadow Transmit Holding Register
UART_FAR	0x0070	W	0x00000000	FIFO Access Register
UART_TFR	0x0074	W	0x00000000	Transmit FIFO Read
UART_RFW	0x0078	W	0x00000000	Receive FIFO Write
UART_USR	0x007c	W	0x00000000	UART Status Register
UART_TFL	0x0080	W	0x00000000	Transmit FIFO Level
UART_RFL	0x0084	W	0x00000000	Receive FIFO Level
UART_SRR	0x0088	W	0x00000000	Software Reset Register
UART_SRTS	0x008c	W	0x00000000	Shadow Request to Send
UART_SBCR	0x0090	W	0x00000000	Shadow Break Control Register
UART_SDMAM	0x0094	W	0x00000000	Shadow DMA Mode
UART_SFE	0x0098	W	0x00000000	Shadow FIFO Enable
UART_SRT	0x009c	W	0x00000000	Shadow RCVR Trigger
UART_STET	0x00a0	W	0x00000000	Shadow TX Empty Trigger

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Name	Offset	Size	Reset Value	Description
UART_HTX	0x00a4	W	0x00000000	Halt TX
UART_DMASA	0x00a8	W	0x00000000	DMA Software Acknowledge
UART_CPR	0x00f4	W	0x00000000	Component Parameter Register
UART_UCV	0x00f8	W	0x0330372a	UART Component Version
UART_CTR	0x00fc	W	0x44570110	Component Type Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

5.1.2 1.4.2 Detail Register Description

UART_RBR

Address: Operational Base + offset (0x0000)

Receive Buffer Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	data_input Data byte received on the serial input port (sin) in UART mode, or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line Status Register (LCR) is set. If in non-FIFO mode (FIFO_MODE == NONE) or FIFOs are disabled (FCR[0] set to zero), the data in the RBR must be read before the next data arrives, otherwise it is overwritten, resulting in an over-run error. If in FIFO mode (FIFO_MODE != NONE) and FIFOs are enabled (FCR[0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO is preserved, but any incoming data are lost and an over-run error occurs.

UART_THR

Address: Operational Base + offset (0x0000)

Transmit Holding Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x00	data_output Data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set. If in non-FIFO mode or FIFOs are disabled (FCR[0] = 0) and THRE is set, writing a single character to the THR clears the THRE. Any additional writes to the THR before the THRE is set again causes the THR data to be overwritten. If in FIFO mode and FIFOs are enabled (FCR[0] = 1) and THRE is set, x number of characters of data may be written to the THR before the FIFO is full. The number x (default=16) is determined by the value of FIFO Depth that you set during configuration. Any attempt to write data when the FIFO is full results in the write data being lost.

UART_DLL

Address: Operational Base + offset (0x0000)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	baud_rate_divisor_L Lower 8-bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero). The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLH is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

UART_DLH

Address: Operational Base + offset (0x0004)

Divisor Latch (High)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	baud_rate_divisor_H Upper 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART.

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UART_IER

Address: Operational Base + offset (0x0004)

Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	prog_thre_int_en Programmable THRE Interrupt Mode Enable This is used to enable/disable the generation of THRE Interrupt. 0 = disabled 1 = enabled
6:4	RO	0x0	reserved
3	RW	0x0	modem_status_int_en Enable Modem Status Interrupt. This is used to enable/disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt. 0 = disabled 1 = enabled
2	RW	0x0	receive_line_status_int_en Enable Receiver Line Status Interrupt. This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt. 0 = disabled 1 = enabled
1	RW	0x0	trans_hold_empty_int_en Enable Transmit Holding Register Empty Interrupt.
0	RW	0x0	receive_data_available_int_en Enable Received Data Available Interrupt. This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode and FIFOs enabled). These are the second highest priority interrupts. 0 = disabled 1 = enabled

UART_IIR

Address: Operational Base + offset (0x0008)

Interrupt Identification Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:6	RO	0x0	fifos_en FIFOs Enabled. This is used to indicate whether the FIFOs are enabled or disabled. 00 = disabled 11 = enabled
5:4	RO	0x0	reserved
3:0	RO	0x0	int_id Interrupt ID This indicates the highest priority pending interrupt which can be one of the following types: 0000 = modem status 0001 = no interrupt pending 0010 = THR empty 0100 = received data available 0110 = receiver line status 0111 = busy detect 1100 = character timeout

UART_FCR

Address: Operational Base + offset (0x0008)

FIFO Control Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	WO	0x0	rcvr_trigger RCVR Trigger. This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode it is used to determine when the rts_n signal is de-asserted. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation. The following trigger levels are supported: 00 = 1 character in the FIFO 01 = FIFO 1/4 full 10 = FIFO 1/2 full 11 = FIFO 2 less than ful

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Bit	Attr	Reset Value	Description
5:4	WO	0x0	tx_empty_trigger TX Empty Trigger. This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation. The following trigger levels are supported: 00 = FIFO empty 01 = 2 characters in the FIFO 10 = FIFO 1/4 full 11 = FIFO 1/2 full
3	WO	0x0	dma_mode DMA Mode This determines the DMA signalling mode used for the dma_tx_req_n and dma_rx_req_n output signals when additional DMA handshaking signals are not selected . 0 = mode 0 1 = mode 11100 = character timeout.
2	WO	0x0	xmit_fifo_reset XMIT FIFO Reset. This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request and single signals when additional DMA handshaking signals are selected . Note that this bit is 'self-clearing'. It is not necessary to clear this bit.
1	WO	0x0	rcvr_fifo_reset RCVR FIFO Reset. This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request and single signals when additional DMA handshaking signals are selected. Note that this bit is 'self-clearing'. It is not necessary to clear this bit.
0	WO	0x0	fifo_en FIFO Enable. FIFO Enable. This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. Whenever the value of this bit is changed both the XMIT and RCVR controller portion of FIFOs is reset.

UART_LCR

Address: Operational Base + offset (0x000c)

Line Control Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RW	0x0	div_lat_access Divisor Latch Access Bit. Writeable only when UART is not busy (USR[0] is zero), always readable. This bit is used to enable reading and writing of the Divisor Latch register (DLL and DLH) to set the baud rate of the UART. This bit must be cleared after initial baud rate setup in order to access other registers.
6	RW	0x0	break_ctrl Break Control Bit. This is used to cause a break condition to be transmitted to the receiving device. If set to one the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared. If MCR[6] set to one, the sir_out_n line is continuously pulsed. When in Loopback Mode, the break condition is internally looped back to the receiver and the sir_out_n line is forced low.
5	RO	0x0	reserved
4	RW	0x0	even_parity_sel Even Parity Select. Writeable only when UART is not busy (USR[0] is zero), always readable. This is used to select between even and odd parity, when parity is enabled (PEN set to one). If set to one, an even number of logic 1s is transmitted or checked. If set to zero, an odd number of logic 1s is transmitted or checked.
3	RW	0x0	parity_en Parity Enable. Writeable only when UART is not busy (USR[0] is zero), always readable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively. 0 = parity disabled 1 = parity enabled
2	RW	0x0	stop_bits_num Number of stop bits. Writeable only when UART is not busy (USR[0] is zero), always readable. This is used to select the number of stop bits per character that the peripheral transmits and receives. If set to zero, one stop bit is transmitted in the serial data.If set to one and the data bits are set to 5 (LCR[1:0] set to zero) one and a half stop bits is transmitted. Otherwise, twostop bits are transmitted. Note that regardless of the number of stop bits selected, the receiver checks only the first stop bit. 0 = 1 stop bit 1 = 1.5 stop bits when DLS (LCR[1:0]) is zero, else 2 stop bit.

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Bit	Attr	Reset Value	Description
1:0	RW	0x0	data_length_sel Data Length Select. Writeable only when UART is not busy (USR[0] is zero), always readable. This is used to select the number of data bits per character that the peripheral transmits and receives. The number of bit that may be selected areas follows: 00 = 5 bits 01 = 6 bits 10 = 7 bits 11 = 8 bits

UART_MCR

Address: Operational Base + offset (0x0010)

Modem Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	sir_mode_en SIR Mode Enable. SIR Mode Enable. This is used to enable/disable the IrDA SIR Mode . 0 = IrDA SIR Mode disabled 1 = IrDA SIR Mode enabled
5	RW	0x0	auto_flow_ctrl_en Auto Flow Control Enable. 0 = Auto Flow Control Mode disabled 1 = Auto Flow Control Mode enabled
4	RW	0x0	loopback LoopBack Bit. This is used to put the UART into a diagnostic mode for test purposes.
3	RW	0x0	out2 OUT2. This is used to directly control the user-designated Output2 (out2_n) output. The value written to this location is inverted and driven out on out2_n, that is: 0 = out2_n de-asserted (logic 1) 1 = out2_n asserted (logic 0)

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Bit	Attr	Reset Value	Description
2	RW	0x0	out1 OUT1 This is used to directly control the user-designated Output2 (out2_n) output. The value written to this location is inverted and driven out on out2_n, that is: 1'b0: out2_n de-asserted (logic 1) 1'b1: out2_n asserted (logic 0)
1	RW	0x0	req_to_send Request to Send. This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data.
0	RW	0x0	data_terminal_ready Data Terminal Ready. This is used to directly control the Data Terminal Ready (dtr_n) output. The value written to this location is inverted and driven out on dtr_n, that is: 0 = dtr_n de-asserted (logic 1) 1 = dtr_n asserted (logic 0)

UART_LSR

Address: Operational Base + offset (0x0014)

Line Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RO	0x0	receiver_fifo_error Receiver FIFO Error bit. This bit is relevant FIFOs are enabled (FCR[0] set to one). This is used to indicate if there is at least one parity error, framing error, or break indication in the FIFO. 0 = no error in RX FIFO 1 = error in RX FIFO
6	RO	0x0	trans_empty Transmitter Empty bit. Transmitter Empty bit. If FIFOs enabled (FCR[0] set to one), this bit is set whenever the Transmitter Shift Register and the FIFO are both empty. If FIFOs are disabled, this bit is set whenever the Transmitter Holding Register and the Transmitter Shift Register are both empty.

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Bit	Attr	Reset Value	Description
5	RO	0x0	trans_hold_reg_empty Transmit Holding Register Empty bit. If THRE mode is disabled (IER[7] set to zero) and regardless of FIFO's being implemented/enabled or not, this bit indicates that the THR or TX FIFO is empty. This bit is set whenever data is transferred from the THR or TX FIFO to the transmitter shift register and no new data has been written to the THR or TX FIFO. This also causes a THRE Interrupt to occur, if the THRE Interrupt is enabled. If IER[7] set to one and FCR[0] set to one respectively, the functionality is switched to indicate the transmitter FIFO is full, and no longer controls THRE interrupts, which are then controlled by the FCR[5:4] threshold setting.
4	RO	0x0	break_int Break Interrupt bit. This is used to indicate the detection of a break sequence on the serial input data.
3	RO	0x0	framing_error Framing Error bit. This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data.
2	RO	0x0	parity_eror Parity Error bit. This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set.
1	RO	0x0	overrun_error Overrun error bit. This is used to indicate the occurrence of an overrun error. This occurs if a new data character was received before the previous data was read.
0	RO	0x0	data_ready Data Ready bit. This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO. 0 = no data ready 1 = data ready

UART_MSR

Address: Operational Base + offset (0x0018)

Modem Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7	RO	0x0	data_carrior_detect Data Carrier Detect. This is used to indicate the current state of the modem control line dcd_n.
6	RO	0x0	ring_indicator Ring Indicator. This is used to indicate the current state of the modem control line ri_n.
5	RO	0x0	data_set_ready Data Set Ready. This is used to indicate the current state of the modem control line dsr_n.
4	RO	0x0	clear_to_send Clear to Send. This is used to indicate the current state of the modem control line cts_n.
3	RO	0x0	delta_data_carrier_detect Delta Data Carrier Detect. This is used to indicate that the modem control line dcd_n has changed since the last time the MSR was read.
2	RO	0x0	trailing_edge_ring_indicator Trailing Edge of Ring Indicator. Trailing Edge of Ring Indicator. This is used to indicate that a change on the input ri_n (from an active-low to an inactive-high state) has occurred since the last time the MSR was read.
1	RO	0x0	delta_data_set_ready Delta Data Set Ready. This is used to indicate that the modem control line dsr_n has changed since the last time the MSR was read.
0	RO	0x0	delta_clear_to_send Delta Clear to Send. This is used to indicate that the modem control line cts_n has changed since the last time the MSR was read.

UART_SCR

Address: Operational Base + offset (0x001c)

Scratchpad Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	temp_store_space This register is for programmers to use as a temporary storage space.

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UART_SRBR

Address: Operational Base + offset (0x0030)

Shadow Receive Buffer Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	shadow_rbr This is a shadow register for the RBR and has been allocated sixteen 32-bit locations so as to accommodate burst accesses from the master. This register contains the data byte received on the serial input port (sin) in UART mode or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line status Register (LSR) is set. If FIFOs are disabled (FCR[0] set to zero), the data in the RBR must be read before the next data arrives, otherwise it is overwritten, resulting in an overrun error. If FIFOs are enabled (FCR[0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO are preserved, but any incoming data is lost. An overrun error also occurs.

UART_STHR

Address: Operational Base + offset (0x006c)

Shadow Transmit Holding Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	shadow_thr This is a shadow register for the THR.

UART_FAR

Address: Operational Base + offset (0x0070)

FIFO Access Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x0	fifo_access_test_en This register is use to enable a FIFO access mode for testing, so that the receive FIFO can be written by the master and the transmit FIFO can be read by the master when FIFOs are implemented and enabled. When FIFOs are not enabled it allows the RBR to be written by the master and the THR to be read by the master. 0 = FIFO access mode disabled 1 = FIFO access mode enabled

UART_TFR

Address: Operational Base + offset (0x0074)

Transmit FIFO Read

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RO	0x00	trans_fifo_read Transmit FIFO Read. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one).When FIFOs are implemented and enabled, reading this register gives the data at the top of the transmit FIFO. Each consecutive read pops the transmit FIFO and gives the next data value that is currently at the top of the FIFO.

UART_RFW

Address: Operational Base + offset (0x0078)

Receive FIFO Write

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	WO	0x0	receive_fifo_framing_error Receive FIFO Framing Error. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one).
8	WO	0x0	receive_fifo_parity_error Receive FIFO Parity Error. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one).

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Bit	Attr	Reset Value	Description
7:0	WO	0x00	receive_fifo_write Receive FIFO Write Data. These bits are only valid when FIFO access mode is enabled (FAR[0] is set to one). When FIFOs are enabled, the data that is written to the RFWD is pushed into the receive FIFO. Each consecutive write pushes the new data to the next write location in the receive FIFO. When FIFOs not enabled, the data that is written to the RFWD is pushed into the RBR.

UART_USR

Address: Operational Base + offset (0x007c)

UART Status Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RO	0x0	receive_fifo_full Receive FIFO Full. This is used to indicate that the receive FIFO is completely full. 0 = Receive FIFO not full 1 = Receive FIFO Full This bit is cleared when the RX FIFO is no longer full.
3	RO	0x0	receive_fifo_not_empty Receive FIFO Not Empty. This is used to indicate that the receive FIFO contains one or more entries. 0 = Receive FIFO is empty 1 = Receive FIFO is not empty This bit is cleared when the RX FIFO is empty.
2	RO	0x0	trasn_fifo_empty Transmit FIFO Empty. This is used to indicate that the transmit FIFO is completely empty. 0 = Transmit FIFO is not empty 1 = Transmit FIFO is empty This bit is cleared when the TX FIFO is no longer empty
1	RO	0x0	trans_fifo_not_full Transmit FIFO Not Full. This is used to indicate that the transmit FIFO in not full. 0 = Transmit FIFO is full 1 = Transmit FIFO is not full This bit is cleared when the TX FIFO is full.

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Bit	Attr	Reset Value	Description
0	RO	0x0	uart_busy UART Busy. UART Busy. This is indicates that a serial transfer is in progress, when cleared indicates that the UART is idle or inactive. 0 = UART is idle or inactivenot busy 1 = UART is busy (actively transferring data)

UART_TFL

Address: Operational Base + offset (0x0080)

Transmit FIFO Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x00	trans_fifo_level Transmit FIFO Level. This is indicates the number of data entries in the transmit FIFO.

UART_RFL

Address: Operational Base + offset (0x0084)

Receive FIFO Level

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RO	0x00	receive_fifo_level Receive FIFO Level. This is indicates the number of data entries in the receive FIFO.

UART_SRR

Address: Operational Base + offset (0x0088)

Software Reset Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	WO	0x0	xmit_fifo_reset XMIT FIFO Reset. This is a shadow register for the XMIT FIFO Reset bit (FCR[2]).
1	WO	0x0	rcvr_fifo_reset RCVR FIFO Reset. This is a shadow register for the RCVR FIFO Reset bit (FCR[1]).

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Bit	Attr	Reset Value	Description
0	WO	0x0	uart_reset UART Reset. This asynchronously resets the UART and synchronously removes the reset assertion. For a two clock implementation both pclk and sclk domains are reset.

UART_SRTS

Address: Operational Base + offset (0x008c)

Shadow Request to Send

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_req_to_send Shadow Request to Send. This is a shadow register for the RTS bit (MCR[1]), this can be used to remove the burden of having to performing a read-modify-write on the MCR.

UART_SBCR

Address: Operational Base + offset (0x0090)

Shadow Break Control Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_break_ctrl Shadow Break Control Bit. This is a shadow register for the Break bit (LCR[6]), this can be used to remove the burden of having to performing a read modify write on the LCR.

UART_SDMAM

Address: Operational Base + offset (0x0094)

Shadow DMA Mode

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_dma_mode Shadow DMA Mode. This is a shadow register for the DMA mode bit (FCR[3]).

UART_SFE

Address: Operational Base + offset (0x0098)

Shadow FIFO Enable

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_fifo_en Shadow FIFO Enable. Shadow FIFO Enable. This is a shadow register for the FIFO enable bit (FCR[0]).

UART_SRT

Address: Operational Base + offset (0x009c)

Shadow RCVR Trigger

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_rcvr_trigger Shadow RCVR Trigger. This is a shadow register for the RCVR trigger bits (FCR[7:6]).

UART_STET

Address: Operational Base + offset (0x00a0)

Shadow TX Empty Trigger

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	shadow_tx_empty_trigger Shadow TX Empty Trigger. This is a shadow register for the TX empty trigger bits (FCR[5:4]).

UART_HTX

Address: Operational Base + offset (0x00a4)

Halt TX

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	halt_tx_en This register is use to halt transmissions for testing, so that the transmit FIFO can be filled by the master when FIFOs are implemented and enabled. 0 = Halt TX disabled 1 = Halt TX enabled

UART_DMASA

Address: Operational Base + offset (0x00a8)

DMA Software Acknowledge

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Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	WO	0x0	dma_software_ack This register is use to perform a DMA software acknowledge if a transfer needs to be terminated due to an error condition.

UART_CPR

Address: Operational Base + offset (0x00f4)

Component Parameter Register

UART_CPR is UART0's own unique register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:16	RO	0x00	FIFO_MODE 0x00 = 0 0x01 = 16 0x02 = 32 to 0x80 = 2048 0x81- 0xff = reserved
15:14	RO	0x0	reserved
13	RO	0x0	DMA_EXTRA 0 = FALSE 1 = TRUE
12	RO	0x0	UART_ADD_ENCODED_PARAMS 0 = FALSE 1 = TRUE
11	RO	0x0	SHADOW 0 = FALSE 1 = TRUE
10	RO	0x0	FIFO_STAT 0 = FALSE 1 = TRUE
9	RO	0x0	FIFO_ACCESS 0 = FALSE 1 = TRUE
8	RO	0x0	NEW_FEAT 0 = FALSE 1 = TRUE
7	RO	0x0	SIR_LP_MODE 0 = FALSE 1 = TRUE

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Bit	Attr	Reset Value	Description
6	RO	0x0	SIR_MODE 0 = FALSE 1 = TRUE
5	RO	0x0	THRE_MODE 0 = FALSE 1 = TRUE
4	RO	0x0	AFCE_MODE 0 = FALSE 1 = TRUE
3:2	RO	0x0	reserved
1:0	RO	0x0	APB_DATA_WIDTH 00 = 8 bits 01 = 16 bits 10 = 32 bits 11 = reserved

UART_UCV

Address: Operational Base + offset (0x00f8)

UART Component Version

Bit	Attr	Reset Value	Description
31:0	RO	0x0330372a	ver ASCII value for each number in the version

UART_CTR

Address: Operational Base + offset (0x00fc)

Component Type Register

Bit	Attr	Reset Value	Description
31:0	RO	0x44570110	peripheral_id This register contains the peripherals identification code.

1.5 Interface Description

Table 29-2UART Interface Description

Modulepin	Dir	Pad name	IOMUX
UART0 Interface			
uart0_sin	I	IO_UART0rx_GMACtxd1m1_GPIO1B0 vccio4	GRF_GPIO1B_IOMUX[1:0]=2'b01
uart0_sout	O	IO_UART0tx_GMACtxd0m1_GPIO1B1 vccio4	GRF_GPIO1B_IOMUX[3:2]=2'b01
uart0_cts_n	I	IO_UART0ctsn_GMACrxd0m1_GPIO1	GRF_GPIO1B_IOMUX[7:6]=2'b01

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		B3vccio4	
uart0_rts_n	O	IO_UART0rtsn_GMACrxd1m1_GPIO1B2vccio4	GRF_GPIO1B_IOMUX[5:4]=2'b01
UART1 Interface			
uart1_sin	I	IO_TSPd2_CIFdata2_SDMMC0EXTd2_UART1rx_USB3PHYdebug6_GPIO3A6vccio6	GRF_GPIO3A_IOMUX[5:3]= 3'b100
uart1_sout	O	IO_TSPd0_CIFda0_SDMMC0EXTd0_UART1tx_USB3PHYdebug4_GPIO3A4vccio6	GRF_GPIO3A_IOMUX[14:12]=3'b100
uart1_cts_n	I	IO_TSPd3_CIFdata3_SDMMC0EXTd3_UART1ctsn_USB3PHYdebug7_GPIO3A7vccio6	GRF_GPIO3A_IOMUX[7:6]= 3'b100
uart1_rts_n	O	IO_TSPd1_CIFdata1_SDMMC0EXTd1_UART1rtsn_USB3PHYdebug5_GPIO3A5vccio6	GRF_GPIO3A_IOMUX[2:0]=3'b100
UART2m0 Interface			
uart2m0_sin	I	IO_SDMMC0d1_UART2DBGrxm0_GPIO1A1vccio3	GRF_GPIO1A_IOMUX[3:2]=2'b10
uart2m0_sout	O	IO_SDMMC0d0_UART2DBGtxm0_GPIO1A0vccio3	GRF_GPIO1A_IOMUX[1:0]=2'b10
UART2m1 Interface			
uart2m1_sin	I	IO_UART2DBGrxm1_POWERstate1_GPIO2A1vccio5	GRF_GPIO2A_IOMUX[3:2]=2'b01
uart2m1_sout	O	IO_UART2DBGtxm1_POWERstate0_GPIO2A0vccio5	GRF_GPIO2A_IOMUX[1:0]=2'b01

The I/O interface of UART2 can be chosen by setting GRF_CON_IOMUX[0]bit, if this bit is set to 1, UART2 uses the UART2m1 I/O interface.

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1.6 Application Notes

5.1.3 1.6.1 None FIFO Mode Transfer Flow

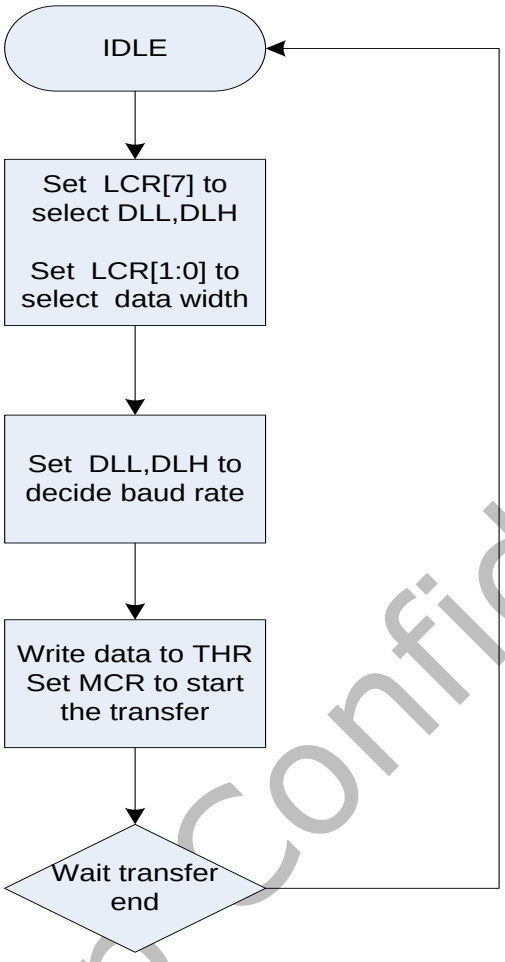

Fig. 29-16UART none fifo mode

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5.1.4 1.6.2 FIFO Mode Transfer Flow

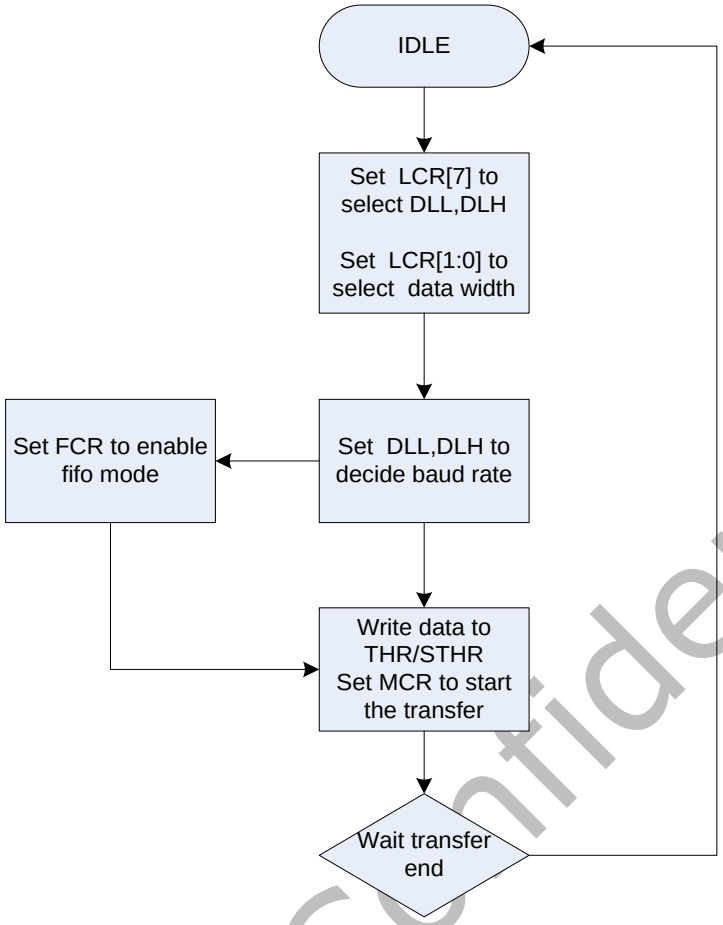

Fig. 29-17UART fifo mode

The UART is an APB slave performing:

Serial-to-parallel conversion on data received from a peripheral device.

Parallel-to-serial conversion on data transmitted to the peripheral device.

The CPU reads and writes data and control/status information through the APB interface. The transmitting and receiving paths are buffered with internal FIFO memories enabling up to 64-bytes to be stored independently in both transmit and receive modes. A baud rate generator can generate a common transmit and receive internal clock input. The baud rates will depend on the internal clock frequency. The UART will also provide transmit, receive and exception interrupts to system. A DMA interface is implemented for improving the system performance.

5.1.5 1.6.3 Baud Rate Calculation

UART clock generation

The following figures shows the UART clock generation.

UART0,UART1 and UART2 source clocks can be selected from three PLL outputs (CODEC PLL/GENERAL PLL/USBPHY_480M). UART clocks can be generated by 1 to 64 division of its source clock, or can be fractionally divided again, or be provided by XIN24M.

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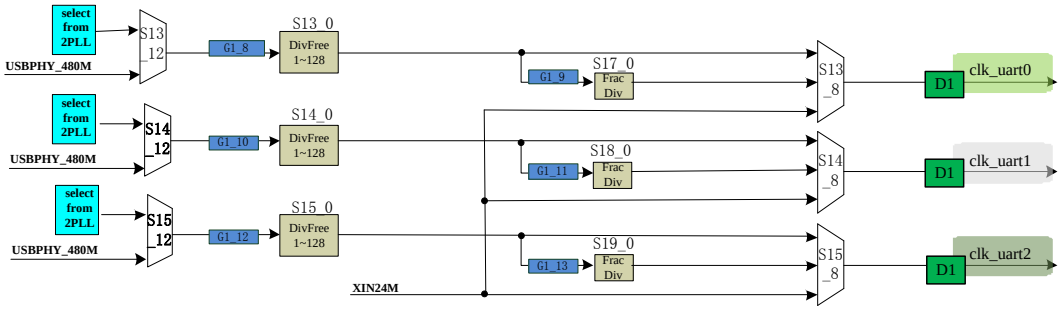

Fig. 29-18UART clock generation

UART baud rate configuration

The following table provides some reference configuration for different UART baud rates.

Table 29-3 UART baud rate configuration

Baud Rate	Reference Configuration
115.2 Kbps	Configure GENERAL PLL to get 1200MHz clock output; Divide 1200MHz clock by 46875/72 to get 1.8432MHz clock; Configure UART_DLL to 1.
460.8 Kbps	Configure GENERAL PLL to get 1200MHz clock output; Divide 1200MHz clock by 46875/288 to get 7.3728MHz clock; Configure UART_DLL to 1.
921.6 Kbps	Configure GENERAL PLL to get 1200MHz clock output; Divide 1200MHz clock by 46875/576 to get 14.7456MHz clock; Configure UART_DLL to 1.
1.5 Mbps	Choose GENERAL PLL to get 1200MHz clock output; Divide 1200MHz clock by 50 to get 24MHz clock; Configure UART_DLL to 1.
3 Mbps	Choose GENERAL PLL to get 1200MHz clock output; Divide 1200MHz clock by 1200/48 to get 48MHz clock; Configure UART_DLL to 1.
4 Mbps	Configure GENERAL PLL to get 1200MHz clock output; Divide 1200MHz clock by 1200/64 to get 64MHz clock; Configure UART_DLL to 1.

5.1.6 1.6.4 CTS_n and RTS_n Polarity Configurable

The polarity of cts_n and rts_n ports can be configured by GRF registers.

● GRF_SOC_CON3[2:0] (grf_uart_cts_sel[2:0]) used to configure the polarity of cts_n.Every bit for one UART, bit2 is for UART2,bit1 is for UART1, bit0 is for UART0

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- GRF_SOC_CON3[5:3] (grf_uart_rts_sel[2:0]) used to configure the polarity of rts_n.Every bit for one UART, bit2 is for UART2,bit1 is for UART1, bit0 is for UART0.
- When grf_uart_cts_sel[*] is configured as 1'b1, cts_n is high active. Otherwise, lowactive.
- When grf_uart_rts_sel[*] is configured as 1'b1, rts_n is high active. Otherwise, lowactive.

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Chapter 31 I2C Interface

6.1 Overview

The Inter-Integrated Circuit (I2C) is a two wired (SCL and SDA), bi-directional serial bus that provides an efficient and simple method of information exchange between devices. This I2C bus controller supports master mode acting as a bridge between AMBA protocol and generic I2C bus system.

I2C Controller supports the following features:

- Item Compatible with I2C-bus
- AMBA APB slave interface
- Supports master mode of I2C bus
- Software programmable clock frequency and transfer rate up to 400Kbit/sec
- Supports 7 bits and 10 bits addressing modes
- Interrupt or polling driven multiple bytes data transfer
- Clock stretching and wait state generation

6.2 Block Diagram

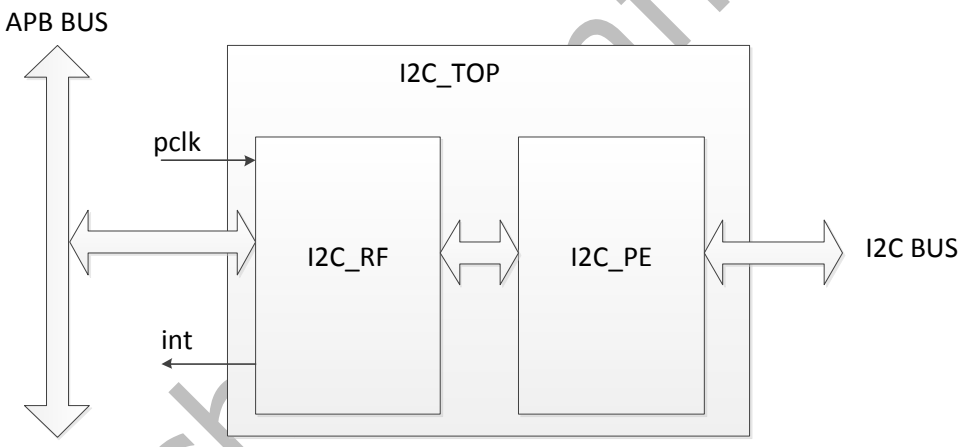

Fig.31-1I2C architecture

6.2.1 I2C_RF

I2C_RF module is used to control the I2C controller operation by the host with APB interface. It implements the register set and the interrupt functionality. The CSR component operates synchronously with the pclk clock.

6.2.2 I2C_PE

I2C_PE module implements the I2C master operation for transmit data to and receive data from other I2C devices. The I2C master controller operates synchronously with the pclk.

6.2.3 I2C_TOP

I2C_TOP module is the top module of the I2C controller.

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6.3 Function Description

This chapter provides a description about the functions and behavior under various conditions.

The I2C controller supports only Masterfunction. Itsupports the 7-bits/10-bits addressing mode and support general call address. The maximum clock frequency and transfer rate can be up to 400Kbit/sec.

The operations of I2C controller is divided to 2 parts and described separately: initialization and master mode programming.

6.3.1 Initialization

The I2C controller is based on AMBA APB bus architecture and usually is part of a SOC. So before I2C operates, some system setting and configuration must be conformed, which includes:

- I2C interrupt connection type: CPU interrupt scheme should be considered. If the I2C interrupt is connected to extra Interrupt Controller module, we need decide the INTC vector.
- I2C Clock Rate: The I2C controller uses the APB clock as the working clock so the APB clock will determine the I2C bus clock. The correct register setting is subject to the system requirement.

6.3.2 Master Mode Programming

- SCL Clock

  When the I2C controller is programmed in Master mode, the SCL frequency is determined by I2C_CLKDIV register. The SCL frequency is calculated by the following formula:

  SCL Divisor = 8*(CLKDIVL + 1 + CLKDIVH + 1)

  SCL = PCLK/ SCLK Divisor

- Data Receiver Register Access

  When the I2C controller received MRXCNT bytes data， CPU can get the data through register RXDATA0 ~ RXDATA7. The controller can receive up to 32 bytes' data in one transaction.

  When MRXCNT register is written, the I2C controller will start to drive SCL to receive data.

- Transmit Transmitter Register

  Data to transmit are written to TXDATA0~7 by CPU. The controller can transmit up to 32 bytes' data in one transaction. The lower byte will be transmitted first.

  When MTXCNT register is written, the I2C controller will start to transmit data.

- Start Command

  Write 1 to I2C_CON[3], the controller will send I2C start command.

- Stop Command

  Write 1 to I2C_CON[4], the controller will send I2C stop command

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- I2C Operation mode

  There are four i2c operation modes.

  - When I2C_CON[2:1] is 2'b00, the controller transmit all valid data in TXDATA0~TXDATA7 byte by byte. The controller will transmit lower byte first.
  - When I2C_CON[2:1] is 2'b01,the controller will transmit device address in MRXADDR first (Write/Read bit = 0) and then transmit device register address in MRXRADDR. After that, the controller will assert restart signal and resend MRXADDR (Write/Read bit = 1). At last, the controller enter receive mode.
  - When I2C_CON[2:1] is 2'b10, the controller is in receive mode, it will trigger clock to read MRXCNT byte data.
  - When I2C_CON[2:1] is 2'b11, the controller will transmit device address in MRXADDR first (Write/Read bit = 1) and then transmit device register address in MRXRADDR . After that, the controller will assert restart signal and resend MRXADDR (Write/Read bit = 1). At last, the controller enter receive mode.

- Read/Write Command
  - When I2C_OPMODE(I2C_CON[2:1]) is 2'b01 or 2'b11, the Read/Write command bit is decided by controller itself.
  - In RX only mode (I2C_CON[2:1] is 2'b10), the Read/Write command bit is decided by MRXADDR[0].
  - In TX only mode (I2C_CON[[2:1] is 2'b00), the Read/Write command bit is decided by TXDATA[0].

- Master Interrupt Condition

  There are 7 interrupt bits in I2C_ISR register related to master mode.

  - Byte transmitted finish interrupt (Bit 0): The bit is asserted when Master completed transmitting a byte.
  - Byte received finish interrupt (Bit 1): The bit is asserted when Master completed receiving a byte.
  - MTXCNT bytes data transmitted finish interrupt (Bit 2): The bit is asserted when Master completed transmitting MTXCNT bytes.
  - MRXCNT bytes data received finish interrupt (Bit 3): The bit is asserted when Master completed receiving MRXCNT bytes.
  - Start interrupt (Bit 4): The bit is asserted when Master finished asserting start command to I2C bus.
  - Stop interrupt (Bit 5): The bit is asserted when Master finished asserting stop command to I2C bus.
  - NAK received interrupt (Bit 6): The bit is asserted when Master received a NAK handshake.

- Last byte acknowledge control
  - If I2C_CON[5] is 1, the I2C controller will transmit NAK handshake to slave when the last byte received in RX only mode.
  - If I2C_CON[5] is 0, the I2C controller will transmit ACK handshake to slave when the last byte received in RX only mode.

- How to handle NAK handshake received
  - If I2C_CON[6] is 1, the I2C controller will stop all transactions when NAK handshake received. And the software should take responsibility to handle the problem.

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- If I2C_CON[6] is 0, the I2C controller will ignore all NAK handshake received.

- I2C controller data transfer waveform
  - Bit transferring
    - Data Validity

      The SDA line must be stable during the high period of SCL, and the data on SDA line can only be changed when SCL is in low state.

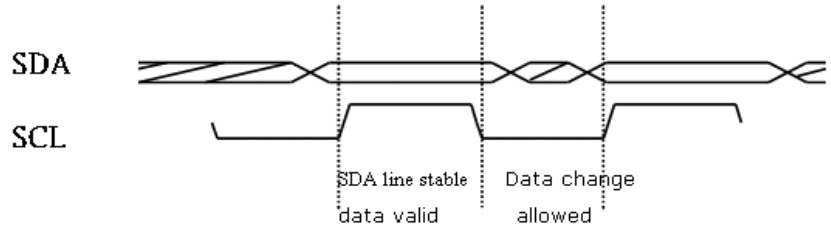

Fig.31-2I2C DATA Validity

    - START and STOP conditions

      START condition occurs when SDA goes low while SCL is in high period. STOP condition is generated when SDA line goes high while SCL is in high state.

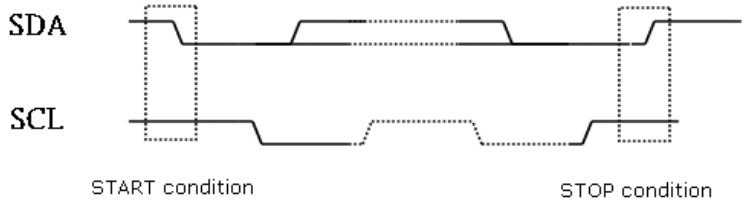

Fig.31-3I2C Start and stop conditions

    - Data transfer
      - Acknowledge

        After a byte of data transferring (clocks labeled as 1~8), in 9th clock the receiver must assert an ACK signal on SDA line, if the receiver pulls SDA line to low, it means "ACK", on the contrary, it's "NOT ACK".

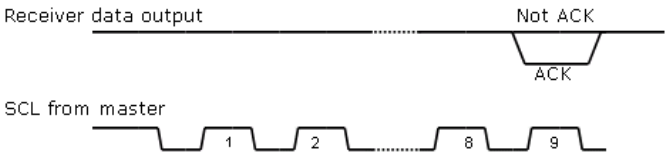

Fig.31-4I2C Acknowledge

      - Byte transfer

        The master own I2C bus might initiate multi byte to transfer to a slave.The transfer starts from a "START" command and ends in a "STOP"command. After every byte transfer, the receiver must reply an ACK to transmitter.

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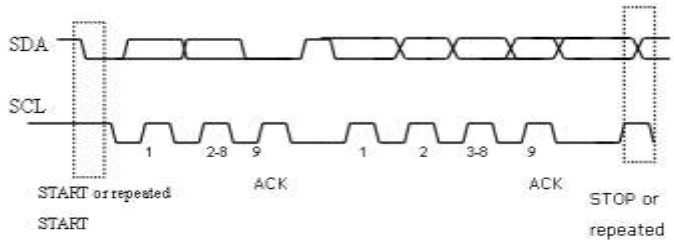

Fig.31-5I2C byte transfer

6.4 Register Description

6.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
RKI2C_CON	0x0000	W	0x00000000	control register
RKI2C_CLKDIV	0x0004	W	0x00000001	clock divider register
RKI2C_MRXADDR	0x0008	W	0x00000000	the slave address accessed for master rx mode
RKI2C_MRXRADDR	0x000c	W	0x00000000	the slave register address accessed for master rx mode
RKI2C_MTXCNT	0x0010	W	0x00000000	master transmit count
RKI2C_MRXCNT	0x0014	W	0x00000000	master rx count
RKI2C_IEN	0x0018	W	0x00000000	interrupt enable register
RKI2C_IPD	0x001c	W	0x00000000	interrupt pending register
RKI2C_FCNT	0x0020	W	0x00000000	finished count
RKI2C_TXDATA0	0x0100	W	0x00000000	I2C tx data register 0
RKI2C_TXDATA1	0x0104	W	0x00000000	I2C tx data register 1
RKI2C_TXDATA2	0x0108	W	0x00000000	I2C tx data register 2
RKI2C_TXDATA3	0x010c	W	0x00000000	I2C tx data register 3
RKI2C_TXDATA4	0x0110	W	0x00000000	I2C tx data register 4
RKI2C_TXDATA5	0x0114	W	0x00000000	I2C tx data register 5
RKI2C_TXDATA6	0x0118	W	0x00000000	I2C tx data register 6
RKI2C_TXDATA7	0x011c	W	0x00000000	I2C tx data register 7
RKI2C_RXDATA0	0x0200	W	0x00000000	I2C rx data register 0
RKI2C_RXDATA1	0x0204	W	0x00000000	I2C rx data register 1
RKI2C_RXDATA2	0x0208	W	0x00000000	I2C rx data register 2
RKI2C_RXDATA3	0x020c	W	0x00000000	I2C rx data register 3
RKI2C_RXDATA4	0x0210	W	0x00000000	I2C rx data register 4
RKI2C_RXDATA5	0x0214	W	0x00000000	I2C rx data register 5
RKI2C_RXDATA6	0x0218	W	0x00000000	I2C rx data register 6
RKI2C_RXDATA7	0x021c	W	0x00000000	I2C rx data register 7

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

6.4.2 Detail Register Description

RKI2C_CON

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Address: Operational Base + offset (0x0000)

control register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	act2nak operation when NAK handshake is received 1'b0: ignored 1'b1: stop transaction
5	RW	0x0	ack last byte acknowledge control in master receive mode 1'b0: ACK 1'b1: NAK
4	RW	0x0	stop stop enable stop enable, when this bit is written to 1, I2C will generate stop signal.
3	RW	0x0	start start enable start enable, when this bit is written to 1, I2C will generate start signal.
2:1	RW	0x0	i2c_mode i2c mode select 2'b00: transmit only 2'b01: transmit address (device + register address) --> restart --> transmit address -> receive only 2'b10: receive only 2'b11: transmit address (device + register address, write/read bit is 1) --> restart --> transmit address (device address) --> receive data
0	RW	0x0	i2c_en i2c module enable 1'b0:not enable 1'b1:enable

RKI2C_CLKDIV

Address: Operational Base + offset (0x0004)

clock divider register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	CLKDIVH scl high level clock count T(SCL_HIGH) = T(PCLK) * (CLKDIVH + 1) * 8

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Bit	Attr	Reset Value	Description
15:0	RW	0x0001	CLKDIVL scl low level clock count T(SCL_LOW) = T(PCLK) * (CLKDIVL + 1) * 8

RKI2C_MRXADDR

Address: Operational Base + offset (0x0008)

the slave address accessed for master rx mode

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	addhvld address high byte valid 1'b0:invalid 1'b1:valid
25	RW	0x0	addmvld address middle byte valid 1'b0:invalid 1'b1:valid
24	RW	0x0	addlvld address low byte valid 1'b0:invalid 1'b1:valid
23:0	RW	0x000000	saddr master address register the lowest bit indicate write or read 24 bits address register

RKI2C_MRXRADDR

Address: Operational Base + offset (0x000c)

the slave register address accessed for master rx mode

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	sraddhvld address high byte valid 1'b0:invalid 1'b1:valid

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Bit	Attr	Reset Value	Description
25	RW	0x0	sraddmvld address middle byte valid 1'b0:invalid 1'b1:valid
24	RW	0x0	sraddlvld address low byte valid 1'b0:invalid 1'b1:valid
23:0	RW	0x000000	sraddr slave register address accessed 24 bits register address

RKI2C_MTXCNT

Address: Operational Base + offset (0x0010)

master transmit count

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	mtxcnt master transmit count 6 bits counter

RKI2C_MRXCNT

Address: Operational Base + offset (0x0014)

masterrx count

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	mrxcnt master rx count 6 bits counter

RKI2C_IEN

Address: Operational Base + offset (0x0018)

interrupt enable register

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Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	nakrcvien NAK handshake received interrupt enable 1'b0:disable 1'b1:enable
5	RW	0x0	stopien stop operation finished interrupt enable 1'b0:disable 1'b1:enable
4	RW	0x0	startien start operation finished interrupt enable 1'b0:disable 1'b1:enable
3	RW	0x0	mbrfien MRXCNT data received finished interrupt enable 1'b0:disable 1'b1:enable
2	RW	0x0	mbtfien MTXCNT data transfer finished interrupt enable 1'b0:disable 1'b1:enable
1	RW	0x0	brfien byte rx finished interrupt enable 1'b0:disable 1'b1:enable
0	RW	0x0	btfien byte tx finished interrupt enable 1'b0:disable 1'b1:enable

RKI2C_IPD

Address: Operational Base + offset (0x001c)

interrupt pending register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	W1C	0x0	nakrcvipd NAK handshake received interrupt pending bit 1'b0:no interrupt available 1'b1:NAK handshake received interrupt appear, write 1 to clear

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Bit	Attr	Reset Value	Description
5	W1C	0x0	stopipd stop operation finished interrupt pending bit 1'b0:no interrupt available 1'b1:stop operation finished interrupt appear, write 1 to clear
4	W1C	0x0	startipd start operation finished interrupt pending bit 1'b0:no interrupt available 1'b1:start operation finished interrupt appear, write 1 to clear
3	W1C	0x0	mbrfipd MRXCNT data received finished interrupt pending bit 1'b0:no interrupt available 1'b1:MRXCNT data received finished interrupt appear, write 1 to clear
2	W1C	0x0	mbtfipd MTXCNT data transfer finished interrupt pending bit 1'b0:no interrupt available 1'b1:MTXCNT data transfer finished interrupt appear, write 1 to clear
1	W1C	0x0	brfipd byte rx finished interrupt pending bit 1'b0:no interrupt available 1'b1:byte rx finished interrupt appear, write 1 to clear
0	W1C	0x0	btfipd byte tx finished interrupt pending bit 1'b0:no interrupt available 1'b1:byte tx finished interrupt appear, write 1 to clear

RKI2C_FCNT

Address: Operational Base + offset (0x0020)

finished count

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RO	0x00	fcnt finished count the count of data which has been transmitted or received for debug purpose

RKI2C_TXDATA0

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Address: Operational Base + offset (0x0100)

I2C tx data register 0

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata0 data0 to be transmitted 32 bits data

RKI2C_TXDATA1

Address: Operational Base + offset (0x0104)

I2C tx data register 1

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata1 data1 to be transmitted 32 bits data

RKI2C_TXDATA2

Address: Operational Base + offset (0x0108)

I2C tx data register 2

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata2 data2 to be transmitted 32 bits data

RKI2C_TXDATA3

Address: Operational Base + offset (0x010c)

I2C tx data register 3

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata3 data3 to be transmitted 32 bits data

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RKI2C_TXDATA4

Address: Operational Base + offset (0x0110)

I2C tx data register 4

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata4 data4 to be transmitted 32 bits data

RKI2C_TXDATA5

Address: Operational Base + offset (0x0114)

I2C tx data register 5

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata5 data5 to be transmitted 32 bits data

RKI2C_TXDATA6

Address: Operational Base + offset (0x0118)

I2C tx data register 6

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata6 data6 to be transmitted 32 bits data

RKI2C_TXDATA7

Address: Operational Base + offset (0x011c)

I2C tx data register 7

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txdata7 data7 to be transmitted 32 bits data

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RKI2C_RXDATA0

Address: Operational Base + offset (0x0200)

I2C rx data register 0

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata0 data0 received 32 bits data

RKI2C_RXDATA1

Address: Operational Base + offset (0x0204)

I2C rx data register 1

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata1 data1 received 32 bits data

RKI2C_RXDATA2

Address: Operational Base + offset (0x0208)

I2C rx data register 2

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata2 data2 received 32 bits data

RKI2C_RXDATA3

Address: Operational Base + offset (0x020c)

I2C rx data register 3

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata3 data3 received 32 bits data

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RKI2C_RXDATA4

Address: Operational Base + offset (0x0210)

I2C rx data register 4

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata4 data4 received 32 bits data

RKI2C_RXDATA5

Address: Operational Base + offset (0x0214)

I2C rx data register 5

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata5 data5 received 32 bits data

RKI2C_RXDATA6

Address: Operational Base + offset (0x0218)

I2C rx data register 6

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata6 data6 received 32 bits data

RKI2C_RXDATA7

Address: Operational Base + offset (0x021c)

I2C rx data register 7

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	rxdata7 data7 received 32 bits data

6.5 Interface Description

Table 31-1I2C Interface Description

Module pin	Direction	Pad name	IOMUX
I2C0 Interface			
i2c0_sda	I/O	IO_I2C0scl_FEPHYled_linkm1_GPIO2D0vccio5	GRF_GPIO2D_IOMUX[3:2]=2'b01
i2c0_scl	I/O	IO_I2C0sda_FEPHYLEDrxm1_FEPHYLEDtxm1_GPIO2D1vccio5	GRF_GPIO2D_IOMUX[1:0]=2'b01
I2C1 Interface			
i2c1_sda	I/O	IO_PWM0_I2C1sda_GPIO2A4vccio5	GRF_GPIO2A_IOMUX[9:8]=2'b10
i2c1_scl	I/O	IO_PWM1_I2C1scl_GPIO2A5vccio5	GRF_GPIO2A_IOMUX[5:4]=2'b10
I2C2 Interface			
i2c2_sda	I/O	IO_ I2C2sda_TSADCshut_GPIO2B5vccio5	GRF_GPIO2B_IOMUX[11:10]=2'b01
i2c2_scl	I/O	IO_ I2C2scl_GPIO2B6vccio5	GRF_GPIO2B_IOMUX[13:12]=2'b01
I2C3 Interface			
i2c3_sda	I/O	IO_HDMIscl_I2C3scl_GPIO0A5pmuio	GRF_GPIO0A_IOMUX[13:12]=2'b10
		IO_I2C3scl5v_HDMISCLpmuio5v	GRF_CON_I2C3_SCL5V=1
i2c3_scl	I/O	IO_HDMIsda_I2C3sda_GPIO0A6pmuio	GRF_GPIO0A_IOMUX[11:10]=2'b10
		IO_I2Csda5v_HDMISDApmuio5v	GRF_CON_I2C3_SDA5V=1

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6.6 Application Notes

The I2C controller core operation flow chart below is to describe how the software configures and performs an I2C transaction through this I2C controller core. Descriptions are divided into 3 sections, transmit only mode, receive only mode, and mix mode. Users are strongly advised to follow

- Transmit only mode (I2C_CON[1:0]=2'b00)

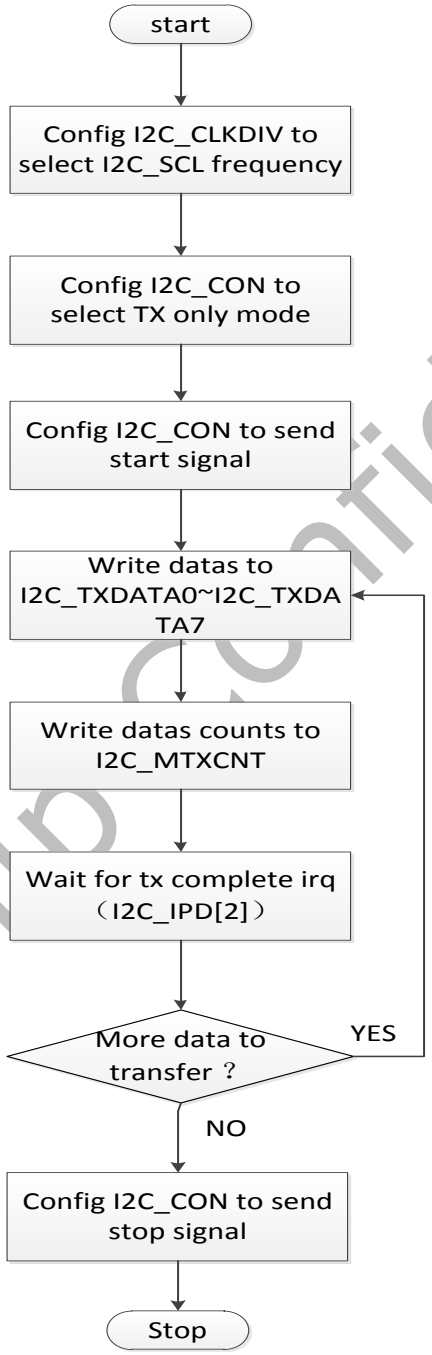

Fig.31-6I2C Flow chat for transmit only mode

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- Receive only mode (I2C_CON[1:0]=2'b10)

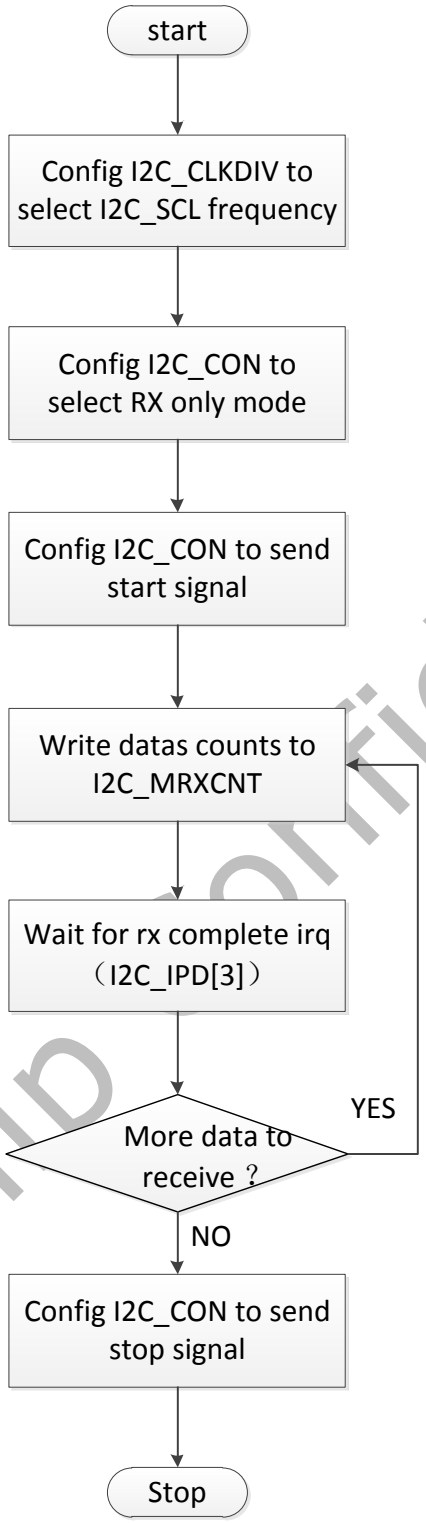

Fig.31-7I2C Flow chat for receive only mode

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- Mix mode (I2C_CON[1:0]=2'b01 or I2C_CON[1:0]=2'b11)

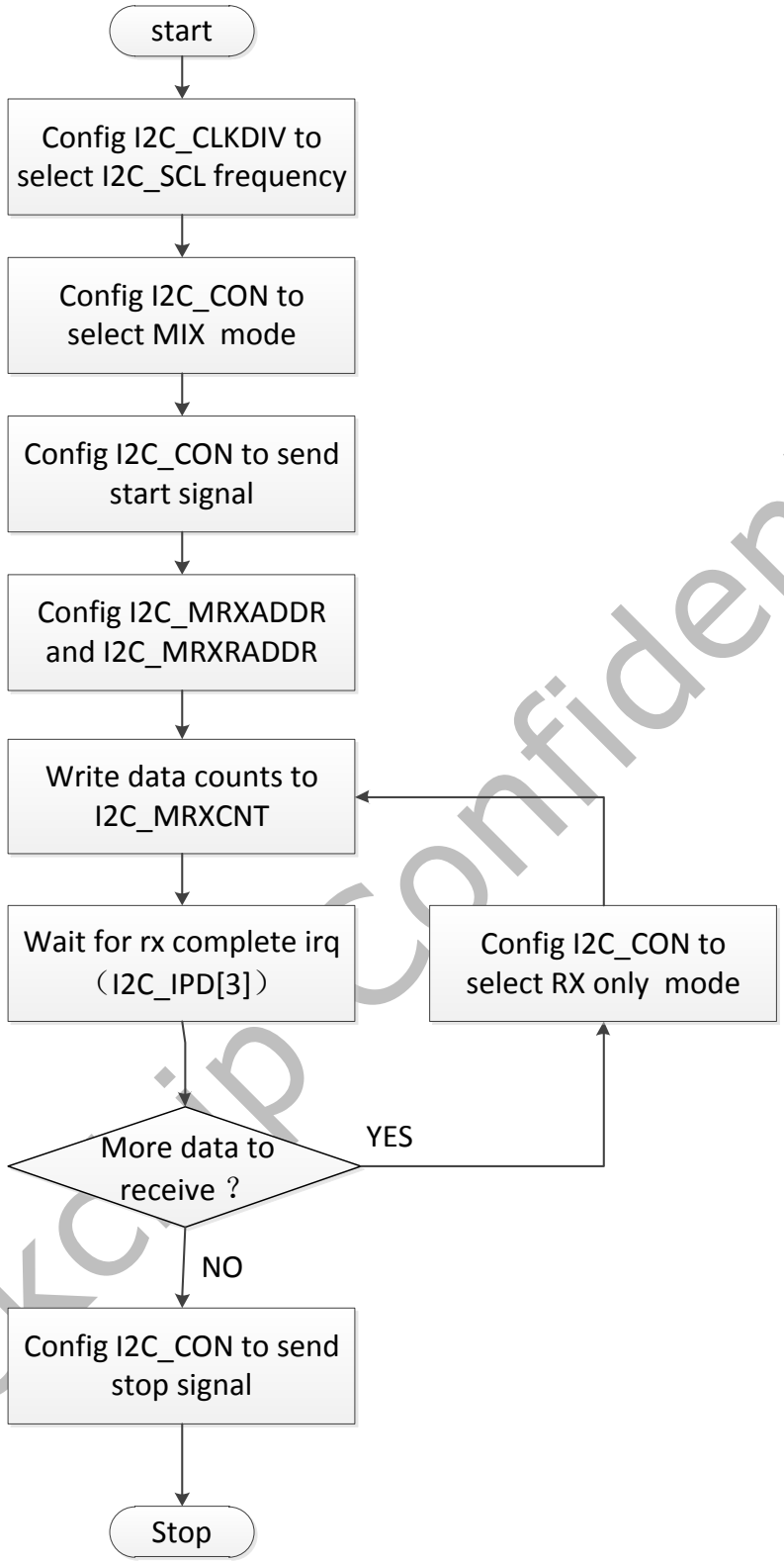

Fig.31-8I2C Flow chat for mix mode

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Chapter 32 GPIO

1.1 Overview

GPIO is a programmable General Purpose Programming I/O peripheral. This component is an APB slave device.GPIO controls the output data and direction of external I/O pads. It also can read back the data on external pads using memory-mapped registers.

GPIO supports the following features:

- 32 bits APB bus width
- 32 independently configurable signals
- Separate data registers and data direction registers for each signal
- Software control for each signal, or for each bit of each signal
- Configurable interrupt mode

1.2 Block Diagram

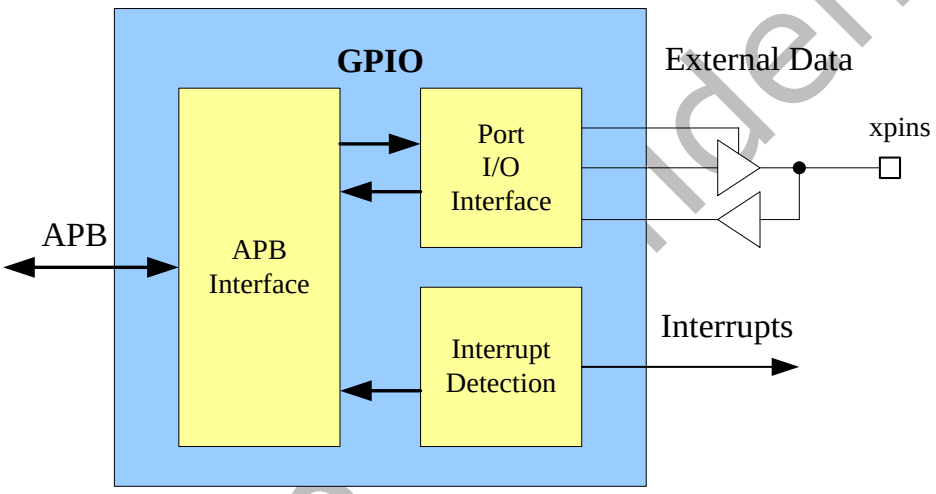

Fig. 31-9GPIO block diagram

Block descriptions:
APB Interface

The APB Interface implements the APB slave operation. Its data bus width is 32 bits.

Port I/O Interface
External data Interface to or from I/O pads.
Interrupt Detection
Interrupt interface to or from interrupt controller.

1.3 Function Description

1.3.1 Operation

Control Mode (software)

Under software control, the data and direction control for the signal are sourced from the data register (GPIO_SWPORTA_DR) and direction control register (GPIO_SWPORTA_DDR).

The direction of the external I/O pad is controlled by a write to the Porta data direction register (GPIO_SWPORTA_DDR). The data written to this memory-mapped register gets mapped onto an output signal, GPIO_PORTA_DDR, of the GPIO peripheral. This output signal controls the direction of an external I/O pad.

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The data written to the Porta data register (GPIO_SWPORTA_DR) drives the output buffer of the I/O pad. External data are input on the external data signal, GPIO_EXT_PORTA. Reading the external signal register(GPIO_EXT_PORTA) shows the value on the signal, regardless of the direction. This register is read-only, meaning that it cannot be written from the APB software interface.

Reading External Signals

The data on the GPIO_EXT_PORTA external signal can always be read. The data on the external GPIO signal is read by an APB read of the memory-mapped register, GPIO_EXT_PORTA.

An APB read to the GPIO_EXT_PORTA register yields a value equal to that which is on the GPIO_EXT_PORTA signal.

Interrupts

Port A can be programmed to accept external signals as interrupt sources on any of the bits of the signal. The type of interrupt is programmable with one of the following settings:

- Active-high and level
- Active-low and level
- Rising edge
- Falling edge

The interrupts can be masked by programming the GPIO_INTMASK register. The interrupt status can be read before masking (called raw status) and after masking.

The interrupts are combined into a single interrupt output signal, which has the same polarity as the individual interrupts. In order to mask the combined interrupt, all individual interrupts have to be masked. The single combined interrupt does not have its own mask bit.

Whenever Port A is configured for interrupts, the data direction must be set to Input. If the data direction register is reprogrammed to Output, then any pending interrupts are not lost. However, no new interrupts are generated.

For edge-detected interrupts, the ISR can clear the interrupt by writing a 1 to the GPIO_PORTA_EOI register for the corresponding bit to disable the interrupt. This write also clears the interrupt status and raw status registers. Writing to the GPIO_PORTA_EOI register has no effect on level-sensitive interrupts. If level-sensitive interrupts cause the processor to interrupt, then the ISR can poll the GPIO_INT_RAWSTATUS register until the interrupt source disappears, or it can write to the GPIO_INTMASK register to mask the interrupt before exiting the ISR. If the ISR exits without masking or disabling the interrupt prior to exiting, then the level-sensitive interrupt repeatedly requests an interrupt until the interrupt is cleared at the source.

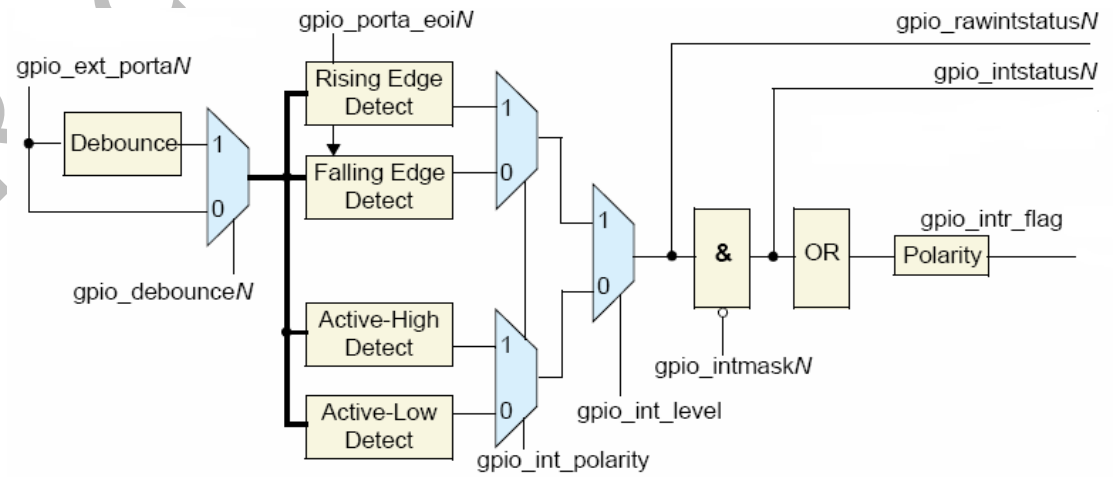

Fig. 31-10 GPIO Interrupt RTL Block Diagram

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Debounce operation

Port A has been configured to include the debounce capability interrupt feature. The external signal can be debounced to remove any spurious glitches that are less than one period of the external debouncing clock.

When input interrupt signals are debounced using a debounce clock (pclk), the signals must be active for a minimum of two cycles of the debounce clock to guarantee that they are registered. Any input pulse widths less than a debounce clock period are bounced. A pulse width between one and two debounce clock widths may or may not propagate, depending on its phase relationship to the debounce clock. If the input pulse spans two rising edges of the debounce clock, it is registered. If it spans only one rising edge, it is not registered.

Synchronization of Interrupt Signals to the System Clock

Interrupt signals are internally synchronized to pclk. Synchronization to pclk must occur for edge-detect signals. With level-sensitive interrupts, synchronization is optional and under software control (GPIO_LS_SYNC).

1.3.2 Programming

Programming Considerations

- Reading from an unused location or unused bits in a particular register always returns zeros. There is no error mechanism in the APB.
- Programming the GPIO registers for interrupt capability, edge-sensitive or level-sensitive interrupts, and interrupt polarity should be completed prior to enabling the interrupts on Port A in order to prevent spurious glitches on the interrupt lines to the interrupt controller.
- Writing to the interrupt clear register clears an edge-detected interrupt and has no effect on a level-sensitive interrupt.

GPIOs' hierarchy in the chip

GPIO0, GPIO1, GPIO2, GPIO3 are in PD_BUS subsystem.

1.4 Register Description

This section describes the control/status registers of the design. Software should read and write these registers using 32-bits accesses. There are 4 GPIOs (GPIO0 ~ GPIO3), and each of them has same register group. Therefore, 4 GPIOs' register groups have 4 different base addresses.

1.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
GPIO_SWPORTA_DR	0x0000	W	0x00000000	Port A data register
GPIO_SWPORTA_DDR	0x0004	W	0x00000000	Port A data direction register
GPIO_INTEN	0x0030	W	0x00000000	Interrupt enable register
GPIO_INTMASK	0x0034	W	0x00000000	Interrupt mask register
GPIO_INTTYPE_LEVEL	0x0038	W	0x00000000	Interrupt level register
GPIO_INT_POLARITY	0x003c	W	0x00000000	Interrupt polarity register
GPIO_INT_STATUS	0x0040	W	0x00000000	Interrupt status of port A
GPIO_INT_RAWSTATUS	0x0044	W	0x00000000	Raw Interrupt status of port A
GPIO_DEBOUNCE	0x0048	W	0x00000000	Debounce enable register
GPIO_PORTA_EOI	0x004c	W	0x00000000	Port A clear interrupt register
GPIO_EXT_PORTA	0x0050	W	0x00000000	Port A external port register

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Name	Offset	Size	Reset Value	Description
GPIO_LS_SYNC	0x0060	W	0x00000000	Level_sensitive synchronization enable register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

1.4.2 Detail Register Description

GPIO_SWPORTA_DR
Address: Operational Base + offset (0x0000)
Port A data register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_swporta_dr Values written to this register are output on the I/O signals for Port A if the corresponding data direction bits for Port A are set to Output mode.The value read back is equal to the last value written to this register.

GPIO_SWPORTA_DDR
Address: Operational Base + offset (0x0004)
Port A data direction register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_swporta_ddr Values written to this register independently control the direction of the corresponding data bit in Port A. 0: Input (default) 1: Output

GPIO_INTEN
Address: Operational Base + offset (0x0030)
Interrupt enable register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_int_en Allows each bit of Port A to be configured for interrupts. Whenever a 1 is written to a bit of this register, it configures the corresponding bit on Port A to become an interrupt; otherwise, Port A operates as a normal GPIO signal. Interrupts are disabled on the corresponding bits of Port A if the corresponding data direction register is set to Output. 0: Configure Port A bit as normal GPIO signal (default) 1: Configure Port A bit as interrupt

GPIO_INTMASK
Address: Operational Base + offset (0x0034)
Interrupt mask register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_int_mask Controls whether an interrupt on Port A can create an interrupt for the interrupt controller by not masking it. Whenever a 1 is written to a bit in this register, it masks the interrupt generation capability for this signal; otherwise interrupts are allowed through. 0: Interrupt bits are unmasked (default) 1: Mask interrupt

GPIO_INTTYPE_LEVEL
Address: Operational Base + offset (0x0038)
Interrupt level register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_inttype_level Controls the type of interrupt that can occur on Port A. 0: Level-sensitive (default) 1: Edge-sensitive

GPIO_INT_POLARITY
Address: Operational Base + offset (0x003c)
Interrupt polarity register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_int_polarity Controls the polarity of edge or level sensitivity that can occur on input of Port A. 0: Active-low (default) 1: Active-high

GPIO_INT_STATUS
Address: Operational Base + offset (0x0040)
Interrupt status of port A

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	gpio_int_status Interrupt status of Port A

GPIO_INT_RAWSTATUS
Address: Operational Base + offset (0x0044)
Raw Interrupt status of port A

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	gpio_int_rawstatus Raw interrupt of status of Port A (premasking bits

GPIO_DEBOUNCE
Address: Operational Base + offset (0x0048)
Debounce enable register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	gpio_debounce Controls whether an external signal that is the source of an interrupt needs to be debounced to remove any spurious glitches. Writing a 1 to a bit in this register enables the debouncing circuitry. A signal must be valid for two periods of an external clock before it is internally processed. 0: No debounce (default) 1: Enable debounce

GPIO_PORTA_EOI
Address: Operational Base + offset (0x004c)
Port A clear interrupt register

Bit	Attr	Reset Value	Description
31:0	WO	0x00000000	gpio_porta_eoi Controls the clearing of edge type interrupts from Port A. When a 1 is written into a corresponding bit of this register, the interrupt is cleared. All interrupts are cleared when Port A is not configured for interrupts. 0: No interrupt clear (default) 1: Clear interrupt

GPIO_EXT_PORTA
Address: Operational Base + offset (0x0050)
Port A external port register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	gpio_ext_porta When Port A is configured as Input, then reading this location reads the values on the signal. When the data direction of Port A is set as Output, reading this location reads the data register for Port A.

GPIO_LS_SYNC
Address: Operational Base + offset (0x0060)
Level_sensitive synchronization enable register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	gpio_ls_sync Writing a 1 to this register results in all level-sensitive interrupts being synchronized to pclk_intr. 0: No synchronization to pclk_intr (default) 1: Synchronize to pclk_intr

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1.5 Interface Description

Table 31-2 GPIO interface description

Module Pin	Dir	Pad Name	IOMUX Setting
GPIO0 Interface			
gpio0_porta[7:0]	I/O	GPIO0_A[7:0]	GRF_GPIO0A_IOMUX[15:0]=16'h0
gpio0_porta[15:8]	I/O	GPIO0_B[7:0]	GRF_GPIO0B_IOMUX[15:0]=16'h0
gpio0_porta[23:16]	I/O	GPIO0_C[7:0]	GRF_GPIO0C_IOMUX[15:0]=16'h0
gpio0_porta[31:24]	I/O	GPIO0_D[7:0]	GRF_GPIO0D_IOMUX[15:0]=16'h0
GPIO1 Interface			
gpio1_porta[7:0]	I/O	GPIO1_A[7:0]	GRF_GPIO1A_IOMUX[15:0]=16'h0
gpio1_porta[15:8]	I/O	GPIO1_B[7:0]	GRF_GPIO1B_IOMUX[15:0]=16'h0
gpio1_porta[23:16]	I/O	GPIO1_C[7:0]	GRF_GPIO1C_IOMUX[15:0]=16'h0
gpio1_porta[31:24]	I/O	GPIO1_D[7:0]	GRF_GPIO1D_IOMUX[15:0]=16'h0
GPIO2 Interface			
gpio2_porta[7:0]	I/O	GPIO2_A[7:0]	GRF_GPIO2A_IOMUX[15:0]=16'h0
gpio2_porta[15:8]	I/O	GPIO2_B[7:0]	GRF_GPIO2BL_IOMUX[15:0]=16'h0 GRF_GPIO2BH_IOMUX[15:0]=16'h0
gpio2_porta[23:16]	I/O	GPIO2_C[7:0]	GRF_GPIO2CL_IOMUX[15:0]=16'h0 GRF_GPIO2CH_IOMUX[15:0]=16'h0
gpio2_porta[31:24]	I/O	GPIO2_D[7:0]	GRF_GPIO2D_IOMUX[15:0]=16'h0
GPIO3 Interface			
gpio3_porta[7:0]	I/O	GPIO3_A[7:0]	GRF_GPIO3AL_IOMUX[15:0]=16'h0 GRF_GPIO3AH_IOMUX[15:0]=16'h0
gpio3_porta[15:8]	I/O	GPIO3_B[7:0]	GRF_GPIO3BL_IOMUX[15:0]=16'h0 GRF_GPIO3BH_IOMUX[15:0]=16'h0
gpio3_porta[23:16]	I/O	GPIO3_C[7:0]	GRF_GPIO3C_IOMUX[15:0]=16'h0
gpio3_porta[31:24]	I/O	GPIO3_D[7:0]	GRF_GPIO3D_IOMUX[15:0]=16'h0

1.6 Application Notes

Steps to set GPIO's direction

- Write GPIO_SWPORT_DDR[x] as 1 to set this gpio as output direction and Write GPIO_SWPORT_DDR[x] as 0 to set this gpio as input direction.
- Default GPIO's direction is input direction.

Steps to set GPIO's level

- Write GPIO_SWPORT_DDR[x] as 1 to set this gpio as output direction.
- Write GPIO_SWPORT_DR[x] as v to set this GPIO's value.

Steps to get GPIO's level

- Write GPIO_SWPORT_DDR[x] as 0 to set this gpio as input direction.
- Read from GPIO_EXT_PORT[x] to get GPIO's value

Steps to set GPIO as interrupt source

- Write GPIO_SWPORT_DDR[x] as 0 to set this gpio as input direction.
- Write GPIO_INTTYPE_LEVEL[x] as v1 and write GPIO_INT_POLARITY[x] as v2 to set interrupt type
- Write GPIO_INTEN[x] as 1 to enable GPIO's interrupt

Note: Please switch iomux to GPIO mode first!

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Chapter 33 RKTIMER

1.1 Overview

Timer is a programmable timer peripheral. This component is an APB slave device. In RK3228H there are 6 Timers and 2 Secure Timers(STimer).

Timer5 and STimer0~1 count up from zero to a programmed value and generate an interrupt when the counter reaches the programmed value.

Timer0~4 count down from a programmed value to zero and generate an interrupt when the counter reaches zero.

Timer supports the following features:

- Timer0~5 is used for no-secure, STimer0~1 is used for secure.
- Two operation modes: free-running and user-defined count.
- Maskable for each individual interrupt.

1.2 Block Diagram

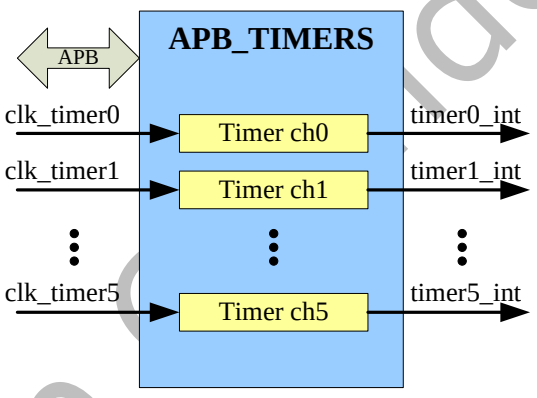

Fig. 31-11 Timer Block Diagram

The above figure shows the architecture of the APB timers (include six programmable timer channels) that in the bus subsystem. The Stimers that in the bus subsystem only include two programmable timer channels.

1.3 Function Description

1.3.1 Timer clock

TIMER0~ TIMER5 and STIMER0~1 are in the pd_bus subsystem. The timer clock is 24MHz OSC.

1.3.2 Programming sequence

1. Initialize the timer by the TIMERn_CONTROLREG (0≤n≤5) register:
- Disable the timer by writing a "0" to the timer enable bit (bit 0). Accordingly, the timer_en output signal is de-asserted.
- Program the timer mode—user-defined or free-running—by writing a "0" or "1" respectively, to the timer mode bit (bit 1).
- Set the interrupt mask as either masked or not masked by writing a "0" or "1" respectively, to the timer interrupt mask bit (bit 2).

2. Load the timer count value into the TIMERn_LOAD_COUNT1 (0≤n≤5) and TIMERn_

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LOAD_COUNT0 (0≤n≤5) register.

3. Enable the timer by writing a "1" to bit 0 of TIMERn_CONTROLREG (0≤n≤5).

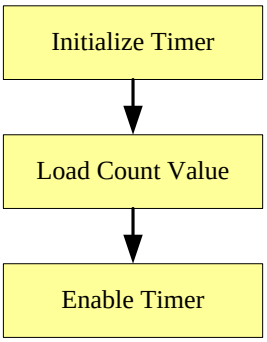

Fig. 31-12 Timer Usage Flow

1.3.3 Loading a timer count value

For the descending Timers(Timer0~4).The initial value for each timer—that is, the value from which it counts down—is loaded into the timer using the load count register (TIMERn_LOAD_COUNT1 and TIMERn_ LOAD_COUNT0). Two events can cause a timer to load the initial value from its load count register:

- Timer is enabled after reset or disabled.
- Timer counts down to 0, when timer is configured into free-running mode.

For the incremental Timers(Timer5 and STimer0~1).The initial value for each timer is zero. The count register will count up to the value loaded in the register TIMERn_LOAD_COUNT1 and TIMERn_ LOAD_COUNT0. Two events can cause a timer to load zero:

- Timer is enabled after reset or disabled.
- Timer counts up to the value stored in TIMERn_LOAD_COUNT1 and TIMERn_ LOAD_COUNT0, when timer is configured into free-running mode.

1.3.4 Timer mode selection

- User-defined count mode – Timer loads TIMERn_LOAD_COUNT1 and TIMERn_ LOAD_ COUNT0 registers (for descending timers) or zero (for incremental timers) as initial value. When the timer counts down to 0 (for descending timers) or counts up to the value in TIMERn_LOAD_COUNT1 and TIMERn_ LOAD_ COUNT0 (for incremental timers),it will not automatically reload the count register. User need to disable timer firstly and follow the programming sequence to make timer work again.
- Free-running mode – Timer loads the TIMERn_LOAD_COUNT1 and TIMERn_LOAD_COUNT0(for descending timers) or zero (for incremental timers)register as initial value. Timer will automatically reload the count register, when timer counts down to 0 (for descending timers) or counts up to the value in TIMERn_LOAD_COUNT1 and TIMERn_LOAD_COUNT0 (for incremental timers).

1.4 Register Description

This section describes the control/status registers of the design. Software should read and write these registers using 32-bits accesses.

1.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
TIMER_n_LOAD_COUNT0	0x0000	W	0x00000000	Timer n Load Count Register
TIMER_n_LOAD_COUNT1	0x0004	W	0x00000000	Timer n Load Count Register
TIMER_n_CURRENT_VALUE0	0x0008	W	0x00000000	Timer n Current Value Register

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Name	Offset	Size	Reset Value	Description
TIMER_n_CURRENT_VALUE1	0x000c	W	0x00000000	Timer n Current Value Register
TIMER_n_CONTROLREG	0x0010	W	0x00000000	Timer n Control Register
TIMER_n_INTSTATUS	0x0018	W	0x00000000	Timer Interrupt Stauts Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

1.4.2 Detail Register Description

TIMER_n_LOAD_COUNT0
Address: Operational Base + offset (0x00)
Timer n Load Count Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	load_count_low bits Low 32 bits value to be loaded into Timer n. This is the value from which counting commences.

TIMER_n_LOAD_COUNT1
Address: Operational Base + offset (0x04)
Timer n Load Count Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	load_count_high bits High 32 bits value to be loaded into Timer n. This is the value from which counting commences.

TIMER_n_CURRENT_VALUE0
Address: Operational Base + offset (0x08)
Timer n Current Value Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	current_cnt_lowbits Low 32 bits of current value of timer n.

TIMER_n_CURRENT_VALUE1
Address: Operational Base + offset (0x0c)
Timer n Current Value Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	current_cnt_highbits High 32 bits of current value of timer n.

TIMER_n_CONTROLREG
Address: Operational Base + offset (0x10)
Timer n Control Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	int_en Timer interrupt mask 0: mask 1: not mask

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Bit	Attr	Reset Value	Description
1	RW	0x0	timer_mode Timer mode. 0: free-running mode 1: user-defined count mode
0	RW	0x0	timer_en Timer enable. 0: disable 1: enable

TIMER_n_INTSTATUS
Address: Operational Base + offset (0x18)
Timer Interrupt Stauts Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	int_pd This register contains the interrupt status for timer n. Write 1 to this register will clear the interrupt.

1.5 Application Notes

In the chip, the timer_clk is from 24MHz OSC, asynchronous to the pclk. When user disables the timer enables bit (bit 0 of TIMERn_CONTROLREG (0≤n≤5)), the timer en output signal is de-asserted, and timer_clk will stop. When user enables the timer, the timer_en signal is asserted and timer_clk will start running.
The application is only allowed to re-config registers when timer_en is low.

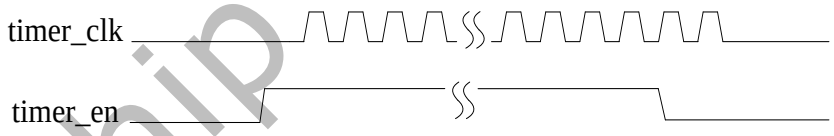

Fig. 31-13 Timing between timer_en and timer_clk

Please refer to function description section for the timer usage flow.

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Chapter 34 Pulse Width Modulation (PWM)

9.1 Overview

The pulse-width modulator (PWM) feature is very common in embedded systems. It provides a way to generate a pulse periodic waveform for motor control or can act as a digital-to-analog converter with some external components.

The PWM Module supports the following features:

- 4-built-in PWM channels
- Configurable to operate in capture mode
  - Measures the high/low polarity effective cycles of this input waveform
  - Generates a single interrupt at the transition of input waveform polarity
  - 32-bit high polarity capture register
  - 32-bit low polarity capture register
  - 32-bit current value register
  - The capture result of channel 3 can be stored in a FIFO. The depths of FIFO is 8, and the data in FIFO can be read through DMA. It also supports timeout interrupt when the data in FIFO has not been read in a time-threshold.
- Configurable to operate in continuous mode or one-shot mode
  - 32-bit period counter
  - 32-bit duty register
  - 32-bit current value register
  - Configurable PWM output polarity in inactive state and duty period pulse polarity
  - Period and duty cycle are shadow buffered. Change takes effect when the end of the effective period is reached or when the channel is disabled
  - Programmable center or left aligned outputs, and change takes effect when the end of the effective period is reached or when the channel is disabled
  - 8-bit repeat counter for one-shot operation. One-shot operation will produce N + 1 periods of the waveform, where N is the repeat counter value, and generates a single interrupt at the end of operation
  - Continuous mode generates the waveform continuously, and does not generates any interrupts
- pre-scaled operation to bus clock and then further scaled
- Available low-power mode to reduce power consumption when the channel is inactive.

9.2 Block Diagram

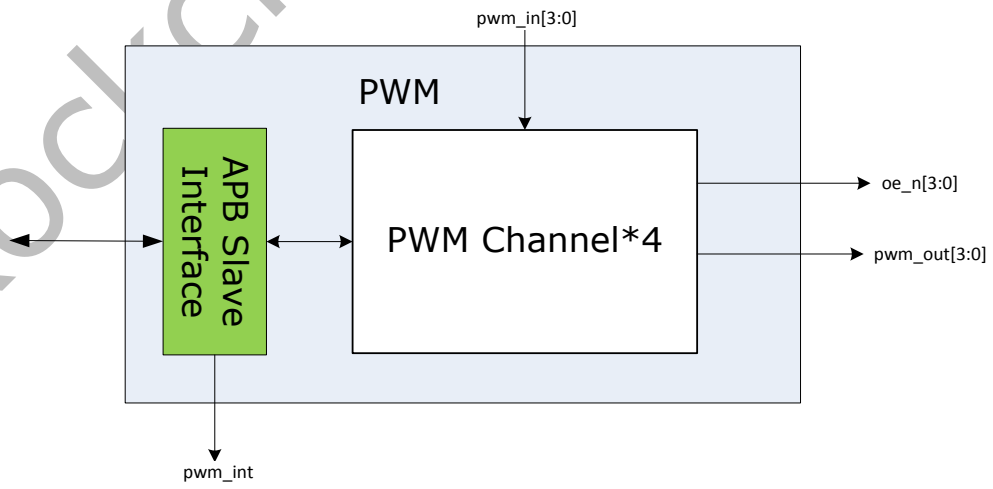

Fig. 34-1 PWM Block Diagram

The host processor gets access to PWM Register Block through the APB slave interface with 32-bit bus width, and asserts the active-high level interrupt. PWM only supports one interrupt output, please refer to interrupt register to know the raw interrupt status when an interrupt is asserted.

PWM Channel is the control logic of PWM module, and controls the operation of PWM module

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according to the configured working mode.

9.3 Function Description

The PWM supports three operation modes: capture mode, one-shot mode and continuous mode. For the one-shot mode and the continuous mode, the PWM output can be configured as the left-aligned mode or the center-aligned mode.

9.3.1 Capture mode

The capture mode is used to measure the PWM channel input waveform high/low effective cycles with the PWM channel clock, and asserts an interrupt when the polarity of the input waveform changes. The number of the high effective cycles is recorded in the PWMx_PERIOD_HPC register, while the number of the low effective cycles is recorded in the PWMx_DUTY_LPC register.

Notes: the PWM input waveform is doubled buffered when the PWM channel is working in order to filter unexpected shot-time polarity transition, and therefore the interrupt is asserted several cycles after the input waveform polarity changes, and so does the change of the values of PWMx_PERIOD_HPC and PWMx_DUTY_LPC.

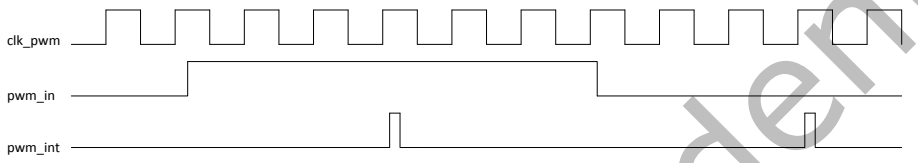

Fig. 34-2 PWM Capture Mode

9.3.2 Continuous mode

The PWM channel generates a series of the pulses continuously as expected once the channel is enabled with continuous mode.

In the continuous mode, the PWM output waveforms can be in one form of the two output mode: left-aligned mode or center-aligned mode.

For the left-aligned output mode, the PWM channel firstly starts the duty cycle with the configured duty polarity (PWMx_CTRL.duty_pol). Once duty cycle number (PWMx_DUTY_LPC) is reached, the output is switched to the opposite polarity. After the period number (PWMx_PERIOD_HPC) is reached, the output is again switched to the opposite polarity to start another period of desired pulse.

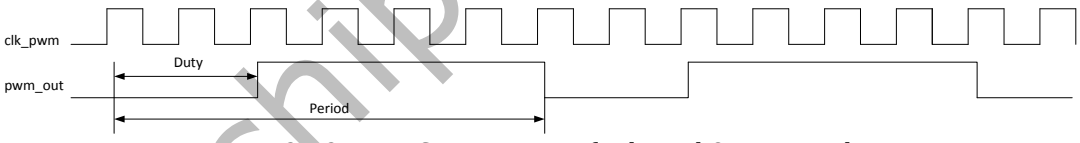

Fig. 34-3 PWM Continuous Left-aligned Output Mode

For the center-aligned output mode, the PWM channel firstly starts the duty cycle with the configured duty polarity (PWMx_CTRL.duty_pol). Once one half of duty cycle number (PWMx_DUTY_LPC) is reached, the output is switched to the opposite polarity. Then if there is one half of duty cycle left for the whole period, the output is again switched to the opposite polarity. Finally after the period number (PWMx_PERIOD_HPC) is reached, the output starts another period of desired pulse.

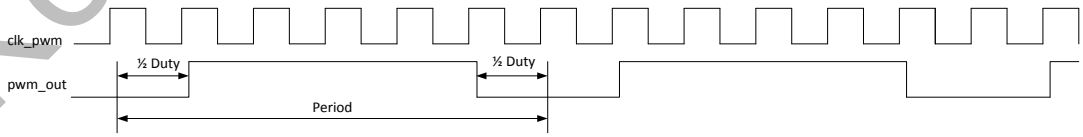

Fig. 34-4 PWM Continuous Center-aligned Output Mode

Once disable the PWM channel, the channel stops generating the output waveforms and output polarity is fixed as the configured inactive polarity (PWMx_CTRL.inactive_pol).

9.3.3 One-shot mode

Unlike the continuous mode, the PWM channel generates the output waveforms within the configured periods (PWM_CTRL.rpt + 1), and then stops. At the same times, an interrupt is asserted to inform that the operation has been finished.

There are also two output modes for the one-shot mode: the left-aligned mode and the center-aligned mode.

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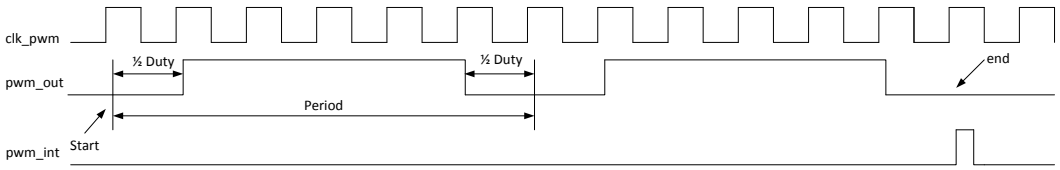

Fig. 34-5 PWM One-shot Center-aligned Output Mode

9.4 Register Description

9.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
PWM_PWM0_CNT	0x0000	W	0x00000000	PWM Channel 0 Counter Register
PWM_PWM0_PERIOD_HPR	0x0004	W	0x00000000	PWM Channel 0 Period Register/High Polarity Capture Register
PWM_PWM0_DUTY_LPR	0x0008	W	0x00000000	PWM Channel 0 Duty Register/Low Polarity Capture Register
PWM_PWM0_CTRL	0x000c	W	0x00000000	PWM Channel 0 Control Register
PWM_PWM1_CNT	0x0010	W	0x00000000	PWM Channel 1 Counter Register
PWM_PWM1_PERIOD_HPR	0x0014	W	0x00000000	PWM Channel 1 Period Register/High Polarity Capture Register
PWM_PWM1_DUTY_LPR	0x0018	W	0x00000000	PWM Channel 1 Duty Register/Low Polarity Capture Register
PWM_PWM1_CTRL	0x001c	W	0x00000000	PWM Channel 1 Control Register
PWM_PWM2_CNT	0x0020	W	0x00000000	PWM Channel 2 Counter Register
PWM_PWM2_PERIOD_HPR	0x0024	W	0x00000000	PWM Channel 2 Period Register/High Polarity Capture Register
PWM_PWM2_DUTY_LPR	0x0028	W	0x00000000	PWM Channel 2 Duty Register/Low Polarity Capture Register
PWM_PWM2_CTRL	0x002c	W	0x00000000	PWM Channel 2 Control Register
PWM_PWM3_CNT	0x0030	W	0x00000000	PWM Channel 3 Counter Register
PWM_PWM3_PERIOD_HPR	0x0034	W	0x00000000	PWM Channel 3 Period Register/High Polarity Capture Register
PWM_PWM3_DUTY_LPR	0x0038	W	0x00000000	PWM Channel 3 Duty Register/Low Polarity Capture Register
PWM_PWM3_CTRL	0x003c	W	0x00000000	PWM Channel 3 Control Register
PWM_INTSTS	0x0040	W	0x00000000	Interrupt Status Register
PWM_INT_EN	0x0044	W	0x00000000	Interrupt Enable Register
PWM_PWM_FIFO_CTRL	0x0050	W	0x00000000	PWM Channel 3 FIFO Mode Control Register
PWM_PWM_FIFO_INTSTS	0x0054	W	0x00000000	FIFO Interrupts Status Register
PWM_PWM_FIFO_TOUTTHR	0x0058	W	0x00000000	FIFO Timeout Threshold Register
PWM_PWM_FIFO	0x0060	W	0x00000000	FIFO Register

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*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

9.4.2 Detail Register Description

PWM_PWM0_CNT
Address: Operational Base + offset (0x0000)
PWM Channel 0 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 0 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM0_PERIOD_HPR
Address: Operational Base + offset (0x0004)
PWM Channel 0 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_HPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM0_DUTY_LPR
Address: Operational Base + offset (0x0008)
PWM Channel 0 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM0_CTRL
Address: Operational Base + offset (0x000c)

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PWM Channel 0 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This field defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7	RO	0x0	reserved
6	RW	0x0	conlock pwm configure lock pwm period and duty lock to previous configuration 0: disable lock 1: enable lock
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode

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Bit	Attr	Reset Value	Description
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt . 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked in the one-shot mode, this bit will be cleared at the end of operation

PWM_PWM1_CNT
Address: Operational Base + offset (0x0010)
PWM Channel 1 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 1 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM1_PERIOD_HPR
Address: Operational Base + offset (0x0014)
PWM Channel 1 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_HPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM1_DUTY_LPR
Address: Operational Base + offset (0x0018)
PWM Channel 1 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM1_CTRL
Address: Operational Base + offset (0x001c)
PWM Channel 1 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This field defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7	RO	0x0	reserved
6	RW	0x0	conlock pwm configure lock pwm period and duty lock to previous configuration 1: enable lock
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive

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Bit	Attr	Reset Value	Description
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked in the one-shot mode, this bit will be cleared at the end of operation

PWM_PWM2_CNT
Address: Operational Base + offset (0x0020)
PWM Channel 2 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 2 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM2_PERIOD_HPR
Address: Operational Base + offset (0x0024)
PWM Channel 2 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_HPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM2_DUTY_LPR
Address: Operational Base + offset (0x0028)
PWM Channel 2 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM2_CTRL
Address: Operational Base + offset (0x002c)
PWM Channel 2 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.
23:16	RW	0x00	scale Scale Factor This fields defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6	RW	0x0	conlock pwm configure lock pwm period and duty lock to previous configuration 1: enable lock
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt. 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked in the one-shot mode, this bit will be cleared at the end of operation

PWM_PWM3_CNT

Address: Operational Base + offset (0x0030)

PWM Channel 3 Counter Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CNT Timer Counter The 32-bit indicates current value of PWM Channel 3 counter. The counter runs at the rate of PWM clock. The value ranges from 0 to (2^32-1).

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PWM_PWM3_PERIOD_HPR
Address: Operational Base + offset (0x0034)
PWM Channel 3 Period Register/High Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	PERIOD_HPR Output Waveform Period/Input Waveform High Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the period of the output waveform. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the bit [31:1] is taken into account and bit [0] always considered as 0. If PWM is operated at the capture mode, this value indicates the effective high polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM3_DUTY_LPR
Address: Operational Base + offset (0x0038)
PWM Channel 3 Duty Register/Low Polarity Capture Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	DUTY_LPR Output Waveform Duty Cycle/Input Waveform Low Polarity Cycle If PWM is operated at the continuous mode or one-shot mode, this value defines the duty cycle of the output waveform. The PWM starts the output waveform with duty cycle. Note that, if the PWM is operated at the center-aligned mode, the period should be an even one, and therefore only the [31:1] is taken into account. If PWM is operated at the capture mode, this value indicates the effective low polarity cycles of input waveform. This value is based on the PWM clock. The value ranges from 0 to (2^32-1).

PWM_PWM3_CTRL
Address: Operational Base + offset (0x003c)
PWM Channel 3 Control Register

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rpt Repeat Counter This field defines the repeated effective periods of output waveform in one-shot mode. The value N means N+1 repeated effective periods.

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Bit	Attr	Reset Value	Description
23:16	RW	0x00	scale Scale Factor This field defines the scale factor applied to prescaled clock. The value N means the clock is divided by 2*N. If N is 0, it means that the clock is divided by 512(2*256).
15	RO	0x0	reserved
14:12	RW	0x0	prescale Prescale Factor This field defines the prescale factor applied to input clock. The value N means that the input clock is divided by 2^N.
11:10	RO	0x0	reserved
9	RW	0x0	clk_sel Clock Source Select 0: non-scaled clock is selected as PWM clock source. It means that the prescale clock is directly used as the PWM clock source 1: scaled clock is selected as PWM clock source
8	RW	0x0	lp_en Low Power Mode Enable 0: disabled 1: enabled When PWM channel is inactive state and Low Power Mode is enabled, the path to PWM Clock prescale module is blocked to reduce power consumption.
7	RO	0x0	reserved
6	RW	0x0	conlock pwm configure lock pwm period and duty lock to previous configuration 1: enable lock
5	RW	0x0	output_mode PWM Output mode 0: left aligned mode 1: center aligned mode
4	RW	0x0	inactive_pol Inactive State Output Polarity This defines the output waveform polarity when PWM channel is in inactive state. The inactive state means that PWM finishes the complete waveform in one-shot mode or PWM channel is disabled. 0: negative 1: positive

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Bit	Attr	Reset Value	Description
3	RW	0x0	duty_pol Duty Cycle Output Polarity This defines the polarity for duty cycle. PWM starts the output waveform with duty cycle. 0: negative 1: positive
2:1	RW	0x0	pwm_mode PWM Operation Mode 00: One shot mode. PWM produces the waveform within the repeated times defined by PWMx_CTRL_rpt 01: Continuous mode. PWM produces the waveform continuously 10: Capture mode. PWM measures the cycles of high/low polarity of input waveform. 11: reserved
0	RW	0x0	pwm_en PWM channel enable 0: disabled 1: enabled. If the PWM is worked in the one-shot mode, this bit will be cleared at the end of operation

PWM_INTSTS
Address: Operational Base + offset (0x0040)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RO	0x0	CH3_Pol Channel 3 Interrupt Polarity Flag This bit is used in capture mode in order to identify the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM3_PERIOD_HPR to know the effective high cycle of Channel 3 input waveform. Otherwise, please refer to PWM3_PERIOD_LPR to know the effective low cycle of Channel 3 input waveform. Write 1 to CH3_IntSts will clear this bit.
10	RO	0x0	CH2_Pol Channel 2 Interrupt Polarity Flag This bit is used in capture mode in order to identify the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM2_PERIOD_HPR to know the effective high cycle of Channel 2 input waveform. Otherwise, please refer to PWM2_PERIOD_LPR to know the effective low cycle of Channel 2 input waveform. Write 1 to CH2_IntSts will clear this bit.

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Bit	Attr	Reset Value	Description
9	RO	0x0	CH1_Pol Channel 1 Interrupt Polarity Flag This bit is used in capture mode in order to identify the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM1_PERIOD_HPR to know the effective high cycle of Channel 1 input waveform. Otherwise, please refer to PWM1_PERIOD_LPR to know the effective low cycle of Channel 1 input waveform. Write 1 to CH1_IntSts will clear this bit.
8	RO	0x0	CH0_Pol Channel 0 Interrupt Polarity Flag This bit is used in capture mode in order to identify the transition of the input waveform when interrupt is generated. When bit is 1, please refer to PWM0_PERIOD_HPR to know the effective high cycle of Channel 0 input waveform. Otherwise, please refer to PWM0_PERIOD_LPR to know the effective low cycle of Channel 0 input waveform. Write 1 to CH0_IntSts will clear this bit.
7:4	RO	0x0	reserved
3	R/WSC	0x0	CH3_IntSts Channel 3 Interrupt Status 0: Channel 3 Interrupt not generated 1: Channel 3 Interrupt generated
2	W1C	0x0	CH2_IntSts Channel 2 Interrupt Status 0: Channel 2 Interrupt not generated 1: Channel 2 Interrupt generated
1	W1C	0x0	CH1_IntSts Channel 1 Interrupt Status 0: Channel 1 Interrupt not generated 1: Channel 1 Interrupt generated
0	W1C	0x0	CH0_IntSts Channel 0 Raw Interrupt Status 0: Channel 0 Interrupt not generated 1: Channel 0 Interrupt generated

PWM_INT_EN
Address: Operational Base + offset (0x0044)
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	CH3_Int_en Channel 3 Interrupt Enable 0: Channel 3 Interrupt disabled 1: Channel 3 Interrupt enabled

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Bit	Attr	Reset Value	Description
2	RW	0x0	CH2_Int_en Channel 2 Interrupt Enable 0: Channel 2 Interrupt disabled 1: Channel 2 Interrupt enabled
1	RW	0x0	CH1_Int_en Channel 1 Interrupt Enable 0: Channel 1 Interrupt disabled 1: Channel 1 Interrupt enabled
0	RW	0x0	CH0_Int_en Channel 0 Interrupt Enable 0: Channel 0 Interrupt disabled 1: Channel 0 Interrupt enabled

PWM_PWM_FIFO_CTRL
Address: Operational Base + offset (0x0050)
PWM Channel 3 FIFO Mode Control Register

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	timeout_en fifo timeout enable
8	RW	0x0	dma_mode_en dma mode enable 1'b1: enable 1'b0: disable
7	RO	0x0	reserved
6:4	RW	0x0	almost_full_watermark Almost full Watermark level
3	RW	0x0	watermark_int_en Watermark full interrupt
2	RW	0x0	overflow_int_en FIFO Overflow Interrupt Enable When high, an interrupt asserts when the channel 3
1	RW	0x0	full_int_en FIFO Full Interrupt Enable When high, an interrupt asserts when the channel 3 FIFO is full.
0	RW	0x0	fifo_mode_sel FIFO MODE Sel When high, PWM FIFO mode is activated

PWM_PWM_FIFO_INTSTS
Address: Operational Base + offset (0x0054)
FIFO Interrupts Status Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RO	0x0	fifo_empty_status FIFO empty Status This bit indicates the FIFO is empty
3	W1C	0x0	timieout_intsts Timeout interrupt Timeout interrupt
2	W1C	0x0	fifo_watermark_full_intsts FIFO Watermark Full Interrupt Status This bit indicates the FIFO is Watermark Full
1	W1C	0x0	fifo_overflow_intsts FIFO Overflow Interrupt Status This bit indicates the FIFO is overflow
0	W1C	0x0	fifo_full_intsts FIFO Full Interrupt Status This bit indicates the FIFO is full

PWM_PWM_FIFO_TOUTTHR
Address: Operational Base + offset (0x0058)
FIFO Timeout Threshold Register

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:0	RW	0x00000	timeout_threshold FIFO Timeout value(unit pwmclk)

PWM_PWM_FIFO
Address: Operational Base + offset (0x0060)
FIFO Register

Bit	Attr	Reset Value	Description
31	RO	0x0	pol Polarity This bit indicates the polarity of the lower 31-bit counter. 0: Low 1: High
30:0	RO	0x00000000	cycle_cnt High/Low Cycle Counter This 31-bit counter indicates the effective cycles of high/low waveform.

9.5 Interface Description

Table 34-1 PWM Interface Description

Module Pin	Direction	Pad Name	IOMUX Setting
PWM0	I/O	IO_PWM0_I2C1sda_GPIO2A4vccio5	GRF_GPIO2A_IOMUX[9:8]=2'b01

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PWM1	I/O	IO_PWM1_I2C1scl _GPIO2A5vccio5	GRF_GPIO2A_IOMUX[11:10]=2'b01
PWM2	I/O	IO_PWM2_GPIO2A6v ccio5	GRF_GPIO2A_IOMUX[13:12]=2'b01
PWM3	I/O	IO_PWMir_POWERsta te2_GPIO2A2vccio5	GRF_GPIO2A_IOMUX[5:4]=2'b01

Notes: I=input, O=output, I/O=input/output, bidirectional.

9.6 Application Notes

9.6.1 PWM Capture Mode Standard Usage Flow

1. Set PWMx_CTRL.pwm_en to '0' to disable the PWM channel.
2. Choose the prescale factor and the scale factor for pclk by programming PWMx_CTRL.prescale and PWMx_CTRL.scale, and select the clock needed by setting PWMx_CTRL.clk_sel.
3. Configure the channel to work in the capture mode.
4. Enable the INT_EN.chx_int_en to enable the interrupt generation.
5. Enable the channel by writing '1' to PWMx_CTRL.pwm_en bit to start the channel.
6. When an interrupt is asserted, refer to INTSTS register to know the raw interrupt status. If the corresponding polarity flag is set, turn to PWMx_PERIOD_HPC register to know the effective high cycles of input waveforms, otherwise turn to PWMx_DUTY_LPC register to know the effective low cycles.
7. Write '0' to PWMx_CTRL.pwm_en to disable the channel.

9.6.2 PWM Capture DMA Mode Standard Usage Flow

1. Set PWMx_CTRL.pwm_en to '0' to disable the PWM channel.
2. Choose the prescale factor and the scale factor for pclk by programming PWMx_CTRL.prescale and PWMx_CTRL.scale, and select the clock needed by setting PWMx_CTRL.clk_sel.
3. Configure the channel 3 to work in the capture mode.
4. Configure the PWM_FIFO_CTRL.dma_mode_en and PWM_FIFO_CTRL.fifo_mode_sel to enable the DMA mode. Configure PWM_FIFO_CTRL.almost_full_watermark at appropriate value.
5. Configure DMAC to tansfer data from PWM to DDR.
6. Enable the channel by writing '1' to PWMx_CTRL.pwm_en bit to start the channel.
7. When an dma_req is asserted, DMAC transfer the data of effective high cycles and low cycles of input waveforms to DDR.
8. Write '0' to PWMx_CTRL.pwm_en to disable the channel.

9.6.3 PWM One-shot Mode/Continuous Standard Usage Flow

1. Set PWMx_CTRL.pwm_en to '0' to disable the PWM channel.
2. Choose the prescale factor and the scale factor for pclk by programming PWMx_CTRL.prescale and PWMx_CTRL.scale, and select the clock needed by setting PWMx_CTRL.clk_sel.
3. Choose the output mode by setting PWMx_CTRL.output_mode, and set the duty polarity and inactive polarity by programming PWMx_CTRL.duty_pol and PWMx_CTRL.inactive_pol.
4. Set the PWMx_CTRL.rpt if the channel is desired to work in the one-shot mode.
5. Configure the channel to work in the one-shot mode or the continuous mode.
6. Enable the INT_EN.chx_int_en to enable the interrupt generation if if the channel is desired to work in the one-shot mode.
7. If the channel is working in the one-shot mode, an interrupt is asserted after the end of operation, and the PWMx_CTRL.pwm_en is automatically cleared. Whatever mode the channel is working in, write '0' to PWMx_CTRL.pwm_en bit to disable the PWM channel.

9.6.4 Low-power mode

Setting PWMx_CTRL.lp_en to '1' makes the channel enter the low-power mode. When the PWM channel is inactive, the APB bus clock to the clock prescale module is gated in order to

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reduce the power consumption. It is recommended to disable the channel before entering the low-power mode, and quit the low-power mode before enabling the channel.

9.6.5 Other notes

When the channel is active to produce waveforms, it is free to program the PWMx_PERIOD_HPC and PWMx_DUTY_LPC register. The change will not take effect immediately until the current period ends.
An active channel can be changed to another operation mode without disable the PWM channel. However, during the transition of the operation mode there may be some irregular output waveforms. So does changing the clock division factor when the channel is active.

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Chapter 35 WDT

1.1 Overview

Watchdog Timer (WDT) is an APB slave peripheral that can be used to prevent system lockup that may becaused by conflicting parts or programs in a SoC.The WDT would generate interrupt or reset signal when it's counter reaches zero, then a reset controller would reset the system.

WDT supports the following features:

- 32 bits APB bus width
- WDT counter's clock is pclk
- 32 bits WDT counter width
- Counter counts down from a preset value to 0 to indicate the occurrence of a timeout
- WDT can perform two types of operations when timeout occurs:
  - Generate a system reset
  - First generate an interrupt and if this is not cleared by the service routine by the time a second timeout occurs then generate a system reset
- Programmable reset pulse length
- Total 16 defined-ranges of main timeout period

1.2 Block Diagram

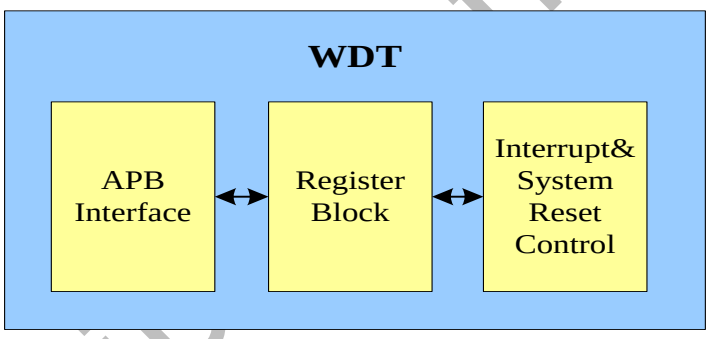

Fig. 34-6WDT block diagram

Block Descriptions:

- APB Interface

The APB Interface implements the APB slave operation. Its data bus width is 32 bits.

- Register Block

A register block that read coherence for the current count register.

- Interrupt & system reset control

An interrupt/system reset generation block is comprised of a decrementing counter and control logic.

1.3 Function Description

1.3.1 Operation

Counter

The WDT counts from a preset (timeout) value in descending order to zero. When the counter reaches zero, depending on the output response mode selected, either a system reset or an interrupt occurs. When the counter reaches zero, it wraps to the selected timeout value and continues decrementing. The user can restart the counter to its initial value. This is programmed by writing to the restart register at any time. The process of restarting the watchdog counter is sometimes referred as kicking the dog. As a safety feature to prevent accidental restarts, the value 0x76 must be written to the Current Counter Value Register

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(WDT_CRR).

Interrupts

The WDT can be programmed to generate an interrupt (and then a system reset) when a timeout occurs. When a 1 is written to the response mode field (RMOD, bit 1) of the Watchdog Timer Control Register (WDT_CR), the WDT generates an interrupt. If it is not cleared by the time a second timeout occurs, then it generates a system reset. If a restart occurs at the same time the watchdog counter reaches zero, an interrupt is not generated.

System Resets

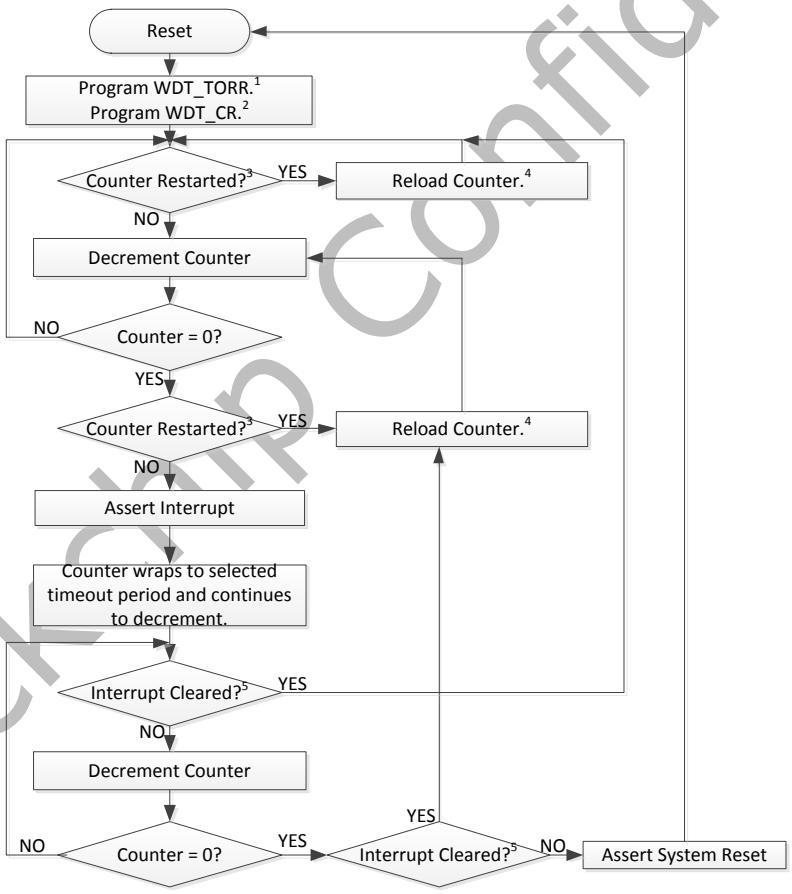
When a 0 is written to the output response mode field (RMOD, bit 1) of the Watchdog Timer Control Register (WDT_CR), the WDT generates a system reset when a timeout occurs.

Reset Pulse Length

The reset pulse length is the number of pclk cycles for which a system reset is asserted. When a system reset is generated, it remains asserted for the number of cycles specified by the reset pulse length or until the system is reset. A counter restart has no effect on the system reset once it has been asserted.

1.3.2 Programming sequence

Operation Flow Chart (Response mode=1)



1. Select required timeout period.
2. Set reset pulse length, response mode, and enable WDT.
3. Write 0x76 to WDT_CRR.
4. Starts back to selected timeout period.
5. Can clear by reading WDT_EOI or restarting (kicking) the counter by writing 0x76 to WDT_CRR.

Fig. 34-7WDT Operation Flow

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1.4 Register Description

This section describes the control/status registers of the design.

1.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
WDT_CR	0x0000	W	0x0000000a	Control Register
WDT_TORR	0x0004	W	0x00000000	Timeout range Register
WDT_CCVR	0x0008	W	0x00000000	Current counter value Register
WDT_CRR	0x000c	W	0x00000000	Counter restart Register
WDT_STAT	0x0010	W	0x00000000	Interrupt status Register
WDT_EOI	0x0014	W	0x00000000	Interrupt clear Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

1.4.2 Detail Register Description

WDT_CR
Address: Operational Base + offset (0x0000)
Control Register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:2	RW	0x2	rst_pluse_lenth Reset pulse length. This is used to select the number of pclk cycles for which the system reset stays asserted. 000: 2 pclk cycles 001: 4 pclk cycles 010: 8 pclk cycles 011: 16 pclk cycles 100: 32 pclk cycles 101: 64 pclk cycles 110: 128 pclk cycles 111: 256 pclk cycles
1	RW	0x1	resp_mode Response mode. Selects the output response generated to a timeout. 0: Generate a system reset. 1: First generate an interrupt and if it is not cleared by the time a second timeout occurs then generate a system reset.
0	RW	0x0	wdt_en WDT enable Writable when the configuration parameter WDT_ALWAYS_EN=0, otherwise,it is readable. This bit is used to enable and disable the DW_apb_wdt. When disabled, the counter dose not decrement .Thus, no interrupt or system reset are generated. Once this bit has been enabled, it can be cleared only by a system reset. 0: WDT disabled; 1: WDT enabled.

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WDT_TORR
Address: Operational Base + offset (0x0004)
Timeout range Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:0	RW	0x0	timeout_period Timeout period. This field is used to select the timeout period from which the watchdog counter restarts. A change of the timeout period takes effect only after the next counter restart (kick). The range of values available for a 32-bit watchdog counter are: 0000: 0x0000ffff 0001: 0x0001ffff 0010: 0x0003ffff 0011: 0x0007ffff 0100: 0x000fffff 0101: 0x001fffff 0110: 0x003fffff 0111: 0x007fffff 1000: 0x00ffffff 1001: 0x01ffffff 1010: 0x03ffffff 1011: 0x07ffffff 1100: 0x0fffffff 1101: 0x1fffffff 1110: 0x3fffffff 1111: 0x7fffffff

WDT_CCVR
Address: Operational Base + offset (0x0008)
Current counter value Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	cur_cnt Current counter value This register, when read, is the current value of the internal counter. This value is read coherently whenever it is read

WDT_CRR
Address: Operational Base + offset (0x000c)
Counter restart Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	W1C	0x00	cnt_restart Counter restart This register is used to restart the WDT counter. As a safety feature to prevent accidental restarts, the value 0x76 must be written. A restart also clears the WDT interrupt. Reading this register returns zero.

WDT_STAT
Address: Operational Base + offset (0x0010)
Interrupt status Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	wdt_status This register shows the interrupt status of the WDT. 1: Interrupt is active regardless of polarity; 0: Interrupt is inactive.

WDT_EOI
Address: Operational Base + offset (0x0014)
Interrupt clear Register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RC	0x0	wdt_int_clr Clears the watchdog interrupt. This can be used to clear the interrupt without restarting the watchdog counter.

1.5 Application Notes

Please refer to the function description section

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Chapter 36 EFUSE

11.1 Overview

In this device, there are two eFuse. Both of them are organized as 32 bits by 32 one-time programmable electrical fuses with random access interface.

It is a type of non-volatile memory fabricated in standard CMOS logic process. The main features are as follows:

- Programming condition : VQPS_EFUSE = 1.5V±10%
- Program time : 10us±0.2us .
- Read condition : VQPS_EFUSE = 0V
- Provide standby mode

11.2 Block Diagram

In the following diagram, all the signals except power supply VDD_EFUSE, VSS_EFUSE and VQPS_EFUSE are controlled by registers. For detailed description, please refer to detailed register descriptions.

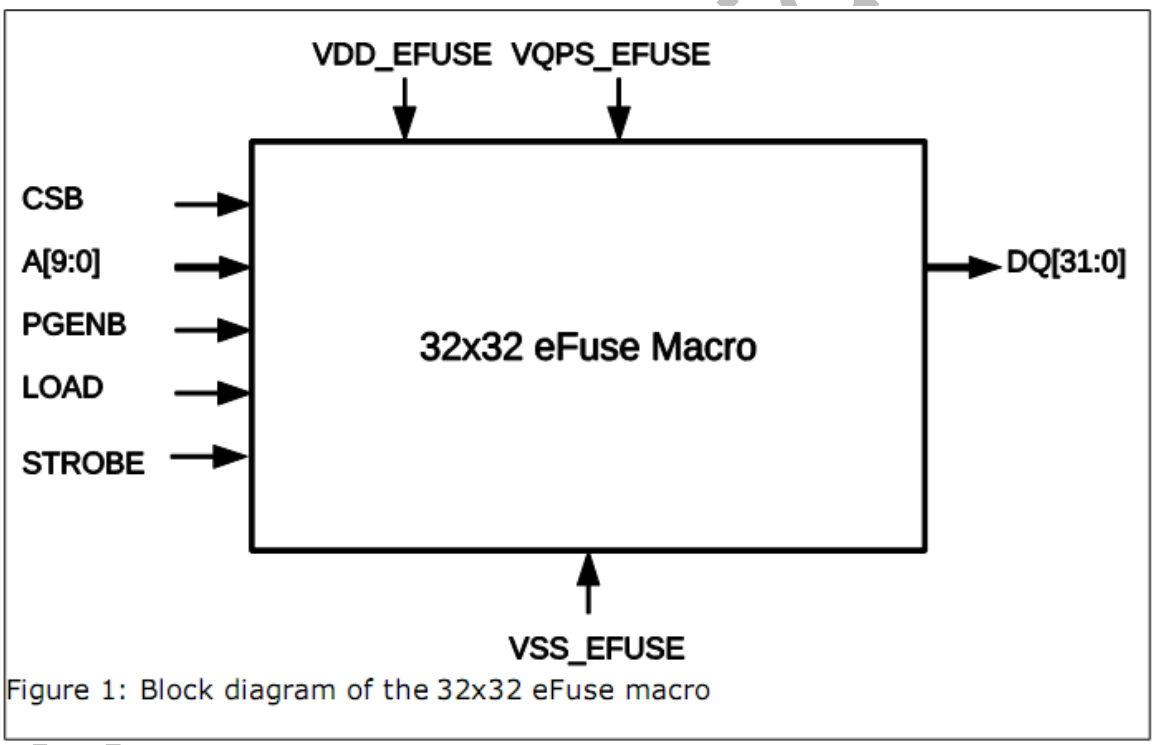

Fig.1-1 eFuse block diagram

11.3 Function Description

eFuse has three operation modes. They are defined as standby, read and programming.

Program (PGM) Mode

In order to enter programming mode, the following conditions need to be satisfied: VQPS_EFUSE is at high voltage, LOAD signal is low, PGENB signal is low, and CSB signal is low. All bits can be individually programmed (one at a time) with the proper address selected, the STROBE signal high and the address bits satisfying setup and hold time with respect to STROBE.

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Read Mode

In order to enter read mode the following conditions need to be satisfied: VQPS_EFUSE is at ground, the LOAD signal is high, the PGENB signal is high, and the CSB is low. An entire 8-bit word of data can be read in one read operation with STROBE being high and a proper address selected (address signals A5~A7 are "don't cares").

Standby Mode

Standby is defined when the macro is not being programmed or read. The conditions for standby mode are: the LOAD signal is low, the STROBE signal is low, the CSB signal is high and PGENB is high.

Table 1-1 list of allowed modes

Signals/Supplies					Mode
VQPS_EFUSE	CSB	PGENB	LOAD	STROBE*	
High	Low	Low	Low	Low to High	Programming
Low	Low	High	High	Low to High	Read
Low	High	High	Low	Low	Standby

11.4 Register Description

11.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
EFUSE_EFUSE_MOD	0x0000	W	0x00000006	EFUSE Mode Control Register
EFUSE_EFUSE_RD_MASK_S	0x0004	W	0x00000000	EFUSE Read Mask control In Secure Mode
EFUSE_EFUSE_PG_MASK_S	0x0008	W	0x00000000	EFUSE Program Mask control In Secure Mode
EFUSE_EFUSE_RD_MASK_NS	0x000c	W	0x00000000	EFUSE Read Mask control In Non-Secure Mode
EFUSE_EFUSE_PG_MASK_NS	0x0010	W	0x00000000	EFUSE Program Mask control In Non-Secure Mode
EFUSE_EFUSE_INT_CON	0x0014	W	0x00000000	EFUSE Interrupt Control
EFUSE_EFUSE_INT_STATUS	0x0018	W	0x00000000	EFUSE Interrupt Status
EFUSE_EFUSE_USER_CTRL	0x001c	W	0x00000009	EFUSE User Mode Control
EFUSE_EFUSE_DOUT	0x0020	W	0x00000000	EFUSE Data Out
EFUSE_EFUSE_AUTO_CTRL	0x0024	W	0x00000000	EFUSE Auto Mode Control
EFUSE_T_CSB_P	0x0028	W	0x000f0000	EFUSE CSB timing control in Program mode
EFUSE_T_PGENB_P	0x002c	W	0x00000000	EFUSE PGENB timing control in Program mode
EFUSE_T_LOAD_P	0x0030	W	0x00000000	EFUSE LOAD timing control in Program mode

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Name	Offset	Size	Reset Value	Description
EFUSE_T_ADDR_P	0x0034	W	0x00000000	EFUSE Address timing control in Program mode
EFUSE_T_STROBE_P	0x0038	W	0x00000000	EFUSE STROBE timing control in Program mode
EFUSE_T_CSB_R	0x003c	W	0x00000000	EFUSE CSB timing control in Read mode
EFUSE_T_PGENB_R	0x0040	W	0x00000000	EFUSE PGENB timing control in Read mode
EFUSE_T_LOAD_R	0x0044	W	0x00000000	EFUSE LOAD timing control in Read mode
EFUSE_T_ADDR_R	0x0048	W	0x00000000	EFUSE ADDR timing control in Read mode
EFUSE_T_STROBE_R	0x004c	W	0x00000000	EFUSE STROBE timing control in Read mode
EFUSE_REVISION	0x0050	W	0x00000010	EFUSE Design Revision

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

11.4.2 Detail Register Description

EFUSE_EFUSE_MOD

Address: Operational Base + offset (0x0000)

EFUSE Mode Control Register

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	efuse_rd_enb_user efuse read enable in user mode 0: disable 1: enable
5	RW	0x0	efuse_pg_enb_user efuse program enable in user mode 0: disable 1: enable
4	RW	0x0	strobe_pol STROBE polarity 0: Active HIGH 1: Active LOW
3	RW	0x0	load_pol LOAD polarity 0: Active HIGH 1: Active LOW

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Bit	Attr	Reset Value	Description
2	RW	0x1	pgenb_pol PGENB polarity 0: Active HIGH 1: Active LOW
1	R/W SC	0x1	csb_pol CSB polarity 0: Active HIGH 1: Active LOW
0	R/W SC	0x0	work_mod EFUSE controller working mode 0: auto_mode 1: user_mode

EFUSE_EFUSE_RD_MASK_S

Address: Operational Base + offset (0x0004)

EFUSE Read Mask control In Secure Mode

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	efuse_rd_mask_s efuse read mask enable in secure mode 0: disable 1: enable

EFUSE_EFUSE_PG_MASK_S

Address: Operational Base + offset (0x0008)

EFUSE Program Mask control In Secure Mode

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	efuse_pg_mask_s efuse program mask enable in secure mode 0: disable 1: enable

EFUSE_EFUSE_RD_MASK_NS

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Address: Operational Base + offset (0x000c)

EFUSE Read Mask control In Non-Secure Mode

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	W1C	0x0	efuse_rd_mask_ns efuse read mask enable in non-secure mode 0: disable 1: enable

EFUSE_EFUSE_PG_MASK_NS

Address: Operational Base + offset (0x0010)

EFUSE Program Mask control In Non-Secure Mode

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	efuse_pg_mask_ns efuse program mask enable in non-secure mode 0: disable 1: enable

EFUSE_EFUSE_INT_CON

Address: Operational Base + offset (0x0014)

EFUSE Interrupt Control

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	user_s_pg_mask_int_en user_s_pg_mask interrupt enable 0: disable 1: enable
11	RW	0x0	user_s_rd_mask_int_en user_s_rd_mask interrupt enable 0: disable 1: enable
10	RW	0x0	user_ns_pg_mask_int_en user_ns_pg_mask interrupt enable 0: disable 1: enable

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Bit	Attr	Reset Value	Description
9	RW	0x0	user_ns_rd_mask_int_en user_ns_rd_mask interrupt enable 0: disable 1: enable
8	RW	0x0	auto_s_pg_mask_int_en auto_s_pg_mask interrupt enable 0: disable 1: enable
7	RW	0x0	auto_s_rd_mask_int_en auto_s_rd_mask interrupt enable 0: disable 1: enable
6	RW	0x0	auto_ns_pg_mask_int_en auto_ns_pg_mask interrupt enable 0: disable 1: enable
5	RW	0x0	auto_ns_rd_mask_int_en auto_ns_rd_mask interrupt enable 0: disable 1: enable
4	RW	0x0	user_s_access_ns_err_int_en user_s_access_ns_err interrupt enable 0: disable 1: enable
3	RW	0x0	user_ns_access_s_err_int_en user_ns_access_s_err interrupt enable 0: disable 1: enable
2	RW	0x0	auto_s_access_ns_err_int_en auto_s_access_ns_err interrupt enable 0: disable 1: enable
1	RW	0x0	auto_ns_access_s_err_int_en auto_ns_access_s_err interrupt enable 0: disable 1: enable
0	RO	0x0	finish_int_en finish interrupt enable 0: disable 1: enable

EFUSE_EFUSE_INT_STATUS

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Address: Operational Base + offset (0x0018)

EFUSE Interrupt Status

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	W1C	0x0	user_s_pg_mask_int_status user_s_pg_mask_int status bit
11	W1C	0x0	user_s_rd_mask_int_status user_s_rd_mask_int status bit
10	W1C	0x0	user_ns_pg_mask_int_status user_ns_pg_mask_int status bit
9	W1C	0x0	user_ns_rd_mask_int_status user_ns_rd_mask_int status bit
8	W1C	0x0	auto_s_pg_mask_int_status auto_s_pg_mask_int status bit
7	W1C	0x0	auto_s_rd_mask_int_status auto_s_rd_mask_int status bit
6	W1C	0x0	auto_ns_pg_mask_int_status auto_ns_pg_mask_int status bit
5	W1C	0x0	auto_ns_rd_mask_int_status auto_ns_rd_mask_int status bit
4	W1C	0x0	user_s_access_ns_err_int_status user_s_access_ns_err_int status bits
3	W1C	0x0	user_ns_access_s_err_int_status user_ns_access_s_err_int status bit
2	W1C	0x0	auto_s_access_ns_err_int_status auto_s_access_ns_err_int status bit
1	W1C	0x0	auto_ns_access_s_err_int_status auto_ns_access_s_err_int status bit
0	W1C	0x0	finish_int_status finish_int status bit

EFUSE_EFUSE_USER_CTRL

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Address: Operational Base + offset (0x001c)

EFUSE User Mode Control

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	efuse_addr_user efuse_addr bit control in user mode
15:4	RO	0x0	reserved
3	RW	0x1	efuse_pgenb_user efuse_pgenb bit control in user mode
2	RW	0x0	efuse_load_user efuse_load bit control in user mode
1	RW	0x0	efuse_strobe_user efuse_strobe bit control in user mode
0	RO	0x1	efuse_csb_user efuse_csb bit control in user mode

EFUSE_EFUSE_DOUT

Address: Operational Base + offset (0x0020)

EFUSE Data Out

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	efuse_dout efuse data out

EFUSE_EFUSE_AUTO_CTRL

Address: Operational Base + offset (0x0024)

EFUSE Auto Mode Control

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25:16	RW	0x000	efuse_addr_auto efuse address in auto mode
15:2	RO	0x0	reserved
1	RW	0x0	p_r_mode program and read control 0: programming mode 1: read mode

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Bit	Attr	Reset Value	Description
0	R/W SC	0x0	enb enable of auto mode 0: disable 1: enable Note, this bit is clear auto

EFUSE_T_CSB_P

Address: Operational Base + offset (0x0028)

EFUSE CSB timing control in Program mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RO	0xf	t_csb_p_s csbstart delay time in programming mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_csb_p_l lasted time in programming mode

EFUSE_T_PGENB_P

Address: Operational Base + offset (0x002c)

EFUSE PGENB timing control in Program mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_pgenb_p_s pgenb start delay time in programming mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_pgenb_p_l pgenb lasted time in programming mode

EFUSE_T_LOAD_P

Address: Operational Base + offset (0x0030)

EFUSE LOAD timing control in Program mode

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_load_p_s load start delay time in programming mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_load_p_l load lasted time in programming mode

EFUSE_T_ADDR_P

Address: Operational Base + offset (0x0034)

EFUSE Address timing control in Program mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_addr_p_s address start delay time in programming mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_addr_p_l address lasted time in programming mode

EFUSE_T_STROBE_P

Address: Operational Base + offset (0x0038)

EFUSE STROBE timing control in Program mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_strobe_p_s strobe start delay time in programming mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_strobe_p_l strobe lasted time in programming mode

EFUSE_T_CSB_R

Address: Operational Base + offset (0x003c)

EFUSE CSB timing control in Read mode

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Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_csb_r_s csb start delay time in read mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_csb_r_l csb lasted time in read mode

EFUSE_T_PGENB_R

Address: Operational Base + offset (0x0040)

EFUSE PGENB timing control in Read mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_pgenb_r_s pgenb start delay time in read mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_pgenb_r_l pgenb lasted time in read mode

EFUSE_T_LOAD_R

Address: Operational Base + offset (0x0044)

EFUSE LOAD timing control in Read mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_load_r_s load start delay time in read mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_load_r_l load lasted time in read mode

EFUSE_T_ADDR_R

Address: Operational Base + offset (0x0048)

EFUSE ADDR timing control in Read mode

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Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_addr_r_s address start delay time in read mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_addr_r_l address lasted time in read mode

EFUSE_T_STROBE_R

Address: Operational Base + offset (0x004c)

EFUSE STROBE timing control in Read mode

Bit	Attr	Reset Value	Description
31:20	RO	0x0	reserved
19:16	RW	0x0	t_strobe_r_s strobe start delay time in read mode
15:10	RO	0x0	reserved
9:0	RW	0x000	t_strobe_r_l strobe lasted time in read mode

EFUSE_REVISION

Address: Operational Base + offset (0x0050)

EFUSE Design Revision

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x10	revision efuse design revsion

11.5 Timing Diagram

- **When efuse32×32 is in program(PGM) mode.**

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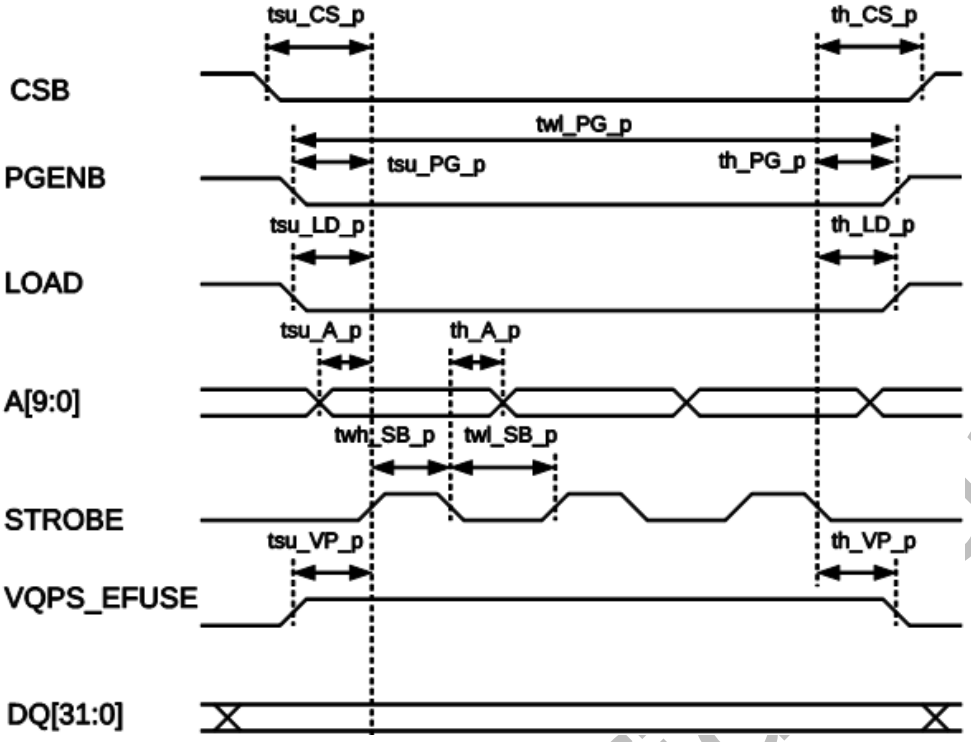

Fig.1 -2 efuse32×32 timing diagram in program mode

- **When efuse32×8 is in read mode.**

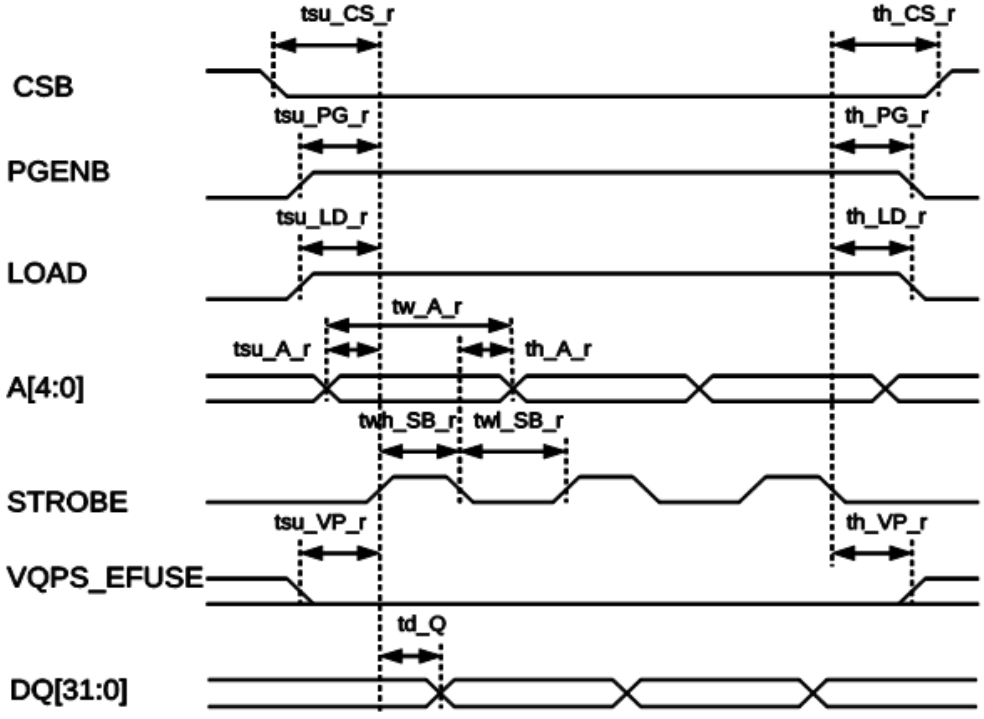

Fig.1-3 efuse32×32 timing diagram in read mode

The following table has shows the detailed value for timing parameters in the above diagram.

Table 36-2 eFuse timing parameters list

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Mode	Item	Description	Min	Typ	Max	Unit
Read Mode	twh_SB_r	Pulse width high of STROBE read strobe	20		-	ns
	twl_SB_r	Pulse width low of STROBE read strobe	15		-	ns
	tsu_A_r	A[9:0] to STROBE setup time in read mode	25		-	ns
	th_A_r	A[9:0] to STROBE hold time in read mode	3		-	ns
	tw_A_r	A[9:0] pulse width while LOAD high in read mode	48		100	ns
	tsu_CS_r	CSB to STROBE setup time in read mode	16		-	ns
	th_CS_r	CSB to STROBE hold time in read mode	6		-	ns
	tsu_PG_r	PGENB to STROBE setup time in read mode	14		-	ns
	th_PG_r	PGENB to STROBE hold time in read mode	10		-	ns
	tsu_LD_r	LOAD to STROBE setup time in read mode	10		-	ns
	th_LD_r	LOAD to STROBE hold time in read mode	7		-	ns
	tsu_VP_r	VQPS_EFUSE to STROBE setup time in read mode	20		-	ns
	th_VP_r	VQPS_EFUSE to STROBE hold time in read mode	20		-	ns
	td_Q	DQ[31:0] delay time after STROBE high	0		8	ns
PGM Mode	twh_SB_p	Pulse width high of STROBE PGM strobe	9.8	10	10.2	us
	twl_SB_p	Pulse width low of STROBE PGM strobe	15		-	ns
	tsu_A_p	A[9:0] to STROBE setup time in PGM mode	12		-	ns
	th_A_p	A[9:0] to STROBE hold time in PGM mode	3		-	ns
	tsu_CS_p	CSB to STROBE setup time in PGM mode	16		-	ns
	th_CS_p	CSB to STROBE hold time in PGM mode	6		-	ns
	tsu_PG_p	PGENB to STROBE setup time in PGM mode	14		-	ns
	th_PG_p	PGENB to STROBE hold time in PGM mode	10		-	ns
	twl_PG_p	PGENB pulse width low (cumulative) in PGM mode	-		100	ms
	tsu_LD_p	LOAD to STROBE setup time in PGM mode	10		-	ns
	th_LD_p	LOAD to STROBE hold time in PGM mode	7		-	ns
	tsu_VP_p	VQPS_EFUSE to STROBE setup time in PGM mode	20		-	ns
	th_VP_p	VQPS_EFUSE to STROBE hold time in PGM mode	20		-	ns

11.6 Application Notes

During usage of efuse, customers must pay more attention to the following items:

1. In condition of program(PGM) mode, VQPS_EFUSE= 1.5V±10%.

2. Q0~Q7 will be reset to "0" once CSB at high.

3. No data access allowed at the rising edge of CSB.

4. All the program timing for each signal must be more than the value defined in the timing table. Please refer to the timing diagram of READ_MODE and PGM_MODE as well as the function description.

Configuration steps:

1. set csb(EFUSE_CTRL[0]), pgenb(EFUSE_CTRL[3]), load(EFUSE_CTRL[2]) at proper value.

2. set addr(EFUSE_CTRL[15:6]).

3. set strobe(EFUSE_CTRL[1]).

4. stay for enough cycle. (Satisfy the timing parameter)

5. dis-assert strobe(EFUSE_CTRL[1]).

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6. set csb(EFUSE_CTRL[0]), pgenb(EFUSE_CTRL[3]), load(EFUSE_CTRL[2]) at proper value.
7. read efuse_data(EFUSE_DOUT).

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Chapter 37 OTP

1.7 Overview

The One Time Programmable Controller (OTPC) is used for communication with the OTP subsystem to achieve the controlling command and receive the returning data. The configuration and command information are written from a master(CPU) over the APB bus to the OTPC and converted to standard form and transmitted to the OTP. The data from OTP can be stored in the registers of OTPC for the master (CPU) to read back.

OTP Controller supports the following features:

- Support secure APB interface
- Support non-secure APB interface
- Support OTP SBPI master interface
- Support OTP user master interface
- Support two programmable working clock for SBPI and user interface
- Support one interrupt output
- Support two busy signals
- SBPI master:
  - Support configurable device ID
  - Support maximum 64 consecutive valid command
  - Support CS automatic de-assert
  - Support CS manual de-assert
  - Support maximum 64B consecutive reading and storage
  - Support reading MISO and FLAG status by the APB bus
- User interface master:
  - Support software configurable DCTRL
  - Support single reading

1.8 Block Diagram

This section provides a description about the functions and behavior under various conditions. The OTP Controller comprises with:

- AMBA APB interface
- SBPI interface
- USER interface
- REG FILE
- SBPI FSM
- USER FSM

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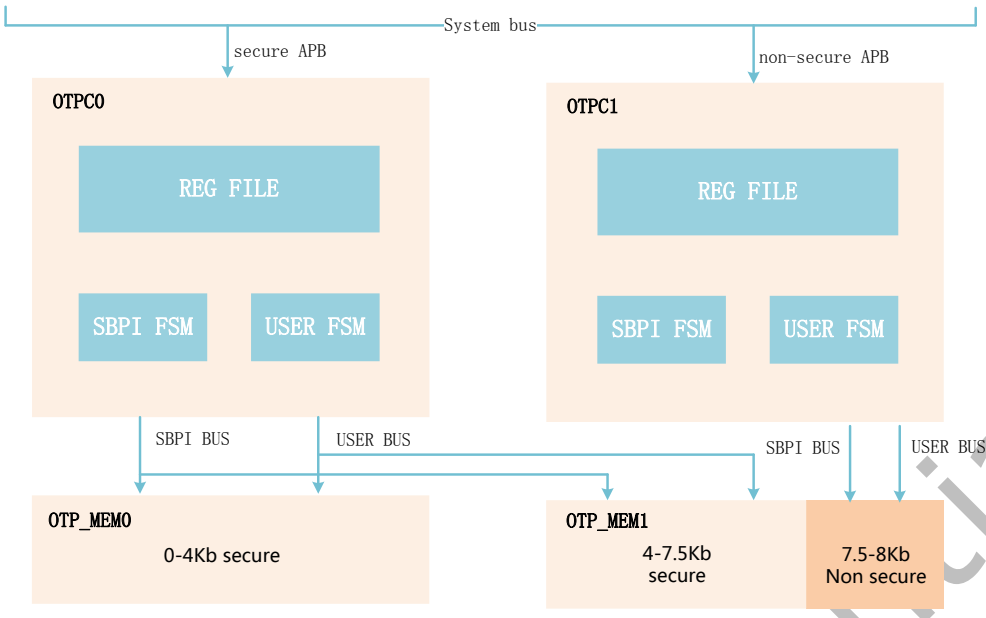

Fig. 36-4 OTP Architecture

APB INTERFACE

The host processor accesses data, control, and status information on the OTPC through the APB interface including secure and non-secure.

Register file

Be responsible for the main OTPC functionality including control, status and interrupt generation.

SBPI FSM & USER FSM

The two FSMs are used for converting command to standard form and receiving data from SBPI and USER interface.

SBPI BUS & USER BUS

SBPI bus and USER bus are used for the transmission of command and the reception of data.

OTP MEMORY

There are two pieces of memory and each of them is 4Kb. The all 4Kb of MEM_0 is secure. The 0-3.5Kb of MEM_1 is secure while the remaining 0.5Kb is non-secure.

1.9 Function Description

SBPI Interface Protocol

The Serial-Parallel Interface, SBPI, defines a half-duplex serial and byte-parallel protocol in which instruction and data are transferred between SBPI agents. Attentively, OTPC only support byte-parallel mode and data are transferred between OTPC and SHF_AP. The SHF_AP is a synthesizable RTL block that interfaces with the SHF OTP memory and Integrated Power Supply (IPS) blocks.

The following description presents the SBPI communication and control protocol for byte-parallel mode. For byte-parallel mode, SP is held LOW.

A typical byte-parallel data transfer frame shown in Figure 1-2 consists of Start-of-Frame (SOF), a Frame Body, and an End-of-Frame (EOF). The IDLE time between frames is used to select the agent (ID).

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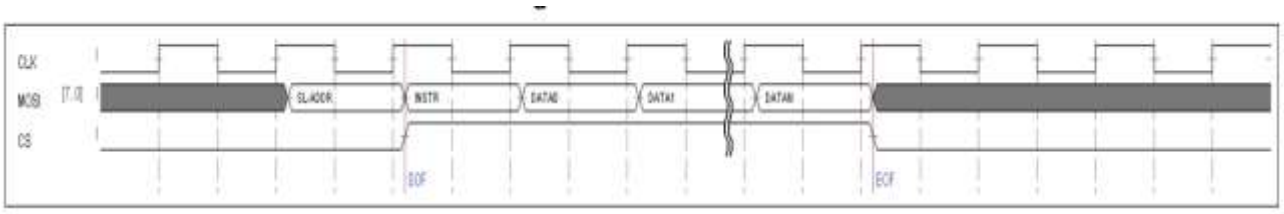

Fig. 36-5 OTP SBPI protocol

- SOF: A frame is started in a selected agent by the first cycle where CKE is HIGH, CS is HIGH and CLK rises; CS is made HIGH in a clock cycle following agent ID selection.
- EOF: A frame ends when CS is made LOW.
- BODY: The frame body consists of one or more clock cycles depending on instruction type. There are up to 3 phases for instruction sent as part of the frame:
  - Argument phase: 1 or more cycles to transfer needed arguments for the instruction to the target slave register
  - Action phase: number of cycles required by the execution engine to provide the required sequence of operations
  - Result phase: number of cycles required to transfer data from the slave's registers to the master

USER Interface Protocol

When DCTRL is asserted LOW, the OTP access is under SBPI control while the data outputs Q and QP remain active and contain the results of the most recent operation. If no new read or data processing is performed using the DAP, the Q outputs contain the last read data.

The user interface is enabled by asserting DCTRL to HIGH. However, this does not disable the SBPI interface. Once DCTRL is asserted HIGH, the SHF address bus and the access strobe signals CK are under user control and the outputs Q and QP contain data read from OTP.

The user read cycle is controlled by the CK access strobe pulse width. A read cycle to the addressed word is initiated on the rising edge of the CK read strobe signal, when the OTP's WE input is LOW (sourced internally from the DAP of the SHF_AP) and the OTP select input, SEL is HIGH. The address A[8:0] and select SEL inputs are latched on the rising edge of CK access strobe signal. The data outputs Q[7:0] and QP[7:0], become valid following the subsequent falling edge of the CK access strobe signal, or after the tACC if the SHF internal timer is enabled. If an output data bit does not change state during a read operation, there is no intermediate transition at the output during the access time.

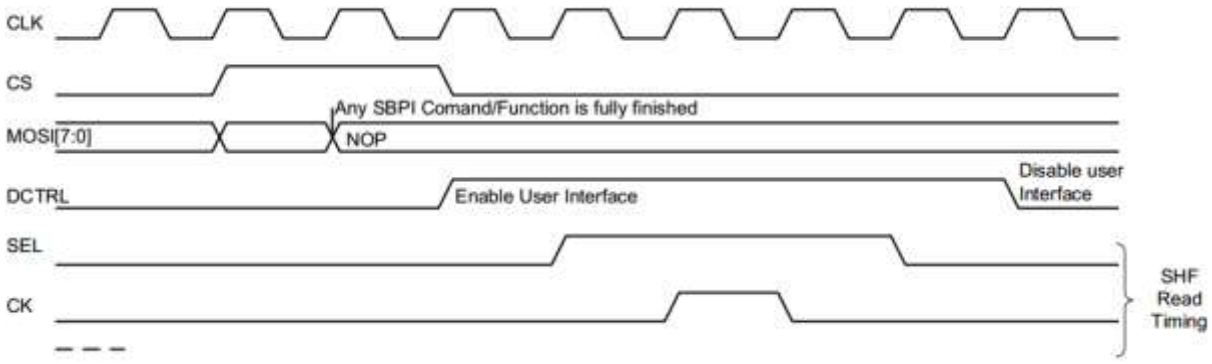

Fig. 36-3 OTP USER protocol

OTP BOOT Function

The BOOT function verifies power supply by performing ROM read operations, loads special registers to drive QSR signals (optional), and sets the default read mode into DAP. The user host controller may need to reload the DAP or PMC registers for the subsequent BIST, PROGRAM or READ operations. We can consult with SHF and IPS data sheets for correct mode

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input signal settings for read and program operations.

After both VCC and VDD have reached operating range levels, the RSTn should be released and the BOOT function can be invoked by issuing a START instruction to the PMC from the host function via the SPBI interface, since the PMC and DAP are preset for BOOT operation.

The BOOT function's purpose is to reliably set the internal Q_SR[n+7:0], Q_RR[n-1:0], and Q_RRP[7:0] registers with user defined content read from pre-defined OTP locations once supplies have reached a suitable level. The internal Q_SR[n+7:0] register bits are all preset to 1'b1 when RSTn is asserted LOW. The internal Q_RR[n-1:0] and Q_RRP[7:0] register bits are all preset to 1'b0 when RSTn is asserted LOW. The BOOT operation sets these internal registers with user content by first performing multiple "read and verify" operations of test ROM locations, until all ROM locations are read correctly. The BOOT operation then loads the internal Q_SR[n+7:0], Q_RR[n-1:0] and Q_RRP[7:0] registers with data read from the pre-defined SHF OTP locations.

The SHF_AP output signal FLAG can be used by the host controller to determine when the BOOT operation is completed in order to issue a STOP instruction to the PMC to terminate the BOOT function. At this point, the OTP can be accessed through the SBPI interface or, by asserting DCTRL to HIGH, the OTP can be accessed through the user interface to perform OTP read operations.

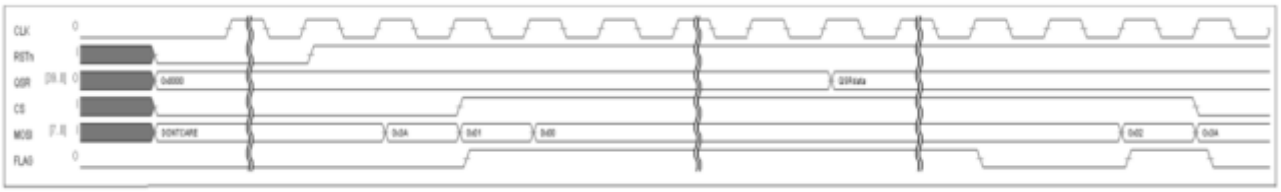

Fig. 36-4 OTP BOOT command

- 0x3A on MOSI = PMC SBPI address
- 0x00 on MOSI = PMC NOP instruction
- 0x01 on MOSI = PMC START instruction
- 0x02 on MOSI = PMC STOP instruction

OTP BIST Function

The Array Clean BIST function provides an array clean checking and bit repair capabilities. It is activated similarly to the PROG function. The user host controller may need to provide control parameters using register configurations before initiating the BIST function, select BIST in the PMC register PMC_CTRL_STATUS and then issue the START instruction. The PMC_CTRL_STATUS register and the FLAG output signal will indicate routine completion and status. By default, the BIST can be configured to run through the entire address space and to attempt to repair any bad bits in the array. Alternatively the BIST can be run through a selected portion of the address space, with or without bit repair.

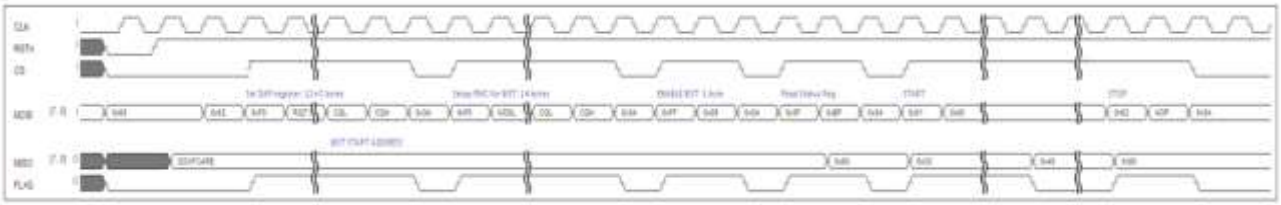

Fig. 36-5 OTP BIST command

OTP PROGRAM Function

In order to initiate programming the OTP, the host (user) must reload DAP and PMC registers with proper program and verify configurations, write the initial address and a data word into DAP, and select the program function in PMC. The programming starts when the host issues a START instruction to the PMC. The PMC takes over control of the SBPI bus to the DAP and

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asserts the FLAG HIGH, indicating that the programming operation is active. When the PMC reaches the end of the programming operation, the FLAG signal will be asserted LOW. The host can also monitor the PMC's PMC_CTRL_STATUS register via the MISO bus while the programming operation is underway. The PMC_CTRL_STATUS register indicates the end of programming operation, (same as the FLAG) and also indicates if the program was successful using error codes. The host must issue the STOP instruction to the PMC which sets the FLAG bit back to HIGH and terminates the programming function. It then re-gains control over the whole SBPI bus, following which the next data word can be written into the DAP register. The OTP address can be incremented automatically using a PMC setting, unless the host overrides it. The ECC parity bits are added automatically to the data provided by the user, and programmed into the OTP, unless the ECC is disabled.

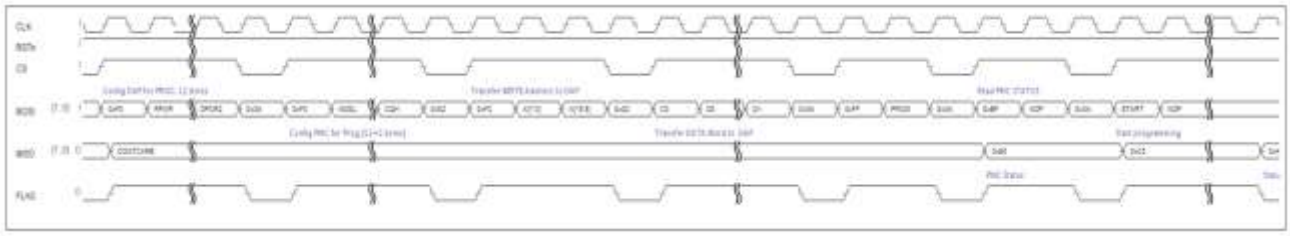

Fig. 36-6 OTP PROG command

OTP ECC Function

The SHF_AP is provisioned with the ability to protect the OTP data word with an Error Correction Code (ECC) able to correct single bit errors. The SHF_AP is provisioned with the ability to protect the OTP data word with an Error Correction Code (ECC) able to correct single bit errors. The ECC circuit is using 4-bit Hamming Code to generate six protection (parity) bits for the data before programming and to correct single bit errors when reading data from the OTP: 8bit of user data can be protected.

Table. 36-1 4bit Hamming code from 8bit data

	P1	P2	D0	P4	D1	D2	D3	P8	D4	D5	D6	D7
S	1	2	3	4	5	6	7	8	9	10	11	12
	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100
P1			X		X		X		X		X	
P2			X			X	X			X	X	
P4					X	X	X					X
P8									X	X	X	X

- Q[7:0]={D7,D6,D5,D4,D3,D2,D1,D0};
- QP[0]=P1=D0⊕D1⊕D3⊕D4⊕D6;
- QP[1]=P2=D0⊕D2⊕D3⊕D5⊕D6;
- QP[2]=P4=D1⊕D2⊕D3⊕D7;
- QP[3]=P8=D4⊕D5⊕D6⊕D7;

When programming the OTP using the PMC, the parity bits can be added automatically to the word being programmed.

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1.10 Register Description

This section describes the control/status registers of the design. There are 2 OTPCs in RK3228H, and each one has its own base address.

1.10.1 Registers Summary

Name	Offset	Size	Reset Value	Description
OTPC_SECURE_MASK_ADDR_BOTTOM	0x0000	W	0x00000000	secure access min address
OTPC_SECURE_MASK_ADDR_TOP	0x0004	W	0x00000000	secure access max address
OTPC_NONSECURE_MASK_ADDR_BOTTOM	0x0008	W	0x00000000	nonsecure access min address
OTPC_NONSECURE_MASK_ADDR_TOP	0x000c	W	0x00000000	nonsecure access max address
OTPC_SECURE_CTRL0	0x0010	W	0x000007ff	OTPC sercure control register0
OTPC_SECURE_CTRL1	0x0014	W	0x000001ff	OTPC sercure control register1
OTPC_SBPI_CTRL	0x0020	W	0x00000000	OTPC SBPI control register
OTPC_SBPI_CMD_VALID_PRELOAD	0x0024	W	0x00000000	OTPC SBPI command preload register
OTPC_SBPI_CS_VALID_PRELOAD	0x0028	W	0x00000000	OTPC SBPI CS valid parameter preload register
OTPC_SBPI_STATUS	0x002c	W	0x00000000	OTPC SBPI status register
OTPC_USER_CTRL	0x0100	W	0x00000000	OTPC USER control register
OTPC_USER_ADDR	0x0104	W	0x00000000	OTPC USER reading address register
OTPC_USER_ENABLE	0x0108	W	0x00000000	OTPC USER enable register
OTPC_USER_STATUS	0x0110	W	0x00000000	OTPC USER status register
OTPC_USER_QP	0x0120	W	0x00000000	OTPC USER QP storage register
OTPC_USER_Q	0x0124	W	0x00000000	OTPC USER Q storage register
OTPC_USER_QSR	0x0128	W	0x00000000	OTPC USER QSR storage register
OTPC_USER_QRR	0x012c	W	0x00000000	OTPC USER QRR storage register
OTPC_SBPI_CMD_BASE	0x1000	W	0x00000000	OTPC SBPI command base register
OTPC_SBPI_READ_DATA_BASE	0x2000	W	0x00000000	OTPC SBPI reading data base register
OTPC_INT_CON	0x0300	W	0x00000000	OTPC interrupt register
OTPC_INT_STATUS	0x0304	W	0x00000000	OTPC interrupt status register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

1.10.2 Detail Register Description

OTPC_SECURE_MASK_ADDR_BOTTOM

Address: Operational Base + offset (0x0000)

secure access min address

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Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_secure_mask_addr_bottom secure must access address >= this value

OTPC_SECURE_MASK_ADDR_TOP

Address: Operational Base + offset (0x0004)

secure access max address

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_secure_mask_addr_top secure must access address <= this value

OTPC_NONSECURE_MASK_ADDR_BOTTOM

Address: Operational Base + offset (0x0008)

nonsecure access min address

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_nonsecure_mask_addr_bottom nonsecure must access address >= this value

OTPC_NONSECURE_MASK_ADDR_TOP

Address: Operational Base + offset (0x000c)

nonsecure access max address

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_nonsecure_mask_addr_top nonsecure must access address <= this value

OTPC_SECURE_CTRL0

Address: Operational Base + offset (0x0010)

OTPC sercure control register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	RW	0x1	otpc_sbpi_read_data_protect protect bit for OTPC_SBPI_READ_DATA 1'b0: non-secure can not access OTPC_SBPI_READ_DATA 1'b1: non-secure can access OTPC_SBPI_READ_DATA
9	RW	0x1	otpc_sbpi_cmd_protect protect bit for OTPC_SBPI_CMD 1'b0: non-secure can not access OTPC_SBPI_CMD 1'b1: non-secure can access OTPC_SBPI_CMD
8	RW	0x1	otpc_sbpi_cs_valid_preload_protect protect bit for OTPC_SBPI_CS_VALID_PRELOAD 1'b0: non-secure can not access OTPC_SBPI_CS_VALID_PRELOAD 1'b1: non-secure can access OTPC_SBPI_CS_VALID_PRELOAD
7	RW	0x1	otpc_sbpi_cmd_valid_preload_protect protect bit for OTPC_SBPI_CMD_VALID_PRELOAD 1'b0: non-secure can not access OTPC_SBPI_CMD_VALID_PRELOAD 1'b1: non-secure can access OTPC_SBPI_CMD_VALID_PRELOAD
6	RW	0x1	otpc_sbpi_status_protect protect bit for OTPC_SBPI_STATUS 1'b0: non-secure can not access OTPC_SBPI_STATUS 1'b1: non-secure can access OTPC_SBPI_STATUS
5	RW	0x1	otpc_secure_ctrl1_protect protect bit for OTPC_SECURE_CTRL1 1'b0: non-secure can not access OTPC_SECURE_CTRL1 1'b1: non-secure can access OTPC_SECURE_CTRL1
4	RW	0x1	otpc_secure_ctrl0_protect protect bit for OTPC_SECURE_CTRL0 1'b0: non-secure can not access OTPC_SECURE_CTRL0 1'b1: non-secure can access OTPC_SECURE_CTRL0
3	RW	0x1	otpc_nonsecure_mask_addr_top_protect protect bit for OTPC_NONSECURE_MASK_ADDR_TOP 1'b0: non-secure can not access OTPC_NONSECURE_MASK_ADDR_TOP 1'b1: non-secure can access OTPC_NONSECURE_MASK_ADDR_TOP
2	RW	0x1	otpc_nonsecure_mask_addr_bottom_protect protect bit for OTPC_NONSECURE_MASK_ADDR_BOTTOM 1'b0: non-secure can not access OTPC_NONSECURE_MASK_ADDR_BOTTOM 1'b1: non-secure can access OTPC_NONSECURE_MASK_ADDR_BOTTOM

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Bit	Attr	Reset Value	Description
1	RW	0x1	otpc_secure_mask_addr_top_protect protect bit for OTPC_SECURE_MASK_ADDR_TOP 1'b0: non-secure can not access OTPC_SECURE_MASK_ADDR_TOP 1'b1: non-secure can access OTPC_SECURE_MASK_ADDR_TOP
0	RW	0x1	otpc_secure_mask_addr_bottom_protect protect bit for OTPC_SECURE_MASK_ADDR_BOTTOM 1'b0: non-secure can not access OTPC_SECURE_MASK_ADDR_BOTTOM 1'b1: non-secure can access OTPC_SECURE_MASK_ADDR_BOTTOM

OTPC_SECURE_CTRL1

Address: Operational Base + offset (0x0014)

OTPC sercure control register0

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15	RW	0x0	otpc_int_status_protect protect bit for OTPC_INT_STATUS 1'b0: non-secure can not access OTPC_INT_STATUS 1'b1: non-secure can access OTPC_INT_STATUS
14	RW	0x0	otpc_int_con_protect protect bit for OTPC_INT_CON 1'b0: non-secure can not access OTPC_INT_CON 1'b1: non-secure can access OTPC_INT_CON
13:9	RO	0x0	reserved
8	RW	0x1	otpc_user_addr_mask_protect protect bit for OTPC_USER_ADDR_MASK 1'b0: non-secure can not access OTPC_USER_ADDR_MASK 1'b1: non-secure can access OTPC_USER_ADDR_MASK
7	RW	0x1	otpc_user_qrr_protect protect bit for OTPC_USER_QRR 1'b0: non-secure can not access OTPC_USER_QRR 1'b1: non-secure can access OTPC_USER_QRR
6	RW	0x1	otpc_user_qsr_protect protect bit for OTPC_USER_QSR 1'b0: non-secure can not access OTPC_USER_QSR 1'b1: non-secure can access OTPC_USER_QSR

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Bit	Attr	Reset Value	Description
5	RW	0x1	otpc_user_q_protect protect bit for OTPC_USER_Q 1'b0: non-secure can not access OTPC_USER_Q 1'b1: non-secure can access OTPC_USER_Q
4	RW	0x1	otpc_user_qp_protect protect bit for OTPC_USER_QP 1'b0: non-secure can not access OTPC_USER_QP 1'b1: non-secure can access OTPC_USER_QP
3	RW	0x1	otpc_user_status_protect protect bit for OTPC_USER_STATUS 1'b0: non-secure can not access OTPC_USER_STATUS 1'b1: non-secure can access OTPC_USER_STATUS
2	RW	0x1	otpc_user_enable_protect protect bit for OTPC_USER_ENABLE 1'b0: non-secure can not access OTPC_USER_ENABLE 1'b1: non-secure can access OTPC_USER_ENABLE
1	RW	0x1	otpc_user_addr_protect protect bit for OTPC_USER_ADDR 1'b0: non-secure can not access OTPC_USER_ADDR 1'b1: non-secure can access OTPC_USER_ADDR
0	RW	0x1	otpc_user_ctrl_protect protect bit for OTPC_USER_CTRL 1'b0: non-secure can not access OTPC_USER_CTRL 1'b1: non-secure can access OTPC_USER_CTRL

OTPC_SBPI_CTRL

Address: Operational Base + offset (0x0020)

OTPC SBPI control register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:8	RW	0x00	sbpi_device_id device id value, user to choose a device
7:4	RO	0x0	reserved
3	R/WSC	0x0	sbpi_cs_deassert write 1 to this bit will deassert cs this bit will be selfclear, do not write 0 to this bit
2	RW	0x0	sbpi_cs_auto 1'b0: cs deassert only under software control 1'b1: cs deassert under software control and will be automatically deassert when a cs_counter reach 0

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Bit	Attr	Reset Value	Description
1	RW	0x0	sbpi_sp sp control of sbpi bus 1'b0: parallel mode 1'b1: serial mode(controller not support)
0	R/W SC	0x0	sbpi_enable write 1 to this register enable sbpi FSM enable It will be selfclear to 0, software do not write 0 to this bit

OTPC_SBPI_CMD_VALID_PRELOAD

Address: Operational Base + offset (0x0024)

OTPC SBPI command preload register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_sbpi_cmd_valid_preload a value define number of sbpi valid command

OTPC_SBPI_CS_VALID_PRELOAD

Address: Operational Base + offset (0x0028)

OTPC SBPI CS valid parameter preload register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_sbpi_cs_valid_preload a value define number of cs valid cycles, when sbpi_cs_auto is set to 1 of OTPC_SBPI_CTRL

OTPC_SBPI_STATUS

Address: Operational Base + offset (0x002c)

OTPC SBPI status register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RO	0x0	SP SP value of SBPI bus
21	RO	0x0	CS CS value of SBPI bus
20:13	RO	0x00	MOSI MOSI value of SBPI bus
12:5	RO	0x00	MISO MISO value of SBPI bus

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Bit	Attr	Reset Value	Description
4	RO	0x0	FLAG FLAG state of SBPI interface
3:1	RO	0x0	sbpi_current_state FSM states of SBPI
0	RO	0x0	sbpi_busy sbpi_busy status

OTPC_USER_CTRL

Address: Operational Base + offset (0x0100)

OTPC USER control register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	user_pd 1'b0: PD of user interface will be set to 0 1'b1: PD of user interface will be set to 1
0	RW	0x0	user_dctrl 1'b0: DCTRL of user interface will be set to 0 1'b1: DCTRL of user interface will be set to 1

OTPC_USER_ADDR

Address: Operational Base + offset (0x0104)

OTPC USER reading address register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:0	RW	0x0000	otpc_user_addr 16bit A of User interface

OTPC_USER_ENABLE

Address: Operational Base + offset (0x0108)

OTPC USER enable register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15:1	RO	0x0	reserved
0	RW	0x0	otpc_user_enable write 1 to enable USER FSM will be selfclear, do not write 0 to this bit

OTPC_USER_STATUS

Address: Operational Base + offset (0x0110)

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OTPC USER status register

Bit	Attr	Reset Value	Description
31:23	RO	0x0	reserved
22	RO	0x0	DCTRL DCTRL of USER interface
21:6	RO	0x0000	A A of USER interface
5	RO	0x0	PD PD of USER interface
4	RO	0x0	SEL SEL of USER interface
3:1	RO	0x0	user_current_state state of USER FSM
0	RO	0x0	user_busy user_busy indication

OTPC_USER_QP

Address: Operational Base + offset (0x0120)

OTPC USER QP storage register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	QP QP value of USER interface

OTPC_USER_Q

Address: Operational Base + offset (0x0124)

OTPC USER Q storage register

Bit	Attr	Reset Value	Description
31:24	RO	0x0	reserved
23:0	RW	0x000000	Q Q value of USER interface

OTPC_USER_QSR

Address: Operational Base + offset (0x0128)

OTPC USER QSR storage register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	QSR QSR value of USER interface

OTPC_USER_QRR

Address: Operational Base + offset (0x012c)

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OTPC USER QRR storage register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	QRR QRR value of USER interface

OTPC_SBPI_CMD_BASE

Address: Operational Base + offset (0x1000)

OTPC SBPI command base register

Bit	Attr	Reset Value	Description
31:9	RO	0x0	reserved
8	RW	0x0	read_bit 1'b0: indicate this command is NOT a read command 1'b1: indicate this command is a read command
7:0	RW	0x00	command_value contain value of the command

OTPC_SBPI_READ_DATA_BASE

Address: Operational Base + offset (0x2000)

OTPC SBPI reading data base register

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	Field0000 Abstract Field0000 Description

OTPC_INT_CON

Address: Operational Base + offset (0x0300)

OTPC interrupt register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	write_mask 16 bit write mask for lower bits
15	RW	0x0	otpc_global_int_enable 1'b0 : diable all interrupt
14:5	RO	0x0	reserved
4	RW	0x0	nonsecure_access_error_interrupt_enable 1'b0 : disable nonsecure access error interrupt 1'b1 : enable nonsecure access error interrupt
3	RW	0x0	secure_access_error_int_enable 1'b0 : disable secure access error interrupt 1'b1 : enable secure access error interrupt

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Bit	Attr	Reset Value	Description
2	RW	0x0	user_done_int_enable 1'b0 : disable user done interrupt 1'b1 : enable user done interrupt
1	RW	0x0	sbpi_done_int_enable 1'b0 : disable sbpi done interrupt 1'b1 : enable sbpi done interrupt
0	RW	0x0	sbpi_flag_detect_int_enable 1'b0 : disable sbpi flag detect interrupt 1'b1 : enable sbpi flag detect interrupt

OTPC_INT_STATUS

Address: Operational Base + offset (0x0304)

OTPC interrupt status register

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	nonsecure_access_error_int_status indicate a nonsecure access try to read an address which is illegal
3	RW	0x0	secure_access_error_int_status indicate a secure access try to read an address which is illegal
2	RW	0x0	user_done_int_status indicate a user done interrupt status
1	RW	0x0	sbpi_done_int_status indicate a sbpi done status
0	R/WSC	0x0	sbpi_flag_detect_int_status indicate detecting a flag negedge

1.11 Application Notes

12.1.1 GRF Register Summary

GRF Register	Register Description
SGRF_SOC_CON5[11]	OTP CKE enable selection 1b'1:enable 1'b0:disable
SGRF_SOC_CON5[10]	OTP secure or non-secure selection 1'b1:secure 1'b0:non-secure
SGRF_SOC_CON5[9]	OTP memory selection 1'b1:select SHF 1 1'b0:select SHF 0

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GRF_SOC_CON7[2]	OTP USER clock mux 1'b1:select 12M 1'b0:select 24M

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Chapter 38 GMAC Ethernet Interface

13.1 Overview

The GMAC Ethernet Controller provides a complete Ethernet interface from processor to a Reduced Media Independent Interface (RMII) and Reduced Gigabit Media Independent Interface (RGMII) compliant Ethernet PHY.
The GMAC includes a DMA controller. The DMA controller efficiently moves packet data from microprocessor's RAM, formats the data for an IEEE 802.3-2002 compliant packet and transmits the data to an Ethernet Physical Interface (PHY). It also efficiently moves packet data from RXFIFO to microprocessor's RAM.

13.1.1 Feature

- Supports 10/100/1000-Mbps data transfer rates with the RGMII interfaces
- Supports 10/100-Mbps data transfer rates with the RMII interfaces
- Supports both full-duplex and half-duplex operation
  - Supports CSMA/CD Protocol for half-duplex operation
  - Supports packet bursting and frame extension in 1000 Mbps half-duplex operation
  - Supports IEEE 802.3x flow control for full-duplex operation
  - Optional forwarding of received pause control frames to the user application in full-duplex operation
  - Back-pressure support for half-duplex operation
  - Automatic transmission of zero-quanta pause frame on de-assertion of flow control input in full-duplex operation
- Preamble and start-of-frame data (SFD) insertion in Transmit, and deletion in Receive paths
- Automatic CRC and pad generation controllable on a per-frame basis
- Options for Automatic Pad/CRC Stripping on receive frames
- Programmable frame length to support Standard Ethernet frames
- Programmable InterFrameGap (40-96 bit times in steps of 8)
- Supports a variety of flexible address filtering modes:
  - 64-bit Hash filter (optional) for multicast and uni-cast (DA) addresses
  - Option to pass all multicast addressed frames
  - Promiscuous mode support to pass all frames without any filtering for network monitoring
  - Passes all incoming packets (as per filter) with a status report
- Separate 32-bit status returned for transmission and reception packets
- Supports IEEE 802.1Q VLAN tag detection for reception frames
- MDIO Master interface for PHY device configuration and management
- Support detection of LAN wake-up frames and AMD Magic Packet frames
- Support checksum off-load for received IPv4 and TCP packets encapsulated by the Ethernet frame
- Support checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagrams
- Comprehensive status reporting for normal operation and transfers with errors
- Support per-frame Transmit/Receive complete interrupt control
- Supports 4-KB receive FIFO depths on reception.
- Supports 2-KB FIFO depth on transmission
- Automatic generation of PAUSE frame control or backpressure signal to the GMAC core based on Receive FIFO-fill (threshold configurable) level
- Handles automatic retransmission of Collision frames for transmission
- Discards frames on late collision, excessive collisions, excessive deferral and underrun conditions
- AXI interface to any CPU or memory
- Software can select the type of AXI burst (fixed and variable length burst) in the AXI Master interface
- Supports internal loopback on the RGMII/RMII for debugging

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- Debug status register that gives status of FSMs in Transmit and Receive data-paths and FIFO fill-levels.

13.2 Block Diagram

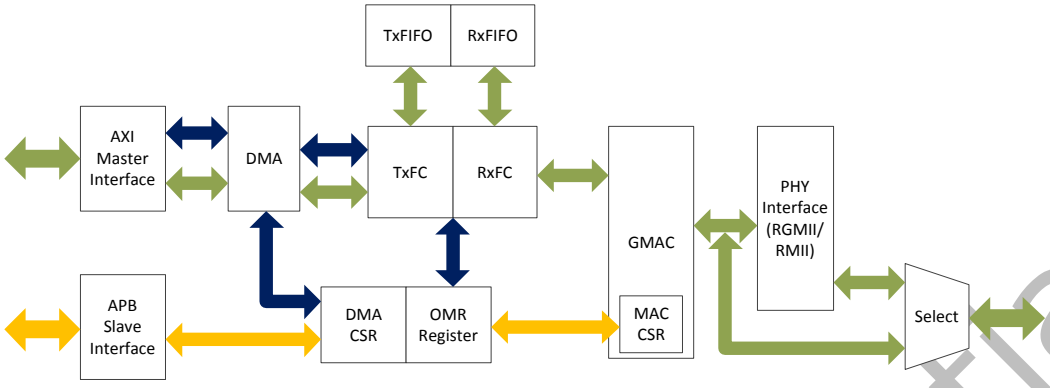

Fig.38-1GMACArchitecture

The GMAC is broken up into multiple separate functional units. These blocks are interconnected in the MAC module. The block diagram shows the general flow of data and control signals between these blocks.

The GMAC transfers data to system memory through the AXI master interface. The host CPU uses the APB Slave interface to access the GMAC subsystem's control and status registers (CSRs).

The GMAC supports the PHY interfaces of reduced GMII (RGMII) and reduced MII (RMII).

The Transmit FIFO (Tx FIFO) buffers data read from system memory by the DMA before transmission by the GMAC Core. Similarly, the Receive FIFO (Rx FIFO) stores the Ethernet frames received from the line until they are transferred to system memory by the DMA. These are asynchronous FIFOs, as they also transfer the data between the application clock and the GMAC line clocks.

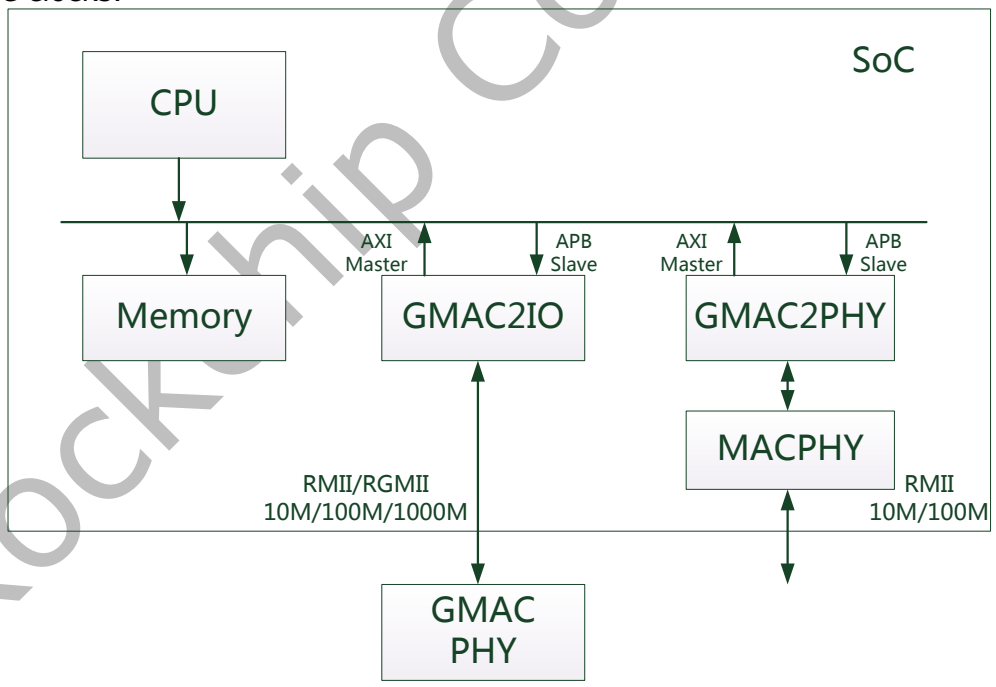

Fig.38-2 GMAC Architecture

There are two independent GMAC controllers named GMAC2IO and GMAC2PHY:

- GMAC2IO Supports 10/100/1000-Mbps data transfer rates with the RGMII interfaces and Supports 10/100-Mbps data transfer rates with the RMII interfaces
- GMAC2PHY Supports 10/100-Mbps data transfer rates with the RMII interfaces

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13.3 Function Description

13.3.1 Frame Structure

Data frames transmitted shall have the frame format shown in Fig. 25-2.

Fig. 38-1MAC Block Diagram

The preamble <preamble> begins a frame transmission. The bit value of the preamble field consists of 7 octets with the following bit values:
10101010 10101010 10101010 10101010 10101010 10101010 10101010
The SFD (start frame delimiter) <sfd> indicates the start of a frame and follows the preamble.The bit value is 10101011.
The data in a well formed frame shall consist of N octet's data.

13.3.2 RMII Interface timing diagram

The Reduced Media Independent Interface (RMII) specification reduces the pin count between Ethernet PHYs and Switch ASICs (only in 10/100 mode). According to the IEEE 802.3u standard, an MII contains 16 pins for data and control. In devices incorporating multiple MAC or PHY interfaces (such as switches), the number of pins adds significant cost with increase in port count. The RMII specification addresses this problem by reducing the pin count to 7 for each port - a 62.5% decrease in pin count.
The RMII module is instantiated between the GMAC and the PHY. This helps translation of the MAC's MII into the RMII. The RMII block has the following characteristics:

- Supports 10-Mbps and 100-Mbps operating rates. It does not support 1000-Mbps operation.
- Two clock references are sourced externally or CRU, providing independent, 2-bit wide transmit and receive paths.

Transmit Bit Ordering

Each nibble from the MII must be transmitted on the RMII a di-bit at a time with the order of di-bit transmission shown in Fig.1-3. The lower order bits (D1 and D0) are transmitted first followed by higher order bits (D2 and D3).

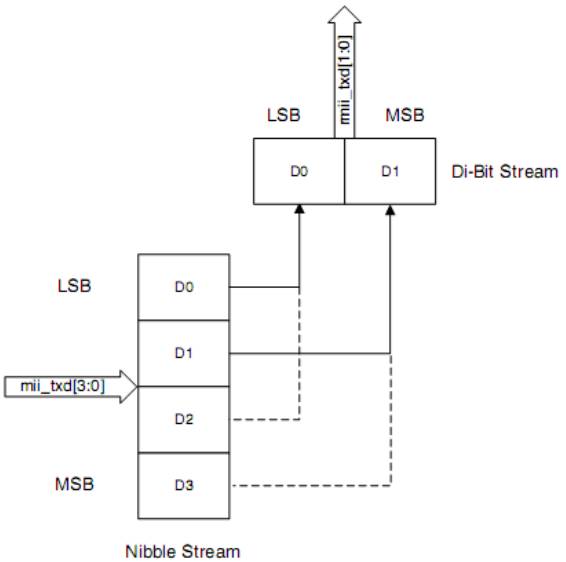

Fig. 38-2 RMII transmission bit ordering

RMII Transmit Timing Diagrams

Fig.1-4 through 1-7 show MII-to-RMII transaction timing.The clk_rmii_i (REF_CLK) frequency is 50MHz in RMII interface.In 10Mb/s mode, as the REF_CLK frequency is 10 times as the data rate, the value on rmii_txd_o[1:0] (TXD[1:0]) shall be valid such that TXD[1:0] may be sampled every 10th cycle,regard-less of the starting cycle within the group and yield the correct frame data.

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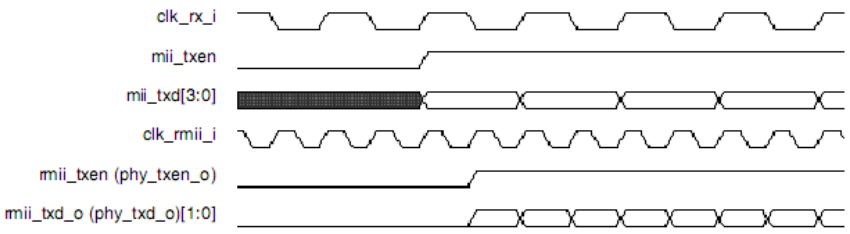

Fig. 38-3Start of MII and RMII transmission in 100-Mbps mode

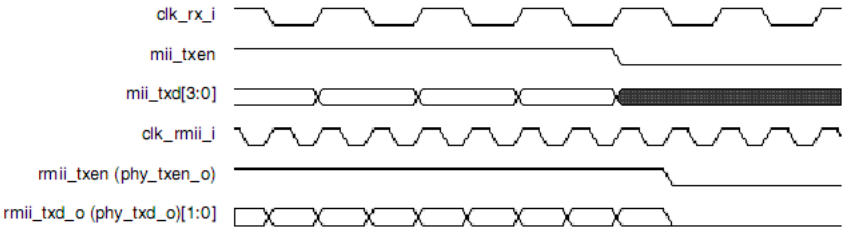

Fig. 38-4End of MII and RMII Transmission in 100-Mbps Mode

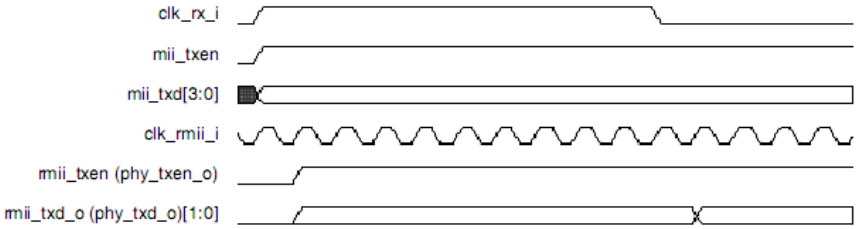

Fig. 38-5Start of MII and RMII Transmission in 10-Mbps Mode

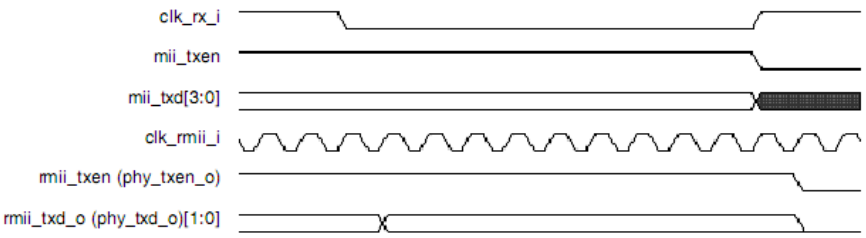

Fig. 38-6End of MII and RMII Transmission in 10-Mbps Mode

Receive Bit Ordering

Each nibble is transmitted to the MII from the di-bit received from the RMII in the nibble transmission order shown in Fig.1-8. The lower order bits (D0 and D1) are received first, followed by the higher order bits (D2 and D3).

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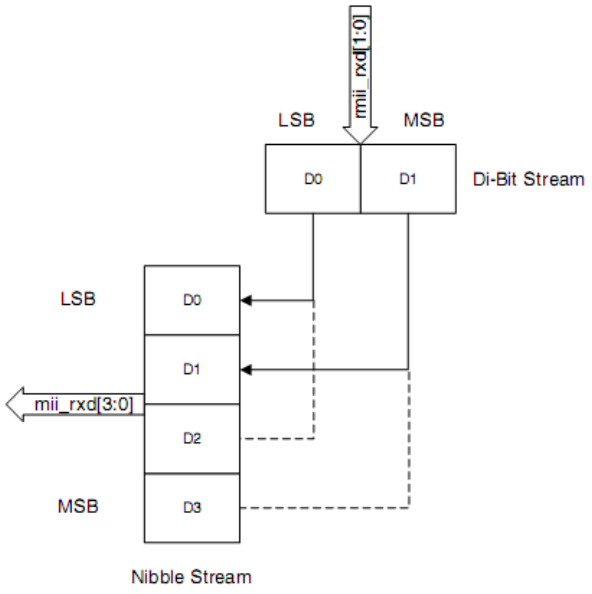

Fig. 38-7 RMII receive bit ordering

13.3.3 RGMII interface

The Reduced Gigabit Media Independent Interface (RGMII) specification reduces the pin count of the interconnection between the GMAC 10/100/1000 controller and the PHY for GMII and MII interfaces. To achieve this, the data path and control signals are reduced and multiplexed together with both the edges of the transmission and receive clocks. For gigabit operation the clocks operate at 125 MHz; for 10/100 operation, the clock rates are 2.5 MHz/25 MHz.
In the GMAC 10/100/1000 controller, the RGMII module is instantiated between the GMAC core's GMII and the PHY to translate the control and data signals between the GMII and RGMII protocols.
The RGMII block has the following characteristics:

- Supports 10-Mbps, 100-Mbps, and 1000-Mbps operation rates.
- For the RGMII block, no extra clock is required because both the edges of the incoming clocks are used.
- The RGMII block extracts the in-band (link speed, duplex mode and link status) status signals from the PHY and provides them to the GMAC core logic for link detection.

13.3.4 Management Interface

The MAC management interface provides a simple, two-wire, serial interface to connect the GMAC and a managed PHY, for the purposes of controlling the PHY and gathering status from the PHY. The management interface consists of a pair of signals that transport the management information across the MII bus: MDIO and MDC.
The GMAC initiates the management write/read operation. The clock gmii_mdc_o(MDC) is a divided clock from the application clock pclk_gmac. The divide factor depends on the clock range setting in the GMII address register. Clock range is set as follows:

Selection	pclk_gmac	MDC Clock
0000	60-100 MHz	pclk_gmac/42
0001	100-150 MHz	pclk_gmac/62
0010	20-35 MHz	pclk_gmac/16
0011	35-60 MHz	pclk_gmac/26
0100	150-250 MHz	pclk_gmac/102
0101	250-300 MHz	pclk_gmac/124
0110, 0111	Reserved	

The MDC is the derivative of the application clock pclk_gmac. The management operation is performed through the gmii_mdi_i, gmii_mdo_o and gmii_mdo_o_e signals. A three-state buffer is implemented in the PAD.
The frame structure on the MDIO line is shown below.

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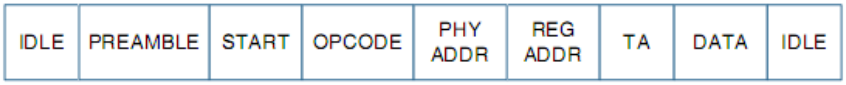

Fig. 38-8 MDIO frame structure

IDLE: The mdio line is three-state; there is no clock on gmii_mdc_o
PREAMBLE: 32 continuous bits of value 1
START: Start-of-frame is 2'b01
OPCODE: 2'b10 for read and 2'b01 for write
PHY ADDR: 5-bit address select for one of 32 PHYs
REG ADDR: Register address in the selected PHY
TA: Turnaround is 2'bZ0 for read and 2'b10 for Write
DATA: Any 16-bit value. In a write operation, the GMAC drives mdio; in a read operation, PHY drives it.

13.3.5 Power Management Block

Power management (PMT) supports the reception of network (remote) wake-up frames and Magic Packet frames. PMT does not perform the clock gate function, but generates interrupts for wake-up frames and Magic Packets received by the GMAC. The PMT block sits on the receiver path of the GMAC and is enabled with remote wake-up frame enable and Magic Packet enable. These enables are in the PMT control and status register and are programmed by the application.
When the power down mode is enabled in the PMT, then all received frames are dropped by the core and they are not forwarded to the application. The core comes out of the power down mode only when either a Magic Packet or a Remote Wake-up frame is received and the corresponding detection is enabled.

Remote Wake-Up Frame Detection

When the GMAC is in sleep mode and the remote wake-up bit is enabled in register GMAC_PMT_CTRL_STA (0x002C), normal operation is resumed after receiving a remote wake-up frame. The application writes all eight wake-up filter registers, by performing a sequential write to address (0028). The application enables remote wake-up by writing a 1 to bit 2 of the register GMAC_PMT_CTRL_STA.
PMT supports four programmable filters that allow support of different receive frame patterns. If the incoming frame passes the address filtering of Filter Command, and if Filter CRC-16 matches the incoming examined pattern, then the wake-up frame is received.
Filter_offset (minimum value 12, which refers to the 13th byte of the frame) determines the offset from which the frame is to be examined. Filter Byte Mask determines which bytes of the frame must be examined. The thirty-first bit of Byte Mask must be set to zero.
The remote wake-up CRC block determines the CRC value that is compared with Filter CRC-16. The wake-up frame is checked only for length error, FCS error, dribble bit error, GMII error, collision, and to ensure that it is not a runt frame. Even if the wake-up frame is more than 512 bytes long, if the frame has a valid CRC value, it is considered valid. Wake-up frame detection is updated in the register GMAC_PMT_CTRL_STA for every remote Wake-up frame received. A PMT interrupt to the application triggers a read to the GMAC_PMT_CTRL_STA register to determine reception of a wake-up frame.

Magic Packet Detection

The Magic Packet frame is based on a method that uses Advanced Micro Device's Magic Packet technology to power up the sleeping device on the network. The GMAC receives a specific packet of information, called a Magic Packet, addressed to the node on the network.
Only Magic Packets that are addressed to the device or a broadcast address will be checked to determine whether they meet the wake-up requirements. Magic Packets that pass the address filtering (unicast or broadcast) will be checked to determine whether they meet the remote Wake-on-LAN data format of 6 bytes of all ones followed by a GMAC Address appearing 16 times.
The application enables Magic Packet wake-up by writing a 1 to Bit 1 of the register GMAC_PMT_CTRL_STA. The PMT block constantly monitors each frame addressed to the node for a specific Magic Packet pattern. Each frame received is checked for a 48'hFF_FF_FF_FF_FF_FF pattern following the destination and source address field. The PMT block then checks the frame for 16 repetitions of the GMAC address without any breaks or

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interruptions. In case of a break in the 16 repetitions of the address, the 48'hFF_FF_FF_FF_FF_FF pattern is scanned for again in the incoming frame. The 16 repetitions can be anywhere in the frame, but must be preceded by the synchronization stream (48'hFF_FF_FF_FF_FF_FF). The device will also accept a multicast frame, as long as the 16 duplications of the GMAC address are detected.
If the MAC address of a node is 48'h00_11_22_33_44_55, then the GMAC scans for the data sequence:
Destination Address Source Address ……. FF FFFFFFFFF
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55
…CRC
Magic Packet detection is updated in the PMT Control and Status register for Magic Packet received. A PMT interrupt to the Application triggers a read to the PMT CSR to determine whether a Magic Packet frame has been received.

13.3.6 MAC Management Counters

The counters in the MAC Management Counters (MMC) module can be viewed as an extension of the register address space of the CSR module. The MMC module maintains a set of registers for gathering statistics on the received and transmitted frames. These include a control register for controlling the behavior of the registers, two 32-bit registers containing interrupts generated (receive and transmit), and two 32-bit registers containing masks for the Interrupt register (receive and transmit). These registers are accessible from the Application through the MAC Control Interface (MCI). Non-32-bit accesses are allowed as long as the address is word-aligned.
The organization of these registers is shown in Register Description. The MMCs are accessed using transactions, in the same way the CSR address space is accessed. The Register Description in this chapter describe the various counters and list the address for each of the statistics counters. This address will be used for Read/Write accesses to the desired transmit/receive counter.
The MMC module gathers statistics on encapsulated IPv4, IPv6, TCP, UDP, or ICMP payloads in received Ethernet frames.

13.4 Register Description

13.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
GMAC_MAC_CONF	0x0000	W	0x00000000	MAC Configuration Register
GMAC_MAC_FRM_FILT	0x0004	W	0x00000000	MAC Frame Filter
GMAC_HASH_TAB_HI	0x0008	W	0x00000000	Hash Table High Register
GMAC_HASH_TAB_LO	0x000c	W	0x00000000	Hash Table Low Register
GMAC_GMII_ADDR	0x0010	W	0x00000000	GMII Address Register
GMAC_GMII_DATA	0x0014	W	0x00000000	GMII Data Register
GMAC_FLOW_CTRL	0x0018	W	0x00000000	Flow Control Register
GMAC_VLAN_TAG	0x001c	W	0x00000000	VLAN Tag Register
GMAC_DEBUG	0x0024	W	0x00000000	Debug register
GMAC_PMT_CTRL_STA	0x002c	W	0x00000000	PMT Control and Status Register
GMAC_INT_STATUS	0x0038	W	0x00000000	Interrupt Status Register
GMAC_INT_MASK	0x003c	W	0x00000000	Interrupt Mask Register
GMAC_MAC_ADDR0_HI	0x0040	W	0x0000ffff	MAC Address0 High Register
GMAC_MAC_ADDR0_LO	0x0044	W	0xffffffff	MAC Address0 Low Register
GMAC_AN_CTRL	0x00c0	W	0x00000000	AN Control Register

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Name	Offset	Size	Reset Value	Description
GMAC_AN_STATUS	0x00c4	W	0x00000008	AN Status Register
GMAC_AN_ADV	0x00c8	W	0x000001e0	Auto Negotiation Advertisement Register
GMAC_AN_LINK_PART_AB	0x00cc	W	0x00000000	Auto Negotiation Link Partner Ability Register
GMAC_AN_EXP	0x00d0	W	0x00000000	Auto Negotiation Expansion Register
GMAC_INTF_MODE_STA	0x00d8	W	0x00000000	RGMII Status Register
GMAC_MMC_CTRL	0x0100	W	0x00000000	MMC Control Register
GMAC_MMC_RX_INTR	0x0104	W	0x00000000	MMC Receive Interrupt Register
GMAC_MMC_TX_INTR	0x0108	W	0x00000000	MMC Transmit Interrupt Register
GMAC_MMC_RX_INT_MSK	0x010c	W	0x00000000	MMC Receive Interrupt Mask Register
GMAC_MMC_TX_INT_MSK	0x0110	W	0x00000000	MMC Transmit Interrupt Mask Register
GMAC_MMC_TXOCTETCNT_GB	0x0114	W	0x00000000	MMC TX OCTET Good and Bad Counter
GMAC_MMC_TXFRMCNT_GB	0x0118	W	0x00000000	MMC TX Frame Good and Bad Counter
GMAC_MMC_TXUNDFLWERR	0x0148	W	0x00000000	MMC TX Underflow Error
GMAC_MMC_TXCARERR	0x0160	W	0x00000000	MMC TX Carrier Error
GMAC_MMC_TXOCTETCNT_G	0x0164	W	0x00000000	MMC TX OCTET Good Counter
GMAC_MMC_TXFRMCNT_G	0x0168	W	0x00000000	MMC TX Frame Good Counter
GMAC_MMC_RXFRMCNT_GB	0x0180	W	0x00000000	MMC RX Frame Good and Bad Counter
GMAC_MMC_RXOCTETCNT_GB	0x0184	W	0x00000000	MMC RX OCTET Good and Bad Counter
GMAC_MMC_RXOCTETCNT_G	0x0188	W	0x00000000	MMC RX OCTET Good Counter
GMAC_MMC_RXMCFRMCNT_G	0x0190	W	0x00000000	MMC RX Multicast Frame Good Counter
GMAC_MMC_RXCRCERR	0x0194	W	0x00000000	MMC RX Carrier
GMAC_MMC_RXLENERR	0x01c8	W	0x00000000	MMC RX Length Error
GMAC_MMC_RXFIFOOVRFLW	0x01d4	W	0x00000000	MMC RX FIFO Overflow
GMAC_MMC_IPC_INT_MSK	0x0200	W	0x00000000	MMC Receive Checksum Offload Interrupt Mask Register
GMAC_MMC_IPC_INTR	0x0208	W	0x00000000	MMC Receive Checksum Offload Interrupt Register
GMAC_MMC_RXIPV4GFRM	0x0210	W	0x00000000	MMC RX IPV4 Good Frame
GMAC_MMC_RXIPV4HDERRFRM	0x0214	W	0x00000000	MMC RX IPV4 Head Error Frame

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Name	Offset	Size	Reset Value	Description
GMAC_MMC_RXIPV6GFRM	0x0224	W	0x00000000	MMC RX IPV6 Good Frame
GMAC_MMC_RXIPV6HDERRFRM	0x0228	W	0x00000000	MMC RX IPV6 Head Error Frame
GMAC_MMC_RXUDPERRFRM	0x0234	W	0x00000000	MMC RX UDP Error Frame
GMAC_MMC_RXTCPERRFRM	0x023c	W	0x00000000	MMC RX TCP Error Frame
GMAC_MMC_RXICMPERRFRM	0x0244	W	0x00000000	MMC RX ICMP Error Frame
GMAC_MMC_RXIPV4HDERROCT	0x0254	W	0x00000000	MMC RX OCTET IPV4 Head Error
GMAC_MMC_RXIPV6HDERROCT	0x0268	W	0x00000000	MMC RX OCTET IPV6 Head Error
GMAC_MMC_RXUDPERROCT	0x0274	W	0x00000000	MMC RX OCTET UDP Error
GMAC_MMC_RXTCPERROCT	0x027c	W	0x00000000	MMC RX OCTET TCP Error
GMAC_MMC_RXICMPERROCT	0x0284	W	0x00000000	MMC RX OCTET ICMP Error
GMAC_BUS_MODE	0x1000	W	0x00020101	Bus Mode Register
GMAC_TX_POLL_DEMAND	0x1004	W	0x00000000	Transmit Poll Demand Register
GMAC_RX_POLL_DEMAND	0x1008	W	0x00000000	Receive Poll Demand Register
GMAC_RX_DESC_LIST_ADDR	0x100c	W	0x00000000	Receive Descriptor List Address Register
GMAC_TX_DESC_LIST_ADDR	0x1010	W	0x00000000	Transmit Descriptor List Address Register
GMAC_STATUS	0x1014	W	0x00000000	Status Register
GMAC_OP_MODE	0x1018	W	0x00000000	Operation Mode Register
GMAC_INT_ENA	0x101c	W	0x00000000	Interrupt Enable Register
GMAC_OVERFLOW_CNT	0x1020	W	0x00000000	Missed Frame and Buffer Overflow Counter Register
GMAC_REC_INT_WDT_TIMER	0x1024	W	0x00000000	Receive Interrupt Watchdog Timer Register
GMAC_AXI_BUS_MODE	0x1028	W	0x00110001	AXI Bus Mode Register
GMAC_AXI_STATUS	0x102c	W	0x00000000	AXI Status Register
GMAC_CUR_HOST_TX_DESC	0x1048	W	0x00000000	Current Host Transmit Descriptor Register
GMAC_CUR_HOST_RX_DESC	0x104c	W	0x00000000	Current Host Receive Descriptor Register
GMAC_CUR_HOST_TX_BUF_ADDR	0x1050	W	0x00000000	Current Host Transmit Buffer Address Register
GMAC_CUR_HOST_RX_BUF_ADDR	0x1054	W	0x00000000	Current Host Receive Buffer Address Register

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

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13.4.2 Detail Register Description

GMAC_MAC_CONF
Address: Operational Base + offset (0x0000)
MAC Configuration Register

Bit	Attr	Reset Value	Description
31:25	RO	0x0	reserved
24	RW	0x0	TC Transmit Configuration in RGMII When set, this bit enables the transmission of duplex mode, link speed, and link up/down information to the PHY in the RGMII ports. When this bit is reset, no such information is driven to the PHY.
23	RW	0x0	WD Watchdog Disable When this bit is set, the GMAC disables the watchdog timer on the receiver, and can receive frames of up to 16,384 bytes. When this bit is reset, the GMAC allows no more than 2,048 bytes (10,240 if JE is set high) of the frame being received and cuts off any bytes received after that.
22	RW	0x0	JD Jabber Disable When this bit is set, the GMAC disables the jabber timer on the transmitter, and can transfer frames of up to 16,384 bytes. When this bit is reset, the GMAC cuts off the transmitter if the application sends out more than 2,048 bytes of data (10,240 if JE is set high) during transmission.
21	RW	0x0	BE Frame Burst Enable When this bit is set, the GMAC allows frame bursting during transmission in GMII Half-Duplex mode.
20	RO	0x0	reserved
19:17	RW	0x0	IFG Inter-Frame Gap These bits control the minimum IFG between frames during transmission. 3'b000: 96 bit times 3'b001: 88 bit times 3'b010: 80 bit times ... 3'b111: 40 bit times
16	RW	0x0	DCRS Disable Carrier Sense During Transmission When set high, this bit makes the MAC transmitter ignore the (G)MII CRS signal during frame transmission in Half-Duplex mode. This request results in no errors generated due to Loss of Carrier or No Carrier during such transmission. When this bit is low, the MAC transmitter generates such errors due to Carrier Sense and will even abort the transmissions.

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Bit	Attr	Reset Value	Description
15	RW	0x0	PS Port Select Selects between GMII and MII: 1'b0: GMII (1000 Mbps) 1'b1: MII (10/100 Mbps)
14	RW	0x0	FES Speed Indicates the speed in Fast Ethernet (MII) mode: 1'b0: 10 Mbps 1'b1: 100 Mbps
13	RW	0x0	DO Disable Receive Own When this bit is set, the GMAC disables the reception of frames when the gmii_txen_o is asserted in Half-Duplex mode. When this bit is reset, the GMAC receives all packets that are given by the PHY while transmitting.
12	RW	0x0	LM Loopback Mode When this bit is set, the GMAC operates in loopback mode at GMII/MII. The (G)MII Receive clock input (clk_rx_i) is required for the loopback to work properly, as the Transmit clock is not looped-back internally.
11	RW	0x0	DM Duplex Mode When this bit is set, the GMAC operates in a Full-Duplex mode where it can transmit and receive simultaneously. This bit is RO with default value of 1'b1 in Full-Duplex-only configuration.
10	RW	0x0	IPC Checksum Offload When this bit is set, the GMAC calculates the 16-bit one's complement of the one's complement sum of all received Ethernet frame payloads. It also checks whether the IPv4 Header checksum (assumed to be bytes 25-26 or 29-30 (VLAN-tagged) of the received Ethernet frame) is correct for the received frame and gives the status in the receive status word. The GMAC core also appends the 16-bit checksum calculated for the IP header datagram payload (bytes after the IPv4 header) and appends it to the Ethernet frame transferred to the application (when Type 2 COE is deselected). When this bit is reset, this function is disabled. When Type 2 COE is selected, this bit, when set, enables IPv4 checksum checking for received frame payloads TCP/UDP/ICMP headers. When this bit is reset, the COE function in the receiver is disabled and the corresponding PCE and IP HCE status bits are always cleared.

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Bit	Attr	Reset Value	Description
9	RW	0x0	DR Disable Retry When this bit is set, the GMAC will attempt only 1 transmission. When a collision occurs on the GMII/MII, the GMAC will ignore the current frame transmission and report a Frame Abort with excessive collision error in the transmit frame status. When this bit is reset, the GMAC will attempt retries based on the settings of BL.
8	RW	0x0	LUD Link Up/Down Indicates whether the link is up or down during the transmission of configuration in RGMII interface: 1'b0: Link Down 1'b1: Link Up
7	RW	0x0	ACS Automatic Pad/CRC Stripping When this bit is set, the GMAC strips the Pad/FCS field on incoming frames only if the length's field value is less than or equal to 1,500 bytes. All received frames with length field greater than or equal to 1,501 bytes are passed to the application without stripping the Pad/FCS field. When this bit is reset, the GMAC will pass all incoming frames to the Host unmodified.
6:5	RW	0x0	BL Back-Off Limit The Back-Off limit determines the random integer number (r) of slot time delays (4,096 bit times for 1000 Mbps and 512 bit times for 10/100 Mbps) the GMAC waits before rescheduling a transmission attempt during retries after a collision. This bit is applicable only to Half-Duplex mode and is reserved (RO) in Full-Duplex-only configuration. 2'b00: k = min (n, 10) 2'b01: k = min (n, 8) 2'b10: k = min (n, 4) 2'b11: k = min (n, 1), Where n = retransmission attempt. The random integer r takes the value in the range 0 = r < 2^k

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Bit	Attr	Reset Value	Description
4	RW	0x0	DC Deferral Check When this bit is set, the deferral check function is enabled in the GMAC. The GMAC will issue a Frame Abort status, along with the excessive deferral error bit set in the transmit frame status when the transmission state machine is deferred for more than 24,288 bit times in 10/100-Mbps mode. If the Core is configured for 1000 Mbps operation, the threshold for deferral is 155,680 bits times. Deferral begins when the transmitter is ready to transmit, but is prevented because of an active CRS (carrier sense) signal on the GMII/MII. Defer time is not cumulative. If the transmitter defers for 10,000 bit times, then transmits, collides, backs off, and then has to defer again after completion of back-off, the deferral timer resets to 0 and restarts. When this bit is reset, the deferral check function is disabled and the GMAC defers until the CRS signal goes inactive.
3	RW	0x0	TE Transmitter Enable When this bit is set, the transmission state machine of the GMAC is enabled for transmission on the GMII/MII. When this bit is reset, the GMAC transmit state machine is disabled after the completion of the transmission of the current frame, and will not transmit any further frames.
2	RW	0x0	RE Receiver Enable When this bit is set, the receiver state machine of the GMAC is enabled for receiving frames from the GMII/MII. When this bit is reset, the GMAC receive state machine is disabled after the completion of the reception of the current frame, and will not receive any further frames from the GMII/MII.
1:0	RO	0x0	reserved

GMAC_MAC_FRM_FILT
Address: Operational Base + offset (0x0004)
MAC Frame Filter

Bit	Attr	Reset Value	Description
31	RW	0x0	RA Receive All When this bit is set, the GMAC Receiver module passes to the Application all frames received irrespective of whether they pass the address filter. The result of the SA/DA filtering is updated (pass or fail) in the corresponding bits in the Receive Status Word. When this bit is reset, the Receiver module passes to the Application only those frames that pass the SA/DA address filter.
30:11	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
10	RW	0x0	HPF Hash or Perfect Filter When set, this bit configures the address filter to pass a frame if it matches either the perfect filtering or the hash filtering as set by HMC or HUC bits. When low and if the HUC/HMC bit is set, the frame is passed only if it matches the Hash filter.
9	RW	0x0	SAF Source Address Filter Enable The GMAC core compares the SA field of the received frames with the values programmed in the enabled SA registers. If the comparison matches, then the SAMatch bit of RxStatus Word is set high. When this bit is set high and the SA filter fails, the GMAC drops the frame. When this bit is reset, then the GMAC Core forwards the received frame to the application and with the updated SA Match bit of the RxStatus depending on the SA address comparison.
8	RW	0x0	SAIF SA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the SA address comparison. The frames whose SA matches the SA registers will be marked as failing the SA Address filter. When this bit is reset, frames whose SA does not match the SA registers will be marked as failing the SA Address filter.
7:6	RW	0x0	PCF Pass Control Frames These bits control the forwarding of all control frames (including unicast and multicast PAUSE frames). Note that the processing of PAUSE control frames depends only on RFE of Register GMAC_FLOW_CTRL[2]. 2'b00: GMAC filters all control frames from reaching the application. 2'b01: GMAC forwards all control frames except PAUSE control frames to application even if they fail the Address filter. 2'b10: GMAC forwards all control frames to application even if they fail the Address Filter. 2'b11: GMAC forwards control frames that pass the Address Filter.
5	RW	0x0	DBF Disable Broadcast Frames When this bit is set, the AFM module filters all incoming broadcast frames. When this bit is reset, the AFM module passes all received broadcast frames.

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Bit	Attr	Reset Value	Description
4	RW	0x0	PM Pass All Multicast When set, this bit indicates that all received frames with a multicast destination address (first bit in the destination address field is '1') are passed. When reset, filtering of multicast frame depends on HMC bit.
3	RW	0x0	DAIF DA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the DA address comparison for both unicast and multicast frames. When reset, normal filtering of frames is performed.
2	RW	0x0	HMC Hash Multicast When set, MAC performs destination address filtering of received multicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for multicast frames, that is, it compares the DA field with the values programmed in DA registers.
1	RW	0x0	HUC Hash Unicast When set, MAC performs destination address filtering of unicast frames according to the hash table. When reset, the MAC performs a perfect destination address filtering for unicast frames, that is, it compares the DA field with the values programmed in DA registers.
0	RW	0x0	PR Promiscuous Mode When this bit is set, the Address Filter module passes all incoming frames regardless of its destination or source address. The SA/DA Filter Fails status bits of the Receive Status Word will always be cleared when PR is set.

GMAC_HASH_TAB_HI
Address: Operational Base + offset (0x0008)
Hash Table High Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HTH Hash Table High This field contains the upper 32 bits of Hash table

GMAC_HASH_TAB_LO
Address: Operational Base + offset (0x000c)
Hash Table Low Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	HTL Hash Table Low This field contains the lower 32 bits of Hash table

GMAC_GMII_ADDR
Address: Operational Base + offset (0x0010)
GMII Address Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:11	RW	0x00	PA Physical Layer Address This field tells which of the 32 possible PHY devices are being accessed
10:6	RW	0x00	GR GMII Register These bits select the desired GMII register in the selected PHY device

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Bit	Attr	Reset Value	Description
5:2	RW	0x0	CR APB Clock Range The APB Clock Range selection determines the frequency of the MDC clock as per the pclk_gmac frequency used in your design. The suggested range of pclk_gmac frequency applicable for each value below (when Bit[5] = 0) ensures that the MDC clock is approximately between the frequency range 1.0 MHz - 2.5 MHz. **Selection pclk_gmacMDC Clock** 0000 60-100 MHz pclk_gmac/42 0001 100-150 MHz pclk_gmac/62 0010 20-35 MHz pclk_gmac/16 0011 35-60 MHz pclk_gmac/26 0100 150-250 MHz pclk_gmac/102 0101 250-300 MHz pclk_gmac/124 0110, 0111 Reserved When bit 5 is set, you can achieve MDC clock of frequency higher than the IEEE802.3 specified frequency limit of 2.5 MHz and program a clock divider of lower value. For example, when pclk_gmac is of frequency 100 MHz and you program these bits as "1010", then the resultant MDC clock will be of 12.5 MHz which is outside the limit of IEEE 802.3 specified range. Please program the values given below only if the interfacing chips supports faster MDC clocks. **Selection MDC Clock** 1000 pclk_gmac/4 1001 pclk_gmac/6 1010 pclk_gmac/8 1011 pclk_gmac/10 1100 pclk_gmac/12 1101 pclk_gmac/14 1110 pclk_gmac/16 1111 pclk_gmac/18
1	RW	0x0	GW GMII Write When set, this bit tells the PHY that this will be a Write operation using register GMAC_GMII_DATA. If this bit is not set, this will be a Read operation, placing the data in register GMAC_GMII_DATA.

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Bit	Attr	Reset Value	Description
0	W1C	0x0	GB GMII Busy This bit should read a logic 0 before writing to Register GMII_ADDR and Register GMII_DATA. This bit must also be set to 0 during a Write to Register GMII_ADDR. During a PHY register access, this bit will be set to 1'b1 by the Application to indicate that a Read or Write access is in progress. Register GMII_DATA (GMII Data) should be kept valid until this bit is cleared by the GMAC during a PHY Write operation. The Register GMII_DATA is invalid until this bit is cleared by the GMAC during a PHY Read operation. The Register GMII_ADDR (GMII Address) should not be written to until this bit is cleared.

GMAC_GMII_DATA
Address: Operational Base + offset (0x0014)
GMII Data Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	GD GMII Data This contains the 16-bit data value read from the PHY after a Management Read operation or the 16-bit data value to be written to the PHY before a Management Write operation.

GMAC_FLOW_CTRL
Address: Operational Base + offset (0x0018)
Flow Control Register

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	PT Pause Time This field holds the value to be used in the Pause Time field in the transmit control frame. If the Pause Time bits is configured to be double-synchronized to the (G)MII clock domain, then consecutive writes to this register should be performed only after at least 4 clock cycles in the destination clock domain.
15:8	RO	0x0	reserved
7	RW	0x0	DZPQ Disable Zero-Quanta Pause When set, this bit disables the automatic generation of Zero-Quanta Pause Control frames on the de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i/mti_flowctrl_i). When this bit is reset, normal operation with automatic Zero-Quanta Pause Control frame generation is enabled.
6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	PLT Pause Low Threshold This field configures the threshold of the PAUSE timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of PAUSE Frame. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot-times), and PLT = 01, then a second PAUSE frame is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot-times after the first PAUSE frame is transmitted. **Selection Threshold** 00 Pause time minus 4 slot times 01 Pause time minus 28 slot times 10 Pause time minus 144 slot times 11 Pause time minus 256 slot times Slot time is defined as time taken to transmit 512 bits (64 bytes) on the GMII/MII interface.
3	RW	0x0	UP Unicast Pause Frame Detect When this bit is set, the GMAC will detect the Pause frames with the station's unicast address specified in MAC Address0 High Register and MAC Address0 Low Register, in addition to the detecting Pause frames with the unique multicast address. When this bit is reset, the GMAC will detect only a Pause frame with the unique multicast address specified in the 802.3x standard.
2	RW	0x0	RFE Receive Flow Control Enable When this bit is set, the GMAC will decode the received Pause frame and disable its transmitter for a specified (Pause Time) time. When this bit is reset, the decode function of the Pause frame is disabled.
1	RW	0x0	TFE Transmit Flow Control Enable In Full-Duplex mode, when this bit is set, the GMAC enables the flow control operation to transmit Pause frames. When this bit is reset, the flow control operation in the GMAC is disabled, and the GMAC will not transmit any Pause frames. In Half-Duplex mode, when this bit is set, the GMAC enables the back-pressure operation. When this bit is reset, the backpressure feature is disabled.

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Bit	Attr	Reset Value	Description
0	RW	0x0	FCB_BPA Flow Control Busy/Backpressure Activate This bit initiates a Pause Control frame in Full-Duplex mode and activates the backpressure function in Half-Duplex mode if TFE bit is set. In Full-Duplex mode, this bit should be read as 1'b0 before writing to the register GMAC_FLOW_CTRL. To initiate a pause control frame, the application must set this bit to 1'b1. During a transfer of the control frame, this bit will continue to be set to signify that a frame transmission is in progress. After the completion of Pause control frame transmission, the GMAC will reset this bit to 1'b0. The register GMAC_FLOW_CTRL should not be written to until this bit is cleared. In Half-Duplex mode, when this bit is set (and TFE is set), then backpressure is asserted by the GMAC Core. During backpressure, when the GMAC receives a new frame, the transmitter starts sending a JAM pattern resulting in a collision. This control register bit is logically OR'ed with the mti_flowctrl_i input signal for the backpressure function.

GMAC_VLAN_TAG
Address: Operational Base + offset (0x001c)
VLAN Tag Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	ETV Enable 12-Bit VLAN Tag Comparison When this bit is set, a 12-bit VLAN identifier, rather than the complete 16-bit VLAN tag, is used for comparison and filtering. Bits[11:0] of the VLAN tag are compared with the corresponding field in the received VLAN-tagged frame. When this bit is reset, all 16 bits of the received VLAN frame's fifteenth and sixteenth bytes are used for comparison.
15:0	RW	0x0000	VL VLAN Tag Identifier for Receive Frames This contains the 802.1Q VLAN tag to identify VLAN frames, and is compared to the fifteenth and sixteenth bytes of the frames being received for VLAN frames. Bits[15:13] are the User Priority, Bit[12] is the Canonical Format Indicator (CFI) and bits[11:0] are the VLAN tag's VLAN Identifier (VID) field. When the ETV bit is set, only the VID (Bits[11:0]) is used for comparison. If VL (VL[11:0] if ETV is set) is all zeros, the GMAC does not check the fifteenth and sixteenth bytes for VLAN tag comparison, and declares all frames with a Type field value of 0x8100 to be VLAN frames.

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GMAC_DEBUG
Address: Operational Base + offset (0x0024)
Debug register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	TFIFO3 When high, it indicates that the MTL TxStatus FIFO is full and hence the MTL will not be accepting any more frames for transmission.
24	RW	0x0	TFIFO2 When high, it indicates that the MTL TxFIFO is not empty and has some data left for transmission.
23	RO	0x0	reserved
22	RW	0x0	TFIFO1 When high, it indicates that the MTL TxFIFO Write Controller is active and transferring data to the TxFIFO.
21:20	RW	0x0	TFIFOSTA This indicates the state of the TxFIFO read Controller: 2'b00: IDLE state 2'b01: READ state (transferring data to MAC transmitter) 2'b10: Waiting for TxStatus from MAC transmitter 2'b11: Writing the received TxStatus or flushing the TxFIFO
19	RW	0x0	PAUSE When high, it indicates that the MAC transmitter is in PAUSE condition (in full-duplex only) and hence will not schedule any frame for transmission
18:17	RW	0x0	TSAT This indicates the state of the MAC Transmit Frame Controller module: 2'b00: IDLE 2'b01: Waiting for Status of previous frame or IFG/backoff period to be over 2'b10: Generating and transmitting a PAUSE control frame (in full duplex mode) 2'b11: Transferring input frame for transmission
16	RW	0x0	TACT When high, it indicates that the MAC GMII/MII transmit protocol engine is actively transmitting data and not in IDLE state.
15:10	RO	0x0	reserved
9:8	RW	0x0	RFIFO This gives the status of the RxFIFO Fill-level: 2'b00: RxFIFO Empty 2'b01: RxFIFO fill-level below flow-control de-activate threshold 2'b10: RxFIFO fill-level above flow-control activate threshold 2'b11: RxFIFO Full
7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:5	RW	0x0	RFIFORD It gives the state of the RxFIFO read Controller: 2'b00: IDLE state 2'b01: Reading frame data 2'b10: Reading frame status (or time-stamp) 2'b11: Flushing the frame data and Status
4	RW	0x0	RFIFOWR When high, it indicates that the MTL RxFIFO Write Controller is active and transferring a received frame to the FIFO.
3	RO	0x0	reserved
2:1	RW	0x0	ACT When high, it indicates the active state of the small FIFO Read and Write controllers respectively of the MAC receive Frame Controller module
0	RW	0x0	RDB When high, it indicates that the MAC GMII/MII receive protocol engine is actively receiving data and not in IDLE state.

GMAC_PMT_CTRL_STA
Address: Operational Base + offset (0x002c)
PMT Control and Status Register

Bit	Attr	Reset Value	Description
31	W1C	0x0	WFFRPR Wake-Up Frame Filter Register Pointer Reset When set, resets the Remote Wake-up Frame Filter register pointer to 3'b000. It is automatically cleared after 1 clock cycle.
30:10	RO	0x0	reserved
9	RW	0x0	GU Global Unicast When set, enables any unicast packet filtered by the GMAC (DAF) address recognition to be a wake-up frame.
8:7	RO	0x0	reserved
6	RC	0x0	WFR Wake-Up Frame Received When set, this bit indicates the power management event was generated due to reception of a wake-up frame. This bit is cleared by a read into this register.
5	RC	0x0	MPR Magic Packet Received When set, this bit indicates the power management event was generated by the reception of a Magic Packet. This bit is cleared by a read into this register.
4:3	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
2	RW	0x0	WFE Wake-Up Frame Enable When set, enables generation of a power management event due to wake-up frame reception.
1	RW	0x0	MPE Magic Packet Enable When set, enables generation of a power management event due to Magic Packet reception.
0	R/WSC	0x0	PD Power Down When set, all received frames will be dropped. This bit is cleared automatically when a magic packet or Wake-Up frame is received, and Power-Down mode is disabled. Frames received after this bit is cleared are forwarded to the application. This bit must only be set when either the Magic Packet Enable or Wake-Up Frame Enable bit is set high.

GMAC_INT_STATUS
Address: Operational Base + offset (0x0038)
Interrupt Status Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RO	0x0	MRCOIS MMC Receive Checksum Offload Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Receive Checksum Offload Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared.
6	RO	0x0	MTIS MMC Transmit Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Transmit Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is only valid when the optional MMC module is selected during configuration.
5	RO	0x0	MRIS MMC Receive Interrupt Status This bit is set high whenever an interrupt is generated in the MMC Receive Interrupt Register. This bit is cleared when all the bits in this interrupt register are cleared. This bit is only valid when the optional MMC module is selected during configuration.
4	RO	0x0	MIS MMC Interrupt Status This bit is set high whenever any of bits 7:5 is set high and cleared only when all of these bits are low. This bit is valid only when the optional MMC module is selected during configuration.

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Bit	Attr	Reset Value	Description
3	RO	0x0	PIS PMT Interrupt Status This bit is set whenever a Magic packet or Wake-on-LAN frame is received in Power-Down mode). This bit is cleared when both bits[6:5] are cleared due to a read operation to the register GMAC_PMT_CTRL_STA.
2:1	RO	0x0	reserved
0	RO	0x0	RIS RGMII Interrupt Status This bit is set due to any change in value of the Link Status of RGMII interface. This bit is cleared when the user makes a read operation the RGMII Status register.

GMAC_INT_MASK
Address: Operational Base + offset (0x003c)
Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	PIM PMT Interrupt Mask This bit when set, will disable the assertion of the interrupt signal due to the setting of PMT Interrupt Status bit in Register GMAC_INT_STATUS.
2:1	RO	0x0	reserved
0	RW	0x0	RIM RGMII Interrupt Mask This bit when set, will disable the assertion of the interrupt signal due to the setting of RGMII Interrupt Status bit in Register GMAC_INT_STATUS.

GMAC_MAC_ADDR0_HI
Address: Operational Base + offset (0x0040)
MAC Address0 High Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0xffff	A47_A32 MAC Address0 [47:32] This field contains the upper 16 bits (47:32) of the 6-byte first MAC address. This is used by the MAC for filtering for received frames and for inserting the MAC address in the Transmit Flow Control (PAUSE) Frames.

GMAC_MAC_ADDR0_LO
Address: Operational Base + offset (0x0044)
MAC Address0 Low Register

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Bit	Attr	Reset Value	Description
31:0	RW	0xffffffff	A31_A0 MAC Address0 [31:0] This field contains the lower 32 bits of the 6-byte first MAC address. This is used by the MAC for filtering for received frames and for inserting the MAC address in the Transmit Flow Control (PAUSE) Frames.

GMAC_AN_CTRL
Address: Operational Base + offset (0x00c0)
AN Control Register

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	ANE Auto-Negotiation Enable When set, will enable the GMAC to perform auto-negotiation with the link partner. Clearing this bit will disable auto-negotiation.
11:10	RO	0x0	reserved
9	R/W SC	0x0	RAN Restart Auto-Negotiation When set, will cause auto-negotiation to restart if the ANE is set. This bit is self-clearing after auto-negotiation starts. This bit should be cleared for normal operation.
8:0	RO	0x0	reserved

GMAC_AN_STATUS
Address: Operational Base + offset (0x00c4)
AN Status Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RO	0x0	ANC Auto-Negotiation Complete When set, this bit indicates that the auto-negotiation process is completed. This bit is cleared when auto-negotiation is reinitiated.
4	RO	0x0	reserved
3	RO	0x1	ANA Auto-Negotiation Ability This bit is always high, because the GMAC supports auto-negotiation.
2	R/W SC	0x0	LS Link Status When set, this bit indicates that the link is up. When cleared, this bit indicates that the link is down.
1:0	RO	0x0	reserved

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GMAC_AN_ADV
Address: Operational Base + offset (0x00c8)
Auto Negotiation Advertisement Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	NP Next Page Support This bit is tied to low, because the GMAC does not support the next page.
14	RO	0x0	reserved
13:12	RW	0x0	RFE Remote Fault Encoding These 2 bits provide a remote fault encoding, indicating to a link partner that a fault or error condition has occurred.
11:9	RO	0x0	reserved
8:7	RW	0x3	PSE Pause Encoding These 2 bits provide an encoding for the PAUSE bits, indicating that the GMAC is capable of configuring the PAUSE function as defined in IEEE 802.3x.
6	RW	0x1	HD Half-Duplex This bit, when set high, indicates that the GMAC supports Half-Duplex. This bit is tied to low (and RO) when the GMAC is configured for Full-Duplex-only operation.
5	RW	0x1	FD Full-Duplex This bit, when set high, indicates that the GMAC supports Full-Duplex.
4:0	RO	0x0	reserved

GMAC_AN_LINK_PART_AB
Address: Operational Base + offset (0x00cc)
Auto Negotiation Link Partner Ability Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	NP Next Page Support When set, this bit indicates that more next page information is available. When cleared, this bit indicates that next page exchange is not desired.

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Bit	Attr	Reset Value	Description
14	RO	0x0	ACK Acknowledge When set, this bit is used by the auto-negotiation function to indicate that the link partner has successfully received the GMAC's base page. When cleared, it indicates that a successful receipt of the base page has not been achieved.
13:12	RO	0x0	RFE Remote Fault Encoding These 2 bits provide a remote fault encoding, indicating a fault or error condition of the link partner.
11:9	RO	0x0	reserved
8:7	RO	0x0	PSE Pause Encoding These 2 bits provide an encoding for the PAUSE bits, indicating that the link partner's capability of configuring the PAUSE function as defined in IEEE 802.3x.
6	RO	0x0	HD Half-Duplex When set, this bit indicates that the link partner has the ability to operate in Half-Duplex mode. When cleared, the link partner does not have the ability to operate in Half-Duplex mode.
5	RO	0x0	FD Full-Duplex When set, this bit indicates that the link partner has the ability to operate in Full-Duplex mode. When cleared, the link partner does not have the ability to operate in Full-Duplex mode.
4:0	RO	0x0	reserved

GMAC_AN_EXP
Address: Operational Base + offset (0x00d0)
Auto Negotiation Expansion Register

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RO	0x0	NPA Next Page Ability This bit is tied to low, because the GMAC does not support next page function.
1	RO	0x0	NPR New Page Received When set, this bit indicates that a new page has been received by the GMAC. This bit will be cleared when read.
0	RO	0x0	reserved

GMAC_INTF_MODE_STA
Address: Operational Base + offset (0x00d8)

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RGMII Status Register

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RO	0x0	LST Link Status Indicates whether the link is up (1'b1) or down (1'b0)
2:1	RO	0x0	LSD Link Speed Indicates the current speed of the link: 2'b00: 2.5 MHz 2'b01: 25 MHz 2'b10: 125 MHz
0	RW	0x0	LM Link Mode Indicates the current mode of operation of the link: 1'b0: Half-Duplex mode 1'b1: Full-Duplex mode

GMAC_MMC_CTRL

Address: Operational Base + offset (0x0100)

MMC Control Register

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	FHP Full-Half preset When low and bit4 is set, all MMC counters get preset to almost-half value. All octet counters get preset to 0x7FFF_F800 (half - 2K Bytes) and all frame-counters gets preset to 0x7FFF_FFF0 (half - 16) When high and bit4 is set, all MMC counters get preset to almost-full value. All octet counters get preset to 0xFFFF_F800 (full - 2K Bytes) and all frame-counters gets preset to 0xFFFF_FFF0 (full - 16)
4	R/W SC	0x0	CP Counters Preset When set, all counters will be initialized or preset to almost full or almost half as per Bit5 above. This bit will be cleared automatically after 1 clock cycle. This bit along with bit5 is useful for debugging and testing the assertion of interrupts due to MMC counter becoming half-full or full.

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Bit	Attr	Reset Value	Description
3	RW	0x0	MCF MMC Counter Freeze When set, this bit freezes all the MMC counters to their current value. (None of the MMC counters are updated due to any transmitted or received frame until this bit is reset to 0. If any MMC counter is read with the Reset on Read bit set, then that counter is also cleared in this mode.)
2	RW	0x0	ROR Reset on Read When set, the MMC counters will be reset to zero after Read (self-clearing after reset). The counters are cleared when the least significant byte lane (bits[7:0]) is read.
1	RW	0x0	CSR Counter Stop Rollover When set, counter after reaching maximum value will not roll over to zero
0	R/WSC	0x0	CR Counters Reset When set, all counters will be reset. This bit will be cleared automatically after 1 clock cycle

GMAC_MMC_RX_INTR
Address: Operational Base + offset (0x0104)
MMC Receive Interrupt Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	INT21 The bit is set when the rxfifooverflow counter reaches half the maximum value, and also when it reaches the maximum value.
20:19	RO	0x0	reserved
18	RC	0x0	INT18 The bit is set when the rxlengtherror counter reaches half the maximum value, and also when it reaches the maximum value.
17:6	RO	0x0	reserved
5	RW	0x0	INT5 The bit is set when the rxcrcerror counter reaches half the maximum value, and also when it reaches the maximum value.
4	RC	0x0	INT4 The bit is set when the rxmulticastframes_g counter reaches half the maximum value, and also when it reaches the maximum value.
3	RO	0x0	reserved
2	RC	0x0	INT2 The bit is set when the rxoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.

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Bit	Attr	Reset Value	Description
1	RC	0x0	INT1 The bit is set when the rxoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.
0	RC	0x0	INT0 The bit is set when the rxframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_TX_INTR
Address: Operational Base + offset (0x0108)
MMC Transmit Interrupt Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RC	0x0	INT21 The bit is set when the txframecount_g counter reaches half the maximum value, and also when it reaches the maximum value.
20	RC	0x0	INT20 The bit is set when the txoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
19	RC	0x0	INT19 The bit is set when the txcarriererror counter reaches half the maximum value, and also when it reaches the maximum value.
18:14	RO	0x0	reserved
13	RC	0x0	INT13 The bit is set when the txunderflowerror counter reaches half the maximum value, and also when it reaches the maximum value.
12:2	RO	0x0	reserved
1	RC	0x0	INT1 The bit is set when the txframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.
0	RC	0x0	INT0 The bit is set when the txoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_RX_INT_MSK
Address: Operational Base + offset (0x010c)
MMC Receive Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	INT21 Setting this bit masks the interrupt when the rxfifooverflow counter reaches half the maximum value, and also when it reaches the maximum value.
20:19	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
18	RW	0x0	INT18 Setting this bit masks the interrupt when the rxlengtherror counter reaches half the maximum value, and also when it reaches the maximum value.
17:6	RO	0x0	reserved
5	RW	0x0	INT5 Setting this bit masks the interrupt when the rxcrcerror counter reaches half the maximum value, and also when it reaches the maximum value.
4	RW	0x0	INT4 Setting this bit masks the interrupt when the rxmulticastframes_g counter reaches half the maximum value, and also when it reaches the maximum value.
3	RO	0x0	reserved
2	RW	0x0	INT2 Setting this bit masks the interrupt when the rxoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
1	RW	0x0	INT1 Setting this bit masks the interrupt when the rxoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.
0	RW	0x0	INT0 Setting this bit masks the interrupt when the rxframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_TX_INT_MSK
Address: Operational Base + offset (0x0110)
MMC Transmit Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:22	RO	0x0	reserved
21	RW	0x0	INT21 Setting this bit masks the interrupt when the txframecount_g counter reaches half the maximum value, and also when it reaches the maximum value.
20	RW	0x0	INT20 Setting this bit masks the interrupt when the txoctetcount_g counter reaches half the maximum value, and also when it reaches the maximum value.
19	RW	0x0	INT19 Setting this bit masks the interrupt when the txcarriererror counter reaches half the maximum value, and also when it reaches the maximum value.
18:14	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
13	RW	0x0	INT13 Setting this bit masks the interrupt when the txunderflowerror counter reaches half the maximum value, and also when it reaches the maximum value.
12:2	RO	0x0	reserved
1	RW	0x0	INT1 Setting this bit masks the interrupt when the txframecount_gb counter reaches half the maximum value, and also when it reaches the maximum value.
0	RW	0x0	INT0 Setting this bit masks the interrupt when the txoctetcount_gb counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_TXOCTETCNT_GB
Address: Operational Base + offset (0x0114)
MMC TX OCTET Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txoctetcount_gb Number of bytes transmitted, exclusive of preamble and retried bytes, in good and bad frames.

GMAC_MMC_TXFRMCNT_GB
Address: Operational Base + offset (0x0118)
MMC TX Frame Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txframecount_gb Number of good and bad frames transmitted, exclusive of retried frames.

GMAC_MMC_TXUNDFLWERR
Address: Operational Base + offset (0x0148)
MMC TX Underflow Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txunderflowerror Number of frames aborted due to frame underflow error.

GMAC_MMC_TXCARERR
Address: Operational Base + offset (0x0160)
MMC TX Carrier Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txcarriererror Number of frames aborted due to carrier sense error (no carrier or loss of carrier).

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GMAC_MMC_TXOCTETCNT_G
Address: Operational Base + offset (0x0164)
MMC TX OCTET Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txoctetcount_g Number of bytes transmitted, exclusive of preamble, in good frames only.

GMAC_MMC_TXFRMCNT_G
Address: Operational Base + offset (0x0168)
MMC TX Frame Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	txframecount_g Number of good frames transmitted.

GMAC_MMC_RXFRMCNT_GB
Address: Operational Base + offset (0x0180)
MMC RX Frame Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxframecount_gb Number of good and bad frames received.

GMAC_MMC_RXOCTETCNT_GB
Address: Operational Base + offset (0x0184)
MMC RX OCTET Good and Bad Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxoctetcount_gb Number of bytes received, exclusive of preamble, in good and bad frames.

GMAC_MMC_RXOCTETCNT_G
Address: Operational Base + offset (0x0188)
MMC RX OCTET Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxoctetcount_g Number of bytes received, exclusive of preamble, only in good frames.

GMAC_MMC_RXMCFRMCNT_G
Address: Operational Base + offset (0x0190)
MMC RX Mulitcast Frame Good Counter

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxmulticastframes_g Number of good multicast frames received.

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GMAC_MMC_RXCRCERR
Address: Operational Base + offset (0x0194)
MMC RX Carrier

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxcrcerror Number of frames received with CRC error.

GMAC_MMC_RXLENERR
Address: Operational Base + offset (0x01c8)
MMC RX Length Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxlengtherror Number of frames received with length error (Length type field ≠frame size), for all frames with valid length field.

GMAC_MMC_RXFIFOOVRFLW
Address: Operational Base + offset (0x01d4)
MMC RX FIFO Overflow

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxfifooverflow Number of missed received frames due to FIFO overflow.

GMAC_MMC_IPC_INT_MSK
Address: Operational Base + offset (0x0200)
MMC Receive Checksum Offload Interrupt Mask Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29	RW	0x0	INT29 Setting this bit masks the interrupt when the rxicmp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
28	RO	0x0	reserved
27	RW	0x0	INT27 Setting this bit masks the interrupt when the rxtcp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
26	RO	0x0	reserved
25	RW	0x0	INT25 Setting this bit masks the interrupt when the rxudp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
24:23	RO	0x0	reserved
22	RW	0x0	INT22 Setting this bit masks the interrupt when the rxipv6_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.

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Bit	Attr	Reset Value	Description
21:18	RO	0x0	reserved
17	RW	0x0	INT17 Setting this bit masks the interrupt when the rxipv4_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
16:14	RO	0x0	reserved
13	RW	0x0	INT13 Setting this bit masks the interrupt when the rxicmp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
12	RO	0x0	reserved
11	RW	0x0	INT11 Setting this bit masks the interrupt when the rxtcp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
10	RO	0x0	reserved
9	RW	0x0	INT9 Setting this bit masks the interrupt when the rxudp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
8:7	RO	0x0	reserved
6	RW	0x0	INT6 Setting this bit masks the interrupt when the rxipv6_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
5	RW	0x0	INT5 Setting this bit masks the interrupt when the rxipv6_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.
4:2	RO	0x0	reserved
1	RW	0x0	INT1 Setting this bit masks the interrupt when the rxipv4_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
0	RW	0x0	INT0 Setting this bit masks the interrupt when the rxipv4_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.

GMAC_MMC_IPC_INTR
Address: Operational Base + offset (0x0208)
MMC Receive Checksum Offload Interrupt Register

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
29	RC	0x0	INT29 The bit is set when the rxicmp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
28	RO	0x0	reserved
27	RC	0x0	INT27 The bit is set when the rxtcp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
26	RO	0x0	reserved
25	RC	0x0	INT25 The bit is set when the rxudp_err_octets counter reaches half the maximum value, and also when it reaches the maximum value.
24:23	RO	0x0	reserved
22	RC	0x0	INT22 The bit is set when the rxipv6_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
21:18	RO	0x0	reserved
17	RC	0x0	INT17 The bit is set when the rxipv4_hdrerr_octets counter reaches half the maximum value, and also when it reaches the maximum value.
16:14	RO	0x0	reserved
13	RC	0x0	INT13 The bit is set when the rxicmp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
12	RO	0x0	reserved
11	RC	0x0	INT11 The bit is set when the rxtcp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
10	RO	0x0	reserved
9	RC	0x0	INT9 The bit is set when the rxudp_err_frms counter reaches half the maximum value, and also when it reaches the maximum value.
8:7	RO	0x0	reserved
6	RC	0x0	INT6 The bit is set when the rxipv6_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
5	RC	0x0	INT5 The bit is set when the rxipv6_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.
4:2	RO	0x0	reserved
1	RC	0x0	INT1 The bit is set when the rxipv4_hdrerr_frms counter reaches half the maximum value, and also when it reaches the maximum value.
0	RC	0x0	INT0 The bit is set when the rxipv4_gd_frms counter reaches half the maximum value, and also when it reaches the maximum value.

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GMAC_MMC_RXIPV4GFRM
Address: Operational Base + offset (0x0210)
MMC RX IPV4 Good Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv4_gd_frms Number of good IPv4 datagrams received with the TCP, UDP, or ICMP payload

GMAC_MMC_RXIPV4HDERRFRM
Address: Operational Base + offset (0x0214)
MMC RX IPV4 Head Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv4_hdrerr_frms Number of IPv4 datagrams received with header (checksum, length, or version mismatch) errors

GMAC_MMC_RXIPV6GFRM
Address: Operational Base + offset (0x0224)
MMC RX IPV6 Good Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv6_gd_frms Number of good IPv6 datagrams received with TCP, UDP, or ICMP payloads.

GMAC_MMC_RXIPV6HDERRFRM
Address: Operational Base + offset (0x0228)
MMC RX IPV6 Head Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv6_hdrerr_frms Number of IPv6 datagrams received with header errors (length or version mismatch).

GMAC_MMC_RXUDPERRFRM
Address: Operational Base + offset (0x0234)
MMC RX UDP Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxudp_err_frms Number of good IP datagrams whose UDP payload has a checksum error.

GMAC_MMC_RXTCPERRFRM
Address: Operational Base + offset (0x023c)
MMC RX TCP Error Frame

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxtcp_err_frms Number of good IP datagrams whose TCP payload has a checksum error.

GMAC_MMC_RXICMPERRFRM
Address: Operational Base + offset (0x0244)
MMC RX ICMP Error Frame

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxicmp_err_frms Number of good IP datagrams whose ICMP payload has a checksum error.

GMAC_MMC_RXIPV4HDERROCT
Address: Operational Base + offset (0x0254)
MMC RX OCTET IPV4 Head Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv4_hdrerr_octets Number of bytes received in IPv4 datagrams with header errors (checksum, length, version mismatch). The value in the Length field of IPv4 header is used to update this counter.

GMAC_MMC_RXIPV6HDERROCT
Address: Operational Base + offset (0x0268)
MMC RX OCTET IPV6 Head Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxipv6_hdrerr_octets Number of bytes received in IPv6 datagrams with header errors (length, version mismatch). The value in the IPv6 header's Length field is used to update this counter.

GMAC_MMC_RXUDPERROCT
Address: Operational Base + offset (0x0274)
MMC RX OCTET UDP Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxudp_err_octets Number of bytes received in a UDP segment that had checksum errors.

GMAC_MMC_RXTCPERROCT
Address: Operational Base + offset (0x027c)
MMC RX OCTET TCP Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxtcp_err_octets Number of bytes received in a TCP segment with checksum errors.

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GMAC_MMC_RXICMPERROCT
Address: Operational Base + offset (0x0284)
MMC RX OCTET ICMP Error

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	rxicmp_err_octets Number of bytes received in an ICMP segment with checksum errors.

GMAC_BUS_MODE
Address: Operational Base + offset (0x1000)
Bus Mode Register

Bit	Attr	Reset Value	Description
31:26	RO	0x0	reserved
25	RW	0x0	AAL Address-Aligned Beats When this bit is set high and the FB bit equals 1, the AXI interface generates all bursts aligned to the start address LS bits. If the FB bit equals 0, the first burst (accessing the data buffer's start address) is not aligned, but subsequent bursts are aligned to the address.
24	RW	0x0	PBL_Mode 8xPBL Mode When set high, this bit multiplies the PBL value programmed (bits [22:17] and bits [13:8]) eight times. Thus the DMA will transfer data in to a maximum of 8, 16, 32, 64, 128, and 256 beats depending on the PBL value.
23	RW	0x0	USP Use Separate PBL When set high, it configures the RxDMA to use the value configured in bits [22:17] as PBL while the PBL value in bits [13:8] is applicable to TxDMA operations only. When reset to low, the PBL value in bits [13:8] is applicable for both DMA engines.
22:17	RW	0x01	RPBL RxDMA PBL These bits indicate the maximum number of beats to be transferred in one RxDMA transaction. This will be the maximum value that is used in a single block Read/Write. The RxDMA will always attempt to burst as specified in RPBL each time it starts a Burst transfer on the host bus. RPBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value will result in undefined behavior. These bits are valid and applicable only when USP is set high.

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Bit	Attr	Reset Value	Description
16	RW	0x0	FB Fixed Burst This bit controls whether the AXI Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 or INCR16 during start of normal burst transfers. When reset, the AXI will use SINGLE and INCR burst transfer operations.
15:14	RO	0x0	reserved
13:8	RW	0x01	PBL Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA transaction. This will be the maximum value that is used in a single block Read/Write. The DMA will always attempt to burst as specified in PBL each time it starts a Burst transfer on the host bus. PBL can be programmed with permissible values of 1, 2, 4, 8, 16, and 32. Any other value will result in undefined behavior. When USP is set high, this PBL value is applicable for TxDMA transactions only. The PBL values have the following limitations. The maximum number of beats (PBL) possible is limited by the size of the Tx FIFO and Rx FIFO in the MTL layer and the data bus width on the DMA. The FIFO has a constraint that the maximum beat supported is half the depth of the FIFO, except when specified (as given below). For different data bus widths and FIFO sizes, the valid PBL range (including x8 mode) is provided in the following table. If the PBL is common for both transmit and receive DMA, the minimum Rx FIFO and Tx FIFO depths must be considered. Do not program out-of-range PBL values, because the system may not behave properly. For TxFIFO, valid PBL range in full duplex mode and duplex mode is 128 or less. For RxFIFO, valid PBL range in full duplex mode is all.
7	RO	0x0	reserved
6:2	RW	0x00	DSL Descriptor Skip Length This bit specifies the number of dword to skip between two unchained descriptors. The address skipping starts from the end of current descriptor to the start of next descriptor. When DSL value equals zero, then the descriptor table is taken as contiguous by the DMA, in Ring mode.
1	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
0	R/W SC	0x1	SWR Software Reset When this bit is set, the MAC DMA Controller resets all GMAC Subsystem internal registers and logic. It is cleared automatically after the reset operation has completed in all of the core clock domains. Read a 0 value in this bit before re-programming any register of the core. Note: The reset operation is completed only when all the resets in all the active clock domains are de-asserted. Hence it is essential that all the PHY inputs clocks (applicable for the selected PHY interface) are present for software reset completion.

GMAC_TX_POLL_DEMAND
Address: Operational Base + offset (0x1004)
Transmit Poll Demand Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TPD Transmit Poll Demand When these bits are written with any value, the DMA reads the current descriptor pointed to by Register GMAC_CUR_HOST_TX_DESC. If that descriptor is not available (owned by Host), transmission returns to the Suspend state and DMA Register GMAC_STATUS[2] is asserted. If the descriptor is available, transmission resumes.

GMAC_RX_POLL_DEMAND
Address: Operational Base + offset (0x1008)
Receive Poll Demand Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	RPD Receive Poll Demand When these bits are written with any value, the DMA reads the current descriptor pointed to by Register GMAC_CUR_HOST_RX_DESC. If that descriptor is not available (owned by Host), reception returns to the Suspended state and Register GMAC_STATUS[7] is not asserted. If the descriptor is available, the Receive DMA returns to active state.

GMAC_RX_DESC_LIST_ADDR
Address: Operational Base + offset (0x100c)
Receive Descriptor List Address Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	SRL Start of Receive List This field contains the base address of the First Descriptor in the Receive Descriptor list. The LSB bits [1/2/3:0] for 32/64/128-bit bus width) will be ignored and taken as all-zero by the DMA internally. Hence these LSB bits are Read Only.

GMAC_TX_DESC_LIST_ADDR
Address: Operational Base + offset (0x1010)
Transmit Descriptor List Address Register

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	STL Start of Transmit List This field contains the base address of the First Descriptor in the Transmit Descriptor list. The LSB bits [1/2/3:0] for 32/64/128-bit bus width) will be ignored and taken as all-zero by the DMA internally. Hence these LSB bits are Read Only.

GMAC_STATUS
Address: Operational Base + offset (0x1014)
Status Register

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RO	0x0	GPI GMAC PMT Interrupt This bit indicates an interrupt event in the GMAC core's PMT module. The software must read the corresponding registers in the GMAC core to get the exact cause of interrupt and clear its source to reset this bit to 1'b0. The interrupt signal from the GMAC subsystem (sbd_intr_o) is high when this bit is high.
27	RO	0x0	GMI GMAC MMC Interrupt This bit reflects an interrupt event in the MMC module of the GMAC core. The software must read the corresponding registers in the GMAC core to get the exact cause of interrupt and clear the source of interrupt to make this bit as 1'b0. The interrupt signal from the GMAC subsystem (sbd_intr_o) is high when this bit is high.
26	RO	0x0	GLI GMAC Line interface Interrupt This bit reflects an interrupt event in the GMAC Core's PCS or RGMII interface block. The software must read the corresponding registers in the GMAC core to get the exact cause of interrupt and clear the source of interrupt to make this bit as 1'b0. The interrupt signal from the GMAC subsystem (sbd_intr_o) is high when this bit is high.

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Bit	Attr	Reset Value	Description
25:23	RO	0x0	EB Error Bits These bits indicate the type of error that caused a Bus Error (e.g., error response on the AXI interface). Valid only with Fatal Bus Error bit (Register GMAC_STATUS[13]) set. This field does not generate an interrupt. Bit 23: 1'b1 Error during data transfer by TxDMA 1'b0 Error during data transfer by RxDMA Bit 24: 1'b1 Error during read transfer 1'b0 Error during write transfer Bit 25: 1'b1 Error during descriptor access 1'b0 Error during data buffer access
22:20	RO	0x0	TS Transmit Process State These bits indicate the Transmit DMA FSM state. This field does not generate an interrupt. 3'b000: Stopped; Reset or Stop Transmit Command issued. 3'b001: Running; Fetching Transmit Transfer Descriptor. 3'b010: Running; Waiting for status. 3'b011: Running; Reading Data from host memory buffer and queuing it to transmit buffer (Tx FIFO). 3'b100: TIME_STAMP write state. 3'b101: Reserved for future use. 3'b110: Suspended; Transmit Descriptor Unavailable or Transmit Buffer Underflow. 3'b111: Running; Closing Transmit Descriptor.
19:17	RO	0x0	RS Receive Process State These bits indicate the Receive DMA FSM state. This field does not generate an interrupt. 3'b000: Stopped: Reset or Stop Receive Command issued. 3'b001: Running: Fetching Receive Transfer Descriptor. 3'b010: Reserved for future use. 3'b011: Running: Waiting for receive packet. 3'b100: Suspended: Receive Descriptor Unavailable. 3'b101: Running: Closing Receive Descriptor. 3'b110: TIME_STAMP write state. 3'b111: Running: Transferring the receive packet data from receive buffer to host memory.

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Bit	Attr	Reset Value	Description
16	W1C	0x0	NIS Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in Register OP_MODE: Register GMAC_STATUS[0]: Transmit Interrupt Register GMAC_STATUS[2]: Transmit Buffer Unavailable Register GMAC_STATUS[6]: Receive Interrupt Register GMAC_STATUS[14]: Early Receive Interrupt Only unmasked bits affect the Normal Interrupt Summary bit. This is a sticky bit and must be cleared (by writing a 1 to this bit) each time a corresponding bit that causes NIS to be set is cleared.
15	W1C	0x0	AIS Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in Register OP_MODE: Register GMAC_STATUS[1]: Transmit Process Stopped Register GMAC_STATUS[3]: Transmit Jabber Timeout Register GMAC_STATUS[4]: Receive FIFO Overflow Register GMAC_STATUS[5]: Transmit Underflow Register GMAC_STATUS[7]: Receive Buffer Unavailable Register GMAC_STATUS[8]: Receive Process Stopped Register GMAC_STATUS[9]: Receive Watchdog Timeout Register GMAC_STATUS[10]: Early Transmit Interrupt Register GMAC_STATUS[13]: Fatal Bus Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared.
14	W1C	0x0	ERI Early Receive Interrupt This bit indicates that the DMA had filled the first data buffer of the packet. Receive Interrupt Register GMAC_STATUS[6] automatically clears this bit.
13	W1C	0x0	FBI Fatal Bus Error Interrupt This bit indicates that a bus error occurred, as detailed in [25:23]. When this bit is set, the corresponding DMA engine disables all its bus accesses.
12:11	RO	0x0	reserved
10	W1C	0x0	ETI Early Transmit Interrupt This bit indicates that the frame to be transmitted was fully transferred to the MTL Transmit FIFO.

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Bit	Attr	Reset Value	Description
9	W1C	0x0	RWT Receive Watchdog Timeout This bit is asserted when a frame with a length greater than 2,048 bytes is received.
8	W1C	0x0	RPS Receive Process Stopped This bit is asserted when the Receive Process enters the Stopped state.
7	W1C	0x0	RU Receive Buffer Unavailable This bit indicates that the Next Descriptor in the Receive List is owned by the host and cannot be acquired by the DMA. Receive Process is suspended. To resume processing Receive descriptors, the host should change the ownership of the descriptor and issue a Receive Poll Demand command. If no Receive Poll Demand is issued, Receive Process resumes when the next recognized incoming frame is received. Register GMAC_STATUS[7] is set only when the previous Receive Descriptor was owned by the DMA.
6	W1C	0x0	RI Receive Interrupt This bit indicates the completion of frame reception. Specific frame status information has been posted in the descriptor. Reception remains in the Running state.
5	W1C	0x0	UNF Transmit Underflow This bit indicates that the Transmit Buffer had an Underflow during frame transmission. Transmission is suspended and an Underflow Error TDES0[1] is set.
4	W1C	0x0	OVF Receive Overflow This bit indicates that the Receive Buffer had an Overflow during frame reception. If the partial frame is transferred to application, the overflow status is set in RDES0[11].
3	W1C	0x0	TJT Transmit Jabber Timeout This bit indicates that the Transmit Jabber Timer expired, meaning that the transmitter had been excessively active. The transmission process is aborted and placed in the Stopped state. This causes the Transmit Jabber Timeout TDES0[14] flag to assert.

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Bit	Attr	Reset Value	Description
2	W1C	0x0	TU Transmit Buffer Unavailable This bit indicates that the Next Descriptor in the Transmit List is owned by the host and cannot be acquired by the DMA. Transmission is suspended. Bits[22:20] explain the Transmit Process state transitions. To resume processing transmit descriptors, the host should change the ownership of the bit of the descriptor and then issue a Transmit Poll Demand command.
1	W1C	0x0	TPS Transmit Process Stopped This bit is set when the transmission is stopped.
0	W1C	0x0	TI Transmit Interrupt This bit indicates that frame transmission is finished and TDES1[31] is set in the First Descriptor.

GMAC_OP_MODE
Address: Operational Base + offset (0x1018)
Operation Mode Register

Bit	Attr	Reset Value	Description
31:27	RO	0x0	reserved
26	RW	0x0	DT Disable Dropping of TCP/IP Checksum Error Frames When this bit is set, the core does not drop frames that only have errors detected by the Receive Checksum Offload engine. Such frames do not have any errors (including FCS error) in the Ethernet frame received by the MAC but have errors in the encapsulated payload only. When this bit is reset, all error frames are dropped if the FEF bit is reset.
25	RW	0x0	RSF Receive Store and Forward When this bit is set, the MTL only reads a frame from the Rx FIFO after the complete frame has been written to it, ignoring RTC bits. When this bit is reset, the Rx FIFO operates in Cut-Through mode, subject to the threshold specified by the RTC bits.
24	RW	0x0	DFF Disable Flushing of Received Frames When this bit is set, the RxDMA does not flush any frames due to the unavailability of receive descriptors/buffers as it does normally when this bit is reset.
23:22	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
21	RW	0x0	TSF Transmit Store and Forward When this bit is set, transmission starts when a full frame resides in the MTL Transmit FIFO. When this bit is set, the TTC values specified in Register GMAC_OP_MODE[16:14] are ignored. This bit should be changed only when transmission is stopped.
20	W1C	0x0	FTF Flush Transmit FIFO When this bit is set, the transmit FIFO controller logic is reset to its default values and thus all data in the Tx FIFO is lost/flushed. This bit is cleared internally when the flushing operation is completed fully. The Operation Mode register should not be written to until this bit is cleared. The data which is already accepted by the MAC transmitter will not be flushed. It will be scheduled for transmission and will result in underflow and runt frame transmission. Note: The flush operation completes only after emptying the TxFIFO of its contents and all the pending Transmit Status of the transmitted frames are accepted by the host. In order to complete this flush operation, the PHY transmit clock (clk_tx_i) is required to be active.
19:17	RO	0x0	reserved
16:14	RW	0x0	TTC Transmit Threshold Control These three bits control the threshold level of the MTL Transmit FIFO. Transmission starts when the frame size within the MTL Transmit FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are also transmitted. These bits are used only when the TSF bit (Bit 21) is reset. 3'b000: 64 3'b001: 128 3'b010: 192 3'b011: 256 3'b100: 40 3'b101: 32 3'b110: 24 3'b111: 16

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Bit	Attr	Reset Value	Description
13	RW	0x0	ST Start/Stop Transmission Command When this bit is set, transmission is placed in the Running state, and the DMA checks the Transmit List at the current position for a frame to be transmitted. Descriptor acquisition is attempted either from the current position in the list, which is the Transmit List Base Address set by Register GMAC_TX_DESC_LIST_ADDR, or from the position retained when transmission was stopped previously. If the current descriptor is not owned by the DMA, transmission enters the Suspended state and Transmit Buffer Unavailable (Register GMAC_STATUS[2]) is set. The Start Transmission command is effective only when transmission is stopped. If the command is issued before setting DMA Register TX_DESC_LIST_ADDR, then the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current frame. The Next Descriptor position in the Transmit List is saved, and becomes the current position when transmission is restarted. The stop transmission command is effective only the transmission of the current frame is complete or when the transmission is in the Suspended state.
12:11	RW	0x0	RFD Threshold for deactivating flow control (in both HD and FD) These bits control the threshold (Fill-level of Rx FIFO) at which the flow-control is de-asserted after activation. 2'b00: Full minus 1 KB 2'b01: Full minus 2 KB 2'b10: Full minus 3 KB 2'b11: Full minus 4 KB Note that the de-assertion is effective only after flow control is asserted.
10:9	RW	0x0	RFA Threshold for activating flow control (in both HD and FD) These bits control the threshold (Fill level of Rx FIFO) at which flow control is activated. 2'b00: Full minus 1 KB 2'b01: Full minus 2 KB 2'b10: Full minus 3 KB 2'b11: Full minus 4 KB Note that the above only applies to Rx FIFOs of 4 KB or more when the EFC bit is set high.
8	RW	0x0	EFC Enable HW flow control When this bit is set, the flow control signal operation based on fill-level of Rx FIFO is enabled. When reset, the flow control operation is disabled.

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Bit	Attr	Reset Value	Description
7	RW	0x0	FEF Forward Error Frames When this bit is reset, the Rx FIFO drops frames with error status (CRC error, collision error, GMII_ER, giant frame, watchdog timeout, overflow). However, if the frame's start byte (write) pointer is already transferred to the read controller side (in Threshold mode), then the frames are not dropped. When FEF is set, all frames except runt error frames are forwarded to the DMA. But when RxFIFO overflows when a partial frame is written, then such frames are dropped even when FEF is set.
6	RW	0x0	FUF Forward Undersized Good Frames When set, the Rx FIFO will forward Undersized frames (frames with no Error and length less than 64 bytes) including pad-bytes and CRC). When reset, the Rx FIFO will drop all frames of less than 64 bytes, unless it is already transferred due to lower value of Receive Threshold (e.g., RTC = 01).
5	RO	0x0	reserved
4:3	RW	0x0	RTC Receive Threshold Control These two bits control the threshold level of the MTL Receive FIFO. Transfer (request) to DMA starts when the frame size within the MTL Receive FIFO is larger than the threshold. In addition, full frames with a length less than the threshold are transferred automatically. Note that value of 11 is not applicable if the configured Receive FIFO size is 128 bytes. These bits are valid only when the RSF bit is zero, and are ignored when the RSF bit is set to 1. 2'b00: 64 2'b01: 32 2'b10: 96 2'b11: 128
2	RW	0x0	OSF Operate on Second Frame When this bit is set, this bit instructs the DMA to process a second frame of Transmit data even before status for first frame is obtained.

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Bit	Attr	Reset Value	Description
1	RW	0x0	SR Start/Stop Receive When this bit is set, the Receive process is placed in the Running state. The DMA attempts to acquire the descriptor from the Receive list and processes incoming frames. Descriptor acquisition is attempted from the current position in the list, which is the address set by register GMAC_RX_DESC_LIST_ADDR or the position retained when the Receive process was previously stopped. If no descriptor is owned by the DMA, reception is suspended and Receive Buffer Unavailable (Register GMAC_STATUS[7]) is set. The Start Receive command is effective only when reception has stopped. If the command was issued before setting register GMAC_RX_DESC_LIST_ADDR, DMA behavior is unpredictable. When this bit is cleared, RxDMA operation is stopped after the transfer of the current frame. The next descriptor position in the Receive list is saved and becomes the current position after the Receive process is restarted. The Stop Receive command is effective only when the Receive process is in either the Running (waiting for receive packet) or in the Suspended state.
0	RO	0x0	reserved

GMAC_INT_ENA
Address: Operational Base + offset (0x101c)
Interrupt Enable Register

Bit	Attr	Reset Value	Description
31:17	RO	0x0	reserved
16	RW	0x0	NIE Normal Interrupt Summary Enable When this bit is set, a normal interrupt is enabled. When this bit is reset, a normal interrupt is disabled. This bit enables the following bits: Register GMAC_STATUS[0]: Transmit Interrupt Register GMAC_STATUS[2]: Transmit Buffer Unavailable Register GMAC_STATUS[6]: Receive Interrupt Register GMAC_STATUS[14]: Early Receive Interrupt

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Bit	Attr	Reset Value	Description
15	RW	0x0	AIE Abnormal Interrupt Summary Enable When this bit is set, an Abnormal Interrupt is enabled. When this bit is reset, an Abnormal Interrupt is disabled. This bit enables the following bits Register GMAC_STATUS[1]: Transmit Process Stopped Register GMAC_STATUS[3]: Transmit Jabber Timeout Register GMAC_STATUS[4]: Receive Overflow Register GMAC_STATUS[5]: Transmit Underflow Register GMAC_STATUS[7]: Receive Buffer Unavailable Register GMAC_STATUS[8]: Receive Process Stopped Register GMAC_STATUS[9]: Receive Watchdog Timeout Register GMAC_STATUS[10]: Early Transmit Interrupt Register GMAC_STATUS[13]: Fatal Bus Error
14	RW	0x0	ERE Early Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Early Receive Interrupt is enabled. When this bit is reset, Early Receive Interrupt is disabled.
13	RW	0x0	FBE Fatal Bus Error Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), the Fatal Bus Error Interrupt is enabled. When this bit is reset, Fatal Bus Error Enable Interrupt is disabled.
12:11	RO	0x0	reserved
10	RW	0x0	ETE Early Transmit Interrupt Enable When this bit is set with an Abnormal Interrupt Summary Enable (BIT 15), Early Transmit Interrupt is enabled. When this bit is reset, Early Transmit Interrupt is disabled.
9	RW	0x0	RWE Receive Watchdog Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), the Receive Watchdog Timeout Interrupt is enabled. When this bit is reset, Receive Watchdog Timeout Interrupt is disabled.
8	RW	0x0	RSE Receive Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Receive Stopped Interrupt is enabled. When this bit is reset, Receive Stopped Interrupt is disabled.

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Bit	Attr	Reset Value	Description
7	RW	0x0	RUE Receive Buffer Unavailable Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Receive Buffer Unavailable Interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable Interrupt is disabled
6	RW	0x0	RIE Receive Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Receive Interrupt is enabled. When this bit is reset, Receive Interrupt is disabled.
5	RW	0x0	UNE Underflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Transmit Underflow Interrupt is enabled. When this bit is reset, Underflow Interrupt is disabled.
4	RW	0x0	OVE Overflow Interrupt Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Receive Overflow Interrupt is enabled. When this bit is reset, Overflow Interrupt is disabled
3	RW	0x0	TJE Transmit Jabber Timeout Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Transmit Jabber Timeout Interrupt is enabled. When this bit is reset, Transmit Jabber Timeout Interrupt is disabled.
2	RW	0x0	TUE Transmit Buffer Unavailable Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Transmit Buffer Unavailable Interrupt is enabled. When this bit is reset, Transmit Buffer Unavailable Interrupt is disabled.
1	RW	0x0	TSE Transmit Stopped Enable When this bit is set with Abnormal Interrupt Summary Enable (BIT 15), Transmission Stopped Interrupt is enabled. When this bit is reset, Transmission Stopped Interrupt is disabled.
0	RW	0x0	TIE Transmit Interrupt Enable When this bit is set with Normal Interrupt Summary Enable (BIT 16), Transmit Interrupt is enabled. When this bit is reset, Transmit Interrupt is disabled.

GMAC_OVERFLOW_CNT

Address: Operational Base + offset (0x1020)

Missed Frame and Buffer Overflow Counter Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28	RC	0x0	FIFO_overflow_bit Overflow bit for FIFO Overflow Counter
27:17	RC	0x000	Frame_miss_number Indicates the number of frames missed by the application This counter is incremented each time the MTL asserts the sideband signal mtl_rxoverflow_o. The counter is cleared when this register is read with mci_be_i[2] at 1'b1.
16	RC	0x0	Miss_frame_overflow_bit Overflow bit for Missed Frame Counter
15:0	RC	0x0000	Frame_miss_number_2 Indicates the number of frames missed by the controller due to the Host Receive Buffer being unavailable. This counter is incremented each time the DMA discards an incoming frame. The counter is cleared when this register is read with mci_be_i[0] at 1'b1.

GMAC_REC_INT_WDT_TIMER
Address: Operational Base + offset (0x1024)
Receive Interrupt Watchdog Timer Register

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x00	RIWT RI Watchdog Timer count Indicates the number of system clock cycles multiplied by 256 for which the watchdog timer is set. The watchdog timer gets triggered with the programmed value after the RxDMA completes the transfer of a frame for which the RI status bit is not set due to the setting in the corresponding descriptor RDES1[31]. When the watch-dog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when RI bit is set high due to automatic setting of RI as per RDES1[31] of any received frame.

GMAC_AXI_BUS_MODE
Address: Operational Base + offset (0x1028)
AXI Bus Mode Register

Bit	Attr	Reset Value	Description
31	RW	0x0	EN_LPI Enable LPI (Low Power Interface) When set to 1, enable the LPI (Low Power Interface) supported by the GMAC and accepts the LPI request from the AXI System Clock controller. When set to 0, disables the Low Power Mode and always denies the LPI request from the AXI System Clock controller.

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Bit	Attr	Reset Value	Description
30	RW	0x0	UNLCK_ON_MGK_RWK Unlock on Magic Packet or Remote Wake Up When set to 1, enables it to request coming out of Low Power mode only when Magic Packet or Remote Wake Up Packet is received. When set to 0, enables it requests to come out of Low Power mode when any frame is received.
29:22	RO	0x0	reserved
21:20	RW	0x1	WR_OSR_LMT AXI Maximum Write Out Standing Request Limit This value limits the maximum outstanding request on the AXI write interface. Maximum outstanding requests = WR_OSR_LMT+1
19:18	RO	0x0	reserved
17:16	RW	0x1	RD_OSR_LMT AXI Maximum Read Out Standing Request Limit This value limits the maximum outstanding request on the AXI read interface. Maximum outstanding requests = RD_OSR_LMT+1
15:13	RO	0x0	reserved
12	RO	0x0	AXI_AAL Address-Aligned Beats This bit is read-only bit and reflects the AAL bit (register GMAC_BUS_MODE[25]). When this bit set to 1, it performs address-aligned burst transfers on both read and write channels.
11:4	RO	0x0	reserved
3	RW	0x0	BLEN16 AXI Burst Length 16 When this bit is set to 1, or when UNDEF is set to 1, it is allowed to select a burst length of 16.
2	RW	0x0	BLEN8 AXI Burst Length 8 When this bit is set to 1, or when UNDEF is set to 1, it is allowed to select a burst length of 8.
1	RW	0x0	BLEN4 AXI Burst Length 4 When this bit is set to 1, or when UNDEF is set to 1, it is allowed to select a burst length of 4.

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Bit	Attr	Reset Value	Description
0	RO	0x1	UNDEF AXI Undefined Burst Length This bit is read-only bit and indicates the complement (invert) value of FB bit in register GMAC_BUS_MODE[16]. When this bit is set to 1, it is allowed to perform any burst length equal to or below the maximum allowed burst length as programmed in bits[7:1]; When this bit is set to 0, it is allowed to perform only fixed burst lengths as indicated by BLEN256/128/64/32/16/8/4, or a burst length of 1.

GMAC_AXI_STATUS
Address: Operational Base + offset (0x102c)
AXI Status Register

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RO	0x0	RD_CH_STA When high, it indicates that AXI Master's read channel is active and transferring data.
0	RO	0x0	WR_CH_STA When high, it indicates that AXI Master's write channel is active and transferring data.

GMAC_CUR_HOST_TX_DESC
Address: Operational Base + offset (0x1048)
Current Host Transmit Descriptor Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HTDAP Host Transmit Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_CUR_HOST_RX_DESC
Address: Operational Base + offset (0x104c)
Current Host Receive Descriptor Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HRDAP Host Receive Descriptor Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_CUR_HOST_TX_Buf_ADDR
Address: Operational Base + offset (0x1050)
Current Host Transmit Buffer Address Register

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HTBAP Host Transmit Buffer Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

GMAC_CUR_HOST_RX_BUF_ADDR
Address: Operational Base + offset (0x1054)
Current Host Receive Buffer Adderss Register

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	HRBAP Host Receive Buffer Address Pointer Cleared on Reset. Pointer updated by DMA during operation.

13.5 Interface Description

Table 38-1 M0 RMII Interface Description

Module pin	Direction	Pad name	IOMUX setting
RMII interface			
mac_clk	I/O	IO_GMACclkm0_GPIO0D0vccio1	GPIO0D_IOMUX_SEL[1:0]=2'b01
mac_txen	O	IO_GMACtxenm0_GPIO0B4vccio1	GPIO0B_IOMUX_SEL[9:8]=2'b01
mac_txd1	O	IO_GMACtxd1m0_GPIO0C0vccio1	GPIO0C_IOMUX_SEL[1:0]=2'b01
mac_txd0	O	IO_GMACtxd0m0_GPIO0C1vccio1	GPIO0C_IOMUX_SEL[3:2]=2'b01
mac_rxdv	I	IO_GMACrxdvm0_GPIO0D1vccio1	GPIO0D_IOMUX_SEL[3:2]=2'b01
mac_rxer	I	IO_GMACrxerm0_GPIO0B5vccio1	GPIO0B_IOMUX_SEL[11:10]=2'b01
mac_rxd1	I	IO_GMACrxd1m0_GPIO0B6vccio1	GPIO0B_IOMUX_SEL[13:12]=2'b01
mac_rxd0	I	IO_GMACrxd0m0_GPIO0B7vccio1	GPIO0B_IOMUX_SEL[15:14]=2'b01
Management interface			
mac_mdio	I/O	IO_GMACmdiom0_GPIO0B3vccio1	GPIO0B_IOMUX_SEL[7:6]=2'b01
mac_mdc	O	IO_GMACmdcm0_GPIO0C3vccio1	GPIO0C_IOMUX_SEL[7:6]=2'b01

Table 38-2 M0 RGMII Interface Description

Module pin	Direction	Pad name	IOMUX setting
RGMII/RMII interface			
mac_clk	I/O	IO_GMACclkm0_GPIO0D0vccio1	GPIO0D_IOMUX_SEL[1:0]=2'b01
mac_txclk	O	IO_GMACtxclkm0_GPIO0B0vccio1	GPIO0B_IOMUX_SEL[1:0]=2'b01
mac_txen	O	IO_GMACtxenm0_GPIO0B4vccio1	GPIO0B_IOMUX_SEL[9:8]=2'b01
mac_txd3	O	IO_GMACtxd3m0_GPIO0C7vccio1	GPIO0C_IOMUX_SEL[15:14]=2'b01
mac_txd2	O	IO_GMACtxd2m0_GPIO0C6vccio1	GPIO0C_IOMUX_SEL[13:12]=2'b01
mac_txd1	O	IO_GMACtxd1m0_GPIO0C0vccio1	GPIO0C_IOMUX_SEL[1:0]=2'b01
mac_txd0	O	IO_GMACtxd0m0_GPIO0C1vccio1	GPIO0C_IOMUX_SEL[3:2]=2'b01
mac_rxclk	I	IO_GMACrxclkm0_GPIO0B2vccio1	GPIO0B_IOMUX_SEL[5:4]=2'b01
mac_rxdv	I	IO_GMACrxdvm0_GPIO0D1vccio1	GPIO0D_IOMUX_SEL[3:2]=2'b01
mac_rxd3	I	IO_GMACrxd3m0_GPIO0C4vccio1	GPIO0C_IOMUX_SEL[9:8]=2'b01
mac_rxd2	I	IO_GMACrxd2m0_GPIO0C5vccio1	GPIO0C_IOMUX_SEL[11:10]=2'b01
mac_rxd1	I	IO_GMACrxd1m0_GPIO0B6vccio1	GPIO0B_IOMUX_SEL[13:12]=2'b01
mac_rxd0	I	IO_GMACrxd0m0_GPIO0B7vccio1	GPIO0B_IOMUX_SEL[15:14]=2'b01
mac_crs	I	IO_GMACcrsm0_GPIO0B1vccio1	GPIO0B_IOMUX_SEL[3:2]=2'b01
mac_col	I	IO_GMACcolm0_GPIO0C2vccio1	GPIO0C_IOMUX_SEL[5:4]=2'b01
Management interface			

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mac_mdio	I/O	IO_GMACmdiom0_GPIO0B3vccio1	GPIO0B_IOMUX_SEL[7:6]=2'b01
mac_mdc	O	IO_GMACmdcm0_GPIO0C3vccio1	GPIO0C_IOMUX_SEL[7:6]=2'b01

Table 38-3 M1 RMII Interface Description

Module pin	Direction	Pad name	IOMUX setting
RMII interface			
mac_clk	I/O	IO_I2S2mclk_GMACclkm1_GPIO1C5vccio4	GPIO1C_IOMUX_SEL[11:10]=2'b10
mac_txen	O	IO_I2S2sdom0_GMACtxenm1_PDMsdi2m1_GPIO1D1vccio4	GPIO1D_IOMUX_SEL[3:2]=2'b10
mac_txd1	O	IO_UART0rx_GMACtxd1m1_GPIO1B0vccio4	GPIO1B_IOMUX_SEL[1:0]=2'b10
mac_txd0	O	IO_UART0tx_GMACtxd0m1_GPIO1B1vccio4	GPIO1B_IOMUX_SEL[3:2]=2'b10
mac_rxdv	I	IO_I2S2sclkm0_GMACrxdvm1_PDMclkm1_GPIO1C6vccio4	GPIO1C_IOMUX_SEL[13:12]=2'b10
mac_rxer	I	IO_I2S2sdim0_GMACrxerm1_PDMsdi1m1_GPIO1D0vccio4	GPIO1D_IOMUX_SEL[1:0]=2'b10
mac_rxd1	I	IO_UART0rtsn_GMACrxd1m1_GPIO1B2vccio4	GPIO1B_IOMUX_SEL[5:4]=2'b10
mac_rxd0	I	IO_UART0ctsn_GMACrxd0m1_GPIO1B3vccio4	GPIO1B_IOMUX_SEL[7:6]=2'b10
Management interface			
mac_mdio	I/O	IO_SDMMC1detn_GMACmdiom1_PDMfsyncm1_GPIO1C3vccio4	GPIO1C_IOMUX_SEL[7:6]=2'b10
mac_mdc	O	IO_I2S2lrcktxm0_GMACmdcm1_PDMsdi0m1_GPIO1C7vccio4	GPIO1C_IOMUX_SEL[15:14]=2'b10

Table 38-4 M1 RGMII Interface Description

Module pin	Direction	Pad name	IOMUX setting
RGMII/RMII interface			
mac_clk	I/O	IO_I2S2mclk_GMACclkm1_GPIO1C5vccio4	GPIO1C_IOMUX_SEL[11:10]=2'b10
mac_txclk	O	IO_SDMMC1clkout_GMACtxclkm1_GPIO1B4vccio4	GPIO1B_IOMUX_SEL[9:8]=2'b10
mac_txen	O	IO_I2S2sdom0_GMACtxenm1_PDMsdi2m1_GPIO1D1vccio4	GPIO1D_IOMUX_SEL[3:2]=2'b10
mac_txd3	O	IO_SDMMC1d2_GMACtxd3m1_GPIO1C0vccio4	GPIO1C_IOMUX_SEL[1:0]=2'b10
mac_txd2	O	IO_SDMMC1d3_GMACtxd2m1_GPIO1C1vccio4	GPIO1C_IOMUX_SEL[3:2]=2'b10
mac_txd1	O	IO_UART0rx_GMACtxd1m1_GPIO1B0vccio4	GPIO1B_IOMUX_SEL[1:0]=2'b10
mac_txd0	O	IO_UART0tx_GMACtxd0m1_GPIO1B1vccio4	GPIO1B_IOMUX_SEL[3:2]=2'b10
mac_rxclk	I	IO_SDMMC1cmd_GMACrxclkm1_GPIO1B5vccio4	GPIO1B_IOMUX_SEL[11:10]=2'b10
mac_rxdv	I	IO_I2S2sclkm0_GMACrxdvm1_PDMclkm1_GPIO1C6vccio4	GPIO1C_IOMUX_SEL[13:12]=2'b10
mac_rxd3	I	IO_SDMMC1d0_GMACrxd3m1_GPIO1B6vccio4	GPIO1B_IOMUX_SEL[13:12]=2'b10
mac_rxd2	I	IO_SDMMC1d1_GMACrxd2m1_GPIO1B7vccio4	GPIO1B_IOMUX_SEL[15:14]=2'b10
mac_rxd1	I	IO_UART0rtsn_GMACrxd1m1_GPIO1B2vccio4	GPIO1B_IOMUX_SEL[5:4]=2'b10
mac_rxd0	I	IO_UART0ctsn_GMACrxd0m1_GPIO1B3vccio4	GPIO1B_IOMUX_SEL[7:6]=2'b10
mac_crs	I	IO_SDMMC1pwren_GMACcrsm1_GPIO1C2vccio4	GPIO1C_IOMUX_SEL[5:4]=2'b10
mac_col	I	IO_SDMMC1wp_GMACcolm1_GPIO1C4vccio4	GPIO1C_IOMUX_SEL[9:8]=2'b10
Management interface			
mac_mdio	I/O	IO_SDMMC1detn_GMACmdiom1_PDMfsyncm1_GPIO1C3vccio4	GPIO1C_IOMUX_SEL[7:6]=2'b10

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mac_mdc	O	IO_I2S2lrcktxm0_GMACmdcm1_PDMsdi0m1_GPIO1C7vccio4	GPIO1C_IOMUX_SEL[15:14]=2'b10

Notes: I=input, O=output, I/O=input/output, bidirectional

13.6 Application Notes

13.6.1 Descriptors

The DMA in GMAC can communicate with Host driver through descriptor lists and data buffers. The DMA transfers data frames received by the core to the Receive Buffer in the Host memory, and Transmit data frames from the Transmit Buffer in the Host memory. Descriptors that reside in the Host memory act as pointers to these buffers.

There are two descriptor lists; one for reception, and one for transmission. The base address of each list is written into DMA Registers RX_DESC_LIST_ADDR and TX_DESC_LIST_ADDR, respectively. A descriptor list is forward linked (either implicitly or explicitly). The last descriptor may point back to the first entry to create a ring structure. Explicit chaining of descriptors is accomplished by setting the second address chained in both Receive and Transmit descriptors (RDES1[24] and TDES1[24]). The descriptor lists resides in the Host physical memory address space. Each descriptor can point to a maximum of two buffers. This enables two buffers to be used, physically addressed, rather than contiguous buffers in memory.

A data buffer resides in the Host physical memory space, and consists of an entire frame or part of a frame, but cannot exceed a single frame. Buffers contain only data, buffer status is maintained in the descriptor. Data chaining refers to frames that span multiple data buffers. However, a single descriptor cannot span multiple frames. The DMA will skip to the next frame buffer when end-of-frame is detected. Data chaining can be enabled or disabled

The descriptor ring and chain structure is shown in following figure.

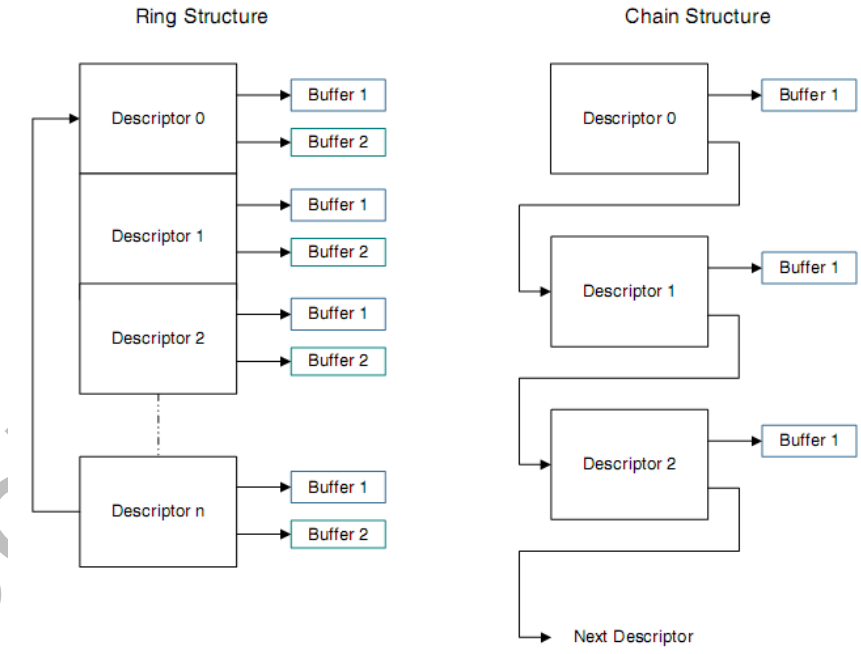

Fig. 38-9Descriptor Ring and Chain Structure

Each descriptor contains two buffers, two byte-count buffers, and two address pointers, which enable the adapter port to be compatible with various types of memory management schemes. The descriptor addresses must be aligned to the bus width used (Word/Dword/Lword for 32/64/128-bit buses).

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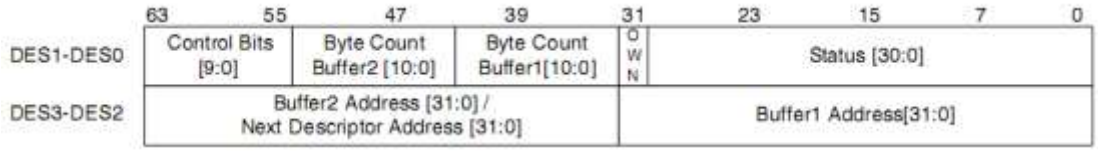

Fig. 38-10Rx/Tx Descriptors definition

13.6.2 Receive Descriptor

The GMAC Subsystem requires at least two descriptors when receiving a frame. The Receive state machine of the DMAalways attempts to acquire an extra descriptor in anticipation of an incoming frame. (The size of the incoming frame is unknown). Before the RxDMA closes a descriptor, it will attempt to acquire the next descriptor even if no frames are received.
In a single descriptor (receive) system, the subsystem will generate a descriptor error if the receive buffer is unable to accommodate the incoming frame and the next descriptor is not owned by the DMA. Thus, the Host is forced to increase either its descriptor pool or the buffer size. Otherwise, the subsystem starts dropping all incoming frames.

Receive Descriptor 0 (RDES0)
RDES0 contains the received frame status, the frame length, and the descriptor ownership information.

Table 38-3Receive Descriptor 0

Bit	Description
31	OWN: Own Bit When set, this bit indicates that the descriptor is owned by the DMA of the GMAC Subsystem. When this bit is reset, this bit indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame reception or when the buffers that are associated with this descriptor are full.
30	AFM: Destination Address Filter Fail When set, this bit indicates a frame that failed in the DA Filter in the GMAC Core.
29:16	FL: Frame Length These bits indicate the byte length of the received frame that was transferred to host memory (including CRC). This field is valid when Last Descriptor (RDES0[8]) is set and either the Descriptor Error (RDES0[14]) or Overflow Error bits are reset. The frame length also includes the two bytes appended to the Ethernet frame when IP checksum calculation (Type 1) is enabled and the received frame is not a MAC control frame. This field is valid when Last Descriptor (RDES0[8]) is set. When the Last Descriptor and Error Summary bits are not set, this field indicates the accumulated number of bytes that have been transferred for the current frame.
15	ES: Error Summary Indicates the logical OR of the following bits: • RDES0[0]: Payload Checksum Error • RDES0[1]: CRC Error • RDES0[3]: Receive Error • RDES0[4]: Watchdog Timeout • RDES0[6]: Late Collision • RDES0[7]: IPC Checksum • RDES0[11]: Overflow Error • RDES0[14]: Descriptor Error This field is valid only when the Last Descriptor (RDES0[8]) is set.
14	DE: Descriptor Error When set, this bit indicates a frame truncation caused by a frame that does not fit within the current descriptor buffers, and that the DMA does not own the Next Descriptor. The frame is truncated. This field is valid only when the Last Descriptor (RDES0[8]) is set

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13	SAF: Source Address Filter Fail When set, this bit indicates that the SA field of frame failed the SA Filter in the GMAC Core.
12	LE: Length Error When set, this bit indicates that the actual length of the frame received and that the Length/ Type field does not match. This bit is valid only when the Frame Type (RDES0[5]) bit is reset. Length error status is not valid when CRC error is present.
11	OE: Overflow Error When set, this bit indicates that the received frame was damaged due to buffer overflow.
10	VLAN: VLAN Tag When set, this bit indicates that the frame pointed to by this descriptor is a VLAN frame tagged by the GMAC Core.
9	FS: First Descriptor When set, this bit indicates that this descriptor contains the first buffer of the frame. If the size of the first buffer is 0, the second buffer contains the beginning of the frame. If the size of the second buffer is also 0, the next Descriptor contains the beginning of the frame.
8	LS: Last Descriptor When set, this bit indicates that the buffers pointed to by this descriptor are the last buffers of the frame.
7	IPC Checksum Error/Giant Frame When IP Checksum Engine is enabled, this bit, when set, indicates that the 16-bit IPv4 Header checksum calculated by the core did not match the received checksum bytes. The Error Summary bit[15] is NOT set when this bit is set in this mode.
6	LC: Late Collision When set, this bit indicates that a late collision has occurred while receiving the frame in Half-Duplex mode.
5	FT: Frame Type When set, this bit indicates that the Receive Frame is an Ethernet-type frame (the LT field is greater than or equal to 16'h0600). When this bit is reset, it indicates that the received frame is an IEEE802.3 frame. This bit is not valid for Runt frames less than 14 bytes.
4	RWT: Receive Watchdog Timeout When set, this bit indicates that the Receive Watchdog Timer has expired while receiving the current frame and the current frame is truncated after the Watchdog Timeout.
3	RE: Receive Error When set, this bit indicates that the gmii_rxer_i signal is asserted while gmii_rxdv_i is asserted during frame reception. This error also includes carrier extension error in GMII and Half-duplex mode. Error can be of less/no extension, or error (rxd≠ 0f) during extension.
2	DE: Dribble Bit Error When set, this bit indicates that the received frame has a non-integer multiple of bytes (odd nibbles). This bit is valid only in MII Mode.
1	CE: CRC Error When set, this bit indicates that a Cyclic Redundancy Check (CRC) Error occurred on the received frame. This field is valid only when the Last Descriptor (RDES0[8]) is set.
0	Rx MAC Address/Payload Checksum Error When set, this bit indicates that the Rx MAC Address registers value (1 to 15) matched the frame's DA field. When reset, this bit indicates that the Rx MAC Address Register 0 value matched the DA field. If Full Checksum Offload Engine is enabled, this bit, when set, indicates the TCP, UDP, or ICMP checksum the core calculated does not match the received encapsulated TCP, UDP, or ICMP segment's Checksum field. This bit is also set when the received number of payload bytes does not match the value indicated in the Length field of the encapsulated IPv4 or IPv6 datagram in the received Ethernet frame.

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Receive Descriptor 1 (RDES1)

RDES1 contains the buffer sizes and other bits that control the descriptor chain/ring.

Table 38-4Receive Descriptor 1

Bit	Description
31	Disable Interrupt on Completion When set, this bit will prevent the setting of the RI (CSR5[6]) bit of the GMAC_STATUS Register for the received frame that ends in the buffer pointed to by this descriptor. This, in turn, will disable the assertion of the interrupt to Host due to RI for that frame.
30:26	Reserved.
25	RER: Receive End of Ring When set, this bit indicates that the descriptor list reached its final descriptor. The DMA returns to the base address of the list, creating a Descriptor Ring.
24	RCH: Second Address Chained When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When RDES1[24] is set, RBS2 (RDES1[21-11]) is a "don't care" value. RDES1[25] takes precedence over RDES1[24].
23:22	Reserved.
21:11	RBS2: Receive Buffer 2 Size These bits indicate the second data buffer size in bytes. The buffer size must be a multiple of 8 depending upon the bus widths (64), even if the value of RDES3 (buffer2 address pointer) is not aligned to bus width. In the case where the buffer size is not a multiple of 8, the resulting behavior is undefined. This field is not valid if RDES1[24] is set.
10:0	RBS1: Receive Buffer 1 Size Indicates the first data buffer size in bytes. The buffer size must be a multiple of 8 depending upon the bus widths (64), even if the value of RDES2 (buffer1 address pointer) is not aligned. In the case where the buffer size is not a multiple of 8, the resulting behavior is undefined. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or next descriptor depending on the value of RCH (Bit 24).

Receive Descriptor 2 (RDES2)

RDES2 contains the address pointer to the first data buffer in the descriptor.

Table 38-5Receive Descriptor 2

Bit	Description
31:0	Buffer 1 Address Pointer These bits indicate the physical address of Buffer 1. There are no limitations on the buffer address alignment except for the following condition: The DMA uses the configured value for its address generation when the RDES2 value is used to store the start of frame. Note that the DMA performs a write operation with the RDES2[2:0] bits as 0 during the transfer of the start of frame but the frame data is shifted as per the actual Buffer address pointer. The DMA ignores RDES2[2:0] (corresponding to bus width of 64) if the address pointer is to a buffer where the middle or last part of the frame is stored.

Receive Descriptor 3 (RDES3)

RDES3 contains the address pointer either to the second data buffer in the descriptor or to the next descriptor.

Table 38-6Receive Descriptor 3

Bit	Description
31:0	Buffer 2 Address Pointer (Next Descriptor Address) These bits indicate the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (RDES1[24]) bit is set, this address contains the pointer to the physical memory where the

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	Next Descriptor is present. If RDES1[24] is set, the buffer (Next Descriptor) address pointer must be bus width-aligned (RDES3[2:0] = 0, corresponding to a bus width of 64. LSBs are ignored internally.) However, when RDES1[24] is reset, there are no limitations on the RDES3 value, except for the following condition: The DMA uses the configured value for its buffer address generation when the RDES3 value is used to store the start of frame. The DMA ignores RDES3[2:0] (corresponding to a bus width of 64) if the address pointer is to a buffer where the middle or last part of the frame is stored.

13.6.3 Transmit Descriptor

The descriptor addresses must be aligned to the bus width used (64). Each descriptor is provided with two buffers, two byte-count buffers, and two address pointers, which enable the adapter port to be compatible with various types of memory-management schemes.

Transmit Descriptor 0 (TDES0)

TDES0 contains the transmitted frame status and the descriptor ownership information.

Table 38-7 Transmit Descriptor 0

Bit	Description
31	OWN: Own Bit When set, this bit indicates that the descriptor is owned by the DMA. When this bit is reset, this bit indicates that the descriptor is owned by the Host. The DMA clears this bit either when it completes the frame transmission or when the buffers allocated in the descriptor are empty. The ownership bit of the First Descriptor of the frame should be set after all subsequent descriptors belonging to the same frame have been set. This avoids a possible race condition between fetching a descriptor and the driver setting an ownership bit.
30:17	Reserved.
16	IHE: IP Header Error When set, this bit indicates that the Checksum Offload engine detected an IP header error and consequently did not modify the transmitted frame for any checksum insertion.
15	ES: Error Summary Indicates the logical OR of the following bits: • TDES0[14]: Jabber Timeout • TDES0[13]: Frame Flush • TDES0[11]: Loss of Carrier • TDES0[10]: No Carrier • TDES0[9]: Late Collision • TDES0[8]: Excessive Collision • TDES0[2]: Excessive Deferral • TDES0[1]: Underflow Error
14	JT: Jabber Timeout When set, this bit indicates the GMAC transmitter has experienced a jabber time-out.
13	FF: Frame Flushed When set, this bit indicates that the DMA/MTL flushed the frame due to a SW flush command given by the CPU.
12	PCE: Payload Checksum Error This bit, when set, indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload. This failure can be either due to insufficient bytes, as indicated by the IP Header's Payload Length field, or the MTL starting to forward the frame to the MAC transmitter in Store-and-Forward mode without the checksum having been calculated yet. This second error condition only occurs when the Transmit FIFO depth is less than the length of the Ethernet frame being transmitted: to avoid deadlock, the MTL starts

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	forwarding the frame when the FIFO is full, even in Store-and-Forward mode.
11	LC: Loss of Carrier When set, this bit indicates that Loss of Carrier occurred during frame transmission. This is valid only for the frames transmitted without collision and when the GMAC operates in Half-Duplex Mode.
10	NC: No Carrier When set, this bit indicates that the carrier sense signal form the PHY was not asserted during transmission.
9	LC: Late Collision When set, this bit indicates that frame transmission was aborted due to a collision occurring after the collision window (64 byte times including Preamble in RMII Mode and 512 byte times including Preamble and Carrier Extension in RGMII Mode). Not valid if Underflow Error is set.
8	EC: Excessive Collision When set, this bit indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current frame. If the DR (Disable Retry) bit in the GMAC Configuration Register is set, this bit is set after the first collision and the transmission of the frame is aborted.
7	VF: VLAN Frame When set, this bit indicates that the transmitted frame was a VLAN-type frame.
6:3	CC: Collision Count This 4-bit counter value indicates the number of collisions occurring before the frame was transmitted. The count is not valid when the Excessive Collisions bit (TDES0[8]) is set.
2	ED: Excessive Deferral When set, this bit indicates that the transmission has ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000-Mbps mode) if the Deferral Check (DC) bit is set high in the GMAC Control Register.
1	UF: Underflow Error When set, this bit indicates that the GMAC aborted the frame because data arrived late from the Host memory. Underflow Error indicates that the DMA encountered an empty Transmit Buffer while transmitting the frame. The transmission process enters the suspended state and sets both Transmit Underflow (Register GMAC_STATUS[5]) and Transmit Interrupt (Register GMAC_STATUS [0]).
0	DB: Deferred Bit When set, this bit indicates that the GMAC defers before transmission because of the presence of carrier. This bit is valid only in Half-Duplex mode.

Transmit Descriptor 1 (TDES1)

TDES1 contains the buffer sizes and other bits which control the descriptor chain/ring and the frame being transferred.

Table 38-8 Transmit Descriptor 1

Bit	Description
31	IC: Interrupt on Completion When set, this bit sets Transmit Interrupt (Register 5[0]) after the present frame has been transmitted.
30	LS: Last Segment When set, this bit indicates that the buffer contains the last segment of the frame.
29	FS: First Segment When set, this bit indicates that the buffer contains the first segment of a frame.
28:27	CIC: Checksum Insertion Control These bits control the insertion of checksums in Ethernet frames that encapsulate TCP, UDP, or ICMP over IPv4 or IPv6 as described below. • 2'b00: Do nothing. Checksum Engine is bypassed • 2'b01: Insert IPv4 header checksum. Use this value to insert IPv4 header checksum when the frame encapsulates an IPv4 datagram.

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	• 2'b10: Insert TCP/UDP/ICMP checksum. The checksum is calculated over the TCP, UDP, or ICMP segment only and the TCP, UDP, or ICMP pseudo-header checksum is assumed to be present in the corresponding input frame's Checksum field. An IPv4 header checksum is also inserted if the encapsulated datagram conforms to IPv4. • 2'b11: Insert a TCP/UDP/ICMP checksum that is fully calculated in this engine. In other words, the TCP, UDP, or ICMP pseudo-header is included in the checksum calculation, and the input frame's corresponding Checksum field has an all-zero value. An IPv4 Header checksum is also inserted if the encapsulated datagram conforms to IPv4. The Checksum engine detects whether the TCP, UDP, or ICMP segment is encapsulated in IPv4 or IPv6 and processes its data accordingly.
26	DC: Disable CRC When set, the GMAC does not append the Cyclic Redundancy Check (CRC) to the end of the transmitted frame. This is valid only when the first segment (TDES1[29]).
25	TER: Transmit End of Ring When set, this bit indicates that the descriptor list reached its final descriptor. The returns to the base address of the list, creating a descriptor ring.
24	TCH: Second Address Chained When set, this bit indicates that the second address in the descriptor is the Next Descriptor address rather than the second buffer address. When TDES1[24] is set, TBS2 (TDES1[21–11]) are "don't care" values. TDES1[25] takes precedence over TDES1[24].
23	DP: Disable Padding When set, the GMAC does not automatically add padding to a frame shorter than 64 bytes. When this bit is reset, the DMA automatically adds padding and CRC to a frame shorter than 64 bytes and the CRC field is added despite the state of the DC (TDES1[26]) bit. This is valid only when the first segment (TDES1[29]) is set.
22	Reserved.
21:11	TBS2: Transmit Buffer 2 Size These bits indicate the Second Data Buffer in bytes. This field is not valid if TDES1[24] is set.
10:0	TBS1: Transmit Buffer 1 Size These bits indicate the First Data Buffer byte size. If this field is 0, the DMA ignores this buffer and uses Buffer 2 or next descriptor depending on the value of TCH (Bit 24).

Transmit Descriptor 2 (TDES2)

TDES2 contains the address pointer to the first buffer of the descriptor.

Table 38-9 Transmit Descriptor 2

Bit	Description
31:0	Buffer 1 Address Pointer These bits indicate the physical address of Buffer 1. There is no limitation on the buffer address alignment.

Transmit Descriptor 3 (TDES3)

TDES3 contains the address pointer either to the second buffer of the descriptor or the next descriptor.

Table 38-10 Transmit Descriptor 3

Bit	Description
31:0	Buffer 2 Address Pointer (Next Descriptor Address) Indicates the physical address of Buffer 2 when a descriptor ring structure is used. If the Second Address Chained (TDES1[24]) bit is set, this address contains the pointer to the physical memory where the Next Descriptor is present. The buffer address pointer must be aligned to the bus width only when TDES1[24] is set. (LSBs are ignored internally.)

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13.6.4 Programming Guide

DMA Initialization – Descriptors

The following operations must be performed to initialize the DMA.

1. Provide a software reset. This will reset all of the GMAC internal registers and logic. (GMAC_OP_MODE[0]).
2. Wait for the completion of the reset process (poll GMAC_OP_MODE[0], which is only cleared after the reset operation is completed).
3. Program the following fields to initialize the Bus Mode Register by setting values in register GMAC_BUS_MODE
   a. Mixed Burst and AAL
   b. Fixed burst or undefined burst
   c. Burst length values and burst mode values.
   d. Descriptor Length (only valid if Ring Mode is used)
   e. Tx and Rx DMA Arbitration scheme
4. Program the AXI Interface options in the register GMAC_BUS_MODE
   a. If fixed burst-length is enabled, then select the maximum burst-length possible on the AXI bus (Bits[7:1])
5. A proper descriptor chain for transmit and receive must be created. It should also ensure that the receive descriptors are owned by DMA (bit 31 of descriptor should be set). When OSF mode is used, at least two descriptors are required.
6. Software should create three or more different transmit or receive descriptors in the chain before reusing any of the descriptors.
7. Initialize receive and transmit descriptor list address with the base address of transmit and receive descriptor (register GMAC_RX_DESC_LIST_ADDR and GMAC_TX_DESC_LIST_ADDR).
8. Program the following fields to initialize the mode of operation by setting values in register GMAC_OP_MODE
   a. Receive and Transmit Store And Forward
   b. Receive and Transmit Threshold Control (RTC and TTC)
   c. Hardware Flow Control enable
   d. Flow Control Activation and De-activation thresholds for MTL Receive and Transmit FIFO (RFA and RFD)
   e. Error Frame and undersized good frame forwarding enable
   f. OSF Mode
9. Clear the interrupt requests, by writing to those bits of the status register (interrupt bits only) which are set. For example, by writing 1 into bit 16 - normal interrupt summary will clear this bit (register GMAC_STATUS).
10. Enable the interrupts by programming the interrupt enable register GMAC_INT_ENA.
11. Start the Receive and Transmit DMA by setting SR (bit 1) and ST (bit 13) of the control register GMAC_OP_MODE.

MAC Initialization

The following MAC Initialization operations can be performed after the DMA initialization sequence. If the MAC Initialization is done before the DMA is set-up, then enable the MAC receiver (last step below) only after the DMA is active. Otherwise, received frames will fill the RxFIFO and overflow.

1. Program the register GMAC_GMII_ADDR for controlling the management cycles for external PHY, for example, Physical Layer Address PA (bits 15-11). Also set bit 0 (GMII Busy) for writing into PHY and reading from PHY.
2. Read the 16-bit data of (GMAC_GMII_DATA) from the PHY for link up, speed of operation, and mode of operation, by specifying the appropriate address value in registerGMAC_GMII_ADDR (bits 15-11).
3. Provide the MAC address registers (GMAC_MAC_ADDR0_HI and GMAC_MAC_ADDR0_LO).
4. If Hash filtering is enabled in your configuration, program the Hash filter register (GMAC_HASH_TAB_HI and GMAC_HASH_TAB_LO).
5. Program the following fields to set the appropriate filters for the incoming frames in register GMAC_MAC_FRM_FILT

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a. Receive All
b. Promiscuous mode
c. Hash or Perfect Filter
d. Unicast, Multicast, broad cast and control frames filter settings etc.

6. Program the following fields for proper flow control in register GMAC_FLOW_CTRL.
a. Pause time and other pause frame control bits
b. Receive and Transmit Flow control bits
c. Flow Control Busy/Backpressure Activate

7. Program the Interrupt Mask register bits, as required, and if applicable, for your configuration.
8. Program the appropriate fields in register GMAC_MAC_CONF for example, Inter-frame gap while transmission, jabber disable, etc. Based on the Auto-negotiation you can set the Duplex mode (bit 11), port select (bit 15), etc.
9. Set the bits Transmit enable (TE bit-3) and Receive Enable (RE bit-2) in register GMAC_MAC_CONF.

Normal Receive and Transmit Operation

For normal operation, the following steps can be followed.

- For normal transmit and receive interrupts, read the interrupt status. Then poll the descriptors, reading the status of the descriptor owned by the Host (either transmit or receive).
- On completion of the above step, set appropriate values for the descriptors, ensuring that transmit and receive descriptors are owned by the DMA to resume the transmission and reception of data.
- If the descriptors were not owned by the DMA (or no descriptor is available), the DMA will go into SUSPEND state. The transmission or reception can be resumed by freeing the descriptors and issuing a poll demand by writing 0 into the Tx/Rx poll demand register (GMAC_TX_POLL_DEMAND and GMAC_RX_POLL_DEMAND).
- The values of the current host transmitter or receiver descriptor address pointer can be read for the debug process (GMAC_CUR_HOST_TX_DESC and GMAC_CUR_HOST_RX_DESC).
- The values of the current host transmit buffer address pointer and receive buffer address pointer can be read for the debug process (GMAC_CUR_HOST_TX_Buf_ADDR and GMAC_CUR_HOST_RX_BUF_ADDR).

Stop and Start Operation

When the transmission is required to be paused for some time then the following steps can be followed.
1. Disable the Transmit DMA (if applicable), by clearing ST (bit 13) of the control register GMAC_OP_MODE.
2. Wait for any previous frame transmissions to complete. This can be checked by reading the appropriate bits of MAC Debug register.
3. Disable the MAC transmitter and MAC receiver by clearing the bits Transmit enable (TE bit-3) and Receive Enable (RE bit-2) in register GMAC_MAC_CONF.
4. Disable the Receive DMA (if applicable), after making sure the data in the RX FIFO is transferred to the system memory (by reading the register GMAC_DEBUG).
5. Make sure both the TX FIFO and RX FIFO are empty.
6. To re-start the operation, start the DMAs first, before enabling the MAC Transmitter and Receiver.

13.6.5 Clock Architecture

In RMII mode, reference clock and TX/RX clock can be from CRU or external OSC as following figure.
The mux select rmii_speed is GRF_SOC_CON1[11].

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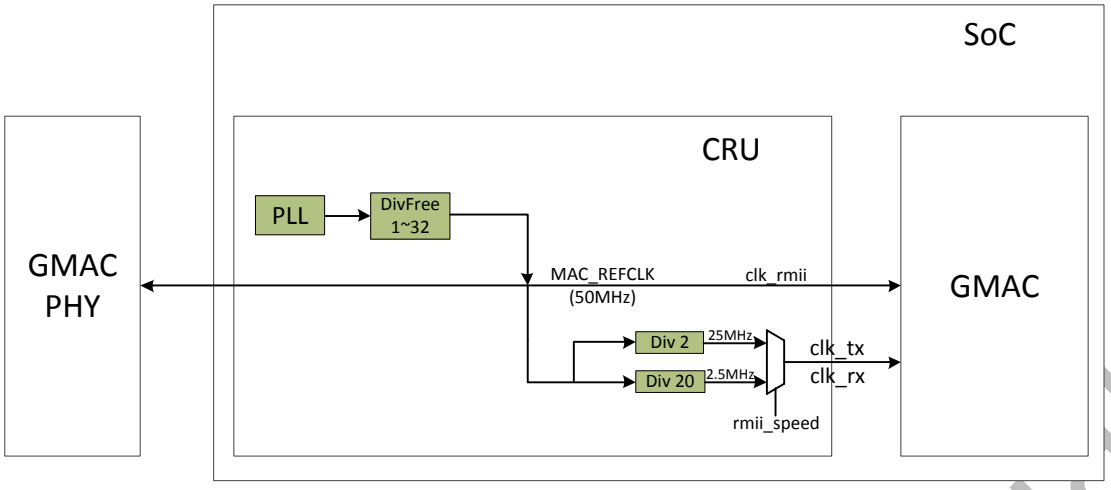

Fig. 38-11 RMII clock architecture when clock source from CRU

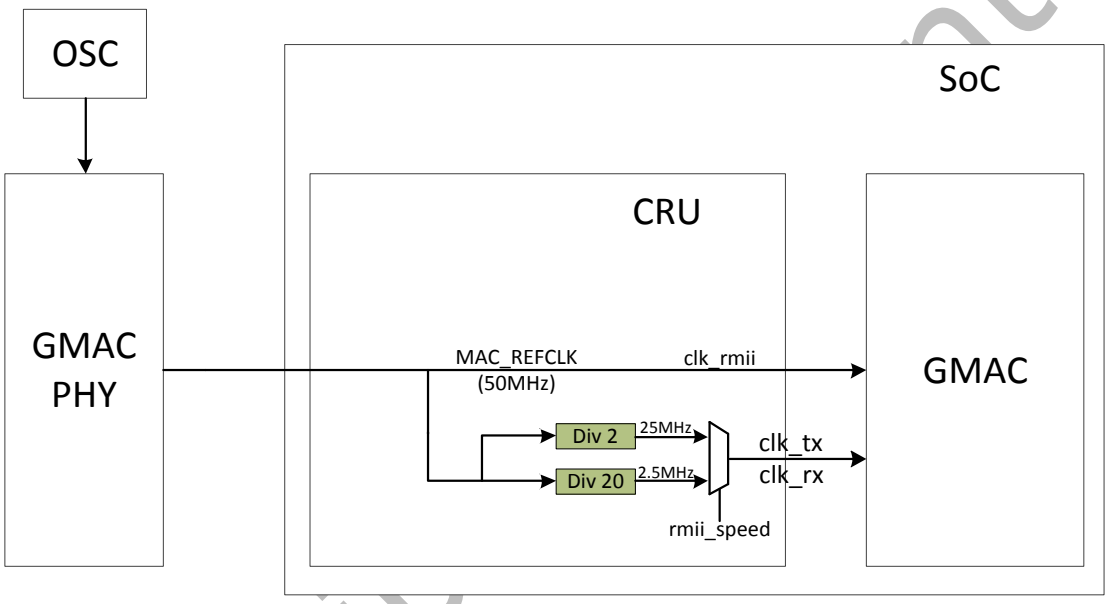

Fig. 38-12 RMII clock architecture when clock source from external OSC

In RGMII mode, clock architecture only supports that TX clock source is from CRU as following figure.

In order to dynamically adjust the timing between TX/RX clocks with data, deleyline is integrated in TX and RX clock path. Register GRF_SOC_CON3[15:14] can enable the deleylines, and GRF_SOC_CON3[13:0] is used to determine the delay length. There are 100 deley elements in each delayline.

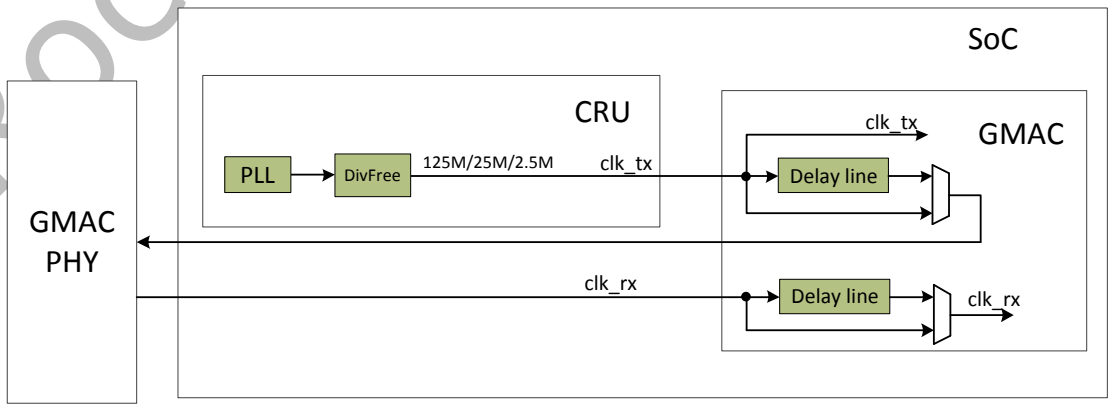

Fig. 38-13 RGMII clock architecture when clock source from CRU

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13.6.6 Remote Wake-Up Frame Filter Register

The register wkupfmfilter_reg, address (028H), loads the Wake-up Frame Filter register. To load values in a Wake-up Frame Filter register, the entire register (wkupfmfilter_reg) must be written. The wkupfmfilter_reg register is loaded by sequentially loading the eight register values in address (028) for wkupfmfilter_reg0, wkupfmfilter_reg1, ..., wkupfmfilter_reg7, respectively. Wkupfmfilter_reg is read in the same way.
The internal counter to access the appropriate wkupfmfilter_reg is incremented when lane3 (or lane 0 in big-endian) is accessed by the CPU. This should be kept in mind if you are accessing these registers in byte or half-word mode.

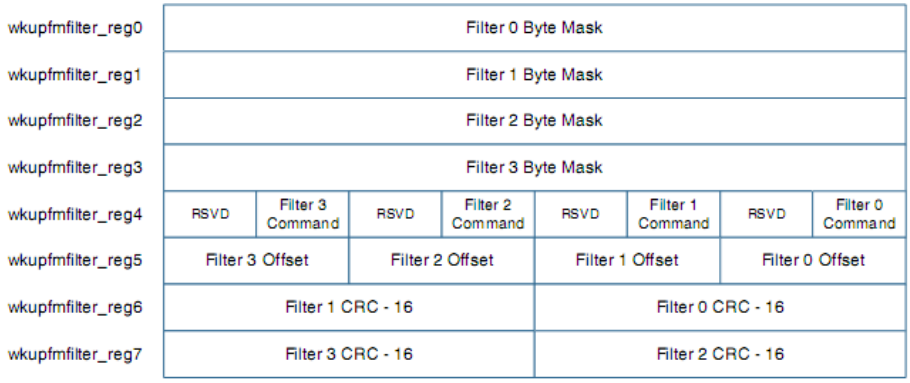

Fig. 38-14Wake-Up Frame Filter Register

Filter i Byte Mask
This register defines which bytes of the frame are examined by filter i (0, 1, 2, and 3) in order to determine whether or not the frame is a wake-up frame. The MSB (thirty-first bit) must be zero. Bit j [30:0] is the Byte Mask. If bit j (byte number) of the Byte Mask is set, then Filter i Offset + j of the incoming frame is processed by the CRC block; otherwise Filter i Offset + j is ignored.
Filter i Command
This 4-bit command controls the filter i operation. Bit 3 specifies the address type, defining the pattern's destination address type. When the bit is set, the pattern applies to only multicast frames; when the bit is reset, the pattern applies only to unicast frame. Bit 2 and Bit 1 are reserved. Bit 0 is the enable for filter i; if Bit 0 is not set, filter i is disabled.
Filter i Offset
This register defines the offset (within the frame) from which the frames are examined by filter i. This 8-bit pattern-offset is the offset for the filter i first byte to examined. The minimum allowed is 12, which refers to the 13th byte of the frame (offset value 0 refers to the first byte of the frame).
Filter i CRC-16
This register contains the CRC_16 value calculated from the pattern, as well as the byte mask programmed to the wake-up filter register block.

13.6.7 System Consideration During Power-Down

GMAC neither gates nor stops clocks when Power-Down mode is enabled. Power saving by clock gating must be done outside the core by the CRU. The receive data path must be clocked with clk_rx_i during Power-Down mode, because it is involved in magic packet/wake-on-LAN frame detection. However, the transmit path and the APB path clocks can be gated off during Power-Down mode.
The PMT interrupt is asserted when a valid wake-up frame is received. This interrupt is generated in the clk_rx domain.
The recommended power-down and wake-up sequence is as follows.
1. Disable the Transmit DMA (if applicable) and wait for any previous frame transmissions to complete. These transmissions can be detected when Transmit Interrupt (TI - Register GMAC_STATUS[0]) is received.
2. Disable the MAC transmitter and MAC receiver by clearing the appropriate bits in the MAC

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Configuration register.
3. Wait until the Receive DMA empties all the frames from the Rx FIFO (a software timer may be required).
4. Enable Power-Down mode by appropriately configuring the PMT registers.
5. Enable the MAC Receiver and enter Power-Down mode.
6. Gate the APB and transmit clock inputs to the core (and other relevant clocks in the system) to reduce power and enter Sleep mode.
7. On receiving a valid wake-up frame, the GMAC asserts the PMT interrupt signal and exits Power-Down mode.
8. On receiving the interrupt, the system must enable the APB and transmit clock inputs to the core.
9. Read the register GMAC_PMT_CTRL_STA to clear the interrupt, then enable the other modules in the system and resume normal operation.

13.6.8 GRF Register Summary

GMAC2IO	
GRF Register	**Register Description**
GRF_MAC_CON0[6:0]	RGMII TX clock delayline value
GRF_MAC_CON0[13:7]	RGMII RX clock delayline value
GRF_MAC_CON1[0]	RGMII TX clock delayline enable 1'b1: enable 1'b0: disable
GRF_MAC_CON1[1]	RGMII RX clock delayline enable 1'b1: enable 1'b0: disable
GRF_MAC_CON1[2]	GMACspeed 1'b1: 100-Mbps 1'b0: 10-Mbps
GRF_MAC_CON1[3]	GMAC transmit flow control When set high, instructs the GMAC to transmit PAUSE Control frames in Full-duplex mode. In Half-duplex mode, the GMAC enables the Back-pressure function until this signal is made low again
GRF_MAC_CON1[6:4]	PHY interface select 3'b001: RGMII 3'b100: RMII All others: Reserved
GRF_MAC_CON1[7]	RMII clock selection 1'b1: 25MHz 1'b0: 2.5MHz
GRF_MAC_CON1[9]	RMII mode selection 1'b1: RMII mode 1’b0: Reserved
GRF_MAC_CON1[10]	GMAC clock source selection 1’b1:clock from external OSC 1’b0:clock from CRU
GRF_MAC_CON1[12:11]	RGMII clock selection 2'b00: 125MHz 2'b11: 25MHz 2'b10: 2.5MHz
GRF_CON_IOMUX[2]	GMAC IO selection 1’b1:select M1 1’b0:select M0
GRF_CON_IOMUX[10]	GMAC M1 channel select 1’b1:M1’s outputs come from M0’s pad when set GRF_CON_IOMUX[2] high

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	1’b0:GMAC controller connect M1 directly when set GRF_CON_IOMUX[2] high
GMAC2PHY	
GRF Register	**Register Description**
GRF_MAC_CON2[2]	GMACspeed 1'b1: 100-Mbps 1'b0: 10-Mbps
GRF_MAC_CON2[3]	GMAC transmit flow control When set high, instructs the GMAC to transmit PAUSE Control frames in Full-duplex mode. In Half-duplex mode, the GMAC enables the Back-pressure function until this signal is made low again
GRF_MAC_CON2[6:4]	PHY interface select 3'b001: RGMII 3'b100: RMII All others: Reserved
GRF_MAC_CON2[7]	RMII clock selection 1'b1: 25MHz 1'b0: 2.5MHz
GRF_MAC_CON2[9]	RMII mode selection 1'b1: RMII mode 1’b0: Reserved
GRF_MAC_CON2[10]	GMAC clock source selection 1’b1:clock from external OSC 1’b0:clock from CRU

13.6.9 GMAC2IO Channel Description

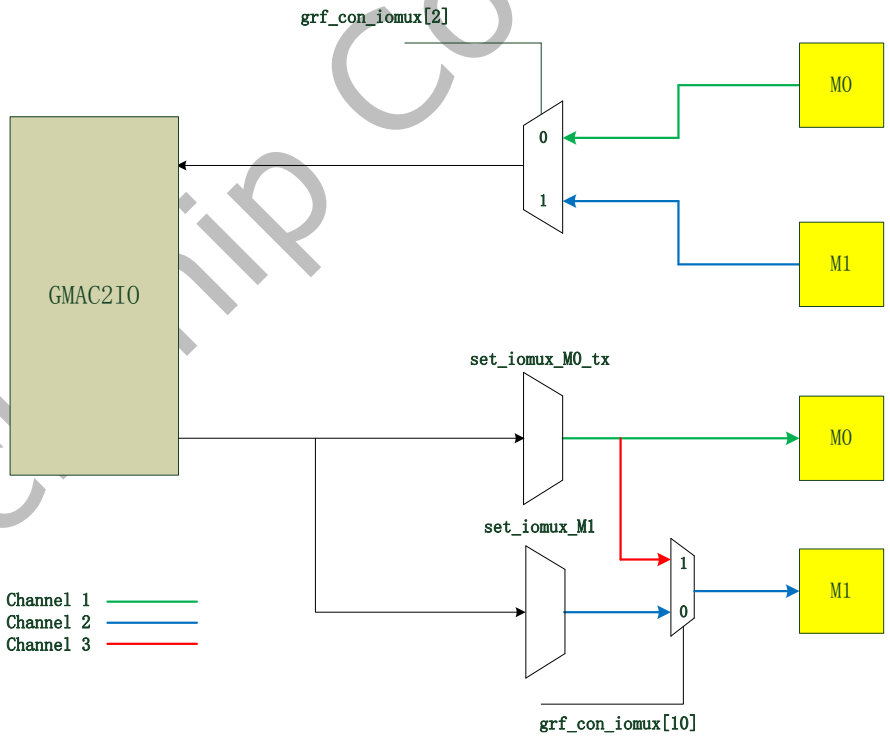

Fig. 38-16 gmac2io channel architecture

There are 3 different channels between GMAC controller and IO. The set_iomux_M0 and set_iomux_M1 in the upper figure means a series of IOMUX settings in table 1-1,1-2,1-3 and 1-4.

1. setting GRF_CON_IOMUX[2] low , GRF_CON_IOMUX[10] low and set_iomux_M0;
2. setting GRF_CON_IOMUX[2] high, GRF_CON_IOMUX[10] low and set_iomux_M1;
3. setting GRF_CON_IOMUX[2] high, GRF_CON_IOMUX[10] high , set_iomux_M1 and set_iomux_M0_tx;

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Chapter 39 MACPHY

1.6 Overview

The 10 Base-T/100 Base-TX MACPHY is a low-power Ethernet Physical Layer (PHY) transceiver with variable I/O voltage that is fully compliant with the IEEE 802.3 and 802.3u standards. The PHY works in both 10Base-T and 100Base-TX Fast Ethernet modes, supports 100Base-FX, and is fully compliant with the IEEE 802.3/802.3u standards. The core has register-selectable configuration-options, which define the functionality of the core.

It supports following features:

- Integrated IEEE 802.3/802.3u conpliant 10/100Mbps Ethernet PHY
- Supporting both full and half duplex for either 10 or 100 Mb/s data rate
- Auto MDIX capable
- Supports wake-on-LAN, EEE
- 100Base-FX support
- RMII interface
- Supports auto-negotiation
- On-board diagnosics
- Supporting LED's polarity configurable

1.7 Block Diagram

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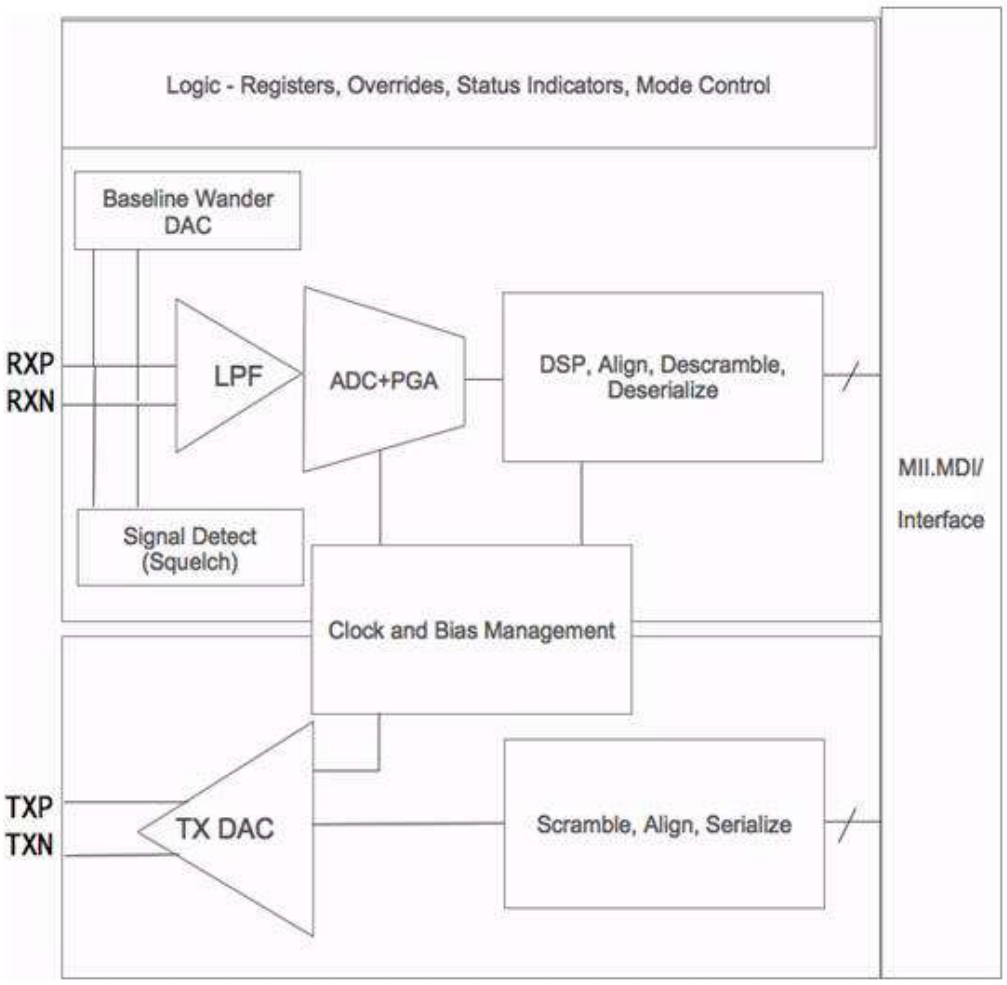

Fig. 38-15 AFE Block Diagram

1.8 Function Description

Functionally, the core PHY can be divided into the following sections:

- 100Base-TX transmit and receive
- 10Base-T transmit and receive
- Logic to handle auto-negotiation and management control

A block diagram of the analog-front-end is shown above. The functions of 100Base-T/FX, 10Base-T, and the related logic will be described thereafter. The complete set of power-modes provided by the standard are supported, including Energy-Detect and Idle modes for both 100- and 10BaseT modes.

100Base-TX and 100Base-TX Transmit and Receive

The transmitter is implemented using a Class-A current steering architecture. The transmitter drives a scrambled MLT3 data into the 100-ohm impedance. The transmitter implements a 6b DAC.

For the 100BaseT receive function, the MLT-3 from the cable is fed into the PHY through a low-pass-filter, and a 6b ADC samples the incoming data. A programmable gain is implemented in the ADC. Baseline wander is corrected using a small DAC.

The ADC outputs are then processed in the digital domain where they are equalized by a DSP block. The DSP also controls the recovered clock by selecting one of eight phases from the PLL output. This clock is used to recover the NRZI data-stream and descrambling takes place. The data are then aligned and decoded prior to being passed in 4b nibbles to the MAC through the MII interface (clocked at the controller rate).

Logic exists to detect if the receive data is valid and without errors (unexpected code groups within a frame).

In 100M FX mode, several sections of the signal path can be powered down and greatly simplified. For example, the scrambling/de-scrambling functions are disabled. However, a special signal detect is required to indicate the presence of an FX signal.

Fault detection monitoring is implemented according to Section 24.3.2.1 of the 802.3u standard.

Various over-ride and bypass mechanisms exist from within the register map

10Base-T Transmit and Receive

The transmitter serializes 2.5MHz nibbles (from the MAC) onto a 10M NRZ steam. This stream is then encoded using a Manchester encoding scheme and sent to the analog Class-A transmitter which drives the magnetics that ultimately drive the load. If no data is being sent, the link drives normal link pulses to the receiver, so that lock is not lost.

The receiver receives the encoded stream from the cable, and the analog signal is filtered and checked using a squelch circuit. The receiver recovers the clock and data to recreate the NRZI stream after confirming that the data is valid encoded data. Polarity is identified and corrected as necessary (observable through register interface). Then stream is deserialized ascent to the MAC interface at 2.5MHz.

Logic to handle auto-negotiation and management control

The PHY automatically determines the capabilities of the link partners according to the specification put forth in clause 28 of IEEE 802.3 standard. Information is passed back/forth using the SMI (serial management interface) bus and the results can be read via status registers. This is independent of the controller. The highest priority is given to 100M Full Duplex, then 100M Half Duplex, 10M Full Duplex, and 10M Half Duplex.

If the device the PHY is connected to lacks the capability, then a capability known as Parallel Detection will be employed – the speed will be determined by 100 MLT-3 Symbols or 10M Normal Link Pulses and that fact will be reflected in the management registers.

Auto-negotiation can be restarted at any time through a register setting. The normal transmission will be suspended for a time while the logic determines the right setting.

Additionally, the PHY implements the Next Page function which allows the Auto-Negotiation infrastructure to transmit arbitrary information using acknowledge-based handshaking. Pages can either be predefined codes/instructions or unformatted bits.

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Functional Modes

- **Normal mode**

In normal mode the PHY is fully operational.
When no packets are coming in and no packets need to be transmitted, the PHY is in Idle mode. The PHY must be in Idle mode for WOL detection.

- **Energy Detect**

In the Energy Detect mode the PHY is powered down, except for the register interface, the SQUELCH circuit and the ENERGYON logic. The ENERGYON logic is used to detect the presence of valid energy from 100Base-TX, 10Base-T or Auto-negotiation signals.
When energy is received (link pulses or packet) the PHY power on. It automatically resets itself into the state it had prior to power-down and asserts the interrupt output if enabled ("intreq"). The first and possibly the second packet to activate the PHY will be lost. Therefore this mode cannot be used for Wake-on-Lan detection. Energy Detect power down mode cannot be used when Auto-negotiation is enabled (Bit 0.12 must be cleared).

- **PLL-Only**

In the PLL-Only mode, only the PLL is active. This mode is required to have the 500 MHz clock from the PLL available in the system, while the rest of the PHY is not used (output "clk_out_500"). This clock can be used in a RGMII clocking scheme when the embedded PHY is bypassed and the MAC is operating in Gbps mode.

- **True Power Down**

In True Power Down mode the entire PHY is powered down except the SMI register interface. Access to the SMI registers is still possible, provided the external input clock ("clk_in") is active (25 MHz or 50 MHz). This is required in order to power up the PHY again via a register access.
The True Power Down mode is different from the General Power Down mode that is mentioned in the standard. This General Power Down mode corresponds to the PLL-Only mode, described in the table item above (please note that the standard does not say that the PLL is on in the so-called General Power Down mode).

- **Disabled**

In the Disabled mode the entire PHY is disabled. Even register access is not possible anymore.

- **General Power-Down**

This power-down is controlled by register. In this mode the entire PHY, except the management interface, is powered-down and stays in that condition as long as the relevant bit is held high. When it is cleared, the PHY powers up and is automatically reset. General power down mode requires external clock of 25/50MHz to be always present. In this case PLL can be shut off.

WOL (Wakeup on LAN)

The PHY should be operating in normal mode for WOL detection. The circuit monitors the network traffic for the special packets called the "Magic Packet". If the magic packet is received, it must further check to see if the magic packet's content has the correct information (CRC check on the received magic packet). If the magic packet contents are correct then the circuit issue an interrupt to the system host controller to wake up the reset of the circuitry and get the client out of the sleep mode.
A Magic packet is a frame addressed to the device, either a unicast to the programmed address or a broadcast with anywhere in a payload containing 48 '1's followed by 16 repetitions of the client's 48-bit MAC address, possibly followed by four or six byte password
The WoL design monitors all the packets transmitted from the Ethernet PHY and generates the WoL interrupt and the status for the following packet conditions.

- Detection of Magic Packet with Perfect DA
- Detection of Magic Packet with Broadcast DA
- Detection of only Magic Packet (Debug)

Refer to register specification for details on control.
The interrupt can be masked through the register 30 (Interrupt mask register) and interrupt status is available on register29 (interrupt status register).
MAC/Host controller should program the 48 bit MAC Address (Destination address) to the MAC address register.

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EEE-LPI functional overview

The EEE-LPI operation is as shown below.

Ts - The period of time that the Tx sends the sleep code on detecting the LPI code on the Tx MII/RMII interface and the Rx detects the sleep code and indicates the MII/RMII that it entering the LPI mode.

Figure: EEE-LPI function overview diagram

Tq – The period of the time that the Tx enters the quiet state by shutting down the link before sending the refresh code and the Rx-detects and enters the quiet state to shut down PMA modules and the MII interface clock to save the power.

Tr - The period of the time that the Tx sends the refresh after the quiet state to make sure that the link is still alive in LPI mode and Rx checks for the refresh code after the Quiet period to confirm that the link is OK.

The 4B/5B code groups to indicate the SLEEP-REFRESH: LPI code

The /P/ code-group is used to indicate LPI.

PCS code-group [4:0] 4 3 2 1 0	Name	MII (TXD/RXD) <3:0> 3 2 1 0	Interpretation
1 1 1 1 1	I	undefined	IDLE; used as inter-stream fill code
0 0 0 0 0	P	0 0 0 1	SLEEP: LPI code only for the EEE capability. Otherwise, Invalid code; refer to Table 22–1 and Table 22–2

The SLEEP code-group (/P/) is used to delineate the boundary of an LPI sequence and to deliver a refresh signal to maintain clock synchronization and verify the link status. The SLEEP code-groups are emitted from, and interpreted by, the PCS.

Reset

- Power-Down reset:

Automatically activated when the PHY comes out of power-down mode. The internal power-down reset is extended by 256us after exiting the power-down mode to allow the PLL to stabilize before the logic is released from reset.

- Hardware reset

Connected to the reset_n input. If the reset_n input is driven by an external source, it should be held LOW for at least 100us to ensure that the core is properly reset.

- Software reset

Activated by writing a register. This signal is self- clearing. After the register-write, internal logic extends the reset by 256us to allow PLL-stabilization before releasing the logic from reset.

Clocking

The "clk_in" input must be 25 MHz or 50 MHz.

For RMII mode "clk_in" should be 50MHz.

The PLL inside the AFE requires 25 MHz as input. The incoming clock is divided by 2 or not in the AFE, depending on the input frequency. The selection is based on the value of the "cfg_clk_freq" input.

The PLL generates the 500 MHz "clk_out_500" output clock. Furthermore it generates a TX clock and a RX clock for the digital part. The frequency of the TX clock is either 100 MHz (10BT mode) or 125 MHz (100Base-TX mode). The frequency of the RX clock is 125 MHz.

LED's

The PHY provides six LED signals (active low). The definition of the LED's is largely driven by

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industry standard practices and backward compatibility.
The Transmit activity LED output is driven low when CRS is active (high) at transmit. When CRS becomes inactive, the Transmit Activity LED output is extended by 128ms.
The Receive activity LED output is driven low when CRS is active (high) at receive. When CRS becomes inactive, the Receive Activity LED output is extended by 128ms.In Loopback mode, this LED is not active.
The Link LED output is driven low whenever the PHY detects a valid link. The use of the 10Mbps or 100Mbps link test status is determined by the condition of the internally determined speed selection.
The 100M Speed LED output is driven low when the operating speed is 100Mbit/s or during Auto-negotiation. This LED will go inactive when the operating speed is 10Mbit/s or power down mode
The 10M Speed LED output is driven low when the operating speed is 10Mbit/s. This LED will go inactive when the operating speed is 100Mbit/s or during power down mode.
The Full-Duplex LED output is driven low when the link is operating in Full-Duplex mode.

1.9 Register Description

Total number of registers are limited to address 0 to 31 as defined in the standard. To address more than 32 address space, indirect addressing scheme is used.
The register set is defined into two groups.
Direct addressable registers – Direct access from SMI/MDIO interface from MAC:

- MDIO register set

Indirect addressable registers -

- Analog register set
- WOL registers set

MDIO register set

Register #	Description	Group
0	Basic Control Register	Basic
1	Basic Status Register	Basic
2	PHY Identifier 1	Extended
3	PHY Identifier 2	Extended
4	Auto-Negotiation Advertisement Register	Extended
5	Auto-Negotiation Link Partner Ability Register (BP/NP)	Extended
6	Auto-Negotiation Expansion Register	Extended
7	Next Page Transmit Register	Extended
16	Silicon Revision Register	Vendor-specific
17	Mode Control/Status Register	Vendor-specific

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18	Special Modes	Vendor-specific
19	SMII configuration status register	Vendor-specific
20	TSTCNTL – Testability/Configuration Control	Vendor-specific
21	TSTREAD1 – Testability data Read for LSB	Vendor-specific
22	TSTREAD2 – Testability data Read for MSB	Vendor-specific
23	TSTWRITE – Testability/Configuration data Write	Vendor-specific
24	Power down control register	Vendor-specific
27	Control / Status Indication Register	Vendor-specific
28	Special internal testability controls	Vendor-specific
29	Interrupt Source Register	Vendor-specific
30	Interrupt Mask Register	Vendor-specific
31	PHY Special Control/Status Register	Vendor-specific

MACPHY_BASIC_CONTROL
Address: 00
Basic Control Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	R/W SC	0x0	reset 1= software reset. Bit is self-clearing.
14	RW	0x0	loopback 1 = loopback mode, 0 = normal operation
13	RW	0x0	speed_select 1 = 100Mbps, 0 = 10Mbps. Ignored if Auto Negotiation is enabled Set by cfg_mode[2:0]
12	RW	0x0	auto_neg_en Field0000 Abstract Auto-Negotiation Enable 1 = enable auto-negotiate process (overrides 0.13 and 0.8) 0 = disable auto-negotiate process Set by cfg_mode[2:0]

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Bit	Attr	Reset Value	Description
11	RW	0x0	power_down 1 = General power down mode 0 = normal
10	RW	0x0	isolate 1 = electrical isolation of PHY from MII 0 = normal operation Set by cfg_mode[2:0]
9	R/W SC	0x0	restart_auto_neg restart_auto_negotiation 1 = restart auto-negotiate process 0 = normal operation. Bit is self-clearing.
8	RW	0x0	duplex_mode 1 = full duplex, 0 = half duplex. Ignored if Auto Negotiation is enabled Set by cfg_mode[2:0]
7	RW	0x0	collision_test 1 = enable COL test, 0 = disable COL test
6:0	RO	0x0	reserved

MACPHY_BASIC_STATUS
Address: 01
Basic Status Register

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	bt4 Field0000 Abstract 100Base-T4 1 = T4 able 0 = no T4 ability
14	RO	0x1	bt100_full_duplex 100Base-TX Full Duplex 1 = TX with full duplex 0 = no TX full duplex ability
13	RO	0x1	bt100_half_duplex 100Base-TX Half Duplex 1 = TX with half duplex 0 = no TX half duplex ability
12	RO	0x1	bt10_full_duplex 10Base-T Full Duplex 1 = 10Mbps with full duplex 0 = no 10Mbps with full duplex ability
11	RO	0x1	bt10_half_duplex 10Base-T Half Duplex 1 = 10Mbps with half duplex 0 = no 10Mbps with half duplex ability

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Bit	Attr	Reset Value	Description
10:6	RO	0x0	reserved
5	RO	0x0	auto_negotiate_complete 1 = auto-negotiate process completed 0 = auto-negotiate process not completed
4	RC	0x0	remote_fault 1 = remote fault condition detected 0 = no remote fault
3	RO	0x1	auto_negotiate_ability 1 = able to perform auto-negotiation function 0 = unable to perform auto-negotiation function
2	RW	0x0	link_status 1 = link is up 0 = link is down
1	RC	0x0	jabber_detect 1 = jabber condition detected 0 = no jabber condition detected
0	RO	0x1	extended_capabilities 1 = supports extended capabilities registers 0 = does not support extended capabilities registers

MACPHY_PHY_IDENTIFIER1
Address: 02
PHY Identifier 1

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	phy_id_num PHY ID number Assigned to the 3rd through 18th bits of the Organizationally Unique Identifier (OUI), respectively. default:cfg_phy_id[15:0]

MACPHY_PHY_IDENTIFIER2
Address: 03
PHY Identifier 2

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:10	RW	0x00	phy_id_num PHY ID number Assigned to the 19th through 24th bits of the OUI default:cfg_phy_id[21:16]
9:4	RW	0x00	mod_num Model number Six bit manufacturer's model number default:cfg_model_nr[5:0]

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	rev_num Revision number Four bit manufacturer's revision number. default:cfg_rev_nr[3:0]

MACPHY_AUTO_NEG_ADV
Address: 04
AUTO NEGOTIATION ADVERTISEMENT

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	next_page 1 = next page capable
14	RO	0x0	reserved
13	RW	0x0	remote_fault 1 = remote fault detected 0 = no remote fault
12	RO	0x0	reserved
11:10	RW	0x0	pause_operation 00 = No PAUSE 01 = -Asymmetric PAUSE toward link partner 10 = Symmetric PAUSE 11 = Both Symmetric PAUSE and Asymmetric PAUSE toward local device
9	RO	0x0	base100_t4 100Base-T4 1 = T4 able, 0 = no T4 ability This core does not support 100Base-T4.
8	RW	0x0	bt100_full_duplex 100Base-TX Full Duplex 1 = TX with full duplex, 0 = no TX full duplex ability
7	RW	0x0	bt100_mod 100Base-TX 1 = TX able, 0 = no TX ability default:cfg_mode[2:0]
6	RW	0x0	bt10_full_duplex 10Base-T Full Duplex 1 = 10Mbps with full duplex 0 = no 10Mbps with full duplex ability default:cfg_mode[2:0]
5	RW	0x0	bt10_mod 10Base-T 1 = 10Mbps able, 0 = no 10Mbps ability default:cfg_mode[2:0]

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Bit	Attr	Reset Value	Description
4:0	RW	0x01	selector_field [00001] = IEEE 802.3

MACPHY_AUTO_NEG_LINK
Address: 05

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RO	0x0	next_page 1 = next page capable, 0 = no next page ability This part does not support next page ability.
14	RO	0x0	acknowledge 1 = link code word received from partner 0 = link code word not yet received
13	RO	0x0	remote_fault 1 = remote fault detected, 0 = no remote fault
12:11	RO	0x0	reserved
10	RO	0x0	pause_operation 1 = Pause Operation is supported by remote MAC, 0 = Pause Operation is not supported by remote MAC
9	RW	0x0	base100_t4 1 = TX with full duplex, 0 = no TX full duplex ability
8	RW	0x0	bt100_full_duplex 100Base-TX Full Duplex 100Base-TX Full Duplex 1 = TX with full duplex, 0 = no TX full duplex ability
7	RO	0x0	bt100_mod 100Base-TX 1 = TX able 0 = no TX ability
6	RO	0x0	bt10_full_duplex 10Base-T Full Duplex 1 = 10Mbps with full duplex 0 = no 10Mbps with full duplex ability
5	RO	0x0	bt10_mod 1 = 10Mbps able, 0 = no 10Mbps ability
4:0	RW	0x00	selector_field [00001] = IEEE 802.3

MACPHY_AUTO_NEG_EXP
Address: 06
AUTO NEGOTIATION EXPANSION

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RC	0x0	parallel_detection_fault 1 = fault detected by parallel detection logic 0 = no fault detected by parallel detection logic
3	RO	0x0	link_partner_next Link Partner Next Page Able 1 = link partner has next page ability 0 = link partner does not have next page ability
2	RO	0x1	next_page_able 1 = local device has next page ability 0 = local device does not have next page ability
1	RC	0x0	page_received 1 = new page received 0 = new page not yet received
0	RO	0x0	link_parter_auto Link Partner Auto- Negotiation Able Link Partner Auto- Negotiation Able 1 = link partner has auto-negotiation ability 0 = link partner does not have auto-negotiation ability

MACPHY_NEXT_PAGE_TRANSMIT
Address: 07

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	next_page 0 =last page. 1 =additional Next Page(s) will follow.
14	RO	0x0	reserved
13	RW	0x0	message_page 0 = Unformatted Page. 1 = Message Page.
12	RW	0x0	acknowledge_2 0 = cannot comply with message. 1 = will comply with message.
11	RW	0x0	toggle 0= previous value of the transmitted Link Code Word equaled logic one. 1= previous value of the transmitted Link Code Word equaled logic zero
10:0	RW	0x001	code_field Message/Unformatted Code field 11 bit code word to be transmitted as next page received from Link Partner

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MACPHY_SILICON_REVISION
Address: 16

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:6	RO	0x1	silicon_revision Four bit silicon revision identifier
5:0	RO	0x0	reserved

MACPHY_MODE_CONTROL_STATUS
Address: 17

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13	RW	0x0	edpwrdown Enable the Energy Detect Power-Down mode: 0 = Energy Detect Power-Down is disabled 1 = Energy Detect Power-Down is enabled
12	RO	0x0	reserved
11	RW	0x0	lowsqen The Low_Squelch signal is equal to LOWSQEN AND EDPWRDOWN. Low_Squelch = 1 implies a lower threshold (more sensitive). Low_Squelch = 0 implies a higher threshold (less sensitive).
10	RW	0x0	mdprebp Management Data Preamble Bypass: 0 = detect SMI packets with Preamble 1 = detect SMI packets without preamble
9	RW	0x0	farloopback Force the module to the FAR Loop Back mode, i.e. all the received packets are sent back
8	RW	0x0	fastest Auto-Negotiation Test Mode 0 = normal operation 1 = activates test mode
7	RW	0x0	automdix_en AutoMDIX enable bit 1= Auto detect MDI/MDIX mode 0= Manual set of MDI/MDIX mode according to bit MDI mode Set by cfg_auto mdix_ mode input
6	RW	0x0	mdi_mode Manual set of MDI / MDIX mode When AutoMDIX_en bit is not set this bit sets the MDI/MDIX mode: 0 = MDI mode 1 = MDIX mode
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4	RW	0x0	dcd_pat_gen When in test mode, 1= Enables DCD measuring pattern generation
3	RW	0x0	phyadbp 1= PHY disregards PHY address in SMI access write.
2	RW	0x0	force_good_link_status 0= normal operation; 1= force 100TX- link active; Note: this bit should be set only during lab testing
1	RO	0x0	energyon ENERGYON - indicates whether energy is detected on the line. After the initial link up, If the Reg17 [3] is set to "1", then it qualifies the energy loss with 256ms timer window to goes to "0" and to enter in to power down mode. Reset to "0" by hardware reset, unaffected by SW reset. Works when AN is disabled.
0	RO	0x0	reserved

MACPHY_SPECIAL_MODES
Address: 18

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:14	RW	0x0	miimode MII Mode: set the mode of the MII: 00= MII interface. 01= RMII interface 10= SMII interface 11= Reserved Set by cfg_mii_mode[1:0]
13	RW	0x0	clkselfreq Clock In Selected Frequency. Set the requested input clock frequency. This bit drives signal that goes to external logic of the Core and select the desired frequency of the input clock: 0 - the clock frequency is 25MHz set by the cfg_clk_freq signal
12	RW	0x0	dspbp DSP Bypass mode. Used only in special lab tests.
11	RW	0x0	sqbp SQUELCH Bypass mode.
10	RW	0x0	fxmode Enables 100Base-FX mode. When enabled, MODE should be set to "011" (100M full-duplex) or "010" (100M half duplex) only.
9	RW	0x0	pllbp PLL Bypass mode. 1= PLL Bypass mode 0= PLL normal mode

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Bit	Attr	Reset Value	Description
8	RW	0x0	adcbp ADC Bypass mode. (debug mode) set by the cfg_adc_bp signal
7:5	RW	0x0	mode PHY Mode of operation. set by the cfg_mode[2:0] signals
4:0	RW	0x00	phyadd PHY Address. The PHY Address is used for the SMI address and for the initialization of the Cipher (Scrambler) key. set by the cfg_phy_add signals

MACPHY_SMII_CONFIGURATION_STATUS
Address: 19

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:14	RW	0x0	tx_fifo_depth Transmitter FIFO depth
13:12	RW	0x0	rx_fifo_depth Receiver FIFO depth
11	RW	0x0	smii_source_sync 1= SMII source Synchronous mode 0= SMII normal mode Set by cfg_smii source synch strap
10	RW	0x0	lpbk_10hd_smii_dis In SMII mode, 10 BT half duplex 1= Disables loopback 0= Enables loopback
9:8	RO	0x0	reserved
7	RC	0x0	t_el_buf_ovf Transmitter Elastic overflow
6	RW	0x0	t_el_buf_udf Transmitter Elastic underflow
5	RC	0x0	r_el_buf_ovf Receiver Elastic overflow
4	RC	0x0	r_el_buf_udf Receiver Elastic underflow
3:0	RO	0x0	reserved

MACPHY_TSTCNTL
Address: 20

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15	RW	0x0	read When setting this bit to "1", the content of the register that is selected by the READ ADDRESS will be latched to the TSTREAD1/2 registers. This bit is self_cleared
14	RW	0x0	write When setting this bit to "1", the register that is selected by the WRITE ADDRESS is going to be written with the data from the TSTWRITE register. This bit is self-cleared.
13	RO	0x0	reserved
12:11	RW	0x0	reg_bank_sel Selects register bank 00 = Selects analog and DSP reg bank 01 = Selects WOL register bank 11 = Selects BIST register bank
10	RW	0x0	test_mode Enable the Testability/Configuration mode: 0 = Testability/Configuration mode disabled 1 = Testability/Configuration mode enabled
9:5	RW	0x00	read_address The address of the Testability/Configuration register that will be latched into the TSTREAD1 and TSTREAD2 registers
4:0	RW	0x00	write_address Field0000 Abstract The address of the Testability/Configuration register that will be written

MACPHY_TST_READ1
Address: 21

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RO	0x0000	read_data When reading registers with a size of less than 16 bits, this register contain the register data, starting from bit 0. When reading registers with a size of more than 16 bits, this register contain the less significant 16 bits of the register data

MACPHY_TST_READ2
Address: 22

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RO	0x0000	read_data When reading registers with a size of less than 16 bits, this register contains zero. When reading registers with a size of more than 16 bits, this register contain the most significant 16 bits of the register data

MACPHY_TST_WRITE
Address: 23

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	write_data This field contains the data that will be written to a specific register on the "Programming" transaction

MACPHY_POWER_DOWN_CONTROL_REGISTER
Address: 24

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	true_power_down 0 = Normal operation 1 = PHY is put in True power down mode Set by cfg_power_down
0	RW	0x0	power_up_reset Controls pulse generation on output pin “powerup_reset” 0 = Power up reset disable 1 = Power up reset enable Set by cfg_powerup_reset

MACPHY_SPECIAL_CONTRL_STATUS_INDICATION
Address: 27

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12	RW	0x0	swrst_fast 1= Accelerates SW reset counter from 256 uS to 10 uS for production testing.
11	RW	0x0	sqeoff Disable the SQE test (Heartbeat): 0 = SQE test is enabled. 1 = SQE test is disabled.
10:6	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
5	RW	0x0	fefien Far End Fault Indication (FEFI) Enable: 0= FEFI generation and detection are disabled 1= FEFI generation and detection are enabled (Default when FXMODE configuration input is HIGH during reset)
4	RO	0x0	xpol Polarity state of the 10Base-T: 0 = Normal polarity 1 = Reversed polarity
3:0	RO	0x0	reserved

MACPHY_SPECIAL_INTERNAL_TESTABILITY_CONTROLS
Address: 28

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1	RW	0x0	pattern_code Codes the pattern to be transmitted when bit 3 is set. Used in production testing.
0	RW	0x0	pattern_gen_en When High enables pattern generation for Transmitter testing

MACPHY_INTERRUPT_SOURCE_FLAGS
Address: 29

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RO	0x0	INT11 1= WOL on correct magic packet only (Debug purpose only) 0= Not source of interrupt
10	RO	0x0	INT10 1= WOL on correct magic packet and DA for broadcast address 0= Not source of interrupt
9	RO	0x0	INT9 1= WOL on magic packet receive and correct DA 0= Not source of interrupt
8	RO	0x0	INT8 1= SMII elastic buffer overflow/underflow 0= not source of interrupt
7	RO	0x0	INT7 1= ENERGYON generated 0= not source of interrupt
6	RO	0x0	INT6 1= Auto-Negotiation complete 0= not source of interrupt

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Bit	Attr	Reset Value	Description
5	RO	0x0	INT5 1= Remote Fault Detected 0= not source of interrupt
4	RO	0x0	INT4 1= Link Down (link status negated) 0= not source of interrupt
3	RO	0x0	INT3 1= Auto-Negotiation LP Acknowledge 0= not source of interrupt
2	RO	0x0	INT2 1= Parallel Detection Fault 0= not source of interrupt
1	RO	0x0	INT1 1= Auto-Negotiation Page Received 0= not source of interrupt
0	RO	0x0	reserved

MACPHY_INTERRUPT_MASK
Address: 30

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11:9	RW	0x0	wol_interrupt_mask_bits
8:0	RW	0x000	mask_bits

MACPHY_PHY_SPECIAL_CONTROL_STATUS
Address: 31

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	bprmg Bypass Remove Glitch - for Production Testing
14:13	RO	0x0	reserved
12	RO	0x0	autodone Auto-negotiation done indication: 0= Auto-negotiation is not done or disabled (or not active) 1= Auto-negotiation is done
11:7	RO	0x0	reserved
6	RW	0x1	enable_4b5b 0= Bypass encoder/decoder. 1= enable 4B5B encoding/decoding. MAC Interface must be configured in MII mode.
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:2	RW	0x0	peed_Indication HCDSPEED value: [001]=10Mbps half-duplex [101]=10Mbps full-duplex [010]=100Base-TX half-duplex [110]=100Base-TX full-duplex
1	RO	0x0	reserved
0	RW	0x0	scramble_disable 0 = enable data scrambling 1 = disable data scrambling

Analog and DSP control registers

Total of 4 SMI registers are used for the different testability options:

Register	SMI Address [Decimal]	Size[bits]	Direction[R/W]
TSTCNTL	20	16	W
TSTREAD1	21	16	R
TSTREAD2	22	16	R
TSTWRITE	23	16	R

Procedure to enable access to Analog and DSP register bank
TSTMODE bit in register 20 (TSTCNTL) should be toggled from 0->1->0->1
Default value of TSTCNTL is zero.
Steps:
1. TSTCNTL = 0x0400
2. TSTCNTL = 0x0000
3. TSTCNTL = 0x0400
If system reset or soft reset is applied same steps should be followed to enable register access.
At any point of time if write to TSTCNTL with TSTMODE =0, access to DSP registers are disabled.
If mode switch happens from 10BT to 100BT, all DSP/AFE registers are set to default values. So any AFE/DSP registers have to be re-initialized in this case.
Procedure to write into Analog and DSP register bank
1. Write the data to TSTWRITE register.
2. Write to the TSTCNTL with the WRITE ADDRESS set to the register address and the WRITE bit set to "1" (and TSTMODE bit is "1"). Bit 11 ("REG_BANK_SEL") should be "00"
Procedure to Read into Analog and DSP register bank
1. Write to the TSTCNTL with the READ ADDRESS set to the register address and the READ bit set to "1".

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2. Read the data from the TSTREAD1 register and if needed (if the register that has been read is larger then 16 bits) read also the data from TSTREAD2 register.

Register List and description

Register name	Write Address [hex]	Register type	Description
MACPHY_DCBLW	00	WO	DC/BLW Canceller Configuration
MACPHY_DBGCNTL	01	WO	Debug control option register
Reserved	02-10		Reserved
MACPHY_CONFIG	11	WO	DSP Configuration options
MACPHY_A1_CONFIG	12	WO	Analog Test Point Configuration
MACPHY_A2_CONFIG	13	WO	Analog Receiver Configuration
MACPHY_A3_CONFIG	14	RW	Analog Receiver Common Mode and DAC Configuration
MACPHY_A4_CONFIG	15	RW	Central Bias Configuration 1
MACPHY_A5_CONFIG	16	RW	Transmitter Configuration
MACPHY_A6_CONFIG	17	RW	PLL Configuration 1
MACPHY_A7_CONFIG	18	WO	Transmitter and Analog Test bus Configuration
MACPHY_A8_CONFIG	1A	WO	PLL Configuration 2
MACPHY_A9_CONFIG	1B	WO	Squelch and RC Calibration Configuration
MACPHY_A10_CONFIG	1C	WO	Central Bias Configuration 2
MACPHY_A11_CONFIG	1D	WO	Analog Spare Register
MACPHY_GAIN1	1E	WO	Gain Register 1

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MACPHY_GAIN2	1F	WO	Gain Register 2

MACPHY_DCBLW
Address: 0x00

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	dacpd BLW DAC power down: If (DACPD_SRC) == 1 then from Reg 0 –BLW DAC is enabled 1 –BLW DAC is powered down.
14:12	RW	0x1	blw_avg_mu
11	RW	0x0	dacpd_src PD control for BLW DAC
10:8	RW	0x3	blw_filt_mu
7	RW	0x0	blw_polarity_100 Direction of BLW correction
6:4	RW	0x5	blwmu_100 Correction Coefficient
3	RW	0x0	dac_test_en Enable Bypass the BLW outputs from "dbg_bp_in"pins (dac_bypass_en =1) or from pattern generation
2	RW	0x1	blw_en_100 BLW Canceling enable
1:0	RW	0x0	blw_window_100

MACPHY_DBGCNTL
Address: 0x01

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:12	RW	0x0	monitor Define which set of signals is connected to the external pins (see 1.4.5) 0000 –Set0 (normal mode) xxxx (except 0000) –Set1-15

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Bit	Attr	Reset Value	Description
11:9	RW	0x0	clk_outm Set the state of the CLK_OUT pin: 000 –High-impedance. 001 –The internal 125MHz receive clock (RX_125) is connected to this pin. 010 –The internal 20MHz receive clock (CLK20) is connected to this pin. 011 –The internal 20MHz transmit clock (TX20) is connected to this pin. 100 –The MII's 25MHz transmit clock (TX_CLK) is connected to this pin. 101 –The internal 25MHz receive clock (RX_25) is connected to this pin. (others) –reserved
8:7	RW	0x0	freezem Set the Freeze mode of the DSP: 00 –No freeze –this is the Continuous mode of the chip. 01 –Freeze Immediately. At this mode the DSP clock is stopped. All other clocks in the chip are remained toggling. 10 –reserved 11 –reserved
6:3	RO	0x0	reserved
2	RW	0x0	tstpts Enable to tri_state (high impedance) the TSTPx pins: 0 –normal mode 1 –The pins are forced into tri-state.
1	RW	0x0	nrziout Enable to output the NRZI stream, during 100Base-TX mode, on the DPECL driver (during 100Base-FX mode, this bit is disregarded): 0 –no signal is generated on the DEPCL output 1 –NRZI signal is generated on the DEPCL output
0	RO	0x0	reserved

MACPHY_CONFIG
Address: 0x11

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10	RW	0x0	phasesrc 0 -CDR Enable
9:2	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
1	RW	0x0	mlt2en When set to "0"–enable Differential MLT-3 to NRZI conversion. When set to "1"–enable normal MLT-3 to NRZI conversion
0	RW	0x0	diffoff When set to "0"–enable the input differentiation mode When set to "1"–disable the input differentiation mode

MACPHY_A1_CONFIG
Address: 0x12

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:14	RO	0x0	analog_id
13	RW	0x0	dac_bypass_en Option to Bypass the BLW outputs from CO_BPIN
12:11	RW	0x0	rcosctst RC Calibration Test Control 00 –Tri-State. 01 –VL select. 10 –CLKOUT select. 11 –VH select.
10	RW	0x0	sqtstsqvrp Enable Squelch SQ Vrefp on ATP: 0 –Test disabled. 1 –Test Squelch SQ Vrefp on ATP
9	RW	0x0	sqtstdavrp Enable Squelch DATA Vrefp ATP: 0 –Test disabled. 1 –Test Squelch DATA Vrefp on ATP
8	RW	0x0	sqtstvcom Squelch Test mode, Enable VCOM on ATP: 0 –Test disabled. 1 –Test Squelch VCOM on ATP
7:6	RW	0x0	dacatpc DAC Analog Test Point Control: 00 –ATP is HIGH-Z 01 –Drive Negative DAC Current on ATP. 10 –Drive Positive DAC Current on ATP. 11 –Drive DAC Common-mode Current on ATP

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Bit	Attr	Reset Value	Description
5	RW	0x0	rxtstib10 Test Receiver 10uA Bias Current on ATP: 0 –Test is disabled. 1 –Drive IB10u on ATP
4	RW	0x0	rxtstadcvrn Test ADC negative reference level: 0 –Test is disabled 1 –Drive ADC positive reference level on ATP
3	RW	0x0	rxtstadcvrp Test ADC positive reference level: 0 –Test is disabled 1 –Drive ADC negative reference level on ATP
2:0	RW	0x0	txtest Transmitter Test modes through ATP

MACPHY_A2_CONFIG
Address: 0x13

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	rxdasel Receiver data output selector: 0 –ADC Output on RXDA[5:0] 1 –Raw Data on RXDA[5:0]
14:12	RW	0x3	rxitrim Receiver reference current trimming
11:10	RW	0x1	rxvmidadj Receiver Vmid adjust control bits
9	RW	0x1	adc_data_rate 0 –ADC data and clock at 125MHz (Functional mode) 1 –ADC data and clock at 62.5MHz (Test mode at half the clock rate)
8	RW	0x1	rxclkinv Invert Receiver Clock control bit: 0 –RX Clock not inverted. 1 –RX Clock inverted.
7	RW	0x0	rxlpfbyp Receiver Low-Pass-Filter bypass: 0 –Low-Pass-Filter is enabled 1 –Low-Pass-Filter is bypassed
6	RW	0x0	rxprepga Receiver Pre-PGA gain: Valid only when RX_SRC=1 0 –x0.278 1 –x0.416

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Bit	Attr	Reset Value	Description
5	RW	0x0	rxmdixen Transmitter MDIX mode enable: Valid only when RX_SRC=1 0 –MDI - Receiver connected to RXP/N. 1 –MDIX - Receiver connected to TXP/N.
4	RW	0x0	rxpd 1- Receiver power down, Valid only when RX_SRC=1
3:1	RW	0x0	rxgain Receiver PGA gain: Valid only when RX_SRC=1 000 –x1.000 001 –x1.250 010 –x1.375 011 –x1.500 100 –x1.625 101 –x1.750 110 –x1.875 111 –x2.000
0	RW	0x0	rx_src Receiver Source: 0 –Receiver block inputs are controlled by the digital state-machines (normal mode). 1 –Receiver block inputs are controlled by register A2_CONFIG bits

MACPHY_A3_CONFIG
Address: 0x14

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:12	RW	0x0	rxftune Receiver LPF POLE tuning: If RXFTUNE_SRC == 1 then from reg. 0000 –nominal frequency - 35%. 0101 –nominal frequency 1111 –nominal frequency + 70%
11	RW	0x0	rxftune_src Field0007 Abstract RX tune Source: 0 –RX tune block inputs are controlled by the digital state-machines (normal mode). 1 –RX tune block inputs are controlled by register A3_CONFIG bits
10	RW	0x0	dacstrbinv Invert DACDA strobe signal

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Bit	Attr	Reset Value	Description
9:8	RW	0x1	dacpole DAC Pole trimming: 00 –nominal -30% 01 –nominal 10 –nominal +30% 11 –nominal +30%
7:5	RW	0x0	dacitrim DAC reference current selector: 000 –-75% 001 –-50% 010 –-25% 011 –Nominal 100 –+25% 101 –+50% 110 –+75% 111 –+100%
4	RO	0x0	reserved
3	RW	0x0	rxcmcntrl3 Receiver CMFB Control bit 3, 0 –Select ATPL, 1- Select ATPH
2	RW	0x0	rxcmcntrl2 Receiver CMFB Control bit 2 (not used)
1	RW	0x0	rxcmcntrl1 Receiver CMFB Control bit 1, 1 –Nominal pulldown current, 0 –Extended pulldown current
0	RW	0x0	rxcmfbpd Receiver Common-mode feedback power-down. Valid only when RX_SRC=1 0 –Common-mode feedback power-on. 1 –Common-mode feedback power-off.

MACPHY_A4_CONFIG
Address: 0x15

Bit	Attr	Reset Value	Description
31:15	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
14:12	RW	0x4	cbrgfbadj Adjust CB reference generator feedback voltage: 000 –1.125 V 001 –1.150 V 010 –1.175volt 011 –1.2 V 100 –1.225 V 101 –1.250 V 110 –1.275 V 111 –1.3 V
11	RO	0x0	reserved
10:8	RW	0x4	cbrxvradj Adjust RX voltage reference: 000 –1 V 001 –1.025 V 010 –1.050 V 011 –1.075 V 100 –1.1 V 101 –1.125 V
7	RW	0x0	dbpd Central bias power down (valid only if CB_SRC=1)
6:4	RW	0x1	cbrxcmadj Adjust receiver common-mode reference voltage: x00 –0.90 v x01 –0.575 v x10 –0.60 v x11 –0.625 v
3:1	RW	0x6	cbtxvbadj Transmitter NMOS bias ON selector: x00 -> 2 (default value) x01 -> 2.25V x10 -> tx_cm_in (external bias voltage from ATP) x11 -> 2.5V
0	RW	0x0	cb_src CB Source: 0 –CB block inputs are controlled by the digital statemachines (normal mode). 1 –CB block inputs are controlled by register A4_CONFIG bits

MACPHY_A5_CONFIG
Address: 0x16

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15	RW	0x1	txclassa Defines Class of Transmitter: 0 –Class B (Code 00 is used to drive zero) 1 –Class A (Code 11 is used to drive zero)
14:12	RW	0x0	txclkphsel TXCLK Phase Selector (For Multi Port Application): 000 –0ns shift. 001 –1ns shift. 010 –2ns shift. 011 –3ns shift. 100 –4ns shift. 101 –5ns shift. 110 –6ns shift. 111 –7ns shift.
11	RW	0x0	txclkinv Invert Transmitter Clock control bit
10	RW	0x1	txclk100en Transmitter Clock 100/125Mhz control bit. Valid only when TX_SRC=1 (smooth mux) 1 –TX Clock is 125Mhz (100BT mode). 0 –TX Clock is 100Mhz (10BT mode).
9	RW	0x0	txsel10 Transmitter mode select (Amplitude control): Valid only when TX_SRC=1 0 –100BT mode. 1 –10BT mode.
8	RW	0x0	txmdixen Transmitter MDIX mode enable: Valid only when TX_SRC=1 0 –MDI - Transmitter connected to TXP/N. 1 –MDIX- Transmitter connected to RXP/N.
7	RW	0x0	txpd Transmitter channel power down Valid only when TX_SRC=1
6:4	RW	0x0	rxslope_10 Transmitter slope control: 111 –Transmit 10BaseT (2/2/2/2)
3:1	RW	0x3	txslope_100 Transmitter slope control: 000 –Transmit 100BaseT (1/1/1/1)
0	RW	0x0	tx_src XMTR Source: 0 –XMTR block inputs are controlled by the digital state-machines (normal mode). 1 –XMTR block inputs are controlled by register A5_CONFIG bits

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MACPHY_A6_CONFIG
Address: 0x17

Bit	Attr	Reset Value	Description
31:14	RO	0x0	reserved
13:12	RW	0x0	pllintratio PLL Receiver Interpolation ratio.[N/A]
11:9	RO	0x0	reserved
8:5	RW	0x6	pllchtrim PLL external charge pump bias current trimmer. (Valid when non-self bias technique is applied –see PLLCHBIASSEL) 0000 –minimum. 1000 –Nominal. 1111 –Maximum.
4	RW	0x0	pllchbiassel Select Bias technique for PLL 0 –Select non self-bias technique. 1 –Select self-bias technique.
3	RW	0x0	pllrstvcopd Disable reset VCO circuitry. 0 –Reset VCO circuit enabled 1 –Reset VCO circuit disabled.
2	RW	0x0	pllcpoff PLL test VCO –Drive External VCO voltage: 0 –Test mode disabled 1 –PLL Charge-pump output is High-Z
1	RW	0x0	pllpd PLL power down control: Valid only when PLL_SRC=1 0 –PLL is enabled 1 –PLL is powered down
0	RW	0x0	pll_src PLL Source: 0 –PLL block inputs are controlled by the digital statemachines (normal mode). 1 –PLL block inputs are controlled by register A6_CONFIG

MACPHY_A7_CONFIG
Address: 0x18

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:13	RW	0x0	pclktst Port Clock Select on ATP 000 –Disabled 001 –TXPH[0] 010 –TXPH[1] 011 –TXPH[2] 100 –TXPH[3] 101 –TXPH[4] 110 –RXCLK 111 –Disabled
12	RW	0x0	pclktsten Port Clock Test Mode Enable 0 –Test disabled. 1 –Test Clocks on ATP.
11:10	RW	0x0	pcbvtst Enables Internal Voltage references on ATP: 00 –ATP is floating. 01 –Internal band-gap voltage driven on ATP 10 –Internal rx adcref voltage driven on ATP 11 –Internal rx cmref voltage driven on ATP
9:8	RW	0x0	pcbctst Enable Internal Current references on ATP: 00 –Test is disabled. 01 –Internal PTAT current driven on ATP. 10 –Internal Const current driven on ATP. 11 –Internal IR Const current driven on ATP
7:6	RW	0x0	pcbextv Enable External Reference Voltage Thru ATP: 00 –Test is disabled. 01 –External RX Vcm. 10 –External RX Vref. 11 –External TX Vref.
5	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
4:0	RW	0x00	txamp Transmitter amplitude Control 00000 0000 –1.573V 0001 –1.617V 0010 –1.6566V 0011 –1.706V 0100 –1.744V 0101 –1.792V 0110 –1.833V 0111 –1.88V 1000 –1.924V 1001 –1.967V 1010 –2.006V 1011 –2.053V 1100 –2.093V 1101 –2.1344V 1110 –2.175V 1111 –2.225V TXAMP[4] - Reserved

MACPHY_A8_CONFIG
Address: 0x1A

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:6	RW	0x0	pllclktst PLL clock test, select clock on ATP: NOT USED
5	RW	0x0	pllclktsten PLL clock test enable on ATP (according to PLLCLKTST[2:0]: 0 –Test is disabled (High-Z) 1 –Enable receiver clock on ATP
4	RW	0x0	plltstvcoen PLL test VCO –Observe/Drive VCO voltage: 0 –Normal 1 –VCO voltage can be observed/driven thru ATP
3:2	RW	0x0	pllrcfactor Trimming of resistor, depends on PLLBIASSEL PLLBIASSEL=1, 00 –4K, 01 –3.6K, 10 -> 3.2K, 11 -> 2.8K PLLBIASSEL=0, 00 –2.4K,01-2K, 10 –1.6K, 11 -> 1.2K
1:0	RO	0x0	reserved

MACPHY_A9_CONFIG
Address: 0x1B

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	rccal_src RC Calibration Source: 0 –RC Calibration block inputs are controlled by the digital state-machines (normal mode). 1 –RC Calibration block inputs are controlled by register A4_CONFIG bits
14	RW	0x0	rcoscpd RC Calibration power down: If RCCAL_SRC == 1 then 0 –RC Calibration circuit not powered down. 1 –RC Calibration circuit powered down.
13	RW	0x0	rcoscmfrq Select RCCLK frequency: If RCCAL_SRC == 1 then 0 –Nominal 1 –Non Controllable
12:9	RO	0x0	reserved
8	RW	0x0	sqfilterbp Squelch Low-Pass-Filter bypass enable
7:6	RW	0x0	sqadjlpf Adjust Squelch LPF poles: 00 –+1 steps 01 –+1 steps 10 –Nominal 11 –-1 steps
5:4	RW	0x0	sqadjref Adjust Squelch reference voltage: 00 –-2 steps. 01 –-1 steps. 10 –Nominal 11 –+1 steps
3	RW	0x0	sqlvth Squelch extended length mode (lower comparator threshold): 0 –Normal 1 –Enable the extended length option
2	RW	0x0	sqpd_x TXP/N Squelch power down –MDIX: 0 –TXP/N Squelch is enabled 1 –TXP/N Squelch is powered down
1	RW	0x0	sqpd RXP/N Squelch power down: 0 –RXP/N Squelch is enabled 1 –RXP/N Squelch is powered down

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Bit	Attr	Reset Value	Description
0	RW	0x0	sq_src Squelch Source: 0 –Squelch blocks inputs are controlled by the digital state-machines (normal mode). 1 –Squelch blocks inputs are controlled by register A4_CONFIG bits

MACPHY_A10_CONFIG
Address: 0x1C

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9	RW	0x0	txda_bypass_en Option to Bypass the TXDA to analog from CO_BPIN
8	RW	0x0	atplen Select LOW Analog Test bus on ATP: (see ATP arch.) 0 –Channel not selected 1 –Channel selected
7	RW	0x0	atphen Select HIGH Analog Test bus on ATP: (see ATP arch.) 0 –Channel not selected 1 –Channel selected
6	RW	0x0	cbchkvbg Test internal Bandgap on ATP: 0 –(Band-gap 1.24volt). 1 –ATP
5:4	RW	0x0	cbextc Enables external bias Current: 00 –I bias generated internally 01 –External PTAT bias current thru ATP 10 –External Const bias current thru ATP 11 –External IPTR bias current thru ATP
3	RW	0x0	cbextvbg Enables external Bandgap reference: 0 –Voltage reference is generated internally (Band-gap 1.24volt). 1 –Drive external band-gap reference thru ATP
2:0	RO	0x0	reserved

MACPHY_A11_CONFIG
Address: 0x1D

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:8	RW	0x0a	cspare Spare Analog Register (Read/Write), Central Bias
7:1	RW	0x1b	pspare Spare Analog Register (Read/Write), PORT
0	RW	0x0	bp_zeta_mode Bypass Zeta mode for the LPF 0 –Zeta mode enable (RXPREPGA control the Zeta as: RXPREPGA = 0 - Zeta = 0.66 (No hig) RXPREPGA =1 - Zeta = 0.33 (Hig) 1 –Zeta mode disable

MACPHY_GAIN1
Address: 0x1E

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	rxprepga_3 RXPREPGA value for Gain=3
14:12	RW	0x0	rxgain_3 RXGAIN value for Gain=3
11	RW	0x0	rxprepga_2 RXPREPGA value for Gain=2
10:8	RW	0x0	rxgain_2 RXGAIN value for Gain=2
7	RW	0x0	rxprepga_1 RXPREPGA value for Gain=1
6:4	RW	0x0	rxgain_1 RXGAIN value for Gain=1
3:1	RW	0x0	rxprepga_0 RXPREPGA value for Gain=0
0	RW	0x0	rxgain_0 RXGAIN value for Gain=0

MACPHY_GAIN2
Address: 0x1F

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	rxprepga_7 RXPREPGA value for Gain=7
14:12	RW	0x0	rxgain_7 RXGAIN value for Gain=7
11	RW	0x0	rxprepga_6 RXPREPGA value for Gain=6

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Bit	Attr	Reset Value	Description
10:8	RW	0x0	rxgain_6 RXGAIN value for Gain=6
7	RW	0x0	rxprepga_5 RXPREPGA value for Gain=5
6:4	RW	0x0	rxgain_5 RXGAIN value for Gain=5
3:1	RW	0x0	rxprepga_4 RXPREPGA value for Gain=4
0	RW	0x0	rxgain_4 RXGAIN value for Gain=4

WOL & LPI register bank

Programming register TSTCNTL [11] (reg20) selects between DSP or PCS_WOL indirect addressing. Programming to '01' makes it PCS_WOL & '00' makes it DSP regbank.
Procedure to write into WOL register bank
1. Write the data to TSTWRITE register.
2. Write to the TSTCNTL with the WRITE ADDRESS set to the register address and the WRITE bit set to "1". Bit 11 ("REG_BANK_SEL") should be "01"
Procedure to Read into WOL register bank
1. Write to the TSTCNTL with the READ ADDRESS set to the register address and the READ bit set to "1". Bit 11 ("REG_BANK_SEL") should be "01"
2. Read the data from the TSTREAD1 register.

Register Name	Read Address	Description
MACPHY_MAC_DA_ADDRS_15_TO_0	0x0	6-Bytes MAC Address
MACPHY_MAC_DA_ADDRS_31_TO_16	0x1	6-Bytes MAC Address
MACPHY_MAC_DA_ADDRS_47_TO_32	0x2	6-Bytes MAC Address
MACPHY_WOL_CONTROL	0x3	WOL control register
MACPHY_EPHY_LPI_CNTRL	0x4	EEE LPI control register
MACPHY_EPHY_LPI_RX_TW_TS_TIMER	0x5	EEE LPI receiver wait & sleep timer count [7:0] - sleep timer count [12:8] - wait timer count
MACPHY_EPHY_LPI_RX_TI_TQ_TIMER	0x6	EEE LPI receiver quiet & idle timer count [7:0] - quiet timer count

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		[12:8] - idle timer count
MACPHY_EPHY_LPI_RX_LINK_FAIL_TIMER	0x7	EEE LPI receiver link fail timer count
MACPHY_EPHY_LPI_TX_TIMER	0x8	EEE LPI transmitter sleep & quiet timer count
RESERVED	0x9	Unused
MACPHY_WOL_STATUS	0xA	WOL status bits
MACPHY_EPHY_LPI_RX_WAKE_ERR_CNTR	0xB	EEE LPI receiver wake error count
MACPHY_EPHY_LPI_STATUS	0xC	EEE LPI transmit and receiver status
RESERVED	0xD-0x1F	Unused

MACPHY_MAC_DA_ADDRS_15_TO_0
Address: 0x00

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	mac_da_addrs_15_to_0 Two bytes of MAC address

MACPHY_MAC_DA_ADDRS_31_TO_16
Address: 0x01

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mac_da_addrs_31_to_16 Two bytes of MAC address

MACPHY_MAC_DA_ADDRS_47_TO_32
Address: 0x02

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	mac_da_addrs_47_to_32 Two bytes of MAC address

MACPHY_WOL_CONTROL
Address: 0x03

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Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4	RW	0x0	wol_check_sa_unicast_en Enable the unicast check on the SA address This configuration is used along with the 'i_wol_mag_pkt_da_en'and the 'i_wol_mag_pkt_broadcast_en'config register bits 1 - enable 0 –disable
3	RW	0x0	wol_mag_pkt_en Enable Wake of LAN for only Magic packet detection 1 - enable 0 –disable
2	RW	0x0	wol_mag_pkt_broadcast_en Enable Wake of LAN for Magic packet detection with broadcast DA 1 - enable 0 –disable
1	RW	0x0	wol_mag_pkt_da_en Enable Wake of LAN for Magic packet detection with perfect DA 1 - enable 0 - disable
0	RW	0x0	wol_en Enable the Wake of LAN detection logic 1 = Enable 0 = Disable

MACPHY_EPHY_LPI_CNTRL
Address: 0x04

Bit	Attr	Reset Value	Description
31:12	RO	0x0	reserved
11	RW	0x0	ephy_10bte_mode_enable 1 = Enable EEE 10BTe mode in AFE 0 = Disable EEE 10BTe mode in AFE
10	RW	0x0	ephy_lpi_rx_mode_se Mode selection to AFE 100BT signal detect circuit 1 = Normal mode If input is <200mv peak to peak the detector output is LOW If input is >100mv peak to peak the detector output is HIGH 0 = LPI mode If input is <200mv peak to peak the detector output is LOW If input is >400mv peak to peak the detector output is HIGH
9	RW	0x0	ephy_lpi_rx_100bt_sig_det_enable 1 = Enable 100BT signal detect circuit in AFE for LPI mode operation in RX side 0 = Disable 100BT signal detect circuit in AFE

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Bit	Attr	Reset Value	Description
8	RW	0x0	ephy_lpi_rx_wake_err_cntr_clr Clears the wake error counter status register. It is self-clearing bit.
7:3	RW	0x00	ephy_lpi_rx_clk_stop_en_range This register range should be programmed between d'10 to d'31 indicating that that the Rx clock to MII interface should be killed on detection of LPI mode after this programmable count. This register is valid only when the ephy_lpi_rx_clk_stop register bit is enabled. (Default value = 'hA)
2	RW	0x0	ephy_lpi_rx_clk_stop 1 = Gate the Rx- clock to MII once the LPI is detected from the Rx-PHY 0 = Do no gate the Rx-clock during LPI mode
1	RW	0x0	ephy_lpi_rx_enable 1 = LPI mode is enabled in RX-side 0 = LPI mode is disabled in RX-side
0	RW	0x0	ephy_lpi_tx_enable 1 = LPI mode is enabled in TX-side 0 = LPI mode is disabled in TX-side

MACPHY_EPHY_LPI_RX_TW_TS_TIMER
Address: 0x05

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	ephy_lpi_rx_tw_timer The LPI-Rx wait counter which is decremented internally with 1 μs event to generate the timer done status. The clock frequency which is used to generate the 1 μs event is 25 MHz (40 ns period)
7:0	RW	0x00	ephy_lpi_rx_ts_timer The LPI Sleep or Refresh counter which is decremented internally with 1 μs event to generate the timer done status. The clock fre

MACPHY_EPHY_LPI_RX_TI_TQ_TIMER
Address: 0x06

Bit	Attr	Reset Value	Description
31:13	RO	0x0	reserved
12:8	RW	0x00	ephy_lpi_rx_ti_timer The LPI-Rx idle counter which is decremented internally with Rx clock event of 25 MHz clock frequency (40 ns period)
7:0	RW	0x00	ephy_lpi_rx_tq_timer The LPI-Rx quiet counter which is decremented internally with 100 μs event to generate the timer done status. The clock frequency which is used to generate the 1 μs event is 25 MHz (40 ns period)

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MACPHY_EPHY_LPI_RX_LINK_FAIL_TIMER
Address: 0x07

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	ephy_lpi_rx_link_fail_timer The LPI-Rx link fail timer counter which is decremented internally with 1 µs event to generate the timer done status. The clock frequency which is used to generate the 1 µs event is 25 MHz (40 ns period)

MACPHY_EPHY_LPI_TX_TIMER
Address:0x08

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	ephy_lpi_tx_tq_timer Quiet timer value
7:0	RW	0x00	ephy_lpi_tx_ts_tr_timer Sleep/refresh timer value

MACPHY_WOL_STATUS
Address: 0x0A

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	wol_mag_pkt_sts Set to "1" on reception of only Magic packet (debug)
1	RW	0x0	wol_mag_pkt_broadcast_sts Set to "1" on reception of Magic packet with broadcast DA match
0	RW	0x0	wol_mag_pkt_da_sts Set to "1" on reception of Magic packet with perfect programmed DA match

MACPHY_EPHY_LPI_RX_WAKE_ERR_CNTR
Address: 0xB

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	ephy_lpi_rx_wake_err_cntr The LPI_rx wake error counter status. This counter clears on setting the register bit ephy_lpi_rx_wake_err_cntr_clr. The counter held all ones in case of overflow

MACPHY_EPHY_LPI_STATUS

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Address: 0xC

Bit	Attr	Reset Value	Description
31:11	RO	0x0	reserved
10:8	RW	0x0	ephy_lpi_tx_state PCS-Tx FSM state
7	RW	0x0	ephy_lpi_tx_detected 1 = EPHY-Tx detected the LPI mode and entering the LPI state 0 = EPHY-Tx not detected the LPI mode
6	RW	0x0	ephy_lpi_tx_quiet_state 1 = EPHY-TX LPI state machine is in quiet state in which power down action has been taken 0 = LPI-FSM is not in quiet state
5	RW	0x0	ephy_lpi_rx_quiet_state 1 = EPHY-RX LPI state machine is in quiet state in which power down action has been taken 0 = LPI-FSM is not in quiet state
4	RW	0x0	ephy_lpi_rx_detected 1 = EPHY-Rx detected the LPI mode and entering the LPI state 0 = EPHY-Rx not detected the LPI mode
3:0	RW	0x0	ephy_pcs_rx_fsm_state PCS-Rx FSM state

BIST registers

Procedure to write into BIST register bank

1. Write the data to TSTWRITE register.
2. Write to the TSTCNTL with the WRITE ADDRESS set to the register address and the WRITE bit set to "1". Bit 11 ("REG_BANK_SEL") should be "11"

Procedure to Read into WOL register bank

1. Write to the TSTCNTL with the READ ADDRESS set to the register address and the READ bit set to "1". Bit 11 ("REG_BANK_SEL") should be "11"
2. Read the data from the TSTREAD1 register.

Register Name	Read Address	Description
BIST_CTRL	0x00	BIST Controls
BIST_STATUS	0x01	BIST status bits
BIST_PREAMBLE_SFD_IPG_LENGTH	0x02	Preamble, SFD & IPG length
BIST_PACKET_DATA_LENGTH	0x03	Packet data length
BIST_DA_0	0x04	Lower 16 bits of DA value

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BIST_DA_1	0x05	Middle 16 bits of DA value
BIST_DA_2	0x06	Higher 16 bits of DA value
BIST_SA_0	0x07	Lower 16 bits of SA value
BIST_SA_1	0x08	Middle 16 bits of SA value
BIST_SA_2	0x09	Higher 16 bits of SA value
BIST_NIBBLE_CNT	0x0A	Programmable total count of nibble sent by the Bist generator
BIST_ERROR_COUNT	0x0B	Bit error count
BIST_RECEIVED_NIBBLE_CNT	0x0C	Total received nibble count
BIST_MISC_CTRL	0x11	BIST Controls

MACPHY_BIST_CTRL
Address: 0x00

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15	RW	0x0	bist_measure_freeze Freezes the Bist measure & is applicable only when bist_tx_en_sequence_en is set to high
14	RW	0x0	bist_da_sa_fcs_en 0 –Disable sending FCS & DA SA 1 –Enable sending FCS & DA SA
13:8	RW	0x00	bist_tx_en_cnt Counts the number of tx_en before asserting rxdv_detect_error when rxdv is not detected in bist_checker (see bit rxdv_detect_error). This register is internally multiplied by 4
7	RW	0x0	bist_tx_en_sequence_en 1 - Generation of bist signals is continuos 0 - Generation of bist signals is governed by register bit bist_tx_en_cnt

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Bit	Attr	Reset Value	Description
6:4	RW	0x0	bist_prbs_select Different type of PRBS is selected 000 –PRBS4 001 –PRBS5 010 –PRBS6 011 –PRBS7 100 –PRBS8 101 –PRBS9 110 –PRBS12 111 –PRBS13
3:2	RW	0x0	bist_prbs_constant_data_sel 00 - Constant data is sent (4'h6) 01 - Divide by4 clock pattern is sent 10 - Ramp data is sent 11 - PRBS output is sent (see bist_prbs_select)
1	RW	0x0	bist_valid This is valid only when bist_enable is set high 0 –Send zeros on txd, tx_er & tx_en 1 –Send internal generator output
0	RW	0x0	bist_enable Selects between bist generator output or from MAC for PCS interface 0 –From Mac (txd, tx_er & tx_en) 1 - From internal bist generator

MACPHY_BIST_STATUS
Address: 0x01

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x0	rx_er_detect_error Indicates rx_er is detected when rxdv is high
2	RW	0x0	rxdv_detect_error Indicates the rxdv is not detected even after programable tx_en i Indicates the rxdv is not detected even after programable tx_en is sent(see bits bist_tx_en_cnt)
1	RO	0x0	bist_fail Indicates the bist failure. It is asserted either check on – 1. Bit error is found 2. RXDV is not detected 3. Error injection is failed (not received expected errors) 4. RX_ER detected when RXDV is high 5. Mismatch between number of transmitted & received nibbles

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Bit	Attr	Reset Value	Description
0	RO	0x0	bist_done Indicates the completion of bist check. It is asserted either check on – 1. Bit error count is saturated to 0xFFFF 2. Programmable wait time is over after the completion of all packets are sent by the Bist generator 3. Forced by programming bist_measure_freeze

MACPHY_BIST_PREAMBLE_SFD_IPG_LENGTH
Address: 0x02

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:8	RW	0x00	bist_ipg_length Length of IPG in bytes (96 bits as per IEEE standard)
7:4	RW	0x0	bist_sfd_length Length of SFD in bytes
3:0	RW	0x0	bist_preamble_length Length of preamble in bytes

MACPHY_BIST_PACKET_DATA_LENGTH
Address: 0x03

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:13	RW	0x0	wait_after_tx_en_cnt Number of 1us wait after the completion of all packets & assert bist_done to the checker. It is effective when bist_tx_en_sequence_en is set to zero. It is internally multiply by 2.
12:0	RW	0x0000	bist_data_length Length of Packet data in bytes Field0000 Description

MACPHY_BIST_DA_0
Address: 0x04

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_da_0 15:0] of DA

MACPHY_BIST_DA_1
Address: 0x05

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Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_da_1 [31:16] of DA

MACPHY_BIST_DA_2
Address: 0x06

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_da_2 [47:32] of DA

MACPHY_BIST_SA_0
Address: 0x07

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_sa_0 [15:0] of SA

MACPHY_BIST_SA_1
Address: 0x08

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_sa_1 31:16] of SA

MACPHY_BIST_SA_2
Address: 0x09

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_sa_2 47:32] of SA

MACPHY_BIST_NIBBLE_CNT
Address: 0x0A

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
15:0	RW	0x0000	bist_nibble_cnt Number of nibbles is transmitted by the Bist generator. Formula for the calculation of this register: (bist_data_length*2) *(tx_en_cnt * 4). This is compared against received nibble count at the Bist checker & asserts bist_fail if there is mismatch.

MACPHY_BIST_ERROR_COUNT
Address: 0x0B

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_err_count Indicates the bit error count when the word counter reached bist_word_cnt (Word counter reaches first)

MACPHY_BIST_RECEIVED_NIBBLE_CNT
Address: 0x0C

Bit	Attr	Reset Value	Description
31:16	RO	0x0	reserved
15:0	RW	0x0000	bist_received_nibble_cnt Indicates the total count of nibble received

MACPHY_BIST_MISC_CTRL
Address: 0x11

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RW	0x0	bist_error_inject_en Error on one nibble (4 bit error) per packet is injected when this bit is set to high. It should be used NOT with sequential mode.

1.10 Interface Description

Table 38-11 Macphy Interface Description

Module pin	Direction	Pad name	IOMUX setting
LED interface			
macphy_link	I/O	IO_FEPHYled_linkm0_GPIO0D4vccio1	GPIO0D_IOMUX_SEL[9:8]=2'b01
macphy_rx/tx	I/O	IO_FEPHYLEDrxm0_FEPHYLEDtxm0_GPIO0D5vccio1	GPIO0D_IOMUX_SEL[11:10]=2'b01:rx GPIO0D_IOMUX_SEL[11:10]=2'b10:tx
macphy_speed10/duplex	I/O	IO_FEPHYLEDspeed10_FEPHYLEDduplex_SDMMC0pwrenm1_GPIO0D6vccio1	GPIO0D_IOMUX_SEL[13:12]=2'b01:speed 10

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			GPIO0D_IOMUX_SEL[13:12]=2'b 10:duplex
macphy_spe ed100	I/O	IO_FEPHYLEDspeed100_GPIO0D7vccio1	GPIO0D_IOMUX_SEL[15:14]=2'b 01

1.11 Application Notes

14.1.1 GRF Register Summary

GRF Register	Register Description
GRF_MACPHY_CON1[10]	LED's speed 100 polarity selection 1'b1:active high 1'b0:active low
GRF_MACPHY_CON1[11]	LED's speed 10 polarity selection 1'b1:active high 1'b0:active low
GRF_MACPHY_CON1[12]	LED's link polarity selection 1'b1:active high 1'b0:active low
GRF_MACPHY_CON1[13]	LED's duplex polarity selection 1'b1:active high 1'b0:active low
GRF_MACPHY_CON1[14]	LED's rx polarity selection 1'b1:active high 1'b0:active low
GRF_MACPHY_CON1[15]	LED's tx polarity selection 1'b1:active high 1'b0:active low

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Chapter 40 SAR-ADC

15.1 Overview

The ADC is a 6-channel signal-ended 10-bit Successive Approximation Register (SAR) A/D Converter. It uses the supply and ground as it reference which avoid use of any external reference. It converts the analog input signal into 10-bit binary digital codes at maximum conversion rate of 1MSPS with 13MHz A/D converter clock.

15.2 Block Diagram

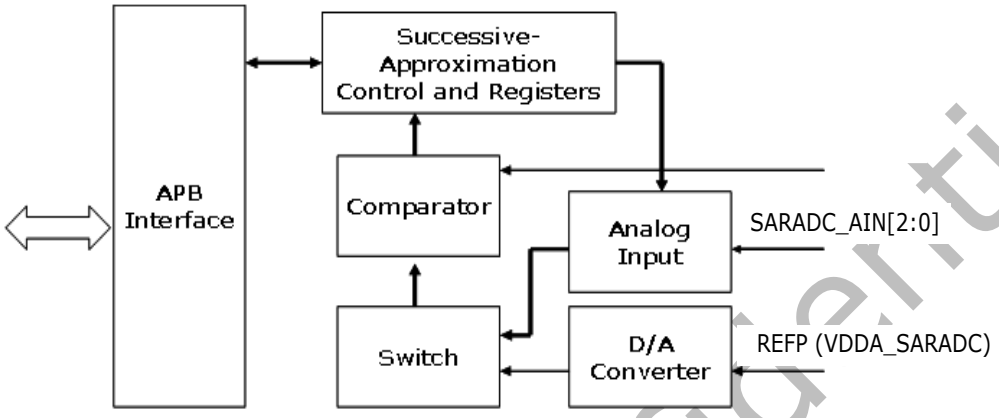

Fig. 40-1 SAR-ADC block diagram

Successive-Approximate Register and Control Logic Block

This block is exploited to realize binary search algorithm, storing the intermediate result and generate control signal for analog block.

Comparator Block

This block compares the analog input SARADC_AIN[2:0] with the voltage generated from D/A Converter, and output the comparison result to SAR and Control Logic Block for binary search. Three level amplifiers are employed in this comparator to provide enough gain.

15.3 Function Description

15.3.1 APB Interface

In RK3228H, SAR-ADC works at single-sample operation mode.

This mode is useful to sample an analog input when there is a gap between two samples to be converted. In this mode START is asserted only on the rising edge of CLKIN where conversion is needed. At the end of every conversion EOC signal is made high and valid output data is available at the rising edge of EOC. The detailed timing diagram will be shown in the following.

15.4 Register description

15.4.1 Registers Summary

Name	Offset	Size	Reset Value	Description
SARADC_DATA	0x0000	W	0x00000000	This register contains the data after A/D Conversion.
SARADC_STAS	0x0004	W	0x00000000	The status register of A/D Converter.
SARADC_CTRL	0x0008	W	0x00000000	The control register of A/D Converter.
SARADC_DLY_PU_SOC	0x000c	W	0x00000000	delay between power up and start command

*Notes:Size:**B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access*

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15.4.2 Detail Register Description

SARADC_DATA

Address: Operational Base + offset (0x0000)
This register contains the data after A/D Conversion.

Bit	Attr	Reset Value	Description
31:10	RO	0x0	reserved
9:0	RO	0x000	adc_data A/D value of the last conversion (DOUT[9:0]).

SARADC_STAS

Address: Operational Base + offset (0x0004)
The status register of A/D Converter.

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
0	RO	0x0	adc_status ADC status (EOC) 0: ADC stop 1: Conversion in progress

SARADC_CTRL

Address: Operational Base + offset (0x0008)
The control register of A/D Converter.

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6	RW	0x0	int_status Interrupt status. This bit will be set to 1 when end-of-conversion. Set 0 to clear the interrupt.
5	RW	0x0	int_en Interrupt enable. 0: Disable 1: Enable
4	RO	0x0	reserved
3	RW	0x0	adc_power_ctrl ADC power down control bit 0: ADC power down; 1: ADC power up and reset. start signal will be asserted (DLY_PU_SOC + 2) sclk clock period later after power up

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Bit	Attr	Reset Value	Description
2:0	RW	0x0	adc_input_src_sel ADC input source selection(CH_SEL[2:0]). 000 : Input source 0 (SARADC_AIN[0]) 001 : Input source 1 (SARADC_AIN[1]) 010 : Input source 2 (SARADC_AIN[2]) 011 : Input source 3 (SARADC_AIN[3]) 100 : Input source 4 (SARADC_AIN[4]) 101 : Input source 5 (SARADC_AIN[5]) Others : Reserved

SARADC_DLY_PU_SOC
Address: Operational Base + offset (0x000c)
delay between power up and start command

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x00	DLY_PU_SOC delay between power up and start command The start signal will be asserted (DLY_PU_SOC + 2) sclk clock period later after power up

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15.5 Timing Diagram

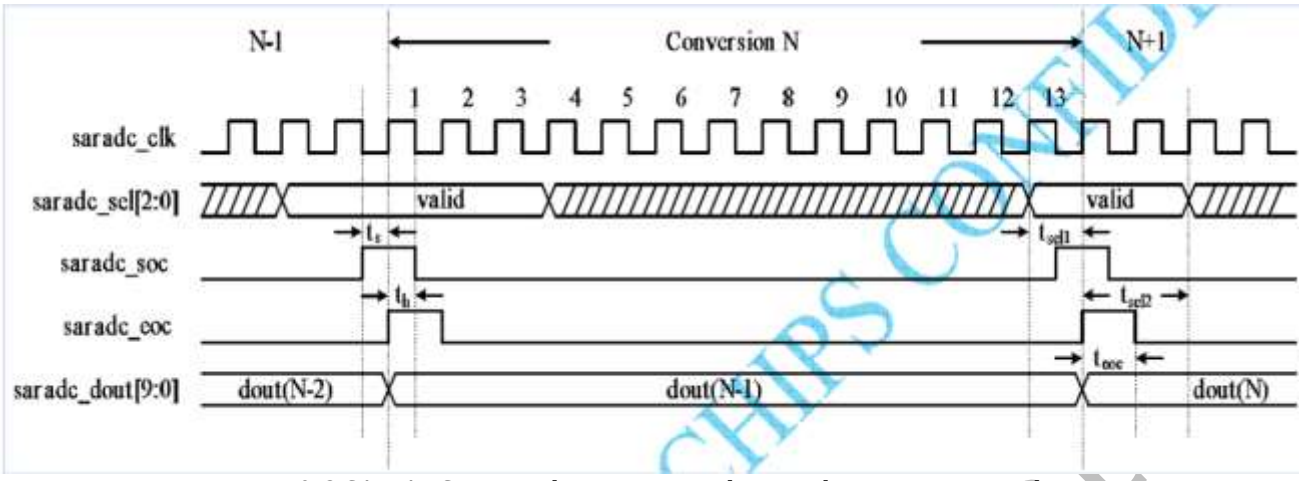

Fig. 40-2 SAR-ADC timing diagram in single-sample conversion mode

15.6 Application Notes

Steps of adc conversion:

- Write SARADC_CTRL[3] as 0 to power down adc converter.
- Write SARADC_CTRL[2:0] as n to select adc channel(n).
- Write SARADC_CTRL[5] as 1 to enable adc interrupt.
- Write SARADC_CTRL[3] as 1 to power up adc converter.
- Wait for adc interrupt or poll SARADC_STAS register to assert whether the conversion is completed
- Read the conversion result from SARADC_DATA[9:0]

Note: The A/D converter was designed to operate at maximum 1MHZ.

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Chapter 4 HDCP22 Controller

16.1 Overview

The HDCP22 controller is an Embedded Security Module (ESM). This module is a self-contained module that can be integrated with HDMI cores to ensure DCP robustness rules for HDCP.

The ESM has a small MCU inside. It reads ESM Image for operation through the AXI interface.

The HDCP22 Controller supports following features:

- Supports HDCP Revision 2.2
- Bus Interface Features:
  - An AXI port for instruction fetch and shared R/W memory
  - An Identity Interface for secret keys used to secure the ESM
  - An Entropy Interface to a true random number generator
  - A Host Port Interface for an external host processor to send commands and get status from the ESM.

    (APB Interface)
- The Tx ports contain interfaces to a HDMI controller, where each port is factory configurable for HDMI support. The HDMI portion of the port contains the following interfaces:
  - HDMI video data and associated control
  - I2C interface for access to the remote end of the HDCP link
  - A GPIO interface for direct hardware signaling between the ESM and the controller
- MMU:
  - 4k/64k page size
  - TLB pre-fetch

16.2 Block Diagram

The following figure illustrates the HDMI controller connections with external interfaces. The ESM executes firmware from system memory external to the ESM which must be processed with the supplied tools as the firmware image is encrypted to the keys you created to present on the ID Interface. A host library (supplied in source code) is compiled with your application software running on the host processor to send commands to the ESM and monitor its status for results.

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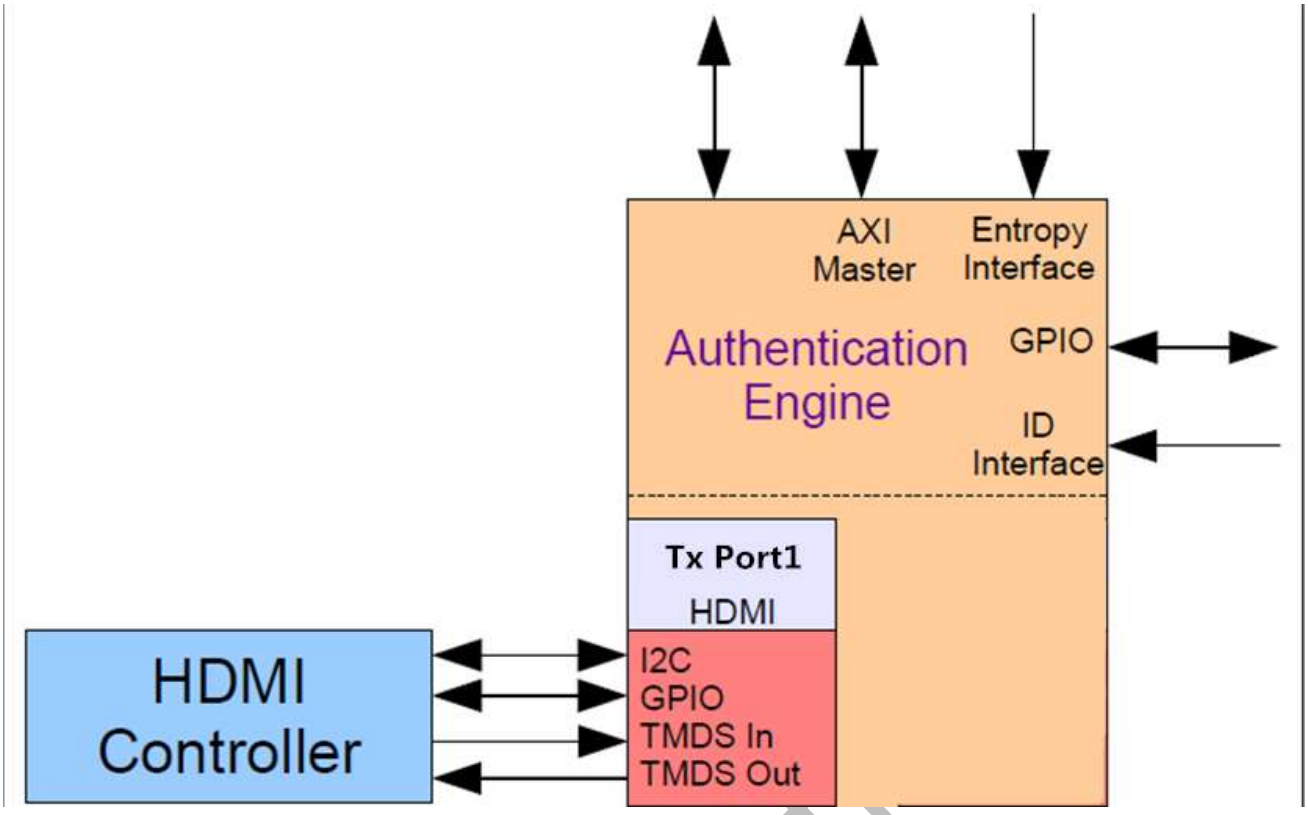

Fig. 4-1 hdcp22 Controller Block Diagram

The embedded security module is composed of two parts:

- The Authentication Engine (AE) contains a controller that runs the HDCP authentication process.
- The Content Encryption Engine (CEE) is an ESM component that encrypts data for the HDCP transmitter.

The following figure represents how the ESM is typically integrated in a SoC.

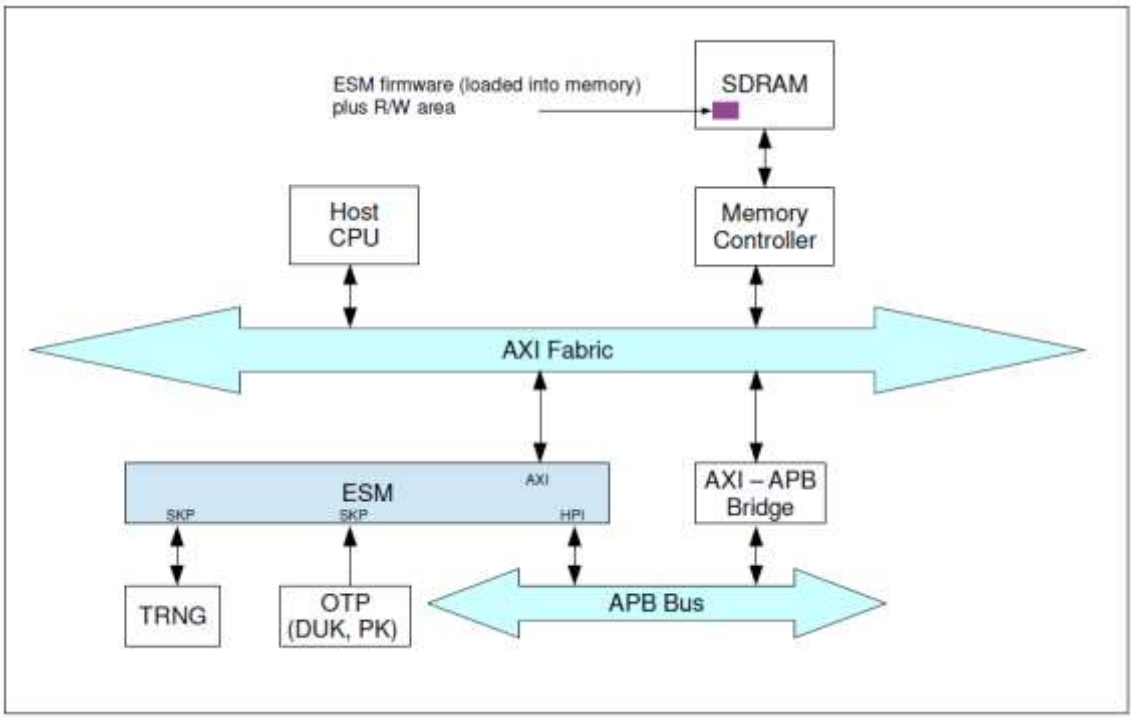

Fig. 4-2 Typical ESM Integration in a SoC

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16.3 Function Description

16.3.1 AXI interface

The ESM requires continuous access to its firmware for operation, which is accessed through the AXI interface and usually placed a reserved portion of SDRAM. It also uses this interface for communicating messages with the external host processor to pass content such as SRM data, pairing information, and debug/logging information.

The ESM requires access to relatively fast and low latency external memory as it randomly accesses its firmware periodically to perform operations. For AXI transactions the average initial access latency may be up to 1000ns.

Note: *The ESM still functions in higher latency systems; however, the ESM may experience intermittent errors, such as failing to authenticate, as there are fixed timeout requirements during the authentication process which must be met. A system with persistent high latency memory accesses may prevent the ESM from fetching its firmware in a timely fashion resulting in protocol timeouts.Following Table summarizes the various requirements the ESM places on the RK3399's memory, assuming an AXI interface clock of 261 MHz and an average latency of 1000 ns.*

Table 4-1 ESM average memory bandwidth usage

Phase	Duration (ms)	Appropriate Bandwidth (Mbytes/sec)	Description
Bootup	100 (maximum)	5.0	Initial startup of ESM to Idle state.
Tx Authentication	525ms	1.5	ESM transmitter authentication phase including capability check.
Rx Authentication	525 ms	1.0	ESM receiver authentication phase.
Idle, authenticated	n/a	< 1.0, 4.0	ESM is either in the Idle state waiting to start authentication or in the authenticated state. The second number is when the ESM has debug logging enabled.

The external host memory required by the ESM must be physically contiguous and allocated before the ESM can be started. The ESM Image and R/W memory buffer addresses must be 4K aligned otherwise the ESM cannot properly access the memory. Following table describes the requirements.

Table 4-2 ESM average memory bandwidth usage

Size (bytes)	ESM Access	Purpose
~256K	Read Only	Contains the ESM firmware. This size varies depending on the particular configuration of the ESM you are using.

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Size (bytes)	ESM Access	Purpose
16K + p*n*256 (Tx) (p is the number of transmitters and n is the number of paired devices supported)	Read / Write	Bidirectional communication buffer to exchange data with the Host. Messages such as SRM updates, pairing data, and logging information are placed in this area by the host and the ESM. The number of pairs supported by this parameter is defined in the configuration file.

In RK3399 the provided ESM host application library allocates a small amount of the processor's memory space, which is accessible by the ESM. Communication with this interface is provided by the ESM host application library.

16.3.2 HPI Interface

The external host issues commands to the ESM through the HPI which supports a limited set of commands such as:

- Enabling and disabling of security on the link
- Enabling debugging/logging
- Retrieving pairing information
- Updating the SRM
- ESM status

Communication with this interface is provided by the ESM host application library.

16.3.3 APB interface (HPI)

The external host issues commands to the ESM through the HPI. The ESM supports a limited set of commands such as:

- Enabling and disabling of security on the link
- Enabling debugging/logging
- Retrieving pairing information
- Updating the SRM
- ESM status

The external host must issue a SRM list to the ESM when one is available, according to the requirements of DCP LLC.

Communication with this interface is provided by the ESM host application library.

16.3.4 Identity Interface

This interface contains the secret keys the ESM requires to implement its security. The KPf and

DUK signals on this interface must be secure from tampering and external user observation as it is used by the ESM to create session keys and other cryptographic secrets required for operation.

When integrating the ESM into your design, you must take to ensure that these requirements are met otherwise the operation of the ESM can be compromised.

Table 4-3 Secure Key for ESM

Description	Purpose

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Description	Purpose
Platform Key (PKf)	A customer created secret random key used to decrypt the ESM firmware. The KPf is intended to be common across a deployed model of a product.
Device Unique Key (DUK)	A customer created secret random key used to decrypt the DCP key data and to encrypt and decrypt content unique to the ESM such as the pairing message content. It is strongly recommended this value should be unique for every unit.
Device Number (Devnum)	This is a non-secret device number and is currently not used.

Note: *The PKf and DUK must be kept confidential as exposing either one compromises the security of the ESM.*

16.3.5 Entropy Interface

This interface connects to a True Random Number Generator (TRNG). The nonce data on this interface must be secure from tampering and external user observation as it is used by the ESM to create session keys and other cryptographic secrets required for operation.

Table 4-4 Entropy Interface

Description	Purpose
Entropy Valid	This is the nonce data from the TRNG. The value is used to seed the ESM's PRNG which is used to generate keys for the authentication process and session keys. The value must be truly random and not generated by another PRNG, as per the HDCP for HDMI specification, Section 2.13 (Random Number Generation).

16.3.6 GPIO Interface

This interface contains a number of general purpose inputs and outputs which can be used by the host to monitor the status of the ESM.

16.3.7 HDMI Interface

The HDMI interface contains logic to apply content protection to the Data and Video Islands as the TMDS data passes through the ESM.

Table 4-5 Entropy Interface

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Description	Purpose
I2C	A standard I2C serial interface for HDCP protocol communication with the remote device, mapped by the controller onto the DDC interface.
GPIO	Direct signals between the controller and the ESM to indicate the status of the ESM to the controller, and to signal the ESM
TMDS In	TMDS data output from the controller. For a Tx this is the unencrypted data, for the Rx this is the encrypted data.
TMDS Out	TMDS data input to the controller. For a Tx this is the encrypted data, for the Rx this is the decrypted data.

16.3.8 HDCP22 Controller Behavior

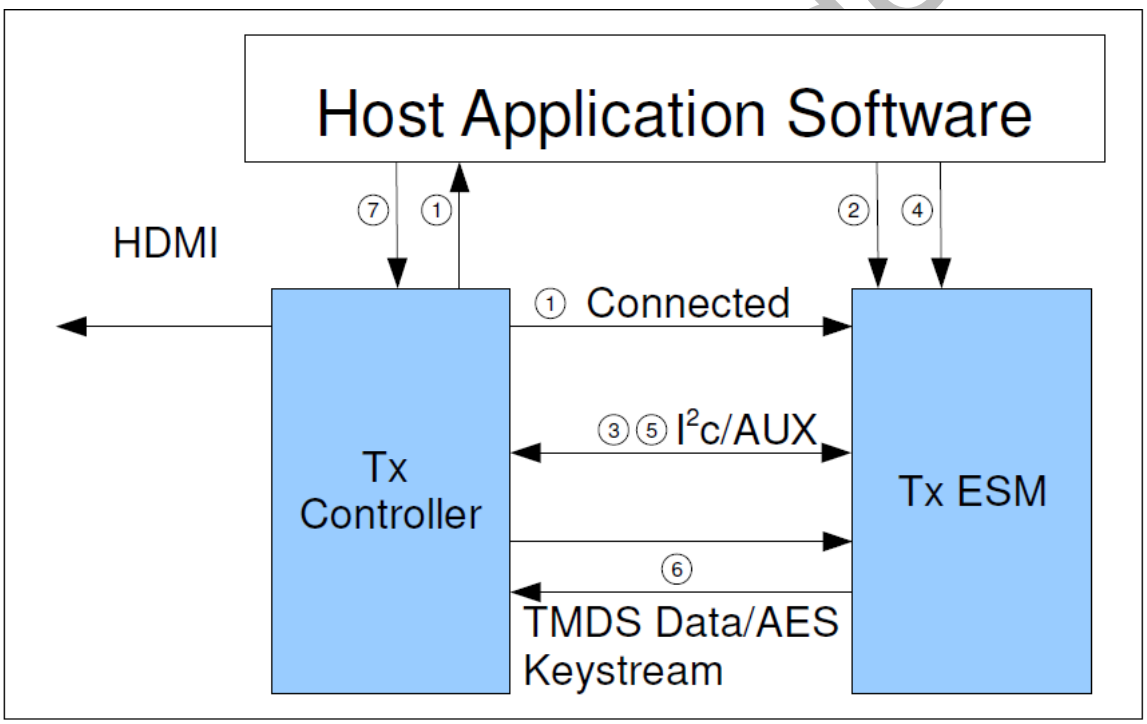

Fig. 4-3 Transmitter Authentication

Hdcp22 controller in RK3399 is a transmitter, it works as following steps:

1. The controller asserts the Connected signal to the ESM. The application software1 receives the notification from the controller. For HDMI operation the ESM outputs a fixed pattern (the BSOD value) on the TMDS data output.

2. The application software issues the HLC_HDCPTX_SetCapability() command which causes the ESM to perform a capability check with the receiver. Before this point in time the ESM does not know the capability of the Rx so the port's capable GPIO output signal is not asserted.

3. The ESM queries the receiver. If the receiver does not support HDCP 2.2 then the port's not_capable GPIO output is asserted. The application software should then check if the receiver supports HDCP 1.4 (or no link protection) and take appropriate action; the ESM is no

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longer required. If the receiver supports HDCP 2.2 then the capable GPIO output is asserted and the ESM transitions to a pre-authorized state. For HDMI operation this means unencrypted low value content is permitted on the video path (TMDS interface).

4. If the receiver supports HDCP 2.2 the application software issues the HLC_HDCP_Authenticate() command to the ESM. As a security feature the ESM enforces a timeout (set in the configuration file) after the receiver indicates it is HDCP 2.2 capable to the time the HLC_HDCP_Authenticate() command must be issued. If you exceed this timeout you must repeat the process from step #2. (When the port is configured for HDMI it will switch its TMDS output back to the fixed pattern output).

5. The ESM initiates authentication with the receiver and successfully completes authentication (or not). The application software and the HDMI controller are appropriately notified (authenticated and authentication_failed outputs asserted accordingly).

6. If authentication is successful the ESM starts to encrypt the TMDS data (HDMI).

7. The application software can now send high value content securely on the link.

Following successful authentication the ESM periodically checks the status of the connected receiver. If a response is not received, or the receiver requests re-authentication the ESM immediately outputs a fixed pattern on the TMDS output (HDMI), asserts the outputs authentication_failed along with a reath_req, and returns to the Idle state. To re-authenticate you must call HLC_HDCP_Authenticate() to drop authentication and then call the function again to start authentication.

16.3.9 HDCP22 Software Initialization

An external host controller (ARM Core or MCU) allocates memory, initializes the ESM, and enables the ESM controller. The general reset and startup sequence is:

a. Configure clocks for hdcp controller.
b. Load PKF & DUK from effuse to gasket.
c. The ESM reset is de-asserted and the ESM enters a reset state.
d. The host allocates contiguous memory for the firmware and R/W areas. The buffers are 4K aligned.
e. The host configures the ESM's HPI interface, defining the physical memory pointer for the ESM firmware.
f. The host copies the ESM firmware from persistent storage to the firmware memory location.
g. The host enables the ESM controller, putting it in a running state. At this point the ESM initiates transactions on the AXI bus to fetch its firmware.
h. ESM asserts a GPIO to indicate it is booted to indicate it can accept commands. The status can also be monitored from the HPI interface.

The ESM host application library provides the startup sequence from steps d through h.

16.3.10 Autostart

The autostart feature is an option which permits a transmitter or receiver port (configurable per port) on the ESM to autonomously take over the detection of a connection and run through the authentication process without intervention from the external host. In the event of an error the port automatically retries up to a specified number of attempts (configuration based) before idling requiring intervention from the host.

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For a transmitter this feature automates the capability checking and authentication along with any reauthentication which may be required due to link errors. For a receiver this feature enables the port to immediately respond to authentication requests without host intervention.

16.3.11 Key Usage

There are two secret, 128 bit keys you are responsible for creating for use by the ESM.

The Platform Key (KPf) is used by the ESM to decrypt its firmware. It is recommended this key be common for all ESMs deployed for a particular product as it permits a common (encrypted) firmware image to be deployed.

The Device Unique Key (DUK) is used by the ESM to protect secrets unique to the device such as HDCP receiver keys and locally generated data such as pairing information. Although it is not required, it is strongly recommended this key be unique for every different unit deployed in a product so that if the kPf is and DUK are compromised, device specific secrets for only that unit are exposed. Using a common DUK for all devices could permit device specific secrets for all devices to be exposed.

As shipped from the factory, the firmware cannot be used as-is and must be processed with tools that are provided with the ESM software package. Figure 1-3 illustrates the tools and the flow that are required to produce encrypted ESM firmware. You edit the firmware.aic file supplied with the firmware image and replace the factory test/simulation values with your secret KPf and DUK keys.

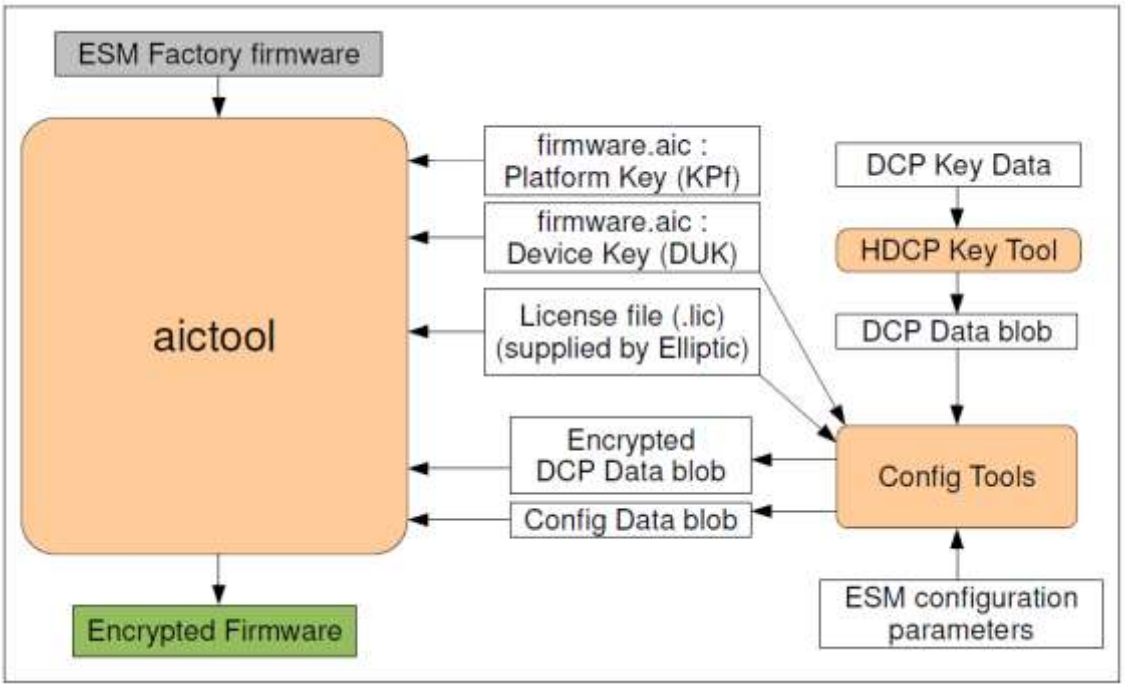

Fig. 4-4 ESM Image tool flow

The DCP Tx key data is common for all transmitter ports whereas the DCP Rx key data is unique for each receiver port.

To facilitate manufacturing it is highly desirable to build and deploy a common software image for all units. If your ESM only contains transmitter ports then it is possible to do this, however if you also have receiver ports you are required by DCP to have a unique receiver key for each

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receiver in your design.

16.4 Application Notes

16.4.1 Host Application Software

The ESM package contains a series of tools and a run-time library as bellow:

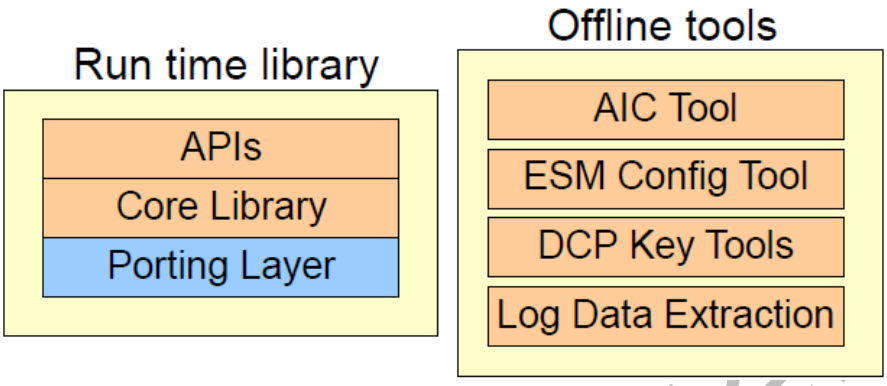

Fig. 4-5 ESM software

16.4.2 Run Time Library

This library is provided as source code and is used by the host application processor to control the ESM.

Setting the configurable options of the ESM is not provided through APIs for security reasons. It is instead embedded into the ESM's Image as part of the process of building a deployable image.

The APIs provide a minimal interface to the ESM, which include the following:

- Load/Initialize firmware image
- Start authentication to secure a link
- Terminate an authenticated link
- Collect logging data
- Get/Set pairing information
- Update SRM data
- Status updates

Sample code is provided and demonstrates how to use the various APIs. In addition, a software API reference guide is included and provides details on the API calls and their parameters.

16.4.3 Offline Tools

There are three separate offline tools that are required to build a deployable image for the ESM as shown in Figure ESM Image tool flow and one tool for logging information. All of these tools are developed to run on a 32 bit Linux system.

16.4.4 Configuration Tool

The troot_base_config_interface tool is used to convert the various configuration files into a data blob to be used by the aictool which creates the encrypted ESM firmware. There are a number of different configuration files which the tools reads, with each configuration file containing a mix of factory configured parameters, and user-adjustable parameters.

This tool also requires the firmware.aic file in order to generate the data blob.

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User Parameters for Configuration Files

The following table indicates which parameters in the configuration files are user-adjustable and other non-described parameters should not be changed. A basic description of the parameter is included below, see the configuration file's comments for a more complete description.

Table 4-6 Configuration files

File Name	Parameter	Description
troot_base_config.cfg	[LOGGING_MODE]	Logging on/off
	[LOGGING_LEVEL]	Sets the verbosity of the output log messages.
	[RESEED_TIMEOUT]	Specifies the time the ESM will wait for the TRNG to supply a new nonce before reporting an error
	[RESEED_AUTO_INTERVAL]	By default the ESM will periodically query the TRNG for a new nonce at the period specified.
	[CPU_FREQ]	Sets the frequency of the esm_clk input. Mandatory to change.
troot_hdcpmgr_tx.cfg	[SRM_VERSION]	Denotes whether the SRM is to be checked or not and the minimum version number permitted.
troot_hdcptx.cfg	[I2C_FREQ]	(HDMI) Sets the default I2C frequency. Can be changed via an API.
	[I2C_USE_SHORT_READ]	(HDMI) Sets whether the interface uses short reads for status requests. Can be changed via an API.
	[CEE_BSOD]	(HDMI) Sets the 24 bit fixed pattern value output by the ESM. See the description following this table for more details.
	[AUTO_START]	Sets the default startup state of the Autostart feature
	[HDCP_TIMEOUTS]	Sets whether the ESM enforces the HDCP timeout periods. Should be

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File Name	Parameter	Description
		enabled only for testing and/or debugging.
	[CEE_FRAME_CNT_DEBUG]	(HDMI) The ESM emits periodic log messages once authenticated, based on the number of VSYNC pulses counted by this parameter.
	[PAIRING_ENABLED]	Sets whether pairing will be enabled or not
	[MAX_PAIRING_DEVICES]	Sets the maximum number of paired devices supported per port. Note the external R/W memory requirements increase based on this setting, and the number of Tx ports in your design.
	[MEM_PAIRING_SIZE]	Sets the amount of R/W memory the pairing data will occupy.
	[CAPABLE_BYPASS_TIME]	(HDMI) Sets the maximum amount of time the ESM will enable low value content after a capability check before switching back to a fixed pattern output. See HLC_HDCPTX_SetCapability in the API guide.
	[LVC_TIMEOUT]	(HDMI) Sets the maximum amount of time the ESM will enable low value content without a capability check. See HLC_HDCPTX_EnableLowValueContent in the API guide.
	[LVC_EXPIRED_CNT]	(HDMI) Sets the maximum number of LVC timeouts permitted.

The [CEE_BSOD] is a 32 bit decimal value which parameter represents fixed data output on the TMDS data bus if the ESM encounters an authentication issue or detects attempts have been made to tamper with it. The value is only relevant for ports configured for HDMI. This 32 bit value is output as follows:

Table 4-7 ESM BSOD Output Mapping

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BSOD Bit Positions	HDMI Controller TMDS Data Channel
7..0	O_tmds_ch0 [7..0]
15..8	O_tmds_ch1 [7..0]
23..16	O_tmds_ch2 [7..0]
32..24	Not used

16.4.5 HDCP Key Tools

This tool (hdcpkeys) is used to create a DCP key data file, which is required to build the encrypted DCP data blob needed for creating the encrypted ESM firmware. The tool reads a configuration file that contains the Rx or Tx key information from DCP in clear text and outputs a structured data file for the configuration interface tool to encrypt with the appropriate secret keys.

The tools can also accept the DCP published key files themselves.

16.4.6 AIC Tool

The aictool tool is used to create the encrypted ESM firmware. To build the encrypted firmware, you must first create the data files for the DCP keys and the configuration. These files, along with an input configuration file for this tool are used to build an image that can be used by the ESM. In addition to specifying the Platform and Device Unique Keys in the configuration file, there are additional keys that also must be specified as noted below.

Content Randomization

There are two separate values in the firmware.aic file, IK and IVc (128 bits and 96 bits respectively), which you must randomly generate to create a new encrypted ESM firmware image. Failure to do this puts secret data in the ESM firmware at risk of being compromised.

16.4.7 Log Data Extraction

The logging information from the ESM is output in a proprietary format. This tool is used to produce readable output of captured logging data content.

The Host Library contains a core library component which includes all the functions necessary to use the ESM. The library is built using portable C code and uses various abstraction functions to allow for migration to different platforms. Before using the ESM in your system you must properly implement the abstraction components specific to your platform.

There are two distinct abstraction components used by the Host Library, the System Abstraction Layer and the Host Library Driver (HLD) component. The System Abstraction Layer (or Common Component) defines system level functions used by the Host Library, such as malloc, memcpy, etc.

These functions are linked into the Host Library when it is built.

The Host Library Driver (HLD) component establishes a communication between the library and the physical ESM hardware, including managing the memory required for the ESM Image

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and the read/write area. This is a run-time plug-in module that is defined in your application and specified as a parameter to the ESM initialization function.

The figure below illustrates the abstraction components used by the Host Library package. This figure demonstrates one example where there are three separate ESMs in the system, requiring (typically) three separate HLD instantiations. Consult the ESM Host Library API guide for more details on the HLD implementation.

16.4.8 Host Library

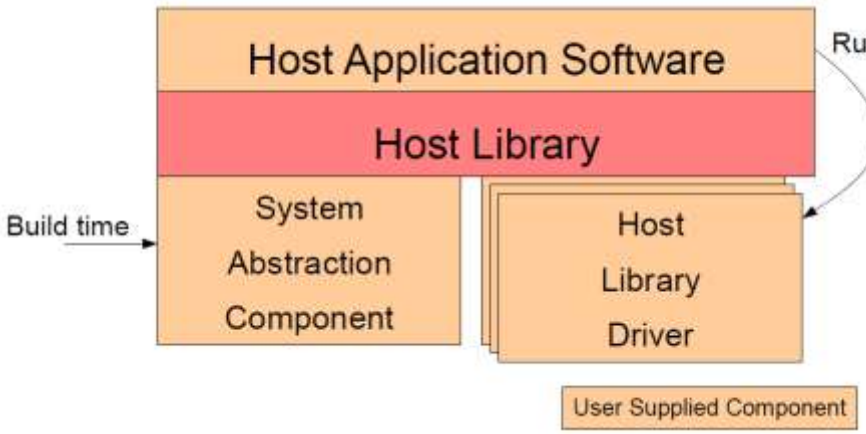

Fig. 4-6 Host Library Layers

The Host Library provides a number of sample applications using the ESM with the library. The applications demonstrate using the library in a User Space Linux application:

- Repeater/Converter (Linux) [Planned in later releases – not currently available]
- Transmitter (Linux)

All the sample applications demonstrate API usage, loading the ESM firmware, and running a specific ESM application.

The sample applications provided as a Linux User Space application communicate to the ESM hardware through a Linux Kernel driver which is implemented as a Host Library Driver (HLD) plug-in.

elliptic	
├── doc	Host Library Documentation
├── host_library	Host Library source code and includes
└── samples	Sample applications and driver applications

Fig. 4-7 Development Tree Overview

Transmitter

There is a sample transmitter applications provided in the samples folder:

hdcp_tx.c is a Linux based example and utilizes file IO and threads.

Linux OS

Sample applications are targeted for a Linux User Space application environment. In order to use them a custom kernel driver is used and the HLD plug-in for the host library was made to

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call IOCTLs which mapped to the appropriate interfaces in the kernel driver module to access the ESM.

To build for this platform run the makefile from the esm_host_lib folder:

make samples

The implementation of the platform specific APIs (for the Linux example) are in the following files:

./host_library/cal/system/sample_platform/elliptic_common.c

./host_library/cal/system/sample_platform/elliptic_log.c

./host_library/include/elliptic_platform_specific_system_type.h

./host_library/include/elliptic_platform_specific_system.h

You can change these implementations as needed for your specific platform.

Debugging Suggestions

The host library driver interface is relatively straightforward to implement as the host library itself takes care of the required protocols to communicate with the ESM. The detailed HPI interface and command protocols are considered proprietary information, however, the information presented below can be followed if you are having difficulty getting the ESM operating in your environment.

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Chapter 41 HDMI TX

16.5 Overview

HDMI TX is fully compliant with HDMI 1.4a and 2.0a specification. It offers a simple implementation for consumer electronics like DVD/player/recorder and camcorder. HDMI TX consists of one HMDI transmitter controller and one HMDI transmitter PHY.

It supports following features:

- Video formats:
  - All CEA-861-E video formats up to 1080p at 60 Hz and 720p/1080i at 120 Hz
  - HDMI 1.4b video formats
    - All CEA-861-E video formats up to 1080p at 120 Hz
    - HDMI 1.4b 4K x 2K video formats
    - HDMI 1.4b 3D video modes with up to 340 MHz (TMDS clock)
  - HDMI 2.0 video formats, All CEA-861-F video formats
- Colorimetry, 24-bit RGB 4:4:4
- Pixel clock from 13.5 MHz up to 600 MHz
- Up to 192 kHz IEC60958 audio sampling rate
- Flexible synchronous enable per clock domain to set functional power down modes
- AMBA APB 3.0 register access
- I2C DDC, EDID block read mode
- SCDC I2C DDC access
- TMDS Scrambler to enable support for 2160p@60Hz with RGB 4:4:4
- Integrated CEC hardware engine
- Support multi-channels PCM or compressed audio transmission (32-192kHz Fs) from I2S/SPDIF, using IEC60958 and IEC 61937

16.6 Block Diagram

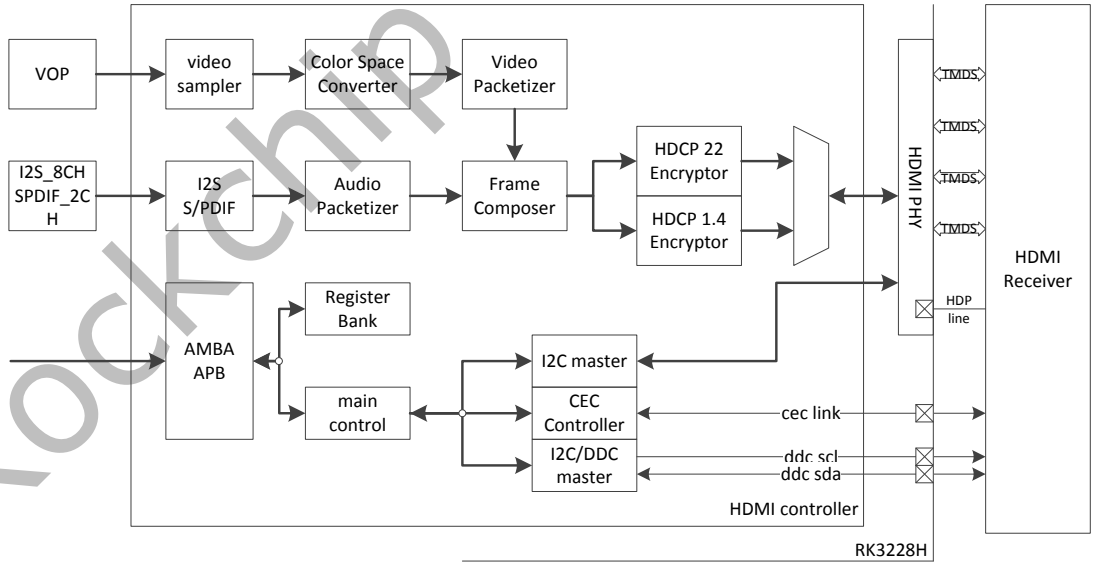

Fig. 41-1 HDMI TX Block Diagram

16.7 Function Description

16.7.1 IOMUX

Customer should notice that CEC channel use IO_HDMIcec_GPIO0A3pmuio for 3V and IO_HDMICEC5Vgpio_HDMICECpmuio5v for 5V.

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DDC channel use IO_HDMIscl_I2C3scl_GPIO0A5pmuio and
IO_HDMIsda_I2C3sda_GPIO0A6pmuio in 3v mode, or
IO_HDMISCL5Vgpio_I2C3scl5v_HDMISCLpmuio5v and
IO_HDMISDA5Vgpio_I2C3sda5v_HDMISDApmuio5v in 5v mode.

The hpd use IO_HDMIhpd_GPIO0A4pmuio in 3v mode or
IO_HDMIHPD5Vgpio_HDMIHPDpmuio5v in 5v mode.

16.7.2 Pixel Clock

When using HDMI as display solution, customers can set pixel clock source to CRU or HDMI PHY PCLK for VOP and HDMI controller.Here HDMI PHY PCLk is recommended.

16.7.3 Video Data Processing

The video processing contains video format timings, pixel encodings (RGB to YCbCr, or YCbCr to RGB), colorimetry and corresponding requirements. This function is implemented by some functional blocks, Video Capture block, Color Space Conversion block, and Deep Color block.

The input video pixels can be encoded in either RGB, YCBCR 4:4:4 or YCBCR 4:2:2 formats by Color Space Conversion block.

The input Video data can have a pixel size of 24bits. The deep color block is used to deal with different pixel size. Video at the default 24-bit color depth is carried at a TMDS clock rate equal to the pixel clock rate. Higher color depths are carried using a correspondingly higher TMDS clock rate. HDMI Transmitter support video formats with TMDS rates below 25MHz (e.g. 13.5MHz for 480i/NTSC) that can be transmitted using a pixel-repetition scheme by setting relative registers.

Color Space Conversion

HDMI Transmitter Color space conversion (CSC) is responsible for carrying out the following video color space conversion functions:

- RGB to/from YCbCr
- 4:2:2 to/from 4:4:4 up (pixel repetition or linear interpolation)/down-converter
- Limited to/from full quantization range conversion
- The CSC supports all the timings reported in the CEA-861-D specification and the following pixel modes:
- RGB 444 and YCbCr 444: 24, 30, 36, and 48 bits
- YCbCr 422: 16, 20, and 24 bits

The color space conversion matrix is ruled by the following equations listed in below figure.

$$out_1 = (X_1 \times in_1/4096 + X_2 \times in_2/4096 + X_3 \times in_3/4096 + X_4) \times 2^{scale}$$
$$out_2 = (Y_1 \times in_1/4096 + Y_2 \times in_2/4096 + Y_3 \times in_3/4096 + Y_4) \times 2^{scale}$$
$$out_3 = (Z_1 \times in_1/4096 + Z_2 \times in_2/4096 + Z_3 \times in_3/4096 + Z_4) \times 2^{scale}$$

Fig. 41-2 HDMI Color Space Conversion Matrix Equations

Note: Color Space Conversion to and from YCrCb 4:2:0 is not supported.

16.7.4 Audio Data Processing

The HDMI TX audio process contain audio clock regeneration, placement of audio samples within packets, packet timing control, audio sample rates setting, and channel/speaker assignments. This function is implemented by Audio Capture blocks

The Audio Capture support either SPDIF or four channel I2S input. SPDIF input supports audio

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sampling rates from 32 to 192 KHz. The I2S input supports from 2-channel to 8-channel audio up to 192 KHz.

The scheme of audio processing as shown in the figure below:

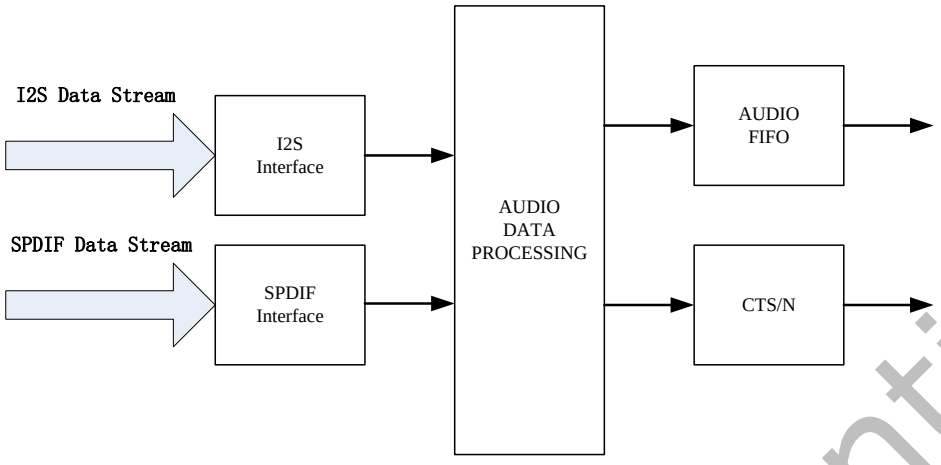

Fig. 41-3 HDMI Audio Data Processing Diagram

1.I2S

The function of this module is to implement I2S audio input feature. The incoming audio stream is captured, processed then transmitted into the TMDS link. Four I2S inputs also allow transmission of DVD-Audio and decoded Dolby Digital to A/V Receivers and high-end displays. The interface supports from 2-channel to 8-channel audio up to 192 kHz. The I2S pins must also be coherent with mclk. The appropriate registers must be configured to describe the format of audio being input. This information is passed over the HDMI link in the CEA-861D Audio Info (AI) packets. Table shows the I2S 8 channel audio formats that are supported for each of the video formats.

Table 41-1 HDMI TX I2S 2 Channel Audio Sampling Frequency

Video Format	32kHz	44.1kHz	48kHz	88.2kHz	96kHz	176.4kHz	192kHz
720x480p /720x576p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1440x480i/ 1440x576i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
720p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080p	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 41-2 HDMI TX I2S 8 Channel Audio Sampling Frequency

Video Format	32kHz	44.1kHz	48kHz	88.2kHz	96kHz	176.4kHz	192kHz
720x480p /720x576p	Yes	Yes	Yes	No	No	No	No

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1440x480i/ 1440x576i	Yes	Yes	Yes	Yes	No	No	No
720p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080p	Yes	Yes	Yes	Yes	Yes	Yes	Yes

2.SPDIF

The function of this module is to implement SPDIF audio input feature. The incoming audio stream is captured, processed then transmitted into the TMDS link. SPDIF stream can carry 2-channel uncompressed PCM data (IEC 60958) or a compressed bit stream for multi-channel (IEC 61937) formats. The audio data capture logic forms the audio data into packets in accordance with the HDMI specification. SPDIF input supports audio sampling rates from 32 to 192 KHz. The following shows the SPDIF audio formats that are supported for each of the video formats

Table 41-3 HDMI SPDIF Sampling Frequency at Each Video Format

Video Format	32kHz	44.1kHz	48kHz	88.2kHz	96kHz	176.4kHz	192kHz
720x480p /720x576p	Yes	Yes	Yes	Yes	Yes	No	No
1440x480i/ 1440x576i	Yes	Yes	Yes	Yes	Yes	No	No
720p	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080i	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1080p	Yes	Yes	Yes	Yes	Yes	Yes	Yes

3.Audio Sample Clock Capture and Regeneration

Audio data is carried across the HDMI link, which is driven by a TMDS clock running at a rate corresponding to the video pixel rate, does not retain the original audio sample clock. The task of recreating this clock at the Sink is called Audio Clock Regeneration.

The HDMI Transmitter determine the fractional relationship between the TMDS clock and an audio reference clock (128 audio sample rate [fs]) and pass the numerator and denominator of that fraction to the HDMI Sink across the HDMI link. The Sink then re-create the audio clock from the TMDS clock by using a clock divider and a clock multiplier.

The exact relationship between the two clocks will be.

$128 \cdot f_s = f_{TMDS_clock} * N / CTS.$

The scheme of the Audio Sample Clock Capture and Regeneration as shown below:

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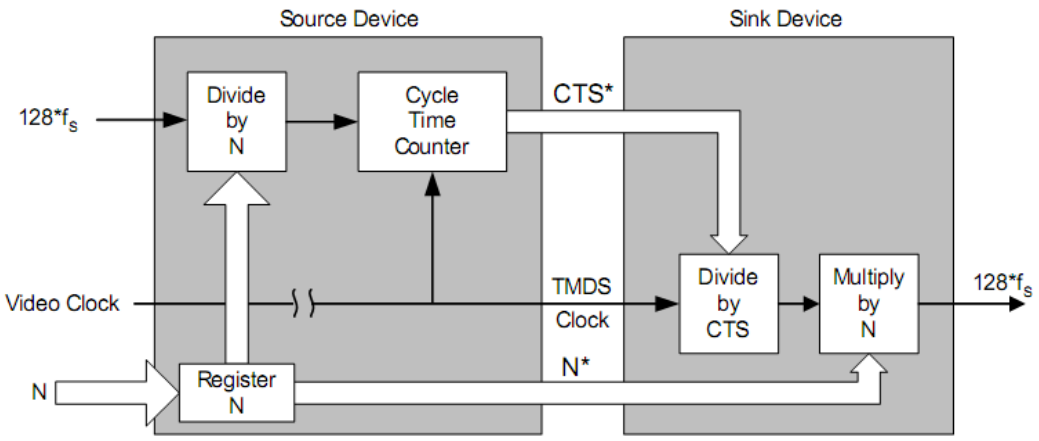

Fig. 41-4 HDMI Audio Clock Regeneration Model

Because there is no audio clock carried through the HDMI link, only the TMDS clock is used. Software sets the CTS/N with a value taken form the below table, which shows the CTS and N value for the supported standard. All other TMDS clocks are not supported; The TMDS clocks divided or multiplied by 1,001 coefficients are not supported.

Table 41-4 HDMI CTS and N table

	TMDS Clock (MHz)													
	25.2		27		54		74.25		148.5		297		597	
Fs (kHZ)	N	CTS	N	CTS	N	CTS	N	CTS	N	CTS	N	CTS	N	CTS
32	4096	25200	4096	27000	4096	54000	4096	74250	4096	148500	3072	222750	3072	445500
44.1	6272	28000	6272	30000	6272	60000	6272	82500	6272	165000	4704	247500	9408	990000
48	6144	25200	6144	27000	6144	54000	6144	74250	6144	148500	5120	247500	6144	495000
88.2	12544	28000	12544	30000	12544	60000	12544	82500	12544	165000	9408	247500	18816	990000
96	12288	25200	12288	27000	12288	54000	12288	74250	12288	148500	10240	247500	12288	495000
176.4	25088	28000	25088	30000	25088	60000	25088	82500	25088	165000	18816	247500	37632	990000
192	24576	25200	24576	27000	24576	54000	24576	74250	24576	148500	20480	247500	24576	4950000

16.7.5 DDC

The DDC functional block is used for configuration and status exchange between the HDMI Source and HDMI Sink. HMDI Transmitter Controller has I2C Master Interface for DDC transactions. It enables for host controller to read EDID, HDCP authentication by issuing simple register access. The I2C bus speed is limited by DDC specification. DDC bus access frequency can be controlled.

16.7.6 EDID

Extended Display Identification Data (EDID) was created by VESA to enable plug and play capabilities of monitors. This data, which is stored in the sink device, describes video formats that the DTV Monitor is capable of receiving and rendering. The information is supplied to the source device, over the interface, upon the request of the source device. The source device then chooses its output format, taking into account the format of the original video stream and the formats supported by the DTV Monitor. The function of this module is to implement EDID feature.

16.7.7 HDCP

HDMI Transmitter has a capability for HDCP authentication by hardware. The function of this module is to implement HDCP encryption feature. This feature can be turned on or off

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depending on register setting.

RK3399 supports up to HDCP 2.2. HDCP 1.4 is done by HDMI Transmitter itself. HDCP22 is descripted in another chapter.

16.7.8 Hot Plug Detect

HDMI Transmitter has a capability for detecting the Sink plug in or plug out, and launch an interrupt and registers state indicating for software controlling.

16.7.9 TMDS encoder

The TMDS encoder converts the 2/4/8 bits data into the 10 bit DC-balanced TMDS data.

HDMI TX put the TMDS encoding on the audio /video /aux data received from the HDCP XOR mask. This data is output onto three TMDS differential data lines along with a TMDS differential clock.

16.7.10 CEC

The CEC functional block provides high-level control functions between all of the various audiovisual products in a user's environment through one line.

16.8 Register Description

The address offset of the HDMI TX is 0xff980000,it contains 16 address section. The offset of the table of Register Summary must multiple with 4 when software configure it。Like the Interrupt registers，its base address is 0x0100.If we want to configure it, its real address is 0xff980000+0x0100*4.

16.8.1 Register Summary

Name	Offset	Size	Reset Value	Description
Identification Registers	0x0000	B		Identification releated registers
Interrupt registers	0x0100	B		Interrupt releated registers
Video Sampler registers	0x0200	B		Video Sampler registers
Video Packetizer registers	0x0800	B		Video Packetizer registers
Frame Composer Registers	0x1000	B		Frame Composer Registers
HDMI Source PHY Registers	0x3000	B		HDMI Source PHY Registers
I2C Master PHY Registers	0x3020	B		I2C Master PHY Registers
Audio Sampler Registers	0x3100	B		Audio Sampler Registers
Main Controller Registers	0x4000	B		Main Controller Registers
Color Space Converter Registers	0x4100	B		Color Space Converter Registers

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Name	Offset	Size	Reset Value	Description
HDCP Encryption Engine Registers	0x5000	B		HDCP Encryption Engine Registers
HDCP BKSV Registers	0x7800	B		HDCP BKSV Registers
HDCP AN Registers	0x7805	B		HDCP AN Registers
Encrypted DPK Embedded Storage Registers	0x780E	B		Encrypted DPK Embedded Storage Registers
CEC Engine Registers	0x7D00	B		CEC Engine Registers
I2C Master Registers	0x7E00	B		I2C Master Registers for E-DDC/SCDC

Identification Registers

Identification Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: Identification

Register	Offset	Description
design_id	0x0	Design Identification Register
revision_id	0x1	Revision Identification Register
product_id0	0x2	Product Identification Register 0
product_id1	0x3	Product Identification Register 1
config0_id	0x4	Configuration Identification Register 0
config1_id	0x5	Configuration Identification Register 1
config2_id	0x6	Configuration Identification Register 2
config3_id	0x7	Configuration Identification Register 3

design_id

Description: Design Identification Register

Size: 8 bits

Offset: 0x0

Bits	Name	Attr	Description
7:0	design_id	R	Design ID code fixed by HDMI that Identifies the instantiated DWC_hdmi_tx controller. For example, DWC_hdmi_tx 2.11a, DESIGN_ID = 21 Value After Reset: 0x21

revision_id

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Description: Revision Identification Register

Size: 8 bits

Offset: 0x1

Bits	Name	Attr	Description
7:0	revision_id	R	Revision ID code fixed by HDMI that Identifies the instantiated DWC_hdmi_tx controller. Value After Reset: 0x1a

product_id0

Description: Product Identification Register 0

Size: 8 bits

Offset: 0x2

Bits	Name	Attr	Description
7:0	product_id0	R	This one byte fixed code Identifies HDMI 's product line ("A0h" for DWC_hdmi_tx products). Value After Reset: 0xa0

product_id1

Description: Product Identification Register 1

Size: 8 bits

Offset: 0x3

Bits	Name	Attr	Description
7:6	product_id1_hdcp	R	These bits identify a HDMI Controller with HDCP encryption according to HDMI product line. Value After Reset: "(HDCP== 1) ? 3 : 0"
5:2			Reserved for future use.
1	product_id1_rx	R	This bit Identifies HDMI 's DWC_hdmi_rx Controller according to HDMI product line. Value After Reset: 0x0
0	product_id1_tx	R	This bit Identifies H Controller according to HDMI product line. Value After Reset: 0x1

config0_id

Description: Configuration Identification Register 0

Size: 8 bits

Offset: 0x4

Bits	Name	Attr	Description
7	prepen	R	Indicates if it is possible to use internal pixel

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			repetition Value After Reset: "(HDMI_TX_INTPREPEN== 1) ? 1 : 0"
6			Reserved for future use.
5	audspdif	R	Indicates if the SPDIF audio interface is present Value After Reset: "(SPDIFPORTS== 1) ? 1 : 0"
4	audi2s	R	Indicates if I2S interface is present Value After Reset: "(I2SPORTS== 1) ? 1 : 0"
3	hdmi14	R	Indicates if HDMI 1.4 features are present Value After Reset: "(HDMI_TX_14== 1) ? 1 : 0"
2	csc	R	Indicates if Color Space Conversion block is present Value After Reset: "(CSC== 1) ? 1 : 0"
1	cec	R	Indicates if CEC is present Value After Reset: "(CEC== 1) ? 1 : 0"
0	hdcp	R	Indicates if HDCP is present Value After Reset: "(HDCP== 1) ? 1 : 0"

config1_id

Description: Configuration Identification Register 1

Size: 8 bits

Offset: 0x5

Bits	Name	Attr	Description
7	hdcp22_snps	R	Indicates if HDCP 2.2 SNPS solution is present Value After Reset: (HTX_HDCP22_SNPS== 1) ? 1 : 0
6	hdcp22_ext	R	Indicates if external HDCP 2.2 interface support is present Value After Reset: "(HTX_HDCP22_EXTERNAL== 1) ? 1 : 0"
5	hdmi20	R	Indicates if HDMI 2.0 features are present Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0"
4:2			Reserved for future use.
1	confapb	R	Indicates that configuration interface is APB interface Value After Reset: 0x1
0			Reserved for future use.

config2_id

Description: Configuration Identification Register 2

Size: 8 bits

Offset: 0x6

Bits	Name	Attr	Description
7:0	phytype	R	Indicates the type of PHY interface selected: 0x00: Legacy PHY (HDMI Tx PHY) 0xF2: PHY GEN2 (HDMI 3D TX PHY) 0xE2: PHY GEN2 (HDMI 3D TX PHY) + HEAC PHY 0xC2: PHY MHL COMBO (MHL+HDMI 2.0 TX PHY) 0xB2: PHY MHL COMBO (MHL+HDMI 2.0 TX PHY) + HEAC PHY 0xF3: PHY HDMI 20 (HDMI 2.0 TX PHY)

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			0xE3: PHY HDMI 20 (HDMI 2.0 TX PHY) + HEAC PHY 0xFE: External PHY Value After Reset: "(PHY_HDMI20==1) ? ((HDMI_HEAC_PHY_EN==1)? 0xE3 : 0xF3) : (PHY_MHL_COMBO==1) ? ((HDMI_HEAC_PHY_EN==1)? 0xB2 : 0xC2) : (PHY_GEN2==1) ? ((HDMI_HEAC_PHY_EN==1)? 0xE2 : 0xF2) : (PHY_EXTERNAL==1)? 0xFE : 0x00"

config3_id

Description: Configuration Identification Register 3

Size: 8 bits

Offset: 0x7

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	confahbauddma	R	Indicates that the audio interface is AHB AUD DMA Value After Reset: "(AHBAUDDMAIF== 1) ? 1 : 0"
0	confgpaud	R	Indicates that the audio interface is Generic Parallel Audio (GPAUD) Value After Reset: "(GPAUDPORTS== 1) ? 1 : 0"

Interrupt Registers

Interrupt Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: Interrupt

Register	Offset	Description
ih_fc_stat0	0x100	Frame Composer Interrupt Status Register 0 (Packet Interrupts)
ih_fc_stat1	0x101	Frame Composer Interrupt Status Register 1 (Packet Interrupts)
ih_fc_stat2	0x102	Frame Composer Interrupt Status Register 2 (Packet Interrupts)
ih_as_stat0	0x103	Audio Sampler Interrupt Status Register (FIFO Threshold, Underflow and Overflow Interrupts)
ih_phy_stat0	0x104	PHY Interface Interrupt Status Register (RXSENSE, PLL Lock and HPD Interrupts)
ih_i2cm_stat0	0x105	E-DDC I2C Master Interrupt Status Register (Done and Error Interrupts)
ih_cec_stat0	0x106	CEC Interrupt Status Register (Functional Operation Interrupts)
ih_vp_stat0	0x107	Video Packetizer Interrupt Status Register (FIFO Full and Empty Interrupts)
ih_i2cmphy_stat0	0x108	PHY GEN2 I2C Master Interrupt Status Register (Done and Error Interrupts)
ih_ahbdmaaud_stat0	0x109	AHB Audio DMA Interrupt Status Register (Functional Operation, Buffer Full and Empty...

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Register	Offset	Description
ih_decode	0x170	Interruption Handler Decode Assist Register
ih_mute_fc_stat0	0x180	Frame Composer Interrupt Mute Control Register 0
ih_mute_fc_stat1	0x181	Frame Composer Interrupt Mute Control Register 1
ih_mute_fc_stat2	0x182	Frame Composer Interrupt Mute Control Register 2
ih_mute_as_stat0	0x183	Audio Sampler Interrupt Mute Control Register
ih_mute_phy_stat0	0x184	PHY Interface Interrupt Mute Control Register
ih_mute_i2cm_stat0	0x185	E-DDC I2C Master Interrupt Mute Control Register
ih_mute_cec_stat0	0x186	CEC Interrupt Mute Control Register
ih_mute_vp_stat0	0x187	Video Packetizer Interrupt Mute Control Register
ih_mute_i2cmphy_stat0	0x188	PHY GEN2 I2C Master Interrupt Mute Control Register
ih_mute_ahbdmaaud_stat0	0x189	AHB Audio DMA Interrupt Mute Control Register
ih_mute	0x1ff	Global Interrupt Mute Control Register

ih_fc_stat0

Description: Frame Composer Interrupt Status Register 0 (Packet Interrupts)

Size: 8 bits

Offset: 0x100

Bits	Name	Attr	Description
7	AUDI	R/W1C	Active after successful transmission of an Audio InfoFrame packet. Value After Reset: 0x0
6	ACP	R/W1C	Active after successful transmission of an Audio Content Protection packet. Value After Reset: 0x0
5	HBR	R/W1C	Active after successful transmission of an Audio HBR packet. Value After Reset: 0x0
4	MAS	R/W1C	Active after successful transmission of an MultiStream Audio packet Value After Reset: 0x0
3	NVBI	R/W1C	Active after successful transmission of an NTSC VBI packet Value After Reset: 0x0
2	AUDS	R/W1C	Active after successful transmission of an Audio Sample packet. Due to high number of audio sample packets transmitted, this interrupt is by default masked at frame composer. Value After Reset: 0x0
1	ACR	R/W1C	Active after successful transmission of an Audio Clock Regeneration (N/ CTS transmission) packet. Value After Reset: 0x0
0	NULL	R/W1C	Active after successful transmission of an Null packet. Due to high number of audio sample packets transmitted, this interrupt is by default

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			masked at frame composer. Value After Reset: 0x0

ih_fc_stat1

Description: Frame Composer Interrupt Status Register 1 (Packet Interrupts)

Size: 8 bits

Offset: 0x101

Bits	Name	Attr	Description
7	GMD	R/W1C	Active after successful transmission of an Gamut metadata packet. Value After Reset: 0x0
6	ISCR1	R/W1C	Active after successful transmission of an International Standard Recording Code 1 packet. Value After Reset: 0x0
5	ISCR2	R/W1C	Active after successful transmission of an International Standard Recording Code 2 packet Value After Reset: 0x0
4	VSD	R/W1C	Active after successful transmission of an Vendor Specific Data InfoFrame packet. Value After Reset: 0x0
3	SPD	R/W1C	Active after successful transmission of an Source Product Descriptor InfoFrame packet. Value After Reset: 0x0
2	AMP	R/W1C	Active after successful transmission of an Audio Metadata packet Value After Reset: 0x0
1	AVI	R/W1C	Active after successful transmission of an AVI InfoFrame packet. Value After Reset: 0x0
0	GCP	R/W1C	Active after successful transmission of an General Control Packet. Value After Reset: 0x0

ih_fc_stat2

Description: Frame Composer Interrupt Status Register 2 (Packet Interrupts)

Size: 8 bits

Offset: 0x102

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	DRM	R/W1C	Active after successful transmission of an DRM packet Value After Reset: 0x0
1	LowPriority_overflow	R/W1C	Frame Composer low priority packet queue descriptor overflow indication Value After Reset: 0x0
0	HighPriority_overflow	R/W1C	Frame Composer high priority packet queue

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			descriptor overflow indication Value After Reset: 0x0

ih_as_stat0

Description: Audio Sampler Interrupt Status Register (FIFO Threshold, Underflow and Overflow Interrupts)

Size: 8 bits

Offset: 0x103

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	fifo_underrun	R/W1C	Indicates an underrun on the audio FIFO Value After Reset: 0x0
3	fifo_overrun	R/W1C	Indicates an overrun on the audio FIFO. Value After Reset: 0x0
2	Aud_fifo_underflow_thr	R/W1C	Audio Sampler audio FIFO empty threshold (four samples) indication for the legacy HBR audio interface. For AHB_DMA, this bit indicates that the number of samples in the FIFO is equal to (or less) than the number of active audio channels. This bit is not relevant for I2S, SPDIF, and GPA interfaces. Value After Reset: 0x0
1	Aud_fifo_underflow	R/W1C	Audio Sampler audio FIFO empty indication. Value After Reset: 0x0
0	Aud_fifo_overflow	R/W1C	Audio Sampler audio FIFO full indication. Value After Reset: 0x0

ih_phy_stat0

Description: PHY Interface Interrupt Status Register (RXSENSE, PLL Lock and HPD Interrupts)

Size: 8 bits

Offset: 0x104

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	RX_SENSE_3	R/W1C	TX PHY RX_SENSE indication for driver 3. You may need to mask or change polarity of this interrupt after it has become active. Value After Reset: 0x0
4	RX_SENSE_2	R/W1C	TX PHY RX_SENSE indication for driver 2. You may need to mask or change polarity of this interrupt after it has become active. Value After Reset: 0x0
3	RX_SENSE_1	R/W1C	TX PHY RX_SENSE indication for driver 1. You may need to mask or change polarity of this interrupt after it has become active. Value After Reset: 0x0

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2	RX_SENSE_0	R/W1C	TX PHY RX_SENSE indication for driver 0. You may need to mask or change polarity of this interrupt after it has become active. Value After Reset: 0x0
1	TX_PHY_LOCK	R/W1C	TX PHY PLL lock indication. Value After Reset: 0x0
0	HPD	R/W1C	HDMI Hot Plug Detect indication. You may need to mask or change polarity of this interrupt after it has become active. Value After Reset: 0x0

ih_i2cm_stat0

Description: E-DDC I2C Master Interrupt Status Register (Done and Error Interrupts)

Size: 8 bits

Offset: 0x105

Bits	Name	Attr	Description
7:3			Reserved for future use.
2	scdc_readreq	R/W1C	I2C Master SCDC read request indication. Value After Reset: 0x0
1	I2Cmasterdone	R/W1C	I2C Master done indication Value After Reset: 0x0
0	I2Cmastererror	R/W1C	I2C Master error indication Value After Reset: 0x0

ih_cec_stat0

Description: CEC Interrupt Status Register (Functional Operation Interrupts)

Size: 8 bits

Offset: 0x106

Bits	Name	Attr	Description
7			Reserved for future use.
6	WAKEUP	R/W1C	CEC Wake-up indication Value After Reset: 0x0
5	ERROR_FOLLOW	R/W1C	CEC Error Follow indication Value After Reset: 0x0
4	ERROR_INITIATOR	R/W1C	CEC Error Initiator indication Value After Reset: 0x0
3	ARB_LOST	R/W1C	CEC Arbitration Lost indication Value After Reset: 0x0
2	NACK	R/W1C	CEC Not Acknowledge indication Value After Reset: 0x0
1	EOM	R/W1C	CEC End of Message Indication Value After Reset: 0x0
0	DONE	R/W1C	CEC Done Indication Value After Reset: 0x0

ih_vp_stat0

Description: Video Packetizer Interrupt Status Register (FIFO Full and Empty Interrupts)

Size: 8 bits

Offset: 0x107

Bits	Name	Attr	Description
7	fifofullrepet	R/W1C	Video Packetizer pixel repeater FIFO full interrupt Value After Reset: 0x0
6	fifoemptyrepet	R/W1C	Video Packetizer pixel repeater FIFO empty

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			interrupt Value After Reset: 0x0
5	fifofullpp	R/W1C	Video Packetizer pixel packing FIFO full interrupt Value After Reset: 0x0
4	fifoemptypp	R/W1C	Video Packetizer pixel packing FIFO empty interrupt Value After Reset: 0x0
3	fifofullremap	R/W1C	Video Packetizer pixel YCC 422 re-mapper FIFO full interrupt Value After Reset: 0x0
2	fifoemptyremap	R/W1C	Video Packetizer pixel YCC 422 re-mapper FIFO empty interrupt Value After Reset: 0x0
1:0		R/W1C	Reserved and read as zero

ih_i2cmphy_stat0

Description: PHY GEN2 I2C Master Interrupt Status Register (Done and Error Interrupts)
Size: 8 bits
Offset: 0x108

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	I2Cmphydone	R/W1C	I2C Master PHY done indication Value After Reset: 0x0
0	I2Cmphyerror	R/W1C	I2C Master PHY error indication Value After Reset: 0x0

ih_ahbdmaaud_stat0

Description: AHB Audio DMA Interrupt Status Register (Functional Operation, Buffer Full and Empty Interrupts)
Size: 8 bits
Offset: 0x109

Bits	Name	Attr	Description
7			Reserved for future use.
6	ahbdmaaud_intbuffoverrun	R/W1C	AHB audio DMA Buffer overrun interruption Value After Reset: 0x0
5	ahbdmaaud_interror	R/W1C	AHB audio DMA error interrupt Value After Reset: 0x0
4	ahbdmaaud_intlostownership	R/W1C	AHB audio DMA lost ownership interrupt Value After Reset: 0x0
3	ahbdmaaud_intretrysplit	R/W1C	AHB audio DMA RETRY/SPLIT interrupt Value After Reset: 0x0
2	ahbdmaaud_intdone	R/W1C	AHB audio DMA done interrupt Value After Reset: 0x0
1	ahbdmaaud_intbufffull	R/W1C	AHB audio DMA Buffer full interrupt Value After Reset: 0x0
0	ahbdmaaud_intbuffempty	R/W1C	AHB audio DMA Buffer empty interrupt

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			Value After Reset: 0x0

ih_decode

Description: Interruption Handler Decode Assist Register

Size: 8 bits

Offset: 0x170

Bits	Name	Attr	Description
7	ih_fc_stat0	R	Interruption active at the ih_fc_stat0 register Value After Reset: 0x0
6	ih_fc_stat1	R	Interruption active at the ih_fc_stat1 register Value After Reset: 0x0
5	ih_fc_stat2_vp	R	Interruption active at the ih_fc_stat2 or ih_vp_stat0 register Value After Reset: 0x0
4	ih_as_stat0	R	Interruption active at the ih_as_stat0 register Value After Reset: 0x0
3	ih_phy	R	Interruption active at the ih_phy_stat0 or ih_i2cmphy_stat0 register Value After Reset: 0x0
2	ih_i2cm_stat0	R	Interruption active at the ih_i2cm_stat0 register Value After Reset: 0x0
1	ih_cec_stat0	R	Interruption active at the ih_cec_stat0 register Value After Reset: 0x0
0	ih_ahbdmaaud_stat0	R	Interruption active at the ih_ahbdmaaud_stat0 register Value After Reset: 0x0

ih_mute_fc_stat0

Description: Frame Composer Interrupt Mute Control Register 0

Size: 8 bits

Offset: 0x180

Bits	Name	Attr	Description
7	AUDI	R/W	When set to 1, mutes ih_fc_stat0[7] Value After Reset: 0x0
6	ACP	R/W	When set to 1, mutes ih_fc_stat0[6] Value After Reset: 0x0
5	HBR	R/W	When set to 1, mutes ih_fc_stat0[5] Value After Reset: 0x0
4	MAS	R/W	When set to 1, mutes ih_fc_stat0[4]. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0"
3	NVBI	R/W	When set to 1, mutes ih_fc_stat0[3]. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0"
2	AUDS	R/W	When set to 1, mutes ih_fc_stat0[2] Value After Reset: 0x0

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Bits	Name	Attr	Description
1	ACR	R/W	When set to 1, mutes ih_fc_stat0[1] Value After Reset: 0x0
0	NULL	R/W	When set to 1, mutes ih_fc_stat0[0] Value After Reset: 0x0

ih_mute_fc_stat1

Description: Frame Composer Interrupt Mute Control Register 1

Size: 8 bits

Offset: 0x181

Bits	Name	Attr	Description
7	GMD	R/W	When set to 1, mutes ih_fc_stat1[7] Value After Reset: 0x0
6	ISCR1	R/W	When set to 1, mutes ih_fc_stat1[6] Value After Reset: 0x0
5	ISCR2	R/W	When set to 1, mutes ih_fc_stat1[5] Value After Reset: 0x0
4	VSD	R/W	When set to 1, mutes ih_fc_stat1[4] Value After Reset: 0x0
3	SPD	R/W	When set to 1, mutes ih_fc_stat1[3] Value After Reset: 0x0
2	AMP	R/W	When set to 1, mutes ih_fc_stat1[2]. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0" Exists: HDMI_TX_20==1
1	AVI	R/W	When set to 1, mutes ih_fc_stat1[1] Value After Reset: 0x0
0	GCP	R/W	When set to 1, mutes ih_fc_stat1[0] Value After Reset: 0x0

ih_mute_fc_stat2

Description: Frame Composer Interrupt Mute Control Register 2

Size: 8 bits

Offset: 0x182

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	DRM	R/W	When set to 1, mutes ih_fc_stat2[4]. Value After Reset: (HDMI_TX_20== 1) ? 1 : 0
3:2			Reserved for future use
1	LowPriority_overflow	R/W	When set to 1, mutes ih_fc_stat2[1] Value After Reset: 0x0
0	HighPriority_overflow	R/W	When set to 1, mutes ih_fc_stat2[0] Value After Reset: 0x0

ih_mute_as_stat0

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Description: Audio Sampler Interrupt Mute Control Register
Size: 8 bits
Offset: 0x183

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	fifo_underrun	R/W	When set to 1, mutes ih_as_stat0[4] Value After Reset: 0x1
3	fifo_overrun	R/W	When set to 1, mutes ih_as_stat0[3] Value After Reset: 0x1
2	Aud_fifo_underflow_thr	R/W	When set to 1, mutes ih_as_stat0[2] Value After Reset: 0x0
1	Aud_fifo_underflow	R/W	When set to 1, mutes ih_as_stat0[1] Value After Reset: 0x0
0	Aud_fifo_overflow	R/W	When set to 1, mutes ih_as_stat0[0] Value After Reset: 0x0

ih_mute_phy_stat0
Description: PHY Interface Interrupt Mute Control Register
Size: 8 bits
Offset: 0x184

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	RX_SENSE_3	R/W	When set to 1, mutes ih_phy_stat0[5] Value After Reset: 0x0
4	RX_SENSE_2	R/W	When set to 1, mutes ih_phy_stat0[4] Value After Reset: 0x0
3	RX_SENSE_1	R/W	When set to 1, mutes ih_phy_stat0[3] Value After Reset: 0x0
2	RX_SENSE_0	R/W	When set to 1, mutes ih_phy_stat0[2] Value After Reset: 0x0
1	TX_PHY_LOCK	R/W	When set to 1, mutes ih_phy_stat0[1] Value After Reset: 0x0
0	HPD	R/W	When set to 1, mutes ih_phy_stat0[0] Value After Reset: 0x0

ih_mute_i2cm_stat0
Description: E-DDC I2C Master Interrupt Mute Control Register
Size: 8 bits
Offset: 0x185

Bits	Name	Attr	Description
7:3			Reserved for future use.
2	scdc_readreq	R/W	When set to 1, mutes ih_i2cm_stat0[2] Value After Reset: 0x1
1	I2Cmasterdone	R/W	When set to 1, mutes ih_i2cm_stat0[1] Value After Reset: 0x0
0	I2Cmastererror	R/W	When set to 1, mutes ih_i2cm_stat0[0]

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			Value After Reset: 0x0

ih_mute_cec_stat0

Description: CEC Interrupt Mute Control Register

Size: 8 bits

Offset: 0x186

Bits	Name	Attr	Description
7			Reserved for future use.
6	WAKEUP	R/W	When set to 1, mutes ih_cec_stat0[6] Value After Reset: 0x0
5	ERROR_FOLLOW	R/W	When set to 1, mutes ih_cec_stat0[5] Value After Reset: 0x0
4	ERROR_INITIATOR	R/W	When set to 1, mutes ih_cec_stat0[4] Value After Reset: 0x0
3	ARB_LOST	R/W	When set to 1, mutes ih_cec_stat0[3] Value After Reset: 0x0
2	NACK	R/W	When set to 1, mutes ih_cec_stat0[2] Value After Reset: 0x0
1	EOM	R/W	When set to 1, mutes ih_cec_stat0[1] Value After Reset: 0x0
0	DONE	R/W	When set to 1, mutes ih_cec_stat0[0] Value After Reset: 0x0

ih_mute_vp_stat0

Description: Video Packetizer Interrupt Mute Control Register

Size: 8 bits

Offset: 0x187

Bits	Name	Attr	Description
7	fifofullrepet	R/W	When set to 1, mutes ih_vp_stat0[7] Value After Reset: 0x0
6	fifoemptyrepet	R/W	When set to 1, mutes ih_vp_stat0[6] Value After Reset: 0x0
5	fifofullpp	R/W	When set to 1, mutes ih_vp_stat0[5] Value After Reset: 0x0
4	fifoemptypp	R/W	When set to 1, mutes ih_vp_stat0[4] Value After Reset: 0x0
3	fifofullremap	R/W	When set to 1, mutes ih_vp_stat0[3] Value After Reset: 0x0
2	fifoemptyremap	R/W	When set to 1, mutes ih_vp_stat0[2] Value After Reset: 0x0
1	spare_2	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0
0	spare_1	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0

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ih_mute_i2cmphy_stat0

Description: PHY GEN2 I2C Master Interrupt Mute Control Register

Size: 8 bits

Offset: 0x188

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	I2Cmphydone	R/W	When set to 1, mutes ih_i2cmphy_stat0[1] Value After Reset: 0x0
0	I2Cmphyerror	R/W	When set to 1, mutes ih_i2cmphy_stat0[0] Value After Reset: 0x0

ih_mute_ahbdmaaud_stat0

Description: AHB Audio DMA Interrupt Mute Control Register

Size: 8 bits

Offset: 0x189

Bits	Name	Attr	Description
7			Reserved for future use.
6	ahbdmaaud_intbuffoverrun	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[6] Value After Reset: 0x1
5	ahbdmaaud_interror	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[5] Value After Reset: 0x0
4	ahbdmaaud_intlostownership	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[4] Value After Reset: 0x0
3	ahbdmaaud_intretrysplit	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[3] Value After Reset: 0x0
2	ahbdmaaud_intdone	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[2] Value After Reset: 0x0
1	ahbdmaaud_intbufffull	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[1] Value After Reset: 0x0
0	ahbdmaaud_intbuffempty	R/W	When set to 1, mutes ih_ahbdmaaud_stat0[0] Value After Reset: 0x0

ih_mute

Description: Global Interrupt Mute Control Register

Size: 8 bits

Offset: 0x1ff

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	mute_wakeup_interrupt	R/W	When set to 1, mutes the main interrupt output port. The sticky bit interrupts continue with their state accessible through the configuration bus, only the main interrupt line is muted. Value After Reset: 0x1
0	mute_all_interrupt	R/W	When set to 1, mutes the main interrupt line (where all interrupts are ORed). The sticky bit interrupts continue with their state; only the main interrupt

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			line is muted. Value After Reset: 0x1

VideoSampler Registers

Video Sampler Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: VideoSampler

Register	Offset	Description
tx_invid0	0x200	Video Input Mapping and Internal Data Enable Configuration Register
tx_instuffing	0x201	Video Input Stuffing Enable Register
tx_gydata0	0x202	Video Input gy Data Channel Stuffing Register 0
tx_gydata1	0x203	Video Input gy Data Channel Stuffing Register 1
tx_rcrdata0	0x204	Video Input rcr Data Channel Stuffing Register 0
tx_rcrdata1	0x205	Video Input rcr Data Channel Stuffing Register 1
tx_bcbdata0	0x206	Video Input bcb Data Channel Stuffing Register 0
tx_bcbdata1	0x207	Video Input bcb Data Channel Stuffing Register 1

tx_invid0

Description: Video Input Mapping and Internal Data Enable Configuration Register

Size: 8 bits

Offset: 0x200

Bits	Name	Attr	Description
7	internal_de_generator	R/W	Internal data enable (DE) generator enable. If data enable is not available for the input video, set this bit to one to activate the internal data enable generator. Attention: This feature only works for input video modes that have native repetition (such as, all CEA videos). No desired pixel repetition can be used with this feature because these configurations only affect the Frame Composer and not this block. The DE Generator does not work for the following conditions: Transmission of video with CEA VIC 39 Transmission of 3D video using the field alternative structure Value After Reset: 0x0
6:5			Reserved for future use.
4:0	video_mapping	R/W	Video Input mapping (color space/color depth): 0x01: RGB 4:4:4/8 bits 0x03: RGB 4:4:4/10 bits 0x05: RGB 4:4:4/12 bits 0x07: RGB 4:4:4/16 bits 0x09: YCbCr 4:4:4 or 4:2:0/8 bits 0x0B: YCbCr 4:4:4 or 4:2:0/10 bits 0x0D: YCbCr 4:4:4 or

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Bits	Name	Attr	Description
			4:2:0/12 bits 0x0F: YCbCr 4:4:4 or 4:2:0/16 bits 0x16: YCbCr 4:2:2/8 bits 0x14: YCbCr 4:2:2/10 bits 0x12: YCbCr 4:2:2/12 bits 0x17: YCbCr 4:4:4 (IPI)/8 bits 0x18: YCbCr 4:4:4 (IPI)/10 bits 0x19: YCbCr 4:4:4 (IPI)/12 bits 0x1A: YCbCr 4:4:4 (IPI)/16 bits 0x1B: YCbCr 4:2:2 (IPI)/12 bits 0x1C: YCbCr 4:2:0 (IPI)/8 bits 0x1D: YCbCr 4:2:0 (IPI)/10 bits 0x1E: YCbCr 4:2:0 (IPI)/12 bits 0x1F: YCbCr 4:2:0 (IPI)/16 bits Value After Reset: 0x1

tx_instuffing

Description: Video Input Stuffing Enable Register

Size: 8 bits

Offset: 0x201

Bits	Name	Attr	Description
7:3			Reserved for future use.
2	bcbdata_stuffing	R/W	0b: When the dataen signal is low, the value in the bcbdata[15:0] output is the one sampled from the corresponding input data. 1b: When the dataen signal is low, the value in the bcbdata[15:0] output is given by the values in the TX_BCBDTA0 and TX_BCBDATA1 registers. Value After Reset: 0x0
1	rcrdata_stuffing	R/W	0b: When the dataen signal is low, the value in the rcrdata[15:0] output is the one sampled from the corresponding input data. 1b: When the dataen signal is low, the value in the rcrdata[15:0] output is given by the values in TX_RCRDTA0 and TX_RCRDATA1 registers. Value After Reset: 0x0
0	gydata_stuffing	R/W	0b: When the dataen signal is low, the value in the gydata[15:0] output is the one sampled from the corresponding input data. 1b: When the dataen signal is low, the value in the gydata[15:0] output is given by the values in TX_GYDTA0 and TX_GYDATA1 registers. Value After Reset: 0x0

tx_gydata0

Description: Video Input gy Data Channel Stuffing Register 0

Size: 8 bits

Offset: 0x202

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Bits	Name	Attr	Description
7:0	gydata	R/W	This register defines the value of gydata[7:0] when TX_INSTUFFING[0] (gydata_stuffing) is set to 1b. Value After Reset: 0x0

tx_gydata1

Description: Video Input gy Data Channel Stuffing Register 1

Size: 8 bits

Offset: 0x203

Bits	Name	Attr	Description
7:0	gydata	R/W	This register defines the value of gydata[15:8] when TX_INSTUFFING[0] (gydata_stuffing) is set to 1b. Value After Reset: 0x0

tx_rcrdata0

Description: Video Input rcr Data Channel Stuffing Register 0

Size: 8 bits

Offset: 0x204

Bits	Name	Attr	Description
7:0	rcrdata	R/W	This register defines the value of rcrydata[7:0] when TX_INSTUFFING[1] (rcrdata_stuffing) is set to 1b. Value After Reset: 0x0

tx_rcrdata1

Description: Video Input rcr Data Channel Stuffing Register 1

Size: 8 bits

Offset: 0x205

Bits	Name	Attr	Description
7:0	rcrdata	R/W	This register defines the value of rcrydata[15:8] when TX_INSTUFFING[1] (rcrdata_stuffing) is set to 1b. Value After Reset: 0x0

tx_bcbdata0

Description: Video Input bcb Data Channel Stuffing Register 0

Size: 8 bits

Offset: 0x206

Bits	Name	Attr	Description
7:0	bcbdata	R/W	This register defines the value of bcbdata[7:0] when TX_INSTUFFING[2] (bcbdata_stuffing) is set to 1b. Value After Reset: 0x0

tx_bcbdata1

Description: Video Input bcb Data Channel Stuffing Register 1

Size: 8 bits

Offset: 0x207

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Bits	Name	Attr	Description
7:0	bcbdata	R/W	This register defines the value of bcbdata[15:8] when TX_INSTUFFING[2] (bcbdata_stuffing) is set to 1b. Value After Reset: 0x0

VideoPacketizer Registers

Video Packetizer Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
vp_status	0x800	Video Packetizer Packing Phase Status Register
vp_pr_cd	0x801	Video Packetizer Pixel Repetition and Color Depth Register
vp_stuff	0x802	Video Packetizer Stuffing and Default Packing Phase Register
vp_remap	0x803	Video Packetizer YCC422 Remapping Register
vp_conf	0x804	Video Packetizer Output and Enable Configuration Register
vp_mask	0x807	Video Packetizer Interrupt Mask Register

vp_status

Description: Video Packetizer Packing Phase Status Register
Size: 8 bits
Offset: 0x800

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	packing_phase	R	Read only register that holds the "packing phase" output of the Video Packetizer block. Value After Reset: 0x0

vp_pr_cd

Description: Video Packetizer Pixel Repetition and Color Depth Register
Size: 8 bits
Offset: 0x801

Bits	Name	Attr	Description
7:4	color_depth	R/W	The Color depth configuration is described as the following, with the action stated corresponding to color_depth[3:0]: 0000b: 24 bits per pixel video (8 bits per component). 8-bit packing mode. 0001b-0011b: Reserved. Not used. 0100b: 24 bits per pixel video (8 bits per component). 8-bit packing mode. 0101b: 30 bits per pixel video (10 bits per component). 10-bit packing mode. 0110b: 36 bits per pixel video (12 bits per component). 12-bit packing mode. 0111b: 48 bits per pixel video (16 bits per component). 16-bit packing mode. Other: Reserved. Not used. Value After Reset: 0x0

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Bits	Name	Attr	Description
3:0	desired_pr_factor	R/W	Desired pixel repetition factor configuration. The configured value sets H13T PHY PLL to multiply pixel clock by the factor in order to obtain the desired repetition clock. For the CEA modes some are already defined with pixel repetition in the input video. So for CEA modes this shall be always 0. Shall only be used if the user wants to do pixel repetition using H13TCTRL controller. The action is stated corresponding to desired_pr_factor[3:0]: 0000b: No pixel repetition (pixel sent only once) 0001b: Pixel sent two times (pixel repeated once) 0010b: Pixel sent three times 0011b: Pixel sent four times 0100b: Pixel sent five times 0101b: Pixel sent six times 0110b: Pixel sent seven times 0111b: Pixel sent eight times 1000b: Pixel sent nine times 1001b: Pixel sent 10 times Other: Reserved. Not used Value After Reset: 0x0

vp_stuff

Description: Video Packetizer Stuffing and Default Packing Phase Register

Size: 8 bits

Offset: 0x802

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	idefault_phase	R/W	Controls the default phase packing machine used according to HDMI 1.4b specification: "If the transmitted video format has timing such that the phase of the first pixel of every Video Data Period corresponds to pixel packing phase 0 (e.g. 10P0, 12P0, 16P0), the Source may set the Default_Phase bit in the GCP. The Sink may use this bit to optimize its filtering or handling of the PP field." This means that for 10-bit mode the Htotal must be dividable by 4; for 12- bit mode, the Htotal must be divisible by 2. Value After Reset: 0x0
4	ifix_pp_to_last	R/W	Reserved. Controls packing machine strategy Value After Reset: 0x0
3	icx_goto_p0_st	R/W	Reserved. Controls packing machine strategy Value After Reset: 0x0
2	ycc422_stuffing	R/W	YCC 422 remap stuffing control. For horizontal

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Bits	Name	Attr	Description
			blanking, the action is stated corresponding to ycc422_stuffing: 0b: YCC 422 remap block in direct mode (input blanking data goes directly to output). 1b: YCC 422 remap block in stuffing mode. When "de" goes to low the outputs are fixed to 0x00. Value After Reset: 0x0
1	pp_stuffing	R/W	Pixel packing stuffing control. The action is stated corresponding to pp_stuffing: 0b: Pixel packing block in direct mode (input blanking data goes directly to output). 1b: Pixel packing block in stuffing mode. When "de_rep" goes to low the outputs are fixed to 0x00. Value After Reset: 0x0

Fields for Register: vp_stuff (Continued)

Bits	Name	Attr	Description
0	pr_stuffing	R/W	Pixel repeater stuffing control. The action is stated corresponding to pp_stuffing: 0b: Pixel repeater block in direct mode (input blanking data goes directly to output). 1b: Pixel repeater block in stuffing mode. When "de" goes to low the outputs are fixed to 0x00. Value After Reset: 0x0

vp_remap

Description: Video Packetizer YCC422 Remapping Register

Size: 8 bits

Offset: 0x803

Bits	Name	Attr	Description
7:2			Reserved for future use.
1:0	ycc422_size	R/W	YCC 422 remap input video size ycc422_size[1:0] 00b: YCC 422 16-bit input video (8 bits per component) 01b: YCC 422 20-bit input video (10 bits per component) 10b: YCC 422 24-bit input video (12 bits per component) 11b: Reserved. Not used Value After Reset: 0x0

vp_conf

Description: Video Packetizer Output and Enable Configuration Register

Size: 8 bits

Offset: 0x804

Bits	Name	Attr	Description
7			Reserved for future use.

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Bits	Name	Attr	Description
6	bypass_en	R/W	When set to 1'b1, Pixel packing enable. When set to 1b'0, the pixel packing block is controlled by pp_en. Value After Reset: 0x0
5	pp_en	R/W	Pixel packing enable. When set to 0, the pixel packing block is disabled if bypass_en is 1'b0. Value After Reset: 0x1
4	pr_en	R/W	Pixel repeater enable. When set to 0, the pixel repetition block is disabled. Value After Reset: 0x0
3	ycc422_en	R/W	YCC 422 select enable. Disabling forces bypass module to output always zeros. Value After Reset: 0x0
2	bypass_select	R/W	bypass_select 0b: Data from pixel repeater block 1b: Data from input of Video Packetizer block Value After Reset: 0x1
1	output_selector	R/W	When set to 1'b1, Data from pixel packing block. Value After Reset: 0x0
0	output_selector_0	R/W	Video Packetizer output selection 0b: Data from pixel packing block 1b: Data from YCC422 remap block Value After Reset: 0x0

vp_mask

Description: Video Packetizer Interrupt Mask Register

Size: 8 bits

Offset: 0x807

Bits	Name	Attr	Description
7	ointfullrepet	R/W	Mask bit for Video Packetizer pixel repeater FIFO full Value After Reset: 0x0
6	ointemptyrepet	R/W	Mask bit for Video Packetizer pixel repeater FIFO empty Value After Reset: 0x0
5	ointfullpp	R/W	Mask bit for Video Packetizer pixel packing FIFO full Value After Reset: 0x0
4	ointemptypp	R/W	Mask bit for Video Packetizer pixel packing FIFO empty Value After Reset: 0x0
3	ointfullremap	R/W	Mask bit for Video Packetizer pixel YCC 422 re-mapper FIFO full Value After Reset: 0x0
2	ointemptyremap	R/W	Mask bit for Video Packetizer pixel YCC 422 re-mapper FIFO empty Value After Reset: 0x0
1	spare_2	R/W	Reserved as "spare" bit with no associated functionality.

			Value After Reset: 0x0
0	spare_1	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0

FrameComposer Registers

Frame Composer Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: FrameComposer

Register	Offset	Description
fc_invidconf	0x1000	Frame Composer Input Video Configuration and HDCP Keepout Register
fc_inhactiv0	0x1001	Frame Composer Input Video HActive Pixels Register 0
fc_inhactiv1	0x1002	Frame Composer Input Video HActive Pixels Register 1
fc_inhblank0	0x1003	Frame Composer Input Video HBlank Pixels Register 0
fc_inhblank1	0x1004	Frame Composer Input Video HBlank Pixels Register 1
fc_invactiv0	0x1005	Frame Composer Input Video VActive Pixels Register 0
fc_invactiv1	0x1006	Frame Composer Input Video VActive Pixels Register 1
fc_invblank	0x1007	Frame Composer Input Video VBlank Pixels Register
fc_hsyncindelay0	0x1008	Frame Composer Input Video HSync Front Porch Register 0
fc_hsyncindelay1	0x1009	Frame Composer Input Video HSync Front Porch Register 1
fc_hsyncinwidth0	0x100a	Frame Composer Input Video HSync Width Register 0
fc_hsyncinwidth1	0x100b	Frame Composer Input Video HSync Width Register 1
fc_vsyncindelay	0x100c	Frame Composer Input Video VSync Front Porch Register
fc_vsyncinwidth	0x100d	Frame Composer Input Video VSync Width Register
fc_infreq0	0x100e	Frame Composer Input Video Refresh Rate Register 0
fc_infreq1	0x100f	Frame Composer Input Video Refresh Rate Register 1
fc_infreq2	0x1010	Frame Composer Input Video Refresh Rate Register 2
fc_ctrldur	0x1011	Frame Composer Control Period Duration Register
fc_exctrldur	0x1012	Frame Composer Extended Control Period Duration Register
fc_exctrlspac	0x1013	Frame Composer Extended Control Period Maximum Spacing Register
fc_ch0pream	0x1014	Frame Composer Channel 0 Non-Preamble Data Register
fc_ch1pream	0x1015	Frame Composer Channel 1 Non-Preamble Data Register
fc_ch2pream	0x1016	Frame Composer Channel 2 Non-Preamble Data Register
fc_aviconf3	0x1017	Frame Composer AVI Packet Configuration Register 3
fc_gcp	0x1018	Frame Composer GCP Packet Configuration Register
fc_aviconf0	0x1019	Frame Composer AVI Packet Configuration Register 0
fc_aviconf1	0x101a	Frame Composer AVI Packet Configuration Register 1

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Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_aviconf2	0x101b	Frame Composer AVI Packet Configuration Register 2
fc_avivid	0x101c	Frame Composer AVI Packet VIC Register
fc_avietb[0:1]	0x101d + (i * 0x1)	Frame Composer AVI Packet End of Top Bar Register Array
fc_avisbb[0:1]	0x101f + (i * 0x1)	Frame Composer AVI Packet Start of Bottom Bar Register Array
fc_avielb[0:1]	0x1021 + (i * 0x1)	Frame Composer AVI Packet End of Left Bar Register Array
fc_avisrb[0:1]	0x1023 + (i * 0x1)	Frame Composer AVI Packet Start of Right Bar Register Array
fc_audiconf0	0x1025	Frame Composer AUD Packet Configuration Register 0
fc_audiconf1	0x1026	Frame Composer AUD Packet Configuration Register 1
fc_audiconf2	0x1027	Frame Composer AUD Packet Configuration Register 2
fc_audiconf3	0x1028	Frame Composer AUD Packet Configuration Register 3
fc_vsdieeeid2	0x1029	Frame Composer VSI Packet Data IEEE Register 2
fc_vsdsize	0x102a	Frame Composer VSI Packet Data Size Register
fc_vsdieeeid1	0x1030	Frame Composer VSI Packet Data IEEE Register 1
fc_vsdieeeid0	0x1031	Frame Composer VSI Packet Data IEEE Register 0
fc_vsdpayload[0:23]	0x1032 + (i * 0x1)	Frame Composer VSI Packet Data Payload Register Array
fc_spdvendorname[0:7]	0x104a + (i * 0x1)	Frame Composer SPD Packet Data Vendor Name Register Array
fc_spdproductname[0:15]	0x1052 + (i * 0x1)	Frame Composer SPD packet Data Product Name Register Array
fc_spddeviceinf	0x1062	Frame Composer SPD Packet Data Source Product Descriptor Register
fc_audsconf	0x1063	Frame Composer Audio Sample Flat and Layout Configuration Register
fc_audsstat	0x1064	Frame Composer Audio Sample Flat and Layout Status Register
fc_audsv	0x1065	Frame Composer Audio Sample Validity Flag Register
fc_audsu	0x1066	Frame Composer Audio Sample User Flag Register
fc_audschnl0	0x1067	Frame Composer Audio Sample Channel Status Configuration Register 0
fc_audschnl1	0x1068	Frame Composer Audio Sample Channel Status Configuration Register 1

Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_audschnl2	0x1069	Frame Composer Audio Sample Channel Status Configuration Register 2
fc_audschnl3	0x106a	Frame Composer Audio Sample Channel Status Configuration Register 3
fc_audschnl4	0x106b	Frame Composer Audio Sample Channel Status Configuration Register 4

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Register	Offset	Description
fc_audschnl5	0x106c	Frame Composer Audio Sample Channel Status Configuration Register 5
fc_audschnl6	0x106d	Frame Composer Audio Sample Channel Status Configuration Register 6
fc_audschnl7	0x106e	Frame Composer Audio Sample Channel Status Configuration Register 7
fc_audschnl8	0x106f	Frame Composer Audio Sample Channel Status Configuration Register 8
fc_ctrlqhigh	0x1073	Frame Composer Number of High Priority Packets Attended Configuration Register
fc_ctrlqlow	0x1074	Frame Composer Number of Low Priority Packets Attended Configuration Register
fc_acp0	0x1075	Frame Composer ACP Packet Type Configuration Register 0
fc_acp16	0x1082	Frame Composer ACP Packet Body Configuration Register 16
fc_acp15	0x1083	Frame Composer ACP Packet Body Configuration Register 15
fc_acp14	0x1084	Frame Composer ACP Packet Body Configuration Register 14
fc_acp13	0x1085	Frame Composer ACP Packet Body Configuration Register 13
fc_acp12	0x1086	Frame Composer ACP Packet Body Configuration Register 12
fc_acp11	0x1087	Frame Composer ACP Packet Body Configuration Register 11
fc_acp10	0x1088	Frame Composer ACP Packet Body Configuration Register 10
fc_acp9	0x1089	Frame Composer ACP Packet Body Configuration Register 9
fc_acp8	0x108a	Frame Composer ACP Packet Body Configuration Register 8
fc_acp7	0x108b	Frame Composer ACP Packet Body Configuration Register 7
fc_acp6	0x108c	Frame Composer ACP Packet Body Configuration Register 6

Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_acp5	0x108d	Frame Composer ACP Packet Body Configuration Register 5
fc_acp4	0x108e	Frame Composer ACP Packet Body Configuration Register 4
fc_acp3	0x108f	Frame Composer ACP Packet Body Configuration Register 3
fc_acp2	0x1090	Frame Composer ACP Packet Body Configuration

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Register	Offset	Description
		Register 2
fc_acp1	0x1091	Frame Composer ACP Packet Body Configuration Register 1
fc_iscr1_0	0x1092	Frame Composer ISRC1 Packet Status, Valid, and Continue Configuration Register
fc_iscr1_16	0x1093	Frame Composer ISRC1 Packet Body Register 16
fc_iscr1_15	0x1094	Frame Composer ISRC1 Packet Body Register 15
fc_iscr1_14	0x1095	Frame Composer ISRC1 Packet Body Register 14
fc_iscr1_13	0x1096	Frame Composer ISRC1 Packet Body Register 13
fc_iscr1_12	0x1097	Frame Composer ISRC1 Packet Body Register 12
fc_iscr1_11	0x1098	Frame Composer ISRC1 Packet Body Register 11
fc_iscr1_10	0x1099	Frame Composer ISRC1 Packet Body Register 10
fc_iscr1_9	0x109a	Frame Composer ISRC1 Packet Body Register 9
fc_iscr1_8	0x109b	Frame Composer ISRC1 Packet Body Register 8
fc_iscr1_7	0x109c	Frame Composer ISRC1 Packet Body Register 7
fc_iscr1_6	0x109d	Frame Composer ISRC1 Packet Body Register 6
fc_iscr1_5	0x109e	Frame Composer ISRC1 Packet Body Register 5
fc_iscr1_4	0x109f	Frame Composer ISRC1 Packet Body Register 4
fc_iscr1_3	0x10a0	Frame Composer ISRC1 Packet Body Register 3
fc_iscr1_2	0x10a1	Frame Composer ISRC1 Packet Body Register 2
fc_iscr1_1	0x10a2	Frame Composer ISRC1 Packet Body Register 1
fc_iscr2_15	0x10a3	Frame Composer ISRC2 Packet Body Register 15
fc_iscr2_14	0x10a4	Frame Composer ISRC2 Packet Body Register 14
fc_iscr2_13	0x10a5	Frame Composer ISRC2 Packet Body Register 13
fc_iscr2_12	0x10a6	Frame Composer ISRC2 Packet Body Register 12
fc_iscr2_11	0x10a7	Frame Composer ISRC2 Packet Body Register 11
fc_iscr2_10	0x10a8	Frame Composer ISRC2 Packet Body Register 10
fc_iscr2_9	0x10a9	Frame Composer ISRC2 Packet Body Register 9
fc_iscr2_8	0x10aa	Frame Composer ISRC2 Packet Body Register 8
fc_iscr2_7	0x10ab	Frame Composer ISRC2 Packet Body Register 7

Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_iscr2_6	0x10ac	Frame Composer ISRC2 Packet Body Register 6
fc_iscr2_5	0x10ad	Frame Composer ISRC2 Packet Body Register 5
fc_iscr2_4	0x10ae	Frame Composer ISRC2 Packet Body Register 4
fc_iscr2_3	0x10af	Frame Composer ISRC2 Packet Body Register 3
fc_iscr2_2	0x10b0	Frame Composer ISRC2 Packet Body Register 2
fc_iscr2_1	0x10b1	Frame Composer ISRC2 Packet Body Register 1
fc_iscr2_0	0x10b2	Frame Composer ISRC2 Packet Body Register 0
fc_datauto0	0x10b3	Frame Composer Data Island Auto Packet Scheduling Register 0 Configures the Frame Composer RDRB(1)/ Manual(0)...
fc_datauto1	0x10b4	Frame Composer Data Island Auto Packet Scheduling Register 1 Configures the Frame Composer (FC)...
fc_datauto2	0x10b5	Frame Composer Data Island Auto packet scheduling

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Register	Offset	Description
		Register 2 Configures the Frame Composer (FC)...
fc_datman	0x10b6	Frame Composer Data Island Manual Packet Request Register Requests to the Frame Composer the data...
fc_datauto3	0x10b7	Frame Composer Data Island Auto Packet Scheduling Register 3 Configures the Frame Composer Automatic(1)/ RDRB(0)...
fc_rdrb0	0x10b8	Frame Composer Round Robin ACR Packet Insertion Register 0 Configures the Frame Composer (FC) RDRB...
fc_rdrb1	0x10b9	Frame Composer Round Robin ACR Packet Insertion Register 1 Configures the Frame Composer (FC) RDRB...
fc_rdrb2	0x10ba	Frame Composer Round Robin AUDI Packet Insertion Register 2 Configures the Frame Composer (FC)...
fc_rdrb3	0x10bb	Frame Composer Round Robin AUDI Packet Insertion Register 3 Configures the Frame Composer (FC)...
fc_rdrb4	0x10bc	Frame Composer Round Robin GCP Packet Insertion Register 4 Configures the Frame Composer (FC) RDRB...
fc_rdrb5	0x10bd	Frame Composer Round Robin GCP Packet Insertion Register 5 Configures the Frame Composer (FC) RDRB...
fc_rdrb6	0x10be	Frame Composer Round Robin AVI Packet Insertion Register 6 Configures the Frame Composer (FC) RDRB...
fc_rdrb7	0x10bf	Frame Composer Round Robin AVI Packet Insertion Register 7 Configures the Frame Composer (FC) RDRB...
fc_rdrb8	0x10c0	Frame Composer Round Robin AMP Packet Insertion Register 8

Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_rdrb9	0x10c1	Frame Composer Round Robin AMP Packet Insertion Register 9
fc_rdrb10	0x10c2	Frame Composer Round Robin NTSC VBI Packet Insertion Register 10
fc_rdrb11	0x10c3	Frame Composer Round Robin NTSC VBI Packet Insertion Register 11
fc_rdrb12	0x10c4	Frame Composer Round Robin DRM Packet Insertion Register 12
fc_rdrb13	0x10c5	Frame Composer Round Robin DRM Packet Insertion Register 13
fc_mask0	0x10d2	Frame Composer Packet Interrupt Mask Register 0
fc_mask1	0x10d6	Frame Composer Packet Interrupt Mask Register 1

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Register	Offset	Description
fc_mask2	0x10da	Frame Composer High/Low Priority Overflow Interrupt Mask Register 2
fc_prconf	0x10e0	Frame Composer Pixel Repetition Configuration Register
fc_scrambler_ctrl	0x10e1	Frame Composer Scrambler Control
fc_multistream_ctrl	0x10e2	Frame Composer Multi-Stream Audio Control
fc_packet_tx_en	0x10e3	Frame Composer Packet Transmission Control
fc_actspc_hdlr_cfg	0x10e8	Frame Composer Active Space Control
fc_invact_2d_0	0x10e9	Frame Composer Input Video 2D VActive Pixels Register 0
fc_invact_2d_1	0x10ea	Frame Composer Input Video VActive pixels Register 1
fc_gmd_stat	0x1100	Frame Composer GMD Packet Status Register Gamut metadata packet status bit information for no_current_gmd,...
fc_gmd_en	0x1101	Frame Composer GMD Packet Enable Register This register enables Gamut metadata (GMD) packet transmission....
fc_gmd_up	0x1102	Frame Composer GMD Packet Update Register This register performs an GMD packet content update according...
fc_gmd_conf	0x1103	Frame Composer GMD Packet Schedule Configuration Register This register configures the number of...
fc_gmd_hb	0x1104	Frame Composer GMD Packet Profile and Gamut Sequence Configuration Register This register configures...
fc_gmd_pb[0:27]	0x1105 + (i * 0x1)	Frame Composer GMD Packet Body Register Array Configures the GMD packet body of the GMD...
fc_amp_hb1	0x1128	Frame Composer AMP Packet Header Register 1
fc_amp_hb2	0x1129	Frame Composer AMP Packet Header Register 2

Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_amp_pb[0:27]	0x112a + (i * 0x1)	Frame Composer AMP Packet Body Register Array
fc_nvbi_hb1	0x1148	Frame Composer NTSC VBI Packet Header Register 1
fc_nvbi_hb2	0x1149	Frame Composer NTSC VBI Packet Header Register 2
fc_nvbi_pb[0:26]	0x114a + (i * 0x1)	Frame Composer NTSC VBI Packet Body Register Array
fc_drm_up	0x1167	Frame Composer DRM Packet Update Register
fc_drm_hb[0:1]	0x1168 + (i * 0x1)	Frame Composer DRM Packet Header Register Array
fc_drm_pb[0:26]	0x116a + (i * 0x1)	Frame Composer DRM Packet Body Register Array
fc_dbgforce	0x1200	Frame Composer video/audio Force Enable Register This register allows to force the controller to...

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Register	Offset	Description
fc_dbgaud0ch0	0x1201	Frame Composer Audio Data Channel 0 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch0	0x1202	Frame Composer Audio Data Channel 0 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch0	0x1203	Frame Composer Audio Data Channel 0 Register 2 Configures the audio fixed data to be used in channel...
fc_dbgaud0ch1	0x1204	Frame Composer Audio Data Channel 1 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch1	0x1205	Frame Composer Audio Data Channel 1 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch1	0x1206	Frame Composer Audio Data Channel 1 Register 2 Configures the audio fixed data to be used in channel...
fc_dbgaud0ch2	0x1207	Frame Composer Audio Data Channel 2 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch2	0x1208	Frame Composer Audio Data Channel 2 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch2	0x1209	Frame Composer Audio Data Channel 2 Register 2 Configures the audio fixed data to be used in channel...
fc_dbgaud0ch3	0x120a	Frame Composer Audio Data Channel 3 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch3	0x120b	Frame Composer Audio Data Channel 3 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch3	0x120c	Frame Composer Audio Data Channel 3 Register 2 Configures the audio fixed data to be used in channel...

Registers for Address Block: FrameComposer (Continued)

Register	Offset	Description
fc_dbgaud0ch4	0x120d	Frame Composer Audio Data Channel 4 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch4	0x120e	Frame Composer Audio Data Channel 4 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch4	0x120f	Frame Composer Audio Data Channel 4 Register 2

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Register	Offset	Description
		Configures the audio fixed data to be used in channel...
fc_dbgaud0ch5	0x1210	Frame Composer Audio Data Channel 5 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch5	0x1211	Frame Composer Audio Data Channel 5 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch5	0x1212	Frame Composer Audio Data Channel 5 Register 2 Configures the audio fixed data to be used in channel...
fc_dbgaud0ch6	0x1213	Frame Composer Audio Data Channel 6 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch6	0x1214	Frame Composer Audio Data Channel 6 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch6	0x1215	Frame Composer Audio Data Channel 6 Register 2 Configures the audio fixed data to be used in channel...
fc_dbgaud0ch7	0x1216	Frame Composer Audio Data Channel 7 Register 0 Configures the audio fixed data to be used in channel...
fc_dbgaud1ch7	0x1217	Frame Composer Audio Data Channel 7 Register 1 Configures the audio fixed data to be used in channel...
fc_dbgaud2ch7	0x1218	Frame Composer Audio Data Channel 7 Register 2 Configures the audio fixed data to be used in channel...
fc_dbgtmds[0:2]	0x1219 + (i * 0x1)	Frame Composer TMDS Data Channel Register Array Configures the video fixed data to be used in TMDS...

fc_invidconf

Description: Frame Composer Input Video Configuration and HDCP Keepout Register

Size: 8 bits

Offset: 0x1000

Bits	Name	Attr	Description
7	HDCP_keepout	R/W	Start/stop HDCP keepout window generation 1b: Active Value After Reset: 0x0
6	vsync_in_polarity	R/W	Vsync input polarity 1b: Active high 0b: Active low Value After Reset: 0x1
5	hsync_in_polarity	R/W	Hsync input polarity 1b: Active high 0b: Active low

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Bits	Name	Attr	Description
			Value After Reset: 0x1
4	de_in_polarity	R/W	Data enable input polarity 1b: Active high 0b: Active low Value After Reset: 0x1
3	DVI_modez	R/W	Active low 0b: DVI mode selected 1b: HDMI mode selected Value After Reset: 0x0
2			Reserved for future use.
1	r_v_blank_in_osc	R/W	Used for CEA861-D modes with fractional Vblank (for example, modes 5, 6, 7, 10, 11, 20, 21, and 22). For more modes, see the CEA861-D specification. Note: Set this field to 1 for video mode 39, although there is no Vblank oscillation. 1b: Active high Value After Reset: 0x0

Fields for Register: fc_invidconf (Continued)

Bits	Name	Attr	Description
0	in_I_P	R/W	Input video mode: 1b: Interlaced 0b: Progressive Value After Reset: 0x0

fc_inhactiv0

Description: Frame Composer Input Video HActive Pixels Register 0

Size: 8 bits

Offset: 0x1001

Bits	Name	Attr	Description
7:0	H_in_activ	R/W	Input video Horizontal active pixel region width. Number of Horizontal active pixels [0...8191]. Value After Reset: 0x0

fc_inhactiv1

Description: Frame Composer Input Video HActive Pixels Register 1

Size: 8 bits

Offset: 0x1002

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	H_in_activ_13	R/W	Input video Horizontal active pixel region width (0 .. 16383) If the configuration parameter HDMI_TX_20 = True (1), this bit field holds bit 13. Value After Reset: 0x0

Fields for Register: fc_inhactiv1 (Continued)

Bits	Name	Attr	Description
4	H_in_activ_12	R/W	Input video Horizontal active pixel region width (0 . .

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			8191) If configuration parameter HDMI_TX_14 = True (1), this bit field holds bit 12. Value After Reset: 0x0
3:0	H_in_activ	R/W	Input video Horizontal active pixel region width Value After Reset: 0x0

fc_inhblank0

Description: Frame Composer Input Video HBlank Pixels Register 0

Size: 8 bits

Offset: 0x1003

Bits	Name	Attr	Description
7:0	H_in_blank	R/W	Input video Horizontal blanking pixel region width. Number of Horizontal blanking pixels [0...4095]. Value After Reset: 0x0

fc_inhblank1

Description: Frame Composer Input Video HBlank Pixels Register 1

Size: 8 bits

Offset: 0x1004

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:2	H_in_blank_12	R/W	Input video Horizontal blanking pixel region width If configuration parameter HDMI_TX_14 = True (1), this bit field holds bit 12:10 of number of horizontal blanking pixels. Value After Reset: 0x0
1:0	H_in_blank	R/W	Input video Horizontal blanking pixel region width this bit field holds bits 9:8 of number of Horizontal blanking pixels. Value After Reset: 0x0

fc_invactiv0

Description: Frame Composer Input Video VActive Pixels Register 0

Size: 8 bits

Offset: 0x1005

Bits	Name	Attr	Description
7:0	V_in_activ	R/W	Input video Vertical active pixel region width. This bit field holds bits 7:0 of number of Vertical active pixels. Value After Reset: 0x0

fc_invactiv1

Description: Frame Composer Input Video VActive Pixels Register 1

Size: 8 bits

Offset: 0x1006

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Bits	Name	Attr	Description
7:5			Reserved for future use.
4:3	V_in_activ_12_11	R/W	Input video Vertical active pixel region width. If the configuration parameter HDMI_TX_14 = True (1), this bit field holds bits 12:10 of number of Vertical active pixels. Value After Reset: 0x0
2:0	V_in_activ	R/W	Input video Vertical active pixel region width. This bit field holds bits 9:8 of number of Vertical active pixels. Value After Reset: 0x0

fc_invblank

Description: Frame Composer Input Video VBlank Pixels Register

Size: 8 bits

Offset: 0x1007

Bits	Name	Attr	Description
7:0	V_in_blank	R/W	Input video Vertical blanking pixel region width. Number of Vertical blanking lines [0...255]. Value After Reset: 0x0

fc_hsyncindelay0

Description: Frame Composer Input Video HSync Front Porch Register 0

Size: 8 bits

Offset: 0x1008

Bits	Name	Attr	Description
7:0	H_in_delay	R/W	Input video Hsync active edge delay. Integer number of pixel clock cycles from "de" non active edge of the last "de" valid period [0...4095]. Value After Reset: 0x0

fc_hsyncindelay1

Description: Frame Composer Input Video HSync Front Porch Register 1

Size: 8 bits

Offset: 0x1009

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:3	H_in_delay_12	R/W	Input video Horizontal active edge delay. If configuration parameter HDMI_TX_14 = True (1), this bit field holds bit 12. Integer number of pixel clock cycles from "de" non-active edge of the last "de" valid period [0...8191]. Value After Reset: 0x0
2:0	H_in_delay	R/W	Input video Horizontal active edge delay. Value After Reset: 0x0

fc_hsyncinwidth0

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Description: Frame Composer Input Video HSync Width Register 0
Size: 8 bits
Offset: 0x100a

Bits	Name	Attr	Description
7:0	H_in_width	R/W	Input video Hsync active pulse width. Integer number of pixel clock cycles [0...511]. Value After Reset: 0x0

fc_hsyncinwidth1
Description: Frame Composer Input Video HSync Width Register 1
Size: 8 bits
Offset: 0x100b

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	H_in_width_9	R/W	Input video Hsync active pulse width. If configuration parameter HDMI_TX_14 = True (1), then this bit field holds bit 9. Number of Horizontal active pixels [0...1024]. Value After Reset: 0x0
0	H_in_width	R/W	Input video Hsync active pulse width. Value After Reset: 0x0

fc_vsyncindelay
Description: Frame Composer Input Video VSync Front Porch Register
Size: 8 bits
Offset: 0x100c

Bits	Name	Attr	Description
7:0	V_in_delay	R/W	Input video Vsync active edge delay. Integer number of Hsync pulses from "de" non active edge of the last "de" valid period. [0...255]. Value After Reset: 0x0

fc_vsyncinwidth
Description: Frame Composer Input Video VSync Width Register
Size: 8 bits
Offset: 0x100d

Bits	Name	Attr	Description
7:6			Reserved for future use.
5:0	V_in_width	R/W	Description: Input video Vsync active pulse width. Integer number of video lines [0...63]. Value After Reset: 0x0

fc_infreq0
Description: Frame Composer Input Video Refresh Rate Register 0
Size: 8 bits
Offset: 0x100e

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Bits	Name	Attr	Description
7:0	infreq	R/W	Video refresh rate in Hz*1E3 format. This register is provided for debug and informative purposes. The Hdmi_tx does not write any data to this register; the data written by software is not used by the Hdmi_tx. Value After Reset: 0x0

fc_infreq1

Description: Frame Composer Input Video Refresh Rate Register 1

Size: 8 bits

Offset: 0x100f

Bits	Name	Attr	Description
7:0	infreq	R/W	Video refresh rate in Hz*1E3 format. This register is provided for debug and informative purposes. The Hdmi_tx does not write any data to this register; the data written by software is not used by the Hdmi_tx. Value After Reset: 0x0

fc_infreq2

Description: Frame Composer Input Video Refresh Rate Register 2

Size: 8 bits

Offset: 0x1010

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	infreq	R/W	Video refresh rate in Hz*1E3 format. This register is provided for debug and informative purposes. The Hdmi_tx does not write any data to this register; the data written by software is not used by the Hdmi_tx. Value After Reset: 0x0

fc_ctrldur

Description: Frame Composer Control Period Duration Register

Size: 8 bits

Offset: 0x1011

Bits	Name	Attr	Description
7:0	ctrlperiodduration	R/W	Configuration of the control period minimum duration (minimum of 12 pixel clock cycles; refer to HDMI 1.4b specification). Integer number of pixel clocks cycles [0..223]. Value After Reset: 0x0

fc_exctrldur

Description: Frame Composer Extended Control Period Duration Register

Size: 8 bits

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Offset: 0x1012

Bits	Name	Attr	Description
7:0	exctrlperiodduration	R/W	Configuration of the extended control period minimum duration (minimum of 32 pixel clock cycles; refer to HDMI 1.4b specification). Integer number of pixel clocks cycles [0..223]. Value After Reset: 0x0

fc_exctrlspac

Description: Frame Composer Extended Control Period Maximum Spacing Register

Size: 8 bits

Offset: 0x1013

Bits	Name	Attr	Description
7:0	exctrlperiodspacing	R/W	Configuration of the maximum spacing between consecutive extended control periods (maximum of 50ms; refer to the applicable HDMI specification). When using the HDMI 2.0 supported features (HDMI_TX_20 = 1): generated spacing = (1/freq tmds clock)*256*512*(extctrlperiodspacing +1) else generated spacing = (1/freq tmds clock)*256*256*(extctrlperiodspacing +1) Value After Reset: 0x0

fc_ch0pream

Description: Frame Composer Channel 0 Non-Preamble Data Register

Size: 8 bits

Offset: 0x1014

Bits	Name	Attr	Description
7:0	ch0_preamble_filter	R/W	When in control mode, configures 8 bits that fill the channel 0 data lines not used to transmit the preamble (for more clarification, refer to the HDMI 1.4b specification). Value After Reset: 0x0

fc_ch1pream

Description: Frame Composer Channel 1 Non-Preamble Data Register

Size: 8 bits

Offset: 0x1015

Bits	Name	Attr	Description
7:6			Reserved for future use.
5:0	ch1_preamble_filter	R/W	When in control mode, configures 6 bits that fill the channel 1 data lines not used to transmit the preamble (for more clarification, refer to the HDMI 1.4b specification). Value After Reset: 0x0

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fc_ch2pream

Description: Frame Composer Channel 2 Non-Preamble Data Register
Size: 8 bits
Offset: 0x1016

Bits	Name	Attr	Description
7:6			Reserved for future use.
5:0	ch2_preamble_filter	R/W	When in control mode, configures 6 bits that fill the channel 2 data lines not used to transmit the preamble (for more clarification, refer to the HDMI 1.4b specification). Value After Reset: 0x0

fc_aviconf3

Description: Frame Composer AVI Packet Configuration Register 3
Size: 8 bits
Offset: 0x1017

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:2	YQ	R/W	YCC Quantization range according to the CEA specification Value After Reset: 0x0
1:0	CN	R/W	IT content type according to CEA the specification Value After Reset: 0x0

fc_gcp

Description: Frame Composer GCP Packet Configuration Register
Size: 8 bits
Offset: 0x1018

Bits	Name	Attr	Description
7:3			Reserved for future use.
2	default_phase	R/W	Value of "default_phase" in the GCP packet. This data must be equal to the default phase used at Video Packetizer packing machine. Value After Reset: 0x0
1	set_avmute	R/W	Value of "set_avmute" in the GCP packet Once the AVmute is set, the frame composer schedules the GCP packet with AVmute set in the packet scheduler to be sent once (may only be transmitted between the active edge of VSYNC and 384 pixels following this edge). Value After Reset: 0x0
0	clear_avmute	R/W	Value of "clear_avmute" in the GCP packet Value After Reset: 0x0

fc_aviconf0

Description: Frame Composer AVI Packet Configuration Register 0

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Size: 8 bits
Offset: 0x1019

Bits	Name	Attr	Description
7	rgc_ycc_indication_2	R/W	Y2, Bit 2 of rgc_ycc_indication Value After Reset: 0x0
6	active_format_present	R/W	Active format present Value After Reset: 0x0
5:4	scan_information	R/W	Scan information Value After Reset: 0x0
3:2	bar_information	R/W	Bar information data valid Value After Reset: 0x0
1:0	rgc_ycc_indication	R/W	Y1,Y0 RGB or YCC indicator Value After Reset: 0x0

fc_aviconf1
Description: Frame Composer AVI Packet Configuration Register 1
Size: 8 bits
Offset: 0x101a

Bits	Name	Attr	Description
7:6	Colorimetry	R/W	Colorimetry Value After Reset: 0x0
5:4	picture_aspect_ratio	R/W	Picture aspect ratio Value After Reset: 0x0
3:0	active_aspect_ratio	R/W	Active aspect ratio Value After Reset: 0x0

fc_aviconf2
Description: Frame Composer AVI Packet Configuration Register 2
Size: 8 bits
Offset: 0x101b

Bits	Name	Attr	Description
7	it_content	R/W	IT content Value After Reset: 0x0
6:4	extended_colorimetry	R/W	Extended colorimetry Value After Reset: 0x0
3:2	quantization_range	R/W	Quantization range Value After Reset: 0x0
1:0	non_uniform_picture_scaling	R/W	Non-uniform picture scaling Value After Reset: 0x0

fc_avivid
Description: Frame Composer AVI Packet VIC Register
Size: 8 bits
Offset: 0x101c

Bits	Name	Attr	Description
7	fc_avivid_7	R/W	Bit 7 of fc_avivid register Value After Reset: 0x0
6:0	fc_avivid	R/W	Configures the AVI InfoFrame Video Identification code. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_avietb[0:1]

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Description: Frame Composer AVI Packet End of Top Bar Register Array
Size: 8 bits
Offset: 0x101d + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_avietb	R/W	Defines the AVI InfoFrame End of Top Bar value. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_avisbb[0:1]
Description: Frame Composer AVI Packet Start of Bottom Bar Register Array
Size: 8 bits
Offset: 0x101f + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_avisbb	R/W	This register defines the AVI InfoFrame Start of Bottom Bar value. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_avielb[0:1]
Description: Frame Composer AVI Packet End of Left Bar Register Array
Size: 8 bits
Offset: 0x1021 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_avielb	R/W	This register defines the AVI InfoFrame End of Left Bar value. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_avisrb[0:1]
Description: Frame Composer AVI Packet Start of Right Bar Register Array
Size: 8 bits
Offset: 0x1023 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_avisrb	R/W	This register defines the AVI InfoFrame Start of Right Bar value. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_audiconf0
Description: Frame Composer AUD Packet Configuration Register 0
Size: 8 bits
Offset: 0x1025

Bits	Name	Attr	Description
7			Reserved for future use.
6:4	CC	R/W	Channel count

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Bits	Name	Attr	Description
			Value After Reset: 0x0
3:0	CT	R/W	Coding Type Value After Reset: 0x0

fc_audiconf1

Description: Frame Composer AUD Packet Configuration Register 1

Size: 8 bits

Offset: 0x1026

Bits	Name	Attr	Description
7:6			Reserved for future use.
5:4	SS	R/W	Sampling size Value After Reset: 0x0
3			Reserved for future use.
2:0	SF	R/W	Sampling frequency Value After Reset: 0x0

fc_audiconf2

Description: Frame Composer AUD Packet Configuration Register 2

Size: 8 bits

Offset: 0x1027

Bits	Name	Attr	Description
7:0	CA	R/W	Channel allocation Value After Reset: 0x0

fc_audiconf3

Description: Frame Composer AUD Packet Configuration Register 3

Size: 8 bits

Offset: 0x1028

Bits	Name	Attr	Description
7			Reserved for future use.
6:5	LFEPBL	R/W	LFE playback information LFEPBL1, LFEPBL0 LFE playback level as compared to the other channels. Value After Reset: 0x0
4	DM_INH	R/W	Down mix enable Value After Reset: 0x0
3:0	LSV	R/W	Level shift value (for down mixing) Value After Reset: 0x0

fc_vsdieeeid2

Description: Frame Composer VSI Packet Data IEEE Register 2

Size: 8 bits

Offset: 0x1029

Bits	Name	Attr	Description
7:0	IEEE	R/W	This register configures the Vendor Specific InfoFrame IEEE registration identifier. For more information, refer to the CEA- 861-E specification. Value After Reset: 0x0

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fc_vsdsize

Description: Frame Composer VSI Packet Data Size Register

Size: 8 bits

Offset: 0x102a

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:0	VSDSIZE	R/W	Packet size as described in the HDMI Vendor Specific InfoFrame (from the HDMI specification). Value After Reset: 0x1b

fc_vsdieeeid1

Description: Frame Composer VSI Packet Data IEEE Register 1

Size: 8 bits

Offset: 0x1030

Bits	Name	Attr	Description
7:0	IEEE	R/W	This register configures the Vendor Specific InfoFrame IEEE registration identifier. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_vsdieeeid0

Description: Frame Composer VSI Packet Data IEEE Register 0

Size: 8 bits

Offset: 0x1031

Bits	Name	Attr	Description
7:0	IEEE	R/W	This register configures the Vendor Specific InfoFrame IEEE registration identifier. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_vsdpayload[0:23]

Description: Frame Composer VSI Packet Data Payload Register Array

Size: 8 bits

Offset: 0x1032 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_vsdpayload	R/W	Frame Composer VSI Packet Data Payload Register Array Configures the Vendor Specific infoFrame 24 bytes specific payload. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_spdvendorname[0:7]

Description: Frame Composer SPD Packet Data Vendor Name Register Array

Size: 8 bits

Offset: 0x104a + (i * 0x1)

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Bits	Name	Attr	Description
7:0	fc_spdvendorname	R/W	Frame Composer SPD Packet Data Vendor Name Register Array Configures the Source Product Descriptor infoFrame 8 bytes Vendor name. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_spdproductname[0:15]

Description: Frame Composer SPD packet Data Product Name Register Array

Size: 8 bits

Offset: 0x1052 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_spdproductname	R/W	Frame Composer SPD packet Data Product Name Register Array Configures the Source Product Descriptor infoFrame 16 bytes Product name. For more information, refer to the CEA-861-E specification. Value After Reset: 0x0

fc_spddeviceinf

Description: Frame Composer SPD Packet Data Source Product Descriptor Register

Size: 8 bits

Offset: 0x1062

Bits	Name	Attr	Description
7:0	fc_spddeviceinf	R/W	Frame Composer SPD Packet Data Source Product Descriptor Register Value After Reset: 0x0

fc_audsconf

Description: Frame Composer Audio Sample Flat and Layout Configuration Register

Size: 8 bits

Offset: 0x1063

Bits	Name	Attr	Description
7:4	aud_packet_sampflt	R/W	Set the audio packet sample flat value to be sent on the packet. Value After Reset: 0x0
3:1			Reserved for future use.
0	aud_packet_layout	R/W	Set the audio packet layout to be sent in the packet: 1b: layout 1 0b: layout 0 If HDMI_TX_20 is defined and register field fc_multistream_ctrl.fc_mas_packet_en is active, this bit has no effect. Value After Reset: 0x0

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fc_audsstat

Description: Frame Composer Audio Sample Flat and Layout Status Register
Size: 8 bits
Offset: 0x1064

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	packet_sampprs	R	Shows the data sample present indication of the last Audio sample packet sent by the HDMI Tx Controller. This register information is at TMDS clock rate. Value After Reset: 0x0

fc_audsv

Description: Frame Composer Audio Sample Validity Flag Register
Size: 8 bits
Offset: 0x1065

Bits	Name	Attr	Description
7	V3r	R/W	Set validity bit "V" for Channel 3, Right Value After Reset: 0x0
6	V2r	R/W	Set validity bit "V" for Channel 2, Right Value After Reset: 0x0
5	V1r	R/W	Set validity bit "V" for Channel 1, Right Value After Reset: 0x0
4	V0r	R/W	Set validity bit "V" for Channel 0, Right Value After Reset: 0x0
3	V3l	R/W	Set validity bit "V" for Channel 3, Left Value After Reset: 0x0
2	V2l	R/W	Set validity bit "V" for Channel 2, Left Value After Reset: 0x0
1	V1l	R/W	Set validity bit "V" for Channel 1, Left Value After Reset: 0x0
0	V0l	R/W	Set validity bit "V" for Channel 0, Left Value After Reset: 0x0

fc_audsu

Description: Frame Composer Audio Sample User Flag Register
Size: 8 bits
Offset: 0x1066

Bits	Name	Attr	Description
7	U3r	R/W	Set user bit "U" for Channel 3, Right Value After Reset: 0x0
6	U2r	R/W	Set user bit "U" for Channel 2, Right Value After Reset: 0x0
5	U1r	R/W	Set user bit "U" for Channel 1, Right Value After Reset: 0x0
4	U0r	R/W	Set user bit "U" for Channel 0, Right

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Bits	Name	Attr	Description
			Value After Reset: 0x0
3	U3l	R/W	Set user bit "U" for Channel 3, Left Value After Reset: 0x0
2	U2l	R/W	Set user bit "U" for Channel 2, Left Value After Reset: 0x0
1	U1l	R/W	Set user bit "U" for Channel 1, Left Value After Reset: 0x0
0	U0l	R/W	Set user bit "U" for Channel 0, Left Value After Reset: 0x0

fc_audschnl0

Description: Frame Composer Audio Sample Channel Status Configuration Register 0

Size: 8 bits

Offset: 0x1067

Bits	Name	Attr	Description
7:6			Reserved for future use.
5:4	oiec_cgmsa	R/W	CGMS-A Value After Reset: 0x0
3:1			Reserved for future use.
0	oiec_copyright	R/W	IEC Copyright indication Value After Reset: 0x0

fc_audschnl1

Description: Frame Composer Audio Sample Channel Status Configuration Register 1

Size: 8 bits

Offset: 0x1068

Bits	Name	Attr	Description
7:0	oiec_categorycode	R/W	Category code Value After Reset: 0x0

fc_audschnl2

Description: Frame Composer Audio Sample Channel Status Configuration Register 2

Size: 8 bits

Offset: 0x1069

Bits	Name	Attr	Description
7			Reserved for future use.
6:4	oiec_pcmaudiomode	R/W	PCM audio mode Value After Reset: 0x0
3:0	oiec_sourcenumber	R/W	Source number Value After Reset: 0x0

fc_audschnl3

Description: Frame Composer Audio Sample Channel Status Configuration Register 3

Size: 8 bits

Offset: 0x106a

Bits	Name	Attr	Description
7:4	oiec_channelnumcr1	R/W	Channel number for second right sample

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Bits	Name	Attr	Description
			Value After Reset: 0x0
3:0	oiec_channelnumcr0	R/W	Channel number for first right sample Value After Reset: 0x0

fc_audschnl4

Description: Frame Composer Audio Sample Channel Status Configuration Register 4

Size: 8 bits

Offset: 0x106b

Bits	Name	Attr	Description
7:4	oiec_channelnumcr3	R/W	Channel number for fourth right sample Value After Reset: 0x0
3:0	oiec_channelnumcr2	R/W	Channel number for third right sample Value After Reset: 0x0

fc_audschnl5

Description: Frame Composer Audio Sample Channel Status Configuration Register 5

Size: 8 bits

Offset: 0x106c

Bits	Name	Attr	Description
7:4	oiec_channelnumcl1	R/W	Channel number for second left sample Value After Reset: 0x0
3:0	oiec_channelnumcl0	R/W	Channel number for first left sample Value After Reset: 0x0

fc_audschnl6

Description: Frame Composer Audio Sample Channel Status Configuration Register 6

Size: 8 bits

Offset: 0x106d

Bits	Name	Attr	Description
7:4	oiec_channelnumcl3	R/W	Channel number for fourth left sample Value After Reset: 0x0
3:0	oiec_channelnumcl2	R/W	Channel number for third left sample Value After Reset: 0x0

fc_audschnl7

Description: Frame Composer Audio Sample Channel Status Configuration Register 7

Size: 8 bits

Offset: 0x106e

Bits	Name	Attr	Description
7:6	oiec_sampfreq_ext	R/W	Sampling frequency (channel status bits 31 and 30) Value After Reset: 0x0
5:4	oiec_clkaccuracy	R/W	Clock accuracy Value After Reset: 0x0
3:0	oiec_sampfreq	R/W	Sampling frequency Value After Reset: 0x0

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fc_audschnl8
Description: Frame Composer Audio Sample Channel Status Configuration Register 8
Size: 8 bits
Offset: 0x106f

Bits	Name	Attr	Description
7:4	oiec_origsampfreq	R/W	Original sampling frequency Value After Reset: 0x0
3:0	oiec_wordlength	R/W	Word length configuration Value After Reset: 0x0

fc_ctrlqhigh
Description: Frame Composer Number of High Priority Packets Attended Configuration Register
Size: 8 bits
Offset: 0x1073

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:0	onhighattended	R/W	Configures the number of high priority packets or audio sample packets consecutively attended before checking low priority queue status. Valid range is from 5'd1 to 5'd31. Value After Reset: 0xf

fc_ctrlqlow
Description: Frame Composer Number of Low Priority Packets Attended Configuration Register
Size: 8 bits
Offset: 0x1074

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:0	onlowattended	R/W	Configures the number of low priority packets or null packets consecutively attended before checking high priority queue status or audio samples availability. Valid range is from 5'd1 to 5'd31. Value After Reset: 0x3

fc_acp0
Description: Frame Composer ACP Packet Type Configuration Register 0
Size: 8 bits
Offset: 0x1075

Bits	Name	Attr	Description
7:0	acptype	R/W	Configures the ACP packet type. Value After Reset: 0x0

fc_acp16
Description: Frame Composer ACP Packet Body Configuration Register 16
Size: 8 bits

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Offset: 0x1082

Bits	Name	Attr	Description
7:0	fc_acp16	R/W	Frame Composer ACP Packet Body Configuration Register 16 Value After Reset: 0x0

fc_acp15

Description: Frame Composer ACP Packet Body Configuration Register 15

Size: 8 bits

Offset: 0x1083

Bits	Name	Attr	Description
7:0	fc_acp15	R/W	Frame Composer ACP Packet Body Configuration Register 15 Value After Reset: 0x0

fc_acp14

Description: Frame Composer ACP Packet Body Configuration Register 14

Size: 8 bits

Offset: 0x1084

Bits	Name	Attr	Description
7:0	fc_acp14	R/W	Frame Composer ACP Packet Body Configuration Register 14 Value After Reset: 0x0

fc_acp13

Description: Frame Composer ACP Packet Body Configuration Register 13

Size: 8 bits

Offset: 0x1085

Bits	Name	Attr	Description
7:0	fc_acp13	R/W	Frame Composer ACP Packet Body Configuration Register 13 Value After Reset: 0x0

fc_acp12

Description: Frame Composer ACP Packet Body Configuration Register 12

Size: 8 bits

Offset: 0x1086

Bits	Name	Attr	Description
7:0	fc_acp12	R/W	Frame Composer ACP Packet Body Configuration Register 12 Value After Reset: 0x0

fc_acp11

Description: Frame Composer ACP Packet Body Configuration Register 11

Size: 8 bits

Offset: 0x1087

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Bits	Name	Attr	Description
7:0	fc_acp11	R/W	Frame Composer ACP Packet Body Configuration Register 11 Value After Reset: 0x0

fc_acp10

Description: Frame Composer ACP Packet Body Configuration Register 10

Size: 8 bits

Offset: 0x1088

Bits	Name	Attr	Description
7:0	fc_acp10	R/W	Frame Composer ACP Packet Body Configuration Register 10 Value After Reset: 0x0

fc_acp9

Description: Frame Composer ACP Packet Body Configuration Register 9

Size: 8 bits

Offset: 0x1089

Bits	Name	Attr	Description
7:0	fc_acp9	R/W	Frame Composer ACP Packet Body Configuration Register 9 Value After Reset: 0x0

fc_acp8

Description: Frame Composer ACP Packet Body Configuration Register 8

Size: 8 bits

Offset: 0x108a

Bits	Name	Attr	Description
7:0	fc_acp8	R/W	Frame Composer ACP Packet Body Configuration Register 8 Value After Reset: 0x0

fc_acp7

Description: Frame Composer ACP Packet Body Configuration Register 7

Size: 8 bits

Offset: 0x108b

Bits	Name	Attr	Description
7:0	fc_acp7	R/W	Frame Composer ACP Packet Body Configuration Register 7 Value After Reset: 0x0

fc_acp6

Description: Frame Composer ACP Packet Body Configuration Register 6

Size: 8 bits

Offset: 0x108c

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	fc_acp6	R/W	Frame Composer ACP Packet Body Configuration Register 6 Value After Reset: 0x0

fc_acp5

Description: Frame Composer ACP Packet Body Configuration Register 5

Size: 8 bits

Offset: 0x108d

Bits	Name	Attr	Description
7:0	fc_acp5	R/W	Frame Composer ACP Packet Body Configuration Register 5 Value After Reset: 0x0

fc_acp4

Description: Frame Composer ACP Packet Body Configuration Register 4

Size: 8 bits

Offset: 0x108e

Bits	Name	Attr	Description
7:0	fc_acp4	R/W	Frame Composer ACP Packet Body Configuration Register 4 Value After Reset: 0x0

fc_acp3

Description: Frame Composer ACP Packet Body Configuration Register 3

Size: 8 bits

Offset: 0x108f

Bits	Name	Attr	Description
7:0	fc_acp3	R/W	Frame Composer ACP Packet Body Configuration Register 3 Value After Reset: 0x0

fc_acp2

Description: Frame Composer ACP Packet Body Configuration Register 2

Size: 8 bits

Offset: 0x1090

Bits	Name	Attr	Description
7:0	fc_acp2	R/W	Frame Composer ACP Packet Body Configuration Register 2 Value After Reset: 0x0

fc_acp1

Description: Frame Composer ACP Packet Body Configuration Register 1

Size: 8 bits

Offset: 0x1091

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	fc_acp1	R/W	Frame Composer ACP Packet Body Configuration Register 1 Value After Reset: 0x0

fc_iscr1_0

Description: Frame Composer ISRC1 Packet Status, Valid, and Continue Configuration Register

Size: 8 bits

Offset: 0x1092

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:2	isrc_status	R/W	ISRC1 Status signal Value After Reset: 0x0
1	isrc_valid	R/W	ISRC1 Valid control signal Value After Reset: 0x0
0	isrc_cont	R/W	ISRC1 Indication of packet continuation (ISRC2 will be transmitted) Value After Reset: 0x0

fc_iscr1_16

Description: Frame Composer ISRC1 Packet Body Register 16

Size: 8 bits

Offset: 0x1093

Bits	Name	Attr	Description
7:0	fc_iscr1_16	R/W	Frame Composer ISRC1 Packet Body Register 16; configures ISRC1 packet body of the ISRC1 packet Value After Reset: 0x0

fc_iscr1_15

Description: Frame Composer ISRC1 Packet Body Register 15

Size: 8 bits

Offset: 0x1094

Bits	Name	Attr	Description
7:0	fc_iscr1_15	R/W	Frame Composer ISRC1 Packet Body Register 15 Value After Reset: 0x0

fc_iscr1_14

Description: Frame Composer ISRC1 Packet Body Register 14

Size: 8 bits

Offset: 0x1095

Bits	Name	Attr	Description
7:0	fc_iscr1_14	R/W	Frame Composer ISRC1 Packet Body Register 14 Value After Reset: 0x0

fc_iscr1_13

Description: Frame Composer ISRC1 Packet Body Register 13

Size: 8 bits

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Offset: 0x1096

Bits	Name	Attr	Description
7:0	fc_iscr1_13	R/W	Frame Composer ISRC1 Packet Body Register 13 Value After Reset: 0x0

fc_iscr1_12

Description: Frame Composer ISRC1 Packet Body Register 12

Size: 8 bits

Offset: 0x1097

Bits	Name	Attr	Description
7:0	fc_iscr1_12	R/W	Frame Composer ISRC1 Packet Body Register 12 Value After Reset: 0x0

fc_iscr1_11

Description: Frame Composer ISRC1 Packet Body Register 11

Size: 8 bits

Offset: 0x1098

Bits	Name	Attr	Description
7:0	fc_iscr1_11	R/W	Frame Composer ISRC1 Packet Body Register 11 Value After Reset: 0x0

fc_iscr1_10

Description: Frame Composer ISRC1 Packet Body Register 10

Size: 8 bits

Offset: 0x1099

Bits	Name	Attr	Description
7:0	fc_iscr1_10	R/W	Frame Composer ISRC1 Packet Body Register 10 Value After Reset: 0x0

fc_iscr1_9

Description: Frame Composer ISRC1 Packet Body Register 9

Size: 8 bits

Offset: 0x109a

Bits	Name	Attr	Description
7:0	fc_iscr1_9	R/W	Frame Composer ISRC1 Packet Body Register 9 Value After Reset: 0x0

fc_iscr1_8

Description: Frame Composer ISRC1 Packet Body Register 8

Size: 8 bits

Offset: 0x109b

Bits	Name	Attr	Description
7:0	fc_iscr1_8	R/W	Frame Composer ISRC1 Packet Body Register 8 Value After Reset: 0x0

fc_iscr1_7

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Description: Frame Composer ISRC1 Packet Body Register 7
Size: 8 bits
Offset: 0x109c

Bits	Name	Attr	Description
7:0	fc_iscr1_7	R/W	Frame Composer ISRC1 Packet Body Register 7 Value After Reset: 0x0

fc_iscr1_6
Description: Frame Composer ISRC1 Packet Body Register 6
Size: 8 bits
Offset: 0x109d

Bits	Name	Attr	Description
7:0	fc_iscr1_6	R/W	Frame Composer ISRC1 Packet Body Register 6 Value After Reset: 0x0

fc_iscr1_5
Description: Frame Composer ISRC1 Packet Body Register 5
Size: 8 bits
Offset: 0x109e

Bits	Name	Attr	Description
7:0	fc_iscr1_5	R/W	Frame Composer ISRC1 Packet Body Register 5 Value After Reset: 0x0

fc_iscr1_4
Description: Frame Composer ISRC1 Packet Body Register 4
Size: 8 bits
Offset: 0x109f

Bits	Name	Attr	Description
7:0	fc_iscr1_4	R/W	Frame Composer ISRC1 Packet Body Register 4 Value After Reset: 0x0

fc_iscr1_3
Description: Frame Composer ISRC1 Packet Body Register 3
Size: 8 bits
Offset: 0x10a0

Bits	Name	Attr	Description
7:0	fc_iscr1_3	R/W	Frame Composer ISRC1 Packet Body Register 3 Value After Reset: 0x0

fc_iscr1_2
Description: Frame Composer ISRC1 Packet Body Register 2
Size: 8 bits
Offset: 0x10a1

Bits	Name	Attr	Description
7:0	fc_iscr1_2	R/W	Frame Composer ISRC1 Packet Body Register 2 Value After Reset: 0x0

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fc_iscr1_1
Description: Frame Composer ISRC1 Packet Body Register 1
Size: 8 bits
Offset: 0x10a2

Bits	Name	Attr	Description
7:0	fc_iscr1_1	R/W	Frame Composer ISRC1 Packet Body Register 1 Value After Reset: 0x0

fc_iscr2_15
Description: Frame Composer ISRC2 Packet Body Register 15
Size: 8 bits
Offset: 0x10a3

Bits	Name	Attr	Description
7:0	fc_iscr2_15	R/W	Frame Composer ISRC2 Packet Body Register 15; configures the ISRC2 packet body of the ISRC2 packet Value After Reset: 0x0

fc_iscr2_14
Description: Frame Composer ISRC2 Packet Body Register 14
Size: 8 bits
Offset: 0x10a4

Bits	Name	Attr	Description
7:0	fc_iscr2_14	R/W	Frame Composer ISRC2 Packet Body Register 14 Value After Reset: 0x0

fc_iscr2_13
Description: Frame Composer ISRC2 Packet Body Register 13
Size: 8 bits
Offset: 0x10a5

Bits	Name	Attr	Description
7:0	fc_iscr2_13	R/W	Frame Composer ISRC2 Packet Body Register 13 Value After Reset: 0x0

fc_iscr2_12
Description: Frame Composer ISRC2 Packet Body Register 12
Size: 8 bits
Offset: 0x10a6

Bits	Name	Attr	Description
7:0	fc_iscr2_12	R/W	Frame Composer ISRC2 Packet Body Register 12 Value After Reset: 0x0

fc_iscr2_11
Description: Frame Composer ISRC2 Packet Body Register 11
Size: 8 bits
Offset: 0x10a7

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Bits	Name	Attr	Description
7:0	fc_iscr2_11	R/W	Frame Composer ISRC2 Packet Body Register 11 Value After Reset: 0x0

fc_iscr2_10

Description: Frame Composer ISRC2 Packet Body Register 10

Size: 8 bits

Offset: 0x10a8

Bits	Name	Attr	Description
7:0	fc_iscr2_10	R/W	Frame Composer ISRC2 Packet Body Register 10 Value After Reset: 0x0

fc_iscr2_9

Description: Frame Composer ISRC2 Packet Body Register 9

Size: 8 bits

Offset: 0x10a9

Bits	Name	Attr	Description
7:0	fc_iscr2_9	R/W	Frame Composer ISRC2 Packet Body Register 9 Value After Reset: 0x0

fc_iscr2_8

Description: Frame Composer ISRC2 Packet Body Register 8

Size: 8 bits

Offset: 0x10aa

Bits	Name	Attr	Description
7:0	fc_iscr2_8	R/W	Frame Composer ISRC2 Packet Body Register 8 Value After Reset: 0x0

fc_iscr2_7

Description: Frame Composer ISRC2 Packet Body Register 7

Size: 8 bits

Offset: 0x10ab

Bits	Name	Attr	Description
7:0	fc_iscr2_7	R/W	Frame Composer ISRC2 Packet Body Register 7 Value After Reset: 0x0

fc_iscr2_6

Description: Frame Composer ISRC2 Packet Body Register 6

Size: 8 bits

Offset: 0x10ac

Bits	Name	Attr	Description
7:0	fc_iscr2_6	R/W	Frame Composer ISRC2 Packet Body Register 6 Value After Reset: 0x0

fc_iscr2_5

Description: Frame Composer ISRC2 Packet Body Register 5

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Size: 8 bits
Offset: 0x10ad

Bits	Name	Attr	Description
7:0	fc_iscr2_5	R/W	Frame Composer ISRC2 Packet Body Register 5 Value After Reset: 0x0

fc_iscr2_4
Description: Frame Composer ISRC2 Packet Body Register 4
Size: 8 bits
Offset: 0x10ae

Bits	Name	Attr	Description
7:0	fc_iscr2_4	R/W	Frame Composer ISRC2 Packet Body Register 4 Value After Reset: 0x0

fc_iscr2_3
Description: Frame Composer ISRC2 Packet Body Register 3
Size: 8 bits
Offset: 0x10af

Bits	Name	Attr	Description
7:0	fc_iscr2_3	R/W	Frame Composer ISRC2 Packet Body Register 3 Value After Reset: 0x0

fc_iscr2_2
Description: Frame Composer ISRC2 Packet Body Register 2
Size: 8 bits
Offset: 0x10b0

Bits	Name	Attr	Description
7:0	fc_iscr2_2	R/W	Frame Composer ISRC2 Packet Body Register 2 Value After Reset: 0x0

fc_iscr2_1
Description: Frame Composer ISRC2 Packet Body Register 1
Size: 8 bits
Offset: 0x10b1

Bits	Name	Attr	Description
7:0	fc_iscr2_1	R/W	Frame Composer ISRC2 Packet Body Register 1 Value After Reset: 0x0

fc_iscr2_0
Description: Frame Composer ISRC2 Packet Body Register 0
Size: 8 bits
Offset: 0x10b2

Bits	Name	Attr	Description
7:0	fc_iscr2_0	R/W	Frame Composer ISRC2 Packet Body Register 0 Value After Reset: 0x0

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fc_datauto0

Description: Frame Composer Data Island Auto Packet Scheduling Register 0
Configures the Frame Composer RDRB(1)/Manual(0) data island packet insertion for SPD, VSD, ISRC2, ISRC1 and ACP packets. On RDRB mode the described packet scheduling is controlled by registers FC_DATAUTO1 and FC_DATAUTO2, while in Manual mode register FC_DATMAN requests to FC the insertion of the requested packet.
Size: 8 bits
Offset: 0x10b3

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	spd_auto	R/W	Enables SPD automatic packet scheduling Value After Reset: 0x0
3	vsd_auto	R/W	Enables VSD automatic packet scheduling Value After Reset: 0x0
2	iscr2_auto	R/W	Enables ISRC2 automatic packet scheduling Value After Reset: 0x0
1	iscr1_auto	R/W	Enables ISRC1 automatic packet scheduling Value After Reset: 0x0
0	acp_auto	R/W	Enables ACP automatic packet scheduling Value After Reset: 0x0

fc_datauto1

Description: Frame Composer Data Island Auto Packet Scheduling Register 1
Configures the Frame Composer (FC) RDRB frame interpolation for SPD, VSD, ISRC2, ISRC1 and ACP packet insertion on data island when FC is on RDRB mode for the listed packets.
Size: 8 bits
Offset: 0x10b4

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	auto_frame_interpolation	R/W	Packet frame interpolation for automatic packet scheduling Value After Reset: 0x0

fc_datauto2

Description: Frame Composer Data Island Auto packet scheduling Register 2
Configures the Frame Composer (FC) RDRB line interpolation and number of packets in frame for SPD, VSD, ISRC2, ISRC1 and ACP packet insertion on data island when FC is on RDRB mode for the listed packets.
Size: 8 bits
Offset: 0x10b5

Bits	Name	Attr	Description
7:4	auto_frame_packets	R/W	Packets per frame, for automatic packet scheduling Value After Reset: 0x0
3:0	auto_line_spacing	R/W	Packets line spacing, for automatic packet scheduling Value After Reset: 0x0

fc_datman

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Description: Frame Composer Data Island Manual Packet Request Register
Requests to the Frame Composer the data island packet insertion for NULL, SPD, VSD, ISRC2, ISRC1 and ACP packets when FC_DATAUTO0 bit is in manual mode for the packet requested.
Size: 8 bits
Offset: 0x10b6

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	null_tx	W	Null packet Value After Reset: 0x0
4	spd_tx	W	SPD packet Value After Reset: 0x0
3	vsd_tx	W	VSD packet Value After Reset: 0x0
2	iscr2_tx	W	ISRC2 packet Value After Reset: 0x0
1	iscr1_tx	W	ISRC1 packet Value After Reset: 0x0
0	acp_tx	W	ACP packet Value After Reset: 0x0

fc_datauto3
Description: Frame Composer Data Island Auto Packet Scheduling Register 3
Configures the Frame Composer Automatic(1)/RDRB(0) data island packet insertion for AVI, GCP, AUDI and ACR packets. In Automatic mode, the packet is inserted on Vblanking when first line with active Vsync appears.
Size: 8 bits
Offset: 0x10b7

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	nvbi_auto	R/W	Enables NTSC VBI packet insertion Value After Reset: 0x1
4	amp_auto	R/W	Enables AMP packet insertion Value After Reset: 0x1
3	avi_auto	R/W	Enables AVI packet insertion Value After Reset: 0x1
2	gcp_auto	R/W	Enables GCP packet insertion Value After Reset: 0x1
1	audi_auto	R/W	Enables AUDI packet insertion Value After Reset: 0x1
0	acr_auto	R/W	Enables ACR packet insertion Value After Reset: 0x1

fc_rdrb0
Description: Frame Composer Round Robin ACR Packet Insertion Register 0
Configures the Frame Composer (FC) RDRB frame interpolation for ACR packet insertion on data island when FC is on RDRB mode for this packet.
Size: 8 bits
Offset: 0x10b8

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	ACRframeinterpolation	R/W	ACR Frame interpolation Value After Reset: 0x0

fc_rdrb1

Description: Frame Composer Round Robin ACR Packet Insertion Register 1
Configures the Frame Composer (FC) RDRB line interpolation and number of packets in frame for the ACR packet insertion on data island when FC is on RDRB mode this packet.
Size: 8 bits
Offset: 0x10b9

Bits	Name	Attr	Description
7:4	ACRpacketsinframe	R/W	ACR packets in frame Value After Reset: 0x0
3:0	ACRpacketlinespacing	R/W	ACR packet line spacing Value After Reset: 0x0

fc_rdrb2

Description: Frame Composer Round Robin AUDI Packet Insertion Register 2
Configures the Frame Composer (FC) RDRB frame interpolation for AUDI packet insertion on data island when FC is on RDRB mode for this packet.
Size: 8 bits
Offset: 0x10ba

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	AUDIframeinterpolation	R/W	Audio frame interpolation Value After Reset: 0x0

fc_rdrb3

Description: Frame Composer Round Robin AUDI Packet Insertion Register 3
Configures the Frame Composer (FC) RDRB line interpolation and number of packets in frame for the AUDI packet insertion on data island when FC is on RDRB mode this packet.
Size: 8 bits
Offset: 0x10bb

Bits	Name	Attr	Description
7:4	AUDIpacketsinframe	R/W	Audio packets per frame Value After Reset: 0x0
3:0	AUDIpacketlinespacing	R/W	Audio packets line spacing Value After Reset: 0x0

fc_rdrb4

Description: Frame Composer Round Robin GCP Packet Insertion Register 4
Configures the Frame Composer (FC) RDRB frame interpolation for GCP packet insertion on data island when FC is on RDRB mode for this packet.
Size: 8 bits
Offset: 0x10bc

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	GCPframeinterpolation	R/W	Frames interpolated between GCP packets Value After Reset: 0x0

fc_rdrb5

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Description: Frame Composer Round Robin GCP Packet Insertion Register 5
Configures the Frame Composer (FC) RDRB line interpolation and number of packets in frame for the GCP packet insertion on data island when FC is on RDRB mode this packet.
Size: 8 bits
Offset: 0x10bd

Bits	Name	Attr	Description
7:4	GCPpacketsinframe	R/W	GCP packets per frame Value After Reset: 0x0
3:0	GCPpacketlinespacing	R/W	GCP packets line spacing Value After Reset: 0x0

fc_rdrb6
Description: Frame Composer Round Robin AVI Packet Insertion Register 6
Configures the Frame Composer (FC) RDRB frame interpolation for AVI packet insertion on data island when FC is on RDRB mode for this packet.
Size: 8 bits
Offset: 0x10be

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	AVIframeinterpolation	R/W	Frames interpolated between AVI packets Value After Reset: 0x0

fc_rdrb7
Description: Frame Composer Round Robin AVI Packet Insertion Register 7
Configures the Frame Composer (FC) RDRB line interpolation and number of packets in frame for the AVI packet insertion on data island when FC is on RDRB mode this packet.
Size: 8 bits
Offset: 0x10bf

Bits	Name	Attr	Description
7:4	AVIpacketsinframe	R/W	AVI packets per frame Value After Reset: 0x0
3:0	AVIpacketlinespacing	R/W	AVI packets line spacing Value After Reset: 0x0

fc_rdrb8
Description: Frame Composer Round Robin AMP Packet Insertion Register 8
Size: 8 bits
Offset: 0x10c0

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	AMPframeinterpolation	R/W	AMP frame interpolation Value After Reset: 0x0

fc_rdrb9
Description: Frame Composer Round Robin AMP Packet Insertion Register 9
Size: 8 bits
Offset: 0x10c1

Bits	Name	Attr	Description
7:4	AMPpacketsinframe	R/W	AMP packets per frame Value After Reset: 0x0
3:0	AMPpacketlinespacing	R/W	AMP packets line spacing Value After Reset: 0x0

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fc_rdrb10

Description: Frame Composer Round Robin NTSC VBI Packet Insertion Register 10

Size: 8 bits

Offset: 0x10c2

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	NVBIframeinterpolation	R/W	NTSC VBI frame interpolation Value After Reset: 0x0

fc_rdrb11

Description: Frame Composer Round Robin NTSC VBI Packet Insertion Register 11

Size: 8 bits

Offset: 0x10c3

Bits	Name	Attr	Description
7:4	NVBIpacketsinframe	R/W	NTSC VBI packets per frame Value After Reset: 0x0
3:0	NVBIpacketlinespacing	R/W	NTSC VBI packets line spacing Value After Reset: 0x0

fc_rdrb12

Description: Frame Composer Round Robin DRM Packet Insertion Register 12

Size: 8 bits

Offset: 0x10c4

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	DRMframeinterpolation	R/W	Description: DRM frame interpolation Value After Reset: 0x0

fc_rdrb13

Description: Frame Composer Round Robin DRM Packet Insertion Register 13

Size: 8 bits

Offset: 0x10c5

Bits	Name	Attr	Description
7:4	DRMpacketsinframe	R/W	DRM packets per frame Value After Reset: 0x0
3:0	DRMpacketlinespacing	R/W	DRM packets line spacing Value After Reset: 0x0

fc_mask0

Description: Frame Composer Packet Interrupt Mask Register 0

Size: 8 bits

Offset: 0x10d2

Bits	Name	Attr	Description
7	AUDI	R/W	Mask bit for FC_INT0.AUDI interrupt bit Value After Reset: 0x0
6	ACP	R/W	Mask bit for FC_INT0.ACP interrupt bit Value After Reset: 0x0
5	HBR	R/W	Mask bit for FC_INT0.HBR interrupt bit Value After Reset: 0x1
4	MAS	R/W	Mask bit for FC_INT0.MAS interrupt bit. Otherwise, this field is a "spare" bit with no associated

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Bits	Name	Attr	Description
			functionality. Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0"
3	NVBI	R/W	Mask bit for FC_INT0.NVBI interrupt bit. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0"
2	AUDS	R/W	Mask bit for FC_INT0.AUDS interrupt bit Value After Reset: 0x1
1	ACR	R/W	Mask bit for FC_INT0.ACR interrupt bit Value After Reset: 0x0
0	NULL	R/W	Mask bit for FC_INT0.NULL interrupt bit Value After Reset: 0x1

fc_mask1

Description: Frame Composer Packet Interrupt Mask Register 1

Size: 8 bits

Offset: 0x10d6

Bits	Name	Attr	Description
7	GMD	R/W	Mask bit for FC_INT1.GMD interrupt bit Value After Reset: 0x0
6	ISCR1	R/W	Mask bit for FC_INT1.ISRC1 interrupt bit Value After Reset: 0x0
5	ISCR2	R/W	Mask bit for FC_INT1.ISRC2 interrupt bit Value After Reset: 0x0
4	VSD	R/W	Mask bit for FC_INT1.VSD interrupt bit Value After Reset: 0x0
3	SPD	R/W	Mask bit for FC_INT1.SPD interrupt bit Value After Reset: 0x0
2	AMP	R/W	Mask bit for FC_INT1.AMP interrupt bit. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: "(HDMI_TX_20== 1) ? 1 : 0"
1	AVI	R/W	Mask bit for FC_INT1.AVI interrupt bit Value After Reset: 0x0
0	GCP	R/W	Mask bit for FC_INT1.GCP interrupt bit Value After Reset: 0x0

fc_mask2

Description: Frame Composer High/Low Priority Overflow and DRM Interrupt Mask Register 2

Size: 8 bits

Offset: 0x10da

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	DRM	R/W	Mask bit for FC_INT2.DRM interrupt bit. Value After Reset: (HDMI_TX_20== 1) ? 1 : 0

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Bits	Name	Attr	Description
3:2			Reserved for future use.
1	LowPriority_overflow	R/W	Mask bit for FC_INT2.LowPriority_overflow interrupt bit Value After Reset: 0x0
0	HighPriority_overflow	R/W	Mask bit for FC_INT2.HighPriority_overflow interrupt bit Value After Reset: 0x0

fc_prconf

Description: Frame Composer Pixel Repetition Configuration Register
Size: 8 bits
Offset: 0x10e0

Bits	Name	Attr	Description
7:4	incoming_pr_factor	R/W	Configures the input video pixel repetition. For CEA modes, this value must be extracted from the CEA specification for the video mode being input. incoming_pr_factor[3:0] 0000b: No action. Not used. 0001b: No pixel repetition (pixel sent only once) 0010b: Pixel sent two times (pixel repeated once) 0011b: Pixel sent three times 0100b: Pixel sent four times 0101b: Pixel sent five times 0110b: Pixel sent six times 0111b: Pixel sent seven times 1000b: Pixel sent eight times 1001b: Pixel sent nine times 1010b: Pixel sent 10 times Other: Reserved. Not used Value After Reset: 0x1
3:0	output_pr_factor	R/W	Configures the video pixel repetition ratio to be sent on the AVI InfoFrame. This value must be valid according to the HDMI specification. The output_pr_factor = incoming_pr_factor * (desired_pr_factor + 1) – 1. output_pr_factor[3:0] 0000b: No action. Not used. 0001b: Pixel sent two times (pixel repeated once) 0010b: Pixel sent three times 0011b: Pixel sent four times 0100b: Pixel sent five times 0101b: Pixel sent six times 0110b: Pixel sent seven times 0111b: Pixel sent eight times 1000b: Pixel sent nine times 1001b: Pixel sent 10 times Other: Reserved. Not used Note: When working in YCC422 video, the actual repetition of the stream is Incoming_pr_factor * (desired_pr_factor + 1). This calculation is done internally in the H13TCTRL and no hardware overflow protection is available. Care must be taken to avoid this result passes the maximum number of 10 pixels repeated because no HDMI support is available for this in the specification and the H13TPHY does not

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Bits	Name	Attr	Description
			support this higher repetition values. Value After Reset: 0x0

fc_scrambler_ctrl

Description: Frame Composer Scrambler Control

Size: 8 bits

Offset: 0x10e1

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	scrambler_ucp_line	R/W	Debug register. When active (1'b1), the Unscrambled Control Period is generated after each active video line (non-compliant behavior). This is quasi-static field which requires a a mc_swrstzreq.tmdsswrst_req reset request to be performed after the change of this configuration bit. Value After Reset: 0x0
3:1			Reserved for future use.
0	scrambler_on	R/W	When set (1'b1), this field activates the HDMI 2.0 scrambler feature. When disabled (1'b0) the scrambler feature is bypassed, placing Hdmi_tx in HDMI 1.4b compatible mode. To activate the scrambler feature, you must ensure that the quasi-static configuration bit fc_invidconf.HDCP_keepout is set (1'b1) at configuration time, before the required mc_swrstzreq.tmdsswrst_req reset request is issued. This is field can be changed in runtime. Value After Reset: 0x0

fc_multistream_ctrl

Description: Frame Composer Multi-Stream Audio Control

Size: 8 bits

Offset: 0x10e2

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	fc_mas_packet_en	R/W	This field, when set (1'b1), activates the HDMI 2.0 Multi- Stream support. The audio stream present at the input of the Hdmi_tx controller is transported using Multi-Stream Audio Sample Packets. Value After Reset: 0x0

fc_packet_tx_en

Description: Frame Composer Packet Transmission Control

Size: 8 bits

Offset: 0x10e3

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Bits	Name	Attr	Description
7	drm_tx_en	R/W	DRM transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x0
6	nvbi_tx_en	R/W	NTSC VBI transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x0
5	amp_tx_en	R/W	AMP transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x0
4	aut_tx_en	R/W	ACP, SPD, VSIF, ISRC1, and SRC2 packet transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x1
3	audi_tx_en	R/W	AUDI packet transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x1
2	avi_tx_en	R/W	AVI packet transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x1
1	gcp_tx_en	R/W	GCP transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x1
0	acr_tx_en	R/W	ACR packet transmission control 1b: Transmission enabled 0b: Transmission disabled Value After Reset: 0x1

fc_actspc_hdlr_cfg

Description: Frame Composer Active Space Control

Size: 8 bits

Offset: 0x10e8

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	actspc_hdlr_tgl	R/W	Active Space handler control 1b: Active space 1 value is different from Active Space 2 value. Refer to Figure 8-4: 3D Structure of the HDMI 1.4b specification. 0b: Active space not oscillating Value After Reset: 0x0
0	actspc_hdlr_en	R/W	Active Space Handler Control 1b: Fixed active space value mode enabled. During active space, a fixed value of 0xAA is applied to all TMDS channels. 0b: Fixed active space value mode disabled Value After Reset: 0x0

fc_invact_2d_0

Description: Frame Composer Input Video 2D VActive Pixels Register 0

Size: 8 bits

Offset: 0x10e9

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Bits	Name	Attr	Description
7:0	fc_invact_2d_0	R/W	2D Input video vertical active pixel region width. Number of 2D video vertical active lines [7:0]. Value After Reset: 0x0

fc_invact_2d_1

Description: Frame Composer Input Video VActive pixels Register 1

Size: 8 bits

Offset: 0x10ea

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	fc_invact_2d_1	R/W	2D Input video vertical active pixel region width. Number of 2D video vertical active lines [11:8]. Value After Reset: 0x0

fc_gmd_stat

Description: Frame Composer GMD Packet Status Register

Gamut metadata packet status bit information for no_current_gmd, next_gmd_field, gmd_packet_sequence and current_gamut_seq_num. For more information, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x1100

Bits	Name	Attr	Description
7	igmdno_crnt_gbd	R	Gamut scheduling: No current gamut data Value After Reset: 0x0
6	igmddnext_field	R	Gamut scheduling: Gamut Next field Value After Reset: 0x0
5:4	igmdpacket_seq	R	Gamut scheduling: Gamut packet sequence Value After Reset: 0x0
3:0	igmdcurrent_gamut_seq_num	R	Gamut scheduling: Current Gamut packet sequence number Value After Reset: 0x0

fc_gmd_en

Description: Frame Composer GMD Packet Enable Register

This register enables Gamut metadata (GMD) packet transmission. Packets are inserted in the incoming frame, starting in the line where active Vsync indication starts. After enable of GMD packets the outgoing packet is sent with no_current_gmd active indication until update GMD request is performed in the controller.

Size: 8 bits

Offset: 0x1101

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	gmdenabletx	R/W	Gamut Metadata packet transmission enable (1b) Value After Reset: 0x0

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fc_gmd_up

Description: Frame Composer GMD Packet Update Register

This register performs an GMD packet content update according to the configured packet body (FC_GMD_PB0 to FC_GMD_PB27) and packet header (FC_GMD_HB). This active high auto clear register reflects the body and header configurations on the GMD packets sent arbitrating the current_gamut_seq_num, gmd_packet_sequence and next_gmd_field bits on packet to correctly indicate to source the Gamut change to be performed. After enable GMD packets the first update request is also responsible for deactivating the no_current_gmd indication bit. Attention packet update request must only be done after correct configuration of GMD packet body and header registers. Correct affected_gamut_seq_num and gmd_profile configuration is user responsibility and must convey with HDMI 1.4b standard gamut rules.

Size: 8 bits

Offset: 0x1102

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	gmdupdatepacket	W	Gamut Metadata packet update Value After Reset: 0x0

fc_gmd_conf

Description: Frame Composer GMD Packet Schedule Configuration Register

This register configures the number of GMD packets to be inserted per frame (starting always in the line where the active Vsync appears) and the line spacing between the transmitted GMD packets.

Note that for profile P0 (refer to the HDMI 1.4b specification) this register should only indicate one GMD packet to be inserted per video field.

Size: 8 bits

Offset: 0x1103

Bits	Name	Attr	Description
7:4	gmdpacketsinframe	R/W	Number of GMD packets per frame or video field (profile P0) Value After Reset: 0x1
3:0	gmdpacketlinespacing	R/W	Number of line spacing between the transmitted GMD packets Value After Reset: 0x0

fc_gmd_hb

Description: Frame Composer GMD Packet Profile and Gamut Sequence Configuration Register

This register configures the GMD packet header affected_gamut_seq_num and gmd_profile bits. For more information, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x1104

Bits	Name	Attr	Description
7			Reserved for future use.
6:4	gmdgbd_profile	R/W	GMD profile bits. Hdmi_tx only supports Profile 0 (P0) of the Gamut Boundary Description Metadata Profiles described in the HDMI 1.4 Specification (which defines

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Bits	Name	Attr	Description
			four profiles, P0-P4). Value After Reset: 0x0
3:0	gmdaffected_gamut_seq_num	R/W	Affected gamut sequence number Value After Reset: 0x0

fc_gmd_pb[0:27]

Description: Frame Composer GMD Packet Body Register Array Configures the GMD packet body of the GMD packet.

Size: 8 bits

Offset: 0x1105 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_gmd_pb	R/W	Frame Composer GMD Packet Body Register Array Value After Reset: 0x0

fc_amp_hb1

Description: Frame Composer AMP Packet Header Register 1

Size: 8 bits

Offset: 0x1128

Bits	Name	Attr	Description
7:0	fc_amp_hb0	R/W	Frame Composer AMP Packet Header Register 1 Value After Reset: 0x0

fc_amp_hb2

Description: Frame Composer AMP Packet Header Register 2

Size: 8 bits

Offset: 0x1129

Bits	Name	Attr	Description
7:0	fc_amp_hb1	R/W	Frame Composer AMP Packet Header Register 2 Value After Reset: 0x0

fc_amp_pb[0:27]

Description: Frame Composer AMP Packet Body Register Array

Size: 8 bits

Offset: 0x112a + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_amp_pb	R/W	Frame Composer AMP Packet Body Register Array Value After Reset: 0x0

fc_nvbi_hb1

Description: Frame Composer NTSC VBI Packet Header Register 1

Size: 8 bits

Offset: 0x1148

Bits	Name	Attr	description
7:0	fc_nvbi_hb0	R/W	Frame Composer NTSC VBI Packet Header Register 1 Value After Reset: 0x0

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fc_nvbi_hb2

Description: Frame Composer NTSC VBI Packet Header Register 2

Size: 8 bits

Offset: 0x1149

Bits	Name	Attr	Description
7:0	fc_nvbi_hb1	R/W	Frame Composer NTSC VBI Packet Header Register 2 Value After Reset: 0x0

fc_nvbi_pb[0:26]

Description: Frame Composer NTSC VBI Packet Body Register Array

Size: 8 bits

Offset: 0x114a + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_nvbi_pb	R/W	Frame Composer NTSC VBI Packet Body Register Array Value After Reset: 0x0

fc_drm_up

Description: Frame Composer DRM Packet Update Register

This register performs an DRM packet content update according to the configured packet body (FC_DRM_PB0 to FC_DRM_PB27) and packet header (FC_DRM_HB). This active high auto clear register reflects the body and header configurations on the DRM packets change to be performed.

Attention packet update request must only be done after correct configuration of DRM packet body and header registers.

Size: 8 bits

Offset: 0x1167

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	drmpacketupdate	W	DRM packet update Value After Reset: 0x0

fc_drm_hb[0:1]

Description: Frame Composer DRM Packet Header Register Array

Size: 8 bits

Offset: 0x1168 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_drm_hb	R/W	Frame Composer DRM Packet Header Register Array Value After Reset: 0x0

fc_drm_pb[0:26]

Description: Frame Composer DRM Packet Body Register Array

Size: 8 bits

Offset: 0x116a + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_drm_pb	R/W	Frame Composer DRM Packet Body Register Array

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Bits	Name	Attr	Description
			Value After Reset: 0x0

fc_dbgforce

Description: Frame Composer video/audio Force Enable Register
This register allows to force the controller to output audio and video data the values configured in the FC_DBGAUD and FC_DBGTMDS registers.
Size: 8 bits
Offset: 0x1200

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	forceaudio	R/W	Force fixed audio output with FC_DBGAUDxCHx register contents. Value After Reset: 0x0
3:1			Reserved for future use.
0	forcevideo	R/W	Force fixed video output with FC_DBGTMDSx register contents. Value After Reset: 0x0

fc_dbgaud0ch0

Description: Frame Composer Audio Data Channel 0 Register 0
Configures the audio fixed data to be used in channel 0 when in fixed audio selection.
Size: 8 bits
Offset: 0x1201

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch0	R/W	Frame Composer Audio Data Channel 0 Register 0 Value After Reset: 0x0

fc_dbgaud1ch0

Description: Frame Composer Audio Data Channel 0 Register 1
Configures the audio fixed data to be used in channel 0 when in fixed audio selection.
Size: 8 bits
Offset: 0x1202

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch0	R/W	Frame Composer Audio Data Channel 0 Register 1 Value After Reset: 0x0

fc_dbgaud2ch0

Description: Frame Composer Audio Data Channel 0 Register 2
Configures the audio fixed data to be used in channel 0 when in fixed audio selection.
Size: 8 bits
Offset: 0x1203

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch0	R/W	Frame Composer Audio Data Channel 0 Register 2 Value After Reset: 0x0

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fc_dbgaud0ch1

Description: Frame Composer Audio Data Channel 1 Register 0

Configures the audio fixed data to be used in channel 1 when in fixed audio selection.

Size: 8 bits

Offset: 0x1204

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch1	R/W	Frame Composer Audio Data Channel 1 Register 0 Value After Reset: 0x0

fc_dbgaud1ch1

Description: Frame Composer Audio Data Channel 1 Register 1

Configures the audio fixed data to be used in channel 1 when in fixed audio selection.

Size: 8 bits

Offset: 0x1205

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch1	R/W	Frame Composer Audio Data Channel 1 Register 1 Value After Reset: 0x0

fc_dbgaud2ch1

Description: Frame Composer Audio Data Channel 1 Register 2

Configures the audio fixed data to be used in channel 1 when in fixed audio selection.

Size: 8 bits

Offset: 0x1206

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch1	R/W	Frame Composer Audio Data Channel 1 Register 2 Value After Reset: 0x0

fc_dbgaud0ch2

Description: Frame Composer Audio Data Channel 2 Register 0

Configures the audio fixed data to be used in channel 2 when in fixed audio selection.

Size: 8 bits

Offset: 0x1207

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch2	R/W	Frame Composer Audio Data Channel 2 Register 0 Value After Reset: 0x0

fc_dbgaud1ch2

Description: Frame Composer Audio Data Channel 2 Register 1

Configures the audio fixed data to be used in channel 2 when in fixed audio selection.

Size: 8 bits Offset: 0x1208

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch2	R/W	Frame Composer Audio Data Channel 2 Register 1 Value After Reset: 0x0

fc_dbgaud2ch2

Description: Frame Composer Audio Data Channel 2 Register 2

Configures the audio fixed data to be used in channel 2 when in fixed audio selection.

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Size: 8 bits
Offset: 0x1209

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch2	R/W	Frame Composer Audio Data Channel 2 Register 2 Value After Reset: 0x0

fc_dbgaud0ch3
Description: Frame Composer Audio Data Channel 3 Register 0
Configures the audio fixed data to be used in channel 3 when in fixed audio selection.
Size: 8 bits
Offset: 0x120a

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch3	R/W	Frame Composer Audio Data Channel 3 Register 0 Value After Reset: 0x0

fc_dbgaud1ch3
Description: Frame Composer Audio Data Channel 3 Register 1
Configures the audio fixed data to be used in channel 3 when in fixed audio selection.
Size: 8 bits
Offset: 0x120b

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch3	R/W	Frame Composer Audio Data Channel 3 Register 1 Value After Reset: 0x0

fc_dbgaud2ch3
Description: Frame Composer Audio Data Channel 3 Register 2
Configures the audio fixed data to be used in channel 3 when in fixed audio selection.
Size: 8 bits
Offset: 0x120c

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch3	R/W	Frame Composer Audio Data Channel 3 Register 2 Value After Reset: 0x0

fc_dbgaud0ch4
Description: Frame Composer Audio Data Channel 4 Register 0
Configures the audio fixed data to be used in channel 4 when in fixed audio selection.
Size: 8 bits
Offset: 0x120d

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch4	R/W	Frame Composer Audio Data Channel 4 Register 0 Value After Reset: 0x0

fc_dbgaud1ch4
Description: Frame Composer Audio Data Channel 4 Register 1
Configures the audio fixed data to be used in channel 4 when in fixed audio selection.

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Size: 8 bits
Offset: 0x120e

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch4	R/W	Frame Composer Audio Data Channel 4 Register 1 Value After Reset: 0x0

fc_dbgaud2ch4
Description: Frame Composer Audio Data Channel 4 Register 2
Configures the audio fixed data to be used in channel 4 when in fixed audio selection.
Size: 8 bits
Offset: 0x120f

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch4	R/W	Frame Composer Audio Data Channel 4 Register 2 Value After Reset: 0x0

fc_dbgaud0ch5
Description: Frame Composer Audio Data Channel 5 Register 0
Configures the audio fixed data to be used in channel 5 when in fixed audio selection.
Size: 8 bits
Offset: 0x1210

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch5	R/W	Frame Composer Audio Data Channel 5 Register 0 Value After Reset: 0x0

fc_dbgaud1ch5
Description: Frame Composer Audio Data Channel 5 Register 1
Configures the audio fixed data to be used in channel 5 when in fixed audio selection.
Size: 8 bits
Offset: 0x1211

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch5	R/W	Frame Composer Audio Data Channel 5 Register 1 Value After Reset: 0x0

fc_dbgaud2ch5
Description: Frame Composer Audio Data Channel 5 Register 2
Configures the audio fixed data to be used in channel 5 when in fixed audio selection.
Size: 8 bits
Offset: 0x1212

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch5	R/W	Frame Composer Audio Data Channel 5 Register 2 Value After Reset: 0x0

fc_dbgaud0ch6
Description: Frame Composer Audio Data Channel 6 Register 0
Configures the audio fixed data to be used in channel 6 when in fixed audio selection.
Size: 8 bits
Offset: 0x1213

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Bits	Name	Attr	Description
7:0	fc_dbgaud0ch6	R/W	Frame Composer Audio Data Channel 6 Register 0 Value After Reset: 0x0

fc_dbgaud1ch6

Description: Frame Composer Audio Data Channel 6 Register 1

Configures the audio fixed data to be used in channel 6 when in fixed audio selection.

Size: 8 bits

Offset: 0x1214

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch6	R/W	Frame Composer Audio Data Channel 6 Register 1 Value After Reset: 0x0

fc_dbgaud2ch6

Description: Frame Composer Audio Data Channel 6 Register 2

Configures the audio fixed data to be used in channel 6 when in fixed audio selection.

Size: 8 bits

Offset: 0x1215

Bits	Name	Attr	Description
7:0	fc_dbgaud2ch6	R/W	Frame Composer Audio Data Channel 6 Register 2 Value After Reset: 0x0

fc_dbgaud0ch7

Description: Frame Composer Audio Data Channel 7 Register 0

Configures the audio fixed data to be used in channel 7 when in fixed audio selection.

Size: 8 bits

Offset: 0x1216

Bits	Name	Attr	Description
7:0	fc_dbgaud0ch7	R/W	Frame Composer Audio Data Channel 7 Register 0 Value After Reset: 0x0

fc_dbgaud1ch7

Description: Frame Composer Audio Data Channel 7 Register 1

Configures the audio fixed data to be used in channel 7 when in fixed audio selection.

Size: 8 bits

Offset: 0x1217

Bits	Name	Attr	Description
7:0	fc_dbgaud1ch7	R/W	Frame Composer Audio Data Channel 7 Register 1 Value After Reset: 0x0

fc_dbgaud2ch7

Description: Frame Composer Audio Data Channel 7 Register 2

Configures the audio fixed data to be used in channel 7 when in fixed audio selection.

Size: 8 bits

Offset: 0x1218

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Bits	Name	Attr	Description
7:0	fc_dbgaud2ch7	R/W	Frame Composer Audio Data Channel 7 Register 2 Value After Reset: 0x0

fc_dbgtmds[0:2]

Description: Frame Composer TMDS Data Channel Register Array

Configures the video fixed data to be used in TMDS channel x (where x is 0 to 2) when in fixed video selection.

For Channel 0, this equals to set B pixel component value in RGB video or Cb pixel component value in YCbCr.

For Channel 1, this equals set G pixel component value in RGB video or Y pixel component value in YCbCr.

For Channel 2, this equals to set R pixel component value in RGB video or Cr pixel component value in YCbCr.

Size: 8 bits

Offset: 0x1219 + (i * 0x1)

Bits	Name	Attr	Description
7:0	fc_dbgtmds	R/W	Frame Composer TMDS Data Channel 0 Register Value After Reset: 0x0

PHYConfiguration Registers

PHY Configuration Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: PHYConfiguration

Register	Offset	Description
phy_conf0	0x3000	PHY Configuration Register This register holds the power down, data enable polarity, and interface...
phy_tst0	0x3001	PHY Test Interface Register 0 PHY TX mapped test interface (control). For more information, refer...
phy_tst1	0x3002	PHY Test Interface Register 1 PHY TX mapped text interface (data in). For more information, refer...
phy_tst2	0x3003	PHY Test Interface Register 2 PHY TX mapped text interface (data out). For more information, refer...
phy_stat0	0x3004	PHY RXSENSE, PLL Lock, and HPD Status Register This register contains the following active high...
phy_int0	0x3005	PHY RXSENSE, PLL Lock, and HPD Interrupt Register This register contains the interrupt indication...
phy_mask0	0x3006	PHY RXSENSE, PLL Lock, and HPD Mask Register Mask register for generation of PHY_INT0...
phy_pol0	0x3007	PHY RXSENSE, PLL Lock, and HPD Polarity Register Polarity register for generation of PHY_INT0...
PHY_PCLFREQ0	0x3008	PHY Test Interface Register 0
PHY_PCLFREQ1	0x3009	PHY Test Interface Register 1
PHY_PLLCFGFREQ0	0x300a	PHY PLL Test Interface Register 0
PHY_PLLCFGFREQ1	0x300b	PHY PLL Test Interface Register 1

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PHY_PLLCFGFREQ2	0x300c	PHY PLL Test Interface Register 2
phy_i2cm_slave	0x3020	PHY I2C Slave Address Configuration Register
phy_i2cm_address	0x3021	PHY I2C Address Configuration Register This register writes the address for read and write...
phy_i2cm_datao_1	0x3022	PHY I2C Data Write Register 1
phy_i2cm_datao_0	0x3023	PHY I2C Data Write Register 0
phy_i2cm_datai_1	0x3024	PHY I2C Data Read Register 1
phy_i2cm_datai_0	0x3025	PHY I2C Data Read Register 0
phy_i2cm_operation	0x3026	PHY I2C RD/RD_EXT/WR Operation Register This register requests read and write operations from the...
phy_i2cm_int	0x3027	PHY I2C Done Interrupt Register This register contains and configures I2C master PHY done...

Registers for Address Block: PHYConfiguration (Continued)

Register	**Offset**	**Description**
phy_i2cm_ctlint	0x3028	PHY I2C error Interrupt Register This register contains and configures the I2C master PHY error...
phy_i2cm_div	0x3029	PHY I2C Speed control Register This register wets the I2C Master PHY to work in either Fast or...
phy_i2cm_softrstz	0x302a	PHY I2C SW reset control register This register sets the I2C Master PHY software reset.
phy_i2cm_ss_scl_hcnt_1_addr	0x302b	PHY I2C Slow Speed SCL High Level Control Register 1
phy_i2cm_ss_scl_hcnt_0_addr	0x302c	PHY I2C Slow Speed SCL High Level Control Register 0
phy_i2cm_ss_scl_lcnt_1_addr	0x302d	PHY I2C Slow Speed SCL Low Level Control Register 1
phy_i2cm_ss_scl_lcnt_0_addr	0x302e	PHY I2C Slow Speed SCL Low Level Control Register 0
phy_i2cm_fs_scl_hcnt_1_addr	0x302f	PHY I2C Fast Speed SCL High Level Control Register 1
phy_i2cm_fs_scl_hcnt_0_addr	0x3030	PHY I2C Fast Speed SCL High Level Control Register 0
phy_i2cm_fs_scl_lcnt_1_addr	0x3031	PHY I2C Fast Speed SCL Low Level Control Register 1
phy_i2cm_fs_scl_lcnt_0_addr	0x3032	PHY I2C Fast Speed SCL Low Level Control Register 0
phy_i2cm_sda_hold	0x3033	PHY I2C SDA HOLD Control Register
jtag_phy_config	0x3034	PHY I2C/JTAG I/O Configuration Control Register
jtag_phy_tap_tck	0x3035	PHY JTAG Clock Control Register
jtag_phy_tap_in	0x3036	PHY JTAG TAP In Control Register
jtag_phy_tap_out	0x3037	PHY JTAG TAP Out Control Register
jtag_phy_addr	0x3038	PHY JTAG Address Control Register

phy_conf0

Description: PHY Configuration Register

This register holds the power down, data enable polarity, and interface control of the HDMI Source PHY control.

Size: 8 bits

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Offset: 0x3000

Bits	Name	Attr	description
7	PDZ	R/W	Power-down enable (active low 0b). Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
6	ENTMDS	R/W	Enable TMDS drivers, bias, and TMDS digital logic. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
5	svsret	R/W	PHY SVSRET signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
4	pddq	R/W	PHY PDDQ signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: "(PHY_MHL_COMBO== 1) ? 1 : 0"
3	txpwron	R/W	PHY TXPWRON signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
2	enhpdrxsense	R/W	PHY ENHPDRXSENSE signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x1
1	seldataenpol	R/W	Select data enable polarity. Value After Reset: 0x1
0	seldipif	R/W	Select interface control. Value After Reset: 0x0

phy_tst0

Description: PHY Test Interface Register 0

PHY TX mapped test interface (control).

Size: 8 bits

Offset: 0x3001

Bits	Name	Attr	Description
7:6	spare_2	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0
5	testclr	R/W	Test Clear signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
4	testen	R/W	Test Enable signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
3:1	spare_1	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0
0	testclk	R/W	Test Clock signal. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0

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phy_tst1

Description: PHY Test Interface Register 1

PHY TX mapped text interface (data in).

Size: 8 bits

Offset: 0x3002

Bits	Name	Attr	Description
7:0	testdin	R/W	Test Data input Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0

phy_tst2

Description: PHY Test Interface Register 2

PHY TX mapped text interface (data out).

Size: 8 bits

Offset: 0x3003

Bits	Name	Attr	Description
7:0	testdout	R	Test Data output. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0

phy_stat0

Description: PHY RXSENSE, PLL Lock, and HPD Status Register

This register contains the following active high packet sent status indications.

Size: 8 bits

Offset: 0x3004

Bits	Name	Attr	Description
7	RX_SENSE_3	R	Status bit. TX PHY RX_SENSE indication for TMDS channel 3 driver. You may need to mask or change polarity of this interrupt after it has became active. Value After Reset: 0x0
6	RX_SENSE_2	R	Status bit. TX PHY RX_SENSE indication for TMDS channel 2 driver. You may need to mask or change polarity of this interrupt after it has became active. Value After Reset: 0x0
5	RX_SENSE_1	R	Status bit. TX PHY RX_SENSE indication for TMDS channel 1 driver. You may need to mask or change polarity of this interrupt after it has became active. Value After Reset: 0x0
4	RX_SENSE_0	R	Status bit. TX PHY RX_SENSE indication for TMDS channel 0 driver. You may need to mask or change polarity of this interrupt after it has became active. Value After Reset: 0x0
3:2			Reserved for future use.
1	HPD	R	Status bit. HDMI Hot Plug Detect indication. You may need to mask or change polarity of this interrupt after it has became active.

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Bits	Name	Attr	Description
			Value After Reset: 0x0
0	TX_PHY_LOCK	R	Status bit. TX PHY PLL lock indication. You may need to mask or change polarity of this interrupt after it has became active. Value After Reset: 0x0

phy_int0

Description: PHY RXSENSE, PLL Lock, and HPD Interrupt Register

This register contains the interrupt indication of the PHY_STAT0 status interrupts. Interrupt generation is accomplished in the following way:

interrupt = (mask == 1'b0) && (polarity == status);

All these interrupts are forwarded to the Interrupt Handler sticky bit register ih_phy_stat0 and after ORed to a single main interrupt line to micro controller. Assertion of this interrupt implies that data related with the corresponding packet has been sent through the HDMI interface.

Size: 8 bits

Offset: 0x3005

Bits	Name	Attr	Description
7	RX_SENSE_3	R	Interrupt indication bit. TX PHY RX_SENSE indication interruption for TMDS CLK driver. Value After Reset: 0x0
6	RX_SENSE_2	R	Interrupt indication bit. TX PHY RX_SENSE indication interruption for TMDS channel 2 driver. Value After Reset: 0x0
5	RX_SENSE_1	R	Interrupt indication bit. TX PHY RX_SENSE indication interruption for TMDS channel 1 driver. Value After Reset: 0x0
4	RX_SENSE_0	R	Interrupt indication bit. TX PHY RX_SENSE indication interruption for TMDS channel 0 driver. Value After Reset: 0x0
3:2			Reserved for future use.
1	HPD	R	Interrupt indication bit. HDMI Hot Plug Detect indication interrupt. Value After Reset: 0x0
0	TX_PHY_LOCK	R	Interrupt indication bit. TX PHY PLL lock indication interrupt. Value After Reset: 0x0

phy_mask0

Description: PHY RXSENSE, PLL Lock, and HPD Mask Register Mask register for generation of PHY_INT0 interrupts.

Size: 8 bits

Offset: 0x3006

Bits	Name	Attr	Description
7	RX_SENSE_3	R/W	Mask bit for PHY_INT0.RX_SENSE[3] interrupt bit Value After Reset: 0x0

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6	RX_SENSE_2	R/W	Mask bit for PHY_INT0.RX_SENSE[2] interrupt bit Value After Reset: 0x0
5	RX_SENSE_1	R/W	Mask bit for PHY_INT0.RX_SENSE[1] interrupt bit Value After Reset: 0x0
4	RX_SENSE_0	R/W	Mask bit for PHY_INT0.RX_SENSE[0] interrupt bit Value After Reset: 0x0
3:2			Reserved for future use.
1	HPD	R/W	Mask bit for PHY_INT0.HPD interrupt bit Value After Reset: 0x0
0	TX_PHY_LOCK	R/W	Mask bit for PHY_INT0.TX_PHY_LOCK interrupt bit Value After Reset: 0x0

phy_pol0

Description: PHY RXSENSE, PLL Lock, and HPD Polarity Register Polarity register for generation of PHY_INT0 interrupts.

Size: 8 bits

Offset: 0x3007

Bits	Name	Attr	Description
7	RX_SENSE_3	R/W	Polarity bit for PHY_INT0.RX_SENSE[3] interrupt bit Value After Reset: 0x1
6	RX_SENSE_2	R/W	Polarity bit for PHY_INT0.RX_SENSE[2] interrupt bit Value After Reset: 0x1
5	RX_SENSE_1	R/W	Polarity bit for PHY_INT0.RX_SENSE[1] interrupt bit Value After Reset: 0x1
4	RX_SENSE_0	R/W	Polarity bit for PHY_INT0.RX_SENSE[0] interrupt bit Value After Reset: 0x1
3:2			Reserved for future use.
1	HPD	R/W	Polarity bit for PHY_INT0.HPD interrupt bit Value After Reset: 0x1
0	TX_PHY_LOCK	R/W	Polarity bit for PHY_INT0.TX_PHY_LOCK interrupt bit Value After Reset: 0x1

PHY_PCLFREQ0

Description: PHY Test Interface Register 0

Size: 8 bits

Offset: 0x3008

Bits	Name	Attr	Description
7:0	pclk_freq	R/W	Pixel Clock Frequency (pclk_freq[7:0]). Value After Reset: 0x32

PHY_PCLFREQ1

Description: PHY Test Interface Register 1

Size: 8 bits

Offset: 0x3009

Bits	Name	Attr	Description

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7:2			Reserved for future use.
1:0	pclk_freq	R/W	Pixel Clock Frequency (pclk_freq[9:8]). Value After Reset: 0x0

PHY_PLLCFGFREQ0

Description: PHY PLL Test Interface Register 0

Size: 8 bits

Offset: 0x300a

Bits	Name	Attr	Description
7:0	pllcfgfreq	R/W	PLL Configuration Frequency (pllcfgfreq[7:0]). Value After Reset: 0x20

PHY_PLLCFGFREQ1

Description: PHY PLL Test Interface Register 1

Size: 8 bits

Offset: 0x300b

Bits	**Name**	**Attr**	**Description**
7:0	pllcfgfreq	R/W	PLL Configuration Frequency (pllcfgfreq[15:8]). Value After Reset: 0x27

PHY_PLLCFGFREQ2

Description: PHY PLL Test Interface Register 2

Size: 8 bits

Offset: 0x300c

Bits	**Name**	**Attr**	**Description**
7:0	pllcfgfreq	R/W	PLL Configuration Frequency (pllcfgfreq[23:16]). Value After Reset: 0x0

phy_i2cm_slave

Description: PHY I2C Slave Address Configuration Register

Size: 8 bits

Offset: 0x3020

Bits	**Name**	**Attr**	**Description**
7			Reserved for future use.
6:0	slaveaddr	R/W	Slave address to be sent during read and write operations. PHY Gen2 slave address: 7'h69 HEAC PHY slave address: 7'h49 Value After Reset: 0x0

phy_i2cm_address

Description: PHY I2C Address Configuration Register

This register writes the address for read and write operations.

Size: 8 bits

Offset: 0x3021

Bits	**Name**	**Attr**	**Description**

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7:0	address	R/W	Register address for read and write operations Value After Reset: 0x0

phy_i2cm_datao_1
Description: PHY I2C Data Write Register 1
Size: 8 bits
Offset: 0x3022

Bits	Name	Attr	Description
7:0	datao	R/W	Data MSB (datao[15:8]) to be written on register pointed by phy_i2cm_address [7:0]. Value After Reset: 0x0

phy_i2cm_datao_0
Description: PHY I2C Data Write Register 0
Size: 8 bits
Offset: 0x3023

Bits	Name	Attr	Description
7:0	datao	R/W	Data LSB (datao[7:0]) to be written on register pointed by phy_i2cm_address [7:0]. Value After Reset: 0x0

phy_i2cm_datai_1
Description: PHY I2C Data Read Register 1
Size: 8 bits
Offset: 0x3024

Bits	Name	Attr	Description
7:0	datai	R	Data MSB (datai[15:8]) read from register pointed by phy_i2cm_address[7:0]. Value After Reset: 0x0

phy_i2cm_datai_0
Description: PHY I2C Data Read Register 0
Size: 8 bits
Offset: 0x3025

Bits	Name	Attr	Description
7:0	datai	R	Data LSB (datai[7:0]) read from register pointed by phy_i2cm_address[7:0]. Value After Reset: 0x0

phy_i2cm_operation
Description: PHY I2C RD/RD_EXT/WR Operation Register
This register requests read and write operations from the I2C Master PHY. This register can only be written; reading this register always results in 00h. Writing 1'b1 simultaneously to read and write requests is considered a read request.
Size: 8 bits
Offset: 0x3026

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Bits	Name	Attr	Description
7:5			Reserved for future use.
4	wr	W	Write operation request Value After Reset: 0x0
3:1			Reserved for future use.
0	rd	W	Read operation request Value After Reset: 0x0

phy_i2cm_int

Description: PHY I2C Done Interrupt Register

This register contains and configures I2C master PHY done interrupt.

Size: 8 bits

Offset: 0x3027

Bits	Name	Attr	Description
7:4			Reserved for future use.
3	done_pol	R/W	Done interrupt polarity configuration Value After Reset: 0x1
2	done_mask	R/W	Done interrupt mask signal Value After Reset: 0x0
1	done_interrupt	R	Operation done interrupt bit. Only lasts for 1 SFR clock cycle and is auto cleared after it. {done_interrupt = (done_mask==0b) && (done_status==done_pol)} Value After Reset: 0x0
0	done_status	R	Operation done status bit. Marks the end of a read or write operation. Value After Reset: 0x0

phy_i2cm_ctlint

Description: PHY I2C error Interrupt Register

This register contains and configures the I2C master PHY error interrupts.

Size: 8 bits

Offset: 0x3028

Bits	Name	Attr	Description
7	nack_pol	R/W	Not acknowledge error interrupt polarity configuration Value After Reset: 0x1
6	nack_mask	R/W	Not acknowledge error interrupt mask signal Value After Reset: 0x0
5	nack_interrupt	R	Not acknowledge error interrupt bit. Only lasts for one SFR clock cycle and is auto cleared after it. {nack_interrupt = (nack_mask==0b) && (nack_status==nack_pol)}. Note: This bit field is read by the sticky bits present on the ih_i2cmphy_stat0 register. Value After Reset: 0x0
4	nack_status	R	Not acknowledge error status bit. Error on I2C not acknowledge. Note: This bit field is read by the sticky bits present on the ih_i2cmphy_stat0 register. Value After Reset: 0x0

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Bits	Name	Attr	Description
3	arbitration_pol	R/W	Arbitration error interrupt polarity configuration. Value After Reset: 0x1
2	arbitration_mask	R/W	Arbitration error interrupt mask signal. Value After Reset: 0x0
1	arbitration_interrupt	R	Arbitration error interrupt bit {arbitration_interrupt = (arbitration_mask==0b) && (arbitration_status==arbitration_pol)} Note: This bit field is read by the sticky bits present on the ih_i2cmphy_stat0 register. Value After Reset: 0x0

Table 41-5 Fields for Register: phy_i2cm_ctlint (Continued)

Bits	Name	Attr	Description
0	arbitration_status	R	Arbitration error status bit. Error on master I2C protocol arbitration. Only lasts for one SFR clock cycle and is auto cleared after it. Note: This bit field is read by the sticky bits present on the ih_i2cmphy_stat0 register. Value After Reset: 0x0

phy_i2cm_div

Description: PHY I2C Speed control Register

This register wets the I2C Master PHY to work in either Fast or Standard mode.

Size: 8 bits

Offset: 0x3029

Bits	Name	Attr	Description
7:4			Reserved for future use.
3	fast_std_mode	R/W	Sets the I2C Master to work in Fast Mode or Standard Mode: 1: Fast Mode 0: Standard Mode Value After Reset: 0x1
2:0	spare	R/W	Reserved as "spare" register with no associated functionality. Value After Reset: 0x3

phy_i2cm_softrstz

Description: PHY I2C SW reset control register

This register sets the I2C Master PHY software reset.

Size: 8 bits

Offset: 0x302a

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	i2c_softrstz	R/W	I2C Master Software Reset. Active by writing a zero and auto cleared to one in the following cycle. Value After Reset: 0x1

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phy_i2cm_ss_scl_hcnt_1_addr

Description: PHY I2C Slow Speed SCL High Level Control Register 1

Size: 8 bits

Offset: 0x302b

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_hcnt1	R/W	PHY I2C Slow Speed SCL High Level Control Register 1 Value After Reset: 0x0

phy_i2cm_ss_scl_hcnt_0_addr

Description: PHY I2C Slow Speed SCL High Level Control Register 0

Size: 8 bits

Offset: 0x302c

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_hcnt0	R/W	PHY I2C Slow Speed SCL High Level Control Register 0 Value After Reset: 0x6c

phy_i2cm_ss_scl_lcnt_1_addr

Description: PHY I2C Slow Speed SCL Low Level Control Register 1

Size: 8 bits

Offset: 0x302d

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_lcnt1	R/W	PHY I2C Slow Speed SCL Low Level Control Register 1 Value After Reset: 0x0

phy_i2cm_ss_scl_lcnt_0_addr

Description: PHY I2C Slow Speed SCL Low Level Control Register 0

Size: 8 bits

Offset: 0x302e

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_lcnt0	R/W	PHY I2C Slow Speed SCL Low Level Control Register 0 Value After Reset: 0x7f

phy_i2cm_fs_scl_hcnt_1_addr

Description: PHY I2C Fast Speed SCL High Level Control Register 1

Size: 8 bits

Offset: 0x302f

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_hcnt1	R/W	PHY I2C Fast Speed SCL High Level Control Register 1 Value After Reset: 0x0

phy_i2cm_fs_scl_hcnt_0_addr

Description: PHY I2C Fast Speed SCL High Level Control Register 0

Size: 8 bits

Offset: 0x3030

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_hcnt0	R/W	PHY I2C Fast Speed SCL High Level Control Register 0

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Bits	Name	Attr	Description
			Value After Reset: 0x11

phy_i2cm_fs_scl_lcnt_1_addr

Description: PHY I2C Fast Speed SCL Low Level Control Register 1
Size: 8 bits
Offset: 0x3031

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_lcnt1	R/W	PHY I2C Fast Speed SCL Low Level Control Register 1 Value After Reset: 0x0

phy_i2cm_fs_scl_lcnt_0_addr

Description: PHY I2C Fast Speed SCL Low Level Control Register 0
Size: 8 bits
Offset: 0x3032

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_lcnt0	R/W	PHY I2C Fast Speed SCL Low Level Control Register 0 Value After Reset: 0x24

phy_i2cm_sda_hold

Description: PHY I2C SDA HOLD Control Register
Size: 8 bits
Offset: 0x3033

Bits	Name	Attr	Description
7:0	osda_hold	R/W	Defines the number of SFR clock cycles to meet tHD:DAT (300 ns) osda_hold = round_to_high_integer (300 ns / (1/ isfrclk_frequency)) Value After Reset: 0x9

jtag_phy_config

Description: PHY I2C/JTAG I/O Configuration Control Register
Size: 8 bits
Offset: 0x3034

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	i2c_jtagz	R/W	Configures the JTAG PHY interface output pin I2C_JTAGZ to select the PHY configuration interface when in internal control mode (iphy_ext_ctrl=1'b0) or ophyext_jtag_i2c_jtagz when PHY_EXTERNAL=1. 1'b0: JTAG configuration Interface 1'b1: I2C configuration Interface Value After Reset: 0x1
3:1			Reserved for future use.
0	jtag_trst_n	R/W	Configures the JTAG PHY interface output pin JTAG_TRST_N when in internal control mode

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Bits	Name	Attr	Description
			(iphy_ext_ctrl=1'b0) or ophyext_jtag_trst_n when PHY_EXTERNAL=1. Value After Reset: 0x1

jtag_phy_tap_tck

Description: PHY JTAG Clock Control Register

Size: 8 bits

Offset: 0x3035

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	jtag_tck	R/W	Configures the JTAG PHY interface pin JTAG_TCK when in internal control mode (iphy_ext_ctrl=1'b0) or ophyext_jtag_tck when PHY_EXTERNAL=1. Value After Reset: 0x0

jtag_phy_tap_in

Description: PHY JTAG TAP In Control Register

Size: 8 bits

Offset: 0x3036

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	jtag_tms	R/W	Configures the JTAG PHY interface pin JTAG_TMS when in internal control mode (iphy_ext_ctrl=1'b0) or ophyext_jtag_tms when PHY_EXTERNAL=1. Value After Reset: 0x1
3:1			Reserved for future use.
0	jtag_tdi	R/W	Configures the JTAG PHY interface pin JTAG_TDI when in internal control mode (iphy_ext_ctrl=1'b0) or ophyext_jtag_tdi when PHY_EXTERNAL=1. Value After Reset: 0x0

jtag_phy_tap_out

Description: PHY JTAG TAP Out Control Register

Size: 8 bits

Offset: 0x3037

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	jtag_tdo_en	R	Read JTAG PHY interface input pin JTAG_TDO_EN when in internal control mode (iphy_ext_ctrl=1'b0) or iphyext_jtag_tdo_en when PHY_EXTERNAL=1 Value After Reset: 0x0
3:1			Reserved for future use.
0	jtag_tdo	R	Read JTAG PHY interface input pin JTAG_TDO when in internal control mode (iphy_ext_ctrl=1'b0) or iphyext_jtag_tdo when PHY_EXTERNAL=1 Value After

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Bits	Name	Attr	Description
			Reset: 0x0

jtag_phy_addr

Description: PHY JTAG Address Control Register

Size: 8 bits

Offset: 0x3038

Bits	Name	Attr	Description
7:0	jtag_addr	R/W	Configures the JTAG PHY interface pin JTAG_ADDR[7:0] when in internal control mode (iphy_ext_ctrl=1'b0) or iphyext_jtag_addr[7:0] when PHY_EXTERNAL=1 Value After Reset: 0x0

AudioSample Registers

Audio Sample Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: AudioSample

Register	Offset	Description
aud_conf0	0x3100	Audio I2S Software FIFO Reset, Select, and Enable Control Register 0 This register configures the...
aud_conf1	0x3101	Audio I2S Width Configuration Register 1 This register configures the data width of the input...
aud_int	0x3102	I2S FIFO status and interrupts. This register configures the I2S FIFO status and interrupts.
aud_conf2	0x3103	Audio I2S PCUV, NLPCM and HBR configuration Register 2 This register configures the I2S Audio
aud_int1	0x3104	I2S Mask Interrupt Register This register masks the interrupts present in the I2S module.

aud_conf0

Description: Audio I2S Software FIFO Reset, Select, and Enable Control Register 0

This register configures the I2S input enable that indicates which input I2S channels have valid data. It also allows the system processor to reset audio FIFOs upon underflow/overflow error detection.

Size: 8 bits

Offset: 0x3100

Bits	Name	Attr	Description
7	sw_audio_fifo_rst	R/W	Audio FIFOs software reset Writing 0b: no action taken Writing 1b: Resets all audio FIFOs Reading from this register always returns 0b. Note: If a FIFO reset request (via SFR command) lands in the middle of an I2S transaction, the samples become misaligned (left-right sequence lost). As a solution, for each FIFO

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Bits	Name	Attr	Description
			reset, an associated I2S reset must be issued (writing 8'hF7 to MC_SWRSTZ register). Value After Reset: 0x0
6	spare_2	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0
5	i2s_select	R/W	1b: Selects I2S Audio Interface 0b: Selects the second (SPDIF/GPA) interface, in configurations with more that one audio interface (DOUBLE/GDOUBLE) Value After Reset: 0x1
4	spare_1	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0
3:0	I2S_in_en	R/W	Action I2S_in_en[0] - I2Sdata[0] enable I2S_in_en[1] - I2Sdata[1] enable I2S_in_en[2] - I2Sdata[2] enable I2S_in_en[3] - I2Sdata[3] enable Value After Reset: 0xf

aud_conf1

Description: Audio I2S Width Configuration Register 1 This register configures the data width of the input data.

Size: 8 bits

Offset: 0x3101

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:0	I2S_width	R/W	I2S input data width I2S_width[4:0] | Action 00000b-01111b | Not used 10000b | 16 bit data samples at input 10001b | 17 bit data samples at input 10010b | 18 bit data samples at input 10011b | 19 bit data samples at input 10100b | 20 bit data samples at input 10101b | 21 bit data samples at input 10110b | 22 bit data samples at input 10111b | 23 bit data samples at input 11000b | 24 bit data samples at input 11001b-11111b | Not Used Value After Reset: 0x18

aud_int

Description: I2S FIFO status and interrupts.

This register configures the I2S FIFO status and interrupts.

Size: 8 bits

Offset: 0x3102

Bits	Name	Attr	Description

Bits	Name	Attr	Description
7:4			Reserved for future use.
3	fifo_empty_mask	R/W	FIFO empty mask. Value After Reset: 0x0
2	fifo_full_mask	R/W	FIFO full mask. Value After Reset: 0x0
1:0			Reserved for future use.

aud_conf2

Description: Audio I2S PCUV, NLPCM and HBR configuration Register 2

This register configures the I2S Audio Data mapping. By default, audio data mapping is the standard I2S Linear PCM (L-PCM) mapping. You can choose to use the I2S interface to transport HBR or Non- Linear PCM (NL-PCM) audio, by setting the relevant bit in this register.

Size: 8 bits

Offset: 0x3103

Bits	Name	Attr	Description
7:3			Reserved for future use.
2	insert_pcuv	R/W	When set (1'b1), this bit enables the insertion of the PCUV (Parity, Channel Status, User bit and Validity) bits on the incoming audio stream (support limited to Linear PCM audio). If disabled, the incoming audio stream must contain the PCUV bits, mapped according to Databook. Value After Reset: 0x1
1	NLPCM	R/W	I2S NLPCM Mode Enable. When enabled, this bit assumes that PCUV data is included on the I2S audio stream according to the description located in the "I2S Interface" section of Chapter 2, "Functional Description." Value After Reset: 0x0
0	HBR	R/W	I2S HBR Mode Enable. When enabled, the I2S audio stream is transmitted using HBR packets. Value After Reset: 0x0

aud_int1

Description: I2S Mask Interrupt Register

This register masks the interrupts present in the I2S module.

Size: 8 bits

Offset: 0x3104

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	fifo_overrun_mask	R/W	FIFO overrun mask Value After Reset: 0x1
3:0			Reserved for future use.

AudioPacketizer Registers

Audio Packetizer Registers. Follow the link for the register to see a detailed description of the register.

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Registers for Address Block: AudioPacketizer

Register	Offset	Description
aud_n1	0x3200	Audio Clock Regenerator N Value Register 1 For N expected values, refer to the HDMI 1.4b...
aud_n2	0x3201	Audio Clock Regenerator N Value Register 2 For N expected values, refer to the HDMI 1.4b...
aud_n3	0x3202	Audio Clock Regenerator N Value Register 3 For N expected values, refer to the HDMI 1.4b...
aud_cts1	0x3203	Audio Clock Regenerator CTS Value Register 1 For CTS expected values, refer to the HDMI 1.4b...
aud_cts2	0x3204	Audio Clock Regenerator CTS Register 2 For CTS expected values, refer to the HDMI 1.4b...
aud_cts3	0x3205	Audio Clock Regenerator CTS value Register 3. For CTS expected values, refer to the HDMI 1.4b...
aud_inputclkfs	0x3206	Audio Input Clock FS Factor Register
aud_cts_dither	0x3207	Audio CTS Dither Register

aud_n1

Description: Audio Clock Regenerator N Value Register 1 For N expected values, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x3200

Bits	Name	Attr	Description
7:0	AudN	R/W	HDMI Audio Clock Regenerator N value Value After Reset: 0x0

aud_n2

Description: Audio Clock Regenerator N Value Register 2 For N expected values, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x3201

Bits	Name	Attr	Description
7:0	AudN	R/W	HDMI Audio Clock Regenerator N value Value After Reset: 0x0

aud_n3

Description: Audio Clock Regenerator N Value Register 3 For N expected values, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x3202

Bits	Name	Attr	Description
7	ncts_atomic_write	R/W	When set, the new N and CTS values are only used when aud_n1 register is written. If clear, N and CTS data is updated each time a new N or CTS byte is written.

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Bits	Name	Attr	Description
			The following write sequence is recommended: aud_n3 (set bit ncts_atomic_write if desired) aud_cts3 (set CTS_manual and CTS value if desired/enabled) aud_cts2 (required in CTS_manual) aud_cts1 (required in CTS_manual) aud_n3 (bit ncts_atomic_write with same value as in step 1.) aud_n2 aud_n1 For dynamic N/CTS changes, perform only steps from 2-7 or 5-7 depending on the state of CTS_manual. Value After Reset: 0x0
6:4			Reserved for future use.
3:0	AudN	R/W	HDMI Audio Clock Regenerator N value Value After Reset: 0x0

aud_cts1

Description: Audio Clock Regenerator CTS Value Register 1 For CTS expected values, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x3203

Bits	Name	Attr	Description
7:0	AudCTS	R/W	HDMI Audio Clock Regenerator CTS calculated value. This value can be manually set using the CTS_manual (AUD_CTS3) mechanism. Value After Reset: 0x0

aud_cts2

Description: Audio Clock Regenerator CTS Register 2

For CTS expected values, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x3204

Bits	Name	Attr	Description
7:0	AudCTS	R/W	HDMI Audio Clock Regenerator CTS calculated value. This value can be manually set using the CTS_manual (AUD_CTS3) mechanism. Value After Reset: 0x0

aud_cts3

Description: Audio Clock Regenerator CTS value Register 3. For CTS expected values, refer to the HDMI 1.4b specification.

Size: 8 bits

Offset: 0x3205

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:5	Spare_bits	R/W	Reserved as "spare" bits with no associated functionality. Value After Reset: 0x0
4	CTS_manual	R/W	If the CTS_manual bit equals 0b, this registers contains audCTS[19:0] generated by the Cycle time counter according to the specified timing. If the CTS_manual bit equals 1b, this register is configured with the audCTS[7:0] value that is output by the Audio Packetizer. Note: When the General Parallel Audio Interface (GPAUD) is enabled (AUDIO_IF = 6) or the AHB DMA Audio Interface is enabled (AUDIO_IF = 8), writing to these bits has no effect; reading these bits always return 0. Value After Reset: 0x0
3:0	AudCTS	R/W	HDMI Audio Clock Regenerator CTS calculated value. This value can be manually set using the CTS_manual (AUD_CTS3) mechanism. Value After Reset: 0x0

aud_inputclkfs

Description: Audio Input Clock FS Factor Register

Size: 8 bits

Offset: 0x3206

Bits	Name	Attr	Description
7:3			Reserved for future use.
2:0	ifsfactor	R/W	Fs factor configuration: ifsfactor[2:0] | Audio Clock | Action 0 | 128xFs | If you select the Bypass SPDIF DRU unit in coreConsultant, the input audio clock (either I2S or SPDIF according to configuration) is used at the audio packetizer to calculate the CTS value and ACR packet insertion rate. | 256xFs | The input audio clock (I2S only) is divided by 2 and then used at audio packetizer to calculate the CTS value and ACR packet insertion rate. | 512xFs | The input audio clock (either I2S or SPDIF according to configuration) used divided by 4 and then used at the audio packetizer to calculate the CTS value and ACR packet insertion rate. Note: When the SPDIF interface is receiving an HBR audio stream ("Support for HBR over SDPIF" parameter must be enabled), it is required that the selected IFSFACTOR to be set at 512xFs in order to comply with the HDMI ACR requirements for HBR audio streams. | Reserved

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Bits	Name	Attr	Description
			| 64xFs | The input audio clock (I2S only) is multiplied by 2 and then used at the audio packetizer to calculate the CTS value and ACR packet insertion rate. others | 128xFs | If you select the Bypass SPDIF DRU unit in coreConsultant, the input audio clock (either I2S or SPDIF according to configuration) is used at the audio packetizer to calculate the CTS value and ACR packet insertion rate. The SPDIF interface, for non HBR audio, requires that the configured oversampling value to be 128xFs when HTX_SPDIFBYPDRU is enabled and 512xFs if not. When the SPDIF interface is receiving HBR audio (HBR_ON_SPDIF must be enabled), in order to comply with the HDMI ACR requirements for HBR audio streams. Value After Reset: 0x0

aud_cts_dither

Description: Audio CTS Dither Register

Size: 8 bits

Offset: 0x3207

Bits	Name	Attr	Description
7:4	divisor	R/W	Dither divisor (4'd1 if no CTS Dither). This field should be configured with the value of divisor from the HDMI specification. Value After Reset: 0x1
3:0	dividend	R/W	Dither dividend (4'd1 if no CTS Dither). This field should be configured with the value of dividend from the HDMI specification. Value After Reset: 0x1

AudioSampleSPDIF Registers

Audio Sample SPDIF Registers. Follow the link for the register to see a detailed description of the register.

Registers for Address Block: AudioSampleSPDIF

Register	Offset	Description
aud_spdif0	0x3300	Audio SPDIF Software FIFO Reset Control Register 0 This register allows the system processor to...
aud_spdif1	0x3301	Audio SPDIF NLPCM and Width Configuration Register 1 This register configures the SPDIF data...
aud_spdifint	0x3302	Audio SPDIF FIFO Empty/Full Mask Register
aud_spdifint1	0x3303	Audio SPDIF Mask Interrupt Register 1 This register masks interrupts present in the SPDIF...
aud_spdif2	0x3304	Audio SPDIF Enable Confiiguration Register 2 This

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Register	Offset	Description
		register configures the SPDIF input enable that...

aud_spdif0

Description: Audio SPDIF Software FIFO Reset Control Register 0
This register allows the system processor to reset audio FIFOs upon underflow/overflow error detection.
Size: 8 bits
Offset: 0x3300

Bits	Name	Attr	Description
7	sw_audio_fifo_rst	R/W	Audio FIFOs software reset Writing 0b: no action taken Writing 1b: Resets all audio FIFOs Reading from this register always returns 0b. Note: If a FIFO reset request (via register write command) lands in the middle of an SPDIF audio transaction, the samples become misaligned (left-right sequence lost). As a solution, for each FIFO reset, an associated SPDIF reset must be issued (writing 8'hEF to MC_SWRSTZ register). Value After Reset: 0x0
6:0	spare	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0xf

aud_spdif1

Description: Audio SPDIF NLPCM and Width Configuration Register 1 This register configures the SPDIF data width.
Size: 8 bits
Offset: 0x3301

Bits	Name	Attr	Description
7	setnlpcm	R/W	Select Non-Linear (1b) / Linear (0b) PCM mode Value After Reset: 0x0
6	spdif_hbr_mode	R/W	When set to 1'b1, this bit field indicates that the input stream has a High Bit Rate (HBR) to be transmitted in HDMI HBR packets. When clear (1b'0), the audio is transmitted in HDMI AUDS packets.Note:< Otherwise, this field is a "spare" bit with no associated functionality.

aud_spdifint

Description: Audio SPDIF FIFO Empty/Full Mask Register
Size: 8 bits
Offset: 0x3302

Bits	Name	Attr	Description
7:4			Reserved for future use.
3	spdif_fifo_empty_mask	R/W	Description: SPDIF FIFO empty mask

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Bits	Name	Attr	Description
			Value After Reset: 0x0
2	spdif_fifo_full_mask	R/W	Description: SPDIF FIFO full mask Value After Reset: 0x0
1:0			Reserved for future use.

aud_spdifint1

Description: Audio SPDIF Mask Interrupt Register 1

This register masks interrupts present in the SPDIF module.

Size: 8 bits

Offset: 0x3303

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	fifo_overrun_mask	R/W	FIFO overrun mask Value After Reset: 0x1
3:0			Reserved for future use.

aud_spdif2

Description: Audio SPDIF Enable Confiiguration Register 2

This register configures the SPDIF input enable that indicates which input SPDIF channels have valid data.

Size: 8 bits

Offset: 0x3304

Bits	Name	Attr	Description
7:4			Reserved for future use.
3:0	SPDIF_in_en	R/W	Action SPDIF_in_en[0] - ispdifdata[0] enable SPDIF_in_en[1] - ispdifdata[1] enable SPDIF_in_en[2] - ispdifdata[2] enable SPDIF_in_en[3] - ispdifdata[3] enable Value After Reset: 0x1

AudioSampleGP Registers

Audio Sample GP Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
gp_conf0	0x3500	Audio GPA Software FIFO Reset Control Register 0
gp_conf1	0x3501	Audio GPA Channel Enable Configuration Register 1
gp_conf2	0x3502	Audio GPA HBR Enable Register 2
gp_mask	0x3506	Audio GPA FIFO Full and Empty Mask Interrupt Register

gp_conf0

Description: Audio GPA Software FIFO Reset Control Register 0

Size: 8 bits

Offset: 0x3500

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	sw_audio_fifo_rst	R/W	Audio FIFOs software reset

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Bits	Name	Attr	Description
			Writing 0b: no action taken Writing 1b: Resets all audio FIFOs Reading from this register always returns 0b. Note: If a FIFO reset request (via register write command) lands in the middle of an GPAUD audio transaction, the samples become misaligned (left-right sequence lost). As a solution, for each FIFO reset, an associated SPDIF reset must be issued (writing 8'h7F to MC_SWRSTZ register). Value After Reset: 0x0

gp_conf1

Description: Audio GPA Channel Enable Configuration Register 1

Size: 8 bits

Offset: 0x3501

Bits	Name	Attr	Description
7:0	ch_in_en	R/W	Each bit controls the enabling of the respective audio channel. For instance, bit 1, when set (1'b1), the audio Channel 1 is enabled. When cleared, the referred channel is disabled. Value After Reset: 0x0

gp_conf2

Description: Audio GPA HBR Enable Register 2

Size: 8 bits

Offset: 0x3502

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	insert_pcuv	R/W	When set (1'b1), this bit enables the insertion of the PCUV (Parity, Channel Status, User bit and Validity) bits on the incoming audio stream (support limited to Linear PCM audio). If disabled, the incoming audio stream must contain the PCUV bits, mapped according to 2.6.4.2 Data Mapping Examples. Value After Reset: 0x0
0	HBR	R/W	HBR packets enable. The Hdmi_tx sends the HBR packets. This bit is enabled when the audio frequency is higher than 192 kHz. If this bit is enabled, the number of channels configured in GP_CONF1 must be set to 8'hFF. Value After Reset: 0x0

gp_mask

Description: Audio GPA FIFO Full and Empty Mask Interrupt Register

Size: 8 bits

Offset: 0x3506

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Bits	Name	Attr	Description
7:5			Reserved for future use.
4	fifo_overrun_mask	R/W	FIFO overrun mask Value After Reset: 0x1
3:2			Reserved for future use.
1	fifo_empty_mask	R/W	FIFO empty flag mask Value After Reset: 0x0
0	fifo_full_mask	R/W	FIFO full flag mask Value After Reset: 0x0

AudioDMA Registers

Audio DMA Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
ahb_dma_conf0	0x3600	Audio DMA SW FIFO reset and DMA Configuration Register 0 This register contains the software reset...
ahb_dma_start	0x3601	Audio DMA Start Register The start_dma_transaction bit field signals the AHB audio DMA to start...
ahb_dma_stop	0x3602	Audio DMA Stop Register The stop_dma_transaction bit field signals the AHB audio DMA to stop current...
ahb_dma_thrsld	0x3603	Audio DMA FIFO Threshold Register This register defines the FIFO medium threshold occupation value....
ahb_dma_straddr_set0[0:3]	0x3604 + (i * 0x1)	Audio DMA Start Address Set0 Register Array Address offset: i = 0 to 3 These registers define...
ahb_dma_stpaddr_set0[0:3]	0x3608 + (i * 0x1)	Audio DMA Stop Address Set0 Register Array Address offset: i = 0 to 3 This registers define the...
ahb_dma_bstraddr[0:3]	0x360c + (i * 0x1)	Audio DMA Burst Start Address Register Array Address offset: i = 0 to 3 These read-only registers...
ahb_dma_mblength0	0x3610	Audio DMA Burst Length Register 0 This registers holds the length of the current burst operation....
ahb_dma_mblength	0x3611	Audio DMA Burst Length Register 1 This registers holds the length of the current burst operation....
ahb_dma_mask	0x3614	Audio DMA Mask Interrupt Register This register masks each of the interrupts present in the AHB...
ahb_dma_conf	0x3616	Audio DMA Channel Enable Configuration Register 1 In AUDS packet configuration with layout 0 selected,...
ahb_dma_buffmask	0x3619	Audio DMA Buffer Mask Interrupt Register
ahb_dma_mask1	0x361b	Audio DMA Mask Interrupt Register 1 This register masks interrupts present in the AHB audio DMA...
ahb_dma_status	0x361c	Audio DMA Status
ahb_dma_conf2	0x361d	Audio DMA Configuration Register 2
ahb_dma_straddr_set1[0:3]	0x3620 + (i * 0x1)	Audio DMA Start Address Set 1 Register Array Address offset: i = 0 to 3 These registers define...
ahb_dma_stpaddr_set1[0:3]	0x3624	Audio DMA Stop Address Set 1 Register Array Address

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Register	Offset	Description
	+ (i * 0x1)	offset: i = 0 to 3 These registers define...

ahb_dma_conf0

Description: Audio DMA SW FIFO reset and DMA Configuration Register 0

This register contains the software reset bit for the audio FIFOs. It also configures operating modes of the AHB master.

Size: 8 bits

Offset: 0x3600

Bits	Name	Attr	Description
7	sw_fifo_rst	R/W	This is the software reset bit for the audio and FIFO clear. Writing 0'b does not result in any action. Writing 1'b to this register resets all audio FIFOs. Reading from this register always returns 0'b. Value After Reset: 0x0
6	insert_pcuv	R/W	Enables the insertion of PCUV data Value After Reset: 0x0
5			Reserved for future use.
4	hbr	R/W	HBR packet enable The Hdmi_tx sends the HBR packets. This bit must be enabled when transmitting non-linear audio of frequency higher than 192 kHz. If this bit is enabled, the number of channels configured in AHB_DMA_CONF1 is always 8. Value After Reset: 0x0
3	enable_hlock	R/W	Enable request of locked burst AHB mechanism. 1'b: Enables the usage of hlock for master request to arbiter of a locked complete burst. 0'b: Disables request of locked burst AHB mechanism Value After Reset: 0x0
2:1	incr_type	R/W	Selects the preferred burst length size 00'b: Corresponds to INCR4 fixed four beat, incremental AHB burst mode. Only valid when burst_mode is high. 01'b: Corresponds to INCR8 fixed eight beat incremental AHB burst mode. Only valid when burst_mode is high. 10'b: Corresponds to INCR16 fixed 16 beat incremental AHB burst mode. Only valid when burst_mode is high. 11'b: Corresponds to INCR16 fixed 16 beat incremental AHB burst mode. Only valid when burst_mode is high. Value After Reset: 0x0
0	burst_mode	R/W	1'b: Forces the burst mode to be fixed beat, incremental burst mode designated by the incr_type[1:0] signal. 0'b: Normal operation is unspecified length, incremental burst. It corresponds to INCR AHB burst mode. Value After Reset: 0x0

ahb_dma_start

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Description: Audio DMA Start Register
The start_dma_transaction bit field signals the AHB audio DMA to start accessing system memory in order to fetch data samples to store in the FIFO. After the operation starts, a new request for a DMA start is ignored until the DMA is stopped or it reaches the end address. Only in one of these situations is a new start request acknowledged.
The first DMA burst request after start_dma_transaction configuration uses initial_addr[31:0] as ohaddr[31:0] value; mburstlength[8:0] is set to the maximum admissible value. This maximum value is constrained by the size of buffer provided, the instantiated FIFO depth, or/and the number of words up to the next 1 Kbyte boundary.
Size: 8 bits
Offset: 0x3601

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	start_dma_transaction	R/W	Start DMA transaction This register is auto-cleared when the transfer operation is completed (done). Value After Reset: 0x0

ahb_dma_stop
Description: Audio DMA Stop Register
The stop_dma_transaction bit field signals the AHB audio DMA to stop current Attr. After it stops, if a new start DMA operation is requested, the DMA engine restarts the Attr using the initial_addr[31:0], which is programmed at ahb_dma_straddr0 to ahb_dma_straddr3.
Size: 8 bits
Offset: 0x3602

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	stop_dma_transaction	R/W	Stop DMA transaction This register is auto-cleared when the transfer operation is stopped (done). Value After Reset: 0x0

ahb_dma_thrsld
Description: Audio DMA FIFO Threshold Register
This register defines the FIFO medium threshold occupation value.
After the AHB master completes a burst transaction successfully, the FIFO may remain full till the data fetch interface requests samples. Each data fetch operation reduces the number of samples stored in the FIFO by the number of channels enabled.
Therefore, the fifo_threshold[7:0] is the medium number of samples that should be available in the audio FIFO across the DMA operation.
As soon as the number of samples in the FIFO drops lower than the fifo_threshold[7:0], the DMA engine requests a new burst of samples for the AHB master. The length is constrained by the size of buffer provided, the instantiated FIFO depth minus fifo_threshold[7:0], and/or the number of words up to the next 1 kbyte boundary.
Size: 8 bits
Offset: 0x3603

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	fifo_threshold	R/W	FIFO medium threshold occupation value Value After Reset: 0x0

ahb_dma_straddr_set0[0:3]

Description: Audio DMA Start Address Set0 Register Array Address offset: i = 0 to 3

These registers define the initial_addr[31:0] used to initiate the DMA burst read transactions upon start_dma_transaction configuration.

Size: 8 bits

Offset: 0x3604 + (i * 0x1)

Bits	Name	Attr	Description
7:0	initial_addr	R/W	Defines init_addr[7:0] to initiate DMA burst transactions Value After Reset: 0x0

ahb_dma_stpaddr_set0[0:3]

Description: Audio DMA Stop Address Set0 Register Array Address offset: i = 0 to 3

This registers define the final_addr[31:0] used as the final point to the DMA burst read transactions.

Upon start_dma_transaction configuration, the DMA engine starts requesting burst reads from the external system memory. Each burst read can have a maximum theoretical length of 256 words (due to the AMBA AHB specification 1 Kbyte boundary burst limitation).

The DMA engine is responsible for incrementing the burst starting address and defining its corresponding burst length to reach the final_addr[31:0] address. The last burst request issued by the DMA engine takes into account that it should only request data until the final_addr[31:0] address (included) and for that should calculate the correct burst length.

After reaching the final_addr[31:0] address, the done interrupt is active to signal completion of DMA operation.

Size: 8 bits

Offset: 0x3608 + (i * 0x1)

Bits	Name	Attr	Description
7:0	final_addr	R/W	Defines final_addr[7:0] to end DMA burst transactions Value After Reset: 0x0

ahb_dma_bstraddr[0:3]

Description: Audio DMA Burst Start Address Register Array Address offset: i = 0 to 3

These read-only registers compose the start address of the current burst operation.

burst_start_addr[31:0] = haddr[31:0] = initial_addr[31:0] + 16.

Size: 8 bits

Offset: 0x360c + (i * 0x1)

Bits	Name	Attr	Description
7:0	burst_addr	R	Start address for the current burst operation Value After Reset: 0x0

ahb_dma_mblength0

Description: Audio DMA Burst Length Register 0

This registers holds the length of the current burst operation. As an example, if the first burst transaction of the AHB audio DMA is a length of 8, then the second burst should start at

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address ohaddr[31:0] = initial_addr[31:0] + 32.
Size: 8 bits
Offset: 0x3610

Bits	Name	Attr	Description
7:0	mburstlength	R	Requested burst length (mburstlength[7:0]) Value After Reset: 0x0

ahb_dma_mblength1

Description: Audio DMA Burst Length Register 1
This registers holds the length of the current burst operation. As an example, if the first burst transaction of the AHB audio DMA is a length of 8, then the second burst should start at address ohaddr[31:0] = initial_addr[31:0] + 32.
Size: 8 bits
Offset: 0x3611

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	mburstlength	R	Requested burst length Value After Reset: 0x0

ahb_dma_mask

Description: Audio DMA Mask Interrupt Register
This register masks each of the interrupts present in the AHB audio DMA module.
Size: 8 bits
Offset: 0x3614

Bits	Name	Attr	Description
7	done_mask	R/W	DMA end of operation interrupt mask. Active when DMA engine reaches final_addr[15:0] or when stop DMA operation is activated. Value After Reset: 0x1
6	retrysplit_mask	R/W	Retry/split interrupt mask. Active when AHB master receives a RETRY or SPLIT response from slave. Value After Reset: 0x1
5	lostownership_mask	R/W	Master lost ownership interrupt mask when in burst transfer. Active when AHB master loses BUS ownership within the course of a burst transfer. Value After Reset: 0x1
4	error_mask	R/W	Error interrupt mask. Active when slave indicates error through the isresp[1:0]. Value After Reset: 0x1
3			Reserved for future use.
2	fifo_thrempty_mask	R/W	Audio FIFO empty interrupt mask when audio FIFO has less than the number of enabled audio channels. Value After Reset: 0x1
1	fifo_full_mask	R/W	Audio FIFO full interrupt mask. Value After Reset: 0x1
0	fifo_empty_mask	R/W	Audio FIFO empty interrupt mask. Value After Reset: 0x1

ahb_dma_conf1

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Description: Audio DMA Channel Enable Configuration Register 1
In AUDS packet configuration with layout 0 selected, the maximum number of active channels is 2.
Size: 8 bits
Offset: 0x3616

Bits	Name	Attr	Description
7:0	ch_in_en	R/W	Each bit controls the enabling of the respective audio channel. For instance, when bit 1 is set (1'b1) the audio Channel 1 is enabled. When cleared, the referred channel is disabled. Value After Reset: 0x0

ahb_dma_buffmask
Description: Audio DMA Buffer Mask Interrupt Register
Size: 8 bits
Offset: 0x3619

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	mask_fifo_overrun	R/W	Buffer overrun flag mask Value After Reset: 0x1
3:2			Reserved for future use.
1	mask_buff_full	R/W	Buffer full flag mask Value After Reset: 0x1
0	mask_buff_empty	R/W	Buffer empty flag mask Value After Reset: 0x1

ahb_dma_mask1
Description: Audio DMA Mask Interrupt Register 1
This register masks interrupts present in the AHB audio DMA module.
Size: 8 bits
Offset: 0x361b

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	fifo_underrun_mask	R/W	AHB DMA FIFO underrun mask Value After Reset: 0x1
0	fifo_overrun_mask	R/W	AHB DMA FIFO overrun mask Value After Reset: 0x1

ahb_dma_status
Description: Audio DMA Status
Size: 8 bits
Offset: 0x361c

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	autostart_status	R	Indicates the set of start and stop addresses currently used by the AHB audio DMA. If cleared (1'b0), the start and stop addresses configured in the address range 0x3604 to 0x360B are being used. When set (1'b1), the configurations at address range 0x3620 to 0x3627 are being used.

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Bits	Name	Attr	Description
			This bit is always at zero when autostart_enable is cleared (1'b0). Value After Reset: 0x0

ahb_dma_conf2

Description: Audio DMA Configuration Register 2

Size: 8 bits

Offset: 0x361d

Bits	Name	Attr	Description
7:2			Reserved for future use.
1	autostart_loop	R/W	Enables the AHB audio DMA auto-start loop mode Value After Reset: 0x1
0	autostart_enable	R/W	Enables the AHB audio DMA auto-start feature Value After Reset: 0x0

ahb_dma_straddr_set1[0:3]

Description: Audio DMA Start Address Set 1 Register Array Address offset: i = 0 to 3

These registers define the initial_addr_1[31:0] used to initiate the DMA burst read transactions upon start_dma_transaction configuration.

Size: 8 bits

Offset: 0x3620 + (i * 0x1)

Bits	Name	Attr	Description
7:0	initial_addr_1	R/W	Defines init_addr_1[7:0] to initiate DMA burst transactions Value After Reset: 0x0

ahb_dma_stpaddr_set1[0:3]

Description: Audio DMA Stop Address Set 1 Register Array Address offset: i = 0 to 3

These registers define the final_addr_1[31:0] used as the final point to the DMA burst read transactions. Upon start_dma_transaction configuration, the DMA engine starts requesting burst reads from the external system memory. Each burst read can have a maximum theoretical length of 256 words (due to the AMBA AHB specification 1 Kbyte boundary burst limitation).

The DMA engine is responsible for incrementing the burst starting address and defining its corresponding burst length to reach the final_addr[31:0] address. The last burst request issued by the DMA engine takes into account that it should only request data until the final_addr[31:0] address (included) and for that should calculate the correct burst length. After reaching the final_addr_1[31:0] address, the done interrupt is active to indicate completion of the DMA operation.

Size: 8 bits

Offset: 0x3624 + (i * 0x1)

Bits	Name	Attr	Description
7:0	final_addr_1	R/W	Defines final_addr_1[7:0] to end DMA burst transactions Value After Reset: 0x0

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MainController Registers

Main Controller Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
mc_clkdis	0x4001	Main Controller Synchronous Clock Domain Disable Register
mc_swrstzreq	0x4002	Main Controller Software Reset Register Main controller software reset request per clock domain....
mc_opctrl	0x4003	Main Controller HDCP Bypass Control Register
mc_flowctrl	0x4004	Main Controller Feed Through Control Register
mc_phyrstz	0x4005	Main Controller PHY Reset Register
mc_lockonclock	0x4006	Main Controller Clock Present Register
mc_heacphy_rst	0x4007	Main Controller HEAC PHY Reset Register
mc_lockonclock_2	0x4009	Main Controller Clock Present Register 2
mc_swrstzreq_2	0x400a	Main Controller Software Reset Register 2 Main controller software reset request per clock domain....
mc_opsts	0x4010	Main Controller Status Register This register contains the information regarding the status of...
base_sfrdivlow	0x4018	SFR Clock Base Time Register Low
base_sfrdivhigh	0x4019	SFR Clock Base Time Register High

mc_clkdis

Description: Main Controller Synchronous Clock Domain Disable Register
Size: 8 bits
Offset: 0x4001

Bits	Name	Attr	Description
7	h22sclk_disable	R/W	HDCP22 clock synchronous disable signal. When active (1b), simultaneously bypasses HDCP22. Value After Reset: 0x0
6	hdcpclk_disable	R/W	HDCP clock synchronous disable signal. When active (1b), simultaneously bypasses HDCP. Value After Reset: 0x0
5	cecclk_disable	R/W	CEC Engine clock synchronous disable signal. Value After Reset: 0x0
4	cscclk_disable	R/W	Color Space Converter clock synchronous disable signal. Value After Reset: 0x0
3	audclk_disable	R/W	Audio Sampler clock synchronous disable signal. Value After Reset: 0x0
2	prepclk_disable	R/W	Pixel Repetition clock synchronous disable signal. Value After Reset: 0x0
1	tmdsclk_disable	R/W	TMDS clock synchronous disable signal. It is required to perform a write action on one of the following registers: fc_invidconf, fc_inhactiv0, fc_inhactiv1, fc_inhblank0, fc_inhblank1, fc_invactiv0 fc_invactiv1, fc_invblank, fc_hsyncindelay0,

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Bits	Name	Attr	Description
			fc_hsyncindelay1, fc_hsyncinwidth0 fc_hsyncinwidth1, fc_vsyncindelay, fc_vsyncinwidth, fc_ctrldur, fc_exctrldur, fc_exctrlspac Value After Reset: 0x0
0	pixelclk_disable	R/W	Pixel clock synchronous disable signal. Value After Reset: 0x0

mc_swrstzreq

Description: Main Controller Software Reset Register

Main controller software reset request per clock domain. Writing zero to a bit of this register results in an NRZ signal toggle at sfrclk rate to an output signal that indicates a software reset request. This toggle must be used to generate a synchronized reset to de corresponding domain, with at least 1 clock cycle.

Size: 8 bits

Offset: 0x4002

Bits	Name	Attr	Description
7	igpaswrst_req	R/W	GPAUD interface soft reset request. This bit is enabled when the Generic Parallel Audio (GPAUD) interface is enabled (AUDIO_IF = 6). Otherwise, this bit returns zero. Value After Reset: 0x1
6	cecswrst_req	R/W	CEC software reset request. Defaults back to 1b after reset request. Note: After you configure cecswrst_req, set the value of the bit csc_clk_disable of the register mc_clkdis to 1, 0, and then 1 again. Value After Reset: 0x1
5			Reserved for future use.
4	ispdifswrst_req	R/W	SPDIF audio software reset request. Value After Reset: 0x1
3	ii2sswrst_req	R/W	I2S audio software reset request. Value After Reset: 0x1
2	prepswrst_req	R/W	Pixel Repetition software reset request. Value After Reset: 0x1
1	tmdsswrst_req	R/W	TMDS software reset request. It is required to perform a write action on one of the following registers: fc_invidconf, fc_inhactiv0, fc_inhactiv1, fc_inhblank0, fc_inhblank1, fc_invactiv0 fc_invactiv1, fc_invblank, fc_hsyncindelay0, fc_hsyncindelay1, fc_hsyncinwidth0 fc_hsyncinwidth1, fc_vsyncindelay, fc_vsyncinwidth, fc_ctrldur, fc_exctrldur, fc_exctrlspac Value After Reset: 0x1
0	pixelswrst_req	R/W	Pixel software reset request. Value After Reset: 0x1

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mc_opctrl

Description: Main Controller HDCP Bypass Control Register

Size: 8 bits

Offset: 0x4003

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	h22s_ovr_val	R/W	HDCP SNPS 2.2 versus 1.4 switch override value 1'b0: The switch is routed to HDCP 1.4 signals when hdcp22snps_switch_lock is not set to 1'b1. 1'b1: The switch is routed to HDCP 2.2 SNPS signals when hdcp22snps_switch_lock is not set to 1'b1. Value After Reset: 0x1
4	h22s_switch_lck	R/W	HDCP 2.2 SNPS switch lock 1'b0: Enables you to change the direction of the HDCP 2.2 SNPS versus 1.4 switch by using the hdcp22snps_ovr_val. 1'b1: You can still write to hdcp22snps_ovr_val but has no effect over the HDCP 2.2 SNPS versus 1.4 switch, that keeps as it was configured by hdcp22snps_ovr_val at the time the 1'b1 was written to this bit field. Once you set the value to 1'b1, you can change the value back to 1'b0 only by issuing a master reset to the Hdmi_tx. Value After Reset: 0x0
3:1			Reserved for future use.
0	hdcp_block_byp	R/W	Block HDCP bypass mechanism 1'b0: This is the default value. You can write to the hdcp_clkdisable bit of the register mc_clkdis and bypass HDCP by acting on the register mc_clkdis bit 6 (hdcp_clkdisable) 1'b1: You can still write to the hdcp_clkdisable bit of the register mc_clkdis but this action disables the HDCP module and blocks the bypass mechanism. The output data is frozen and the HDMI Tx and RX fail authentication. Once you set the value to 1'b1, you can change the value back to 1'b0 only by issuing a master reset to the Hdmi_tx. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0

mc_flowctrl

Description: Main Controller Feed Through Control Register

Size: 8 bits

Offset: 0x4004

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	Feed_through_off	R/W	Video path Feed Through enable bit: 1b: Color Space Converter is in the video data path. 0b: Color Space Converter is bypassed (not in the video

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Bits	Name	Attr	Description
			data path). Value After Reset: 0x0

mc_phyrstz

Description: Main Controller PHY Reset Register

Size: 8 bits

Offset: 0x4005

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	phyrstz	R/W	HDMI Source PHY active low reset control for PHY GEN1, active high reset control for PHY GEN2. Value After Reset: "(PHY_GEN2== 1) ? 1 : 0"

mc_lockonclock

Description: Main Controller Clock Present Register

Size: 8 bits

Offset: 0x4006

Bits	Name	Attr	Description
7	igpaclk	R/W1C	GPAUD interface clock status. This bit is enabled when the Generic Parallel Audio (GPAUD) interface is enabled (AUDIO_IF = 6). Otherwise, this bit returns zero. This bit indicates the clock is present in the system. It is cleared by writing 1 to this bit. Value After Reset: 0x0
6	pclk	R/W1C	Pixel clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0
5	tclk	R/W1C	TMDS clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0
4	prepclk	R/W1C	Pixel Repetition clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0
3	i2sclk	R/W1C	I2S clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0
2	audiospdifclk	R/W1C	SPDIF clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0
1			Reserved for future use.
0	cecclk	R/W1C	CEC clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0

mc_heacphy_rst

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Description: Main Controller HEAC PHY Reset Register
Size: 8 bits
Offset: 0x4007

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	heacphyrst	R/W	HEAC PHY reset (active high) Value After Reset: 0x1

mc_lockonclock_2
Description: Main Controller Clock Present Register 2
Size: 8 bits
Offset: 0x4009

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	ahbdmaclk	R/W1C	AHB audio DMA clock status. Indicates that the clock is present in the system. Cleared by WR 1 to this position. Value After Reset: 0x0

mc_swrstzreq_2
Description: Main Controller Software Reset Register 2
Main controller software reset request per clock domain. Writing zero to a bit of this register results in a signal toggle that indicates a software reset request. This toggle is used to generate a synchronized reset to the corresponding domain, with one or more clock cycles.
Size: 8 bits
Offset: 0x400a

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	ahbdmaswrst_req	R/W	AHB audio DMA software reset request. Writing 1'b1 does not result in any action. Writing 1'b0 to this register resets all AHB audio logic. Value After Reset: 0x0

mc_opsts
Description: Main Controller Status Register
This register contains the information regarding the status of the HDCP SNPS 2.2 versus 1.4 switch.
Size: 8 bits
Offset: 0x4010

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	h22s_switch_sts	R	HDCP SNPS 2.2 versus 1.4 switch value status. 1'b0: HDCP 1.4 selected 1'b1: HDCP 2.2 selected Value After Reset: 0x1

base_sfrdivlow
Description: SFR Clock Base Time Register Low
Size: 8 bits
Offset: 0x4018

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Bits	Name	Attr	Description
7:0	base_sfrdiv_lo	R/W	SFR clock divider Low This register must be configured with the 8 least-significant bits of the value sfrclk frequency divided by 1000 (for example, for 27 MHz base_sfrdiv[14:0] = 27027). The configured data is used to generate a reference pulse of 1ms period that is needed by several timers within the controller. Value After Reset: 0x93

base_sfrdivhigh

Description: SFR Clock Base Time Register High

Size: 8 bits

Offset: 0x4019

Bits	Name	Attr	Description
7			Reserved for future use.
6:0	base_sfrdiv_hi	R/W	SFR clock divider High This register must be configured with the 7 most-significant bits of the value sfrclk frequency divided by 1000 (for example, for 27 MHz base_sfrdiv[14:0] = 27027). The configured data is used to generate a reference pulse of 1ms period that is needed by several timers within the controller. Value After Reset: 0x69

ColorSpaceConverter Registers

Color Space Converter Registers Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
csc_cfg	0x4100	Color Space Converter Interpolation and Decimation Configuration Register
csc_scale	0x4101	Color Space Converter Scale and Deep Color Configuration Register
csc_coef_a1_msb	0x4102	Color Space Converter Matrix A1 Coefficient Register MSB Notes: - The coefficients used in the...
csc_coef_a1_lsb	0x4103	Color Space Converter Matrix A1 Coefficient Register LSB Notes: - The coefficients used in the...
csc_coef_a2_msb	0x4104	Color Space Converter Matrix A2 Coefficient Register MSB Color Space Conversion A2...
csc_coef_a2_lsb	0x4105	Color Space Converter Matrix A2 Coefficient Register LSB Color Space Conversion A2...
csc_coef_a3_msb	0x4106	Color Space Converter Matrix A3 Coefficient Register MSB Color Space Conversion A3...
csc_coef_a3_lsb	0x4107	Color Space Converter Matrix A3 Coefficient Register LSB Color Space Conversion A3...
csc_coef_a4_msb	0x4108	Color Space Converter Matrix A4 Coefficient Register MSB

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Register	Offset	Description
		Color Space Conversion A4...
csc_coef_a4_lsb	0x4109	Color Space Converter Matrix A4 Coefficient Register LSB Color Space Conversion A4...
csc_coef_b1_msb	0x410a	Color Space Converter Matrix B1 Coefficient Register MSB Color Space Conversion B1...
csc_coef_b1_lsb	0x410b	Color Space Converter Matrix B1 Coefficient Register LSB Color Space Conversion B1...
csc_coef_b2_msb	0x410c	Color Space Converter Matrix B2 Coefficient Register MSB Color Space Conversion B2...
csc_coef_b2_lsb	0x410d	Color Space Converter Matrix B2 Coefficient Register LSB Color Space Conversion B2...
csc_coef_b3_msb	0x410e	Color Space Converter Matrix B3 Coefficient Register MSB Color Space Conversion B3...
csc_coef_b3_lsb	0x410f	Color Space Converter Matrix B3 Coefficient Register LSB Color Space Conversion B3...
csc_coef_b4_msb	0x4110	Color Space Converter Matrix B4 Coefficient Register MSB Color Space Conversion B4...

Registers for Address Block: ColorSpaceConverter (Continued)

Register	Offset	Description
csc_coef_b4_lsb	0x4111	Color Space Converter Matrix B4 Coefficient Register LSB Color Space Conversion B4...
csc_coef_c1_msb	0x4112	Color Space Converter Matrix C1 Coefficient Register MSB Color Space Conversion C1...
csc_coef_c1_lsb	0x4113	Color Space Converter Matrix C1 Coefficient Register LSB Color Space Conversion C1...
csc_coef_c2_msb	0x4114	Color Space Converter Matrix C2 Coefficient Register MSB Color Space Conversion C2...
csc_coef_c2_lsb	0x4115	Color Space Converter Matrix C2 Coefficient Register LSB Color Space Conversion C2...
csc_coef_c3_msb	0x4116	Color Space Converter Matrix C3 Coefficient Register MSB Color Space Conversion C3...
csc_coef_c3_lsb	0x4117	Color Space Converter Matrix C3 Coefficient Register LSB Color Space Conversion C3...
csc_coef_c4_msb	0x4118	Color Space Converter Matrix C4 Coefficient Register MSB Color Space Conversion C4...
csc_coef_c4_lsb	0x4119	Color Space Converter Matrix C4 Coefficient Register LSB Color Space Conversion C4...
csc_limit_up_msb	0x411a	Color Space Converter Matrix Output Up Limit Register MSB For more details, refer to the HDMI 1.4...
csc_limit_up_lsb	0x411b	Color Space Converter Matrix output Up Limit Register LSB For more details, refer to the HDMI 1.4...
csc_limit_dn_msb	0x411c	Color Space Converter Matrix output Down Limit Register MSB For more details, refer to the HDMI...
csc_limit_dn_lsb	0x411d	Color Space Converter Matrix output Down Limit Register LSB For more details, refer to the HDMI...

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csc_cfg

Description: Color Space Converter Interpolation and Decimation Configuration Register
Size: 8 bits
Offset: 0x4100

Bits	Name	Attr	Description
7	csc_limit	R/W	When set (1'b1), the range limitation values defined in registers csc_mat_uplim and csc_mat_dnlim are applied to the output of the Color Space Conversion matrix. This feature ensures that the video output range is always respected, independently of the matrix coefficient configuration or of the video input stream. Value After Reset: 0x0
6	spare_2	R/W	Reserved as "spare" register with no associated functionality. Value After Reset: 0x0
5:4	intmode	R/W	Chroma interpolation configuration: intmode[1:0] | Chroma Interpolation 00 | interpolation disabled 01 | Hu (z) =1 + z^(-1) 10 | Hu(z)=1/ 2 + z^(-11)+1/2 z^(-2) 11 | interpolation disabled Value After Reset: 0x0
3:2	spare_1	R/W	Reserved as "spare" register with no associated functionality. Value After Reset: 0x0
1:0	decmode	R/W	Chroma decimation configuration: decmode[1:0] | Chroma Decimation 00 | decimation disabled 01 | Hd (z) =1 10 | Hd(z)=1/ 4 + 1/2z^(-1)+1/4 z^(-2) 11 | Hd(z)x2^(11)= -5+12z^(-2) - 22z^(-4)+39z^(-8) +109z^(-10) -204z^(-12)+648z^(-14) + 1024z^(-15) +648z^(-16) - 204z^(-18) +109z^(-20)- 65z^(-22) +39z^(-24) -22z^(-26) +12z^(- 28)-5z^(-30) Value After Reset: 0x0

csc_scale

Description: Color Space Converter Scale and Deep Color Configuration Register
Size: 8 bits
Offset: 0x4101

Bits	Name	Attr	Description
7:4	csc_color_depth	R/W	Color space converter color depth configuration:

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Bits	Name	Attr	Description
			csc_colordepth[3:0] | Action 0000 | 24 bit per pixel video (8 bit per component). 0001-0011 | Reserved. Not used. 0100 | 24 bit per pixel video (8 bit per component). 0101 | 30 bit per pixel video (10 bit per component). 0110 | 36 bit per pixel video (12 bit per component). 0111 | 48 bit per pixel video (16 bit per component). other | Reserved. Not used. Value After Reset: 0x0
3:2	spare	R/W	The is a Reserved as "spare" register with no associated functionality. Value After Reset: 0x0
1:0	cscscale	R/W	Defines the cscscale[1:0] scale factor to apply to all coefficients in Color Space Conversion. This scale factor is expressed in the number of left shifts to apply to each of the coefficients, ranging from 0 to 2. Value After Reset: 0x1

csc_coef_a1_msb

Description: Color Space Converter Matrix A1 Coefficient Register MSB Notes:
The coefficients used in the CSC matrix use only 15 bits for the internal computations.
Coefficients are represented in 2's complementary format and stored in two registers:
csc_coef_*_lsb[7:0]: coefficient bits 7 to 0
csc_coef_*_msb[7]: spare bit
csc_coef_*_msb[6:0]: coefficient bits 14 to 8
Examples for standard ITU601 and ITU709 RGB/YCC conversion CSC coefficients exist in the Video Datapath Application Note.
Size: 8 bits
Offset: 0x4102

Bits	Name	Attr	Description
7:0	csc_coef_a1_msb	R/W	Color Space Converter Matrix A1 Coefficient Register MSB Value After Reset: 0x20

csc_coef_a1_lsb

Description: Color Space Converter Matrix A1 Coefficient Register LSB Notes:
The coefficients used in the CSC matrix use only 15 bits for the internal computations.
Coefficients are represented in 2's complementary format and stored in two registers:
csc_coef_*_lsb[7:0]: coefficient bits 7 to 0
csc_coef_*_msb[7]: spare bit
csc_coef_*_msb[6:0]: coefficient bits 14 to 8

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Examples for standard ITU601 and ITU709 RGB/YCC conversion CSC coefficients exist in the Video Datapath Application Note.
Size: 8 bits
Offset: 0x4103

Bits	Name	Attr	Description
7:0	csc_coef_a1_lsb	R/W	Color Space Converter Matrix A1 Coefficient Register LSB Value After Reset: 0x0

csc_coef_a2_msb
Description: Color Space Converter Matrix A2 Coefficient Register MSB Color Space Conversion A2 coefficient.
Size: 8 bits
Offset: 0x4104

Bits	Name	Attr	Description
7:0	csc_coef_a2_msb	R/W	Color Space Converter Matrix A2 Coefficient Register MSB Value After Reset: 0x0

csc_coef_a2_lsb
Description: Color Space Converter Matrix A2 Coefficient Register LSB Color Space Conversion A2 coefficient.
Size: 8 bits
Offset: 0x4105

Bits	Name	Attr	Description
7:0	csc_coef_a2_lsb	R/W	Color Space Converter Matrix A2 Coefficient Register LSB Value After Reset: 0x0

csc_coef_a3_msb
Description: Color Space Converter Matrix A3 Coefficient Register MSB Color Space Conversion A3 coefficient.
Size: 8 bits
Offset: 0x4106

Bits	Name	Attr	Description
7:0	csc_coef_a3_msb	R/W	Color Space Converter Matrix A3 Coefficient Register MSB Value After Reset: 0x0

csc_coef_a3_lsb
Description: Color Space Converter Matrix A3 Coefficient Register LSB Color Space Conversion A3 coefficient.
Size: 8 bits
Offset: 0x4107

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	csc_coef_a3_lsb	R/W	Color Space Converter Matrix A3 Coefficient Register LSB Value After Reset: 0x0

csc_coef_a4_msb

Description: Color Space Converter Matrix A4 Coefficient Register MSB Color Space Conversion A4 coefficient.

Size: 8 bits

Offset: 0x4108

Bits	Name	Attr	Description
7:0	csc_coef_a4_msb	R/W	Color Space Converter Matrix A4 Coefficient Register MSB Value After Reset: 0x0

csc_coef_a4_lsb

Description: Color Space Converter Matrix A4 Coefficient Register LSB Color Space Conversion A4 coefficient.

Size: 8 bits

Offset: 0x4109

Bits	Name	Attr	Description
7:0	csc_coef_a4_lsb	R/W	Color Space Converter Matrix A4 Coefficient Register LSB Value After Reset: 0x0

csc_coef_b1_msb

Description: Color Space Converter Matrix B1 Coefficient Register MSB Color Space Conversion B1 coefficient.

Size: 8 bits

Offset: 0x410a

Bits	Name	Attr	Description
7:0	csc_coef_b1_msb	R/W	Color Space Converter Matrix B1 Coefficient Register MSB Value After Reset: 0x0

csc_coef_b1_lsb

Description: Color Space Converter Matrix B1 Coefficient Register LSB Color Space Conversion B1 coefficient.

Size: 8 bits

Offset: 0x410b

Bits	Name	Attr	Description
7:0	csc_coef_b1_lsb	R/W	Color Space Converter Matrix B1 Coefficient Register LSB Value After Reset: 0x0

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csc_coef_b2_msb

Description: Color Space Converter Matrix B2 Coefficient Register MSB Color Space Conversion B2 coefficient.

Size: 8 bits

Offset: 0x410c

Bits	Name	Attr	Description
7:0	csc_coef_b2_msb	R/W	Color Space Converter Matrix B2 Coefficient Register MSB Value After Reset: 0x20

csc_coef_b2_lsb

Description: Color Space Converter Matrix B2 Coefficient Register LSB Color Space Conversion B2 coefficient.

Size: 8 bits

Offset: 0x410d

Bits	Name	Attr	Description
7:0	csc_coef_b2_lsb	R/W	Color Space Converter Matrix B2 Coefficient Register LSB Value After Reset: 0x0

csc_coef_b3_msb

Description: Color Space Converter Matrix B3 Coefficient Register MSB Color Space Conversion B3 coefficient.

Size: 8 bits

Offset: 0x410e

Bits	Name	Attr	Description
7:0	csc_coef_b3_msb	R/W	Color Space Converter Matrix B3 Coefficient Register MSB Value After Reset: 0x0

csc_coef_b3_lsb

Description: Color Space Converter Matrix B3 Coefficient Register LSB Color Space Conversion B3 coefficient.

Size: 8 bits

Offset: 0x410f

Bits	Name	Attr	Description
7:0	csc_coef_b3_lsb	R/W	Color Space Converter Matrix B3 Coefficient Register LSB Value After Reset: 0x0

csc_coef_b4_msb

Description: Color Space Converter Matrix B4 Coefficient Register MSB Color Space Conversion B4 coefficient.

Size: 8 bits

Offset: 0x4110

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	csc_coef_b4_msb	R/W	Color Space Converter Matrix B4 Coefficient Register MSB Value After Reset: 0x0

csc_coef_b4_lsb

Description: Color Space Converter Matrix B4 Coefficient Register LSB Color Space Conversion B4 coefficient.

Size: 8 bits

Offset: 0x4111

Bits	Name	Attr	Description
7:0	csc_coef_b4_lsb	R/W	Color Space Converter Matrix B4 Coefficient Register LSB Value After Reset: 0x0

csc_coef_c1_msb

Description: Color Space Converter Matrix C1 Coefficient Register MSB Color Space Conversion C1 coefficient.

Size: 8 bits

Offset: 0x4112

Bits	Name	Attr	Description
7:0	csc_coef_c1_msb	R/W	Color Space Converter Matrix C1 Coefficient Register MSB Value After Reset: 0x0

csc_coef_c1_lsb

Description: Color Space Converter Matrix C1 Coefficient Register LSB Color Space Conversion C1 coefficient.

Size: 8 bits

Offset: 0x4113

Bits	Name	Attr	Description
7:0	csc_coef_c1_lsb	R/W	Color Space Converter Matrix C1 Coefficient Register LSB Value After Reset: 0x0

csc_coef_c2_msb

Description: Color Space Converter Matrix C2 Coefficient Register MSB Color Space Conversion C2 coefficient.

Size: 8 bits

Offset: 0x4114

Bits	Name	Attr	Description
7:0	csc_coef_c2_msb	R/W	Color Space Converter Matrix C2 Coefficient Register MSB Value After Reset: 0x0

csc_coef_c2_lsb

Description: Color Space Converter Matrix C2 Coefficient Register LSB Color Space

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Conversion C2 coefficient.
Size: 8 bits
Offset: 0x4115

Bits	Name	Attr	Description
7:0	csc_coef_c2_lsb	R/W	Color Space Converter Matrix C2 Coefficient Register LSB Value After Reset: 0x0

csc_coef_c3_msb
Description: Color Space Converter Matrix C3 Coefficient Register MSB Color Space Conversion C3 coefficient.
Size: 8 bits
Offset: 0x4116

Bits	Name	Attr	Description
7:0	csc_coef_c3_msb	R/W	Color Space Converter Matrix C3 Coefficient Register MSB Value After Reset: 0x20

csc_coef_c3_lsb
Description: Color Space Converter Matrix C3 Coefficient Register LSB Color Space Conversion C3 coefficient.
Size: 8 bits
Offset: 0x4117

Bits	Name	Attr	Description
7:0	csc_coef_c3_lsb	R/W	Color Space Converter Matrix C3 Coefficient Register LSB Value After Reset: 0x0

csc_coef_c4_msb
Description: Color Space Converter Matrix C4 Coefficient Register MSB Color Space Conversion C4 coefficient.
Size: 8 bits
Offset: 0x4118

Bits	Name	Attr	Description
7:0	csc_coef_c4_msb	R/W	Description: Color Space Converter Matrix C4 Coefficient Register MSB Value After Reset: 0x0

csc_coef_c4_lsb
Description: Color Space Converter Matrix C4 Coefficient Register LSB Color Space Conversion C4 coefficient.
Size: 8 bits
Offset: 0x4119

Bits	Name	Attr	Description
7:0	csc_coef_c4_lsb	R/W	Color Space Converter Matrix C4 Coefficient Register LSB

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Bits	Name	Attr	Description
			Value After Reset: 0x0

csc_limit_up_msb

Description: Color Space Converter Matrix Output Up Limit Register MSB
For more details, refer to the HDMI 1.4 specification, paragraph 6.6 Video Quantization Ranges. For an RGB output of 8 bits, the expected limit is 254 (valid range column of Table 6-3), and this register must be configured with 0x00.
Size: 8 bits
Offset: 0x411a

Bits	Name	Attr	Description
7:0	csc_limit_up_msb	R/W	Color Space Converter Matrix Output Upper Limit Register MSB Value After Reset: 0xff

csc_limit_up_lsb

Description: Color Space Converter Matrix output Up Limit Register LSB
For more details, refer to the HDMI 1.4 specification, paragraph 6.6 Video Quantization Ranges. For an RGB output of 8 bits, the expected limit is 254 (valid range column of Table 6-3), and this register must be configured with 0xFE.
Size: 8 bits
Offset: 0x411b

Bits	Name	Attr	Description
7:0	csc_limit_up_lsb	R/W	Color Space Converter Matrix Output Upper Limit Register LSB Value After Reset: 0xff

csc_limit_dn_msb

Description: Color Space Converter Matrix output Down Limit Register MSB
For more details, refer to the HDMI 1.4 specification, paragraph 6.6 Video Quantization Ranges. For an RGB output of 8 bits, the expected limit is 1 (valid range column of Table 6-3), and this register must be configured with 0x00.
Size: 8 bits
Offset: 0x411c

Bits	Name	Attr	Description
7:0	csc_limit_dn_msb	R/W	Color Space Converter Matrix output Down Limit Register MSB Value After Reset: 0x0

csc_limit_dn_lsb

Description: Color Space Converter Matrix output Down Limit Register LSB
For more details, refer to the HDMI 1.4 specification, paragraph 6.6 Video Quantization Ranges. For an RGB output of 8 bits, the expected limit is 1 (valid range column of Table 6-3), and this register must be configured with 0x01.
Size: 8 bits
Offset: 0x411d

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Bits	Name	Attr	Description
7:0	csc_limit_dn_lsb	R/W	Color Space Converter Matrix Output Down Limit Register LSB Value After Reset: 0x0

HDCP Registers

HDCP Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
a_hdcpcfg0	0x5000	HDCP Enable and Functional Control Configuration Register 0
a_hdcpcfg1	0x5001	HDCP Software Reset and Functional Control Configuration Register 1
a_hdcpobs0	0x5002	HDCP Observation Register 0
a_hdcpobs1	0x5003	HDCP Observation Register 1
a_hdcpobs2	0x5004	HDCP Observation Register 2
a_hdcpobs3	0x5005	HDCP Observation Register 3
a_apiintclr	0x5006	HDCP Interrupt Clear Register Write only register, active high and auto cleared, cleans the respective...
a_apiintstat	0x5007	HDCP Interrupt Status Register Read only register, reports the interruption which caused the activation...
a_apiintmsk	0x5008	HDCP Interrupt Mask Register The configuration of this register mask a given setup of interruption,...
a_vidpolcfg	0x5009	HDCP Video Polarity Configuration Register
a_oesswcfg	0x500a	HDCP OESS WOO Configuration Register Pulse width of the encryption enable (CTL3) signal in the...
a_coreverlsb	0x5014	HDCP Controller Version Register LSB Design ID number.
a_corevermsb	0x5015	HDCP Controller Version Register MSB Revision ID number.
a_ksvmemctrl	0x5016	HDCP KSV Memory Control Register The KSVCTRLupd bit is a notification flag. This flag changes polarity...
hdcp_bstatus[0:1]	0x5020 + (i * 0x1)	HDCP BStatus Register Array
hdcp_m0[0:7]	0x5022 + (i * 0x1)	HDCP M0 Register Array
hdcp_ksv[0:634]	0x502a + (i * 0x1)	HDCP KSV Registers
hdcp_vh[0:19]	0x52a5	HDCP SHA-1 VH Registers

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Register	Offset	Description
	+ (i * 0x1)	
hdcp_revoc_size_0	0x52b9	HDCP Revocation KSV List Size Register 0
hdcp_revoc_size_1	0x52ba	HDCP Revocation KSV List Size Register 1
hdcp_revoc_list[0:5059]	0x52bb + (i * 0x1)	HDCP Revocation KSV Registers
hdcpreg_bksv0	0x7800	HDCP KSV Status Register 0

Registers for Address Block: HDCP (Continued)

Register	Offset	Description
hdcpreg_bksv1	0x7801	HDCP KSV Status Register 1
hdcpreg_bksv2	0x7802	HDCP KSV Status Register 2
hdcpreg_bksv3	0x7803	HDCP KSV Status Register 3
hdcpreg_bksv4	0x7804	HDCP KSV Status Register 4
hdcpreg_anconf	0x7805	HDCP AN Bypass Control Register
hdcpreg_an0	0x7806	HDCP Forced AN Register 0
hdcpreg_an1	0x7807	HDCP Forced AN Register 1
hdcpreg_an2	0x7808	HDCP forced AN Register 2
hdcpreg_an3	0x7809	HDCP Forced AN Register 3
hdcpreg_an4	0x780a	HDCP Forced AN Register 4
hdcpreg_an5	0x780b	HDCP Forced AN Register 5
hdcpreg_an6	0x780c	HDCP Forced AN Register 6
hdcpreg_an7	0x780d	HDCP Forced AN Register 7
hdcpreg_rmlctl	0x780e	HDCP Encrypted Device Private Keys Control Register This register is the control register for the...
hdcpreg_rmlsts	0x780f	HDCP Encrypted DPK Status Register The required software configuration sequence is documented in...
hdcpreg_seed0	0x7810	HDCP Encrypted DPK Seed Register 0 This register contains a byte of the HDCP Encrypted DPK seed...
hdcpreg_seed1	0x7811	HDCP Encrypted DPK Seed Register 1 This register contains a byte of the HDCP Encrypted DPK seed...
hdcpreg_dpk0	0x7812	HDCP Encrypted DPK Data Register 0 This register contains an HDCP DPK byte. The required software...
hdcpreg_dpk1	0x7813	HDCP Encrypted DPK Data Register 1 This register contains an HDCP DPK byte. The required software...
hdcpreg_dpk2	0x7814	HDCP Encrypted DPK Data Register 2 This register contains an HDCP DPK byte. The required software...
hdcpreg_dpk3	0x7815	HDCP Encrypted DPK Data Register 3 This register contains an HDCP DPK byte. The required software...
hdcpreg_dpk4	0x7816	HDCP Encrypted DPK Data Register 4 This register contains an HDCP DPK byte. The required software...
hdcpreg_dpk5	0x7817	HDCP Encrypted DPK Data Register 5 This register contains an HDCP DPK byte. The required software...
hdcpreg_dpk6	0x7818	HDCP Encrypted DPK Data Register 6 This register

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Register	Offset	Description
		contains an HDCP DPK byte. The required software...

a_hdcpcfg0

Description: HDCP Enable and Functional Control Configuration Register 0

Size: 8 bits

Offset: 0x5000

Bits	Name	Attr	Description
7	ELVena	R/W	Enables the Enhanced Link Verification from the transmitter's side Value After Reset: 0x0
6	I2Cfastmode	R/W	Enable the I2C fast mode option from the transmitter's side. Value After Reset: 0x0
5	bypencryption	R/W	Bypasses all the data encryption stages Value After Reset: "(HDMI_HDCP_BYPASS== 1) ? 1 : 0"
4	syncricheck	R/W	Configures if the Ri check should be done at every 2s even or synchronously to every 128 encrypted frame. Value After Reset: 0x0
3	avmute	R	This register holds the current AVMUTE state of the Hdmi_tx controller, as expected to be perceived by the connected HDMI/HDCP sink device. Value After Reset: 0x0
2	rxdetect	R/W	Information that a sink device was detected connected to the HDMI port Value After Reset: 0x0
1	en11feature	R/W	Enable the use of features 1.1 from the transmitter's side Value After Reset: 0x0
0	hdmidvi	R/W	Configures the transmitter to operate with a HDMI capable device or with a DVI device. Value After Reset: 0x0

a_hdcpcfg1

Description: HDCP Software Reset and Functional Control Configuration Register 1

Size: 8 bits

Offset: 0x5001

Bits	Name	Attr	Description
7:5	spare	R/W	Reserved as "spare" register with no associated functionality. Value After Reset: 0x0
4	hdcp_lock	R/W	Lock the HDCP bypass and encryption disable mechanisms: 1'b0: The default 1'b0 value enables you to bypass

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Bits	Name	Attr	Description
			HDCP through bit 5 (bypencryption) of the A_HDCPCFG0 register or to disable the encryption through bit 1 (encryptiondisable) of A_HDCPCFG1. 1'b1: You can still write to the bit bypencryption of A_HDCPCFG0 or encryptiondisable bit of A_HDCPCFG1 but you cannot enable the bypass. Once you set the value to 1'b1, you can change the value back to 1'b0 only by issuing a master reset to the Hdmi_tx. Value After Reset: 0x0
3	dissha1check	R/W	Disables the request to the API processor to verify the SHA1 message digest of a received KSV List Value After Reset: 0x0
2	ph2upshftenc	R/W	Enables the encoding of packet header in the tmdsch0 bit[0] with cipher[2] instead of the tmdsch0 bit[2] Note: This bit must always be set to 1 for all PHYs. Value After Reset: 0x0
1	encryptiondisable	R/W	Disable encryption without losing authentication Value After Reset: 0x0
0	swreset	R/W	Software reset signal, active by writing a zero and auto cleared to 1 in the following cycle. Value After Reset: 0x1

a_hdcpobs0

Description: HDCP Observation Register 0

Size: 8 bits

Offset: 0x5002

Bits	Name	Attr	Description
7:4	STATEA	R	Observability register informs in which state the authentication machine is on. Value After Reset: 0x0
3:1	SUBSTATEA	R	Observability register informs in which sub-state the authentication is on. Value After Reset: 0x0
0	hdcpengaged	R	Informs that the current HDMI link has the HDCP protocol fully engaged. Value After Reset: 0x0

a_hdcpobs1

Description: HDCP Observation Register 1

Size: 8 bits

Offset: 0x5003

Bits	Name	Attr	Description
7			Reserved for future use.

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Bits	Name	Attr	Description
6:4	STATEOEG	R	Observability register informs in which state the OESS machine is on. Value After Reset: 0x0
3:0	STATER	R	Observability register informs in which state the revocation machine is on. Value After Reset: 0x0

a_hdcpobs2

Description: HDCP Observation Register 2

Size: 8 bits

Offset: 0x5004

Bits	Name	Attr	Description
7:6			Reserved for future use.
5:3	STATEE	R	Observability register informs in which state the cipher machine is on. Value After Reset: 0x0
2:0	STATEEEG	R	Observability register informs in which state the EESS machine is on. Value After Reset: 0x0

a_hdcpobs3

Description: HDCP Observation Register 3

Size: 8 bits

Offset: 0x5005

Bits	Name	Attr	Description
7	HDMI_RESERVED_1	R	Register read from attached sink device: Bcap(0x40) bit 7. Value After Reset: 0x0
6	REPEATER	R	Register read from attached sink device: Bcap(0x40) bit 6. Value After Reset: 0x0
5	KSV_FIFO_READY	R	Register read from attached sink device: Bcap(0x40) bit 5. Value After Reset: 0x0
4	FAST_I2C	R	Register read from attached sink device: Bcap(0x40) bit 4. Value After Reset: 0x0
3	HDMI_RESERVED_2	R	Register read from attached sink device: Bstatus(0x41) bit 13. Value After Reset: 0x0
2	HDMI_MODE	R	Register read from attached sink device: Bstatus(0x41) bit 12. Value After Reset: 0x0
1	FEATURES_1_1	R	Register read from attached sink device: Bcap(0x40) bit 1.

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Bits	Name	Attr	Description
			Value After Reset: 0x0
0	FAST_REAUTHENTICATION	R	Register read from attached sink device: Bcap(0x40) bit 0. Value After Reset: 0x0

a_apiintclr

Description: HDCP Interrupt Clear Register

Write only register, active high and auto cleared, cleans the respective interruption in the interrupt status register.

Size: 8 bits

Offset: 0x5006

Bits	Name	Attr	Description
7	HDCP_engaged	W	Clears the interruption related to HDCP authentication process successful. Value After Reset: 0x0
6	HDCP_failed	W	Clears the interruption related to HDCP authentication process failed. Value After Reset: 0x0
5	KSVsha1calcdoneint	W	Clears the interruption related to SHA1 verification has been done Value After Reset: 0x0
4	I2Cnack	W	Clears the interruption related to I2C NACK reception. Value After Reset: 0x0
3	Lostarbitration	W	Clears the interruption related to I2C arbitration lost. Value After Reset: 0x0
2	Keepouterrorint	W	Clears the interruption related to keep out window error. Value After Reset: 0x0
1	KSVsha1calcint	W	Clears the interruption related to KSV list update in memory that needs to be SHA1 verified. Value After Reset: 0x0
0	KSVaccessint	W	Clears the interruption related to KSV Attr grant for Read-Write access. Value After Reset: 0x0

a_apiintstat

Description: HDCP Interrupt Status Register

Read only register, reports the interruption which caused the activation of the interruption output pin.

Size: 8 bits

Offset: 0x5007

Bits	Name	Attr	Description
7	HDCP_engaged	R	Notifies that the HDCP authentication process was successful

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Bits	Name	Attr	Description
			Value After Reset: 0x0
6	HDCP_failed	R	Notifies that the HDCP authentication process was failed. Value After Reset: 0x0
5	KSVsha1calcdoneint	R	Notifies that the HDCP13TCTRL controller SHA1 verification has been done. The status ready to be read. Value After Reset: 0x0
4	I2Cnack	R	Notifies that the I2C received a NACK from slave device. Value After Reset: 0x0
3	Lostarbitration	R	Notifies that the I2C lost the arbitration to communicate. Another master gained arbitration. Value After Reset: 0x0
2	Keepouterrorint	R	Notifies that during the keep out window, the ctlout[3:0] bus was used besides control period. Value After Reset: 0x0
1	KSVsha1calcint	R	Notifies that the HDCP13TCTRL controller as updated a KSV list in memory that needs to be SHA1 verified. Value After Reset: 0x0
0	KSVaccessint	R	Notifies that the KSV Attr as been guaranteed for Read-Write access. Value After Reset: 0x0

a_apiintmsk

Description: HDCP Interrupt Mask Register

The configuration of this register mask a given setup of interruption, disabling them from generating interruption pulses in the interruption output pin.

Size: 8 bits

Offset: 0x5008

Bits	Name	Attr	Description
7	HDCP_engaged	R/W	Masks the interruption related to HDCP authentication process successful. Value After Reset: 0x0
6	HDCP_failed	R/W	Masks the interruption related to HDCP authentication process failed. Value After Reset: 0x0
5	KSVsha1calcdoneint	R/W	Masks the interruption related to SHA1 verification has been done Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
4	I2Cnack	R/W	Masks the interruption related to I2C NACK reception. Value After Reset: 0x0
3	Lostarbitration	R/W	Masks the interruption related to I2C arbitration lost.

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Bits	Name	Attr	Description
			Value After Reset: 0x0
2	Keepouterrorint	R/W	Masks the interruption related to keep out window error. Value After Reset: 0x0
1	KSVsha1calcint	R/W	Masks the interruption related to KSV list update in memory that needs to be SHA1 verified. Otherwise, this field is a "spare" bit with no associated functionality. Value After Reset: 0x0
0	KSVaccessint	R/W	Masks the interruption related to KSV Attr grant for Read-Write access. Value After Reset: 0x0

a_vidpolcfg

Description: HDCP Video Polarity Configuration Register

Size: 8 bits

Offset: 0x5009

Bits	Name	Attr	Description
7			Reserved for future use.
6:5	unencryptconf	R/W	Configuration of the color sent when sending unencrypted video data Value After Reset: 0x0
4	dataenpol	R/W	Configuration of the video data enable polarity Value After Reset: 0x0
3	vsyncpol	R/W	Configuration of the video Vertical synchronism polarity Value After Reset: 0x0
2	spare_2	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0
1	hsyncpol	R/W	Configuration of the video Horizontal synchronism polarity. Value After Reset: 0x0
0	spare_1	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x0

a_oesswcfg

Description: HDCP OESS WOO Configuration Register

Pulse width of the encryption enable (CTL3) signal in the HDCP OESS mode. The window of opportunity for the Original Encryption Status Signaling starts at the active edge of the Vertical synchronism and stops after oesswindowoffset[7:0]*4 clock cycles of TMDS clock. According to the HDCP specification, the CTL3 signal must be asserted at least for eight TMDS clock cycles (oesswindowoffset[7:0] must be greater than 1), and it is recommended to transmit a larger pulse width for enhanced link reliability.

Size: 8 bits

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Offset: 0x500a

Bits	Name	Attr	Description
7:0	a_oesswcfg	R/W	HDCP OESS WOO Configuration Register Value After Reset: 0x80

a_coreverlsb

Description: HDCP Controller Version Register LSB Design ID number.

Size: 8 bits

Offset: 0x5014

Bits	Name	Attr	Description
7:0	a_coreverlsb	R	HDCP Controller Version Register LSB Value After Reset: 0x2

a_corevermsb

Description: HDCP Controller Version Register MSB Revision ID number.

Size: 8 bits

Offset: 0x5015

Bits	Name	Attr	Description
7:0	a_corevermsb	R	HDCP Controller Version Register MSB Value After Reset: 0x3

a_ksvmemctrl

Description: HDCP KSV Memory Control Register

The KSVCTRLupd bit is a notification flag. This flag changes polarity whenever the register is written. This flag acts as a trigger to other blocks that processes this data. Upon reset the flag returns to low default value.

Size: 8 bits

Offset: 0x5016

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	KSVsha1status	R	Notification whether the KSV list message digest is correct from the controller: 1'b1 if digest message verification failed 1'b0 if digest message verification succeeded Value After Reset: 0x0
3	SHA1fail	R/W	Notification whether the KSV list message digest is correct. Value After Reset: 0x0
2	KSVCTRLupd	R/W	Set to inform that the KSV list in memory has been analyzed and the response to the Message Digest has been updated if on configurations on software SHA-1 calculation. Value After Reset: 0x0
1	KSVMEMaccess	R	Notification that the KSV Attr as been guaranteed. Value After Reset: 0x0
0	KSVMEMrequest	R/W	Request access to the KSV memory; must be

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Bits	Name	Attr	Description
			de-asserted after the access is completed by the system. Value After Reset: 0x0

hdcp_bstatus[0:1]
Description: HDCP BStatus Register Array
Size: 8 bits
Offset: 0x5020 + (i * 0x1)

Bits	Name	Attr	Description
7:0	bstatus	R/W	HDCP BSTATUS[15:0]. If Attr has not been granted (see register a_ksvmemctrl), the value read will be 8'hff. Value After Reset: 0xff

hdcp_m0[0:7]
Description: HDCP M0 Register Array
Size: 8 bits
Offset: 0x5022 + (i * 0x1)

Bits	Name	Attr	Description
7:0	M0	R/W	HDCP M0[32:0]. If Attr has not been granted (see register a_ksvmemctrl) , the value read will be 8'hff. These values are only available on a configuration that has the SHA1 calculation by software. Value After Reset: 0xff

hdcp_ksv[0:634]
Description: HDCP KSV Registers.
Size: 8 bits
Offset: 0x502a + (i * 0x1)

Bits	Name	Attr	Description
7:0	hdcp_ksv_byte	R/W	Sink KSV FIFO byte, ordered in little endian (byte at address 0x502a belongs to byte 0 of KSV0). If Attr has not been granted (see register a_ksvmemctrl), the value read is 8'hff. In this address space, 635 KSV FIFO bytes are mapped, which allow for 127 KSV values, each with 5 bytes (40 bits). Value After Reset: 0xff

hdcp_vh[0:19]
Description: HDCP SHA-1 VH Registers.
Size: 8 bits
Offset: 0x52a5 + (i * 0x1)

Bits	Name	Attr	Description
7:0	hdcp_vh_byte	R/W	Sink VH' byte, ordered in little endian (byte at

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Bits	Name	Attr	Description
			address 0x525a belongs to byte 0 of VH0). If Attr has not been granted (see register a_ksvmemctrl), the value read is 8'hff. In this address space 20 VH bytes are mapped, which allow for 5 VH values, each with 4 bytes (32bits). Value After Reset: 0xff

hdcp_revoc_size_0

Description: HDCP Revocation KSV List Size Register 0

Size: 8 bits

Offset: 0x52b9

Bits	Name	Attr	Description
7:0	hdcp_revoc_size_0	R/W	Register containing the LSB of KSV list size (ksv_list_size[7:0]). If Attr has not been granted (see register a_ksvmemctrl), the value read is 8'hff. Value After Reset: 0xff

hdcp_revoc_size_1

Description: HDCP Revocation KSV List Size Register 1

Size: 8 bits

Offset: 0x52ba

Bits	Name	Attr	Description
7:0	hdcp_revoc_size_1	R/W	Register containing the MSB of KSV list size (ksv_list_size[15:8]). If Attr has not been granted (see register a_ksvmemctrl), the value read is 8'hff. Value After Reset: 0xff

hdcp_revoc_list[0:5059]

Description: HDCP Revocation KSV Registers.

Size: 8 bits

Offset: 0x52bb + (i * 0x1)

Bits	Name	Attr	Description
7:0	hdcp_revoc_list_ksv_byte	R/W	Revocation KSV byte, ordered in little endian (byte at address 0x52bb belongs to byte 0 of the first revoked KSV). If Attr has not been granted (see register a_ksvmemctrl), the value read is 8'hff. In this address space 5060 revoked KSV bytes are mapped, which allow for 1012 KSV values, each with 5 bytes (40 bits). Value After Reset: 0xff

hdcpreg_bksv0

Description: HDCP KSV Status Register 0

Size: 8 bits

Offset: 0x7800

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Bits	Name	Attr	Description
7:0	hdcpreg_bksv0	R	Contains the value of BKSV[7:0]. Value After Reset: 0x0

hdcpreg_bksv1

Description: HDCP KSV Status Register 1

Size: 8 bits

Offset: 0x7801

Bits	Name	Attr	Description
7:0	hdcpreg_bksv1	R	Description: Contains the value of BKSV[15:8]. Value After Reset: 0x0

hdcpreg_bksv2

Description: HDCP KSV Status Register 2

Size: 8 bits

Offset: 0x7802

Bits	Name	Attr	Description
7:0	hdcpreg_bksv2	R	Contains the value of BKSV[23:16]. Value After Reset: 0x0

hdcpreg_bksv3

Description: HDCP KSV Status Register 3

Size: 8 bits

Offset: 0x7803

Bits	Name	Attr	Description
7:0	hdcpreg_bksv3	R	Contains the value of BKSV[31:24]. Value After Reset: 0x0

hdcpreg_bksv4

Description: HDCP KSV Status Register 4

Size: 8 bits

Offset: 0x7804

Bits	Name	Attr	Description
7:0	hdcpreg_bksv4	R	Contains the value of BKSV[39:32]. Value After Reset: 0x0

hdcpreg_anconf

Description: HDCP AN Bypass Control Register

Size: 8 bits

Offset: 0x7805

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	oanbypass	R/W	When oanbypass=1, the value of AN used in the HDCP engine comes from the hdcpreg_an0 to hdcpreg_an7 registers. When oanbypass=0, the value of AN used in the

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Bits	Name	Attr	Description
			HDCP engine comes from the random number input. Value After Reset: 0x0

hdcpreg_an0

Description: HDCP Forced AN Register 0

Size: 8 bits

Offset: 0x7806

Bits	Name	Attr	Description
7:0	hdcpreg_an0	R/W	Contains the value of AN[7:0] Value After Reset: 0x0

hdcpreg_an1

Description: HDCP Forced AN Register 1

Size: 8 bits

Offset: 0x7807

Bits	Name	Attr	Description
7:0	hdcpreg_an1	R/W	Contains the value of AN[15:8] Value After Reset: 0x0

hdcpreg_an2

Description: HDCP forced AN Register 2

Size: 8 bits

Offset: 0x7808

Bits	Name	Attr	Description
7:0	hdcpreg_an2	R/W	Contains the value of AN[23:16] Value After Reset: 0x0

hdcpreg_an3

Description: HDCP Forced AN Register 3

Size: 8 bits

Offset: 0x7809

Bits	Name	Attr	Description
7:0	hdcpreg_an3	R/W	Contains the value of AN[31:24] Value After Reset: 0x0

hdcpreg_an4

Description: HDCP Forced AN Register 4

Size: 8 bits

Offset: 0x780a

Bits	Name	Attr	Description
7:0	hdcpreg_an4	R/W	Contains the value of AN[39:32] Value After Reset: 0x0

hdcpreg_an5

Description: HDCP Forced AN Register 5

Size: 8 bits

Offset: 0x780b

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Bits	Name	Attr	Description
7:0	hdcpreg_an5	R/W	Contains the value of AN[47:40] Value After Reset: 0x0

hdcpreg_an6

Description: HDCP Forced AN Register 6

Size: 8 bits

Offset: 0x780c

Bits	Name	Attr	Description
7:0	hdcpreg_an6	R/W	Contains the value of AN[55:48] Value After Reset: 0x0

hdcpreg_an7

Description: HDCP Forced AN Register 7

Size: 8 bits

Offset: 0x780d

Bits	Name	Attr	Description
7:0	hdcpreg_an7	R/W	Contains the value of BKSV[63:56] Value After Reset: 0x0

hdcpreg_rmlctl

Description: HDCP Encrypted Device Private Keys Control Register

This register is the control register for the software programmable encrypted DPK embedded storage feature. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."

Size: 8 bits

Offset: 0x780e

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	odpk_decrypt_enable	R/W	When set (1'b1), this bit activates the decryption of the Device Private keys. Value After Reset: 0x0

hdcpreg_rmlsts

Description: HDCP Encrypted DPK Status Register

The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."

Size: 8 bits

Offset: 0x780f

Bits	Name	Attr	Description
7			Reserved for future use.
6	idpk_wr_ok_sts	R	When high (1'b1), it indicates that a DPK write is allowed. Value After Reset: 0x0
5:0	idpk_data_index	R	Current Device Private Key being written plus one.

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			Position 0 is occupied by the AKSV. Value After Reset: 0x0

hdcpreg_seed0

Description: HDCP Encrypted DPK Seed Register 0

This register contains a byte of the HDCP Encrypted DPK seed value used to encrypt the Device Private Keys. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."

Size: 8 bits

Offset: 0x7810

Bits	Name	Attr	Description
7:0	hdcpreg_seed0	W	Least significant byte of the decryption seed value (dpk_decrypt_seed[7:0]). Value After Reset: 0x0

hdcpreg_seed1

Description: HDCP Encrypted DPK Seed Register 1

This register contains a byte of the HDCP Encrypted DPK seed value used to encrypt the Device Private Keys. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."

Size: 8 bits

Offset: 0x7811

Bits	Name	Attr	Description
7:0	hdcpreg_seed1	W	Most significant byte of the decryption seed value (dpk_decrypt_seed[15:8]). Value After Reset: 0x0

hdcpreg_dpk0

Description: HDCP Encrypted DPK Data Register 0

This register contains an HDCP DPK byte. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."

Size: 8 bits

Offset: 0x7812

Bits	Name	Attr	Description
7:0	dpk_data	W	Byte of the encrypted DPK value. dpk[7:0] When this byte is written, a strobe signal is generated that triggers the decryption and/or storage of the DPK word on the DPK internal RAM memory. Value After Reset: 0x0

hdcpreg_dpk1

Description: HDCP Encrypted DPK Data Register 1

This register contains an HDCP DPK byte. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter,

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Section 3.2.4, "Configure HDCP."
Size: 8 bits
Offset: 0x7813

Bits	Name	Attr	Description
7:0	dpk_data	W	Byte of the encrypted DPK value. dpk[15:8] Value After Reset: 0x0

hdcpreg_dpk2
Description: HDCP Encrypted DPK Data Register 2
This register contains an HDCP DPK byte. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."
Size: 8 bits
Offset: 0x7814

Bits	Name	Attr	Description
7:0	dpk_data	W	Byte of the encrypted DPK value. dpk[23:16] Value After Reset: 0x0

hdcpreg_dpk3
Description: HDCP Encrypted DPK Data Register 3
This register contains an HDCP DPK byte. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."
Size: 8 bits
Offset: 0x7815

Bits	Name	Attr	Description
7:0	dpk_data	W	Byte of the encrypted DPK value. dpk[31:24] Value After Reset: 0x0

hdcpreg_dpk4
Description: HDCP Encrypted DPK Data Register 4
This register contains an HDCP DPK byte.
Size: 8 bits
Offset: 0x7816

Bits	Name	Attr	Description
7:0	dpk_data	W	Byte of the encrypted DPK value. dpk[39:32] Value After Reset: 0x0

hdcpreg_dpk5
Description: HDCP Encrypted DPK Data Register 5
This register contains an HDCP DPK byte. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."
Size: 8 bits
Offset: 0x7817

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	dpk_data	W	Contains the value of DPK[x][47:40] Value After Reset: 0x0

hdcpreg_dpk6

Description: HDCP Encrypted DPK Data Register 6

This register contains an HDCP DPK byte. The required software configuration sequence is documented in the Cores HDMI Transmitter User Guide in the "Programming" chapter, Section 3.2.4, "Configure HDCP."

Size: 8 bits

Offset: 0x7818

Bits	Name	Attr	Description
7:0	dpk_data	W	Contains the value of DPK[x][55:48] Value After Reset: 0x0

HDCP22 Registers

HDCP22 Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
hdcp22reg_id	0x7900	HDCP 2.2 Identification Register
hdcp22reg_ctrl	0x7904	HDCP 2.2 Control Register
hdcp22reg_ctrl1	0x7905	HDCP 2.2 Control Register 1
hdcp22reg_sts	0x7908	HDCP 2.2 Status Register
hdcp22reg_mask	0x790c	HDCP 2.2 Interrupt Mask Register
hdcp22reg_stat	0x790d	HDCP 2.2 interrupt Sticky Bit Status Register
hdcp22reg_mute	0x790e	HDCP 2.2 Interrupt Mute Vector

hdcp22reg_id

Description: HDCP 2.2 Identification Register

Size: 8 bits

Offset: 0x7900

Bits	Name	Attr	Description
7:3			Reserved for future use.
2	hdcp22_3rdparty	R	Indicates that External HDCP 2.2 interface is present and 3rd party HDCP 2.2 module is connected to this interface. Value After Reset: "(HTX_HDCP22_EXTERNAL_ELLP== 1) ? 1 : 0"
1	hdcp22_externalif	R	Indicates that External HDCP 2.2 interface is present. Value After Reset: "(HTX_HDCP22_EXTERNAL_NONE== 1) ? 1 : 0"
0			Reserved for future use.

hdcp22reg_ctrl

Description: HDCP 2.2 Control Register

Size: 8 bits

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Offset: 0x7904

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	hpd_ovr_val	R/W	HPD Override Value 1'b0: If hpd_ovr_en is 1'b1 the HPD value to the HDCP 2.2 external interface is set to 1'b0. 1'b1: If hpd_ovr_en is 1'b1 the HPD value to the HDCP 2.2 external interface is set to 1'b1. Value After Reset: 0x0
4	hpd_ovr_en	R/W	HPD Override enable 1'b0: The HPD value to the HDCP 2.2 external interface comes from the PHY as in phy_stat0.HPD. 1'b1: The HPD value to the HDCP 2.2 external interface comes from hpd_ovr_val bit field. Value After Reset: 0x0
3			Reserved for future use.
2	hdcp22_ovr_val	R/W	HDCP 2.2 versus 1.4 switch override value 1'b0: The switch is routed to HDCP 1.4 signals when hdcp22_ovr_en is 1'b1 and hdcp22_switch_lock is not set to 1'b1. 1'b1: The switch is routed to HDCP 2.2 signals when hdcp22_ovr_en is 1'b1 and hdcp22_switch_lock is not set to 1'b1. Value After Reset: 0x0
1	hdcp22_ovr_en	R/W	HDCP 2.2 versus 1.4 switch override enable 1'b0: The switch is automatically controlled by the HDCP 2.2 controller using the ist_hdcp2_capable and ist_hdcp2_not_capable status level indications. If the HDCP 2.2 controller indicates ist_hdcp2_capable at 1'b1, the switch is routed to HDCP 2.2 signals. If the HDCP 2.2 controller indicates ist_hdcp2_not_capable at 1'b1, the switch is routed to HDCP 1.4 signals. 1'b1: The HDCP 2.2 ist_hdcp2_capable and ist_hdcp2_not_capable values are ignored, and the direction of the HDCP 2.2 versus 1.4 switch is directly controlled by the hdcp22_ovr_val. Value After Reset: 0x0
0	hdcp22_switch_lck	R/W	HDCP 2.2 switch lock 1'b0: Enables you to change the direction of the HDCP 2.2 versus 1.4 switch by using the hdcp22_ovr_en and hdcp22_ovr_val. 1'b1: You can still write to hdcp22_ovr_en and hdcp22_ovr_val but has no effect over the HDCP 2.2 versus 1.4 switch, that keeps as it was configured by hdcp22_ovr_en and hdcp22_ovr_val at the time the 1'b1 was written to this bit field.

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Bits	Name	Attr	Description
			Once you set the value to 1'b1, you can change the value back to 1'b0 only by issuing a master reset to the Hdmi_tx. Value After Reset: 0x0

hdcp22reg_ctrl1

Description: HDCP 2.2 Control Register 1

Size: 8 bits

Offset: 0x7905

Bits	Name	Attr	Description
7:4	hdcp22_cd_ovr_val	R/W	HDCP color depth override value, which is sent through the HDCP 2.2 external interface when hdcp22reg_ctrl1.hdcp22_cd_ovr_en is set. For reference on the HDMI allowed values consult the HDMI 1.4b specification, Table 6-1. Value After Reset: 0x0
3	hdcp22_cd_ovr_en	R/W	HDCP 2.2 versus 1.4 color depth override enable: 1'b0: The default 1'b0 value indicates that the color depth sent to the external interface is the one configured in the vp_pr_cd.color_depth register field. 1'b1: Although the used color depth for pixel encoding is defined by the field vp_pr_cd.color_depth register, the color depth sent to the external interface is the one defined in register field hdcp22reg_ctrl1.hdcp22_cd_ovr_val. Value After Reset: 0x0
2			Reserved for future use.
1	hdcp22_avmute_ovr_val	R/W	HDCP AV_MUTE override value, which is sent through the HDCP 2.2 external interface when hdcp22reg_ctrl1.hdcp22_avmute_ovr_en is set. Value After Reset: 0x0
0	hdcp22_avmute_ovr_en	R/W	HDCP 2.2 versus 1.4 avmute override enable 1'b0: The default 1'b0 value indicates that the AVMUTE sent to the external interface is the one configured through register fields fc_gcp.set_avmute and fc_gcp.clear_avmute. 1'b1: Although the GCP packet sends the set_avmute or clear_avmute as configured in register fc_gcp, the AV_MUTE sent to the external interface is the one defined in register field hdcp22reg_ctrl1.hdcp22_avmute_ovr_val. Value After Reset: 0x0

hdcp22reg_sts

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Description: HDCP 2.2 Status Register
Size: 8 bits
Offset: 0x7908

Bits	Name	Attr	Description
7:4			Reserved for future use.
3	hdcp_decrypted_sts	R	Value of HDCP 2.2 ist_hdcp_decrypted line. Provided for debug only Value After Reset: 0x0
2	hdcp22_switch_sts	R	HDCP 2.2 HDCP 2.2 versus 1.4 switch status after lock mechanism (hdcp22reg_ctrl.hdcp22_switch_lock, hdcp22reg_ctrl.hdcp22_ovr_en and hdcp22reg_ctrl.hdcp22_ovr_val). 1'b0: HDCP 1.4 selected 1'b1: HDCP 2.2 selected Value After Reset: 0x0
1	hdcp_avmute_sts	R	HDCP 2.2 AVMUTE external interface status. 1'b0: External HDCP used AVMUTE is clear 1'b1: External HDCP AVMUTE is set (audio/video should be muted) Value After Reset: 0x0
0	hdmi_hpd_sts	R	HDCP 2.2 HPD external interface status after lock mechanism (hdcp22reg_ctrl.hdcp22_switch_lock, hdcp22reg_ctrl.hdcp22_ovr_en and hdcp22reg_ctrl.hdcp22_ovr_val). 1'b0: Sink not detected (HPD line clear) 1'b1: Sink detected (HPD line set) Value After Reset: 0x0

hdcp22reg_mask
Description: HDCP 2.2 Interrupt Mask Register
Size: 8 bits
Offset: 0x790c

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	mask_hdcp_decrypted_chg	R/W	Active high interrupt mask to HDCP 2.2 decrypted value change interrupt status Value After Reset: 0x1
4	mask_hdcp_authentication_fail	R/W	Active high interrupt mask to HDCP 2.2 authentication fail interrupt status Value After Reset: 0x1
3	mask_hdcp_authenticated	R/W	Active high interrupt mask to HDCP 2.2 authenticated interrupt status Value After Reset: 0x1
2	mask_hdcp_authentication_lost	R/W	Active high interrupt mask to HDCP 2.2 authentication lost interrupt status Value After Reset: 0x1
1	mask_hdcp2_not_capable	R/W	Active high interrupt mask to HDCP 2.2 not capable

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Bits	Name	Attr	Description
			rise interrupt status Value After Reset: 0x1
0	mask_hdcp2_capable	R/W	Active high interrupt mask to HDCP 2.2 capable rise interrupt status Value After Reset: 0x1

hdcp22reg_stat

Description: HDCP 2.2 interrupt Sticky Bit Status Register

Size: 8 bits

Offset: 0x790d

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	st_hdcp_decrypted_chg	R/W1C	HDCP 2.2 decrypted value change interrupt status sticky bit. Clear by Write 1 to this bit field Value After Reset: 0x0
4	st_hdcp_authentication_fail	R/W1C	HDCP 2.2 authentication fail interrupt status sticky bit. Clear by Write 1 to this bit field Value After Reset: 0x0
3	st_hdcp_authenticated	R/W1C	HDCP 2.2 authenticated interrupt status sticky bit. Clear by Write 1 to this bit field Value After Reset: 0x0
2	st_hdcp_authentication_lost	R/W1C	HDCP 2.2 authentication lost interrupt status sticky bit. Clear by Write 1 to this bit field Value After Reset: 0x0
1	st_hdcp2_not_capable	R/W1C	HDCP 2.2 not capable rise interrupt status sticky bit. Clear by Write 1 to this bit field Value After Reset: 0x0
0	st_hdcp2_capable	R/W1C	HDCP 2.2 capable rise interrupt status sticky bit. Clear by Write 1 to this bit field Value After Reset: 0x0

hdcp22reg_mute

Description: HDCP 2.2 Interrupt Mute Vector

Size: 8 bits

Offset: 0x790e

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	mute_hdcp_decrypted_chg	R/W	Active high interrupt mute to HDCP 2.2 decrypted value change interrupt status Value After Reset: 0x1
4	mute_hdcp_authentication_fail	R/W	Active high interrupt mute to HDCP 2.2 authentication fail interrupt status Value After Reset: 0x1
3	mute_hdcp_authenticated	R/W	Active high interrupt mute to HDCP 2.2 authenticated interrupt status

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Bits	Name	Attr	Description
			Value After Reset: 0x1
2	mute_hdcp_authentication_lost	R/W	Active high interrupt mute to HDCP 2.2 authentication lost interrupt status Value After Reset: 0x1
1	mute_hdcp2_not_capable	R/W	Active high interrupt mute to HDCP 2.2 not capable rise interrupt status Value After Reset: 0x1
0	mute_hdcp2_capable	R/W	Active high interrupt mute to HDCP 2.2 capable rise interrupt status Value After Reset: 0x1

CEC Registers

CEC Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
cec_ctrl	0x7d00	CEC Control Register This register handles the main control of the CEC initiator.
cec_mask	0x7d02	CEC Interrupt Mask Register This read/write register masks/unmasks the interrupt events. When the...
cec_addr_l	0x7d05	CEC Logical Address Register Low This register indicates the logical address(es) allocated to the...
cec_addr_h	0x7d06	CEC Logical Address Register High This register indicates the logical address(es) allocated to...
cec_tx_cnt	0x7d07	CEC TX Frame Size Register This register indicates the size of the frame in bytes (including header...
cec_rx_cnt	0x7d08	CEC RX Frame Size Register This register indicates the size of the frame in bytes (including header...
cec_tx_data[0:15]	0x7d10 + (i * 0x1)	CEC TX Data Register Array Address offset: i = 0 to 15 These registers (8 bits each) are the buffers...
cec_rx_data[0:15]	0x7d20 + (i * 0x1)	CEC RX Data Register Array Address offset: i =0 to 15 These registers (8 bit each) are the buffers...
cec_lock	0x7d30	CEC Buffer Lock Register
cec_wakeupctrl	0x7d31	CEC Wake-up Control Register After receiving a message in the CEC_RX_DATA1 (OPCODE) registers,...

cec_ctrl

Description: CEC Control Register

This register handles the main control of the CEC initiator.

Size: 8 bits

Offset: 0x7d00

Bits	Name	Attr	Description
7:5			Reserved for future use.
4	standby	R/W	1: CEC controller responds with a NACK to all messages and generates a wakeup status for opcode. It only

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Bits	Name	Attr	Description
			responds with a NACK when the EOM is received. This means only the last block of a frame responds with NACK. The follower sends an ACK to the message when there is only one head block pointed to the follower, if the follower is in the standby mode. 0: CEC controller responds the ACK to all messages. Value After Reset: 0x0
3	bc_nack	R/W	1'b1: Set by software to NACK the received broadcast message. This bit holds until software resets. The broadcasts is answered with 1'b0, indicating the follower reject the message. 1'b0: Reset by software to ACK the received broadcast message. Value After Reset: 0x0
2:1	frame_typ	R/W	2'b00: Signal Free Time = 3-bit periods. Previous attempt to send frame is unsuccessful. 2'b01: Signal Free Time = 5-bit periods. New initiator wants to send a frame. 2'b10: Signal Free Time = 7-bit periods. Present initiator wants to send another frame immediately after its previous frame. (specification CEC 9.1). 2'b11: Illegal value. If software writes this value, hardware sets the value to the default 2'b01. Value After Reset: 0x1
0	send	R/W	1'b1: Set by software to trigger CEC sending a frame as an initiator. This bit keeps at 1'b1 while the transmission is going on. 1'b0: Reset to 1'b0 by hardware when the CEC transmission is done (no matter successful or failed). It can also work as an indicator checked by software to see whether the transmission is finished. Value After Reset: 0x0

cec_mask

Description: CEC Interrupt Mask Register

This read/write register masks/unmasks the interrupt events. When the bit is set to 1 (masked), the corresponding event does not trigger an interrupt signal at the system interface. When the bit is reset to 0, the interrupt event is unmasked.

Size: 8 bits

Offset: 0x7d02

Bits	Name	Attr	Description
7			Reserved for future use.
6	wakeup	R/W	Follower wake-up signal mask Value After Reset: 0x0
5	error_flow	R/W	An error is notified by a follower. Abnormal logic data bit error (for follower) Value After Reset: 0x0

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Bits	Name	Attr	Description
4	error_initiator	R/W	An error is detected on a CEC line (for initiator only). Value After Reset: 0x0
3	arb_lost	R/W	The initiator losses the CEC line arbitration to a second initiator. (specification CEC 9) Value After Reset: 0x0
2	nack	R/W	A frame is not acknowledged in a directly addressed message. Or a frame is negatively acknowledged in a broadcast message (for initiator only) Value After Reset: 0x0
1	eom	R/W	EOM is detected so that the received data is ready in the receiver data buffer (for follower only) Value After Reset: 0x0
0	done	R/W	The current transmission is successful (for initiator only) Value After Reset: 0x0

cec_addr_l

Description: CEC Logical Address Register Low

This register indicates the logical address(es) allocated to the CEC device.

This register is written by software when the logical allocation is finished. Bit value 1 means the corresponding logical address is allocated to this device. Bit value 0 means the corresponding logical address is not allocated to this device.

Size: 8 bits

Offset: 0x7d05

Bits	Name	Attr	Description
7	cec_addr_l_7	R/W	Logical address 7 - Tuner 3 Value After Reset: 0x0
6	cec_addr_l_6	R/W	Logical address 6 - Tuner 2 Value After Reset: 0x0
5	cec_addr_l_5	R/W	Logical address 5 - Audio System Value After Reset: 0x0
4	cec_addr_l_4	R/W	Logical address 4 - Playback Device 1 Value After Reset: 0x0
3	cec_addr_l_3	R/W	Logical address 3 - Tuner 1 Value After Reset: 0x0
2	cec_addr_l_2	R/W	Logical address 2 - Recording Device 2 Value After Reset: 0x0
1	cec_addr_l_1	R/W	Logical address 1 - Recording Device 1 Value After Reset: 0x0
0	cec_addr_l_0	R/W	Logical address 0 - Device TV Value After Reset: 0x0

cec_addr_h

Description: CEC Logical Address Register High

This register indicates the logical address(es) allocated to the CEC device.

This register is written by software when the logical allocation is finished. Bit value 1 means the corresponding logical address is allocated to this device. Bit value 0 means the corresponding logical address is not allocated to this device.

Size: 8 bits

Offset: 0x7d06

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Bits	Name	Attr	Description
7	cec_addr_h_7	R/W	Logical address 15 - Unregistered (as initiator address), Broadcast (as destination address) Value After Reset: 0x1
6	cec_addr_h_6	R/W	Logical address 14 - Free use Value After Reset: 0x0
5	cec_addr_h_5	R/W	Logical address 13 - Reserved Value After Reset: 0x0
4	cec_addr_h_4	R/W	Logical address 12 - Reserved Value After Reset: 0x0
3	cec_addr_h_3	R/W	Logical address 11 - Playback Device 3 Value After Reset: 0x0
2	cec_addr_h_2	R/W	Logical address 10 - Tuner 4 Value After Reset: 0x0
1	cec_addr_h_1	R/W	Logical address 9 - Playback Device 3 Value After Reset: 0x0
0	cec_addr_h_0	R/W	Logical address 8 - Playback Device 2 Value After Reset: 0x0

cec_tx_cnt

Description: CEC TX Frame Size Register

This register indicates the size of the frame in bytes (including header and data blocks), which are available in the transmitter data buffer.

Note: When the value is zero, the CEC controller ignores the send command triggered by software. When the transmission is done (no matter success or not), the current value is held until it is overwritten by software.

Size: 8 bits

Offset: 0x7d07

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:0	cec_tx_cnt	R/W	CEC Transmitter Counter register 5'd0: No data needs to be transmitted 5'd1: Frame size is 1 byte 5'd16: Frame size is 16 bytes Value After Reset: 0x0

cec_rx_cnt

Description: CEC RX Frame Size Register

This register indicates the size of the frame in bytes (including header and data blocks), which are available in the receiver data buffer.

Note: Only after the whole receiving process is finished successfully, the counter is refreshed to the value which indicates the total number of data bytes in the Receiver Data Register.

Size: 8 bits

Offset: 0x7d08

Bits	Name	Attr	Description
7:5			Reserved for future use.
4:0	cec_rx_cnt	R	CEC Receiver Counter register: 5'd0: No data received 5'd1: 1-byte data is received 5'd16: 16-byte data is received Value After Reset: 0x0

cec_tx_data[0:15]

Description: CEC TX Data Register Array Address offset: i = 0 to 15

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These registers (8 bits each) are the buffers used for storing the data waiting for transmission (including header and data blocks).
Size: 8 bits
Offset: 0x7d10 + (i * 0x1)

Bits	Name	Attr	Description
7:0	databyte	R/W	Data byte[x], where x is 0 to 15 Value After Reset: 0x0

cec_rx_data[0:15]
Description: CEC RX Data Register Array Address offset: i =0 to 15
These registers (8 bit each) are the buffers used for storing the received data (including header and data blocks).
Size: 8 bits
Offset: 0x7d20 + (i * 0x1)

Bits	Name	Attr	Description
7:0	databyte	R	Data byte[x], where x is 0 to 15 Value After Reset: 0x0

cec_lock
Description: CEC Buffer Lock Register
Size: 8 bits
Offset: 0x7d30

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	locked_buffer	R/W	When a frame is received, this bit would be active. The CEC controller answers to all the messages with NACK until the CPU writes it to '0'. Value After Reset: 0x0

cec_wakeupctrl
Description: CEC Wake-up Control Register
After receiving a message in the CEC_RX_DATA1 (OPCODE) registers, the CEC engine verifies the message opcode[7:0] against one of the previously defined values to generate the wake-up status:
Wakeupstatus is 1 when:
received opcode is 0x04 and opcode0x04en is 1 or
received opcode is 0x0D and opcode0x0Den is 1 or
received opcode is 0x41 and opcode0x41en is 1 or
received opcode is 0x42 and opcode0x42en is 1 or
received opcode is 0x44 and opcode0x44en is 1 or
received opcode is 0x70 and opcode0x70en is 1 or
received opcode is 0x82 and opcode0x82en is 1 or
received opcode is 0x86 and opcode0x86en is 1
Wakeupstatus is 0 when none of the previous conditions are true.
This formula means that the wake-up status (on CEC_STAT[6] register) is only '1' if the opcode[7:0] received is equal to one of the defined values and the corresponding enable bit of that defined value is set to '1'.
Size: 8 bits
Offset: 0x7d31

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7	opcode0x86en	R/W	OPCODE 0x86 wake up enable Value After Reset: 0x1
6	opcode0x82en	R/W	OPCODE 0x82 wake up enable Value After Reset: 0x1
5	opcode0x70en	R/W	OPCODE 0x70 wake up enable Value After Reset: 0x1
4	opcode0x44en	R/W	OPCODE 0x44 wake up enable Value After Reset: 0x1
3	opcode0x42en	R/W	OPCODE 0x42 wake up enable Value After Reset: 0x1
2	opcode0x41en	R/W	OPCODE 0x41 wake up enable Value After Reset: 0x1
1	opcode0x0Den	R/W	OPCODE 0x0D wake up enable Value After Reset: 0x1
0	opcode0x04en	R/W	OPCODE 0x04 wake up enable Value After Reset: 0x1

EDDC Registers

E-DDC Registers. Follow the link for the register to see a detailed description of the register.

Register	Offset	Description
i2cm_slave	0x7e00	I2C DDC Slave address Configuration Register
i2cm_address	0x7e01	I2C DDC Address Configuration Register
i2cm_datao	0x7e02	I2C DDC Data Write Register
i2cm_datai	0x7e03	I2C DDC Data read Register
i2cm_operation	0x7e04	I2C DDC RD/RD_EXT/WR Operation Register Read and write operation request. This register can only...
i2cm_int	0x7e05	I2C DDC Done Interrupt Register This register configures the I2C master interrupts.
i2cm_ctlint	0x7e06	I2C DDC error Interrupt Register This register configures the I2C master arbitration lost and not...
i2cm_div	0x7e07	I2C DDC Speed Control Register This register configures the division relation between master and...
i2cm_segaddr	0x7e08	I2C DDC Segment Address Configuration Register This register configures the segment address for...
i2cm_softrstz	0x7e09	I2C DDC Software Reset Control Register This register resets the I2C master.
i2cm_segptr	0x7e0a	I2C DDC Segment Pointer Register This register configures the segment pointer for extended RD/WR...
i2cm_ss_scl_hcnt_1_addr	0x7e0b	I2C DDC Slow Speed SCL High Level Control Register 1
i2cm_ss_scl_hcnt_0_addr	0x7e0c	I2C DDC Slow Speed SCL High Level Control Register 0
i2cm_ss_scl_lcnt_1_addr	0x7e0d	I2C DDC Slow Speed SCL Low Level Control Register 1
i2cm_ss_scl_lcnt_0_addr	0x7e0e	I2C DDC Slow Speed SCL Low Level Control Register 0
i2cm_fs_scl_hcnt_1_addr	0x7e0f	I2C DDC Fast Speed SCL High Level Control Register 1
i2cm_fs_scl_hcnt_0_addr	0x7e10	I2C DDC Fast Speed SCL High Level Control Register 0
i2cm_fs_scl_lcnt_1_addr	0x7e11	I2C DDC Fast Speed SCL Low Level Control Register 1

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Register	Offset	Description
i2cm_fs_scl_lcnt_0_addr	0x7e12	I2C DDC Fast Speed SCL Low Level Control Register 0
i2cm_sda_hold	0x7e13	I2C DDC SDA Hold Register
i2cm_scdc_read_update	0x7e14	SCDC Control Register This register configures the SCDC update status read through the I2C master...
i2cm_read_buff0	0x7e20	I2C Master Sequential Read Buffer Register 0
i2cm_read_buff1	0x7e21	I2C Master Sequential Read Buffer Register 1

Registers for Address Block: EDDC (Continued)

Register	Offset	Description
i2cm_read_buff2	0x7e22	I2C Master Sequential Read Buffer Register 2
i2cm_read_buff3	0x7e23	I2C Master Sequential Read Buffer Register 3
i2cm_read_buff4	0x7e24	I2C Master Sequential Read Buffer Register 4
i2cm_read_buff5	0x7e25	I2C Master Sequential Read Buffer Register 5
i2cm_read_buff6	0x7e26	I2C Master Sequential Read Buffer Register 6
i2cm_read_buff7	0x7e27	I2C Master Sequential Read Buffer Register 7
i2cm_scdc_update0	0x7e30	I2C SCDC Read Update Register 0
i2cm_scdc_update1	0x7e31	I2C SCDC Read Update Register 1

i2cm_slave

Description: I2C DDC Slave address Configuration Register

Size: 8 bits

Offset: 0x7e00

Bits	Name	Attr	Description
7			Reserved for future use.
6:0	slaveaddr	R/W	Slave address to be sent during read and write normal operations. Value After Reset: 0x0

i2cm_address

Description: I2C DDC Address Configuration Register

Size: 8 bits

Offset: 0x7e01

Bits	Name	Attr	Description
7:0	address	R/W	Register address for read and write operations Value After Reset: 0x0

i2cm_datao

Description: I2C DDC Data Write Register

Size: 8 bits

Offset: 0x7e02

Bits	Name	Attr	Description
7:0	datao	R/W	Data to be written on register pointed by address[7:0]. Value After Reset: 0x0

i2cm_datai

Description: I2C DDC Data read Register

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Size: 8 bits
Offset: 0x7e03

Bits	Name	Attr	Description
7:0	datai	R	Data read from register pointed by address[7:0]. Value After Reset: 0x0

i2cm_operation
Description: I2C DDC RD/RD_EXT/WR Operation Register
Read and write operation request. This register can only be written; reading this register always results in 00h. Writing 1'b1 simultaneously to rd, rd_ext and wr requests is considered as a read (rd) request.
Size: 8 bits
Offset: 0x7e04

Bits	Name	Attr	Description
7:6			Reserved for future use.
5	busclear	W	Bus clear operation request. Value After Reset: 0x0
4	wr	W	Single byte write operation request. Value After Reset: 0x0
3	rd8_ext	W	Extended sequential read operation request. Eight bytes are read starting at the address defined in register field i2cm_address.address and stored in registers i2cm_read_buffx. Value After Reset: 0x0
2	rd8	W	Sequential read operation request. Eight bytes are read starting at the address defined in the i2cm_address.address register field and stored in the i2cm_read_buffx registers. Value After Reset: 0x0
1	rd_ext	W	After writing 1'b1 to rd_ext bit a extended data read operation is started (E-DDC read operation). Value After Reset: 0x0
0	rd	W	Single byte read operation request Value After Reset: 0x0

i2cm_int
Description: I2C DDC Done Interrupt Register This register configures the I2C master interrupts.
Size: 8 bits
Offset: 0x7e05

Bits	Name	Attr	Description
7			Reserved for future use.
6	read_req_mask	R/W	Read request interruption mask signal. Value After Reset: 0x1
5:3			Reserved for future use.
2	done_mask	R/W	Done interrupt mask signal. Value After Reset: 0x0
1:0			Reserved for future use.

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i2cm_ctlint

Description: I2C DDC error Interrupt Register

This register configures the I2C master arbitration lost and not acknowledge error interrupts.

Size: 8 bits

Offset: 0x7e06

Bits	Name	Attr	Description
7			Reserved for future use.
6	nack_mask	R/W	Not acknowledge error interrupt mask signal. Value After Reset: 0x0
5:3			Reserved for future use.
2	arbitration_mask	R/W	Arbitration error interrupt mask signal. Value After Reset: 0x0
1:0			Reserved for future use.

i2cm_div

Description: I2C DDC Speed Control Register

This register configures the division relation between master and scl clock.

Size: 8 bits

Offset: 0x7e07

Bits	Name	Attr	Description
7:4			Reserved for future use.
3	fast_std_mode	R/W	Sets the I2C Master to work in Fast Mode or Standard Mode: 1: Fast Mode 0: Standard Mode Value After Reset: 0x1
2:0	spare	R/W	Reserved as "spare" bit with no associated functionality. Value After Reset: 0x3

i2cm_segaddr

Description: I2C DDC Segment Address Configuration Register

This register configures the segment address for extended R/W destination and is used for EDID reading operations, particularly for the Extended Data Read Operation for Enhanced DDC.

Size: 8 bits

Offset: 0x7e08

Bits	Name	Attr	Description
7			Reserved for future use.
6:0	seg_addr	R/W	I2C DDC Segment Address Configuration Register Value After Reset: 0x0

i2cm_softrstz

Description: I2C DDC Software Reset Control Register This register resets the I2C master.

Size: 8 bits

Offset: 0x7e09

Bits	Name	Attr	Description
7:1			Reserved for future use.
0	i2c_softrstz	R/W	I2C Master Software Reset. Active by writing a zero and

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Bits	Name	Attr	Description
			auto cleared to one in the following cycle. Value After Reset: 0x1

i2cm_segptr

Description: I2C DDC Segment Pointer Register

This register configures the segment pointer for extended RD/WR request.

Size: 8 bits

Offset: 0x7e0a

Bits	Name	Attr	Description
7:0	segptr	R/W	I2C DDC Segment Pointer Register Value After Reset: 0x0

i2cm_ss_scl_hcnt_1_addr

Description: I2C DDC Slow Speed SCL High Level Control Register 1

Size: 8 bits

Offset: 0x7e0b

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_hcnt1	R/W	I2C DDC Slow Speed SCL High Level Control Register 1 Value After Reset: 0x0

i2cm_ss_scl_hcnt_0_addr

Description: I2C DDC Slow Speed SCL High Level Control Register 0

Size: 8 bits

Offset: 0x7e0c

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_hcnt0	R/W	I2C DDC Slow Speed SCL High Level Control Register 0 Value After Reset: 0x6c

i2cm_ss_scl_lcnt_1_addr

Description: I2C DDC Slow Speed SCL Low Level Control Register 1

Size: 8 bits

Offset: 0x7e0d

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_lcnt1	R/W	I2C DDC Slow Speed SCL Low Level Control Register 1 Value After Reset: 0x0

i2cm_ss_scl_lcnt_0_addr

Description: I2C DDC Slow Speed SCL Low Level Control Register 0

Size: 8 bits

Offset: 0x7e0e

Bits	Name	Attr	Description
7:0	i2cmp_ss_scl_lcnt0	R/W	I2C DDC Slow Speed SCL Low Level Control Register 0 Value After Reset: 0x7f

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i2cm_fs_scl_hcnt_1_addr

Description: I2C DDC Fast Speed SCL High Level Control Register 1

Size: 8 bits

Offset: 0x7e0f

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_hcnt1	R/W	I2C DDC Fast Speed SCL High Level Control Register 1 Value After Reset: 0x0

i2cm_fs_scl_hcnt_0_addr

Description: I2C DDC Fast Speed SCL High Level Control Register 0

Size: 8 bits

Offset: 0x7e10

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_hcnt0	R/W	I2C DDC Fast Speed SCL High Level Control Register 0 Value After Reset: 0x11

i2cm_fs_scl_lcnt_1_addr

Description: I2C DDC Fast Speed SCL Low Level Control Register 1

Size: 8 bits

Offset: 0x7e11

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_lcnt1	R/W	I2C DDC Fast Speed SCL Low Level Control Register 1 Value After Reset: 0x0

i2cm_fs_scl_lcnt_0_addr

Description: I2C DDC Fast Speed SCL Low Level Control Register 0

Size: 8 bits

Offset: 0x7e12

Bits	Name	Attr	Description
7:0	i2cmp_fs_scl_lcnt0	R/W	I2C DDC Fast Speed SCL Low Level Control Register 0 Value After Reset: 0x24

i2cm_sda_hold

Description: I2C DDC SDA Hold Register

Size: 8 bits

Offset: 0x7e13

Bits	Name	Attr	Description
7:0	osda_hold	R/W	Defines the number of SFR clock cycles to meet tHD;DAT (300 ns) osda_hold = round_to_high_integer (300 ns / (1 / isfrclk_frequency)) Value After Reset: 0x9

i2cm_scdc_read_update

Description: SCDC Control Register

This register configures the SCDC update status read through the I2C master interface.

Size: 8 bits

Offset: 0x7e14

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Bits	Name	Attr	Description
7:6			Reserved for future use.
5	updtrd_vsyncpoll_en	R/W	Update read polling enabled. When active (1'b1), an SCDC Update Read is performed at the fall of the active edge of the vertical sync pulse. Value After Reset: 0x0
4	read_request_en	R/W	Read request enabled. When active (1'b1) an SCDC Update Read shall be performed whenever a SCDC read request is detected. Value After Reset: 0x0
3:1			Reserved for future use.
0	read_update	W	When set to 1'b1, a SCDC Update Read is performed and the read data loaded into registers i2cm_scdc_update0 and i2cm_scdc_update1. Value After Reset: 0x0

i2cm_read_buff0
Description: I2C Master Sequential Read Buffer Register 0
Size: 8 bits
Offset: 0x7e20

Bits	Name	Attr	Description
7:0	i2cm_read_buff0	R	Byte 0 of a I2C read buffer sequential read (from address i2cm_address) Value After Reset: 0x0

i2cm_read_buff1
Description: I2C Master Sequential Read Buffer Register 1
Size: 8 bits
Offset: 0x7e21

Bits	Name	Attr	Description
7:0	i2cm_read_buff1	R	Byte 1 of a I2C read buffer sequential read (from address i2cm_address+1) Value After Reset: 0x0

i2cm_read_buff2
Description: I2C Master Sequential Read Buffer Register 2
Size: 8 bits
Offset: 0x7e22

Bits	Name	Attr	Description
7:0	i2cm_read_buff2	R	Byte 2 of a I2C read buffer sequential read (from address i2cm_address+2) Value After Reset: 0x0

i2cm_read_buff3
Description: I2C Master Sequential Read Buffer Register 3
Size: 8 bits
Offset: 0x7e23

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Bits	Name	Attr	Description
7:0	i2cm_read_buff3	R	Byte 3 of a I2C read buffer sequential read (from address i2cm_address+3) Value After Reset: 0x0

i2cm_read_buff4

Description: I2C Master Sequential Read Buffer Register 4

Size: 8 bits

Offset: 0x7e24

Bits	Name	Attr	Description
7:0	i2cm_read_buff4	R	Byte 4 of a I2C read buffer sequential read (from address i2cm_address+4) Value After Reset: 0x0

i2cm_read_buff5

Description: I2C Master Sequential Read Buffer Register 5

Size: 8 bits

Offset: 0x7e25

Bits	Name	Attr	Description
7:0	i2cm_read_buff5	R	Byte 5 of a I2C read buffer sequential read (from address i2cm_address+5) Value After Reset: 0x0

i2cm_read_buff6

Description: I2C Master Sequential Read Buffer Register 6

Size: 8 bits

Offset: 0x7e26

Bits	Name	Attr	Description
7:0	i2cm_read_buff6	R	Byte 6 of a I2C read buffer sequential read (from address i2cm_address+6) Value After Reset: 0x0

i2cm_read_buff7

Description: I2C Master Sequential Read Buffer Register 7

Size: 8 bits

Offset: 0x7e27

Bits	Name	Attr	Description
7:0	i2cm_read_buff7	R	Byte 7 of a I2C read buffer sequential read (from address i2cm_address+7) Value After Reset: 0x0

i2cm_scdc_update0

Description: I2C SCDC Read Update Register 0

Size: 8 bits

Offset: 0x7e30

Bits	Name	Attr	Description

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Bits	Name	Attr	Description
7:0	i2cm_scdc_update0	R	Byte 0 of a SCDC I2C update sequential read Value After Reset: 0x0

i2cm_scdc_update1

Description: I2C SCDC Read Update Register 1

Size: 8 bits

Offset: 0x7e31

Bits	Name	Attr	Description
7:0	i2cm_scdc_update1	R	Byte 1 of a SCDC I2C update sequential read Value After Reset: 0x0

16.9 Application Notes

This chapter describes how to bring up HDMI transmitter in your system. As shown few examples below, these introduce the basically HDMI transmitter application, likes, the Hot Plug Detect, EDID read back, multiple audio format input and different video resolution displaying.

16.9.1 Initial Operation

The default HDMI transmitter is configured to 24bit RGB 1080P resolution video with 8 channel 48K sample I2S format audio input. It is easily for customer to turn on HDMI transmitter without doing more complex operation. Just do the step, reset the HDMI TX.

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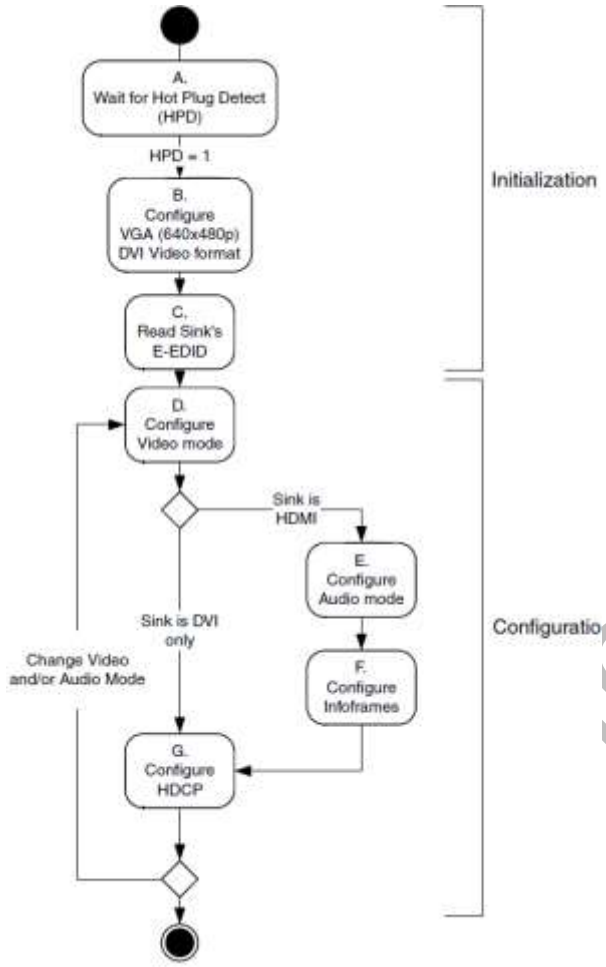

Fig. 41-5 HDMI programming sequence

16.9.2 Step A: Hot Plug Detection

The Hot Plug Detect information notifies the HDMI TX controller when the cable is plugged into a Sink (Receiver) device and when the device is ready to receive video content.

The Hot-Plug Detection indication is a +5V signal on the HDMI cable. The HDMI TX PHY performs level-shifting to a low-level digital signal.

For this step, registers are used on the HDMI TX PHY.

To check the Hot Plug Detect status, the following steps have to be followed:

1. Power-on the HDMI TX PHY HPD Detector. Write 1'b1 in the phy_conf0.enhpdrxsense bit field register.

2. Check the HPD status bit.

- Read the phy_stat0.HPD bit field register.
  - If HPD = 0, the Hot Plug signal is low (no Sink (Receiver) detected).
  - If HPD = 1, the Hot Plug signal is high (Sink (Receiver) detected).

Note: Some receivers may turn off the Hot-Plug Detect indication when the HDMI input is not selected, even when the cable is plugged in.

You can control this function by using the interrupt signal and proper registers from the HDMI transmitter with few operations.

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16.9.3 Step B: Configure Video Mode

Once a receiver is detected, it is necessary to start sending video content in DVI mode. Because some receivers need to receive a video signal (TMDS clock more precisely), it is important to clock all the digital logic and be able to reply on an E-EDID read access (Step C). Following flow is a configuration example of VGA (640x480p).

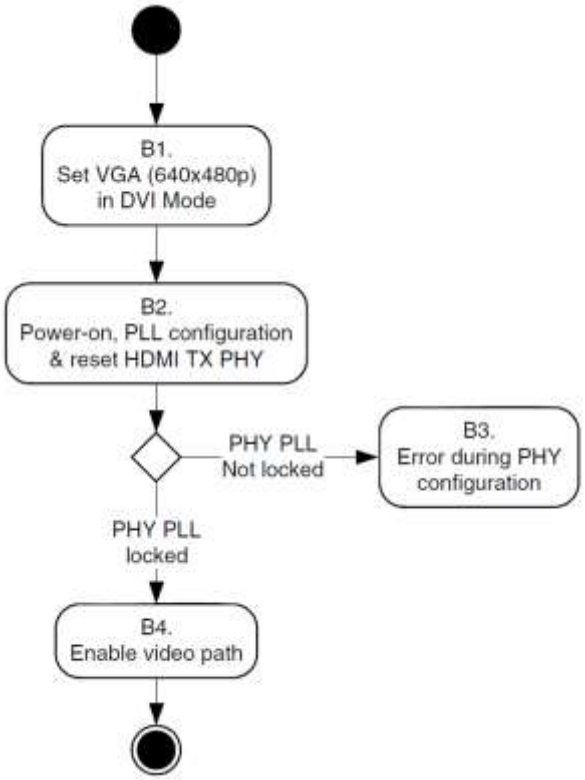

Fig. 41-6 Configure Video Mode

The VGA (640x480p) mode is mandatory for all receivers and transmitters, because it requires the lowest data rate possible on the HDMI specification.

1. Set VGA (640x480p) in DVI mode.

- To select the Video Mapping input RGB444, 8-bits per color components (24-bits RGB), write "1" in the tx_invid0.video_mapping register.
- To configure VGA timing information (CEA mode 1):
  - Write "0" in the fc_invidconf.vsync_in_polarity register.
  - Write "0" in the fc_invidconf.hsync_in_polarity register.
  - Write "1" in the fc_invidconf.de_in_polarity register.
  - Write "0" in the fc_invidconf.r_v_blank_in_osc register.
  - Write "0" in the fc_invidconf.in_I_P register.
  - 640 H active pixels: 0x280
    - Write "2" in the fc_inhactiv1.H_in_activ register.
    - Write "0x80" in the fc_inhactiv0.H_in_activ register.
  - 480 V active pixels: 0x1E0

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- Write "1" in the fc_invactiv1.V_in_activ register.

- Write "0xE0" in the fc_invactiv0.V_in_activ register.

■ 160 H blanking pixels: 0xA0

Write "0xA0" in thefc_inhblank0.H_in_blank.

■ 45 V blanking pixels: 0x2D

Write "0x2D" in the fc_invblank.V_in_blank register.

■ 16 Sync offset: 0x10

Write "0x10" in the fc_hsyncindelay0.H_in_delay register.

■ 10 Vsync offset: 0x0A

Write "0x0A" in the fc_vsyncindelay0.V_in_delay register.

■ 96 HSync pulse width: 0x60

Write "0x60" in the fc_hsyncinwidth0.H_in_width register.

■ 2 VSync pulse width: 0x02

Write "0x02" in the fc_vsyncinwidth0.V_in_width register.

- To select the HDMI mode:

  Write "1" in the fc_invidconf.DVI_modez register.

2. Power-on, configure the PLL, and reset the HDMI TX PHY.

The HDMI TX PHY has an internal PLL that generates TMDS clock from Pixel clock input. The power-on sequence and PLL configuration is part of the HDMI TX PHY documentation.

To configure HDMI TX PHY, you are generally required to:

a. Place the PHY in reset by writing 0x01 in the mc_phyrstz register.

b. Write the desired color depth and the pixel repetition in the vp_pr_cd register.

c. After a PHY-dependent time, it is required to lift the reset by writing 0x00 to the mc_phyrstz register.

d. Clear the pddq, txpwron configuration bits, define the data enable polarity and clear the interface control selection by writing these values in the phy_cfg0 register.

e. Set the PHY slave address by writing 0x69 in the phy_i2cm_slave register.

f. You are required to look up the configuration for your intended video mode and write those values on the PHY I2C interface. The baseline flow to write to the PHY through the I2C interface is:

- Write register address in the phy_i2cm_address register.
- Write data in the phy_i2cm_datao_1 (MSB, [15:8]) and phy_i2cm_datao_0 (LSB, [7:0]) registers.
- Initialize the write operation by writing 8'h10 in the phy_i2cm_operation register.
- Wait for a done interruption from the I2C master.

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g. After all of the required PHY I2C registers have been configured, you now need to set the txpwron bit in the PHY_CONF0 register, leaving the remaining bits in the state defined previously.

3. Handle errors during PHY configuration.

If the PHY is not correctly configured, the PLL does not lock and no activity can happen on the HDMI output.

4. Enable the video path.

a. Set default parameters in the HDMI TX Controller, Control period duration: 12: 0x0C.

Write "0x0C" in the fc_ctrldur.ctrlperiodduration bit field register.

b. Set default parameters in the HDMI TX Controller, Extended Control period duration: 32: 0x20.

Write "0x20" in the fc_exctrldur.exctrlperiodduration bit field register.

c. Set default parameters in the HDMI TX Controller, Max spacing between extended Control period duration: 1: 0x1.

Write "0x01" in the fc_exctrlspac.exctrlperiodspacing bit field register.

d. Set default parameters in the HDMI TX Controller, Preamble filters to fill TMDS data channels.

- Write "0x0B" in the fc_ch0pream.ch0_preamble_filter bit field register.
- Write "0x16" in the fc_ch1pream.ch1_preamble_filter bit field register.
- Write "0x21" in the fc_ch2pream.ch2_preamble_filter bit field register.

e. Enable pixel clock data path:

Write "0" in the mc_clkdis.pixelclk_disable bit field register.

f. Enable TMDS clock data path:

i. Write "0" in the mc_clkdis.tmdsclk_disable bit field register.

ii. Re-Write the VSync pulse width in the fc_vsyncinwidth0.V_in_width bit field register.

After these steps, it is expected to observe a video picture in VGA mode on the receiver side.

16.9.4 Step C: Reading E-EDID

The E-EDID is a memory present on the receiver side. It contains the video and audio capabilities of the receiver. It is important to read this information and parse it in order to configure the transmitter system in accordance to what the receiver supports. The E-EDID is read using the E-DDC channel present on the HDMI cable.

The E-DDC channel protocol is based on I2C, with segment pointer addressing extension (such as, Extended Read). The hdmi tx controller has an I2C Master controller for this purpose.

1. To perform a "normal" read operation

- Set I2C slave address.

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Write i2cm_slave.slaveaddr[6:0] bit field register.

- Set I2C register address.

  Write i2cm_address.address[7:0] bit field register.

- Activate Read operation.

  Write "1" in the i2cm_operation.rd bit field register.

- Wait for interruption.

  Wait for the ii2cmasterdone interrupt in the ih_i2cm_stat0 register.

- Read data result.

  Read data in the i2cm_datai.datai[7:0] bit field.

2. To perform an E-DDC extended read operation

- Set I2C slave address.

  Write i2cm_slave.slaveaddr[6:0] bit field register.

- Set I2C segment address.

  Write the i2cm_segaddr.seg_addr bit field register.

- Set I2C segment pointer.

  Write i2cm_segptr.segptr bit field register.

- Activate Read operation.

  Write "1"in the i2cm_operation.rd_ext bit field register.

- Wait for interruption.

  Wait for the ii2cmasterdone interrupt in the ih_i2cm_stat0 register.

- Read data result.

  Read data in the i2cm_datai.datai[7:0] bit field register.

16.9.5 Step D: Configure Video Mode

The video mode selected has to match what is supported by the source of the video (Transmitter side) and the sink of the video (receiver side). The video capabilities of the receiver are extracted from the E-EDID information (refer to step C).

To configure any video mode, a similar procedure in step B has to be followed.

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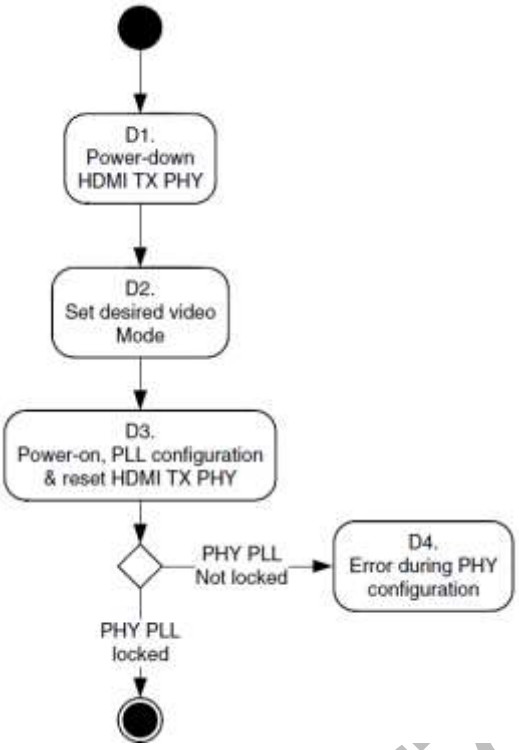

Fig. 41-7 Video Mode Configuration

Before any change is made on the video modes, is it recommended you power-down the HDMI TX PHY to avoid any unexpected behavior on the receiver side.

1. Power-down HDMI TX PHY Detector

- Place the PHY in reset by writing 0x01 in the mc_phyrstz register.
- Then proceed to set SVSRET_MODEZ to 1'b0, PDDQ to 1'b1 and TX_PWRON to 1b0, by writing 2'h12 in the phy_conf0 register.

2. Set desired video mode.

- To select the Video Mapping input mode (RGB444, YCC444, YCC422). Write the video code in the tx_invid0.video_mapping bit field register.
- Set video timing information configuration:
  - Write the fc_invidconf.vsync_in_polarity register.
  - Write the fc_invidconf.hsync_in_polarity register.
  - Write the fc_invidconf.de_in_polarity register.
  - Write the fc_invidconf.r_v_blank_in_osc register.
  - Write the fc_invidconf.in_I_P register.
  - H active pixels

    - Write the fc_inhactiv1.H_in_activ register.

    - Write the fc_inhactiv0.H_in_activ register.

  - V active pixels

    If the desired video mode is 3D, the fc_invact_2d_1, fc_invact_2d_0, and all registers in the following table must be written.

Table 41-6 HDMI 3D structure table

3D STRUCTURE	fc_actspc_hdlr_cfg	
	fc_actspc_hdlr_tgl	fc_actspc_hdlr_en
Frame Packing for interlaced	1	1

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format		
Frame Packing for progressive format	0	1
Field alternative	0	0
Line Alternative	0	0
Side-by-Side(FULL)	0	0
L+depth	0	1
L+depth+graphics+graphics-depth	0	1
Top-and-Bottom	0	0
Reserved for Future Use	0	0
Side-by-Side(FULL)	0	0
Reserved for Future Use	0	0
Not in use	0	0

- Write the fc_invact_2D_1 register with the 2D Vertical video active[11:8].

- Write the fc_invact_2D_0 register with the 2D Vertical video active[7:0]

The following registers must always be written with the full V active of the desired video

regardless of the type of the video mode.

- Write the fc_invactiv1.V_in_activ register.

- Write the fc_invactiv0.V_in_activ register.

■ H blanking pixels

Write the fc_inhblank0.H_in_blank register.

■ V blanking pixels

Write the fc_invblank.V_in_blank register.

■ HSync offset

Write the fc_hsyncindelay0.H_in_delay register.

■ VSync offset

Write the fc_vsyncindelay0.V_in_delay register.

■ HSync pulse width

Write the fc_hsyncinwidth0.H_in_width register.

■ VSync pulse width

Write the fc_vsyncinwidth0.V_in_width register.

- Select DVI or HDMI mode:

  Write "1" for HDMI in the fc_invidconf.DVI_modez bit field register.

3. Power-on, configure the PLL, and reset the HDMI TX PHY (same as Step B2).

The HDMI TX PHY has an internal PLL that generates TMDS clock from Pixel clock input. All the power-on sequence and PLL configuration is part of the HDMI TX PHY documentation.

The HDMI TX Controller provides all registers necessary for the PHY sequence implementation,

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which are presented in the "HDMI PHY Registers".

- Place the PHY in reset by writing 0x01 in the mc_phyrstz register.
- Write in the desired color depth and the pixel repetition in the vp_pr_cd register.
- After a PHY-dependent time, it is required to lift the reset by writing 0x00 to the mc_phyrstz register.
- Clear the pddq, txpwron configuration bits, define the data enable polarity and clear the interface control selection by writing these values in the phy_conf0 register.
- Set the PHY slave address by writing 0x69 to the phy_i2cm_slave register.
- Look up the configuration for your intended video mode and write those values on the PHY I2C interface. The baseline flow to write in the PHY through the I2C interface is

  ■ Write the register address in the phy_i2cm_address register.

  ■ Write data into phy_i2cm_datao_1 (MSB, [7:0]) and phy_i2cm_datao_0 (LSB, [7:0]) registers.

  ■ Initialize the write operation by writing 8'h10 in the phy_i2cm_operation register.

  ■ Wait for a done interrupt from the I2C master.

- After all of the required PHY I2C registers have been configured, you now need to set the phy_conf0.txpwron register, leaving the remaining bits in the state defined previously.

16.9.6 Step E: Audio input configuration

HDMI transmitter audio support either SPDIF or four channel I2S input. SPDIF input supports audio sampling rates from 32 to 192 KHz. The I2S input supports from 2-channel to 8-channel audio up to 192 KHz. The default audio format is I2S input with 8 channels. The audio sample rate is 48K.

- Configure Audio Input Format with I2S Steps:
  ■ Select I2S input.
  Write "1" in the aud_conf0.i2s_select bit field register.
  ■ Enable I2S inputs:
  Write "1" in the aud_conf0.i2s_in_en[3:0] bit field register.
  ■ Set I2S Mode [Standard | Right-justified | Left-justified | Burst1 | Burst2]:
  Write the aud_conf1.i2s_mode[2:0] bit field register.
  ■ Set I2S data width [16 bits up to 24 bits]:
  Write the aud_conf1.i2s_width[4:0] bit field.
- Configure Audio Input Format with SPDIF Steps:
  ■ Select SPDIF input.
  Write "0" in the aud_conf0.i2s_select bit field register.
  ■ Set S/PDIF Linear-PCM or Non-Linear PCM audio samples:
  Write the aud_spdif1.setnlpcm bit field register.
  ■ Set SPDIF data width [16 bits up to 24 bits]:
  Write the aud_spdif1.spdif_width[4:0] bit field.
- Configure Audio Parameters Steps:
  ■ Set Audio input frequency clock FS ratio factor [128 Fs | 256 Fs |512 Fs]:
  Write the aud_inputclkfs.lfsfactor bit field register.
  ■ Set Audio fixed N factor for Audio Clock Regeneration. This factor depends on the audio sampling rate and video mode.
  Write the aud_n1.audN, aud_n2.audN, and aud_n3.audN bit field registers.

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■ Set Audio CTS factor for Audio Clock Regeneration. This factor can be generated automatically or manually.

For Automatic CTS generation

Write "0" on the bit field "CTS_manual", Register 0x3205: AUD_CTS3

For Manual CTS setting

Write "1" in the aud_cts3.CTS_manual register bit field.

Write the aud_cts1.audCTS, aud_cts2.audCTS, aud_cts3.audCTS bit field registers.

■ Enable Audio sampler block:

Write "0" in the mc_clkdis.audclk_disable bit field register.

16.9.7 Step F: Configure Infoframes

This step is only relevant when the HDMI TX controller is configured in HDMI mode (refer to "Step D: Configure Video Mode"). In DVI mode, no Infoframes are transmitted. The configuration of the Infoframes is essential to correctly inform the HDMI Receiver about the content of the video and audio format been transmitted. All the detailed information is provided in the HDMI 1.4b Specification.

16.9.8 Step G: Configure HDCP 1.4

Following flow is an initialization of HDCP 1.4. To detect if the HDMI TX Controller is HDCP capable:

1. Read the product_id1.product_id1 bit field register:

- If product_id1 = 01, HDCP is not present.
- If product_id1 = C1, HDCP is present.

2. Select HDMI mode:

- Write "1" for HDMI in the a_hdcpcfg0.hdmidvi bit field register.

3. Set the Data enable, Hsync and VSync polarity:

Write the dataenpol, vsyncpol, and hsyncpol bit fields of the a_vidpolcfg register.

4. Set encryption on:

- Write "64 (0x40)" for "OesswindowSize" in the a_oesswcfg register.
- Write "0" for "no HDCP bypass" a_hdcpcfg0.BypassEncryption bit field register.
- Write "0" for "HDCP enable" in the a_hdcpcfg1.encryptiondisable bit field register.

5. Reset HDCP engine:

- Write "0" in the a_hdcpcfg1.swreset bit field register.

6. Configure Device Private Keys, which is illustrated in the flow chart Figure 3-6

7. Enable the encryption:

- Write "1" in the a_hdcpcfg0.rxdetect bit field register.

8. When connected to a repeater the SHA-1 calculation is needed to be done by software, which are indicated in the following steps

a. Wait for an interrupt to be triggered (a_apiintstat.KSVSha1calcint).

b. Request access to KSV memory through setting a_ksvmemctrl.KSVMEMrequest to 1'b1

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and pool a_ksvmemctrl.KSVMEMaccess until this value is 1'b1 (access granted).

c. Read VH', M0, Bstatus, and the KSV FIFO.

The data is stored in the revocation memory.

d. Calculate the SHA-1 checksum (VH) over M0, Bstatus, and the KSV FIFO.

e. If the calculated VH equals the VH‘, set a_ksvmemctrl.SHA1fail to 0 and set a_ksvmemctrl.KSVCTRLupd to 1. If the calculated VH is different from VH‘ then set a_ksvmemctrl.SHA1fail to 1 and set a_ksvmemctrl.KSVCTRLupd to 1, forcing the controller to re-authenticate from the beginning.

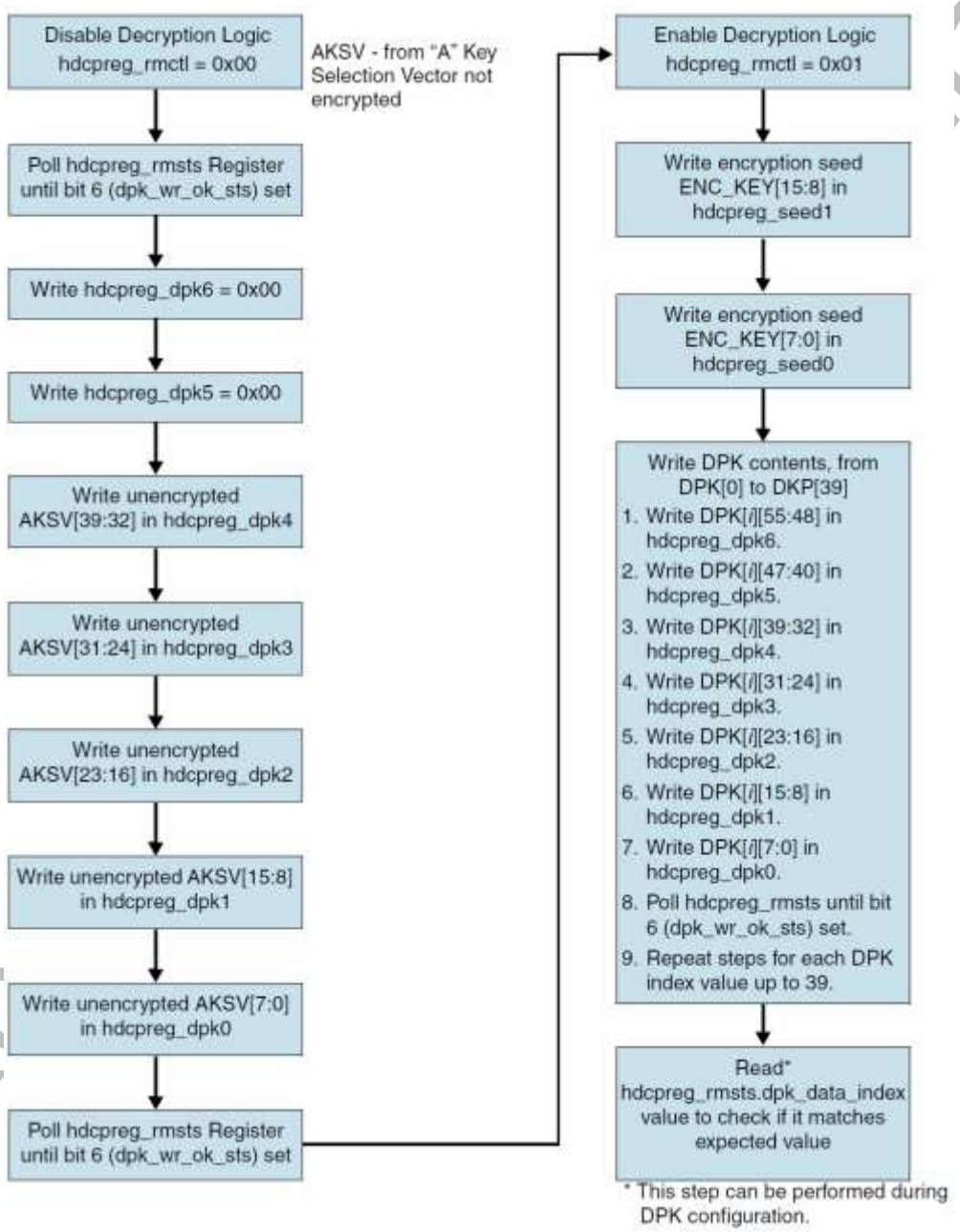

Fig. 41-8 Configuring Device Protection Key Process

16.9.9 CEC OPERATION

The CEC line is used for high-level user control of HDMI-connected devices.The HDMI TX contain CEC TX operations and CEC RX operations.

You can control this function by using the interrupt signal and proper registers from the HDMI transmitter with few operations. The register offset is from 0x7D00.

- Configure The CEC Step:
  - ■ Write the CEC logical address to cec_addr_l,cec_addr_h register
  - ■ Write the size of the frame in bytes which are available in the transmitter data buffer to cec_tx_cnt register
  - ■ Write the desired CEC data(including header and data blocks) to cec_tx_data0 to cec_tx_data15
  - ■ Write 1 to cec_ctrl.send register, to start the cec transmit.

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Chapter 42 HDMI PHY

17.1 Overview

The HDMI Tx PHY is the physical layer of an HDMI digital transmitter (source), capable of encoding and transmitting high-speed data streams carrying RGB video, audio, and control information.
The HDMI TX PHY is compliant with HDMI2.0 specification and optimized up to 3.72Gbps per TMDS link High-Definition applications which supporting 3D display and up to 4Kx2K@60/50Hz Ultra High Definition(UHD) resolution at 10-bit YUV420 video input mode.
The HDMI TX PHY contains programmable circuit with pre-emphasis, slew rate,output voltage swing and resistance values to drive transmit data with optimal signal quality and distance overd the standard HDMI cable to a stand HDMI display device.The TX PHY contains all HDMI componets including all I/O library,high performace wide range PLL, the data symbol conversion/synchronization unit.
Below is the TX PHY features:

- HDMI 2.0/1/4a/b/1.3/1.2/1.1,DVI 1.0 standard compliant transmitter PHY
- Supports data rate from 25M, up to 3.72Gbps over a signle TMDS channel
- Supports TMDS TX with programmable output swing,resistance values and pre-emphasis
- Low power consumption with less than 100mW for PHY an UHD 4K dispalying mode
- Library deliverd to support all major EDA toos and detailed guide to integrate into pad ring/BGA bumps
- 1.8V high speed I/O and 1.0V core power supply
- APB slave interface for internal registers access
- GF28nm SLP process ready with die size less than 0.6mm^2 before shrink for PHY(including IO PADs)

17.2 Architecture

HDMI 2.0 TX PHY is designed to be fully compatible with the HDMI 2.0 TPI source characteristics. It is optimized for high speed (up to 3.72Gbps per TMDS link) HD applications with robust timing, power consumption and small silicon area.
The transmitter PHY integrates low jitter PLL, bandgap reference and 4 TMDS channels to deliver multimedia data. Specialized transmitter driver circuit with tunable pre-emphasis and distance over the standard HDMI cable. Basically, the output driver switches the current supplied by the current source and drives the differential pair line.

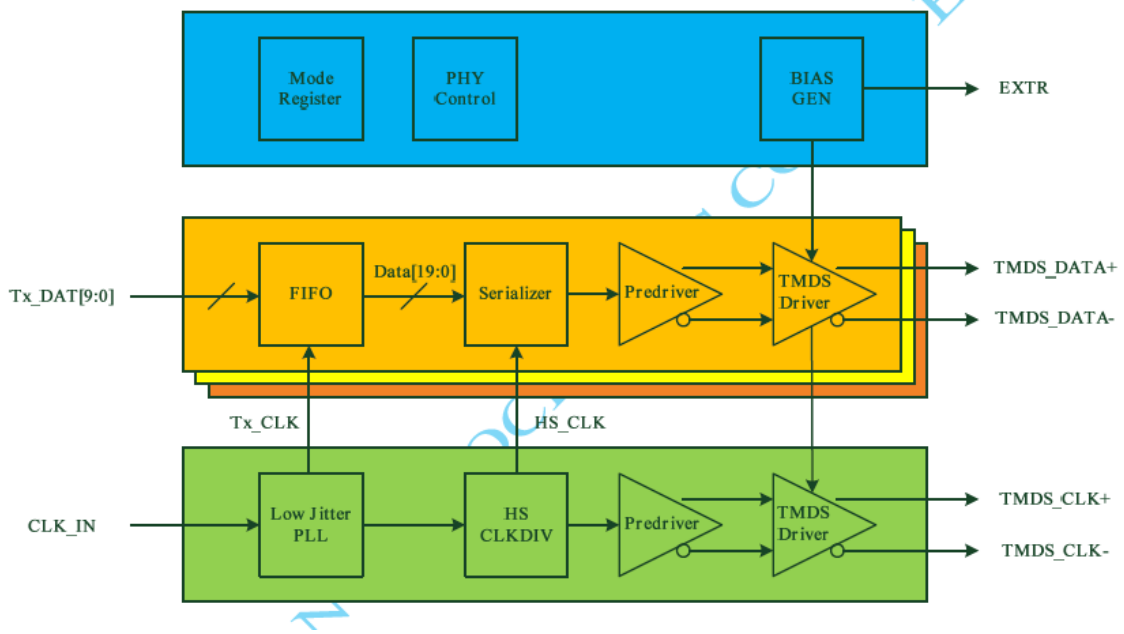

Fig. 42-1 HDMI2.0 TX PHY Block Diagram

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17.3 PLL Description

There are two PLLs in TX PHY: PRE-PLL and POST-PLL.
The PRE-PLL is used to generate tmds clk, tmds character clk, pclk and prepclk. The POST-PLL is only used to generate differential serial clock for drivers which differential frequency is always 5 times of the pin_hd20_tmdsclk.
You should make sure the VCO frequncy at range 750Mhz~3Ghz for both PRE-PLL and POST-PLL.

17.4 Power Control Timing

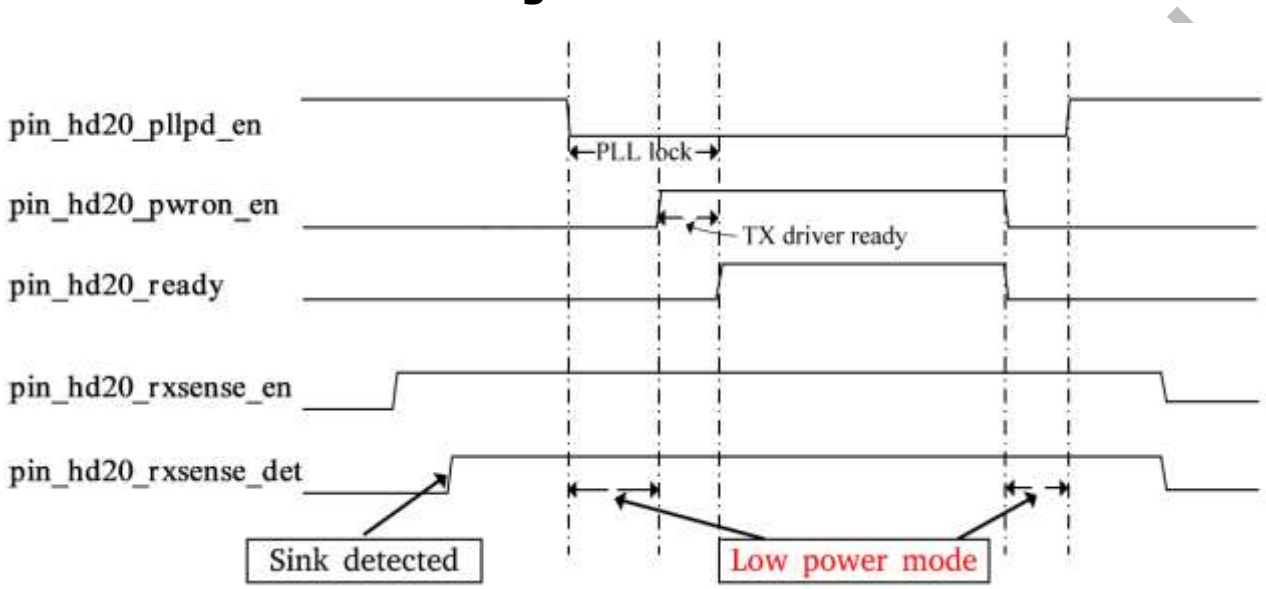

Fig. 42-2 Power relative signals timing

Pin_hd20_pwron_en is the HDMI PHY tx driver's power enable and analog reset signal, which is configured from grf. After pll is power up and ready, this signal must be set from 0 to 1 to reset HDMI PHY tx analog driver. pin_hd20_pllpd_en is PLL power down signal, the PLL is powered down when this pin set high. It is recommended to power down the driver and pll before apb bus configuration, and do not pull the pin_hd20_pwron_en high until pin_hd20_ready rise.

17.5 PHY register description

The base address of the HDMI TX is 0XFF43_0000, please not change the reseaverd bits of PHY registers.
The offset of the table of Register Summary **must multiple with 4** when software configures it.

HDMI_PHY_general_reg
Address: Operational Base + offset (0x0000)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	analog_rst analog resetn 0: reset
6	RW	0x1	dig_rst soft digital reset_n 0: reset
5:4	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
3	RW	0x0	pclk_inv_en pclk invert enable 1: invert 0: not invert
2	RW	0x0	prepclk_inv_en prepclk invert enable 1: invert 0: not invert
1	RW	0x0	tmdsclk_inv_en tmdsclk invert enable 1: invert 0: not invert
0	RW	0x0	prepll_ref_sel pre-pll reference clock select 0: select synchoronous pclk 1: sel crystal oscillater clock

HDMI_PHY_bypass_reg
Address: Operational Base + offset (0x0004)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	bypass input pin_hd20_rxsense_en 1:bypass, use internal register control 0:not bypass, use pin_hd20_rxsense_en control
1	RW	0x0	bypass input pin_hd20_pwron_en 1:bypass, use internal register control 0:not bypass, use pin_hd20_pwron_en control
0	RW	0x0	bypass input pin_hd20_pllpd_en 1:bypass, use internal register control 0:not bypass, use pin_hd20_pllpd _en control

HDMI_PHY_interrupt_reg
Address: Operational Base + offset (0x0008)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	pin_hd20_int_out polarity control 1:interrupt out active high 0:interrupt out active low
6:5	RW	0x0	reserved
4	RW	0x0	bypass input pin_hd20_pdata_en 1:bypass, use internal register control 0:not bypass, use pin_hd20_pdata_en control
3:1	RW	0x0	reserved

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Bit	Attr	Reset Value	Description
0	RW	0x1	TMDS data channels synchronous enable 1:enable 0:disable

HDMI_PHY_interrupt_mask_reg
Address: Operational Base + offset (0x000c)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	Interrrupt mask for pre-PLL lock detection 1:not mask interrupt 0:mask interrupt
4	RW	0x0	Interrrupt mask for post-PLL lock detection 1:not mask interrupt 0:mask interrupt
3	R0	0x0	reserved
2	RW	0x0	Interrrupt mask for Sync Enable detection of TMDS data channel 2 1:not mask interrupt 0:mask interrupt
1	RW	0x0	Interrrupt mask for Sync Enable detection of TMDS data channel 1 1:not mask interrupt 0:mask interrupt
0	RW	0x0	Interrrupt mask for Sync Enable detection of TMDS data channel 0 1:not mask interrupt 0:mask interrupt

HDMI_PHY_interrupt_status_reg
Address: Operational Base + offset (0x0010)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5	RW	0x0	Interrrupt for pre-PLL lock detection 1:interrupt 0:no interrupt
4	RW	0x0	Interrrupt for post-PLL lock detection 1: interrupt 0:no interrupt
3	R0	0x0	reserved
2	RW	0x0	Interrrupt for Sync Enable detection of TMDS data channel 2 1: interrupt 0:no interrupt
1	RW	0x0	Interrrupt for Sync Enable detection of TMDS data channel 1 1: interrupt 0:no interrupt

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Bit	Attr	Reset Value	Description
0	RW	0x0	Interrrupt for Sync Enable detection of TMDS data channel 0 1: interrupt 0:no interrupt

HDMI_PHY_interrupt_esd_mask1_reg
Address: Operational Base + offset (0x0014)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	Interrrupt mask for ESD detection of TMDS clock channel 1:not mask interrupt 0:mask interrupt
3:0	RW	0x0	Interrrupt mask for ESD detection of TMDS data channel 2 1:not mask interrupt 0:mask interrupt

HDMI_PHY_interrupt_esd_status1_reg
Address: Operational Base + offset (0x0018)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	Interrrupt for ESD detection of TMDS clock channel 1: interrupt 0:no interrupt
3:0	RW	0x0	Interrrupt for ESD detection of TMDS data channel 2 1: interrupt 0:no interrupt

HDMI_PHY_interrupt_esd_mask2_reg
Address: Operational Base + offset (0x001c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	Interrrupt mask for ESD detection of TMDS data channel 1 1:not mask interrupt 0:mask interrupt
3:0	RW	0x0	Interrrupt mask for ESD detection of TMDS data channel 0 1:not mask interrupt 0:mask interrupt

HDMI_PHY_interrupt_esd_status2_reg
Address: Operational Base + offset (0x0020)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:4	RW	0x0	Interrrupt for ESD detection of TMDS dada channel 1 1: interrupt 0:no interrupt
3:0	RW	0x0	Interrrupt for ESD detection of TMDS data channel 0 1: interrupt 0:no interrupt

HDMI_PHY_prepll_ctrl_reg
Address: Operational Base + offset (0x0280)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	Pre-PLL bypass enable, if bypass Pre-PLL, the pin_hd20_pclk, pin_hd20_prepclk and pin_hd20_tmdsclk will directly out come from the input reference clock 1:bypass Pre-PLL 0:not bypass Pre-PLL
1	RW	0x0	Enable of pin_hd20_pclk directly divided by 5 from VCO of the pre-PLL 1:enable 0:disable
0	RW	0x0	Pre-PLL power on internal register control 1:power down 0:power on Note, only effective when bypass input pin_hd20_pwron_en

HDMI_PHY_prepll_divider_reg
Address: Operational Base + offset (0x0284)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:0	RW	0x1	Pre-PLL pre divider[5:0]

HDMI_PHY_spread_ctrl_reg
Address: Operational Base + offset (0x0288)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	Selects center spread or down spread 0:Center Spread 1:Down Spread Note, only effective when Spread Spectrum Modulation enabled
6	RW	0x1	Spread Spectrum Modulation enable 1'b0:enable 1'b1:disable

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Bit	Attr	Reset Value	Description
5:4	RW	0x3	Fractional divider enable 2'b00:enable 2'b11:disable others:invalid Note, enable Spread Spectrum Modulation should enable the fractional divider togerther.
3:0	RW	0x0	Pre-PLL feedback divider bit[11:8]

HDMI_PHY_PrePLL_fb_reg
Address: Operational Base + offset (0x028c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x63	Pre-PLL feedback divider bit[7:0]

HDMI_PHY_PrePLL_div_reg
Address: Operational Base + offset (0x0290)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x1	Pre-PLL tmds-dividera 00:divide by 1 01:divide by 2 10:divide by 3 11:divide by 5
3:2	RW	0x1	Pre-PLL tmds-dividerb 00:divide by 1 01:divide by 2 10:divide by 4 11:divide by 8
1:0	RW	0x1	Pre-PLL tmds-dividerc 00:divide by 1 01:divide by 2 10:divide by 4 11:divide by 8

HDMI_PHY_PrePLL_pclkdiv_reg
Address: Operational Base + offset (0x0294)

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
6:5	RW	0x2	Pre-PLL pclk-dividerb 00:divide by 2 01:divide by 3 10:divide by 4 11:divide by 5
4:0	RW	0x1	Pre-PLL pclk-dividera Range from 1 to 31(decimal)

HDMI_PHY_PrePLL_pclkdiv2_reg
Address: Operational Base + offset (0x0298)

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:5	RW	0x2	Pre-PLL pclk-dividerc 00:divide by 1 01:divide by 2 10:divide by 4 11:divide by 8
4:0	RW	0x2	Pre-PLL pclk-dividerd Range from 1 to 31(decimal)

HDMI_PHY_PrePLL_bw1_reg
Address: Operational Base + offset (0x029c)

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x3	Pre-PLL band-width control register 1
3	RO	0x0	reserved
2:0	RW	0x2	Pre-PLL band-width control register 2

HDMI_PHY_PrePLL_bw2_reg
Address: Operational Base + offset (0x02a4)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x1	Pre-PLL band-width control register 3
3:1	RO	0x0	reserved
0	RW	0x0	Pre-PLL lock status 1:locked 2:unlock

HDMI_PHY_PostPLL_divenable_reg
Address: Operational Base + offset (0x02a8)

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3:2	RW	0x3	Post-PLL post divider enable 2'b11:enable 2'b00:disable others:invalid
1	RW	0x1	Post-PLL reference clock selection 1'b1:select tmds_clk as the reference clock 1'b0:select oscillator clock as the reference clock
0	RW	0x0	reserved

HDMI_PHY_PostPLL_fb1_reg
Address: Operational Base + offset (0x02ac)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x3	Post-PLL feedback divider bit[8]
5	RO	0x0	reverved
4:0	RW	0x4	Post-PLL pre-divider

HDMI_PHY_PostPLL_fb2_reg
Address: Operational Base + offset (0x02b0)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x50	Post-PLL feedback divider bit[7:00]

HDMI_PHY_PostPLL_fb2_reg
Address: Operational Base + offset (0x02b4)

Bit	Attr	Reset Value	Description
31:2	RO	0x0	reserved
1:0	RW	0x1	Post-PLL post divider 2'bx0:divide by 2 2'b01:divide by 4 2'b11:divide by 8 Note, post divider only effective when post-PLL post divider enable set.

HDMI_PHY_PostPLL_bw1_reg
Address: Operational Base + offset (0x02b8)

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x3	Post-PLL band-width control register 1

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Bit	Attr	Reset Value	Description
3	RO	0x0	reserved
1:0	RW	0x1	Post-PLL band-width control register 2

HDMI_PHY_PostPLL_bw2_reg
Address: Operational Base + offset (0x02bc)

Bit	Attr	Reset Value	Description
31:1	RO	0x0	reserved
1	RW	0x0	Post-PLL lock status 1:locked 0:unlock

HDMI_PHY_bg_enable_reg
Address: Operational Base + offset (0x02c0)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x1	Band Gap internal enable 1:enable 0:disable Note, only effective when bypass input pin_hd20_pwron_en
1:0	RW	0x0	Band Gap reference circuit control

HDMI_PHY_TMDS_Data_enable_reg
Address: Operational Base + offset (0x02c8)

Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x1	Driver internal enable of TMDS clock channel 1:enable 0:disable Note, only effective when bypass input pin_hd20_pwron_en
2	RW	0x1	Driver internal enable of TMDS data channel 2 1:enable 0:disable Note, only effective when bypass input pin_hd20_pwron_en
1	RW	0x1	Driver internal enable of TMDS data channel 1 1:enable 0:disable Note, only effective when bypass input pin_hd20_pwron_en
0	RW	0x1	Driver internal enable of TMDS data channel 0 1:enable 0:disable Note, only effective when bypass input pin_hd20_pwron_en

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HDMI_PHY_TMDS_Data_phase_sel_ch2_reg
Address: Operational Base + offset (0x02cc)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2:0	RW	0x2	Internal phase selection of TMDS data channel 2

HDMI_PHY_TMDS_Data_phase_sel_ch10_reg
Address: Operational Base + offset (0x02d0)

Bit	Attr	Reset Value	Description
31:7	RO	0x0	reserved
6:4	RW	0x2	Internal phase selection of TMDS data channel 1
3	RO	0x0	reserved
2:0	RW	0x2	Internal phase selection of TMDS data channel 0

HDMI_PHY_swing_clk_reg
Address: Operational Base + offset (0x02d4)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x5	output swing control for TMDS clock channel 5'b00000:minimal swing … 5'b11111:maximal swing

HDMI_PHY_swing_datach2_reg
Address: Operational Base + offset (0x02d8)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x8	output swing control for TMDS data channel 2 5'b00000:minimal swing … 5'b11111:maximal swing

HDMI_PHY_swing_datach1_reg
Address: Operational Base + offset (0x02dc)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x8	output swing control for TMDS data channel 1 5'b00000:minimal swing … 5'b11111:maximal swing

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HDMI_PHY_swing_datach0_reg
Address: Operational Base + offset (0x02e0)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x8	output swing control for TMDS data channel 0 5'b00000:minimal swing … 5'b11111:maximal swing

HDMI_PHY_pre_em_mode_ctrl_reg
Address: Operational Base + offset (0x02e4)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	Pre-emphasis mode control for TMDS data channel 2 0:select full-bit pre-emphasis mode 1:select half-bit pre-emphasis mode
1	RW	0x0	Pre-emphasis mode control for TMDS data channel 1 0:select full-bit pre-emphasis mode 1:select half-bit pre-emphasis mode
0	RW	0x0	Pre-emphasis mode control for TMDS data channel 0 0:select full-bit pre-emphasis mode 1:select half-bit pre-emphasis mode

HDMI_PHY_pre_em_enable_ctrl_reg
Address: Operational Base + offset (0x02e8)

Bit	Attr	Reset Value	Description
31:6	RO	0x0	reserved
5:4	RW	0x0	Pre-emphasis enable signal for TMDS data channel 2 00:turn off pre-emphasis 01 or 10:select pre-emphasis level 1 11: select pre-emphasis level 2
3:2	RW	0x0	Pre-emphasis enable signal for TMDS data channel 1 00:turn off pre-emphasis 01 or 10:select pre-emphasis level 1 11: select pre-emphasis level 2
0	RW	0x0	Pre-emphasis enable signal for TMDS data channel 0 00:turn off pre-emphasis 01 or 10:select pre-emphasis level 1 11: select pre-emphasis level 2

HDMI_PHY_pre_em_amplitude_ch2_reg
Address: Operational Base + offset (0x02ec)

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Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x8	Pre-emphasis amplitude control for TMD data channel 2 5'b00000:select minimum pre-emphasis amplitude … 5'b11111:select maximum pre-emphasis amplitude

HDMI_PHY_pre_em_amplitude_ch1_reg
Address: Operational Base + offset (0x02f0)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x8	Pre-emphasis amplitude control for TMD data channel 1 5'b00000:select minimum pre-emphasis amplitude … 5'b11111:select maximum pre-emphasis amplitude

HDMI_PHY_pre_em_amplitude_ch0_reg
Address: Operational Base + offset (0x02f4)

Bit	Attr	Reset Value	Description
31:5	RO	0x0	reserved
4:0	RW	0x8	Pre-emphasis amplitude control for TMD data channel 0 5'b00000:select minimum pre-emphasis amplitude … 5'b11111:select maximum pre-emphasis amplitude

HDMI_PHY_delay_enable _reg
Address: Operational Base + offset (0x02f8)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	Delay path enable signal of the TMDS clock channel [1] main driver path enable 1:enable 0:disable [0]pre-emphasis driver path enable 1:enable 0:disable

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Bit	Attr	Reset Value	Description
5:4	RW	0x0	Delay path enable signal of the TMDS data channel 2 [1] main driver path enable 1:enable 0:disable [0]pre-emphasis driver path enable 1:enable 0:disable
3:2	RW	0x0	Delay path enable signal of the TMDS data channel 1 [1] main driver path enable 1:enable 0:disable [0]pre-emphasis driver path enable 1:enable 0:disable
1:0	RW	0x0	Delay path enable signal of the TMDS data channel 0 [1] main driver path enable 1:enable 0:disable [0]pre-emphasis driver path enable 1:enable 0:disable

HDMI_PHY_delay_modectrl1_reg
Address: Operational Base + offset (0x02fc)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xa	Delay mode control for the TMDS clock channel [3] first delay path of main driver control 1:select fast signal path 0:select slow signal path [2]second delay path of main driver 1:select fast signal path 0:select slow signal path [1] fist delay path of pre-emphasis driver 1:select fast signal path 0:select slow signal path [0] second delay path of pre-emphasis driver 1:selcet fast signal path 0:select slow signal path

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Bit	Attr	Reset Value	Description
3:0	RW	0xc	Delay mode control for the TMDS data channel 2 [3] first delay path of main driver control 1:select fast signal path 0:select slow signal path [2]second delay path of main driver 1:select fast signal path 0:select slow signal path [1] fist delay path of pre-emphasis driver 1:select fast signal path 0:select slow signal path [0] second delay path of pre-emphasis driver 1:selcet fast signal path 0:select slow signal path

HDMI_PHY_delay_modectrl2_reg
Address: Operational Base + offset (0x0300)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xc	Delay mode control for the TMDS data channel 1 [3] first delay path of main driver control 1:select fast signal path 0:select slow signal path [2]second delay path of main driver 1:select fast signal path 0:select slow signal path [1] fist delay path of pre-emphasis driver 1:select fast signal path 0:select slow signal path [0] second delay path of pre-emphasis driver 1:selcet fast signal path 0:select slow signal path
3:0	RW	0xc	Delay mode control for the TMDS data channel 0 [3] first delay path of main driver control 1:select fast signal path 0:select slow signal path [2]second delay path of main driver 1:select fast signal path 0:select slow signal path [1] fist delay path of pre-emphasis driver 1:select fast signal path 0:select slow signal path [0] second delay path of pre-emphasis driver 1:selcet fast signal path 0:select slow signal path

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HDMI_PHY_delay_modectrl2_reg
Address: Operational Base + offset (0x0300)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xc	Delay mode control for the TMDS data channel 1 [3] first delay path of main driver control 1:select fast signal path 0:select slow signal path [2]second delay path of main driver 1:select fast signal path 0:select slow signal path [1] fist delay path of pre-emphasis driver 1:select fast signal path 0:select slow signal path [0] second delay path of pre-emphasis driver 1:selcet fast signal path 0:select slow signal path
3:0	RW	0xc	Delay mode control for the TMDS data channel 0 [3] first delay path of main driver control 1:select fast signal path 0:select slow signal path [2]second delay path of main driver 1:select fast signal path 0:select slow signal path [1] fist delay path of pre-emphasis driver 1:select fast signal path 0:select slow signal path [0] second delay path of pre-emphasis driver 1:selcet fast signal path 0:select slow signal path

HDMI_PHY_delay_time_ctrl1_reg
Address: Operational Base + offset (0x0304)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xc	Delay time control for slow signal path in TMDS clock channel 4’b0000:select minimum delay time … 4’b1111:select maximum delay time
3:0	RW	0xc	Delay time control for slow signal path in TMDS data channel 2 4’b0000:select minimum delay time … 4’b1111:select maximum delay time

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HDMI_PHY_delay_time_ctrl2_reg
Address: Operational Base + offset (0x0308)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0xc	Delay time control for slow signal path in TMDS data channel 1 4'b0000:select minimum delay time … 4'b1111:select maximum delay time
3:0	RW	0xc	Delay time control for slow signal path in TMDS data channel 0 4'b0000:select minimum delay time … 4'b1111:select maximum delay time

HDMI_PHY_ termination_res_speed
Address: Operational Base + offset (0x0314)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	Bypass auto termination resistor calibration 1: bypass 0: not bypass
6:0	RW	0x1	Auto termination resistor calibration speed control. Freq(configure control) = Freq(sys_clk)/{reg0x0314[6:0], reg0x03180[7:0]}. Typically 100KHz. Assume sys_clk is 100MHz, then this value should be set to 100MHz/100KHz = 1000d. Note: this value only effective when not bypass auto termination resistor calibaration.

HDMI_PHY_ termination_res_speed2
Address: Operational Base + offset (0x0318)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xe	Auto termination resistor calibration speed control. Freq(configure control) = Freq(sys_clk)/{reg0x0314[6:0], reg0x03180[7:0]}. Typically 100KHz. Assume sys_clk is 100MHz, then this value should be set to 100MHz/100KHz = 1000d. Note: this value only effective when not bypass auto termination resistor calibaration.

HDMI_PHY_ termination_cal
Address: Operational Base + offset (0x031c)

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
2:1	RW	0x0	Auto termination resistor calibration setting for all TMDS channels 00:50 Ohm 01:62.5 Ohm 10:75 Ohm 11:100 Ohm
0	RO	0x0	reserved

HDMI_PHY_ESD_clkch_reg
Address: Operational Base + offset (0x0320)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x3	ESM set value of the TMDS clock channel
5:0	RW	0x0	Termination resistor setting for driver of TMDS clock channel.

HDMI_PHY_ESD_datach2_reg
Address: Operational Base + offset (0x0324)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x3	ESM set value of the TMDS data channel 2
5:0	RW	0x0	Termination resistor setting for driver of TMDS data channel 2.

HDMI_PHY_ESD_datach1_reg
Address: Operational Base + offset (0x0328)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x3	ESM set value of the TMDS data channel 1
5:0	RW	0x0	Termination resistor setting for driver of TMDS data channel 1.

HDMI_PHY_ESD_datach0_reg
Address: Operational Base + offset (0x032c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x3	ESM set value of the TMDS data channel 0
5:0	RW	0x0	Termination resistor setting for driver of TMDS data channel 0.

HDMI_PHY_rxsense_enable_reg
Address: Operational Base + offset (0x0330)

Bit	Attr	Reset Value	Description

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Bit	Attr	Reset Value	Description
31:4	RO	0x0	reserved
3	RW	0x1	RX_SENSE detect internal enable of TMDS clock channel 1:enable 0:disable Note, only effective when bypass input pin_hd20_rxsense_en
2	RW	0x1	RX_SENSE detect internal enable of TMDS data channel 2 1:enable 0:disable Note, only effective when bypass input pin_hd20_rxsense_en
1	RW	0x1	RX_SENSE detect internal enable of TMDS data channel 1 1:enable 0:disable Note, only effective when bypass input pin_hd20_rxsense_en
0	RW	0x1	RX_SENSE detect internal enable of TMDS data channel 0 1:enable 0:disable Note, only effective when bypass input pin_hd20_rxsense_en

HDMI_PHY_rxsense_detect_reg
Address: Operational Base + offset (0x0334)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:6	RW	0x0	RX_SENSE detected result of TMDS clock channel 1:rxsense detected 0:rxsense un-detected
5:4	RW	0x0	RX_SENSE detected result of TMDS data channel 2 1:rxsense detected 0:rxsense un-detected
3:2	RW	0x0	RX_SENSE detected result of TMDS data channel 1 1:rxsense detected 0:rxsense un-detected
1:0	RW	0x0	RX_SENSE detected result of TMDS data channel 0 1:rxsense detected 0:rxsense un-detected

HDMI_PHY_rxsense_detect_status1_reg
Address: Operational Base + offset (0x0338)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	RX_SENSE detected status of TMDS clock channel 1:normal status 0:ESD error status

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Bit	Attr	Reset Value	Description
3:0	RW	0x0	RX_SENSE detected status of TMDS data channel 2 1:normal status 0:ESD error status

HDMI_PHY_rxsense_detect_status2_reg
Address: Operational Base + offset (0x033c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x0	RX_SENSE detected status of TMDS data channel 1 1:normal status 0:ESD error status
3:0	RW	0x0	RX_SENSE detected status of TMDS data channel 0 1:normal status 0:ESD error status

HDMI_PHY_spread_spectrum_reg
Address: Operational Base + offset (0x0340)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x0	Selects external wave table for Spread Spectrum Modulation 1:select external wave table 0:select internal wave table Note, only effective when spread spectrum modulation enabled
6:4	RW	0x4	Sets modulation depth(Spread%) 3'b000 = 0 3'b001 = 0.049% 3'b010 = 0.088% 3'b011 = 0.195% 3'b100 = 0.391% 3'b101 = 0.781% 3'b110 = 0.563% 3'b110 = 3.125% Note, only effective when Spread Spectrum Modulation enabled.
3:0	RW	0x6	Divider for the modulation frequency

HDMI_PHY_PLL_fractional1_reg
Address: Operational Base + offset (0x0344)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x0	{reg(0x344)[7:0], reg(0x348)[7:0], reg(0x34c)[7:0]} is the Pre-PLL fractional feedback divider value. Note, if enabled fractional divider or Spread Spectrum Modulation, the Pre-PLL total feedback divider equals this value divide by 2^24 and add the integer feedback value.

HDMI_PHY_PLL_fractional2_reg
Address: Operational Base + offset (0x0348)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x0	{reg(0x344)[7:0], reg(0x348)[7:0], reg(0x34c)[7:0]} is the Pre-PLL fractional feedback divider value. Note, if enabled fractional divider or Spread Spectrum Modulation, the Pre-PLL total feedback divider equals this value divide by 2^24 and add the integer feedback value.

HDMI_PHY_PLL_fractional3_reg
Address: Operational Base + offset (0x034c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x0	{reg(0x344)[7:0], reg(0x348)[7:0], reg(0x34c)[7:0]} is the Pre-PLL fractional feedback divider value. Note, if enabled fractional divider or Spread Spectrum Modulation, the Pre-PLL total feedback divider equals this value divide by 2^24 and add the integer feedback value.

HDMI_PHY_external_wave_table_depth_reg
Address: Operational Base + offset (0x0350)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x7f	External wave table depth, range 0~127 Every time before write/read external wave table, it should write/read this address to clear the base address for internal wave values. Note, only effective when Spread Spectrum Modulation enabled and selected external wave table.

HDMI_PHY_external_wave_values_reg
Address: Operational Base + offset (0x0354)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved

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Bit	Attr	Reset Value	Description
7:0	RW	0x0	External wave values, the offset address auto increased every 1 write/read time up to the external wave table depth.

HDMI_PHY_BandGapDiv_reg
Address: Operational Base + offset (0x0358)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7	RW	0x1	Band-Gap clock divider enable 1:enable 0:disable
6:0	RW	0x0	Band-Gap clock divider value

HDMI_PHY_BandGap_reg
Address: Operational Base + offset (0x035c)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x2	Note: Band-Gap clock is divided from reference clock, default frequency is 12MHz.

HDMI_PHY_refclkcounterlen1_reg
Address: Operational Base + offset (0x0360)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x1	Reference clock domain counter length for sync enable detection

HDMI_PHY_refclkcounterlen2_reg
Address: Operational Base + offset (0x0364)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x40	Reference clock domain counter length for sync enable detection

HDMI_PHY_syncenableclkcounterlen1_reg
Address: Operational Base + offset (0x0368)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0x1	Sync enable clock domain counter length for sync enable detection.

HDMI_PHY_syncenableclkcounterlen2_reg
Address: Operational Base + offset (0x036c)

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Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:0	RW	0xef	Sync enable clock domain counter length for sync enable detection.

HDMI_PHY_clkcountergap_reg
Address: Operational Base + offset (0x0370)

Bit	Attr	Reset Value	Description
31:8	RO	0x0	reserved
7:4	RW	0x8	Reference clock domain counter gap for sync enable detection.
3:0	RW	0x2	Sync enable clock domain error range for sync enable detection.

HDMI_PHY_syncenablestatus_reg
Address: Operational Base + offset (0x0374)

Bit	Attr	Reset Value	Description
31:3	RO	0x0	reserved
2	RW	0x0	Sync Enable Status of TMDS data channel 2. 1:locked 0:un-locked
1	RW	0x0	Sync Enable Status of TMDS data channel 1. 1:locked 0:un-locked
0	RW	0x0	Sync Enable Status of TMDS data channel 0. 1:locked 0:un-locked

17.6 HDMI PHY PLL DESCRIPTION

17.6.1 PLL Requirement

The PLL of HDMI 2.0 TX PHY can use crystal oscillator clock (recommended, 24MHz) or SOC pixel clock (optional) as the reference input clock. The requirements of these clocks are listed in the following table.

Table 42-1 PLL Reference Clock Requirement

Requirement Items	Value
Oscillator Frequency (when selected)	24Mhz
Frequency Tolerance	-300ppm ~ 300ppm
Duty Cycle	40% ~ 60%
Trise and T fall	< 200ps

In order to reduce the jitter of the PLL output clock, it's suggested that the clock path from the oscillator input IO pad to the pin of INNO HDMI 2.0 TX PHY and the power supply should be as short and clean as possible. If this clock tree is longer than 500us, it should insert buffer with lager driving ability and ensure the wire width and space appropriately. What's more, the power supply of the inserted buffers also need be clean.

17.6.2 PRE-PLL OUTPUT CLOCK FREQUENCY EQUATION

The VCO frequency of the pre-PLL($F_{pre\text{-}vco}$) is controlled by pre-pll-pre-divider[5:0] and pre-pll-feedback-divider[11:0] (and pre-pll-fractional-feedback-divider[23:0] if fractional PLL enabled)

$F_{pre\text{-}vco}$ = (F_{ref}/pre-pll-pre-divider[5:0]) * (pre-pll-feedback-divider[11:0] +

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pre-pll-fractional-feedback-divider[23:0]/2^24)

The TMDS clock channel frequency (F_{tx3}) is controllerd by tmds-dividera[1:0] and tmds-dividerb[1:0]

F_{tx3} = $F_{pre\text{-}vco}$/(4*tmds-dividera[1:0] * tmds-dividerb[1:0]),

where tmds-dividera[1:0] = 1,2,3,5;
tmds-dividerb[1:0] = 1,2,4,8

The tmdsclk frequency is controlled by tmds-dividera[1:0] and tmds-dividerc[1:0]

$F_{tmdsclk}$ = $F_{pre\text{-}vco}$/(4 * tmds-dividera[1:0] * tmds-dividerc[1:0])

where tmds-dividera[1:0] = 1,2,3,5;
tmds-dividerb[1:0] = 1,2,4,8

The frequency of the prepclk equation is

$F_{prepclk}$ = (pclk-dividera[4:0] == 1)?{$F_{pre\text{-}vco}$/(pclk-dividerb[1:0]*pclk-dividerc[1:0])}:
{$F_{pre\text{-}vco}$/(pclk-dividera[4:0]*pclk-dividerc[1:0])};

where pclk-dividera[4:0] = 1~31;
pclk-dividerb[4:0] = 2,3,4,5;
pclk-dividerc[4:0] = 1,2,4,8;

The frequency of the pclk if F_{prevco}/5 when enabling pclk directly divide by 5 from VCO set, otherwise pclk freq is calculated by

F_{pclk} = (pclk-dividera[4:0] == 1)? {$F_{pre\text{-}vco}$/(pclk-dividerb[1:0]*pclk-dividerd[4:0])*2}:
{$F_{pre\text{-}vco}$/(pclk-dividera[4:0]*pclk-dividerd[4:0])*2};

where pclk-dividera[4:0] = 1~31;
pclk-dividerb[1:0] = 2,3,4,5;
pclk-dividerc[4:0] = 1~31;

Table 42-2 Reference Pre-PLL Configuration for Main Typical Resolution

Resolutions	8-bit RGB444 480P@60Hz	8-bit RGB444 1080P@60Hz	8-bit RGB444 2160P@30Hz	10-bit YCbCr420 2160P@60Hz
pre-divider	6'd1	6'd1	6'd1	6'd4
feedback-divider	12'd90	12'd99	12'd99	12'd495
tmds-dividera	2'b11	2'b01	2'b01	2'b01
tmds-dividerb	2'b10	2'b01	2'b00	2'b10
tmds-dividerc	2'b10	2'b01	2'b00	2'b00
pclk-dividera	5'd10	5'd1	5'd1	5'd1
pclk-dividerb	2'b11	2'b10	2'b00	2'b11
pclk-dividerc	2'b11	2'b10	2'b10	2'b01
pclk-dividerd	5'd4	5'd2	5'd2	5'd2
enable divided from VCO	1'b0	1'b0	1'b0	1'b1

17.6.3 POST-PLL OUTPUT CLOCK FREQUENCY EQUATION

The post-pll just only used to generate differential serial clock for drivers which differential frequency ($F_{diff\text{-}sclk}$) is always 5 times of the tmdsclk.

Fpost-vco = ($F_{tmdsclk}$/post-pll-pre-divider[4:0])*post-pll-feedback-divider[8:0]

The $F_{diff\text{-}sclk}$ equal $F_{postvco}$ when the post-pll post divider enable is clear (2'b00), otherwise.

$F_{diff\text{-}sclk}$= $F_{post\text{-}vco}$/post-pll-post-divier[1:0]

where post-pll-post-divider = 2,4,8

17.6.4 SPREAD SPECTRUM MODULATION

The HDMI 2.0 TX PHY adds the spread spectrum modulation to the link tmds clock for reducing the EMI.

The feature is as below:

- Support down spread and center spread
- Support 3-bit programmable modulation depth

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- Modulation depth from 490-ppm to 31250-ppm
- Internal triangular modulation wave table and supports programmable external wave table
- Adjustable modulation frequency

Spread Spectrum Modulation used the external wave table steps as below:
The frequency of the reference clock is 24MHz, the default Pre-PLL pre-divider is 6'd1. Assume the external wave table depth is 128, the required modulation is about 32KHz. So the divider for the Spread Spectrum Modulation is

$$24MHz / 6'd1 / 128 / 30KHz = 6.25 \approx 6$$

The step is as below:

- Select external wave table and configure the divider for modulation frequency to 4'd6
  Write 1'b1 to regd0[7] and write 4'd6 to regd0[3:0]
- Configure the external wave table depth to (128-1) and clear the base address.
  Write 8'h7f to regd4
- Write the 128 external wave values
  For(i=0;i<128;i++){ Write external wave value[i] to regd5;}
- Enable Spread Spectrum Modulation
  Write 3'b000 to rega2[6:4]

Note, every once before write/read the external wave table, it should write/read redd4 first to clear the base address, the offset address could auto increase after each write/read regd5.

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